

? Water makes life possible: The cells of your body could not function without water in which to dissolve essential biological molecules. Water is such a good solvent because its molecules (i) have zero net charge; (ii) have zero net charge, but the positive and negative charges are separated; (iii) have nonzero net charge; (iv) do not respond to electric forces; (v) exert repulsive electric forces on each other.



21 Electric Charge and Electric Field

LEARNING OUTCOMES

In this chapter, you'll learn...

- 21.1 The nature of electric charge, and how we know that electric charge is conserved.
- 21.2 How objects become electrically charged.
- 21.3 How to use Coulomb's law to calculate the electric force between charges.
- 21.4 The distinction between electric force and electric field.
- 21.5 How to calculate the electric field due to a collection of charges.
- 21.6 How to use the idea of electric field lines to visualize and interpret electric fields.
- 21.7 How to calculate the properties of electric dipoles.

You'll need to review...

- 1.7–1.10 Vector algebra, including the scalar (dot) product and the vector (cross) product.
- 4.3 Newton's second law.
- 7.5 Stable and unstable equilibria.
- 12.5 Streamlines in fluid flow.

In Chapter 5 we mentioned the four kinds of fundamental forces. To this point the only one of these forces that we have examined in any detail is gravity. Now we are ready to examine the force of *electromagnetism*, which encompasses both electricity and magnetism.

Electromagnetic interactions involve particles that have *electric charge*, an attribute that is as fundamental as mass. Just as objects with mass are accelerated by gravitational forces, so electrically charged objects are accelerated by electric forces. The shock you feel when you scuff your shoes across a carpet and then reach for a metal doorknob is due to charged particles leaping between your finger and the doorknob. Electric currents are simply streams of charged particles flowing within wires in response to electric forces. Even the forces that hold atoms together to form solid matter, and that keep the atoms of solid objects from passing through each other, are fundamentally due to electric interactions between the charged particles within atoms.

We begin our study of electromagnetism in this chapter by examining the nature of electric charge. We'll find that charge is quantized and obeys a conservation principle. When charges are at rest in our frame of reference, they exert *electrostatic* forces on each other. These forces are of tremendous importance in chemistry and biology and have many technological applications. Electrostatic forces are governed by a simple relationship known as *Coulomb's law* and are most conveniently described by using the concept of *electric field*. In later chapters we'll expand our discussion to include electric charges in motion. This will lead us to an understanding of magnetism and, remarkably, of the nature of light.

While the key ideas of electromagnetism are conceptually simple, applying them to practical problems will make use of many of your mathematical skills, especially your knowledge of geometry and integral calculus. For this reason you may find this chapter and those that follow to be more mathematically demanding than earlier chapters. The reward for your extra effort will be a deeper understanding of principles that are at the heart of modern physics and technology.

21.1 ELECTRIC CHARGE

The ancient Greeks discovered as early as 600 B.C. that after they rubbed amber with wool, the amber could attract other objects. Today we say that the amber has acquired a net **electric charge**, or has become *charged*. The word “electric” is derived from the Greek word *elektron*, meaning amber. When you scuff your shoes across a nylon carpet, you become electrically charged, and you can charge a comb by passing it through dry hair.

Plastic rods and fur (real or fake) are particularly good for demonstrating **electrostatics**, the interactions between electric charges that are at rest (or nearly so). After we charge both plastic rods in **Fig. 21.1a** by rubbing them with the piece of fur, we find that the rods repel each other.

When we rub glass rods with silk, the glass rods also become charged and repel each other (Fig. 21.1b). But a charged plastic rod *attracts* a charged glass rod; furthermore, the plastic rod and the fur attract each other, and the glass rod and the silk attract each other (Fig. 21.1c).

These experiments and many others like them have shown that there are exactly two kinds of electric charge: the kind on the plastic rod rubbed with fur and the kind on the glass rod rubbed with silk. Benjamin Franklin (1706–1790) suggested calling these two kinds of charge *negative* and *positive*, respectively, and these names are still used. The plastic rod and the silk have negative charge; the glass rod and the fur have positive charge.

Two positive charges or two negative charges repel each other. A positive charge and a negative charge attract each other.

CAUTION **Electric attraction and repulsion** The attraction and repulsion of two charged objects are sometimes summarized as “Like charges repel, and opposite charges attract.” But “like charges” does *not* mean that the two charges are exactly identical, only that both charges have the same algebraic *sign* (both positive or both negative). “Opposite charges” means that both objects have an electric charge, and those charges have different signs (one positive and the other negative). ▮

Figure 21.1 Experiments in electrostatics. (a) Negatively charged objects repel each other. (b) Positively charged objects repel each other. (c) Positively charged objects and negatively charged objects attract each other.

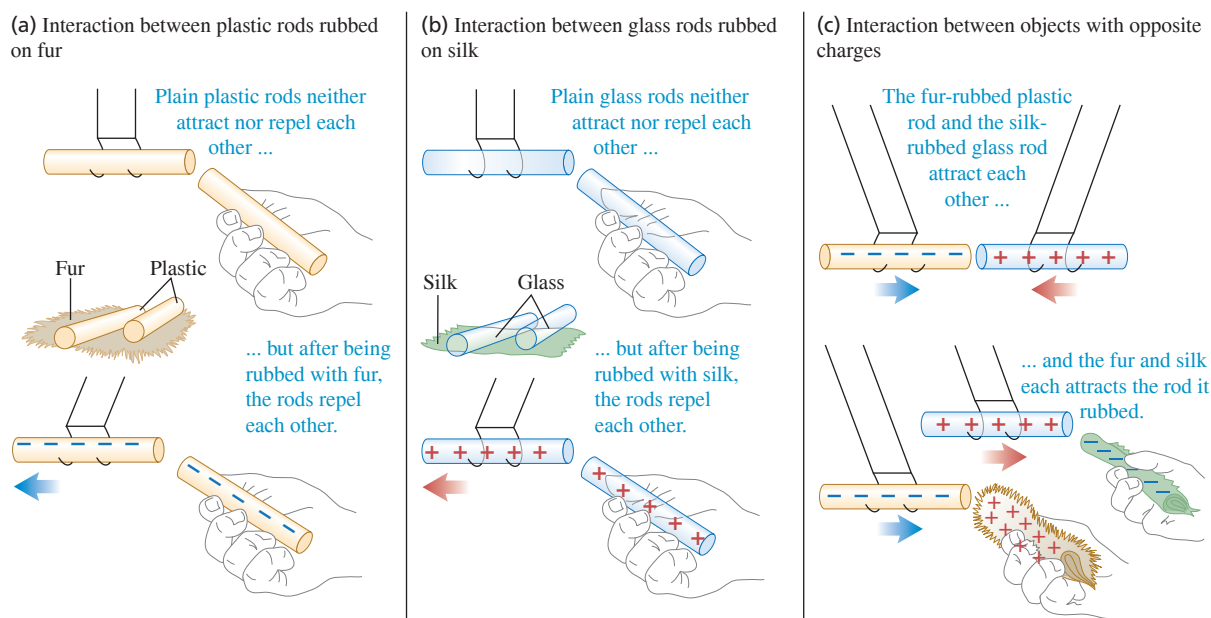
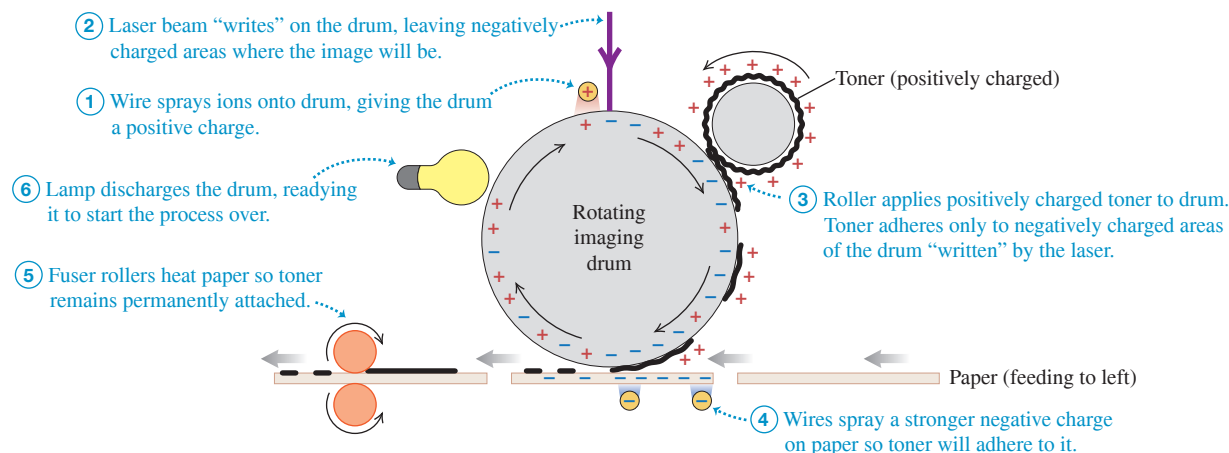


Figure 21.2 Schematic diagram of the operation of a laser printer.



A laser printer (**Fig. 21.2**) utilizes the forces between charged objects. The printer's light-sensitive imaging drum is given a positive charge. As the drum rotates, a laser beam shines on selected areas of the drum, leaving those areas with a *negative* charge. Positively charged particles of toner adhere only to the areas of the drum "written" by the laser. When a piece of paper is placed in contact with the drum, the toner particles stick to the paper and form an image.

Electric Charge and the Structure of Matter

When you charge a rod by rubbing it with fur or silk as in Fig. 21.1, there is no visible change in the appearance of the rod. What, then, actually happens to the rod when you charge it? To answer this question, we must look more closely at the structure of atoms, the building blocks of ordinary matter.

The structure of atoms can be described in terms of three particles: the negatively charged **electron**, the positively charged **proton**, and the uncharged **neutron** (**Fig. 21.3**). The proton and neutron are combinations of other entities called *quarks*, which have charges of $\pm\frac{1}{3}$ and $\pm\frac{2}{3}$ times the electron charge. Isolated quarks have not been observed, and there are theoretical reasons to believe that it is impossible in principle to observe a quark in isolation.

The protons and neutrons in an atom make up a small, very dense core called the **nucleus**, with dimensions of the order of 10^{-15} m. Surrounding the nucleus are the electrons, extending out to distances of the order of 10^{-10} m from the nucleus. If an atom were a few kilometers across, its nucleus would be the size of a tennis ball. The negatively charged electrons are held within the atom by the attractive electric forces exerted on them by the positively charged nucleus. (The protons and neutrons are held within stable atomic nuclei by an attractive interaction, called the *strong nuclear force*, that overcomes the electric repulsion of the protons. The strong nuclear force has a short range, and its effects do not extend far beyond the nucleus.)

The masses of the individual particles, to the precision that they are known (as of this writing), are

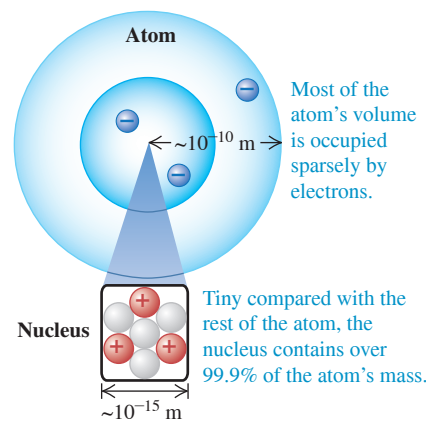
$$\text{Mass of electron} = m_e = 9.10938356(11) \times 10^{-31} \text{ kg}$$

$$\text{Mass of proton} = m_p = 1.672621898(21) \times 10^{-27} \text{ kg}$$

$$\text{Mass of neutron} = m_n = 1.674927471(21) \times 10^{-27} \text{ kg}$$

The numbers in parentheses are the uncertainties in the last two digits. Note that the masses of the proton and neutron are nearly equal and are roughly 2000 times the mass of the electron. Over 99.9% of the mass of any atom is concentrated in its nucleus.

Figure 21.3 The structure of an atom. The particular atom depicted here is lithium (see Fig. 21.4a).



Proton: Positive charge
Mass = 1.673×10^{-27} kg

Neutron: No charge
Mass = 1.675×10^{-27} kg

Electron: Negative charge
Mass = 9.109×10^{-31} kg

The charges of the electron and proton are equal in magnitude.

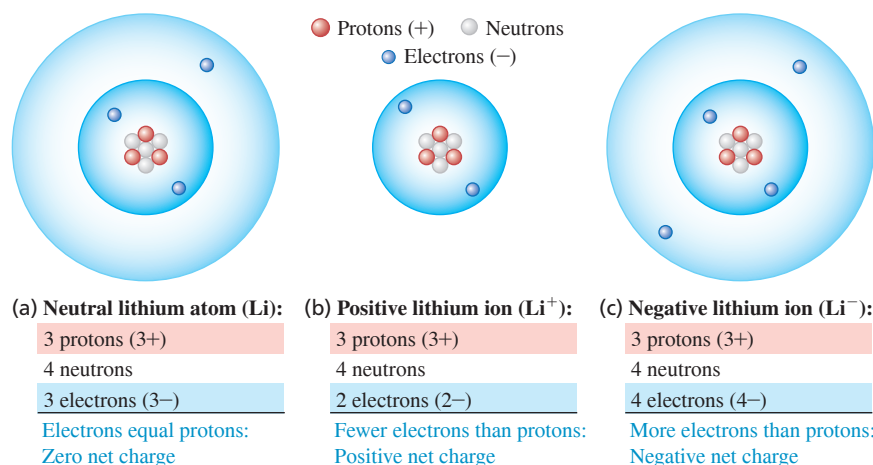


Figure 21.4 (a) A neutral atom has as many electrons as it does protons. (b) A positive ion has a deficit of electrons. (c) A negative ion has an excess of electrons. (The electron “shells” are a schematic representation of the actual electron distribution, a diffuse cloud many times larger than the nucleus.)

The negative charge of the electron has (within experimental error) *exactly* the same magnitude as the positive charge of the proton. In a neutral atom the number of electrons equals the number of protons in the nucleus, and the net electric charge (the algebraic sum of all the charges) is exactly zero (**Fig. 21.4a**). The number of protons or electrons in a neutral atom of an element is called the **atomic number** of the element. If one or more electrons are removed from an atom, what remains is called a **positive ion** (Fig. 21.4b). A **negative ion** is an atom that has *gained* one or more electrons (Fig. 21.4c). This gain or loss of electrons is called **ionization**.

When the total number of protons in a macroscopic object equals the total number of electrons, the total charge is zero and the object as a whole is electrically neutral. To give an object an excess negative charge, we may either *add negative* charges to a neutral object or *remove positive* charges from that object. Similarly, we can create an excess positive charge by either *adding positive* charge or *removing negative* charge. In most cases, negatively charged (and highly mobile) electrons are added or removed, and a “positively charged object” is one that has lost some of its normal complement of electrons. When we speak of the charge of an object, we always mean its *net* charge. The net charge is always a very small fraction (typically no more than 10^{-12}) of the total positive charge or negative charge in the object.

Electric Charge Is Conserved

Implicit in the foregoing discussion are two very important principles, the first of which is:

PRINCIPLE OF CONSERVATION OF CHARGE The algebraic sum of all the electric charges in any closed system is constant.

If we rub together a plastic rod and a piece of fur, both initially uncharged, the rod acquires a negative charge (since it takes electrons from the fur) and the fur acquires a positive charge of the *same* magnitude (since it has lost as many electrons as the rod has gained). Hence the total electric charge on the two objects together does not change. In any charging process, charge is not created or destroyed; it is merely *transferred* from one object to another.

Conservation of charge is thought to be a *universal* conservation law. No experimental evidence for any violation of this principle has ever been observed. Even in high-energy interactions in which particles are created and destroyed, such as the creation of electron–positron pairs, the total charge of any closed system is exactly constant.

Figure 21.5 Most of the forces on this water skier are electric. Electric interactions between adjacent molecules give rise to the force of the water on the ski, the tension in the tow rope, and the resistance of the air on the skier's body. Electric interactions also hold the atoms of the skier's body together. Only one wholly nonelectric force acts on the skier: the force of gravity.



The second important principle is:

The magnitude of charge of the electron or proton is a natural unit of charge.

Every observable amount of electric charge is always an integer multiple of this basic unit. We say that charge is *quantized*. A familiar example of quantization is money. When you pay cash for an item in a store, you have to do it in one-cent increments. Cash can't be divided into amounts smaller than one cent, and electric charge can't be divided into amounts smaller than the charge of one electron or proton. (The quark charges, $\pm\frac{1}{3}$ and $\pm\frac{2}{3}$ of the electron charge, are probably not observable as isolated charges.) Thus the charge on any macroscopic object is always zero or an integer multiple (negative or positive) of the electron charge.

Understanding the electric nature of matter gives us insight into many aspects of the physical world (**Fig. 21.5**). The chemical bonds that hold atoms together to form molecules are due to electric interactions between the atoms. They include the strong ionic bonds that hold sodium and chlorine atoms together to make table salt and the relatively weak bonds between the strands of DNA that record your body's genetic code. When you stand, the normal force exerted on you by the floor arises from electric forces between charged particles in the atoms of your shoes and the atoms of the floor. The tension force in a stretched string and the adhesive force of glue are likewise due to electric interactions of atoms.

TEST YOUR UNDERSTANDING OF SECTION 21.1 Two charged objects repel each other through the electric force. The charges on the objects are (i) one positive and one negative; (ii) both positive; (iii) both negative; (iv) either (ii) or (iii); (v) any of (i), (ii), or (iii).

ANSWER

(iv) Two charged objects repel if their charges are of the same sign (either both positive or both negative).

21.2 CONDUCTORS, INSULATORS, AND INDUCED CHARGES

Some materials permit electric charge to move easily from one region of the material to another, while others do not. For example, **Fig. 21.6a** shows a copper wire supported by a nylon thread. Suppose you touch one end of the wire to a charged plastic rod and attach the other end to a metal ball that is initially uncharged; you then remove the charged rod and the wire. When you bring another charged object up close to the ball (**Figs. 21.6b** and **21.6c**), the ball is attracted or repelled, showing that the ball has become electrically charged. Electric charge has been transferred through the copper wire between the ball and the surface of the plastic rod.

The copper wire is called a **conductor** of electricity. If you repeat the experiment using a rubber band or nylon thread in place of the wire, you find that *no* charge is transferred to the ball. These materials are called **insulators**. Conductors permit the easy movement of charge through them, while insulators do not. (The supporting nylon threads shown in **Fig. 21.6** are insulators, which prevents charge from leaving the metal ball and copper wire.)

As an example, carpet fibers on a dry day are good insulators. As you walk across a carpet, the rubbing of your shoes against the fibers causes charge to build up on you, and this charge remains on you because it can't flow through the insulating fibers. If you then touch a conducting object such as a doorknob, a rapid charge transfer takes place between your finger and the doorknob, and you feel a shock. One way to prevent this is to wind some of the carpet fibers around conducting cores so that any charge that builds up on you can be transferred harmlessly to the carpet. Another solution is to coat the carpet fibers with an antistatic layer that does not easily transfer electrons to or from your shoes; this prevents any charge from building up on you in the first place.

Most metals are good conductors, while most nonmetals are insulators. Within a solid metal such as copper, one or more outer electrons in each atom become detached and can move freely throughout the material, just as the molecules of a gas can move through the spaces between the grains in a bucket of sand. The other electrons remain bound to the positively charged nuclei, which themselves are bound in nearly fixed positions within the material. In an insulator there are no, or very few, free electrons, and electric charge cannot move freely through the material. Some materials called *semiconductors* are intermediate in their properties between good conductors and good insulators.

Charging by Induction

We can charge a metal ball by using a copper wire and an electrically charged plastic rod, as in Fig. 21.6a. In this process, some of the excess electrons on the rod are transferred from it to the ball, leaving the rod with a smaller negative charge. But there is a different technique in which the plastic rod can give another object a charge of *opposite* sign without losing any of its own charge. This process is called charging by **induction**.

Figure 21.7 shows an example of charging by induction. An uncharged metal ball is supported on an insulating stand (Fig. 21.7a). When you bring a negatively charged rod near it, without actually touching it (Fig. 21.7b), the free electrons in the metal ball are repelled by the excess electrons on the rod, and they shift toward the right, away from the rod. They cannot escape from the ball because the supporting stand and the surrounding air are insulators. So we get excess negative charge at the right surface of the ball and a deficiency of negative charge (that is, a net positive charge) at the left surface. These excess charges are called **induced charges**.

Not all of the free electrons move to the right surface of the ball. As soon as any induced charge develops, it exerts forces toward the *left* on the other free electrons. These electrons are repelled by the negative induced charge on the right and attracted toward the positive induced charge on the left. The system reaches an equilibrium state in which the force toward the right on an electron, due to the charged rod, is just balanced by the force toward the left due to the induced charge. If we remove the charged rod, the free electrons shift back to the left, and the original neutral condition is restored.

What happens if, while the plastic rod is nearby, you touch one end of a conducting wire to the right surface of the ball and the other end to the earth (Fig. 21.7c)? The earth is a conductor, and it is so large that it can act as a practically infinite source of extra electrons or sink of unwanted electrons. Some of the negative charge flows through the wire to the earth. Now suppose you disconnect the wire (Fig. 21.7d) and then remove the rod (Fig. 21.7e); a net positive charge is left on the ball. The charge on the negatively charged rod has not changed during this process. The earth acquires a negative charge that is equal in magnitude to the induced positive charge remaining on the ball.

Figure 21.6 Copper is a good conductor of electricity; nylon is a good insulator. (a) The copper wire conducts charge between the metal ball and the charged plastic rod to charge the ball negatively. Afterward, the metal ball is (b) repelled by a negatively charged plastic rod and (c) attracted to a positively charged glass rod.

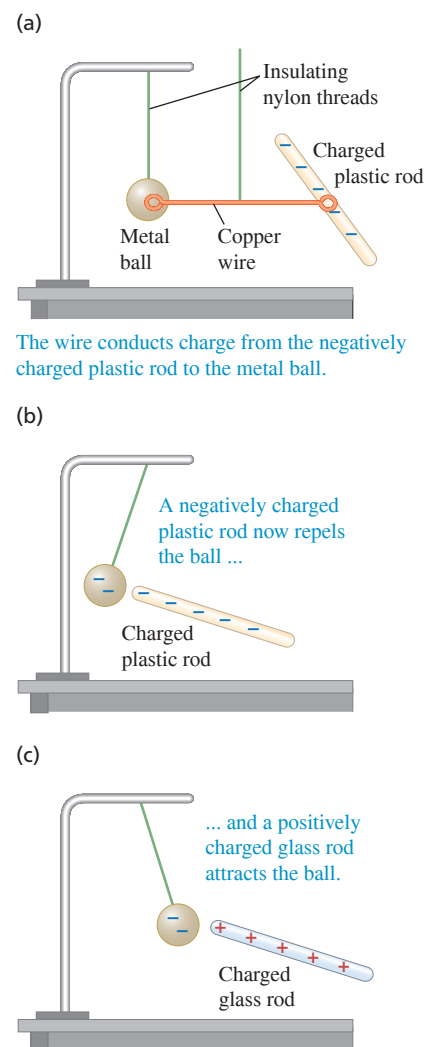


Figure 21.7 Charging a metal ball by induction.

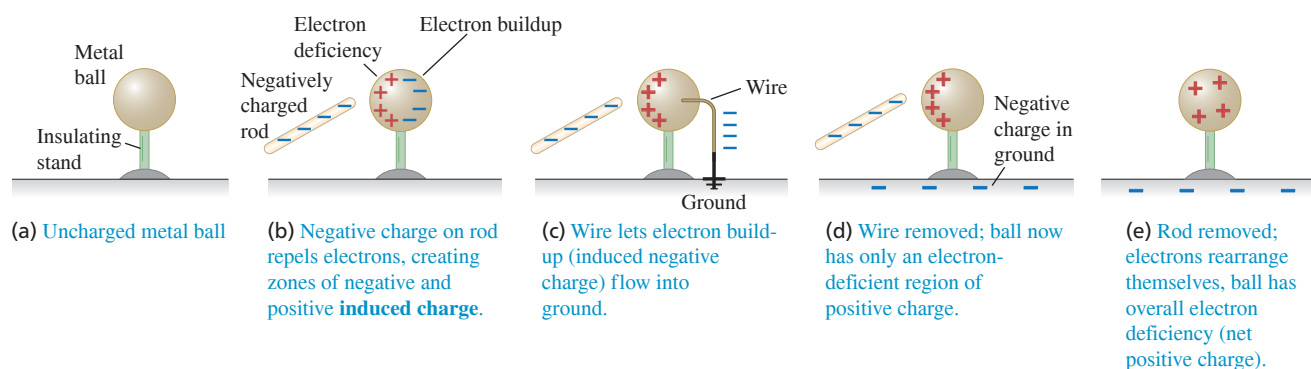
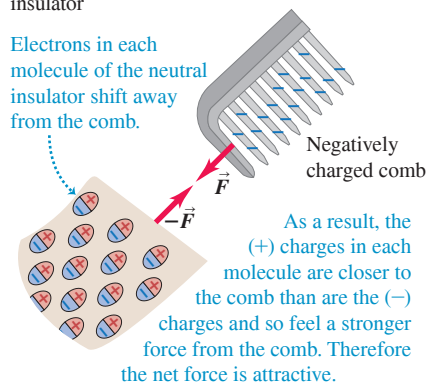


Figure 21.8 The charges within the molecules of an insulating material can shift slightly. As a result, a comb with either sign of charge attracts a neutral insulator. By Newton's third law the neutral insulator exerts an equal-magnitude attractive force on the comb.

(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator



(c) How a positively charged comb attracts an insulator

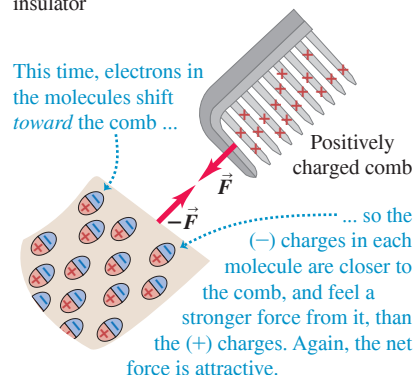
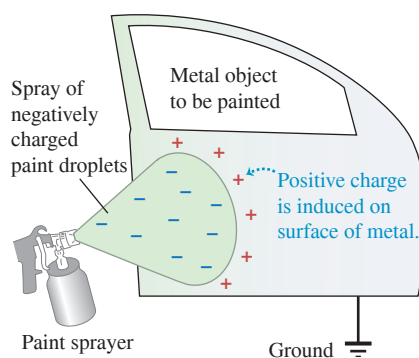


Figure 21.9 The electrostatic painting process (compare Figs. 21.7b and 21.7c). A metal object to be painted is connected to the earth ("ground"), and the paint droplets are given an electric charge as they exit the sprayer nozzle. Induced charges of the opposite sign appear in the object as the droplets approach, just as in Fig. 21.7b, and they attract the droplets to the surface. This process minimizes overspray from clouds of stray paint particles and gives a particularly smooth finish.



Electric Forces on Uncharged Objects

Finally, we note that a charged object can exert forces even on objects that are *not* charged themselves. If you rub a balloon on the rug and then hold the balloon against the ceiling, it sticks, even though the ceiling has no net electric charge. After you electrify a comb by running it through your hair, you can pick up uncharged bits of paper or plastic with it (Fig. 21.8a). How is this possible?

This interaction is an induced-charge effect. Even in an insulator, electric charges can shift back and forth a little when there is charge nearby. This is shown in Fig. 21.8b; the negatively charged plastic comb causes a slight shifting of charge within the molecules of the neutral insulator, an effect called *polarization*. The positive and negative charges in the material are present in equal amounts, but the positive charges are closer to the plastic comb and so feel an attraction that is stronger than the repulsion felt by the negative charges, giving a net attractive force. (In Section 21.3 we'll study how electric forces depend on distance.) Note that a neutral insulator is also attracted to a *positively* charged comb (Fig. 21.8c). Now the charges in the insulator shift in the opposite direction; the negative charges in the insulator are closer to the comb and feel an attractive force that is stronger than the repulsion felt by the positive charges in the insulator. Hence a charged object of *either* sign exerts an attractive force on an uncharged insulator. **Figure 21.9** shows an industrial application of this effect.

TEST YOUR UNDERSTANDING OF SECTION 21.2 You have two lightweight metal spheres, each hanging from an insulating nylon thread. One of the spheres has a net negative charge, while the other sphere has no net charge. (a) If the spheres are close together but do not touch, will they (i) attract each other, (ii) repel each other, or (iii) exert no force on each other? (b) You now allow the two spheres to touch. Once they have touched, will the two spheres (i) attract each other, (ii) repel each other, or (iii) exert no force on each other?

ANSWER

(a) (i) Before the two spheres touch, the negatively charged sphere exerts a repulsive force on the electrons in the other sphere, causing zones of positive and negative induced charge (see Fig. 21.7b). The positive zone is closer to the negatively charged sphere than the negative zone, so there is a net force of attraction that pulls the spheres together, like the comb and insulator in Fig. 21.8b. Once the two metal spheres touch, some of the excess electrons on the negatively charged sphere will flow onto the other sphere (because metals are conductors). Then both spheres will have a net negative charge and will repel each other.

21.3 COULOMB'S LAW

Charles Augustin de Coulomb (1736–1806) studied the interaction forces of charged particles in detail in 1784. He used a torsion balance (**Fig. 21.10a**) similar to the one used 13 years later by Cavendish to study the much weaker gravitational interaction, as we discussed in Section 13.1. For **point charges**, charged objects that are very small in comparison with the distance r between them, Coulomb found that the electric force is proportional to $1/r^2$. That is, when the distance r doubles, the force decreases to one-quarter of its initial value; when the distance is halved, the force increases to four times its initial value.

The electric force between two point charges also depends on the quantity of charge on each object, which we'll denote by q or Q . To explore this dependence, Coulomb divided a charge into two equal parts by placing a small charged spherical conductor into contact with an identical but uncharged sphere; by symmetry, the charge is shared equally between the two spheres. (Note the essential role of the principle of conservation of charge in this procedure.) Thus he could obtain one-half, one-quarter, and so on, of any initial charge. He found that the forces that two point charges q_1 and q_2 exert on each other are proportional to each charge and therefore are proportional to the *product* q_1q_2 of the two charges.

Thus Coulomb established the following law:

COULOMB'S LAW The magnitude of the electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

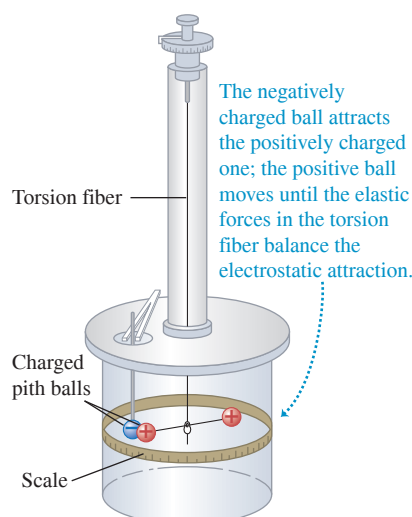
In mathematical terms, the magnitude F of the force that each of two point charges q_1 and q_2 a distance r apart exerts on the other can be expressed as

$$F = k \frac{|q_1 q_2|}{r^2} \quad (21.1)$$

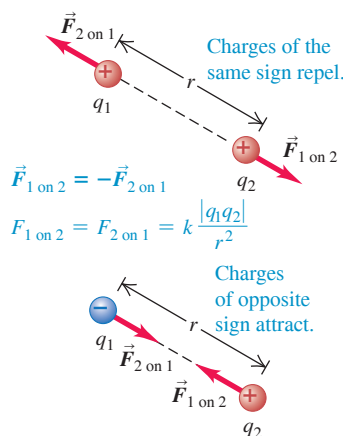
where k is a proportionality constant whose numerical value depends on the system of units used. The absolute value bars are used in Eq. (21.1) because the charges q_1 and q_2 can be either positive or negative, while the force magnitude F is always positive.

The directions of the forces the two charges exert on each other are always along the line joining them. When the charges q_1 and q_2 have the same sign, either both positive or both negative, the forces are repulsive; when the charges have opposite signs, the forces are attractive (Fig. 21.10b). The two forces obey Newton's third law; they are always equal in magnitude and opposite in direction, even when the charges are not equal in magnitude.

(a) A torsion balance of the type used by Coulomb to measure the electric force



(b) Interactions between point charges



BIO APPLICATION Electric Forces, Sweat, and Cystic Fibrosis

One way to test for the genetic disease cystic fibrosis (CF) is to measure the salt content of a person's sweat. Sweat is a mixture of water and ions, including the sodium (Na^+) and chloride (Cl^-) ions that make up ordinary salt (NaCl). When sweat is secreted by epithelial cells, some of the Cl^- ions flow from the sweat back into these cells (a process called reabsorption). The electric attraction between negative and positive charges pulls Na^+ ions along with the Cl^- . Water molecules cannot flow back into the epithelial cells, so sweat on the skin has a low salt content. However, in persons with CF the reabsorption of Cl^- ions is blocked. Hence the sweat of persons with CF is unusually salty, with up to four times the normal concentration of Cl^- and Na^+ .



Figure 21.10 (a) Measuring the electric force between point charges. (b) The electric forces between point charges obey Newton's third law: $\vec{F}_{1 \text{ on } 2} = -\vec{F}_{2 \text{ on } 1}$.

The proportionality of the electric force to $1/r^2$ has been verified with great precision. There is no reason to suspect that the exponent is different from precisely 2. Thus the form of Eq. (21.1) is the same as that of the law of gravitation. But electric and gravitational interactions are two distinct classes of phenomena. Electric interactions depend on electric charges and can be either attractive or repulsive, while gravitational interactions depend on mass and are always attractive (because there is no such thing as negative mass).

Fundamental Electric Constants

The value of the proportionality constant k in Coulomb's law depends on the system of units used. In our study of electricity and magnetism we'll use SI units exclusively. The SI electric units include most of the familiar units such as the volt, the ampere, the ohm, and the watt. (There is *no* British system of electric units.) The SI unit of electric charge is called one **coulomb** (1 C). As of this writing (2018) the value of the constant k in Eq. (21.1) is, to nine significant figures,

$$k = 8.987551787 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2 \cong 8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

Prior to 2018 k was *defined* to have a precise value in terms of the speed of light in vacuum c , whose value is in turn related to the definitions of the second and the meter that we stated in Section 1.3. (We'll see the connection between k and c in Chapter 32 when we study electromagnetic radiation.) With the revisions to the definitions of units in the International System (SI) in 2018, the value of k is no longer exact, but has a very small fractional uncertainty of about 2×10^{-10} . Because this uncertainty is so small, these revisions to the definitions of the units have no effect on the calculations that we present in this book, and don't affect the calculations that you'll do.

In principle we can measure the electric force F between two equal charges q at a measured distance r and use Coulomb's law to determine the charge. Thus we could regard the value of k as an operational definition of the coulomb. For reasons of experimental precision it is better to define the coulomb instead in terms of the most fundamental unit of charge. This is the magnitude of the charge of an electron or a proton, which is denoted by e . The value of e is *defined* to be a certain fraction of a coulomb:

$$e = 1.602176634 \times 10^{-19} \text{ C}$$

So **one coulomb (C) is that amount of charge such that $1.602176634 \times 10^{-19} \text{ C}$ is equal to the magnitude of the charge on an electron or proton.** (An older definition of the coulomb involving the unit of electric *current*, or charge per unit time, was abandoned in 2018.)

One coulomb represents the negative of the total charge of about 6×10^8 electrons. For comparison, a copper cube 1 cm on a side contains about 2.4×10^{24} electrons. About 10^{19} electrons pass through the glowing filament of a flashlight bulb every second.

In SI units we usually write the constant k in Eq. (21.1) as $1/4\pi\epsilon_0$, where ϵ_0 ("epsilon-nought" or "epsilon-zero") is called the **electric constant**. This shorthand simplifies many formulas that we'll encounter in later chapters. From now on, we'll usually write Coulomb's law as

$$\text{Coulomb's law:} \quad \text{Magnitude of electric force between two point charges} \quad F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} \quad \text{Electric constant} \quad \text{Values of the two charges} \quad \text{Distance between the two charges} \quad (21.2)$$

The constants in Eq. (21.2) are approximately

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2 \quad \text{and} \quad \frac{1}{4\pi\epsilon_0} = k = 8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

In examples and problems we'll often use the approximate value

$$\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

In electrostatics problems (problems that involve charges at rest), it's very unusual to encounter charges as large as 1 coulomb. Two 1 C charges separated by 1 m would exert forces on each other of magnitude $9 \times 10^9 \text{ N}$ (about 1 million tons)! The total charge of all the electrons in a copper one-cent coin is even greater, about $1.4 \times 10^5 \text{ C}$, which shows that we can't disturb electric neutrality very much without using enormous forces. More typical values of charge range from about a microcoulomb ($1 \mu\text{C} = 10^{-6} \text{ C}$) to about a nanocoulomb ($1 \text{ nC} = 10^{-9} \text{ C}$).

EXAMPLE 21.1 Electric force versus gravitational force

An α particle (the nucleus of a helium atom) has mass $m = 6.64 \times 10^{-27} \text{ kg}$ and charge $q = +2e = 3.2 \times 10^{-19} \text{ C}$. Compare the magnitude of the electric repulsion between two α ("alpha") particles with that of the gravitational attraction between them.

IDENTIFY and SET UP This problem involves Newton's law for the gravitational force F_g between particles (see Section 13.1) and Coulomb's law for the electric force F_e between point charges. To compare these forces, we make our target variable the ratio F_e/F_g . We use Eq. (21.2) for F_e and Eq. (13.1) for F_g .

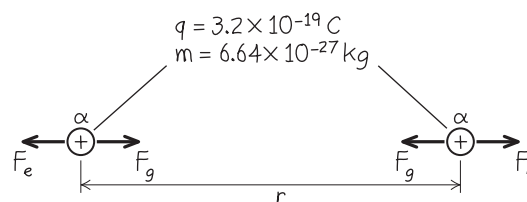
EXECUTE Figure 21.11 shows our sketch. From Eqs. (21.2) and (13.1),

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} \quad F_g = G \frac{m^2}{r^2}$$

These are both inverse-square forces, so the r^2 factors cancel when we take the ratio:

$$\begin{aligned} \frac{F_e}{F_g} &= \frac{1}{4\pi\epsilon_0 G} \frac{q^2}{m^2} \\ &= \frac{9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2}{6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2} \frac{(3.2 \times 10^{-19} \text{ C})^2}{(6.64 \times 10^{-27} \text{ kg})^2} = 3.1 \times 10^{35} \end{aligned}$$

Figure 21.11 Our sketch for this problem.



EVALUATE This astonishingly large number shows that the gravitational force in this situation is completely negligible in comparison to the electric force. This is always true for interactions of atomic and subnuclear particles. But within objects the size of a person or a planet, the positive and negative charges are nearly equal in magnitude, and the net electric force is usually much *smaller* than the gravitational force.

KEYCONCEPT The magnitude of the electric force that two point charges exert on each other is (a) proportional to the product of the magnitudes of their charges and (b) inversely proportional to the square of the distance between them.

Superposition of Forces

Coulomb's law as we have stated it describes only the interaction of two *point* charges. Experiments show that when two charges exert forces simultaneously on a third charge, the total force acting on that charge is the *vector sum* of the forces that the two charges would exert individually. This important property, called the **principle of superposition of forces**, holds for any number of charges. By using this principle, we can apply Coulomb's law to *any* collection of charges. Two of the examples at the end of this section use the superposition principle.

Strictly speaking, Coulomb's law as we have stated it should be used only for point charges *in a vacuum*. If matter is present in the space between the charges, the net force acting on each charge is altered because charges are induced in the molecules of the intervening material. We'll describe this effect later. As a practical matter, though, we can use Coulomb's law unaltered for point charges in air. At normal atmospheric pressure, the presence of air changes the electric force from its vacuum value by only about one part in 2000.

PROBLEM-SOLVING STRATEGY 21.1 Coulomb's Law

IDENTIFY the relevant concepts: Coulomb's law describes the electric force between charged particles.

SET UP the problem using the following steps:

1. Sketch the locations of the charged particles and label each particle with its charge.
2. If the charges do not all lie on a single line, set up an xy -coordinate system.
3. The problem will ask you to find the electric force on one or more particles. Identify which these are.

EXECUTE the solution as follows:

1. For each particle that exerts an electric force on a given particle of interest, use Eq. (21.2) to calculate the magnitude of that force.
2. Using those magnitudes, sketch a free-body diagram showing the electric-force vectors acting on each particle of interest. The force exerted by particle 1 on particle 2 points from particle 2 toward particle 1 if the charges have opposite signs, but points from particle 2 directly away from particle 1 if the charges have the same sign.
3. Use the principle of superposition to calculate the total electric force—a *vector* sum—on each particle of interest. (Review the vector algebra in Sections 1.7 through 1.9. The method of components is often helpful.)

4. Use consistent units; SI units are completely consistent. With $1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$, distances must be in meters, charges in coulombs, and forces in newtons.
5. Some examples and problems in this and later chapters involve *continuous* distributions of charge along a line, over a surface, or throughout a volume. In these cases the vector sum in step 3 becomes a vector *integral*. We divide the charge distribution into infinitesimal pieces, use Coulomb's law for each piece, and integrate to find the vector sum. Sometimes this can be done without actual integration.
6. Exploit any symmetries in the charge distribution to simplify your problem solving. For example, two identical charges q exert zero net electric force on a charge Q midway between them, because the forces on Q have equal magnitude and opposite direction.

EVALUATE your answer: Check whether your numerical results are reasonable. Confirm that the direction of the net electric force agrees with the principle that charges of the same sign repel and charges of opposite sign attract.

EXAMPLE 21.2 Force between two point charges

Two point charges, $q_1 = +25 \text{ nC}$ and $q_2 = -75 \text{ nC}$, are separated by a distance $r = 3.0 \text{ cm}$ (Fig. 21.12a). Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1 .

IDENTIFY and SET UP This problem asks for the electric forces that two charges exert on each other. We use Coulomb's law, Eq. (21.2), to calculate the magnitudes of the forces. The signs of the charges will determine the directions of the forces.

EXECUTE (a) After converting the units of r to meters and the units of q_1 and q_2 to coulombs, Eq. (21.2) gives us

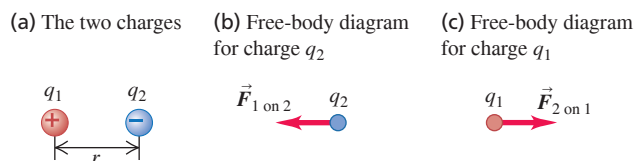
$$\begin{aligned} F_{1 \text{ on } 2} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{|(+25 \times 10^{-9} \text{ C})(-75 \times 10^{-9} \text{ C})|}{(0.030 \text{ m})^2} \\ &= 0.019 \text{ N} \end{aligned}$$

The charges have opposite signs, so the force is attractive (to the left in Fig. 21.12b); that is, the force that acts on q_2 is directed toward q_1 along the line joining the two charges.

(b) Proceeding as in part (a), we have

$$F_{2 \text{ on } 1} = \frac{1}{4\pi\epsilon_0} \frac{|q_2 q_1|}{r^2} = F_{1 \text{ on } 2} = 0.019 \text{ N}$$

Figure 21.12 What force does q_1 exert on q_2 , and what force does q_2 exert on q_1 ? Gravitational forces are negligible.



The attractive force that acts on q_1 is to the right, toward q_2 (Fig. 21.12c).

EVALUATE Even though the charges have different magnitudes, the magnitude of the force that q_2 exerts on q_1 is the same as the magnitude of the force that q_1 exerts on q_2 .

KEYCONCEPT The electric forces that two point charges exert on each other obey Newton's third law: The force that the first charge exerts on the second charge has the same magnitude as the force that the second charge exerts on the first charge, but points in the opposite direction. The forces are repulsive if the two charges are both positive or both negative, and attractive if one charge is positive and the other negative.

EXAMPLE 21.3 Vector addition of electric forces on a line**WITH VARIATION PROBLEMS**

Two point charges are located on the x -axis of a coordinate system: $q_1 = 1.0 \text{ nC}$ is at $x = +2.0 \text{ cm}$, and $q_2 = -3.0 \text{ nC}$ is at $x = +4.0 \text{ cm}$. What is the total electric force exerted by q_1 and q_2 on a charge $q_3 = 5.0 \text{ nC}$ at $x = 0$?

IDENTIFY and SET UP Figure 21.13a shows the situation. To find the total force on q_3 , our target variable, we find the vector sum of the two electric forces on it.

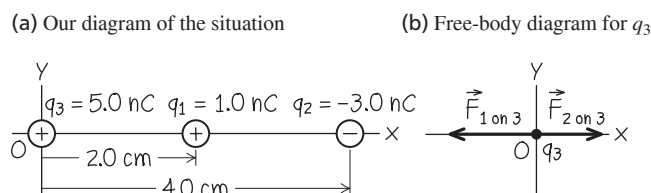
EXECUTE Figure 21.13b is a free-body diagram for q_3 , which is repelled by q_1 (which has the same sign) and attracted to q_2 (which has the opposite sign): $\vec{F}_{1 \text{ on } 3}$ is in the $-x$ -direction and $\vec{F}_{2 \text{ on } 3}$ is in the $+x$ -direction. After unit conversions, we have from Eq. (21.2)

$$\begin{aligned} F_{1 \text{ on } 3} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_3|}{r_{13}^2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(1.0 \times 10^{-9} \text{ C})(5.0 \times 10^{-9} \text{ C})}{(0.020 \text{ m})^2} \\ &= 1.12 \times 10^{-4} \text{ N} = 112 \mu\text{N} \end{aligned}$$

In the same way you can show that $F_{2 \text{ on } 3} = 84 \mu\text{N}$. We thus have $\vec{F}_{1 \text{ on } 3} = (-112 \mu\text{N})\hat{i}$ and $\vec{F}_{2 \text{ on } 3} = (84 \mu\text{N})\hat{i}$. The net force on q_3 is

$$\vec{F}_3 = \vec{F}_{1 \text{ on } 3} + \vec{F}_{2 \text{ on } 3} = (-112 \mu\text{N})\hat{i} + (84 \mu\text{N})\hat{i} = (-28 \mu\text{N})\hat{i}$$

Figure 21.13 Our sketches for this problem.



EVALUATE As a check, note that the magnitude of q_2 is three times that of q_1 , but q_2 is twice as far from q_3 as q_1 . Equation (21.2) then says that $F_{2 \text{ on } 3}$ must be $3/2^2 = 3/4 = 0.75$ as large as $F_{1 \text{ on } 3}$. This agrees with our calculated values: $F_{2 \text{ on } 3}/F_{1 \text{ on } 3} = (84 \mu\text{N})/(112 \mu\text{N}) = 0.75$. Because $F_{2 \text{ on } 3}$ is the weaker force, the direction of the net force is that of $\vec{F}_{1 \text{ on } 3}$ —that is, in the negative x -direction.

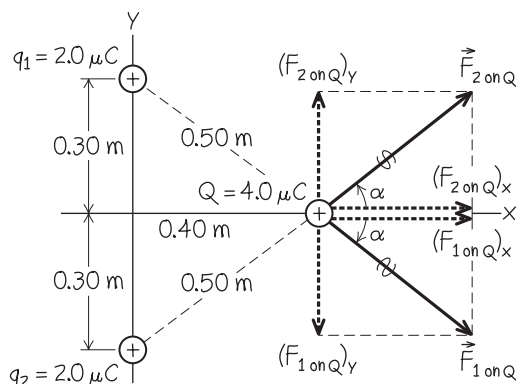
KEYCONCEPT Electric forces obey the principle of superposition of forces: The net electric force on a point charge due to two or more other charges is the vector sum of the individual electric forces that each of the other charges exert on it.

EXAMPLE 21.4 Vector addition of electric forces in a plane**WITH VARIATION PROBLEMS**

Two equal positive charges $q_1 = q_2 = 2.0 \mu\text{C}$ are located at $x = 0, y = 0.30 \text{ m}$ and $x = 0, y = -0.30 \text{ m}$, respectively. What are the magnitude and direction of the total electric force that q_1 and q_2 exert on a third charge $Q = 4.0 \mu\text{C}$ at $x = 0.40 \text{ m}, y = 0$?

IDENTIFY and SET UP As in Example 21.3, we must compute the force that each charge exerts on Q and then find the vector sum of those forces. Figure 21.14 shows the situation. Since the three charges do not all lie on a line, the best way to calculate the forces is to use components.

Figure 21.14 Our sketch for this problem.



EXECUTE Figure 21.14 shows the forces $\vec{F}_{1 \text{ on } Q}$ and $\vec{F}_{2 \text{ on } Q}$ due to the identical charges q_1 and q_2 , which are at equal distances from Q . From Coulomb's law, *both* forces have magnitude

$$\begin{aligned} F_{1 \text{ or } 2 \text{ on } Q} &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(4.0 \times 10^{-6} \text{ C})(2.0 \times 10^{-6} \text{ C})}{(0.50 \text{ m})^2} \\ &= 0.29 \text{ N} \end{aligned}$$

The x -components of the two forces are equal:

$$(F_{1 \text{ or } 2 \text{ on } Q})_x = (F_{1 \text{ or } 2 \text{ on } Q}) \cos \alpha = (0.29 \text{ N}) \frac{0.40 \text{ m}}{0.50 \text{ m}} = 0.23 \text{ N}$$

From symmetry we see that the y -components of the two forces are equal and opposite. Hence their sum is zero and the total force $\vec{F}$ on Q has only an x -component $F_x = 0.23 \text{ N} + 0.23 \text{ N} = 0.46 \text{ N}$. The total force on Q is in the $+x$ -direction, with magnitude 0.46 N .

EVALUATE The total force on Q points neither directly away from q_1 nor directly away from q_2 . Rather, this direction is a compromise that points away from the *system* of charges q_1 and q_2 . Can you see that the total force would *not* be in the $+x$ -direction if q_1 and q_2 were not equal or if the geometrical arrangement of the charges were not so symmetric?

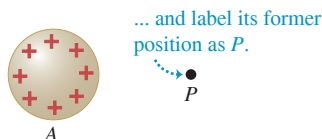
KEYCONCEPT When a point charge experiences electric forces due to two or more other charges, you must use vector addition to find the magnitude and direction of the net electric force on the first charge. This is most easily done by first finding the vector components of each individual electric force.

Figure 21.15 A charged object creates an electric field in the space around it.

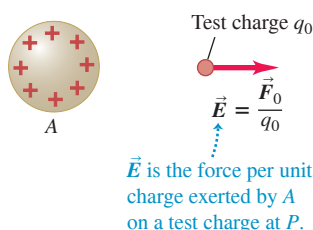
(a) *A* and *B* exert electric forces on each other.



(b) Remove object *B* ...



(c) Object *A* sets up an electric field $\vec{E}$ at point *P*.



BIO APPLICATION Sharks and the "Sixth Sense"

Sharks have the ability to locate prey (such as flounder and other bottom-dwelling fish) that are completely hidden beneath the sand at the bottom of the ocean. They do this by sensing the weak electric fields produced by muscle contractions in their prey. Sharks derive their sensitivity to electric fields (a "sixth sense") from jelly-filled canals in their bodies. These canals end in pores on the shark's skin (shown in this photograph). An electric field as weak as 5×10^{-7} N/C causes charge flow within the canals and triggers a signal in the shark's nervous system. Because the shark has canals with different orientations, it can measure different components of the electric-field vector and hence determine the direction of the field.



TEST YOUR UNDERSTANDING OF SECTION 21.3 Suppose that charge q_2 in Example 21.4 were $-2.0 \mu\text{C}$. In this case, the total electric force on Q would be (i) in the positive x -direction; (ii) in the negative x -direction; (iii) in the positive y -direction; (iv) in the negative y -direction; (v) zero; (vi) none of these.

ANSWER

The force exerted by q_1 on Q is still as in Example 21.4. The magnitude of the force exerted by q_2 on Q is still equal to $F_{1 \text{ on } Q}$, but the direction of the force is now *toward* q_2 at an angle α below the x -axis. Hence the x -components of the two forces cancel while the (negative) y -components add together, and the total electric force is in the negative y -direction.

21.4 ELECTRIC FIELD AND ELECTRIC FORCES

When two electrically charged particles in empty space interact, how does each one know the other is there? We can begin to answer this question, and at the same time reformulate Coulomb's law in a very useful way, by using the concept of *electric field*.

Electric Field

To introduce this concept, let's look at the mutual repulsion of two positively charged objects *A* and *B* (Fig. 21.15a). Suppose *B* has charge q_0 , and let $\vec{F}_0$ be the electric force of *A* on *B*. One way to think about this force is as an "action-at-a-distance" force—that is, as a force that acts across empty space without needing physical contact between *A* and *B*. (Gravity can also be thought of as an "action-at-a-distance" force.) But a more fruitful way to visualize the repulsion between *A* and *B* is as a two-stage process. We first envision that object *A*, as a result of the charge that it carries, somehow *modifies the properties of the space around it*. Then object *B*, as a result of the charge that it carries, senses how space has been modified at its position. The response of object *B* is to experience the force $\vec{F}_0$.

To clarify how this two-stage process occurs, we first consider object *A* by itself: We remove object *B* and label its former position as point *P* (Fig. 21.15b). We say that the charged object *A* produces or causes an **electric field** at point *P* (and at all other points in the neighborhood). This electric field is present at *P* even if there is no charge at *P*; it is a consequence of the charge on object *A* only. If a point charge q_0 is then placed at point *P*, it experiences the force $\vec{F}_0$. We take the point of view that this force is exerted on q_0 by the field at *P* (Fig. 21.15c). Thus the electric field is the intermediary through which *A* communicates its presence to q_0 . Because the point charge q_0 would experience a force at *any* point in the neighborhood of *A*, the electric field that *A* produces exists at all points in the region around *A*.

We can likewise say that the point charge q_0 produces an electric field in the space around it and that this electric field exerts the force $-\vec{F}_0$ on object *A*. For each force (the force of *A* on q_0 and the force of q_0 on *A*), one charge sets up an electric field that exerts a force on the second charge. We emphasize that this is an *interaction* between *two* charged objects. A single charge produces an electric field in the surrounding space, but this electric field cannot exert a net force on the charge that created it; as we discussed in Section 4.3, an object cannot exert a net force on itself. (If this wasn't true, you would be able to lift yourself to the ceiling by pulling up on your belt!)

The electric force on a charged object is exerted by the electric field created by other charged objects.

To find out experimentally whether there is an electric field at a particular point, we place a small charged object, which we call a **test charge**, at the point (Fig. 21.15c). If the test charge experiences an electric force, then there is an electric field at that point. This field is produced by charges other than q_0 .

Force is a vector quantity, so electric field is also a vector quantity. (Note the use of vector signs as well as boldface letters and plus, minus, and equals signs in the following discussion.) We define the *electric field* $\vec{E}$ at a point as the electric force $\vec{F}_0$

experienced by a test charge q_0 at the point, divided by the charge q_0 . That is, the electric field at a certain point is equal to the *electric force per unit charge* experienced by a charge at that point:

$$\text{Electric field} = \vec{E} = \frac{\vec{F}_0}{q_0} \quad (21.3)$$

Electric force on a test charge q_0 due to other charges
Value of test charge

In SI units, in which the unit of force is 1 N and the unit of charge is 1 C, the unit of electric-field magnitude is 1 newton per coulomb (1 N/C).

If the field $\vec{E}$ at a certain point is known, rearranging Eq. (21.3) gives the force $\vec{F}_0$ experienced by a point charge q_0 placed at that point. This force is just equal to the electric field $\vec{E}$ produced at that point by charges other than q_0 , multiplied by the charge q_0 :

$$\vec{F}_0 = q_0 \vec{E} \quad \begin{array}{l} \text{(force exerted on a point charge } q_0 \\ \text{by an electric field } \vec{E}) \end{array} \quad (21.4)$$

The charge q_0 can be either positive or negative. If q_0 is *positive*, the force $\vec{F}_0$ experienced by the charge is in the same direction as $\vec{E}$; if q_0 is *negative*, $\vec{F}_0$ and $\vec{E}$ are in opposite directions (**Fig. 21.16**).

While the electric field concept may be new to you, the basic idea—that one object sets up a field in the space around it and a second object responds to that field—is one that you’ve actually used before. Compare Eq. (21.4) to the familiar expression for the gravitational force $\vec{F}_g$ that the earth exerts on a mass m_0 :

$$\vec{F}_g = m_0 \vec{g} \quad (21.5)$$

In this expression, $\vec{g}$ is the acceleration due to gravity. If we divide both sides of Eq. (21.5) by the mass m_0 , we obtain

$$\vec{g} = \frac{\vec{F}_g}{m_0}$$

Thus $\vec{g}$ can be regarded as the gravitational force per unit mass. By analogy to Eq. (21.3), we can interpret $\vec{g}$ as the *gravitational field*. Thus we treat the gravitational interaction between the earth and the mass m_0 as a two-stage process: The earth sets up a gravitational field $\vec{g}$ in the space around it, and this gravitational field exerts a force given by Eq. (21.5) on the mass m_0 (which we can regard as a *test mass*). The gravitational field $\vec{g}$, or gravitational force per unit mass, is a useful concept because it does not depend on the mass of the object on which the gravitational force is exerted; likewise, the electric field $\vec{E}$, or electric force per unit charge, is useful because it does not depend on the charge of the object on which the electric force is exerted.

CAUTION $\vec{F}_0 = q_0 \vec{E}$ is for **point test charges only** The electric force experienced by a test charge q_0 can vary from point to point, so the electric field can also be different at different points. For this reason, use Eq. (21.4) to find the electric force on a *point* charge only. If a charged object is large enough in size, the electric field $\vec{E}$ may be noticeably different in magnitude and direction at different points on the object, and calculating the net electric force on it can be complicated. ■

Electric Field of a Point Charge

If the source distribution is a point charge q , it is easy to find the electric field that it produces. We call the location of the charge the **source point**, and we call the point P where we are determining the field the **field point**. It is also useful to introduce a *unit vector* $\hat{r}$ that points along the line from source point to field point (**Fig. 21.17a**, next page). This unit vector is equal to the displacement vector $\vec{r}$ from the source point to the field point, divided by the distance $r = |\vec{r}|$ between these two

Figure 21.16 The force $\vec{F}_0 = q_0 \vec{E}$ exerted on a point charge q_0 placed in an electric field $\vec{E}$.

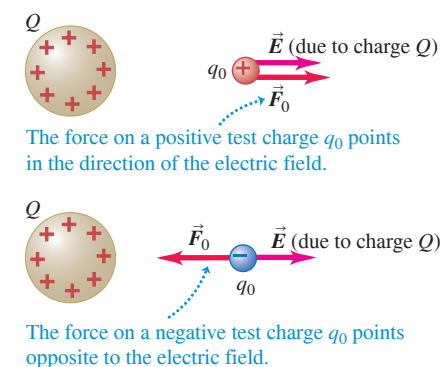
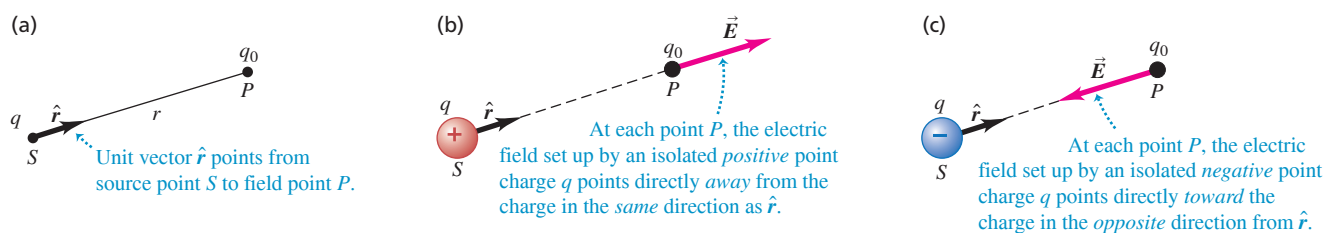


Figure 21.17 The electric field $\vec{E}$ produced at point P by an isolated point charge q at S . Note that in both (b) and (c), $\vec{E}$ is produced by q [see Eq. (21.7)] but acts on the charge q_0 at point P [see Eq. (21.4)].



points; that is, $\hat{r} = \vec{r}/r$. If we place a small test charge q_0 at the field point P , at a distance r from the source point, the magnitude F_0 of the force is given by Coulomb's law, Eq. (21.2):

$$F_0 = \frac{1}{4\pi\epsilon_0} \frac{|qq_0|}{r^2}$$

From Eq. (21.3) the magnitude E of the electric field at P is

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} \quad (\text{magnitude of electric field due to a point charge}) \quad (21.6)$$

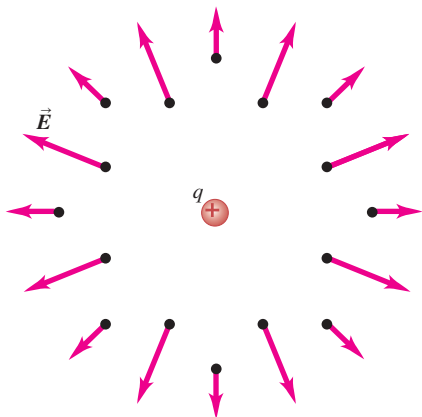
Using the unit vector $\hat{r}$, we can write a vector equation that gives both the magnitude and direction of the electric field $\vec{E}$:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (21.7)$$

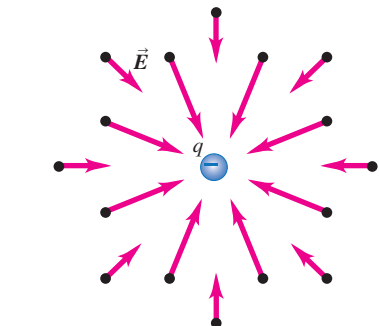
Labels for Eq. (21.7):
 - Electric field due to a point charge: $\vec{E}$
 - Value of point charge: q
 - Unit vector from point charge toward where field is measured: $\hat{r}$
 - Distance from point charge to where field is measured: r
 - Electric constant: $1/(4\pi\epsilon_0)$

Figure 21.18 A point charge q produces an electric field $\vec{E}$ at all points in space. The field strength decreases with increasing distance.

(a) The field produced by a positive point charge points away from the charge.



(b) The field produced by a negative point charge points toward the charge.



By definition, the electric field of a point charge always points away from a positive charge (that is, in the same direction as $\hat{r}$; see Fig. 21.17b) but toward a negative charge (that is, in the direction opposite $\hat{r}$; see Fig. 21.17c).

CAUTION The electric field at a point does not depend on the charge at that point. Equation (21.7) tells you the value of the electric field $\vec{E}$ at a certain point due to a point charge q at a different point. The electric field at a certain point is present whether or not there is a second point charge at that point to experience the field. **I**

We have emphasized calculating the electric field $\vec{E}$ at a certain point. But since $\vec{E}$ can vary from point to point, it is not a single vector quantity but rather an infinite set of vector quantities, one associated with each point in space. This is an example of a **vector field**. Figure 21.18 shows a number of the field vectors produced by a positive or negative point charge. If we use a rectangular (x, y, z) coordinate system, each component of $\vec{E}$ at any point is in general a function of the coordinates (x, y, z) of the point. We can represent the functions as $E_x(x, y, z)$, $E_y(x, y, z)$, and $E_z(x, y, z)$. Another example of a vector field is the velocity $\vec{v}$ of wind currents; the magnitude and direction of $\vec{v}$, and hence its vector components, vary from point to point in the atmosphere.

In some situations the magnitude and direction of the field (and hence its vector components) have the same values everywhere throughout a certain region; we then say that the field is *uniform* in this region. An important example of this is the electric field inside a *conductor*. If there is an electric field within a conductor, the field exerts a force on every charge in the conductor, giving the free charges a net motion. By definition an electrostatic situation is one in which the charges have *no* net motion. We conclude that *in electrostatics the electric field at every point within the material of a conductor must be zero*. (Note that we are not saying that the field is necessarily zero in a *hole* inside a conductor.)

In summary, our description of electric interactions has two parts. First, a given charge distribution acts as a source of electric field. Second, the electric field exerts a force on any

charge that is present in the field. Our analysis often has two corresponding steps: first, calculating the field caused by a source charge distribution; second, looking at the effect of the field in terms of force and motion. The second step often involves Newton's laws as well as the principles of electric interactions. In the next section we show how to calculate fields caused by various source distributions, but first here are three examples of calculating the field due to a point charge and of finding the force on a charge due to a given field $\vec{E}$.

EXAMPLE 21.5 Electric-field magnitude for a point charge

What is the magnitude of the electric field $\vec{E}$ at a field point 2.0 m from a point charge $q = 4.0$ nC?

IDENTIFY and SET UP This problem concerns the electric field due to a point charge. We are given the magnitude of the charge and the distance from the charge to the field point, so we use Eq. (21.6) to calculate the field magnitude E .

EXECUTE From Eq. (21.6),

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} = (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{4.0 \times 10^{-9} \text{ C}}{(2.0 \text{ m})^2} = 9.0 \text{ N/C}$$

EVALUATE Our result $E = 9.0$ N/C means that if we placed a 1.0 C charge at a point 2.0 m from q , it would experience a 9.0 N force. The force on a 2.0 C charge at that point would be $(2.0 \text{ C})(9.0 \text{ N/C}) = 18 \text{ N}$, and so on.

KEYCONCEPT The magnitude of the electric field that a point charge produces at a given point is (a) proportional to the magnitude of the charge and (b) inversely proportional to the square of the distance from the point charge to the point where the field is measured.

EXAMPLE 21.6 Electric-field vector for a point charge

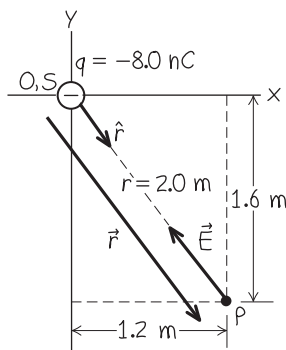
A point charge $q = -8.0$ nC is located at the origin. Find the electric-field vector at the field point $x = 1.2$ m, $y = -1.6$ m.

IDENTIFY and SET UP We must find the electric-field *vector* $\vec{E}$ due to a point charge. **Figure 21.19** shows the situation. We use Eq. (21.7); to do this, we must find the distance r from the source point S (the position of the charge q , which in this example is at the origin O) to the field point P , and we must obtain an expression for the unit vector $\hat{r} = \vec{r}/r$ that points from S to P .

EXECUTE The distance from S to P is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.2 \text{ m})^2 + (-1.6 \text{ m})^2} = 2.0 \text{ m}$$

Figure 21.19 Our sketch for this problem.



The unit vector $\hat{r}$ is then

$$\begin{aligned} \hat{r} &= \frac{\vec{r}}{r} = \frac{x\hat{i} + y\hat{j}}{r} \\ &= \frac{(1.2 \text{ m})\hat{i} + (-1.6 \text{ m})\hat{j}}{2.0 \text{ m}} = 0.60\hat{i} - 0.80\hat{j} \end{aligned}$$

Then, from Eq. (21.7),

$$\begin{aligned} \vec{E} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(-8.0 \times 10^{-9} \text{ C})}{(2.0 \text{ m})^2} (0.60\hat{i} - 0.80\hat{j}) \\ &= (-11 \text{ N/C})\hat{i} + (14 \text{ N/C})\hat{j} \end{aligned}$$

EVALUATE Since q is negative, $\vec{E}$ points from the field point to the charge (the source point), in the direction opposite to $\hat{r}$ (compare Fig. 21.17c). We leave the calculation of the magnitude and direction of $\vec{E}$ to you.

KEYCONCEPT When you calculate the electric field due to a point charge q , the unit vector $\hat{r}$ always points from the charge to the point where the field is measured. If q is positive, the field is in the same direction as $\hat{r}$ (away from the charge); if q is negative, the field is in the direction opposite to $\hat{r}$ (toward the charge.)

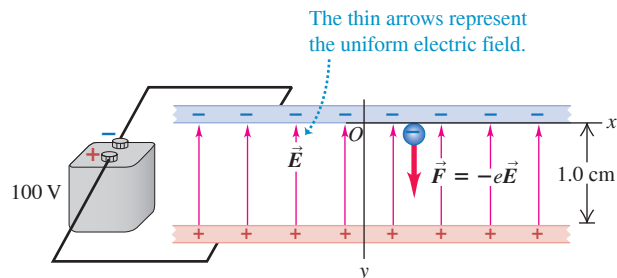
EXAMPLE 21.7 Electron in a uniform field

When the terminals of a battery are connected to two parallel conducting plates with a small gap between them, the resulting charges on the plates produce a nearly uniform electric field $\vec{E}$ between the plates. (In the next section we'll see why this is.) If the plates are 1.0 cm apart and are connected to a 100 volt battery as shown in **Fig. 21.20** (next page), the field

is vertically upward and has magnitude $E = 1.00 \times 10^4$ N/C. (a) If an electron (charge $-e = -1.60 \times 10^{-19}$ C, mass $m = 9.11 \times 10^{-31}$ kg) is released from rest at the upper plate, what is its acceleration? (b) What speed and kinetic energy does it acquire while traveling 1.0 cm to the lower plate? (c) How long does it take to travel this distance?

Continued

Figure 21.20 A uniform electric field between two parallel conducting plates connected to a 100 volt battery. (The separation of the plates is exaggerated in this figure relative to the dimensions of the plates.)



IDENTIFY and SET UP This example involves the relationship between electric field and electric force. It also involves the relationship between force and acceleration, the definition of kinetic energy, and the kinematic relationships among acceleration, distance, velocity, and time. Figure 21.20 shows our coordinate system. We are given the electric field, so we use Eq. (21.4) to find the force on the electron and Newton's second law to find its acceleration. Because the field is uniform, the force is constant and we can use the constant-acceleration formulas from Chapter 2 to find the electron's velocity and travel time. We find the kinetic energy from $K = \frac{1}{2}mv^2$.

EXECUTE (a) Although $\vec{E}$ is upward (in the $+y$ -direction), $\vec{F}$ is downward (because the electron's charge is negative) and so F_y is negative. Because F_y is constant, the electron's acceleration is constant:

$$a_y = \frac{F_y}{m} = \frac{-eE}{m} = \frac{(-1.60 \times 10^{-19} \text{ C})(1.00 \times 10^4 \text{ N/C})}{9.11 \times 10^{-31} \text{ kg}} = -1.76 \times 10^{15} \text{ m/s}^2$$

(b) The electron starts from rest, so its motion is in the y -direction only (the direction of the acceleration). We can find the electron's speed at any position y from the constant-acceleration Eq. (2.13), $v_y^2 = v_{0y}^2 + 2a_y(y - y_0)$. We have $v_{0y} = 0$ and $y_0 = 0$, so at $y = -1.0 \text{ cm} = -1.0 \times 10^{-2} \text{ m}$ we have

$$|v_y| = \sqrt{2a_y y} = \sqrt{2(-1.76 \times 10^{15} \text{ m/s}^2)(-1.0 \times 10^{-2} \text{ m})} = 5.9 \times 10^6 \text{ m/s}$$

The velocity is downward, so $v_y = -5.9 \times 10^6 \text{ m/s}$. The electron's kinetic energy is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(9.11 \times 10^{-31} \text{ kg})(5.9 \times 10^6 \text{ m/s})^2 = 1.6 \times 10^{-17} \text{ J}$$

(c) From Eq. (2.8) for constant acceleration, $v_y = v_{0y} + a_y t$, so

$$t = \frac{v_y - v_{0y}}{a_y} = \frac{(-5.9 \times 10^6 \text{ m/s}) - (0)}{-1.76 \times 10^{15} \text{ m/s}^2} = 3.4 \times 10^{-9} \text{ s}$$

EVALUATE Our results show that in problems concerning subatomic particles such as electrons, many quantities—including acceleration, speed, kinetic energy, and time—will have *very* different values from those typical of everyday objects such as baseballs and automobiles.

KEYCONCEPT If you place a point charge q_0 at a location where the electric field is $\vec{E}$, the electric force on the point charge is $q_0 \vec{E}$. This force is in the same direction as $\vec{E}$ if q_0 is positive; the force is in the direction opposite to $\vec{E}$ if q_0 is negative.

TEST YOUR UNDERSTANDING OF SECTION 21.4 (a) A negative point charge moves along a straight-line path directly toward a stationary positive point charge. Which aspect(s) of the electric force on the negative point charge will remain constant as it moves? (i) Magnitude; (ii) direction; (iii) both magnitude and direction; (iv) neither magnitude nor direction. (b) A negative point charge moves along a circular orbit around a positive point charge. Which aspect(s) of the electric force on the negative point charge will remain constant as it moves? (i) Magnitude; (ii) direction; (iii) both magnitude and direction; (iv) neither magnitude nor direction.

ANSWER

(a) (i) (ii) (iii) (b) (i) (ii) (iii) The electric field $\vec{E}$ produced by a positive point charge points directly away from the charge (see Fig. 21.18a) and has a magnitude that depends on the distance r from the charge to the field point. Hence a second, negative point charge $q < 0$ will feel a force $\vec{F} = q\vec{E}$ that points directly toward the positive charge and has a magnitude that depends on the distance r between the two charges. If the negative charge moves directly toward the positive charge, the direction of the force remains the same but the force magnitude increases as the distance r decreases. If the negative charge moves in a circle around the positive charge, the force magnitude stays the same (because the distance r is constant) but the force direction changes.

21.5 ELECTRIC-FIELD CALCULATIONS

Equation (21.7) gives the electric field caused by a single point charge. But in most realistic situations that involve electric fields and forces, we encounter charge that is *distributed* over space. The charged plastic and glass rods in Fig. 21.1 have electric charge distributed over their surfaces, as does the imaging drum of a laser printer (Fig. 21.2). In this section we'll learn to calculate electric fields caused by various distributions of electric charge. Calculations of this kind are of tremendous importance for technological applications of electric forces. To determine the trajectories of atomic nuclei in an accelerator for cancer

radiotherapy or of charged particles in a semiconductor electronic device, you have to know the detailed nature of the electric field acting on the charges.

The Superposition of Electric Fields

To find the field caused by a charge distribution, we imagine the distribution to be made up of many point charges $q_1, q_2, q_3, \dots$ (This is actually quite a realistic description, since we have seen that charge is carried by electrons and protons that are so small as to be almost pointlike.) At any given point P , each point charge produces its own electric field $\vec{E}_1, \vec{E}_2, \vec{E}_3, \dots$, so a test charge q_0 placed at P experiences a force $\vec{F}_1 = q_0\vec{E}_1$ from charge q_1 , a force $\vec{F}_2 = q_0\vec{E}_2$ from charge q_2 , and so on. From the principle of superposition of forces discussed in Section 21.3, the *total* force $\vec{F}_0$ that the charge distribution exerts on q_0 is the vector sum of these individual forces:

$$\vec{F}_0 = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots = q_0\vec{E}_1 + q_0\vec{E}_2 + q_0\vec{E}_3 + \dots$$

The combined effect of all the charges in the distribution is described by the *total* electric field $\vec{E}$ at point P . From the definition of electric field, Eq. (21.3), this is

$$\vec{E} = \frac{\vec{F}_0}{q_0} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$

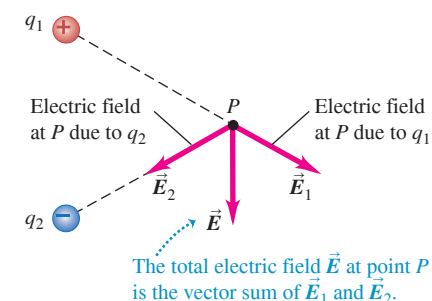
This equation is the mathematical expression of the *principle of superposition of electric fields*, illustrated in **Fig. 21.21**:

PRINCIPLE OF SUPERPOSITION OF ELECTRIC FIELDS The total electric field at a point P is the vector sum of the fields at P due to each point charge in the charge distribution.

When charge is distributed along a line, over a surface, or through a volume, a few additional terms are useful. For a line charge distribution (such as a long, thin, charged plastic rod), we use λ (the Greek letter lambda) to represent the **linear charge density** (charge per unit length, measured in C/m). When charge is distributed over a surface (such as the surface of the imaging drum of a laser printer), we use σ (sigma) to represent the **surface charge density** (charge per unit area, measured in C/m²). And when charge is distributed through a volume, we use ρ (rho) to represent the **volume charge density** (charge per unit volume, C/m³).

Some of the calculations in the following examples may look complex. After you've worked through the examples one step at a time, the process will seem less formidable. We'll use many of the calculational techniques in these examples in Chapter 28 to calculate the *magnetic* fields caused by charges in motion.

Figure 21.21 Illustrating the principle of superposition of electric fields.



PROBLEM-SOLVING STRATEGY 21.2 Electric-Field Calculations

IDENTIFY the relevant concepts: Use the principle of superposition to calculate the electric field due to a discrete or continuous charge distribution.

SET UP the problem using the following steps:

1. Make a drawing showing the locations of the charges and your choice of coordinate axes.
2. On your drawing, indicate the position of the *field point* P (the point at which you want to calculate the electric field $\vec{E}$).

EXECUTE the solution as follows:

1. Use consistent units. Distances must be in meters and charge must be in coulombs. If you are given centimeters or nanocoulombs, don't forget to convert.
2. Distinguish between the source point S and the field point P . The field produced by a point charge always points from S to P if the charge is positive, and from P to S if the charge is negative.

3. Use *vector* addition when applying the principle of superposition; review the treatment of vector addition in Chapter 1 if necessary.
4. Simplify your calculations by exploiting any symmetries in the charge distribution.
5. If the charge distribution is continuous, define a small element of charge that can be considered as a point, find its electric field at P , and find a way to add the fields of all the charge elements by doing an integral. Usually it is easiest to do this for each component of $\vec{E}$ separately, so you may need to evaluate more than one integral. Ensure that the limits on your integrals are correct; especially when the situation has symmetry, don't count a charge twice.

EVALUATE your answer: Check that the direction of $\vec{E}$ is reasonable. If your result for the electric-field magnitude E is a function of position (say, the coordinate x), check your result in any limits for which you know what the magnitude should be. When possible, check your answer by calculating it in a different way.

EXAMPLE 21.8 Field of an electric dipole

WITH VARIATION PROBLEMS

Point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ are 0.100 m apart (**Fig. 21.22**). (Such pairs of point charges with equal magnitude and opposite sign are called *electric dipoles*.) Compute the electric field caused by q_1 , the field caused by q_2 , and the total field (a) at point a ; (b) at point b ; and (c) at point c .

IDENTIFY and SET UP We must find the total electric field at various points due to two point charges. We use the principle of superposition: $\vec{E} = \vec{E}_1 + \vec{E}_2$. Figure 21.22 shows the coordinate system and the locations of the field points a , b , and c .

EXECUTE At each field point, $\vec{E}$ depends on $\vec{E}_1$ and $\vec{E}_2$ there; we first calculate the magnitudes E_1 and E_2 at each field point. At a the magnitude of the field $\vec{E}_{1a}$ caused by q_1 is

$$\begin{aligned} E_{1a} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1|}{r^2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{12 \times 10^{-9} \text{ C}}{(0.060 \text{ m})^2} \\ &= 3.0 \times 10^4 \text{ N/C} \end{aligned}$$

We calculate the other field magnitudes in a similar way. The results are

$$\begin{aligned} E_{1a} &= 3.0 \times 10^4 \text{ N/C} \\ E_{1b} &= 6.8 \times 10^4 \text{ N/C} \\ E_{1c} &= 6.39 \times 10^3 \text{ N/C} \\ E_{2a} &= 6.8 \times 10^4 \text{ N/C} \\ E_{2b} &= 0.55 \times 10^4 \text{ N/C} \\ E_{2c} &= E_{1c} = 6.39 \times 10^3 \text{ N/C} \end{aligned}$$

The *directions* of the corresponding fields are in all cases *away* from the positive charge q_1 and *toward* the negative charge q_2 .

(a) At a , $\vec{E}_{1a}$ and $\vec{E}_{2a}$ are both directed to the right, so

$$\vec{E}_a = E_{1a}\hat{i} + E_{2a}\hat{i} = (9.8 \times 10^4 \text{ N/C})\hat{i}$$

(b) At b , $\vec{E}_{1b}$ is directed to the left and $\vec{E}_{2b}$ is directed to the right, so

$$\vec{E}_b = -E_{1b}\hat{i} + E_{2b}\hat{i} = (-6.2 \times 10^4 \text{ N/C})\hat{i}$$

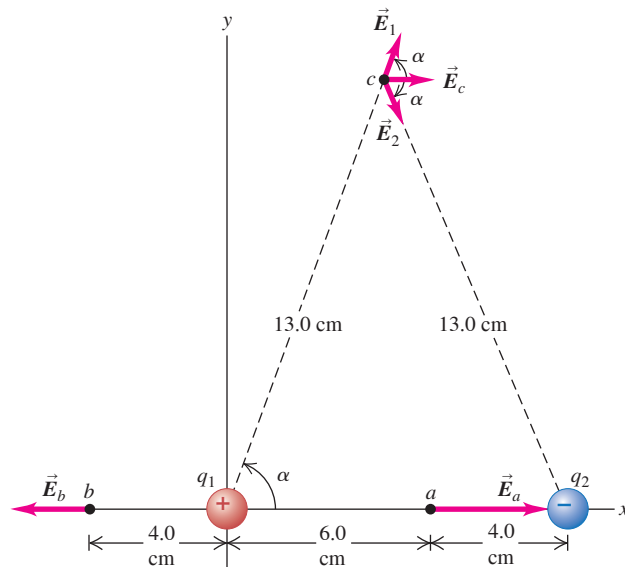
(c) Figure 21.22 shows the directions of $\vec{E}_1$ and $\vec{E}_2$ at c . Both vectors have the same x -component:

$$\begin{aligned} E_{1cx} &= E_{2cx} = E_{1c} \cos \alpha = (6.39 \times 10^3 \text{ N/C}) \left(\frac{5}{13} \right) \\ &= 2.46 \times 10^3 \text{ N/C} \end{aligned}$$

From symmetry, E_{1y} and E_{2y} are equal and opposite, so their sum is zero. Hence

$$\vec{E}_c = 2(2.46 \times 10^3 \text{ N/C})\hat{i} = (4.9 \times 10^3 \text{ N/C})\hat{i}$$

Figure 21.22 Electric field at three points, a , b , and c , set up by charges q_1 and q_2 , which form an electric dipole.



EVALUATE We can also find $\vec{E}_c$ by using Eq. (21.7) for the field of a point charge. The displacement vector $\vec{r}_1$ from q_1 to point c is $\vec{r}_1 = r \cos \alpha \hat{i} + r \sin \alpha \hat{j}$. Hence the unit vector that points from q_1 to point c is $\hat{r}_1 = \vec{r}_1/r = \cos \alpha \hat{i} + \sin \alpha \hat{j}$. By symmetry, the unit vector that points from q_2 to point c has the opposite x -component but the same y -component: $\hat{r}_2 = -\cos \alpha \hat{i} + \sin \alpha \hat{j}$. We can now use Eq. (21.7) to write the fields $\vec{E}_{1c}$ and $\vec{E}_{2c}$ at c in vector form, then find their sum. Since $q_2 = -q_1$ and the distance r to c is the same for both charges,

$$\begin{aligned} \vec{E}_c &= \vec{E}_{1c} + \vec{E}_{2c} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} \hat{r}_1 + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r^2} \hat{r}_2 \\ &= \frac{1}{4\pi\epsilon_0 r^2} (q_1 \hat{r}_1 + q_2 \hat{r}_2) \\ &= \frac{q_1}{4\pi\epsilon_0 r^2} (\hat{r}_1 - \hat{r}_2) \\ &= \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} (2 \cos \alpha \hat{i}) \\ &= 2(9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{12 \times 10^{-9} \text{ C}}{(0.13 \text{ m})^2} \left(\frac{5}{13} \right) \hat{i} \\ &= (4.9 \times 10^3 \text{ N/C})\hat{i} \end{aligned}$$

This is the same as we calculated in part (c).

KEYCONCEPT To find the net electric field at a point due to two or more point charges, first find the field at that point due to each individual charge. Then use vector addition to find the magnitude and direction of the net field at that point.

EXAMPLE 21.9 Field of a ring of charge**WITH VARIATION PROBLEMS**

Charge Q is uniformly distributed around a conducting ring of radius a (Fig. 21.23). Find the electric field at a point P on the ring axis at a distance x from its center.

IDENTIFY and SET UP This is a problem in the superposition of electric fields. Each bit of charge around the ring produces an electric field at an arbitrary point on the x -axis; our target variable is the total field at this point due to all such bits of charge.

EXECUTE We divide the ring into infinitesimal segments ds as shown in Fig. 21.23. In terms of the linear charge density $\lambda = Q/2\pi a$, the charge in a segment of length ds is $dQ = \lambda ds$. Consider two identical segments, one as shown in the figure at $y = a$ and another halfway around the ring at $y = -a$. From Example 21.4, we see that the net force $d\vec{F}$ they exert on a point test charge at P , and thus their net field $d\vec{E}$, are directed along the x -axis. The same is true for any such pair of segments around the ring, so the *net* field at P is along the x -axis: $\vec{E} = E_x \hat{i}$.

To calculate E_x , note that the square of the distance r from a single ring segment to the point P is $r^2 = x^2 + a^2$. Hence the magnitude of this segment's contribution $d\vec{E}$ to the electric field at P is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dQ}{x^2 + a^2}$$

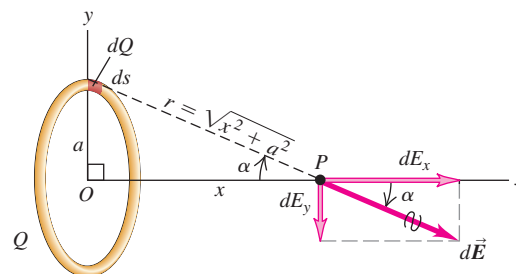
The x -component of this field is $dE_x = dE \cos \alpha$. We know $dQ = \lambda ds$ and Fig. 21.23 shows that $\cos \alpha = x/r = x/(x^2 + a^2)^{1/2}$, so

$$dE_x = dE \cos \alpha = \frac{1}{4\pi\epsilon_0} \frac{dQ}{x^2 + a^2} \frac{x}{\sqrt{x^2 + a^2}} = \frac{1}{4\pi\epsilon_0} \frac{\lambda x}{(x^2 + a^2)^{3/2}} ds$$

To find E_x we integrate this expression over the entire ring—that is, for s from 0 to $2\pi a$ (the circumference of the ring). The integrand has the same value for all points on the ring, so it can be taken outside the integral. Hence we get

$$\begin{aligned} E_x &= \int dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda x}{(x^2 + a^2)^{3/2}} \int_0^{2\pi a} ds \\ &= \frac{1}{4\pi\epsilon_0} \frac{\lambda x}{(x^2 + a^2)^{3/2}} (2\pi a) \end{aligned}$$

Figure 21.23 Calculating the electric field on the axis of a ring of charge. In this figure, the charge is assumed to be positive.



$$\vec{E} = E_x \hat{i} = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}} \hat{i} \quad (21.8)$$

EVALUATE Equation (21.8) shows that $\vec{E} = \mathbf{0}$ at the center of the ring ($x = 0$). This makes sense; charges on opposite sides of the ring push in opposite directions on a test charge at the center, and the vector sum of each such pair of forces is zero. When the field point P is much farther from the ring than the ring's radius, we have $x \gg a$ and the denominator in Eq. (21.8) becomes approximately equal to x^3 . In this limit the electric field at P is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x^2} \hat{i}$$

That is, when the ring is so far away that its radius is negligible in comparison to the distance x , its field is the same as that of a point charge.

KEYCONCEPT To find the vector components of the net electric field at a point due to a continuous distribution of charge, first divide the distribution into infinitesimally small segments. Then find the components of the field at the point due to one such segment. Finally, integrate each component of the field due to a segment over all segments in the charge distribution.

EXAMPLE 21.10 Field of a charged line segment**WITH VARIATION PROBLEMS**

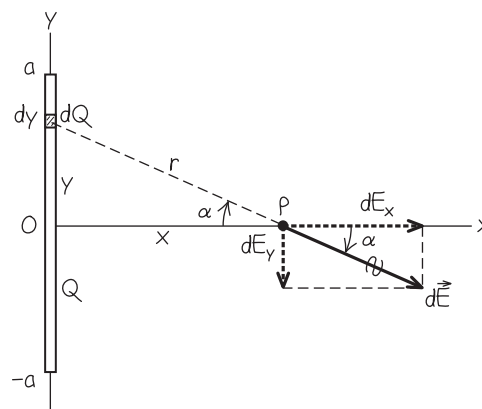
Positive charge Q is distributed uniformly along the y -axis between $y = -a$ and $y = +a$. Find the electric field at point P on the x -axis at a distance x from the origin.

IDENTIFY and SET UP Figure 21.24 shows the situation. As in Example 21.9, we must find the electric field due to a continuous distribution of charge. Our target variable is an expression for the electric field at P as a function of x . The x -axis is a perpendicular bisector of the segment, so we can use a symmetry argument.

EXECUTE We divide the line charge of length $2a$ into infinitesimal segments of length dy . The linear charge density is $\lambda = Q/2a$, and the charge in a segment is $dQ = \lambda dy = (Q/2a)dy$. The distance r from a segment at height y to the field point P is $r = (x^2 + y^2)^{1/2}$, so the magnitude of the field at P due to the segment at height y is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dQ}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{dy}{(x^2 + y^2)}$$

Figure 21.24 Our sketch for this problem.



Continued

Figure 21.24 shows that the x - and y -components of this field are $dE_x = dE \cos \alpha$ and $dE_y = -dE \sin \alpha$, where $\cos \alpha = x/r$ and $\sin \alpha = y/r$. Hence

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{xdy}{(x^2 + y^2)^{3/2}}$$

$$dE_y = -\frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{ydy}{(x^2 + y^2)^{3/2}}$$

To find the total field at P , we must sum the fields from all segments along the line—that is, we must integrate from $y = -a$ to $y = +a$. You should work out the details of the integration (a table of integrals will help). The results are

$$E_x = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \int_{-a}^{+a} \frac{xdy}{(x^2 + y^2)^{3/2}} = \frac{Q}{4\pi\epsilon_0} \frac{1}{x\sqrt{x^2 + a^2}}$$

$$E_y = -\frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \int_{-a}^{+a} \frac{ydy}{(x^2 + y^2)^{3/2}} = 0$$

or, in vector form,

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x\sqrt{x^2 + a^2}} \hat{i} \quad (21.9)$$

$\vec{E}$ points away from the line of charge if λ is positive and toward the line of charge if λ is negative.

EVALUATE Using a symmetry argument as in Example 21.9, we could have guessed that E_y would be zero; if we place a positive test charge at P , the upper half of the line of charge pushes downward on it, and the lower half pushes up with equal magnitude. Symmetry also tells us that the upper and lower halves of the segment contribute equally to the total field at P .

If the segment is very *short* (or the field point is very far from the segment) so that $x \gg a$, we can ignore a in the denominator of Eq. (21.9). Then the field becomes that of a point charge, just as in Example 21.9:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x^2} \hat{i}$$

To see what happens if the segment is very *long* (or the field point is very close to it) so that $a \gg x$, we first rewrite Eq. (21.9) slightly:

$$\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{x\sqrt{(x^2/a^2) + 1}} \hat{i} \quad (21.10)$$

In the limit $a \gg x$ we can ignore x^2/a^2 in the denominator of Eq. (21.10), so

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 x} \hat{i}$$

This is the field of an *infinitely long* line of charge. At any point P at a perpendicular distance r from the line in *any* direction, $\vec{E}$ has magnitude

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{infinite line of charge})$$

Note that this field is proportional to $1/r$ rather than to $1/r^2$ as for a point charge.

There's really no such thing in nature as an infinite line of charge. But when the field point is close enough to the line, there's very little difference between the result for an infinite line and the real-life finite case. For example, if the distance r of the field point from the center of the line is 1% of the length of the line, the value of E differs from the infinite-length value by less than 0.02%.

KEYCONCEPT The electric field due to a symmetrical distribution of charge is most easily calculated at a point of symmetry. Whenever possible, take advantage of the symmetry of the situation to check your results.

EXAMPLE 21.11 Field of a uniformly charged disk

A nonconducting disk of radius R has a uniform positive surface charge density σ . Find the electric field at a point along the axis of the disk a distance x from its center. Assume that x is positive.

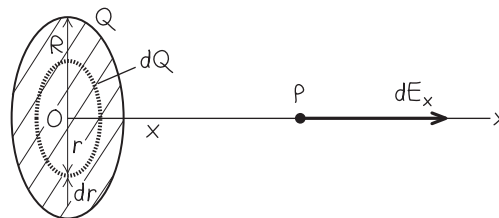
IDENTIFY and SET UP Figure 21.25 shows the situation. We represent the charge distribution as a collection of concentric rings of charge dQ . In Example 21.9 we obtained Eq. (21.8) for the field on the axis of a single uniformly charged ring, so all we need do here is integrate the contributions of our rings.

EXECUTE A typical ring has charge dQ , inner radius r , and outer radius $r + dr$. Its area is approximately equal to its width dr times its circumference $2\pi r$, or $dA = 2\pi r dr$. The charge per unit area is $\sigma = dQ/dA$, so the charge of the ring is $dQ = \sigma dA = 2\pi\sigma r dr$. We use dQ in place of Q in Eq. (21.8), the expression for the field due to a ring that we found in Example 21.9, and replace the ring radius a with r . Then the field component dE_x at point P due to this ring is

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{2\pi\sigma r x dr}{(x^2 + r^2)^{3/2}}$$

To find the total field due to all the rings, we integrate dE_x over r from $r = 0$ to $r = R$ (not from $-R$ to R):

Figure 21.25 Our sketch for this problem.



$$E_x = \int_0^R \frac{1}{4\pi\epsilon_0} \frac{(2\pi\sigma r dr)x}{(x^2 + r^2)^{3/2}} = \frac{\sigma x}{4\epsilon_0} \int_0^R \frac{2r dr}{(x^2 + r^2)^{3/2}}$$

You can evaluate this integral by making the substitution $t = x^2 + r^2$ (which yields $dt = 2r dr$); you can work out the details. The result is

$$E_x = \frac{\sigma x}{2\epsilon_0} \left[-\frac{1}{\sqrt{x^2 + R^2}} + \frac{1}{x} \right]$$

$$= \frac{\sigma}{2\epsilon_0} \left[1 - \frac{1}{\sqrt{(R^2/x^2) + 1}} \right] \quad (21.11)$$

EVALUATE If the disk is very large (or if we are very close to it), so that $R \gg x$, the term $1/\sqrt{(R^2/x^2) + 1}$ in Eq. (21.11) is very much less than 1. Then Eq. (21.11) becomes

$$E = \frac{\sigma}{2\epsilon_0} \quad (21.12)$$

Our final result does not contain the distance x from the plane. Hence the electric field produced by an *infinite* plane sheet of charge is *independent of the distance from the sheet*. The field direction is everywhere perpendicular to the sheet, away from it. There is no such thing as an infinite sheet of charge, but if the dimensions of the sheet are

much larger than the distance x of the field point P from the sheet, the field is very nearly given by Eq. (21.12).

If P is to the *left* of the plane ($x < 0$), the result is the same except that the direction of $\vec{E}$ is to the left instead of the right. If the surface charge density is negative, the directions of the fields on both sides of the plane are toward it rather than away from it.

KEYCONCEPT To find the electric field due to a two-dimensional charge distribution such as a disk, divide the distribution into infinitesimal segments such as rings for which you know the components of the electric field. Then integrate over these segments to find the net field.

EXAMPLE 21.12 Field of two oppositely charged infinite sheets

Two infinite plane sheets with uniform surface charge densities $+\sigma$ and $-\sigma$ are placed parallel to each other with separation d (Fig. 21.26). Find the electric field between the sheets, above the upper sheet, and below the lower sheet.

IDENTIFY and SET UP Equation (21.12) gives the electric field due to a single infinite plane sheet of charge. To find the field due to *two* such sheets, we combine the fields by using the principle of superposition (Fig. 21.26).

EXECUTE From Eq. (21.12), both $\vec{E}_1$ and $\vec{E}_2$ have the same magnitude at all points, independent of distance from either sheet:

$$E_1 = E_2 = \frac{\sigma}{2\epsilon_0}$$

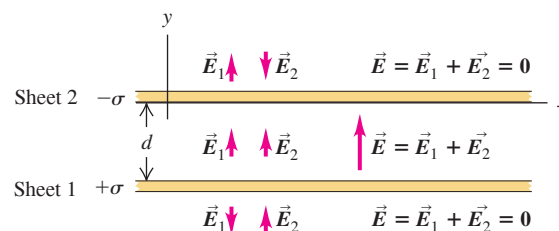
From Example 21.11, $\vec{E}_1$ is everywhere directed away from sheet 1, and $\vec{E}_2$ is everywhere directed toward sheet 2.

Between the sheets, $\vec{E}_1$ and $\vec{E}_2$ reinforce each other; above the upper sheet and below the lower sheet, they cancel each other. Thus the total field is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \begin{cases} 0 & \text{above the upper sheet} \\ \frac{\sigma}{\epsilon_0} \hat{j} & \text{between the sheets} \\ 0 & \text{below the lower sheet} \end{cases}$$

EVALUATE Because we considered the sheets to be infinite, our result does not depend on the separation d . Our result shows that the field between oppositely charged plates is essentially uniform if the plate separation is much smaller than the dimensions of the plates. We actually used this result in Example 21.7 (Section 21.4).

Figure 21.26 Finding the electric field due to two oppositely charged infinite sheets. The sheets are seen edge-on; only a portion of the infinite sheets can be shown!



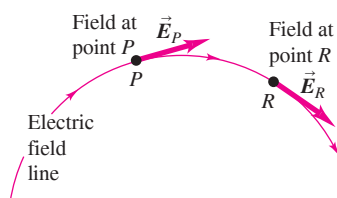
CAUTION Electric fields are not “flows” You may have thought that the field $\vec{E}_1$ of sheet 1 would be unable to “penetrate” sheet 2, and that field $\vec{E}_2$ caused by sheet 2 would be unable to “penetrate” sheet 1. You might conclude this if you think of the electric field as some kind of physical substance that “flows” into or out of charges. But there is no such substance, and the electric fields $\vec{E}_1$ and $\vec{E}_2$ depend on only the individual charge distributions that create them. The *total* field at every point is just the vector sum of $\vec{E}_1$ and $\vec{E}_2$.

KEYCONCEPT An infinite, uniform sheet of charge produces a uniform electric field at all points. If the charge is positive, the field points away from the sheet; if the charge is negative, the field points toward the sheet.

TEST YOUR UNDERSTANDING OF SECTION 21.5 Suppose that the line of charge in Fig. 21.24 (Example 21.10) had charge $+Q$ distributed uniformly between $y = 0$ and $y = +a$ and had charge $-Q$ distributed uniformly between $y = 0$ and $y = -a$. In this situation, the electric field at P would be (i) in the positive x -direction; (ii) in the negative x -direction; (iii) in the positive y -direction; (iv) in the negative y -direction; (v) zero; (vi) none of these.

ANSWER (iv) Think of a pair of segments of length dy , one at coordinate $y > 0$ and the other at coordinate $-y > 0$. The upper segment has a positive charge and produces an electric field $d\vec{E}$ at P that points away from the segment, so this $d\vec{E}$ has a positive x -component and a negative y -component, like the vector $d\vec{E}$ in Fig. 21.24. The lower segment has the same amount of negative charge. It produces a $d\vec{E}$ that has the same magnitude but points *toward* the lower segment, so it has a negative x -component and a negative y -component. By symmetry, the two x -components are equal but opposite, so they cancel. Thus the total electric field has only a negative y -component.

Figure 21.27 The direction of the electric field at any point is tangent to the field line through that point.



21.6 ELECTRIC FIELD LINES

The concept of an electric field can be a little elusive because you can't see an electric field directly. Electric field *lines* can be a big help for visualizing electric fields and making them seem more real. An **electric field line** is an imaginary line or curve drawn through a region of space so that its tangent at any point is in the direction of the electric-field vector at that point. **Figure 21.27** shows the basic idea. (We used a similar concept in our discussion of fluid flow in Section 12.5. A *streamline* is a line or curve whose tangent at any point is in the direction of the velocity of the fluid at that point. However, the similarity between electric field lines and fluid streamlines is a mathematical one only; there is nothing “flowing” in an electric field.) The English scientist Michael Faraday (1791–1867) first introduced the concept of field lines. He called them “lines of force,” but the term “field lines” is preferable.

Electric field lines show the direction of $\vec{E}$ at each point, and their spacing gives a general idea of the *magnitude* of $\vec{E}$ at each point. Where $\vec{E}$ is strong, we draw lines close together; where $\vec{E}$ is weaker, they are farther apart. At any particular point, the electric field has a unique direction, so only one field line can pass through each point of the field. In other words, *field lines never intersect*.

Figure 21.28 shows some of the electric field lines in a plane containing (a) a single positive charge; (b) two equal-magnitude charges, one positive and one negative (a dipole); and (c) two equal positive charges. Such diagrams are called *field maps*; they are cross sections of the actual three-dimensional patterns. The direction of the total electric field at every point in each diagram is along the tangent to the electric field line passing through the point. Arrowheads indicate the direction of the $\vec{E}$ -field vector along each field line. The actual field vectors have been drawn at several points in each pattern. Notice that in general, the magnitude of the electric field is different at different points on a given field line; a field line is *not* a curve of constant electric-field magnitude!

Figure 21.28 shows that field lines are directed *away* from positive charges (since close to a positive point charge, $\vec{E}$ points away from the charge) and *toward* negative charges (since close to a negative point charge, $\vec{E}$ points toward the charge). In regions where the field magnitude is large, such as between the positive and negative charges in Fig. 21.28b, the field lines are drawn close together. In regions where the field magnitude is small, such as between the two positive charges in Fig. 21.28c, the lines are widely separated. In a *uniform* field, the field lines are straight, parallel, and uniformly spaced, as in Fig. 21.20.

Figure 21.28 Electric field lines for three different charge distributions. In general, the magnitude of $\vec{E}$ is different at different points along a given field line.

(a) A single positive charge

(b) Two equal and opposite charges (a dipole)

(c) Two equal positive charges

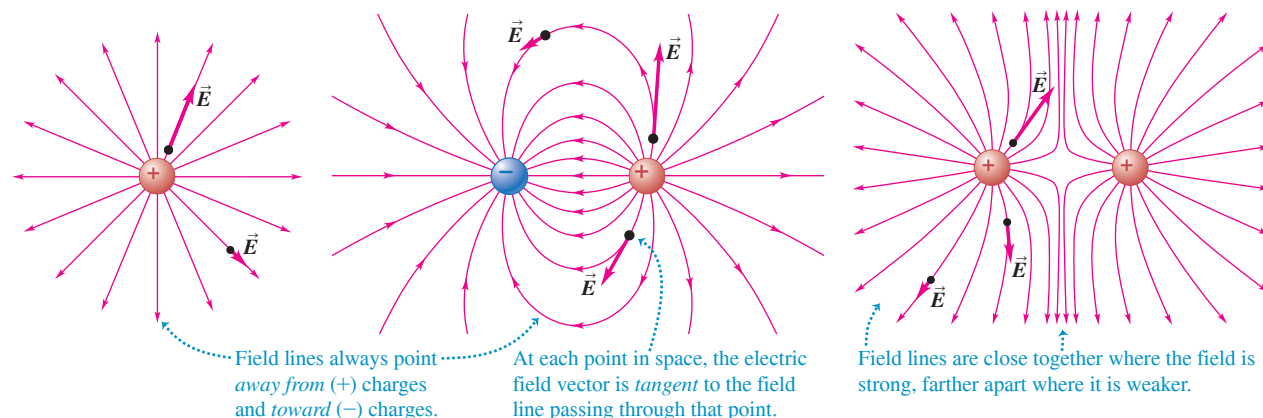


Figure 21.29a is a view from above of a demonstration setup for visualizing electric field lines. In the arrangement shown here, the tips of two positively charged wires are inserted in a container of insulating liquid, and some grass seeds are floated on the liquid. The grass seeds are electrically neutral insulators, but the electric field of the two charged wires causes *polarization* of the grass seeds; there is a slight shifting of the positive and negative charges within the molecules of each seed, like that shown in Fig. 21.8. The positively charged end of each grass seed is pulled in the direction of $\vec{E}$ and the negatively charged end is pulled opposite $\vec{E}$. Hence the long axis of each grass seed tends to orient parallel to the electric field, in the direction of the field line that passes through the position of the seed (Fig. 21.29b).

CAUTION Electric field lines are not trajectories It's a common misconception that if a particle of charge q is in motion where there is an electric field, the particle must move along an electric field line. Because $\vec{E}$ at any point is tangent to the field line that passes through that point, it is true that the force $\vec{F} = q\vec{E}$ on the particle, and hence the particle's acceleration, are tangent to the field line. But we learned in Chapter 3 that when a particle moves on a curved path, its acceleration *cannot* be tangent to the path. In general, the trajectory of a charged particle is *not* the same as a field line. ■

TEST YOUR UNDERSTANDING OF SECTION 21.6 Suppose the electric field lines in a region of space are straight lines. If a charged particle is released from rest in that region, will the trajectory of the particle be along a field line?

ANSWER

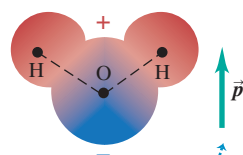
■ **yes** If the field lines are straight, $\vec{E}$ must point in the same direction throughout the region. Hence the force $\vec{F} = q\vec{E}$ on a particle of charge q is always in the same direction. A particle released from rest accelerates in a straight line in the direction of $\vec{F}$, and so its trajectory is a straight line along a field line.

21.7 ELECTRIC DIPOLES

An **electric dipole** is a pair of point charges with equal magnitude and opposite sign (a positive charge q and a negative charge $-q$) separated by a distance d . We introduced electric dipoles in Example 21.8 (Section 21.5); the concept is worth exploring further because many physical systems, from molecules to TV antennas, can be described as electric dipoles. We'll also use this concept extensively in our discussion of dielectrics in Chapter 24.

Figure 21.30a shows a molecule of water (H_2O), which in many ways behaves **?** like an electric dipole. The water molecule as a whole is electrically neutral, but the chemical bonds within the molecule cause a displacement of charge; the result is a net negative charge on the oxygen end of the molecule and a net positive charge on the hydrogen end, forming an electric dipole. The effect is equivalent to shifting one electron only about 4×10^{-11} m (about the radius of a hydrogen atom), but the consequences of this shift are profound. Water is an excellent solvent for ionic substances such as table salt (sodium chloride, NaCl) precisely because the water molecule is an electric dipole (Fig. 21.30b). When dissolved in water, salt dissociates into a positive sodium ion (Na^+)

(a) A water molecule, showing positive charge as red and negative charge as blue



The electric dipole moment $\vec{p}$ is directed from the negative end to the positive end of the molecule.

(b) Various substances dissolved in water



Figure 21.29 (a) Electric field lines produced by two opposite charges. The pattern is formed by grass seeds floating in mineral oil; the charges are on two wires whose tips are inserted into the oil. Compare this pattern with Fig. 21.28b. (b) The electric field causes polarization of the grass seeds, which in turn causes the seeds to align with the field.

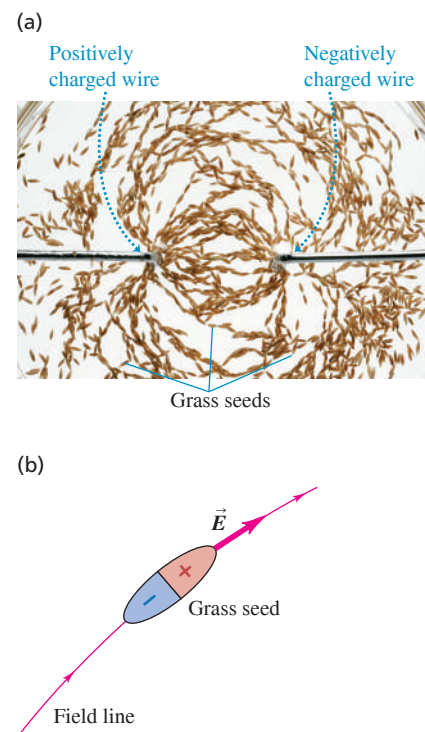
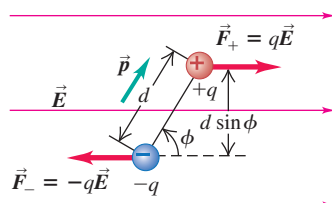


Figure 21.30 (a) A water molecule is an example of an electric dipole. (b) Each test tube contains a solution of a different substance in water. The large electric dipole moment of water makes it an excellent solvent.

and a negative chlorine ion (Cl^-), which tend to be attracted to the negative and positive ends, respectively, of water molecules; this holds the ions in solution. If water molecules were not electric dipoles, water would be a poor solvent, and almost all of the chemistry that occurs in aqueous solutions would be impossible. This includes all of the biochemical reactions that occur in all of the life on earth. In a very real sense, your existence as a living being depends on electric dipoles!

We examine two questions about electric dipoles. First, what forces and torques does an electric dipole experience when placed in an external electric field (that is, a field set up by charges outside the dipole)? Second, what electric field does an electric dipole itself produce?

Figure 21.31 The net force on this electric dipole is zero, but there is a torque directed into the page that tends to rotate the dipole clockwise.



Force and Torque on an Electric Dipole

To start with the first question, let's place an electric dipole in a *uniform* external electric field $\vec{E}$, as shown in Fig. 21.31. Both forces $\vec{F}_+$ and $\vec{F}_-$ on the two charges have magnitude qE , but their directions are opposite, and they add to zero. *The net force on an electric dipole in a uniform external electric field is zero.*

However, the two forces don't act along the same line, so their *torques* don't add to zero. We calculate torques with respect to the center of the dipole. Let the angle between the electric field $\vec{E}$ and the dipole axis be ϕ ; then the lever arm for both $\vec{F}_+$ and $\vec{F}_-$ is $(d/2) \sin \phi$. The torque of $\vec{F}_+$ and the torque of $\vec{F}_-$ both have the same magnitude of $(qE)(d/2) \sin \phi$, and both torques tend to rotate the dipole clockwise (that is, $\vec{\tau}$ is directed into the page in Fig. 21.31). Hence the magnitude of the net torque is twice the magnitude of either individual torque:

$$\tau = (qE)(d \sin \phi) \quad (21.13)$$

where $d \sin \phi$ is the perpendicular distance between the lines of action of the two forces.

The product of the charge q and the separation d is the magnitude of a quantity called the **electric dipole moment**, denoted by p :

$$p = qd \quad (\text{magnitude of electric dipole moment}) \quad (21.14)$$

The units of p are charge times distance ($\text{C} \cdot \text{m}$). For example, the magnitude of the electric dipole moment of a water molecule is $p = 6.13 \times 10^{-30} \text{ C} \cdot \text{m}$.

CAUTION The symbol p has multiple meanings Do not confuse dipole moment with momentum or pressure. There aren't as many letters in the alphabet as there are physical quantities, so some letters are used several times. The context usually makes it clear what we mean, but be careful. ■

We further define the electric dipole moment to be a *vector* quantity $\vec{p}$. The magnitude of $\vec{p}$ is given by Eq. (21.14), and its direction is along the dipole axis from the negative charge to the positive charge as shown in Fig. 21.31.

In terms of p , Eq. (21.13) for the magnitude τ of the torque exerted by the field becomes

$$\text{Magnitude of torque on an electric dipole} \quad \tau = pE \sin \phi \quad \begin{array}{l} \text{Magnitude of electric field } \vec{E} \\ \text{Angle between } \vec{p} \text{ and } \vec{E} \\ \text{Magnitude of electric dipole moment } \vec{p} \end{array} \quad (21.15)$$

Since the angle ϕ in Fig. 21.31 is the angle between the directions of the vectors $\vec{p}$ and $\vec{E}$, this is reminiscent of the expression for the magnitude of the *vector product* discussed in Section 1.10. (You may want to review that discussion.) Hence we can write the torque on the dipole in vector form as

$$\vec{\tau} = \vec{p} \times \vec{E} \quad (21.16)$$

Vector torque on an electric dipole Electric dipole moment Electric field

You can use the right-hand rule for the vector product to verify that in the situation shown in Fig. 21.31, $\vec{\tau}$ is directed into the page. The torque is greatest when $\vec{p}$ and $\vec{E}$ are perpendicular and is zero when they are parallel or antiparallel. The torque always tends to turn $\vec{p}$ to line it up with $\vec{E}$. The position $\phi = 0$, with $\vec{p}$ parallel to $\vec{E}$, is a position of stable equilibrium, and the position $\phi = \pi$, with $\vec{p}$ and $\vec{E}$ antiparallel, is a position of unstable equilibrium. The polarization of a grass seed in the apparatus of Fig. 21.29b gives it an electric dipole moment; the torque exerted by $\vec{E}$ then causes the seed to align with $\vec{E}$ and hence with the field lines.

Potential Energy of an Electric Dipole

When a dipole changes direction in an electric field, the electric-field torque does *work* on it, with a corresponding change in potential energy. The work dW done by a torque τ during an infinitesimal displacement $d\phi$ is given by Eq. (10.19): $dW = \tau d\phi$. Because the torque is in the direction of decreasing ϕ , we must write the torque as $\tau = -pE \sin \phi$, and

$$dW = \tau d\phi = -pE \sin \phi d\phi$$

In a finite displacement from ϕ_1 to ϕ_2 the total work done on the dipole is

$$\begin{aligned} W &= \int_{\phi_1}^{\phi_2} (-pE \sin \phi) d\phi \\ &= pE \cos \phi_2 - pE \cos \phi_1 \end{aligned}$$

The work is the negative of the change of potential energy, just as in Chapter 7: $W = U_1 - U_2$. So a suitable definition of potential energy U for this system is

$$U(\phi) = -pE \cos \phi \quad (21.17)$$

In this expression we recognize the *scalar product* $\vec{p} \cdot \vec{E} = pE \cos \phi$, so we can also write

$$U = -\vec{p} \cdot \vec{E} \quad (21.18)$$

Potential energy for an electric dipole in an electric field Electric field Electric dipole moment

The potential energy has its minimum (most negative) value $U = -pE$ at the stable equilibrium position, where $\phi = 0$ and $\vec{p}$ is parallel to $\vec{E}$. The potential energy is maximum when $\phi = \pi$ and $\vec{p}$ is antiparallel to $\vec{E}$; then $U = +pE$. At $\phi = \pi/2$, where $\vec{p}$ is perpendicular to $\vec{E}$, U is zero. We could define U differently so that it is zero at some other orientation of $\vec{p}$, but our definition is simplest.

Equation (21.18) gives us another way to look at the effect illustrated in Fig. 21.29. The electric field $\vec{E}$ gives each grass seed an electric dipole moment, and the grass seed then aligns itself with $\vec{E}$ to minimize the potential energy.

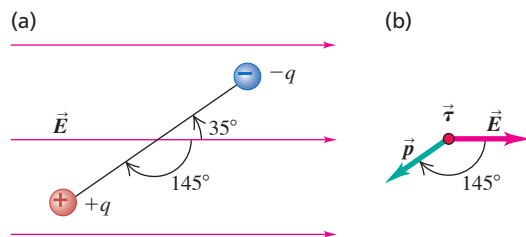
EXAMPLE 21.13 Force and torque on an electric dipoleWITH  **VARIATION PROBLEMS**

Figure 21.32a (next page) shows an electric dipole in a uniform electric field of magnitude $5.0 \times 10^5 \text{ N/C}$ that is directed parallel to the plane of the figure. The charges are $\pm 1.6 \times 10^{-19} \text{ C}$; both lie in the plane and are separated by $0.125 \text{ nm} = 0.125 \times 10^{-9} \text{ m}$. Find (a) the net force exerted by the field on the dipole; (b) the magnitude and direction of the electric dipole moment; (c) the magnitude and direction of the torque; (d) the potential energy of the system in the position shown.

IDENTIFY and SET UP This problem uses the ideas of this section about an electric dipole placed in an electric field. We use the relationship $\vec{F} = q\vec{E}$ for each point charge to find the force on the dipole as a whole. Equation (21.14) gives the dipole moment, Eq. (21.16) gives the torque on the dipole, and Eq. (21.18) gives the potential energy of the system.

EXECUTE (a) The field is uniform, so the forces on the two charges are equal and opposite. Hence the total force on the dipole is zero.

Figure 21.32 (a) An electric dipole. (b) Directions of the electric dipole moment, electric field, and torque ($\vec{\tau}$ points out of the page).



(b) The magnitude p of the electric dipole moment $\vec{p}$ is

$$\begin{aligned} p &= qd = (1.6 \times 10^{-19} \text{ C})(0.125 \times 10^{-9} \text{ m}) \\ &= 2.0 \times 10^{-29} \text{ C} \cdot \text{m} \end{aligned}$$

The direction of $\vec{p}$ is from the negative to the positive charge, 145° clockwise from the electric-field direction (Fig. 21.32b).

(c) The magnitude of the torque is

$$\begin{aligned} \tau &= pE \sin \phi = (2.0 \times 10^{-29} \text{ C} \cdot \text{m})(5.0 \times 10^5 \text{ N/C})(\sin 145^\circ) \\ &= 5.7 \times 10^{-24} \text{ N} \cdot \text{m} \end{aligned}$$

From the right-hand rule for vector products (see Section 1.10), the direction of the torque $\vec{\tau} = \vec{p} \times \vec{E}$ is out of the page. This corresponds to a counterclockwise torque that tends to align $\vec{p}$ with $\vec{E}$.

(d) The potential energy

$$\begin{aligned} U &= -pE \cos \phi \\ &= -(2.0 \times 10^{-29} \text{ C} \cdot \text{m})(5.0 \times 10^5 \text{ N/C})(\cos 145^\circ) \\ &= 8.2 \times 10^{-24} \text{ J} \end{aligned}$$

EVALUATE The charge magnitude, the distance between the charges, the dipole moment, and the potential energy are all very small, but are all typical of molecules.

KEYCONCEPT An electric dipole placed in an electric field tends to orient itself with its dipole moment in the same direction as the field.

BIO APPLICATION A Fish with an Electric Dipole Moment

Unlike the tiger shark (Section 21.4), which senses the electric fields produced by its prey, the African knifefish *Gymnarchus niloticus*—which is nocturnal and has poor vision—hunts other fish by generating its own electric field. It can make its tail negatively charged relative to its head, producing a dipole-like field similar to that in Fig. 21.28b. When a smaller fish ventures into the field, its body alters the field pattern and alerts *G. niloticus* that a meal is present.



In this discussion we have assumed that $\vec{E}$ is uniform, so there is no net force on the dipole. If $\vec{E}$ is not uniform, the forces at the ends may not cancel completely, and the net force may not be zero. Thus an object with zero net charge but an electric dipole moment can experience a net force in a nonuniform electric field. As we mentioned in Section 21.1, an uncharged object can be polarized by an electric field, giving rise to a separation of charge and an electric dipole moment. This is how uncharged objects can experience electrostatic forces (see Fig. 21.8).

Field of an Electric Dipole

Now let's think of an electric dipole as a *source* of electric field. Figure 21.28b shows the general shape of the field due to a dipole. At each point in the pattern the total $\vec{E}$ field is the vector sum of the fields from the two individual charges, as in Example 21.8 (Section 21.5). Try drawing diagrams showing this vector sum for several points.

To get quantitative information about the field of an electric dipole, we have to do some calculating, as illustrated in the next example. Notice the use of the principle of superposition of electric fields to add up the contributions to the field of the individual charges. Also notice that we need to use approximation techniques even for the relatively simple case of a field due to two charges. Field calculations often become very complicated, and computer analysis is typically used to determine the field due to an arbitrary charge distribution.

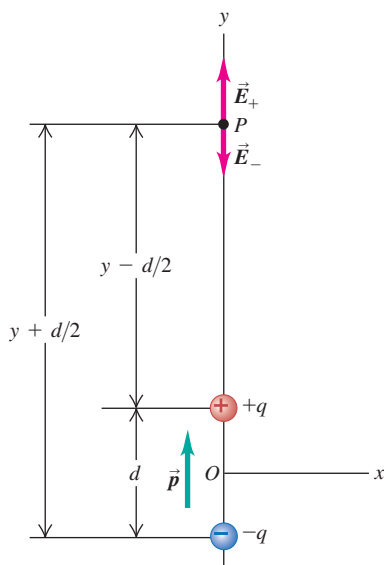
EXAMPLE 21.14 Field of an electric dipole, revisited

WITH VARIATION PROBLEMS

An electric dipole is centered at the origin, with $\vec{p}$ in the direction of the $+y$ -axis (**Fig. 21.33**). Derive an approximate expression for the electric field at a point P on the y -axis for which y is much larger than d . To do this, use the binomial expansion $(1 + x)^n \cong 1 + nx + n(n-1)x^2/2 + \cdots$ (valid for the case $|x| < 1$).

IDENTIFY and SET UP We use the principle of superposition: The total electric field is the vector sum of the field produced by the positive charge and the field produced by the negative charge. At the field point P shown in Fig. 21.33, the field $\vec{E}_+$ of the positive charge has a positive (upward) y -component and the field $\vec{E}_-$ of the negative charge has a negative (downward) y -component. We add these components to find the total field and then apply the approximation that y is much greater than d .

Figure 21.33 Finding the electric field of an electric dipole at a point on its axis.



EXECUTE The total y -component E_y of electric field from the two charges is

$$E_y = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(y - d/2)^2} - \frac{1}{(y + d/2)^2} \right] \\ = \frac{q}{4\pi\epsilon_0 y^2} \left[\left(1 - \frac{d}{2y}\right)^{-2} - \left(1 + \frac{d}{2y}\right)^{-2} \right]$$

We used this same approach in Example 21.8 (Section 21.5). Now the approximation: When we are far from the dipole compared to its size, so $y \gg d$, we have $d/2y \ll 1$. With $n = -2$ and with $d/2y$ replacing x in the binomial expansion, we keep only the first two terms (the terms we discard are much smaller). We then have

$$\left(1 - \frac{d}{2y}\right)^{-2} \cong 1 + \frac{d}{y} \quad \text{and} \quad \left(1 + \frac{d}{2y}\right)^{-2} \cong 1 - \frac{d}{y}$$

Hence E_y is given approximately by

$$E_y \cong \frac{q}{4\pi\epsilon_0 y^2} \left[1 + \frac{d}{y} - \left(1 - \frac{d}{y}\right) \right] = \frac{qd}{2\pi\epsilon_0 y^3} = \frac{p}{2\pi\epsilon_0 y^3}$$

EVALUATE An alternative route to this result is to put the fractions in the first expression for E_y over a common denominator, add, and then approximate the denominator $(y - d/2)^2(y + d/2)^2$ as y^4 . We leave the details to you.

For points P off the coordinate axes, the expressions are more complicated, but at *all* points far away from the dipole (in any direction) the field drops off as $1/r^3$. We can compare this with the $1/r^2$ behavior of a point charge, the $1/r$ behavior of a long line charge, and the independence of r for a large sheet of charge. There are charge distributions for which the field drops off even more quickly. At large distances, the field of an *electric quadrupole*, which consists of two equal dipoles with opposite orientation, separated by a small distance, drops off as $1/r^4$.

KEYCONCEPT The electric field produced by an electric dipole is the vector sum of the fields produced by the positive and negative charges that make up the dipole. At distances r from the dipole that are large compared to the dipole size, the field magnitude is inversely proportional to the cube of r .

TEST YOUR UNDERSTANDING OF SECTION 21.7 An electric dipole is placed in a region of uniform electric field $\vec{E}$, with the electric dipole moment $\vec{p}$ pointing in the direction opposite to $\vec{E}$. Is the dipole (i) in stable equilibrium, (ii) in unstable equilibrium, or (iii) neither? (*Hint:* You may want to review Section 7.5.)

ANSWER

(ii) Equations (21.17) and (21.18) tell us that the potential energy for a dipole in an electric field is $U = -\vec{p} \cdot \vec{E} = -pE \cos \phi$, where ϕ is the angle between the directions of $\vec{p}$ and $\vec{E}$. If $\vec{p}$ and $\vec{E}$ point in opposite directions, so that $\phi = 180^\circ$, we have $\cos \phi = -1$ and $U = +pE$. This is the maximum value that U can have. From our discussion of energy diagrams in Section 7.5, it follows that this is a situation of unstable equilibrium.

CHAPTER 21 SUMMARY

Electric charge, conductors, and insulators: The fundamental quantity in electrostatics is electric charge. There are two kinds of charge, positive and negative. Charges of the same sign repel each other; charges of opposite sign attract. Charge is conserved; the total charge in an isolated system is constant.

All ordinary matter is made of protons, neutrons, and electrons. The positive protons and electrically neutral neutrons in the nucleus of an atom are bound together by the nuclear force; the negative electrons surround the nucleus at distances much greater than the nuclear size. Electric interactions are chiefly responsible for the structure of atoms, molecules, and solids.

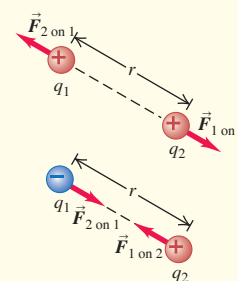
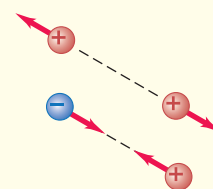
Conductors are materials in which charge moves easily; in insulators, charge does not move easily. Most metals are good conductors; most nonmetals are insulators.

Coulomb's law: For charges q_1 and q_2 separated by a distance r , the magnitude of the electric force on either charge is proportional to the product $q_1 q_2$ and inversely proportional to r^2 . The force on each charge is along the line joining the two charges—repulsive if q_1 and q_2 have the same sign, attractive if they have opposite signs. In SI units the unit of electric charge is the coulomb, abbreviated C. (See Examples 21.1 and 21.2.)

When two or more charges each exert a force on a charge, the total force on that charge is the vector sum of the forces exerted by the individual charges. (See Examples 21.3 and 21.4.)

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} \quad (21.2)$$

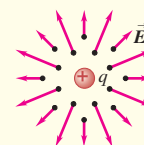
$$\frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$



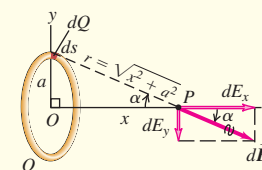
Electric field: Electric field $\vec{E}$, a vector quantity, is the force per unit charge exerted on a test charge at any point. The electric field produced by a point charge is directed radially away from a positive charge and radially toward a negative charge. (See Examples 21.5–21.7.)

$$\vec{E} = \frac{\vec{F}_0}{q_0} \quad (21.3)$$

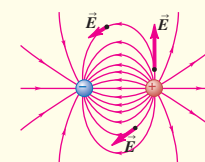
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (21.7)$$



Superposition of electric fields: The electric field $\vec{E}$ of any combination of charges is the vector sum of the fields caused by the individual charges. To calculate the electric field caused by a continuous distribution of charge, divide the distribution into small elements, calculate the field caused by each element, and then carry out the vector sum, usually by integrating. Charge distributions are described by linear charge density λ , surface charge density σ , and volume charge density ρ . (See Examples 21.8–21.12.)



Electric field lines: Field lines provide a graphical representation of electric fields. At any point on a field line, the tangent to the line is in the direction of $\vec{E}$ at that point. The number of lines per unit area (perpendicular to their direction) is proportional to the magnitude of $\vec{E}$ at the point.

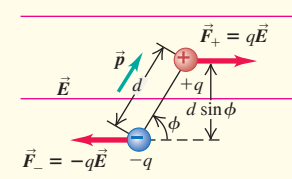


Electric dipoles: An electric dipole is a pair of electric charges of equal magnitude q but opposite sign, separated by a distance d . The electric dipole moment $\vec{p}$ has magnitude $p = qd$. The direction of $\vec{p}$ is from negative toward positive charge. An electric dipole in an electric field $\vec{E}$ experiences a torque $\vec{\tau}$ equal to the vector product of $\vec{p}$ and $\vec{E}$. The magnitude of the torque depends on the angle ϕ between $\vec{p}$ and $\vec{E}$. The potential energy U for an electric dipole in an electric field also depends on the relative orientation of $\vec{p}$ and $\vec{E}$. (See Examples 21.13 and 21.14.)

$$\tau = pE \sin \phi \quad (21.15)$$

$$\vec{\tau} = \vec{p} \times \vec{E} \quad (21.16)$$

$$U = -\vec{p} \cdot \vec{E} \quad (21.18)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 21.3 and 21.4** (Section 21.3) before attempting these problems.

VP21.4.1 Charge $q_1 = +4.00$ nC is on the x -axis at $x = 0$ and charge $q_2 = -1.20$ nC is on the x -axis at $x = 6.00$ cm. Find the total electric force exerted by q_1 and q_2 on a third charge $q_3 = -2.00$ nC on the x -axis at $x = 4.00$ cm.

VP21.4.2 Charge $q_1 = +3.60$ nC is on the x -axis at $x = 0$ and an unknown charge q_2 is on the x -axis at $x = -4.00$ cm. The total electric force that q_1 and q_2 exert on a third charge $q_3 = +2.00$ nC on the x -axis at $x = 4.00$ cm is in the $+x$ -direction and has magnitude 54.0 μ N. Find (a) the magnitude and direction of the force that q_1 exerts on q_3 , (b) the magnitude and direction of the force that q_2 exerts on q_3 , and (c) the charge q_2 .

VP21.4.3 Charge $q_1 = +6.00$ nC is at $x = 0$, $y = 0.200$ m and charge $q_2 = -6.00$ nC is at $x = 0$, $y = -0.200$ m. Find the magnitude and direction of the total electric force exerted by q_1 and q_2 on a third charge $q_3 = -2.50$ nC at $x = 0.150$ m, $y = 0$.

VP21.4.4 A small piece of plastic with charge $q_1 = +4.00$ nC is at $x = 0$, $y = 0.200$ m and a second small piece of plastic with charge $q_2 = -4.00$ nC is at $x = 0$, $y = 0$. Find the x - and y -components of the total electric force exerted by q_1 and q_2 on a third small piece of plastic with charge $q_3 = -1.50$ nC at $x = 0.250$ m, $y = 0$.

Be sure to review **EXAMPLES 21.8, 21.9, and 21.10** (Section 21.5) before attempting these problems.

VP21.10.1 Charge $q_1 = +4.00$ nC is at $x = 0$, $y = 0.200$ m and charge $q_2 = -5.00$ nC is at $x = 0$, $y = 0$. Find the x - and y -components of the total electric field caused by q_1 and q_2 at (a) $x = 0$, $y = 0.100$ m; (b) $x = 0$, $y = 0.400$ m; (c) $x = 0.200$ m, $y = 0$.

VP21.10.2 Charge $q_1 = +1.80$ nC is on the x -axis at $x = 0$ and an unknown charge q_2 is on the x -axis at $x = +4.00$ cm. The total electric field caused by q_1 and q_2 at a point P on the x -axis at $x = +2.00$ cm is in the $+x$ -direction and has magnitude 6.75×10^4 N/C. Find (a) the magnitude and direction of the electric field that q_1 causes at P , (b) the magnitude and direction of the electric field that q_2 causes at P , and (c) the charge q_2 .

VP21.10.3 A simple model of a hydrogen atom is a positive point charge $+e$ (representing the proton) at the center of a ring of radius a

with negative charge $-e$ distributed uniformly around the ring (representing the electron in orbit around the proton). (a) Find the magnitude of the total electric field due to this charge distribution at a point a distance a from the proton and perpendicular to the plane of the ring. (b) Does the field point toward or away from the proton?

VP21.10.4 Positive charge Q is distributed uniformly along a rod of length L that lies along the x -axis from $x = L$ to $x = 2L$. Find (a) how much charge is contained within a segment of the rod of length dx , (b) the x - and y -components of the infinitesimal electric field caused at the origin ($x = 0$) by one such segment at coordinate x , and (c) the x - and y -components of the total electric field caused at the origin by the rod.

Be sure to review **EXAMPLES 21.13 and 21.14** (Section 21.7) before attempting these problems.

VP21.14.1 The electric dipole moment of a water molecule (H_2O) is 6.13×10^{-30} C $\cdot$ m. You place a water molecule in a uniform electric field of magnitude 3.00×10^5 N/C, with the dipole moment vector at an angle of 50.0° from the direction of the field. Calculate (a) the magnitude of the torque on the water molecule and (b) the potential energy for the water molecule.

VP21.14.2 An electric dipole is in a uniform electric field of magnitude 8.50×10^4 N/C. The charges in the dipole are separated by 1.10×10^{-10} m, and the torque on the dipole when its dipole moment is perpendicular to the electric field is 6.60×10^{-26} N $\cdot$ m. Calculate the magnitude of the charges that make up the dipole.

VP21.14.3 You place a molecule in a uniform electric field of magnitude 1.20×10^5 N/C. It requires 4.60×10^{-25} J of work to change the orientation of the molecule's dipole moment from parallel to the field to opposite to the field. Calculate the magnitude of the dipole moment.

VP21.14.4 The ionic molecule potassium bromide (KBr), made up of a positive potassium ion (K^+) of charge $+e = 1.60 \times 10^{-19}$ C and a negative bromine ion (Br^-) of charge $-e = -1.60 \times 10^{-19}$ C, has an electric dipole moment of 3.50×10^{-29} C $\cdot$ m. (a) Calculate the distance between the two ions. (b) At a certain point along the axis that connects the ions, the electric field due to the KBr molecule has magnitude 8.00×10^4 N/C. How far from the center of the molecule is this point?

BRIDGING PROBLEM Calculating Electric Field: Half a Ring of Charge

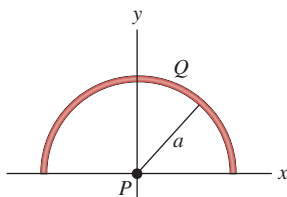
Positive charge Q is uniformly distributed around a semicircle of radius a as shown in **Fig. 21.34**. Find the magnitude and direction of the resulting electric field at point P , the center of curvature of the semicircle.

SOLUTION GUIDE

IDENTIFY and SET UP

1. The target variables are the components of the electric field at P .
2. Divide the semicircle into infinitesimal segments, each of which is a short circular arc of radius a and angle $d\theta$. What is the length of such a segment? How much charge is on a segment?
3. Consider an infinitesimal segment located at an angular position θ on the semicircle, measured from the lower right corner of the semicircle at $x = a$, $y = 0$. (Thus $\theta = \pi/2$ at

Figure 21.34 Charge uniformly distributed around a semicircle.



$x = 0$, $y = a$ and $\theta = \pi$ at $x = -a$, $y = 0$.) What are the x - and y -components of the electric field at P (dE_x and dE_y) produced by just this segment?

EXECUTE

4. Integrate your expressions for dE_x and dE_y from $\theta = 0$ to $\theta = \pi$. The results will be the x -component and y -component of the electric field at P .
5. Use your results from step 4 to find the magnitude and direction of the field at P .

EVALUATE

6. Does your result for the electric-field magnitude have the correct units?
7. Explain how you could have found the x -component of the electric field by using a symmetry argument.
8. What would be the electric field at P if the semicircle were extended to a full circle centered at P ?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q21.1 If you peel two strips of transparent tape off the same roll and immediately let them hang near each other, they will repel each other. If you then stick the sticky side of one to the shiny side of the other and rip them apart, they will attract each other. Give a plausible explanation, involving transfer of electrons between the strips of tape, for this sequence of events.

Q21.2 Two metal spheres are hanging from nylon threads. When you bring the spheres close to each other, they tend to attract. Based on this information alone, discuss all the possible ways that the spheres could be charged. Is it possible that after the spheres touch, they will cling together? Explain.

Q21.3 The electric force between two charged particles becomes weaker with increasing distance. Suppose instead that the electric force were *independent* of distance. In this case, would a charged comb still cause a neutral insulator to become polarized as in Fig. 21.8? Why or why not? Would the neutral insulator still be attracted to the comb? Again, why or why not?

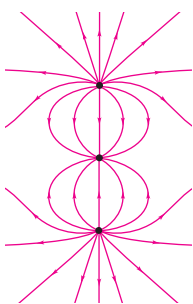
Q21.4 Your clothing tends to cling together after going through the dryer. Why? Would you expect more or less clinging if all your clothing were made of the same material (say, cotton) than if you dried different kinds of clothing together? Again, why? (You may want to experiment with your next load of laundry.)

Q21.5 An uncharged metal sphere hangs from a nylon thread. When a positively charged glass rod is brought close to the metal sphere, the sphere is drawn toward the rod. But if the sphere touches the rod, it suddenly flies away from the rod. Explain why the sphere is first attracted and then repelled.

Q21.6 **BIO** Estimate how many electrons there are in your body. Make any assumptions you feel are necessary, but clearly state what they are. (*Hint*: Most of the atoms in your body have equal numbers of electrons, protons, and neutrons.) What is the combined charge of all these electrons?

Q21.7 **Figure Q21.7** shows some of the electric field lines due to three point charges arranged along the vertical axis. All three charges have the same magnitude. (a) What are the signs of the three charges? Explain your reasoning. (b) At what point(s) is the magnitude of the electric field the smallest? Explain your reasoning. Explain how the fields produced by each individual point charge combine to give a small net field at this point or points.

Figure Q21.7



Q21.8 Good conductors of electricity, such as metals, are typically good conductors of heat; insulators, such as wood, are typically poor conductors of heat. Explain why there is a relationship between conduction of electricity and conduction of heat in these materials.

Q21.9 Suppose that the charge shown in Fig. 21.28a is fixed in position. A small, positively charged particle is then placed at some location and released from rest. Will the trajectory of the particle follow an electric field line? Why or why not? Suppose instead that the particle is placed at some point in Fig. 21.28b and released from rest (the positive and negative charges shown are fixed in position). Will its trajectory follow an electric field line? Again, why or why not? Explain any differences between your answers for the two situations.

Q21.10 Two identical metal objects are mounted on insulating stands. Describe how you could place charges of opposite sign but exactly equal magnitude on the two objects.

Q21.11 Because the charges on the electron and proton have the same absolute value, atoms are electrically neutral. Suppose that this is not precisely true, and the absolute value of the charge of the electron is less than the charge of the proton by 0.00100%. Estimate what the net charge of this textbook would be under these circumstances. Make any assumptions you feel are justified, but state clearly what they are. (*Hint*: Most of the atoms in this textbook have equal numbers of electrons, protons, and neutrons.) What would be the magnitude of the electric force between two textbooks placed 5.0 m apart? Would this force be attractive or repulsive? Discuss how the fact that ordinary matter is stable shows that the absolute values of the charges on the electron and proton must be identical to a *very* high level of accuracy.

Q21.12 If you walk across a nylon rug and then touch a large metal object such as a doorknob, you may get a spark and a shock. Why does this tend to happen more on dry days than on humid days? (*Hint*: See Fig. 21.30.) Why are you less likely to get a shock if you touch a *small* metal object, such as a paper clip?

Q21.13 You have a negatively charged object. How can you use it to place a net negative charge on an insulated metal sphere? To place a net positive charge on the sphere?

Q21.14 When two point charges of equal mass and charge are released on a frictionless table, each has an initial acceleration (magnitude) a_0 . If instead you keep one fixed and release the other one, what will be its initial acceleration: a_0 , $2a_0$, or $a_0/2$? Explain.

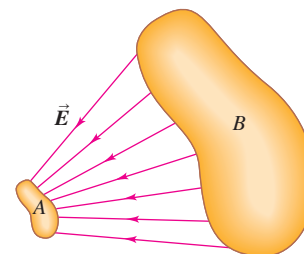
Q21.15 A point charge of mass m and charge Q and another point charge of mass m but charge $2Q$ are released on a frictionless table. If the charge Q has an initial acceleration a_0 , what will be the acceleration of $2Q$: a_0 , $2a_0$, $4a_0$, $a_0/2$, or $a_0/4$? Explain.

Q21.16 A proton is placed in a uniform electric field and then released. Then an electron is placed at this same point and released. Do these two particles experience the same force? The same acceleration? Do they move in the same direction when released?

Q21.17 In Example 21.1 (Section 21.3) we saw that the electric force between two α particles is of the order of 10^{35} times as strong as the gravitational force. So why do we readily feel the gravity of the earth but no electric force from it?

Q21.18 What similarities do electric forces have with gravitational forces? What are the most significant differences?

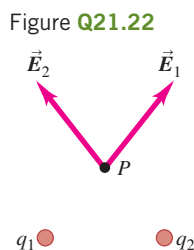
Q21.19 Two irregular objects *A* and *B* carry charges of opposite sign. **Figure Q21.19** shows the electric field lines near each of these objects. (a) Which object is positive, *A* or *B*? How do you know? (b) Where is the electric field stronger, close to *A* or close to *B*? How do you know?



Q21.20 Atomic nuclei are made of protons and neutrons. This shows that there must be another kind of interaction in addition to gravitational and electric forces. Explain.

Q21.21 Sufficiently strong electric fields can cause atoms to become positively ionized—that is, to lose one or more electrons. Explain how this can happen. What determines how strong the field must be to make this happen?

Q21.22 The electric fields at point P due to the positive charges q_1 and q_2 are shown in **Fig. Q21.22**. Does the fact that they cross each other violate the statement in Section 21.6 that electric field lines never cross? Explain.



Q21.23 The air temperature and the velocity of the air have different values at different places in the earth's atmosphere. Is the air velocity a vector field? Why or why not? Is the air temperature a vector field? Again, why or why not?

EXERCISES

Section 21.3 Coulomb's Law

21.1 • Excess electrons are placed on a small lead sphere with mass 8.00 g so that its net charge is $-3.20 \times 10^{-9}\text{ C}$. (a) Find the number of excess electrons on the sphere. (b) How many excess electrons are there per lead atom? The atomic number of lead is 82, and its atomic mass is 207 g/mol .

21.2 • Lightning occurs when there is a flow of electric charge (principally electrons) between the ground and a thundercloud. The maximum rate of charge flow in a lightning bolt is about $20,000\text{ C/s}$; this lasts for $100\text{ }\mu\text{s}$ or less. How much charge flows between the ground and the cloud in this time? How many electrons flow during this time?

21.3 • If a proton and an electron are released when they are $2.0 \times 10^{-10}\text{ m}$ apart (a typical atomic distance), find the initial acceleration of each particle.

21.4 • In Example 21.4, what is the net force (magnitude and direction) on charge q_1 exerted by the other two charges?

21.5 • **BIO Signal Propagation in Neurons.** Neurons are components of the nervous system of the body that transmit signals as electric impulses travel along their length. These impulses propagate when charge suddenly rushes into and then out of a part of the neuron called an *axon*. Measurements have shown that, during the inflow part of this cycle, approximately $5.6 \times 10^{11}\text{ Na}^+$ (sodium ions) per meter, each with charge $+e$, enter the axon. How many coulombs of charge enter a 1.5 cm length of the axon during this process?

21.6 • Two small spheres spaced 20.0 cm apart have equal charge. How many excess electrons must be present on each sphere if the magnitude of the force of repulsion between them is $3.33 \times 10^{-21}\text{ N}$?

21.7 • An average human weighs about 650 N . If each of two average humans could carry 1.0 C of excess charge, one positive and one negative, how far apart would they have to be for the electric attraction between them to equal their 650 N weight?

21.8 • Two small aluminum spheres, each having mass 0.0250 kg , are separated by 80.0 cm . (a) How many electrons does each sphere contain? (The atomic mass of aluminum is 26.982 g/mol , and its atomic number is 13.) (b) How many electrons would have to be removed from one sphere and added to the other to cause an attractive force between the spheres of magnitude $1.00 \times 10^4\text{ N}$ (roughly 1 ton)? Assume that the spheres may be treated as point charges. (c) What fraction of all the electrons in each sphere does this represent?

21.9 • Two small plastic spheres are given positive electric charges. When they are 15.0 cm apart, the repulsive force between them has magnitude 0.220 N . What is the charge on each sphere (a) if the two charges are equal and (b) if one sphere has four times the charge of the other?

21.10 • Two point charges are placed on the x -axis as follows: Charge $q_1 = +4.00\text{ nC}$ is located at $x = 0.200\text{ m}$, and charge $q_2 = +5.00\text{ nC}$ is at $x = -0.300\text{ m}$. What are the magnitude and direction of the total force exerted by these two charges on a negative point charge $q_3 = -6.00\text{ nC}$ that is placed at the origin?

21.11 • In an experiment in space, one proton is held fixed and another proton is released from rest a distance of 2.50 mm away. (a) What is the initial acceleration of the proton after it is released? (b) Sketch qualitative (no numbers!) acceleration–time and velocity–time graphs of the released proton's motion.

21.12 • A negative charge of $-0.550\text{ }\mu\text{C}$ exerts an upward 0.600 N force on an unknown charge that is located 0.300 m directly below the first charge. What are (a) the value of the unknown charge (magnitude and sign); (b) the magnitude and direction of the force that the unknown charge exerts on the $-0.550\text{ }\mu\text{C}$ charge?

21.13 • Three point charges are arranged on a line. Charge $q_3 = +5.00\text{ nC}$ and is at the origin. Charge $q_2 = -3.00\text{ nC}$ and is at $x = +4.00\text{ cm}$. Charge q_1 is at $x = +2.00\text{ cm}$. What is q_1 (magnitude and sign) if the net force on q_3 is zero?

21.14 • In Example 21.4, suppose the point charge on the y -axis at $y = -0.30\text{ m}$ has negative charge $-2.0\text{ }\mu\text{C}$, and the other charges remain the same. Find the magnitude and direction of the net force on Q . How does your answer differ from that in Example 21.4? Explain the differences.

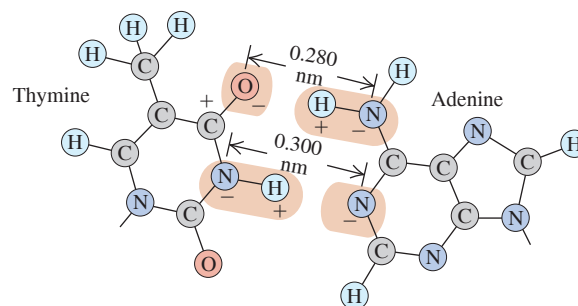
21.15 • In Example 21.3, calculate the net force on charge q_1 .

21.16 • Three point charges are arranged along the x -axis. Charge $q_1 = +3.00\text{ }\mu\text{C}$ is at the origin, and charge $q_2 = -5.00\text{ }\mu\text{C}$ is at $x = 0.200\text{ m}$. Charge $q_3 = -8.00\text{ }\mu\text{C}$. Where is q_3 located if the net force on q_1 is 7.00 N in the $-x$ -direction?

21.17 • Two point charges are located on the y -axis as follows: charge $q_1 = -1.50\text{ nC}$ at $y = -0.600\text{ m}$, and charge $q_2 = +3.20\text{ nC}$ at the origin ($y = 0$). What is the total force (magnitude and direction) exerted by these two charges on a third charge $q_3 = +5.00\text{ nC}$ located at $y = -0.400\text{ m}$?

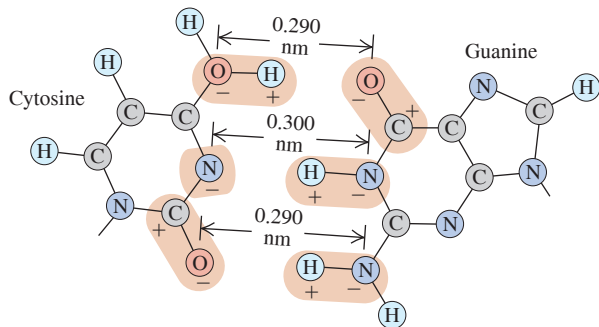
21.18 • **BIO Base Pairing in DNA, I.** The two sides of the DNA double helix are connected by pairs of bases (adenine, thymine, cytosine, and guanine). Because of the geometric shape of these molecules, adenine bonds with thymine and cytosine bonds with guanine. **Figure E21.18** shows the bonding of thymine and adenine. Each charge shown is $\pm e$, and the H–N distance is 0.110 nm . (a) Calculate the *net* force that thymine exerts on adenine. Is it attractive or repulsive? To keep the calculations fairly simple, yet reasonable, consider only the forces due to the O–H–N and the N–H–N combinations, assuming that these two combinations are parallel to each other. Remember, however, that in the O–H–N set, the O^- exerts a force on both the H^+ and the N^+ , and likewise along the N–H–N set. (b) Calculate the force on the electron in the hydrogen atom, which is 0.0529 nm from the proton. Then compare the strength of the bonding force of the electron in hydrogen with the bonding force of the adenine–thymine molecules.

Figure E21.18



21.19 •• BIO Base Pairing in DNA, II. Refer to Exercise 21.18. **Figure E21.19** shows the bonding of cytosine and guanine. The O—H and H—N distances are each 0.110 nm. In this case, assume that the bonding is due only to the forces along the O—H—O, N—H—N, and O—H—N combinations, and assume also that these three combinations are parallel to each other. Calculate the *net* force that cytosine exerts on guanine due to the preceding three combinations. Is this force attractive or repulsive?

Figure E21.19



Section 21.4 Electric Field and Electric Forces

21.20 •• CP A uniform electric field exists in the region between two oppositely charged plane parallel plates. A proton is released from rest at the surface of the positively charged plate and strikes the surface of the opposite plate, 1.60 cm distant from the first, in a time interval of 3.20×10^{-6} s. (a) Find the magnitude of the electric field. (b) Find the speed of the proton when it strikes the negatively charged plate.

21.21 • CP A proton is placed in a uniform electric field of 2.75×10^3 N/C. Calculate (a) the magnitude of the electric force felt by the proton; (b) the proton's acceleration; (c) the proton's speed after $1.00 \mu\text{s}$ in the field, assuming it starts from rest.

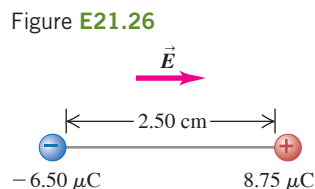
21.22 • A particle has charge -5.00 nC. (a) Find the magnitude and direction of the electric field due to this particle at a point 0.250 m directly above it. (b) At what distance from this particle does its electric field have a magnitude of 12.0 N/C?

21.23 • CP A proton is traveling horizontally to the right at 4.50×10^6 m/s. (a) Find the magnitude and direction of the weakest electric field that can bring the proton uniformly to rest over a distance of 3.20 cm. (b) How much time does it take the proton to stop after entering the field? (c) What minimum field (magnitude and direction) would be needed to stop an electron under the conditions of part (a)?

21.24 • CP An electron is released from rest in a uniform electric field. The electron accelerates vertically upward, traveling 4.50 m in the first $3.00 \mu\text{s}$ after it is released. (a) What are the magnitude and direction of the electric field? (b) Are we justified in ignoring the effects of gravity? Justify your answer quantitatively.

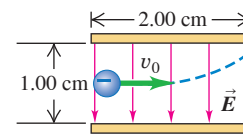
21.25 •• (a) What must the charge (sign and magnitude) of a 1.45 g particle be for it to remain stationary when placed in a downward-directed electric field of magnitude 650 N/C? (b) What is the magnitude of an electric field in which the electric force on a proton is equal in magnitude to its weight?

21.26 •• A $+8.75 \mu\text{C}$ point charge is glued down on a horizontal frictionless table. It is tied to a $-6.50 \mu\text{C}$ point charge by a light, nonconducting 2.50 cm wire. A uniform electric field of magnitude 1.85×10^8 N/C is directed parallel to the wire, as shown in **Fig. E21.26**. (a) Find the tension in the wire. (b) What would the tension be if both charges were negative?



21.27 •• CP An electron is projected with an initial speed $v_0 = 1.60 \times 10^6$ m/s into the uniform field between two parallel plates (**Fig. E21.27**). Assume that the field between the plates is uniform and directed vertically downward and that the field outside the plates is zero. The electron enters the field at a point midway between the plates. (a) If the electron just misses the upper plate as it emerges from the field, find the magnitude of the electric field. (b) Suppose that the electron in Fig. E21.27 is replaced by a proton with the same initial speed v_0 . Would the proton hit one of the plates? If not, what would be the magnitude and direction of its vertical displacement as it exits the region between the plates? (c) Compare the paths traveled by the electron and the proton, and explain the differences. (d) Discuss whether it is reasonable to ignore the effects of gravity for each particle.

Figure E21.27



21.28 •• CP In Exercise 21.27, what is the speed of the electron as it emerges from the field?

21.29 • A point charge is at the origin. With this point charge as the source point, what is the unit vector $\hat{r}$ in the direction of the field point (a) at $x = 0$, $y = -1.35$ m; (b) at $x = 12.0$ cm, $y = 12.0$ cm; (c) at $x = -1.10$ m, $y = 2.60$ m? Express your results in terms of the unit vectors $\hat{i}$ and $\hat{j}$.

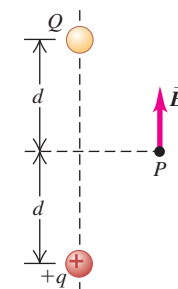
21.30 •• (a) An electron is moving east in a uniform electric field of 1.50 N/C directed to the west. At point A, the velocity of the electron is 4.50×10^5 m/s toward the east. What is the speed of the electron when it reaches point B, 0.375 m east of point A? (b) A proton is moving in the uniform electric field of part (a). At point A, the velocity of the proton is 1.90×10^4 m/s, east. What is the speed of the proton at point B?

Section 21.5 Electric-Field Calculations

21.31 •• A uniform line of charge with length 20.0 cm is along the x -axis, with its midpoint at $x = 0$. Its charge per length is $+4.80$ nC/m. A small sphere with charge $-2.00 \mu\text{C}$ is located at $x = 0$, $y = 5.00$ cm. What are the magnitude and direction of the force that the charged sphere exerts on the line of charge?

21.32 • Two point charges Q and $+q$ (where q is positive) produce the net electric field shown at point P in **Fig. E21.32**. The field points parallel to the line connecting the two charges. (a) What can you conclude about the sign and magnitude of Q ? Explain your reasoning. (b) If the lower charge were negative instead, would it be possible for the field to have the direction shown in the figure? Explain your reasoning.

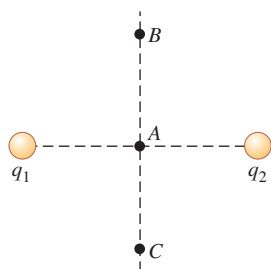
Figure E21.32



21.33 •• A very long line of charge with charge per unit length $+8.00 \mu\text{C}/\text{m}$ is on the x -axis and its midpoint is at $x = 0$. A second very long line of charge with charge per length $-4.00 \mu\text{C}/\text{m}$ is parallel to the x -axis at $y = 10.0$ cm and its midpoint is also at $x = 0$. At what point on the y -axis is the resultant electric field of the two lines of charge equal to zero?

21.34 • The two charges q_1 and q_2 shown in **Fig. E21.34** have equal magnitudes. What is the direction of the net electric field due to these two charges at points A (midway between the charges), B, and C if (a) both charges are negative, (b) both charges are positive, (c) q_1 is positive and q_2 is negative.

Figure E21.34

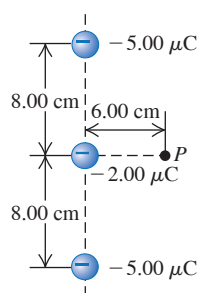


21.35 • A $+2.00 \text{ nC}$ point charge is at the origin, and a second -5.00 nC point charge is on the x -axis at $x = 0.800 \text{ m}$. (a) Find the electric field (magnitude and direction) at each of the following points on the x -axis: (i) $x = 0.200 \text{ m}$; (ii) $x = 1.20 \text{ m}$; (iii) $x = -0.200 \text{ m}$. (b) Find the net electric force that the two charges would exert on an electron placed at each point in part (a).

21.36 • Repeat Exercise 21.35, but now let the charge at the origin be -4.00 nC .

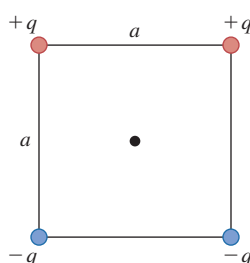
21.37 • Three negative point charges lie along a line as shown in Fig. E21.37. Find the magnitude and direction of the electric field this combination of charges produces at point P , which lies 6.00 cm from the $-2.00 \mu\text{C}$ charge measured perpendicular to the line connecting the three charges.

Figure E21.37



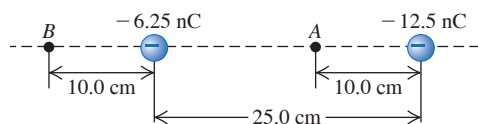
21.38 • A point charge is placed at each corner of a square with side length a . All charges have magnitude q . Two of the charges are positive and two are negative (Fig. E21.38). What is the direction of the net electric field at the center of the square due to the four charges, and what is its magnitude in terms of q and a ?

Figure E21.38



21.39 • Two point charges are separated by 25.0 cm (Fig. E21.39). Find the net electric field these charges produce at (a) point A and (b) point B . (c) What would be the magnitude and direction of the electric force this combination of charges would produce on a proton at A ?

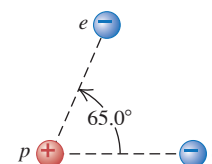
Figure E21.39



21.40 • Point charge $q_1 = -5.00 \text{ nC}$ is at the origin and point charge $q_2 = +3.00 \text{ nC}$ is on the x -axis at $x = 3.00 \text{ cm}$. Point P is on the y -axis at $y = 4.00 \text{ cm}$. (a) Calculate the electric fields $\vec{E}_1$ and $\vec{E}_2$ at point P due to the charges q_1 and q_2 . Express your results in terms of unit vectors (see Example 21.6). (b) Use the results of part (a) to obtain the resultant field at P , expressed in unit vector form.

21.41 • If two electrons are each $1.50 \times 10^{-10} \text{ m}$ from a proton (Fig. E21.41), find the magnitude and direction of the net electric force they will exert on the proton.

Figure E21.41



21.42 • **BIO Electric Field of Axons.** A nerve signal is transmitted through a neuron when an excess of Na^+ ions suddenly enters the axon, a long cylindrical part of the neuron. Axons are approximately $10.0 \mu\text{m}$ in diameter, and measurements show that about $5.6 \times 10^{11} \text{ Na}^+$ ions per meter (each of charge $+e$) enter during this process. Although the axon is a long cylinder, the charge does not all enter everywhere at the same time. A plausible model would be a series of point charges moving along the axon. Consider a 0.10 mm length of the axon and model it as a point charge. (a) If the charge that enters each meter of the axon gets distributed uniformly along it, how many coulombs of charge enter a 0.10 mm length of the axon? (b) What electric field (magnitude and direction) does the sudden influx of charge produce at the surface of the body if the axon is 5.00 cm below the skin? (c) Certain sharks can respond to electric fields as weak as $1.0 \mu\text{N/C}$. How far from this segment of axon could a shark be and still detect its electric field?

21.43 • In a rectangular coordinate system a positive point charge $q = 6.00 \times 10^{-9} \text{ C}$ is placed at the point $x = +0.150 \text{ m}$, $y = 0$, and an identical point charge is placed at $x = -0.150 \text{ m}$, $y = 0$. Find the x - and y -components, the magnitude, and the direction of the electric field at the following points: (a) the origin; (b) $x = 0.300 \text{ m}$, $y = 0$; (c) $x = 0.150 \text{ m}$, $y = -0.400 \text{ m}$; (d) $x = 0$, $y = 0.200 \text{ m}$.

21.44 • A point charge $q_1 = -4.00 \text{ nC}$ is at the point $x = 0.600 \text{ m}$, $y = 0.800 \text{ m}$, and a second point charge $q_2 = +6.00 \text{ nC}$ is at the point $x = 0.600 \text{ m}$, $y = 0$. Calculate the magnitude and direction of the net electric field at the origin due to these two point charges.

21.45 • Three parallel sheets of charge, large enough to be treated as infinite sheets, are perpendicular to the x -axis. Sheet A has surface charge density $\sigma_A = +8.00 \text{ nC/m}^2$. Sheet B is 4.00 cm to the right of sheet A and has surface charge density $\sigma_B = -4.00 \text{ nC/m}^2$. Sheet C is 4.00 cm to the right of sheet B , so is 8.00 cm to the right of sheet A , and has surface charge density $\sigma_C = +6.00 \text{ nC/m}^2$. What are the magnitude and direction of the resultant electric field at a point that is midway between sheets B and C , or 2.00 cm from each of these two sheets?

21.46 • Point charge $q_1 = -5.00 \text{ nC}$ is on the x -axis at $x = -0.400 \text{ m}$. Point P is on the x -axis at $x = +0.200 \text{ m}$. Point charge q_2 is at the origin. What are the sign and magnitude of q_2 if the resultant electric field at point P is zero?

21.47 • Point charge A is on the x -axis at $x = -3.00 \text{ cm}$. At $x = 1.00 \text{ cm}$ on the x -axis its electric field is 2700 N/C . Point charge B is also on the x -axis, at $x = 5.00 \text{ cm}$. The absolute magnitude of charge B is twice that of A . Find the magnitude and direction of the total electric field at the origin if (a) both A and B are positive; (b) both are negative; (c) A is positive and B is negative; (d) A is negative and B is positive.

21.48 • A very long, straight wire has charge per unit length $3.20 \times 10^{-10} \text{ C/m}$. At what distance from the wire is the electric-field magnitude equal to 2.50 N/C ?

21.49 • A ring-shaped conductor with radius $a = 2.50 \text{ cm}$ has a total positive charge $Q = +0.125 \text{ nC}$ uniformly distributed around it (see Fig. 21.23). The center of the ring is at the origin of coordinates O . (a) What is the electric field (magnitude and direction) at point P , which is on the x -axis at $x = 40.0 \text{ cm}$? (b) A point charge $q = -2.50 \mu\text{C}$ is placed at P . What are the magnitude and direction of the force exerted by the charge q on the ring?

21.50 •• A straight, nonconducting plastic wire 8.50 cm long carries a charge density of $+175 \text{ nC/m}$ distributed uniformly along its length. It is lying on a horizontal tabletop. (a) Find the magnitude and direction of the electric field this wire produces at a point 6.00 cm directly above its midpoint. (b) If the wire is now bent into a circle lying flat on the table, find the magnitude and direction of the electric field it produces at a point 6.00 cm directly above its center.

Section 21.7 Electric Dipoles

21.51 • Point charges $q_1 = -4.5 \text{ nC}$ and $q_2 = +4.5 \text{ nC}$ are separated by 3.1 mm, forming an electric dipole. (a) Find the electric dipole moment (magnitude and direction). (b) The charges are in a uniform electric field whose direction makes an angle of 36.9° with the line connecting the charges. What is the magnitude of this field if the torque exerted on the dipole has magnitude $7.2 \times 10^{-9} \text{ N} \cdot \text{m}$?

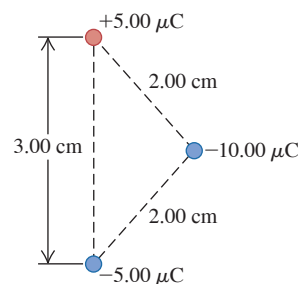
21.52 • The ammonia molecule (NH_3) has a dipole moment of $5.0 \times 10^{-30} \text{ C} \cdot \text{m}$. Ammonia molecules in the gas phase are placed in a uniform electric field $\vec{E}$ with magnitude $1.6 \times 10^6 \text{ N/C}$. (a) What is the change in electric potential energy when the dipole moment of a molecule changes its orientation with respect to $\vec{E}$ from parallel to perpendicular? (b) At what absolute temperature T is the average translational kinetic energy $\frac{3}{2}kT$ of a molecule equal to the change in potential energy calculated in part (a)? (Note: Above this temperature, thermal agitation prevents the dipoles from aligning with the electric field.)

21.53 • Torque on a Dipole. An electric dipole with dipole moment $\vec{p}$ is in a uniform external electric field $\vec{E}$. (a) Find the orientations of the dipole for which the torque on the dipole is zero. (b) Which of the orientations in part (a) is stable, and which is unstable? (Hint: Consider a small rotation away from the equilibrium position and see what happens.) (c) Show that for the stable orientation in part (b), the dipole's own electric field tends to oppose the external field.

21.54 • The dipole moment of the water molecule (H_2O) is $6.17 \times 10^{-30} \text{ C} \cdot \text{m}$. Consider a water molecule located at the origin whose dipole moment $\vec{p}$ points in the $+x$ -direction. A chlorine ion (Cl^-), of charge $-1.60 \times 10^{-19} \text{ C}$, is located at $x = 3.00 \times 10^{-9} \text{ m}$. Find the magnitude and direction of the electric force that the water molecule exerts on the chlorine ion. Is this force attractive or repulsive? Assume that x is much larger than the separation d between the charges in the dipole, so that the approximate expression for the electric field along the dipole axis derived in Example 21.14 can be used.

21.55 • Three charges are at the corners of an isosceles triangle as shown in Fig. E21.55. The $\pm 5.00 \mu\text{C}$ charges form a dipole. (a) Find the force (magnitude and direction) the $-10.00 \mu\text{C}$ charge exerts on the dipole. (b) For an axis perpendicular to the line connecting the $\pm 5.00 \mu\text{C}$ charges at the midpoint of this line, find the torque (magnitude and direction) exerted on the dipole by the $-10.00 \mu\text{C}$ charge.

Figure E21.55



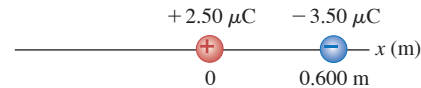
21.56 • An electric dipole with a dipole moment of magnitude p is placed at various orientations in an electric field $\vec{E}$ that is directed to the left. (a) What orientation of the dipole will result in maximum torque directed into the page? What then is the electric potential energy? (b) What orientation of the dipole will give zero torque and maximum electric potential energy? What type of equilibrium is this: stable, unstable, or neutral?

PROBLEMS

21.57 ••• Four identical charges Q are placed at the corners of a square of side L . (a) In a free-body diagram, show all of the forces that act on one of the charges. (b) Find the magnitude and direction of the total force exerted on one charge by the other three charges.

21.58 ••• Two charges are placed on the x -axis: one, of $2.50 \mu\text{C}$, at the origin and the other, of $-3.50 \mu\text{C}$, at $x = 0.600 \text{ m}$ (Fig. P21.58). Find the position on the x -axis where the net force on a small charge $+q$ would be zero.

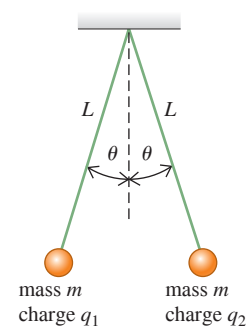
Figure P21.58



21.59 •• A charge $q_1 = +5.00 \text{ nC}$ is placed at the origin of an xy -coordinate system, and a charge $q_2 = -2.00 \text{ nC}$ is placed on the positive x -axis at $x = 4.00 \text{ cm}$. (a) If a third charge $q_3 = +6.00 \text{ nC}$ is now placed at the point $x = 4.00 \text{ cm}$, $y = 3.00 \text{ cm}$, find the x - and y -components of the total force exerted on this charge by the other two. (b) Find the magnitude and direction of this force.

21.60 •• CP Two identical spheres with mass m are hung from silk threads of length L (Fig. P21.60). The spheres have the same charge, so $q_1 = q_2 = q$. The radius of each sphere is very small compared to the distance between the spheres, so they may be treated as point charges. Show that if the angle θ is small, the equilibrium separation d between the spheres is $d = (q^2 L / 2\pi\epsilon_0 mg)^{1/3}$. (Hint: If θ is small, then $\tan \theta \approx \sin \theta$.)

Figure P21.60

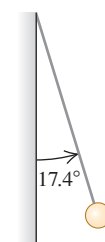


21.61 ••• CP Two small spheres with mass $m = 15.0 \text{ g}$ are hung by silk threads of length $L = 1.20 \text{ m}$ from a common point (Fig. P21.60). When the spheres are given equal quantities of negative charge, so that $q_1 = q_2 = q$, each thread hangs at $\theta = 25.0^\circ$ from the vertical. (a) Draw a diagram showing the forces on each sphere. Treat the spheres as point charges. (b) Find the magnitude of q . (c) Both threads are now shortened to length $L = 0.600 \text{ m}$, while the charges q_1 and q_2 remain unchanged. What new angle will each thread make with the vertical? (Hint: This part of the problem can be solved numerically by using trial values for θ and adjusting the values of θ until a self-consistent answer is obtained.)

21.62 •• CP A small sphere with charge $q = 5.00 \mu\text{C}$ and mass 0.500 g is traveling horizontally toward the east at a height of 60.0 cm above the ground. The sphere has a speed of 2.00 m/s as it enters a region of uniform electric field with magnitude E . What is E if the sphere has a speed of 5.00 m/s just before it strikes the ground?

21.63 •• CP A small 12.3 g plastic ball is tied to a very light 28.6 cm string that is attached to the vertical wall of a room (Fig. P21.63). A uniform horizontal electric field exists in this room. When the ball has been given an excess charge of $-1.11 \mu\text{C}$, you observe that it remains suspended, with the string making an angle of 17.4° with the wall. Find the magnitude and direction of the electric field in the room.

Figure P21.63



21.64 •• CP A small sphere with positive charge q and mass m is released from rest in a uniform electric field $\vec{E}$ that is directed vertically upward. The magnitude of the field is large enough for the sphere to travel upward when it is released. How long does it take the sphere to travel upward a distance d after being released from rest? Give your answer in terms of q , m , d , E , and the acceleration due to gravity, g .

21.65 •• If we rub a balloon on our hair, the balloon sticks to a wall or ceiling. This is because the rubbing transfers electrons from our hair to the balloon, giving it a net negative charge. When the balloon is placed near the ceiling, the extra electrons in it repel nearby electrons in the ceiling, creating a separation of charge on the ceiling, with positive charge closer to the balloon. Model the interaction as two point-like charges of equal magnitude and opposite signs, separated by a distance of $500\ \mu\text{m}$. Neglect the more distant negative charges on the ceiling. (a) A typical balloon has a mass of $4\ \text{g}$. Estimate the minimum magnitude of charge the balloon requires to stay attached to the ceiling. (b) Since a balloon sticks handily to the ceiling after being rubbed, assume that it has attained 10 times the estimated minimum charge. Estimate the number of electrons that were transferred to the balloon by the process of rubbing.

21.66 •• An American penny is 97.5% zinc and 2.5% copper and has a mass of $2.5\ \text{g}$. (a) Use the approximation that a penny is pure zinc, which has an atomic mass of $65.38\ \text{g/mol}$, to estimate the number of electrons in a penny. (Each zinc atom has 30 electrons.) (b) Estimate the net charge on all of the electrons in one penny. (c) The net positive charge on all of the protons in a penny has the same magnitude as the charge on the electrons. Estimate the force on either of two objects with this net magnitude of charge if the objects are separated by $2\ \text{cm}$. (d) Estimate the number of leaves on an oak tree that is 60 feet tall. (e) Imagine a forest filled with such trees, arranged in a square lattice, each $10\ \text{m}$ distant from its neighbors. Estimate how large such a forest would need to be to include as many leaves as there are electrons in one penny. (f) How does that area compare to the surface area of the earth?

21.67 •• Two particles having charges $q_1 = 0.500\ \text{nC}$ and $q_2 = 8.00\ \text{nC}$ are separated by a distance of $1.20\ \text{m}$. At what point along the line connecting the two charges is the total electric field due to the two charges equal to zero?

21.68 •• A $-3.00\ \text{nC}$ point charge is on the x -axis at $x = 1.20\ \text{m}$. A second point charge, Q , is on the x -axis at $-0.600\ \text{m}$. What must be the sign and magnitude of Q for the resultant electric field at the origin to be (a) $45.0\ \text{N/C}$ in the $+x$ -direction, (b) $45.0\ \text{N/C}$ in the $-x$ -direction?

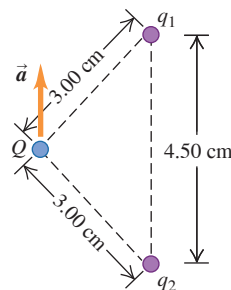
21.69 ••• A charge $+Q$ is located at the origin, and a charge $+4Q$ is at distance d away on the x -axis. Where should a third charge, q , be placed, and what should be its sign and magnitude, so that all three charges will be in equilibrium?

21.70 •• A charge of $-3.00\ \text{nC}$ is placed at the origin of an xy -coordinate system, and a charge of $2.00\ \text{nC}$ is placed on the y -axis at $y = 4.00\ \text{cm}$. (a) If a third charge, of $5.00\ \text{nC}$, is now placed at the point $x = 3.00\ \text{cm}$, $y = 4.00\ \text{cm}$, find the x - and y -components of the total force exerted on this charge by the other two charges. (b) Find the magnitude and direction of this force.

21.71 • Three identical point charges q are placed at each of three corners of a square of side L . Find the magnitude and direction of the net force on a point charge $-3q$ placed (a) at the center of the square and (b) at the vacant corner of the square. In each case, draw a free-body diagram showing the forces exerted on the $-3q$ charge by each of the other three charges.

21.72 ••• Two point charges q_1 and q_2 are held in place $4.50\ \text{cm}$ apart. Another point charge $Q = -1.75\ \mu\text{C}$, of mass $5.00\ \text{g}$, is initially located $3.00\ \text{cm}$ from both of these charges (Fig. P21.72) and released from rest. You observe that the initial acceleration of Q is $324\ \text{m/s}^2$ upward, parallel to the line connecting the two point charges. Find q_1 and q_2 .

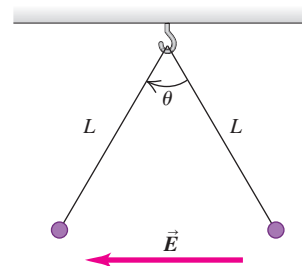
Figure P21.72



21.73 •• CP Strength of the Electric Force. Imagine two $1.0\ \text{g}$ bags of protons, one at the earth's north pole and the other at the south pole. (a) How many protons are in each bag? (b) Calculate the gravitational attraction and the electric repulsion that each bag exerts on the other. (c) Are the forces in part (b) large enough for you to feel if you were holding one of the bags?

21.74 ••• CP Two tiny spheres of mass $6.80\ \text{mg}$ carry charges of equal magnitude, $72.0\ \text{nC}$, but opposite sign. They are tied to the same ceiling hook by light strings of length $0.530\ \text{m}$. When a horizontal uniform electric field E that is directed to the left is turned on, the spheres hang at rest with the angle θ between the strings equal to 58.0° (Fig. P21.74). (a) Which ball (the one on the right or the one on the left) has positive charge? (b) What is the magnitude E of the field?

Figure P21.74



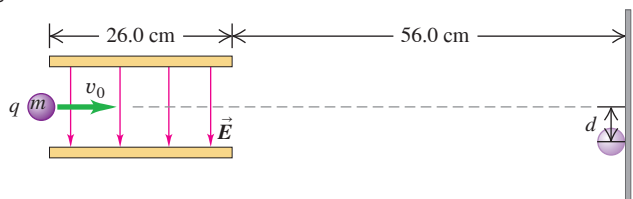
21.75 •• CP Consider a model of a hydrogen atom in which an electron is in a circular orbit of radius $r = 5.29 \times 10^{-11}\ \text{m}$ around a stationary proton. What is the speed of the electron in its orbit?

21.76 ••• CP A disk with radius R and uniform positive charge density σ lies horizontally on a tabletop. A small plastic sphere with mass M and positive charge Q hovers motionless above the center of the disk, suspended by the Coulomb repulsion due to the charged disk. (a) What is the magnitude of the net upward force on the sphere as a function of the height z above the disk? (b) At what height h does the sphere hover? Express your answer in terms of the dimensionless constant $v \equiv 2\epsilon_0 M g / (Q \sigma)$. (c) If $M = 100\ \text{g}$, $Q = 1\ \mu\text{C}$, $R = 5\ \text{cm}$, and $\sigma = 10\ \text{nC/cm}^2$, what is h ?

21.77 •• CP A proton is projected into a uniform electric field that points vertically upward and has magnitude E . The initial velocity of the proton has a magnitude v_0 and is directed at an angle α below the horizontal. (a) Find the maximum distance h_{max} that the proton descends vertically below its initial elevation. Ignore gravitational forces. (b) After what horizontal distance d does the proton return to its original elevation? (c) Sketch the trajectory of the proton. (d) Find the numerical values of h_{max} and d if $E = 500\ \text{N/C}$, $v_0 = 4.00 \times 10^5\ \text{m/s}$, and $\alpha = 30.0^\circ$.

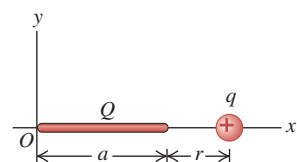
21.78 ••• A small object with mass m , charge q , and initial speed $v_0 = 5.00 \times 10^3$ m/s is projected into a uniform electric field between two parallel metal plates of length 26.0 cm (**Fig. P21.78**). The electric field between the plates is directed downward and has magnitude $E = 800$ N/C. Assume that the field is zero outside the region between the plates. The separation between the plates is large enough for the object to pass between the plates without hitting the lower plate. After passing through the field region, the object is deflected downward a vertical distance $d = 1.25$ cm from its original direction of motion and reaches a collecting plate that is 56.0 cm from the edge of the parallel plates. Ignore gravity and air resistance. Calculate the object's charge-to-mass ratio, q/m .

Figure P21.78



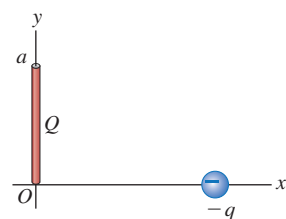
21.79 •• CALC Positive charge Q is distributed uniformly along the x -axis from $x = 0$ to $x = a$. A positive point charge q is located on the positive x -axis at $x = a + r$, a distance r to the right of the end of Q (**Fig. P21.79**). (a) Calculate the x - and y -components of the electric field produced by the charge distribution Q at points on the positive x -axis where $x > a$. (b) Calculate the force (magnitude and direction) that the charge distribution Q exerts on q . (c) Show that if $r \gg a$, the magnitude of the force in part (b) is approximately $Qq/4\pi\epsilon_0 r^2$. Explain why this result is obtained.

Figure P21.79



21.80 •• CALC Positive charge Q is distributed uniformly along the positive y -axis between $y = 0$ and $y = a$. A negative point charge $-q$ lies on the positive x -axis, a distance x from the origin (**Fig. P21.80**). (a) Calculate the x - and y -components of the electric field produced by the charge distribution Q at points on the positive x -axis. (b) Calculate the x - and y -components of the force that the charge distribution Q exerts on q . (c) Show that if $x \gg a$, $F_x \cong -Qq/4\pi\epsilon_0 x^2$ and $F_y \cong +Qqa/8\pi\epsilon_0 x^3$. Explain why this result is obtained.

Figure P21.80



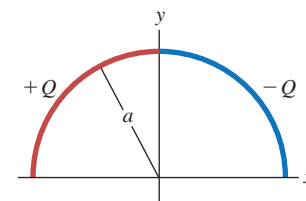
21.81 ••• A uniformly charged disk like the disk in Fig. 21.25 has radius 2.50 cm and carries a total charge of 7.0×10^{-12} C. (a) Find the electric field (magnitude and direction) on the x -axis at $x = 20.0$ cm. (b) Show that for $x \gg R$, Eq. (21.11) becomes $E = Q/4\pi\epsilon_0 x^2$, where Q is the total charge on the disk. (c) Is the magnitude of the electric field you calculated in part (a) larger or smaller than the electric field 20.0 cm from a point charge that has the same total charge as this disk? In terms of the approximation used in part (b) to derive $E = Q/4\pi\epsilon_0 x^2$ for a point charge from Eq. (21.11), explain why this is so. (d) What is the percent difference between the electric fields produced by the finite disk and by a point charge with the same charge at $x = 20.0$ cm and at $x = 10.0$ cm?

21.82 •• CP A small sphere with mass m carries a positive charge q and is attached to one end of a silk fiber of length L . The other end of the fiber is attached to a large vertical insulating sheet that has a positive surface charge density σ . Show that when the sphere is in equilibrium, the fiber makes an angle equal to $\arctan(q\sigma/2mg\epsilon_0)$ with the vertical sheet.

21.83 •• CALC Negative charge $-Q$ is distributed uniformly around a quarter-circle of radius a that lies in the first quadrant, with the center of curvature at the origin. Find the x - and y -components of the net electric field at the origin.

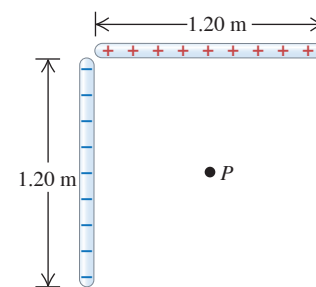
21.84 •• CALC A semicircle of radius a is in the first and second quadrants, with the center of curvature at the origin. Positive charge $+Q$ is distributed uniformly around the left half of the semicircle, and negative charge $-Q$ is distributed uniformly around the right half of the semicircle (**Fig. P21.84**). What are the magnitude and direction of the net electric field at the origin produced by this distribution of charge?

Figure P21.84



21.85 •• Two 1.20 m non-conducting rods meet at a right angle. One rod carries $+2.50 \mu\text{C}$ of charge distributed uniformly along its length, and the other carries $-2.50 \mu\text{C}$ distributed uniformly along it (**Fig. P21.85**). (a) Find the magnitude and direction of the electric field these rods produce at point P , which is 60.0 cm from each rod. (b) If an electron is released at P , what are the magnitude and direction of the net force that these rods exert on it?

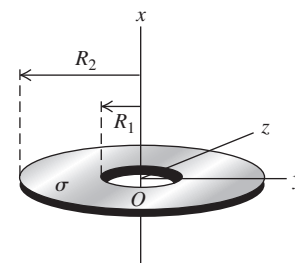
Figure P21.85



21.86 • Two very large parallel sheets are 5.00 cm apart. Sheet A carries a uniform surface charge density of $-8.80 \mu\text{C}/\text{m}^2$, and sheet B, which is to the right of A, carries a uniform charge density of $-11.6 \mu\text{C}/\text{m}^2$. Assume that the sheets are large enough to be treated as infinite. Find the magnitude and direction of the net electric field these sheets produce at a point (a) 4.00 cm to the right of sheet A; (b) 4.00 cm to the left of sheet A; (c) 4.00 cm to the right of sheet B.

21.87 •• CP A thin disk with a circular hole at its center, called an *annulus*, has inner radius R_1 and outer radius R_2 (**Fig. P21.87**). The disk has a uniform positive surface charge density σ on its surface. (a) Determine the total electric charge on the annulus. (b) The annulus lies in the yz -plane, with its center at the origin. For an arbitrary point on the x -axis (the axis of the annulus), find the magnitude and direction of the electric field $\vec{E}$. Consider points both above and below the annulus. (c) Show that at points on the x -axis that are sufficiently close to the origin, the magnitude of the electric field is approximately proportional to the distance between the center of the annulus and the point. How close is "sufficiently close"? (d) A point particle with mass m and negative charge $-q$ is free to move along the x -axis (but cannot move off the axis). The particle is originally placed at rest at $x = 0.01 R_1$ and released. Find the frequency of oscillation of the particle. (*Hint:* Review Section 14.2. The annulus is held stationary.)

Figure P21.87

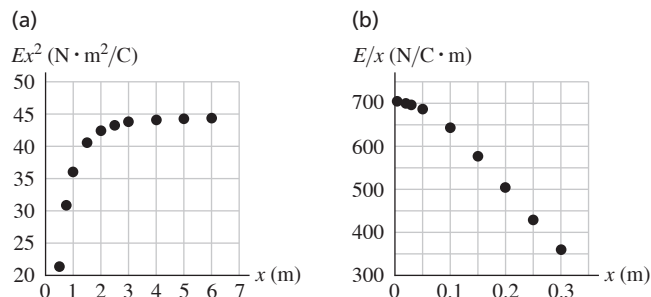


21.88 ••• DATA CP Design of an Inkjet Printer. Inkjet printers can be described as either continuous or drop-on-demand. In a continuous inkjet printer, letters are built up by squirting drops of ink at the paper from a rapidly moving nozzle. You are part of an engineering group working on the design of such a printer. Each ink drop will have a mass of 1.4×10^{-8} g. The drops will leave the nozzle and travel toward the paper at 50 m/s in a horizontal direction, passing through a charging unit that gives each drop a positive charge q by removing some electrons from it. The drops will then pass between parallel deflecting plates, 2.0 cm long, where there is a uniform vertical electric field with magnitude 8.0×10^4 N/C. Your team is working on the design of the charging unit that places the charge on the drops. (a) If a drop is to be deflected 0.30 mm by the time it reaches the end of the deflection plates, what magnitude of charge must be given to the drop? How many electrons must be removed from the drop to give it this charge? (b) If the unit that produces the stream of drops is redesigned so that it produces drops with a speed of 25 m/s, what q value is needed to achieve the same 0.30 mm deflection?

21.89 •• DATA Two small spheres, each carrying a net positive charge, are separated by 0.400 m. You have been asked to perform measurements that will allow you to determine the charge on each sphere. You set up a coordinate system with one sphere (charge q_1) at the origin and the other sphere (charge q_2) at $x = +0.400$ m. Available to you are a third sphere with net charge $q_3 = 4.00 \times 10^{-6}$ C and an apparatus that can accurately measure the location of this sphere and the net force on it. First you place the third sphere on the x -axis at $x = 0.200$ m; you measure the net force on it to be 4.50 N in the $+x$ -direction. Then you move the third sphere to $x = +0.600$ m and measure the net force on it now to be 3.50 N in the $+x$ -direction. (a) Calculate q_1 and q_2 . (b) What is the net force (magnitude and direction) on q_3 if it is placed on the x -axis at $x = -0.200$ m? (c) At what value of x (other than $x = \pm \infty$) could q_3 be placed so that the net force on it is zero?

21.90 ••• DATA Positive charge Q is distributed uniformly around a very thin conducting ring of radius a , as in Fig. 21.23. You measure the electric field E at points on the ring axis, at a distance x from the center of the ring, over a wide range of values of x . (a) Your results for the larger values of x are plotted in Fig. P21.90a as Ex^2 versus x . Explain why the quantity Ex^2 approaches a constant value as x increases. Use Fig. P21.90a to calculate the net charge Q on the ring. (b) Your results for smaller values of x are plotted in Fig. P21.90b as E/x versus x . Explain why E/x approaches a constant value as x approaches zero. Use Fig. P21.90b to calculate a .

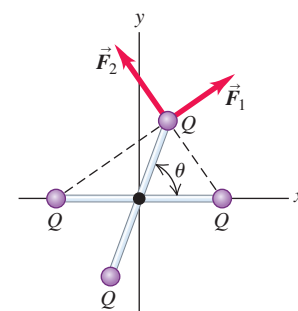
Figure P21.90



21.91 ••• CP Consider an infinite flat sheet with positive charge density σ in which a circular hole of radius R has been cut out. The sheet lies in the xy -plane with the origin at the center of the hole. The sheet is parallel to the ground, so that the positive z -axis describes the “upward” direction. If a particle of mass m and negative charge $-q$ sits at rest at the center of the hole and is released, the particle, constrained to the z -axis, begins to fall. As it drops farther beneath the sheet, the upward electric force increases. For a sufficiently low value of m , the upward electrical attraction eventually exceeds the particle’s weight and the particle will slow, come to a stop, and then rise back to its original position. This sequence of events will repeat indefinitely. (a) What is the electric field at a depth Δ beneath the origin along the negative z -axis? (b) What is the maximum mass $m_{\max}$ that would prevent the particle from falling indefinitely? (c) If $m < m_{\max}$, how much work is done by the electric field as the particle drops from $z = 0$ to $z = -\Delta$? (d) In this same interval, how much work is done by gravity? (e) To what ultimate depth $\Delta_{\max}$ will the particle drop? Express your answer in terms of the dimensionless parameter $\alpha \equiv m/m_{\max}$. (Use the work–energy theorem to find the speed as a function of depth, and then solve for the depth at which the speed is zero.) (f) What is the particle’s speed at depth $\Delta \leq \Delta_{\max}$? (g) If the sheet has charge density 1.00 nC/cm², the radius of the hole is $R = 10.0$ cm, and the particle has mass 25.0 g and charge 1.00 μ C, what are $m_{\max}$ and $\Delta_{\max}$?

21.92 ••• CP Two rigid insulating rods, each of length L and negligible mass, are connected at their centers by a frictionless hinge. One rod is fixed horizontally atop a vertical shaft, while the other rod is free to pivot about the hinge horizontally. A small ball with charge Q and mass M is affixed to each end of each rod. When the rods are perpendicular, there is no net torque on the second rod. We may describe the configuration of this system by the angle θ between the two rods. The first rod lies on the x -axis and the hinge sits at the origin. The second rod is rotated clockwise as seen from above in Fig. 21.92. (a) Determine the force $\vec{F}_1$ exerted by the charge on the left side of the fixed rod on the charge at the upper right side of the movable rod, as a function of θ . (b) Determine the force $\vec{F}_2$ exerted by the charge on the right side of the fixed rod on the charge at the upper right side of the movable rod, as a function of θ . (c) Determine the torque about the pivot point associated with the force $\vec{F}_1$. (d) Determine the torque about the pivot point associated with the force $\vec{F}_2$. (e) Given that the net torque exerted on the movable rod has contributions from the charges at both ends, write a formula for the net torque on the movable rod. (f) The equilibrium configuration at which the torque vanishes is $\theta = \pi/2$. Deviations from equilibrium may be parameterized as $\theta = \pi/2 - \epsilon$. Using power series expansions (see Appendix B), derive the torque for small ϵ , keeping only the lowest order term. (g) For small amplitudes, this system describes a torsional oscillator. Using the previous result, write an expression for the frequency of the small oscillations.

Figure P21.92



CHALLENGE PROBLEMS

21.93 ••• CP CALC Two thin rods, each with length L and total charge $+Q$, are parallel and separated by a distance a . The first rod has one end at the origin and its other end on the positive y -axis. The second rod has its lower end on the positive x -axis. (a) Explain why the y -component of the net force on the second rod vanishes. (b) Determine the x -component of the differential force dF_2 exerted on a small portion of the second rod, with length dy_2 and position y_2 , by the first rod. (This requires integrating over differential portions of the first rod, parameterized by dy_1 .) (c) Determine the net force $\vec{F}_2$ on the second rod by integrating dF_{2x} over the second rod. (d) Show that in

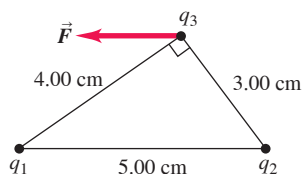
the limit $a \gg L$ the force determined in part (c) becomes $\frac{1}{4\pi\epsilon_0} \frac{Q^2}{a^2} \hat{i}$.

(e) Determine the external work required to move the second rod from very far away to the position $x = a$, provided the first rod is held fixed at $x = 0$. This describes the potential energy of the original configuration. (f) Suppose $L = 50.0$ cm, $a = 10.0$ cm, $Q = 10.0$ μC , and $m = 500$ g. If the two rods are released from the original configuration, they will fly apart and ultimately achieve a particular *relative speed*. What is that relative speed?

21.94 ••• CP CALC An insulating rigid rod of length $2a$ and negligible mass is attached at its center to a pivot at the origin and is free to rotate in the xy -plane. A small ball with mass M and charge Q is attached to one end of the rod. A second small ball with mass M and no charge is attached to the other end. A constant electric field $\vec{E} = -E\hat{i}$ is present in the region $y > 0$, while the region $y < 0$ has a vanishing electric field. Define $\vec{r}$ as the vector that points from the center of the rod to the charged end of the rod, and θ as the angle between $\vec{r}$ and the positive x -axis. The rod is oriented so that $\theta = 0$ and is given an infinitesimal nudge in the direction of increasing θ . (a) Write an expression for the vector $\vec{r}$. (b) Determine the torque $\vec{\tau}$ about the center of the rod when $0 \leq \theta \leq \pi$. (c) Determine the torque on the rod about its center when $\pi \leq \theta \leq 2\pi$. (d) What is the moment of inertia I of the system about the z -axis? (e) The potential energy $U(\theta)$ is determined by $\tau = -dU/d\theta$. Use this equation to write an expression for $U(\theta)$ over the range $0 \leq \theta \leq 4\pi$, using the convention that $U(0) = 0$. Make sure that $U(\theta)$ is continuous. (f) The angular velocity of the rod is $\omega = \omega(\theta)$. Using $\tau = I d^2\theta/dt^2$, show that the energy $\frac{1}{2}I\omega^2 + U(\theta)$ is conserved. (g) Using energy conservation, determine an expression for the angular velocity at the n th time the positive charge crosses the negative y -axis.

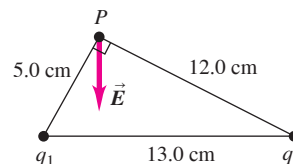
21.95 ••• Three charges are placed as shown in **Fig. P21.95**. The magnitude of q_1 is 2.00 μC , but its sign and the value of the charge q_2 are not known. Charge q_3 is $+4.00$ μC , and the net force $\vec{F}$ on q_3 is entirely in the negative x -direction. (a) Considering the different possible signs of q_1 , there are four possible force diagrams representing the forces $\vec{F}_1$ and $\vec{F}_2$ that q_1 and q_2 exert on q_3 . Sketch these four possible force configurations. (b) Using the sketches from part (a) and the direction of $\vec{F}$, deduce the signs of the charges q_1 and q_2 . (c) Calculate the magnitude of q_2 . (d) Determine F , the magnitude of the net force on q_3 .

Figure P21.95



21.96 ••• Two charges are placed as shown in **Fig. P21.96**. The magnitude of q_1 is 3.00 μC , but its sign and the value of the charge q_2 are not known. The direction of the net electric field $\vec{E}$ at point P is entirely in the negative y -direction. (a) Considering the different possible signs of q_1 and q_2 , four possible diagrams could represent the electric fields $\vec{E}_1$ and $\vec{E}_2$ produced by q_1 and q_2 . Sketch the four possible electric-field configurations. (b) Using the sketches from part (a) and the direction of $\vec{E}$, deduce the signs of q_1 and q_2 . (c) Determine the magnitude of $\vec{E}$.

Figure P21.96



21.97 ••• CALC Two thin rods of length L lie along the x -axis, one between $x = \frac{1}{2}a$ and $x = \frac{1}{2}a + L$ and the other between $x = -\frac{1}{2}a$ and $x = -\frac{1}{2}a - L$. Each rod has positive charge Q distributed uniformly along its length. (a) Calculate the electric field produced by the second rod at points along the positive x -axis. (b) Show that the magnitude of the force that one rod exerts on the other is

$$F = \frac{Q^2}{4\pi\epsilon_0 L^2} \ln \left[\frac{(a+L)^2}{a(a+2L)} \right]$$

(c) Show that if $a \gg L$, the magnitude of this force reduces to $F = Q^2/4\pi\epsilon_0 a^2$. (Hint: Use the expansion $\ln(1+z) = z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \dots$, valid for $|z| \ll 1$. Carry all expansions to at least order L^2/a^2 .) Interpret this result.

MCAT-STYLE PASSAGE PROBLEMS

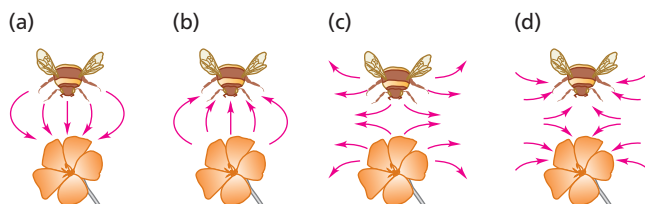
BIO Electric Bees. Flying insects such as bees may accumulate a small positive electric charge as they fly. In one experiment, the mean electric charge of 50 bees was measured to be (30 ± 5) pC per bee. Researchers also observed the electrical properties of a plant consisting of a flower atop a long stem. The charge on the stem was measured as a positively charged bee approached, landed, and flew away. Plants are normally electrically neutral, so the measured net electric charge on the stem was zero when the bee was very far away. As the bee approached the flower, a small net positive charge was detected in the stem, even before the bee landed. Once the bee landed, the whole plant became positively charged, and this positive charge remained on the plant after the bee flew away. By creating artificial flowers with various charge values, experimenters found that bees can distinguish between charged and uncharged flowers and may use the positive electric charge left by a previous bee as a cue indicating whether a plant has already been visited (in which case, little pollen may remain).

21.98 Consider a bee with the mean electric charge found in the experiment. This charge represents roughly how many missing electrons? (a) 1.9×10^8 ; (b) 3.0×10^8 ; (c) 1.9×10^{18} ; (d) 3.0×10^{18} .

21.99 What is the best explanation for the observation that the electric charge on the stem became positive as the charged bee approached (before it landed)? (a) Because air is a good conductor, the positive charge on the bee's surface flowed through the air from bee to plant. (b) Because the earth is a reservoir of large amounts of charge, positive ions were drawn up the stem from the ground toward the charged bee. (c) The plant became electrically polarized as the charged bee approached. (d) Bees that had visited the plant earlier deposited a positive charge on the stem.

21.100 After one bee left a flower with a positive charge, that bee flew away and another bee with the same amount of positive charge flew close to the plant. Which diagram in **Fig. P21.100** (next page) best represents the electric field lines between the bee and the flower?

Figure P21.100



21.101 In a follow-up experiment, a charge of $+40 \text{ pC}$ was placed at the center of an artificial flower at the end of a 30-cm-long stem. Bees were observed to approach no closer than 15 cm from the center of this flower before they flew away. This observation suggests that the smallest external electric field to which bees may be sensitive is closest to which of these values? (a) 2.4 N/C ; (b) 16 N/C ; (c) $2.7 \times 10^{-10} \text{ N/C}$; (d) $4.8 \times 10^{-10} \text{ N/C}$.

ANSWERS

Chapter Opening Question ?

(ii) Water molecules have a permanent electric dipole moment: One end of the molecule has a positive charge and the other end has a negative charge. These ends attract negative and positive ions, respectively, holding the ions apart in solution. Water is less effective as a solvent for materials whose molecules do not ionize (called *nonionic* substances), such as oils.

Key Example VARIATION Problems

VP21.4.1 $(-98.9 \mu\text{N})\hat{i}$

VP21.4.2 (a) $40.5 \mu\text{N}$ in the $+x$ -direction (b) $13.6 \mu\text{N}$ in the $+x$ -direction (c) $+4.82 \text{ nC}$

VP21.4.3 $3.45 \mu\text{N}$ in the $+y$ -direction

VP21.4.4 $F_x = +0.452 \mu\text{N}$, $F_y = +0.329 \mu\text{N}$

VP21.10.1 (a) $E_x = 0$, $E_y = -8.09 \times 10^3 \text{ N/C}$ (b) $E_x = 0$, $E_y = +6.18 \times 10^2 \text{ N/C}$ (c) $E_x = -8.06 \times 10^2 \text{ N/C}$, $E_y = -3.18 \times 10^2 \text{ N/C}$

VP21.10.2 (a) $4.05 \times 10^4 \text{ N/C}$ in the $+x$ -direction (b) $2.70 \times 10^4 \text{ N/C}$ in the $+x$ -direction (c) -1.20 nC

VP21.10.3 (a) $E = \frac{e}{4\pi\epsilon_0 a^2} \left(1 - \frac{1}{2\sqrt{2}} \right)$ (b) away from the proton

VP21.10.4 (a) $dq = (Q/L)dx$ (b) $dE_x = -\frac{dq}{4\pi\epsilon_0 x^2} = -\frac{Qdx}{4\pi\epsilon_0 Lx^2}$, $dE_y = 0$ (c) $E_x = -\frac{Q}{8\pi\epsilon_0 L^2}$, $E_y = 0$

VP21.14.1 (a) $1.41 \times 10^{-24} \text{ N} \cdot \text{m}$ (b) $-1.18 \times 10^{-24} \text{ J}$

VP21.14.2 $7.06 \times 10^{-21} \text{ C}$

VP21.14.3 $1.92 \times 10^{-30} \text{ C} \cdot \text{m}$

VP21.14.4 (a) $2.19 \times 10^{-10} \text{ m}$ (b) $1.99 \times 10^{-8} \text{ m}$

Bridging Problem

$E = 2kQ/\pi a^2$ in the $-y$ -direction

? This child acquires an electric charge by touching the charged metal shell. The charged hairs on the child's head repel and stand out. What would happen if the child stood *inside* a large, charged metal shell? She would acquire (i) the same sign of charge as on the shell, and her hairs would stand out; (ii) the opposite sign of charge as on the shell, and her hairs would stand out; (iii) no charge, and her hairs would be relaxed; (iv) any of these, depending on the amount of charge on the shell.



22 Gauss's Law

LEARNING OUTCOMES

In this chapter, you'll learn...

- 22.1** How you can determine the amount of charge within a closed surface by examining the electric field on the surface.
- 22.2** What is meant by electric flux, and how to calculate it.
- 22.3** How Gauss's law relates the electric flux through a closed surface to the charge enclosed by the surface.
- 22.4** How to use Gauss's law to calculate the electric field due to a symmetric charge distribution.
- 22.5** Where the charge is located on a charged conductor.

You'll need to review...

- 21.4–21.6** Electric fields and their properties.

The discussion of Gauss's law in this section is based on and inspired by the innovative ideas of Ruth W. Chabay and Bruce A. Sherwood in *Electric and Magnetic Interactions* (John Wiley & Sons, 1994).

In physics, an important tool for simplifying problems is the *symmetry properties* of systems. Many physical systems have symmetry; for example, a cylindrical body doesn't look any different after you've rotated it around its axis, and a charged metal sphere looks just the same after you've turned it about any axis through its center.

In this chapter we'll use symmetry ideas along with a new principle, called *Gauss's law*, to simplify electric-field calculations. For example, the field of a straight-line or plane-sheet charge distribution, which we derived in Section 21.5 by using some fairly strenuous integrations, can be obtained in a few steps with the help of Gauss's law. But Gauss's law is more than just a way to make certain calculations easier. Indeed, it is a fundamental statement about the relationship between electric charges and electric fields. Among other things, Gauss's law can help us understand how electric charge distributes itself over conducting bodies.

Here's what Gauss's law is all about. Given any general distribution of charge, we surround it with an imaginary surface that encloses the charge. Then we look at the electric field at various points on this imaginary surface. Gauss's law is a relationship between the field at *all* the points on the surface and the total charge enclosed within the surface. This may sound like a rather indirect way of expressing things, but it turns out to be a tremendously useful relationship. In the next several chapters, we'll make frequent use of the insights that Gauss's law provides into the character of electric fields.

22.1 CHARGE AND ELECTRIC FLUX

In Chapter 21 we asked the question, "Given a charge distribution, what is the electric field produced by that distribution at a point P ?" We saw that the answer could be found by representing the distribution as an assembly of point charges, each of which produces an electric field $\vec{E}$ given by Eq. (21.7). The total field at P is then the vector sum of the fields due to all the point charges.

But there is an alternative relationship between charge distributions and electric fields. To discover this relationship, let's stand the question of Chapter 21 on its head and ask,

“If the electric-field pattern is known in a given region, what can we determine about the charge distribution in that region?”

Here’s an example. Consider the box shown in **Fig. 22.1a**, which may or may not contain electric charge. We’ll imagine that the box is made of a material that has no effect on any electric fields; it’s of the same breed as the massless rope and the frictionless incline. Better still, let the box represent an *imaginary* surface that may or may not enclose some charge. We’ll refer to the box as a **closed surface** because it completely encloses a volume. How can you determine how much (if any) electric charge lies within the box?

Knowing that a charge distribution produces an electric field and that an electric field exerts a force on a test charge, you move a test charge q_0 around the vicinity of the box. By measuring the force $\vec{F}$ experienced by the test charge at different positions, you make a three-dimensional map of the electric field $\vec{E} = \vec{F}/q_0$ outside the box. In the case shown in Fig. 22.1b, the map turns out to be the same as that of the electric field produced by a positive point charge (Fig. 21.28a). From the details of the map, you can find the exact value of the point charge inside the box.

To determine the contents of the box, we actually need to measure $\vec{E}$ on only the *surface* of the box. In **Fig. 22.2a** there is a single *positive* point charge inside the box, and in Fig. 22.2b there are two such charges. The field patterns on the surfaces of the boxes are different in detail, but in each case the electric field points *out* of the box. Figures 22.2c and 22.2d show cases with one and two *negative* point charges, respectively, inside the box. Again, the details of $\vec{E}$ are different for the two cases, but the electric field points *into* each box.

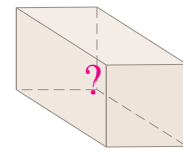
Electric Flux and Enclosed Charge

In Section 21.4 we mentioned the analogy between electric-field vectors and the velocity vectors of a fluid in motion. This analogy can be helpful, even though an electric field does not actually “flow.” Using this analogy, in Figs. 22.2a and 22.2b, in which the electric-field vectors point out of the surface, we say that there is an **outward electric flux**. (The word “flux” comes from a Latin word meaning “flow.”) In Figs. 22.2c and 22.2d the $\vec{E}$ vectors point into the surface, and the electric flux is **inward**.

Figure 22.2 suggests a simple relationship: Positive charge inside the box goes with an outward electric flux through the box’s surface, and negative charge inside goes with an inward electric flux. What happens if there is *zero* charge inside the box? In **Fig. 22.3a** (next page) the box is empty and $\vec{E} = \mathbf{0}$ everywhere, so there is no electric flux into or out of the box. In Fig. 22.3b, one positive and one negative point charge of equal magnitude are enclosed within the box, so the *net* charge inside the box is zero. There is an electric field, but it “flows into” the box on half of its surface and “flows out of” the box on the other half. Hence there is *no net* electric flux into or out of the box.

Figure 22.1 How can you measure the charge inside a box without opening it?

(a) A box containing an unknown amount of charge



(b) Using a test charge outside the box to probe the amount of charge inside the box

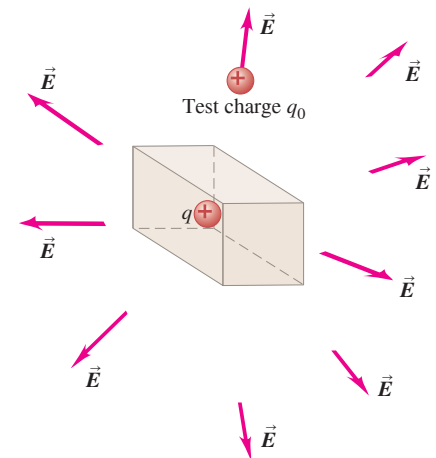
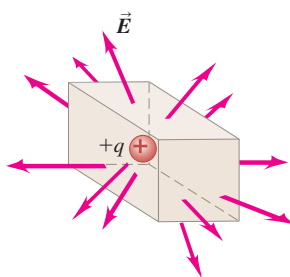
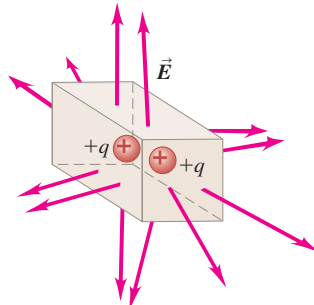


Figure 22.2 The electric field on the surface of boxes containing (a) a single positive point charge, (b) two positive point charges, (c) a single negative point charge, or (d) two negative point charges.

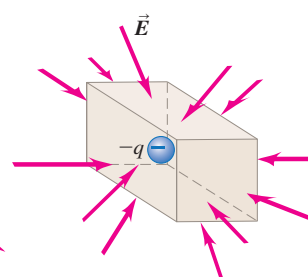
(a) Positive charge inside box, outward flux



(b) Positive charges inside box, outward flux



(c) Negative charge inside box, inward flux



(d) Negative charges inside box, inward flux

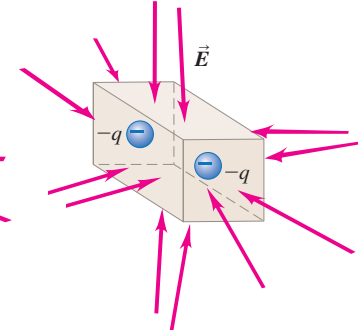


Figure 22.3 Three cases in which there is zero *net* charge inside a box and no net electric flux through the surface of the box. (a) An empty box with $\vec{E} = 0$. (b) A box containing one positive and one equal-magnitude negative point charge. (c) An empty box immersed in a uniform electric field.

(a) No charge inside box, zero flux

(b) Zero *net* charge inside box, inward flux cancels outward flux.

(c) No charge inside box, inward flux cancels outward flux.

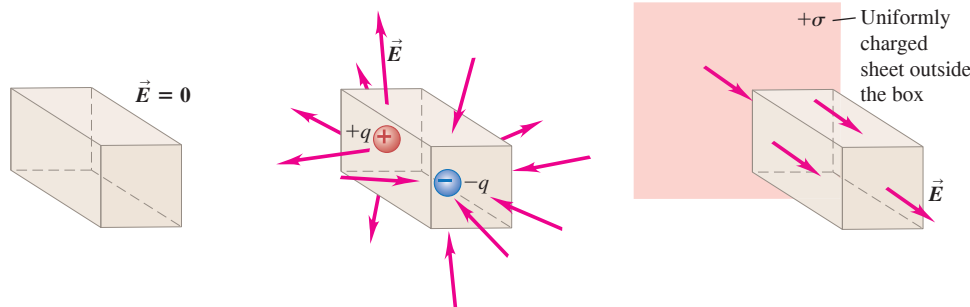
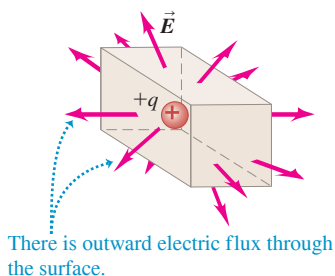
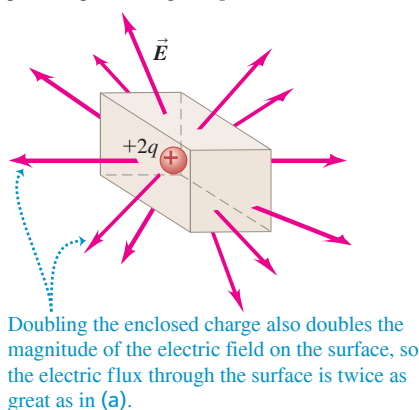


Figure 22.4 Three boxes, each of which encloses a positive point charge.

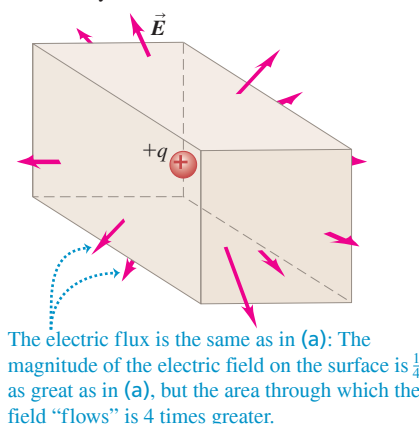
(a) A box containing a positive point charge $+q$



(b) The same box as in (a), but containing a positive point charge $+2q$



(c) The same positive point charge $+q$, but enclosed by a box with twice the dimensions



The box is again empty in Fig. 22.3c. However, there is charge present *outside* the box; the box has been placed with one end parallel to a uniformly charged infinite sheet, which produces a uniform electric field perpendicular to the sheet (see Example 21.11 of Section 21.5). On one end of the box, $\vec{E}$ points into the box; on the opposite end, $\vec{E}$ points out of the box; and on the sides, $\vec{E}$ is parallel to the surface and so points neither into nor out of the box. As in Fig. 22.3b, the inward electric flux on one part of the box exactly compensates for the outward electric flux on the other part. So in all of the cases shown in Fig. 22.3, there is no *net* electric flux through the surface of the box, and no *net* charge is enclosed in the box.

Figures 22.2 and 22.3 demonstrate a connection between the *sign* (positive, negative, or zero) of the *net* charge enclosed by a closed surface and the *sense* (outward, inward, or none) of the net electric flux through the surface. There is also a connection between the *magnitude* of the net charge inside the closed surface and the *strength* of the net "flow" of $\vec{E}$ over the surface. In both **Figs. 22.4a** and **22.4b** there is a single point charge inside the box, but in Fig. 22.4b the magnitude of the charge is twice as great, and so $\vec{E}$ is everywhere twice as great in magnitude as in Fig. 22.4a. If we keep in mind the fluid-flow analogy, this means that the net outward electric flux is also twice as great in Fig. 22.4b as in Fig. 22.4a. This suggests that the net electric flux through the surface of the box is *directly proportional* to the magnitude of the net charge enclosed by the box.

This conclusion is independent of the size of the box. In Fig. 22.4c the point charge $+q$ is enclosed by a box with twice the linear dimensions of the box in Fig. 22.4a. The magnitude of the electric field of a point charge decreases with distance according to $1/r^2$, so the average magnitude of $\vec{E}$ on each face of the large box in Fig. 22.4c is just $\frac{1}{4}$ of the average magnitude on the corresponding face in Fig. 22.4a. But each face of the large box has exactly four times the area of the corresponding face of the small box. Hence the outward electric flux is the *same* for the two boxes if we *define* electric flux as follows: For each face of the box, take the product of the average perpendicular component of $\vec{E}$ and the area of that face; then add up the results from all faces of the box. With this definition the net electric flux due to a single point charge inside the box is independent of the size of the box and depends only on the net charge inside the box.

To summarize, for the special cases of a closed surface in the shape of a rectangular box and charge distributions made up of point charges or infinite charged sheets, we have found:

1. Whether there is a net outward or inward electric flux through a closed surface depends on the sign of the enclosed charge.
2. Charges *outside* the surface do not give a net electric flux through the surface.
3. The net electric flux is directly proportional to the net amount of charge enclosed within the surface but is otherwise independent of the size of the closed surface.

These observations are a qualitative statement of *Gauss's law*.

Do these observations hold true for other kinds of charge distributions and for closed surfaces of arbitrary shape? The answer to these questions will prove to be yes. But to explain why this is so, we need a precise mathematical statement of what we mean by electric flux. We develop this in the next section.

TEST YOUR UNDERSTANDING OF SECTION 22.1 If all of the dimensions of the box in Fig. 22.2a are increased by a factor of 3, how will the electric flux through the box change?

(i) The flux will be $3^2 = 9$ times greater; (ii) the flux will be 3 times greater; (iii) the flux will be unchanged; (iv) the flux will be $\frac{1}{3}$ as great; (v) the flux will be $(\frac{1}{3})^2 = \frac{1}{9}$ as great; (vi) not enough information is given to decide.

ANSWER

(iii) Each part of the surface of the box will be three times farther from the charge $+q$, so the electric field will be $(\frac{1}{3})^2 = \frac{1}{9}$ as strong. But the area of the box will increase by a factor of $3^2 = 9$. Hence the electric flux will be multiplied by a factor of $(\frac{1}{9})(9) = 1$. In other words, the flux will be unchanged.

22.2 CALCULATING ELECTRIC FLUX

In the preceding section we introduced the concept of *electric flux*. We used this to give a rough qualitative statement of Gauss's law: The net electric flux through a closed surface is directly proportional to the net charge inside that surface. To be able to make full use of this law, we need to know how to *calculate* electric flux. To do this, let's again make use of the analogy between an electric field $\vec{E}$ and the field of velocity vectors $\vec{v}$ in a flowing fluid. (Again, keep in mind that this is only an analogy; an electric field is *not* a flow.)

Flux: Fluid-Flow Analogy

Figure 22.5 shows a fluid flowing steadily from left to right. Let's examine the volume flow rate dV/dt (in, say, cubic meters per second) through the wire rectangle with area A . When the area is perpendicular to the flow velocity $\vec{v}$ (Fig. 22.5a) and the flow velocity is the same at all points in the fluid, the volume flow rate dV/dt is the area A multiplied by the flow speed v :

$$\frac{dV}{dt} = vA$$

When the rectangle is tilted at an angle ϕ (Fig. 22.5b) so that its face is not perpendicular to $\vec{v}$, the area that counts is the silhouette area that we see when we look in the direction of $\vec{v}$. This area, which is outlined in red and labeled $A_{\perp}$ in Fig. 22.5b, is the *projection* of the area A onto a surface perpendicular to $\vec{v}$. Two sides of the projected rectangle have the same length as the original one, but the other two are foreshortened by a factor of $\cos \phi$, so the projected area $A_{\perp}$ is equal to $A \cos \phi$. Then the volume flow rate through A is

$$\frac{dV}{dt} = vA \cos \phi$$

If $\phi = 90^\circ$, $dV/dt = 0$; the wire rectangle is edge-on to the flow, and no fluid passes through the rectangle.

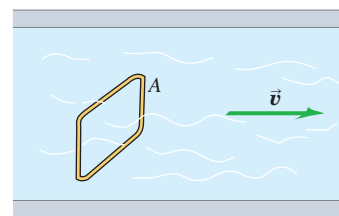
Also, $v \cos \phi$ is the component of the vector $\vec{v}$ perpendicular to the plane of the area A . Calling this component $v_{\perp}$, we can rewrite the volume flow rate as

$$\frac{dV}{dt} = v_{\perp} A$$

We can express the volume flow rate more compactly by using the concept of *vector area* $\vec{A}$, a vector quantity with magnitude A and a direction perpendicular to the plane of the area we are describing. The vector area $\vec{A}$ describes both the size of an area and its

Figure 22.5 The volume flow rate of fluid through the wire rectangle (a) is vA when the area of the rectangle is perpendicular to $\vec{v}$ and (b) is $vA \cos \phi$ when the rectangle is tilted at an angle ϕ .

(a) A wire rectangle in a fluid



(b) The wire rectangle tilted by an angle ϕ

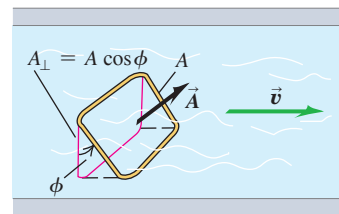
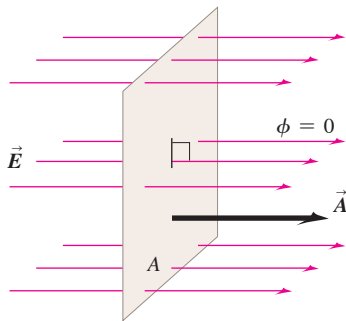
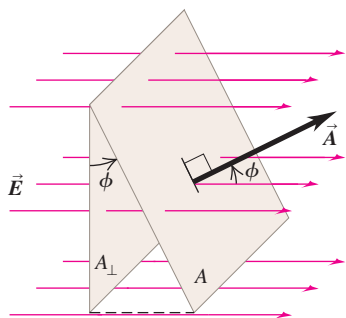


Figure 22.6 A flat surface in a uniform electric field. The electric flux Φ_E through the surface equals the scalar product of the electric field $\vec{E}$ and the area vector $\vec{A}$.

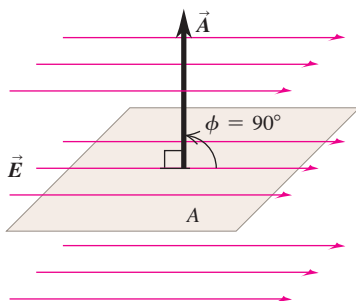
- (a) Surface is face-on to electric field:
- $\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$.



- (b) Surface is tilted from a face-on orientation by an angle ϕ :
- The angle between $\vec{E}$ and $\vec{A}$ is ϕ .
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \phi$.



- (c) Surface is edge-on to electric field:
- $\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90^\circ$).
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos 90^\circ = 0$.



orientation in space. In terms of $\vec{A}$, we can write the volume flow rate of fluid through the rectangle in Fig. 22.5b as a scalar (dot) product:

$$\frac{dV}{dt} = \vec{v} \cdot \vec{A}$$

Flux of a Uniform Electric Field

Using the analogy between electric field and fluid flow, we now define electric flux in the same way as we have just defined the volume flow rate of a fluid; we simply replace the fluid velocity $\vec{v}$ by the electric field $\vec{E}$. The symbol that we use for electric flux is Φ_E (the capital Greek letter phi; the subscript E is a reminder that this is *electric* flux). Consider first a flat area A perpendicular to a uniform electric field $\vec{E}$ (Fig. 22.6a). We define the electric flux through this area to be the product of the field magnitude E and the area A :

$$\Phi_E = EA$$

Roughly speaking, we can picture Φ_E in terms of the field lines passing through A . Increasing the area means that more lines of $\vec{E}$ pass through the area, increasing the flux; a stronger field means more closely spaced lines of $\vec{E}$ and therefore more lines per unit area, so again the flux increases.

If the area A is flat but not perpendicular to the field $\vec{E}$, then fewer field lines pass through it. In this case the area that counts is the silhouette area that we see when looking in the direction of $\vec{E}$. This is the area $A_\perp$ in Fig. 22.6b and is equal to $A \cos \phi$ (compare to Fig. 22.5b). We generalize our definition of electric flux for a uniform electric field to

$$\Phi_E = EA \cos \phi \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface}) \quad (22.1)$$

Since $E \cos \phi$ is the component of $\vec{E}$ perpendicular to the area, we can rewrite Eq. (22.1) as

$$\Phi_E = E_\perp A \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface}) \quad (22.2)$$

In terms of the vector area $\vec{A}$ perpendicular to the area, we can write the electric flux as the scalar product of $\vec{E}$ and $\vec{A}$:

$$\Phi_E = \vec{E} \cdot \vec{A} \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface}) \quad (22.3)$$

Equations (22.1), (22.2), and (22.3) express the electric flux for a *flat* surface and a *uniform* electric field in different but equivalent ways. The SI unit for electric flux is $1 \text{ N} \cdot \text{m}^2/\text{C}$. Note that if the area is edge-on to the field, $\vec{E}$ and $\vec{A}$ are perpendicular and the flux is zero (Fig. 22.6c).

We can represent the direction of a vector area $\vec{A}$ by using a *unit vector* $\hat{n}$ perpendicular to the area; $\hat{n}$ stands for “normal.” Then

$$\vec{A} = A\hat{n} \quad (22.4)$$

A surface has two sides, so there are two possible directions for $\hat{n}$ and $\vec{A}$. We must always specify which direction we choose. In Section 22.1 we related the charge inside a *closed* surface to the electric flux through the surface. With a closed surface we’ll always choose the direction of $\hat{n}$ to be *outward*, and we’ll speak of the flux *out of* a closed surface. Thus what we called “outward electric flux” in Section 22.1 corresponds to a *positive* value of Φ_E , and what we called “inward electric flux” corresponds to a *negative* value of Φ_E .

Flux of a Nonuniform Electric Field

What happens if the electric field $\vec{E}$ isn't uniform but varies from point to point over the area A ? Or what if A is part of a curved surface? Then we divide A into many small elements dA , each of which has a unit vector $\hat{n}$ perpendicular to it and a vector area $d\vec{A} = \hat{n} dA$. We calculate the electric flux through each element and integrate the results to obtain the total flux:

$$\Phi_E = \int E \cos \phi \, dA = \int E_{\perp} dA = \int \vec{E} \cdot d\vec{A} \quad (22.5)$$

Labels in the diagram:
 - Electric flux through a surface: Φ_E
 - Angle between $\vec{E}$ and normal to surface: ϕ
 - Magnitude of electric field $\vec{E}$: E
 - Component of $\vec{E}$ perpendicular to surface: $E_{\perp}$
 - Element of surface area: dA
 - Vector element of surface area: $d\vec{A}$

We call this integral the **surface integral** of the component $E_{\perp}$ over the area, or the surface integral of $\vec{E} \cdot d\vec{A}$. In specific problems, one form of the integral is sometimes more convenient than another. Example 22.3 at the end of this section illustrates the use of Eq. (22.5).

In Eq. (22.5) the electric flux $\int E_{\perp} dA$ is equal to the *average* value of the perpendicular component of the electric field, multiplied by the area of the surface. This is the same definition of electric flux that we were led to in Section 22.1, now expressed more mathematically. In the next section we'll see the connection between the total electric flux through *any* closed surface, no matter what its shape, and the amount of charge enclosed within that surface.

BIO APPLICATION Flux Through a Basking Shark's Mouth

Unlike aggressive carnivorous sharks such as great whites, a basking shark feeds passively on plankton in the water that passes through the shark's gills as it swims. To survive on these tiny organisms requires a huge flux of water through a basking shark's immense mouth, which can be up to a meter across. The water flux—the product of the shark's speed through the water and the area of its mouth—can be up to $0.5 \text{ m}^3/\text{s}$ (500 liters per second, or almost 5×10^5 gallons per hour). In a similar way, the flux of electric field through a surface depends on the magnitude of the field and the area of the surface (as well as the relative orientation of the field and surface).

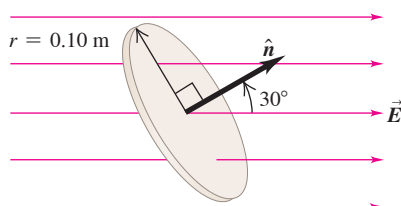


EXAMPLE 22.1 Electric flux through a disk

A disk of radius 0.10 m is oriented with its normal unit vector $\hat{n}$ at 30° to a uniform electric field $\vec{E}$ of magnitude $2.0 \times 10^3 \text{ N/C}$ (Fig. 22.7). (Since this isn't a closed surface, it has no "inside" or "outside." That's why we have to specify the direction of $\hat{n}$ in the figure.) (a) What is the electric flux through the disk? (b) What is the flux through the disk if it is turned so that $\hat{n}$ is perpendicular to $\vec{E}$? (c) What is the flux through the disk if $\hat{n}$ is parallel to $\vec{E}$?

IDENTIFY and SET UP This problem is about a flat surface in a uniform electric field, so we can apply the ideas of this section. We calculate the electric flux from Eq. (22.1).

Figure 22.7 The electric flux Φ_E through a disk depends on the angle between its normal $\hat{n}$ and the electric field $\vec{E}$.



EXECUTE (a) The area is $A = \pi(0.10 \text{ m})^2 = 0.0314 \text{ m}^2$ and the angle between $\vec{E}$ and $\vec{A} = A\hat{n}$ is $\phi = 30^\circ$, so from Eq. (22.1),

$$\Phi_E = EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(\cos 30^\circ) = 54 \text{ N} \cdot \text{m}^2/\text{C}$$

(b) The normal to the disk is now perpendicular to $\vec{E}$, so $\phi = 90^\circ$, $\cos \phi = 0$, and $\Phi_E = 0$.

(c) The normal to the disk is parallel to $\vec{E}$, so $\phi = 0$ and $\cos \phi = 1$:

$$\Phi_E = EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(1) = 63 \text{ N} \cdot \text{m}^2/\text{C}$$

EVALUATE Our answer to part (b) is smaller than that to part (a), which is in turn smaller than that to part (c). Is all this as it should be?

KEYCONCEPT To find the electric flux of a uniform electric field $\vec{E}$ through a flat surface, multiply the area of the surface by the component of $\vec{E}$ normal (perpendicular) to the surface.

EXAMPLE 22.2 Electric flux through a cube

WITH VARIATION PROBLEMS

An imaginary cubical surface of side L is in a region of uniform electric field $\vec{E}$. Find the electric flux through each face of the cube and the total flux through the cube when (a) it is oriented with two of its faces perpendicular to $\vec{E}$ (Fig. 22.8a) and (b) the cube is turned by an angle θ about a vertical axis (Fig. 22.8b).

IDENTIFY and SET UP Since $\vec{E}$ is uniform and each of the six faces of the cube is flat, we find the flux Φ_{Ei} through each face from Eqs. (22.3) and (22.4). The total flux through the cube is the sum of the six individual fluxes.

EXECUTE (a) Figure 22.8a shows the unit vectors $\hat{n}_1$ through $\hat{n}_6$ for each face; each unit vector points *outward* from the cube's closed surface. The angle between $\vec{E}$ and $\hat{n}_1$ is 180° , the angle between $\vec{E}$ and $\hat{n}_2$ is 0° , and the angle between $\vec{E}$ and each of the other four unit vectors is 90° . Each face of the cube has area L^2 , so the fluxes through the faces are

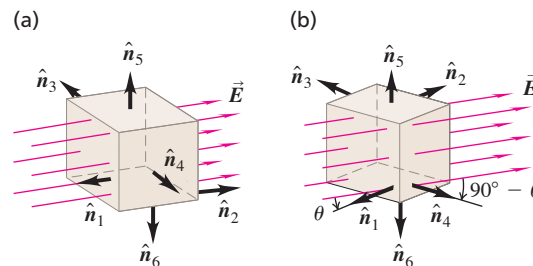
$$\begin{aligned}\Phi_{E1} &= \vec{E} \cdot \hat{n}_1 A = EL^2 \cos 180^\circ = -EL^2 \\ \Phi_{E2} &= \vec{E} \cdot \hat{n}_2 A = EL^2 \cos 0^\circ = +EL^2 \\ \Phi_{E3} &= \Phi_{E4} = \Phi_{E5} = \Phi_{E6} = EL^2 \cos 90^\circ = 0\end{aligned}$$

The flux is negative on face 1, where $\vec{E}$ is directed into the cube, and positive on face 2, where $\vec{E}$ is directed out of the cube. The total flux through the cube is

$$\begin{aligned}\Phi_E &= \Phi_{E1} + \Phi_{E2} + \Phi_{E3} + \Phi_{E4} + \Phi_{E5} + \Phi_{E6} \\ &= -EL^2 + EL^2 + 0 + 0 + 0 + 0 = 0\end{aligned}$$

(b) The field $\vec{E}$ is directed into faces 1 and 3, so the fluxes through them are negative; $\vec{E}$ is directed out of faces 2 and 4, so the fluxes through them are positive. We find

Figure 22.8 Electric flux of a uniform field $\vec{E}$ through a cubical box of side L in two orientations.



$$\begin{aligned}\Phi_{E1} &= \vec{E} \cdot \hat{n}_1 A = EL^2 \cos (180^\circ - \theta) = -EL^2 \cos \theta \\ \Phi_{E2} &= \vec{E} \cdot \hat{n}_2 A = +EL^2 \cos \theta \\ \Phi_{E3} &= \vec{E} \cdot \hat{n}_3 A = EL^2 \cos (90^\circ + \theta) = -EL^2 \sin \theta \\ \Phi_{E4} &= \vec{E} \cdot \hat{n}_4 A = EL^2 \cos (90^\circ - \theta) = +EL^2 \sin \theta \\ \Phi_{E5} &= \Phi_{E6} = EL^2 \cos 90^\circ = 0\end{aligned}$$

The total flux $\Phi_E = \Phi_{E1} + \Phi_{E2} + \Phi_{E3} + \Phi_{E4} + \Phi_{E5} + \Phi_{E6}$ through the surface of the cube is again zero.

EVALUATE We came to the same conclusion in our discussion of Fig. 22.3c: There is zero net flux of a uniform electric field through a closed surface that contains no electric charge.

KEYCONCEPT When calculating the net electric flux through a closed three-dimensional surface, always choose the normal at every location on the surface to point outward. The net flux equals the sum of the fluxes through each part of the surface.

EXAMPLE 22.3 Electric flux through a sphere

WITH VARIATION PROBLEMS

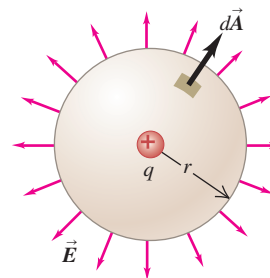
A point charge $q = +3.0 \mu\text{C}$ is surrounded by an imaginary sphere of radius $r = 0.20 \text{ m}$ centered on the charge (Fig. 22.9). Find the resulting electric flux through the sphere.

IDENTIFY and SET UP The surface is not flat and the electric field is not uniform, so to calculate the electric flux (our target variable) we must use the general definition, Eq. (22.5). Because the sphere is centered on the point charge, at any point on the spherical surface, $\vec{E}$ is directed out of the sphere perpendicular to the surface. The positive direction for both $\hat{n}$ and $E_\perp$ is outward, so $E_\perp = E$ and the flux through a surface element dA is $\vec{E} \cdot d\vec{A} = E dA$. This greatly simplifies the integral in Eq. (22.5).

EXECUTE We must evaluate the integral of Eq. (22.5), $\Phi_E = \int E dA$. At any point on the sphere of radius r the electric field has the same magnitude $E = q/4\pi\epsilon_0 r^2$. Hence E can be taken outside the integral, which becomes $\Phi_E = E \int dA = EA$, where A is the area of the spherical surface: $A = 4\pi r^2$. Hence the total flux through the sphere is

$$\begin{aligned}\Phi_E &= EA = \frac{q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{q}{\epsilon_0} \\ &= \frac{3.0 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 3.4 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

Figure 22.9 Electric flux through a sphere centered on a point charge.



EVALUATE The radius r of the sphere cancels out of the result for Φ_E . We would have obtained the same flux with a sphere of radius 2.0 m or 200 m . We came to essentially the same conclusion in our discussion of Fig. 22.4 in Section 22.1, where we considered rectangular closed surfaces of two different sizes enclosing a point charge. There we found that the flux of $\vec{E}$ was independent of the size of the surface; the same result holds true for a spherical surface. Indeed, the flux through *any* surface enclosing a single point charge is independent of the shape or size of the surface, as we'll soon see.

KEYCONCEPT If a closed surface is curved, calculating the net electric flux through the surface requires integrating the normal component of the electric field over the entire surface. It greatly simplifies the integration if the field is symmetrical and the surface used has the same symmetry.

TEST YOUR UNDERSTANDING OF SECTION 22.2 Rank the following surfaces in order from most positive to most negative electric flux: (i) a flat rectangular surface with vector area $\vec{A} = (6.0 \text{ m}^2)\hat{i}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{j}$; (ii) a flat circular surface with vector area $\vec{A} = (3.0 \text{ m}^2)\hat{j}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{i} + (2.0 \text{ N/C})\hat{j}$; (iii) a flat square surface with vector area $\vec{A} = (3.0 \text{ m}^2)\hat{i} + (7.0 \text{ m}^2)\hat{j}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{i} - (2.0 \text{ N/C})\hat{j}$; (iv) a flat oval surface with vector area $\vec{A} = (3.0 \text{ m}^2)\hat{i} - (7.0 \text{ m}^2)\hat{j}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{i} - (2.0 \text{ N/C})\hat{j}$.

ANSWER

In each case the electric field is uniform, so the flux is $\Phi_E = \vec{E} \cdot \vec{A}$. We use the relationships for the scalar products of unit vectors: $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = 1$, $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i} = 0$. In case (i) we have $\Phi_E = (4.0 \text{ N/C})(6.0 \text{ m}^2)\hat{j} \cdot \hat{i} = 0$ (the electric field and vector area are perpendicular, so there is zero flux). In case (ii) we have $\Phi_E = [(4.0 \text{ N/C})\hat{i} + (2.0 \text{ N/C})\hat{j}] \cdot [(3.0 \text{ m}^2)\hat{j}] = (2.0 \text{ N/C})(3.0 \text{ m}^2) = 6.0 \text{ N} \cdot \text{m}^2/\text{C}$. Similarly, in case (iii) we have $\Phi_E = [(4.0 \text{ N/C})\hat{i} + (2.0 \text{ N/C})\hat{j}] \cdot [(3.0 \text{ m}^2)\hat{i} + (7.0 \text{ m}^2)\hat{j}] = (4.0 \text{ N/C})(3.0 \text{ m}^2) + (2.0 \text{ N/C})(7.0 \text{ m}^2) = 12.0 \text{ N} \cdot \text{m}^2/\text{C} + 14.0 \text{ N} \cdot \text{m}^2/\text{C} = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$. In case (iv) we have $\Phi_E = [(4.0 \text{ N/C})\hat{i} - (2.0 \text{ N/C})\hat{j}] \cdot [(3.0 \text{ m}^2)\hat{i} - (7.0 \text{ m}^2)\hat{j}] = (4.0 \text{ N/C})(3.0 \text{ m}^2) - (2.0 \text{ N/C})(-7.0 \text{ m}^2) = 12.0 \text{ N} \cdot \text{m}^2/\text{C} + 14.0 \text{ N} \cdot \text{m}^2/\text{C} = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$. We have $\Phi_E = 0$ for (i), $\Phi_E = 6.0 \text{ N} \cdot \text{m}^2/\text{C}$ for (ii), $\Phi_E = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$ for (iii), and $\Phi_E = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$ for (iv). The ranking from most positive to most negative electric flux is (i), (ii), (iii), (iv).

22.3 GAUSS'S LAW

Gauss's law is an alternative to Coulomb's law. While completely equivalent to Coulomb's law, Gauss's law provides a different way to express the relationship between electric charge and electric field. It was formulated by Carl Friedrich Gauss (1777–1855), one of the greatest mathematicians of all time (**Fig. 22.10**).

Point Charge Inside a Spherical Surface

Gauss's law states that the total electric flux through any closed surface (a surface enclosing a definite volume) is proportional to the total (net) electric charge inside the surface. In Section 22.1 we observed this relationship qualitatively; now we'll develop it more rigorously. We'll start with the field of a single positive point charge q . The field lines radiate out equally in all directions. We place this charge at the center of an imaginary spherical surface with radius R . The magnitude E of the electric field at every point on the surface is given by

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

At each point on the surface, $\vec{E}$ is perpendicular to the surface, and its magnitude is the same at every point, as in Example 22.3 (Section 22.2). The total electric flux is the product of the field magnitude E and the total area $A = 4\pi R^2$ of the sphere:

$$\Phi_E = EA = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} (4\pi R^2) = \frac{q}{\epsilon_0} \quad (22.6)$$

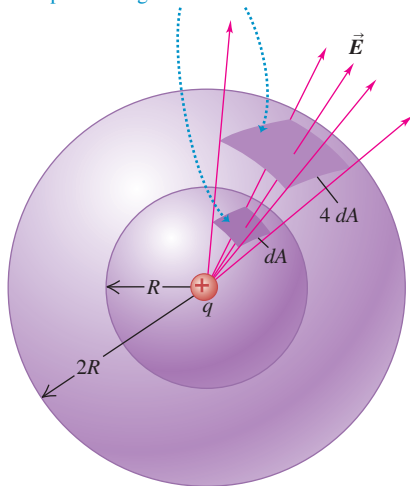
The flux is independent of the radius R of the sphere. It depends on only the charge q enclosed by the sphere.



Figure 22.10 Carl Friedrich Gauss helped develop several branches of mathematics, including differential geometry, real analysis, and number theory. The “bell curve” of statistics is one of his inventions. Gauss also made state-of-the-art investigations of the earth's magnetism and calculated the orbit of the first asteroid to be discovered.

Figure 22.11 Projection of an element of area dA of a sphere of radius R onto a concentric sphere of radius $2R$. The projection multiplies each linear dimension by 2, so the area element on the larger sphere is $4dA$.

The same number of field lines and the same flux pass through both of these area elements.



We can also interpret this result in terms of field lines. **Figure 22.11** shows two spheres with radii R and $2R$ centered on the point charge q . Every field line that passes through the smaller sphere also passes through the larger sphere, so the total flux through each sphere is the same.

What is true of the entire sphere is also true of any portion of its surface. In Fig. 22.11 an area dA is outlined on the sphere of radius R and projected onto the sphere of radius $2R$ by drawing lines from the center through points on the boundary of dA . The area projected on the larger sphere is clearly $4dA$. But since the electric field due to a point charge is inversely proportional to r^2 , the field magnitude is $\frac{1}{4}$ as great on the sphere of radius $2R$ as on the sphere of radius R . Hence the electric flux is the same for both areas and is independent of the radius of the sphere.

Point Charge Inside a Nonspherical Surface

We can extend this projection technique to nonspherical surfaces. Instead of a second sphere, let us surround the sphere of radius R by a surface of irregular shape, as in **Fig. 22.12a**. Consider a small element of area dA on the irregular surface; we note that this area is *larger* than the corresponding element on a spherical surface at the same distance from q . If a normal to dA makes an angle ϕ with a radial line from q , two sides of the area projected onto the spherical surface are foreshortened by a factor $\cos \phi$ (Fig. 22.12b). The other two sides are unchanged. Thus the electric flux through the spherical surface element is equal to the flux $E dA \cos \phi$ through the corresponding irregular surface element.

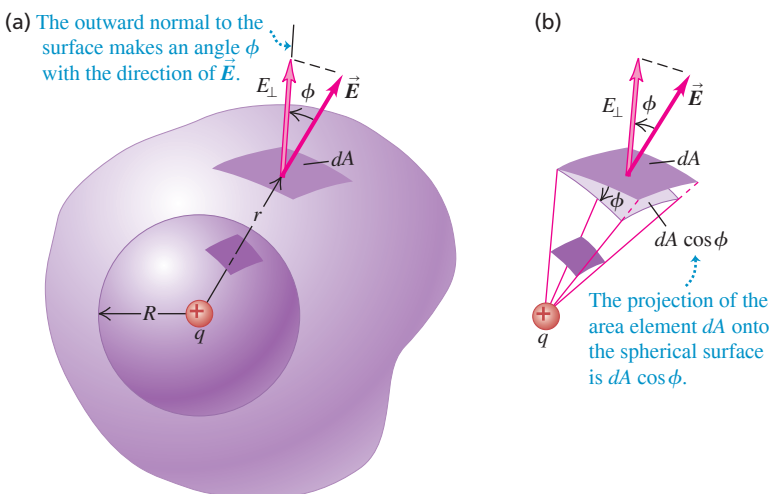
We can divide the entire irregular surface into elements dA , compute the electric flux $E dA \cos \phi$ for each, and sum the results by integrating, as in Eq. (22.5). Each of the area elements projects onto a corresponding spherical surface element. Thus the *total* electric flux through the irregular surface, given by any of the forms of Eq. (22.5), must be the same as the total flux through a sphere, which Eq. (22.6) shows is equal to q/ϵ_0 . Thus, for the irregular surface,

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0} \quad (22.7)$$

Equation (22.7) holds for a surface of *any* shape or size, provided only that it is a *closed* surface enclosing the charge q . The circle on the integral sign reminds us that the integral is always taken over a *closed* surface.

The area elements $d\vec{A}$ and the corresponding unit vectors $\hat{n}$ always point *out of* the volume enclosed by the surface. The electric flux is then positive in areas where the electric field points out of the surface and negative where it points inward. Also, $E_{\perp}$ is positive at

Figure 22.12 Calculating the electric flux through a nonspherical surface.



points where $\vec{E}$ points out of the surface and negative at points where $\vec{E}$ points into the surface.

If the point charge in Fig. 22.12 is negative, the $\vec{E}$ field is directed radially *inward*; the angle ϕ is then greater than 90° , its cosine is negative, and the integral in Eq. (22.7) is negative. But since q is also negative, Eq. (22.7) holds.

For a closed surface enclosing *no* charge,

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = 0$$

This is the mathematical statement that when a region contains no charge, any field lines caused by charges *outside* the region that enter on one side must leave again on the other side. (In Section 22.1 we came to the same conclusion by considering the special case of a rectangular box in a uniform field.) **Figure 22.13** illustrates this point. *Electric field lines can begin or end inside a region of space only when there is charge in that region.*

General Form of Gauss's Law

Now comes the final step in obtaining the general form of Gauss's law. Suppose the surface encloses not just one point charge q but several charges $q_1, q_2, q_3, \dots$. The total (resultant) electric field $\vec{E}$ at any point is the vector sum of the $\vec{E}$ fields of the individual charges. Let Q_{encl} be the *total* charge enclosed by the surface: $Q_{\text{encl}} = q_1 + q_2 + q_3 + \dots$. Also let $\vec{E}$ be the *total* field at the position of the surface area element $d\vec{A}$, and let $E_\perp$ be its component perpendicular to the plane of that element (that is, parallel to $d\vec{A}$). Then we can write an equation like Eq. (22.7) for each charge and its corresponding field and add the results. When we do, we obtain the general statement of Gauss's law:

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$ (22.8)

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant

The total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by ϵ_0 .

Using the definition of Q_{encl} and the various ways to express electric flux given in Eq. (22.5), we can express Gauss's law in the following equivalent forms:

Various forms of Gauss's law:

$$\Phi_E = \oint E \cos \phi \, dA = \oint E_\perp \, dA = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

Electric flux through a closed surface

Magnitude of electric field $\vec{E}$

Angle between $\vec{E}$ and normal to surface

Component of $\vec{E}$ perpendicular to surface

Element of surface area

Vector element of surface area

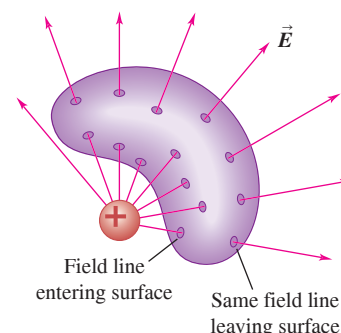
Total charge enclosed by surface

Electric constant

As in Eq. (22.5), the various forms of the integral all express the same thing, the total electric flux through the Gaussian surface, in different terms. One form is sometimes more convenient than another.

As an example, **Fig. 22.14a** (next page) shows a spherical Gaussian surface of radius r around a positive point charge $+q$. The electric field points out of the Gaussian surface, so at every point on the surface $\vec{E}$ is in the same direction as $d\vec{A}$, $\phi = 0$, and $E_\perp$ is equal to the field magnitude $E = q/4\pi\epsilon_0 r^2$. Since E is the same at all points on the surface,

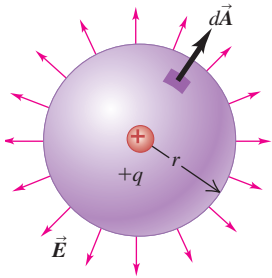
Figure 22.13 A point charge *outside* a closed surface that encloses no charge. If an electric field line from the external charge enters the surface at one point, it must leave at another.



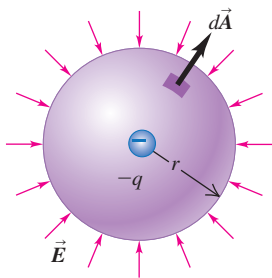
CAUTION **Gaussian surfaces are imaginary** Remember that the closed surface in Gauss's law is *imaginary*; there need not be any material object at the position of the surface. We often refer to a closed surface used in Gauss's law as a **Gaussian surface**.

Figure 22.14 Spherical Gaussian surfaces around (a) a positive point charge and (b) a negative point charge.

(a) Gaussian surface around positive charge: positive (outward) flux



(b) Gaussian surface around negative charge: negative (inward) flux



we can take it outside the integral in Eq. (22.9). Then the remaining integral is $\int dA = A = 4\pi r^2$, the area of the sphere. Hence Eq. (22.9) becomes

$$\Phi_E = \oint E_{\perp} dA = \oint \left(\frac{q}{4\pi\epsilon_0 r^2} \right) dA = \frac{q}{4\pi\epsilon_0 r^2} \oint dA = \frac{q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{q}{\epsilon_0}$$

The enclosed charge Q_{encl} is just the charge $+q$, so this agrees with Gauss's law. If the Gaussian surface encloses a *negative* point charge as in Fig. 22.14b, then $\vec{E}$ points *into* the surface at each point in the direction opposite $d\vec{A}$. Then $\phi = 180^\circ$ and $E_{\perp}$ is equal to the negative of the field magnitude: $E_{\perp} = -E = -|q|/4\pi\epsilon_0 r^2 = -q/4\pi\epsilon_0 r^2$. Equation (22.9) then becomes

$$\Phi_E = \oint E_{\perp} dA = \oint \left(\frac{-q}{4\pi\epsilon_0 r^2} \right) dA = \frac{-q}{4\pi\epsilon_0 r^2} \oint dA = \frac{-q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{-q}{\epsilon_0}$$

This again agrees with Gauss's law because the enclosed charge in Fig. 22.14b is $Q_{\text{encl}} = -q$.

In Eqs. (22.8) and (22.9), Q_{encl} is always the algebraic sum of all the positive and negative charges enclosed by the Gaussian surface, and $\vec{E}$ is the *total* field at each point on the surface. Also note that in general, this field is caused partly by charges inside the surface and partly by charges outside. But as Fig. 22.13 shows, the outside charges do *not* contribute to the total (net) flux through the surface. So Eqs. (22.8) and (22.9) are correct even when there are charges outside the surface that contribute to the electric field at the surface. When $Q_{\text{encl}} = 0$, the total flux through the Gaussian surface must be zero, even though some areas may have positive flux and others may have negative flux (see Fig. 22.3b).

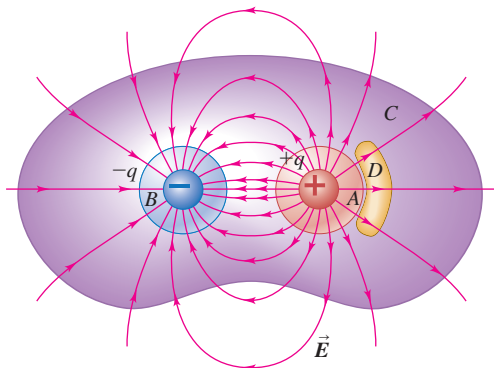
Gauss's law is the definitive answer to the question we posed at the beginning of Section 22.1: "If the electric-field pattern is known in a given region, what can we determine about the charge distribution in that region?" It provides a relationship between the electric field on a closed surface and the charge distribution within that surface. But in some cases we can use Gauss's law to answer the reverse question: "If the charge distribution is known, what can we determine about the electric field that the charge distribution produces?" Gauss's law may seem like an unappealing way to address this question, since it may look as though evaluating the integral in Eq. (22.8) is a hopeless task. Sometimes it is, but other times it is surprisingly easy. Here's an example in which *no* integration is involved at all; we'll work out several more examples in the next section.

CONCEPTUAL EXAMPLE 22.4 Electric flux and enclosed charge

WITH VARIATION PROBLEMS

Figure 22.15 shows the field produced by two point charges $+q$ and $-q$ (an electric dipole). Find the electric flux through each of the closed surfaces A , B , C , and D .

Figure 22.15 The net number of field lines leaving a closed surface is proportional to the total charge enclosed by that surface.



SOLUTION Gauss's law, Eq. (22.8), says that the total electric flux through a closed surface is equal to the total enclosed charge divided by ϵ_0 . In Fig. 22.15, surface A (shown in red) encloses the positive charge, so $Q_{\text{encl}} = +q$; surface B (in blue) encloses the negative charge, so $Q_{\text{encl}} = -q$; surface C (in purple) encloses *both* charges, so $Q_{\text{encl}} = +q + (-q) = 0$; and surface D (in yellow) encloses no charges, so $Q_{\text{encl}} = 0$. Hence, without having to do any integration, we have $\Phi_{EA} = +q/\epsilon_0$, $\Phi_{EB} = -q/\epsilon_0$, and $\Phi_{EC} = \Phi_{ED} = 0$. These results depend only on the charges enclosed within each Gaussian surface, not on the precise shapes of the surfaces.

We can draw similar conclusions by examining the electric field lines. All the field lines that cross surface A are directed out of the surface, so the flux through A must be positive. Similarly, the flux through B must be negative since all of the field lines that cross that surface point inward. For both surface C and surface D , there are as many field lines pointing into the surface as there are field lines pointing outward, so the flux through each of these surfaces is zero.

KEYCONCEPT The *net* electric flux Φ_E through a closed surface is always proportional to the *net* amount of charge q that the surface encloses. If q is zero, Φ_E is zero. These statements (Gauss's law) are true no matter what the shape of the surface.

3. If $\vec{E}$ is *perpendicular* (normal) at every point to a surface with area A , if it points *outward* from the interior of the surface, and if it has the same *magnitude* at every point on the surface, then $E_{\perp} = E = \text{constant}$, and $\int E_{\perp} dA$ over that surface is equal to EA . (If $\vec{E}$ is inward, then $E_{\perp} = -E$ and $\int E_{\perp} dA = -EA$.) This should be the case for part or all of your Gaussian surface. If $\vec{E}$ is tangent to a surface at every point, then $E_{\perp} = 0$ and the integral over that surface is zero. This may be the case for parts of a cylindrical Gaussian surface. If $\vec{E} = 0$ at every point on a surface, the integral is zero.
4. Even when there is *no* charge within a Gaussian surface, the field at any given point on the surface is not necessarily zero. In that case, however, the total electric flux through the surface is always zero.

5. The flux integral $\oint E_{\perp} dA$ can be approximated as the difference between the numbers of electric lines of force leaving and entering the Gaussian surface. In this sense the flux gives the sign of the enclosed charge, but is only proportional to it; zero flux corresponds to zero enclosed charge.
6. Once you have evaluated $\oint E_{\perp} dA$, use Eq. (22.9) to solve for your target variable.

EVALUATE your answer: If your result is a *function* that describes how the magnitude of the electric field varies with position, ensure that it makes sense.

EXAMPLE 22.5 Field of a charged conducting sphere

WITH VARIATION PROBLEMS

We place a total positive charge q on a solid conducting sphere with radius R (Fig. 22.18). Find $\vec{E}$ at any point inside or outside the sphere.

IDENTIFY and SET UP As we discussed earlier in this section, all of the charge must be on the surface of the sphere. The charge is free to move on the conductor, and there is no preferred position on the surface; the charge is therefore distributed *uniformly* over the surface, and the system is spherically symmetric. To exploit this symmetry, we take as our Gaussian surface a sphere of radius r centered on the conductor. We can calculate the field inside or outside the conductor by taking $r < R$ or $r > R$, respectively. In either case, the point at which we want to calculate $\vec{E}$ lies on the Gaussian surface.

EXECUTE The spherical symmetry means that the direction of the electric field must be radial; that's because there is no preferred direction parallel to the surface, so $\vec{E}$ can have no component parallel to the surface. There is also no preferred orientation of the sphere, so the field magnitude E can depend only on the distance r from the center and must have the same value at all points on the Gaussian surface.

For $r > R$ the entire conductor is within the Gaussian surface, so the enclosed charge is q . The area of the Gaussian surface is $4\pi r^2$, and $\vec{E}$ is uniform over the surface and perpendicular to it at each point. The flux integral $\oint E_{\perp} dA$ is then just $E(4\pi r^2)$, and Eq. (22.8) gives

$$E(4\pi r^2) = \frac{q}{\epsilon_0} \quad \text{and} \\ E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad (\text{outside a charged conducting sphere})$$

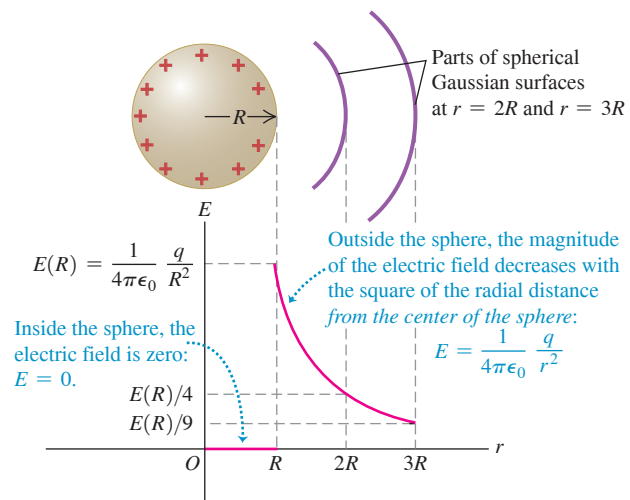
This expression is the same as that for a point charge; outside the charged sphere, its field is the same as though the entire charge were concentrated at its center. Just outside the surface of the sphere, where $r = R$,

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \quad (\text{at the surface of a charged conducting sphere})$$

CAUTION **Flux can be positive or negative** Remember that we have chosen the charge q to be *positive*. If the charge is negative, the electric field is radially *inward* instead of radially outward, and the electric flux through the Gaussian surface is negative. The electric-field magnitudes outside and at the surface of the sphere are given by the same expressions as above, except that q denotes the *magnitude* (absolute value) of the charge. ■

For $r < R$ we again have $E(4\pi r^2) = Q_{\text{encl}}/\epsilon_0$. But now our Gaussian surface (which lies entirely within the conductor) encloses *no* charge, so $Q_{\text{encl}} = 0$. The electric field inside the conductor is therefore zero.

Figure 22.18 Calculating the electric field of a conducting sphere with positive charge q . Outside the sphere, the field is the same as if all of the charge were concentrated at the center of the sphere.



EVALUATE We already knew that $\vec{E} = 0$ inside a solid conductor (whether spherical or not) when the charges are at rest. Figure 22.18 shows E as a function of the distance r from the center of the sphere. Note that in the limit as $R \rightarrow 0$, the sphere becomes a point charge; there is then only an “outside,” and the field is everywhere given by $E = q/4\pi\epsilon_0 r^2$. Thus we have deduced Coulomb’s law from Gauss’s law. (In Section 22.3 we deduced Gauss’s law from Coulomb’s law; the two laws are equivalent.)

We can also use this method for a conducting spherical *shell* (a spherical conductor with a concentric spherical hole inside) if there is no charge inside the hole. We use a spherical Gaussian surface with radius r less than the radius of the hole. If there *were* a field inside the hole, it would have to be radial and spherically symmetric as before, so $E = Q_{\text{encl}}/4\pi\epsilon_0 r^2$. But now there is no enclosed charge, so $Q_{\text{encl}} = 0$ and $E = 0$ inside the hole.

Can you use this same technique to find the electric field in the region between a charged sphere and a concentric hollow conducting sphere that surrounds it?

KEYCONCEPT To apply Gauss’s law to a charge distribution with spherical symmetry, use a spherical Gaussian surface. At all points on such a surface, the electric field has the same magnitude and is normal to the surface.

EXAMPLE 22.6 Field of a uniform line charge

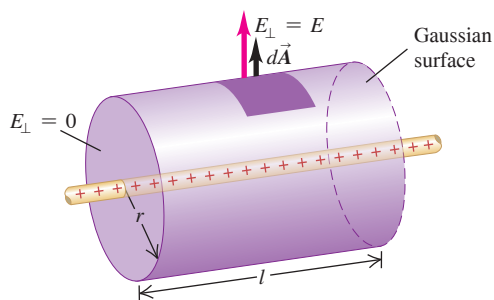
WITH  **VARIATION PROBLEMS**

Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ (assumed positive). Find the electric field by using Gauss's law.

IDENTIFY and SET UP We found in Example 21.10 (Section 21.5) that the field $\vec{E}$ of a uniformly charged, infinite wire is radially outward if λ is positive and radially inward if λ is negative, and that the field magnitude E depends on only the radial distance from the wire. This suggests that we use a *cylindrical* Gaussian surface, of radius r and arbitrary length l , coaxial with the wire and with its ends perpendicular to the wire (Fig. 22.19).

EXECUTE The flux through the flat ends of our Gaussian surface is zero because the radial electric field is parallel to these ends, and so $\vec{E} \cdot \hat{n} = 0$. On the cylindrical part of our surface we have $\vec{E} \cdot \hat{n} = E_{\perp} = E$ everywhere. (If λ were negative, we would have $\vec{E} \cdot \hat{n} = E_{\perp} = -E$ everywhere.) The area of the cylindrical surface is $2\pi rl$, so the flux through it—and hence the total flux Φ_E through

Figure 22.19 A coaxial cylindrical Gaussian surface is used to find the electric field outside an infinitely long, charged wire.



the Gaussian surface—is $EA = 2\pi rlE$. The total enclosed charge is $Q_{\text{encl}} = \lambda l$, and so from Gauss's law, Eq. (22.8),

$$\Phi_E = 2\pi rlE = \frac{\lambda l}{\epsilon_0} \quad \text{and}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \quad (\text{field of an infinite line of charge})$$

We found this same result in Example 21.10 with *much* more effort.

If λ is *negative*, $\vec{E}$ is directed radially inward, and in the above expression for E we must interpret λ as the absolute value of the charge per unit length.

EVALUATE We saw in Example 21.10 that the *entire* charge on the wire contributes to the field at any point, and yet we consider only that part of the charge $Q_{\text{encl}} = \lambda l$ within the Gaussian surface when we apply Gauss's law. There's nothing inconsistent here; it takes the entire charge to give the field the properties that allow us to calculate Φ_E so easily, and Gauss's law always applies to the enclosed charge only. If the wire is short, the symmetry of the infinite wire is lost, and E is not uniform over a coaxial, cylindrical Gaussian surface. Gauss's law then *cannot* be used to find Φ_E ; we must solve the problem the hard way, as in Example 21.10.

We can use the Gaussian surface in Fig. 22.19 to show that the field outside a long, uniformly charged cylinder is the same as though all the charge were concentrated on a line along its axis (see Problem 22.39). We can also calculate the electric field in the space between a charged cylinder and a coaxial hollow conducting cylinder surrounding it (see Problem 22.37).

KEYCONCEPT To apply Gauss's law to a charge distribution with cylindrical symmetry, such as an infinite line or cylinder of charge, use a cylindrical Gaussian surface. At all points on the curved sides of the surface, the electric field has the same magnitude and is normal to the surface. On the flat ends of the cylindrical surface, the electric field is parallel to the surface, so the flux through the ends is zero.

EXAMPLE 22.7 Field of an infinite plane sheet of charge

WITH  **VARIATION PROBLEMS**

Use Gauss's law to find the electric field caused by a thin, flat, infinite sheet with a uniform positive surface charge density σ .

IDENTIFY and SET UP In Example 21.11 (Section 21.5) we found that the field $\vec{E}$ of a uniformly charged infinite sheet is normal to the sheet, and that its magnitude is independent of the distance from the sheet. To take advantage of these symmetry properties, we use a cylindrical Gaussian surface with ends of area A and with its axis perpendicular to the sheet of charge (Fig. 22.20).

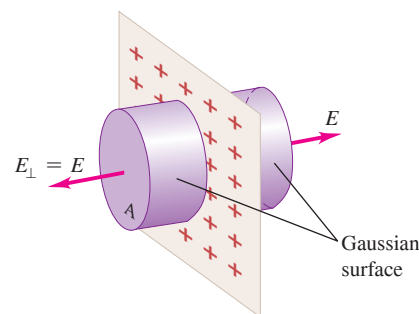
EXECUTE The flux through the cylindrical part of our Gaussian surface is zero because $\vec{E} \cdot \hat{n} = 0$ everywhere. The flux through each flat end of the surface is $+EA$ because $\vec{E} \cdot \hat{n} = E_{\perp} = E$ everywhere, so the total flux through both ends—and hence the total flux Φ_E through the Gaussian surface—is $+2EA$. The total enclosed charge is $Q_{\text{encl}} = \sigma A$, and so from Gauss's law,

$$2EA = \frac{\sigma A}{\epsilon_0} \quad \text{and}$$

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{field of an infinite sheet of charge})$$

If σ is negative, $\vec{E}$ is directed *toward* the sheet, the flux through the Gaussian surface in Fig. 22.20 is negative, and σ in the expression $E = \sigma/2\epsilon_0$ denotes the magnitude (absolute value) of the charge density.

Figure 22.20 A cylindrical Gaussian surface is used to find the field of an infinite plane sheet of charge.



EVALUATE We got the same result for the field of an infinite sheet of charge in Example 21.11 (Section 21.5). That calculation was much more complex and involved a fairly challenging integral. Thanks to the favorable symmetry, Gauss's law makes it much easier to solve this problem.

KEYCONCEPT To apply Gauss's law to a charge distribution with planar symmetry, such as an infinite sheet of charge, use a cylindrical Gaussian surface. At all points on the flat ends of the surface, the electric field has the same magnitude and is normal to the surface. On the curved sides of the cylindrical surface, the electric field is parallel to the surface, so the flux through the sides is zero.

EXAMPLE 22.8 Field between oppositely charged parallel conducting plates**WITH VARIATION PROBLEMS**

Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field in the region between the plates.

IDENTIFY and SET UP Figure 22.21a shows the field. Because opposite charges attract, most of the charge accumulates at the opposing faces of the plates. A small amount of charge resides on the *outer* surfaces of the plates, and there is some spreading or “fringing” of the field at the edges. But if the plates are very large in comparison to the distance between them, the amount of charge on the outer surfaces is negligibly small, and the fringing can be ignored except near the edges. In this case we can assume that the field is uniform in the interior region between the plates, as in Fig. 22.21b, and that the charges are distributed uniformly over the opposing surfaces. To exploit this symmetry, we can use the shaded Gaussian surfaces S_1 , S_2 , S_3 , and S_4 . These surfaces are cylinders with flat ends of area A ; one end of each surface lies *within* a plate.

EXECUTE The left-hand end of surface S_1 is within the positive plate 1. Since the field is zero within the volume of any solid conductor under electrostatic conditions, there is no electric flux through this end. The electric field between the plates is perpendicular to the right-hand end, so on that end, $E_\perp$ is equal to E and the flux is EA ; this is positive, since $\vec{E}$ is directed out of the Gaussian surface. There is no flux through the side walls of the cylinder, since these walls are parallel to $\vec{E}$. So the total

flux integral in Gauss's law is EA . The net charge enclosed by the cylinder is σA , so Eq. (22.8) yields $EA = \sigma A/\epsilon_0$; we then have

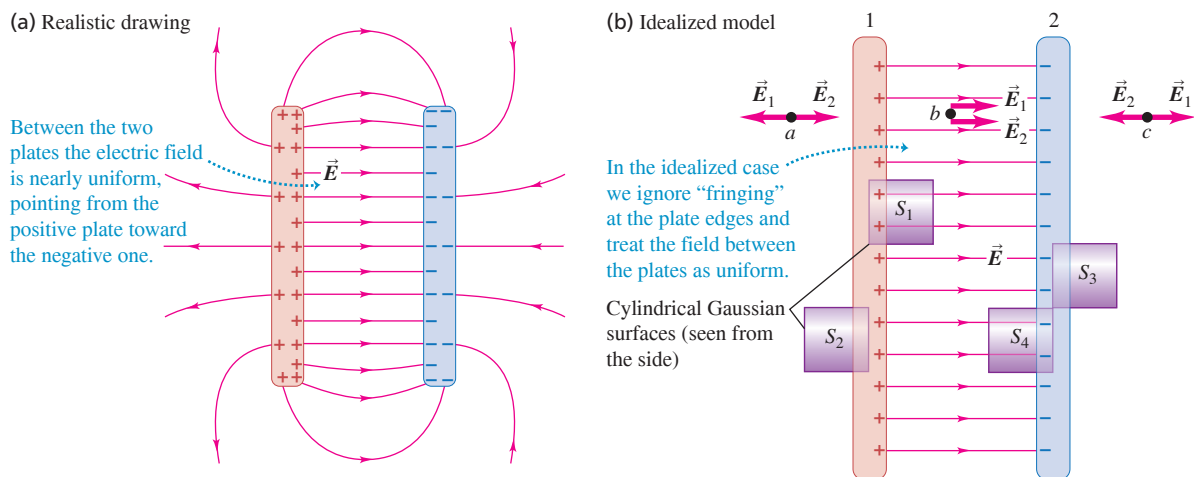
$$E = \frac{\sigma}{\epsilon_0} \quad (\text{field between oppositely charged conducting plates})$$

The field is uniform and perpendicular to the plates, and its magnitude is independent of the distance from either plate. The Gaussian surface S_4 yields the same result. Surfaces S_2 and S_3 yield $E = 0$ to the left of plate 1 and to the right of plate 2, respectively. We leave these calculations to you.

EVALUATE We obtained the same results in Example 21.12 by using the principle of superposition of electric fields. The fields due to the two sheets of charge (one on each plate) are $\vec{E}_1$ and $\vec{E}_2$; from Example 22.7, both of these have magnitude $\sigma/2\epsilon_0$. The total electric field at any point is the vector sum $\vec{E} = \vec{E}_1 + \vec{E}_2$. At points a and c in Fig. 22.21b, $\vec{E}_1$ and $\vec{E}_2$ point in opposite directions, and their sum is zero. At point b , $\vec{E}_1$ and $\vec{E}_2$ are in the same direction; their sum has magnitude $E = \sigma/\epsilon_0$, just as we found by using Gauss's law.

KEYCONCEPT The electric field between two large, parallel, oppositely charged conducting plates is uniform in the region between the plates. The field is zero in the region outside the plates as well as in the interiors of the plates. The charge on either plate lies on the surface that faces the other plate.

Figure 22.21 Electric field between oppositely charged parallel plates.

**EXAMPLE 22.9** Field of a uniformly charged sphere**WITH VARIATION PROBLEMS**

Positive electric charge Q is distributed uniformly *throughout the volume* of an *insulating* sphere with radius R . Find the magnitude of the electric field at a point P a distance r from the center of the sphere.

IDENTIFY and SET UP As in Example 22.5, the system is spherically symmetric. Hence we can use the conclusions of that example about the direction and magnitude of $\vec{E}$. To make use of the spherical symmetry, we choose as our Gaussian surface a sphere with radius r , concentric with the charge distribution.

EXECUTE From symmetry, the direction of $\vec{E}$ is radial at every point on the Gaussian surface, so $E_\perp = E$ and the field magnitude E is the same at every point on the surface. Hence the total electric flux through the Gaussian surface is the product of E and the total area of the surface $A = 4\pi r^2$ —that is, $\Phi_E = 4\pi r^2 E$.

The amount of charge enclosed within the Gaussian surface depends on r . To find E *inside* the sphere, we choose $r < R$. The volume

charge density ρ is the charge Q divided by the volume of the entire charged sphere of radius R :

$$\rho = \frac{Q}{4\pi R^3/3}$$

The volume V_{encl} enclosed by the Gaussian surface is $\frac{4}{3}\pi r^3$, so the total charge Q_{encl} enclosed by that surface is

$$Q_{\text{encl}} = \rho V_{\text{encl}} = \left(\frac{Q}{4\pi R^3/3} \right) \left(\frac{4}{3}\pi r^3 \right) = Q \frac{r^3}{R^3}$$

Then Gauss's law, Eq. (22.8), becomes

$$4\pi r^2 E = \frac{Q}{\epsilon_0} \frac{r^3}{R^3} \quad \text{and} \quad E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3} \quad (\text{field inside a uniformly charged sphere})$$

The field magnitude is proportional to the distance r of the field point from the center of the sphere (see the graph of E versus r in Fig. 22.22).

To find E outside the sphere, we take $r > R$. This surface encloses the entire charged sphere, so $Q_{\text{encl}} = Q$, and Gauss's law gives

$$4\pi r^2 E = \frac{Q}{\epsilon_0} \quad \text{and}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (\text{field outside a uniformly charged sphere})$$

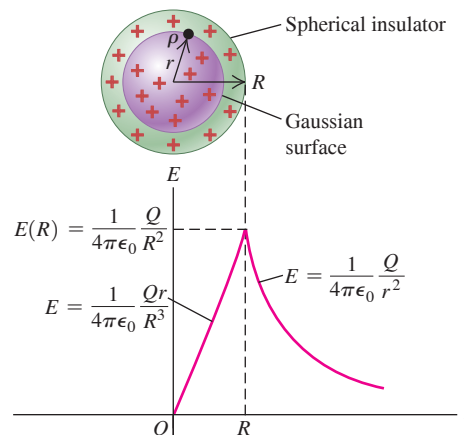
The field outside *any* spherically symmetric charged body varies as $1/r^2$, as though the entire charge were concentrated at the center. This is graphed in Fig. 22.22.

If the charge is *negative*, $\vec{E}$ is radially *inward* and in the expressions for E we interpret Q as the absolute value of the charge.

EVALUATE Notice that if we set $r = R$ in either expression for E , we get the same result $E = Q/4\pi\epsilon_0 R^2$ for the magnitude of the field at the surface of the sphere. This is because the magnitude E is a *continuous* function of r . By contrast, for the charged conducting sphere of Example 22.5 the electric-field magnitude is *discontinuous* at $r = R$ (it jumps from $E = 0$ just inside the sphere to $E = Q/4\pi\epsilon_0 R^2$ just outside the sphere). In general, the electric field $\vec{E}$ is discontinuous in magnitude, direction, or both wherever there is a *sheet* of charge, such as at the surface of a charged conducting sphere (Example 22.5), at the surface of an infinite charged sheet (Example 22.7), or at the surface of a charged conducting plate (Example 22.8).

The approach used here can be applied to *any* spherically symmetric distribution of charge, even if it is not radially uniform, as it was here.

Figure 22.22 The magnitude of the electric field of a uniformly charged insulating sphere. Compare this with the field for a conducting sphere (see Fig. 22.18).



Such charge distributions occur within many atoms and atomic nuclei, so Gauss's law is useful in atomic and nuclear physics.

KEYCONCEPT For any spherically symmetric distribution of charge, the field at a distance r from the center of the distribution is the same as if all the charge interior to r were concentrated in a point at the center. The charge at distances greater than r from the center has no effect on the field at r .

EXAMPLE 22.10 Charge on a hollow sphere

WITH VARIATION PROBLEMS

A thin-walled, hollow sphere of radius 0.250 m has an unknown charge distributed uniformly over its surface. At a distance of 0.300 m from the center of the sphere, the electric field points radially inward and has magnitude 1.80×10^2 N/C. How much charge is on the sphere?

IDENTIFY and SET UP The charge distribution is spherically symmetric. As in Examples 22.5 and 22.9, it follows that the electric field is radial everywhere and its magnitude is a function of only the radial distance r from the center of the sphere. We use a spherical Gaussian surface that is concentric with the charge distribution and has radius $r = 0.300$ m. Our target variable is $Q_{\text{encl}} = q$.

EXECUTE The charge distribution is the same as if the charge were on the surface of a 0.250-m-radius conducting sphere. Hence we can borrow the results of Example 22.5. We note that the electric field here is directed toward the sphere, so that q must be *negative*. Furthermore, the electric field is directed into the Gaussian surface, so that $E_{\perp} = -E$ and $\Phi_E = \oint E_{\perp} dA = -E(4\pi r^2)$.

By Gauss's law, the flux is equal to the charge q on the sphere (all of which is enclosed by the Gaussian surface) divided by ϵ_0 . Solving for q , we find

$$q = -E(4\pi\epsilon_0 r^2) = -(1.80 \times 10^2 \text{ N/C})(4\pi) \\ \times (8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.300 \text{ m})^2 \\ = -1.80 \times 10^{-9} \text{ C} = -1.80 \text{ nC}$$

EVALUATE To determine the charge, we had to know the electric field at *all* points on the Gaussian surface so that we could calculate the flux integral. This was possible here because the charge distribution is highly symmetric. If the charge distribution is irregular or lacks symmetry, Gauss's law is not very useful for calculating the charge distribution from the field, or vice versa.

KEYCONCEPT The electric field outside a spherical distribution of charge is the same as if all of the charge in the distribution were concentrated at its center.

TEST YOUR UNDERSTANDING OF SECTION 22.4 You place a known amount of charge Q on the irregularly shaped conductor shown in Fig. 22.17. If you know the size and shape of the conductor, can you use Gauss's law to calculate the electric field at an arbitrary position outside the conductor?

ANSWER

no You might be tempted to draw a Gaussian surface that is an enlarged version of the conductor, with the same shape and placed so that it completely encloses the conductor. While you know the flux through this Gaussian surface (by Gauss's law, it's $\Phi_E = Q/\epsilon_0$), the direction of the electric field need not be perpendicular to the surface and the magnitude of the field need not be the same at all points on the surface. It's not possible to do the flux integral $\oint \vec{E} \cdot d\vec{A}$, and we can't calculate the electric field. Gauss's law is useful for calculating the electric field only when the charge distribution is highly symmetric.

BIO APPLICATION Charge**Distribution Inside a Nerve Cell**

The interior of a human nerve cell contains both positive potassium ions (K^+) and negatively charged protein molecules (Pr^-). Potassium ions can flow out of the cell through the cell membrane, but the much larger protein molecules cannot. The result is that the interior of the cell has a net negative charge. (The fluid outside the cell has a positive charge that balances this.) The fluid within the cell is a good conductor, so the Pr^- molecules distribute themselves on the outer surface of the fluid—that is, on the inner surface of the cell membrane, which is an insulator. This is true no matter what the shape of the cell.

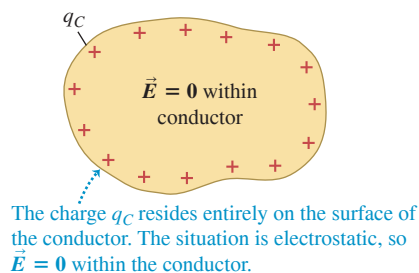
**22.5 CHARGES ON CONDUCTORS**

We have learned that in an electrostatic situation (in which there is no net motion of charge) the electric field at every point within a conductor is zero and any excess charge on a solid conductor is located entirely on its surface (**Fig. 22.23a**). But what if there is a *cavity* inside the conductor (**Fig. 22.23b**)? If there is no charge within the cavity, we can use a Gaussian surface such as *A* (which lies completely within the material of the conductor) to show that the *net* charge on the *surface of the cavity* must be zero, because $\vec{E} = 0$ everywhere on the Gaussian surface. In fact, we can prove in this situation that there can't be any charge *anywhere* on the cavity surface. We'll postpone detailed proof of this statement until Chapter 23.

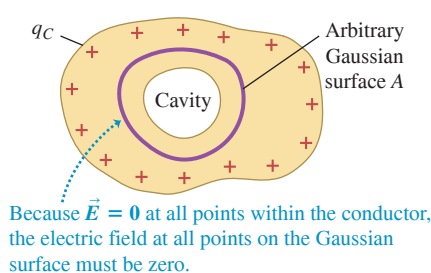
Suppose we place a small body with a charge q inside a cavity within a conductor (**Fig. 22.23c**). The conductor is uncharged and is insulated from the charge q . Again $\vec{E} = 0$ everywhere on surface *A*, so according to Gauss's law the *total* charge inside this surface must be zero. Therefore there must be a charge $-q$ distributed on the surface of the cavity, drawn there by the charge q inside the cavity. The *total* charge on the conductor must remain zero, so a charge $+q$ must appear either on its outer surface or inside the material. But we showed that in an electrostatic situation there can't be any excess charge within the material of a conductor. So we conclude that charge $+q$ must appear on the outer surface. By the same reasoning, if the conductor originally had a charge q_C , then the total charge on the outer surface must be $q_C + q$ after charge q is inserted into the cavity.

Figure 22.23 Finding the electric field within a charged conductor.

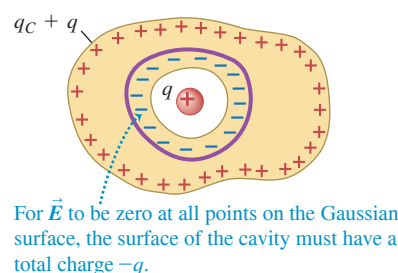
(a) Solid conductor with charge q_C



(b) The same conductor with an internal cavity



(c) An isolated charge q placed in the cavity

**CONCEPTUAL EXAMPLE 22.11 A conductor with a cavity****WITH VARIATION PROBLEMS**

A conductor with a cavity carries a total charge of $+7$ nC. Within the cavity, insulated from the conductor, is a point charge of -5 nC. How much charge is on each surface (inner and outer) of the conductor?

SOLUTION **Figure 22.24** shows the situation. If the charge in the cavity is $q = -5$ nC, the charge on the inner cavity surface must be $-q = -(-5 \text{ nC}) = +5$ nC. The conductor carries a *total* charge of $+7$ nC, none of which is in the interior of the material. If $+5$ nC is on the inner surface of the cavity, then there must be $(+7 \text{ nC}) - (+5 \text{ nC}) = +2$ nC on the outer surface of the conductor.

KEYCONCEPT If there is excess charge at rest on a conductor, all of that charge must lie on the surface of the conductor and the electric field inside the solid conductor must be zero. If there is a cavity inside the conductor, the net charge on the cavity walls equals the amount of charge enclosed by the cavity.

Figure 22.24 Our sketch for this problem. There is zero electric field inside the bulk conductor and hence zero flux through the Gaussian surface shown, so the charge on the cavity wall must be the opposite of the point charge.

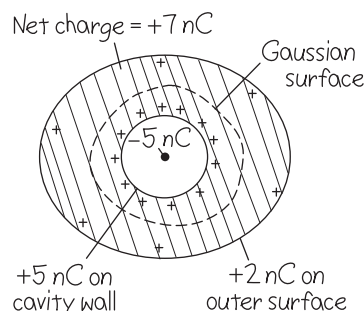
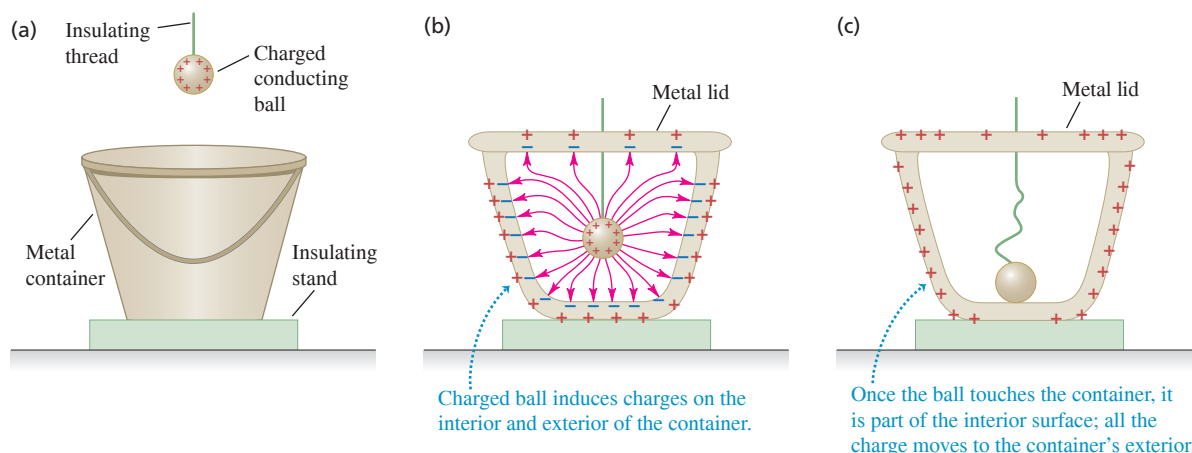


Figure 22.25 (a) A charged conducting ball suspended by an insulating thread outside a conducting container on an insulating stand. (b) The ball is lowered into the container, and the lid is put on. (c) The ball is touched to the inner surface of the container.



Testing Gauss's Law Experimentally

Figure 22.25 shows a historic experiment designed to test Gauss's law. We mount a conducting container on an insulating stand. The container is initially uncharged. Then we hang a charged metal ball from an insulating thread (Fig. 22.25a), lower it into the container, and put the lid on (Fig. 22.25b). Charges are induced on the walls of the container, as shown. But now we let the ball *touch* the inner wall (Fig. 22.25c). The surface of the ball becomes part of the cavity surface. The situation is now the same as Fig. 22.23b; if Gauss's law is correct, the net charge on the cavity surface must be zero. Thus the ball must lose all its charge. Finally, we pull the ball out; we find that it has indeed lost all its charge.

This experiment was performed in the 19th century by the English scientist Michael Faraday, using a metal icepail with a lid, and it is called **Faraday's icepail experiment**. The result confirms the validity of Gauss's law and therefore of Coulomb's law. Faraday's result was significant because Coulomb's experimental method, using a torsion balance and dividing of charges, was not very precise; it is very difficult to confirm the $1/r^2$ dependence of the electrostatic force by direct force measurements. By contrast, experiments like Faraday's test the validity of Gauss's law, and therefore of Coulomb's law, with much greater precision. Modern versions of this experiment have shown that the exponent 2 in the $1/r^2$ of Coulomb's law does not differ from precisely 2 by more than 10^{-16} . So there is no reason to believe it is anything other than exactly 2.

The same principle behind Faraday's icepail experiment is used in a *Van de Graaff electrostatic generator* (Fig. 22.26). A charged belt continuously produces a buildup of charge on the inside of a conducting shell. By Gauss's law, there can never be any charge on the inner surface of this shell, so the charge is immediately carried away to the outside surface of the shell. As a result, the charge on the shell and the electric field around it can become very large very rapidly. The Van de Graaff generator is used as an accelerator of charged particles and for physics demonstrations.

This principle also forms the basis for *electrostatic shielding*. Suppose we have a very sensitive electronic instrument that we want to protect from stray electric fields that might cause erroneous measurements. We surround the instrument with a conducting box, or we line the walls, floor, and ceiling of the room with a conducting material such as sheet copper. The external electric field redistributes the free electrons in the conductor, leaving a net positive charge on the outer surface in some regions and a net negative charge in others (Fig. 22.27, next page). This charge distribution causes an additional electric field such that the *total* field at every point inside the box is zero, as Gauss's law says it must be. The charge distribution on the box also alters the shapes of the field lines near the box, as the figure shows. Such a setup is often called a *Faraday cage*. The same

Figure 22.26 Cutaway view of the essential parts of a Van de Graaff electrostatic generator. The electron sink at the bottom draws electrons from the belt, giving it a positive charge; at the top the belt attracts electrons away from the conducting shell, giving the shell a positive charge.

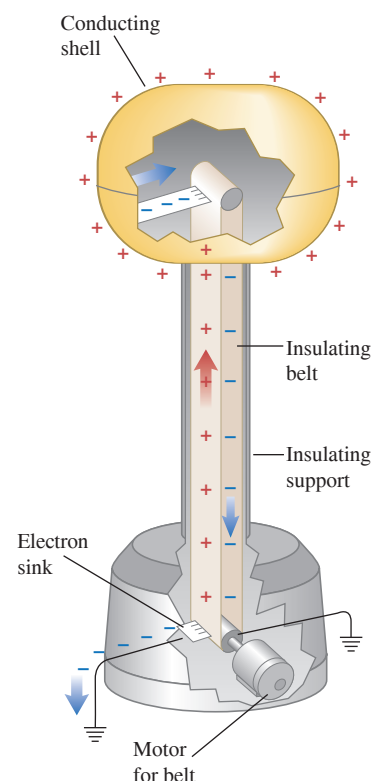


Figure 22.27 (a) A conducting box (a Faraday cage) immersed in a uniform electric field. The field of the induced charges on the box combines with the uniform field to give zero total field inside the box. (b) This person is inside a Faraday cage, and so is protected from the powerful electric discharge.

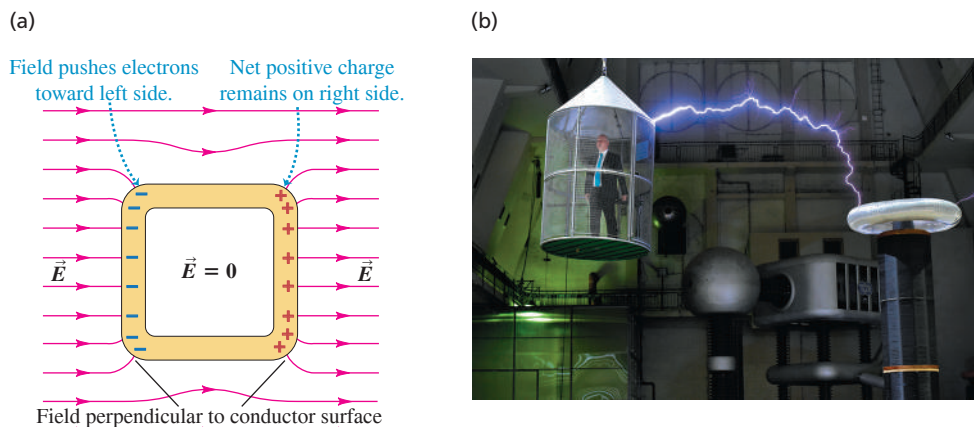
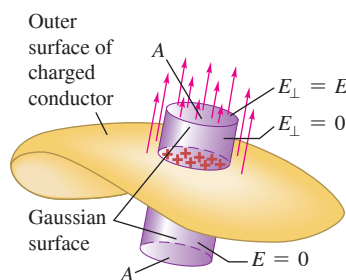


Figure 22.28 The field just outside a charged conductor is perpendicular to the surface, and its perpendicular component $E_{\perp}$ is equal to σ/ϵ_0 .



APPLICATION Why Lightning Bolts Are Vertical

Our planet is a good conductor, and its surface has a negative charge. Hence, the electric field in the atmosphere above the surface points generally downward, toward the negative charge and perpendicular to the surface (see Example 22.12). The negative charge is balanced by positive charges in the atmosphere. In a lightning storm, the vertical electric field becomes great enough to cause charges to flow vertically through the air. The air is excited and ionized by the passage of charge through it, producing a visible lightning bolt.



physics tells you that one of the safest places to be in a lightning storm is inside a car; if the car is struck by lightning, the charge tends to remain on the metal skin of the vehicle, and little or no electric field is produced inside the passenger compartment. Keep in mind this basic principle, which is a consequence of Gauss's law: *Charges outside a conductor have no effect on the interior of that conductor, even if the conductor has a cavity inside.*

Field at the Surface of a Conductor

Finally, we note that there is a direct relationship between the $\vec{E}$ field at a point just outside any conductor and the surface charge density σ at that point. In general, σ varies from point to point on the surface. We'll show in Chapter 23 that at any such point, the direction of $\vec{E}$ is always *perpendicular* to the surface. (You can see this effect in Fig. 22.27a.)

To find a relationship between σ at any point on the surface and the perpendicular component of the electric field at that point, we construct a Gaussian surface in the form of a small cylinder (Fig. 22.28). One end face, with area A , lies within the conductor and the other lies just outside. The electric field is zero at all points within the conductor. Outside the conductor the component of $\vec{E}$ perpendicular to the side walls of the cylinder is zero, and over the end face the perpendicular component is equal to $E_{\perp}$. (If σ is positive, the electric field points out of the conductor and $E_{\perp}$ is positive; if σ is negative, the field points inward and $E_{\perp}$ is negative.) Hence the total flux through the surface is $E_{\perp}A$. The charge enclosed within the Gaussian surface is σA , so from Gauss's law, $E_{\perp}A = (\sigma A)/\epsilon_0$ and

$$\text{Electric field at surface of a conductor; } \vec{E} \text{ perpendicular to surface } E_{\perp} = \frac{\sigma}{\epsilon_0} \quad (22.10)$$

Surface charge density
Electric constant

This agrees with our result for the field at the surface of a charged conducting plate (Example 22.8). We can also verify this for a charged conducting sphere. In Example 22.5 (Section 22.4) we showed that the electric field just outside the surface of a conducting sphere with radius R and total charge q is

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

The surface charge density is uniform and equal to q divided by the surface area of the sphere:

$$\sigma = \frac{q}{4\pi R^2}$$

Comparing these two expressions, we see that $E = \sigma/\epsilon_0$, which verifies Eq. (22.10).

We showed in Example 22.8 that the field magnitude between two infinite flat oppositely charged conducting plates also equals σ/ϵ_0 . In this case the field magnitude E is the same at *all* distances from the plates, but in all other cases E decreases with increasing distance from the surface.

EXAMPLE 22.12 Electric field of the earth

WITH  **ARIATION PROBLEMS**

The earth (a conductor) has a net electric charge. The resulting electric field near the surface has an average value of about 150 N/C, directed toward the center of the earth. (a) What is the corresponding surface charge density? (b) What is the *total* surface charge of the earth?

IDENTIFY and SET UP We are given the electric-field magnitude at the surface of the conducting earth. We can calculate the surface charge density σ from Eq. (22.10). The total charge Q on the earth's surface is then the product of σ and the earth's surface area.

EXECUTE (a) The direction of the field means that σ is negative (corresponding to $\vec{E}$ being directed *into* the surface, so $E_\perp$ is negative). From Eq. (22.10),

$$\begin{aligned}\sigma &= \epsilon_0 E_\perp = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(-150 \text{ N/C}) \\ &= -1.33 \times 10^{-9} \text{ C/m}^2 = -1.33 \text{ nC/m}^2\end{aligned}$$

(b) The earth's surface area is $4\pi R_E^2$, where $R_E = 6.38 \times 10^6 \text{ m}$ is the radius of the earth (see Appendix F). The total charge Q is the product $4\pi R_E^2 \sigma$, or

$$\begin{aligned}Q &= 4\pi(6.38 \times 10^6 \text{ m})^2(-1.33 \times 10^{-9} \text{ C/m}^2) \\ &= -6.8 \times 10^5 \text{ C} = -680 \text{ kC}\end{aligned}$$

EVALUATE You can check our result in part (b) by using the result of Example 22.5. Solving for Q , we find

$$\begin{aligned}Q &= 4\pi\epsilon_0 R^2 E_\perp \\ &= \frac{1}{9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2} (6.38 \times 10^6 \text{ m})^2 (-150 \text{ N/C}) \\ &= -6.8 \times 10^5 \text{ C}\end{aligned}$$

One electron has a charge of $-1.60 \times 10^{-19} \text{ C}$. Hence this much excess negative electric charge corresponds to there being $(-6.8 \times 10^5 \text{ C})/(-1.60 \times 10^{-19} \text{ C}) = 4.2 \times 10^{24}$ excess electrons on the earth, or about 7 moles of excess electrons. This is compensated by an equal *deficiency* of electrons in the earth's upper atmosphere, so the combination of the earth and its atmosphere is electrically neutral.

KEYCONCEPT The electric field $\vec{E}$ at the surface of a charged conductor is normal to the surface. The normal component of $\vec{E}$ at any point on the surface is equal to σ/ϵ_0 , where σ is the surface charge density (charge per unit area) at that point. The field points out of the surface if σ is positive and into the surface if σ is negative.

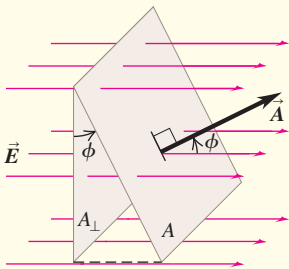
TEST YOUR UNDERSTANDING OF SECTION 22.5 A hollow conducting sphere has no net charge. There is a positive point charge q at the center of the spherical cavity within the sphere. You connect a conducting wire from the outside of the sphere to ground. Will you measure an electric field outside the sphere?

ANSWER **no** Before you connect the wire to the sphere, the presence of the point charge will induce a charge $-q$ on the inner surface of the hollow sphere and a charge q on the outer surface (the net charge on the sphere is zero). There will be an electric field outside the sphere due to the charge on the outer surface. Once you touch the conducting wire to the sphere, however, electrons will flow from ground to the outer surface of the sphere to neutralize the charge there (see Fig. 21.7c). As a result the sphere will have no charge on its outer surface and no electric field outside.

CHAPTER 22
 SUMMARY

Electric flux: Electric flux is a measure of the “flow” of electric field through a surface. It is equal to the product of an area element and the perpendicular component of $\vec{E}$, integrated over a surface. (See Examples 22.1–22.3.)

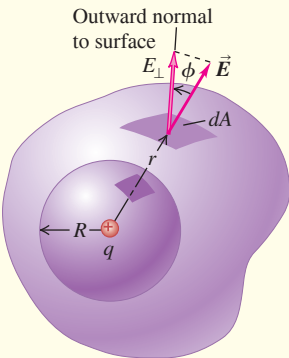
$$\begin{aligned}\Phi_E &= \int E \cos \phi \, dA \\ &= \int E_{\perp} \, dA = \int \vec{E} \cdot d\vec{A}\end{aligned}\tag{22.5}$$



Gauss’s law: Gauss’s law states that the total electric flux through a closed surface, which can be written as the surface integral of the component of $\vec{E}$ normal to the surface, equals a constant times the total charge Q_{encl} enclosed by the surface. Gauss’s law is logically equivalent to Coulomb’s law, but its use greatly simplifies problems with a high degree of symmetry. (See Examples 22.4–22.10.)

When excess charge is placed on a solid conductor and is at rest, it resides entirely on the surface, and $\vec{E} = \mathbf{0}$ everywhere in the material of the conductor. (See Examples 22.11 and 22.12.)

$$\begin{aligned}\Phi_E &= \oint E \cos \phi \, dA \\ &= \oint E_{\perp} \, dA = \oint \vec{E} \cdot d\vec{A} \\ &= \frac{Q_{\text{encl}}}{\epsilon_0}\end{aligned}\tag{22.8), (22.9)}$$



Electric field of various symmetric charge distributions: The following table lists electric fields caused by several symmetric charge distributions. In the table, q , Q , λ , and σ refer to the *magnitudes* of the quantities.

Charge Distribution	Point in Electric Field	Electric-Field Magnitude
Single point charge q	Distance r from q	$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
Charge q on surface of conducting sphere with radius R	Outside sphere, $r > R$	$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
	Inside sphere, $r < R$	$E = 0$
Infinite wire, charge per unit length λ	Distance r from wire	$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$
Infinite conducting cylinder with radius R , charge per unit length λ	Outside cylinder, $r > R$	$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$
	Inside cylinder, $r < R$	$E = 0$
Solid insulating sphere with radius R , charge Q distributed uniformly throughout volume	Outside sphere, $r > R$	$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
	Inside sphere, $r < R$	$E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3}$
Infinite sheet of charge with uniform charge per unit area σ	Any point	$E = \frac{\sigma}{2\epsilon_0}$
Two oppositely charged conducting plates with surface charge densities $+\sigma$ and $-\sigma$	Any point between plates	$E = \frac{\sigma}{\epsilon_0}$
Charged conductor	Just outside the conductor	$E = \frac{\sigma}{\epsilon_0}$



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 22.2 and 22.3** (Section 22.2) and **EXAMPLE 22.4** (Section 22.3) before attempting these problems.

VP22.4.1 An imaginary cubical surface of side L has its edges parallel to the x -, y - and z -axes, one corner at the point $x = 0, y = 0, z = 0$ and the opposite corner at the point $x = L, y = L, z = L$. The cube is in a region of uniform electric field $\vec{E} = E_1\hat{i} + E_2\hat{j}$, where E_1 and E_2 are positive constants. Calculate the electric flux through (a) the cube face in the plane $x = 0$, (b) the cube face in the plane $x = L$, (c) the cube face in the plane $y = 0$, (d) the cube face in the plane $y = L$, (e) the cube face in the plane $z = 0$, (f) the cube face in the plane $z = L$, and (g) the entire cubical surface. For each face the normal points out of the cube.

VP22.4.2 A point charge q is at the point $x = 0, y = 0, z = 0$. An imaginary hemispherical surface is made by starting with a spherical surface of radius R centered on the point $x = 0, y = 0, z = 0$ and cutting off the half in the region $z < 0$. The normal to this surface points out of the hemisphere, away from its center. Calculate the electric flux through the hemisphere if (a) $q = 8.00$ nC and $R = 0.100$ m; (b) $q = -4.00$ nC and $R = 0.100$ m; (c) $q = -4.00$ nC and $R = 0.200$ m.

VP22.4.3 An imaginary cubical surface with sides of length 5.00 cm has a point charge $q = +6.00$ nC at its center. Calculate the electric flux (a) through the entire closed cubical surface and (b) through one of the six faces of the cube.

VP22.4.4 An imaginary spherical surface of radius 5.00 cm is centered on the point $x = 0, y = 0, z = 0$. Calculate the net electric flux through the surface if the following charges are present: (a) a point charge $q_1 = +3.00$ nC at $x = 1.00$ cm, $y = 1.00$ cm, $z = 0$; (b) the charge in part (a) plus a point charge $q_2 = -8.00$ nC at $x = 2.00$ cm, $y = 0, z = -4.00$ cm; (c) the charges in part (b) plus a point charge $q_3 = +2.00$ nC at $x = 4.00$ cm, $y = -2.00$ cm, $z = 3.00$ cm.

Be sure to review **EXAMPLES 22.5, 22.6, 22.7, 22.8, 22.9, and 22.10** (Section 22.4) before attempting these problems.

VP22.10.1 A sphere of radius 5.00 cm carries charge $+3.00$ nC. Calculate the electric-field magnitude at a distance 4.00 cm from the center of the sphere and at a distance 6.00 cm from the center of the sphere if the sphere is (a) a solid insulator with the charge spread uniformly throughout its volume and (b) a solid conductor.

VP22.10.2 You place a point charge $q = -4.00$ nC a distance of 9.00 cm from an infinitely long, thin wire that has linear charge density 3.00×10^{-9} C/m. What is the magnitude of the electric force that the wire exerts on the point charge? Is the force attractive or repulsive?

VP22.10.3 A point charge $q = +1.80$ nC is 8.00 cm from a thin, flat, infinite sheet that has a uniform surface charge density. If the total electric field is zero at a point halfway between the point charge and the sheet, what is the surface charge density of the sheet? Is this positive or negative?

VP22.10.4 Two large parallel conducting plates carry charges of equal magnitude and opposite charge. When you place a point charge $q = +3.60$ nC between the plates, the force on the point charge is 22.0 μ N. What is the magnitude of the surface charge density on either plate?

Be sure to review **EXAMPLES 22.11 and 22.12** (Section 22.5) before attempting these problems.

VP22.12.1 A solid block of copper, which is a good conductor, has a cavity in its interior. Within the cavity, insulated from the conductor, is a point charge of $+3.00$ nC. The block of copper itself has an excess charge of -8.00 nC. How much charge is on (a) the outer surface of the block and (b) the surface of the cavity?

VP22.12.2 A solid sphere of silver, which is a good conductor, has a spherical cavity at its center. There is a point charge at the center of the cavity. The silver sphere has a charge of $+9.00$ nC on its outer surface and a charge of -2.00 nC on the surface of the cavity. (a) What is the value of the point charge? (b) If the point charge moved to a different position within the cavity (not at the center), would this affect the total charge on the surface of the cavity or the total charge on the outer surface of the sphere?

VP22.12.3 In an electricity demonstration at the Deutsches Museum in Munich, Germany, a person sits inside a metal sphere of radius 0.90 m. Charge is applied to the sphere until the magnitude of the electric field on the outside surface of the sphere is 3.0×10^5 N/C. When the demonstration is over, the person climbs out unharmed. During the demonstration, what are the magnitudes of (a) the electric field inside the sphere, (b) the total charge on the sphere, and (c) the surface charge density on the outside of the sphere?

VP22.12.4 A spherical conductor of radius 0.330 m has a spherical cavity of radius 0.120 m at its center. The conductor carries a total charge of -6.00 nC; in addition, at the center of the spherical cavity is a point charge of $+4.00$ nC. Find (a) the total charge on the surface of the cavity, (b) the total charge on the outer surface of the conductor, (c) the magnitude of the electric field just inside the surface of the cavity, and (d) the magnitude of the electric field just outside the outer surface of the conductor.

BRIDGING PROBLEM Electric Field Inside a Hydrogen Atom

A hydrogen atom is made up of a proton of charge $+Q = 1.60 \times 10^{-19}$ C and an electron of charge $-Q = -1.60 \times 10^{-19}$ C. The proton may be regarded as a point charge at $r = 0$, the center of the atom. The motion of the electron causes its charge to be “smeared out” into a spherical distribution around the proton (**Fig. 22.29**), so that the electron is equivalent to a charge per unit volume of $\rho(r) = -(Q/\pi a_0^3)e^{-2r/a_0}$, where $a_0 = 5.29 \times 10^{-11}$ m is called the *Bohr radius*. (a) Find the total amount of the hydrogen atom’s charge that is enclosed within a sphere with radius r centered on the proton. (b) Find the electric field (magnitude and direction) caused by the charge of the hydrogen atom as a function of r .

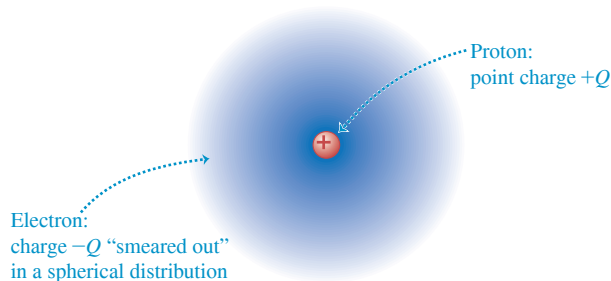
(c) Make a graph as a function of r of the ratio of the electric-field magnitude E to the magnitude of the field due to the proton alone.

SOLUTION GUIDE

IDENTIFY and SET UP

1. The charge distribution in this problem is spherically symmetric, as in Example 22.9, so you can solve it with Gauss’s law.
2. The charge within a sphere of radius r includes the proton charge $+Q$ plus the portion of the electron charge distribution

Figure 22.29 The charge distribution in a hydrogen atom.



that lies within the sphere. The difference from Example 22.9 is that the electron charge distribution is *not* uniform, so the charge enclosed within a sphere of radius r is *not* simply the charge density multiplied by the volume $4\pi r^3/3$ of the sphere. Instead, you'll have to do an integral.

3. Consider a thin spherical shell centered on the proton, with radius r' and infinitesimal thickness dr' . Since the shell is so thin, every point within the shell is at essentially the same radius from the proton. Hence the amount of electron charge within this shell is equal to the electron charge density $\rho(r')$ at this

radius multiplied by the volume dV of the shell. What is dV in terms of r' ?

4. The total electron charge within a radius r equals the integral of $\rho(r')dV$ from $r' = 0$ to $r' = r$. Set up this integral (but don't solve it yet), and use it to write an expression for the total charge (including the proton) within a sphere of radius r .

EXECUTE

5. Integrate your expression from step 4 to find the charge within radius r . (*Hint:* Integrate by substitution: Change the integration variable from r' to $x = 2r'/a_0$. You can use integration by parts to calculate the integral $\int x^2 e^{-x} dx$, or you can look it up in a table of integrals or on the Web.)
6. Use Gauss's law and your results from step 5 to find the electric field at a distance r from the proton.
7. Find the ratio referred to in part (c) and graph it versus r . (You'll actually find it simplest to graph this function versus the quantity r/a_0 .)

EVALUATE

8. How do your results for the enclosed charge and the electric-field magnitude behave in the limit $r \rightarrow 0$? In the limit $r \rightarrow \infty$? Explain your results.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q22.1 A rubber balloon has a single point charge in its interior. Does the electric flux through the balloon depend on whether or not it is fully inflated? Explain your reasoning.

Q22.2 Suppose that in Fig. 22.15 both charges were positive. What would be the fluxes through each of the four surfaces in the example?

Q22.3 In Fig. 22.15, suppose a third point charge were placed outside the purple Gaussian surface C . Would this affect the electric flux through any of the surfaces A , B , C , or D in the figure? Why or why not?

Q22.4 A certain region of space bounded by an imaginary closed surface contains no charge. Is the electric field always zero everywhere on the surface? If not, under what circumstances is it zero on the surface?

Q22.5 A spherical Gaussian surface encloses a point charge q . If the point charge is moved from the center of the sphere to a point away from the center, does the electric field at a point on the surface change? Does the total flux through the Gaussian surface change? Explain.

Q22.6 You find a sealed box on your doorstep. You suspect that the box contains several charged metal spheres packed in insulating material. How can you determine the total net charge inside the box without opening the box? Or isn't this possible?

Q22.7 A solid copper sphere has a net positive charge. The charge is distributed uniformly over the surface of the sphere, and the electric field inside the sphere is zero. Then a negative point charge outside the sphere is brought close to the surface of the sphere. Is all the net charge on the sphere still on its surface? If so, is this charge still distributed uniformly over the surface? If it is not uniform, how is it distributed? Is the electric field inside the sphere still zero? In each case justify your answers.

Q22.8 If the electric field of a point charge were proportional to $1/r^3$ instead of $1/r^2$, would Gauss's law still be valid? Explain your reasoning. (*Hint:* Consider a spherical Gaussian surface centered on a single point charge.)

Q22.9 In a conductor, one or more electrons from each atom are free to roam throughout the volume of the conductor. Does this contradict the statement that any excess charge on a solid conductor must reside on its surface? Why or why not?

Q22.10 You charge up the Van de Graaff generator shown in Fig. 22.26, and then bring an identical but uncharged hollow conducting sphere near it, without letting the two spheres touch. Sketch the distribution of charges on the second sphere. What is the net flux through the second sphere? What is the electric field inside the second sphere?

Q22.11 A lightning rod is a rounded copper rod mounted on top of a building and welded to a heavy copper cable running down into the ground. Lightning rods are used to protect houses and barns from lightning; the lightning current runs through the copper rather than through the building. Why? Why should the end of the rod be rounded?

Q22.12 A solid conductor has a cavity in its interior. Would the presence of a point charge inside the cavity affect the electric field outside the conductor? Why or why not? Would the presence of a point charge outside the conductor affect the electric field inside the cavity? Again, why or why not?

Q22.13 Explain this statement: “In a static situation, the electric field at the surface of a conductor can have no component parallel to the surface because this would violate the condition that the charges on the surface are at rest.” Would this statement be valid for the electric field at the surface of an *insulator*? Explain your answer and the reason for any differences between the cases of a conductor and an insulator.

Q22.14 In a certain region of space, the electric field $\vec{E}$ is uniform. (a) Use Gauss’s law to prove that this region of space must be electrically neutral; that is, the volume charge density ρ must be zero. (b) Is the converse true? That is, in a region of space where there is no charge, must $\vec{E}$ be uniform? Explain.

Q22.15 (a) In a certain region of space, the volume charge density ρ has a uniform positive value. Can $\vec{E}$ be uniform in this region? Explain. (b) Suppose that in this region of uniform positive ρ there is a “bubble” within which $\rho = 0$. Can $\vec{E}$ be uniform within this bubble? Explain.

Q22.16 A negative charge $-Q$ is placed inside the cavity of a hollow metal solid. The outside of the solid is grounded by connecting a conducting wire between it and the earth. Is any excess charge induced on the inner surface of the metal? Is there any excess charge on the outside surface of the metal? Why or why not? Would someone outside the solid measure an electric field due to the charge $-Q$? Is it reasonable to say that the grounded conductor has *shielded* the region outside the conductor from the effects of the charge $-Q$? In principle, could the same thing be done for gravity? Why or why not?

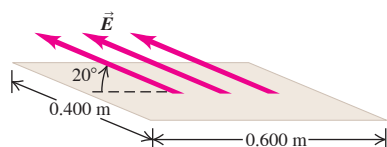
EXERCISES

Section 22.2 Calculating Electric Flux

22.1 • A flat sheet of paper of area 0.250 m^2 is oriented so that the normal to the sheet is at an angle of 60° to a uniform electric field of magnitude 14 N/C . (a) Find the magnitude of the electric flux through the sheet. (b) Does the answer to part (a) depend on the shape of the sheet? Why or why not? (c) For what angle ϕ between the normal to the sheet and the electric field is the magnitude of the flux through the sheet (i) largest and (ii) smallest? Explain your answers.

22.2 •• A flat sheet is in the shape of a rectangle with sides of lengths 0.400 m and 0.600 m . The sheet is immersed in a uniform electric field of magnitude 90.0 N/C that is directed at 20° from the plane of the sheet (Fig. E22.2). Find the magnitude of the electric flux through the sheet.

Figure E22.2



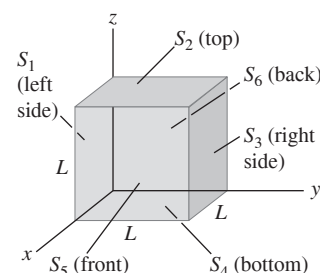
22.3 • You measure an electric field of $1.25 \times 10^6 \text{ N/C}$ at a distance of 0.150 m from a point charge. There is no other source of electric field in the region other than this point charge. (a) What is the electric flux through the surface of a sphere that has this charge at its center and that has radius 0.150 m ? (b) What is the magnitude of this charge?

22.4 • It was shown in Example 21.10 (Section 21.5) that the electric field due to an infinite line of charge is perpendicular to the line and has magnitude $E = \lambda/2\pi\epsilon_0 r$. Consider an imaginary cylinder with radius $r = 0.250 \text{ m}$ and length $l = 0.400 \text{ m}$ that has an infinite line of positive charge running along its axis. The charge per unit length on the line is $\lambda = 3.00 \mu\text{C/m}$. (a) What is the electric flux through the cylinder due to this infinite line of charge? (b) What is the flux through the cylinder if its radius is increased to $r = 0.500 \text{ m}$? (c) What is the flux through the cylinder if its length is increased to $l = 0.800 \text{ m}$?

22.5 • A uniform electric field makes an angle of 60.0° with a flat surface. The area of the surface is $6.66 \times 10^{-4} \text{ m}^2$. The resulting electric flux through the surface is $4.44 \text{ N} \cdot \text{m}^2/\text{C}$. Calculate the magnitude of the electric field.

22.6 • The cube in Fig. E22.6 has sides of length $L = 10.0 \text{ cm}$. The electric field is uniform, has magnitude $E = 4.00 \times 10^3 \text{ N/C}$, and is parallel to the xy -plane at an angle of 53.1° measured from the $+x$ -axis toward the $+y$ -axis. (a) What is the electric flux through each of the six cube faces S_1, S_2, S_3, S_4, S_5 , and S_6 ? (b) What is the total electric flux through all faces of the cube?

Figure E22.6

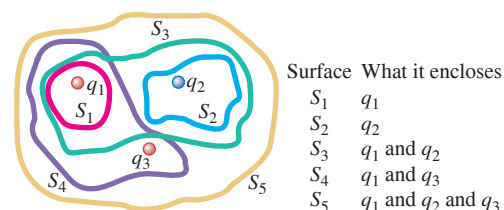


Section 22.3 Gauss’s Law

22.7 • A charge of 87.6 pC is uniformly distributed on the surface of a thin sheet of insulating material that has a total area of 29.2 cm^2 . A Gaussian surface encloses a portion of the sheet of charge. If the flux through the Gaussian surface is $5.00 \text{ N} \cdot \text{m}^2/\text{C}$, what area of the sheet is enclosed by the Gaussian surface?

22.8 • The three small spheres shown in Fig. E22.8 carry charges $q_1 = 4.00 \text{ nC}$, $q_2 = -7.80 \text{ nC}$, and $q_3 = 2.40 \text{ nC}$. Find the net electric flux through each of the following closed surfaces shown in cross section in the figure: (a) S_1 ; (b) S_2 ; (c) S_3 ; (d) S_4 ; (e) S_5 . (f) Do your answers to parts (a)–(e) depend on how the charge is distributed over each small sphere? Why or why not?

Figure E22.8



22.9 •• A charged paint is spread in a very thin uniform layer over the surface of a plastic sphere of diameter 12.0 cm , giving it a charge of $-49.0 \mu\text{C}$. Find the electric field (a) just inside the paint layer; (b) just outside the paint layer; (c) 5.00 cm outside the surface of the paint layer.

22.10 • A point charge $q_1 = 4.00 \text{ nC}$ is located on the x -axis at $x = 2.00 \text{ m}$, and a second point charge $q_2 = -6.00 \text{ nC}$ is on the y -axis at $y = 1.00 \text{ m}$. What is the total electric flux due to these two point charges through a spherical surface centered at the origin and with radius (a) 0.500 m , (b) 1.50 m , (c) 2.50 m ?

22.11 • BIO As discussed in Section 22.5, human nerve cells have a net negative charge and the material in the interior of the cell is a good conductor. If a cell has a net charge of -8.65 pC , what are the magnitude and direction (inward or outward) of the net flux through the cell boundary?

22.12 • Electric Fields in an Atom. The nuclei of large atoms, such as uranium, with 92 protons, can be modeled as spherically symmetric spheres of charge. The radius of the uranium nucleus is approximately $7.4 \times 10^{-15} \text{ m}$. (a) What is the electric field this nucleus produces just outside its surface? (b) What magnitude of electric field does it produce at the distance of the electrons, which is about $1.0 \times 10^{-10} \text{ m}$? (c) The electrons can be modeled as forming a uniform shell of negative charge. What net electric field do they produce at the location of the nucleus?

Section 22.4 Applications of Gauss's Law and

Section 22.5 Charges on Conductors

22.13 •• Two very long uniform lines of charge are parallel and are separated by 0.300 m . Each line of charge has charge per unit length $+5.20 \text{ } \mu\text{C}/\text{m}$. What magnitude of force does one line of charge exert on a 0.0500 m section of the other line of charge?

22.14 •• A solid metal sphere with radius 0.450 m carries a net charge of 0.250 nC . Find the magnitude of the electric field (a) at a point 0.100 m outside the surface of the sphere and (b) at a point inside the sphere, 0.100 m below the surface.

22.15 • (a) A conducting sphere has charge Q and radius R . If the electric field of the sphere at a distance $r = 2R$ from the center of the sphere is 1400 N/C , what is the electric field of the sphere at $r = 4R$? (b) A very long conducting cylinder of radius R has charge per unit length λ . Let r be the perpendicular distance from the axis of the cylinder. If the electric field of the cylinder at $r = 2R$ is 1400 N/C , what is the electric field at $r = 4R$? (c) A very large uniform sheet of charge has surface charge density σ . If the electric field of the sheet has a value of 1400 N/C at a perpendicular distance d from the sheet, what is the electric field of the sheet at a distance of $2d$ from the sheet?

22.16 • Some planetary scientists have suggested that the planet Mars has an electric field somewhat similar to that of the earth, producing a net electric flux of $-3.63 \times 10^{16} \text{ N} \cdot \text{m}^2/\text{C}$ at the planet's surface. Calculate: (a) the total electric charge on the planet; (b) the electric field at the planet's surface (refer to the astronomical data in Appendix F or the inside back cover); (c) the charge density on Mars, assuming all the charge is uniformly distributed over the planet's surface.

22.17 •• A very long uniform line of charge has charge per unit length $4.80 \text{ } \mu\text{C}/\text{m}$ and lies along the x -axis. A second long uniform line of charge has charge per unit length $-2.40 \text{ } \mu\text{C}/\text{m}$ and is parallel to the x -axis at $y = 0.400 \text{ m}$. What is the net electric field (magnitude and direction) at the following points on the y -axis: (a) $y = 0.200 \text{ m}$ and (b) $y = 0.600 \text{ m}$?

22.18 •• The electric field 0.400 m from a very long uniform line of charge is 840 N/C . How much charge is contained in a 2.00 cm section of the line?

22.19 •• A hollow, conducting sphere with an outer radius of 0.250 m and an inner radius of 0.200 m has a uniform surface charge density of $+6.37 \times 10^{-6} \text{ C}/\text{m}^2$. A charge of $-0.500 \text{ } \mu\text{C}$ is now introduced at the center of the cavity inside the sphere. (a) What is the new charge density on the outside of the sphere? (b) Calculate the strength of the electric field just outside the sphere. (c) What is the electric flux through a spherical surface just inside the inner surface of the sphere?

22.20 • A solid insulating sphere has total charge Q and radius R . The sphere's charge is distributed uniformly throughout its volume. Let r be the radial distance measured from the center of the sphere. If $E = 800 \text{ N/C}$ at $r = R/2$, what is E at $r = 2R$?

22.21 •• The electric field at a distance of 0.145 m from the surface of a solid insulating sphere with radius 0.355 m is 1750 N/C . (a) Assuming the sphere's charge is uniformly distributed, what is the charge density inside it? (b) Calculate the electric field inside the sphere at a distance of 0.200 m from the center.

22.22 •• A point charge of $-3.00 \text{ } \mu\text{C}$ is located in the center of a spherical cavity of radius 6.50 cm that, in turn, is at the center of an insulating charged solid sphere. The charge density in the solid is $\rho = 7.35 \times 10^{-4} \text{ C}/\text{m}^3$. Calculate the electric field inside the solid at a distance of 9.50 cm from the center of the cavity.

22.23 •• CP An electron is released from rest at a distance of 0.300 m from a large insulating sheet of charge that has uniform surface charge density $+2.90 \times 10^{-12} \text{ C}/\text{m}^2$. (a) How much work is done on the electron by the electric field of the sheet as the electron moves from its initial position to a point 0.050 m from the sheet? (b) What is the speed of the electron when it is 0.050 m from the sheet?

22.24 •• Charge Q is distributed uniformly throughout the volume of an insulating sphere of radius $R = 4.00 \text{ cm}$. At a distance of $r = 8.00 \text{ cm}$ from the center of the sphere, the electric field due to the charge distribution has magnitude $E = 940 \text{ N/C}$. What are (a) the volume charge density for the sphere and (b) the electric field at a distance of 2.00 cm from the sphere's center?

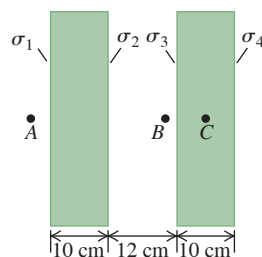
22.25 • A conductor with an inner cavity, like that shown in Fig. 22.23c, carries a total charge of $+5.00 \text{ nC}$. The charge within the cavity, insulated from the conductor, is -6.00 nC . How much charge is on (a) the inner surface of the conductor and (b) the outer surface of the conductor?

22.26 • A square insulating sheet 80.0 cm on a side is held horizontally. The sheet has 4.50 nC of charge spread uniformly over its area. (a) Estimate the electric field at a point 0.100 mm above the center of the sheet. (b) Estimate the electric field at a point 100 m above the center of the sheet. (c) Would the answers to parts (a) and (b) be different if the sheet were made of a conducting material? Why or why not?

22.27 • An infinitely long cylindrical conductor has radius R and uniform surface charge density σ . (a) In terms of σ and R , what is the charge per unit length λ for the cylinder? (b) In terms of σ , what is the magnitude of the electric field produced by the charged cylinder at a distance $r > R$ from its axis? (c) Express the result of part (b) in terms of λ and show that the electric field outside the cylinder is the same as if all the charge were on the axis. Compare your result to the result for a line of charge in Example 22.6 (Section 22.4).

22.28 • Two very large, nonconducting plastic sheets, each 10.0 cm thick, carry uniform charge densities $\sigma_1, \sigma_2, \sigma_3$, and σ_4 on their surfaces (Fig. E22.28). These surface charge densities have the values $\sigma_1 = -6.00 \text{ } \mu\text{C}/\text{m}^2$, $\sigma_2 = +5.00 \text{ } \mu\text{C}/\text{m}^2$, $\sigma_3 = +2.00 \text{ } \mu\text{C}/\text{m}^2$, and $\sigma_4 = +4.00 \text{ } \mu\text{C}/\text{m}^2$. Use Gauss's law to find the magnitude and direction of the electric field at the following points, far from the edges of these sheets: (a) point A, 5.00 cm from the left face of the left-hand sheet; (b) point B, 1.25 cm from the inner surface of the right-hand sheet; (c) point C, in the middle of the right-hand sheet.

Figure E22.28



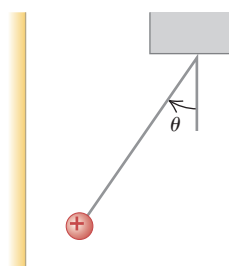
PROBLEMS

22.29 •• CP At time $t = 0$ a proton is a distance of 0.360 m from a very large insulating sheet of charge and is moving parallel to the sheet with speed 9.70×10^2 m/s. The sheet has uniform surface charge density 2.34×10^{-9} C/m². What is the speed of the proton at $t = 5.00 \times 10^{-8}$ s?

22.30 •• CP A very small object with mass 8.20×10^{-9} kg and positive charge 6.50×10^{-9} C is projected directly toward a very large insulating sheet of positive charge that has uniform surface charge density 5.90×10^{-8} C/m². The object is initially 0.400 m from the sheet. What initial speed must the object have in order for its closest distance of approach to the sheet to be 0.100 m?

22.31 •• CP A small sphere with mass 4.00×10^{-6} kg and charge 5.00×10^{-8} C hangs from a thread near a very large, charged insulating sheet (Fig. P22.31). The charge density on the surface of the sheet is uniform and equal to -2.50×10^{-9} C/m². Find the angle of the thread.

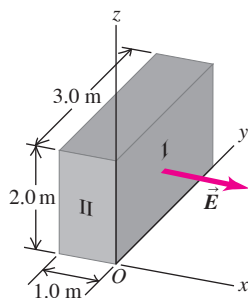
Figure P22.31



22.32 •• A cube has sides of length $L = 0.300$ m. One corner is at the origin (Fig. E22.6). The nonuniform electric field is given by $\vec{E} = (-5.00 \text{ N/C} \cdot \text{m})x\hat{i} + (3.00 \text{ N/C} \cdot \text{m})z\hat{k}$. (a) Find the electric flux through each of the six cube faces S_1, S_2, S_3, S_4, S_5 , and S_6 . (b) Find the total electric charge inside the cube.

22.33 • The electric field $\vec{E}$ in Fig. P22.33 is everywhere parallel to the x -axis, so the components E_y and E_z are zero. The x -component of the field E_x depends on x but not on y or z . At points in the yz -plane (where $x = 0$), $E_x = 125$ N/C. (a) What is the electric flux through surface I in Fig. P22.33? (b) What is the electric flux through surface II? (c) The volume shown is a small section of a very large insulating slab 1.0 m thick. If there is a total charge of -24.0 nC within the volume shown, what are the magnitude and direction of $\vec{E}$ at the face opposite surface I? (d) Is the electric field produced by charges only within the slab, or is the field also due to charges outside the slab? How can you tell?

Figure P22.33



22.34 •• CALC In a region of space there is an electric field $\vec{E}$ that is in the z -direction and that has magnitude $E = [964 \text{ N/(C} \cdot \text{m)}]x$. Find the flux for this field through a square in the xy -plane at $z = 0$ and with side length 0.350 m. One side of the square is along the $+x$ -axis and another side is along the $+y$ -axis.

22.35 • Negative charge $-Q$ is distributed uniformly over the surface of a thin spherical insulating shell with radius R . Calculate the force (magnitude and direction) that the shell exerts on a positive point charge q located a distance (a) $r > R$ from the center of the shell (outside the shell); (b) $r < R$ from the center of the shell (inside the shell).

22.36 • A long line carrying a uniform linear charge density $+50.0 \mu\text{C/m}$ runs parallel to and 10.0 cm from the surface of a large, flat plastic sheet that has a uniform surface charge density of $-100 \mu\text{C/m}^2$ on one side. Find the location of all points where an α particle would feel no force due to this arrangement of charged objects.

22.37 • The Coaxial Cable. A long coaxial cable consists of an inner cylindrical conductor with radius a and an outer coaxial cylinder with inner radius b and outer radius c . The outer cylinder is mounted on insulating supports and has no net charge. The inner cylinder has a uniform positive charge per unit length λ . Calculate the electric field (a) at any point between the cylinders a distance r from the axis and (b) at any point outside the outer cylinder. (c) Graph the magnitude of the electric field as a function of the distance r from the axis of the cable, from $r = 0$ to $r = 2c$. (d) Find the charge per unit length on the inner surface and on the outer surface of the outer cylinder.

22.38 • A very long conducting tube (hollow cylinder) has inner radius a and outer radius b . It carries charge per unit length $+\alpha$, where α is a positive constant with units of C/m. A line of charge lies along the axis of the tube. The line of charge has charge per unit length $+\alpha$. (a) Calculate the electric field in terms of α and the distance r from the axis of the tube for (i) $r < a$; (ii) $a < r < b$; (iii) $r > b$. Show your results in a graph of E as a function of r . (b) What is the charge per unit length on (i) the inner surface of the tube and (ii) the outer surface of the tube?

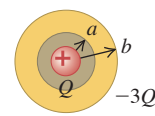
22.39 • A very long, solid cylinder with radius R has positive charge uniformly distributed throughout it, with charge per unit volume ρ . (a) Derive the expression for the electric field inside the volume at a distance r from the axis of the cylinder in terms of the charge density ρ . (b) What is the electric field at a point outside the volume in terms of the charge per unit length λ in the cylinder? (c) Compare the answers to parts (a) and (b) for $r = R$. (d) Graph the electric-field magnitude as a function of r from $r = 0$ to $r = 3R$.

22.40 • A Sphere in a Sphere. A solid conducting sphere carrying charge q has radius a . It is inside a concentric hollow conducting sphere with inner radius b and outer radius c . The hollow sphere has no net charge. (a) Derive expressions for the electric-field magnitude in terms of the distance r from the center for the regions $r < a$, $a < r < b$, $b < r < c$, and $r > c$. (b) Graph the magnitude of the electric field as a function of r from $r = 0$ to $r = 2c$. (c) What is the charge on the inner surface of the hollow sphere? (d) On the outer surface? (e) Represent the charge of the small sphere by four plus signs. Sketch the field lines of the system within a spherical volume of radius $2c$.

22.41 • A solid conducting sphere with radius R that carries positive charge Q is concentric with a very thin insulating shell of radius $2R$ that also carries charge Q . The charge Q is distributed uniformly over the insulating shell. (a) Find the electric field (magnitude and direction) in each of the regions $0 < r < R$, $R < r < 2R$, and $r > 2R$. (b) Graph the electric-field magnitude as a function of r .

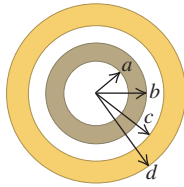
22.42 • A conducting spherical shell with inner radius a and outer radius b has a positive point charge Q located at its center. The total charge on the shell is $-3Q$, and it is insulated from its surroundings (Fig. P22.42). (a) Derive expressions for the electric-field magnitude E in terms of the distance r from the center for the regions $r < a$, $a < r < b$, and $r > b$. What is the surface charge density (b) on the inner surface of the conducting shell; (c) on the outer surface of the conducting shell? (d) Sketch the electric field lines and the location of all charges. (e) Graph E as a function of r .

Figure P22.42



22.43 • Concentric Spherical Shells. A small conducting spherical shell with inner radius a and outer radius b is concentric with a larger conducting spherical shell with inner radius c and outer radius d (Fig. P22.43). The inner shell has total charge $+2q$, and the outer shell has charge $+4q$. (a) Calculate the electric field $\vec{E}$ (magnitude and direction) in terms of q and the distance r from the common center of the two shells for (i) $r < a$; (ii) $a < r < b$; (iii) $b < r < c$; (iv) $c < r < d$; (v) $r > d$. Graph the radial component of $\vec{E}$ as a function of r . (b) What is the total charge on the (i) inner surface of the small shell; (ii) outer surface of the small shell; (iii) inner surface of the large shell; (iv) outer surface of the large shell?

Figure P22.43



22.44 • Repeat Problem 22.43, but now let the outer shell have charge $-2q$. The inner shell still has charge $+2q$.

22.45 ••• CALC An insulating hollow sphere has inner radius a and outer radius b . Within the insulating material the volume charge density is given by $\rho(r) = \alpha/r$, where α is a positive constant. (a) In terms of α and a , what is the magnitude of the electric field at a distance r from the center of the shell, where $a < r < b$? (b) A point charge q is placed at the center of the hollow space, at $r = 0$. In terms of α and a , what value must q have (sign and magnitude) in order for the electric field to be constant in the region $a < r < b$, and what then is the value of the constant field in this region?

22.46 •• CP Thomson's Model of the Atom. Early in the 20th century, a leading model of the structure of the atom was that of English physicist J. J. Thomson (the discoverer of the electron). In Thomson's model, an atom consisted of a sphere of positively charged material in which were embedded negatively charged electrons, like chocolate chips in a ball of cookie dough. Consider such an atom consisting of one electron with mass m and charge $-e$, which may be regarded as a point charge, and a uniformly charged sphere of charge $+e$ and radius R . (a) Explain why the electron's equilibrium position is at the center of the nucleus. (b) In Thomson's model, it was assumed that the positive material provided little or no resistance to the electron's motion. If the electron is displaced from equilibrium by a distance less than R , show that the resulting motion of the electron will be simple harmonic, and calculate the frequency of oscillation. (*Hint:* Review the definition of SHM in Section 14.2. If it can be shown that the net force on the electron is of this form, then it follows that the motion is simple harmonic. Conversely, if the net force on the electron does not follow this form, the motion is not simple harmonic.) (c) By Thomson's time, it was known that excited atoms emit light waves of only certain frequencies. In his model, the frequency of emitted light is the same as the oscillation frequency of the electron(s) in the atom. What radius would a Thomson-model atom need for it to produce red light of frequency 4.57×10^{14} Hz? Compare your answer to the radii of real atoms, which are of the order of 10^{-10} m (see Appendix F). (d) If the electron were displaced from equilibrium by a distance greater than R , would the electron oscillate? Would its motion be simple harmonic? Explain your reasoning. (*Historical note:* In 1910, the atomic nucleus was discovered, proving the Thomson model to be incorrect. An atom's positive charge is not spread over its volume, as Thomson supposed, but is concentrated in the tiny nucleus of radius 10^{-14} to 10^{-15} m.)

22.47 •• CALC A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

$$\rho(r) = \rho_0 \left(1 - \frac{r}{R} \right) \quad \text{for } r \leq R$$

$$\rho(r) = 0 \quad \text{for } r \geq R$$

where $\rho_0 = 3Q/\pi R^3$ is a positive constant. (a) Show that the total charge contained in the charge distribution is Q . (b) Show that the electric field in the region $r \geq R$ is identical to that produced by a point charge Q at $r = 0$. (c) Obtain an expression for the electric field in the region $r \leq R$. (d) Graph the electric-field magnitude E as a function of r . (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

22.48 •• (a) How many excess electrons must be distributed uniformly within the volume of an isolated plastic sphere 30.0 cm in diameter to produce an electric field of magnitude 1390 N/C just outside the surface of the sphere? (b) What is the electric field at a point 10.0 cm outside the surface of the sphere?

22.49 •• CALC A very long insulating cylinder has radius R and carries positive charge distributed throughout its volume. The charge distribution has cylindrical symmetry but varies with perpendicular distance from the axis of the cylinder. The volume charge density is $\rho(r) = \alpha(1 - r/R)$, where α is a constant with units C/m^3 and r is the perpendicular distance from the center line of the cylinder. Derive an expression, in terms of α and R , for $E(r)$, the electric field as a function of r . Do this for $r < R$ and also for $r > R$. Do your results agree for $r = R$?

22.50 •• CALC A solid insulating sphere has radius R and carries positive charge distributed throughout its volume. The charge distribution has spherical symmetry but varies with radial distance r from the center of the sphere. The volume charge density is $\rho(r) = \rho_0(1 - r/R)$, where ρ_0 is a constant with units of C/m^3 . (a) Derive an expression for the electric field as a function of r for $r < R$. (b) Repeat part (a) for $r > R$. (c) At what value of r , in terms of R , does the electric field have its maximum value?

22.51 • CALC A Nonuniformly Charged Slab. A slab of insulating material has thickness $2d$ and is oriented so that its faces are parallel to the yz -plane and given by the planes $x = d$ and $x = -d$. The y - and z -dimensions of the slab are very large compared to d , so treat them as infinite. The slab has charge density given by $\rho(x) = \rho_0(x/d)^2$, where ρ_0 is a positive constant. Using Gauss's law, find the electric field due to the slab (magnitude and direction) at all points in space.

22.52 • CALC A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

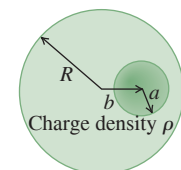
$$\rho(r) = \rho_0 \left(1 - \frac{4r}{3R} \right) \quad \text{for } r \leq R$$

$$\rho(r) = 0 \quad \text{for } r \geq R$$

where ρ_0 is a positive constant. (a) Find the total charge contained in the charge distribution. Obtain an expression for the electric field in the region (b) $r \geq R$; (c) $r \leq R$. (d) Graph the electric-field magnitude E as a function of r . (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

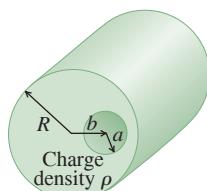
22.53 • (a) An insulating sphere with radius a has a uniform charge density ρ . The sphere is not centered at the origin but at $\vec{r} = \vec{b}$. Show that the electric field inside the sphere is given by $\vec{E} = \rho(\vec{r} - \vec{b})/3\epsilon_0$. (b) An insulating sphere of radius R has a spherical hole of radius a located within its volume and centered a distance b from the center of the sphere, where $a < b < R$ (a cross section of the sphere is shown in Fig. P22.53). The solid part of the sphere has a uniform volume charge density ρ . Find the magnitude and direction of the electric field $\vec{E}$ inside the hole, and show that $\vec{E}$ is uniform over the entire hole. [*Hint:* Use the principle of superposition and the result of part (a).]

Figure P22.53



22.54 • A very long, solid insulating cylinder has radius R ; bored along its entire length is a cylindrical hole with radius a . The axis of the hole is a distance b from the axis of the cylinder, where $a < b < R$ (Fig. P22.54). The solid material of the cylinder has a uniform volume charge density ρ . Find the magnitude and direction of the electric field $\vec{E}$ inside the hole, and show that $\vec{E}$ is uniform over the entire hole. (Hint: See Problem 22.53.)

Figure P22.54



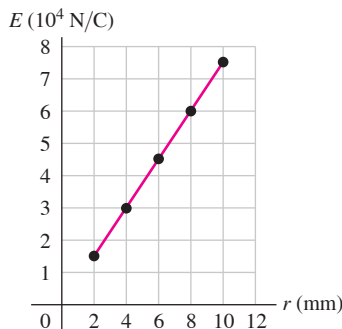
22.55 •• DATA In one experiment the electric field is measured for points at distances r from a uniform line of charge that has charge per unit length λ and length l , where $l \gg r$. In a second experiment the electric field is measured for points at distances r from the center of a uniformly charged insulating sphere that has volume charge density ρ and radius $R = 8.00$ mm, where $r > R$. The results of the two measurements are listed in the table, but you aren't told which set of data applies to which experiment:

r (cm)	E (10^5 N/C)	
	Measurement A	Measurement B
1.00	2.72	5.45
1.50	1.79	2.42
2.00	1.34	1.34
2.50	1.07	0.861
3.00	0.902	0.605
3.50	0.770	0.443
4.00	0.677	0.335

For each set of data, draw two graphs: one for Er^2 versus r and one for Er versus r . (a) Use these graphs to determine which data set, A or B, is for the uniform line of charge and which set is for the uniformly charged sphere. Explain your reasoning. (b) Use the graphs in part (a) to calculate λ for the uniform line of charge and ρ for the uniformly charged sphere.

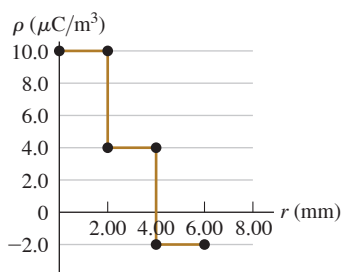
22.56 •• DATA The electric field is measured for points at distances r from the center of a uniformly charged insulating sphere that has volume charge density ρ and radius R , where $r < R$ (Fig. P22.56). Calculate ρ .

Figure P22.56



22.57 •• DATA The volume charge density ρ for a spherical charge distribution of radius $R = 6.00$ mm is not uniform. Figure P22.57 shows ρ as a function of the distance r from the center of the distribution. Calculate the electric field at these values of r : (i) 1.00 mm; (ii) 3.00 mm; (iii) 5.00 mm; (iv) 7.00 mm.

Figure P22.57



22.58 •• BIO As a honeybee flies, the passing air strips electrons from its hairs, giving the bee a net positive charge. Since flowers are negatively charged, pollen then jumps onto a bee even if the bee does not physically touch the pollen particles. (a) Estimate the diameter of the central disk of a daisy. (b) If a bee has had 75,000 electrons stripped by the air, what is its net charge? (c) If this bee lands at the edge of the daisy's central disk, determine its electric field at the far edge of the disk. Treat the bee as a thin-walled hollow sphere with its net charge distributed uniformly over its surface. (d) A pollen particle requires a force of 10 pN to dislodge from a stamen. Estimate the net charge on a pollen particle at the far end of the disk required for the particle to dislodge and jump to the bee.

22.59 •• CP CALC A very long insulating cylinder with radius R_{cylinder} has nonuniform positive charge density $\rho = (1 - r/R_{\text{cylinder}})\rho_0$, where ρ_0 is constant and r is measured radially from the axis of the cylinder. A particle with charge $-Q$ and mass M orbits the cylinder at a constant distance $R_{\text{orbit}} > R_{\text{cylinder}}$. (a) What is the linear charge density λ of the tube, in terms of R_{cylinder} and ρ_0 ? (b) Determine the period of the motion in terms of R_{orbit} . (Hint: Use Gauss's law to determine the electric field, and therefore the electric force felt by the particle, that acts centripetally.)

22.60 •• CP (a) Show that the component of the electric force normal to any flat surface with a uniform charge density σ is given by $F_{\perp} = \sigma \Phi_E$, where Φ_E is the electric flux through that surface due to an external electric field. (b) An insulating hemisphere with radius R and charge Q distributed uniformly over its flat, circular surface lies above a large plane with uniform charge density σ . The axis of the hemisphere is oriented vertically. For what mass M could the hemisphere remain stationary? (c) If the hemisphere and the plane share the same charge density of $100 \mu\text{C}/\text{m}^2$ and the hemisphere has a radius of 3.00 cm, what would be its upward acceleration if its mass were 100 g?

CHALLENGE PROBLEMS

22.61 ••• CALC A uniformly charged insulating sphere with radius r and charge $+Q$ lies at the center of a thin-walled hollow cylinder with radius $R > r$ and length $L > 2r$. The cylinder is non-conducting and carries no net charge. (a) Determine the outward electric flux through the rounded "side" of the cylinder, excluding the circular end caps. (Hint: Choose a cylindrical coordinate system with the axis of the cylinder as its z -axis and the center of the charged sphere as its origin. Note that an area element on the cylinder has magnitude $dA = 2\pi R dz$.) (b) Determine the electric flux upward through the circular cap at the top of the cylinder. (c) Determine the electric flux downward through the circular cap at the bottom of the cylinder. (d) Add the results from parts (a)–(c) to determine the outward electric flux through the closed cylinder. (e) Show that your result is consistent with Gauss's law.

22.62 ••• CP CALC A region in space contains a total positive charge Q that is distributed spherically such that the volume charge density $\rho(r)$ is given by

$$\begin{aligned} \rho(r) &= 3\alpha r/2R & \text{for } r \leq R/2 \\ \rho(r) &= \alpha[1 - (r/R)^2] & \text{for } R/2 \leq r \leq R \\ \rho(r) &= 0 & \text{for } r \geq R \end{aligned}$$

Here α is a positive constant having units of C/m^3 . (a) Determine α in terms of Q and R . (b) Using Gauss's law, derive an expression for the magnitude of the electric field as a function of r . Do this separately for all three regions. Express your answers in terms of Q . (c) What fraction of the total charge is contained within the region $R/2 \leq r \leq R$? (d) What is the magnitude of $\vec{E}$ at $r = R/2$? (e) If an electron with charge $q' = -e$ is released from rest at any point in any of the three regions, the resulting motion will be oscillatory but not simple harmonic. Why?

MCAT-STYLE PASSAGE PROBLEMS

Space Radiation Shielding. One of the hazards facing humans in space is space radiation: high-energy charged particles emitted by the sun. During a solar flare, the intensity of this radiation can reach lethal levels. One proposed method of protection for astronauts on the surface of the moon or Mars is an array of large, electrically charged spheres placed high above areas where people live and work. The spheres would produce a strong electric field $\vec{E}$ to deflect the charged particles that make up space radiation. The spheres would be similar in construction to a Mylar balloon, with a thin, electrically conducting layer on the outside surface on which a net positive or negative charge would be placed. A typical sphere might be 5 m in diameter.

22.63 Suppose that to repel electrons in the radiation from a solar flare, each sphere must produce an electric field $\vec{E}$ of magnitude 1×10^6 N/C at 25 m from the center of the sphere. What net charge on each sphere is needed? (a) -0.07 C; (b) -8 mC; (c) -80 μ C; (d) -1×10^{-20} C.

22.64 What is the magnitude of $\vec{E}$ just outside the surface of such a sphere? (a) 0; (b) 10^6 N/C; (c) 10^7 N/C; (d) 10^8 N/C.

22.65 What is the direction of $\vec{E}$ just outside the surface of such a sphere? (a) Tangent to the surface of the sphere; (b) perpendicular to the surface, pointing toward the sphere; (c) perpendicular to the surface, pointing away from the sphere; (d) there is no electric field just outside the surface.

22.66 Which statement is true about $\vec{E}$ inside a negatively charged sphere as described here? (a) It points from the center of the sphere to the surface and is largest at the center. (b) It points from the surface to the center of the sphere and is largest at the surface. (c) It is zero. (d) It is constant but not zero.

ANSWERS

Chapter Opening Question ?

(iii) The electric field inside a cavity within a conductor is zero, so there would be no electric effect on the child. (See Section 22.5.)

Key Example **✓** VARIATION Problems

VP22.4.1 (a) $-E_1 L^2$ (b) $+E_1 L^2$ (c) $-E_2 L^2$ (d) $+E_2 L^2$ (e) 0 (f) 0 (g) 0

VP22.4.2 (a) $+452$ N $\cdot$ m²/C (b) -226 N $\cdot$ m²/C (c) -226 N $\cdot$ m²/C

VP22.4.3 (a) $+678$ N $\cdot$ m²/C (b) $+113$ N $\cdot$ m²/C

VP22.4.4 (a) $+339$ N $\cdot$ m²/C (b) -565 N $\cdot$ m²/C (c) -565 N $\cdot$ m²/C

VP22.10.1 (a) 8.63×10^3 N/C at 4.00 cm, 7.49×10^3 N/C at 6.00 cm
(b) 0 at 4.00 cm, 7.49×10^3 N/C at 6.00 cm

VP22.10.2 2.42 μ N, attractive

VP22.10.3 1.79×10^{-7} C/m², positive

VP22.10.4 5.41×10^{-8} C/m²

VP22.12.1 (a) -5.00 nC (b) -3.00 nC

VP22.12.2 (a) $+2.00$ nC (b) neither would be affected

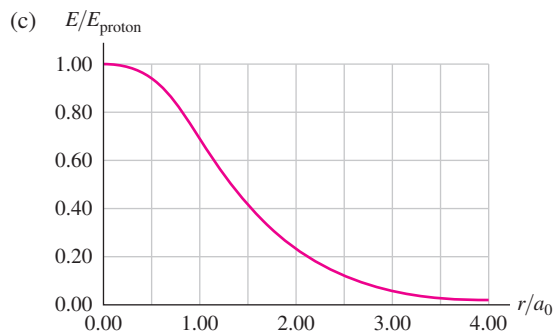
VP22.12.3 (a) 0 (b) 2.7×10^{-5} C (c) 2.7×10^{-6} C/m²

VP22.12.4 (a) $+4.00$ nC (b) -2.00 nC (c) 2.50×10^3 N/C
(d) 1.65×10^2 N/C

Bridging Problem

(a) $Q(r) = Qe^{-2r/a_0} [2(r/a_0)^2 + 2(r/a_0) + 1]$

(b) $E = \frac{kQe^{-2r/a_0}}{r^2} [2(r/a_0)^2 + 2(r/a_0) + 1]$





? In one type of welding, electric charge flows between the welding tool and the metal pieces that are to be joined. This produces a glowing arc whose high temperature fuses the pieces together. Why must the tool be held close to the metal pieces? (i) To maximize the potential difference between tool and pieces; (ii) to minimize this potential difference; (iii) to maximize the electric field between tool and pieces; (iv) to minimize this electric field; (v) more than one of these.

23 Electric Potential

This chapter is about energy associated with electrical interactions. Every time you turn on a light, use a mobile phone, or make toast in a toaster, you are using electrical energy, an indispensable ingredient of our technological society. In Chapters 6 and 7 we introduced the concepts of *work* and *energy* in the context of mechanics; now we'll combine these concepts with what we've learned about electric charge, electric forces, and electric fields. Just as we found for many problems in mechanics, using energy ideas makes it easier to solve a variety of problems in electricity.

When a charged particle moves in an electric field, the field exerts a force that can do *work* on the particle. This work can be expressed in terms of electric potential energy. Just as gravitational potential energy depends on the height of a mass above the earth's surface, electric potential energy depends on the position of the charged particle in the electric field. We'll use a new concept called *electric potential*, or simply *potential* to describe electric potential energy. In circuits, a difference in potential from one point to another is often called *voltage*. The concepts of potential and voltage are crucial to understanding how electric circuits work and have equally important applications to electron beams used in cancer radiotherapy, high-energy particle accelerators, and many other devices.

23.1 ELECTRIC POTENTIAL ENERGY

The concepts of work, potential energy, and conservation of energy proved to be extremely useful in our study of mechanics. In this section we'll show that these concepts are just as useful for understanding and analyzing electrical interactions.

Let's begin by reviewing three essential points from Chapters 6 and 7. First, when a force $\vec{F}$ acts on a particle that moves from point a to point b , the work $W_{a \rightarrow b}$ done by the force is given by a *line integral*:

$$W_{a \rightarrow b} = \int_a^b \vec{F} \cdot d\vec{l} = \int_a^b F \cos \phi \, dl \quad (\text{work done by a force}) \quad (23.1)$$

where $d\vec{l}$ is an infinitesimal displacement along the particle's path and ϕ is the angle between $\vec{F}$ and $d\vec{l}$ at each point along the path.

LEARNING OUTCOMES

In this chapter, you'll learn...

- 23.1** How to calculate the electric potential energy of a collection of charges.
- 23.2** The meaning and significance of electric potential.
- 23.3** How to calculate the electric potential that a collection of charges produces at a point in space.
- 23.4** How to use equipotential surfaces to visualize how the electric potential varies in space.
- 23.5** How to use electric potential to calculate the electric field.

You'll need to review...

- 7.1–7.4** Conservative forces and potential energy.
- 21.1–21.6** Electric force and electric fields.
- 22.4, 22.5** Applications of Gauss's law.

Second, if the force $\vec{F}$ is *conservative*, as we defined the term in Section 7.3, the work done by $\vec{F}$ can always be expressed in terms of a **potential energy** U . When the particle moves from a point where the potential energy is U_a to a point where it is U_b , the change in potential energy is $\Delta U = U_b - U_a$ and

$$W_{a \rightarrow b} = U_a - U_b = -(U_b - U_a) = -\Delta U \quad (23.2)$$

Work done by a conservative force
 Potential energy at initial position
 Potential energy at final position
 Negative of change in potential energy

Figure 23.1 The work done on a baseball moving in a uniform gravitational field.

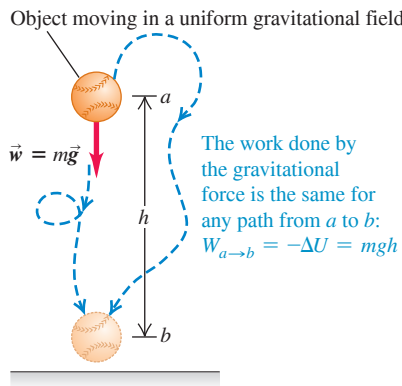
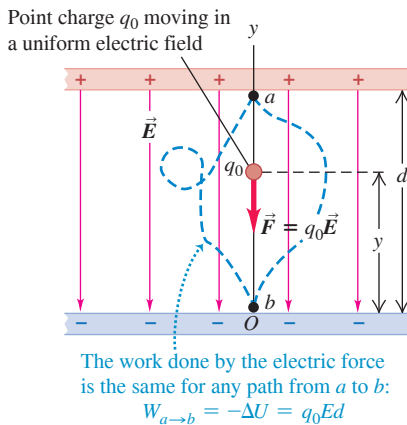


Figure 23.2 The work done on a point charge moving in a uniform electric field. Compare with Fig. 23.1.



When $W_{a \rightarrow b}$ is positive, U_a is greater than U_b , ΔU is negative, and the potential energy *decreases*. That's what happens when a baseball falls from a high point (a) to a lower point (b) under the influence of the earth's gravity; the force of gravity does positive work, and the gravitational potential energy decreases (**Fig. 23.1**). When a tossed ball is moving upward, the gravitational force does negative work during the ascent, and the potential energy increases.

Third, the work–energy theorem says that the change in kinetic energy $\Delta K = K_b - K_a$ during a displacement equals the *total* work done on the particle. If only conservative forces do work, then Eq. (23.2) gives the total work, and $K_b - K_a = -(U_b - U_a)$. We usually write this as

$$K_a + U_a = K_b + U_b \quad (23.3)$$

That is, the total mechanical energy (kinetic plus potential) is *conserved* under these circumstances.

Electric Potential Energy in a Uniform Field

Let's look at an electrical example of these concepts. In **Fig. 23.2** a pair of large, charged, parallel metal plates sets up a uniform, downward electric field with magnitude E . The field exerts a downward force with magnitude $F = q_0 E$ on a positive test charge q_0 . As the charge moves downward a distance d from point a to point b , the force on the test charge is constant and independent of its location. So the work done by the electric field is the product of the force magnitude and the component of displacement in the (downward) direction of the force:

$$W_{a \rightarrow b} = Fd = q_0 E d \quad (23.4)$$

This work is positive, since the force is in the same direction as the net displacement of the test charge.

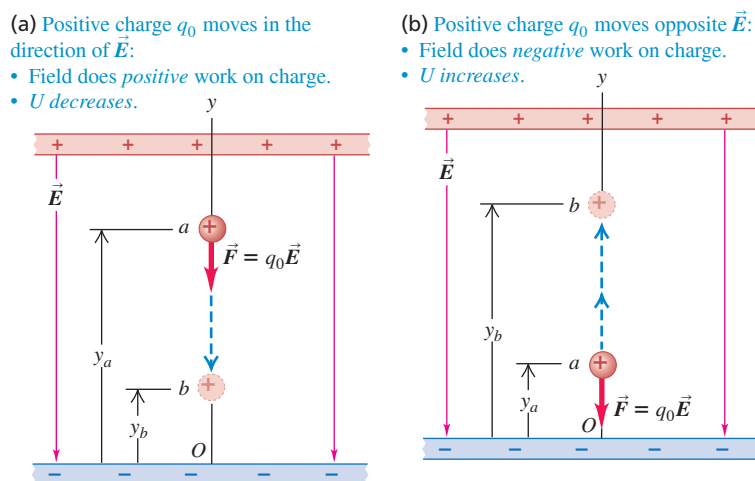
The y -component of the electric force, $F_y = -q_0 E$, is constant, and there is no x - or z -component. This is exactly analogous to the gravitational force on a mass m near the earth's surface; for this force, there is a constant y -component $F_y = -mg$ and the x - and z -components are zero. Because of this analogy, we can conclude that the force exerted on q_0 by the uniform electric field in Fig. 23.2 is *conservative*, just as is the gravitational force. This means that the work $W_{a \rightarrow b}$ done by the field is independent of the path the particle takes from a to b . We can represent this work with a *potential-energy* function U , just as we did for gravitational potential energy in Section 7.1. The potential energy for the gravitational force $F_y = -mg$ was $U = mgy$; hence the potential energy for the electric force $F_y = -q_0 E$ is

$$U = q_0 E y \quad (23.5)$$

When the test charge moves from height y_a to height y_b , the work done on the charge by the field is given by

$$W_{a \rightarrow b} = -\Delta U = -(U_b - U_a) = -(q_0 E y_b - q_0 E y_a) = q_0 E (y_a - y_b) \quad (23.6)$$

Figure 23.3 A positive charge moving (a) in the direction of the electric field $\vec{E}$ and (b) in the direction opposite $\vec{E}$.



When y_a is greater than y_b (Fig. 23.3a), the positive test charge q_0 moves downward, in the same direction as $\vec{E}$; the displacement is in the same direction as the force $\vec{F} = q_0\vec{E}$, so the field does positive work and U decreases. [In particular, if $y_a - y_b = d$ as in Fig. 23.2, Eq. (23.6) gives $W_{a \rightarrow b} = q_0Ed$, in agreement with Eq. (23.4).] When y_a is less than y_b (Fig. 23.3b), the positive test charge q_0 moves upward, in the opposite direction to $\vec{E}$; the displacement is opposite the force, the field does negative work, and U increases.

If the test charge q_0 is *negative*, the potential energy increases when it moves with the field and decreases when it moves against the field (Fig. 23.4).

Whether the test charge is positive or negative, the following general rules apply: U *increases* if the test charge q_0 moves in the direction *opposite* the electric force $\vec{F} = q_0\vec{E}$ (Figs. 23.3b and 23.4a); U *decreases* if q_0 moves in the *same* direction as $\vec{F} = q_0\vec{E}$ (Figs. 23.3a and 23.4b). This is the same behavior as for gravitational potential energy, which increases if a mass m moves upward (opposite the direction of the gravitational force) and decreases if m moves downward (in the same direction as the gravitational force).

CAUTION Electric potential energy The relationship between electric potential energy change and motion in an electric field is an important one that we'll use often, but that takes some effort to understand. Studying the preceding paragraph as well as Figs. 23.3 and Fig. 23.4 carefully now will help you tremendously later!

Electric Potential Energy of Two Point Charges

The idea of electric potential energy isn't restricted to the special case of a uniform electric field. Indeed, we can apply this concept to a point charge in *any* electric field caused by a static charge distribution. Recall from Chapter 21 that we can represent any charge distribution as a collection of point charges. Therefore it's useful to calculate the work done on a test charge q_0 moving in the electric field caused by a single, stationary point charge q .

We'll consider first a displacement along the *radial* line in Fig. 23.5. The force on q_0 is given by Coulomb's law, and its radial component is

$$F_r = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \quad (23.7)$$

If q and q_0 have the same sign (+ or -), the force is repulsive and F_r is positive; if the two charges have opposite signs, the force is attractive and F_r is negative. The force is *not*

Figure 23.4 A negative charge moving (a) in the direction of the electric field $\vec{E}$ and (b) in the direction opposite $\vec{E}$. Compare with Fig. 23.3.

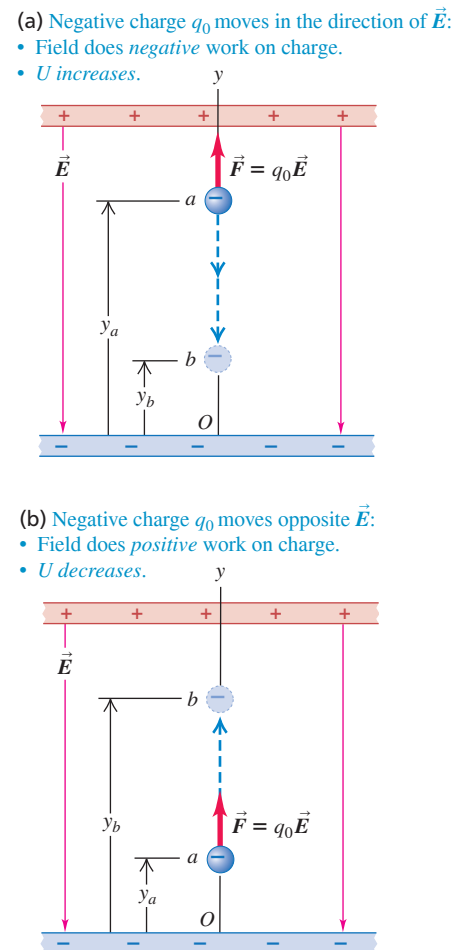


Figure 23.5 Test charge q_0 moves along a straight line extending radially from charge q . As it moves from a to b , the distance varies from r_a to r_b .

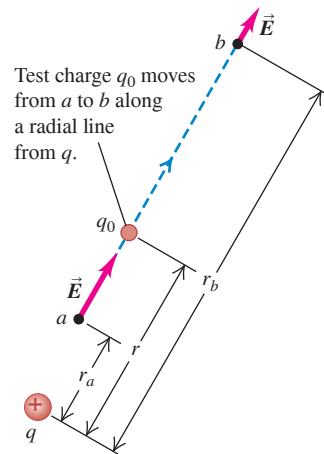
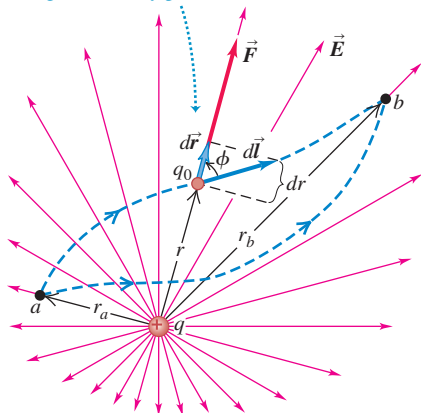


Figure 23.6 The work done on charge q_0 by the electric field of charge q does not depend on the path taken, but only on the distances r_a and r_b .

Test charge q_0 moves from a to b along an arbitrary path.



constant during the displacement, and we must integrate to calculate the work $W_{a \rightarrow b}$ done on q_0 by this force as q_0 moves from a to b :

$$W_{a \rightarrow b} = \int_{r_a}^{r_b} F_r dr = \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} dr = \frac{qq_0}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right) \quad (23.8)$$

The work done by the electric force for this path depends on only the endpoints.

Now let's consider a more general displacement (Fig. 23.6) in which a and b do not lie on the same radial line. From Eq. (23.1) the work done on q_0 during this displacement is given by

$$W_{a \rightarrow b} = \int_{r_a}^{r_b} F \cos \phi dl = \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \cos \phi dl$$

But Fig. 23.6 shows that $\cos \phi dl = dr$. That is, the work done during a small displacement $d\vec{l}$ depends only on the change dr in the distance r between the charges, which is the radial component of the displacement. Thus Eq. (23.8) is valid even for this more general displacement; the work done on q_0 by the electric field $\vec{E}$ produced by q depends only on r_a and r_b , not on the details of the path. Also, if q_0 returns to its starting point a by a different path, the total work done in the round-trip displacement is zero (the integral in Eq. (23.8) is from r_a back to r_a). These are the needed characteristics for a conservative force, as we defined it in Section 7.3. Thus the force on q_0 is a conservative force.

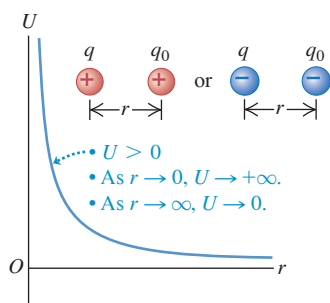
We see that Eqs. (23.2) and (23.8) are consistent if we define the potential energy to be $U_a = qq_0/4\pi\epsilon_0 r_a$ when q_0 is a distance r_a from q , and to be $U_b = qq_0/4\pi\epsilon_0 r_b$ when q_0 is a distance r_b from q . Thus

$$\text{Electric potential energy of two point charges} \quad U = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r} \quad (23.9)$$

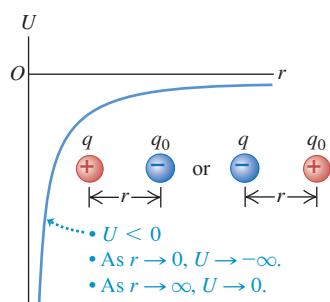
Electric constant
Values of two charges
Distance between two charges

Figure 23.7 Graphs of the potential energy U of two point charges q and q_0 versus their separation r .

(a) q and q_0 have the same sign.



(b) q and q_0 have opposite signs.



Equation (23.9) is valid no matter what the signs of the charges q and q_0 . The potential energy is positive if the charges q and q_0 have the same sign (Fig. 23.7a) and negative if they have opposite signs (Fig. 23.7b).

CAUTION Electric potential energy vs. electric force Don't confuse Eq. (23.9) for the potential energy of two point charges with Eq. (23.7) for the radial component of the electric force that one charge exerts on the other. Potential energy U is proportional to $1/r$, while the force component F_r is proportional to $1/r^2$.

Potential energy is always defined relative to some reference point where $U = 0$. In Eq. (23.9), U is zero when q and q_0 are infinitely far apart and $r = \infty$. Therefore U represents the work that would be done on the test charge q_0 by the field of q if q_0 moved from an initial distance r to infinity. If q and q_0 have the same sign, the interaction is repulsive, this work is positive, and U is positive at any finite separation (Fig. 23.7a). If the charges have opposite signs, the interaction is attractive, the work done is negative, and U is negative (Fig. 23.7b).

We emphasize that the potential energy U given by Eq. (23.9) is a *shared* property of the two charges. If the distance between q and q_0 is changed from r_a to r_b , the change in potential energy is the same whether q is held fixed and q_0 is moved or q_0 is held fixed and q is moved. For this reason, we never use the phrase “the electric potential energy of a point charge.” (Likewise, if a mass m is at a height h above the earth's surface, the gravitational potential energy is a shared property of the mass m and the earth. We emphasized this in Sections 7.1 and 13.3.)

Equation (23.9) also holds if the charge q_0 is outside a spherically symmetric charge *distribution* with total charge q ; the distance r is from q_0 to the center of the distribution. That's because Gauss's law tells us that the electric field outside such a distribution is the same as if all of its charge q were concentrated at its center (see Example 22.9 in Section 22.4).

EXAMPLE 23.1 Conservation of energy with electric forces**WITH VARIATION PROBLEMS**

A positron (the electron's antiparticle) has mass 9.11×10^{-31} kg and charge $q_0 = +e = +1.60 \times 10^{-19}$ C. Suppose a positron moves in the vicinity of an α (alpha) particle, which has charge $q = +2e = 3.20 \times 10^{-19}$ C and mass 6.64×10^{-27} kg. The α particle's mass is more than 7000 times that of the positron, so we assume that the α particle remains at rest. When the positron is 1.00×10^{-10} m from the α particle, it is moving directly away from the α particle at 3.00×10^6 m/s. (a) What is the positron's speed when the particles are 2.00×10^{-10} m apart? (b) What is the positron's speed when it is very far from the α particle? (c) Suppose the initial conditions are the same but the moving particle is an electron (with the same mass as the positron but charge $q_0 = -e$). Describe the subsequent motion.

IDENTIFY and SET UP The electric force between a positron (or an electron) and an α particle is conservative, so the total mechanical energy (kinetic plus potential) is conserved. Equation (23.9) gives the potential energy U at any separation r : The potential-energy function for parts (a) and (b) looks like that of Fig. 23.7a, and the function for part (c) looks like that of Fig. 23.7b. We are given the positron speed $v_a = 3.00 \times 10^6$ m/s when the separation between the particles is $r_a = 1.00 \times 10^{-10}$ m. In parts (a) and (b) we use Eqs. (23.3) and (23.9) to find the speed for $r = r_b = 2.00 \times 10^{-10}$ m and $r = r_c \rightarrow \infty$, respectively. In part (c) we replace the positron with an electron and reconsider the problem.

EXECUTE (a) Both particles have positive charge, so the positron speeds up as it moves away from the α particle. From the energy-conservation equation, Eq. (23.3), the final kinetic energy is

$$K_b = \frac{1}{2}mv_b^2 = K_a + U_a - U_b$$

In this expression,

$$\begin{aligned} K_a &= \frac{1}{2}mv_a^2 = \frac{1}{2}(9.11 \times 10^{-31} \text{ kg})(3.00 \times 10^6 \text{ m/s})^2 \\ &= 4.10 \times 10^{-18} \text{ J} \\ U_a &= \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r_a} = (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \\ &\quad \times \frac{(3.20 \times 10^{-19} \text{ C})(1.60 \times 10^{-19} \text{ C})}{1.00 \times 10^{-10} \text{ m}} \\ &= 4.61 \times 10^{-18} \text{ J} \\ U_b &= \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r_b} = 2.30 \times 10^{-18} \text{ J} \end{aligned}$$

Hence the positron kinetic energy and speed at $r = r_b = 2.00 \times 10^{-10}$ m are

$$\begin{aligned} K_b &= \frac{1}{2}mv_b^2 \\ &= (4.10 \times 10^{-18} \text{ J}) + (4.61 \times 10^{-18} \text{ J}) - (2.30 \times 10^{-18} \text{ J}) \\ &= 6.41 \times 10^{-18} \text{ J} \\ v_b &= \sqrt{\frac{2K_b}{m}} = \sqrt{\frac{2(6.41 \times 10^{-18} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} \\ &= 3.8 \times 10^6 \text{ m/s} \end{aligned}$$

(b) When the positron and α particle are very far apart so that $r = r_c \rightarrow \infty$, the final potential energy U_c approaches zero. Again from energy conservation, the final kinetic energy and speed of the positron in this case are

$$\begin{aligned} K_c &= K_a + U_a - U_c = (4.10 \times 10^{-18} \text{ J}) + (4.61 \times 10^{-18} \text{ J}) - 0 \\ &= 8.71 \times 10^{-18} \text{ J} \\ v_c &= \sqrt{\frac{2K_c}{m}} = \sqrt{\frac{2(8.71 \times 10^{-18} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} = 4.4 \times 10^6 \text{ m/s} \end{aligned}$$

(c) The electron and α particle have opposite charges, so the force is attractive and the electron slows down as it moves away. Changing the moving particle's sign from $+e$ to $-e$ means that the initial potential energy is now $U_a = -4.61 \times 10^{-18}$ J, which makes the total mechanical energy *negative*:

$$\begin{aligned} K_a + U_a &= (4.10 \times 10^{-18} \text{ J}) - (4.61 \times 10^{-18} \text{ J}) \\ &= -0.51 \times 10^{-18} \text{ J} \end{aligned}$$

The total mechanical energy would have to be positive for the electron to move infinitely far away from the α particle. Like a rock thrown upward at low speed from the earth's surface, it will reach a maximum separation $r = r_d$ from the α particle before reversing direction. At this point its speed and its kinetic energy K_d are zero, so at separation r_d we have

$$\begin{aligned} U_d &= K_a + U_a - K_d = (-0.51 \times 10^{-18} \text{ J}) - 0 \\ U_d &= \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r_d} = -0.51 \times 10^{-18} \text{ J} \\ r_d &= \frac{1}{U_d} \frac{qq_0}{4\pi\epsilon_0} \\ &= \frac{(9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)}{-0.51 \times 10^{-18} \text{ J}} (3.20 \times 10^{-19} \text{ C})(-1.60 \times 10^{-19} \text{ C}) \\ &= 9.0 \times 10^{-10} \text{ m} \end{aligned}$$

For $r_b = 2.00 \times 10^{-10}$ m we have $U_b = -2.30 \times 10^{-18}$ J, so the electron kinetic energy and speed at this point are

$$\begin{aligned} K_b &= \frac{1}{2}mv_b^2 \\ &= (4.10 \times 10^{-18} \text{ J}) + (-4.61 \times 10^{-18} \text{ J}) - (-2.30 \times 10^{-18} \text{ J}) \\ &= 1.79 \times 10^{-18} \text{ J} \\ v_b &= \sqrt{\frac{2K_b}{m}} = \sqrt{\frac{2(1.79 \times 10^{-18} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} = 2.0 \times 10^6 \text{ m/s} \end{aligned}$$

EVALUATE Both particles behave as expected as they move away from the α particle: The positron speeds up, and the electron slows down and eventually turns around. How fast would an electron have to be moving at $r_a = 1.00 \times 10^{-10}$ m to travel infinitely far from the α particle? (*Hint*: See Example 13.5 in Section 13.3.)

KEYCONCEPT The electrostatic forces that two charged particles exert on each other are conservative. The associated electric potential energy U is proportional to the product of the two charges and inversely proportional to the distance r between them. As r approaches infinity, U approaches zero.

Figure 23.8 The potential energy associated with a charge q_0 at point a depends on the other charges q_1 , q_2 , and q_3 and on their distances r_1 , r_2 , and r_3 from point a .

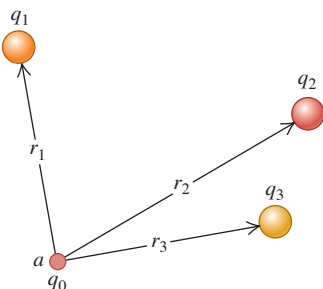
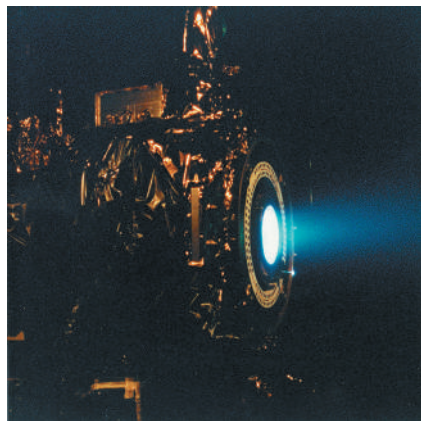


Figure 23.9 This ion engine for spacecraft uses electric forces to eject a stream of positive xenon ions (Xe^+) at speeds in excess of 30 km/s. The thrust produced is very low (about 0.09 newton) but can be maintained continuously for days, in contrast to chemical rockets, which produce a large thrust for a short time (see Fig. 8.34). Such ion engines have been used for maneuvering interplanetary spacecraft.



Electric Potential Energy with Several Point Charges

Suppose the electric field $\vec{E}$ in which charge q_0 moves is caused by *several* point charges q_1 , q_2 , q_3 , $\dots$ at distances r_1 , r_2 , r_3 , $\dots$ from q_0 , as in Fig. 23.8. For example, q_0 could be a positive ion moving in the presence of other ions (Fig. 23.9). The total electric field at each point is the *vector sum* of the fields due to the individual charges, and the total work done on q_0 during any displacement is the sum of the contributions from the individual charges. From Eq. (23.9) we conclude that the potential energy associated with the test charge q_0 at point a in Fig. 23.8 is the *algebraic sum* (not a vector sum):

$$U = \frac{q_0}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \dots \right) = \frac{q_0}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} \quad (23.10)$$

Electric potential energy of point charge q_0 and collection of charges $q_1, q_2, q_3, \dots$
 Electric constant Distances from q_0 to $q_1, q_2, q_3, \dots$

When q_0 is at a different point b , the potential energy is given by the same expression, but r_1 , r_2 , $\dots$ are the distances from q_1 , q_2 , $\dots$ to point b . The work done on charge q_0 when it moves from a to b along any path is equal to the difference $U_a - U_b$ between the potential energies when q_0 is at a and at b .

We can represent *any* charge distribution as a collection of point charges, so Eq. (23.10) shows that we can always find a potential-energy function for *any* static electric field. It follows that **for every electric field due to a static charge distribution, the force exerted by that field is conservative.**

Equations (23.9) and (23.10) define U to be zero when distances r_1 , r_2 , $\dots$ are infinite—that is, when the test charge q_0 is very far away from all the charges that produce the field. As with any potential-energy function, the point where $U = 0$ is arbitrary; we can always add a constant to make U equal zero at any point we choose. In electrostatics problems it's usually simplest to choose this point to be at infinity. When we analyze electric circuits in Chapters 25 and 26, other choices will be more convenient.

Equation (23.10) gives the potential energy associated with the presence of the test charge q_0 in the $\vec{E}$ field produced by q_1 , q_2 , q_3 , $\dots$. But there is also potential energy involved in assembling these charges. If we start with charges q_1 , q_2 , q_3 , $\dots$ all separated from each other by infinite distances and then bring them together so that the distance between q_i and q_j is r_{ij} , the *total* potential energy U is the sum of the potential energies of interaction for each pair of charges. We can write this as

$$U = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}} \quad (23.11)$$

This sum extends over all *pairs* of charges; we don't let $i = j$ (because that would be an interaction of a charge with itself), and we include only terms with $i < j$ to make sure that we count each pair only once. Thus, to account for the interaction between q_3 and q_4 , we include a term with $i = 3$ and $j = 4$ but not a term with $i = 4$ and $j = 3$.

Interpreting Electric Potential Energy

As a final comment, here are two viewpoints on electric potential energy. We have defined it in terms of the work done *by the electric field* on a charged particle moving in the field, just as in Chapter 7 we defined potential energy in terms of the work done by gravity or by a spring. When a particle moves from point a to point b , the work done on it by the electric field is $W_{a \rightarrow b} = U_a - U_b$. Thus the potential-energy difference $U_a - U_b$ equals *the work that is done by the electric force when the particle moves from a to b* . When U_a is greater than U_b , the field does positive work on the particle as it “falls” from a point of higher potential energy (a) to a point of lower potential energy (b).

An alternative but equivalent viewpoint is to consider how much work we would have to do to “raise” a particle from a point b where the potential energy is U_b to a point a where it has a greater value U_a (pushing two positive charges closer together, for example). To move the particle slowly (so as not to give it any kinetic energy), we need to exert an additional external force $\vec{F}_{\text{ext}}$ that is equal and opposite to the electric-field force and does positive work. The potential-energy difference $U_a - U_b$ is then defined as *the work that must be done by an external force to move the particle slowly from b to a against the electric force*. Because $\vec{F}_{\text{ext}}$ is the negative of the electric-field force and the displacement is in the opposite direction, this definition of the potential difference $U_a - U_b$ is equivalent to that given above. This alternative viewpoint also works if U_a is less than U_b , corresponding to “lowering” the particle; an example is moving two positive charges away from each other. In this case, $U_a - U_b$ is again equal to the work done by the external force, but now this work is negative.

We’ll use both of these viewpoints in the next section to interpret what is meant by electric *potential*, or potential energy per unit charge.

EXAMPLE 23.2 A system of point charges

WITH VARIATION PROBLEMS

Two point charges are at fixed positions on the x -axis, $q_1 = -e$ at $x = 0$ and $q_2 = +e$ at $x = a$. (a) Find the work that must be done by an external force to bring a third point charge $q_3 = +e$ from infinity to $x = 2a$. (b) Find the total potential energy of the system of three charges.

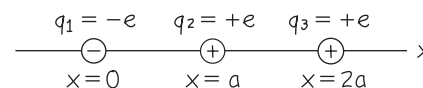
IDENTIFY and SET UP Figure 23.10 shows the final arrangement of the three charges. In part (a) we need to find the work W that must be done on q_3 by an external force $\vec{F}_{\text{ext}}$ to bring q_3 in from infinity to $x = 2a$. We do this by using Eq. (23.10) to find the potential energy associated with q_3 in the presence of q_1 and q_2 . In part (b) we use Eq. (23.11), the expression for the potential energy of a collection of point charges, to find the total potential energy of the system.

EXECUTE (a) The work W equals the difference between (i) the potential energy U associated with q_3 when it is at $x = 2a$ and (ii) the potential energy when it is infinitely far away. The second of these is zero, so the work required is equal to U . The distances between the charges are $r_{13} = 2a$ and $r_{23} = a$, so from Eq. (23.10),

$$W = U = \frac{q_3}{4\pi\epsilon_0} \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right) = \frac{+e}{4\pi\epsilon_0} \left(\frac{-e}{2a} + \frac{+e}{a} \right) = \frac{+e^2}{8\pi\epsilon_0 a}$$

This is positive, just as we should expect. If we bring q_3 in from infinity along the $+x$ -axis, it is attracted by q_1 but is repelled more strongly by q_2 . Hence we must do positive work to push q_3 to the position at $x = 2a$.

Figure 23.10 Our sketch of the situation after the third charge has been brought in from infinity.



(b) From Eq. (23.11), the total potential energy of the three-charge system is

$$\begin{aligned} U &= \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}} = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{(-e)(e)}{a} + \frac{(-e)(e)}{2a} + \frac{(e)(e)}{a} \right] = \frac{-e^2}{8\pi\epsilon_0 a} \end{aligned}$$

EVALUATE Our negative result in part (b) means that the system has lower potential energy than it would if the three charges were infinitely far apart. An external force would have to do *negative* work to bring the three charges from infinity to assemble this entire arrangement and would have to do *positive* work to move the three charges back to infinity.

KEYCONCEPT To find the total potential energy of a system of charges, sum the electric potential energies of all the distinct pairs of charges in the system.

TEST YOUR UNDERSTANDING OF SECTION 23.1 Consider the system of three point charges in Example 21.4 (Section 21.3) and shown in Fig. 21.14. (a) What is the sign of the total potential energy of this system? (i) Positive; (ii) negative; (iii) zero. (b) What is the sign of the total amount of work you would have to do to move these charges infinitely far from each other? (i) Positive; (ii) negative; (iii) zero.

ANSWER

(a) (i) The three charges q_1 , q_2 , and q_3 are all positive, so all three of the terms in the sum in Eq. (23.11)— $q_1 q_2 / r_{12}$, $q_1 q_3 / r_{13}$, and $q_2 q_3 / r_{23}$ —are positive. Hence the total electric potential energy U is positive. This means that it would take positive work to bring the three charges from infinity to the positions shown in Fig. 21.14, and hence *negative* work to move the three charges from these positions back to infinity.

23.2 ELECTRIC POTENTIAL

In Section 23.1 we looked at the potential energy U associated with a test charge q_0 in an electric field. Now we want to describe this potential energy on a “per unit charge” basis, just as electric field describes the force per unit charge on a charged particle in the field. This leads us to the concept of *electric potential*, often called simply *potential*. This concept is very useful in calculations involving energies of charged particles. It also facilitates many electric-field calculations because electric potential is closely related to the electric field $\vec{E}$. When we need to determine an electric field, it is often easier to determine the potential first and then find the field from it.

Potential is *potential energy per unit charge*. We define the potential V at any point in an electric field as the potential energy U *per unit charge* associated with a test charge q_0 at that point:

$$V = \frac{U}{q_0} \quad \text{or} \quad U = q_0 V \quad (23.12)$$

Potential energy and charge are both scalars, so potential is a scalar. From Eq. (23.12) its units are the units of energy divided by those of charge. The SI unit of potential, called one **volt** (1 V) in honor of the Italian electrical experimenter Alessandro Volta (1745–1827), equals 1 joule per coulomb:

$$1 \text{ V} = 1 \text{ volt} = 1 \text{ J/C} = 1 \text{ joule/coulomb}$$

Let’s put Eq. (23.2), which equates the work done by the electric force during a displacement from a to b to the quantity $-\Delta U = -(U_b - U_a)$, on a “work per unit charge” basis. We divide this equation by q_0 , obtaining

$$\frac{W_{a \rightarrow b}}{q_0} = -\frac{\Delta U}{q_0} = -\left(\frac{U_b}{q_0} - \frac{U_a}{q_0}\right) = -(V_b - V_a) = V_a - V_b \quad (23.13)$$

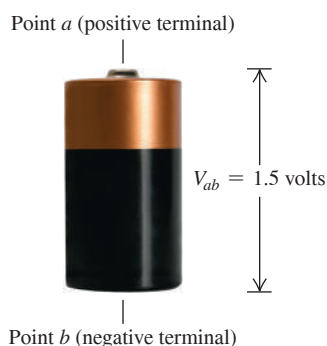
where $V_a = U_a/q_0$ is the potential energy per unit charge at point a and similarly for V_b . We call V_a and V_b the *potential at point a* and *potential at point b* , respectively. Thus the work done per unit charge by the electric force when a charged object moves from a to b is equal to the potential at a minus the potential at b .

The difference $V_a - V_b$ is called the *potential of a with respect to b* ; we sometimes abbreviate this difference as $V_{ab} = V_a - V_b$ (note the order of the subscripts). This is often called the potential difference between a and b , but that’s ambiguous unless we specify which is the reference point. In electric circuits, which we’ll analyze in later chapters, the potential difference between two points is often called **voltage** (Fig. 23.11). Equation (23.13) then states: **V_{ab} , the potential (in V) of a with respect to b , equals the work (in J) done by the electric force when a UNIT (1 C) charge moves from a to b .**

Another way to interpret the potential difference V_{ab} in Eq. (23.13) is to use the alternative viewpoint mentioned at the end of Section 23.1. In that viewpoint, $U_a - U_b$ is the amount of work that must be done by an *external* force to move a particle of charge q_0 slowly from b to a against the electric force. The work that must be done *per unit charge* by the external force is then $(U_a - U_b)/q_0 = V_a - V_b = V_{ab}$. In other words: **V_{ab} , the potential (in V) of a with respect to b , equals the work (in J) that must be done to move a UNIT (1 C) charge slowly from b to a against the electric force.**

An instrument that measures the difference of potential between two points is called a *voltmeter*. (In Chapter 26 we’ll discuss how these devices work.) Voltmeters that can measure a potential difference of 1 μV are common, and sensitivities down to 10^{-12} V can be attained.

Figure 23.11 The voltage of this battery equals the difference in potential $V_{ab} = V_a - V_b$ between its positive terminal (point a) and its negative terminal (point b).



Calculating Electric Potential

To find the potential V due to a single point charge q , we divide Eq. (23.9) by q_0 :

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (23.14)$$

Electric potential due to a point charge V is equal to the electric constant $4\pi\epsilon_0$ times the value of point charge q divided by the distance from point charge to where potential is measured r .

If q is positive, the potential that it produces is positive at all points; if q is negative, it produces a potential that is negative everywhere. In either case, V is equal to zero at $r = \infty$, an infinite distance from the point charge. Note that potential, like electric field, is independent of the test charge q_0 that we use to define it.

Similarly, we divide Eq. (23.10) by q_0 to find the potential due to a collection of point charges:

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} \quad (23.15)$$

Electric potential due to a collection of point charges V is equal to the electric constant $4\pi\epsilon_0$ times the sum of the value of i th point charge q_i divided by the distance from i th point charge to where potential is measured r_i .

Just as the electric field due to a collection of point charges is the *vector* sum of the fields produced by each charge, the electric potential due to a collection of point charges is the *scalar* sum of the potentials due to each charge. When we have a continuous distribution of charge along a line, over a surface, or through a volume, we divide the charge into elements dq , and the sum in Eq. (23.15) becomes an integral:

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \quad (23.16)$$

Electric potential due to a continuous distribution of charge V is equal to the electric constant $4\pi\epsilon_0$ times the integral over charge distribution of the charge element dq divided by the distance from charge element to where potential is measured r .

We'll work out several examples of such cases. The potential defined by Eqs. (23.15) and (23.16) is zero at points that are infinitely far away from *all* the charges. Later we'll encounter cases in which the charge distribution itself extends to infinity. We'll find that in such cases we cannot set $V = 0$ at infinity, and we'll need to exercise care in using and interpreting Eqs. (23.15) and (23.16).

CAUTION What is electric potential? Before getting too involved in the details of how to calculate electric potential, remind yourself what potential is. The electric *potential* at a certain point is the potential energy per *unit* charge placed at that point. That's why potential is measured in joules per coulomb, or volts. Keep in mind, too, that there doesn't have to be a charge at a given point for a potential V to exist at that point. (In the same way, an electric field can exist at a given point even if there's no charge there to respond to it.)

Finding Electric Potential from Electric Field

When we are given a collection of point charges, Eq. (23.15) is usually the easiest way to calculate the potential V . But in some problems in which the electric field is known or can be found easily, it is easier to determine V from $\vec{E}$. The force $\vec{F}$ on a test charge q_0 can be written as $\vec{F} = q_0\vec{E}$, so from Eq. (23.1) the work done by the electric force as the test charge moves from a to b is given by

$$W_{a \rightarrow b} = \int_a^b \vec{F} \cdot d\vec{l} = \int_a^b q_0 \vec{E} \cdot d\vec{l}$$

BIO APPLICATION Electrocardiography

The electrodes used in an electrocardiogram—EKG or ECG for short—measure the potential differences (typically no greater than $1 \text{ mV} = 10^{-3} \text{ V}$) between different parts of the patient's skin. These are indicative of the potential differences between regions of the heart, and so provide a sensitive way to detect any abnormalities in the electrical activity that drives cardiac function.



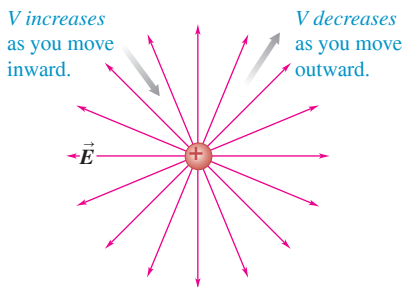
If we divide this by q_0 and compare the result with Eq. (23.13), we find

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E \cos \phi \, dl \quad (23.17)$$

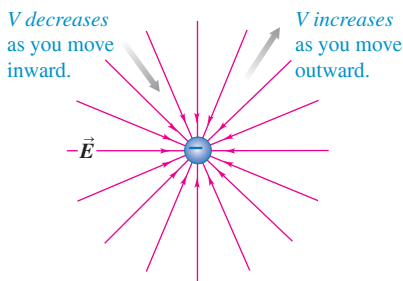
Electric potential difference $V_a - V_b$ Integral along path from a to b Displacement $d\vec{l}$ Angle between $\vec{E}$ and $d\vec{l}$
 Scalar product of electric field and displacement vector Electric-field magnitude

Figure 23.12 If you move in the direction of $\vec{E}$, electric potential V decreases; if you move in the direction opposite $\vec{E}$, V increases.

(a) A positive point charge



(b) A negative point charge



The value of $V_a - V_b$ is independent of the path taken from a to b , just as the value of $W_{a \rightarrow b}$ is independent of the path. To interpret Eq. (23.17), remember that $\vec{E}$ is the electric force per unit charge on a test charge. If the line integral $\int_a^b \vec{E} \cdot d\vec{l}$ is positive, the electric field does positive work on a positive test charge as it moves from a to b . In this case the electric potential energy decreases as the test charge moves, so the potential energy per unit charge decreases as well; hence V_b is less than V_a and $V_a - V_b$ is positive.

As an illustration, consider a positive point charge (Fig. 23.12a). The electric field is directed away from the charge, and $V = q/4\pi\epsilon_0 r$ is positive at any finite distance from the charge. If you move away from the charge, in the direction of $\vec{E}$, you move toward lower values of V ; if you move toward the charge, in the direction opposite $\vec{E}$, you move toward greater values of V . For the negative point charge in Fig. 23.12b, $\vec{E}$ is directed toward the charge and $V = q/4\pi\epsilon_0 r$ is negative at any finite distance from the charge. In this case, if you move toward the charge, you are moving in the direction of $\vec{E}$ and in the direction of decreasing (more negative) V . Moving away from the charge, in the direction opposite $\vec{E}$, moves you toward increasing (less negative) values of V . The general rule, valid for any electric field, is: Moving *with* the direction of $\vec{E}$ means moving in the direction of *decreasing* V , and moving *against* the direction of $\vec{E}$ means moving in the direction of *increasing* V .

Also, a positive test charge q_0 experiences an electric force in the direction of $\vec{E}$, toward lower values of V ; a negative test charge experiences a force opposite $\vec{E}$, toward higher values of V . Thus a positive charge tends to “fall” from a high-potential region to a lower-potential region. The opposite is true for a negative charge.

Notice that Eq. (23.17) can be rewritten as

$$V_a - V_b = - \int_b^a \vec{E} \cdot d\vec{l} \quad (23.18)$$

This has a negative sign compared to the integral in Eq. (23.17), and the limits are reversed; hence Eqs. (23.17) and (23.18) are equivalent. But Eq. (23.18) has a slightly different interpretation. To move a unit charge slowly against the electric force, we must apply an *external* force per unit charge equal to $-\vec{E}$, equal and opposite to the electric force per unit charge $\vec{E}$. Equation (23.18) says that $V_a - V_b = V_{ab}$, the potential of a with respect to b , equals the work done per unit charge by this external force to move a unit charge from b to a . This is the same alternative interpretation we discussed under Eq. (23.13).

Equations (23.17) and (23.18) show that the unit of potential difference (1 V) is equal to the unit of electric field (1 N/C) multiplied by the unit of distance (1 m). Hence the unit of electric field can be expressed as 1 volt per meter (1 V/m) as well as 1 N/C:

$$1 \text{ V/m} = 1 \text{ volt/meter} = 1 \text{ N/C} = 1 \text{ newton/coulomb}$$

In practice, the volt per meter is the usual unit of electric-field magnitude.

Electron Volts

The magnitude e of the electron charge can be used to define a unit of energy that is useful in many calculations with atomic and nuclear systems. When a particle with charge q moves from a point where the potential is V_b to a point where it is V_a , the change in the potential energy U is

$$U_a - U_b = q(V_a - V_b) = qV_{ab}$$

If charge q equals the magnitude e of the electron charge, 1.602×10^{-19} C, and the potential difference is $V_{ab} = 1$ V = 1 J/C, the change in energy is

$$U_a - U_b = (1.602 \times 10^{-19} \text{ C})(1 \text{ J/C}) = 1.602 \times 10^{-19} \text{ J}$$

This quantity of energy is defined to be 1 **electron volt** (1 eV):

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

The multiples meV, keV, MeV, GeV, and TeV are often used.

CAUTION Electron volts vs. volts Remember that the electron volt is a unit of energy, *not* a unit of potential or potential difference!

When a particle with charge e moves through a potential difference of 1 volt, the change in potential *energy* is 1 eV. If the charge is some multiple of e —say, Ne —the change in potential energy in electron volts is N times the potential difference in volts. For example, when an alpha particle, which has charge $2e$, moves between two points with a potential difference of 1000 V, the change in potential energy is $2(1000 \text{ eV}) = 2000 \text{ eV}$. To confirm this, we write

$$\begin{aligned} U_a - U_b &= qV_{ab} = (2e)(1000 \text{ V}) = (2)(1.602 \times 10^{-19} \text{ C})(1000 \text{ V}) \\ &= 3.204 \times 10^{-16} \text{ J} = 2000 \text{ eV} \end{aligned}$$

Although we defined the electron volt in terms of *potential* energy, we can use it for *any* form of energy, such as the kinetic energy of a moving particle. When we speak of a “one-million-electron-volt proton,” we mean a proton with a kinetic energy of one million electron volts (1 MeV), equal to $(10^6)(1.602 \times 10^{-19} \text{ J}) = 1.602 \times 10^{-13} \text{ J}$. The Large Hadron Collider near Geneva, Switzerland, is designed to accelerate protons to a kinetic energy of 7 TeV ($7 \times 10^{12} \text{ eV}$).

EXAMPLE 23.3 Electric force and electric potential

A proton (charge $+e = 1.602 \times 10^{-19}$ C) moves a distance $d = 0.50$ m in a straight line between points a and b in a linear accelerator. The electric field is uniform along this line, with magnitude $E = 1.5 \times 10^7$ V/m = 1.5×10^7 N/C in the direction from a to b . Determine (a) the force on the proton; (b) the work done on it by the field; (c) the potential difference $V_a - V_b$.

IDENTIFY and SET UP This problem uses the relationship between electric field and electric force. It also uses the relationship among force, work, and potential-energy difference. We are given the electric field, so it is straightforward to find the electric force on the proton. Calculating the work is also straightforward because $\vec{E}$ is uniform, so the force on the proton is constant. Once the work is known, we find $V_a - V_b$ from Eq. (23.13).

EXECUTE (a) The force on the proton is in the same direction as the electric field, and its magnitude is

$$\begin{aligned} F &= qE = (1.602 \times 10^{-19} \text{ C})(1.5 \times 10^7 \text{ N/C}) \\ &= 2.4 \times 10^{-12} \text{ N} \end{aligned}$$

(b) The force is constant and in the same direction as the displacement, so the work done on the proton is

$$\begin{aligned} W_{a \rightarrow b} &= Fd = (2.4 \times 10^{-12} \text{ N})(0.50 \text{ m}) \\ &= 1.2 \times 10^{-12} \text{ J} \\ &= (1.2 \times 10^{-12} \text{ J}) \frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} \\ &= 7.5 \times 10^6 \text{ eV} = 7.5 \text{ MeV} \end{aligned}$$

(c) From Eq. (23.13) the potential difference is the work per unit charge, which is

$$\begin{aligned} V_a - V_b &= \frac{W_{a \rightarrow b}}{q} = \frac{1.2 \times 10^{-12} \text{ J}}{1.602 \times 10^{-19} \text{ C}} \\ &= 7.5 \times 10^6 \text{ J/C} = 7.5 \times 10^6 \text{ V} = 7.5 \text{ MV} \end{aligned}$$

We can get this same result even more easily by remembering that 1 electron volt equals 1 volt multiplied by the charge e . The work done is $7.5 \times 10^6 \text{ eV}$ and the charge is e , so the potential difference is $(7.5 \times 10^6 \text{ eV})/e = 7.5 \times 10^6 \text{ V}$.

EVALUATE We can check our result in part (c) by using Eq. (23.17) or (23.18). The angle ϕ between the constant field $\vec{E}$ and the displacement is zero, so Eq. (23.17) becomes

$$V_a - V_b = \int_a^b E \cos \phi \, dl = \int_a^b E \, dl = E \int_a^b dl$$

The integral of dl from a to b is just the distance d , so we again find

$$V_a - V_b = Ed = (1.5 \times 10^7 \text{ V/m})(0.50 \text{ m}) = 7.5 \times 10^6 \text{ V}$$

KEYCONCEPT The potential difference V_{ab} between point a and point b , equal to the difference $V_a - V_b$ of the potentials V at the two points, is the amount of work the electric force does on a unit charge as it moves from a to b . If you move in the direction of the electric field, V decreases (V_{ab} is positive); if you move opposite to the direction of the electric field, V increases (V_{ab} is negative).

BIO APPLICATION Electron Volts and Cancer Radiotherapy

One way to destroy a cancerous tumor is to aim high-energy electrons directly at it. Each electron has a kinetic energy of 4 to 20 million electron volts, or MeV (1 MeV = 10^6 eV), and transfers its energy to the tumor through collisions with the tumor's atoms. Electrons in this energy range can penetrate only a few centimeters into a patient, which makes them useful for treating superficial tumors, such as those on the skin or lips.



EXAMPLE 23.4 Potential due to two point charges**WITH VARIATION PROBLEMS**

An electric dipole consists of point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ placed 10.0 cm apart (**Fig. 23.13**). Compute the electric potentials at points a , b , and c .

IDENTIFY and SET UP This is the same arrangement as in Example 21.8 (Section 21.5), in which we calculated the electric *field* at each point by doing a *vector* sum. Here our target variable is the electric *potential* V at three points, which we find by doing the *algebraic* sum in Eq. (23.15).

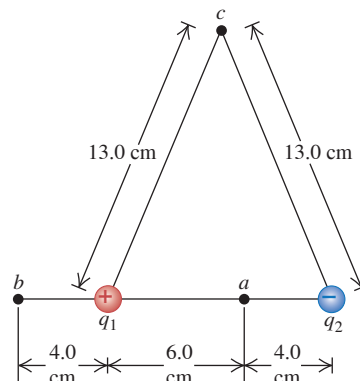
EXECUTE At point a we have $r_1 = 0.060 \text{ m}$ and $r_2 = 0.040 \text{ m}$, so Eq. (23.15) becomes

$$\begin{aligned} V_a &= \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{12 \times 10^{-9} \text{ C}}{0.060 \text{ m}} \\ &\quad + (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(-12 \times 10^{-9} \text{ C})}{0.040 \text{ m}} \\ &= 1800 \text{ N} \cdot \text{m}/\text{C} + (-2700 \text{ N} \cdot \text{m}/\text{C}) \\ &= 1800 \text{ V} + (-2700 \text{ V}) = -900 \text{ V} \end{aligned}$$

In a similar way you can show that the potential at point b (where $r_1 = 0.040 \text{ m}$ and $r_2 = 0.140 \text{ m}$) is $V_b = 1930 \text{ V}$ and that the potential at point c (where $r_1 = r_2 = 0.130 \text{ m}$) is $V_c = 0$.

EVALUATE Let's confirm that these results make sense. Point a is closer to the -12 nC charge than to the $+12 \text{ nC}$ charge, so the potential at a is negative. The potential is positive at point b , which is closer to the $+12 \text{ nC}$ charge than the -12 nC charge. Finally, point c is equidistant from the $+12 \text{ nC}$ charge and the -12 nC charge, so the potential there is zero. (The potential is also equal to zero at a point infinitely far from both charges.)

Figure 23.13 What are the potentials at points a , b , and c due to this electric dipole?



Comparing this example with Example 21.8 shows that it's much easier to calculate electric potential (a scalar) than electric field (a vector). We'll take advantage of this simplification whenever possible.

KEYCONCEPT The electric potential caused at a point P by a single point charge is proportional to the charge and inversely proportional to the distance from the point charge to P . The total electric potential at P due to a system of charges is the sum of the potentials at P due to each individual charge.

EXAMPLE 23.5 Potential and potential energy**WITH VARIATION PROBLEMS**

Compute the potential energy associated with a $+4.0 \text{ nC}$ point charge if it is placed at points a , b , and c in Fig. 23.13.

IDENTIFY and SET UP The potential energy U associated with a point charge q at a location where the electric potential is V is $U = qV$. We use the values of V from Example 23.4.

EXECUTE At the three points we find

$$\begin{aligned} U_a &= qV_a = (4.0 \times 10^{-9} \text{ C})(-900 \text{ J/C}) = -3.6 \times 10^{-6} \text{ J} \\ U_b &= qV_b = (4.0 \times 10^{-9} \text{ C})(1930 \text{ J/C}) = 7.7 \times 10^{-6} \text{ J} \\ U_c &= qV_c = 0 \end{aligned}$$

All of these values correspond to U and V being zero at infinity.

EVALUATE Note that *zero* net work is done on the 4.0 nC charge if it moves from point c to infinity *by any path*. In particular, let the path be along the perpendicular bisector of the line joining the other two charges q_1 and q_2 in Fig. 23.13. As shown in Example 21.8 (Section 21.5), at points on the bisector, the direction of $\vec{E}$ is perpendicular to the bisector. Hence the force on the 4.0 nC charge is perpendicular to the path, and no work is done in any displacement along it.

KEYCONCEPT If you place a point charge q at a point P , the associated electric potential energy is equal to q multiplied by the value at P of the electric potential due to all *other* charges.

EXAMPLE 23.6 Finding potential by integration

By integrating the electric field as in Eq. (23.17), find the potential at a distance r from a point charge q .

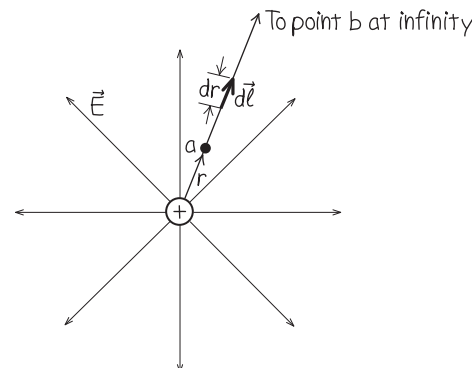
IDENTIFY and SET UP We let point a in Eq. (23.17) be at distance r and let point b be at infinity (Fig. 23.14). As usual, we choose the potential to be zero at an infinite distance from the charge q .

EXECUTE To carry out the integral, we can choose any path we like between points a and b . The most convenient path is a radial line as shown in Fig. 23.14, so that $d\vec{l}$ is in the radial direction and has magnitude dr . Writing $d\vec{l} = \hat{r}dr$, we have from Eq. (23.17)

$$\begin{aligned} V - 0 &= V = \int_r^\infty \vec{E} \cdot d\vec{l} \\ &= \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \cdot \hat{r} dr = \int_r^\infty \frac{q}{4\pi\epsilon_0 r^2} dr \\ &= -\frac{q}{4\pi\epsilon_0 r} \Big|_r^\infty = 0 - \left(-\frac{q}{4\pi\epsilon_0 r} \right) = \frac{q}{4\pi\epsilon_0 r} \end{aligned}$$

EVALUATE Our result agrees with Eq. (23.14) and is correct for positive or negative q .

Figure 23.14 Calculating the potential by integrating $\vec{E}$ for a single point charge.



KEYCONCEPT The potential difference $V_{ab} = V_a - V_b$ between point a and point b equals the integral of the electric field along a path from a to b . This integral does not depend on the path taken between the two points.

EXAMPLE 23.7 Moving through a potential difference**WITH VARIATION PROBLEMS**

In Fig. 23.15 a dust particle with mass $m = 5.0 \times 10^{-9} \text{ kg} = 5.0 \mu\text{g}$ and charge $q_0 = 2.0 \text{ nC}$ starts from rest and moves in a straight line from point a to point b . What is its speed v at point b ?

IDENTIFY and SET UP Only the conservative electric force acts on the particle, so the total mechanical energy is conserved: $K_a + U_a = K_b + U_b$. We get the potential energies U from the corresponding potentials V from Eq. (23.12): $U_a = q_0 V_a$ and $U_b = q_0 V_b$.

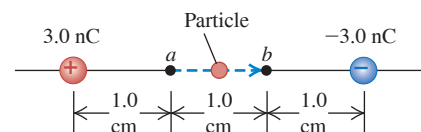
EXECUTE We have $K_a = 0$ and $K_b = \frac{1}{2}mv^2$. We substitute these and our expressions for U_a and U_b into the energy-conservation equation, then solve for v . We find

$$\begin{aligned} 0 + q_0 V_a &= \frac{1}{2}mv^2 + q_0 V_b \\ v &= \sqrt{\frac{2q_0(V_a - V_b)}{m}} \end{aligned}$$

We calculate the potentials from Eq. (23.15), $V = q/4\pi\epsilon_0 r$:

$$\begin{aligned} V_a &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \left(\frac{3.0 \times 10^{-9} \text{ C}}{0.010 \text{ m}} + \frac{(-3.0 \times 10^{-9} \text{ C})}{0.020 \text{ m}} \right) \\ &= 1350 \text{ V} \\ V_b &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \left(\frac{3.0 \times 10^{-9} \text{ C}}{0.020 \text{ m}} + \frac{(-3.0 \times 10^{-9} \text{ C})}{0.010 \text{ m}} \right) \\ &= -1350 \text{ V} \\ V_a - V_b &= (1350 \text{ V}) - (-1350 \text{ V}) = 2700 \text{ V} \end{aligned}$$

Figure 23.15 The particle moves from point a to point b ; its acceleration is not constant.



Finally,

$$v = \sqrt{\frac{2(2.0 \times 10^{-9} \text{ C})(2700 \text{ V})}{5.0 \times 10^{-9} \text{ kg}}} = 46 \text{ m/s}$$

EVALUATE Our result makes sense: The positive dust particle speeds up as it moves away from the $+3.0 \text{ nC}$ charge and toward the -3.0 nC charge. To check unit consistency in the final line of the calculation, note that $1 \text{ V} = 1 \text{ J/C}$, so the numerator under the radical has units of J or $\text{kg} \cdot \text{m}^2/\text{s}^2$.

KEYCONCEPT When a particle of charge q moves from a point of higher potential to a point of lower potential, its potential energy *decreases* if q is positive; its potential energy *increases* if q is negative.

TEST YOUR UNDERSTANDING OF SECTION 23.2 If the electric *potential* at a certain point is zero, does the electric *field* at that point have to be zero? (*Hint:* Consider point c in Examples 23.4 and 21.8.)

ANSWER

no If $V = 0$ at a certain point, $\vec{E}$ does *not* have to be zero at that point. An example is point c in Figs. 21.23 and 23.13, for which there is an electric field in the $+x$ -direction (see Example 21.9 in Section 21.5) even though $V = 0$ (see Example 23.4). This isn't a surprising result because V and $\vec{E}$ are quite different quantities: V is the net amount of work required to bring a unit charge from infinity to the point in question, whereas $\vec{E}$ is the electric force that acts on a unit charge when it arrives at that point.

23.3 CALCULATING ELECTRIC POTENTIAL

When calculating the potential due to a charge distribution, we usually follow one of two routes. If we know the charge distribution, we can use Eq. (23.15) or (23.16). Or if we know how the electric field depends on position, we can use Eq. (23.17), defining the potential to be zero at some convenient place. Some problems require a combination of these approaches.

As you read through these examples, compare them with the related examples of calculating electric *field* in Section 21.5. You'll see how much easier it is to calculate scalar electric potentials than vector electric fields. The moral is clear: Whenever possible, solve problems by means of an energy approach (using electric potential and electric potential energy) rather than a dynamics approach (using electric fields and electric forces).

PROBLEM-SOLVING STRATEGY 23.1 Calculating Electric Potential

IDENTIFY *the relevant concepts:* Remember that electric potential is *potential energy per unit charge*.

SET UP *the problem* using the following steps:

1. Make a drawing showing the locations and values of the charges (which may be point charges or a continuous distribution of charge) and your choice of coordinate axes.
2. Indicate on your drawing the position of the point at which you want to calculate the electric potential V . Sometimes this position will be an arbitrary one (say, a point a distance r from the center of a charged sphere).

EXECUTE *the solution* as follows:

1. To find the potential due to a collection of point charges, use Eq. (23.15). If you are given a continuous charge distribution, devise a way to divide it into infinitesimal elements and use Eq. (23.16). Carry out the integration, using appropriate limits to include the entire charge distribution.
2. If you are given the electric field, or if you can find it from any of the methods presented in Chapters 21 or 22, it may be easier

to find the potential difference between points a and b from Eq. (23.17) or (23.18). When appropriate, make use of your freedom to define V to be zero at some convenient place, and choose this place to be point b . (For point charges, this will usually be at infinity. For other distributions of charge—especially those that themselves extend to infinity—it may be necessary to define V_b to be zero at some finite distance from the charge distribution.) Then the potential at any other point, say a , can be found from Eq. (23.17) or (23.18) with $V_b = 0$.

3. Although potential V is a *scalar* quantity, you may have to use components of the vectors $\vec{E}$ and $d\vec{l}$ when you use Eq. (23.17) or (23.18) to calculate V .

EVALUATE *your answer:* Check whether your answer agrees with your intuition. If your result gives V as a function of position, graph the function to see whether it makes sense. If you know the electric field, you can make a rough check of your result for V by verifying that V decreases if you move in the direction of $\vec{E}$.

EXAMPLE 23.8 A charged conducting sphere**WITH VARIATION PROBLEMS**

A solid conducting sphere of radius R has a total charge q . Find the electric potential everywhere, both outside and inside the sphere.

IDENTIFY and SET UP In Example 22.5 (Section 22.4) we used Gauss's law to find the electric *field* at all points for this charge distribution. We can use that result to determine the potential.

EXECUTE From Example 22.5, the field *outside* the sphere is the same as if the sphere were removed and replaced by a point charge q . We take $V = 0$ at infinity, as we did for a point charge. Then the potential at a point outside the sphere at a distance r from its center is the same as that due to a point charge q at the center:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

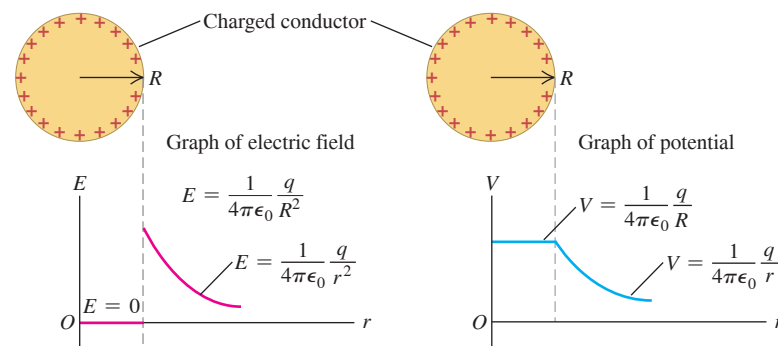
The potential at the surface of the sphere is $V_{\text{surface}} = q/4\pi\epsilon_0 R$.

Inside the sphere, $\vec{E}$ is zero everywhere. Hence no work is done on a test charge that moves from any point to any other point inside the sphere. Thus the potential is the same at every point inside the sphere and is equal to its value $q/4\pi\epsilon_0 R$ at the surface.

EVALUATE Figure 23.16 shows the field and potential for a positive charge q . In this case the electric field points radially away from the sphere. As you move away from the sphere, in the direction of $\vec{E}$, V decreases (as it should).

KEYCONCEPT If there is excess charge at rest on a solid spherical conductor, all of the charge lies on the surface of the conductor, and the electric field and electric potential outside the sphere are the same as if all the charge were concentrated at the center of the sphere. Everywhere inside the sphere the field is zero and the potential has the same value as at the surface.

Figure 23.16 Electric-field magnitude E and potential V at points inside and outside a positively charged spherical conductor.

**Ionization and Corona Discharge**

The results of Example 23.8 have numerous practical consequences. One consequence relates to the maximum potential to which a conductor in air can be raised. This potential is limited because air molecules become *ionized*, and air becomes a conductor, at an electric-field magnitude of about 3×10^6 V/m. Assume for the moment that q is positive. When we compare the expressions in Example 23.8 for the potential V_{surface} and field magnitude E_{surface} at the surface of a charged conducting sphere, we note that $V_{\text{surface}} = E_{\text{surface}} R$. Thus, if E_m represents the electric-field magnitude at which air becomes conductive (known as the *dielectric strength* of air), then the maximum potential V_m to which a spherical conductor can be raised is

$$V_m = RE_m$$

For a conducting sphere 1 cm in radius in air, $V_m = (10^{-2} \text{ m})(3 \times 10^6 \text{ V/m}) = 30,000 \text{ V}$. No amount of “charging” could raise the potential of a conducting sphere of this size in air higher than about 30,000 V; attempting to raise the potential further by adding extra charge would cause the surrounding air to become ionized and conductive, and the extra added charge would leak into the air.

To attain even higher potentials, high-voltage machines such as Van de Graaff generators use spherical terminals with very large radii (see Fig. 22.26 and the photograph that opens Chapter 22). For example, a terminal of radius $R = 2 \text{ m}$ has a maximum potential $V_m = (2 \text{ m})(3 \times 10^6 \text{ V/m}) = 6 \times 10^6 \text{ V} = 6 \text{ MV}$.

Figure 23.17 The metal mast at the top of the Empire State Building acts as a lightning rod. It is struck by lightning as many as 500 times each year.



Our result in Example 23.8 also explains what happens with a charged conductor with a very *small* radius of curvature, such as a sharp point or thin wire. Because the maximum potential is proportional to the radius, even relatively small potentials applied to sharp points in air produce sufficiently high fields just outside the point to ionize the surrounding air, making it become a conductor. The resulting current and its associated glow (visible in a dark room) are called *corona discharge*. Laser printers and photocopying machines use corona discharge from fine wires to spray charge on the imaging drum (see Fig. 21.2).

A large-radius conductor is used in situations where it's important to *prevent* corona discharge. An example is the blunt end of a metal lightning rod (Fig. 23.17). If there is an excess charge in the atmosphere, as happens during thunderstorms, a substantial charge of the opposite sign can build up on this blunt end. As a result, when the atmospheric charge is discharged through a lightning bolt, it tends to be attracted to the charged lightning rod rather than to other structures that could be damaged. (A conducting wire connecting the lightning rod to the ground then allows the acquired charge to dissipate harmlessly.) A lightning rod with a sharp end would allow less charge buildup and hence be less effective.

EXAMPLE 23.9 Oppositely charged parallel plates

WITH VARIATION PROBLEMS

Find the potential at any height y between the two oppositely charged parallel plates discussed in Section 23.1 (Fig. 23.18).

IDENTIFY and SET UP We discussed this situation in Section 23.1. From Eq. (23.5), we know the electric *potential energy* U for a test charge q_0 is $U = q_0 E y$. (We set $y = 0$ and $U = 0$ at the bottom plate.) We use Eq. (23.12), $U = q_0 V$, to find the electric *potential* V as a function of y .

EXECUTE The potential $V(y)$ at coordinate y is the potential energy per unit charge:

$$V(y) = \frac{U(y)}{q_0} = \frac{q_0 E y}{q_0} = E y$$

The potential decreases as we move in the direction of $\vec{E}$ from the upper to the lower plate. At point a , where $y = d$ and $V(y) = V_a$,

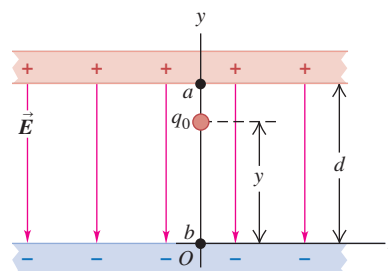
$$V_a - V_b = E d$$

and

$$E = \frac{V_a - V_b}{d} = \frac{V_{ab}}{d}$$

where V_{ab} is the potential of the positive plate with respect to the negative plate. That is, the electric field equals the potential difference between the plates divided by the distance between them. For a given potential difference V_{ab} , the smaller the distance d between the two plates, the greater the magnitude E of the electric field. (This relationship between E and V_{ab} holds *only* for the planar geometry we have described. It does *not* work for situations such as concentric cylinders or spheres in which the electric field is not uniform.)

Figure 23.18 The charged parallel plates from Fig. 23.2.



EVALUATE Our result shows that $V = 0$ at the bottom plate (at $y = 0$). This is consistent with our choice that $U = q_0 V = 0$ for a test charge placed at the bottom plate.

CAUTION “Zero potential” is arbitrary You might think that if a conducting object has zero potential, it must also have zero net charge. But that just isn't so! As an example, the plate at $y = 0$ in Fig. 23.18 has zero potential ($V = 0$) but has a nonzero charge per unit area $-\sigma$. There's nothing special about the place where potential is zero; we *define* this place to be wherever we want it to be.

KEYCONCEPT In a region of uniform electric field $\vec{E}$, such as between two oppositely charged parallel plates, the electric potential decreases linearly with position as you move in the direction of $\vec{E}$.

EXAMPLE 23.10 An infinite line charge or charged conducting cylinder**WITH VARIATION PROBLEMS**

Find the potential at a distance r from a very long line of charge with linear charge density (charge per unit length) λ .

IDENTIFY and SET UP In both Example 21.10 (Section 21.5) and Example 22.6 (Section 22.4) we found that the electric field at a radial distance r from a long straight-line charge (**Fig. 23.19a**) has only a radial component $E_r = \lambda/2\pi\epsilon_0 r$. We use this expression to find the potential by integrating $\vec{E}$ as in Eq. (23.17).

EXECUTE Since the field has only a radial component, we have $\vec{E} \cdot d\vec{l} = E_r dr$. Hence from Eq. (23.17) the potential of any point a with respect to any other point b , at radial distances r_a and r_b from the line of charge, is

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E_r dr = \frac{\lambda}{2\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}$$

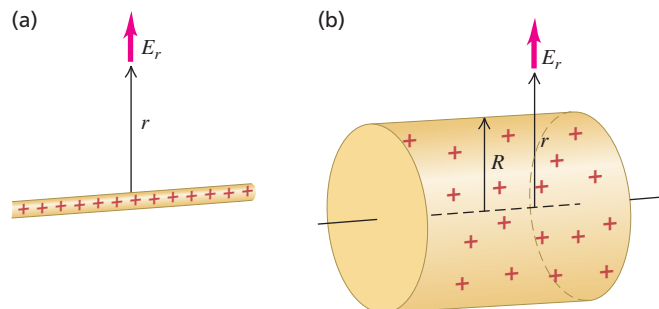
If we take point b at infinity and set $V_b = 0$, we find that V_a is *infinite* for any finite distance r_a from the line charge: $V_a = (\lambda/2\pi\epsilon_0) \ln(\infty/r_a) = \infty$. This is *not* a useful way to define V for this problem! The difficulty is that the charge distribution itself extends to infinity.

Instead, as recommended in Problem-Solving Strategy 23.1, we set $V_b = 0$ at point b at an arbitrary but *finite* radial distance r_0 . Then the potential $V = V_a$ at point a at a radial distance r is given by $V - 0 = (\lambda/2\pi\epsilon_0) \ln(r_0/r)$, or

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

EVALUATE According to our result, if λ is positive, then V decreases as r increases. This is as it should be: V decreases as we move in the direction of $\vec{E}$.

Figure 23.19 Electric field outside (a) a long, positively charged wire and (b) a long, positively charged cylinder.



From Example 22.6, the expression for E_r with which we started also applies outside a long, charged conducting cylinder with charge per unit length λ (**Fig. 23.19b**). Hence our result also gives the potential for such a cylinder, but only for values of r (the distance from the cylinder axis) equal to or greater than the radius R of the cylinder. If we choose r_0 to be the radius R , so that $V = 0$ when $r = R$, then at any point for which $r > R$,

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R}{r}$$

Inside the cylinder, $\vec{E} = 0$, and V has the same value (zero) as on the cylinder's surface.

KEYCONCEPT The electric potential outside a very long cylindrical distribution of charge or line of charge decreases logarithmically with the distance from the axis of the distribution or line.

EXAMPLE 23.11 A ring of charge**WITH VARIATION PROBLEMS**

Electric charge Q is distributed uniformly around a thin ring of radius a (**Fig. 23.20**). Find the potential at a point P on the ring axis at a distance x from the center of the ring.

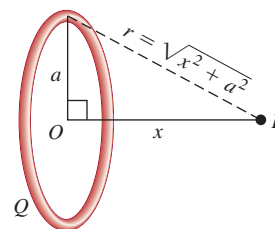
IDENTIFY and SET UP We divide the ring into infinitesimal segments and use Eq. (23.16) to find V . All parts of the ring (and therefore all elements of the charge distribution) are at the same distance from P .

EXECUTE Figure 23.20 shows that the distance from each charge element dq to P is $r = \sqrt{x^2 + a^2}$. Hence we can take the factor $1/r$ outside the integral in Eq. (23.16), and

$$\begin{aligned} V &= \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \\ &= \frac{1}{4\pi\epsilon_0} \frac{1}{\sqrt{x^2 + a^2}} \int dq \\ &= \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + a^2}} \end{aligned}$$

EVALUATE When x is much larger than a , our expression for V becomes approximately $V = Q/4\pi\epsilon_0 x$, which is the potential at a distance x from a point charge Q . Very far from a charged ring, its electric potential looks like that of a point charge. We drew a similar conclusion about the electric *field* of a ring in Example 21.9 (Section 21.5).

Figure 23.20 All the charge in a ring of charge Q is the same distance r from a point P on the ring axis.



We know the electric field at all points along the x -axis from Example 21.9 (Section 21.5), so we can also find V along this axis by integrating $\vec{E} \cdot d\vec{l}$ as in Eq. (23.17).

KEYCONCEPT To find the electric potential at a point due to a continuous distribution of charge, first divide the distribution into infinitesimally small segments. Then find the potential at the point due to one such segment. Finally, integrate over all segments in the charge distribution.

EXAMPLE 23.12 Potential of a line of charge**WITH VARIATION PROBLEMS**

Positive electric charge Q is distributed uniformly along a line of length $2a$ lying along the y -axis between $y = -a$ and $y = +a$ (**Fig. 23.21**). Find the electric potential at a point P on the x -axis at a distance x from the origin.

IDENTIFY and SET UP This is the situation of Example 21.10 (Section 21.5), where we found an expression for the electric field $\vec{E}$ at an arbitrary point on the x -axis. We can find V at point P by using Eq. (23.16) to integrate over the charge distribution. Unlike the situation in Example 23.11, each charge element dQ is a *different* distance from point P , so the integration will take a little more effort.

EXECUTE As in Example 21.10, the element of charge dQ corresponding to an element of length dy on the rod is $dQ = (Q/2a)dy$. The distance from dQ to P is $\sqrt{x^2 + y^2}$, so the contribution dV that the charge element makes to the potential at P is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{dy}{\sqrt{x^2 + y^2}}$$

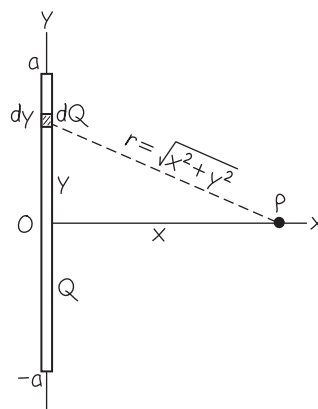
To find the potential at P due to the entire rod, we integrate dV over the length of the rod from $y = -a$ to $y = a$:

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \int_{-a}^a \frac{dy}{\sqrt{x^2 + y^2}}$$

You can look up the integral in a table. The final result is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \ln \left(\frac{\sqrt{a^2 + x^2} + a}{\sqrt{a^2 + x^2} - a} \right)$$

Figure 23.21 Our sketch for this problem.



EVALUATE We can check our result by letting x approach infinity. In this limit the point P is infinitely far from all of the charge, so we expect V to approach zero; you can verify that it does.

We know the electric field at all points along the x -axis from Example 21.10. We invite you to use this information to find V along this axis by integrating $\vec{E}$ as in Eq. (23.17).

KEYCONCEPT The electric potential due to a symmetrical distribution of charge is most easily calculated at a point of symmetry. Whenever possible, take advantage of the symmetry of the situation to check your results.

TEST YOUR UNDERSTANDING OF SECTION 23.3 If the electric *field* at a certain point is zero, does the electric *potential* at that point have to be zero? (*Hint:* Consider the center of the ring in Examples 23.11 and 21.9.)

ANSWER

no If $\vec{E} = 0$ at a certain point, V does *not* have to be zero at that point. An example is point O at the center of the charged ring in Figs. 21.23 and 23.21. From Example 21.9 (Section 21.5), the electric field is zero at O because the electric-field contributions from different parts of the ring completely cancel. From Example 23.11, however, the potential at O is *not* zero. This point corresponds to $x = 0$, so $V = (1/4\pi\epsilon_0)(Q/a)$. This value of V corresponds to the work that would have to be done to move a unit positive test charge along a path from infinity to point O ; it is nonzero because the charged ring repels the test charge, so positive work must be done to move the test charge toward the ring.

23.4 EQUIPOTENTIAL SURFACES

Field lines (see Section 21.6) help us visualize electric fields. In a similar way, the potential at various points in an electric field can be represented graphically by *equipotential surfaces*. These use the same fundamental idea as topographic maps like those used by hikers and mountain climbers (**Fig. 23.22**). On a topographic map, contour lines are drawn through points that are all at the same elevation. Any number of these could be drawn, but typically only a few contour lines are shown at equal spacings of elevation. If a mass m is moved over the terrain along such a contour line, the gravitational potential energy mgy does not change because the elevation y is constant. Thus contour lines on a topographic map are really curves of constant gravitational potential energy. Contour lines are close together where the terrain is steep and there are large changes in elevation over a small horizontal distance; the contour lines are farther apart where the terrain

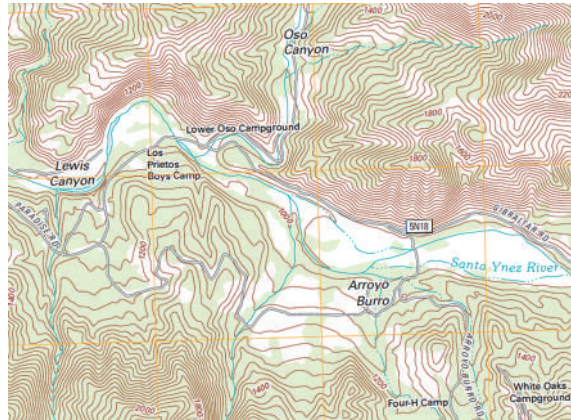


Figure 23.22 Contour lines on a topographic map are curves of constant elevation and hence of constant gravitational potential energy.

is gently sloping. A ball allowed to roll downhill will experience the greatest downhill gravitational force where contour lines are closest together.

By analogy to contour lines on a topographic map, an **equipotential surface** is a three-dimensional surface on which the *electric potential* V is the same at every point. If a test charge q_0 is moved from point to point on such a surface, the *electric potential energy* q_0V remains constant. In a region where an electric field is present, we can construct an equipotential surface through any point. In diagrams we usually show only a few representative equipotentials, often with equal potential differences between adjacent surfaces. No point can be at two different potentials, so equipotential surfaces for different potentials can never touch or intersect.

Equipotential Surfaces and Field Lines

Because potential energy does not change as a test charge moves over an equipotential surface, the electric field can do no work on such a charge. It follows that $\vec{E}$ must be perpendicular to the surface at every point so that the electric force $q_0\vec{E}$ is always perpendicular to the displacement of a charge moving on the surface. **Field lines and equipotential surfaces are always mutually perpendicular.** In general, field lines are curves, and equipotentials are curved surfaces. For the special case of a *uniform* field, in which the field lines are straight, parallel, and equally spaced, the equipotentials are parallel *planes* perpendicular to the field lines.

Figure 23.23 shows three arrangements of charges. The field lines in the plane of the charges are represented by red lines, and the intersections of the equipotential surfaces with this plane (that is, cross sections of these surfaces) are shown as blue lines. The actual equipotential surfaces are three-dimensional. At each crossing of an equipotential and a field line, the two are perpendicular.

Figure 23.23 Cross sections of equipotential surfaces (blue lines) and electric field lines (red lines) for assemblies of point charges. There are equal potential differences between adjacent surfaces. Compare these diagrams to those in Fig. 21.28, which showed only the electric field lines.

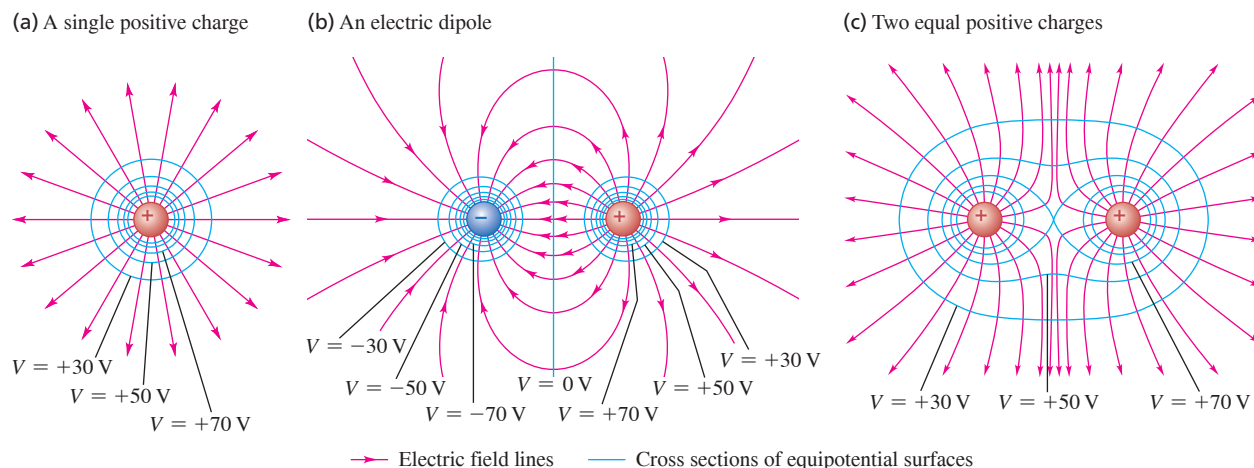


Figure 23.24 When charges are at rest, a conducting surface is always an equipotential surface. Field lines are perpendicular to a conducting surface.

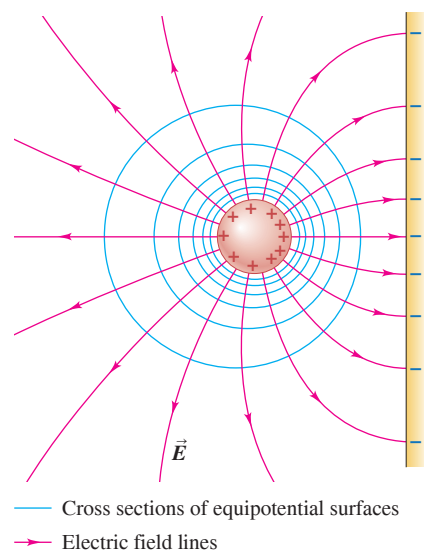


Figure 23.25 At all points on a conductor's surface, the electric field must be perpendicular to the surface. If $\vec{E}$ had a tangential component, a charge could move in a loop as shown—which is impossible because the electric force is conservative.

An impossible electric field

If the electric field just outside a conductor had a tangential component $E_{\parallel}$, a charge could move in a loop with net work done. Hence the field just outside a conductor can have only a perpendicular component.

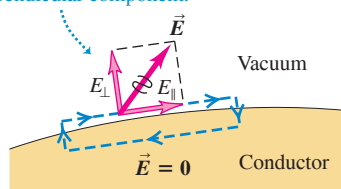
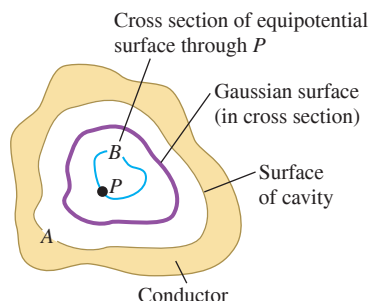


Figure 23.26 A cavity in a conductor. If the cavity contains no charge, every point in the cavity is at the same potential, the electric field is zero everywhere in the cavity, and there is no charge anywhere on the surface of the cavity.



In Fig. 23.23 we have drawn equipotentials so that there are equal potential differences between adjacent surfaces. In regions where the magnitude of $\vec{E}$ is large, the equipotential surfaces are close together because the field does a relatively large amount of work on a test charge in a relatively small displacement. This is the case near the point charge in Fig. 23.23a or between the two point charges in Fig. 23.23b; note that in these regions the field lines are also closer together. This is directly analogous to the downhill force of gravity being greatest in regions on a topographic map where contour lines are close together. Conversely, in regions where the field is weaker, the equipotential surfaces are farther apart; this happens at larger radii in Fig. 23.23a, to the left of the negative charge or the right of the positive charge in Fig. 23.23b, and at greater distances from both charges in Fig. 23.23c. (It may appear that two equipotential surfaces intersect at the center of Fig. 23.23c, in violation of the rule that this can never happen. In fact this is a single figure-8-shaped equipotential surface.)

CAUTION E need not be constant over an equipotential surface On a given equipotential surface, the potential V has the same value at every point. In general, however, the electric-field magnitude E is *not* the same at all points on an equipotential surface. For instance, on equipotential surface “ $V = -30$ V” in Fig. 23.23b, E is less to the left of the negative charge than it is between the two charges. On the figure-8-shaped equipotential surface in Fig. 23.23c, $E = 0$ at the middle point halfway between the two charges; at any other point on this surface, E is nonzero.

Equipotentials and Conductors

Here's an important statement about equipotential surfaces: **When all charges are at rest, the surface of a conductor is always an equipotential surface.** Since the electric field $\vec{E}$ is always perpendicular to an equipotential surface, we can prove this statement by proving that **when all charges are at rest, the electric field just outside a conductor must be perpendicular to the surface at every point** (Fig. 23.24). We know that $\vec{E} = 0$ everywhere inside the conductor; otherwise, charges would move. In particular, at any point just inside the surface the component of $\vec{E}$ tangent to the surface is zero. It follows that the tangential component of $\vec{E}$ is also zero just *outside* the surface. If it were not, a charge could move around a rectangular path partly inside and partly outside (Fig. 23.25) and return to its starting point with a net amount of work having been done on it. This would violate the conservative nature of electrostatic fields, so the tangential component of $\vec{E}$ just outside the surface must be zero at every point on the surface. Thus $\vec{E}$ is perpendicular to the surface at each point, proving our statement.

It also follows that **when all charges are at rest, the entire solid volume of a conductor is at the same potential.** Equation (23.17) states that the potential difference between two points a and b within the conductor's solid volume, $V_a - V_b$, is equal to the line integral $\int_a^b \vec{E} \cdot d\vec{l}$ of the electric field from a to b . Since $\vec{E} = 0$ everywhere inside the conductor, the integral is guaranteed to be zero for any two such points a and b . Hence the potential is the same for any two points within the solid volume of the conductor. We describe this by saying that the solid volume of the conductor is an *equipotential volume*.

We can now prove a theorem that we quoted without proof in Section 22.5. The theorem is as follows: In an electrostatic situation, if a conductor contains a cavity and if no charge is present inside the cavity, then there can be no net charge *anywhere* on the surface of the cavity. This means that if you're inside a charged conducting box, you can safely touch any point on the inside walls of the box without being shocked. To prove this theorem, we first prove that *every point in the cavity is at the same potential*. In Fig. 23.26 the conducting surface A of the cavity is an equipotential surface, as we have just proved. Suppose point P in the cavity is at a different potential; then we can construct a different equipotential surface B including point P .

Now consider a Gaussian surface, shown in Fig. 23.26, between the two equipotential surfaces. Because of the relationship between $\vec{E}$ and the equipotentials, we know that the field at every point between the equipotentials is from A toward B , or else at every point it is from B toward A , depending on which equipotential surface is at higher potential.

In either case the flux through this Gaussian surface is certainly not zero. But then Gauss's law says that the charge enclosed by the Gaussian surface cannot be zero. This contradicts our initial assumption that there is *no* charge in the cavity. So the potential at *P* *cannot* be different from that at the cavity wall.

The entire region of the cavity must therefore be at the same potential. But for this to be true, *the electric field inside the cavity must be zero everywhere*. Finally, Gauss's law shows that the electric field at any point on the surface of a conductor is proportional to the surface charge density σ at that point. We conclude that *the surface charge density on the wall of the cavity is zero at every point*. This chain of reasoning may seem tortuous, but it is worth careful study.

CAUTION Equipotential surfaces vs. Gaussian surfaces Don't confuse equipotential surfaces with the Gaussian surfaces we encountered in Chapter 22. Gaussian surfaces have relevance only when we are using Gauss's law, and we can choose *any* Gaussian surface that's convenient. We *cannot* choose equipotential surfaces; the shape is determined by the charge distribution. ■

TEST YOUR UNDERSTANDING OF SECTION 23.4 Would the shapes of the equipotential surfaces in Fig. 23.23 change if the sign of each charge were reversed?

ANSWER

If the positive charges in Fig. 23.23 were replaced by negative charges, and vice versa, the equipotential surfaces would be the same but the sign of the potential would be reversed. For example, the surfaces in Fig. 23.23b with potential $V = +30$ V and $V = -50$ V would have potential $V = -30$ V and $V = +50$ V, respectively.

23.5 POTENTIAL GRADIENT

Electric field and potential are closely related. Equation (23.17), restated here, expresses one aspect of that relationship:

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l}$$

If we know $\vec{E}$ at various points, we can use this equation to calculate potential differences. In this section we show how to turn this around; if we know the potential V at various points, we can use it to determine $\vec{E}$. Regarding V as a function of the coordinates (x, y, z) of a point in space, we'll show that the components of $\vec{E}$ are related to the *partial derivatives* of V with respect to x , y , and z .

In Eq. (23.17), $V_a - V_b$ is the potential of a with respect to b —that is, the change of potential encountered on a trip from b to a . We can write this as

$$V_a - V_b = \int_b^a dV = - \int_a^b dV$$

where dV is the infinitesimal change of potential accompanying an infinitesimal element $d\vec{l}$ of the path from b to a . Comparing to Eq. (23.17), we have

$$- \int_a^b dV = \int_a^b \vec{E} \cdot d\vec{l}$$

These two integrals must be equal for *any* pair of limits a and b , and for this to be true the *integrands* must be equal. Thus, for *any* infinitesimal displacement $d\vec{l}$,

$$-dV = \vec{E} \cdot d\vec{l}$$

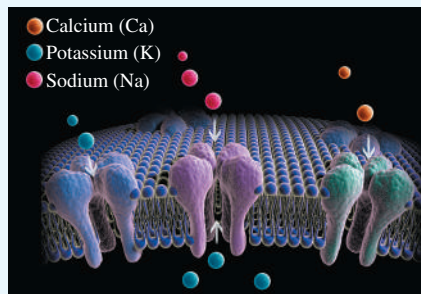
To interpret this expression, we write $\vec{E}$ and $d\vec{l}$ in terms of their components: $\vec{E} = \hat{i} E_x + \hat{j} E_y + \hat{k} E_z$ and $d\vec{l} = \hat{i} dx + \hat{j} dy + \hat{k} dz$. Then

$$-dV = E_x dx + E_y dy + E_z dz$$

Suppose the displacement is parallel to the x -axis, so $dy = dz = 0$. Then $-dV = E_x dx$ or $E_x = -(dV/dx)_{y,z \text{ constant}}$, where the subscript reminds us that only x varies in the

BIO APPLICATION Potential Gradient Across a Cell Membrane

The interior of a human cell is at a lower electric potential V than the exterior. (The potential difference when the cell is inactive is about -70 mV in neurons and about -95 mV in skeletal muscle cells.) Hence there is a potential gradient $\vec{\nabla}V$ that points from the *interior* to the *exterior* of the cell membrane, and an electric field $\vec{E} = -\vec{\nabla}V$ that points from the *exterior* to the *interior*. This field affects how ions flow into or out of the cell through special channels in the membrane.



derivative; recall that V is in general a function of x , y , and z . But this is just what is meant by the partial derivative $\partial V/\partial x$. The y - and z -components of $\vec{E}$ are related to the corresponding derivatives of V in the same way, so

Electric field components found from potential: $E_x = -\frac{\partial V}{\partial x}$ $E_y = -\frac{\partial V}{\partial y}$ $E_z = -\frac{\partial V}{\partial z}$ (23.19)

Each electric field component ... equals the negative of the corresponding partial derivative of electric potential function V .

This is consistent with the units of electric field being V/m. In terms of unit vectors we can write

Electric field vector found from potential: $\vec{E} = -\left(\hat{i}\frac{\partial V}{\partial x} + \hat{j}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right)$ (23.20)

Partial derivatives of electric potential function V

The following operation is called the **gradient** of the function f :

$$\vec{\nabla}f = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)f \quad (23.21)$$

The operator denoted by $\vec{\nabla}$ is called “grad” or “del.” Thus in vector notation,

$$\vec{E} = -\vec{\nabla}V \quad (23.22)$$

This is read “ $\vec{E}$ is the negative of the gradient of V ” or “ $\vec{E}$ equals negative grad V .” The quantity $\vec{\nabla}V$ is called the *potential gradient*.

At each point, the potential gradient $\vec{\nabla}V$ points in the direction in which V *increases* most rapidly with a change in position. Hence at each point the direction of $\vec{E} = -\vec{\nabla}V$ is the direction in which V *decreases* most rapidly and is always perpendicular to the equipotential surface through the point. This agrees with our observation in Section 23.2 that moving in the direction of the electric field means moving in the direction of decreasing potential.

Equation (23.22) doesn’t depend on the particular choice of the zero point for V . If we were to change the zero point, the effect would be to change V at every point by the same amount; the derivatives of V would be the same.

If $\vec{E}$ has a radial component E_r with respect to a point or an axis and r is the distance from the point or axis, the relationship corresponding to Eqs. (23.19) is

$$E_r = -\frac{\partial V}{\partial r} \quad (\text{radial electric field component}) \quad (23.23)$$

Often we can compute the electric field caused by a charge distribution in either of two ways: directly, by adding the $\vec{E}$ fields of point charges, or by first calculating the potential and then taking its gradient to find the field. The second method is often easier because potential is a *scalar* quantity, requiring at worst the integration of a scalar function. Electric field is a *vector* quantity, requiring computation of components for each element of charge and a separate integration for each component. Thus, quite apart from its fundamental significance, potential offers a very useful computational technique in field calculations. In the next two examples, a knowledge of V is used to find the electric field.

We stress once more that if we know $\vec{E}$ as a function of position, we can calculate V from Eq. (23.17) or (23.18), and if we know V as a function of position, we can calculate $\vec{E}$ from Eq. (23.19), (23.20), or (23.23). Deriving V from $\vec{E}$ requires integration, and deriving $\vec{E}$ from V requires differentiation.

EXAMPLE 23.13 Potential and field of a point charge

From Eq. (23.14) the potential at a radial distance r from a point charge q is $V = q/4\pi\epsilon_0 r$. Find the vector electric field from this expression for V .

IDENTIFY and SET UP This problem uses the general relationship between the electric potential as a function of position and the electric-field vector. By symmetry, the electric field here has only a radial component E_r . We use Eq. (23.23) to find this component.

EXECUTE From Eq. (23.23),

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

so the vector electric field is

$$\vec{E} = \hat{r}E_r = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

EVALUATE Our result agrees with Eq. (21.7), as it must.

An alternative approach is to ignore the radial symmetry, write the radial distance as $r = \sqrt{x^2 + y^2 + z^2}$, and take the derivatives of V with respect to x , y , and z as in Eq. (23.20). We find

$$\begin{aligned} \frac{\partial V}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{x^2 + y^2 + z^2}} \right) = -\frac{1}{4\pi\epsilon_0} \frac{qx}{(x^2 + y^2 + z^2)^{3/2}} \\ &= -\frac{qx}{4\pi\epsilon_0 r^3} \end{aligned}$$

and similarly

$$\frac{\partial V}{\partial y} = -\frac{qy}{4\pi\epsilon_0 r^3} \quad \frac{\partial V}{\partial z} = -\frac{qz}{4\pi\epsilon_0 r^3}$$

Then from Eq. (23.20),

$$\begin{aligned} \vec{E} &= - \left[\hat{i} \left(-\frac{qx}{4\pi\epsilon_0 r^3} \right) + \hat{j} \left(-\frac{qy}{4\pi\epsilon_0 r^3} \right) + \hat{k} \left(-\frac{qz}{4\pi\epsilon_0 r^3} \right) \right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \end{aligned}$$

This approach gives us the same answer, but with more effort. Clearly it's best to exploit the symmetry of the charge distribution whenever possible.

KEYCONCEPT If you know the electric potential as a function of position in a region of space, you can find the electric field in that region by calculating the negative gradient of the potential.

EXAMPLE 23.14 Potential and field of a ring of charge

In Example 23.11 (Section 23.3) we found that for a ring of charge with radius a and total charge Q , the potential at a point P on the ring's symmetry axis a distance x from the center (see Fig. 23.20) is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + a^2}}$$

Find the electric field at P .

IDENTIFY and SET UP We are given V as a function of x along the x -axis, and we wish to find the electric field at a point on this axis. From the symmetry of the charge distribution, the electric field along the symmetry (x -) axis of the ring can have only an x -component. We find it by using the first of Eqs. (23.19).

EXECUTE The x -component of the electric field is

$$E_x = -\frac{\partial V}{\partial x} = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}}$$

EVALUATE This agrees with our result in Example 21.9.

CAUTION Don't use expressions where they don't apply In this example, V is not a function of y or z on the ring axis, so $\partial V/\partial y = \partial V/\partial z = 0$ and $E_y = E_z = 0$. But that does not mean that it's true *everywhere*; our expressions for V and E_x are valid *on the ring axis only*. If we had an expression for V valid at *all* points in space, we could use it to find the components of $\vec{E}$ at any point by using Eqs. (23.19). **|**

KEYCONCEPT If you know the electric potential as a function of position only along a line, you can find the component of the electric field parallel to that line by calculating the negative partial derivative of the potential with respect to position on that line. You cannot find the electric field at points off the line in this way.

TEST YOUR UNDERSTANDING OF SECTION 23.5 In a certain region of space the potential is given by $V = A + Bx + Cy^3 + Dxy$, where A , B , C , and D are positive constants. Which of these statements about the electric field $\vec{E}$ in this region of space is correct? (There may be more than one correct answer.) (i) Increasing the value of A will increase the value of $\vec{E}$ at all points; (ii) increasing the value of A will decrease the value of $\vec{E}$ at all points; (iii) $\vec{E}$ has no z -component; (iv) the electric field is zero at the origin ($x = 0$, $y = 0$, $z = 0$).

ANSWER

| (iii) From Eqs. (23.19), the components of the electric field are $E_x = -\partial V/\partial x = -(B + Dy)$, $E_y = -\partial V/\partial y = -(3Cy^2 + Dx)$, and $E_z = -\partial V/\partial z = 0$. The value of A has no effect, which means that we can add a constant to the electric potential at all points without changing $\vec{E}$ or the potential difference between two points. The potential does not depend on z , so the z -component of $\vec{E}$ is zero. Note that at the origin the electric field is not zero because it has a nonzero x -component: $E_x = -B$, $E_y = 0$, $E_z = 0$.

CHAPTER 23 SUMMARY

Electric potential energy: The electric force caused by any collection of charges at rest is a conservative force. The work W done by the electric force on a charged particle moving in an electric field can be represented by the change in a potential-energy function U .

The electric potential energy for two point charges q and q_0 depends on their separation r . The electric potential energy for a charge q_0 in the presence of a collection of charges q_1, q_2, q_3 depends on the distance from q_0 to each of these other charges. (See Examples 23.1 and 23.2.)

$$W_{a \rightarrow b} = U_a - U_b$$

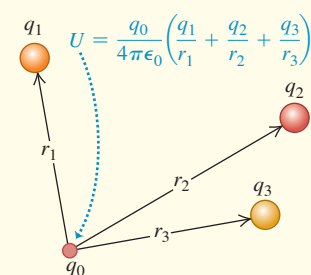
$$U = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$$

(two point charges)

$$U = \frac{q_0}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \cdots \right)$$

$$= \frac{q_0}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

(q_0 in presence of other point charges)

$$(23.2) \quad U = \frac{q_0}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right)$$


Electric potential: Potential, denoted by V , is potential energy per unit charge. The potential difference between two points equals the amount of work per charge that would be required to move a positive test charge between those points. The potential V due to a quantity of charge can be calculated by summing (if the charge is a collection of point charges) or by integrating (if the charge is a distribution). (See Examples 23.3, 23.4, 23.5, 23.7, 23.11, and 23.12.)

The potential difference between two points a and b , also called the potential of a with respect to b , is given by the line integral of $\vec{E}$. The potential at a given point can be found by first finding $\vec{E}$ and then carrying out this integral. (See Examples 23.6, 23.8, 23.9, and 23.10.)

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

(due to a point charge)

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

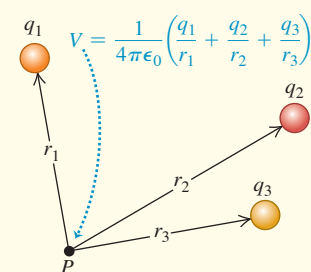
(due to a collection of point charges)

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

(due to a charge distribution)

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l}$$

$$= \int_a^b E \cos \phi \, dl$$

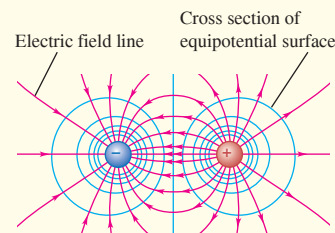
$$(23.14) \quad V = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right)$$


$$(23.15)$$

$$(23.16)$$

$$(23.17)$$

Equipotential surfaces: An equipotential surface is a surface on which the potential has the same value at every point. At a point where a field line crosses an equipotential surface, the two are perpendicular. When all charges are at rest, the surface of a conductor is always an equipotential surface and all points in the interior of a conductor are at the same potential. When a cavity within a conductor contains no charge, the entire cavity is an equipotential region and there is no surface charge anywhere on the surface of the cavity.



Finding electric field from electric potential: If the potential V is known as a function of the coordinates x , y , and z , the components of electric field $\vec{E}$ at any point are given by partial derivatives of V . (See Examples 23.13 and 23.14.)

$$E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z} \quad (23.19)$$

$$\vec{E} = -\left(\hat{i} \frac{\partial V}{\partial x} + \hat{j} \frac{\partial V}{\partial y} + \hat{k} \frac{\partial V}{\partial z} \right) \quad (23.20)$$

(vector form)



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 23.1 and 23.2** (Section 23.1) before attempting these problems.

VP23.2.1 An electron (charge $-e = -1.60 \times 10^{-19}$ C) is at rest at a distance 1.00×10^{-10} m from the center of a nucleus of lead (charge $+82e = +1.31 \times 10^{-17}$ C). You may treat both objects as point charges. (a) How much work would you have to do to move the electron to a distance 5.00×10^{-10} m from the center of the lead nucleus? (b) If you release the electron from the position in (a), what will be its kinetic energy when it is a distance 8.00×10^{-12} m from the center of the lead nucleus? (You can assume that the lead nucleus remains at rest, since it is much more massive than the electron.)

VP23.2.2 In a nuclear physics experiment, a proton (mass 1.67×10^{-27} kg, charge $+e = +1.60 \times 10^{-19}$ C) is fired directly at a target nucleus of unknown charge. (You can treat both objects as point charges, and assume that the nucleus remains at rest.) When it is far from its target, the proton has speed 2.50×10^6 m/s. The proton comes momentarily to rest at a distance 5.29×10^{-13} m from the center of the target nucleus, then flies back in the direction from which it came. (a) What is the electric potential energy of the proton and nucleus when they are 5.29×10^{-13} m apart? (b) What is the charge of the target nucleus?

VP23.2.3 Two point charges are at fixed positions on the y -axis: $q_1 = +e$ at $y = 0$ and $q_2 = -e$ at $y = a$. Find (a) the work you must do to bring a third charge $q_3 = -e$ from infinity to being at rest at $y = -2a$ and (b) the total potential energy of the system of three charges.

VP23.2.4 A point charge $q_1 = +5.00$ nC is at the fixed position $x = 0$, $y = 0$, $z = 0$. You find that you must do 8.10×10^{-6} J of work to bring a second point charge from infinity to the position $x = +4.00$ cm, $y = 0$, $z = 0$. (a) What is the value of the second charge? (b) Once the second charge is in place, how much additional work would you have to do to bring a third point charge $q_3 = +2.00$ nC from infinity to the position $x = 0$, $y = +3.00$ cm, $z = 0$?

Be sure to review **EXAMPLES 23.3, 23.4, 23.5, and 23.7** (Section 23.2) before attempting these problems.

VP23.7.1 In a certain region of space the electric field is uniform and given by $\vec{E} = (5.00 \times 10^2 \text{ V/m})\hat{i}$. If the electric potential at the point $x = 0$, $y = 0$, $z = 0$ is equal to V_0 , find the potential difference $V_0 - V_P$ for each of the following points P : (a) $x = +5.00$ cm, $y = 0$, $z = 0$; (b) $x = +3.00$ cm, $y = +4.00$ cm, $z = 0$; (c) $x = 0$, $y = +5.00$ cm, $z = 0$; (d) $x = -5.00$ cm, $y = 0$, $z = 0$.

VP23.7.2 You have two identical point charges of $+6.0$ nC each, one at the position $x = 3.0$ cm, $y = 0$, $z = 0$ and the other at $x = -3.0$ cm, $y = 0$, $z = 0$. Find the electric potential due to these charges at (a) $x = 0$, $y = 0$, $z = 0$; (b) $x = 0$, $y = 4.0$ cm, $z = 0$; (c) $x = 8.0$ cm, $y = 0$, $z = 0$. Take $V = 0$ at infinity. (d) At which (if any) of these points is the electric field equal to zero? If the field is zero at a certain point, is the potential necessarily zero there as well?

VP23.7.3 One point charge of $+5.90$ nC is at the position $x = 3.00$ cm, $y = 0$, $z = 0$, and one point charge of -5.90 nC is at $x = -3.00$ cm, $y = 0$, $z = 0$. Find the electric potential due to these charges at (a) $x = 3.00$ cm, $y = 8.00$ cm, $z = 0$; (b) $x = -3.00$ cm, $y = 8.00$ cm, $z = 0$. Take $V = 0$ at infinity. (c) A proton (mass 1.67×10^{-27} kg, charge $+e = +1.60 \times 10^{-19}$ C) follows a path from point (b) to point (a). If its speed at (b) is 3.00×10^5 m/s, what is its speed at (a)?

VP23.7.4 In Example 22.9 (Section 22.4) we found that the electric field *inside* a uniformly charged sphere with positive charge Q and radius R points radially outward from the center of the sphere and has magnitude $E = Qr/4\pi\epsilon_0 R^3$ at a distance r from the center. (a) By integrating the electric field, find the potential difference between the center of the sphere and its surface. (b) Which is at higher potential: the center or the surface?

Be sure to review **EXAMPLES 23.8, 23.9, 23.10, 23.11, and 23.12** (Section 23.3) before attempting these problems.

VP23.12.1 A conducting sphere of radius R carries positive charge q . Calculate the amount of work that would be required to move a small positive test charge q_0 slowly (a) from $r = 5R$ to $r = 3R$; (b) from $r = 2R$ to $r = 7R$; (c) from $r = 4R$ to $r = R/2$. In each case assume that the presence of q_0 has no effect on how the charge q is distributed over the sphere.

VP23.12.2 For the two oppositely charged parallel plates in Fig. 23.18, the potential difference between the plates is $V_{ab} = 24.0$ V, $V = 0$ at the bottom plate, and the distance between the plates is $d = 4.50$ mm. (a) What is the electric potential at a point 3.00 mm above the lower plate? (b) What is the potential energy for a dust particle of mass 5.00×10^{-9} kg and charge $+2.00$ nC at the position in part (a)? (c) If the particle in part (b) is released from rest, will it move toward the upper plate or the lower plate? What will be its speed when it hits that plate?

VP23.12.3 The uniformly charged ring shown in Fig. 23.20 has charge $+5.00$ nC and radius 2.50 cm. A point charge $q = +3.00$ nC of mass 4.00×10^{-9} kg is on the ring axis 8.00 cm from the center of the ring and is moving toward the center at 60.0 m/s. Take $V = 0$ at infinity. What is the potential energy of the point charge (a) at 8.00 cm from the ring's center and (b) at the ring's center? (c) What are the kinetic energy and speed of the point charge when it reaches the ring's center?

VP23.12.4 Positive charge Q is distributed uniformly along a rod of length L that lies along the x -axis from $x = L$ to $x = 2L$. (a) How much charge is contained within a segment of the rod of length dx ? (b) Integrate to find the total electric potential at the origin ($x = 0$) due to the rod.

BRIDGING PROBLEM A Point Charge and a Line of Charge

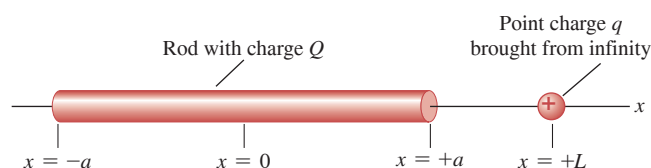
Positive electric charge Q is distributed uniformly along a thin rod of length $2a$. The rod lies along the x -axis between $x = -a$ and $x = +a$ (Fig. 23.27). Calculate how much work you must do to bring a positive point charge q from infinity to the point $x = +L$ on the x -axis, where $L > a$.

SOLUTION GUIDE

IDENTIFY and SET UP

1. In this problem you must first calculate the potential V at $x = +L$ due to the charged rod. You can then find the change in potential energy involved in bringing the point charge q from infinity (where $V = 0$) to $x = +L$.
2. To find V , divide the rod into infinitesimal segments of length dx' . How much charge is on such a segment? Consider one such

Figure 23.27 How much work must you do to bring point charge q in from infinity?



segment located at $x = x'$, where $-a \leq x' \leq a$. What is the potential dV at $x = +L$ due to this segment?

3. The total potential at $x = +L$ is the integral of dV , including contributions from all of the segments for x' from $-a$ to $+a$. Set up this integral.

EXECUTE

4. Integrate your expression from step 3 to find the potential V at $x = +L$. A simple, standard substitution will do the trick; use a table of integrals only as a last resort.
5. Use your result from step 4 to find the potential energy for a point charge q placed at $x = +L$.
6. Use your result from step 5 to find the work you must do to bring the point charge from infinity to $x = +L$.

EVALUATE

7. What does your result from step 5 become in the limit $a \rightarrow 0$? Does this make sense?
8. Suppose the point charge q were negative rather than positive. How would this affect your result in step 4? In step 5?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q23.1 A student asked, “Since electrical potential is always proportional to potential energy, why bother with the concept of potential at all?” How would you respond?

Q23.2 The potential (relative to a point at infinity) midway between two charges of equal magnitude and opposite sign is zero. Is it possible to bring a test charge from infinity to this midpoint in such a way that no work is done in any part of the displacement? If so, describe how it can be done. If it is not possible, explain why.

Q23.3 Is it possible to have an arrangement of two point charges separated by a finite distance such that the electric potential energy of the arrangement is the same as if the two charges were infinitely far apart? Why or why not? What if there are three charges? Explain.

Q23.4 Since potential can have any value you want depending on the choice of the reference level of zero potential, how does a voltmeter know what to read when you connect it between two points?

Q23.5 If $\vec{E}$ is zero everywhere along a certain path that leads from point A to point B, what is the potential difference between those two points? Does this mean that $\vec{E}$ is zero everywhere along *any* path from A to B? Explain.

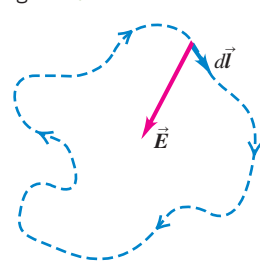
Q23.6 If $\vec{E}$ is zero throughout a certain region of space, is the potential necessarily also zero in this region? Why or why not? If not, what *can* be said about the potential?

Q23.7 Which way do electric field lines point, from high to low potential or from low to high? Explain.

Q23.8 (a) Take $V = 0$ at infinity. If the potential is zero at a point, is the electric field necessarily zero at that point? (b) If the electric field is zero at a point, is the potential necessarily zero there? Prove your answers, using simple examples.

Q23.9 If you carry out the integral of the electric field $\int \vec{E} \cdot d\vec{l}$ for a *closed* path like that shown in Fig. Q23.9, the integral will *always* be equal to zero, independent of the shape of the path and independent of where charges may be located relative to the path. Explain why.

Figure Q23.9



Q23.10 The potential difference between the two terminals of an AA battery (used in flashlights and portable stereos) is 1.5 V. If two AA batteries are placed end to end with the positive terminal of one battery touching the negative terminal of the other, what is the potential difference between the terminals at the exposed ends of the combination? What if the two positive terminals are touching each other? Explain your reasoning.

Q23.11 It is easy to produce a potential difference of several thousand volts between your body and the floor by scuffing your shoes across a nylon carpet. When you touch a metal doorknob, you get a mild shock. Yet contact with a power line of comparable voltage would probably be fatal. Why is there a difference?

Q23.12 If the electric potential at a single point is known, can $\vec{E}$ at that point be determined? If so, how? If not, why not?

Q23.13 Because electric field lines and equipotential surfaces are always perpendicular, two equipotential surfaces can never cross; if they did, the direction of $\vec{E}$ would be ambiguous at the crossing points. Yet two equipotential surfaces appear to cross at the center of Fig. 23.23c. Explain why there is no ambiguity about the direction of $\vec{E}$ in this particular case.

Q23.14 A uniform electric field is directed due east. Point B is 2.00 m west of point A , point C is 2.00 m east of point A , and point D is 2.00 m south of A . For each point, B , C , and D , is the potential at that point larger, smaller, or the same as at point A ? Give the reasoning behind your answers.

Q23.15 We often say that if point A is at a higher potential than point B , A is at positive potential and B is at negative potential. Does it necessarily follow that a point at positive potential is positively charged, or that a point at negative potential is negatively charged? Illustrate your answers with clear, simple examples.

Q23.16 A conducting sphere is to be charged by bringing in positive charge a little at a time until the total charge is Q . The total work required for this process is alleged to be proportional to Q^2 . Is this correct? Why or why not?

Q23.17 In electronics it is customary to define the potential of ground (thinking of the earth as a large conductor) as zero. Is this consistent with the fact that the earth has a net electric charge that is not zero?

Q23.18 A conducting sphere is placed between two charged parallel plates such as those shown in Fig. 23.2. Does the electric field inside the sphere depend on precisely where between the plates the sphere is placed? What about the electric potential inside the sphere? Do the answers to these questions depend on whether or not there is a net charge on the sphere? Explain your reasoning.

Q23.19 A conductor that carries a net charge Q has a hollow, empty cavity in its interior. Does the potential vary from point to point within the material of the conductor? What about within the cavity? How does the potential inside the cavity compare to the potential within the material of the conductor?

Q23.20 A high-voltage dc power line falls on a car, so the entire metal body of the car is at a potential of 10,000 V with respect to the ground. What happens to the occupants (a) when they are sitting in the car and (b) when they step out of the car? Explain your reasoning.

Q23.21 When a thunderstorm is approaching, sailors at sea sometimes observe a phenomenon called “St. Elmo’s fire,” a bluish flickering light at the tips of masts. What causes this? Why does it occur at the tips of masts? Why is the effect most pronounced when the masts are wet? (*Hint:* Seawater is a good conductor of electricity.)

Q23.22 A positive point charge is placed near a very large conducting plane. A professor of physics asserted that the field caused by this configuration is the same as would be obtained by removing the plane and placing a negative point charge of equal magnitude in the mirror-image position behind the initial position of the plane. Is this correct? Why or why not? (*Hint:* Inspect Fig. 23.23b.)

EXERCISES

Section 23.1 Electric Potential Energy

23.1 • A point charge $q_1 = +2.40 \mu\text{C}$ is held stationary at the origin. A second point charge $q_2 = -4.30 \mu\text{C}$ moves from the point $x = 0.150 \text{ m}$, $y = 0$ to the point $x = 0.250 \text{ m}$, $y = 0.250 \text{ m}$. How much work is done by the electric force on q_2 ?

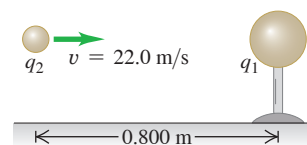
23.2 • A point charge q_1 is held stationary at the origin. A second charge q_2 is placed at point a , and the electric potential energy of the pair of charges is $+5.4 \times 10^{-8} \text{ J}$. When the second charge is moved to point b , the electric force on the charge does $-1.9 \times 10^{-8} \text{ J}$ of work. What is the electric potential energy of the pair of charges when the second charge is at point b ?

23.3 • • **Energy of the Nucleus.** How much work is needed to assemble an atomic nucleus containing three protons (such as Li) if we model it as an equilateral triangle of side $2.00 \times 10^{-15} \text{ m}$ with a proton at each vertex? Assume the protons started from very far away.

23.4 • • (a) How much work would it take to push two protons very slowly from a separation of $2.00 \times 10^{-10} \text{ m}$ (a typical atomic distance) to $3.00 \times 10^{-15} \text{ m}$ (a typical nuclear distance)? (b) If the protons are both released from rest at the closer distance in part (a), how fast are they moving when they reach their original separation?

23.5 • • A small metal sphere, carrying a net charge of $q_1 = -2.80 \mu\text{C}$, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -7.80 \mu\text{C}$ and mass 1.50 g, is projected toward q_1 . When the two spheres are 0.800 m apart, q_2 is moving toward q_1 with speed 22.0 m/s (**Fig. E23.5**). Assume that the two spheres can be treated as point charges. You can ignore the force of gravity. (a) What is the speed of q_2 when the spheres are 0.400 m apart? (b) How close does q_2 get to q_1 ?

Figure E23.5



23.6 • • **BIO Energy of DNA Base Pairing.** (See Exercise 21.18.)

(a) Calculate the electric potential energy of the adenine–thymine bond, using the same combinations of molecules ($\text{O}-\text{H}-\text{N}$ and $\text{N}-\text{H}-\text{N}$) as in Exercise 21.18. (b) Compare this energy with the potential energy of the proton–electron pair in the hydrogen atom.

23.7 • • Two protons, starting several meters apart, are aimed directly at each other with speeds of $2.00 \times 10^5 \text{ m/s}$, measured relative to the earth. Find the maximum electric force that these protons will exert on each other.

23.8 • • Three equal $1.20 \mu\text{C}$ point charges are placed at the corners of an equilateral triangle with sides 0.400 m long. What is the potential energy of the system? (Take as zero the potential energy of the three charges when they are infinitely far apart.)

23.9 • • Two protons are released from rest when they are 0.750 nm apart. (a) What is the maximum speed they will reach? When does this speed occur? (b) What is the maximum acceleration they will achieve? When does this acceleration occur?

23.10 • • Four electrons are located at the corners of a square 10.0 nm on a side, with an alpha particle at its midpoint. How much work is needed to move the alpha particle to the midpoint of one of the sides of the square?

Section 23.2 Electric Potential

23.11 • • **CALC** Points a and b lie in a region where the y -component of the electric field is $E_y = \alpha + \beta/y^2$. The constants in this expression have the values $\alpha = 600 \text{ N/C}$ and $\beta = 5.00 \text{ N} \cdot \text{m}^2/\text{C}$. Points a and b are on the $+y$ -axis. Point a is at $y = 2.00 \text{ cm}$ and point b is at $y = 3.00 \text{ cm}$. What is the potential difference $V_a - V_b$ between these two points and which point, a or b , is at higher potential?

23.12 • An object with charge $q = -6.00 \times 10^{-9} \text{ C}$ is placed in a region of uniform electric field and is released from rest at point A . After the charge has moved to point B , 0.500 m to the right, it has kinetic energy $3.00 \times 10^{-7} \text{ J}$. (a) If the electric potential at point A is $+30.0 \text{ V}$, what is the electric potential at point B ? (b) What are the magnitude and direction of the electric field?

23.13 • A small particle has charge $-5.00 \mu\text{C}$ and mass $2.00 \times 10^{-4} \text{ kg}$. It moves from point A , where the electric potential is $V_A = +200 \text{ V}$, to point B , where the electric potential is $V_B = +800 \text{ V}$. The electric force is the only force acting on the particle. The particle has speed 5.00 m/s at point A . What is its speed at point B ? Is it moving faster or slower at B than at A ? Explain.

23.14 • A particle with charge $+4.20 \text{ nC}$ is in a uniform electric field $\vec{E}$ directed to the left. The charge is released from rest and moves to the left; after it has moved 6.00 cm, its kinetic energy is $+2.20 \times 10^{-6} \text{ J}$. What are (a) the work done by the electric force, (b) the potential of the starting point with respect to the end point, and (c) the magnitude of $\vec{E}$?

23.15 • A charge of 28.0 nC is placed in a uniform electric field that is directed vertically upward and has a magnitude of $4.00 \times 10^4 \text{ V/m}$. What work is done by the electric force when the charge moves (a) 0.450 m to the right; (b) 0.670 m upward; (c) 2.60 m at an angle of 45.0° downward from the horizontal?

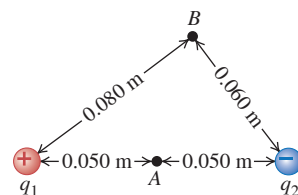
23.16 • Two stationary point charges $+3.00 \text{ nC}$ and $+2.00 \text{ nC}$ are separated by a distance of 50.0 cm . An electron is released from rest at a point midway between the two charges and moves along the line connecting the two charges. What is the speed of the electron when it is 10.0 cm from the $+3.00 \text{ nC}$ charge?

23.17 •• Point charges $q_1 = +2.00 \mu\text{C}$ and $q_2 = -2.00 \mu\text{C}$ are placed at adjacent corners of a square for which the length of each side is 3.00 cm . Point a is at the center of the square, and point b is at the empty corner closest to q_2 . Take the electric potential to be zero at a distance far from both charges. (a) What is the electric potential at point a due to q_1 and q_2 ? (b) What is the electric potential at point b ? (c) A point charge $q_3 = -5.00 \mu\text{C}$ moves from point a to point b . How much work is done on q_3 by the electric forces exerted by q_1 and q_2 ? Is this work positive or negative?

23.18 • Two point charges of equal magnitude Q are held a distance d apart. Consider only points on the line passing through both charges; take $V = 0$ at infinity. (a) If the two charges have the same sign, find the location of all points (if there are any) at which (i) the potential is zero (is the electric field zero at these points?), and (ii) the electric field is zero (is the potential zero at these points?). (b) Repeat part (a) for two point charges having opposite signs.

23.19 • Two point charges $q_1 = +2.40 \text{ nC}$ and $q_2 = -6.50 \text{ nC}$ are

Figure E23.19



0.100 m apart. Point A is midway between them; point B is 0.080 m from q_1 and 0.060 m from q_2 (Fig. E23.19). Take the electric potential to be zero at infinity. Find (a) the potential at point A ; (b) the potential at point B ; (c) the work done by the electric field on a charge

of 2.50 nC that travels from point B to point A .

23.20 •• (a) An electron is to be accelerated from $3.00 \times 10^6 \text{ m/s}$ to $8.00 \times 10^6 \text{ m/s}$. Through what potential difference must the electron pass to accomplish this? (b) Through what potential difference must the electron pass if it is to be slowed from $8.00 \times 10^6 \text{ m/s}$ to a halt?

23.21 •• Points A and B lie within a region of space where there is a uniform electric field that has no x - or z -component; only the y -component E_y is nonzero. Point A is at $y = 8.00 \text{ cm}$ and point B is at $y = 15.0 \text{ cm}$. The potential difference between B and A is $V_B - V_A = +12.0 \text{ V}$, so point B is at higher potential than point A . (a) Is E_y positive or negative? (b) What is the magnitude of the electric field? (c) Point C has coordinates $x = 5.00 \text{ cm}$, $y = 5.00 \text{ cm}$. What is the potential difference between points B and C ?

23.22 • At a certain distance from a point charge, the potential and electric-field magnitude due to that charge are 4.98 V and 16.2 V/m , respectively. (Take $V = 0$ at infinity.) (a) What is the distance to the point charge? (b) What is the magnitude of the charge? (c) Is the electric field directed toward or away from the point charge?

23.23 • A uniform electric field has magnitude E and is directed in the negative x -direction. The potential difference between point a (at $x = 0.60 \text{ m}$) and point b (at $x = 0.90 \text{ m}$) is 240 V . (a) Which point, a or b , is at the higher potential? (b) Calculate the value of E . (c) A negative point charge $q = -0.200 \mu\text{C}$ is moved from b to a . Calculate the work done on the point charge by the electric field.

23.24 •• A small sphere with charge $q = -5.00 \mu\text{C}$ is moving in a uniform electric field that has no y - or z -component. The only force on the sphere is the force exerted by the electric field. Point A is on the x -axis at $x = -0.400 \text{ m}$, and point B is at the origin. At point A the sphere has kinetic energy $K_A = 8.00 \times 10^{-4} \text{ J}$, and at point B its kinetic energy is $K_B = 3.00 \times 10^{-4} \text{ J}$. (a) What is the potential difference $V_{AB} = V_A - V_B$? Which point, A or B , is at higher potential? (b) What are the magnitude and direction of the electric field?

23.25 •• Identical point charges q_1 and q_2 each have positive charge $+6.00 \mu\text{C}$. Charge q_1 is held fixed on the x -axis at $x = +0.400 \text{ m}$, and q_2 is held fixed on the x -axis at $x = -0.400 \text{ m}$. A small sphere has charge $Q = -0.200 \mu\text{C}$ and mass 12.0 g . The sphere is initially very far from the origin. It is released from rest and moves along the y -axis toward the origin. (a) As the sphere moves from very large y to $y = 0$, how much work is done on it by the resultant force exerted by q_1 and q_2 ? (b) If the only force acting on the sphere is the force exerted by the point charges, what is its speed when it reaches the origin?

Section 23.3 Calculating Electric Potential

23.26 •• A solid conducting sphere of radius 5.00 cm carries a net charge. To find the value of the charge, you measure the potential difference $V_{AB} = V_A - V_B$ between point A , which is 8.00 cm from the center of the sphere, and point B , which is a distance r from the center of the sphere. You repeat these measurements for several values of $r > 8.00 \text{ cm}$. When you plot your data as V_{AB} versus $1/r$, the values lie close to a straight line with slope $-18.0 \text{ V} \cdot \text{m}$. What does your data give for the net charge on the sphere? Is the net charge positive or negative?

23.27 •• A thin spherical shell with radius $R_1 = 3.00 \text{ cm}$ is concentric with a larger thin spherical shell with radius $R_2 = 5.00 \text{ cm}$. Both shells are made of insulating material. The smaller shell has charge $q_1 = +6.00 \text{ nC}$ distributed uniformly over its surface, and the larger shell has charge $q_2 = -9.00 \text{ nC}$ distributed uniformly over its surface. Take the electric potential to be zero at an infinite distance from both shells. (a) What is the electric potential due to the two shells at the following distance from their common center: (i) $r = 0$; (ii) $r = 4.00 \text{ cm}$; (iii) $r = 6.00 \text{ cm}$? (b) What is the magnitude of the potential difference between the surfaces of the two shells? Which shell is at higher potential: the inner shell or the outer shell?

23.28 • A total electric charge of 3.50 nC is distributed uniformly over the surface of a metal sphere with a radius of 24.0 cm . If the potential is zero at a point at infinity, find the value of the potential at the following distances from the center of the sphere: (a) 48.0 cm ; (b) 24.0 cm ; (c) 12.0 cm .

23.29 •• A uniformly charged, thin ring has radius 15.0 cm and total charge $+24.0 \text{ nC}$. An electron is placed on the ring's axis a distance 30.0 cm from the center of the ring and is constrained to stay on the axis of the ring. The electron is then released from rest. (a) Describe the subsequent motion of the electron. (b) Find the speed of the electron when it reaches the center of the ring.

23.30 • A solid conducting sphere has net positive charge and radius $R = 0.400 \text{ m}$. At a point 1.20 m from the center of the sphere, the electric potential due to the charge on the sphere is 24.0 V . Assume that $V = 0$ at an infinite distance from the sphere. What is the electric potential at the center of the sphere?

23.31 • Charge $Q = 5.00 \mu\text{C}$ is distributed uniformly over the volume of an insulating sphere that has radius $R = 12.0 \text{ cm}$. A small sphere with charge $q = +3.00 \mu\text{C}$ and mass $6.00 \times 10^{-5} \text{ kg}$ is projected toward the center of the large sphere from an initial large distance. The large sphere is held at a fixed position and the small sphere can be treated as a point charge. What minimum speed must the small sphere have in order to come within 8.00 cm of the surface of the large sphere?

23.32 •• An infinitely long line of charge has linear charge density $5.00 \times 10^{-12} \text{ C/m}$. A proton (mass $1.67 \times 10^{-27} \text{ kg}$, charge $+1.60 \times 10^{-19} \text{ C}$) is 18.0 cm from the line and moving directly toward the line at $3.50 \times 10^3 \text{ m/s}$. (a) Calculate the proton's initial kinetic energy. (b) How close does the proton get to the line of charge?

23.33 •• A very long insulating cylindrical shell of radius 6.00 cm carries charge of linear density $8.50 \mu\text{C/m}$ spread uniformly over its outer surface. What would a voltmeter read if it were connected between (a) the surface of the cylinder and a point 4.00 cm above the surface, and (b) the surface and a point 1.00 cm from the central axis of the cylinder?

23.34 •• A very long insulating cylinder of charge of radius 2.50 cm carries a uniform linear density of 15.0 nC/m . If you put one probe of a voltmeter at the surface, how far from the surface must the other probe be placed so that the voltmeter reads 175 V ?

23.35 •• A very small sphere with positive charge $q = +8.00 \mu\text{C}$ is released from rest at a point 1.50 cm from a very long line of uniform linear charge density $\lambda = +3.00 \mu\text{C/m}$. What is the kinetic energy of the sphere when it is 4.50 cm from the line of charge if the only force on it is the force exerted by the line of charge?

23.36 • CP Two large, parallel conducting plates carrying opposite charges of equal magnitude are separated by 2.20 cm . (a) If the surface charge density for each plate has magnitude 47.0 nC/m^2 , what is the magnitude of $\vec{E}$ in the region between the plates? (b) What is the potential difference between the two plates? (c) If the separation between the plates is doubled while the surface charge density is kept constant at the value in part (a), what happens to the magnitude of the electric field and to the potential difference?

23.37 • Two large, parallel, metal plates carry opposite charges of equal magnitude. They are separated by 45.0 mm , and the potential difference between them is 360 V . (a) What is the magnitude of the electric field (assumed to be uniform) in the region between the plates? (b) What is the magnitude of the force this field exerts on a particle with charge $+2.40 \text{ nC}$? (c) Use the results of part (b) to compute the work done by the field on the particle as it moves from the higher-potential plate to the lower. (d) Compare the result of part (c) to the change of potential energy of the same charge, computed from the electric potential.

23.38 • BIO Electrical Sensitivity of Sharks. Certain sharks can detect an electric field as weak as $1.0 \mu\text{V/m}$. To grasp how weak this field is, if you wanted to produce it between two parallel metal plates by connecting an ordinary 1.5 V AA battery across these plates, how far apart would the plates have to be?

23.39 • The electric field at the surface of a charged, solid, copper sphere with radius 0.200 m is 3800 N/C , directed toward the center of the sphere. What is the potential at the center of the sphere, if we take the potential to be zero infinitely far from the sphere?

23.40 •• (a) How much excess charge must be placed on a copper sphere 25.0 cm in diameter so that the potential of its center is 3.75 kV ? Take the point where $V = 0$ to be infinitely far from the sphere. (b) What is the potential of the sphere's surface?

Section 23.4 Equipotential Surfaces and

Section 23.5 Potential Gradient

23.41 •• CALC A metal sphere with radius r_a is supported on an insulating stand at the center of a hollow, metal, spherical shell with radius r_b . There is charge $+q$ on the inner sphere and charge $-q$ on the outer spherical shell. (a) Calculate the potential $V(r)$ for (i) $r < r_a$; (ii) $r_a < r < r_b$; (iii) $r > r_b$. (Hint: The net potential is the sum of the potentials due to the individual spheres.) Take V to be zero when r is infinite. (b) Show that the potential of the inner sphere with respect to the outer is

$$V_{ab} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right)$$

(c) Use Eq. (23.23) and the result from part (a) to show that the electric field at any point between the spheres has magnitude

$$E(r) = \frac{V_{ab}}{(1/r_a - 1/r_b) r^2}$$

(d) Use Eq. (23.23) and the result from part (a) to find the electric field at a point outside the larger sphere at a distance r from the center, where $r > r_b$. (e) Suppose the charge on the outer sphere is not $-q$ but a negative charge of different magnitude, say $-Q$. Show that the answers for parts (b) and (c) are the same as before but the answer for part (d) is different.

23.42 • A very large plastic sheet carries a uniform charge density of -6.00 nC/m^2 on one face. (a) As you move away from the sheet along a line perpendicular to it, does the potential increase or decrease? How do you know, without doing any calculations? Does your answer depend on where you choose the reference point for potential? (b) Find the spacing between equipotential surfaces that differ from each other by 1.00 V . What type of surfaces are these?

23.43 • CALC In a certain region of space, the electric potential is $V(x, y, z) = Axy - Bx^2 + Cy$, where A , B , and C are positive constants. (a) Calculate the x -, y -, and z -components of the electric field. (b) At which points is the electric field equal to zero?

23.44 • CALC In a certain region of space the electric potential is given by $V = +Ax^2y - Bxy^2$, where $A = 5.00 \text{ V/m}^3$ and $B = 8.00 \text{ V/m}^3$. Calculate the magnitude and direction of the electric field at the point in the region that has coordinates $x = 2.00 \text{ m}$, $y = 0.400 \text{ m}$, and $z = 0$.

23.45 • A metal sphere with radius $r_a = 1.20 \text{ cm}$ is supported on an insulating stand at the center of a hollow, metal, spherical shell with radius $r_b = 9.60 \text{ cm}$. Charge $+q$ is put on the inner sphere and charge $-q$ on the outer spherical shell. The magnitude of q is chosen to make the potential difference between the spheres 500 V , with the inner sphere at higher potential. (a) Use the result of Exercise 23.41(b) to calculate q . (b) With the help of the result of Exercise 23.41(a), sketch the equipotential surfaces that correspond to 500 , 400 , 300 , 200 , 100 , and 0 V . (c) In your sketch, show the electric field lines. Are the electric field lines and equipotential surfaces mutually perpendicular? Are the equipotential surfaces closer together when the magnitude of $\vec{E}$ is largest?

PROBLEMS

23.46 • CP A point charge $q_1 = +5.00 \mu\text{C}$ is held fixed in space. From a horizontal distance of 6.00 cm , a small sphere with mass $4.00 \times 10^{-3} \text{ kg}$ and charge $q_2 = +2.00 \mu\text{C}$ is fired toward the fixed charge with an initial speed of 40.0 m/s . Gravity can be neglected. What is the acceleration of the sphere at the instant when its speed is 25.0 m/s ?

23.47 ••• A point charge $q_1 = 4.00 \text{ nC}$ is placed at the origin, and a second point charge $q_2 = -3.00 \text{ nC}$ is placed on the x -axis at $x = +20.0 \text{ cm}$. A third point charge $q_3 = 2.00 \text{ nC}$ is to be placed on the x -axis between q_1 and q_2 . (Take as zero the potential energy of the three charges when they are infinitely far apart.) (a) What is the potential energy of the system of the three charges if q_3 is placed at $x = +10.0 \text{ cm}$? (b) Where should q_3 be placed to make the potential energy of the system equal to zero?

23.48 •• A point charge $+8.00 \text{ nC}$ is on the $-x$ -axis at $x = -0.200 \text{ m}$, and a point charge -4.00 nC is on the $+x$ -axis at $x = 0.200 \text{ m}$. (a) In addition to $x = \pm\infty$, at what point on the x -axis is the resultant field of the two charges equal to zero? (b) Let $V = 0$ at $x = \pm\infty$. At what two other points on the x -axis is the total electric potential due to the two charges equal to zero? (c) Is $E = 0$ at either of the points in part (b) where $V = 0$? Explain.

23.49 •• A very long uniform line of charge with charge per unit length $\lambda = +5.00 \mu\text{C}/\text{m}$ lies along the x -axis, with its midpoint at the origin. A very large uniform sheet of charge is parallel to the xy -plane; the center of the sheet is at $z = +0.600 \text{ m}$. The sheet has charge per unit area $\sigma = +8.00 \mu\text{C}/\text{m}^2$, and the center of the sheet is at $x = 0$, $y = 0$. Point A is on the z -axis at $z = +0.300 \text{ m}$, and point B is on the z -axis at $z = -0.200 \text{ m}$. What is the potential difference $V_{AB} = V_A - V_B$ between points A and B ? Which point, A or B , is at higher potential?

23.50 ••• A small sphere with mass $5.00 \times 10^{-7} \text{ kg}$ and charge $+7.00 \mu\text{C}$ is released from rest a distance of 0.400 m above a large horizontal insulating sheet of charge that has uniform surface charge density $\sigma = +8.00 \text{ pC}/\text{m}^2$. Using energy methods, calculate the speed of the sphere when it is 0.100 m above the sheet.

23.51 •• A gold nucleus has a radius of $7.3 \times 10^{-15} \text{ m}$ and a charge of $+79e$. Through what voltage must an alpha particle, with charge $+2e$, be accelerated so that it has just enough energy to reach a distance of $2.0 \times 10^{-14} \text{ m}$ from the surface of a gold nucleus? (Assume that the gold nucleus remains stationary and can be treated as a point charge.)

23.52 •• CP A proton and an alpha particle are released from rest when they are 0.225 nm apart. The alpha particle (a helium nucleus) has essentially four times the mass and two times the charge of a proton. Find the maximum *speed* and maximum *acceleration* of each of these particles. When do these maxima occur: just following the release of the particles or after a very long time?

23.53 • A particle with charge $+7.60 \text{ nC}$ is in a uniform electric field directed to the left. Another force, in addition to the electric force, acts on the particle so that when it is released from rest, it moves to the right. After it has moved 8.00 cm , the additional force has done $6.50 \times 10^{-5} \text{ J}$ of work and the particle has $4.35 \times 10^{-5} \text{ J}$ of kinetic energy. (a) What work was done by the electric force? (b) What is the potential of the starting point with respect to the end point? (c) What is the magnitude of the electric field?

23.54 •• Identical charges $q = +5.00 \mu\text{C}$ are placed at opposite corners of a square that has sides of length 8.00 cm . Point A is at one of the empty corners, and point B is at the center of the square. A charge $q_0 = -3.00 \mu\text{C}$ is placed at point A and moves along the diagonal of the square to point B . (a) What is the magnitude of the net electric force on q_0 when it is at point A ? Sketch the placement of the charges and the direction of the net force. (b) What is the magnitude of the net electric force on q_0 when it is at point B ? (c) How much work does the electric force do on q_0 during its motion from A to B ? Is this work positive or negative? When it goes from A to B , does q_0 move to higher potential or to lower potential?

23.55 •• CALC A vacuum tube diode consists of concentric cylindrical electrodes, the negative cathode and the positive anode. Because of the accumulation of charge near the cathode, the electric potential between the electrodes is given by $V(x) = Cx^{4/3}$ where x is the distance from the cathode and C is a constant, characteristic of a particular diode and operating conditions. Assume that the distance between the cathode and anode is 13.0 mm and the potential difference between electrodes is 240 V . (a) Determine the value of C . (b) Obtain a formula for the electric field between the electrodes as a function of x . (c) Determine the force on an electron when the electron is halfway between the electrodes.

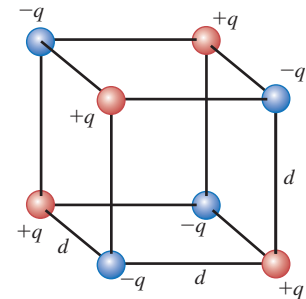
23.56 •• When you scuff your feet on a carpet, you gain electrons and become negatively charged. If you then place your finger near a metallic surface, such as a doorknob, an electric field develops between your finger and the doorknob. As your finger gets closer to the surface, the magnitude of the electric field increases. When it exceeds a threshold of $3 \text{ MV}/\text{m}$, the air breaks down, creating a small “lightning” strike, which you feel as a shock. (a) Estimate the distance between your finger and a doorknob at the point you feel the shock. (b) Using this estimate and the

threshold field strength of $3 \text{ MV}/\text{m}$, find the potential between your finger and the doorknob. (c) At this (small) distance, we can treat the tip of your finger and the end of the doorknob as infinite planar surfaces with opposite charge density. Estimate that density. (d) Estimate the effective area of your fingertip as presented to the doorknob. (e) Use these results to estimate the magnitude of charge on your finger. (f) Assuming that your net excess charge has built up on your finger, attracted there by the doorknob, estimate the number of electrons that you lost while scuffing your feet.

23.57 •• An Ionic Crystal.

Figure P23.57 shows eight point charges arranged at the corners of a cube with sides of length d . The values of the charges are $+q$ and $-q$, as shown. This is a model of one cell of a cubic ionic crystal. In sodium chloride (NaCl), for instance, the positive ions are Na^+ and the negative ions are Cl^- . (a) Calculate the potential energy U of this arrangement. (Take as zero the potential energy of the eight charges when they are infinitely far apart.) (b) In part (a), you should have found that $U < 0$. Explain the relationship between this result and the observation that such ionic crystals exist in nature.

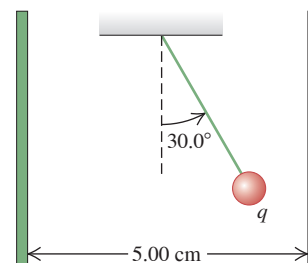
Figure P23.57



23.58 •• Electrical power is transmitted to our homes by overhead wires strung between poles. A typical residential utility line has a maximum potential of 22 kV relative to ground. We can treat this potential as constant and model it as generated by a net charge distributed on the wire. (a) Estimate the height of an electrical transmission line. (b) Treat the electrical wire as a long conducting cylinder with a radius of 2 cm . If the potential between the surface of the wire and a position directly beneath the wire is 22 kV , what is the linear charge density on the wire? (c) Use your estimate of the linear charge density to estimate the strength of the electric field on the ground beneath the wire.

23.59 •• CP A small sphere with mass 1.50 g hangs by a thread between two very large parallel vertical plates 5.00 cm apart (**Fig. P23.59**). The plates are insulating and have uniform surface charge densities $+\sigma$ and $-\sigma$. The charge on the sphere is $q = 8.90 \times 10^{-6} \text{ C}$. What potential difference between the plates will cause the thread to assume an angle of 30.0° with the vertical?

Figure P23.59



23.60 •• Two spherical shells have a common center. The inner shell has radius $R_1 = 5.00 \text{ cm}$ and charge $q_1 = +3.00 \times 10^{-6} \text{ C}$; the outer shell has radius $R_2 = 15.0 \text{ cm}$ and charge $q_2 = -5.00 \times 10^{-6} \text{ C}$. Both charges are spread uniformly over the shell surface. What is the electric potential due to the two shells at the following distances from their common center: (a) $r = 2.50 \text{ cm}$; (b) $r = 10.0 \text{ cm}$; (c) $r = 20.0 \text{ cm}$? Take $V = 0$ at a large distance from the shells.

23.61 • CALC Coaxial Cylinders. A long metal cylinder with radius a is supported on an insulating stand on the axis of a long, hollow, metal tube with radius b . The positive charge per unit length on the inner cylinder is λ , and there is an equal negative charge per unit length on the outer cylinder. (a) Calculate the potential $V(r)$ for (i) $r < a$; (ii) $a < r < b$; (iii) $r > b$. (Hint: The net potential is the sum of the potentials due to the individual conductors.) Take $V = 0$ at $r = b$. (b) Show that the potential of the inner cylinder with respect to the outer is

$$V_{ab} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}$$

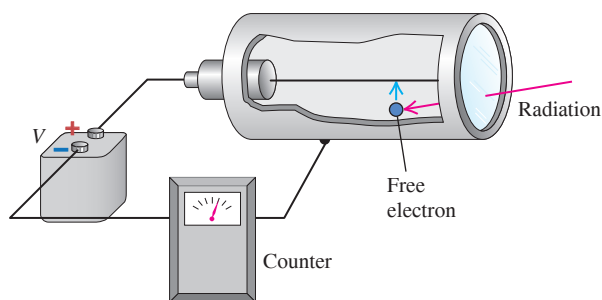
(c) Use Eq. (23.23) and the result from part (a) to show that the electric field at any point between the cylinders has magnitude

$$E(r) = \frac{V_{ab}}{\ln(b/a)} \frac{1}{r}$$

(d) What is the potential difference between the two cylinders if the outer cylinder has no net charge?

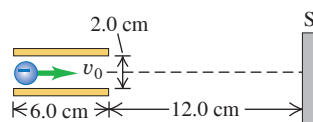
23.62 •• A Geiger counter detects radiation such as alpha particles by using the fact that the radiation ionizes the air along its path. A thin wire lies on the axis of a hollow metal cylinder and is insulated from it (Fig. P23.62). A large potential difference is established between the wire and the outer cylinder, with the wire at higher potential; this sets up a strong electric field directed radially outward. When ionizing radiation enters the device, it ionizes a few air molecules. The free electrons produced are accelerated by the electric field toward the wire and, on the way there, ionize many more air molecules. Thus a current pulse is produced that can be detected by appropriate electronic circuitry and converted to an audible “click.” Suppose the radius of the central wire is $145 \mu\text{m}$ and the radius of the hollow cylinder is 1.80 cm . What potential difference between the wire and the cylinder produces an electric field of $2.00 \times 10^4 \text{ V/m}$ at a distance of 1.20 cm from the axis of the wire? (The wire and cylinder are both very long in comparison to their radii, so the results of Problem 23.61 apply.)

Figure P23.62



23.63 • CP Deflection in a CRT. Cathode-ray tubes (CRTs) were often found in oscilloscopes and computer monitors. In Fig. P23.63 an electron with an initial speed of $6.50 \times 10^6 \text{ m/s}$ is projected along the axis midway between the deflection plates of a cathode-ray tube. The potential difference between the two plates is 22.0 V and the lower plate is the one at higher potential. (a) What is the force (magnitude and direction) on the electron when it is between the plates? (b) What is the acceleration of the electron (magnitude and direction) when acted on by the force in part (a)? (c) How far below the axis has the electron moved when it reaches the end of the plates? (d) At what angle with the axis is it moving as it leaves the plates? (e) How far below the axis will it strike the fluorescent screen S?

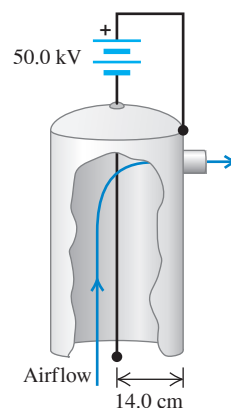
Figure P23.63



23.64 •• CP Deflecting Plates of an Oscilloscope. The vertical deflecting plates of a typical classroom oscilloscope are a pair of parallel square metal plates carrying equal but opposite charges. Typical dimensions are about 3.0 cm on a side, with a separation of about 5.0 mm . The potential difference between the plates is 25.0 V . The plates are close enough that we can ignore fringing at the ends. Under these conditions: (a) how much charge is on each plate, and (b) how strong is the electric field between the plates? (c) If an electron is ejected at rest from the negative plate, how fast is it moving when it reaches the positive plate?

23.65 •• Electrostatic precipitators use electric forces to remove pollutant particles from smoke, in particular in the smokestacks of coal-burning power plants. One form of precipitator consists of a vertical, hollow, metal cylinder with a thin wire, insulated from the cylinder, running along its axis (Fig. P23.65). A large potential difference is established between the wire and the outer cylinder, with the wire at lower potential. This sets up a strong radial electric field directed inward. The field produces a region of ionized air near the wire. Smoke enters the precipitator at the bottom, ash and dust in it pick up electrons, and the charged pollutants are accelerated toward the outer cylinder wall by the electric field. Suppose the radius of the central wire is $90.0 \mu\text{m}$, the radius of the cylinder is 14.0 cm , and a potential difference of 50.0 kV is established between the wire and the cylinder. Also assume that the wire and cylinder are both very long in comparison to the cylinder radius, so the results of Problem 23.61 apply. (a) What is the magnitude of the electric field midway between the wire and the cylinder wall? (b) What magnitude of charge must a $30.0 \mu\text{g}$ ash particle have if the electric field computed in part (a) is to exert a force ten times the weight of the particle?

Figure P23.65



23.66 •• CALC A disk with radius R has uniform surface charge density σ . (a) By regarding the disk as a series of thin concentric rings, calculate the electric potential V at a point on the disk's axis a distance x from the center of the disk. Assume that the potential is zero at infinity. (Hint: Use the result of Example 23.11 in Section 23.3.) (b) Calculate $-\partial V/\partial x$. Show that the result agrees with the expression for E_x calculated in Example 21.11 (Section 21.5).

23.67 ••• CALC Self-Energy of a Sphere of Charge. A solid sphere of radius R contains a total charge Q distributed uniformly throughout its volume. Find the energy needed to assemble this charge by bringing infinitesimal charges from far away. This energy is called the “self-energy” of the charge distribution. (Hint: After you have assembled a charge q in a sphere of radius r , how much energy would it take to add a spherical shell of thickness dr having charge dq ? Then integrate to get the total energy.)

23.68 • CALC A thin insulating rod is bent into a semicircular arc of radius a , and a total electric charge Q is distributed uniformly along the rod. Calculate the potential at the center of curvature of the arc if the potential is assumed to be zero at infinity.

23.69 •• Charge $Q = +4.00 \mu\text{C}$ is distributed uniformly over the volume of an insulating sphere that has radius $R = 5.00 \text{ cm}$. What is the potential difference between the center of the sphere and the surface of the sphere?

23.70 • (a) If a spherical raindrop of radius 0.650 mm carries a charge of -3.60 pC uniformly distributed over its volume, what is the potential at its surface? (Take the potential to be zero at an infinite distance from the raindrop.) (b) Two identical raindrops, each with radius and charge specified in part (a), collide and merge into one larger raindrop. What is the radius of this larger drop, and what is the potential at its surface, if its charge is uniformly distributed over its volume?

23.71 • CALC Electric charge is

distributed uniformly along a thin rod of length a , with total charge Q . Take the potential to be zero at infinity. Find the potential at the following points (**Fig. P23.71**):

- (a) point P , a distance x to the right of the rod, and (b) point R , a distance y above the right-hand end of the rod. (c) In parts (a) and (b), what does your result reduce to as x or y becomes much larger than a ?

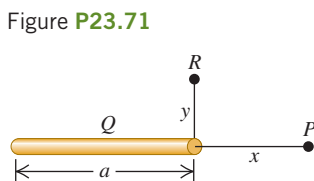
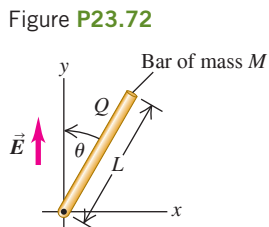


Figure P23.72

23.72 •• CP CALC A rigid bar with mass M , length L , and a uniformly distributed positive charge Q is free to pivot about the origin in the presence of a spatially uniform electric field $\vec{E} = E\hat{j}$, as shown in **Fig. P23.72**. Assume that the electric forces are large enough for gravity to be neglected. (a) Write the potential $V(y)$ due to the electric field as a function of y , using the convention $V(0) = V_0$, where V_0 is a constant to be determined. (b) Determine the electric potential energy U of the system as a function of θ and V_0 . (*Hint: $U = \int V dq$.*) (c) For what value of V_0 does the potential energy vanish when $\theta = 0^\circ$? (d) If the bar is released from rest at the position $\theta = 90^\circ$, what is the angular speed of the bar as it passes the position $\theta = 0^\circ$? (e) If the bar starts from rest at a small value of θ , with what frequency will the bar oscillate around the position $\theta = 0^\circ$?



23.73 •• CP A helium nucleus, also known as an α (alpha) particle, consists of two protons and two neutrons and has a diameter of 10^{-15} m = 1 fm. The protons, with a charge of $+e$, are subject to a repulsive Coulomb force. Since the neutrons have zero charge, there must be an attractive force that counteracts the electric repulsion and keeps the protons from flying apart. This so-called strong force plays a central role in particle physics. (a) As a crude model, assume that an α particle consists of two pointlike protons attracted by a Hooke's-law spring with spring constant k , and ignore the neutrons. Assume further that in the absence of other forces, the spring has an equilibrium separation of zero. Write an expression for the potential energy when the protons are separated by distance d . (b) Minimize this potential to find the equilibrium separation d_0 in terms of e and k . (c) If $d_0 = 1.00$ fm, what is the value of k ? (d) How much energy is stored in this system, in terms of electron volts? (e) A proton has a mass of 1.67×10^{-27} kg. If the spring were to break, the α particle would disintegrate and the protons would fly off in opposite directions. What would be their ultimate speed?

23.74 • A metal sphere with radius R_1 has a charge Q_1 . Take the electric potential to be zero at an infinite distance from the sphere. (a) What are the electric field and electric potential at the surface of the sphere? This sphere is now connected by a long, thin conducting wire to another sphere of radius R_2 that is several meters from the first sphere. Before the connection is made, this second sphere is uncharged. After electrostatic equilibrium has been reached, what are (b) the total charge on each sphere; (c) the electric potential at the surface of each sphere; (d) the electric field at the surface of each sphere? Assume that the amount of charge on the wire is much less than the charge on each sphere.

23.75 • An alpha particle with kinetic energy 9.50 MeV (when far away) collides head-on with a lead nucleus at rest. What is the distance of closest approach of the two particles? (Assume that the lead nucleus remains stationary and may be treated as a point charge. The atomic number of lead is 82. The alpha particle is a helium nucleus, with atomic number 2.)

23.76 • CALC The electric potential V in a region of space is given by

$$V(x, y, z) = A(x^2 - 3y^2 + z^2)$$

where A is a constant. (a) Derive an expression for the electric field $\vec{E}$ at any point in this region. (b) The work done by the field when a $1.50 \mu\text{C}$ test charge moves from the point $(x, y, z) = (0, 0, 0.250 \text{ m})$ to the origin is measured to be $6.00 \times 10^{-5} \text{ J}$. Determine A . (c) Determine the electric field at the point $(0, 0, 0.250 \text{ m})$. (d) Show that in every plane parallel to the xz -plane the equipotential contours are circles. (e) What is the radius of the equipotential contour corresponding to $V = 1280 \text{ V}$ and $y = 2.00 \text{ m}$?

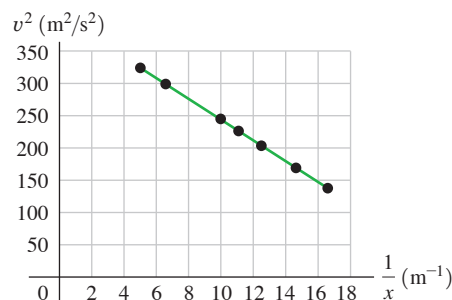
23.77 •• DATA The electric potential in a region that is within 2.00 m of the origin of a rectangular coordinate system is given by $V = Ax^l + By^m + Cz^n + D$, where A, B, C, D, l, m , and n are constants. The units of A, B, C , and D are such that if x, y , and z are in meters, then V is in volts. You measure V and each component of the electric field at four points and obtain these results:

Point	$(x, y, z) \text{ (m)}$	$V \text{ (V)}$	$E_x \text{ (V/m)}$	$E_y \text{ (V/m)}$	$E_z \text{ (V/m)}$
1	(0, 0, 0)	10.0	0	0	0
2	(1.00, 0, 0)	4.0	12.0	0	0
3	(0, 1.00, 0)	6.0	0	12.0	0
4	(0, 0, 1.00)	8.0	0	0	12.0

- (a) Use the data in the table to calculate A, B, C, D, l, m , and n . (b) What are V and the magnitude of E at the points $(0, 0, 0)$, $(0.50 \text{ m}, 0.50 \text{ m}, 0.50 \text{ m})$, and $(1.00 \text{ m}, 1.00 \text{ m}, 1.00 \text{ m})$?

23.78 •• DATA A small, stationary sphere carries a net charge Q . You perform the following experiment to measure Q : From a large distance you fire a small particle with mass $m = 4.00 \times 10^{-4} \text{ kg}$ and charge $q = 5.00 \times 10^{-8} \text{ C}$ directly at the center of the sphere. The apparatus you are using measures the particle's speed v as a function of the distance x from the sphere. The sphere's mass is much greater than the mass of the projectile particle, so you assume that the sphere remains at rest. All of the measured values of x are much larger than the radius of either object, so you treat both objects as point particles. You plot your data on a graph of v^2 versus $(1/x)$ (**Fig. P23.78**). The straight line $v^2 = 400 \text{ m}^2/\text{s}^2 - [(15.75 \text{ m}^3/\text{s}^2)/x]$ gives a good fit to the data points. (a) Explain why the graph is a straight line. (b) What is the initial speed v_0 of the particle when it is very far from the sphere? (c) What is Q ? (d) How close does the particle get to the sphere? Assume that this distance is much larger than the radii of the particle and sphere, so continue to treat them as point particles and to assume that the sphere remains at rest.

Figure P23.78



23.79 ••• DATA The Millikan Oil-Drop Experiment. The charge of an electron was first measured by the American physicist Robert Millikan during 1909–1913. In his experiment, oil was sprayed in very fine drops (about 10^{-4} mm in diameter) into the space between two parallel horizontal plates separated by a distance d . A potential difference V_{AB} was maintained between the plates, causing a downward electric field between them. Some of the oil drops acquired a negative charge because of frictional effects or because of ionization of the surrounding air by x rays or radioactivity. The drops were observed through a microscope. (a) Show that an oil drop of radius r at rest between the plates remained at rest if the magnitude of its charge was

$$q = \frac{4\pi}{3} \frac{\rho r^3 g d}{V_{AB}}$$

where ρ is oil's density. (Ignore the buoyant force of the air.) By adjusting V_{AB} to keep a given drop at rest, Millikan determined the charge on that drop, provided its radius r was known. (b) Millikan's oil drops were much too small to measure their radii directly. Instead, Millikan determined r by cutting off the electric field and measuring the *terminal speed* v_t of the drop as it fell. (We discussed terminal speed in Section 5.3.) The viscous force F on a sphere of radius r moving at speed v through a fluid with viscosity η is given by Stokes's law: $F = 6\pi\eta rv$. When a drop fell at v_t , the viscous force just balanced the drop's weight $w = mg$. Show that the magnitude of the charge on the drop was

$$q = 18\pi \frac{d}{V_{AB}} \sqrt{\frac{\eta^3 v_t^3}{2\rho g}}$$

(c) You repeat the Millikan oil-drop experiment. Four of your measured values of V_{AB} and v_t are listed in the table:

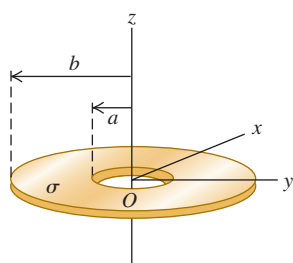
Drop	1	2	3	4
V_{AB} (V)	9.16	4.57	12.32	6.28
v_t (10^{-5} m/s)	2.54	0.767	4.39	1.52

In your apparatus, the separation d between the horizontal plates is 1.00 mm. The density of the oil you use is 824 kg/m^3 . For the viscosity η of air, use the value $1.81 \times 10^{-5} \text{ N}\cdot\text{s/m}^2$. Assume that $g = 9.80 \text{ m/s}^2$. Calculate the charge q of each drop. (d) If electric charge is *quantized* (that is, exists in multiples of the magnitude of the charge of an electron), then the charge on each drop is $-ne$, where n is the number of excess electrons on each drop. (All four drops in your table have negative charge.) Drop 2 has the smallest magnitude of charge observed in the experiment, for all 300 drops on which measurements were made, so assume that its charge is due to an excess charge of one electron. Determine the number of excess electrons n for each of the other three drops. (e) Use $q = -ne$ to calculate e from the data for each of the four drops, and average these four values to get your best experimental value of e .

CHALLENGE PROBLEMS

23.80 ••• CALC An annulus with an inner radius of a and an outer radius of b has charge density σ and lies in the xy -plane with its center at the origin, as shown in Fig. P23.80. (a) Using the convention that the potential vanishes at infinity, determine the potential at all points on the z -axis. (b) Determine the electric field at all points on the z -axis by differentiating the potential. (c) Show that in the limit $a \rightarrow 0$, $b \rightarrow \infty$ the

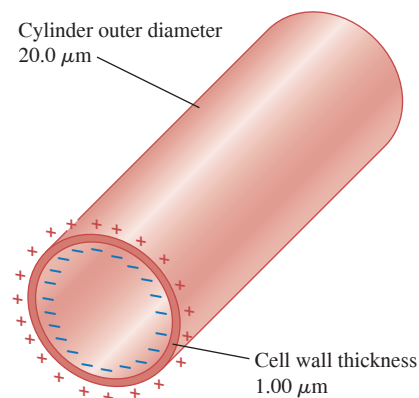
Figure P23.80



electric field reproduces the result obtained in Example 22.7 for an infinite plane sheet of charge. (d) If $a = 5.00 \text{ cm}$, $b = 10.0 \text{ cm}$ and the total charge on the annulus is $1.00 \mu\text{C}$, what is the potential at the origin? (e) If a particle with mass 1.00 g (much less than the mass of the annulus) and charge $1.00 \mu\text{C}$ is placed at the origin and given the slightest nudge, it will be projected along the z -axis. In this case, what will be its ultimate speed?

23.81 ••• BIO A heart cell can be modeled as a cylindrical shell that is $100 \mu\text{m}$ long, with an outer diameter of $20.0 \mu\text{m}$ and a cell wall thickness of $1.00 \mu\text{m}$, as shown in Fig. P23.81. Potassium ions move across the cell wall, depositing positive charge on the outer surface and leaving a net negative charge on the inner surface. During the so-called resting phase, the inside of the cell has a potential that is 90.0 mV lower than the potential on its outer surface. (a) If the net charge of the cell is zero, what is the magnitude of the total charge on either cell wall membrane? Ignore edge effects and treat the cell as a very long cylinder. (b) What is the magnitude of the electric field just inside the cell wall? (c) In a subsequent depolarization event, sodium ions move through channels in the cell wall, so that the inner membrane becomes positively charged. At the end of this event, the inside of the cell has a potential that is 20.0 mV higher than the potential outside the cell. If we model this event by charge moving from the outer membrane to the inner membrane, what magnitude of charge moves across the cell wall during this event? (d) If this were done entirely by the motion of sodium ions, Na^+ , how many ions have moved?

Figure P23.81



23.82 ••• CALC A hollow, thin-walled insulating cylinder of radius R and length L (like the cardboard tube in a roll of toilet paper) has charge Q uniformly distributed over its surface. (a) Calculate the electric potential at all points along the axis of the tube. Take the origin to be at the center of the tube, and take the potential to be zero at infinity. (b) Show that if $L \ll R$, the result of part (a) reduces to the potential on the axis of a ring of charge of radius R . (See Example 23.11 in Section 23.3.) (c) Use the result of part (a) to find the electric field at all points along the axis of the tube.

23.83 ••• CP In experiments in which atomic nuclei collide, head-on collisions like that described in Problem 23.75 do happen, but “near misses” are more common. Suppose the alpha particle in that problem is not “aimed” at the center of the lead nucleus but has an initial non-zero angular momentum (with respect to the stationary lead nucleus) of magnitude $L = p_0 b$, where p_0 is the magnitude of the particle's initial momentum and $b = 1.00 \times 10^{-12} \text{ m}$. What is the distance of closest approach? Repeat for $b = 1.00 \times 10^{-13} \text{ m}$ and $b = 1.00 \times 10^{-14} \text{ m}$.

MCAT-STYLE PASSAGE PROBLEMS

Materials Analysis with Ions. *Rutherford backscattering spectrometry* (RBS) is a technique used to determine the structure and composition of materials. A beam of ions (typically helium ions) is accelerated to high energy and aimed at a sample. By analyzing the distribution and energy of the ions that are scattered from (that is, deflected by collisions with) the atoms in the sample, researchers can determine the sample's composition. To accelerate the ions to high energies, a *tandem electrostatic accelerator* may be used. In this device, negative ions (He^-) start at a potential $V = 0$ and are accelerated by a high positive voltage at the midpoint of the accelerator. The high voltage produces a constant electric field in the acceleration tube through which the ions move. When accelerated ions reach the midpoint, the electrons are stripped off, turning the negative ions into doubly positively charged ions (He^{++}). These positive ions are then repelled from the midpoint by the high positive voltage there and continue to accelerate to the far end of the accelerator, where again $V = 0$.

23.84 For a particular experiment, helium ions are to be given a kinetic energy of 3.0 MeV. What should the voltage at the center of the accelerator be, assuming that the ions start essentially at rest? (a) -3.0 MV; (b) $+3.0$ MV; (c) $+1.5$ MV; (d) $+1.0$ MV.

23.85 A helium ion (He^{++}) that comes within about 10 fm of the center of the nucleus of an atom in the sample may induce a nuclear reaction instead of simply scattering. Imagine a helium ion with a kinetic energy of 3.0 MeV heading straight toward an atom at rest in the sample. Assume that the atom stays fixed. What minimum charge can the nucleus of the atom have such that the helium ion gets no closer than 10 fm from the center of the atomic nucleus? ($1 \text{ fm} = 1 \times 10^{-15} \text{ m}$, and e is the magnitude of the charge of an electron or a proton.) (a) $2e$; (b) $11e$; (c) $20e$; (d) $22e$.

23.86 The maximum voltage at the center of a typical tandem electrostatic accelerator is 6.0 MV. If the distance from one end of the acceleration tube to the midpoint is 12 m, what is the magnitude of the average electric field in the tube under these conditions? (a) 41,000 V/m; (b) 250,000 V/m; (c) 500,000 V/m; (d) 6,000,000 V/m.

ANSWERS

Chapter Opening Question ?

(iii) A large, constant potential difference V_{ab} is maintained between the welding tool (a) and the metal pieces to be welded (b). For a given potential difference between two conductors a and b , the smaller the distance d separating the conductors, the greater is the magnitude E of the field between them. Hence d must be small in order for E to be large enough to ionize the gas between the conductors (see Section 23.3) and produce an arc through this gas.

Key Example VARIATION Problems

VP23.2.1 (a) $1.51 \times 10^{-16} \text{ J}$ (b) $2.32 \times 10^{-15} \text{ J}$

VP23.2.2 (a) $5.22 \times 10^{-15} \text{ J}$ (b) $1.92 \times 10^{-18} \text{ C}$

VP23.2.3 (a) $-e^2/24\pi\epsilon_0 a$ (b) $-7e^2/24\pi\epsilon_0 a$

VP23.2.4 (a) $7.21 \times 10^{-9} \text{ C}$ (b) $5.59 \times 10^{-6} \text{ J}$

VP23.7.1 (a) $+25.0 \text{ V}$ (b) $+15.0 \text{ V}$ (c) 0 (d) -25.0 V

VP23.7.2 (a) $3.6 \times 10^3 \text{ V}$ (b) $2.2 \times 10^3 \text{ V}$ (c) $1.6 \times 10^3 \text{ V}$

(d) point (a); no

VP23.7.3 (a) $+133 \text{ V}$ (b) -133 V (c) $1.98 \times 10^5 \text{ m/s}$

VP23.7.4 (a) $Q/8\pi\epsilon_0 R$ (b) the center

VP23.12.1 (a) $qq_0/30\pi\epsilon_0 R$ (b) $-5qq_0/56\pi\epsilon_0 R$ (c) $3qq_0/16\pi\epsilon_0 R$

VP23.12.2 (a) $+16.0 \text{ V}$ (b) $3.20 \times 10^{-8} \text{ J}$ (c) lower, 3.58 m/s

VP23.12.3 (a) $1.61 \times 10^{-6} \text{ J}$ (b) $5.39 \times 10^{-6} \text{ J}$ (c) $K = 3.42 \times 10^{-6} \text{ J}$, $v = 41.3 \text{ m/s}$

VP23.12.4 (a) $dq = (Q/L)dx$ (b) $(Q/4\pi\epsilon_0 L)\ln 2$

Bridging Problem

$$\frac{qQ}{8\pi\epsilon_0 a} \ln \left(\frac{L+a}{L-a} \right)$$



? In flash photography, the energy used to make the flash is stored in a capacitor, which consists of two closely spaced conductors that carry opposite charges. If the amount of charge on the conductors is doubled, by what factor does the stored energy increase? (i) $\sqrt{2}$; (ii) 2; (iii) $2\sqrt{2}$; (iv) 4; (v) 8.

24 Capacitance and Dielectrics

When you stretch the rubber band of a slingshot or pull back the string of an archer's bow, you are storing mechanical energy as elastic potential energy. A capacitor is a device that stores *electric* potential energy and electric charge. To make a capacitor, just insulate two conductors from each other. To store energy in this device, transfer charge from one conductor to the other so that one has a negative charge and the other has an equal amount of positive charge. Work must be done to move the charges through the resulting potential difference between the conductors, and the work done is stored as electric potential energy.

Capacitors have a tremendous number of practical applications in devices such as electronic flash units for photography, mobile phones, airbag sensors for cars, and radio and television receivers. We'll encounter many of these applications in later chapters (particularly Chapter 31, in which we'll see the crucial role played by capacitors in the alternating-current circuits that pervade our technological society). In this chapter, however, our emphasis is on the fundamental properties of capacitors. For a particular capacitor, the ratio of the charge on each conductor to the potential difference between the conductors is a constant, called the *capacitance*. The capacitance depends on the sizes and shapes of the conductors and on the insulating material (if any) between them. Compared to the case in which there is only vacuum between the conductors, the capacitance increases when an insulating material (a *dielectric*) is present. This happens because a redistribution of charge, called *polarization*, takes place within the insulating material. Studying polarization will give us added insight into the electrical properties of matter.

Capacitors also give us a new way to think about electric potential energy. The energy stored in a charged capacitor is related to the electric field in the space between the conductors. We'll see that electric potential energy can be regarded as being stored *in the field itself*. The idea that the electric field is itself a storehouse of energy is at the heart of the theory of electromagnetic waves and our modern understanding of the nature of light, to be discussed in Chapter 32.

LEARNING OUTCOMES

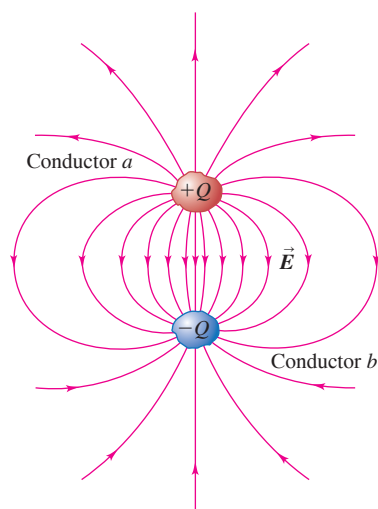
In this chapter, you'll learn...

- 24.1** The nature of capacitors, and how to calculate a quantity that measures their ability to store charge.
- 24.2** How to analyze capacitors connected in a network.
- 24.3** How to calculate the amount of energy stored in a capacitor.
- 24.4** What dielectrics are, and how they make capacitors more effective.
- 24.5** How a dielectric inside a charged capacitor becomes polarized.
- 24.6** How to use Gauss's laws when dielectrics are present.

You'll need to review...

- 21.2, 21.5, 21.7** Polarization; field of charged conductors; electric dipoles.
- 22.3–22.5** Gauss's law.
- 23.3, 23.4** Potential for charged conductors; potential due to a cylindrical charge distribution.

Figure 24.1 Any two conductors a and b insulated from each other form a capacitor.



24.1 CAPACITORS AND CAPACITANCE

Any two conductors separated by an insulator (or a vacuum) form a **capacitor** (Fig. 24.1). In most practical applications, each conductor initially has zero net charge and electrons are transferred from one conductor to the other; this is called *charging* the capacitor. Then the two conductors have charges with equal magnitude and opposite sign, and the *net* charge on the capacitor as a whole remains zero. We'll assume throughout this chapter that this is the case. When we say that a capacitor has charge Q , or that a charge Q is *stored* on the capacitor, we mean that the conductor at higher potential has charge $+Q$ and the conductor at lower potential has charge $-Q$ (assuming that Q is positive). Keep this in mind in the following discussion and examples.

In circuit diagrams a capacitor is represented by either of these symbols:



The vertical lines (straight or curved) represent the conductors, and the horizontal lines represent wires connected to either conductor. One common way to charge a capacitor is to connect these two wires to opposite terminals of a battery. Once the charges Q and $-Q$ are established on the conductors, the battery is disconnected. This gives a fixed *potential difference* V_{ab} between the conductors (that is, the potential of the positively charged conductor a with respect to the negatively charged conductor b) that is just equal to the voltage of the battery.

The electric field at any point in the region between the conductors is proportional to the magnitude Q of charge on each conductor. It follows that the potential difference V_{ab} between the conductors is also proportional to Q . If we double the magnitude of charge on each conductor, the charge density at each point doubles, the electric field at each point doubles, and the potential difference between conductors doubles; however, the *ratio* of charge to potential difference does not change. This ratio is called the **capacitance** C of the capacitor:

$$\text{Capacitance of a capacitor } C = \frac{Q}{V_{ab}} \quad (24.1)$$

Q ← Magnitude of charge on each conductor
 V_{ab} ← Potential difference between conductors (a has charge $+Q$, b has charge $-Q$)

The SI unit of capacitance is called one **farad** (1 F), in honor of the 19th-century English physicist Michael Faraday. From Eq. (24.1), one farad is equal to one *coulomb per volt* (1 C/V):

$$1 \text{ F} = 1 \text{ farad} = 1 \text{ C/V} = 1 \text{ coulomb/volt}$$

CAUTION Capacitance vs. coulombs Don't confuse the symbol C for capacitance (which is always in italics) with the abbreviation C for coulombs (which is never italicized).

The greater the capacitance C of a capacitor, the greater the magnitude Q of charge on either conductor for a given potential difference V_{ab} and hence the greater the amount of stored energy. (Remember that potential is potential energy per unit charge.) Thus *capacitance is a measure of the ability of a capacitor to store energy*. We'll see that the capacitance depends only on the shapes, sizes, and relative positions of the conductors and on the nature of the insulator between them. (For special types of insulating materials, the capacitance *does* depend on Q and V_{ab} . We won't discuss these materials, however.)

Calculating Capacitance: Capacitors in Vacuum

We can calculate the capacitance C of a given capacitor by finding the potential difference V_{ab} between the conductors for a given magnitude of charge Q and then using Eq. (24.1). For now we'll consider only *capacitors in vacuum*; that is, empty space separates the conductors that make up the capacitor.

The simplest form of capacitor consists of two parallel conducting plates, each with area A , separated by a distance d that is small in comparison with their dimensions (Fig. 24.2a). When the plates are charged, the electric field is almost completely localized in the region between the plates (Fig. 24.2b). As we discussed in Example 22.8 (Section 22.4), the field between such plates is essentially *uniform*, and the charges on the plates are uniformly distributed over their opposing surfaces. We call this arrangement a **parallel-plate capacitor**.

We found the electric-field magnitude E for this arrangement in Example 21.12 (Section 21.5) by using the principle of superposition of electric fields and again in Example 22.8 (Section 22.4) by using Gauss's law. It would be a good idea to review those examples. We found that $E = \sigma/\epsilon_0$, where σ is the magnitude (absolute value) of the surface charge density on each plate. This is equal to the magnitude of the total charge Q on each plate divided by the area A of the plate, or $\sigma = Q/A$, so the field magnitude E can be expressed as

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

The field is uniform and the distance between the plates is d , so the potential difference (voltage) between the two plates is

$$V_{ab} = Ed = \frac{1}{\epsilon_0} \frac{Qd}{A}$$

Thus

$$\text{Capacitance of a parallel-plate capacitor in vacuum} \quad C = \frac{Q}{V_{ab}} = \epsilon_0 \frac{A}{d} \quad (24.2)$$

Magnitude of charge on each plate
Area of each plate
Distance between plates
Electric constant

Potential difference between plates

The capacitance depends on only the geometry of the capacitor; it is directly proportional to the area A of each plate and inversely proportional to their separation d . The quantities A and d are constants for a given capacitor, and ϵ_0 is a universal constant. Thus in vacuum the capacitance C is a constant independent of the charge on the capacitor or the potential difference between the plates. If one of the capacitor plates is flexible, C changes as the plate separation d changes. This is the operating principle of a condenser microphone (Fig. 24.3).

When matter is present between the plates, its properties affect the capacitance. We'll return to this topic in Section 24.4. Meanwhile, we remark that if the space contains air at atmospheric pressure instead of vacuum, the capacitance differs from the prediction of Eq. (24.2) by less than 0.06%.

In Eq. (24.2), if A is in square meters and d in meters, then C is in farads. The units of the electric constant ϵ_0 are $\text{C}^2/\text{N} \cdot \text{m}^2$, so

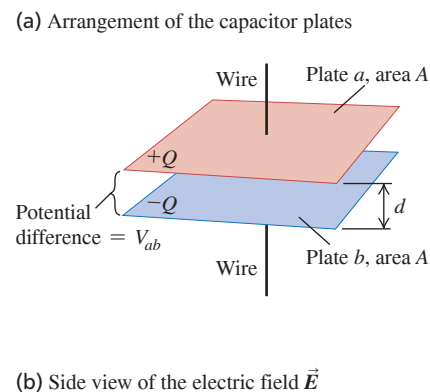
$$1 \text{ F} = 1 \text{ C}^2/\text{N} \cdot \text{m} = 1 \text{ C}^2/\text{J}$$

Because $1 \text{ V} = 1 \text{ J/C}$ (energy per unit charge), this equivalence agrees with our definition $1 \text{ F} = 1 \text{ C/V}$. Finally, we can express the units of ϵ_0 as $1 \text{ C}^2/\text{N} \cdot \text{m}^2 = 1 \text{ F/m}$, so

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

This relationship is useful in capacitance calculations, and it also helps us to verify that Eq. (24.2) is dimensionally consistent.

Figure 24.2 A charged parallel-plate capacitor.

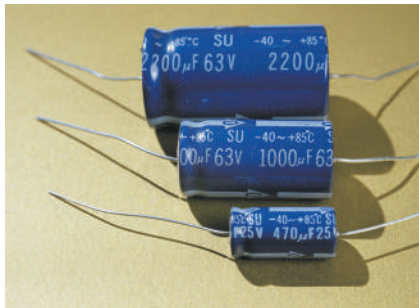


When the separation of the plates is small compared to their size, the fringing of the field is slight.

Figure 24.3 Inside a condenser microphone is a capacitor with one rigid plate and one flexible plate. The two plates are kept at a constant potential difference V_{ab} . Sound waves cause the flexible plate to move back and forth, varying the capacitance C and causing charge to flow to and from the capacitor in accordance with the relationship $C = Q/V_{ab}$. Thus a sound wave is converted to a charge flow that can be amplified and recorded digitally.



Figure 24.4 A commercial capacitor is labeled with the value of its capacitance. For these capacitors, $C = 2200 \mu\text{F}$, $1000 \mu\text{F}$, and $470 \mu\text{F}$.



One farad is a very large capacitance, as the following example shows. In many applications the most convenient units of capacitance are the *microfarad* ($1 \mu\text{F} = 10^{-6} \text{ F}$) and the *picofarad* ($1 \text{ pF} = 10^{-12} \text{ F}$). For example, the flash unit in a point-and-shoot camera uses a capacitor of a few hundred microfarads (Fig. 24.4), while capacitances in a radio tuning circuit are typically from 10 to 100 picofarads.

For *any* capacitor in vacuum, the capacitance C depends only on the shapes, dimensions, and separation of the conductors that make up the capacitor. If the conductor shapes are more complex than those of the parallel-plate capacitor, the expression for capacitance is more complicated than in Eq. (24.2). In the following examples we show how to calculate C for two other conductor geometries.

CAUTION **Capacitor charge and potential difference** Saying that a capacitor carries charge Q means that one conductor has positive charge $+Q$ and the other conductor has negative charge $-Q$. The *net* charge on the capacitor is zero! Furthermore, the positively charged conductor is always at higher potential than the negatively charged conductor. If the potential difference V_{ab} between the conductors is 500 V, the potential of the positive conductor is 500 V greater than that of the negative conductor. !

EXAMPLE 24.1 Size of a 1 F capacitor

The parallel plates of a 1.0 F capacitor are 1.0 mm apart. What is their area?

IDENTIFY and SET UP This problem uses the relationship among the capacitance C , plate separation d , and plate area A (our target variable) for a parallel-plate capacitor. We solve Eq. (24.2) for A .

EXECUTE From Eq. (24.2),

$$A = \frac{Cd}{\epsilon_0} = \frac{(1.0 \text{ F})(1.0 \times 10^{-3} \text{ m})}{8.85 \times 10^{-12} \text{ F/m}} = 1.1 \times 10^8 \text{ m}^2$$

EVALUATE This corresponds to a square about 10 km (about 6 miles) on a side! The volume of such a capacitor would be at least

$Ad = 1.1 \times 10^5 \text{ m}^3$, equivalent to that of a cube about 50 m on a side. In fact, it's possible to make 1 F capacitors a few *centimeters* on a side. The trick is to have an appropriate substance between the plates rather than a vacuum, so that (among other things) the plate separation d can be greatly reduced. We'll explore this further in Section 24.4.

KEYCONCEPT The capacitance of a capacitor is a measure of how much charge its plates will hold for a given potential difference (voltage) between the plates. Capacitance is proportional to the area of the plates and inversely proportional to the distance between them.

EXAMPLE 24.2 Properties of a parallel-plate capacitor

WITH VARIATION PROBLEMS

The plates of a parallel-plate capacitor in vacuum are 5.00 mm apart and 2.00 m^2 in area. A 10.0 kV potential difference is applied across the capacitor. Compute (a) the capacitance; (b) the charge on each plate; and (c) the magnitude of the electric field between the plates.

IDENTIFY and SET UP We are given the plate area A , the plate spacing d , and the potential difference $V_{ab} = 1.00 \times 10^4 \text{ V}$ for this parallel-plate capacitor. Our target variables are the capacitance C , the charge Q on each plate, and the electric-field magnitude E . We use Eq. (24.2) to calculate C and then use Eq. (24.1) and V_{ab} to find Q . We use $E = Q/\epsilon_0 A$ to find E .

EXECUTE (a) From Eq. (24.2),

$$\begin{aligned} C &= \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12} \text{ F/m}) \frac{(2.00 \text{ m}^2)}{5.00 \times 10^{-3} \text{ m}} \\ &= 3.54 \times 10^{-9} \text{ F} = 0.00354 \mu\text{F} \end{aligned}$$

(b) The charge on the capacitor is

$$\begin{aligned} Q &= CV_{ab} = (3.54 \times 10^{-9} \text{ C/V})(1.00 \times 10^4 \text{ V}) \\ &= 3.54 \times 10^{-5} \text{ C} = 35.4 \mu\text{C} \end{aligned}$$

The plate at higher potential has charge $+35.4 \mu\text{C}$, and the other plate has charge $-35.4 \mu\text{C}$.

(c) The electric-field magnitude is

$$\begin{aligned} E &= \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A} = \frac{3.54 \times 10^{-5} \text{ C}}{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.00 \text{ m}^2)} \\ &= 2.00 \times 10^6 \text{ N/C} \end{aligned}$$

EVALUATE We can also find E by recalling that the electric field is equal in magnitude to the potential gradient [Eq. (23.22)]. The field between the plates is uniform, so

$$E = \frac{V_{ab}}{d} = \frac{1.00 \times 10^4 \text{ V}}{5.00 \times 10^{-3} \text{ m}} = 2.00 \times 10^6 \text{ V/m}$$

(Remember that $1 \text{ N/C} = 1 \text{ V/m}$.)

KEYCONCEPT The electric field between the plates of a parallel-plate capacitor is uniform. To find its magnitude E , divide the potential difference between the plates by their separation; E is also proportional to the ratio of the charge on the capacitor plates to the plate's area.

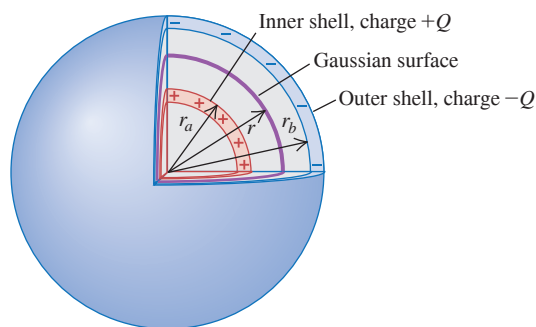
EXAMPLE 24.3 A spherical capacitor**WITH VARIATION PROBLEMS**

Two concentric spherical conducting shells are separated by vacuum (**Fig. 24.5**). The inner shell has total charge $+Q$ and outer radius r_a , and the outer shell has charge $-Q$ and inner radius r_b . Find the capacitance of this spherical capacitor.

IDENTIFY and SET UP By definition, the capacitance C is the magnitude Q of the charge on either sphere divided by the potential difference V_{ab} between the spheres. We first find V_{ab} , and then use Eq. (24.1) to find the capacitance $C = Q/V_{ab}$.

EXECUTE Using a Gaussian surface such as that shown in **Fig. 24.5**, we found in Example 22.5 (Section 22.4) that the charge on a conducting sphere produces zero field *inside* the sphere, so the outer sphere makes no contribution to the field between the spheres. Therefore the electric field *and* the electric potential between the shells are the same as those outside a charged conducting sphere with charge $+Q$. We considered that problem in Example 23.8 (Section 23.3), so the same result applies here: The potential at any point between the spheres is $V = Q/4\pi\epsilon_0 r$.

Figure 24.5 A spherical capacitor.



Hence the potential of the inner (positive) conductor at $r = r_a$ with respect to that of the outer (negative) conductor at $r = r_b$ is

$$\begin{aligned} V_{ab} &= V_a - V_b = \frac{Q}{4\pi\epsilon_0 r_a} - \frac{Q}{4\pi\epsilon_0 r_b} \\ &= \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right) = \frac{Q}{4\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b} \end{aligned}$$

The capacitance is then

$$C = \frac{Q}{V_{ab}} = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$$

As an example, if $r_a = 9.5$ cm and $r_b = 10.5$ cm,

$$\begin{aligned} C &= 4\pi(8.85 \times 10^{-12} \text{ F/m}) \frac{(0.095 \text{ m})(0.105 \text{ m})}{0.010 \text{ m}} \\ &= 1.1 \times 10^{-10} \text{ F} = 110 \text{ pF} \end{aligned}$$

EVALUATE We can relate our expression for C to that for a parallel-plate capacitor. The quantity $4\pi r_a r_b$ is intermediate between the areas $4\pi r_a^2$ and $4\pi r_b^2$ of the two spheres; in fact, it's the *geometric mean* of these two areas, which we can denote by A_{gm} . The distance between spheres is $d = r_b - r_a$, so we can write $C = 4\pi\epsilon_0 r_a r_b / (r_b - r_a) = \epsilon_0 A_{\text{gm}} / d$. This has the same form as for parallel plates: $C = \epsilon_0 A / d$. If the distance between spheres is very small in comparison to their radii, their capacitance is the same as that of parallel plates with the same area and spacing.

KEYCONCEPT To find the capacitance of any capacitor made up of two conductors, first calculate the potential difference V_{ab} between the two conductors when they carry charges $+Q$ and $-Q$. The capacitance is equal to Q divided by V_{ab} .

EXAMPLE 24.4 A cylindrical capacitor**WITH VARIATION PROBLEMS**

Two long, coaxial cylindrical conductors are separated by vacuum (**Fig. 24.6**). The inner cylinder has outer radius r_a and linear charge density $+\lambda$. The outer cylinder has inner radius r_b and linear charge density $-\lambda$. Find the capacitance per unit length for this capacitor.

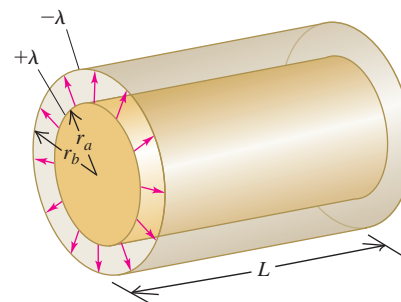
IDENTIFY and SET UP As in Example 24.3, we use the definition of capacitance, $C = Q/V_{ab}$. We use the result of Example 23.10 (Section 23.3) to find the potential difference V_{ab} between the cylinders, and find the charge Q on a length L of the cylinders from the linear charge density. We then find the corresponding capacitance C from Eq. (24.1). Our target variable is this capacitance divided by L .

EXECUTE As in Example 24.3, the potential V between the cylinders is not affected by the presence of the charged outer cylinder. Hence our result in Example 23.10 for the potential outside a charged conducting cylinder also holds in this example for potential in the space between the cylinders:

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

Here r_0 is the arbitrary, *finite* radius at which $V = 0$. We take $r_0 = r_b$, the radius of the inner surface of the outer cylinder. Then the potential at the outer surface of the inner cylinder (at which $r = r_a$) is just the potential V_{ab} of the inner (positive) cylinder a with respect to the outer (negative) cylinder b :

Figure 24.6 A long cylindrical capacitor. The linear charge density λ is assumed to be positive in this figure. The magnitude of charge in a length L of either cylinder is λL .



$$V_{ab} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}$$

If λ is positive as in **Fig. 24.6**, then V_{ab} is positive as well: The inner cylinder is at higher potential than the outer.

Continued

The total charge Q in a length L is $Q = \lambda L$, so from Eq. (24.1) the capacitance C of a length L is

$$C = \frac{Q}{V_{ab}} = \frac{\lambda L}{\frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}} = \frac{2\pi\epsilon_0 L}{\ln(r_b/r_a)}$$

The capacitance per unit length is

$$\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln(r_b/r_a)}$$

Substituting $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m} = 8.85 \text{ pF/m}$, we get

$$\frac{C}{L} = \frac{55.6 \text{ pF/m}}{\ln(r_b/r_a)}$$

EVALUATE The capacitance of coaxial cylinders is determined entirely by their dimensions, just as for parallel-plate and spherical capacitors. Ordinary coaxial cables are made like this but with an insulating material instead of vacuum between the conductors. A typical cable used for connecting a television to a cable TV feed has a capacitance per unit length of 69 pF/m.

KEYCONCEPT The capacitance per unit length of two long, coaxial, conducting cylinders depends on the ratio of the inner radius of the outer conductor to the outer radius of the inner conductor.

Figure 24.7 An assortment of commercially available capacitors.



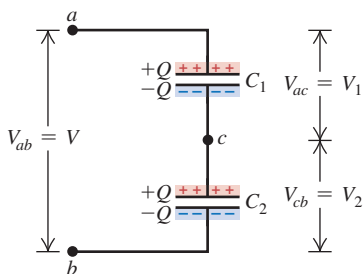
Figure 24.8 A series connection of two capacitors.

(a) Two capacitors in series

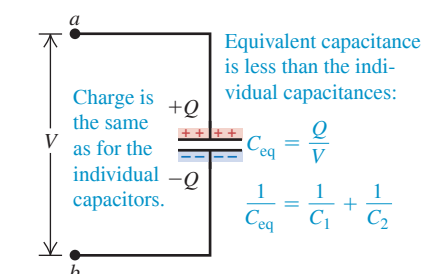
Capacitors in series:

- The capacitors have the same charge Q .
- Their potential differences add:

$$V_{ac} + V_{cb} = V_{ab}$$



(b) The equivalent single capacitor



TEST YOUR UNDERSTANDING OF SECTION 24.1 A capacitor has vacuum in the space between the conductors. If you double the amount of charge on each conductor, what happens to the capacitance? (i) It increases; (ii) it decreases; (iii) it remains the same; (iv) the answer depends on the size or shape of the conductors.

ANSWER

(iii) The capacitance does not depend on the value of the charge Q . Doubling Q causes the potential difference V_{ab} to double, so the capacitance $C = Q/V_{ab}$ remains the same. These statements are true no matter what the geometry of the capacitor.

24.2 CAPACITORS IN SERIES AND PARALLEL

Capacitors are manufactured with certain standard capacitances and working voltages (Fig. 24.7). However, these standard values may not be the ones you actually need in a particular application. You can obtain the values you need by combining capacitors; many combinations are possible, but the simplest combinations are a series connection and a parallel connection.

Capacitors in Series

Figure 24.8a is a schematic diagram of a **series connection**. Two capacitors are connected in series (one after the other) by conducting wires between points a and b . Both capacitors are initially uncharged. When a constant positive potential difference V_{ab} is applied between points a and b , the capacitors become charged; the figure shows that the charge on *all* conducting plates has the same magnitude. To see why, note first that the top plate of C_1 acquires a positive charge Q . The electric field of this positive charge pulls negative charge up to the bottom plate of C_1 until all of the field lines that begin on the top plate end on the bottom plate. This requires that the bottom plate have charge $-Q$. These negative charges had to come from the top plate of C_2 , which becomes positively charged with charge $+Q$. This positive charge then pulls negative charge $-Q$ from the connection at point b onto the bottom plate of C_2 . The total charge on the lower plate of C_1 and the upper plate of C_2 together must always be zero because these plates aren't connected to anything except each other. Thus *in a series connection the magnitude of charge on all plates is the same*.

Referring to Fig. 24.8a, we can write the potential differences between points a and c , c and b , and a and b as

$$V_{ac} = V_1 = \frac{Q}{C_1}, \quad V_{cb} = V_2 = \frac{Q}{C_2}, \quad V_{ab} = V = V_1 + V_2 = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

and so

$$\frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} \quad (24.3)$$

Following a common convention, we use the symbols V_1 , V_2 , and V to denote the potential differences V_{ac} (across the first capacitor), V_{cb} (across the second capacitor), and V_{ab} (across the entire combination of capacitors), respectively.

The **equivalent capacitance** C_{eq} of the series combination is defined as the capacitance of a *single* capacitor for which the charge Q is the same as for the combination, when the potential difference V is the same. In other words, the combination can be replaced by an *equivalent capacitor* of capacitance C_{eq} . For such a capacitor, shown in Fig. 24.8b,

$$C_{eq} = \frac{Q}{V} \quad \text{or} \quad \frac{1}{C_{eq}} = \frac{V}{Q} \quad (24.4)$$

Combining Eqs. (24.3) and (24.4), we find

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

We can extend this analysis to any number of capacitors in series. We find the following result for the *reciprocal* of the equivalent capacitance:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \quad (24.5)$$

Capacitors in series: Equivalent capacitance of series combination Capacitances of individual capacitors

The reciprocal of the equivalent capacitance of a series combination equals the sum of the reciprocals of the individual capacitances. In a series connection the equivalent capacitance is always *less than* any individual capacitance.

CAUTION Capacitors in series The magnitude of charge is the same on all plates of all the capacitors in a series combination; however, the potential differences of the individual capacitors are *not* the same unless their individual capacitances are the same. The potential differences of the individual capacitors add to give the total potential difference across the series combination: $V_{\text{total}} = V_1 + V_2 + V_3 + \cdots$!

Capacitors in Parallel

The arrangement shown in **Fig. 24.9a** is called a **parallel connection**. Two capacitors are connected in parallel between points a and b . In this case the upper plates of the two capacitors are connected by conducting wires to form an equipotential surface, and the lower plates form another. Hence *in a parallel connection the potential difference for all individual capacitors is the same* and is equal to $V_{ab} = V$. The charges Q_1 and Q_2 are not necessarily equal, however, since charges can reach each capacitor independently from the source (such as a battery) of the voltage V_{ab} . The charges are

$$Q_1 = C_1 V \quad \text{and} \quad Q_2 = C_2 V$$

The *total* charge Q of the combination, and thus the total charge on the equivalent capacitor, is

$$Q = Q_1 + Q_2 = (C_1 + C_2)V$$

so

$$\frac{Q}{V} = C_1 + C_2 \quad (24.6)$$

The parallel combination is equivalent to a single capacitor with the same total charge $Q = Q_1 + Q_2$ and potential difference V as the combination (Fig. 24.9b). The equivalent capacitance of the combination, C_{eq} , is the same as the capacitance Q/V of this single equivalent capacitor. So from Eq. (24.6),

$$C_{eq} = C_1 + C_2$$

APPLICATION Touch Screens and Capacitance

The touch screen on a mobile phone, a tablet, or (as shown here) a medical device uses the physics of capacitors. Behind the screen are two parallel layers, one behind the other, of thin strips of a transparent conductor such as indium tin oxide. A voltage is maintained between the two layers. The strips in one layer are oriented perpendicular to those in the other layer; the points where two strips overlap act as a grid of capacitors. When you bring your finger (a conductor) up to a point on the screen, your finger and the front conducting layer act like a second capacitor in series at that point. The circuitry attached to the conducting layers detects the location of the capacitance change, and so detects where you touched the screen.

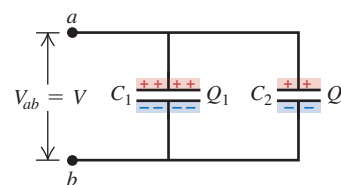


Figure 24.9 A parallel connection of two capacitors.

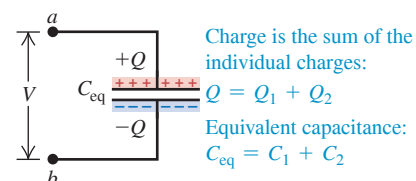
(a) Two capacitors in parallel

Capacitors in parallel:

- The capacitors have the same potential V .
- The charge on each capacitor depends on its capacitance: $Q_1 = C_1 V$, $Q_2 = C_2 V$.



(b) The equivalent single capacitor



In the same way we can show that for any number of capacitors in parallel,

$$\text{Capacitors in parallel: } C_{\text{eq}} = C_1 + C_2 + C_3 + \cdots \quad (24.7)$$

Equivalent capacitance of parallel combination Capacitances of individual capacitors

The equivalent capacitance of a parallel combination equals the *sum* of the individual capacitances. In a parallel connection the equivalent capacitance is always *greater than* any individual capacitance.

CAUTION Capacitors in parallel The potential differences are the same for all capacitors in a parallel combination; however, the charges on individual capacitors are *not* the same unless their individual capacitances are the same. The charges on the individual capacitors add to give the total charge on the parallel combination: $Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \cdots$. [Compare these statements to those in the “Caution” paragraph following Eq. (24.5).]

PROBLEM-SOLVING STRATEGY 24.1 Equivalent Capacitance

IDENTIFY *the relevant concepts:* The concept of equivalent capacitance is useful whenever two or more capacitors are connected.

SET UP *the problem* using the following steps:

1. Make a drawing of the capacitor arrangement.
2. Identify all groups of capacitors that are connected in series or in parallel.
3. Keep in mind that when we say a capacitor “has charge Q ,” we mean that the plate at higher potential has charge $+Q$ and the other plate has charge $-Q$.

EXECUTE *the solution* as follows:

1. Use Eq. (24.5) to find the equivalent capacitance of capacitors connected in series, as in Fig. 24.8. Such capacitors each have the *same charge* if they were uncharged before they were connected; that charge is the same as that on the equivalent capacitor. The potential difference across the combination is the sum of the potential differences across the individual capacitors.

2. Use Eq. (24.7) to find the equivalent capacitance of capacitors connected in parallel, as in Fig. 24.9. Such capacitors all have the *same potential difference* across them; that potential difference is the same as that across the equivalent capacitor. The total charge on the combination is the sum of the charges on the individual capacitors.
3. After replacing all the series or parallel groups you initially identified, you may find that more such groups reveal themselves. Replace those groups by using the same procedure as above until no more replacements are possible. If you then need to find the charge or potential difference for an individual original capacitor, you may have to retrace your steps.

EVALUATE *your answer:* Check whether your result makes sense. If the capacitors are connected in series, the equivalent capacitance C_{eq} must be *smaller* than any of the individual capacitances. If the capacitors are connected in parallel, C_{eq} must be *greater* than any of the individual capacitances.

EXAMPLE 24.5 Capacitors in series and in parallel

WITH VARIATION PROBLEMS

In Figs. 24.8 and 24.9, let $C_1 = 6.0 \mu\text{F}$, $C_2 = 3.0 \mu\text{F}$, and $V_{ab} = 18 \text{ V}$. Find the equivalent capacitance and the charge and potential difference for each capacitor when the capacitors are connected (a) in series (see Fig. 24.8) and (b) in parallel (see Fig. 24.9).

IDENTIFY and SET UP In both parts of this example a target variable is the equivalent capacitance C_{eq} , which is given by Eq. (24.5) for the series combination in part (a) and by Eq. (24.7) for the parallel combination in part (b). In each part we find the charge and potential difference from the definition of capacitance, Eq. (24.1), and the rules outlined in Problem-Solving Strategy 24.1.

EXECUTE (a) From Eq. (24.5) for a series combination,

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{6.0 \mu\text{F}} + \frac{1}{3.0 \mu\text{F}} \quad C_{\text{eq}} = 2.0 \mu\text{F}$$

The charge Q on each capacitor in series is the same as that on the equivalent capacitor:

$$Q = C_{\text{eq}}V = (2.0 \mu\text{F})(18 \text{ V}) = 36 \mu\text{C}$$

The potential difference across each capacitor is inversely proportional to its capacitance:

$$V_{ac} = V_1 = \frac{Q}{C_1} = \frac{36 \mu\text{C}}{6.0 \mu\text{F}} = 6.0 \text{ V}$$

$$V_{cb} = V_2 = \frac{Q}{C_2} = \frac{36 \mu\text{C}}{3.0 \mu\text{F}} = 12.0 \text{ V}$$

(b) From Eq. (24.7) for a parallel combination,

$$C_{\text{eq}} = C_1 + C_2 = 6.0 \mu\text{F} + 3.0 \mu\text{F} = 9.0 \mu\text{F}$$

The potential difference across each of the capacitors is the same as that across the equivalent capacitor, 18 V. The charge on each capacitor is directly proportional to its capacitance:

$$Q_1 = C_1V = (6.0 \mu\text{F})(18 \text{ V}) = 108 \mu\text{C}$$

$$Q_2 = C_2V = (3.0 \mu\text{F})(18 \text{ V}) = 54 \mu\text{C}$$

EVALUATE As expected, the equivalent capacitance C_{eq} for the series combination in part (a) is less than either C_1 or C_2 , while that for the parallel combination in part (b) is greater than either C_1 or C_2 . For two capacitors in series, as in part (a), the charge is the same on either capacitor and the *larger* potential difference appears across the capacitor with the *smaller* capacitance. Furthermore, the sum of the potential differences across the individual capacitors in series equals the potential difference across the equivalent capacitor: $V_{ac} + V_{cb} = V_{ab} = 18 \text{ V}$. By contrast, for two capacitors in parallel, as in part (b), each capacitor has the same potential difference and the *larger* charge appears on the capacitor with the *larger* capacitance. Can you show that the total

charge $Q_1 + Q_2$ on the parallel combination is equal to the charge $Q = C_{eq}V$ on the equivalent capacitor?

KEYCONCEPT When capacitors are combined in *series*, the charge is the same on each capacitor but the potential differences across the capacitors may not be the same. The reciprocal of the equivalent capacitance of a series combination equals the sum of the reciprocals of the individual capacitances. When capacitors are combined in *parallel*, the potential difference across each capacitor is the same but the charges on the capacitors may not be the same. The equivalent capacitance of a parallel combination equals the sum of the individual capacitances.

EXAMPLE 24.6 A capacitor network

WITH VARIATION PROBLEMS

Find the equivalent capacitance of the five-capacitor network shown in Fig. 24.10a.

IDENTIFY and SET UP These capacitors are neither all in series nor all in parallel. We can, however, identify portions of the arrangement that *are* either in series or parallel. We combine these as described in Problem-Solving Strategy 24.1 to find the net equivalent capacitance, using Eq. (24.5) for series connections and Eq. (24.7) for parallel connections.

EXECUTE The caption of Fig. 24.10 outlines our procedure. We first use Eq. (24.5) to replace the $12 \mu\text{F}$ and $6 \mu\text{F}$ series combination by its equivalent capacitance C' :

$$\frac{1}{C'} = \frac{1}{12 \mu\text{F}} + \frac{1}{6 \mu\text{F}} \quad C' = 4 \mu\text{F}$$

This gives us the equivalent combination of Fig. 24.10b. Now we see three capacitors in parallel, and we use Eq. (24.7) to replace them with their equivalent capacitance C'' :

$$C'' = 3 \mu\text{F} + 11 \mu\text{F} + 4 \mu\text{F} = 18 \mu\text{F}$$

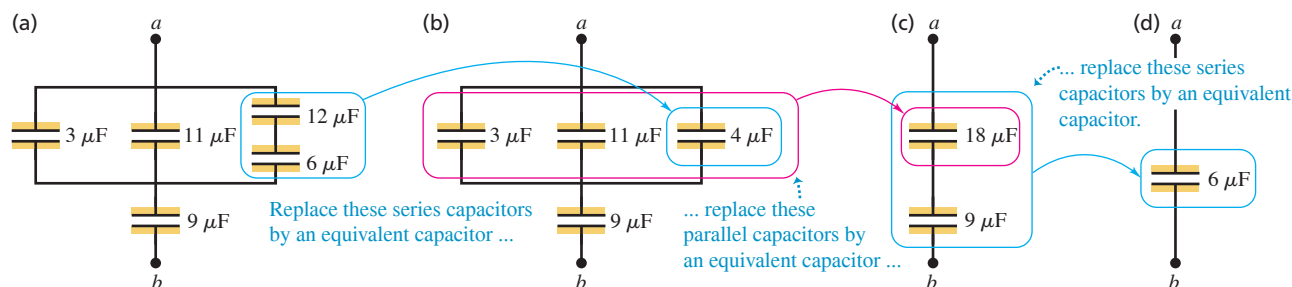
This gives us the equivalent combination of Fig. 24.10c, which has two capacitors in series. We use Eq. (24.5) to replace them with their equivalent capacitance C_{eq} , which is our target variable (Fig. 24.10d):

$$\frac{1}{C_{eq}} = \frac{1}{18 \mu\text{F}} + \frac{1}{9 \mu\text{F}} \quad C_{eq} = 6 \mu\text{F}$$

EVALUATE If the potential difference across the entire network in Fig. 24.10a is $V_{ab} = 9.0 \text{ V}$, the net charge on the network is $Q = C_{eq}V_{ab} = (6 \mu\text{F})(9.0 \text{ V}) = 54 \mu\text{C}$. Can you find the charge on, and the voltage across, each of the five individual capacitors?

KEYCONCEPT To analyze any network of capacitors, first identify which portions of the network are in series and which are in parallel. Then use the rules for series and parallel capacitors to determine the equivalent capacitance of the network.

Figure 24.10 (a) A capacitor network between points a and b . (b) The $12 \mu\text{F}$ and $6 \mu\text{F}$ capacitors in series in (a) are replaced by an equivalent $4 \mu\text{F}$ capacitor. (c) The $3 \mu\text{F}$, $11 \mu\text{F}$, and $4 \mu\text{F}$ capacitors in parallel in (b) are replaced by an equivalent $18 \mu\text{F}$ capacitor. (d) Finally, the $18 \mu\text{F}$ and $9 \mu\text{F}$ capacitors in series in (c) are replaced by an equivalent $6 \mu\text{F}$ capacitor.



TEST YOUR UNDERSTANDING OF SECTION 24.2 (a) How should you connect a $4 \mu\text{F}$ capacitor and an $8 \mu\text{F}$ capacitor so that the $4 \mu\text{F}$ capacitor has a greater potential difference across it than the $8 \mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel. (b) How should you connect them so that the $4 \mu\text{F}$ capacitor has a greater charge than the $8 \mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel.

ANSWER (i) In a series connection the two capacitors carry the same charge Q but have different potential differences $V_{ab} = Q/C$; the capacitor with the smaller capacitance C has the greater potential difference. In a parallel connection the two capacitors have the same potential difference V_{ab} but carry different charges $Q = CV_{ab}$; the capacitor with the larger capacitance C has the greater charge. Hence a $4 \mu\text{F}$ capacitor will have a greater potential difference than an $8 \mu\text{F}$ capacitor if the two are connected in series. The $4 \mu\text{F}$ capacitor cannot carry more charge than the $8 \mu\text{F}$ capacitor no matter how they are connected. In a series connection they will carry the same charge, and in a parallel connection the $8 \mu\text{F}$ capacitor will carry more charge.

24.3 ENERGY STORAGE IN CAPACITORS AND ELECTRIC-FIELD ENERGY

Many of the most important applications of capacitors depend on their ability to store energy. The electric potential energy stored in a charged capacitor is just equal to the amount of work required to charge it—that is, to separate opposite charges and place them on different conductors. When the capacitor is discharged, this stored energy is recovered as work done by electrical forces.

We can calculate the potential energy U of a charged capacitor by calculating the work W required to charge it. Suppose that when we are done charging the capacitor, the final charge is Q and the final potential difference is V . From Eq. (24.1) these quantities are related by

$$V = \frac{Q}{C}$$

Let q and v be the charge and potential difference, respectively, at an intermediate stage during the charging process; then $v = q/C$. At this stage the work dW required to transfer an additional element of charge dq is

$$dW = v dq = \frac{q dq}{C}$$

The total work W needed to increase the capacitor charge q from zero to Q is

$$W = \int_0^Q dW = \frac{1}{C} \int_0^Q q dq = \frac{Q^2}{2C} \quad (\text{work to charge a capacitor}) \quad (24.8)$$

This is also the total work done by the electric field on the charge when the capacitor discharges. Then q decreases from an initial value Q to zero as the elements of charge dq “fall” through potential differences v that vary from V down to zero.

If we define the potential energy of an *uncharged* capacitor to be zero, then W in Eq. (24.8) is equal to the potential energy U of the charged capacitor. The final stored charge is $Q = CV$, so we can express U (which is equal to W) as

$$\text{Potential energy stored in a capacitor} \quad U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV \quad (24.9)$$

Magnitude of charge on each plate
Capacitance Potential difference between plates

When Q is in coulombs, C in farads (coulombs per volt), and V in volts (joules per coulomb), U is in joules.

The last form of Eq. (24.9), $U = \frac{1}{2} QV$, shows that the total work W required to charge the capacitor is equal to the total charge Q multiplied by the *average* potential difference $\frac{1}{2} V$ during the charging process.

The expression $U = \frac{1}{2} (Q^2/C)$ in Eq. (24.9) shows that a charged capacitor is the electrical analog of a stretched spring with elastic potential energy $U = \frac{1}{2} kx^2$. The charge Q is analogous to the elongation x , and the *reciprocal* of the capacitance, $1/C$, is analogous to the force constant k . The energy supplied to a capacitor in the charging process is analogous to the work we do on a spring when we stretch it.

Equations (24.8) and (24.9) tell us that capacitance measures the ability of a capacitor to store both energy and charge. If a capacitor is charged by connecting it to a battery or other source that provides a fixed potential difference V , then increasing the value of C gives a greater charge $Q = CV$ and a greater amount of stored energy $U = \frac{1}{2} CV^2$. If instead the goal is to transfer a given quantity of charge Q from one conductor to another, Eq. (24.8) shows that the work W required is inversely proportional to C ; the greater the capacitance, the easier it is to give a capacitor a fixed amount of charge.

Applications of Capacitors: Energy Storage

Most practical applications of capacitors take advantage of their ability to store and release energy. In an electronic flash unit used in photography, the energy stored in a capacitor (see Fig. 24.4) is released when the button is pressed to take a photograph. This provides a conducting path from one capacitor plate to the other through the flash tube. Once this path is established, the stored energy is rapidly converted into a brief but intense flash of light. An extreme example of the same principle is the Z machine at Sandia National Laboratories in New Mexico, which is used in experiments in controlled nuclear fusion (**Fig. 24.11**). A bank of charged capacitors releases more than a million joules of energy in just a few billionths of a second. For that brief space of time, the power output of the Z machine is 2.9×10^{14} W, or about 80 times the power output of all the electric power plants on earth combined!

In other applications, the energy is released more slowly. Springs in the suspension of an automobile help smooth out the ride by absorbing the energy from sudden jolts and releasing that energy gradually; in an analogous way, a capacitor in an electronic circuit can smooth out unwanted variations in voltage due to power surges. We'll discuss these circuits in detail in Chapter 26.

Electric-Field Energy

We can charge a capacitor by moving electrons directly from one plate to another. This requires doing work against the electric field between the plates. Thus we can think of the energy as being stored *in the field* in the region between the plates. To see this, let's find the energy *per unit volume* in the space between the plates of a parallel-plate capacitor with plate area A and separation d . We call this the **energy density**, denoted by u . From Eq. (24.9) the total stored potential energy is $\frac{1}{2}CV^2$ and the volume between the plates is Ad ; hence

$$u = \text{Energy density} = \frac{\frac{1}{2}CV^2}{Ad} \quad (24.10)$$

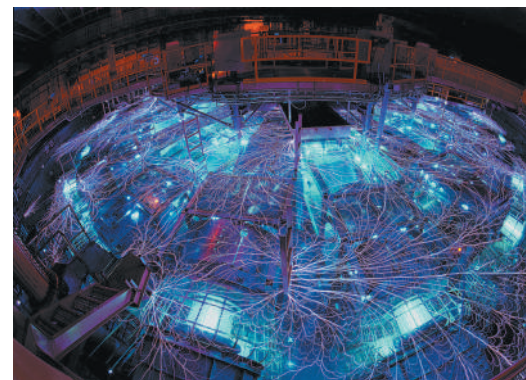
From Eq. (24.2) the capacitance C is given by $C = \epsilon_0 A/d$. The potential difference V is related to the electric-field magnitude E by $V = Ed$. If we use these expressions in Eq. (24.10), the geometric factors A and d cancel, and we find

$$\text{Electric energy density in a vacuum} \quad u = \frac{1}{2}\epsilon_0 E^2 \quad \text{Magnitude of electric field} \quad \text{Electric constant} \quad (24.11)$$

Although we have derived this relationship for a parallel-plate capacitor only, it turns out to be valid for any capacitor in vacuum and indeed *for any electric-field configuration in vacuum*. This result has an interesting implication. We think of vacuum as space with no matter in it, but vacuum can nevertheless have electric fields and therefore energy. Thus “empty” space need not be truly empty after all. We'll use this idea and Eq. (24.11) in Chapter 32 in connection with the energy transported by electromagnetic waves.

CAUTION **Electric-field energy is electric potential energy** It's a common misconception that electric-field energy is a new kind of energy, different from the electric potential energy described before. This is *not* the case; it is simply a different way of interpreting electric potential energy. We can regard the energy of a given system of charges as being a shared property of all the charges, or we can think of the energy as being a property of the electric field that the charges create. Either interpretation leads to the same value of the potential energy. ■

Figure 24.11 The Z machine uses a large number of capacitors in parallel to give a tremendous equivalent capacitance C (see Section 24.2). Hence a large amount of energy $U = \frac{1}{2}CV^2$ can be stored with even a modest potential difference V . The arcs shown here are produced when the capacitors discharge their energy into a target, which is no larger than a spool of thread. This heats the target to a temperature higher than 2×10^9 K.



EXAMPLE 24.7 Transferring charge and energy between capacitors**WITH VARIATION PROBLEMS**

We connect a capacitor $C_1 = 8.0 \mu\text{F}$ to a power supply, charge it to a potential difference $V_0 = 120 \text{ V}$, and disconnect the power supply (**Fig. 24.12**). Switch S is open. (a) What is the charge Q_0 on C_1 ? (b) What is the energy stored in C_1 ? (c) Capacitor $C_2 = 4.0 \mu\text{F}$ is initially uncharged. We close switch S . After charge no longer flows, what is the potential difference across each capacitor, and what is the charge on each capacitor? (d) What is the final energy of the system?

IDENTIFY and SET UP In parts (a) and (b) we find the charge Q_0 and stored energy U_{initial} for the single charged capacitor C_1 from Eqs. (24.1) and (24.9), respectively. After we close switch S , one wire connects the upper plates of the two capacitors and another wire connects the lower plates; the capacitors are now connected in parallel. In part (c) we use the character of the parallel connection to determine how Q_0 is shared between the two capacitors. In part (d) we again use Eq. (24.9) to find the energy stored in capacitors C_1 and C_2 ; the energy of the system is the sum of these values.

EXECUTE (a) The initial charge Q_0 on C_1 is

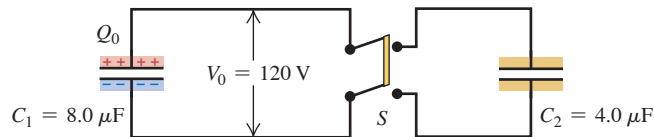
$$Q_0 = C_1 V_0 = (8.0 \mu\text{F})(120 \text{ V}) = 960 \mu\text{C}$$

(b) The energy initially stored in C_1 is

$$U_{\text{initial}} = \frac{1}{2} Q_0 V_0 = \frac{1}{2} (960 \times 10^{-6} \text{ C})(120 \text{ V}) = 0.058 \text{ J}$$

(c) When we close the switch, the positive charge Q_0 is distributed over the upper plates of both capacitors and the negative charge $-Q_0$ is distributed over the lower plates. Let Q_1 and Q_2 be the magnitudes of the final charges on the capacitors. Conservation of charge requires that $Q_1 + Q_2 = Q_0$. The potential difference V between the plates is the same for both capacitors because they are connected in parallel, so the charges are $Q_1 = C_1 V$ and $Q_2 = C_2 V$. We now have three independent equations relating the three unknowns Q_1 , Q_2 , and V . Solving these, we find

Figure 24.12 When the switch S is closed, the charged capacitor C_1 is connected to an uncharged capacitor C_2 . The center part of the switch is an insulating handle; charge can flow only between the two upper terminals and between the two lower terminals.



$$V = \frac{Q_0}{C_1 + C_2} = \frac{960 \mu\text{C}}{8.0 \mu\text{F} + 4.0 \mu\text{F}} = 80 \text{ V}$$

$$Q_1 = 640 \mu\text{C} \quad Q_2 = 320 \mu\text{C}$$

(d) The final energy of the system is

$$U_{\text{final}} = \frac{1}{2} Q_1 V + \frac{1}{2} Q_2 V = \frac{1}{2} Q_0 V \\ = \frac{1}{2} (960 \times 10^{-6} \text{ C})(80 \text{ V}) = 0.038 \text{ J}$$

EVALUATE The final energy is less than the initial energy; the difference was converted to energy of some other form. The conductors become a little warmer because of their resistance, and some energy is radiated as electromagnetic waves. We'll study the behavior of capacitors in more detail in Chapters 26 and 31.

KEYCONCEPT There are three useful ways to write the amount of electric potential energy stored in a charged capacitor: in terms of the charge Q on the plates and the potential difference V between the plates, in terms of Q and the capacitance C , and in terms of V and C .

EXAMPLE 24.8 Electric-field energy**WITH VARIATION PROBLEMS**

(a) What is the magnitude of the electric field required to store 1.00 J of electric potential energy in a volume of 1.00 m^3 in vacuum? (b) If the field magnitude is 10 times larger than that, how much energy is stored per cubic meter?

IDENTIFY and SET UP We use the relationship between the electric-field magnitude E and the energy density u . In part (a) we use the given information to find u ; then we use Eq. (24.11) to find the corresponding value of E . In part (b), Eq. (24.11) tells us how u varies with E .

EXECUTE (a) The desired energy density is $u = 1.00 \text{ J/m}^3$. Then from Eq. (24.11),

$$E = \sqrt{\frac{2u}{\epsilon_0}} = \sqrt{\frac{2(1.00 \text{ J/m}^3)}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2}} \\ = 4.75 \times 10^5 \text{ N/C} = 4.75 \times 10^5 \text{ V/m}$$

(b) Equation (24.11) shows that u is proportional to E^2 . If E increases by a factor of 10, u increases by a factor of $10^2 = 100$, so the energy density becomes $u = 100 \text{ J/m}^3$.

EVALUATE Dry air can sustain an electric field of about $3 \times 10^6 \text{ V/m}$ without experiencing *dielectric breakdown*, which we'll discuss in Section 24.4. There we'll see that field magnitudes in practical insulators can be even larger than this.

KEYCONCEPT Energy is stored in any region of space where an electric field is present. The electric energy density (energy per unit volume) at a point is proportional to the square of the electric-field magnitude at that point.

EXAMPLE 24.9 Two ways to calculate energy stored in a capacitor**WITH VARIATION PROBLEMS**

The spherical capacitor described in Example 24.3 (Section 24.1) has charges $+Q$ and $-Q$ on its inner and outer conductors. Find the electric potential energy stored in the capacitor (a) by using the capacitance C found in Example 24.3 and (b) by integrating the electric-field energy density u .

IDENTIFY and SET UP We can determine the energy U stored in a capacitor in two ways: in terms of the work done to put the charges on the two conductors, and in terms of the energy in the electric field between the conductors. The descriptions are equivalent, so they must give us the same result. In Example 24.3 we found the capacitance C and the field magnitude E in the space between the conductors. (The electric field is zero inside the inner sphere and is also zero outside the inner surface of the outer sphere, because a Gaussian surface with radius $r < r_a$ or $r > r_b$ encloses zero net charge. Hence the energy density is nonzero only in the space between the spheres, $r_a < r < r_b$.) In part (a) we use Eq. (24.9) to find U . In part (b) we use Eq. (24.11) to find u , which we integrate over the volume between the spheres to find U .

EXECUTE (a) From Example 24.3, the spherical capacitor has capacitance

$$C = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$$

where r_a and r_b are the radii of the inner and outer conducting spheres, respectively. From Eq. (24.9) the energy stored in this capacitor is

$$U = \frac{Q^2}{2C} = \frac{Q^2}{8\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b}$$

(b) The electric field in the region $r_a < r < r_b$ between the two conducting spheres has magnitude $E = Q/4\pi\epsilon_0 r^2$. The energy density in this region is

$$u = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 r^2} \right)^2 = \frac{Q^2}{32\pi^2\epsilon_0 r^4}$$

The energy density is *not* uniform; it decreases rapidly with increasing distance from the center of the capacitor. To find the total electric-field energy, we integrate u (the energy per unit volume) over the region $r_a < r < r_b$. We divide this region into spherical shells of radius r , surface area $4\pi r^2$, thickness dr , and volume $dV = 4\pi r^2 dr$. Then

$$\begin{aligned} U &= \int u dV = \int_{r_a}^{r_b} \left(\frac{Q^2}{32\pi^2\epsilon_0 r^4} \right) 4\pi r^2 dr \\ &= \frac{Q^2}{8\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{Q^2}{8\pi\epsilon_0} \left(-\frac{1}{r_b} + \frac{1}{r_a} \right) = \frac{Q^2}{8\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b} \end{aligned}$$

EVALUATE Electric potential energy can be associated with either the *charges*, as in part (a), or the *field*, as in part (b); the calculated amount of stored energy is the same in either case.

KEYCONCEPT An alternative way to calculate the electric potential energy stored in a capacitor is to first find the electric field at each point in the space between the conducting plates. From this, find the electric energy density at each point. Finally, integrate the electric energy density over the entire volume where the electric field is present.

TEST YOUR UNDERSTANDING OF SECTION 24.3 You want to connect a $4\ \mu\text{F}$ capacitor and an $8\ \mu\text{F}$ capacitor. With which type of connection will the $4\ \mu\text{F}$ capacitor have a greater amount of *stored energy* than the $8\ \mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel.

ANSWER

(i) Capacitors connected in series carry the same charge Q . To compare the amount of energy stored, we use the expression $U = Q^2/2C$ from Eq. (24.9); it shows that the capacitor with the *smaller* capacitance ($C = 4\ \mu\text{F}$) has more stored energy in a series combination. By contrast, capacitors in parallel have the same potential difference V , so to compare them we use $U = \frac{1}{2}CV^2$ from Eq. (24.9). It shows that in a parallel combination, the capacitor with the *larger* capacitance ($C = 8\ \mu\text{F}$) has more stored energy. (If we had instead used $U = \frac{1}{2}CV^2$ to analyze the series combination, we would have to account for the different potential differences across the two capacitors. Likewise, using $U = Q^2/2C$ to study the parallel combination would require us to account for the different charges on the capacitors.)

24.4 DIELECTRICS

Most capacitors have a nonconducting material, or **dielectric**, between their conducting plates. A common type of capacitor uses long strips of metal foil for the plates, separated by strips of plastic sheet such as Mylar. A sandwich of these materials is rolled up, forming a unit that can provide a capacitance of several microfarads in a compact package (**Fig. 24.13**).

Placing a solid dielectric between the plates of a capacitor serves three functions. First, it solves the mechanical problem of maintaining two large metal sheets at a very small separation without actual contact.

Second, using a dielectric increases the maximum possible potential difference between the capacitor plates. Any insulating material, when subjected to a sufficiently large electric field, experiences a partial ionization that permits conduction through it (Section 23.3).

Figure 24.13 A common type of capacitor uses dielectric sheets to separate the conductors.

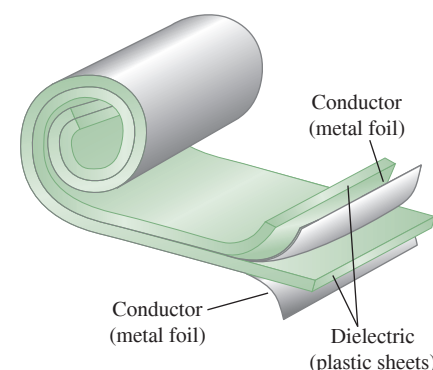
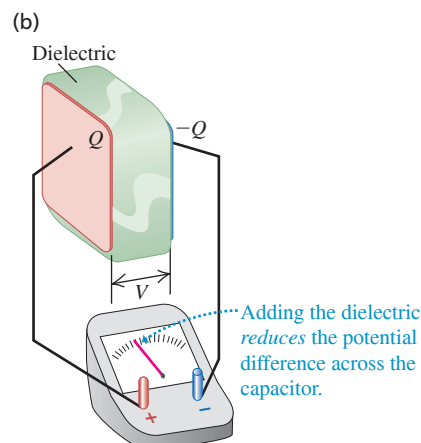
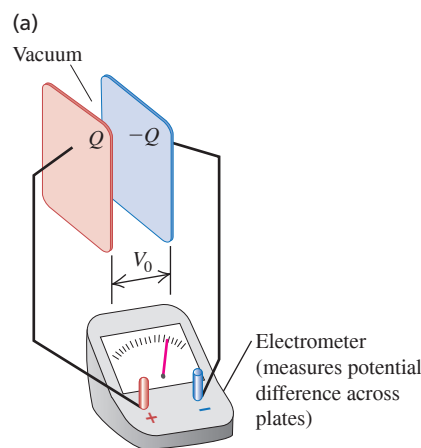


Figure 24.14 Effect of a dielectric between the plates of a parallel-plate capacitor. (a) With a given charge, the potential difference is V_0 . (b) With the same charge but with a dielectric between the plates, the potential difference V is smaller than V_0 .



This phenomenon is called **dielectric breakdown**. Many dielectric materials can tolerate stronger electric fields without breakdown than can air. Thus using a dielectric allows a capacitor to sustain a higher potential difference V and so store greater amounts of charge and energy.

Third, the capacitance of a capacitor of given dimensions is *greater* when there is a dielectric material between the plates than when there is vacuum. We can demonstrate this effect with the aid of a sensitive *electrometer*, a device that measures the potential difference between two conductors without letting any appreciable charge flow from one to the other. **Figure 24.14a** shows an electrometer connected across a charged capacitor, with magnitude of charge Q on each plate and potential difference V_0 . When we insert an uncharged sheet of dielectric, such as glass, paraffin, or polystyrene, between the plates, experiment shows that the potential difference *decreases* to a smaller value V (Fig. 24.14b). When we remove the dielectric, the potential difference returns to its original value V_0 , showing that the original charges on the plates have not changed.

The original capacitance C_0 is given by $C_0 = Q/V_0$, and the capacitance C with the dielectric present is $C = Q/V$. The charge Q is the same in both cases, and V is less than V_0 , so we conclude that the capacitance C with the dielectric present is *greater* than C_0 . When the space between plates is completely filled by the dielectric, the ratio of C to C_0 (equal to the ratio of V_0 to V) is called the **dielectric constant** of the material, K :

$$K = \frac{C}{C_0} \quad (\text{definition of dielectric constant}) \quad (24.12)$$

When the charge is constant, $Q = C_0 V_0 = CV$ and $C/C_0 = V_0/V$. In this case,

$$V = \frac{V_0}{K} \quad (\text{when } Q \text{ is constant}) \quad (24.13)$$

With the dielectric present, the potential difference for a given charge Q is *reduced* by a factor K .

The dielectric constant K is a pure number. Because C is always greater than C_0 , K is always greater than unity. **Table 24.1** gives some representative values of K . For vacuum, $K = 1$ by definition. For air at ordinary temperatures and pressures, K is about 1.0006; this is so nearly equal to 1 that for most purposes an air capacitor is equivalent to one in vacuum. Note that while water has a very large value of K , it is usually not a very practical dielectric for use in capacitors. The reason is that while pure water is a very poor conductor, it is also an excellent ionic solvent. Any ions that are dissolved in the water will cause charge to flow between the capacitor plates, so the capacitor discharges.

CAUTION Dielectric constant vs. electric constant Don't confuse the *dielectric* constant K with the *electric* constant ϵ_0 . The value of K is a pure number with no units and is different for different materials (see Table 24.1). By contrast, ϵ_0 is a universal constant with units $C^2/N \cdot m^2$ or F/m . ■

TABLE 24.1 Values of Dielectric Constant K at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas®	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon®	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar®	3.1	Strontium titanate	310

No real dielectric is a perfect insulator. Hence there is always some *leakage current* between the charged plates of a capacitor with a dielectric. We tacitly ignored this effect in Section 24.2 when we derived expressions for the equivalent capacitances of capacitors in series, Eq. (24.5), and in parallel, Eq. (24.7). But if a leakage current flows for a long enough time to substantially change the charges from the values we used to derive Eqs. (24.5) and (24.7), those equations may no longer be accurate.

Induced Charge and Polarization

When a dielectric material is inserted between the plates while the charge is kept constant, the potential difference between the plates decreases by a factor K . Therefore the electric field between the plates must decrease by the same factor. If E_0 is the vacuum value and E is the value with the dielectric, then

$$E = \frac{E_0}{K} \quad (\text{when } Q \text{ is constant}) \quad (24.14)$$

Since the electric-field magnitude is smaller when the dielectric is present, the surface charge density (which causes the field) must be smaller as well. The surface charge on the conducting plates does not change, but an *induced* charge of the opposite sign appears on each surface of the dielectric (Fig. 24.15). The dielectric was originally electrically neutral and is still neutral; the induced surface charges arise as a result of *redistribution* of positive and negative charges within the dielectric material, a phenomenon called **polarization**. We first encountered polarization in Section 21.2, and we suggest that you reread the discussion of Fig. 21.8. We'll assume that the induced surface charge is *directly proportional* to the electric-field magnitude E in the material; this is indeed the case for many common dielectrics. (This direct proportionality is analogous to Hooke's law for a spring.) In that case, K is a constant for any particular material. When the electric field is very strong or if the dielectric is made of certain crystalline materials, the relationship between induced charge and the electric field can be more complex; we won't consider such cases here.

We can derive a relationship between this induced surface charge and the charge on the plates. Let σ_i be the magnitude of the charge per unit area induced on the surfaces of the dielectric (the induced surface charge density). The magnitude of the surface charge density on the capacitor plates is σ , as usual. Then the *net* surface charge on each side of the capacitor has magnitude $(\sigma - \sigma_i)$; see Fig. 24.15b. As we found in Example 21.12 (Section 21.5) and in Example 22.8 (Section 22.4), the field between the plates is related to the net surface charge density by $E = \sigma_{\text{net}}/\epsilon_0$. Without and with the dielectric, respectively,

$$E_0 = \frac{\sigma}{\epsilon_0} \quad E = \frac{\sigma - \sigma_i}{\epsilon_0} \quad (24.15)$$

Using these expressions in Eq. (24.14) and rearranging the result, we find

$$\sigma_i = \sigma \left(1 - \frac{1}{K} \right) \quad (\text{induced surface charge density}) \quad (24.16)$$

This equation shows that when K is very large, σ_i is nearly as large as σ . In this case, σ_i nearly cancels σ , and the field and potential difference are much smaller than their values in vacuum.

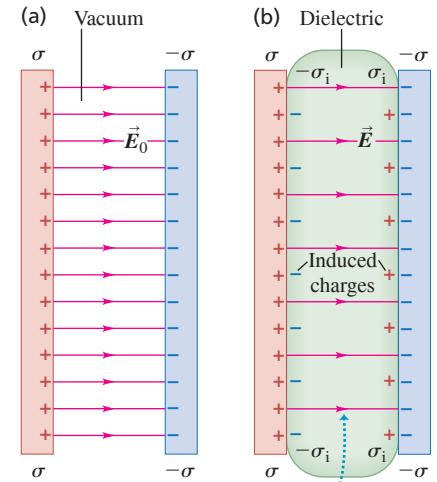
The product $K\epsilon_0$ is called the **permittivity** of the dielectric, denoted by ϵ :

$$\epsilon = K\epsilon_0 \quad (\text{definition of permittivity}) \quad (24.17)$$

In terms of ϵ we can express the electric field within the dielectric as

$$E = \frac{\sigma}{\epsilon} \quad (24.18)$$

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

APPLICATION Capacitors in the Toolbox

Several practical devices rely on the way a capacitor responds to a change in dielectric constant. One example is an electric stud finder, used to locate metal segments hidden behind a wall's surface. It consists of a metal plate with associated circuitry. The plate acts as one half of a capacitor; the wall acts as the other half. If the stud finder moves over a metal segment, the effective dielectric constant for the capacitor changes, which changes the capacitance and triggers a signal.



Then

$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d} \quad (24.19)$$

Capacitance of a parallel-plate capacitor, dielectric between plates: C
 Dielectric constant: K
 Area of each plate: A
 Distance between plates: d
 Capacitance without dielectric: C_0
 Electric constant: ϵ_0
 Permittivity: $\epsilon = K\epsilon_0$

We can repeat the derivation of Eq. (24.11) for the energy density u in an electric field for the case in which a dielectric is present. The result is

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2 \quad (24.20)$$

Electric energy density in a dielectric: u
 Dielectric constant: K
 Electric constant: ϵ_0
 Permittivity: $\epsilon = K\epsilon_0$
 Magnitude of electric field: E

CAUTION Capacitance is independent of charge or voltage Equation (24.19) shows that the capacitance C of a capacitor depends on only the capacitor geometry (the size, shape, and position of its plates) and whether or not a dielectric is present. The capacitance does *not* depend on the amount of charge placed on the plates or on the voltage between the plates! ■

In empty space, where $K = 1$, $\epsilon = \epsilon_0$ and Eqs. (24.19) and (24.20) reduce to Eqs. (24.2) and (24.11), respectively, for a parallel-plate capacitor in vacuum. For this reason, ϵ_0 is sometimes called the “permittivity of free space” or the “permittivity of vacuum.” Because K is a pure number, ϵ and ϵ_0 have the same units, $C^2/N \cdot m^2$ or F/m .

Equation (24.19) shows that extremely high capacitances can be obtained with plates that have a large surface area A and are separated by a small distance d by a dielectric with a large value of K . In an *electrolytic double-layer capacitor*, tiny carbon granules adhere to each plate: A is the combined surface area of the granules, which can be tremendous. The plates with granules attached are separated by a very thin dielectric sheet. Such a capacitor can have a capacitance of 5000 F yet fit in the palm of your hand (compare Example 24.1 in Section 24.1).

PROBLEM-SOLVING STRATEGY 24.2 Dielectrics

IDENTIFY the relevant concepts: The relationships in this section are useful whenever there is an electric field in a dielectric, such as a dielectric between charged capacitor plates. Typically you must relate the potential difference V_{ab} between the plates, the electric-field magnitude E in the capacitor, the charge density σ on the capacitor plates, and the induced charge density σ_i on the surfaces of the capacitor.

SET UP the problem using the following steps:

1. Make a drawing of the situation.
2. Identify the target variables, and choose which equations from this section will help you solve for those variables.

EXECUTE the solution as follows:

1. In problems such as the next example, it is easy to get lost in a blizzard of formulas. Ask yourself at each step what kind of quantity each symbol represents. For example, distinguish clearly

between charges and charge densities, and between electric fields and electric potential differences.

2. Check for consistency of units. Distances must be in meters. A microfarad is 10^{-6} farad, and so on. Don't confuse the numerical value of ϵ_0 with the value of $1/4\pi\epsilon_0$. Electric-field magnitude can be expressed in both N/C and V/m . The units of ϵ_0 are $C^2/N \cdot m^2$ or F/m .

EVALUATE your answer: With a dielectric present, (a) the capacitance is greater than without a dielectric; (b) for a given charge on the capacitor, the electric field and potential difference are less than without a dielectric; and (c) the magnitude of the induced surface charge density σ_i on the dielectric is less than that of the charge density σ on the capacitor plates.

EXAMPLE 24.10 A capacitor with and without a dielectric

WITH VARIATION PROBLEMS

Suppose the parallel plates in Fig. 24.15 each have an area of 2000 cm^2 ($2.00 \times 10^{-1} \text{ m}^2$) and are 1.00 cm ($1.00 \times 10^{-2} \text{ m}$) apart. We connect the capacitor to a power supply, charge it to a potential difference $V_0 = 3.00 \text{ kV}$, and disconnect the power supply. We then insert a sheet of insulating plastic material between the plates, completely filling the space between them. We find that the potential difference decreases to 1.00 kV while the charge on each capacitor

plate remains constant. Find (a) the original capacitance C_0 ; (b) the magnitude of charge Q on each plate; (c) the capacitance C after the dielectric is inserted; (d) the dielectric constant K of the dielectric; (e) the permittivity ϵ of the dielectric; (f) the magnitude of the induced charge Q_i on each face of the dielectric; (g) the original electric field E_0 between the plates; and (h) the electric field E after the dielectric is inserted.

IDENTIFY and SET UP This problem uses most of the relationships we have discussed for capacitors and dielectrics. (Energy relationships are treated in Example 24.11.) Most of the target variables can be obtained in several ways. The methods used below are a sample; we encourage you to think of others and compare your results.

EXECUTE (a) With vacuum between the plates, we use Eq. (24.19) with $K = 1$:

$$C_0 = \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12} \text{ F/m}) \frac{2.00 \times 10^{-1} \text{ m}^2}{1.00 \times 10^{-2} \text{ m}} \\ = 1.77 \times 10^{-10} \text{ F} = 177 \text{ pF}$$

(b) From the definition of capacitance, Eq. (24.1),

$$Q = C_0 V_0 = (1.77 \times 10^{-10} \text{ F})(3.00 \times 10^3 \text{ V}) \\ = 5.31 \times 10^{-7} \text{ C} = 0.531 \mu\text{C}$$

(c) When the dielectric is inserted, Q is unchanged but the potential difference decreases to $V = 1.00 \text{ kV}$. Hence from Eq. (24.1), the new capacitance is

$$C = \frac{Q}{V} = \frac{5.31 \times 10^{-7} \text{ C}}{1.00 \times 10^3 \text{ V}} = 5.31 \times 10^{-10} \text{ F} = 531 \text{ pF}$$

(d) From Eq. (24.12), the dielectric constant is

$$K = \frac{C}{C_0} = \frac{5.31 \times 10^{-10} \text{ F}}{1.77 \times 10^{-10} \text{ F}} = \frac{531 \text{ pF}}{177 \text{ pF}} = 3.00$$

Alternatively, from Eq. (24.13),

$$K = \frac{V_0}{V} = \frac{3000 \text{ V}}{1000 \text{ V}} = 3.00$$

(e) With K from part (d) in Eq. (24.17), the permittivity is

$$\epsilon = K\epsilon_0 = (3.00)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2) = 2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2$$

(f) Multiplying both sides of Eq. (24.16) by the plate area A gives the induced charge $Q_i = \sigma_i A$ in terms of the charge $Q = \sigma A$ on each plate:

$$Q_i = Q \left(1 - \frac{1}{K} \right) = (5.31 \times 10^{-7} \text{ C}) \left(1 - \frac{1}{3.00} \right) = 3.54 \times 10^{-7} \text{ C}$$

(g) Since the electric field between the plates is uniform, its magnitude is the potential difference divided by the plate separation:

$$E_0 = \frac{V_0}{d} = \frac{3000 \text{ V}}{1.00 \times 10^{-2} \text{ m}} = 3.00 \times 10^5 \text{ V/m}$$

(h) After the dielectric is inserted,

$$E = \frac{V}{d} = \frac{1000 \text{ V}}{1.00 \times 10^{-2} \text{ m}} = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.18),

$$E = \frac{\sigma}{\epsilon} = \frac{Q}{\epsilon A} = \frac{5.31 \times 10^{-7} \text{ C}}{(2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.00 \times 10^{-1} \text{ m}^2)} \\ = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.15),

$$E = \frac{\sigma - \sigma_i}{\epsilon_0} = \frac{Q - Q_i}{\epsilon_0 A} \\ = \frac{(5.31 - 3.54) \times 10^{-7} \text{ C}}{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.00 \times 10^{-1} \text{ m}^2)} \\ = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.14),

$$E = \frac{E_0}{K} = \frac{3.00 \times 10^5 \text{ V/m}}{3.00} = 1.00 \times 10^5 \text{ V/m}$$

EVALUATE Inserting the dielectric increased the capacitance by a factor of $K = 3.00$ and reduced the electric field between the plates by a factor of $1/K = 1/3.00$. It did so by developing induced charges on the faces of the dielectric of magnitude $Q(1 - 1/K) = Q(1 - 1/3.00) = 0.667Q$.

KEYCONCEPT Adding a dielectric (with dielectric constant K) that fills the space between the plates of a capacitor increases the capacitance by a factor of K . For a given amount of charge, adding the dielectric also reduces both the potential difference between the plates and the electric field at each point by a factor of $1/K$.

EXAMPLE 24.11 Energy storage with and without a dielectric

WITH VARIATION PROBLEMS

Find the energy stored in the electric field of the capacitor in Example 24.10 and the energy density, both before and after the dielectric sheet is inserted.

IDENTIFY and SET UP We consider the ideas of energy stored in a capacitor and of electric-field energy density. We use Eq. (24.9) to find the stored energy and Eq. (24.20) to find the energy density.

EXECUTE From Eq. (24.9), the stored energies U_0 and U without and with the dielectric in place are

$$U_0 = \frac{1}{2} C_0 V_0^2 = \frac{1}{2} (1.77 \times 10^{-10} \text{ F})(3000 \text{ V})^2 = 7.97 \times 10^{-4} \text{ J} \\ U = \frac{1}{2} C V^2 = \frac{1}{2} (5.31 \times 10^{-10} \text{ F})(1000 \text{ V})^2 = 2.66 \times 10^{-4} \text{ J}$$

The final energy is one-third of the original energy.

Equation (24.20) gives the energy densities without and with the dielectric:

$$u_0 = \frac{1}{2} \epsilon_0 E_0^2 = \frac{1}{2} (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(3.00 \times 10^5 \text{ N/C})^2 \\ = 0.398 \text{ J/m}^3 \\ u = \frac{1}{2} \epsilon E^2 = \frac{1}{2} (2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2)(1.00 \times 10^5 \text{ N/C})^2 \\ = 0.133 \text{ J/m}^3$$

The energy density with the dielectric is one-third of the original energy density.

Continued

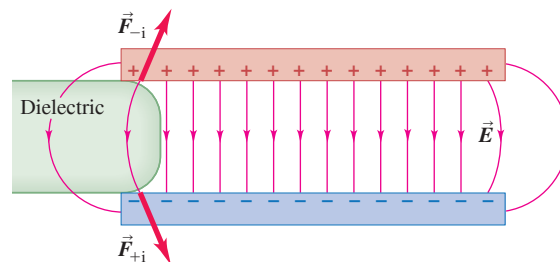
EVALUATE We can check our answer for u_0 by noting that the volume between the plates is $V_{\text{between}} = (0.200 \text{ m}^2)(0.0100 \text{ m}) = 0.00200 \text{ m}^3$. Since the electric field between the plates is uniform, u_0 is uniform as well and the energy density is just the stored energy divided by the volume:

$$u_0 = \frac{U_0}{V_{\text{between}}} = \frac{7.97 \times 10^{-4} \text{ J}}{0.00200 \text{ m}^3} = 0.398 \text{ J/m}^3$$

This agrees with our earlier answer. You can use the same approach to check our result for u .

In general, when a dielectric is inserted into a capacitor while the charge on each plate remains the same, the permittivity ϵ increases by a factor of K (the dielectric constant), and the electric field E and the energy density $u = \frac{1}{2}\epsilon E^2$ decrease by a factor of $1/K$. Where does the energy go? The answer lies in the fringing field at the edges of a real parallel-plate capacitor. As **Fig. 24.16** shows, that field tends to pull the dielectric into the space between the plates, doing work on it as it does so. We could attach a spring to the left end of the dielectric in Fig. 24.16 and use this force to stretch the spring. Because work is done by the field, the field energy density decreases.

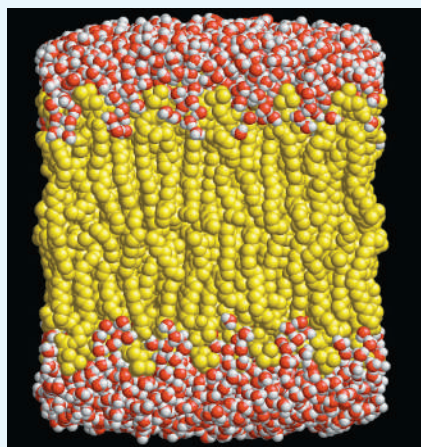
Figure 24.16 The fringing field at the edges of the capacitor exerts forces $\vec{F}_{-i}$ and $\vec{F}_{+i}$ on the negative and positive induced surface charges of a dielectric, pulling the dielectric into the capacitor.



KEYCONCEPT Adding a dielectric (with dielectric constant K) that fills the space between the plates of a capacitor reduces the electric field, the electric energy density, and the total stored energy, all by a factor of $1/K$.

BIO APPLICATION Dielectric Cell Membrane

The membrane of a living cell behaves like a dielectric between the plates of a capacitor. The membrane is made of two sheets of lipid molecules, with their water-insoluble ends in the middle and their water-soluble ends (shown in red) on the outer surfaces. Conductive fluids on either side of the membrane (water with negative ions inside the cell, water with positive ions outside) act as charged capacitor plates, and the nonconducting membrane acts as a dielectric with K of about 10. The potential difference V across the membrane is about 0.07 V and the membrane thickness d is about $7 \times 10^{-9} \text{ m}$, so the electric field $E = V/d$ in the membrane is about 10^7 V/m —close to the dielectric strength of the membrane. If the membrane were made of air, V and E would be larger by a factor of $K \approx 10$ and dielectric breakdown would occur.



Dielectric Breakdown

We mentioned earlier that when a dielectric is subjected to a sufficiently strong electric field, *dielectric breakdown* takes place and the dielectric becomes a conductor. This occurs when the electric field is so strong that electrons are ripped loose from their molecules and crash into other molecules, liberating even more electrons. This avalanche of moving charge forms a spark or arc discharge. Lightning is a dramatic example of dielectric breakdown in air.

Because of dielectric breakdown, capacitors always have maximum voltage ratings. When a capacitor is subjected to excessive voltage, an arc may form through a layer of dielectric, burning or melting a hole in it. This arc creates a conducting path (a short circuit) between the conductors. If a conducting path remains after the arc is extinguished, the device is rendered permanently useless as a capacitor.

The maximum electric-field magnitude that a material can withstand without the occurrence of breakdown is called its **dielectric strength**. This quantity is affected significantly by temperature, trace impurities, small irregularities in the metal electrodes, and other factors that are difficult to control. For this reason we can give only approximate figures for dielectric strengths. The dielectric strength of dry air is about $3 \times 10^6 \text{ V/m}$. **Table 24.2** lists the dielectric strengths of a few common insulating materials. All of the values are substantially greater than the value for air. For example, a layer of polycarbonate 0.01 mm thick (about the smallest practical thickness) has 10 times the dielectric strength of air and can withstand a maximum voltage of about $(3 \times 10^7 \text{ V/m})(1 \times 10^{-5} \text{ m}) = 300 \text{ V}$.

TABLE 24.2 Dielectric Constant and Dielectric Strength of Some Insulating Materials

Material	Dielectric Constant, K	Dielectric Strength, E_m (V/m)
Polycarbonate	2.8	3×10^7
Polyester	3.3	6×10^7
Polypropylene	2.2	7×10^7
Polystyrene	2.6	2×10^7
Pyrex® glass	4.7	1×10^7

TEST YOUR UNDERSTANDING OF SECTION 24.4 The space between the plates of an isolated parallel-plate capacitor is filled by a slab of dielectric with dielectric constant K . The two plates of the capacitor have charges Q and $-Q$. You pull out the dielectric slab. If the charges do not change, how does the energy in the capacitor change when you remove the slab? (i) It increases; (ii) it decreases; (iii) it remains the same.

ANSWER

(i) Here Q remains the same, so we use $U = Q^2/2C$ from Eq. (24.9). Removing the dielectric lowers the capacitance by a factor of $1/K$; since U is inversely proportional to C , the stored energy *increases* by a factor of K . It takes work to pull the dielectric slab out of the capacitor because the fringing field tries to pull the slab back in (Fig. 24.16). The work that you do goes into the energy stored in the capacitor.

24.5 MOLECULAR MODEL OF INDUCED CHARGE

In Section 24.4 we discussed induced surface charges on a dielectric in an electric field. Now let's look at how these surface charges can arise. If the material were a *conductor*, the answer would be simple. Conductors contain charge that is free to move, and when an electric field is present, some of the charge redistributes itself on the surface so that there is no electric field inside the conductor. But an ideal dielectric has *no* charges that are free to move, so how can a surface charge occur?

To understand this, we have to look again at rearrangement of charge at the *molecular* level. Some molecules, such as H_2O and N_2O , have equal amounts of positive and negative charges but a lopsided distribution, with excess positive charge concentrated on one side of the molecule and negative charge on the other. As we described in Section 21.7, such an arrangement is called an *electric dipole*, and the molecule is called a *polar molecule*. When no electric field is present in a gas or liquid with polar molecules, the molecules are oriented randomly (Fig. 24.17a). In an electric field, however, they tend to orient themselves as in Fig. 24.17b, as a result of the electric-field torques described in Section 21.7. Because of thermal agitation, the alignment of the molecules with $\vec{E}$ is not perfect.

Even a molecule that is *not* ordinarily polar *becomes* a dipole when it is placed in an electric field because the field pushes the positive charges in the molecules in the direction of the field and pushes the negative charges in the opposite direction. This causes a redistribution of charge within the molecule (Fig. 24.18). Such dipoles are called *induced* dipoles.

With either polar or nonpolar molecules, the redistribution of charge caused by the field leads to the formation of a layer of charge on each surface of the dielectric material (Fig. 24.19). These layers are the surface charges described in Section 24.4; their surface charge density is denoted by σ_i . The charges are *not* free to move indefinitely, as they

Figure 24.19 Polarization of a dielectric in an electric field $\vec{E}$ gives rise to thin layers of bound charges on the surfaces, creating surface charge densities σ_i and $-\sigma_i$. The sizes of the molecules are greatly exaggerated for clarity.

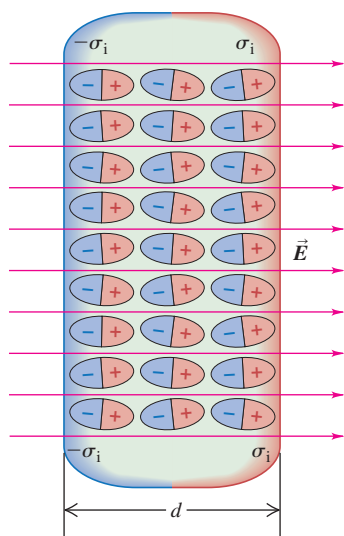


Figure 24.17 Polar molecules (a) without and (b) with an applied electric field $\vec{E}$.

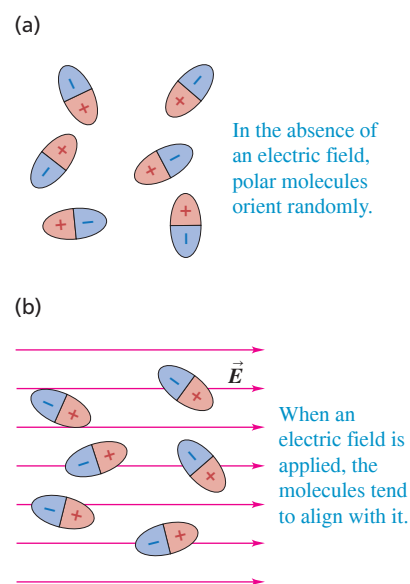
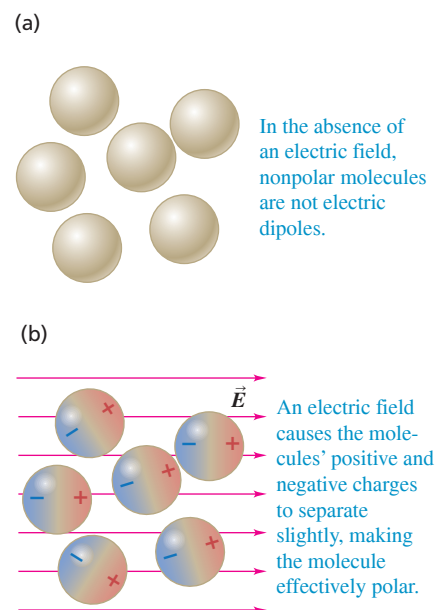


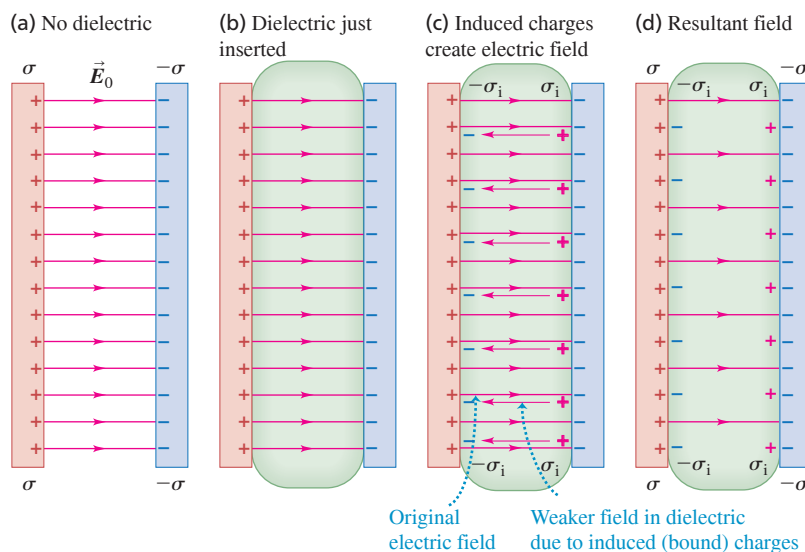
Figure 24.18 Nonpolar molecules (a) without and (b) with an applied electric field $\vec{E}$.



APPLICATION Smartphones, Capacitors, and Dielectrics A typical smartphone contains several hundred miniature capacitors, each of which uses a ceramic dielectric between its plates to boost its capacitance. These capacitors are used in every subsystem of the smartphone, including the main processor, the dynamic random access memory, the on-off and volume controls, the microphone, and the touch screen (see page 787).



Figure 24.20 (a) Electric field of magnitude E_0 between two charged plates. (b) Introduction of a dielectric of dielectric constant K . (c) The induced surface charges and their field. (d) Resultant field of magnitude E_0/K .

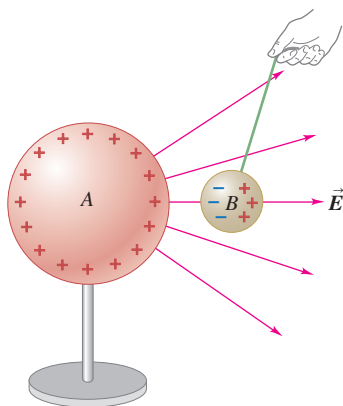


would be in a conductor, because each charge is bound to a molecule. They are in fact called **bound charges** to distinguish them from the **free charges** that are added to and removed from the conducting capacitor plates. In the interior of the material the net charge per unit volume remains zero. As we have seen, this redistribution of charge is called *polarization*, and we say that the material is *polarized*.

The four parts of Fig. 24.20 show the behavior of a slab of dielectric when it is inserted in the field between a pair of oppositely charged capacitor plates. Figure 24.20a shows the original field. Figure 24.20b is the situation after the dielectric has been inserted but before any rearrangement of charges has occurred. Figure 24.20c shows by thinner arrows the additional field set up in the dielectric by its induced surface charges. This field is *opposite* to the original field, but it is not great enough to cancel the original field completely because the charges in the dielectric are not free to move indefinitely. The resultant field in the dielectric, shown in Fig. 24.20d, is therefore decreased in magnitude. In the field-line representation, some of the field lines leaving the positive plate go through the dielectric, while others terminate on the induced charges on the faces of the dielectric.

As we discussed in Section 21.2, polarization is also the reason a charged object, such as an electrified plastic rod, can exert a force on an *uncharged* object such as a bit of paper or a pith ball. **Figure 24.21** shows an uncharged dielectric sphere B in the radial field of a positively charged object A . The induced positive charges on B experience a force toward the right, while the force on the induced negative charges is toward the left. The negative charges are closer to A , and thus are in a stronger field, than are the positive charges. The force toward the left is stronger than that toward the right, and B is attracted toward A , even though its net charge is zero. The attraction occurs whether the sign of A 's charge is positive or negative (see Fig. 21.8). Furthermore, the effect is not limited to dielectrics; an uncharged conducting object would be attracted in the same way.

Figure 24.21 A neutral sphere B in the radial electric field of a positively charged sphere A is attracted to the charge because of polarization.



TEST YOUR UNDERSTANDING OF SECTION 24.5 A parallel-plate capacitor has charges Q and $-Q$ on its two plates. A dielectric slab with $K = 3$ is then inserted into the space between the plates as shown in Fig. 24.20. Rank the following electric-field magnitudes in order from largest to smallest. (i) The field before the slab is inserted; (ii) the resultant field after the slab is inserted; (iii) the field due to the bound charges.

ANSWER

(i), (iii), (ii) Equation (24.14) says that if E_0 is the initial electric-field magnitude (before the dielectric slab is inserted), then the resultant field magnitude after the slab is inserted is $E_0/K = E_0/3$. The magnitude of the resultant field equals the difference between the initial field magnitude and the magnitude of the field due to the bound charges (see Fig. 24.20). Hence $E_0 - E_i = E_0/3$ and $E_i = 2E_0/3$.

24.6 GAUSS'S LAW IN DIELECTRICS

We can extend the analysis of Section 24.4 to reformulate Gauss's law in a form that is particularly useful for dielectrics. **Figure 24.22** is a close-up view of the left capacitor plate and left surface of the dielectric in Fig. 24.15b. Let's apply Gauss's law to the rectangular box shown in cross section by the purple line; the surface area of the left and right sides is A . The left side is embedded in the conductor that forms the left capacitor plate, and so the electric field everywhere on that surface is zero. The right side is embedded in the dielectric, where the electric field has magnitude E , and $E_{\perp} = 0$ everywhere on the other four sides. The total charge enclosed, including both the charge on the capacitor plate and the induced charge on the dielectric surface, is $Q_{\text{encl}} = (\sigma - \sigma_i)A$, so Gauss's law gives

$$EA = \frac{(\sigma - \sigma_i)A}{\epsilon_0} \quad (24.21)$$

This equation is not very illuminating as it stands because it relates two unknown quantities: E inside the dielectric and the induced surface charge density σ_i . But now we can use Eq. (24.16), developed for this same situation, to simplify this equation by eliminating σ_i . Equation (24.16) is

$$\sigma_i = \sigma \left(1 - \frac{1}{K}\right)$$

or

$$\sigma - \sigma_i = \frac{\sigma}{K}$$

Combining this with Eq. (24.21), we get

$$EA = \frac{\sigma A}{K\epsilon_0}$$

or

$$KEA = \frac{\sigma A}{\epsilon_0} \quad (24.22)$$

Equation (24.22) says that the flux of $K\vec{E}$, not $\vec{E}$, through the Gaussian surface in Fig. 24.22 is equal to the enclosed *free* charge σA divided by ϵ_0 . It turns out that for *any* Gaussian surface, whenever the induced charge is proportional to the electric field in the material, we can rewrite Gauss's law as

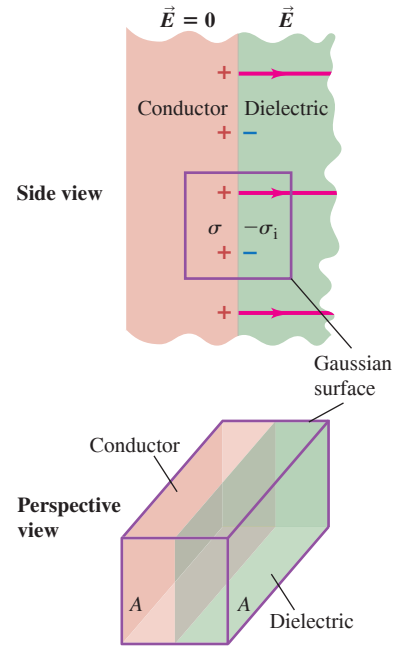
$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \quad (24.23)$$

Gauss's law in a dielectric:
Dielectric constant
Total free charge enclosed by surface
Electric constant

Surface integral of $K\vec{E}$ over a closed surface

where $Q_{\text{encl-free}}$ is the total *free* charge (not bound charge) enclosed by the Gaussian surface. The significance of these results is that the right sides contain only the *free* charge on the conductor, not the bound (induced) charge. In fact, although we have not proved it, Eq. (24.23) remains valid even when different parts of the Gaussian surface are embedded in dielectrics having different values of K , provided that the value of K in each dielectric is independent of the electric field (usually the case for electric fields that are not too strong) and that we use the appropriate value of K for each point on the Gaussian surface.

Figure 24.22 Gauss's law with a dielectric. This figure shows a close-up of the left-hand capacitor plate in Fig. 24.15b. The Gaussian surface is a rectangular box that lies half in the conductor and half in the dielectric.



EXAMPLE 24.12 A spherical capacitor with dielectric

Use Gauss's law to find the capacitance of the spherical capacitor of Example 24.3 (Section 24.1) if the volume between the shells is filled with an insulating oil with dielectric constant K .

IDENTIFY and SET UP The spherical symmetry of the problem is not changed by the presence of the dielectric, so as in Example 24.3, we use a concentric spherical Gaussian surface of radius r between the shells. Since a dielectric is present, we use Gauss's law in the form of Eq. (24.23).

EXECUTE From Eq. (24.23),

$$\oint \vec{K}\vec{E} \cdot d\vec{A} = \oint KE dA = KE \oint dA = (KE)(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi K\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon r^2}$$

where $\epsilon = K\epsilon_0$. Compared to the case in which there is vacuum between the shells, the electric field is reduced by a factor of $1/K$. The

potential difference V_{ab} between the shells is reduced by the same factor, and so the capacitance $C = Q/V_{ab}$ is *increased* by a factor of K , just as for a parallel-plate capacitor when a dielectric is inserted. Using the result of Example 24.3, we find that the capacitance with the dielectric is

$$C = \frac{4\pi K\epsilon_0 r_a r_b}{r_b - r_a} = \frac{4\pi\epsilon r_a r_b}{r_b - r_a}$$

EVALUATE If the dielectric fills the volume between the two conductors, the capacitance is just K times the value with no dielectric. The result is more complicated if the dielectric only partially fills this volume.

KEYCONCEPT Gauss's law can be written in a modified way if a dielectric is present. If the volume of space being studied is filled with a dielectric that has dielectric constant K , then the flux of the *product* of K and the electric field through a closed surface is proportional to the amount of *free* charge enclosed by the surface.

TEST YOUR UNDERSTANDING OF SECTION 24.6 A single point charge q is embedded in a very large block of dielectric of dielectric constant K . At a point inside the dielectric a distance r from the point charge, what is the magnitude of the electric field? (i) $q/4\pi\epsilon_0 r^2$; (ii) $Kq/4\pi\epsilon_0 r^2$; (iii) $q/4\pi K\epsilon_0 r^2$; (iv) none of these.

ANSWER

(iii) Equation (24.23) shows that this situation is the same as an isolated point charge in vacuum but with $\vec{E}$ replaced by $K\vec{E}$. Hence KE at the point of interest is equal to $q/4\pi\epsilon_0 r^2$, and so $E = q/4\pi K\epsilon_0 r^2$. As in Example 24.12, filling the space with a dielectric reduces the electric field by a factor of $1/K$.

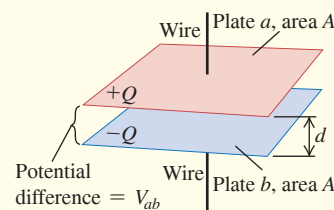
CHAPTER 24 SUMMARY

Capacitors and capacitance: A capacitor is any pair of conductors separated by an insulating material. When the capacitor is charged, there are charges of equal magnitude Q and opposite sign on the two conductors, and the potential V_{ab} of the positively charged conductor with respect to the negatively charged conductor is proportional to Q . The capacitance C is defined as the ratio of Q to V_{ab} . The SI unit of capacitance is the farad (F): $1 \text{ F} = 1 \text{ C/V}$.

A parallel-plate capacitor consists of two parallel conducting plates, each with area A , separated by a distance d . If they are separated by vacuum, the capacitance depends on only A and d . For other geometries, the capacitance can be found by using the definition $C = Q/V_{ab}$. (See Examples 24.1–24.4.)

$$C = \frac{Q}{V_{ab}} \quad (24.1)$$

$$C = \frac{Q}{V_{ab}} = \epsilon_0 \frac{A}{d} \quad (24.2)$$



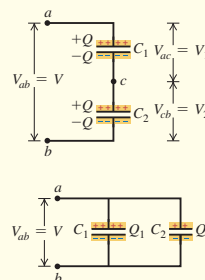
Capacitors in series and parallel: When capacitors with capacitances $C_1, C_2, C_3, \dots$ are connected in series, the reciprocal of the equivalent capacitance C_{eq} equals the sum of the reciprocals of the individual capacitances. When capacitors are connected in parallel, the equivalent capacitance C_{eq} equals the sum of the individual capacitances. (See Examples 24.5 and 24.6.)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots \quad (24.5)$$

(capacitors in series)

$$C_{eq} = C_1 + C_2 + C_3 + \dots \quad (24.7)$$

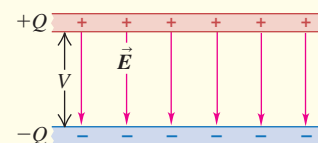
(capacitors in parallel)



Energy in a capacitor: The energy U required to charge a capacitor C to a potential difference V and a charge Q is equal to the energy stored in the capacitor. This energy can be thought of as residing in the electric field between the conductors; the energy density u (energy per unit volume) is proportional to the square of the electric-field magnitude. (See Examples 24.7–24.9.)

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV \quad (24.9)$$

$$u = \frac{1}{2}\epsilon_0 E^2 \quad (24.11)$$



Dielectrics: When the space between the conductors is filled with a dielectric material, the capacitance increases by a factor K , the dielectric constant of the material. The quantity $\epsilon = K\epsilon_0$ is the permittivity of the dielectric. For a fixed amount of charge on the capacitor plates, induced charges on the surface of the dielectric decrease the electric field and potential difference between the plates by the same factor K . The surface charge results from polarization, a microscopic rearrangement of charge in the dielectric. (See Example 24.10.)

Under sufficiently strong electric fields, dielectrics become conductors, a situation called dielectric breakdown. The maximum field that a material can withstand without breakdown is called its dielectric strength.

In a dielectric, the expression for the energy density is the same as in vacuum but with ϵ_0 replaced by $\epsilon = K\epsilon_0$. (See Example 24.11.)

Gauss's law in a dielectric has almost the same form as in vacuum, with two key differences: $\vec{E}$ is replaced by $K\vec{E}$ and Q_{encl} is replaced by $Q_{\text{encl-free}}$, which includes only the free charge (not bound charge) enclosed by the Gaussian surface. (See Example 24.12.)

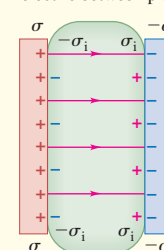
$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d} \quad (24.19)$$

(parallel-plate capacitor filled with dielectric)

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2 \quad (24.20)$$

$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \quad (24.23)$$

Dielectric between plates



Chapter 24 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 24.2, 24.3, and 24.4** (Section 24.1) before attempting these problems.

VP24.4.1 The plates of a parallel-plate capacitor in vacuum are 3.70 mm apart and 2.75 m² in area. When you apply a certain potential difference across the capacitor, the surface charge density on the positive plate is 1.40×10^{-5} C/m². Calculate (a) the capacitance of the capacitor and (b) the potential difference.

VP24.4.2 One model of a parallel-plate capacitor in vacuum is advertised as having a capacitance of 8.30 nF. When you disassemble one such capacitor, you find that its plates are 4.50 m² in area. (a) What is the spacing between this capacitor's plates? (b) When the potential difference between the plates is 3.00 kV, what are the charge on the capacitor and the magnitude of the electric field between the plates?

VP24.4.3 A spherical capacitor has vacuum between its conducting shells and a capacitance of 125 pF. The outer shell has inner radius 9.00 cm. (a) What is the outer radius of the inner shell? If the potential of the inner shell is 355 V higher than the potential of the outer shell, what are (b) the capacitor charge and (c) the surface charge densities on the inner and outer shells?

VP24.4.4 A cylindrical capacitor has a capacitance per length of 69.0 pF/m. (a) What is the ratio of the inner radius of the outer cylinder to the outer radius of the inner cylinder? (b) If the outer conductor carries charge -8.62 nC per meter of length, what is the potential difference between the two conductors? Which conductor is at higher potential?

Be sure to review **EXAMPLES 24.5 and 24.6** (Section 24.2) and **EXAMPLES 24.7, 24.8, and 24.9** (Section 24.3) before attempting these problems.

VP24.9.1 You have three capacitors that you wish to arrange in a circuit: $C_1 = 1.00$ μ F, $C_2 = 2.50$ μ F, and $C_3 = 5.00$ μ F. Find their equivalent capacitance if (a) they are connected in series; (b) they are connected in parallel; (c) C_1 and C_2 are connected in parallel, and that combination is connected to C_3 in series; (d) C_1 and C_2 are connected in series, and that combination is connected to C_3 in parallel.

VP24.9.2 You have two capacitors that you wish to use in an energy-storage device: $C_1 = 1.50$ μ F and $C_2 = 6.00$ μ F. (a) How much energy is stored in each capacitor if each has charge 4.50×10^{-4} C? Which has greater stored energy? (b) How much energy is stored in each capacitor if each has potential difference 125 V? Which has greater stored energy?

VP24.9.3 You connect two capacitors, $C_1 = 1.50$ μ F and $C_2 = 4.00$ μ F. (a) How much energy is stored in the combination if the potential difference across the combination is 145 V and the capacitors are connected in series? In parallel? In which case is the stored energy greater? (b) How much energy is stored in the combination if the charge on the combination is 4.00×10^{-4} C and the capacitors are connected in series? In parallel? In which case is the stored energy greater?

VP24.9.4 The plates of a parallel-plate capacitor in vacuum are 1.40 mm apart and 2.45 m² in area. The charge on the capacitor is 1.25×10^{-5} C. Calculate (a) the surface charge density on the positive plate, (b) the electric field magnitude between the plates, (c) the electric energy density between the plates, and (d) the total energy stored in the field between the plates.

Be sure to review **EXAMPLES 24.10 and 24.11** (Section 24.4) before attempting these problems.

VP24.11.1 The two plates of a parallel-plate capacitor each have area 0.460 m^2 , are 3.00 mm apart, and initially have vacuum between them. A power supply is attached to the capacitor, charging it to 4.00 kV , and is then disconnected. A dielectric sheet is then inserted that fills the space between the plates. The potential difference between the plates decreases to 2.50 kV , and the charge on each plate remains constant. Find (a) the original capacitance, (b) the charge on the capacitor plates, (c) the dielectric constant of the dielectric, (d) the new capacitance after you insert the dielectric, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

VP24.11.2 The two plates of a parallel-plate capacitor each have area 1.20 m^2 , are 4.50 mm apart, and initially have vacuum between them. You connect a power supply to the capacitor, charging it to 3.50 kV . Keeping the power supply connected, you insert a sheet of dielectric with dielectric constant 2.50 that fills the space between the plates.

Find (a) the original capacitance, (b) the original charge on the capacitor plates, (c) the new capacitance once you insert the dielectric, (d) the new charge on the capacitor plates, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

VP24.11.3 With vacuum between its plates, a parallel-plate capacitor has capacitance $4.50 \mu\text{F}$. You attach a power supply to the capacitor, charging it to 2.80 kV , and then disconnect it. You then insert a dielectric sheet that fills the space between the plates. The potential difference between the plates decreases to 1.20 kV , and the charge on each plate remains constant. Find the energy stored in the capacitor (a) before you insert the sheet and (b) after you insert the sheet.

VP24.11.4 With vacuum between its plates, a parallel-plate capacitor has capacitance $5.60 \mu\text{F}$. You attach a power supply to the capacitor, charging it to 1.50 kV . Keeping the power supply connected, you insert a dielectric sheet with dielectric constant 2.85 that fills the space between the plates. Find the energy stored in the capacitor (a) before the sheet is inserted and (b) after the sheet is inserted.

BRIDGING PROBLEM Electric-Field Energy and Capacitance of a Conducting Sphere

A solid conducting sphere of radius R carries a charge Q . Calculate the electric-field energy density at a point a distance r from the center of the sphere for (a) $r < R$ and (b) $r > R$. (c) Calculate the total electric-field energy associated with the charged sphere. (d) How much work is required to assemble the charge Q on the sphere? (e) Use the result of part (c) to find the capacitance of the sphere. (You can think of the second conductor as a hollow conducting shell of infinite radius.)

SOLUTION GUIDE

IDENTIFY and SET UP

1. You know the electric field for this situation at all values of r from Example 22.5 (Section 22.4). You'll use this to find the electric-field energy density u and total electric-field energy U . You can then find the capacitance from the relationship $U = Q^2/2C$.
2. To find U , consider a spherical shell of radius r and thickness dr that has volume $dV = 4\pi r^2 dr$. (It will help to make a drawing of such a shell concentric with the conducting sphere.) The energy stored in this volume is $u dV$, and the total energy is the integral of $u dV$ from $r = 0$ to $r \rightarrow \infty$. Set up this integral.

EXECUTE

3. Find u for $r < R$ and for $r > R$. (Hint: What is the field inside a solid conductor?)
4. Substitute your results from step 3 into the expression from step 2. Then calculate the integral to find the total electric-field energy U .
5. Use your understanding of the energy stored in a charge distribution to find the work required to assemble the charge Q .
6. Find the capacitance of the sphere.

EVALUATE

7. Where is the electric-field energy density greatest? Where is it least?
8. How would the results be affected if the solid sphere were replaced by a hollow conducting sphere of the same radius R ?
9. You can find the potential difference between the sphere and infinity from $C = Q/V$. Does this agree with the result of Example 23.8 (Section 23.3)?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q24.1 Equation (24.2) shows that the capacitance of a parallel-plate capacitor becomes larger as the plate separation d decreases. However, there is a practical limit to how small d can be made, which places limits on how large C can be. Explain what sets the limit on d . (Hint: What happens to the magnitude of the electric field as $d \rightarrow 0$?)

Q24.2 Suppose several different parallel-plate capacitors are charged up by a constant-voltage source. Thinking of the actual movement and position of the charges on an atomic level, why does it make sense that the capacitances are proportional to the surface areas of the plates? Why does it make sense that the capacitances are *inversely* proportional to the distance between the plates?

Q24.3 Suppose the two plates of a capacitor have different areas. When the capacitor is charged by connecting it to a battery, do the charges on the two plates have equal magnitude, or may they be different? Explain your reasoning.

Q24.4 To store the maximum amount of energy in a parallel-plate capacitor with a given battery (voltage source), would it be better to have the plates far apart or close together?

Q24.5 In the parallel-plate capacitor of Fig. 24.2, suppose the plates are pulled apart so that the separation d is much larger than the size of the plates. (a) Is it still accurate to say that the electric field between the plates is uniform? Why or why not? (b) In the situation shown in Fig. 24.2, the potential difference between the plates is $V_{ab} = Qd/\epsilon_0 A$. If the plates are pulled apart as described above, is V_{ab} more or less than this formula would indicate? Explain your reasoning. (c) With the plates pulled apart as described above, is the capacitance more than, less than, or the same as that given by Eq. (24.2)? Explain your reasoning.

Q24.6 A parallel-plate capacitor is charged by being connected to a battery and is kept connected to the battery. The separation between the plates is then doubled. How does the electric field change? The charge on the plates? The total energy? Explain.

Q24.7 A parallel-plate capacitor is charged by being connected to a battery and is then disconnected from the battery. The separation between the plates is then doubled. How does the electric field change? The potential difference? The total energy? Explain.

Q24.8 Two parallel-plate capacitors, identical except that one has twice the plate separation of the other, are charged by the same voltage source. Which capacitor has a stronger electric field between the plates? Which capacitor has a greater charge? Which has greater energy density? Explain your reasoning.

Q24.9 The charged plates of a capacitor attract each other, so to pull the plates farther apart requires work by some external force. What becomes of the energy added by this work? Explain.

Q24.10 You have two capacitors and want to connect them across a voltage source (battery) to store the maximum amount of energy. Should they be connected in series or in parallel?

Q24.11 As shown in Table 24.1, water has a very large dielectric constant $K = 80.4$. Why do you think water is not commonly used as a dielectric in capacitors?

Q24.12 Is dielectric strength the same thing as dielectric constant? Explain any differences between the two quantities. Is there a simple relationship between dielectric strength and dielectric constant (see Table 24.2)?

Q24.13 A capacitor made of aluminum foil strips separated by Mylar film was subjected to excessive voltage, and the resulting dielectric breakdown melted holes in the Mylar. After this, the capacitance was found to be about the same as before, but the breakdown voltage was much less. Why?

Q24.14 Suppose you bring a slab of dielectric close to the gap between the plates of a charged capacitor, preparing to slide it between the plates. What force will you feel? What does this force tell you about the energy stored between the plates once the dielectric is in place, compared to before the dielectric is in place?

Q24.15 The freshness of fish can be measured by placing a fish between the plates of a capacitor and measuring the capacitance. How does this work? (*Hint:* As time passes, the fish dries out. See Table 24.1.)

Q24.16 Electrolytic capacitors use as their dielectric an extremely thin layer of nonconducting oxide between a metal plate and a conducting solution. Discuss the advantage of such a capacitor over one constructed using a solid dielectric between the metal plates.

Q24.17 In terms of the dielectric constant K , what happens to the electric flux through the Gaussian surface shown in Fig. 24.22 when the dielectric is inserted into the previously empty space between the plates? Explain.

Q24.18 A parallel-plate capacitor is connected to a power supply that maintains a fixed potential difference between the plates. (a) If a sheet of dielectric is then slid between the plates, what happens to (i) the electric field between the plates, (ii) the magnitude of charge on each plate, and (iii) the energy stored in the capacitor? (b) Now suppose that before the dielectric is inserted, the charged capacitor is disconnected from the power supply. In this case, what happens to (i) the electric field between the plates, (ii) the magnitude of charge on each plate, and (iii) the energy stored in the capacitor? Explain any differences between the two situations.

Q24.19 Liquid dielectrics that have polar molecules (such as water) always have dielectric constants that decrease with increasing temperature. Why?

Q24.20 A conductor is an extreme case of a dielectric, since if an electric field is applied to a conductor, charges are free to move within the conductor to set up “induced charges.” What is the dielectric constant of a perfect conductor? Is it $K = 0$, $K \rightarrow \infty$, or something in between? Explain your reasoning.

Q24.21 The two plates of a capacitor are given charges $\pm Q$. The capacitor is then disconnected from the charging device so that the charges on the plates can’t change, and the capacitor is immersed in a tank of oil. Does the electric field between the plates increase, decrease, or stay the same? Explain your reasoning. How can this field be measured?

EXERCISES

Section 24.1 Capacitors and Capacitance

24.1 • The plates of a parallel-plate capacitor are 2.50 mm apart, and each carries a charge of magnitude 80.0 nC. The plates are in vacuum. The electric field between the plates has a magnitude of 4.00×10^6 V/m. What is (a) the potential difference between the plates; (b) the area of each plate; (c) the capacitance?

24.2 • The plates of a parallel-plate capacitor are 3.28 mm apart, and each has an area of 9.82 cm^2 . Each plate carries a charge of magnitude 4.35×10^{-8} C. The plates are in vacuum. What is (a) the capacitance; (b) the potential difference between the plates; (c) the magnitude of the electric field between the plates?

24.3 • A parallel-plate air capacitor of capacitance 245 pF has a charge of magnitude $0.148 \mu\text{C}$ on each plate. The plates are 0.328 mm apart. (a) What is the potential difference between the plates? (b) What is the area of each plate? (c) What is the electric-field magnitude between the plates? (d) What is the surface charge density on each plate?

24.4 • A $5.00 \mu\text{F}$ parallel-plate capacitor is connected to a 12.0 V battery. After the capacitor is fully charged, the battery is disconnected without loss of any of the charge on the plates. (a) A voltmeter is connected across the two plates without discharging them. What does it read? (b) What would the voltmeter read if (i) the plate separation were doubled; (ii) the radius of each plate were doubled but their separation was unchanged?

24.5 • A $10.0 \mu\text{F}$ parallel-plate capacitor with circular plates is connected to a 12.0 V battery. (a) What is the charge on each plate? (b) How much charge would be on the plates if their separation were doubled while the capacitor remained connected to the battery? (c) How much charge would be on the plates if the capacitor were connected to the 12.0 V battery after the radius of each plate was doubled without changing their separation?

24.6 • A 5.00 pF , parallel-plate, air-filled capacitor with circular plates is to be used in a circuit in which it will be subjected to potentials of up to 1.00×10^2 V. The electric field between the plates is to be no greater than 1.00×10^4 N/C. As a budding electrical engineer for Live-Wire Electronics, your tasks are to (a) design the capacitor by finding what its physical dimensions and separation must be; (b) find the maximum charge these plates can hold.

24.7 • A parallel-plate air capacitor is to store charge of magnitude 240.0 pC on each plate when the potential difference between the plates is 42.0 V. (a) If the area of each plate is 6.80 cm^2 , what is the separation between the plates? (b) If the separation between the two plates is double the value calculated in part (a), what potential difference is required for the capacitor to store charge of magnitude 240.0 pC on each plate?

24.8 • A cylindrical capacitor consists of a solid inner conducting core with radius 0.250 cm, surrounded by an outer hollow conducting tube. The two conductors are separated by air, and the length of the cylinder is 12.0 cm. The capacitance is 36.7 pF. (a) Calculate the inner radius of the hollow tube. (b) When the capacitor is charged to 125 V, what is the charge per unit length λ on the capacitor?

24.9 • A capacitor is made from two hollow, coaxial, iron cylinders, one inside the other. The inner cylinder is negatively charged and the outer is positively charged; the magnitude of the charge on each is 10.0 pC. The inner cylinder has radius 0.50 mm, the outer one has radius 5.00 mm, and the length of each cylinder is 18.0 cm. (a) What is the capacitance? (b) What applied potential difference is necessary to produce these charges on the cylinders?

24.10 •• A cylindrical capacitor has an inner conductor of radius 2.2 mm and an outer conductor of radius 3.5 mm. The two conductors are separated by vacuum, and the entire capacitor is 2.8 m long. (a) What is the capacitance per unit length? (b) The potential of the inner conductor is 350 mV higher than that of the outer conductor. Find the charge (magnitude and sign) on both conductors.

24.11 •• A spherical capacitor contains a charge of 3.30 nC when connected to a potential difference of 220 V . If its plates are separated by vacuum and the inner radius of the outer shell is 4.00 cm , calculate: (a) the capacitance; (b) the radius of the inner sphere; (c) the electric field just outside the surface of the inner sphere.

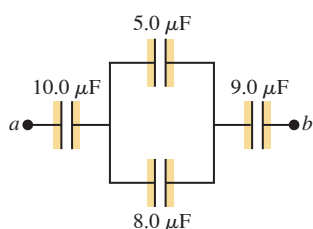
24.12 • A spherical capacitor is formed from two concentric, spherical, conducting shells separated by vacuum. The inner sphere has radius 15.0 cm and the capacitance is 116 pF . (a) What is the radius of the outer sphere? (b) If the potential difference between the two spheres is 220 V , what is the magnitude of charge on each sphere?

Section 24.2 Capacitors in Series and Parallel

24.13 •• You measure the capacitance C_1 of a capacitor by doing the following: First connect capacitors C_1 and C_2 in series to a power supply that provides a voltage V that can be varied. The capacitance of C_2 is known to be $3.00 \mu\text{F}$. Then vary the applied voltage V , and for each value of V measure the voltage V_2 across C_2 . After plotting your data as V_2 versus V , you find that the data fall close to a straight line that has slope 0.650 . What is the capacitance C_1 ?

24.14 • Figure E24.14 shows a system of four capacitors, where the potential difference across ab is 50.0 V . (a) Find the equivalent capacitance of this system between a and b . (b) How much charge is stored by this combination of capacitors? (c) How much charge is stored in each of the $10.0 \mu\text{F}$ and the $9.0 \mu\text{F}$ capacitors?

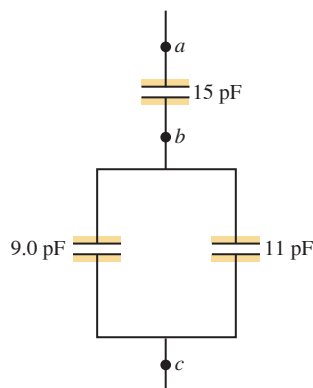
Figure E24.14



24.15 • BIO Electric Eels. Electric eels and electric fish generate large potential differences that are used to stun enemies and prey. These potentials are produced by cells that each can generate 0.10 V . We can plausibly model such cells as charged capacitors. (a) How should these cells be connected (in series or in parallel) to produce a total potential of more than 0.10 V ? (b) Using the connection in part (a), how many cells must be connected together to produce the 500 V surge of the electric eel?

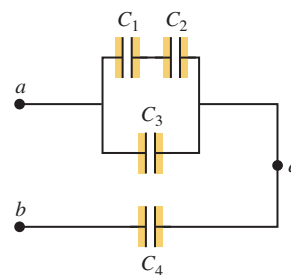
24.16 • For the system of capacitors shown in Fig. E24.16, find the equivalent capacitance (a) between b and c , and (b) between a and c .

Figure E24.16



24.17 • In Fig. E24.17, each capacitor has $C = 4.00 \mu\text{F}$ and $V_{ab} = +28.0 \text{ V}$. Calculate (a) the charge on each capacitor; (b) the potential difference across each capacitor; (c) the potential difference between points a and d .

Figure E24.17

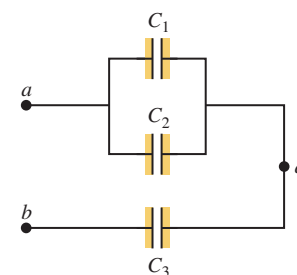


24.18 • In Fig. 24.8a, let $C_1 = 3.00 \mu\text{F}$, $C_2 = 5.00 \mu\text{F}$, and $V_{ab} = +64.0 \text{ V}$. Calculate (a) the charge on each capacitor and (b) the potential difference across each capacitor.

24.19 • In Fig. 24.9a, let $C_1 = 3.00 \mu\text{F}$, $C_2 = 5.00 \mu\text{F}$, and $V_{ab} = +52.0 \text{ V}$. Calculate (a) the charge on each capacitor and (b) the potential difference across each capacitor.

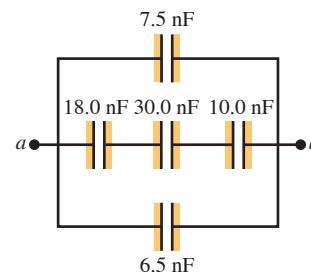
24.20 • In Fig. E24.20, $C_1 = 6.00 \mu\text{F}$, $C_2 = 3.00 \mu\text{F}$, and $C_3 = 5.00 \mu\text{F}$. The capacitor network is connected to an applied potential V_{ab} . After the charges on the capacitors have reached their final values, the charge on C_2 is $30.0 \mu\text{C}$. (a) What are the charges on capacitors C_1 and C_3 ? (b) What is the applied voltage V_{ab} ?

Figure E24.20



24.21 •• For the system of capacitors shown in Fig. E24.21, a potential difference of 25 V is maintained across ab . (a) What is the equivalent capacitance of this system between a and b ? (b) How much charge is stored by this system? (c) How much charge does the 6.5 nF capacitor store? (d) What is the potential difference across the 7.5 nF capacitor?

Figure E24.21



24.22 •• A capacitor is formed from two concentric spherical conducting shells separated by vacuum. The inner sphere has radius 12.5 cm , and the outer sphere has radius 14.8 cm . A potential difference of 120 V is applied to the capacitor. (a) What is the energy density at $r = 12.6 \text{ cm}$, just outside the inner sphere? (b) What is the energy density at $r = 14.7 \text{ cm}$, just inside the outer sphere? (c) For a parallel-plate capacitor the energy density is uniform in the region between the plates, except near the edges of the plates. Is this also true for a spherical capacitor?

Section 24.3 Energy Storage in Capacitors and Electric-Field Energy

24.23 • A $5.80 \mu\text{F}$, parallel-plate, air capacitor has a plate separation of 5.00 mm and is charged to a potential difference of 400 V . Calculate the energy density in the region between the plates, in units of J/m^3 .

24.24 • A parallel-plate air capacitor has a capacitance of 920 pF . The charge on each plate is $3.90 \mu\text{C}$. (a) What is the potential difference between the plates? (b) If the charge is kept constant, what will be the potential difference if the plate separation is doubled? (c) How much work is required to double the separation?

24.25 • An air capacitor is made from two flat parallel plates 1.50 mm apart. The magnitude of charge on each plate is $0.0180\text{ }\mu\text{C}$ when the potential difference is 200 V. (a) What is the capacitance? (b) What is the area of each plate? (c) What maximum voltage can be applied without dielectric breakdown? (Dielectric breakdown for air occurs at an electric-field strength of $3.0 \times 10^6\text{ V/m}$.) (d) When the charge is $0.0180\text{ }\mu\text{C}$, what total energy is stored?

24.26 •• A parallel-plate vacuum capacitor has 8.38 J of energy stored in it. The separation between the plates is 2.30 mm. If the separation is decreased to 1.15 mm, what is the energy stored (a) if the capacitor is disconnected from the potential source so the charge on the plates remains constant, and (b) if the capacitor remains connected to the potential source so the potential difference between the plates remains constant?

24.27 • You have two identical capacitors and an external potential source. (a) Compare the total energy stored in the capacitors when they are connected to the applied potential in series and in parallel. (b) Compare the maximum amount of charge stored in each case. (c) Energy storage in a capacitor can be limited by the maximum electric field between the plates. What is the ratio of the electric field for the series and parallel combinations?

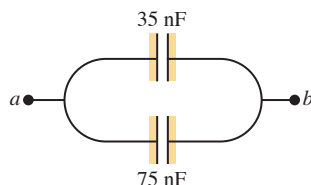
24.28 • For the capacitor network shown in **Fig. E24.28**, the potential difference across ab is 48 V. Find (a) the total charge stored in this network; (b) the charge on each capacitor; (c) the total energy stored in the network; (d) the energy stored in each capacitor; (e) the potential differences across each capacitor.

Figure E24.28



24.29 • For the capacitor network shown in **Fig. E24.29**, the potential difference across ab is 220 V. Find (a) the total charge stored in this network; (b) the charge on each capacitor; (c) the total energy stored in the network; (d) the energy stored in each capacitor; (e) the potential difference across each capacitor.

Figure E24.29



Section 24.4 Dielectrics

24.30 • Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in parallel to a battery that has a voltage of 36.0 V; $C_1 = 4.00\text{ }\mu\text{F}$ and $C_2 = 6.00\text{ }\mu\text{F}$. (a) What is the total positive charge stored in the two capacitors? (b) While the capacitors remain connected to the battery, a dielectric with dielectric constant 5.00 is inserted between the plates of capacitor C_1 , completely filling the space between them. Then what is the total positive charge stored on the two capacitors? Does the insertion of the dielectric cause the total charge stored to increase or decrease?

24.31 • A $12.5\text{ }\mu\text{F}$ capacitor is connected to a power supply that keeps a constant potential difference of 24.0 V across the plates. A piece of material having a dielectric constant of 3.75 is placed between the plates, completely filling the space between them. (a) How much energy is stored in the capacitor before and after the dielectric is inserted? (b) By how much did the energy change during the insertion? Did it increase or decrease?

24.32 • A parallel-plate capacitor has capacitance $C_0 = 8.00\text{ pF}$ when there is air between the plates. The separation between the plates is 1.50 mm. (a) What is the maximum magnitude of charge Q that can be placed on each plate if the electric field in the region between the plates is not to exceed $3.00 \times 10^4\text{ V/m}$? (b) A dielectric with $K = 2.70$ is inserted between the plates of the capacitor, completely filling the volume between the plates. Now what is the maximum magnitude of charge on each plate if the electric field between the plates is not to exceed $3.00 \times 10^4\text{ V/m}$?

24.33 • Two parallel plates have equal and opposite charges. When the space between the plates is evacuated, the electric field is $E = 3.20 \times 10^5\text{ V/m}$. When the space is filled with dielectric, the electric field is $E = 2.50 \times 10^5\text{ V/m}$. (a) What is the charge density on each surface of the dielectric? (b) What is the dielectric constant?

24.34 •• Two identical air-filled parallel-plate capacitors C_1 and C_2 , each with capacitance C , are connected in series to a battery that has voltage V . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of one of the capacitors, completely filling the space between them. Let U_0 be the total energy stored in the two capacitors without the dielectric and U be the total energy stored after the dielectric is inserted. In terms of K , what is the ratio U/U_0 ? Does the total stored energy increase, decrease, or stay the same after the dielectric is inserted?

24.35 •• Two identical air-filled parallel-plate capacitors C_1 and C_2 are connected in series to a battery that has voltage V . The charge on each capacitor is Q_0 . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of capacitor C_1 , completely filling the space between them. In terms of K and Q_0 , what is the charge on capacitor C_1 after the dielectric is inserted? Does the charge on C_1 increase, decrease, or stay the same?

24.36 •• Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in series to a battery that has voltage V ; $C_1 = 3.00\text{ }\mu\text{F}$ and $C_2 = 6.00\text{ }\mu\text{F}$. The electric field between the plates of capacitor C_2 is E_{02} . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K = 4$ is inserted between the plates of capacitor C_1 , completely filling the space between them. After the dielectric is inserted in C_1 , the electric field between the plates of capacitor C_2 is E_2 . (a) What is the ratio E_2/E_{02} ? When the dielectric is inserted into C_1 , does the electric field in C_2 increase, decrease, or remain the same? (b) Repeat the calculation in part (a) for the two capacitors connected to the battery in parallel.

24.37 • The dielectric to be used in a parallel-plate capacitor has a dielectric constant of 3.60 and a dielectric strength of $1.60 \times 10^7\text{ V/m}$. The capacitor is to have a capacitance of $1.25 \times 10^{-9}\text{ F}$ and must be able to withstand a maximum potential difference of 5500 V. What is the minimum area the plates of the capacitor may have?

24.38 •• BIO Potential in Human Cells. Some cell walls in the human body have a layer of negative charge on the inside surface and a layer of positive charge of equal magnitude on the outside surface. Suppose that the charge density on either surface is $\pm 0.50 \times 10^{-3}\text{ C/m}^2$, the cell wall is 5.0 nm thick, and the cell-wall material is air. (a) Find the magnitude of $\vec{E}$ in the wall between the two layers of charge. (b) Find the potential difference between the inside and the outside of the cell. Which is at the higher potential? (c) A typical cell in the human body has a volume of 10^{-16} m^3 . Estimate the total electric-field energy stored in the wall of a cell of this size. (*Hint:* Assume that the cell is spherical, and calculate the volume of the cell wall.) (d) In reality, the cell wall is made up, not of air, but of tissue with a dielectric constant of 5.4. Repeat parts (a) and (b) in this case.

24.39 • A constant potential difference of 12 V is maintained between the terminals of a $0.25\text{ }\mu\text{F}$, parallel-plate, air capacitor. (a) A sheet of Mylar is inserted between the plates of the capacitor, completely filling the space between the plates. When this is done, how much additional charge flows onto the positive plate of the capacitor (see Table 24.1)? (b) What is the total induced charge on either face of the Mylar sheet? (c) What effect does the Mylar sheet have on the electric field between the plates? Explain how you can reconcile this with the increase in charge on the plates, which acts to *increase* the electric field.

24.40 •• Polystyrene has dielectric constant 2.6 and dielectric strength 2.0×10^7 V/m. A piece of polystyrene is used as a dielectric in a parallel-plate capacitor, filling the volume between the plates. (a) When the electric field between the plates is 80% of the dielectric strength, what is the energy density of the stored energy? (b) When the capacitor is connected to a battery with voltage 500.0 V, the electric field between the plates is 80% of the dielectric strength. What is the area of each plate if the capacitor stores 0.200 mJ of energy under these conditions?

24.41 • When a 360 nF air capacitor ($1 \text{ nF} = 10^{-9} \text{ F}$) is connected to a power supply, the energy stored in the capacitor is $1.85 \times 10^{-5} \text{ J}$. While the capacitor is kept connected to the power supply, a slab of dielectric is inserted that completely fills the space between the plates. This increases the stored energy by $2.32 \times 10^{-5} \text{ J}$. (a) What is the potential difference between the capacitor plates? (b) What is the dielectric constant of the slab?

24.42 • A parallel-plate capacitor has capacitance $C = 12.5 \text{ pF}$ when the volume between the plates is filled with air. The plates are circular, with radius 3.00 cm. The capacitor is connected to a battery, and a charge of magnitude 25.0 pC goes onto each plate. With the capacitor still connected to the battery, a slab of dielectric is inserted between the plates, completely filling the space between the plates. After the dielectric has been inserted, the charge on each plate has magnitude 45.0 pC. (a) What is the dielectric constant K of the dielectric? (b) What is the potential difference between the plates before and after the dielectric has been inserted? (c) What is the electric field at a point midway between the plates before and after the dielectric has been inserted?

Section 24.6 Gauss's Law in Dielectrics

24.43 • A parallel-plate capacitor has the volume between its plates filled with plastic with dielectric constant K . The magnitude of the charge on each plate is Q . Each plate has area A , and the distance between the plates is d . (a) Use Gauss's law as stated in Eq. (24.23) to calculate the magnitude of the electric field in the dielectric. (b) Use the electric field determined in part (a) to calculate the potential difference between the two plates. (c) Use the result of part (b) to determine the capacitance of the capacitor. Compare your result to Eq. (24.12).

24.44 • A parallel-plate capacitor has plates with area 0.0225 m^2 separated by 1.00 mm of Teflon. (a) Calculate the charge on the plates when they are charged to a potential difference of 12.0 V. (b) Use Gauss's law [Eq. (24.23)] to calculate the electric field inside the Teflon. (c) Use Gauss's law to calculate the electric field if the voltage source is disconnected and the Teflon is removed.

PROBLEMS

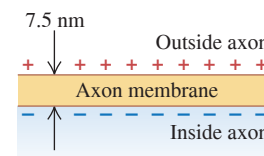
24.45 • Electronic flash units for cameras contain a capacitor for storing the energy used to produce the flash. In one such unit, the flash lasts for $\frac{1}{675} \text{ s}$ with an average light power output of $2.70 \times 10^5 \text{ W}$. (a) If the conversion of electrical energy to light is 95% efficient (the rest of the energy goes to thermal energy), how much energy must be stored in the capacitor for one flash? (b) The capacitor has a potential difference between its plates of 125 V when the stored energy equals the value calculated in part (a). What is the capacitance?

24.46 • A parallel-plate air capacitor is made by using two plates 12 cm square, spaced 3.7 mm apart. It is connected to a 12 V battery. (a) What is the capacitance? (b) What is the charge on each plate? (c) What is the electric field between the plates? (d) What is the energy stored in the capacitor? (e) If the battery is disconnected and then the plates are pulled apart to a separation of 7.4 mm, what are the answers to parts (a)–(d)?

24.47 ••• In one type of computer keyboard, each key holds a small metal plate that serves as one plate of a parallel-plate, air-filled capacitor. When the key is depressed, the plate separation decreases and the capacitance increases. Electronic circuitry detects the change in capacitance and thus detects that the key has been pressed. In one particular keyboard, the area of each metal plate is 42.0 mm^2 , and the separation between the plates is 0.700 mm before the key is depressed. (a) Calculate the capacitance before the key is depressed. (b) If the circuitry can detect a change in capacitance of 0.250 pF, how far must the key be depressed before the circuitry detects its depression?

24.48 ••• BIO Cell Membranes. Cell membranes (the walled enclosure around a cell) are typically about 7.5 nm thick. They are partially permeable to allow charged material to pass in and out, as needed. Equal but opposite charge densities build up on the inside and outside faces of such a membrane, and these charges prevent additional charges from passing through the cell wall. We can model a cell membrane as a parallel-plate capacitor, with the membrane itself containing proteins embedded in an organic material to give the membrane a dielectric constant of about 10. (See Fig. P24.48.) (a) What is the capacitance per square centimeter of such a cell wall? (b) In its normal resting state, a cell has a potential difference of 85 mV across its membrane. What is the electric field inside this membrane?

Figure P24.48

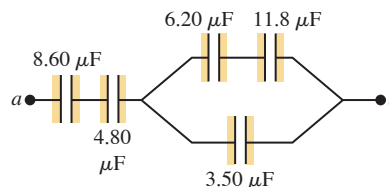


24.49 •• A $20.0 \mu\text{F}$ capacitor is charged to a potential difference of 800 V. The terminals of the charged capacitor are then connected to those of an uncharged $10.0 \mu\text{F}$ capacitor. Compute (a) the original charge of the system, (b) the final potential difference across each capacitor, (c) the final energy of the system, and (d) the decrease in energy when the capacitors are connected.

24.50 •• When lightning strikes a car, the metallic outer shell, which is insulated from the ground by its rubber tires, attains a high voltage. We can estimate how much charge is deposited by roughly modeling the car as a spherical capacitor with the outer radius taken to infinity. (a) Determine the capacitance of a sphere in terms of its radius, either by considering the potential on a sphere relative to infinity as a function of its charge, or by considering a spherical capacitor as the outer shell becomes very large. (These methods provide the same result.) (b) Estimate the radius of a sphere that corresponds to the size of a car. (c) Determine the corresponding capacitance. (d) A typical lightning strike could deliver a 100 MV potential to a car. Then what net charge would be deposited on the car?

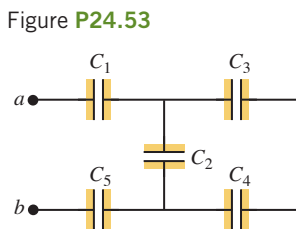
24.51 • For the capacitor network shown in Fig. P24.51, the potential difference across ab is 12.0 V. Find (a) the total energy stored in this network and (b) the energy stored in the $4.80 \mu\text{F}$ capacitor.

Figure P24.51



24.52 •• In Fig. E24.17, $C_1 = 6.00 \mu\text{F}$, $C_2 = 3.00 \mu\text{F}$, $C_3 = 4.00 \mu\text{F}$, and $C_4 = 8.00 \mu\text{F}$. The capacitor network is connected to an applied potential difference V_{ab} . After the charges on the capacitors have reached their final values, the voltage across C_3 is 40.0 V. What are (a) the voltages across C_1 and C_2 , (b) the voltage across C_4 , and (c) the voltage V_{ab} applied to the network?

24.53 • In Fig. P24.53, $C_1 = C_5 = 8.4 \mu\text{F}$ and $C_2 = C_3 = C_4 = 4.2 \mu\text{F}$. The applied potential is $V_{ab} = 220 \text{ V}$. (a) What is the equivalent capacitance of the network between points a and b ? (b) Calculate the charge on each capacitor and the potential difference across each capacitor.



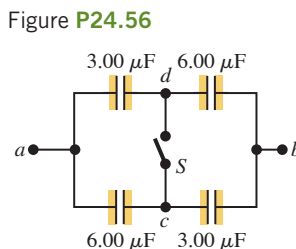
24.54 •• **CP** After combing your hair on a dry day, some of your hair stands up, forced away from your head by electrostatic repulsion. (a) Estimate the length L of your hair. (b) Using the average linear mass density of hair, which is $65 \mu\text{g}/\text{cm}$, estimate the mass m of one of your hairs. (c) Estimate the number N of hairs that stand after combing. (d) Assume that the comb has taken away a charge $-2Q$, and that your hair has therefore gained an amount of charge $2Q$. Assume further that half of this charge resides next to your head and the other half is distributed at the ends of the N strands that stand up. Assume the electrostatic force that lifted a hair was twice its weight. Show that this leads to

$$2mg = \frac{1}{4\pi\epsilon_0} \frac{Q^2/N}{L^2}$$

Use this equation to estimate the charge Q that resides on your head. (e) If your head were a sphere with radius R , it would have a capacitance of $4\pi\epsilon_0 R$. Estimate the radius of your head; then use your estimate to determine your head's capacitance. (f) Use your result to estimate the potential attained due to combing. (The surprising result illustrates an interesting feature of static electricity.)

24.55 •• In Fig. E24.20, $C_1 = 3.00 \mu\text{F}$ and $V_{ab} = 150 \text{ V}$. The charge on capacitor C_1 is $150 \mu\text{C}$ and the charge on C_3 is $450 \mu\text{C}$. What are the values of the capacitances of C_2 and C_3 ?

24.56 • The capacitors in Fig. P24.56 are initially uncharged and are connected, as in the diagram, with switch S open. The applied potential difference is $V_{ab} = +210 \text{ V}$. (a) What is the potential difference V_{cd} ? (b) What is the potential difference across each capacitor after switch S is closed? (c) How much charge flowed through the switch when it was closed?

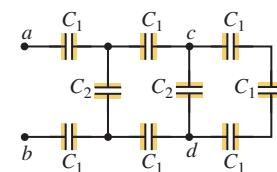


24.57 •• Three capacitors having capacitances of 8.4 , 8.4 , and $4.2 \mu\text{F}$ are connected in series across a 36 V potential difference. (a) What is the charge on the $4.2 \mu\text{F}$ capacitor? (b) What is the total energy stored in all three capacitors? (c) The capacitors are disconnected from the potential difference without allowing them to discharge. They are then reconnected in parallel with each other, with the positively charged plates connected together. What is the voltage across each capacitor in the parallel combination? (d) What is the total energy now stored in the capacitors?

24.58 • **Capacitance of a Thundercloud.** The charge center of a thundercloud, drifting 3.0 km above the earth's surface, contains 20 C of negative charge. Assuming the charge center has a radius of 1.0 km , and modeling the charge center and the earth's surface as parallel plates, calculate: (a) the capacitance of the system; (b) the potential difference between charge center and ground; (c) the average strength of the electric field between cloud and ground; (d) the electrical energy stored in the system.

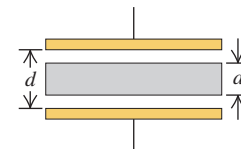
24.59 •• In Fig. P24.59, each capacitance C_1 is $6.9 \mu\text{F}$, and each capacitance C_2 is $4.6 \mu\text{F}$. (a) Compute the equivalent capacitance of the network between points a and b . (b) Compute the charge on each of the three capacitors nearest a and b when $V_{ab} = 420 \text{ V}$. (c) With 420 V across a and b , compute V_{cd} .

Figure P24.59



24.60 •• An air capacitor is made by using two flat plates, each with area A , separated by a distance d . Then a metal slab having thickness a (less than d) and the same shape and size as the plates is inserted between them, parallel to the plates and not touching either plate (Fig. P24.60).

Figure P24.60



(a) What is the capacitance of this arrangement? (b) Express the capacitance as a multiple of the capacitance C_0 when the metal slab is not present. (c) Discuss what happens to the capacitance in the limits $a \rightarrow 0$ and $a \rightarrow d$.

24.61 •• A potential difference $V_{ab} = 48.0 \text{ V}$ is applied across the capacitor network of Fig. E24.17. If $C_1 = C_2 = 4.00 \mu\text{F}$ and $C_4 = 8.00 \mu\text{F}$, what must the capacitance C_3 be if the network is to store $2.90 \times 10^{-3} \text{ J}$ of electrical energy?

24.62 • **CALC** The inner cylinder of a long, cylindrical capacitor has radius r_a and linear charge density $+\lambda$. It is surrounded by a coaxial cylindrical conducting shell with inner radius r_b and linear charge density $-\lambda$ (see Fig. 24.6). (a) What is the energy density in the region between the conductors at a distance r from the axis? (b) Integrate the energy density calculated in part (a) over the volume between the conductors in a length L of the capacitor to obtain the total electric-field energy per unit length. (c) Use Eq. (24.9) and the capacitance per unit length calculated in Example 24.4 (Section 24.1) to calculate U/L . Does your result agree with that obtained in part (b)?

24.63 •• **CP** Dielectric elastomers are used to create voltage-dependent capacitors. These elastomers contract when subject to the stresses induced by Coulomb forces between capacitor plates. Consider a pair of parallel plates with area A separated by distance d_0 when no voltage is applied. A material with dielectric constant K and Young's modulus Y fills the space between the plates. (a) What compressive force is applied to the dielectric when the plates have opposite charges Q and $-Q$? (b) The distance between the plates is $d = d_0 - sQ^2$, where s is a squeezing coefficient. Determine s in terms of the parameters specified. (c) A capacitor of this sort has plate area 1.00 cm^2 and noncharged separation distance $d_0 = 0.400 \text{ mm}$. The plates are separated by a silicone dielectric elastomer with dielectric constant $K = 3.00$ and Young's modulus 0.0100 GPa . What is the squeezing coefficient in this case? (d) If each plate has charge of magnitude $0.700 \mu\text{C}$, what is the applied voltage? (e) What new voltage would double the amount of charge on the plates?

24.64 •• A parallel-plate capacitor is made from two plates 12.0 cm on each side and 4.50 mm apart. Half of the space between these plates contains only air, but the other half is filled with Plexiglas® of dielectric constant 3.40 (Fig. P24.64). An

Figure P24.64

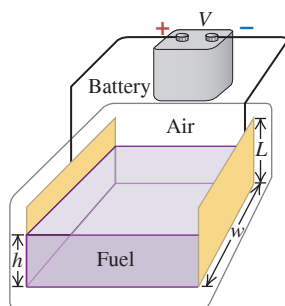


18.0 V battery is connected across the plates. (a) What is the capacitance of this combination? (Hint: Can you think of this capacitor as equivalent to two capacitors in parallel?) (b) How much energy is stored in the capacitor? (c) If we remove the Plexiglas but change nothing else, how much energy will be stored in the capacitor?

24.65 •• CP CALC A metallic circular plate with radius r is fixed to a tabletop. An identical circular plate supported from above by a cable is fixed in place a distance d above the first plate. Assume that d is much smaller than r . The two plates are attached by wires to a battery that supplies voltage V . (a) What is the tension in the cable? Neglect the weight of the plate. (b) The upper plate is slowly raised to a new height $2d$. Determine the work done by the cable by integrating $\int_d^{2d} F(z) dz$, where $F(z)$ is the cable tension when the plates are separated by a distance z . (c) Compute the energy stored in the electric field before the top plate was raised. (d) Compute the energy stored in the electric field after the top plate was raised. (e) Is the work done by the cable equal to the change in the stored electrical energy? If not, why not?

24.66 •• A fuel gauge uses a capacitor to determine the height of the fuel in a tank. The effective dielectric constant K_{eff} changes from a value of 1 when the tank is empty to a value of K , the dielectric constant of the fuel, when the tank is full. The appropriate electronic circuitry can determine the effective dielectric constant of the combined air and fuel between the capacitor plates. Each of the two rectangular plates has a width w and a length L (Fig. P24.66). The height of the fuel between the plates is h .

Figure P24.66



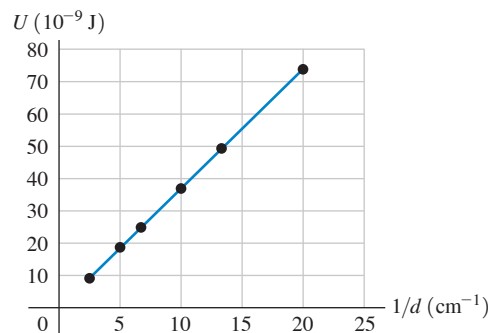
You can ignore any fringing effects. (a) Derive an expression for K_{eff} as a function of h . (b) What is the effective dielectric constant for a tank $\frac{1}{4}$ full, $\frac{1}{2}$ full, and $\frac{3}{4}$ full if the fuel is gasoline ($K = 1.95$)? (c) Repeat part (b) for methanol ($K = 33.0$). (d) For which fuel is this fuel gauge more practical?

24.67 •• DATA Your electronics company has several identical capacitors with capacitance C_1 and several others with capacitance C_2 . You must determine the values of C_1 and C_2 but don't have access to C_1 and C_2 individually. Instead, you have a network with C_1 and C_2 connected in series and a network with C_1 and C_2 connected in parallel. You have a 200.0 V battery and instrumentation that measures the total energy supplied by the battery when it is connected to the network. When the parallel combination is connected to the battery, 0.180 J of energy is stored in the network. When the series combination is connected, 0.0400 J of energy is stored. You are told that C_1 is greater than C_2 . (a) Calculate C_1 and C_2 . (b) For the series combination, does C_1 or C_2 store more charge, or are the values equal? Does C_1 or C_2 store more energy, or are the values equal? (c) Repeat part (b) for the parallel combination.

24.68 •• DATA You are designing capacitors for various applications. For one application, you want the maximum possible stored energy. For another, you want the maximum stored charge. For a third application, you want the capacitor to withstand a large applied voltage without dielectric breakdown. You start with an air-filled parallel-plate capacitor that has $C_0 = 6.00$ pF and a plate separation of 2.50 mm. You then consider the use of each of the dielectric materials listed in Table 24.2. In each application, the dielectric will fill the volume between the plates, and the electric field between the plates will be 50% of the dielectric strength given in the table. (a) For each of the five materials given in the table, calculate the energy stored in the capacitor. Which dielectric allows the maximum stored energy? (b) For each material, what is the charge Q stored on each plate of the capacitor? (c) For each material, what is the voltage applied across the capacitor? (d) Is one dielectric material in the table your best choice for all three applications?

24.69 •• DATA You are conducting experiments with an air-filled parallel-plate capacitor. You connect the capacitor to a battery with voltage 24.0 V. Initially the separation d between the plates is 0.0500 cm. In one experiment, you leave the battery connected to the capacitor, increase the separation between the plates, and measure the energy stored in the capacitor for each value of d . In a second experiment, you make the same measurements but disconnect the battery before you change the plate separation. One set of your data is given in Fig. P24.69, where you have plotted the stored energy U versus $1/d$. (a) For which experiment does this data set apply: the first (battery remains connected) or the second (battery disconnected before d is changed)? Explain. (b) Use the data plotted in Fig. P24.69 to calculate the area A of each plate. (c) For which case, the battery connected or the battery disconnected, is there more energy stored in the capacitor when $d = 0.400$ cm? Explain.

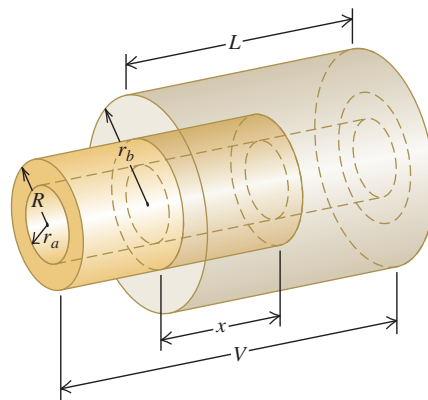
Figure P24.69



CHALLENGE PROBLEMS

24.70 •• A thin-walled, hollow, conducting cylinder with radius r_b is concentric with a solid conducting cylinder with radius $r_a < r_b$. Each has length L . The two cylinders are attached by conducting wires to the terminals of a battery that supplies potential V . A solid cylindrical shell, with inner radius r_a and outer radius $R < r_b$, made of a material with dielectric constant K , slides between the conducting cylinders, as shown in Fig. P24.70. By changing the insertion distance x , we can alter the capacitance seen by the battery and therefore alter the amount of charge stored in this device. (a) Determine the capacitance as a function of x . (b) If $L = 10.0$ cm, $r_a = 1.00$ cm, $r_b = 4.00$ cm, $R = 3.00$ cm, and $K = 3.21$, what is the capacitance when $x = 0$? (c) What is the capacitance when $x = L$? (d) What value of x results in 6.00 nC of charge on the positively charged cylinder plate when $V = 1.00$ kV?

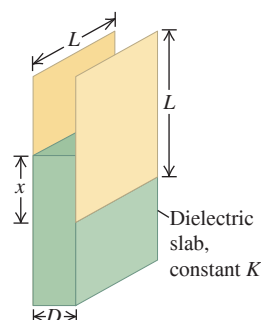
Figure P24.70



24.71 ••• CALC Two conducting plates with area A are separated by distance d . Between the plates is a material with a dielectric constant that varies linearly from a value of unity next to one plate to K next to the other plate. (a) What is the capacitance of this device? [Hint: We can envision this as a continuum of capacitors with differential plate separation connected in series. The reciprocal of the capacitance of a differential slice is then $dx/(K(x)\epsilon_0 A)$, where $K(x)$ is the dielectric constant specific to that locale.] (b) Show that this result matches the expected result when $K \rightarrow 1$.

24.72 ••• Two square conducting plates with sides of length L are separated by a distance D . A dielectric slab with constant K with dimensions $L \times L \times D$ is inserted a distance x into the space between the plates, as shown in Fig. P24.72. (a) Find the capacitance C of this system. (b) Suppose that the capacitor is connected to a battery that maintains a constant potential difference V between the plates. If the dielectric slab is inserted an additional distance dx into the space between the plates, show that the change in stored energy is

Figure P24.72



$$dU = + \frac{(K - 1)\epsilon_0 V^2 L}{2D} dx$$

(c) Suppose that before the slab is moved by dx , the plates are disconnected from the battery, so that the charges on the plates remain constant. Determine the magnitude of the charge on each plate, and then show that when the slab is moved dx farther into the space between the plates, the stored energy changes by an amount that is the *negative* of the expression for dU given in part (b). (d) If F is the force exerted on the slab by the charges on the plates, then dU should equal the work done *against* this force to move the slab a distance dx . Thus $dU = -Fdx$. Show that applying this expression to the result of part (b) suggests that the electric force on the slab pushes it *out* of the capacitor, while the result of part (c) suggests that the force pulls the slab *into* the capacitor. (e) Figure 24.16 shows that the force in fact pulls the slab into the capacitor. Explain why the result of part (b) gives an incorrect answer for the direction of this force, and calculate the magnitude of the force. (This method does not require knowledge of the nature of the fringing field.)

MCAT-STYLE PASSAGE PROBLEMS

BIO The Electric Egg. Upon fertilization, the eggs of many species undergo a rapid change in potential difference across their outer membrane. This change affects the physiological development of the eggs. The potential difference across the membrane is called the *membrane potential*, V_m , which is the potential inside the membrane minus the potential outside it. The membrane potential arises when enzymes use the energy available in ATP to expel three sodium ions (Na^+) actively and accumulate two potassium ions (K^+) inside the membrane—making the interior less positively charged than the exterior. For a sea urchin egg, V_m is about -70 mV; that is, the potential inside is 70 mV less than that outside. The egg membrane behaves as a capacitor with a capacitance of about $1 \mu\text{F}/\text{cm}^2$. The membrane of the unfertilized egg is *selectively permeable* to K^+ ; that is, K^+ can readily pass through certain channels in the membrane, but other ions cannot. When a sea urchin egg is fertilized, Na^+ channels in the membrane open, Na^+ enters the egg, and V_m rapidly increases to $+30$ mV, where it remains for several minutes. The concentration of Na^+ is about 30 mmol/L in the egg's interior but 450 mmol/L in the surrounding seawater. The K^+ concentration is about 200 mmol/L inside but 10 mmol/L outside. A useful constant that connects electrical and chemical units is the *Faraday number*, which has a value of approximately 10^5 C/mol; that is, Avogadro's number (a mole) of monovalent ions, such as Na^+ or K^+ , carries a charge of 10^5 C.

24.73 How many moles of Na^+ must move per unit area of membrane to change V_m from -70 mV to $+30$ mV, if we assume that the membrane behaves purely as a capacitor? (a) 10^{-4} mol/ cm^2 ; (b) 10^{-9} mol/ cm^2 ; (c) 10^{-12} mol/ cm^2 ; (d) 10^{-14} mol/ cm^2 .

24.74 Suppose that the egg has a diameter of $200 \mu\text{m}$. What fractional change in the internal Na^+ concentration results from the fertilization-induced change in V_m ? Assume that Na^+ ions are distributed throughout the cell volume. The concentration increases by (a) 1 part in 10^4 ; (b) 1 part in 10^5 ; (c) 1 part in 10^6 ; (d) 1 part in 10^7 .

24.75 Suppose that the change in V_m was caused by the entry of Ca^{2+} instead of Na^+ . How many Ca^{2+} ions would have to enter the cell per unit membrane to produce the change? (a) Half as many as for Na^+ ; (b) the same as for Na^+ ; (c) twice as many as for Na^+ ; (d) cannot say without knowing the inside and outside concentrations of Ca^{2+} .

24.76 What is the minimum amount of work that must be done by the cell to restore V_m to -70 mV? (a) 3 mJ; (b) 3 μJ ; (c) 3 nJ; (d) 3 pJ.

ANSWERS

Chapter Opening Question ?

(iv) Equation (24.9) shows that the energy stored in a capacitor with capacitance C and charge Q is $U = Q^2/2C$. If Q is doubled, the stored energy increases by a factor of $2^2 = 4$. Note that if the value of Q is too great, the electric-field magnitude inside the capacitor will exceed the dielectric strength of the material between the plates and dielectric breakdown will occur (see Section 24.4). This puts a practical limit on the amount of energy that can be stored.

Key Example VARIATION Problems

VP24.4.1 (a) 6.58×10^{-9} F (b) 5.85 kV

VP24.4.2 (a) 4.80 mm (b) $Q = 2.49 \times 10^{-5}$ C, $E = 6.25 \times 10^5$ V/m

VP24.4.3 (a) 8.33 cm (b) 4.44×10^{-8} C

(c) $\sigma_{\text{inner}} = +5.09 \times 10^{-7}$ C/ m^2 , $\sigma_{\text{outer}} = -4.36 \times 10^{-7}$ C/ m^2

VP24.4.4 (a) 2.24 (b) 125 V; inner conductor

VP24.9.1 (a) 0.625 μF (b) 8.50 μF (c) 2.06 μF (d) 5.71 μF

VP24.9.2 (a) $U_1 = 0.0675$ J, $U_2 = 0.0169$ J; C_1 (b) $U_1 = 0.0117$ J, $U_2 = 0.0469$ J; C_2

VP24.9.3 (a) series 0.0115 J, parallel 0.0578 J; parallel (c) series 0.0733 J, parallel 0.0145 J; series

VP24.9.4 (a) 5.10×10^{-6} C/ m^2 (b) 5.76×10^5 V/m (c) 1.47 J/ m^3 (d) 5.04×10^{-3} J

VP24.11.1 (a) 1.36 nF (b) 5.43×10^{-6} C (c) 1.60 (d) 2.17 nF (e) 2.04×10^{-6} C

VP24.11.2 (a) 2.36 nF (b) 8.26×10^{-6} C (c) 5.90 nF (d) 2.07×10^{-5} C (e) 1.24×10^{-5} C

VP24.11.3 (a) 17.6 J (b) 7.56 J

VP24.11.4 (a) 6.30 J (b) 18.0 J

Bridging Problem

(a) 0

(b) $Q^2/32\pi^2\epsilon_0 r^4$

(c) $Q^2/8\pi\epsilon_0 R$

(d) $Q^2/8\pi\epsilon_0 R$

(e) $C = 4\pi\epsilon_0 R$

? The needle of a magnetic compass points north. This alignment is due to (i) a magnetic force on the needle; (ii) a magnetic torque on the needle; (iii) the magnetic field that the needle itself produces; (iv) both (i) and (ii); (v) all of (i), (ii), and (iii).



27 Magnetic Field and Magnetic Forces

LEARNING OUTCOMES

In this chapter, you'll learn...

- 27.1 The properties of magnets, and how magnets interact with each other.
- 27.2 The nature of the force that a moving charged particle experiences in a magnetic field.
- 27.3 How magnetic field lines are different from electric field lines.
- 27.4 How to analyze the motion of a charged particle in a magnetic field.
- 27.5 Some practical applications of magnetic fields in chemistry and physics.
- 27.6 How to analyze magnetic forces on current-carrying conductors.
- 27.7 How current loops behave when placed in a magnetic field.
- 27.8 How direct-current motors work.
- 27.9 How magnetic forces give rise to the Hall effect.

You'll need to review...

- 1.10 Vector product of two vectors.
- 3.4, 5.4 Uniform circular motion.
- 10.1 Torque.
- 21.6, 21.7 Electric field lines and electric dipole moment.
- 22.2, 22.3 Electric flux and Gauss's law.
- 25.1 Electric current.
- 26.3 Galvanometers.

Everybody uses magnetic forces. They are at the heart of electric motors, microwave ovens, loudspeakers, computer printers, and disk drives. The most familiar examples of magnetism are permanent magnets, which attract unmagnetized iron objects and can also attract or repel other magnets. A compass needle aligning itself with the earth's magnetism is an example of this interaction. But the *fundamental* nature of magnetism is the interaction of moving electric charges. Unlike electric forces, which act on electric charges whether they are moving or not, magnetic forces act only on *moving* charges.

We saw in Chapter 21 that the electric force arises in two stages: (1) a charge produces an electric field in the space around it, and (2) a second charge responds to this field. Magnetic forces also arise in two stages. First, a *moving* charge or a collection of moving charges (that is, an electric current) produces a *magnetic* field. Next, a second current or moving charge responds to this magnetic field, and so experiences a magnetic force.

In this chapter we study the second stage in the magnetic interaction—that is, how moving charges and currents *respond* to magnetic fields. In particular, we'll see how to calculate magnetic forces and torques, and we'll discover why magnets can pick up iron objects like paper clips. In Chapter 28 we'll complete our picture of the magnetic interaction by examining how moving charges and currents *produce* magnetic fields.

27.1 MAGNETISM

Magnetic phenomena were first observed at least 2500 years ago in fragments of magnetized iron ore found near the ancient city of Magnesia (now Manisa, in western Turkey). These fragments were what are now called **permanent magnets**; you probably have several permanent magnets on your refrigerator door at home. Permanent magnets were found to exert forces on each other as well as on pieces of iron that were not magnetized. It was discovered that when an iron rod is brought in contact with a natural magnet, the rod also becomes magnetized. When such a rod is floated on water or suspended by a

string from its center, it tends to line itself up in a north-south direction. The needle of an ordinary compass is just such a piece of magnetized iron.

Before the relationship of magnetic interactions to moving charges was understood, the interactions of permanent magnets and compass needles were described in terms of *magnetic poles*. If a bar-shaped permanent magnet, or *bar magnet*, is free to rotate, one end points north. This end is called a *north pole* or *N pole*; the other end is a *south pole* or *S pole*. Opposite poles attract each other, and like poles repel each other (Fig. 27.1). An object that contains iron but is not itself magnetized (that is, it shows no tendency to point north or south) is attracted by *either* pole of a permanent magnet (Fig. 27.2). This is the attraction that acts between a magnet and the unmagnetized steel door of a refrigerator. By analogy to electric interactions, we describe the interactions in Figs. 27.1 and 27.2 by saying that a bar magnet sets up a *magnetic field* in the space around it and a second object responds to that field. A compass needle tends to align with the magnetic field at the needle's position.

The earth itself is a magnet. Its north geographic pole is close to a magnetic *south* pole, which is why the north pole of a compass needle points north. The earth's magnetic axis is not quite parallel to its geographic axis (the axis of rotation), so a compass reading deviates somewhat from geographic north. This deviation, which varies with location, is called *magnetic declination* or *magnetic variation*. Also, the magnetic field is not horizontal at most points on the earth's surface; its angle up or down is called *magnetic inclination*. At the magnetic poles the magnetic field is vertical.

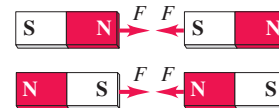
Figure 27.3 is a sketch of the earth's magnetic field. The lines, called *magnetic field lines*, show the direction that a compass would point at each location; they are discussed in detail in Section 27.3. The direction of the field at any point can be defined as the direction of the force that the field would exert on a magnetic north pole. In Section 27.2 we'll describe a more fundamental way to define the direction and magnitude of a magnetic field.

Magnetic Poles Versus Electric Charge

The concept of magnetic poles may appear similar to that of electric charge, and north and south poles may seem analogous to positive and negative charges. But the analogy can be misleading. While isolated positive and negative charges exist, there is *no* experimental evidence that one isolated magnetic pole exists; poles always appear in pairs. If a bar

Figure 27.1 (a) Two bar magnets attract when opposite poles (N and S, or S and N) are next to each other. (b) The bar magnets repel when like poles (N and N, or S and S) are next to each other.

(a) Opposite poles attract.



(b) Like poles repel.

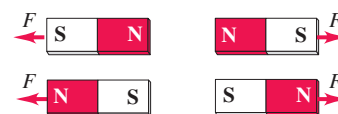
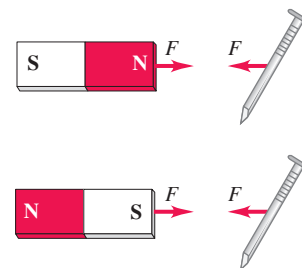


Figure 27.2 (a) Either pole of a bar magnet attracts an unmagnetized object that contains iron, such as a nail. (b) A real-life example of this effect.

(a)



(b)

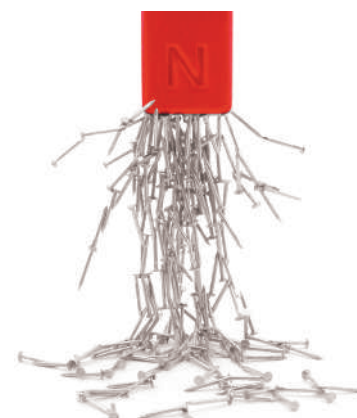


Figure 27.3 A sketch of the earth's magnetic field. The field, which is caused by currents in the earth's molten core, changes with time; geologic evidence shows that it reverses direction entirely at irregular intervals of 10^4 to 10^6 years.

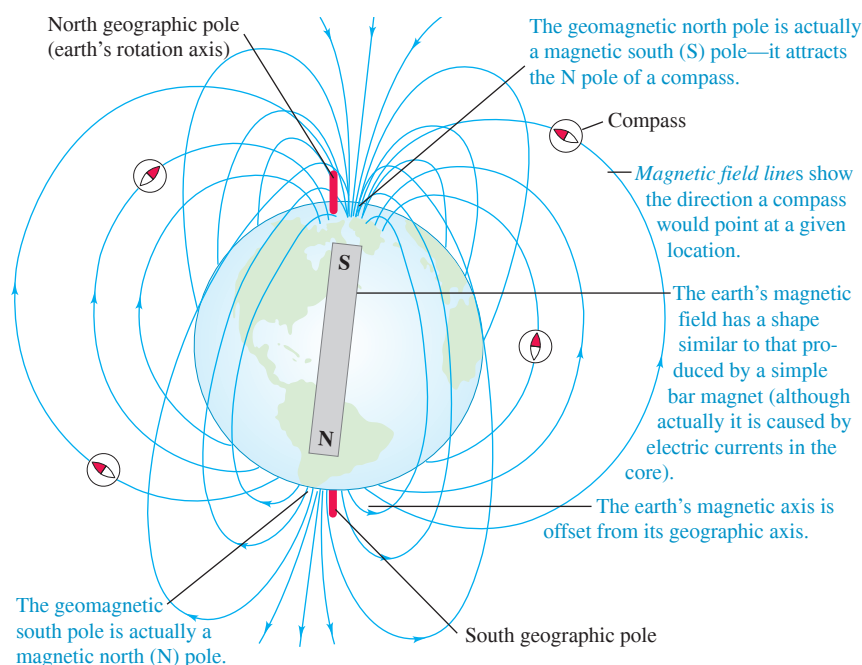


Figure 27.4 Breaking a bar magnet. Each piece has a north and south pole, even if the pieces are different sizes. (The smaller the piece, the weaker its magnetism.)

In contrast to electric charges, magnetic poles always come in pairs and can't be isolated.

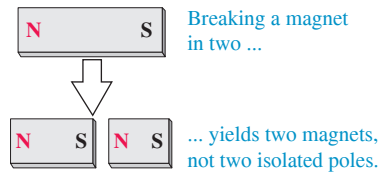
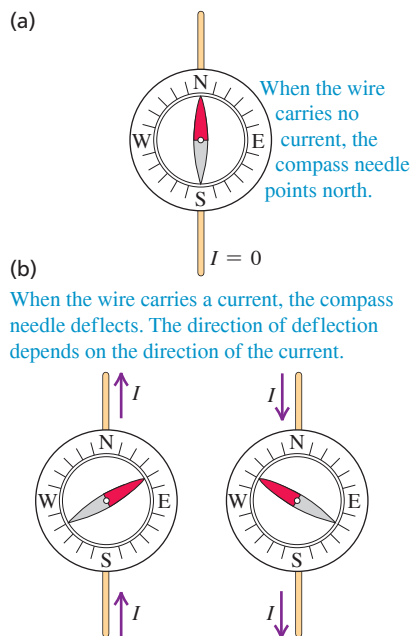


Figure 27.5 In Oersted's experiment, a compass is placed directly over a horizontal wire (here viewed from above).



BIO APPLICATION Spiny Lobsters and Magnetic Compasses Although the Caribbean spiny lobster (*Panulirus argus*) has a relatively simple nervous system, it is remarkably sensitive to magnetic fields. It has an internal magnetic “compass” that allows it to distinguish north, east, south, and west. This lobster can also sense small differences in the earth's magnetic field from one location to another and may use these differences to help it navigate.



magnet is broken in two, each broken end becomes a pole (**Fig. 27.4**). The existence of an isolated magnetic pole, or **magnetic monopole**, would have sweeping implications for theoretical physics. Extensive searches for magnetic monopoles have been carried out, but so far without success.

The first evidence of the relationship of magnetism to moving charges was discovered in 1820 by the Danish scientist Hans Christian Oersted. He found that a compass needle was deflected by a current-carrying wire (**Fig. 27.5**). Similar investigations were carried out in France by André Ampère. A few years later, Michael Faraday in England and Joseph Henry in the United States discovered that moving a magnet near a conducting loop can cause a current in the loop. We now know that the magnetic forces between two objects shown in Figs. 27.1 and 27.2 are fundamentally due to interactions between moving electrons in the atoms of the objects. (There are also *electric* interactions between the two objects, but these are far weaker than the magnetic interactions because the objects are electrically neutral.) Inside a magnetized object such as a permanent magnet, the motion of certain of the atomic electrons is *coordinated*; in an unmagnetized object these motions are not coordinated. (We'll describe these motions further in Section 27.7 and see how the interactions shown in Figs. 27.1 and 27.2 come about.)

Electric and magnetic interactions prove to be intimately connected. Over the next several chapters we'll develop the unifying principles of electromagnetism, culminating in the expression of these principles in *Maxwell's equations*. These equations represent the synthesis of electromagnetism, just as Newton's laws of motion are the synthesis of mechanics, and like Newton's laws they represent a towering achievement of the human intellect.

TEST YOUR UNDERSTANDING OF SECTION 27.1 Suppose you cut off the part of the compass needle shown in Fig. 27.5a that is painted gray. You discard this part, drill a hole in the remaining red part, and place the red part on the pivot at the center of the compass. Will the red part still swing when a current is applied as in Fig. 27.5b?

ANSWER **yes** When a magnet is cut apart, each part has a north and south pole (see Fig. 27.4). Hence the small red part behaves much like the original, full-sized compass needle.

27.2 MAGNETIC FIELD

To introduce the concept of magnetic field properly, let's review our formulation of *electric* interactions in Chapter 21, where we introduced the concept of *electric* field. We represented electric interactions in two steps:

1. A distribution of electric charge creates an electric field $\vec{E}$ in the surrounding space.
2. The electric field exerts a force $\vec{F} = q\vec{E}$ on any other charge q that is present in the field.

We can describe magnetic interactions in a similar way:

1. A moving charge or a current creates a **magnetic field** in the surrounding space (in addition to its *electric* field).
2. The magnetic field exerts a force $\vec{F}$ on any other moving charge or current that is present in the field.

In this chapter we'll concentrate on the *second* aspect of the interaction: Given the presence of a magnetic field, what force does it exert on a moving charge or a current? In Chapter 28 we'll come back to the problem of how magnetic fields are *created* by moving charges and currents.

Like electric field, magnetic field is a *vector field*—that is, a vector quantity associated with each point in space. We'll use the symbol $\vec{B}$ for magnetic field. At any position the direction of $\vec{B}$ is defined as the direction in which the north pole of a compass needle tends to point. The arrows in Fig. 27.3 suggest the direction of the earth's magnetic field; for any magnet, $\vec{B}$ points out of its north pole and into its south pole.

Magnetic Forces on Moving Charges

There are four key characteristics of the magnetic force on a moving charge. First, its magnitude is proportional to the magnitude of the charge. If a $1\text{ }\mu\text{C}$ charge and a $2\text{ }\mu\text{C}$ charge move through a given magnetic field with the same velocity, experiments show that the force on the $2\text{ }\mu\text{C}$ charge is twice as great as the force on the $1\text{ }\mu\text{C}$ charge. Second, the magnitude of the force is also proportional to the magnitude, or “strength,” of the field; if we double the magnitude of the field (for example, by using two identical bar magnets instead of one) without changing the charge or its velocity, the force doubles.

A third characteristic is that the magnetic force depends on the particle’s velocity. This is quite different from the electric-field force, which is the same whether the charge is moving or not. A charged particle at rest experiences *no* magnetic force. And fourth, we find by experiment that the magnetic force $\vec{F}$ *does not* have the same direction as the magnetic field $\vec{B}$ but instead is always *perpendicular* to both $\vec{B}$ and the velocity $\vec{v}$. The magnitude F of the force is proportional to the component of $\vec{v}$ perpendicular to the field; when that component is zero (that is, when $\vec{v}$ and $\vec{B}$ are parallel or antiparallel), the force is zero.

Figure 27.6 shows these relationships. The direction of $\vec{F}$ is always perpendicular to the plane containing $\vec{v}$ and $\vec{B}$. Its magnitude is given by

$$F = |q|v_{\perp}B = |q|vB\sin\phi \quad (27.1)$$

where $|q|$ is the magnitude of the charge and ϕ is the angle measured from the direction of $\vec{v}$ to the direction of $\vec{B}$, as shown in the figure.

This description does not specify the direction of $\vec{F}$ completely; there are always two directions, opposite to each other, that are both perpendicular to the plane of $\vec{v}$ and $\vec{B}$. To complete the description, we use the same right-hand rule that we used to define the vector product in Section 1.10. (It would be a good idea to review that section before you go on.) Draw the vectors $\vec{v}$ and $\vec{B}$ with their tails together, as in **Fig. 27.7a**. Imagine turning $\vec{v}$ until it points in the direction of $\vec{B}$ (turning through the smaller of the two possible angles). Wrap the fingers of your right hand around the line perpendicular to the plane of $\vec{v}$ and $\vec{B}$ so that they curl around with the sense of rotation from $\vec{v}$ to $\vec{B}$. Your thumb then points in the direction of the force $\vec{F}$ on a *positive* charge.

This discussion shows that the force on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is given, both in magnitude and in direction, by

$$\text{Magnetic force on a moving charged particle} \quad \vec{F} = q\vec{v} \times \vec{B} \quad (27.2)$$

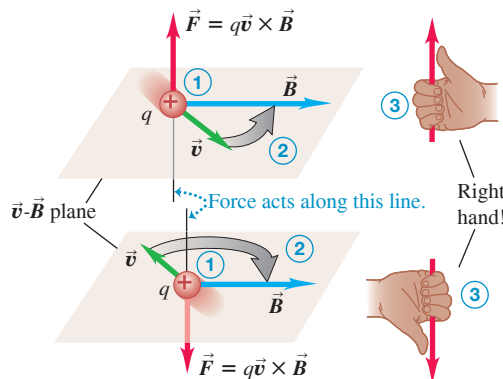
Particle's charge Particle's velocity Magnetic field

Figure 27.7 Finding the direction of the magnetic force on a moving charged particle.

(a)

Right-hand rule for the direction of magnetic force on a **positive** charge moving in a magnetic field:

- ① Place the $\vec{v}$ and $\vec{B}$ vectors tail to tail.
- ② Imagine turning $\vec{v}$ toward $\vec{B}$ in the $\vec{v}$ - $\vec{B}$ plane (through the smaller angle).
- ③ The force acts along a line perpendicular to the $\vec{v}$ - $\vec{B}$ plane. Curl the fingers of your *right hand* around this line in the same direction you rotated $\vec{v}$. Your thumb now points in the direction the force acts.



(b)

If the charge is negative, the direction of the force is **opposite** to that given by the right-hand rule.

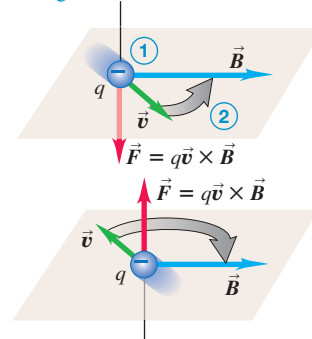
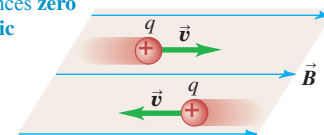


Figure 27.6 The magnetic force $\vec{F}$ acting on a positive charge q moving with velocity $\vec{v}$ is perpendicular to both $\vec{v}$ and the magnetic field $\vec{B}$. For given values of speed v and magnetic field strength B , the force is greatest when $\vec{v}$ and $\vec{B}$ are perpendicular.

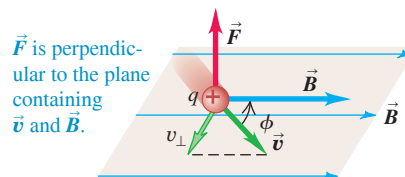
(a)

A charge moving **parallel** to a magnetic field experiences **zero magnetic force**.



(b)

A charge moving at an angle ϕ to a magnetic field experiences a magnetic force with magnitude $F = |q|v_{\perp}B = |q|vB\sin\phi$.



(c)

A charge moving **perpendicular** to a magnetic field experiences a **maximal magnetic force** with magnitude $F_{\max} = qvB$.

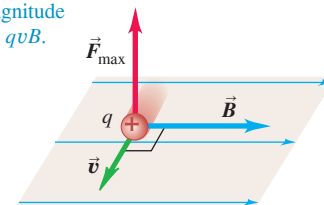
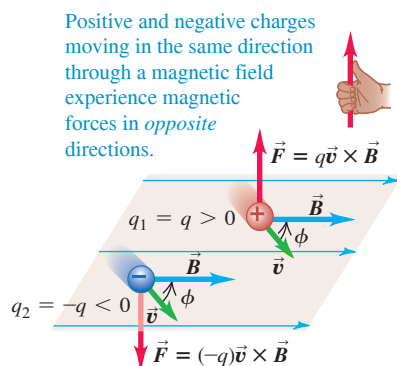


Figure 27.8 Reversing the sign of a moving charge reverses the direction of the magnetic force that acts on it.



BIO APPLICATION Magnetic Fields of the Body

All living cells are electrically active, and the feeble electric currents within your body produce weak but measurable magnetic fields. The fields produced by skeletal muscles have magnitudes less than 10^{-10} T, about one-millionth as strong as the earth's magnetic field. Your brain produces magnetic fields that are far weaker, only about 10^{-12} T.



Figure 27.9 Determining the direction of a magnetic field by using a cathode-ray tube. Because electrons have a negative charge, the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ in part (b) points opposite to the direction given by the right-hand rule (see Fig. 27.7b).

This is the first of several vector products we'll encounter in our study of magnetic-field relationships. It's important to note that Eq. (27.2) was *not* deduced theoretically; it is an observation based on *experiment*.

Equation (27.2) is valid for both positive and negative charges. When q is negative, the direction of the force $\vec{F}$ is opposite to that of $\vec{v} \times \vec{B}$ (Fig. 27.7b). If two charges with equal magnitude and opposite sign move in the same $\vec{B}$ field with the same velocity (Fig. 27.8), the forces have equal magnitude and opposite direction. Figures 27.6, 27.7, and 27.8 show several examples of the relationships of the directions of $\vec{F}$, $\vec{v}$, and $\vec{B}$ for both positive and negative charges. Be sure you understand the relationships shown in these figures.

Equation (27.1) gives the magnitude of the magnetic force $\vec{F}$ in Eq. (27.2). Since ϕ is the angle between the directions of vectors $\vec{v}$ and $\vec{B}$, we may interpret $B \sin \phi$ as the component of $\vec{B}$ perpendicular to $\vec{v}$ —that is, $B_{\perp}$. With this notation the force magnitude is

$$F = |q|vB_{\perp} \quad (27.3)$$

This form may be more convenient, especially in problems involving *currents* rather than individual particles. We'll discuss forces on currents later in this chapter.

From Eq. (27.1) the *units* of B must be the same as the units of F/qv . Therefore the SI unit of B is equivalent to $1 \text{ N} \cdot \text{s}/\text{C} \cdot \text{m}$, or, since one ampere is one coulomb per second ($1 \text{ A} = 1 \text{ C/s}$), $1 \text{ N/A} \cdot \text{m}$. This unit is called the **tesla** (abbreviated T), in honor of Nikola Tesla (1856–1943), the prominent Serbian-American scientist and inventor:

$$1 \text{ tesla} = 1 \text{ T} = 1 \text{ N/A} \cdot \text{m}$$

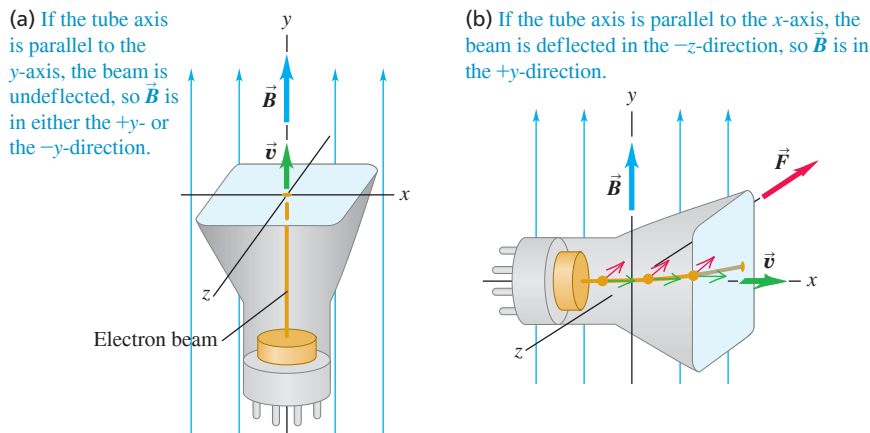
Another unit of B , the **gauss** ($1 \text{ G} = 10^{-4} \text{ T}$), is also in common use.

The magnetic field of the earth is of the order of 10^{-4} T or 1 G. Magnetic fields of the order of 10 T occur in the interior of atoms and are important in the analysis of atomic spectra. The largest steady magnetic field that can be produced at present in the laboratory is about 45 T. Some pulsed-current electromagnets can produce fields of the order of 120 T for millisecond time intervals.

Measuring Magnetic Fields with Test Charges

To explore an unknown magnetic field, we can measure the magnitude and direction of the force on a *moving* test charge and then use Eq. (27.2) to determine $\vec{B}$. The electron beam in a cathode-ray tube, such as that in an older television set (not a flat-screen TV), is a convenient device for this. The electron gun shoots out a narrow beam of electrons at a known speed. If there is no force to deflect the beam, it strikes the center of the screen.

If a magnetic field is present, in general the electron beam is deflected. But if the beam is parallel or antiparallel to the field, then $\phi = 0$ or π in Eq. (27.1) and $F = 0$; there is no force and hence no deflection. If we find that the electron beam is not deflected when its direction is parallel to a certain axis as in Fig. 27.9a, the $\vec{B}$ vector must point either up or down along that axis.



If we then turn the tube 90° (Fig. 27.9b), $\phi = \pi/2$ in Eq. (27.1) and the magnetic force is maximum; the beam is deflected in a direction perpendicular to the plane of $\vec{B}$ and $\vec{v}$. The direction and magnitude of the deflection determine the direction and magnitude of $\vec{B}$. We can perform additional experiments in which the angle between $\vec{B}$ and $\vec{v}$ is between 0° and 90° to confirm Eq. (27.1). We note that the electron has a negative charge; the force in Fig. 27.9b is opposite in direction to the force on a positive charge.

When a charged particle moves through a region of space where *both* electric and magnetic fields are present, both fields exert forces on the particle. The total force $\vec{F}$ is the vector sum of the electric and magnetic forces:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (27.4)$$

CAUTION **No perpendicular field component, no magnetic force** When a charged particle moves in a magnetic field $\vec{B}$, the particle experiences a magnetic force *only* if there is a component of $\vec{B}$ that is perpendicular to the particle's velocity. If $\vec{B}$ and the velocity are in the same direction (as in Fig. 27.9a) or in opposite directions, there is no perpendicular component of $\vec{B}$ and hence no magnetic force. ■

PROBLEM-SOLVING STRATEGY 27.1 Magnetic Forces

IDENTIFY *the relevant concepts:* The equation $\vec{F} = q\vec{v} \times \vec{B}$ allows you to determine the magnetic force on a moving charged particle.

SET UP *the problem* using the following steps:

1. Draw the velocity $\vec{v}$ and magnetic field $\vec{B}$ with their tails together so that you can visualize the plane that contains them.
2. Determine the angle ϕ between $\vec{v}$ and $\vec{B}$.
3. Identify the target variables.

EXECUTE *the solution* as follows:

1. Use Eq. (27.2), $\vec{F} = q\vec{v} \times \vec{B}$, to express the magnetic force. Equation (27.1) gives the magnitude of the force, $F = qvB \sin \phi$.
2. Remember that $\vec{F}$ is perpendicular to the plane containing $\vec{v}$ and $\vec{B}$. The right-hand rule (see Fig. 27.7) gives the direction of $\vec{v} \times \vec{B}$. If q is negative, $\vec{F}$ is *opposite* to $\vec{v} \times \vec{B}$.

EVALUATE *your answer:* Whenever possible, solve the problem in two ways to confirm that the results agree. Do it directly from the geometric definition of the vector product. Then find the components of the vectors in some convenient coordinate system and calculate the vector product from the components. Verify that the results agree.

EXAMPLE 27.1 Magnetic force on a proton

WITH ✓ **ARIATION PROBLEMS**

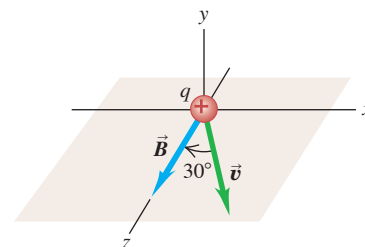
A beam of protons ($q = 1.6 \times 10^{-19}$ C) moves at 3.0×10^5 m/s through a uniform 2.0 T magnetic field directed along the positive z -axis (Fig. 27.10). The velocity of each proton lies in the xz -plane and is directed at 30° to the $+z$ -axis. Find the force on a proton.

IDENTIFY and SET UP This problem uses the expression $\vec{F} = q\vec{v} \times \vec{B}$ for the magnetic force $\vec{F}$ on a moving charged particle. The target variable is $\vec{F}$.

EXECUTE The charge is positive, so the force is in the same direction as the vector product $\vec{v} \times \vec{B}$. From the right-hand rule, this direction is along the negative y -axis. The magnitude of the force, from Eq. (27.1), is

$$\begin{aligned} F &= qvB \sin \phi \\ &= (1.6 \times 10^{-19} \text{ C})(3.0 \times 10^5 \text{ m/s})(2.0 \text{ T})(\sin 30^\circ) \\ &= 4.8 \times 10^{-14} \text{ N} \end{aligned}$$

Figure 27.10 Directions of $\vec{v}$ and $\vec{B}$ for a proton in a magnetic field.



Continued

EVALUATE To check our result, we evaluate the force by using vector language and Eq. (27.2). We have

$$\begin{aligned}\vec{v} &= (3.0 \times 10^5 \text{ m/s})(\sin 30^\circ)\hat{i} + (3.0 \times 10^5 \text{ m/s})(\cos 30^\circ)\hat{k} \\ \vec{B} &= (2.0 \text{ T})\hat{k} \\ \vec{F} &= q\vec{v} \times \vec{B} \\ &= (1.6 \times 10^{-19} \text{ C})(3.0 \times 10^5 \text{ m/s})(2.0 \text{ T}) \\ &\quad \times (\sin 30^\circ\hat{i} + \cos 30^\circ\hat{k}) \times \hat{k} \\ &= (-4.8 \times 10^{-14} \text{ N})\hat{j}\end{aligned}$$

(Recall that $\hat{i} \times \hat{k} = -\hat{j}$ and $\hat{k} \times \hat{k} = \mathbf{0}$.) We again find that the force is in the negative y -direction with magnitude $4.8 \times 10^{-14} \text{ N}$.

If the beam consists of *electrons* rather than protons, the charge is negative ($q = -1.6 \times 10^{-19} \text{ C}$) and the direction of the force is reversed. The force is now directed along the *positive* y -axis, but the magnitude is the same as before, $F = 4.8 \times 10^{-14} \text{ N}$.

KEYCONCEPT A charged particle moving with velocity $\vec{v}$ through a magnetic field $\vec{B}$ experiences a magnetic force that depends on its charge, its speed, the magnitude of $\vec{B}$, and the angle between $\vec{v}$ and $\vec{B}$. The force is zero if the directions of $\vec{v}$ and $\vec{B}$ are the same or opposite; otherwise, the force is nonzero and in a direction perpendicular to both $\vec{v}$ and $\vec{B}$.

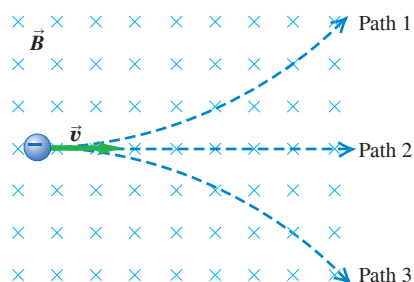
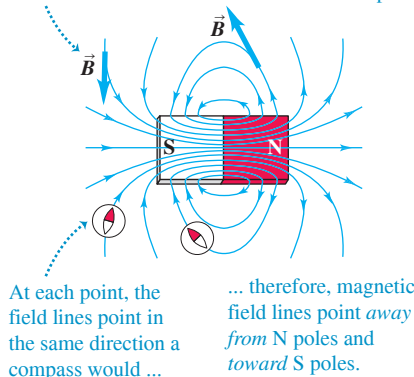


Figure 27.11 Magnetic field lines of a permanent magnet. Note that the field lines pass through the interior of the magnet.

At each point, the field line is tangent to the magnetic-field vector $\vec{B}$.

The more densely the field lines are packed, the stronger the field is at that point.



At each point, the field lines point in the same direction a compass would ...

... therefore, magnetic field lines point away from N poles and toward S poles.

TEST YOUR UNDERSTANDING OF SECTION 27.2 The accompanying figure shows a uniform magnetic field $\vec{B}$ directed into the plane of the paper (shown by the blue $\times$'s). A particle with a negative charge moves in the plane. Which path—1, 2, or 3—does the particle follow?

ANSWER

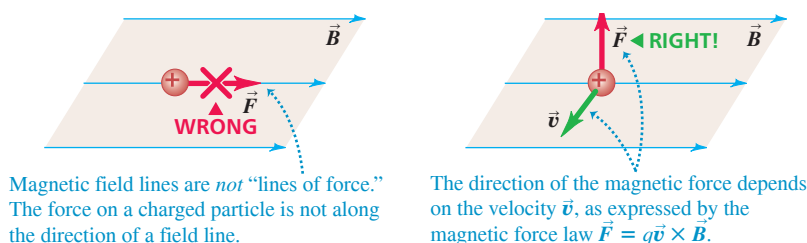
path 3 Applying the right-hand rule to the vectors $\vec{v}$ (which points to the right) and $\vec{B}$ (which points into the plane of the figure) says that the force $\vec{F} = q\vec{v} \times \vec{B}$ on a positive charge would point upward. Since the charge is negative, the force points downward and the particle follows a trajectory that curves downward.

27.3 MAGNETIC FIELD LINES AND MAGNETIC FLUX

We can represent any magnetic field by **magnetic field lines**, just as we did for the earth's magnetic field in Fig. 27.3. The idea is the same as for the electric field lines we introduced in Section 21.6. We draw the lines so that the line through any point is tangent to the magnetic-field vector $\vec{B}$ at that point (**Fig. 27.11**). Just as with electric field lines, we draw only a few representative lines; otherwise, the lines would fill up all of space. Where adjacent field lines are close together, the field magnitude is large; where these field lines are far apart, the field magnitude is small. Also, because the direction of $\vec{B}$ at each point is unique, field lines never intersect.

CAUTION **Magnetic field lines are not “lines of force”** Unlike electric field lines, magnetic field lines *do not* point in the direction of the force on a charge (**Fig. 27.12**). Equation (27.2) shows that the force on a moving charged particle is always perpendicular to the magnetic field and hence to the magnetic field line that passes through the particle's position. The direction of the force depends on the particle's velocity and the sign of its charge, so just looking at magnetic field lines cannot tell you the direction of the force on an arbitrary moving charged particle. Magnetic field lines *do* have the direction that a compass needle would point at each location; this may help you visualize them. **I**

Figure 27.12 Magnetic field lines are *not* “lines of force.”

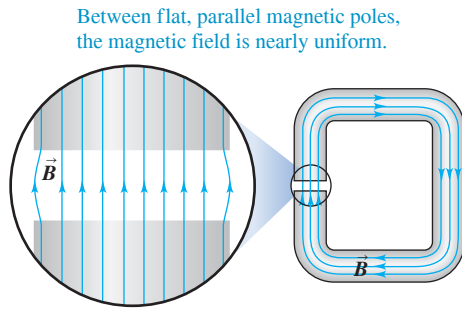


Magnetic field lines are *not* “lines of force.” The force on a charged particle is not along the direction of a field line.

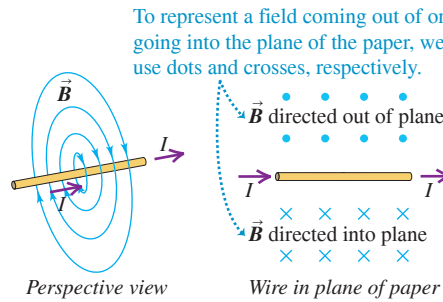
The direction of the magnetic force depends on the velocity $\vec{v}$, as expressed by the magnetic force law $\vec{F} = q\vec{v} \times \vec{B}$.

Figure 27.13 Magnetic field lines produced by some common sources of magnetic field.

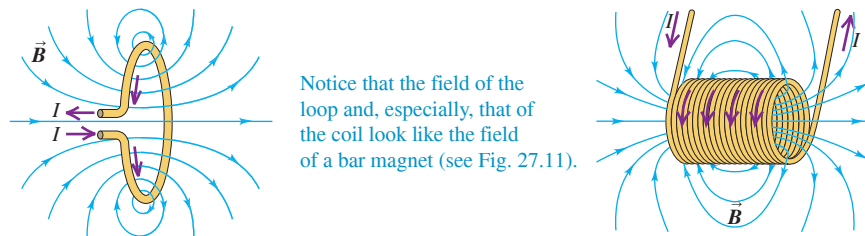
(a) Magnetic field of a C-shaped magnet



(b) Magnetic field of a straight current-carrying wire



(c) Magnetic fields of a current-carrying loop and a current-carrying coil (solenoid)



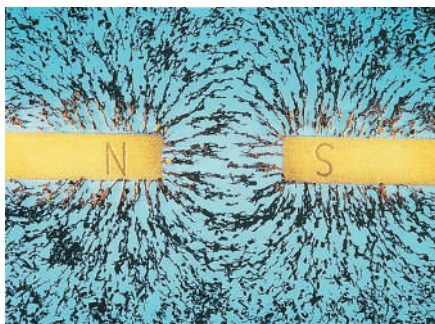
Figures 27.11 and 27.13 show magnetic field lines produced by several common sources of magnetic field. In the gap between the poles of the magnet shown in **Fig. 27.13a**, the field lines are approximately straight, parallel, and equally spaced, showing that the magnetic field in this region is approximately *uniform* (that is, constant in magnitude and direction).

Because magnetic-field patterns are three-dimensional, it's often necessary to draw magnetic field lines that point into or out of the plane of a drawing. To do this we use a dot ($\bullet$) to represent a vector directed out of the plane and a cross ($\times$) to represent a vector directed into the plane (Fig. 27.13b). To remember these, think of a dot as the head of an arrow coming directly toward you, and think of a cross as the feathers of an arrow flying directly away from you.

Iron filings, like compass needles, tend to align with magnetic field lines. Hence they provide an easy way to visualize field lines (**Fig. 27.14**).

Figure 27.14 (a) Like little compass needles, iron filings line up tangent to magnetic field lines. (b) Drawing of field lines for the situation shown in (a).

(a)



(b)

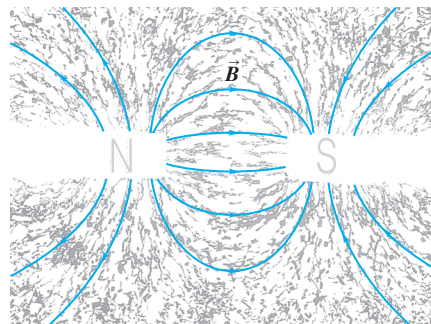
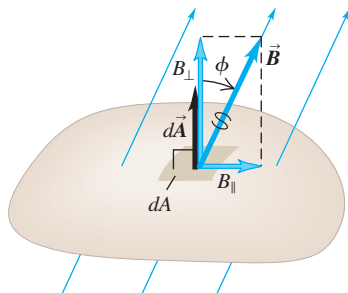


Figure 27.15 The magnetic flux through an area element dA is defined to be $d\Phi_B = B_{\perp} dA$.



Magnetic Flux and Gauss's Law for Magnetism

We define the **magnetic flux** Φ_B through a surface just as we defined electric flux in connection with Gauss's law in Section 22.2. We can divide any surface into elements of area dA (Fig. 27.15). For each element we determine $B_{\perp}$, the component of $\vec{B}$ normal to the surface at the position of that element, as shown. From the figure, $B_{\perp} = B \cos \phi$, where ϕ is the angle between the direction of $\vec{B}$ and a line perpendicular to the surface. (Be careful not to confuse ϕ with Φ_B .) In general, this component varies from point to point on the surface. We define the magnetic flux $d\Phi_B$ through this area as

$$d\Phi_B = B_{\perp} dA = B \cos \phi dA = \vec{B} \cdot d\vec{A} \quad (27.5)$$

The *total* magnetic flux through the surface is the sum of the contributions from the individual area elements:

$$\text{Magnetic flux through a surface } \Phi_B = \int B \cos \phi dA = \int B_{\perp} dA = \int \vec{B} \cdot d\vec{A} \quad (27.6)$$

Magnitude of magnetic field $\vec{B}$ Component of $\vec{B}$ perpendicular to surface
 Angle between $\vec{B}$ and normal to surface Element of surface area Vector element of surface area

(Review the concepts of vector area and surface integral in Section 22.2.)

Magnetic flux is a *scalar* quantity. If $\vec{B}$ is uniform over a plane surface with total area A , then $B_{\perp}$ and ϕ are the same at all points on the surface, and

$$\Phi_B = B_{\perp} A = BA \cos \phi \quad (27.7)$$

If $\vec{B}$ is also perpendicular to the surface (parallel to the area vector), then $\cos \phi = 1$ and Eq. (27.7) reduces to $\Phi_B = BA$. We'll use the concept of magnetic flux extensively during our study of electromagnetic induction in Chapter 29.

The SI unit of magnetic flux is equal to the unit of magnetic field (1 T) times the unit of area (1 m^2). This unit is called the **weber** (1 Wb), in honor of the German physicist Wilhelm Weber (1804–1891):

$$1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$$

Also, $1 \text{ T} = 1 \text{ N/A} \cdot \text{m}$, so

$$1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2 = 1 \text{ N} \cdot \text{m/A}$$

In Gauss's law the total *electric* flux through a closed surface is proportional to the total electric charge enclosed by the surface. For example, if the closed surface encloses an electric dipole, the total electric flux is zero because the total charge is zero. (You may want to review Section 22.3 on Gauss's law.) By analogy, if there were such a thing as a single magnetic charge (magnetic monopole), the total *magnetic* flux through a closed surface would be proportional to the total magnetic charge enclosed. But we have mentioned that no magnetic monopole has ever been observed, despite intensive searches. This leads us to **Gauss's law for magnetism**:

CAUTION **Magnetic field lines have no ends** Unlike electric field lines, which begin and end on electric charges, magnetic field lines *never* have endpoints; such a point would indicate the presence of a monopole. You might be tempted to draw magnetic field lines that begin at the north pole of a magnet and end at a south pole. But as Fig. 27.11 shows, a magnet's field lines continue through the interior of the magnet. Like all other magnetic field lines, they form closed loops. **I**

Gauss's law for magnetism:

The total magnetic flux through *any* closed surface ...

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad \dots \text{equals zero.} \quad (27.8)$$

You can verify Gauss's law for magnetism by examining Figs. 27.11 and 27.13; if you draw a closed surface anywhere in any of the field maps shown in those figures, you'll see that every field line that enters the surface also exits from it; the net flux through the surface is zero. It also follows from Eq. (27.8) that magnetic field lines always form closed loops.

For Gauss's law, which always deals with *closed* surfaces, the vector area element $d\vec{A}$ in Eq. (27.6) always points *out of* the surface. However, some applications of *magnetic* flux involve an *open* surface with a boundary line; there is then an ambiguity of sign in Eq. (27.6) because of the two possible choices of direction for $d\vec{A}$. In these cases we choose one of the two sides of the surface to be the “positive” side and use that choice consistently.

If the element of area dA in Eq. (27.5) is at right angles to the field lines, then $B_{\perp} = B$; calling the area $dA_{\perp}$, we have

$$B = \frac{d\Phi_B}{dA_{\perp}} \quad (27.9)$$

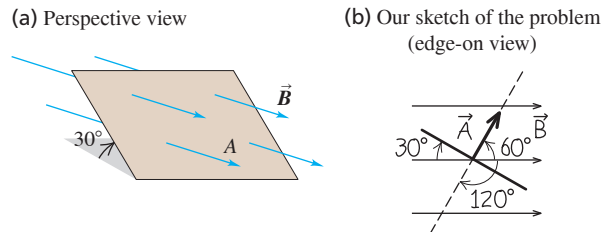
That is, the magnitude of the magnetic field is equal to *flux per unit area* across an area at right angles to the magnetic field. For this reason, magnetic field $\vec{B}$ is sometimes called **magnetic flux density**.

EXAMPLE 27.2 Magnetic flux calculations

Figure 27.16a is a perspective view of a flat surface with area 3.0 cm^2 in a uniform magnetic field $\vec{B}$. The magnetic flux through this surface is $+0.90 \text{ mWb}$. Find the magnitude of the magnetic field and the direction of the area vector $\vec{A}$.

IDENTIFY and SET UP Our target variables are the field magnitude B and the direction of the area vector. Because $\vec{B}$ is uniform, B and ϕ are the same at all points on the surface. Hence we can use Eq. (27.7), $\Phi_B = BA \cos \phi$.

Figure 27.16 (a) A flat area A in a uniform magnetic field $\vec{B}$. (b) The area vector $\vec{A}$ makes a 60° angle with $\vec{B}$. (If we had chosen $\vec{A}$ to point in the opposite direction, ϕ would have been 120° and the magnetic flux Φ_B would have been negative.)



EXECUTE The area A is $3.0 \times 10^{-4} \text{ m}^2$; the direction of $\vec{A}$ is perpendicular to the surface, so ϕ could be either 60° or 120° . But Φ_B , B , and A are all positive, so $\cos \phi$ must also be positive. This rules out 120° , so $\phi = 60^\circ$ (Fig. 27.16b). Hence we find

$$B = \frac{\Phi_B}{A \cos \phi} = \frac{0.90 \times 10^{-3} \text{ Wb}}{(3.0 \times 10^{-4} \text{ m}^2)(\cos 60^\circ)} = 6.0 \text{ T}$$

EVALUATE In many problems we are asked to calculate the flux of a given magnetic field through a given area. This example is somewhat different: It tests your understanding of the definition of magnetic flux.

KEYCONCEPT If a uniform magnetic field $\vec{B}$ is present over a surface, the magnetic flux through that surface equals the area of the surface multiplied by the component of $\vec{B}$ normal (perpendicular) to the surface. The net magnetic flux Φ_B through a closed surface is always zero.

TEST YOUR UNDERSTANDING OF SECTION 27.3 Imagine moving along the axis of the current-carrying loop in Fig. 27.13c, starting at a point well to the left of the loop and ending at a point well to the right of the loop. (a) How would the magnetic field strength vary as you moved along this path? (i) It would be the same at all points along the path; (ii) it would increase and then decrease; (iii) it would decrease and then increase. (b) Would the magnetic field direction vary as you moved along the path?

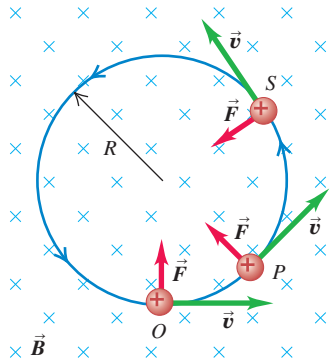
ANSWER

(a) (iii) The magnitude of $\vec{B}$ would increase as you moved to the right, reaching a maximum as you passed through the plane of the loop. As you moved beyond the plane of the loop, the field magnitude would decrease. You can tell this from the spacing of the field lines: The closer the field lines, the stronger the field. The direction of the field would be to the right at all points along the path, since the path is along a field line and the direction of $\vec{B}$ at any point is tangent to the field line through that point.

Figure 27.17 A charged particle moves in a plane perpendicular to a uniform magnetic field $\vec{B}$.

(a) The orbit of a charged particle in a uniform magnetic field

A charge moving at right angles to a uniform $\vec{B}$ field moves in a circle at constant speed because $\vec{F}$ and $\vec{v}$ are always perpendicular to each other.



(b) An electron beam (seen as a white arc) curving in a magnetic field

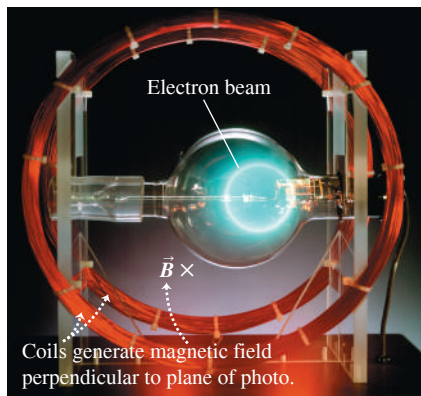
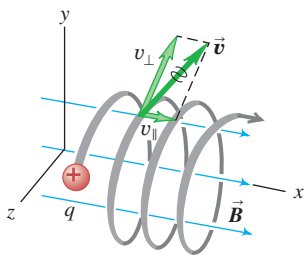


Figure 27.18 The general case of a charged particle moving in a uniform magnetic field $\vec{B}$. The magnetic field does no work on the particle, so its speed and kinetic energy remain constant.

This particle's motion has components both parallel ($v_{\parallel}$) and perpendicular ($v_{\perp}$) to the magnetic field, so it moves in a helical path.



27.4 MOTION OF CHARGED PARTICLES IN A MAGNETIC FIELD

When a charged particle moves in a magnetic field, it is acted on by the magnetic force given by Eq. (27.2), and the motion is determined by Newton's laws. **Figure 27.17a** shows a simple example. A particle with positive charge q is at point O , moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ directed into the plane of the figure. The vectors $\vec{v}$ and $\vec{B}$ are perpendicular, so the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ has magnitude $F = qvB$ and a direction as shown in the figure. The force is *always* perpendicular to $\vec{v}$, so it cannot change the *magnitude* of the velocity, only its direction. To put it differently, the magnetic force never has a component parallel to the particle's motion, so the magnetic force can never do *work* on the particle. This is true even if the magnetic field is not uniform.

Motion of a charged particle under the action of a magnetic field alone is always motion with constant speed.

Using this principle, we see that in Fig. 27.17a the magnitudes of both $\vec{F}$ and $\vec{v}$ are constant. As the particle of mass m moves from O to P to S , the directions of force and velocity change but their magnitudes stay the same. So the particle is acted on by a constant-magnitude force that is always at right angles to the velocity of the particle. It follows from our discussion of circular motion in Sections 3.4 and 5.4 that the particle moves in a *circle* of radius R with constant speed v . The centripetal acceleration is v^2/R and only the magnetic force acts, so from Newton's second law,

$$F = |q|vB = m \frac{v^2}{R} \quad (27.10)$$

We solve Eq. (27.10) for R :

$$\text{Radius of a circular orbit in a magnetic field} \quad R = \frac{\text{Particle's mass} \times \text{Particle's speed}}{\text{Magnetic-field magnitude} \times \text{Particle's charge}} = \frac{mv}{|q|B} \quad (27.11)$$

If the charge q is negative, the particle moves *clockwise* around the orbit in Fig. 27.17a. Figure 27.17b shows a beam of negatively charged electrons following just such an orbit.

The angular speed ω of the particle can be found from Eq. (9.13), $v = R\omega$. Combining this with Eq. (27.11), we get

$$\omega = \frac{v}{R} = v \frac{|q|B}{mv} = \frac{|q|B}{m} \quad (27.12)$$

The number of revolutions per unit time is $f = \omega/2\pi$. This frequency f is independent of the radius R of the path. It is called the **cyclotron frequency**; in a particle accelerator called a *cyclotron*, particles moving in nearly circular paths are given a boost twice each revolution, increasing their energy and their orbital radii but not their angular speed or frequency. Similarly, one type of *magnetron*, a common source of microwave radiation for microwave ovens and radar systems, emits radiation with a frequency equal to the frequency of circular motion of electrons in a vacuum chamber between the poles of a magnet.

If the direction of the initial velocity is *not* perpendicular to the field, the velocity *component* parallel to the field is constant because there is no force parallel to the field. Then the particle moves in a helix (**Fig. 27.18**). The radius of the helix is given by Eq. (27.11), where v is now the component of velocity perpendicular to the $\vec{B}$ field.

Figure 27.19 A magnetic bottle. Particles near either end of the region experience a magnetic force toward the center of the region. This is one way of containing an ionized gas that has a temperature of the order of 10^6 K, which would vaporize any material container.

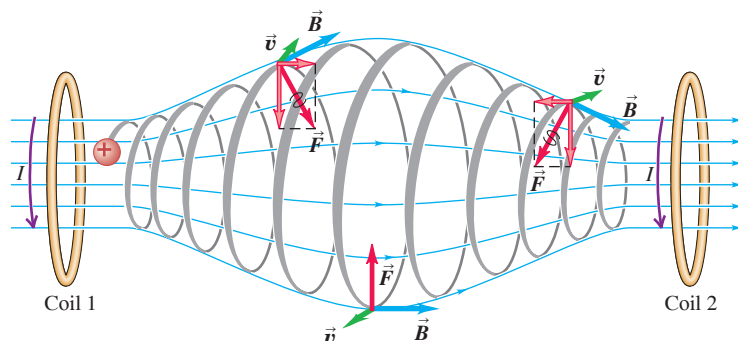
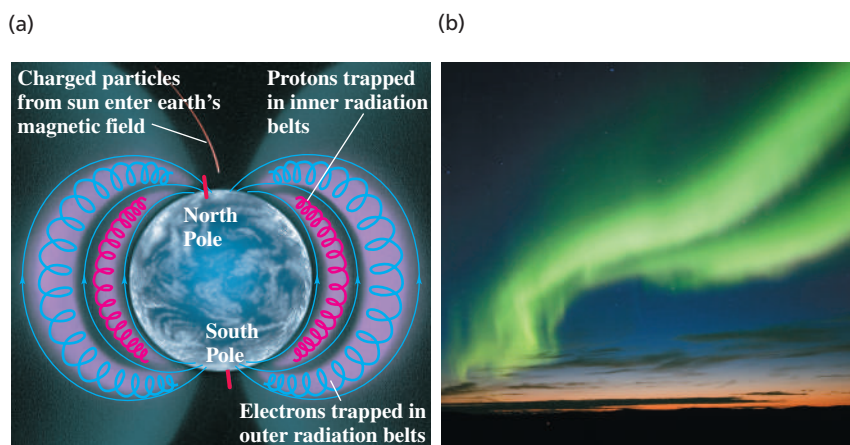


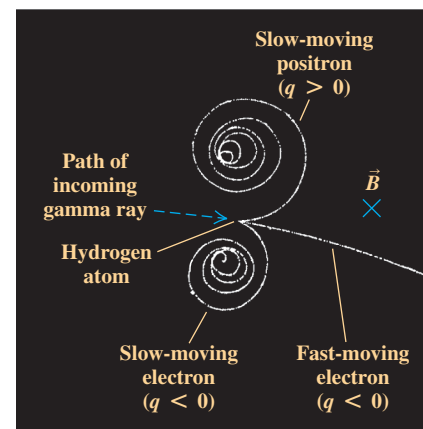
Figure 27.20 (a) The Van Allen radiation belts around the earth. Near the poles, charged particles from these belts can enter the atmosphere, producing the aurora borealis (“northern lights”) and aurora australis (“southern lights”). (b) A photograph of the aurora borealis.



Motion of a charged particle in a nonuniform magnetic field is more complex. **Figure 27.19** shows a field produced by two circular coils separated by some distance. Particles near either coil experience a magnetic force toward the center of the region; particles with appropriate speeds spiral repeatedly from one end of the region to the other and back. Because charged particles can be trapped in such a magnetic field, it is called a *magnetic bottle*. This technique is used to confine very hot plasmas with temperatures of the order of 10^6 K. In a similar way the earth's nonuniform magnetic field traps charged particles coming from the sun in doughnut-shaped regions around the earth, as shown in **Fig. 27.20**. These regions, called the *Van Allen radiation belts*, were discovered in 1958 from data obtained by instruments aboard the Explorer I satellite.

Magnetic forces on charged particles play an important role in studies of elementary particles. **Figure 27.21** shows a chamber filled with liquid hydrogen and with a magnetic field directed into the plane of the photograph. A high-energy gamma ray dislodges an electron from a hydrogen atom, sending it off at high speed and creating a visible track. The track shows the electron curving downward due to the magnetic force. The energy of the collision also produces another electron and a *positron* (a positively charged electron). Because of their opposite charges, the trajectories of the electron and the positron curve in opposite directions. As these particles plow through the liquid hydrogen, they collide with other charged particles, losing energy and speed. As a result, the radius of curvature decreases as suggested by Eq. (27.11). (The electron's speed is comparable to the speed of light, so Eq. (27.11) isn't directly applicable here.) Similar experiments allow physicists to determine the mass and charge of newly discovered particles.

Figure 27.21 This bubble chamber image shows the result of a high-energy gamma ray (which does not leave a track) that collides with an electron in a hydrogen atom. This electron flies off to the right at high speed. Some of the energy in the collision is transformed into a second electron and a positron (a positively charged electron). A magnetic field is directed into the plane of the image, which makes the positive and negative particles curve off in different directions.



PROBLEM-SOLVING STRATEGY 27.2 Motion in Magnetic Fields

IDENTIFY the relevant concepts: In analyzing the motion of a charged particle in electric and magnetic fields, you'll apply Newton's second law of motion, $\Sigma \vec{F} = m\vec{a}$, with the net force given by $\Sigma \vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. Often other forces such as gravity can be ignored. Many of the problems are similar to the trajectory and circular-motion problems in Sections 3.3, 3.4, and 5.4; it would be a good idea to review those sections.

SET UP the problem using the following steps:

1. Determine the target variable(s).
2. Often the use of components is the most efficient approach. Choose a coordinate system and then express all vector quantities in terms of their components in this system.

EXECUTE the solution as follows:

1. If the particle moves perpendicular to a uniform magnetic field, the trajectory is a circle with a radius and angular speed given by Eqs. (27.11) and (27.12), respectively.
2. If your calculation involves a more complex trajectory, use $\Sigma \vec{F} = m\vec{a}$ in component form: $\Sigma F_x = ma_x$, and so forth. This approach is particularly useful when both electric and magnetic fields are present.

EVALUATE your answer: Check whether your results are reasonable.

EXAMPLE 27.3 Electron motion in a magnetron

WITH VARIATION PROBLEMS

A magnetron in a microwave oven emits electromagnetic waves with frequency $f = 2450$ MHz. What magnetic field strength is required for electrons to move in circular paths with this frequency?

IDENTIFY and SET UP The problem refers to circular motion as shown in Fig. 27.17a. We use Eq. (27.12) to solve for the field magnitude B .

EXECUTE The angular speed that corresponds to the frequency f is $\omega = 2\pi f = (2\pi)(2450 \times 10^6 \text{ s}^{-1}) = 1.54 \times 10^{10} \text{ s}^{-1}$. Then from Eq. (27.12),

$$B = \frac{m\omega}{|q|} = \frac{(9.11 \times 10^{-31} \text{ kg})(1.54 \times 10^{10} \text{ s}^{-1})}{1.60 \times 10^{-19} \text{ C}} = 0.0877 \text{ T}$$

EVALUATE This is a moderate field strength, easily produced with a permanent magnet. Incidentally, 2450 MHz electromagnetic waves are useful for heating and cooking food because they are strongly absorbed by water molecules.

KEYCONCEPT A charged particle that travels perpendicular to a uniform magnetic field $\vec{B}$ undergoes uniform circular motion. The frequency of the motion is independent of the speed of the particle: It depends on only the magnetic field strength and the particle's charge and mass.

EXAMPLE 27.4 Helical particle motion in a magnetic field

WITH VARIATION PROBLEMS

In a situation like that shown in Fig. 27.18, the charged particle is a proton ($q = 1.60 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$) and the uniform, 0.500 T magnetic field is directed along the x -axis. At $t = 0$ the proton has velocity components $v_x = 1.50 \times 10^5 \text{ m/s}$, $v_y = 0$, and $v_z = 2.00 \times 10^5 \text{ m/s}$. Only the magnetic force acts on the proton. (a) At $t = 0$, find the force on the proton and its acceleration. (b) Find the radius of the resulting helical path, the angular speed of the proton, and the pitch of the helix (the distance traveled along the helix axis per revolution).

IDENTIFY and SET UP The magnetic force is $\vec{F} = q\vec{v} \times \vec{B}$; Newton's second law gives the resulting acceleration. Because $\vec{F}$ is perpendicular to $\vec{v}$, the proton's speed does not change. Hence Eq. (27.11) gives the radius of the helical path if we replace v with the velocity component perpendicular to $\vec{B}$. Equation (27.12) gives the angular speed ω , which yields the time T for one revolution (the period). Given the velocity component parallel to the magnetic field, we can then determine the pitch.

EXECUTE (a) With $\vec{B} = B\hat{i}$ and $\vec{v} = v_x\hat{i} + v_z\hat{k}$, Eq. (27.2) yields

$$\begin{aligned}\vec{F} &= q\vec{v} \times \vec{B} = q(v_x\hat{i} + v_z\hat{k}) \times B\hat{i} = qv_zB\hat{j} \\ &= (1.60 \times 10^{-19} \text{ C})(2.00 \times 10^5 \text{ m/s})(0.500 \text{ T})\hat{j} \\ &= (1.60 \times 10^{-14} \text{ N})\hat{j}\end{aligned}$$

(Recall: $\hat{i} \times \hat{i} = \mathbf{0}$ and $\hat{k} \times \hat{i} = \hat{j}$.) The resulting acceleration is

$$\vec{a} = \frac{\vec{F}}{m} = \frac{1.60 \times 10^{-14} \text{ N}}{1.67 \times 10^{-27} \text{ kg}}\hat{j} = (9.58 \times 10^{12} \text{ m/s}^2)\hat{j}$$

(b) Since $v_y = 0$, the component of velocity perpendicular to $\vec{B}$ is v_z ; then from Eq. (27.11),

$$\begin{aligned}R &= \frac{mv_z}{|q|B} = \frac{(1.67 \times 10^{-27} \text{ kg})(2.00 \times 10^5 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(0.500 \text{ T})} \\ &= 4.18 \times 10^{-3} \text{ m} = 4.18 \text{ mm}\end{aligned}$$

From Eq. (27.12) the angular speed is

$$\omega = \frac{|q|B}{m} = \frac{(1.60 \times 10^{-19} \text{ C})(0.500 \text{ T})}{1.67 \times 10^{-27} \text{ kg}} = 4.79 \times 10^7 \text{ rad/s}$$

The period is $T = 2\pi/\omega = 2\pi/(4.79 \times 10^7 \text{ s}^{-1}) = 1.31 \times 10^{-7} \text{ s}$. The pitch is the distance traveled along the x -axis in this time, or

$$\begin{aligned}v_x T &= (1.50 \times 10^5 \text{ m/s})(1.31 \times 10^{-7} \text{ s}) \\ &= 0.0197 \text{ m} = 19.7 \text{ mm}\end{aligned}$$

EVALUATE Although the magnetic force has a tiny magnitude, it produces an immense acceleration because the proton mass is so small. Note that the pitch of the helix is almost five times greater than the radius R , so this helix is much more "stretched out" than that shown in Fig. 27.18.

KEYCONCEPT The most general path that a charged particle follows in a uniform magnetic field $\vec{B}$ is a helix: straight-line, constant-speed motion parallel to $\vec{B}$ combined with uniform circular motion perpendicular to $\vec{B}$.

TEST YOUR UNDERSTANDING OF SECTION 27.4 (a) If you double the speed of the charged particle in Fig. 27.17a while keeping the magnetic field the same (as well as the charge and the mass), how does this affect the radius of the trajectory? (i) The radius is unchanged; (ii) the radius is twice as large; (iii) the radius is four times as large; (iv) the radius is $\frac{1}{2}$ as large; (v) the radius is $\frac{1}{4}$ as large. (b) How does this affect the time required for one complete circular orbit? (i) The time is unchanged; (ii) the time is twice as long; (iii) the time is four times as long; (iv) the time is $\frac{1}{2}$ as long; (v) the time is $\frac{1}{4}$ as long.

ANSWER

(a) (i) The radius of the orbit as given by Eq. (27.11) is directly proportional to the speed, so doubling the particle speed causes the radius to double as well. The particle has to travel far to complete one orbit but is traveling at double the speed, so the time for one orbit is unchanged. This result also follows from Eq. (27.12), which states that the angular speed ω is independent of the linear speed v . Hence the time per orbit, $T = 2\pi/\omega$, likewise does not depend on v .

27.5 APPLICATIONS OF MOTION OF CHARGED PARTICLES

This section describes several applications of the principles introduced in this chapter. Study them carefully, watching for applications of Problem-Solving Strategy 27.2 (Section 27.4).

Velocity Selector

In a beam of charged particles produced by a heated cathode or a radioactive material, not all particles move with the same speed. Many applications, however, require a beam in which all the particle speeds are the same. Particles of a specific speed can be selected from the beam by using an arrangement of electric and magnetic fields called a *velocity selector*. In **Fig. 27.22a** a charged particle with mass m , charge q , and speed v enters a region of space where the electric and magnetic fields are perpendicular to the particle's velocity and to each other. The electric field $\vec{E}$ is to the left, and the magnetic field $\vec{B}$ is into the plane of the figure. If q is positive, the electric force is to the left, with magnitude qE , and the magnetic force is to the right, with magnitude qvB . For given field magnitudes E and B , for a particular value of v the electric and magnetic forces will be equal in magnitude; the total force is then zero, and the particle travels in a straight line with constant velocity. This will be the case if $qE = qvB$ (Fig. 27.22b), so the speed v for which there is no deflection is

$$v = \frac{E}{B} \quad (27.13)$$

Only particles with speeds equal to E/B can pass through without being deflected by the fields. By adjusting E and B appropriately, we can select particles having a particular speed for use in other experiments. Because q divides out in Eq. (27.13), a velocity selector for positively charged particles also works for electrons or other negatively charged particles.

Thomson's e/m Experiment

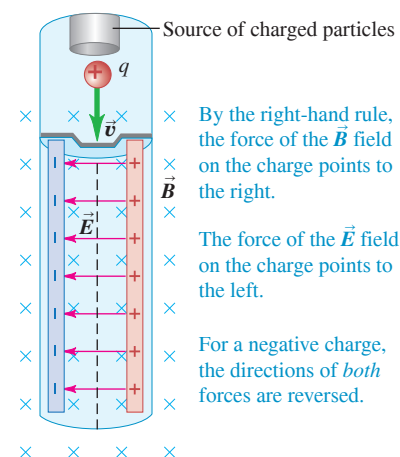
In one of the landmark experiments in physics at the end of the 19th century, J. J. Thomson (1856–1940) used the idea just described to measure the ratio of charge to mass for the electron. For this experiment, carried out in 1897 at the Cavendish Laboratory in Cambridge, England, Thomson used the apparatus shown in **Fig. 27.23** (next page). In a highly evacuated glass container, electrons from the hot cathode are accelerated and formed into a beam by a potential difference V between the two anodes A and A' . The speed v of the electrons is determined by the accelerating potential V . The gained kinetic energy $\frac{1}{2}mv^2$ equals the lost electric potential energy eV , where e is the magnitude of the electron charge:

$$\frac{1}{2}mv^2 = eV \quad \text{or} \quad v = \sqrt{\frac{2eV}{m}} \quad (27.14)$$

The electrons pass between the plates P and P' and strike the screen at the end of the tube, which is coated with a material that fluoresces (glows) at the point of impact. The

Figure 27.22 (a) A velocity selector for charged particles uses perpendicular $\vec{E}$ and $\vec{B}$ fields. Only charged particles with $v = E/B$ move through undeflected. (b) The electric and magnetic forces on a positive charge. Both forces are reversed if the charge is negative.

(a) Schematic diagram of velocity selector



(b) Free-body diagram for a positive particle

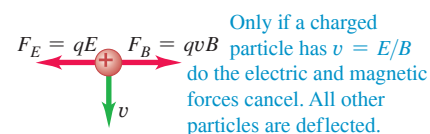
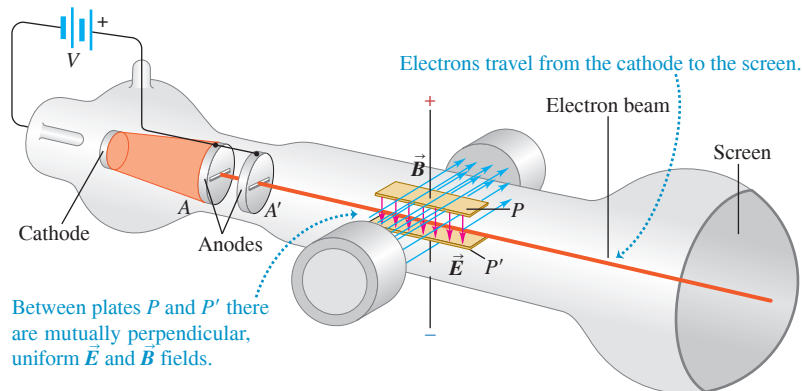


Figure 27.23 Thomson's apparatus for measuring the ratio e/m for the electron.

electrons pass straight through the plates when Eq. (27.13) is satisfied; combining this with Eq. (27.14), we get

$$\frac{E}{B} = \sqrt{\frac{2eV}{m}} \quad \text{so} \quad \frac{e}{m} = \frac{E^2}{2VB^2} \quad (27.15)$$

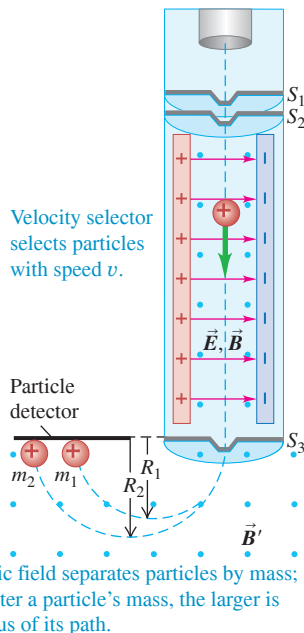
All the quantities on the right side can be measured, so the ratio e/m of charge to mass can be determined. It is *not* possible to measure e or m separately by this method, only their ratio.

The most significant aspect of Thomson's e/m measurements was that he found a *single value* for this quantity. It did not depend on the cathode material, the residual gas in the tube, or anything else about the experiment. This independence showed that the particles in the beam, which we now call electrons, are a common constituent of all matter. Thus Thomson is credited with the first discovery of a subatomic particle, the electron.

The most precise value of e/m available as of this writing is

$$e/m = 1.758820024(11) \times 10^{11} \text{ C/kg}$$

Figure 27.24 Bainbridge's mass spectrometer utilizes a velocity selector to produce particles with uniform speed v . In the region of magnetic field B' , particles with greater mass ($m_2 > m_1$) travel in paths with larger radius ($R_2 > R_1$).



In this expression, (11) indicates the likely uncertainty in the last two digits, 24.

Fifteen years after Thomson's experiments, the American physicist Robert Millikan succeeded in measuring the charge of the electron precisely (see Challenge Problem 23.81). This value, together with the value of e/m , enables us to determine the *mass* of the electron. The most precise value available at present is

$$m = 9.10938356(11) \times 10^{-31} \text{ kg}$$

Mass Spectrometers

Techniques similar to Thomson's e/m experiment can be used to measure masses of ions and thus measure atomic and molecular masses. In 1919, Francis Aston (1877–1945), a student of Thomson's, built the first of a family of instruments called **mass spectrometers**. A variation built by Bainbridge is shown in **Fig. 27.24**. Positive ions from a source pass through the slits S_1 and S_2 , forming a narrow beam. Then the ions pass through a velocity selector with crossed $\vec{E}$ and $\vec{B}$ fields, as we have described, to block all ions except those with speeds v equal to E/B . Finally, the ions pass into a region with a magnetic field $\vec{B}'$ perpendicular to the figure, where they move in circular arcs with radius R determined by Eq. (27.11): $R = mv/qB'$. Ions with different masses strike the detector at different points, and the values of R can be measured. We assume that each ion has lost one electron, so the net charge of each ion is just $+e$. With everything known in this equation except m , we can compute the mass m of the ion.

One of the earliest results from this work was the discovery that neon has two species of atoms, with atomic masses 20 and 22 g/mol. We now call these species **isotopes** of the element. Later experiments have shown that many elements have several isotopes—atoms with identical chemical behaviors but different masses due to differing numbers of neutrons in their nuclei. This is just one of the many applications of mass spectrometers in chemistry and physics.

EXAMPLE 27.5 An e/m demonstration experiment**WITH VARIATION PROBLEMS**

You set out to reproduce Thomson's e/m experiment with an accelerating potential of 150 V and a deflecting electric field of magnitude $6.0 \times 10^6 \text{ N/C}$. (a) How fast do the electrons move? (b) What magnetic-field magnitude will yield zero beam deflection? (c) With this magnetic field, how will the electron beam behave if you increase the accelerating potential above 150 V?

IDENTIFY and SET UP This is the situation shown in Fig. 27.23. We use Eq. (27.14) to determine the electron speed and Eq. (27.13) to determine the required magnetic field B .

EXECUTE (a) From Eq. (27.14), the electron speed v is

$$v = \sqrt{2(e/m)V} = \sqrt{2(1.76 \times 10^{11} \text{ C/kg})(150 \text{ V})} \\ = 7.27 \times 10^6 \text{ m/s} = 0.024c$$

(b) From Eq. (27.13), the required field strength is

$$B = \frac{E}{v} = \frac{6.0 \times 10^6 \text{ N/C}}{7.27 \times 10^6 \text{ m/s}} = 0.83 \text{ T}$$

(c) Increasing the accelerating potential V increases the electron speed v . In Fig. 27.23 this doesn't change the upward electric force eE , but it increases the downward magnetic force evB . Therefore the electron beam will turn *downward* and will hit the end of the tube below the undeflected position.

EVALUATE The required magnetic field is relatively large because the electrons are moving fairly rapidly (2.4% of the speed of light). If the maximum available magnetic field is less than 0.83 T, the electric field strength E would have to be reduced to maintain the desired ratio E/B in Eq. (27.15).

KEYCONCEPT A charged particle can experience both electric and magnetic forces if both an electric field $\vec{E}$ and a magnetic field $\vec{B}$ are present. These forces can cancel if $\vec{E}$, $\vec{B}$, and the particle's velocity are all mutually perpendicular and the particle travels at one particular speed [Eq. (27.14)].

EXAMPLE 27.6 Finding leaks in a vacuum system**WITH VARIATION PROBLEMS**

There is almost no helium in ordinary air, so helium sprayed near a leak in a vacuum system will quickly show up in the output of a vacuum pump connected to such a system. You are designing a leak detector that uses a mass spectrometer to detect He^+ ions (charge $+e = +1.60 \times 10^{-19} \text{ C}$, mass $6.65 \times 10^{-27} \text{ kg}$). Ions emerge from the velocity selector with a speed of $1.00 \times 10^5 \text{ m/s}$. They are curved in a semicircular path by a magnetic field B' and are detected at a distance of 10.16 cm from the slit S_3 in Fig. 27.24. Calculate the magnitude of the magnetic field B' .

IDENTIFY and SET UP After it passes through the slit, the ion follows a circular path as described in Section 27.4 (see Fig. 27.17). We solve Eq. (27.11) for B' .

EXECUTE The distance given is the *diameter* of the semicircular path shown in Fig. 27.24, so the radius is $R = \frac{1}{2}(10.16 \times 10^{-2} \text{ m})$. From Eq. (27.11), $R = mv/qB'$, we get

$$B' = \frac{mv}{qR} = \frac{(6.65 \times 10^{-27} \text{ kg})(1.00 \times 10^5 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(5.08 \times 10^{-2} \text{ m})} = 0.0818 \text{ T}$$

EVALUATE Helium leak detectors are widely used with high-vacuum systems. Our result shows that only a small magnetic field is required, so leak detectors can be relatively compact.

KEYCONCEPT The radius of the circular path that a charged particle follows in a uniform magnetic field depends on the particle's charge, mass, and speed. This dependence can be used to separate ions according to their mass.

TEST YOUR UNDERSTANDING OF SECTION 27.5 In Example 27.6 He^+ ions with charge $+e$ move at $1.00 \times 10^5 \text{ m/s}$ in a straight line through a velocity selector. Suppose the He^+ ions were replaced with He^{2+} ions, in which both electrons have been removed from the helium atom and the ion charge is $+2e$. At what speed must the He^{2+} ions travel through the same velocity selector in order to move in a straight line? (i) $4.00 \times 10^5 \text{ m/s}$; (ii) $2.00 \times 10^5 \text{ m/s}$; (iii) $1.00 \times 10^5 \text{ m/s}$; (iv) $0.50 \times 10^5 \text{ m/s}$; (v) $0.25 \times 10^5 \text{ m/s}$.

ANSWER is required is that the particles (in this case, ions) have a nonzero charge. selector does not depend on the magnitude or sign of the charge or the mass of the particle. All that

27.6 MAGNETIC FORCE ON A CURRENT-CARRYING CONDUCTOR

What makes an electric motor work? Within the motor are conductors that carry currents (that is, whose charges are in motion), as well as magnets that exert forces on the moving charges. Hence there is a magnetic force on each current-carrying conductor, and these forces make the motor turn. The d'Arsonval galvanometer (Section 26.3) also uses magnetic forces on conductors.

We can compute the force on a current-carrying conductor starting with the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on a single moving charge. **Figure 27.25** shows a straight segment of a conducting wire, with length l and cross-sectional area A ; the current is from bottom to

Figure 27.25 Forces on a moving positive charge in a current-carrying conductor.

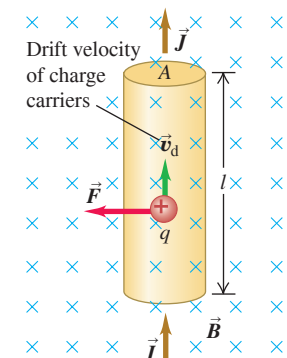


Figure 27.26 A straight wire segment of length $\vec{l}$ carries a current I in the direction of $\vec{l}$. The magnetic force on this segment is perpendicular to both $\vec{l}$ and the magnetic field $\vec{B}$.

Force $\vec{F}$ on a straight wire carrying a positive current and oriented at an angle ϕ to a magnetic field $\vec{B}$:

- Magnitude is $F = I l B_{\perp} = I l B \sin \phi$.
- Direction of $\vec{F}$ is given by the right-hand rule.

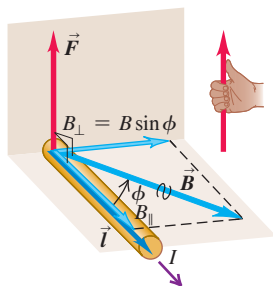
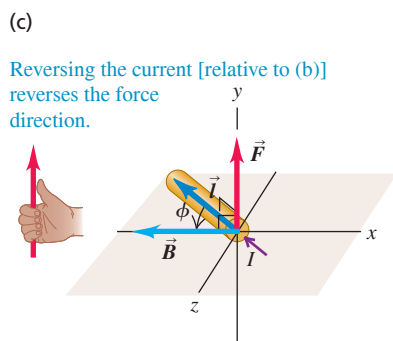
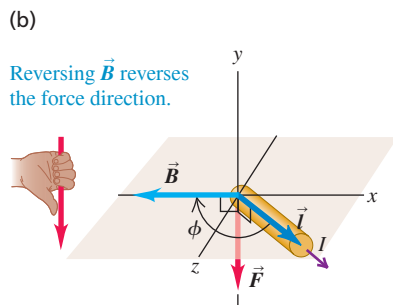
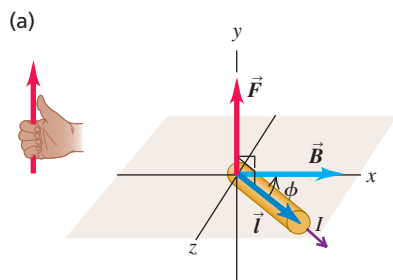


Figure 27.27 Magnetic field $\vec{B}$, length $\vec{l}$, and force $\vec{F}$ vectors for a straight wire carrying a current I .



top. The wire is in a uniform magnetic field $\vec{B}$, perpendicular to the plane of the diagram and directed *into* the plane. Let's assume first that the moving charges are positive. Later we'll see what happens when they are negative.

The drift velocity $\vec{v}_d$ is upward, perpendicular to $\vec{B}$. The average force on each charge is $\vec{F} = q\vec{v}_d \times \vec{B}$, directed to the left as shown in the figure; since $\vec{v}_d$ and $\vec{B}$ are perpendicular, the magnitude of the force is $F = qv_d B$.

We can derive an expression for the *total* force on all the moving charges in a length l of conductor with cross-sectional area A by using the same language we used in Eqs. (25.2) and (25.3) of Section 25.1. The number of charges per unit volume, or charge concentration, is n ; a segment of conductor with length l has volume Al and contains a number of charges equal to nAl . The total force $\vec{F}$ on *all* the moving charges in this segment has magnitude

$$F = (nAl)(qv_d B) = (nqv_d A)(lB) \quad (27.16)$$

From Eq. (25.3) the current density is $J = nqv_d$. The product JA is the total current I , so we can rewrite Eq. (27.16) as

$$F = IlB \quad (27.17)$$

If the $\vec{B}$ field is not perpendicular to the wire but makes an angle ϕ with it, as in Fig. 27.26, we handle the situation the same way we did in Section 27.2 for a single charge. Only the component of $\vec{B}$ perpendicular to the wire (and to the drift velocities of the charges) exerts a force; this component is $B_{\perp} = B \sin \phi$. The magnetic force on the wire segment is then

$$F = IlB_{\perp} = IlB \sin \phi \quad (27.18)$$

The force is always perpendicular to both the conductor and the field, with the direction determined by the same right-hand rule we used for a moving positive charge (Fig. 27.26). Hence this force can be expressed as a vector product, like the force on a single moving charge. We represent the segment of wire with a vector $\vec{l}$ along the wire in the direction of the current; then the force $\vec{F}$ on this segment is

$$\text{Magnetic force on a straight wire segment } \vec{F} = I \vec{l} \times \vec{B} \quad (27.19)$$

Current
Vector length of segment (points in current direction)

Figure 27.27 illustrates the directions of $\vec{B}$, $\vec{l}$, and $\vec{F}$ for several cases.

If the conductor is not straight, we can divide it into infinitesimal segments $d\vec{l}$. The force $d\vec{F}$ on each segment is

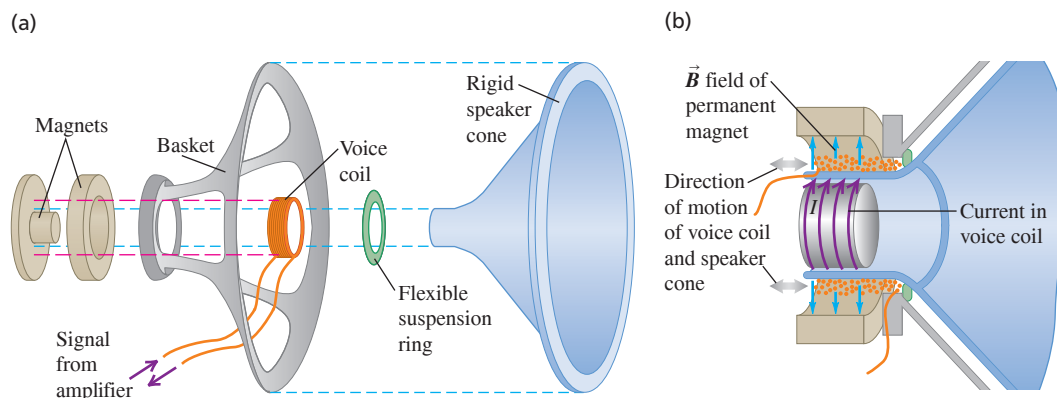
$$\text{Magnetic force on an infinitesimal wire segment } d\vec{F} = I d\vec{l} \times \vec{B} \quad (27.20)$$

Current
Vector length of segment (points in current direction)

Then we can integrate this expression along the wire to find the total force on a conductor of any shape. The integral is a *line integral*, the same mathematical operation we have used to define work (Section 6.3) and electric potential (Section 23.2).

CAUTION **Current is not a vector** Recall from Section 25.1 that the current I is not a vector. The direction of current flow is described by $d\vec{l}$, not I . If the conductor is curved, I is the same at all points along its length, but $d\vec{l}$ changes direction—it is always tangent to the conductor. ■

Figure 27.28 (a) Components of a loudspeaker. (b) The permanent magnet creates a magnetic field that exerts forces on the current in the voice coil; for a current I in the direction shown, the force is to the right. If the electric current in the voice coil oscillates, the speaker cone attached to the voice coil oscillates at the same frequency.



Finally, what happens when the moving charges are negative, such as electrons in a metal? Then in Fig. 27.25 an upward current corresponds to a downward drift velocity. But because q is now negative, the direction of the force $\vec{F}$ is the same as before. Thus Eqs. (27.17) through (27.20) are valid for *both* positive and negative charges and even when *both* signs of charge are present at once. This happens in some semiconductor materials and in ionic solutions.

A common application of the magnetic forces on a current-carrying wire is found in loudspeakers (Fig. 27.28). The radial magnetic field created by the permanent magnet exerts a force on the voice coil that is proportional to the current in the coil; the direction of the force is either to the left or to the right, depending on the direction of the current. The signal from the amplifier causes the current to oscillate in direction and magnitude. The coil and the speaker cone to which it is attached respond by oscillating with an amplitude proportional to the amplitude of the current in the coil. Turning up the volume knob on the amplifier increases the current amplitude and hence the amplitudes of the cone's oscillation and of the sound wave produced by the moving cone.

EXAMPLE 27.7 Magnetic force on a straight conductor

WITH VARIATION PROBLEMS

A straight horizontal copper rod carries a current of 50.0 A from west to east in a region between the poles of a large electromagnet. In this region there is a horizontal magnetic field toward the northeast (that is, 45° north of east) with magnitude 1.20 T. (a) Find the magnitude and direction of the force on a 1.00 m section of rod. (b) While keeping the rod horizontal, how should it be oriented to maximize the magnitude of the force? What is the force magnitude in this case?

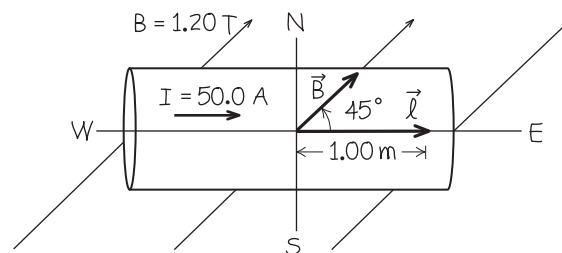
IDENTIFY and SET UP Figure 27.29 shows the situation. This is a straight wire segment in a uniform magnetic field, as in Fig. 27.26. Our target variables are the force $\vec{F}$ on the segment and the angle ϕ for which the force magnitude F is greatest. We find the magnitude of the magnetic force from Eq. (27.18) and the direction from the right-hand rule.

EXECUTE (a) The angle ϕ between the directions of current and field is 45° . From Eq. (27.18) we obtain

$$F = I l B \sin \phi = (50.0 \text{ A})(1.00 \text{ m})(1.20 \text{ T})(\sin 45^\circ) = 42.4 \text{ N}$$

The direction of the force is perpendicular to the plane of the current and the field, both of which lie in the horizontal plane. Thus the force must be vertical; the right-hand rule shows that it is vertically *upward* (out of the plane of the figure).

Figure 27.29 Our sketch of the copper rod as seen from overhead.



(b) From $F = I l B \sin \phi$, F is maximum for $\phi = 90^\circ$, so that $\vec{l}$ and $\vec{B}$ are perpendicular. To keep $\vec{F} = \vec{l} \times \vec{B}$ upward, we rotate the rod *clockwise* by 45° from its orientation in Fig. 27.29, so that the current runs toward the southeast. Then $F = I l B = (50.0 \text{ A})(1.00 \text{ m})(1.20 \text{ T}) = 60.0 \text{ N}$.

EVALUATE We check the result in part (a) by using Eq. (27.19) to calculate the force vector. If we use a coordinate system with the

Continued

x -axis pointing east, the y -axis north, and the z -axis upward, we have $\vec{l} = (1.00 \text{ m})\hat{i}$, $\vec{B} = (1.20 \text{ T})[(\cos 45^\circ)\hat{i} + (\sin 45^\circ)\hat{j}]$, and

$$\begin{aligned}\vec{F} &= \vec{l} \times \vec{B} \\ &= (50.0 \text{ A})(1.00 \text{ m})\hat{i} \times (1.20 \text{ T})[(\cos 45^\circ)\hat{i} + (\sin 45^\circ)\hat{j}] \\ &= (42.4 \text{ N})\hat{k}\end{aligned}$$

Note that the maximum upward force of 60.0 N can hold the conductor in midair against the force of gravity—that is, *magnetically levitate* the conductor—if its weight is 60.0 N and its mass is $m = w/g =$

$(60.0 \text{ N})/(9.8 \text{ m/s}^2) = 6.12 \text{ kg}$. Magnetic levitation is used in some high-speed trains to suspend the train over the tracks. Eliminating rolling friction in this way allows the train to achieve speeds of over 400 km/h.

KEYCONCEPT A straight, current-carrying conductor in a magnetic field $\vec{B}$ experiences a magnetic force that depends on its current, its length, the magnitude of $\vec{B}$, and the angle between the current direction and $\vec{B}$. The force is zero if the directions of the current and $\vec{B}$ are the same or opposite; otherwise, the force is nonzero and in a direction perpendicular to both the current and $\vec{B}$.

EXAMPLE 27.8 Magnetic force on a curved conductor

In Fig. 27.30 the magnetic field $\vec{B}$ is uniform and perpendicular to the plane of the figure, pointing out of the page. The conductor, carrying current I to the left, has three segments: (1) a straight segment with length L perpendicular to the plane of the figure, (2) a semicircle with radius R , and (3) another straight segment with length L parallel to the x -axis. Find the total magnetic force on this conductor.

IDENTIFY and SET UP The magnetic field $\vec{B} = B\hat{k}$ is uniform, so we find the forces $\vec{F}_1$ and $\vec{F}_3$ on the straight segments (1) and (3) from Eq. (27.19). We divide the curved segment (2) into infinitesimal straight segments and find the corresponding force $d\vec{F}_2$ on each straight segment from Eq. (27.20). We then integrate to find $\vec{F}_2$. The total magnetic force on the conductor is then $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$.

EXECUTE For segment (1), $\vec{L} = -L\hat{k}$. Hence from Eq. (27.19), $\vec{F}_1 = \vec{I}\vec{L} \times \vec{B} = \mathbf{0}$. For segment (3), $\vec{L} = -L\hat{i}$, so $\vec{F}_3 = \vec{I}\vec{L} \times \vec{B} = I(-L\hat{i}) \times (B\hat{k}) = ILB\hat{j}$.

For the curved segment (2), Fig. 27.30 shows a segment $d\vec{l}$ with length $dl = R d\theta$, at angle θ . The right-hand rule shows that the

direction of $d\vec{l} \times \vec{B}$ is radially outward from the center; make sure you can verify this. Because $d\vec{l}$ and $\vec{B}$ are perpendicular, the magnitude dF_2 of the force on the segment $d\vec{l}$ is $dF_2 = I dl B = I(R d\theta)B$. The components of the force on this segment are

$$dF_{2x} = IR d\theta B \cos \theta \quad dF_{2y} = IR d\theta B \sin \theta$$

To find the components of the total force, we integrate these expressions with respect to θ from $\theta = 0$ to $\theta = \pi$ to take in the whole semicircle. The results are

$$\begin{aligned}F_{2x} &= IRB \int_0^\pi \cos \theta d\theta = 0 \\ F_{2y} &= IRB \int_0^\pi \sin \theta d\theta = 2IRB\end{aligned}$$

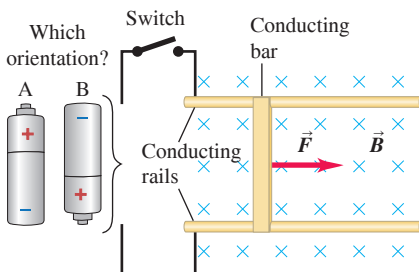
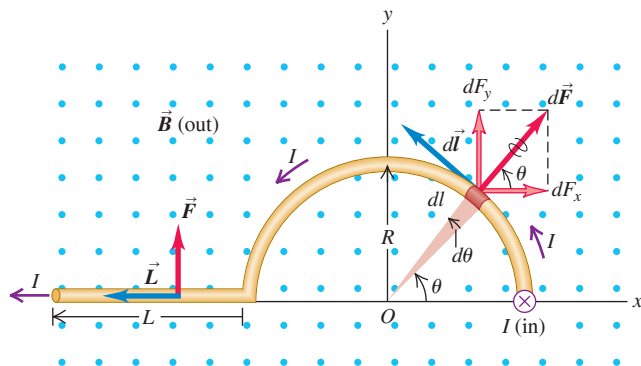
Hence $\vec{F}_2 = 2IRB\hat{j}$. Finally, adding the forces on all three segments, we find that the total force is in the positive y -direction:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \mathbf{0} + 2IRB\hat{j} + ILB\hat{j} = IB(2R + L)\hat{j}$$

EVALUATE We could have predicted from symmetry that the x -component of $\vec{F}_2$ would be zero: On the right half of the semicircle the x -component of the force is positive (to the right) and on the left half it is negative (to the left); the positive and negative contributions to the integral cancel. The result is that $\vec{F}_2$ is the force that would be exerted if we replaced the semicircle with a *straight* segment of length $2R$ along the x -axis. Do you see why?

KEYCONCEPT To find the magnetic force on a curved, current-carrying conductor in a magnetic field, first divide the conductor into infinitesimally small straight segments. Then find the force on one such segment. Finally, integrate over all segments in the conductor to find the net force.

Figure 27.30 What is the total magnetic force on the conductor?



TEST YOUR UNDERSTANDING OF SECTION 27.6 The accompanying figure shows a top view of two conducting rails on which a conducting bar can slide. A uniform magnetic field is directed perpendicular to the plane of the figure as shown. A battery is to be connected to the two rails so that when the switch is closed, current will flow through the bar and cause a magnetic force to push the bar to the right. In which orientation, A or B, should the battery be placed in the circuit?

ANSWER

A This orientation will cause current to flow clockwise around the circuit and hence through the conducting bar in the direction from the top to the bottom of the figure. From the right-hand rule, the magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ on the bar will then point to the right.

27.7 FORCE AND TORQUE ON A CURRENT LOOP

Current-carrying conductors usually form closed loops, so it is worthwhile to use the results of Section 27.6 to find the *total* magnetic force and torque on a conductor in the form of a loop. Many practical devices make use of the magnetic force or torque on a conducting loop, including loudspeakers (see Fig. 27.28) and galvanometers (see Section 26.3). Hence the results of this section are of substantial practical importance. These results will also help us understand the behavior of bar magnets described in Section 27.1.

As an example, let's look at a rectangular current loop in a uniform magnetic field. We can represent the loop as a series of straight line segments. We'll find that the *total force* on the loop is zero but that there can be a net *torque* acting on the loop, with some interesting properties.

Figure 27.31a shows a rectangular loop of wire with side lengths a and b . A line perpendicular to the plane of the loop (i.e., a *normal* to the plane) makes an angle ϕ with the direction of the magnetic field $\vec{B}$, and the loop carries a current I . The wires leading the current into and out of the loop and the source of emf are omitted to keep the diagram simple.

The force $\vec{F}$ on the right side of the loop (length a) is to the right, in the $+x$ -direction as shown. On this side, $\vec{B}$ is perpendicular to the current direction, and the force on this side has magnitude

$$F = IaB \quad (27.21)$$

A force $-\vec{F}$ with the same magnitude but opposite direction acts on the opposite side of the loop, as shown in the figure.

The sides with length b make an angle $(90^\circ - \phi)$ with the direction of $\vec{B}$. The forces on these sides are the vectors $\vec{F}'$ and $-\vec{F}'$; their magnitude F' is given by

$$F' = IbB \sin(90^\circ - \phi) = IbB \cos \phi$$

The lines of action of both forces lie along the y -axis.

The *total* force on the loop is zero because the forces on opposite sides cancel out in pairs.

The net force on a current loop in a uniform magnetic field is zero. However, the net torque is not in general equal to zero.

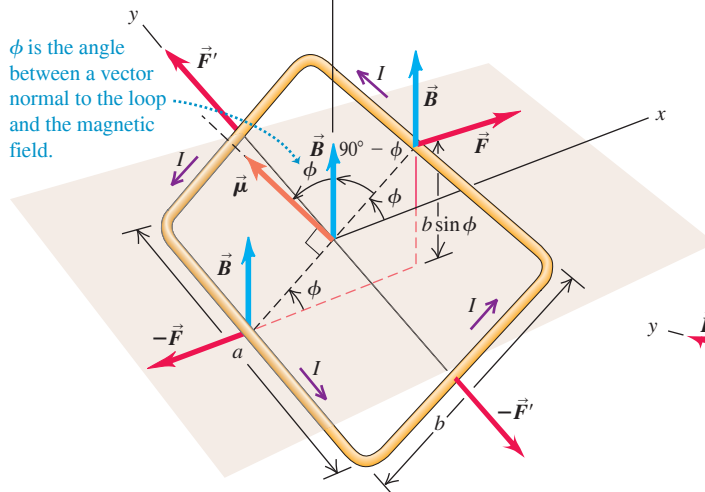
(You may find it helpful to review the discussion of torque in Section 10.1.) The two forces $\vec{F}'$ and $-\vec{F}'$ in Fig. 27.31a lie along the same line and so give rise to zero net torque with

Figure 27.31 Finding the torque on a current-carrying loop in a uniform magnetic field.

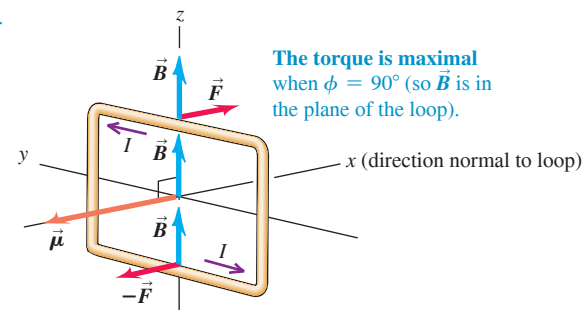
(a)

The two pairs of forces acting on the loop cancel, so no net force acts on the loop.

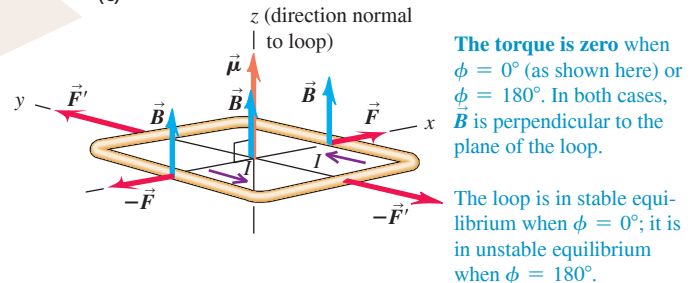
However, the forces on the a sides of the loop ($\vec{F}$ and $-\vec{F}$) produce a torque $\tau = (Iba)(b \sin \phi)$ on the loop.



(b)

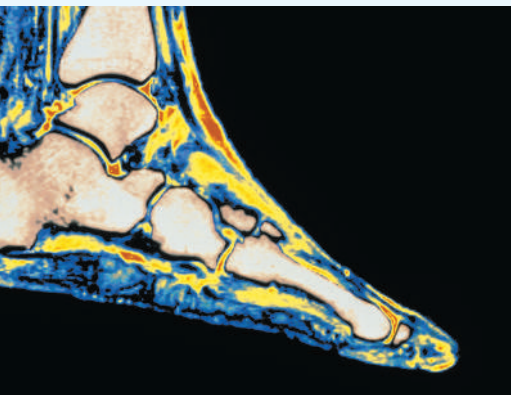


(c)



BIO APPLICATION Magnetic Resonance Imaging

In magnetic resonance imaging (MRI), a patient is placed in a strong magnetic field. Each hydrogen nucleus in the patient acts like a miniature current loop with a magnetic dipole moment that tends to align with the applied field. Radio waves of just the right frequency then flip these magnetic moments out of alignment. The extent to which the radio waves are absorbed is proportional to the amount of hydrogen present. This makes it possible to image details in hydrogen-rich soft tissue that cannot be seen in x-ray images. (X rays are superior to MRI for imaging bone, which is hydrogen deficient.)



respect to any point. The two forces $\vec{F}$ and $-\vec{F}$ lie along different lines, and each gives rise to a torque about the y-axis. According to the right-hand rule for determining the direction of torques, the vector torques due to $\vec{F}$ and $-\vec{F}$ are both in the $+y$ -direction; hence the net vector torque $\vec{\tau}$ is in the $+y$ -direction as well. The moment arm for each of these forces (equal to the perpendicular distance from the rotation axis to the line of action of the force) is $(b/2) \sin \phi$, so the torque due to each force has magnitude $F(b/2) \sin \phi$. If we use Eq. (27.21) for F , the magnitude of the net torque is

$$\tau = 2F(b/2) \sin \phi = (IBA)(b \sin \phi) \quad (27.22)$$

The torque is greatest when $\phi = 90^\circ$, $\vec{B}$ is in the plane of the loop, and the normal to this plane is perpendicular to $\vec{B}$ (Fig. 27.31b). The torque is zero when ϕ is 0° or 180° and the normal to the loop is parallel or antiparallel to the field (Fig. 27.31c). The value $\phi = 0^\circ$ is a stable equilibrium position because the torque is zero there, and when the loop is rotated slightly from this position, the resulting torque tends to rotate it back toward $\phi = 0^\circ$. The position $\phi = 180^\circ$ is an *unstable* equilibrium position; if displaced slightly from this position, the loop tends to move farther away from $\phi = 180^\circ$. Figure 27.31 shows rotation about the y-axis, but because the net force on the loop is zero, Eq. (27.22) for the torque is valid for *any* choice of axis. The torque always tends to rotate the loop in the direction of *decreasing* ϕ —that is, toward the stable equilibrium position $\phi = 0^\circ$.

The area A of the loop is equal to ab , so we can rewrite Eq. (27.22) as

$$\tau = IBA \sin \phi \quad (27.23)$$

Magnitude of magnetic torque on a current loop τ Current Magnetic-field magnitude Angle between normal to loop plane and field direction
Area of loop

The product IA is called the **magnetic dipole moment** or **magnetic moment** of the loop, for which we use the symbol μ (the Greek letter mu):

$$\mu = IA \quad (27.24)$$

It is analogous to the electric dipole moment introduced in Section 21.7. In terms of μ , the magnitude of the torque on a current loop is

$$\tau = \mu B \sin \phi \quad (27.25)$$

where ϕ is the angle between the normal to the loop (the direction of the vector area $\vec{A}$) and $\vec{B}$. A current loop, or any other object that experiences a magnetic torque given by Eq. (27.25), is also called a **magnetic dipole**.

Magnetic Torque: Vector Form

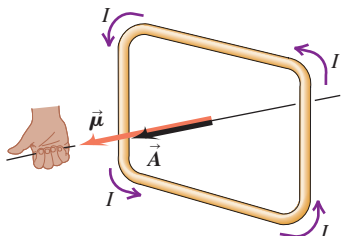
We can also define a vector magnetic moment $\vec{\mu}$ with magnitude IA : This is shown in Fig. 27.31. The direction of $\vec{\mu}$ is defined to be perpendicular to the plane of the loop, with a sense determined by a right-hand rule, as shown in **Fig. 27.32**. Wrap the fingers of your right hand around the perimeter of the loop in the direction of the current. Then extend your thumb so that it is perpendicular to the plane of the loop; its direction is the direction of $\vec{\mu}$ (and of the vector area $\vec{A}$ of the loop). The torque is greatest when $\vec{\mu}$ and $\vec{B}$ are perpendicular and is zero when they are parallel or antiparallel. In the stable equilibrium position, $\vec{\mu}$ and $\vec{B}$ are parallel.

Finally, we can express this interaction in terms of the torque vector $\vec{\tau}$, which we used for *electric*-dipole interactions in Section 21.7. From Eq. (27.25) the magnitude of $\vec{\tau}$ is equal to the magnitude of $\vec{\mu} \times \vec{B}$, and reference to Fig. 27.31 shows that the directions are also the same. So we have

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (27.26)$$

Vector magnetic torque on a current loop $\vec{\tau}$ Magnetic dipole moment Magnetic field

Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



This result is directly analogous to the result we found in Section 21.7 for the torque exerted by an *electric* field $\vec{E}$ on an *electric* dipole with dipole moment $\vec{p}$.

Potential Energy for a Magnetic Dipole

When a magnetic dipole changes orientation in a magnetic field, the field does work on it. In an infinitesimal angular displacement $d\phi$, the work dW is given by $\tau d\phi$, and there is a corresponding change in potential energy. As the above discussion suggests, the potential energy U is least when $\vec{\mu}$ and $\vec{B}$ are parallel and greatest when they are antiparallel. To find an expression for U as a function of orientation, note that the torque on an *electric* dipole in an *electric* field is $\vec{\tau} = \vec{p} \times \vec{E}$; we found in Section 21.7 that the corresponding potential energy is $U = -\vec{p} \cdot \vec{E}$. The torque on a *magnetic* dipole in a *magnetic* field is $\vec{\tau} = \vec{\mu} \times \vec{B}$, so we can conclude immediately that

$$\text{Potential energy for a magnetic dipole in a magnetic field} \quad U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \phi \quad (27.27)$$

Magnetic dipole moment
Angle between $\vec{\mu}$ and $\vec{B}$
Magnetic field

With this definition, U is zero when the magnetic dipole moment is perpendicular to the magnetic field ($\phi = 90^\circ$); then $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos 90^\circ = 0$.

Magnetic Torque: Loops and Coils

Although we have derived Eqs. (27.21) through (27.27) for a rectangular current loop, all these relationships are valid for a plane loop of any shape at all. Any planar loop may be approximated as closely as we wish by a very large number of rectangular loops, as shown in **Fig. 27.33**. If these loops all carry equal currents in the same clockwise sense, then the forces and torques on the sides of two loops adjacent to each other cancel, and the only forces and torques that do not cancel are due to currents around the boundary. Thus all the above relationships are valid for a plane current loop of any shape, with the magnetic moment $\vec{\mu} = I\vec{A}$.

We can also generalize this whole formulation to a coil consisting of N planar loops close together; the effect is simply to multiply each force, the magnetic moment, the torque, and the potential energy by a factor of N .

An arrangement of particular interest is the **solenoid**, a helical winding of wire, such as a coil wound on a circular cylinder (**Fig. 27.34**). If the windings are closely spaced, the solenoid can be approximated by a number of circular loops lying in planes at right angles to its long axis. The total torque on a solenoid in a magnetic field is simply the sum of the torques on the individual turns. For a solenoid with N turns in a uniform field B , the magnetic moment is $\mu = NIA$ and

$$\tau = NIAB \sin \phi \quad (27.28)$$

where ϕ is the angle between the axis of the solenoid and the direction of the field. The magnetic-moment vector $\vec{\mu}$ is along the solenoid axis. The torque is greatest when the solenoid axis is perpendicular to the magnetic field and zero when they are parallel. The effect of this torque is to tend to rotate the solenoid into a position where its axis is parallel to the field. Solenoids are also useful as *sources* of magnetic field, as we'll discuss in Chapter 28.

The d'Arsonval galvanometer, described in Section 26.3, makes use of a magnetic torque on a coil carrying a current. As **Fig. 26.14** shows, the magnetic field is not uniform but is *radial*, so the side thrusts on the coil are always perpendicular to its plane. Thus the angle ϕ in Eq. (27.28) is always 90° , and the magnetic torque is directly proportional to the current, no matter what the orientation of the coil. A restoring torque proportional to the angular displacement of the coil is provided by two hairsprings, which also serve as current leads to the coil. When current is supplied to the coil, it rotates along with its attached pointer until the restoring spring torque just balances the magnetic torque. Thus the pointer deflection is proportional to the current.

Figure 27.33 The collection of rectangles exactly matches the irregular plane loop in the limit as the number of rectangles approaches infinity and the width of each rectangle approaches zero.

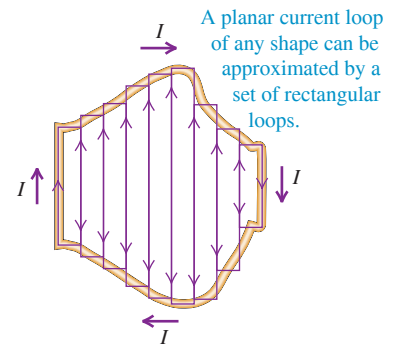
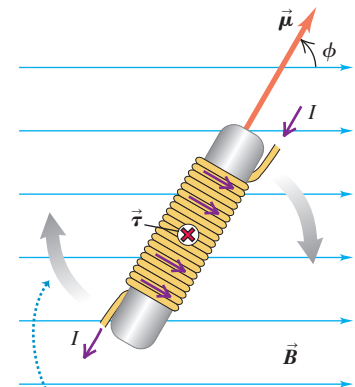


Figure 27.34 The torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on this solenoid in a uniform magnetic field is directed straight into the page. An actual solenoid has many more turns, wrapped closely together.



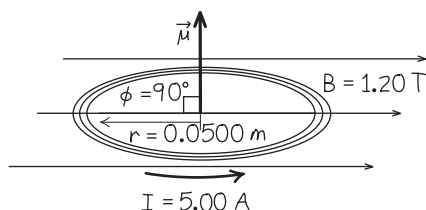
The torque tends to make the solenoid rotate clockwise in the plane of the page, aligning magnetic moment $\vec{\mu}$ with field $\vec{B}$.

EXAMPLE 27.9 Magnetic torque on a circular coil

A circular coil 0.0500 m in radius, with 30 turns of wire, lies in a horizontal plane. It carries a counterclockwise (as viewed from above) current of 5.00 A. The coil is in a uniform 1.20 T magnetic field directed toward the right. Find the magnitudes of the magnetic moment and the torque on the coil.

IDENTIFY and SET UP This problem uses the definition of magnetic moment and the expression for the torque on a magnetic dipole in a magnetic field. **Figure 27.35** shows the situation. Equation (27.24) gives the magnitude μ of the magnetic moment of a single turn of wire; for N turns, the magnetic moment is N times greater. Equation (27.25) gives the magnitude τ of the torque.

Figure 27.35 Our sketch for this problem.



EXECUTE The area of the coil is $A = \pi r^2$. From Eq. (27.24), the total magnetic moment of all 30 turns is

$$\mu_{\text{total}} = NIA = 30(5.00 \text{ A})\pi(0.0500 \text{ m})^2 = 1.18 \text{ A} \cdot \text{m}^2$$

The angle ϕ between the direction of $\vec{B}$ and the direction of $\vec{\mu}$ (which is along the normal to the plane of the coil) is 90° . From Eq. (27.25), the torque on the coil is

$$\begin{aligned}\tau &= \mu_{\text{total}} B \sin \phi = (1.18 \text{ A} \cdot \text{m}^2)(1.20 \text{ T})(\sin 90^\circ) \\ &= 1.41 \text{ N} \cdot \text{m}\end{aligned}$$

EVALUATE The torque tends to rotate the right side of the coil down and the left side up, into a position where the normal to its plane is parallel to $\vec{B}$.

KEYCONCEPT A current-carrying coil in a magnetic field $\vec{B}$ experiences a magnetic torque. The torque tends to orient the coil so its magnetic dipole moment $\vec{\mu}$ (a vector perpendicular to the plane of the coil, with magnitude equal to the product of the number of turns, the current, and the coil area) is in the same direction as $\vec{B}$.

EXAMPLE 27.10 Potential energy for a coil in a magnetic field

If the coil in Example 27.9 rotates from its initial orientation to one in which its magnetic moment $\vec{\mu}$ is parallel to $\vec{B}$, what is the change in potential energy?

IDENTIFY and SET UP Equation (27.27) gives the potential energy for each orientation. The initial position is as shown in Fig. 27.35, with $\phi_1 = 90^\circ$. In the final orientation, the coil has been rotated 90° clockwise so that $\vec{\mu}$ and $\vec{B}$ are parallel, so the angle between these vectors is $\phi_2 = 0$.

EXECUTE From Eq. (27.27), the potential energy change is

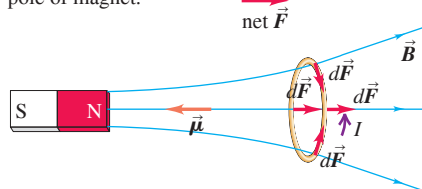
$$\begin{aligned}\Delta U &= U_2 - U_1 = -\mu B \cos \phi_2 - (-\mu B \cos \phi_1) \\ &= -\mu B (\cos \phi_2 - \cos \phi_1) \\ &= -(1.18 \text{ A} \cdot \text{m}^2)(1.20 \text{ T})(\cos 0^\circ - \cos 90^\circ) = -1.41 \text{ J}\end{aligned}$$

EVALUATE The potential energy decreases because the rotation is in the direction of the magnetic torque that we found in Example 27.9.

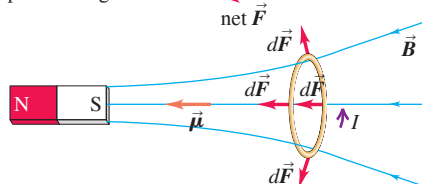
KEYCONCEPT The potential energy of a current-carrying coil in a magnetic field $\vec{B}$ depends on the magnitudes and directions of $\vec{B}$ and of the coil's magnetic dipole moment $\vec{\mu}$. The potential energy is minimum when $\vec{\mu}$ and $\vec{B}$ are in the same direction.

Figure 27.36 Forces on current loops in a nonuniform $\vec{B}$ field. In each case the axis of the bar magnet is perpendicular to the plane of the loop and passes through the center of the loop.

(a) Net force on this coil is away from north pole of magnet.



(b) Net force on same coil is toward south pole of magnet.

**Magnetic Dipole in a Nonuniform Magnetic Field**

We have seen that a current loop (that is, a magnetic dipole) experiences zero net force in a uniform magnetic field. **Figure 27.36** shows two current loops in the *nonuniform* $\vec{B}$ field of a bar magnet; in both cases the net force on the loop is *not* zero. In Fig. 27.36a the magnetic moment $\vec{\mu}$ is in the direction opposite to the field, and the force $d\vec{F} = I d\vec{l} \times \vec{B}$ on a segment of the loop has both a radial component and a component to the right. When these forces are summed to find the net force $\vec{F}$ on the loop, the radial components cancel so that the net force is to the right, away from the magnet. Note that in this case the force is toward the region where the field lines are farther apart and the field magnitude B is less. The polarity of the bar magnet is reversed in Fig. 27.36b, so $\vec{\mu}$ and $\vec{B}$ are parallel; now the net force on the loop is to the left, toward the region of greater field magnitude near the magnet. Later in this section we'll use these observations to explain why bar magnets can pick up unmagnetized iron objects.

Magnetic Dipoles and How Magnets Work

The behavior of a solenoid in a magnetic field (see Fig. 27.34) resembles that of a bar magnet or compass needle; if free to turn, both the solenoid and the magnet orient themselves with their axes parallel to the $\vec{B}$ field. In both cases this is due to the interaction of moving electric charges with a magnetic field; the difference is that in a bar magnet the motion of charge occurs on the microscopic scale of the atom.

Think of an electron as being like a spinning ball of charge. In this analogy the circulation of charge around the spin axis is like a current loop, and so the electron has a net magnetic moment. (This analogy, while helpful, is inexact; an electron isn't really a spinning sphere. A full explanation of the origin of an electron's magnetic moment involves quantum mechanics, which is beyond our scope here.) In an iron atom a substantial fraction of the electron magnetic moments align with each other, and the atom has a nonzero magnetic moment. (By contrast, the atoms of most elements have little or no net magnetic moment.) In an unmagnetized piece of iron there is no overall alignment of the magnetic moments of the atoms; their vector sum is zero, and the net magnetic moment is zero (**Fig. 27.37a**). But in an iron bar magnet the magnetic moments of many of the atoms are parallel, and there is a substantial net magnetic moment $\vec{\mu}$ (**Fig. 27.37b**). If the magnet is placed in a magnetic field $\vec{B}$, the field exerts a torque given by Eq. (27.26) that tends to align $\vec{\mu}$ with $\vec{B}$ (**Fig. 27.37c**). A bar magnet tends to align with a $\vec{B}$ field so that a line from the south pole to the north pole of the magnet is in the direction of $\vec{B}$; hence the real significance of a magnet's north and south poles is that they represent the head and tail, respectively, of the magnet's dipole moment $\vec{\mu}$.

The torque experienced by a current loop in a magnetic field also explains how an unmagnetized iron object like that in Fig. 27.37a becomes magnetized. If an unmagnetized iron paper clip is placed next to a powerful magnet, the magnetic moments of the paper clip's atoms tend to align with the $\vec{B}$ field of the magnet. When the paper clip is removed, its atomic dipoles tend to remain aligned, and the paper clip has a net magnetic moment. The paper clip can be demagnetized by being dropped on the floor or heated; the added internal energy jostles and re-randomizes the atomic dipoles.

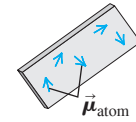
The magnetic-dipole picture of a bar magnet explains the attractive and repulsive forces between bar magnets shown in Fig. 27.1. The magnetic moment $\vec{\mu}$ of a bar magnet points from its south pole to its north pole, so the current loops in Figs. 27.36a and 27.36b are both equivalent to a magnet with its north pole on the left. Hence the situation in Fig. 27.36a is equivalent to two bar magnets with their north poles next to each other; the resultant force is repulsive, as in Fig. 27.1b. In Fig. 27.36b we again have the equivalent of two bar magnets end to end, but with the south pole of the left-hand magnet next to the north pole of the right-hand magnet. The resultant force is attractive, as in Fig. 27.1a.

Finally, we can explain how a magnet can attract an unmagnetized iron object (see Fig. 27.2). It's a two-step process. First, the atomic magnetic moments of the iron tend to align with the $\vec{B}$ field of the magnet, so the iron acquires a net magnetic dipole moment $\vec{\mu}$ parallel to the field. Second, the nonuniform field of the magnet attracts the magnetic dipole. **Figure 27.38a** shows an example. The north pole of the magnet is closer to the nail (which contains iron), and the magnetic dipole produced in the nail is equivalent to a loop with a current that circulates in a direction opposite to that shown in Fig. 27.36a. Hence the net magnetic force on the nail is opposite to the force on the loop in Fig. 27.36a, and the nail is attracted toward the magnet. Changing the polarity of the magnet, as in Fig. 27.38b, reverses the directions of both $\vec{B}$ and $\vec{\mu}$. The situation is now equivalent to that shown in Fig. 27.36b; like the loop in that figure, the nail is attracted toward the magnet. Hence a previously unmagnetized object containing iron is attracted to *either* pole of a magnet. By contrast, objects made of brass, aluminum, or wood hardly respond at all to a magnet; the atomic magnetic dipoles of these materials, if present at all, have less tendency to align with an external field.

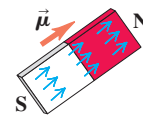
Our discussion of how magnets and pieces of iron interact has just scratched the surface of a diverse subject known as *magnetic properties of materials*. We'll discuss these properties in more depth in Section 28.8.

Figure 27.37 (a) An unmagnetized piece of iron. (Only a few representative atomic moments are shown.) (b) A magnetized piece of iron (bar magnet). The net magnetic moment of the bar magnet points from its south pole to its north pole. (c) A bar magnet in a magnetic field.

(a) Unmagnetized iron; magnetic moments are oriented randomly.



(b) In a bar magnet, the magnetic moments are aligned.



(c) A magnetic field creates a torque on the bar magnet that tends to align its dipole moment with the $\vec{B}$ field.

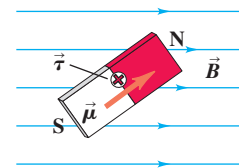
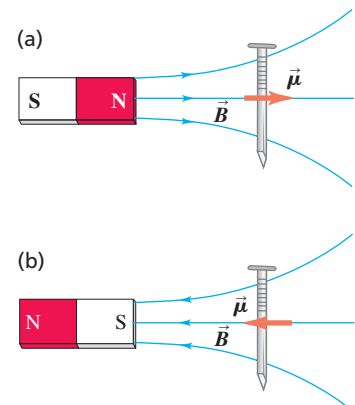


Figure 27.38 A bar magnet attracts an unmagnetized iron nail in two steps. First, the $\vec{B}$ field of the bar magnet gives rise to a net magnetic moment in the nail. Second, because the field of the bar magnet is not uniform, this magnetic dipole is attracted toward the magnet. The attraction is the same whether the nail is closer to (a) the magnet's north pole or (b) the magnet's south pole.



TEST YOUR UNDERSTANDING OF SECTION 27.7 Figure 27.13c depicts the magnetic field lines due to a circular current-carrying loop. (a) What is the direction of the magnetic moment of this loop? (b) Which side of the loop is equivalent to the north pole of a magnet, and which side is equivalent to the south pole?

ANSWER

(a) to the right; (b) North pole on the right, south pole on the left. If you wrap the fingers of your right hand around the coil in the direction of the current, your right thumb points to the right (perpendicular to the plane of the coil). This is the direction of the magnetic moment $\vec{\mu}$. The magnetic moment points from the south pole to the north pole, so the right side of the loop is equivalent to a north pole and the left side is equivalent to a south pole.

27.8 THE DIRECT-CURRENT MOTOR

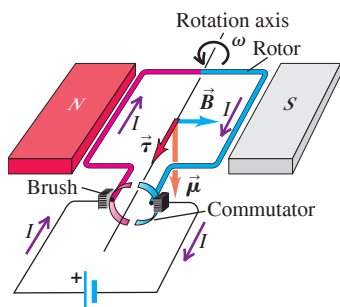
Electric motors play an important role in contemporary society. In a motor a magnetic torque acts on a current-carrying conductor, and electric energy is converted to mechanical energy. As an example, let's look at a simple type of direct-current (dc) motor, shown in **Fig. 27.39**.

The moving part of the motor is the *rotor*, a length of wire formed into an open-ended loop and free to rotate about an axis. The ends of the rotor wires are attached to circular conducting segments that form a *commutator*. In Fig. 27.39a, each of the two commutator segments makes contact with one of the terminals, or *brushes*, of an external circuit that includes a source of emf. This causes a current to flow into the rotor on one side, shown in red, and out of the rotor on the other side, shown in blue. Hence the rotor is a current loop with a magnetic moment $\vec{\mu}$. The rotor lies between opposing poles of a permanent magnet, so there is a magnetic field $\vec{B}$ that exerts a torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on the rotor. For the rotor orientation shown in Fig. 27.39a the torque causes the rotor to turn counterclockwise, in the direction that will align $\vec{\mu}$ with $\vec{B}$.

In Fig. 27.39b the rotor has rotated by 90° from its orientation in Fig. 27.39a. If the current through the rotor were constant, the rotor would now be in its equilibrium orientation; it would simply oscillate around this orientation. But here's where the commutator comes into play; each brush is now in contact with *both* segments of the commutator. There is no potential difference between the commutators, so at this instant no current flows through the rotor, and the magnetic moment is zero. The rotor continues to rotate counterclockwise because of its inertia, and current again flows through the rotor as in Fig. 27.39c. But now current enters on the *blue* side of the rotor and exits on the *red* side, just the opposite of the situation in Fig. 27.39a. While the direction of the current has reversed with respect

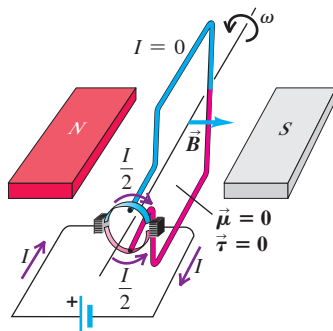
Figure 27.39 Schematic diagram of a simple dc motor. The rotor is a wire loop that is free to rotate about an axis; the rotor ends are attached to the two curved conductors that form the commutator. (The rotor halves are colored red and blue for clarity.) The commutator segments are insulated from one another.

(a) Brushes are aligned with commutator segments.



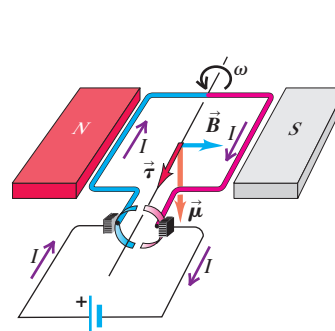
- Current flows into the red side of the rotor and out of the blue side.
- Therefore the magnetic torque causes the rotor to spin counterclockwise.

(b) Rotor has turned 90° .



- Each brush is in contact with both commutator segments, so the current bypasses the rotor altogether.
- No magnetic torque acts on the rotor.

(c) Rotor has turned 180° .



- The brushes are again aligned with commutator segments. This time the current flows into the blue side of the rotor and out of the red side.
- Therefore the magnetic torque again causes the rotor to spin counterclockwise.

to the rotor, the rotor itself has rotated 180° and the magnetic moment $\vec{\mu}$ is in the same direction with respect to the magnetic field. Hence the magnetic torque $\vec{\tau}$ is in the same direction in Fig. 27.39c as in Fig. 27.39a. Thanks to the commutator, the current reverses after every 180° of rotation, so the torque is always in the direction to rotate the rotor counterclockwise. When the motor has come “up to speed,” the average magnetic torque is just balanced by an opposing torque due to air resistance, friction in the rotor bearings, and friction between the commutator and brushes.

The simple motor shown in Fig. 27.39 has only a single turn of wire in its rotor. In practical motors, the rotor has many turns; this increases the magnetic moment and the torque so that the motor can spin larger loads. The torque can also be increased by using a stronger magnetic field, which is why many motor designs use electromagnets instead of a permanent magnet. Another drawback of the simple design in Fig. 27.39 is that the magnitude of the torque rises and falls as the rotor spins. This can be remedied by having the rotor include several independent coils of wire oriented at different angles (Fig. 27.40).

Power for Electric Motors

Because a motor converts electric energy to mechanical energy or work, it requires electric energy input. If the potential difference between its terminals is V_{ab} and the current is I , then the power input is $P = V_{ab}I$. Even if the motor coils have negligible resistance, there must be a potential difference between the terminals if P is to be different from zero. This potential difference results principally from magnetic forces exerted on the currents in the conductors of the rotor as they rotate through the magnetic field. The associated electromotive force $\mathcal{E}$ is called an *induced* emf; it is also called a *back* emf because its sense is opposite to that of the current. In Chapter 29 we’ll study induced emfs resulting from motion of conductors in magnetic fields.

In a *series* motor the rotor is connected in series with the electromagnet that produces the magnetic field; in a *shunt* motor they are connected in parallel. In a series motor with internal resistance r , V_{ab} is greater than $\mathcal{E}$, and the difference is the potential drop Ir across the internal resistance. That is,

$$V_{ab} = \mathcal{E} + Ir \quad (27.29)$$

Because the magnetic force is proportional to velocity, $\mathcal{E}$ is *not* constant but is proportional to the speed of rotation of the rotor.

EXAMPLE 27.11 A series dc motor

A dc motor with its rotor and field coils connected in series has an internal resistance of $2.00\ \Omega$. When running at full load on a 120 V line, it draws a 4.00 A current. (a) What is the emf in the rotor? (b) What is the power delivered to the motor? (c) What is the rate of dissipation of energy in the internal resistance? (d) What is the mechanical power developed? (e) What is the motor’s efficiency? (f) What happens if the machine being driven by the motor jams, so that the rotor suddenly stops turning?

IDENTIFY and SET UP This problem uses the ideas of power and potential drop in a series dc motor. We are given the internal resistance $r = 2.00\ \Omega$, the voltage $V_{ab} = 120\text{ V}$ across the motor, and the current $I = 4.00\text{ A}$ through the motor. We use Eq. (27.29) to determine the emf $\mathcal{E}$ from these quantities. The power delivered to the motor is $V_{ab}I$, the rate of energy dissipation is I^2r , and the power output by the motor is the difference between the power input and the power dissipated. The efficiency e is the ratio of mechanical power output to electric power input.

EXECUTE (a) From Eq. (27.29), $V_{ab} = \mathcal{E} + Ir$, we have

$$120\text{ V} = \mathcal{E} + (4.00\text{ A})(2.00\ \Omega) \quad \text{and so} \quad \mathcal{E} = 112\text{ V}$$

(b) The power delivered to the motor from the source is

$$P_{\text{input}} = V_{ab}I = (120\text{ V})(4.00\text{ A}) = 480\text{ W}$$

(c) The power dissipated in the resistance r is

$$P_{\text{dissipated}} = I^2r = (4.00\text{ A})^2(2.00\ \Omega) = 32\text{ W}$$

(d) The mechanical power output is the electric power input minus the rate of dissipation of energy in the motor’s resistance (assuming that there are no other power losses):

$$P_{\text{output}} = P_{\text{input}} - P_{\text{dissipated}} = 480\text{ W} - 32\text{ W} = 448\text{ W}$$

(e) The efficiency e is the ratio of mechanical power output to electric power input:

$$e = \frac{P_{\text{output}}}{P_{\text{input}}} = \frac{448\text{ W}}{480\text{ W}} = 0.93 = 93\%$$

(f) With the rotor stalled, the back emf $\mathcal{E}$ (which is proportional to rotor speed) goes to zero. From Eq. (27.29) the current becomes

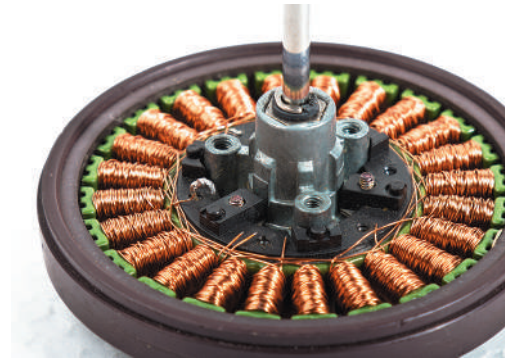
$$I = \frac{V_{ab}}{r} = \frac{120\text{ V}}{2.00\ \Omega} = 60\text{ A}$$

and the power dissipated in the resistance r becomes

$$P_{\text{dissipated}} = I^2r = (60\text{ A})^2(2.00\ \Omega) = 7200\text{ W}$$

Continued

Figure 27.40 This electric motor has multiple current-carrying coils. They interact with permanent magnets on the rotor (not shown) to make the rotor spin around the motor’s shaft. (This design is the reverse of the design in Fig. 27.39, in which the permanent magnets are stationary and the coil rotates.) Because there are multiple coils, the magnetic torque is very nearly constant and the rotor spins at a constant rate. Motors of this kind are used in drones, electric cars, and many other applications.



EVALUATE If this massive overload doesn't blow a fuse or trip a circuit breaker, the coils will quickly melt. When the motor is first turned on, there's a momentary surge of current until the motor picks up speed. This surge causes greater-than-usual voltage drops ($V = IR$) in the power lines supplying the current. Similar effects are responsible for the momentary dimming of lights that can occur in a house when an air conditioner or dishwasher motor starts.

KEYCONCEPT When an electric motor is in operation, a large emf is induced in the rotor coils. To find the mechanical power developed by the motor, multiply this emf by the current in the coils.

TEST YOUR UNDERSTANDING OF SECTION 27.8 In the circuit shown in Fig. 27.39, you add a switch in series with the source of emf so that the current can be turned on and off. When you close the switch and allow current to flow, will the rotor begin to turn no matter what its original orientation?

ANSWER

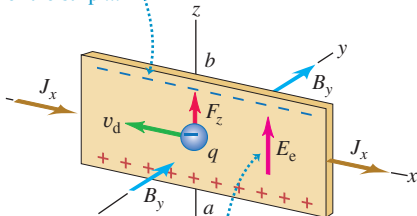
The rotor will not begin to turn when the switch is closed if the rotor is initially oriented as shown in Fig. 27.39b. In this case there is no current through the rotor and hence no magnetic torque. This situation can be remedied by using multiple rotor coils oriented at different angles around the rotation axis. With this arrangement, there is always a magnetic torque no matter what the orientation.

27.9 THE HALL EFFECT

Figure 27.41 Forces on charge carriers in a conductor in a magnetic field.

(a) Negative charge carriers (electrons)

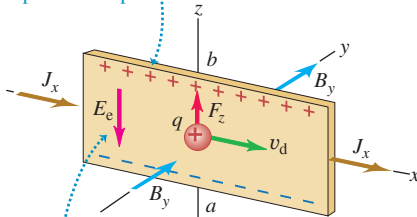
The charge carriers are pushed toward the top of the strip ...



... so point a is at a higher potential than point b.

(b) Positive charge carriers

The charge carriers are again pushed toward the top of the strip ...



... so the polarity of the potential difference is opposite to that for negative charge carriers.

The reality of the forces acting on the moving charges in a conductor in a magnetic field is strikingly demonstrated by the *Hall effect*, an effect analogous to the transverse deflection of an electron beam in a magnetic field in vacuum. (The effect was discovered by the American physicist Edwin Hall in 1879 while he was still a graduate student.) To describe this effect, let's consider a conductor in the form of a flat strip, as shown in **Fig. 27.41**. The current is in the direction of the $+x$ -axis and there is a uniform magnetic field $\vec{B}$ perpendicular to the plane of the strip, in the $+y$ -direction. The drift velocity of the moving charges (charge magnitude $|q|$) has magnitude v_d . Figure 27.41a shows the case of negative charges, such as electrons in a metal, and Fig. 27.41b shows positive charges. In both cases the magnetic force is upward, just as the magnetic force on a conductor is the same whether the moving charges are positive or negative. In either case a moving charge is driven toward the *upper* edge of the strip by the magnetic force $F_z = |q|v_d B$.

If the charge carriers are electrons, as in Fig. 27.41a, an excess negative charge accumulates at the upper edge of the strip, leaving an excess positive charge at its lower edge. This accumulation continues until the resulting transverse electrostatic field $\vec{E}_e$ becomes large enough to cause a force (magnitude $|q|E_e$) that is equal and opposite to the magnetic force (magnitude $|q|v_d B$). After that, there is no longer any net transverse force to deflect the moving charges. This electric field causes a transverse potential difference between opposite edges of the strip, called the *Hall voltage* or the *Hall emf*. The polarity depends on whether the moving charges are positive or negative. Experiments show that for metals the upper edge of the strip in Fig. 27.41a *does* become negatively charged, showing that the charge carriers in a metal are indeed negative electrons.

However, if the charge carriers are *positive*, as in Fig. 27.41b, then *positive* charge accumulates at the upper edge, and the potential difference is *opposite* to the situation with negative charges. Soon after the discovery of the Hall effect in 1879, it was observed that some materials, particularly some *semiconductors*, show a Hall emf opposite to that of the metals, as if their charge carriers were positively charged. We now know that these materials conduct by a process known as *hole conduction*. Within such a material there are locations, called *holes*, that would normally be occupied by an electron but are actually empty. A missing negative charge is equivalent to a positive charge. When an electron moves in one direction to fill a hole, it leaves another hole behind it. The hole migrates in the direction opposite to that of the electron.

In terms of the coordinate axes in Fig. 27.41b, the electrostatic field $\vec{E}_e$ for the positive- q case is in the $-z$ -direction; its z -component E_z is negative. The magnetic field is in the $+y$ -direction, and we write it as B_y . The magnetic force (in the $+z$ -direction) is $qv_d B_y$. The current density J_x is in the $+x$ -direction. In the steady state, when the forces qE_z and $qv_d B_y$ sum to zero,

$$qE_z + qv_d B_y = 0 \quad \text{or} \quad E_z = -v_d B_y$$

This confirms that when q is positive, E_z is negative. From Eq. (25.4),

$$J_x = nqv_d$$

Eliminating v_d between these equations, we find

$$\text{Hall effect: } nq = \frac{-J_x B_y}{E_z} \quad (27.30)$$

Concentration of mobile charge carriers
Current density
Magnetic field
Charge per carrier
Electrostatic field in conductor

Note that this result (as well as the entire derivation) is valid for both positive and negative q . When q is negative, E_z is positive, and conversely.

We can measure J_x , B_y , and E_z , so we can compute the product nq . In both metals and semiconductors, q is equal in magnitude to the electron charge, so the Hall effect permits a direct measurement of n , the concentration of current-carrying charges in the material. The *sign* of the charges is determined by the polarity of the Hall emf, as we have described.

The Hall effect can also be used for a direct measurement of electron drift speed v_d in metals. As we saw in Chapter 25, these speeds are very small, often of the order of 1 mm/s or less. If we move the entire conductor in the opposite direction to the current with a speed equal to the drift speed, then the electrons are at rest with respect to the magnetic field, and the Hall emf disappears. Thus the conductor speed needed to make the Hall emf vanish is equal to the drift speed.

EXAMPLE 27.12 A Hall-effect measurement

You place a strip of copper, 2.0 mm thick and 1.50 cm wide, in a uniform 0.40 T magnetic field as shown in Fig. 27.41a. When you run a 75 A current in the $+x$ -direction, you find that the potential at the bottom of the slab is $0.81 \mu\text{V}$ higher than at the top. From this measurement, determine the concentration of mobile electrons in copper.

IDENTIFY and SET UP This problem describes a Hall-effect experiment. We use Eq. (27.30) to determine the mobile electron concentration n .

EXECUTE First we find the current density J_x and the electric field E_z :

$$J_x = \frac{I}{A} = \frac{75 \text{ A}}{(2.0 \times 10^{-3} \text{ m})(1.50 \times 10^{-2} \text{ m})} = 2.5 \times 10^6 \text{ A/m}^2$$

$$E_z = \frac{V}{d} = \frac{0.81 \times 10^{-6} \text{ V}}{1.5 \times 10^{-2} \text{ m}} = 5.4 \times 10^{-5} \text{ V/m}$$

Then, from Eq. (27.30),

$$n = \frac{-J_x B_y}{qE_z} = \frac{-(2.5 \times 10^6 \text{ A/m}^2)(0.40 \text{ T})}{(-1.60 \times 10^{-19} \text{ C})(5.4 \times 10^{-5} \text{ V/m})} = 11.6 \times 10^{28} \text{ m}^{-3}$$

EVALUATE The actual value of n for copper is $8.5 \times 10^{28} \text{ m}^{-3}$. The difference shows that our simple model of the Hall effect, which ignores quantum effects and electron interactions with the ions, must be used with caution. This example also shows that with good conductors, the Hall emf is very small even with large current densities. In practice, Hall-effect devices for magnetic-field measurements use semiconductor materials, for which moderate current densities give much larger Hall emfs.

KEYCONCEPT In the Hall effect, a current in a conductor flows in a direction perpendicular to a magnetic field $\vec{B}$. The magnetic force on the current produces a charge separation in the conductor. This separation is perpendicular to both the current and $\vec{B}$, and produces a transverse electric field and potential difference in the conductor.

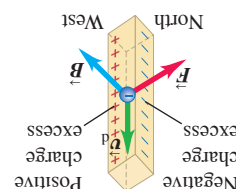
BIO APPLICATION Exercise Machines and the Hall Effect This athlete is making a flywheel spin inside a rowing machine. A gear attached to the flywheel has magnetic teeth, and every time a tooth on the spinning gear passes a current-carrying sensor, the tooth's magnetic field deflects the current and produces a Hall voltage in the sensor. The harder the athlete works, the faster the flywheel spins and the more voltage pulses there are per minute. The voltage information is passed to the rowing machine's display and tells the athlete her power output.



TEST YOUR UNDERSTANDING OF SECTION 27.9 A copper wire of square cross section is oriented vertically. The four sides of the wire face north, south, east, and west. There is a uniform magnetic field directed from east to west, and the wire carries current downward. Which side of the wire is at the highest electric potential? (i) North side; (ii) south side; (iii) east side; (iv) west side.

ANSWER

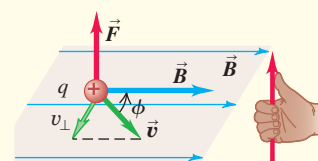
(ii) The mobile charge carriers in copper are negatively charged electrons, which move upward through the wire to give a downward current. From the right-hand rule, the force on a positively charged particle moving upward in a westward-pointing magnetic field would be to the south; hence the force on a negatively charged particle is to the north. The result is an excess of negative charge on the north side of the wire, leaving an excess of positive charge—and hence a higher electric potential—on the south side.



CHAPTER 27 SUMMARY

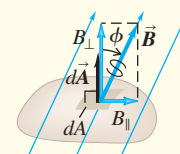
Magnetic forces: Magnetic interactions are fundamentally interactions between moving charged particles. These interactions are described by the vector magnetic field, denoted by $\vec{B}$. A particle with charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ experiences a force $\vec{F}$ that is perpendicular to both $\vec{v}$ and $\vec{B}$. The SI unit of magnetic field is the tesla ($1 \text{ T} = 1 \text{ N/A} \cdot \text{m}$). (See Example 27.1.)

$$\vec{F} = q\vec{v} \times \vec{B} \quad (27.2)$$



Magnetic field lines and flux: A magnetic field can be represented graphically by magnetic field lines. At each point a magnetic field line is tangent to the direction of $\vec{B}$ at that point. Where field lines are close together, the field magnitude is large, and vice versa. Magnetic flux Φ_B through an area is defined in an analogous way to electric flux; its SI unit is the weber ($1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$). The net magnetic flux through any closed surface is zero (Gauss's law for magnetism), so magnetic field lines always close on themselves. (See Example 27.2.)

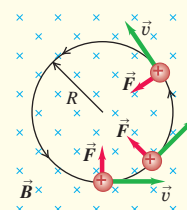
$$\begin{aligned} \Phi_B &= \int B \cos \phi \, dA = \int B_{\perp} dA \\ &= \int \vec{B} \cdot d\vec{A} \end{aligned} \quad (27.6)$$



$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{closed surface}) \quad (27.8)$$

Motion in a magnetic field: The magnetic force is always perpendicular to $\vec{v}$; a particle moving under the action of a magnetic field alone moves with constant speed. In a uniform field, a particle with initial velocity perpendicular to the field moves in a circle with radius R that depends on the magnetic field strength B and the particle mass m , speed v , and charge q . (See Examples 27.3 and 27.4.)

$$R = \frac{mv}{|q|B} \quad (27.11)$$

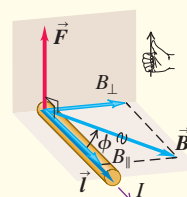


Crossed electric and magnetic fields can be used as a velocity selector. The electric and magnetic forces exactly cancel when $v = E/B$. (See Examples 27.5 and 27.6.)

Magnetic force on a conductor: A straight segment of a conductor carrying current I in a uniform magnetic field $\vec{B}$ experiences a force $\vec{F}$ that is perpendicular to both $\vec{B}$ and the vector $\vec{l}$, which points in the direction of the current and has magnitude equal to the length of the segment. A similar relationship gives the force $d\vec{F}$ on an infinitesimal current-carrying segment $d\vec{l}$. (See Examples 27.7 and 27.8.)

$$\vec{F} = I\vec{l} \times \vec{B} \quad (27.19)$$

$$d\vec{F} = I d\vec{l} \times \vec{B} \quad (27.20)$$

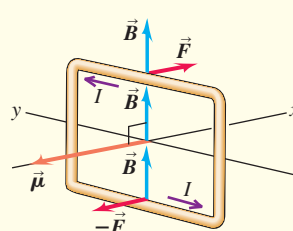


Magnetic torque: A current loop with area A and current I in a uniform magnetic field $\vec{B}$ experiences no net magnetic force, but does experience a magnetic torque of magnitude τ . The vector torque $\vec{\tau}$ can be expressed in terms of the magnetic moment $\vec{\mu} = I\vec{A}$ of the loop, as can the potential energy U of a magnetic moment in a magnetic field $\vec{B}$. The magnetic moment of a loop depends only on the current and the area; it is independent of the shape of the loop. (See Examples 27.9 and 27.10.)

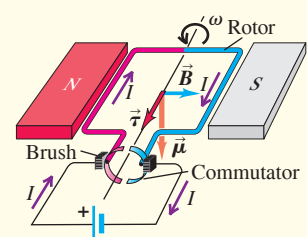
$$\tau = IBA \sin \phi \quad (27.23)$$

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (27.26)$$

$$U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \phi \quad (27.27)$$

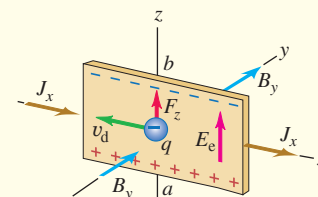


Electric motors: In a dc motor a magnetic field exerts a torque on a current in the rotor. Motion of the rotor through the magnetic field causes an induced emf called a back emf. For a series motor, in which the rotor coil is in series with coils that produce the magnetic field, the terminal voltage is the sum of the back emf and the potential drop Ir across the internal resistance. (See Example 27.11.)



The Hall effect: The Hall effect is a potential difference perpendicular to the direction of current in a conductor, when the conductor is placed in a magnetic field. The Hall potential is determined by the requirement that the associated electric field must just balance the magnetic force on a moving charge. Hall-effect measurements can be used to determine the sign of charge carriers and their concentration n . (See Example 27.12.)

$$nq = \frac{-J_x B_y}{E_z} \quad (27.30)$$



Chapter 27 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 27.1** (Section 27.2) before attempting these problems.

VP27.1.1 An electron (charge -1.60×10^{-19} C) moves at 2.20×10^5 m/s through a uniform 1.55 T magnetic field that points in the $+y$ -direction. The velocity of the electron lies in the xy -plane and is directed at 40.0° to the $+x$ -axis and 50.0° to the $+y$ -axis. Find (a) the magnitude and (b) the direction of the magnetic force on the electron.

VP27.1.2 A charged particle moving in the presence of a 0.750 T magnetic field experiences a magnetic force of magnitude 2.50×10^{-10} N. The particle is moving at 1.95×10^5 m/s at an angle of 36.0° to the direction of the magnetic field. Find the magnitude of the charge on the particle.

VP27.1.3 A particle (charge $+3.20 \times 10^{-19}$ C) with velocity $v_x = 3.75 \times 10^5$ m/s is in the presence of a $\vec{B}$ field with only a z -component. The magnetic force on the particle is 2.10×10^{-15} N in the $-y$ -direction. Find (a) the magnitude and (b) the direction of $\vec{B}$.

VP27.1.4 Highly ionized iron atoms are found in the sun's outer atmosphere, or corona. One such ion has lost 13 electrons and so has charge $+13e = +2.08 \times 10^{-18}$ C. This ion experiences a magnetic force of 2.30×10^{-15} N when it moves at 7.30×10^3 m/s in the xz -plane, 36.9° from the $+x$ -axis and 53.1° from the $+z$ -axis. The magnetic field is in the $+z$ -direction. Find (a) the magnitude of the magnetic field and (b) the direction of the force on the ion.

Be sure to review **EXAMPLES 27.3 and 27.4** (Section 27.4) and **EXAMPLES 27.5 and 27.6** (Section 27.5) before attempting these problems.

VP27.6.1 A typical magnetic field in sunspots (highly magnetized regions on the surface of the sun) is 0.300 T. For a proton (charge $+1.60 \times 10^{-19}$ C, mass 1.67×10^{-27} kg) moving at 1.25×10^4 m/s in a direction perpendicular to such a field, find (a) the radius of its circular orbit, (b) its angular speed in its orbit, and (c) the frequency of its orbital motion.

VP27.6.2 In the situation of the previous problem, suppose the 0.300 T magnetic field is in the $+y$ -direction and the proton's motion is *not* perpendicular to the field: Initially its velocity has components $v_x = 1.00 \times 10^4$ m/s, $v_y = 7.50 \times 10^3$ m/s, $v_z = 0$. Find (a) the radius of the proton's helical path, (b) how far the proton moves along the helix axis per revolution, and (c) the magnitude of the magnetic force on the proton.

VP27.6.3 A velocity selector uses an electric field $\vec{E} = (2.80 \times 10^4 \text{ N/C})\hat{i}$ and a magnetic field $\vec{B} = (0.0350 \text{ T})\hat{k}$. (a) What particle speed will yield zero deflection? (b) In what direction should a charged particle travel through these fields to have zero deflection? (c) A positively charged particle travels through these fields with the speed you found in part (a) and the direction you found in part (b), but the magnetic-field magnitude is now greater than 0.0350 T. In what direction will the particle initially be deflected?

VP27.6.4 You send a beam of oxygen ions through a mass spectrometer like the one shown in Fig. 27.24. The nuclei emerge from the velocity selector at 2.00×10^4 m/s and encounter a magnetic field of magnitude $B' = 0.0500$ T. All the ions have charge $+1.60 \times 10^{-19}$ C, but some are the isotope ^{16}O (mass 2.66×10^{-26} kg) and some are the isotope ^{18}O (mass 2.99×10^{-26} kg). Find the radius of the semicircular path followed by (a) the ^{16}O ions and (b) the ^{18}O ions.

Be sure to review **EXAMPLE 27.7** (Section 27.6) before attempting these problems.

VP27.7.1 A straight wire 0.150 m in length carries a current of 3.50 A in the $+x$ -direction. The wire is in a uniform 0.0136 T magnetic field in the xy -plane that points in a direction 20.0° from the $-x$ -axis and 70.0° from the $+y$ -axis. Find (a) the magnitude and (b) the direction of the magnetic force on the wire.

VP27.7.2 A vertical straight wire 25.0 cm in length carries a current. You do not know either the magnitude of the current or whether the current is moving upward or downward. If there is a uniform horizontal magnetic field of 0.0350 T that points due north, the wire experiences a horizontal magnetic force to the west of 0.0140 N. Find (a) the magnitude and (b) the direction of the current.

VP27.7.3 A straight wire carries a current of 1.20 A. The length vector of the wire is $\vec{l} = (0.200 \text{ m})\hat{i} + (-0.120 \text{ m})\hat{j}$, and the wire is in a uniform magnetic field $\vec{B} = (0.0175 \text{ T})\hat{k}$. Find (a) the x -component, (b) the y -component, (c) the z -component, and (d) the magnitude of the magnetic force on the wire.

VP27.7.4 A straight wire 0.280 m in length carries a current of 3.40 A. What are the *two* angles between the direction of the current and the direction of a uniform 0.0400 T magnetic field for which the magnetic force on the wire has magnitude 0.0250 N?

BRIDGING PROBLEM Magnetic Torque on a Current-Carrying Ring

A circular ring with area 4.45 cm^2 is carrying a current of 12.5 A . The ring, initially at rest, is immersed in a region of uniform magnetic field given by $\vec{B} = (1.15 \times 10^{-2} \text{ T})(12\hat{i} + 3\hat{j} - 4\hat{k})$. The ring is positioned initially such that its magnetic moment is given by $\vec{\mu}_i = \mu(-0.800\hat{i} + 0.600\hat{j})$, where μ is the (positive) magnitude of the magnetic moment. (a) Find the initial magnetic torque on the ring. (b) The ring (which is free to rotate around one diameter) is released and turns through an angle of 90.0° , at which point its magnetic moment is given by $\vec{\mu}_f = -\mu\hat{k}$. Determine the decrease in potential energy. (c) If the moment of inertia of the ring about a diameter is $8.50 \times 10^{-7} \text{ kg} \cdot \text{m}^2$, determine the angular speed of the ring as it passes through the second position.

SOLUTION GUIDE**IDENTIFY and SET UP**

1. The current-carrying ring acts as a magnetic dipole, so you can use the equations for a magnetic dipole in a uniform magnetic field.

2. There are no nonconservative forces acting on the ring as it rotates, so the sum of its rotational kinetic energy (discussed in Section 9.4) and the potential energy is conserved.

EXECUTE

3. Use the vector expression for the torque on a magnetic dipole to find the answer to part (a). (*Hint:* Review Section 1.10.)
4. Find the change in potential energy from the first orientation of the ring to the second orientation.
5. Use your result from step 4 to find the rotational kinetic energy of the ring when it is in the second orientation.
6. Use your result from step 5 to find the ring's angular speed when it is in the second orientation.

EVALUATE

7. If the ring were free to rotate around *any* diameter, in what direction would the magnetic moment point when the ring is in a state of stable equilibrium?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q27.1 Can a charged particle move through a magnetic field without experiencing any force? If so, how? If not, why not?

Q27.2 At any point in space, the electric field $\vec{E}$ is defined to be in the direction of the electric force on a positively charged particle at that point. Why don't we similarly define the magnetic field $\vec{B}$ to be in the direction of the magnetic force on a moving, positively charged particle?

Q27.3 Section 27.2 describes a procedure for finding the direction of the magnetic force using your right hand. If you use the same procedure, but with your left hand, will you get the correct direction for the force? Explain.

Q27.4 The magnetic force on a moving charged particle is always perpendicular to the magnetic field $\vec{B}$. Is the trajectory of a moving charged particle always perpendicular to the magnetic field lines? Explain your reasoning.

Q27.5 A charged particle is fired into a cubical region of space where there is a uniform magnetic field. Outside this region, there is no magnetic field. Is it possible that the particle will remain inside the cubical region? Why or why not?

Q27.6 If the magnetic force does no work on a charged particle, how can it have any effect on the particle's motion? Are there other examples of forces that do no work but have a significant effect on a particle's motion?

Q27.7 A charged particle moves through a region of space with constant velocity (magnitude and direction). If the external magnetic field is zero in this region, can you conclude that the external electric field in the region is also zero? Explain. (By "external" we mean fields other than those produced by the charged particle.) If the external electric field is zero in the region, can you conclude that the external magnetic field in the region is also zero?

Q27.8 How might a loop of wire carrying a current be used as a compass? Could such a compass distinguish between north and south? Why or why not?

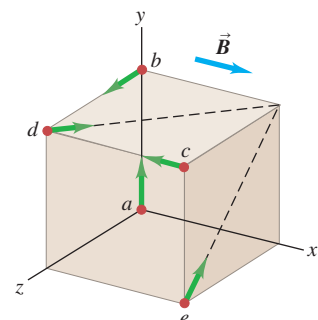
Q27.9 How could the direction of a magnetic field be determined by making only *qualitative* observations of the magnetic force on a straight wire carrying a current?

Q27.10 A loose, floppy loop of wire is carrying current I . The loop of wire is placed on a horizontal table in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the table. This causes the loop of wire to expand into a circular shape while still lying on the table. In a diagram, show all possible orientations of the current I and magnetic field $\vec{B}$ that could cause this to occur. Explain your reasoning.

Q27.11 Several charges enter a uniform magnetic field directed into the page. (a) What path would a positive charge q moving with a velocity of magnitude v follow through the field? (b) What path would a positive charge q moving with a velocity of magnitude $2v$ follow through the field? (c) What path would a negative charge $-q$ moving with a velocity of magnitude v follow through the field? (d) What path would a neutral particle follow through the field?

Q27.12 Each of the lettered points in Figure Q27.12

represents a positive charge q moving with a velocity of magnitude v in the direction indicated. The region in the figure is in a uniform magnetic field $\vec{B}$, parallel to the x -axis and directed toward the right. Which charges experience a force due to $\vec{B}$? What is the direction of the force on each charge?



Q27.13 A student claims that if lightning strikes a metal flagpole, the force exerted by the earth's magnetic field on the current in the pole can be large enough to bend it. Typical lightning currents are of the order of 10^4 to 10^5 A. Is the student's opinion justified? Explain your reasoning.

Q27.14 Could an accelerator be built in which *all* the forces on the particles, for steering and for increasing speed, are magnetic forces? Why or why not?

Q27.15 The magnetic force acting on a charged particle can never do work because at every instant the force is perpendicular to the velocity. The torque exerted by a magnetic field can do work on a current loop when the loop rotates. Explain how these seemingly contradictory statements can be reconciled.

Q27.16 When the polarity of the voltage applied to a dc motor is reversed, the direction of motion does *not* reverse. Why not? How *could* the direction of motion be reversed?

Q27.17 In a Hall-effect experiment, is it possible that *no* transverse potential difference will be observed? Under what circumstances might this happen?

Q27.18 Hall-effect voltages are much greater for relatively poor conductors (such as germanium) than for good conductors (such as copper), for comparable currents, fields, and dimensions. Why?

EXERCISES

Section 27.2 Magnetic Field

27.1 • A particle with a charge of -1.24×10^{-8} C is moving with instantaneous velocity $\vec{v} = (4.19 \times 10^4 \text{ m/s})\hat{i} + (-3.85 \times 10^4 \text{ m/s})\hat{j}$. What is the force exerted on this particle by a magnetic field (a) $\vec{B} = (1.40 \text{ T})\hat{i}$ and (b) $\vec{B} = (1.40 \text{ T})\hat{k}$?

27.2 • A particle of mass 0.195 g carries a charge of -2.50×10^{-8} C. The particle is given an initial horizontal velocity that is due north and has magnitude 4.00×10^4 m/s. What are the magnitude and direction of the minimum magnetic field that will keep the particle moving in the earth's gravitational field in the same horizontal, northward direction?

27.3 • In a 1.25 T magnetic field directed vertically upward, a particle having a charge of magnitude $8.50 \mu\text{C}$ and initially moving northward at 4.75 km/s is deflected toward the east. (a) What is the sign of the charge of this particle? Make a sketch to illustrate how you found your answer. (b) Find the magnetic force on the particle.

27.4 • A particle with mass 1.81×10^{-3} kg and a charge of 1.22×10^{-8} C has, at a given instant, a velocity $\vec{v} = (3.00 \times 10^4 \text{ m/s})\hat{j}$. What are the magnitude and direction of the particle's acceleration produced by a uniform magnetic field $\vec{B} = (1.63 \text{ T})\hat{i} + (0.980 \text{ T})\hat{j}$?

27.5 • An electron is moving in the xy -plane. If at time t a magnetic field $B = 0.200$ T in the $+z$ -direction exerts a force on the electron equal to $F = 5.50 \times 10^{-18}$ N in the $-y$ -direction, what is the velocity (magnitude and direction) of the electron at this instant?

27.6 • An electron moves at 1.40×10^6 m/s through a region in which there is a magnetic field of unspecified direction and magnitude 7.40×10^{-2} T. (a) What are the largest and smallest possible magnitudes of the acceleration of the electron due to the magnetic field? (b) If the actual acceleration of the electron is one-fourth of the largest magnitude in part (a), what is the angle between the electron velocity and the magnetic field?

27.7 • A group of particles is traveling in a magnetic field of unknown magnitude and direction. You observe that a proton moving at 1.50 km/s in the $+x$ -direction experiences a force of 2.25×10^{-16} N in the $+y$ -direction, and an electron moving at 4.75 km/s in the $-z$ -direction experiences a force of 8.50×10^{-16} N in the $+y$ -direction. (a) What are the magnitude and direction of the magnetic field? (b) What are the magnitude and direction of the magnetic force on an electron moving in the $-y$ -direction at 3.20 km/s?

27.8 • CP A particle with charge -5.60 nC is moving in a uniform magnetic field $\vec{B} = -(1.25 \text{ T})\hat{k}$. The magnetic force on the particle is measured to be $\vec{F} = -(3.40 \times 10^{-7} \text{ N})\hat{i} + (7.40 \times 10^{-7} \text{ N})\hat{j}$. (a) Calculate all the components of the velocity of the particle that you can from this information. (b) Are there components of the velocity that are not determined by the measurement of the force? Explain. (c) Calculate the scalar product $\vec{v} \cdot \vec{F}$. What is the angle between $\vec{v}$ and $\vec{F}$?

Section 27.3 Magnetic Field Lines and Magnetic Flux

27.9 • A circular area with a radius of 6.50 cm lies in the xy -plane. What is the magnitude of the magnetic flux through this circle due to a uniform magnetic field $B = 0.230$ T (a) in the $+z$ -direction; (b) at an angle of 53.1° from the $+z$ -direction; (c) in the $+y$ -direction?

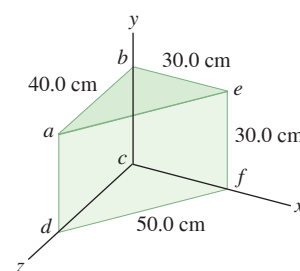
27.10 • A flat, square surface with side length 3.40 cm is in the xy -plane at $z = 0$. Calculate the magnitude of the flux through this surface produced by a magnetic field $\vec{B} = (0.200 \text{ T})\hat{i} + (0.300 \text{ T})\hat{j} - (0.500 \text{ T})\hat{k}$.

27.11 • An open plastic soda bottle with an opening diameter of 2.5 cm is placed on a table. A uniform 1.75 T magnetic field directed upward and oriented 25° from vertical encompasses the bottle. What is the total magnetic flux through the plastic of the soda bottle?

27.12 • A horizontal rectangular surface has dimensions 2.80 cm by 3.20 cm and is in a uniform magnetic field that is directed at an angle of 30.0° above the horizontal. What must the magnitude of the magnetic field be to produce a flux of 3.10×10^{-4} Wb through the surface?

27.13 • The magnetic field $\vec{B}$ in a certain region is 0.128 T, and its direction is that of the $+z$ -axis in **Fig. E27.13**. (a) What is the magnetic flux across the surface $abcd$ in the figure? (b) What is the magnetic flux across the surface $befc$? (c) What is the magnetic flux across the surface $aefd$? (d) What is the net flux through all five surfaces that enclose the shaded volume?

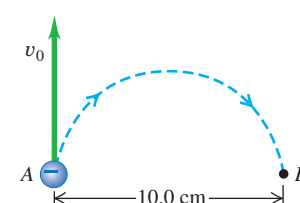
Figure E27.13



Section 27.4 Motion of Charged Particles in a Magnetic Field

27.14 • An electron at point A in **Fig. E27.14** has a speed v_0 of 1.41×10^6 m/s. Find (a) the magnitude and direction of the magnetic field that will cause the electron to follow the semicircular path from A to B, and (b) the time required for the electron to move from A to B.

Figure E27.14



27.15 • Repeat Exercise 27.14 for the case in which the particle is a proton rather than an electron.

27.16 • An alpha particle (a He nucleus, containing two protons and two neutrons and having a mass of 6.64×10^{-27} kg) traveling horizontally at 35.6 km/s enters a uniform, vertical, 1.80 T magnetic field. (a) What is the diameter of the path followed by this alpha particle? (b) What effect does the magnetic field have on the speed of the particle? (c) What are the magnitude and direction of the acceleration of the alpha particle while it is in the magnetic field? (d) Explain why the speed of the particle does not change even though an unbalanced external force acts on it.

27.17 • CP A 150 g ball containing 4.00×10^8 excess electrons is dropped into a 125 m vertical shaft. At the bottom of the shaft, the ball suddenly enters a uniform horizontal magnetic field that has magnitude 0.250 T and direction from east to west. If air resistance is negligibly small, find the magnitude and direction of the force that this magnetic field exerts on the ball just as it enters the field.

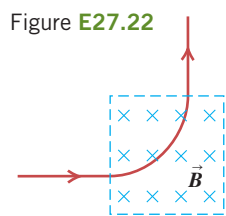
27.18 • BIO Cyclotrons are widely used in nuclear medicine for producing short-lived radioactive isotopes. These cyclotrons typically accelerate H^- (the *hydride* ion, which has one proton and two electrons) to an energy of 5 MeV to 20 MeV. This ion has a mass very close to that of a proton because the electron mass is negligible—about $\frac{1}{2000}$ of the proton's mass. A typical magnetic field in such cyclotrons is 1.9 T. (a) What is the speed of a 5.0 MeV H^- ? (b) If the H^- has energy 5.0 MeV and $B = 1.9$ T, what is the radius of this ion's circular orbit?

27.19 • In an experiment with cosmic rays, a vertical beam of particles that have charge of magnitude $3e$ and mass 12 times the proton mass enters a uniform horizontal magnetic field of 0.250 T and is bent in a semicircle of diameter 95.0 cm, as shown in **Fig. E27.19**. (a) Find the speed of the particles and the sign of their charge. (b) Is it reasonable to ignore the gravity force on the particles? (c) How does the speed of the particles as they enter the field compare to their speed as they exit the field?

27.20 • A deuteron (the nucleus of an isotope of hydrogen) has a mass of 3.34×10^{-27} kg and a charge of $+e$. The deuteron travels in a circular path with a radius of 6.96 mm in a magnetic field with magnitude 2.50 T. (a) Find the speed of the deuteron. (b) Find the time required for it to make half a revolution. (c) Through what potential difference would the deuteron have to be accelerated to acquire this speed?

27.21 • An electron in the beam of a cathode-ray tube is accelerated by a potential difference of 2.00 kV. Then it passes through a region of transverse magnetic field, where it moves in a circular arc with radius 0.180 m. What is the magnitude of the field?

27.22 •• A beam of protons traveling at 1.20 km/s enters a uniform magnetic field, traveling perpendicular to the field. The beam exits the magnetic field, leaving the field in a direction perpendicular to its original direction (**Fig. E27.22**). The beam travels a distance of 1.18 cm while in the field. What is the magnitude of the magnetic field?



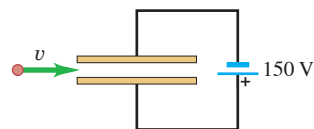
Section 27.5 Applications of Motion of Charged Particles

27.23 • Crossed $\vec{E}$ and $\vec{B}$ Fields. A particle with initial velocity $\vec{v}_0 = (5.85 \times 10^3 \text{ m/s})\hat{j}$ enters a region of uniform electric and magnetic fields. The magnetic field in the region is $\vec{B} = -(1.35 \text{ T})\hat{k}$. Calculate the magnitude and direction of the electric field in the region if the particle is to pass through undeflected, for a particle of charge (a) $+0.640$ nC and (b) -0.320 nC. You can ignore the weight of the particle.

27.24 • (a) What is the speed of a beam of electrons when the simultaneous influence of an electric field of 1.56×10^4 V/m and a magnetic field of 4.62×10^{-3} T, with both fields normal to the beam and to each other, produces no deflection of the electrons? (b) In a diagram, show the relative orientation of the vectors $\vec{v}$, $\vec{E}$, and $\vec{B}$. (c) When the electric field is removed, what is the radius of the electron orbit? What is the period of the orbit?

27.25 •• A 150 V battery is connected across two parallel metal plates of area 28.5 cm^2 and separation 8.20 mm. A beam of alpha particles (charge $+2e$, mass 6.64×10^{-27} kg) is accelerated from rest through a potential difference of 1.75 kV and enters the

Figure E27.25



region between the plates perpendicular to the electric field, as shown in **Fig. E27.25**. What magnitude and direction of magnetic field are needed so that the alpha particles emerge undeflected from between the plates?

27.26 • In the Bainbridge mass spectrometer (see **Fig. 27.22**), the magnetic-field magnitude in the velocity selector is 0.510 T, and ions having a speed of 1.82×10^6 m/s pass through undeflected. (a) What is the electric-field magnitude in the velocity selector? (b) If the separation of the plates is 5.20 mm, what is the potential difference between the plates?

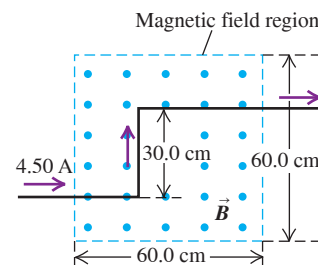
27.27 • Singly ionized (one electron removed) atoms are accelerated and then passed through a velocity selector consisting of perpendicular electric and magnetic fields. The electric field is 155 V/m and the magnetic field is 0.0315 T. The ions next enter a uniform magnetic field of magnitude 0.0175 T that is oriented perpendicular to their velocity. (a) How fast are the ions moving when they emerge from the velocity selector? (b) If the radius of the path of the ions in the second magnetic field is 17.5 cm, what is their mass?

27.28 •• BIO Ancient Meat Eating. The amount of meat in prehistoric diets can be determined by measuring the ratio of the isotopes ^{15}N to ^{14}N in bone from human remains. Carnivores concentrate ^{15}N , so this ratio tells archaeologists how much meat was consumed. For a mass spectrometer that has a path radius of 12.5 cm for ^{12}C ions (mass 1.99×10^{-26} kg), find the separation of the ^{14}N (mass 2.32×10^{-26} kg) and ^{15}N (mass 2.49×10^{-26} kg) isotopes at the detector.

Section 27.6 Magnetic Force on a Current-Carrying Conductor

27.29 •• A long wire carrying 4.50 A of current makes two 90° bends, as shown in **Fig. E27.29**. The bent part of the wire passes through a uniform 0.240 T magnetic field directed as shown in the figure and confined to a limited region of space. Find the magnitude and direction of the force that the magnetic field exerts on the wire.

Figure E27.29

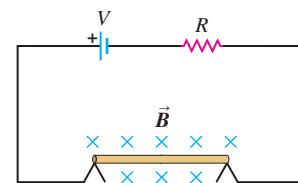


27.30 • A straight, 2.5 m wire carries a typical household current of 1.5 A (in one direction) at a location where the earth's magnetic field is 0.55 gauss from south to north. Find the magnitude and direction of the force that our planet's magnetic field exerts on this wire if it is oriented so that the current in it is running (a) from west to east,

(b) vertically upward, (c) from north to south. (d) Is the magnetic force ever large enough to cause significant effects under normal household conditions?

27.31 • A thin, 50.0-cm-long metal bar with mass 750 g rests on, but is not attached to, two metallic supports in a uniform 0.450 T magnetic field, as shown in **Fig. E27.31**. A battery and a 25.0Ω resistor in series are connected to the supports. (a) What is the highest voltage the battery can have without breaking the circuit at the supports?

Figure E27.31

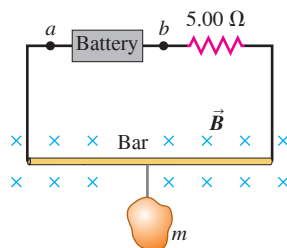


(b) The battery voltage has the maximum value calculated in part (a). If the resistor suddenly gets partially short-circuited, decreasing its resistance to 2.0Ω , find the initial acceleration of the bar.

27.32 •• An electromagnet produces a magnetic field of 0.550 T in a cylindrical region of radius 2.50 cm between its poles. A straight wire carrying a current of 10.8 A passes through the center of this region and is perpendicular to both the axis of the cylindrical region and the magnetic field. What magnitude of force does this field exert on the wire?

27.33 • Magnetic Balance. The circuit shown in **Fig. E27.33** is used to make a magnetic balance to weigh objects. The mass m to be measured is hung from the center of the bar that is in a uniform magnetic field of 1.50 T, directed into the plane of the figure. The battery voltage can be adjusted to vary the current in the circuit. The horizontal bar is 60.0 cm long and is made of extremely lightweight material. It is connected to the battery by thin vertical wires that can support no appreciable tension; all the weight of the suspended mass m is supported by the magnetic force on the bar. A resistor with $R = 5.00\ \Omega$ is in series with the bar; the resistance of the rest of the circuit is much less than this. (a) Which point, a or b , should be the positive terminal of the battery? (b) If the maximum terminal voltage of the battery is 175 V, what is the greatest mass m that this instrument can measure?

Figure E27.33



27.34 • A straight, vertical wire carries a current of 2.60 A downward in a region between the poles of a large superconducting electro-magnet, where the magnetic field has magnitude $B = 0.588\text{ T}$ and is horizontal. What are the magnitude and direction of the magnetic force on a 1.00 cm section of the wire that is in this uniform magnetic field, if the magnetic field direction is (a) east; (b) south; (c) 30.0° south of west?

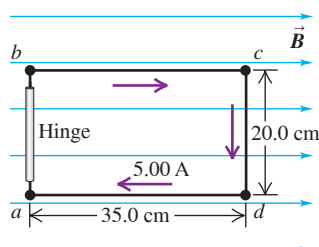
Section 27.7 Force and Torque on a Current Loop

27.35 • A flat circular coil carrying a current of 8.80 A has a magnetic dipole moment of $0.194\text{ A} \cdot \text{m}^2$ to the left. Its area vector $\vec{A}$ is 4.0 cm^2 to the left. (a) How many turns does the coil have? (b) An observer is on the coil's axis to the left of the coil and is looking toward the coil. Does the observer see a clockwise or counterclockwise current? (c) If a huge 45.0 T external magnetic field directed out of the paper is applied to the coil, what torque (magnitude and direction) results?

27.36 •• The plane of a $5.0\text{ cm} \times 8.0\text{ cm}$ rectangular loop of wire is parallel to a 0.19 T magnetic field. The loop carries a current of 6.2 A. (a) What torque acts on the loop? (b) What is the magnetic moment of the loop? (c) What is the maximum torque that can be obtained with the same total length of wire carrying the same current in this magnetic field?

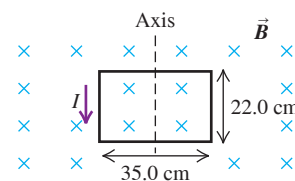
27.37 • The $20.0\text{ cm} \times 35.0\text{ cm}$ rectangular circuit shown in **Fig. E27.37** is hinged along side ab . It carries a clockwise 5.00 A current and is located in a uniform 1.20 T magnetic field oriented perpendicular to two of its sides, as shown. (a) Draw a clear diagram showing the direction of the force that the magnetic field exerts on each segment of the circuit (ab , bc , etc.). (b) Of the four forces you drew in part (a), decide which ones exert a torque about the hinge ab . Then calculate only those forces that exert this torque. (c) Use your results from part (b) to calculate the torque that the magnetic field exerts on the circuit about the hinge axis ab .

Figure E27.37



27.38 • A rectangular coil of wire, 22.0 cm by 35.0 cm and carrying a current of 1.95 A, is oriented with the plane of its loop perpendicular to a uniform 1.50 T magnetic field (**Fig. E27.38**). (a) Calculate the net force and torque that the magnetic field exerts on the coil. (b) The coil is rotated through a 30.0° angle about

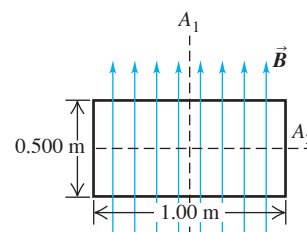
Figure E27.38



the axis shown, with the left side coming out of the plane of the figure and the right side going into the plane. Calculate the net force and torque that the magnetic field now exerts on the coil. (*Hint:* To visualize this three-dimensional problem, make a careful drawing of the coil as viewed along the rotation axis.)

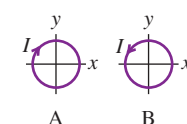
27.39 • CP A uniform rectangular coil of total mass 212 g and dimensions $0.500\text{ m} \times 1.00\text{ m}$ is oriented with its plane parallel to a uniform 3.00 T magnetic field (**Fig. E27.39**). A current of 2.00 A is suddenly started in the coil. (a) About which axis (A_1 or A_2) will the coil begin to rotate? Why? (b) Find the initial angular acceleration of the coil just after the current is started.

Figure E27.39

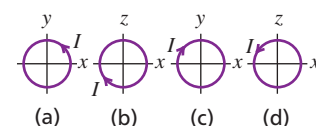


27.40 • Both circular coils A and B (**Fig. E27.40**) have area A and N turns. They are free to rotate about a diameter that coincides with the x -axis. Current I circulates in each coil in the direction shown. There is a uniform magnetic field $\vec{B}$ in the $+z$ -direction. (a) What is the direction of the magnetic moment $\vec{\mu}$ for each coil? (b) Explain why the torque on both coils due to the magnetic field is zero, so the coil is in rotational equilibrium. (c) Use Eq. (27.27) to calculate the potential energy for each coil. (d) For each coil, is the equilibrium stable or unstable? Explain.

Figure E27.40



27.41 • A circular coil with area A and N turns is free to rotate about a diameter that coincides with the x -axis. Current I is circulating in the coil. There is a uniform magnetic field $\vec{B}$ in the positive y -direction. Calculate the magnitude and direction of the torque $\vec{\tau}$ and the value of the potential energy U , as given in Eq. (27.27), when the coil is oriented as shown in parts (a) through (d) of **Fig. E27.41**.

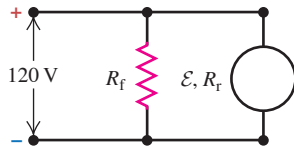


27.42 •• A coil with magnetic moment $1.45\text{ A} \cdot \text{m}^2$ is oriented initially with its magnetic moment antiparallel to a uniform 0.835 T magnetic field. What is the change in potential energy of the coil when it is rotated 180° so that its magnetic moment is parallel to the field?

Section 27.8 The Direct-Current Motor

27.43 •• In a shunt-wound dc motor with the field coils and rotor connected in parallel (Fig. E27.43), the resistance R_f of the field coils is $106\ \Omega$, and the resistance R_r of the rotor is $5.9\ \Omega$. When a potential difference of 120 V is applied to the brushes and the motor is running at full speed delivering mechanical power, the current supplied to it is 4.82 A . (a) What is the current in the field coils? (b) What is the current in the rotor? (c) What is the induced emf developed by the motor? (d) How much mechanical power is developed by this motor?

Figure E27.43

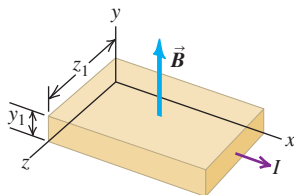


27.44 • A dc motor with its rotor and field coils connected in series has an internal resistance of $3.2\ \Omega$. When the motor is running at full load on a 120 V line, the emf in the rotor is 105 V . (a) What is the current drawn by the motor from the line? (b) What is the power delivered to the motor? (c) What is the mechanical power developed by the motor?

Section 27.9 The Hall Effect

27.45 • Figure E27.45 shows a portion of a silver ribbon with $z_1 = 11.8\text{ mm}$ and $y_1 = 0.23\text{ mm}$, carrying a current of 120 A in the $+x$ -direction. The ribbon lies in a uniform magnetic field, in the y -direction, with magnitude 0.95 T . Apply the simplified model of the Hall effect presented in Section 27.9. If there are 5.85×10^{28} free electrons per cubic meter, find (a) the magnitude of the drift velocity of the electrons in the x -direction; (b) the magnitude and direction of the electric field in the z -direction due to the Hall effect; (c) the Hall emf.

Figure E27.45



27.46 • Let Fig. E27.45 represent a strip of an unknown metal of the same dimensions as those of the silver ribbon in Exercise 27.45. When the magnetic field is 2.29 T and the current is 78.0 A , the Hall emf is found to be $131\ \mu\text{V}$. What does the simplified model of the Hall effect presented in Section 27.9 give for the density of free electrons in the unknown metal?

PROBLEMS

27.47 •• CP A small particle with positive charge $q = +3.75 \times 10^{-4}\text{ C}$ and mass $m = 5.00 \times 10^{-5}\text{ kg}$ is moving in a region of uniform electric and magnetic fields. The magnetic field is $B = 4.00\text{ T}$ in the $+z$ -direction. The electric field is also in the $+z$ -direction and has magnitude $E = 60.0\text{ N/C}$. At time $t = 0$ the particle is on the y -axis at $y = +1.00\text{ m}$ and has velocity $v = 30.0\text{ m/s}$ in the $+x$ -direction. Neglect gravity. (a) What are the x -, y -, and z -coordinates of the particle at $t = 0.0200\text{ s}$? (b) What is the speed of the particle at $t = 0.0200\text{ s}$?

27.48 • A particle with charge $7.26 \times 10^{-8}\text{ C}$ is moving in a region where there is a uniform 0.650 T magnetic field in the $+x$ -direction. At a particular instant, the velocity of the particle has components $v_x = -1.68 \times 10^4\text{ m/s}$, $v_y = -3.11 \times 10^4\text{ m/s}$, and $v_z = 5.85 \times 10^4\text{ m/s}$. What are the components of the force on the particle at this time?

27.49 •• CP Fusion Reactor. If two deuterium nuclei (charge $+e$, mass $3.34 \times 10^{-27}\text{ kg}$) get close enough together, the attraction of the strong nuclear force will fuse them to make an isotope of helium, releasing vast amounts of energy. The range of this force is about 10^{-15} m . This is the principle behind the fusion reactor. The deuterium nuclei are moving much too fast to be contained by physical walls, so they are confined magnetically. (a) How fast would two nuclei have to move so that in a head-on collision they would get close enough to fuse? (Assume their speeds are equal. Treat the nuclei as point charges, and assume that a separation of $1.0 \times 10^{-15}\text{ m}$ is required for fusion.) (b) What strength magnetic field is needed to make deuterium nuclei with this speed travel in a circle of diameter 2.50 m ?

27.50 •• Magnetic Moment of the Hydrogen Atom. In the Bohr model of the hydrogen atom (see Section 39.3), in the lowest energy state the electron orbits the proton at a speed of $2.2 \times 10^6\text{ m/s}$ in a circular orbit of radius $5.3 \times 10^{-11}\text{ m}$. (a) What is the orbital period of the electron? (b) If the orbiting electron is considered to be a current loop, what is the current I ? (c) What is the magnetic moment of the atom due to the motion of the electron?

27.51 •• You measure the charge-to-mass ratio q/m for a particle with positive charge in the following way: The particle starts from rest, is accelerated through a potential difference ΔV , and attains a velocity with magnitude v . It then enters a region of uniform magnetic field $B = 0.200\text{ T}$ that is directed perpendicular to the velocity; the particle moves in a path that is an arc of a circle of radius R . You measure R as a function of ΔV . You plot your data as R^2 (in units of m^2) versus ΔV (in V) and find that the values lie close to a straight line that has slope $1.04 \times 10^{-6}\text{ m}^2/\text{V}$. What is the value of q/m for this particle?

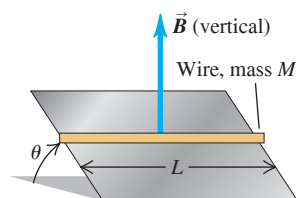
27.52 • The magnetic poles of a small cyclotron produce a magnetic field with magnitude 0.85 T . The poles have a radius of 0.40 m , which is the maximum radius of the orbits of the accelerated particles. (a) What is the maximum energy to which protons ($q = 1.60 \times 10^{-19}\text{ C}$, $m = 1.67 \times 10^{-27}\text{ kg}$) can be accelerated by this cyclotron? Give your answer in electron volts and in joules. (b) What is the time for one revolution of a proton orbiting at this maximum radius? (c) What would the magnetic-field magnitude have to be for the maximum energy to which a proton can be accelerated to be twice that calculated in part (a)? (d) For $B = 0.85\text{ T}$, what is the maximum energy to which alpha particles ($q = 3.20 \times 10^{-19}\text{ C}$, $m = 6.64 \times 10^{-27}\text{ kg}$) can be accelerated by this cyclotron? How does this compare to the maximum energy for protons?

27.53 •• Suppose the electric field between the plates in Fig. 27.22 is $1.88 \times 10^4\text{ V/m}$ and the magnetic field in both regions is 0.682 T . If the source contains the three isotopes of krypton, ^{82}Kr , ^{84}Kr , and ^{86}Kr , and the ions are singly charged, find the distance between the lines formed by the three isotopes on the particle detector. Assume the atomic masses of the isotopes (in atomic mass units) are equal to their mass numbers, 82, 84, and 86. (One atomic mass unit $= 1\text{ u} = 1.66 \times 10^{-27}\text{ kg}$.)

27.54 •• Mass Spectrograph. A mass spectrograph is used to measure the masses of ions, or to separate ions of different masses (see Section 27.5). In one design for such an instrument, ions with mass m and charge q are accelerated through a potential difference V . They then enter a uniform magnetic field that is perpendicular to their velocity, and they are deflected in a semicircular path of radius R . A detector measures where the ions complete the semicircle and from this it is easy to calculate R . (a) Derive the equation for calculating the mass of the ion from measurements of B , V , R , and q . (b) What potential difference V is needed so that singly ionized ^{12}C atoms will have $R = 50.0\text{ cm}$ in a 0.150 T magnetic field? (c) Suppose the beam consists of a mixture of ^{12}C and ^{14}C ions. If v and B have the same values as in part (b), calculate the separation of these two isotopes at the detector. Do you think that this beam separation is sufficient for the two ions to be distinguished? (Make the assumption described in Problem 27.53 for the masses of the ions.)

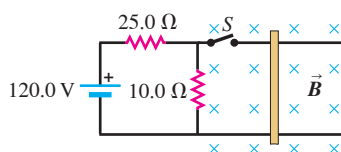
27.55 •• A straight piece of conducting wire with mass M and length L is placed on a frictionless incline tilted at an angle θ from the horizontal (**Fig. P27.55**). There is a uniform, vertical magnetic field $\vec{B}$ at all points (produced by an arrangement of magnets not shown in the figure). To keep the wire from sliding down the incline, a voltage source is attached to the ends of the wire. When just the right amount of current flows through the wire, the wire remains at rest. Determine the magnitude and direction of the current in the wire that will cause the wire to remain at rest. Copy the figure and draw the direction of the current on your copy. In addition, show in a free-body diagram all the forces that act on the wire.

Figure P27.55



27.56 •• CP A 2.60 N metal bar, 0.850 m long and having a resistance of $10.0 \, \Omega$, rests horizontally on conducting wires connecting it to the circuit shown in **Fig. P27.56**. The bar is in a uniform, horizontal, 1.60 T magnetic field and is not attached to the wires in the circuit. What is the acceleration of the bar just after the switch S is closed?

Figure P27.56

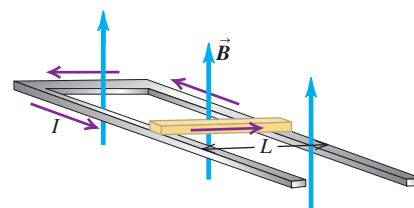


27.57 •• BIO Determining Diet. One method for determining the amount of corn in early Native American diets is the *stable isotope ratio analysis* (SIRA) technique. As corn photosynthesizes, it concentrates the isotope carbon-13, whereas most other plants concentrate carbon-12. Overreliance on corn consumption can then be correlated with certain diseases, because corn lacks the essential amino acid lysine. Archaeologists use a mass spectrometer to separate the ^{12}C and ^{13}C isotopes in samples of human remains. Suppose you use a velocity selector to obtain singly ionized (missing one electron) atoms of speed 8.50 km/s, and you want to bend them within a uniform magnetic field in a semicircle of diameter 25.0 cm for the ^{12}C . The measured masses of these isotopes are $1.99 \times 10^{-26} \, \text{kg}$ (^{12}C) and $2.16 \times 10^{-26} \, \text{kg}$ (^{13}C). (a) What strength of magnetic field is required? (b) What is the diameter of the ^{13}C semicircle? (c) What is the separation of the ^{12}C and ^{13}C ions at the detector at the end of the semicircle? Is this distance large enough to be easily observed?

27.58 •• CP A plastic circular loop has radius R , and a positive charge q is distributed uniformly around the circumference of the loop. The loop is then rotated around its central axis, perpendicular to the plane of the loop, with angular speed ω . If the loop is in a region where there is a uniform magnetic field $\vec{B}$ directed parallel to the plane of the loop, calculate the magnitude of the magnetic torque on the loop.

27.59 •• CP An Electromagnetic Rail Gun. A conducting bar with mass m and length L slides over horizontal rails that are connected to a voltage source. The voltage source maintains a constant current I in the rails and bar, and a constant, uniform, vertical magnetic field $\vec{B}$ fills the region between the rails (**Fig. P27.59**). (a) Find the magnitude and direction of the net force on the conducting bar. Ignore friction, air resistance, and electrical resistance. (b) If the bar has mass m , find the distance d that the bar must move along the rails from rest to attain speed v . (c) It has been suggested that rail guns based on this principle could accelerate payloads into earth orbit or beyond. Find the distance the bar must travel along the rails if it is to reach the escape speed for the earth (11.2 km/s). Let $B = 0.80 \, \text{T}$, $I = 2.0 \times 10^3 \, \text{A}$, $m = 25 \, \text{kg}$, and $L = 50 \, \text{cm}$. For simplicity assume the net force on the object is equal to the magnetic force, as in parts (a) and (b), even though gravity plays an important role in an actual launch in space.

Figure P27.59



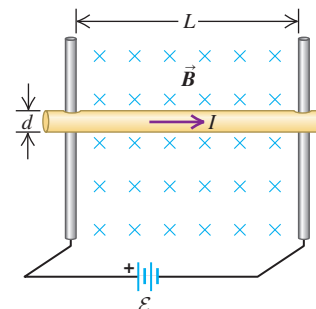
27.60 • A wire 25.0 cm long lies along the z -axis and carries a current of 7.40 A in the $+z$ -direction. The magnetic field is uniform and has components $B_x = -0.242 \, \text{T}$, $B_y = -0.985 \, \text{T}$, and $B_z = -0.336 \, \text{T}$. (a) Find the components of the magnetic force on the wire. (b) What is the magnitude of the net magnetic force on the wire?

27.61 •• CALC Point a is on the $+y$ -axis at $y = +0.200 \, \text{m}$ and point b is on the $+x$ -axis at $x = +0.200 \, \text{m}$. A wire in the shape of a circular arc of radius 0.200 m and centered on the origin goes from a to b and carries current $I = 5.00 \, \text{A}$ in the direction from a to b . (a) If the wire is in a uniform magnetic field $B = 0.800 \, \text{T}$ in the $+z$ -direction, what are the magnitude and direction of the net force that the magnetic field exerts on the wire segment? (b) What are the magnitude and direction of the net force on the wire if the field is $B = 0.800 \, \text{T}$ in the $+x$ -direction?

27.62 •• Ceiling fans use electric motors that involve stationary permanent magnets (called stators) attached to the central hub and with a typical strength of 1.0 T, that supply torque to current-carrying coils (called rotors) fixed to the fan blades. Think of a ceiling fan at your home or workplace. (a) Estimate the diameter of the central hub at the inner edge of the blades. (b) If 12 rotors are fixed around the central hub, and if the diameters of the rotors make up one-third of the circumference of the hub, what is the diameter of a single rotor? (c) A typical ceiling fan supplies $1.4 \, \text{N} \cdot \text{m}$ of torque. If this torque is supplied equally by the 12 rotors in the presence of 1.0 T stator fields parallel to the axis of each rotor, estimate the magnitude of the magnetic moment of each rotor. (d) The rotors are connected in parallel, so each receives $\frac{1}{12}$ of the fan current, which is typically 0.75 A. Estimate the number of windings in each rotor.

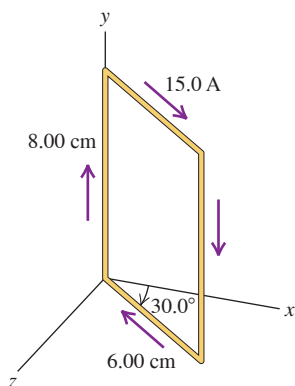
27.63 •• Earth's magnetic field near the ground is typically 0.50 G (0.50 gauss), where $1 \, \text{G} = 10^{-4} \, \text{tesla}$. In temperate northern latitudes, this field is inclined downward at an angle of approximately 45° . Consider the feasibility of using the earth's magnetic field to enable the levitation device shown in **Fig. P27.63** (next page). (a) Estimate the current needed to lift a copper bar with diameter $d = 5.0 \, \text{mm}$ using the horizontal component of the earth's magnetic field. The density of copper is $8900 \, \text{kg/m}^3$. (b) A copper bar with diameter 5.0 mm will melt if it carries a current greater than the fusing current of 900 A. Is it feasible for our device? (c) Estimate the minimum strength of the magnetic field needed to levitate our copper bar. (d) Suppose we use easily obtainable 1.0 T permanent magnets to supply the horizontal magnetic field and suppose our bar has length 10 cm. Estimate how much extra weight we could levitate with our device using a current of 10 A.

Figure P27.63



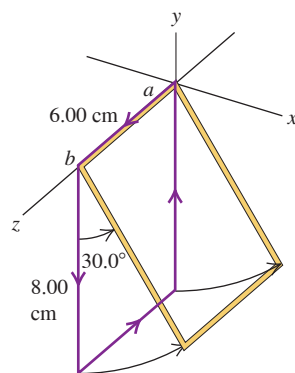
27.64 •• The rectangular loop shown in **Fig. P27.64** is pivoted about the y -axis and carries a current of 15.0 A in the direction indicated. (a) If the loop is in a uniform magnetic field with magnitude 0.48 T in the $+x$ -direction, find the magnitude and direction of the torque required to hold the loop in the position shown. (b) Repeat part (a) for the case in which the field is in the $-z$ -direction. (c) For each of the above magnetic fields, what torque would be required if the loop were pivoted about an axis through its center, parallel to the y -axis?

Figure P27.64



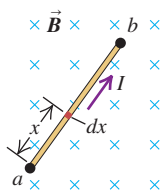
27.65 •• CP The rectangular loop of wire shown in **Fig. P27.65** has a mass of 0.15 g per centimeter of length and is pivoted about side ab on a frictionless axis. The current in the wire is 8.2 A in the direction shown. Find the magnitude and direction of the magnetic field parallel to the y -axis that will cause the loop to swing up until its plane makes an angle of 30.0° with the yz -plane.

Figure P27.65



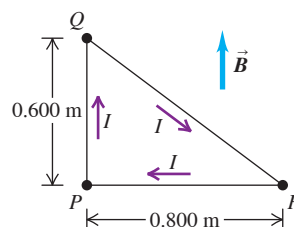
27.66 •• CALC A uniform bar of length L carries a current I in the direction from point a to point b (**Fig. P27.66**). The bar is in a uniform magnetic field that is directed into the page. Consider the torque about an axis perpendicular to the bar at point a that is due to the force that the magnetic field exerts on the bar. (a) Suppose that an infinitesimal section of the bar has length dx and is located a distance x from point a . Calculate the torque $d\tau$ about point a due to the magnetic force on this infinitesimal section. (b) Use $\tau = \int_a^b d\tau$ to calculate the total torque τ on the bar. (c) Show that τ is the same as though all of the magnetic force acted at the midpoint of the bar.

Figure P27.66



27.67 •• The loop of wire shown in **Fig. P27.67** forms a right triangle and carries a current $I = 5.00$ A in the direction shown. The loop is in a uniform magnetic field that has magnitude $B = 3.00$ T and the same direction as the current in side PQ of the loop. (a) Find the force exerted by the magnetic field on each side of the triangle. If the force is not zero, specify its direction. (b) What is the net force on the loop? (c) The loop is pivoted about an axis that lies along side PR . Use the forces calculated in part (a) to calculate the torque on each side of the loop (see Problem 27.66). (d) What is the magnitude of the net torque on the loop? Calculate the net torque from the torques calculated in part (c) and also from Eq. (27.28). Do these two results agree? (e) Is the net torque directed to rotate point Q into the plane of the figure or out of the plane of the figure?

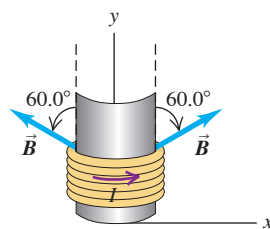
Figure P27.67



27.68 •• Repeat Problem 27.67 but with the magnetic field $B = 3.00$ T directed into the page in **Fig. P27.67**.

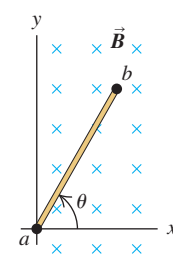
27.69 •• CALC A Voice Coil. It was shown in Section 27.7 that the net force on a current loop in a *uniform* magnetic field is zero. The magnetic force on the voice coil of a loudspeaker (see **Fig. 27.28**) is nonzero because the magnetic field at the coil is not uniform. A voice coil in a loudspeaker has 50 turns of wire and a diameter of 1.56 cm, and the current in the coil is 0.950 A. Assume that the magnetic field at each point of the coil has a constant magnitude of 0.220 T and is directed at an angle of 60.0° outward from the normal to the plane of the coil (**Fig. P27.69**). Let the axis of the coil be in the y -direction. The current in the coil is in the direction shown (counterclockwise as viewed from a point above the coil on the y -axis). Calculate the magnitude and direction of the net magnetic force on the coil.

Figure P27.69



27.70 •• CP A uniform bar has mass 0.0120 kg and is 30.0 cm long. It pivots without friction about an axis perpendicular to the bar at point a (**Fig. P27.70**). The gravitational force on the bar acts in the $-y$ -direction. The bar is in a uniform magnetic field that is directed into the page and has magnitude $B = 0.150$ T. (a) What must be the current I in the bar for the bar to be in rotational equilibrium when it is at an angle $\theta = 30.0^\circ$ above the horizontal? Use your result from Problem 27.66. (b) For the bar to be in rotational equilibrium, should I be in the direction from a to b or b to a ?

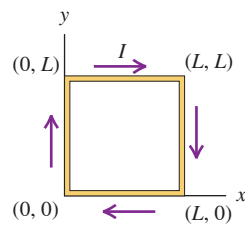
Figure P27.70



27.71 •• CALC Force on a Current Loop in a Nonuniform Magnetic Field.

It was shown in Section 27.7 that the net force on a current loop in a *uniform* magnetic field is zero. But what if $\vec{B}$ is *not* uniform? **Figure P27.71** shows a square loop of wire that lies in the xy -plane. The loop has corners at $(0, 0)$, $(0, L)$, $(L, 0)$, and (L, L) and carries a constant current I in the clockwise direction.

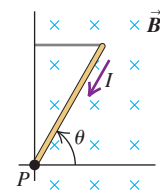
Figure P27.71



The magnetic field has no x -component but has both y - and z -components: $\vec{B} = (B_0 z/L)\hat{j} + (B_0 y/L)\hat{k}$, where B_0 is a positive constant. (a) Sketch the magnetic field lines in the yz -plane. (b) Find the magnitude and direction of the magnetic force exerted on each of the sides of the loop by integrating Eq. (27.20). (c) Find the magnitude and direction of the net magnetic force on the loop.

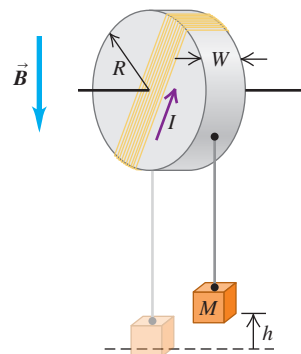
27.72 •• CP The lower end of the thin uniform rod in **Fig. P27.72** is attached to the floor by a frictionless hinge at point P . The rod has mass 0.0840 kg and length 18.0 cm and is in a uniform magnetic field $B = 0.120$ T that is directed into the page. The rod is held at an angle $\theta = 53.0^\circ$ above the horizontal by a horizontal string that connects the top of the rod to the wall. The rod carries a current $I = 12.0$ A in the direction toward P . Calculate the tension in the string. Use your result from Problem 27.66 to calculate the torque due to the magnetic-field force.

Figure P27.72



27.73 ••• CP A magnetic lift uses a cylinder with radius R and width W wrapped by conducting wire N times on a diameter, as shown in **Fig. P27.73**. A uniform external magnetic field $\vec{B}$ points downward while current I flows in the sense indicated. A mass M is hung from the cylinder by a cable attached to its rim on the axis of the coil. The height above the lowest possible position of the mass is h .

Figure P27.73



(a) What is the minimum current I for which the mass can be suspended with $h > 0$? (b) What is the height of the mass $h = h_{\text{top}}$ when the cable attachment rotates one-half turn to the top of the cylinder, in terms of R ? Define a dimensionless parameter $\sigma = 2NIWB/Mg$ and express subsequent results in terms of σ and other given quantities. (c) What is the net torque on the cylinder if $0 \leq h \leq R$? (d) What is the net torque on the cylinder if $R \leq h \leq h_{\text{top}}$? (e) Define potential energy $U(h)$ that is zero when $h = 0$. What is the potential energy for $0 \leq h \leq R$? (f) What is the potential energy for $R \leq h \leq h_{\text{top}}$? (g) If $\sigma > 1$, for what value of h will the mass remain suspended motionless? (h) For what values of σ will the cylinder rotate more than 180° if the mass is released from rest at $h = h_{\text{top}}$? (Hint: This corresponds to $U(h_{\text{top}}) > 0$.)

27.74 ••• CP CALC A particle with charge q and mass m is dropped at time $t = 0$ from rest at its origin in a region of constant magnetic field $\vec{B}$ that points horizontally. What happens? To answer, construct a Cartesian coordinate system with the y -axis pointing downward and the z -axis pointing in the direction of the magnetic field. At time $t \geq 0$ the particle has velocity $\vec{v} = v_x\hat{i} + v_y\hat{j}$. The net force $\vec{F} = F_x\hat{i} + F_y\hat{j}$ on the particle is the vector sum of its weight and the magnetic force. (a) Using Newton's second law, write equations for a_x and a_y , where $\vec{a} = a_x\hat{i} + a_y\hat{j}$ is the acceleration of the particle. (b) Differentiate the second of these

equations with respect to time. Then substitute your expression for $a_x = dv_x/dt$ to determine an equation for dv_y^2/dt^2 in terms of v_y . (c) This result shows that v_y is a simple harmonic oscillator. Use the initial conditions to determine $v_y(t)$. Write your answer in terms of the angular frequency $\omega = qB/m$. Note that $(dv_y/dt)_0 = g$. (d) Substitute your result for $v_y(t)$ into your equation for dv_x/dt . Integrate using the initial conditions to determine $v_x(t)$. (e) Integrate your expressions for $v_x(t)$ and $v_y(t)$ to determine $x(t)$ and $y(t)$. (f) If $m = 1.00$ mg, $q = 19.6$ μC , and $B = 10.0$ T, what maximum vertical distance does the particle drop before returning upward?

27.75 •• A circular loop of wire with area A lies in the xy -plane. As viewed along the z -axis looking in the $-z$ -direction toward the origin, a current I is circulating clockwise around the loop. The torque produced by an external magnetic field $\vec{B}$ is given by $\vec{\tau} = D(4\hat{i} - 3\hat{j})$, where D is a positive constant, and for this orientation of the loop the magnetic potential energy $U = -\vec{\mu} \cdot \vec{B}$ is negative. The magnitude of the magnetic field is $B_0 = 13D/IA$. (a) Determine the vector magnetic moment of the current loop. (b) Determine the components B_x , B_y , and B_z of $\vec{B}$.

27.76 •• DATA You are using a type of mass spectrometer to measure charge-to-mass ratios of atomic ions. In the device, atoms are ionized with a beam of electrons to produce positive ions, which are then accelerated through a potential difference V . (The final speed of the ions is great enough that you can ignore their initial speed.) The ions then enter a region in which a uniform magnetic field $\vec{B}$ is perpendicular to the velocity of the ions and has magnitude $B = 0.250$ T. In this $\vec{B}$ region, the ions move in a semicircular path of radius R . You measure R as a function of the accelerating voltage V for one particular atomic ion:

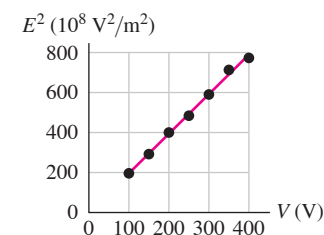
V (kV)	10.0	12.0	14.0	16.0	18.0
R (cm)	19.9	21.8	23.6	25.2	26.8

(a) How can you plot the data points so that they will fall close to a straight line? Explain. (b) Construct the graph described in part (a). Use the slope of the best-fit straight line to calculate the charge-to-mass ratio (q/m) for the ion. (c) For $V = 20.0$ kV, what is the speed of the ions as they enter the $\vec{B}$ region? (d) If ions that have $R = 21.2$ cm for $V = 12.0$ kV are singly ionized, what is R when $V = 12.0$ kV for ions that are doubly ionized?

27.77 •• DATA You are a research scientist working on a high-energy particle accelerator. Using a modern version of the Thomson e/m apparatus, you want to measure the mass of a muon (a fundamental particle that has the same charge as an electron but greater mass). The magnetic field between the two charged plates is 0.340 T. You measure the electric field for zero particle deflection as a function of the accelerating potential V . This potential is large enough that you can assume the initial speed of the muons to be zero.

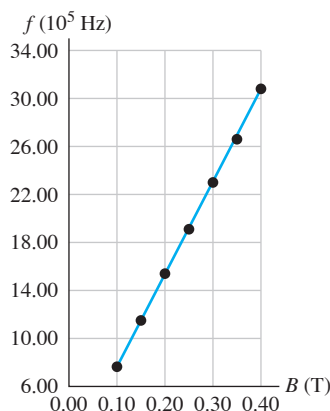
Figure P27.77 is an E^2 -versus- V graph of your data. (a) Explain why the data points fall close to a straight line. (b) Use the graph in **Fig. P27.77** to calculate the mass m of a muon. (c) If the two charged plates are separated by 6.00 mm, what must be the voltage between the plates in order for the electric field between the plates to be 2.00×10^5 V/m? Assume that the dimensions of the plates are much larger than the separation between them. (d) When $V = 400$ V, what is the speed of the muons as they enter the region between the plates?

Figure P27.77



27.78 •• DATA You are a technician testing the operation of a cyclotron. An alpha particle in the device moves in a circular path in a magnetic field $\vec{B}$ that is directed perpendicular to the path of the alpha particle. You measure the number of revolutions per second (the frequency f) of the alpha particle as a function of the magnetic field strength B . **Figure P27.78** shows your results and the best straight-line fit to your data. (a) Use the graph in Fig. P27.78 to calculate the charge-to-mass ratio of the alpha particle, which has charge $+2e$. On the basis of your data, what is the mass of an alpha particle? (b) With $B = 0.300$ T, what are the cyclotron frequencies f of a proton and of an electron? How do these f values compare to the frequency of an alpha particle? (c) With $B = 0.300$ T, what speed and kinetic energy does an alpha particle have if the radius of its path is 12.0 cm?

Figure P27.78



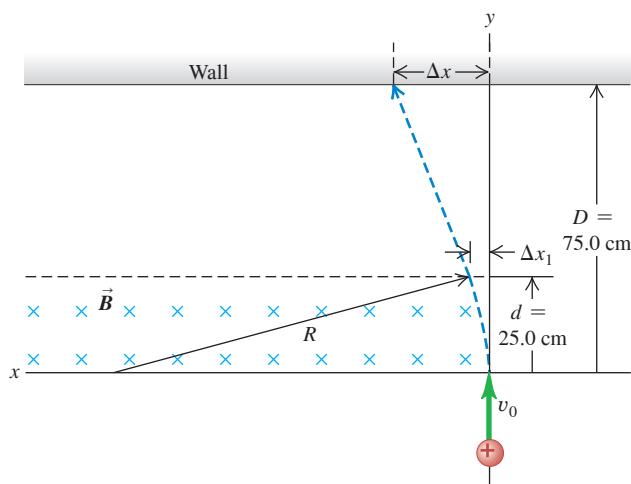
CHALLENGE PROBLEMS

27.79 ••• CALC Determine the magnetic moment $\vec{\mu}$ of a spherical shell with radius R and uniform charge Q rotating with angular speed $\vec{\Omega} = \omega \hat{k}$. Use the following steps: (a) Consider a coordinate system with the origin at the center of the sphere. Parameterize each latitude on the sphere with the angle θ measured from the positive z -axis. There is a circular current loop at each value of θ for $0 \leq \theta \leq \pi$. What is the radius of the loop at latitude θ ? (b) The differential current carried by that loop is $dI = \sigma v dW$, where σ is the charge density of the sphere, v is the tangential speed of the loop, and $dW = R d\theta$ is its differential width. Express dI in terms of σ , R , ω , θ , and $d\theta$. (c) The differential magnetic moment of the loop is $d\mu = A dI$, where A is the area enclosed by the loop. Express $d\mu$ in terms of R , ω , θ , and $d\theta$. (d) Integrate over the sphere to determine the magnetic moment. Express your result as a vector; use the total charge Q rather than the charge density σ . (e) If the sphere is in a uniform magnetic field $\vec{B} = (\sin \alpha \hat{i} + \cos \alpha \hat{j})B$, what is the torque on the sphere?

27.80 ••• An electron traveling from the sun as part of the solar wind strikes the earth's magnetosphere at latitude 80.0° N in a region where the magnetic field has a strength of $15.0 \mu\text{T}$ and is directed toward the earth's center. The electron has a speed of 400 km/s and is directed toward the earth's axis parallel to the equator. Magnetic forces send the electron on a helical trajectory. (a) What is the radius of this helix? (b) With what speed does the electron approach the surface of the earth? (c) If you are looking downward toward the earth from space, is the electron's motion clockwise or counterclockwise? (d) What is the frequency of the motion? (e) This electron strikes the ionosphere, where it is further accelerated by an electric field with strength 20.0 mV/m directed northward parallel to the earth's surface. What is the electron's new speed after it has been deflected 100 km southward by this field? (f) By what factor has its kinetic energy been increased by the electric field?

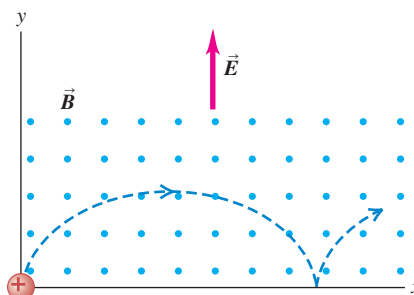
27.81 ••• A particle with charge $2.15 \mu\text{C}$ and mass 3.20×10^{-11} kg is initially traveling in the $+y$ -direction with a speed $v_0 = 1.45 \times 10^5$ m/s. It then enters a region containing a uniform magnetic field that is directed into, and perpendicular to, the page in **Fig. P27.81**. The magnitude of the field is 0.420 T. The region extends a distance of 25.0 cm along the initial direction of travel; 75.0 cm from the point of entry into the magnetic field region is a wall. The length of the field-free region is thus 50.0 cm. When the charged particle enters the magnetic field, it follows a curved path whose radius of curvature is R . It then leaves the magnetic field after a time t_1 , having been deflected a distance Δx_1 . The particle then travels in the field-free region and strikes the wall after undergoing a total deflection Δx . (a) Determine the radius R of the curved part of the path. (b) Determine t_1 , the time the particle spends in the magnetic field. (c) Determine Δx_1 , the horizontal deflection at the point of exit from the field. (d) Determine Δx , the total horizontal deflection.

Figure P27.81



27.82 ••• CP A Cycloidal Path. A particle with mass m and positive charge q starts from rest at the origin shown in **Fig. P27.82**. There is a uniform electric field $\vec{E}$ in the $+y$ -direction and a uniform magnetic field $\vec{B}$ directed out of the page. It is shown in more advanced books that the path is a *cycloid* whose radius of curvature at the top points is twice the y -coordinate at that level. (a) Explain why the path has this general shape and why it is repetitive. (b) Prove that the speed at any point is equal to $\sqrt{2qEy/m}$. (Hint: Use energy conservation.) (c) Applying Newton's second law at the top point and taking as given that the radius of curvature here equals $2y$, prove that the speed at this point is $2E/B$.

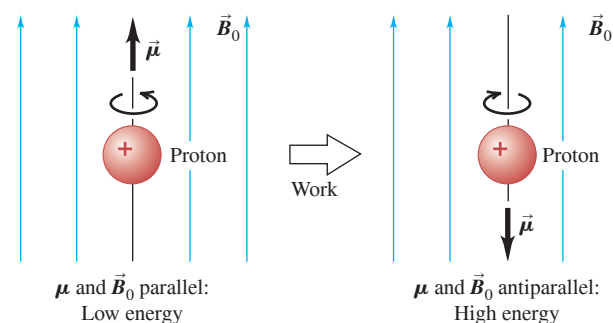
Figure P27.82



MCAT-STYLE PASSAGE PROBLEMS

BIO Magnetic Fields and MRI. *Magnetic resonance imaging* (MRI) is a powerful imaging method that, unlike x-ray imaging, allows sharp images of soft tissue to be made without exposing the patient to potentially damaging radiation. A rudimentary understanding of this method can be achieved by the relatively simple application of the classical (that is, non-quantum) physics of magnetism. The starting point for MRI is *nuclear magnetic resonance* (NMR), a technique that depends on the fact that protons in the atomic nucleus have a magnetic field $\vec{B}$. The origin of the proton's magnetic field is the spin of the proton. Being charged, the spinning proton constitutes an electric current analogous to a wire loop through which current flows. Like the wire loop, the proton has a magnetic moment $\vec{\mu}$; thus it will experience a torque when it is subjected to an external magnetic field $\vec{B}_0$. The magnitude of $\vec{\mu}$ is about 1.4×10^{-26} J/T. The proton can be thought of as being in one of two states, with $\vec{\mu}$ oriented parallel or antiparallel to the applied magnetic field, and work must be done to flip the proton from the low-energy state to the high-energy state, as the accompanying figure below shows.

An important consideration is that the net magnetic field of any nucleus, except for that of hydrogen (which has a proton only), consists of contributions from both protons and neutrons. If a nucleus has an even number of protons and neutrons, they will pair in such a way that half of the protons have spins in one orientation and half have spins in the other orientation. Thus the net magnetic moment of the nucleus is zero. Only nuclei with a net magnetic moment are candidates for MRI. Hydrogen is the atom that is most commonly imaged.



27.83 If a proton is exposed to an external magnetic field of 2 T that has a direction perpendicular to the axis of the proton's spin, what will be the torque on the proton? (a) 0; (b) 1.4×10^{-26} N·m; (c) 2.8×10^{-26} N·m; (d) 0.7×10^{-26} N·m.

27.84 Which of following elements is a candidate for MRI? (a) $^{12}\text{C}_6$; (b) $^{16}\text{O}_8$; (c) $^{40}\text{Ca}_{20}$; (d) $^{31}\text{P}_{15}$.

27.85 The large magnetic fields used in MRI can produce forces on electric currents within the human body. This effect has been proposed as a possible method for imaging “biocurrents” flowing in the body, such as the current that flows in individual nerves. For a magnetic field strength of 2 T, estimate the magnitude of the maximum force on a 1-mm-long segment of a single cylindrical nerve that has a diameter of 1.5 mm. Assume that the entire nerve carries a current due to an applied voltage of 100 mV (that of a typical action potential). The resistivity of the nerve is $0.6 \Omega \cdot \text{m}$. (a) 6×10^{-7} N; (b) 1×10^{-6} N; (c) 3×10^{-4} N; (d) 0.3 N.

ANSWERS

Chapter Opening Question?

(ii) A magnetized compass needle has a magnetic dipole moment along its length, and the earth's magnetic field (which points generally northward) exerts a torque that tends to align that dipole moment with the field. See Section 27.7 for details.

Key Example **VP27.1** VARIATION Problems

VP27.1.1 (a) 4.18×10^{-14} N (b) $-z$ -direction

VP27.1.2 2.91×10^{-15} C

VP27.1.3 (a) 0.0175 T (b) $+z$ -direction

VP27.1.4 (a) 0.189 T (b) $-y$ -direction

VP27.6.1 (a) 0.435 mm (b) 2.87×10^7 rad/s (c) 4.57×10^6 Hz

VP27.6.2 (a) 0.348 mm (b) 1.64 mm (c) 4.80×10^{-16} N

VP27.6.3 (a) 8.00×10^5 m/s (b) $-y$ -direction (c) $-x$ -direction

VP27.6.4 (a) 6.64 cm (b) 7.47 cm

VP27.7.1 (a) 2.44×10^{-3} N (b) $+z$ -direction

VP27.7.2 (a) 1.60 A (b) upward

VP27.7.3 (a) -2.52×10^{-3} N (b) -4.20×10^{-3} N (c) 0 N

(d) 4.90×10^{-3} N

VP27.7.4 41.0° and 139.0°

Bridging Problem

(a) $\tau_x = -1.54 \times 10^{-4}$ N·m,

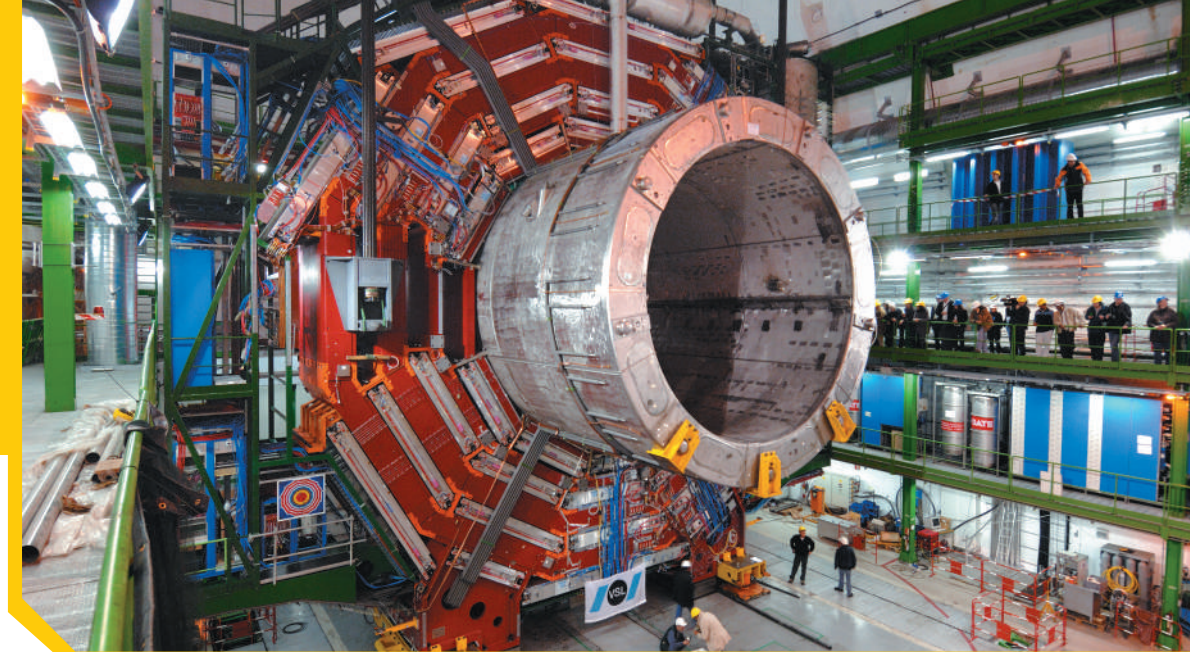
$\tau_y = -2.05 \times 10^{-4}$ N·m,

$\tau_z = -6.14 \times 10^{-4}$ N·m

(b) -7.55×10^{-4} J

(c) 42.1 rad/s

? The immense cylinder in this photograph is a current-carrying coil, or solenoid, that generates a uniform magnetic field in its interior as part of an experiment at CERN, the European Organization for Nuclear Research. If two such solenoids were joined end to end, the magnetic field along their common axis would (i) become four times stronger; (ii) double in strength; (iii) become $\sqrt{2}$ times stronger; (iv) not change; (v) weaken.



28 Sources of Magnetic Field

LEARNING OUTCOMES

In this chapter, you'll learn...

- 28.1** The nature of the magnetic field produced by a single moving charged particle.
- 28.2** How to describe the magnetic field produced by an element of a current-carrying conductor.
- 28.3** How to calculate the magnetic field produced by a long, straight, current-carrying wire.
- 28.4** Why wires carrying current in the same direction attract, while wires carrying opposing currents repel.
- 28.5** How to calculate the magnetic field produced by a current-carrying wire bent into a circle.
- 28.6** What Ampere's law is, and what it tells us about magnetic fields.
- 28.7** How to use Ampere's law to calculate the magnetic field of symmetric current distributions.
- 28.8** How microscopic currents within materials give them their magnetic properties.

You'll need to review...

- 10.5** Angular momentum of a particle.
- 21.3–21.5** Coulomb's law and electric-field calculations.
- 22.4** Solving problems with Gauss's law.
- 27.2–27.9** Magnetic field and magnetic force.

In Chapter 27 we studied the forces exerted on moving charges and on current-carrying conductors in a magnetic field. We didn't worry about how the magnetic field got there; we simply took its existence as a given fact. But how are magnetic fields *created*? We know that both permanent magnets and electric currents in electromagnets create magnetic fields. In this chapter we'll study these sources of magnetic field in detail.

We've learned that a charge creates an electric field and that an electric field exerts a force on a charge. But a *magnetic* field exerts a force on only a *moving* charge. Similarly, we'll see that only *moving* charges *create* magnetic fields. We'll begin our analysis with the magnetic field created by a single moving point charge. We can use this analysis to determine the field created by a small segment of a current-carrying conductor. Once we can do that, we can in principle find the magnetic field produced by *any* shape of conductor.

Then we'll introduce Ampere's law, which plays a role in magnetism analogous to the role of Gauss's law in electrostatics. Ampere's law lets us exploit symmetry properties in relating magnetic fields to their sources.

Moving charged particles within atoms respond to magnetic fields and can also act as sources of magnetic field. We'll use these ideas to understand how certain magnetic materials can be used to intensify magnetic fields as well as why some materials such as iron act as permanent magnets.

28.1 MAGNETIC FIELD OF A MOVING CHARGE

Let's start with the basics, the magnetic field of a single point charge q moving with a constant velocity $\vec{v}$. In practical applications, such as the solenoid shown in the photo that opens this chapter, magnetic fields are produced by tremendous numbers of charged particles moving together in a current. But once we understand how to calculate the magnetic field due to a single point charge, it's a small leap to calculate the field due to a current-carrying wire or collection of wires.

As we did for electric fields, we call the location of the moving charge at a given instant the **source point** and the point P where we want to find the field the **field point**. In Section 21.4 we found that at a field point a distance r from a point charge q , the magnitude of the

electric field $\vec{E}$ caused by the charge is proportional to the charge magnitude $|q|$ and to $1/r^2$, and the direction of $\vec{E}$ (for positive q) is along the line from source point to field point. The corresponding relationship for the *magnetic* field $\vec{B}$ of a point charge q moving with constant velocity has some similarities and some interesting differences.

Experiments show that the magnitude of $\vec{B}$ is also proportional to $|q|$ and to $1/r^2$. But the *direction* of $\vec{B}$ is *not* along the line from source point to field point. Instead, $\vec{B}$ is perpendicular to the plane containing this line and the particle's velocity vector $\vec{v}$, as shown in **Fig. 28.1**. Furthermore, the field *magnitude* B is also proportional to the particle's speed v and to the sine of the angle ϕ . Thus the magnetic-field magnitude at point P is

$$B = \frac{\mu_0}{4\pi} \frac{|q|v \sin \phi}{r^2} \quad (28.1)$$

The quantity μ_0 (read as “mu-nought” or “mu-sub-zero”) is called the **magnetic constant**. The reason for including the factor of 4π will emerge shortly. We did something similar with Coulomb's law in Section 21.3.

Moving Charge: Vector Magnetic Field

We can incorporate both the magnitude and direction of $\vec{B}$ into a single vector equation by using the vector product. To avoid having to say “the direction from the source q to the field point P ” over and over, we introduce a *unit* vector $\hat{r}$ (“r-hat”) that points from the source point to the field point. (We used $\hat{r}$ for the same purpose in Section 21.4.) This unit vector is equal to the vector $\vec{r}$ from the source to the field point divided by its magnitude: $\hat{r} = \vec{r}/r$. Then

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2} \quad (28.2)$$

Magnetic field due to a point charge with constant velocity $\vec{B}$ = $\frac{\mu_0}{4\pi}$ $\frac{q\vec{v} \times \hat{r}}{r^2}$

Magnetic constant μ_0 Charge q Velocity $\vec{v}$ Unit vector from point charge toward where field is measured $\hat{r}$ Distance from point charge to where field is measured r

Figure 28.1 shows the relationship of $\hat{r}$ to P and shows the magnetic field $\vec{B}$ at several points in the vicinity of the charge. At all points along a line through the charge parallel to the velocity $\vec{v}$, the field is zero because $\sin \phi = 0$ at all such points. At any distance r from q , $\vec{B}$ has its greatest magnitude at points lying in the plane perpendicular to $\vec{v}$, because there $\phi = 90^\circ$ and $\sin \phi = 1$. If q is negative, the directions of $\vec{B}$ are opposite to those shown in Fig. 28.1.

Moving Charge: Magnetic Field Lines

A point charge in motion also produces an *electric* field, with field lines that radiate outward from a positive charge. The *magnetic* field lines are completely different. For a point charge moving with velocity $\vec{v}$, the magnetic field lines are *circles* centered on the line of $\vec{v}$ and lying in planes perpendicular to this line. The field-line directions for a positive charge are given by the following *right-hand rule*, one of several that we'll encounter in this chapter: Grasp the velocity vector $\vec{v}$ with your right hand so that your right thumb points in the direction of $\vec{v}$; your fingers then curl around the line of $\vec{v}$ in the same sense as the magnetic field lines, assuming q is positive. Figure 28.1a shows parts of a few field lines; Fig. 28.1b shows some field lines in a plane through q , perpendicular to $\vec{v}$. If the point charge is negative, the directions of the field and field lines are the opposite of those shown in Fig. 28.1.

Equations (28.1) and (28.2) describe the $\vec{B}$ field of a point charge moving with *constant* velocity. If the charge *accelerates*, the field can be much more complicated. We won't need these more complicated results for our purposes. (The moving charged particles that make up a current in a wire accelerate at points where the wire bends and the direction of $\vec{v}$ changes. But because the magnitude v_d of the drift velocity in a conductor is typically very small, the centripetal acceleration v_d^2/r is so small that we can ignore its effects.)

As we discussed in Section 27.2, the unit of B is one tesla (1 T):

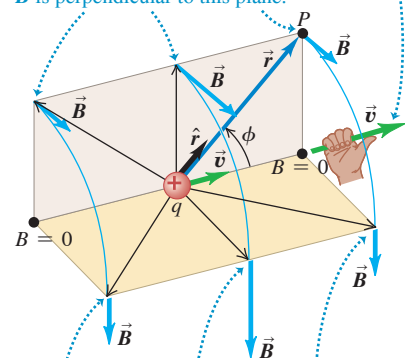
$$1 \text{ T} = 1 \text{ N} \cdot \text{s} / \text{C} \cdot \text{m} = 1 \text{ N} / \text{A} \cdot \text{m}$$

Figure 28.1 (a) Magnetic-field vectors due to a moving positive point charge q . At each point, $\vec{B}$ is perpendicular to the plane of $\vec{r}$ and $\vec{v}$, and its magnitude is proportional to the sine of the angle between them. (b) Magnetic field lines in a plane containing a moving positive charge.

(a) Perspective view

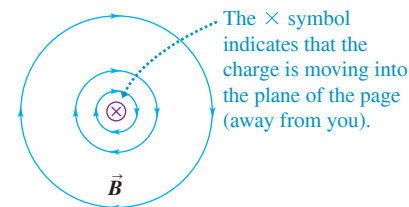
Right-hand rule for the magnetic field due to a positive charge moving at constant velocity: Point the thumb of your right hand in the direction of the velocity. Your fingers now curl around the charge in the direction of the magnetic field lines. (If the charge is negative, the field lines are in the opposite direction.)

For these field points, $\vec{r}$ and $\vec{v}$ both lie in the beige plane, and $\vec{B}$ is perpendicular to this plane.



For these field points, $\vec{r}$ and $\vec{v}$ both lie in the gold plane, and $\vec{B}$ is perpendicular to this plane.

(b) View from behind the charge



Using this with Eq. (28.1) or (28.2), we find that the units of the constant μ_0 are

$$1 \text{ N} \cdot \text{s}^2/\text{C}^2 = 1 \text{ N}/\text{A}^2 = 1 \text{ Wb}/\text{A} \cdot \text{m} = 1 \text{ T} \cdot \text{m}/\text{A}$$

In SI units the numerical value of μ_0 is, to nine significant figures,

$$\begin{aligned}\mu_0 &= 1.25663706 \times 10^{-6} \text{ N} \cdot \text{s}^2/\text{C}^2 = 1.25663706 \times 10^{-6} \text{ Wb}/\text{A} \cdot \text{m} \\ &\cong 4\pi \times 10^{-7} \text{ T} \cdot \text{m}/\text{A}\end{aligned}\quad (28.3)$$

It may seem incredible that μ_0 is equal to $4\pi \times 10^{-7} \text{ T} \cdot \text{m}/\text{A}$ to nine significant figures! In fact this is a consequence of how the ampere and coulomb were previously defined. We'll explore this further in Section 28.4.

Here's another remarkable aspect of the value of the magnetic constant μ_0 . Recall from Section 21.3 that the *electric* constant ϵ_0 that appears in Coulomb's law and Gauss's law has the value $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$. If we multiply ϵ_0 by $\mu_0 = 1.257 \times 10^{-6} \text{ N} \cdot \text{s}^2/\text{C}^2$, take the square root of this product, then take the reciprocal of the result, we get

$$\frac{1}{\sqrt{\epsilon_0 \mu_0}} = 2.998 \times 10^8 \text{ m/s}$$

But $2.998 \times 10^8 \text{ m/s}$ is just the value of the speed of light in vacuum! We conclude that

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad (28.4)$$

This suggests that electric and magnetic fields are intimately related to the nature of light. We'll explore this relationship when we study electromagnetic waves in Chapter 32.

EXAMPLE 28.1 Forces between two moving protons

WITH VARIATION PROBLEMS

Two protons move parallel to the x -axis in opposite directions (**Fig. 28.2**) at the same speed v (small compared to the speed of light c). At the instant shown, find the electric and magnetic forces on the upper proton and compare their magnitudes.

IDENTIFY and SET UP Coulomb's law [Eq. (21.2)] gives the electric force F_E on the upper proton. The magnetic force law [Eq. (27.2)] gives the magnetic force on the upper proton; to use it, we must first use Eq. (28.2) to find the magnetic field that the lower proton produces at the position of the upper proton. The unit vector from the lower proton (the source) to the position of the upper proton is $\hat{r} = \hat{j}$.

EXECUTE From Coulomb's law, the magnitude of the electric force on the upper proton is

$$F_E = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2}$$

The forces are repulsive, and the force on the upper proton is vertically upward (in the $+y$ -direction).

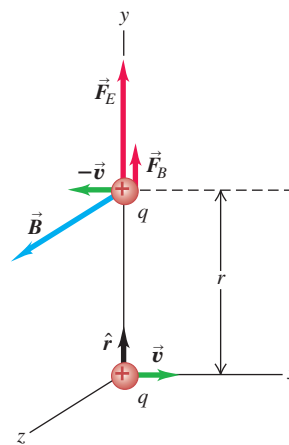
The velocity of the lower proton is $\vec{v} = v\hat{i}$. From the right-hand rule for the cross product $\vec{v} \times \hat{r}$ in Eq. (28.2), the $\vec{B}$ field due to the lower proton at the position of the upper proton is in the $+z$ -direction (see Fig. 28.2). From Eq. (28.2), the field is

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q(v\hat{i}) \times \hat{j}}{r^2} = \frac{\mu_0}{4\pi} \frac{qv}{r^2} \hat{k}$$

The velocity of the upper proton is $-\vec{v} = -v\hat{i}$, so the magnetic force on it is

$$\vec{F}_B = q(-\vec{v}) \times \vec{B} = q(-v\hat{i}) \times \frac{\mu_0}{4\pi} \frac{qv}{r^2} \hat{k} = \frac{\mu_0}{4\pi} \frac{q^2 v^2}{r^2} \hat{j}$$

Figure 28.2 Electric and magnetic forces between two moving protons.



The magnetic interaction in this situation is also repulsive.

The ratio of the force magnitudes is

$$\frac{F_B}{F_E} = \frac{\mu_0 q^2 v^2 / 4\pi r^2}{q^2 / 4\pi \epsilon_0 r^2} = \frac{\mu_0 v^2}{1/\epsilon_0} = \epsilon_0 \mu_0 v^2$$

With the relationship $\epsilon_0 \mu_0 = 1/c^2$, Eq. (28.4), this becomes

$$\frac{F_B}{F_E} = \frac{v^2}{c^2}$$

When v is small in comparison to the speed of light, the magnetic force is much smaller than the electric force.

EVALUATE We have described the velocities, fields, and forces as they are measured by an observer who is stationary in the coordinate system of Fig. 28.2. In a coordinate system that moves with one of the charges, one of the velocities would be zero, so there would be *no* magnetic force. The explanation of this apparent paradox provided one of the paths that led to the special theory of relativity.

KEYCONCEPT A moving charged particle produces a magnetic field $\vec{B}$. The magnitude of $\vec{B}$ at a given point is proportional to the charge magnitude and to the particle's speed, and inversely proportional to the square of the distance from the charge; it also depends on the direction from the particle to that point compared to the direction of the particle's velocity. The direction of $\vec{B}$ is given by a right-hand rule.

TEST YOUR UNDERSTANDING OF SECTION 28.1 (a) If two protons are traveling parallel to each other in the *same* direction and at the same speed, is the magnetic force between them (i) attractive or (ii) repulsive? (b) Is the net force between them (i) attractive, (ii) repulsive, or (iii) zero? (Assume that the protons' speed is much slower than the speed of light.)

ANSWER (a) (i), (b) (ii) The situation is the same as shown in Fig. 28.2 except that the upper proton has velocity $\vec{v}$ rather than $-\vec{v}$. The magnetic field due to the lower proton is the same as shown in Fig. 28.2, but the direction of the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on the upper proton is reversed. Hence the magnetic force is attractive. Since the speed v is small compared to c , the magnetic force is much smaller in magnitude than the repulsive electric force and the net force is still repulsive.

28.2 MAGNETIC FIELD OF A CURRENT ELEMENT

As for electric fields, there is a *principle of superposition of magnetic fields*:

PRINCIPLE OF SUPERPOSITION OF MAGNETIC FIELDS The total magnetic field caused by several moving charges is the vector sum of the fields caused by the individual charges.

We can use this principle with the results of Section 28.1 to find the magnetic field produced by a current in a conductor.

We begin by calculating the magnetic field caused by a short segment $d\vec{l}$ of a current-carrying conductor, as shown in Fig. 28.3a. The volume of the segment is $A dl$, where A is the cross-sectional area of the conductor. If there are n moving charged particles per unit volume, each of charge q , the total moving charge dQ in the segment is

$$dQ = nqA dl$$

The moving charges in this segment are equivalent to a single charge dQ , traveling with a velocity equal to the *drift* velocity $\vec{v}_d$. (Magnetic fields due to the *random* motions of the charges will, on average, cancel out at every point.) From Eq. (28.1) the magnitude of the resulting field $d\vec{B}$ at any field point P is

$$dB = \frac{\mu_0}{4\pi} \frac{|dQ|v_d \sin \phi}{r^2} = \frac{\mu_0}{4\pi} \frac{n|q|v_d A dl \sin \phi}{r^2}$$

But from Eq. (25.2), $n|q|v_d A$ equals the current I in the element. So

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \phi}{r^2} \quad (28.5)$$

Current Element: Vector Magnetic Field

In vector form, using the unit vector $\hat{r}$ as in Section 28.1, we have

$$\vec{d\vec{B}} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} \quad (28.6)$$

Magnetic field due to an infinitesimal current element

Magnetic constant

Current

Vector length of element (points in current direction)

Unit vector from element toward where field is measured

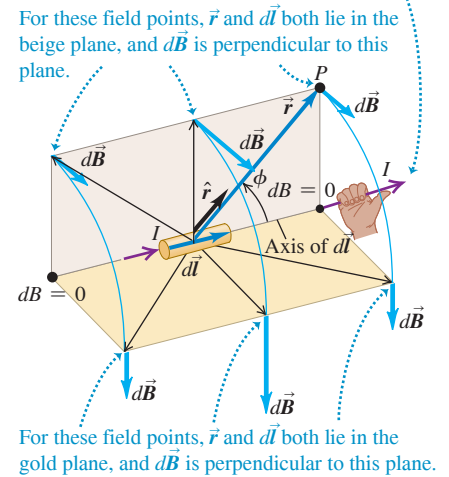
Distance from element to where field is measured

where $d\vec{l}$ is a vector with length dl , in the same direction as the current.

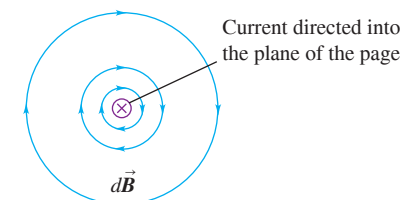
Figure 28.3 (a) Magnetic-field vectors due to a current element $d\vec{l}$. (b) Magnetic field lines in a plane containing the current element $d\vec{l}$. Compare this figure to Fig. 28.1 for the field of a moving point charge.

(a) Perspective view

Right-hand rule for the magnetic field due to a current element: Point the thumb of your right hand in the direction of the current. Your fingers now curl around the current element in the direction of the magnetic field lines.



(b) View along the axis of the current element



APPLICATION Currents and Planetary Magnetism

The earth's magnetic field is caused by currents circulating within its molten, conducting interior. These currents are stirred by our planet's relatively rapid spin (one rotation per 24 hours). The planet Jupiter's internal currents are much stronger: Jupiter is much larger than the earth, has an interior that is mostly liquid hydrogen (which is a very good conductor under high pressures), and spins very rapidly (one rotation per 10 hours). Hence at the same distance from the center of each planet, Jupiter's magnetic field is about 2×10^4 times greater than the earth's field.



Equations (28.5) and (28.6) are called the **law of Biot and Savart** (pronounced “Bee-oh” and “Suh-var”). We can use this law to find the total magnetic field $\vec{B}$ at any point in space due to the current in a complete circuit. To do this, we integrate Eq. (28.6) over all segments $d\vec{l}$ that carry current; symbolically,

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \hat{r}}{r^2} \quad (28.7)$$

In the following sections we'll carry out this vector integration for several examples.

Current Element: Magnetic Field Lines

As Fig. 28.3 shows, the field vectors $d\vec{B}$ and the magnetic field lines of a current element are exactly like those set up by a positive charge dQ moving in the direction of the drift velocity $\vec{v}_d$. The field lines are circles in planes perpendicular to $d\vec{l}$ and centered on the line of $d\vec{l}$. Their directions are given by the same right-hand rule that we introduced for point charges in Section 28.1.

We can't verify Eq. (28.5) or (28.6) directly because we can never experiment with an isolated segment of a current-carrying circuit. What we measure experimentally is the **total** $\vec{B}$ for a complete circuit. But we can still verify these equations indirectly by calculating $\vec{B}$ for various current configurations with Eq. (28.7) and comparing the results with experimental measurements.

If matter is present in the space around a current-carrying conductor, the field at a field point P in its vicinity will have an additional contribution resulting from the **magnetization** of the material. We'll return to this point in Section 28.8. However, unless the material is iron or some other ferromagnetic material, the additional field is small and is usually negligible. Additional complications arise if time-varying electric or magnetic fields are present or if the material is a superconductor; we'll return to these topics later.

PROBLEM-SOLVING STRATEGY 28.1 Magnetic-Field Calculations

IDENTIFY *the relevant concepts:* The law of Biot and Savart [Eqs. (28.5) and (28.6)] allows you to calculate the magnetic field at a field point P due to a current-carrying wire of any shape. The idea is to calculate the field element $d\vec{B}$ at P due to a representative current element in the wire and integrate all such field elements to find the field $\vec{B}$ at P .

SET UP *the problem* using the following steps:

1. Make a diagram showing a representative current element and the field point P .
2. Draw the current element $d\vec{l}$, being careful that it points in the direction of the current.
3. Draw the unit vector $\hat{r}$ directed *from* the current element (the source point) to P .
4. Identify the target variable (usually $\vec{B}$).

EXECUTE *the solution* as follows:

1. Use Eq. (28.5) or (28.6) to express the magnetic field $d\vec{B}$ at P from the representative current element.
2. Add up all the $d\vec{B}$'s using the rules of *vector* addition to find the total field at point P . In some situations the $d\vec{B}$'s at point P have

the same direction for all the current elements; then the magnitude of the total $\vec{B}$ field is the sum of the magnitudes of the $d\vec{B}$'s. But often the $d\vec{B}$'s have different directions for different current elements. Then you have to set up a coordinate system and represent each $d\vec{B}$ in terms of its components. The integral for the total $\vec{B}$ is then expressed in terms of an integral for each component.

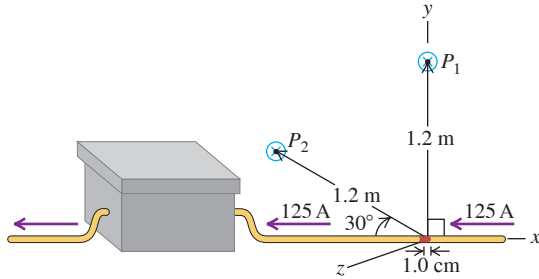
3. Sometimes you can use the symmetry of the situation to prove that one component of $\vec{B}$ must vanish. Always be alert for ways to use symmetry to simplify the problem.
4. Look for ways to use the principle of superposition of magnetic fields. Later in this chapter we'll determine the fields produced by certain simple conductor shapes; if you encounter a conductor of a complex shape that can be represented as a combination of these simple shapes, you can use superposition to find the field of the complex shape. Examples include a rectangular loop and a semi-circle with straight line segments on both sides.

EVALUATE *your answer:* Often your answer will be a mathematical expression for $\vec{B}$ as a function of the position of the field point. Check the answer by examining its behavior in as many limits as you can.

EXAMPLE 28.2 Magnetic field of a current segment**WITH VARIATION PROBLEMS**

A copper wire carries a steady 125 A current to an electroplating tank (Fig. 28.4). Find the magnetic field due to a 1.0 cm segment of this wire at a point 1.2 m away from it, if the point is (a) point P_1 , straight out to the side of the segment, and (b) point P_2 , in the xy -plane and on a line at 30° to the segment.

Figure 28.4 Finding the magnetic field at two points due to a 1.0 cm segment of current-carrying wire (not shown to scale).



IDENTIFY and SET UP Although Eqs. (28.5) and (28.6) apply only to infinitesimal current elements, we may use either of them here because the segment length is much less than the distance to the field point. The current element is shown in red in Fig. 28.4 and points in the $-x$ -direction (the direction of the current), so $d\vec{l} = dl(-\hat{i})$. The unit vector $\hat{r}$ for each field point is directed from the current element toward that point: $\hat{r}$ is in the $+y$ -direction for point P_1 and at an angle of 30° above the $-x$ -direction for point P_2 .

EXECUTE (a) At point P_1 , $\hat{r} = \hat{j}$, so

$$\begin{aligned}\vec{B} &= \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl(-\hat{i}) \times \hat{j}}{r^2} = -\frac{\mu_0}{4\pi} \frac{I dl}{r^2} \hat{k} \\ &= -(10^{-7} \text{ T} \cdot \text{m/A}) \frac{(125 \text{ A})(1.0 \times 10^{-2} \text{ m})}{(1.2 \text{ m})^2} \hat{k} \\ &= -(8.7 \times 10^{-8} \text{ T}) \hat{k}\end{aligned}$$

The direction of $\vec{B}$ at P_1 is into the xy -plane of Fig. 28.4.

(b) At P_2 , the unit vector is $\hat{r} = (-\cos 30^\circ)\hat{i} + (\sin 30^\circ)\hat{j}$. From Eq. (28.6),

$$\begin{aligned}\vec{B} &= \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl(-\hat{i}) \times (-\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j})}{r^2} \\ &= -\frac{\mu_0 I dl \sin 30^\circ}{4\pi r^2} \hat{k} \\ &= -(10^{-7} \text{ T} \cdot \text{m/A}) \frac{(125 \text{ A})(1.0 \times 10^{-2} \text{ m})(\sin 30^\circ)}{(1.2 \text{ m})^2} \hat{k} \\ &= -(4.3 \times 10^{-8} \text{ T}) \hat{k}\end{aligned}$$

The direction of $\vec{B}$ at P_2 is also into the xy -plane of Fig. 28.4.

EVALUATE We can check our results for the direction of $\vec{B}$ by comparing them with Fig. 28.3. The xy -plane in Fig. 28.4 corresponds to the beige plane in Fig. 28.3, but here the direction of the current and hence of $d\vec{l}$ is the reverse of that shown in Fig. 28.3. Consequently the direction of the magnetic field is reversed as well. It follows that the field at points in the xy -plane in Fig. 28.4 must point *into*, not *out of*, that plane, just as we concluded above.

KEYCONCEPT The magnetic field $\vec{B}$ due to a short current segment of length dl is like that of a moving charged particle. The difference is that the product of the moving particle's charge and velocity is replaced by the product of the current and the vector $d\vec{l}$ in the direction of the current.

TEST YOUR UNDERSTANDING OF SECTION 28.2 An infinitesimal current element located at the origin ($x = y = z = 0$) carries current I in the positive y -direction. Rank the following locations in order of the strength of the magnetic field that the current element produces at that location, from largest to smallest value. (i) $x = L$, $y = 0$, $z = 0$; (ii) $x = 0$, $y = L$, $z = 0$; (iii) $x = 0$, $y = 0$, $z = L$; (iv) $x = L/\sqrt{2}$, $y = L/\sqrt{2}$, $z = 0$.

ANSWER

(i) 0, (ii) $1/\sqrt{2}$, (iii) $1/\sqrt{2}$, (iv) $1/\sqrt{2}$.
 $\phi = 45^\circ$, so the values of $\sin \phi$ are (i) 1, (ii) $1/\sqrt{2}$, (iii) $1/\sqrt{2}$, and (iv) 1.
 points the angle is (i) 90° , (ii) 45° , (iii) 45° , and (iv) 90° . For the four
 so the value of dB is proportional to the value of $\sin \phi$. For the four
 All four points are the same distance $r = L$ from the current element.

of the current and a vector from the current element to the field point.
 the element to the field point, and ϕ is the angle between the direction
 $dB = (\mu_0/4\pi)(I dl \sin \phi/r^2)$. In this expression r is the distance from
 field dB due to a current element of length dl carrying current I is
 (i) and (iii) (ii), (iv), (iii) From Eq. (28.5), the magnitude of the

28.3 MAGNETIC FIELD OF A STRAIGHT CURRENT-CARRYING CONDUCTOR

Let's use the law of Biot and Savart to find the magnetic field produced by a straight current-carrying conductor. This result is useful because straight conducting wires are found in essentially all electric and electronic devices. **Figure 28.5** (next page) shows such a conductor with length $2a$ carrying a current I . We'll find $\vec{B}$ at a point a distance x from the conductor on its perpendicular bisector.

We first use the law of Biot and Savart, Eq. (28.5), to find the field $d\vec{B}$ caused by the element of conductor of length $dl = dy$ shown in Fig. 28.5. From the figure, $r = \sqrt{x^2 + y^2}$ and $\sin \phi = \sin(\pi - \phi) = x/\sqrt{x^2 + y^2}$. The right-hand rule for the vector product $d\vec{l} \times \hat{r}$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

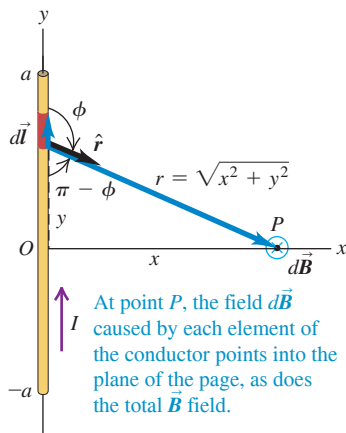
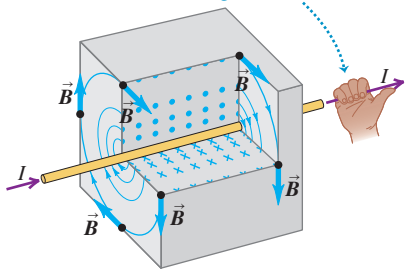


Figure 28.6 Magnetic field around a long, straight, current-carrying conductor. The field lines are circles, with directions determined by the right-hand rule.

Right-hand rule for the magnetic field around a current-carrying wire: Point the thumb of your right hand in the direction of the current. Your fingers now curl around the wire in the direction of the magnetic field lines.



shows that the *direction* of $d\vec{B}$ is into the plane of the figure, perpendicular to the plane; furthermore, the directions of the $d\vec{B}$'s from *all* elements of the conductor are the same. Thus in integrating Eq. (28.7), we can just add the *magnitudes* of the $d\vec{B}$'s, a significant simplification.

Putting the pieces together, we find that the magnitude of the total $\vec{B}$ field is

$$B = \frac{\mu_0 I}{4\pi} \int_{-a}^a \frac{x dy}{(x^2 + y^2)^{3/2}}$$

We can integrate this by trigonometric substitution or by using an integral table:

$$B = \frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}} \quad (28.8)$$

When the length $2a$ of the conductor is much greater than its distance x from point P , we can consider it to be infinitely long. When a is much larger than x , $\sqrt{x^2 + a^2}$ is approximately equal to a ; hence in the limit $a \rightarrow \infty$, Eq. (28.8) becomes

$$B = \frac{\mu_0 I}{2\pi x}$$

The physical situation has axial symmetry about the y -axis. Hence $\vec{B}$ must have the same *magnitude* at all points on a circle centered on the conductor and lying in a plane perpendicular to it, and the *direction* of $\vec{B}$ must be everywhere tangent to such a circle (**Fig. 28.6**). Thus, at all points on a circle of radius r around the conductor, the magnitude B is

$$\text{Magnetic field near a long, straight, current-carrying conductor} \quad B = \frac{\mu_0 I}{2\pi r} \quad \text{Magnetic constant} \quad \text{Current} \quad \text{Distance from conductor} \quad (28.9)$$

The geometry in this case is similar to that of Example 21.10 (Section 21.5), in which we solved the problem of the *electric* field caused by an infinite line of charge. The same integral appears in both problems, and the field magnitudes in both problems are proportional to $1/r$. But the lines of $\vec{B}$ in the magnetic problem have completely different shapes than the lines of $\vec{E}$ in the analogous electrical problem. Electric field lines radiate outward from a positive line charge distribution (inward for negative charges). By contrast, magnetic field lines *encircle* the current that acts as their source. Electric field lines due to charges begin and end at those charges, but magnetic field lines always form closed loops and *never* have endpoints, irrespective of the shape of the current-carrying conductor that sets up the field. As we discussed in Section 27.3, this is a consequence of Gauss's law for magnetism, which states that the total magnetic flux through *any* closed surface is always zero:

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{magnetic flux through any closed surface}) \quad (28.10)$$

Any magnetic field line that enters a closed surface must emerge from that surface.

EXAMPLE 28.3 Magnetic field of a single wire

WITH VARIATION PROBLEMS

A long, straight conductor carries a 1.0 A current. At what distance from the axis of the conductor does the resulting magnetic field have magnitude $B = 0.5 \times 10^{-4}$ T (about that of the earth's magnetic field in Pittsburgh)?

IDENTIFY and SET UP The length of a “long” conductor is much greater than the distance from the conductor to the field point. Hence we can use the ideas of this section. The geometry is the same as that of Fig. 28.6, so we use Eq. (28.9). All of the quantities in this equation are known except the target variable, the distance r .

EXECUTE We solve Eq. (28.9) for r :

$$r = \frac{\mu_0 I}{2\pi B} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(1.0 \text{ A})}{(2\pi)(0.5 \times 10^{-4} \text{ T})} = 4 \times 10^{-3} \text{ m} = 4 \text{ mm}$$

EVALUATE As we saw in Example 26.14, currents of an ampere or more are typical of those found in the wiring of home appliances. This example shows that the magnetic fields produced by these appliances are very weak even very close to the wire; the fields are proportional to $1/r$, so they become even weaker at greater distances.

KEYCONCEPT The magnetic field $\vec{B}$ of a long, straight, current-carrying wire is proportional to the current and inversely proportional to the distance from the wire. The field lines are concentric circles around the wire, with the field direction given by a right-hand rule.

EXAMPLE 28.4 Magnetic field of two wires

WITH  **ARIATION PROBLEMS**

Figure 28.7a is an end-on view of two long, straight, parallel wires perpendicular to the xy -plane, each carrying a current I but in opposite directions. (a) Find $\vec{B}$ at points P_1 , P_2 , and P_3 . (b) Find an expression for $\vec{B}$ at any point on the x -axis to the right of wire 2.

IDENTIFY and SET UP We can find the magnetic fields $\vec{B}_1$ and $\vec{B}_2$ due to wires 1 and 2 at each point by using the ideas of this section. By the superposition principle, the magnetic field at each point is then $\vec{B} = \vec{B}_1 + \vec{B}_2$. We use Eq. (28.9) to find the magnitudes B_1 and B_2 of these fields and the right-hand rule to find the corresponding directions. Figure 28.7a shows $\vec{B}_1$, $\vec{B}_2$, and $\vec{B} = \vec{B}_{\text{total}}$ at each point; you should confirm that the directions and relative magnitudes shown are correct. Figure 28.7b shows some of the magnetic field lines due to this two-wire system.

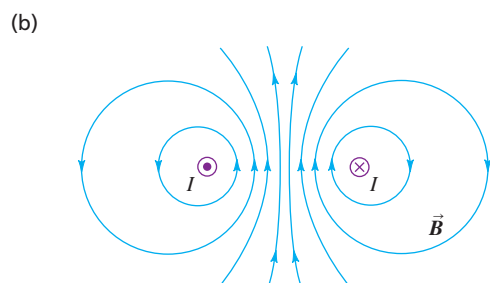
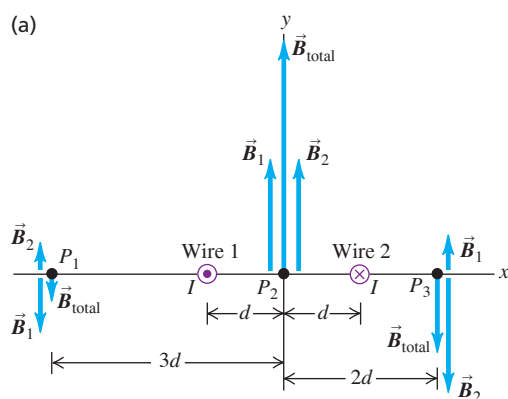
EXECUTE (a) Since point P_1 is a distance $2d$ from wire 1 and a distance $4d$ from wire 2, $B_1 = \mu_0 I / 2\pi(2d) = \mu_0 I / 4\pi d$ and $B_2 = \mu_0 I / 2\pi(4d) = \mu_0 I / 8\pi d$. The right-hand rule shows that $\vec{B}_1$ is in the negative y -direction and $\vec{B}_2$ is in the positive y -direction, so

$$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 = -\frac{\mu_0 I}{4\pi d} \hat{j} + \frac{\mu_0 I}{8\pi d} \hat{j} = -\frac{\mu_0 I}{8\pi d} \hat{j} \quad (\text{point } P_1)$$

At point P_2 , a distance d from both wires, $\vec{B}_1$ and $\vec{B}_2$ are both in the positive y -direction, and both have the same magnitude $B_1 = B_2 = \mu_0 I / 2\pi d$. Hence

$$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{2\pi d} \hat{j} + \frac{\mu_0 I}{2\pi d} \hat{j} = \frac{\mu_0 I}{\pi d} \hat{j} \quad (\text{point } P_2)$$

Figure 28.7 (a) Two long, straight conductors carrying equal currents in opposite directions. The conductors are seen end-on. (b) Map of the magnetic field produced by the two conductors. The field lines are closest together between the conductors, where the field is strongest.



Finally, at point P_3 the right-hand rule shows that $\vec{B}_1$ is in the positive y -direction and $\vec{B}_2$ is in the negative y -direction. This point is a distance $3d$ from wire 1 and a distance d from wire 2, so $B_1 = \mu_0 I / 2\pi(3d) = \mu_0 I / 6\pi d$ and $B_2 = \mu_0 I / 2\pi d$. The total field at P_3 is

$$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{6\pi d} \hat{j} - \frac{\mu_0 I}{2\pi d} \hat{j} = -\frac{\mu_0 I}{3\pi d} \hat{j} \quad (\text{point } P_3)$$

The same technique can be used to find $\vec{B}_{\text{total}}$ at any point; for points off the x -axis, caution must be taken in vector addition, since $\vec{B}_1$ and $\vec{B}_2$ need no longer be simply parallel or antiparallel.

(b) At any point on the x -axis to the right of wire 2 (that is, for $x > d$), $\vec{B}_1$ and $\vec{B}_2$ are in the same directions as at P_3 . Such a point is a distance $x + d$ from wire 1 and a distance $x - d$ from wire 2, so the total field is

$$\begin{aligned} \vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 &= \frac{\mu_0 I}{2\pi(x + d)} \hat{j} - \frac{\mu_0 I}{2\pi(x - d)} \hat{j} \\ &= -\frac{\mu_0 I d}{\pi(x^2 - d^2)} \hat{j} \end{aligned}$$

where we used a common denominator to combine the two terms.

EVALUATE Consider our result from part (b) at a point very far from the wires, so that x is much larger than d . Then the d^2 term in the denominator can be ignored, and the magnitude of the total field is approximately $B_{\text{total}} = \mu_0 I d / \pi x^2$. For one wire, Eq. (28.9) shows that the magnetic field decreases with distance in proportion to $1/x$; for two wires carrying opposite currents, $\vec{B}_1$ and $\vec{B}_2$ partially cancel each other, and so B_{total} decreases more rapidly, in proportion to $1/x^2$. This effect is used in communication systems such as telephone or computer networks. The wiring is arranged so that a conductor carrying a signal in one direction and the conductor carrying the return signal are side by side, as in Fig. 28.7a, or twisted around each other (Fig. 28.8). As a result, the magnetic field due to these signals *outside* the conductors is very small, making it less likely to exert unwanted forces on other information-carrying currents.

Figure 28.8 Computer cables, or cables for audio-video equipment, create little or no magnetic field. This is because within each cable, closely spaced wires carry current in both directions along the length of the cable. The magnetic fields from these opposing currents cancel each other.



KEYCONCEPT Magnetic fields obey the superposition principle: The net magnetic field due to two or more sources is the vector sum of the fields due to the individual sources.

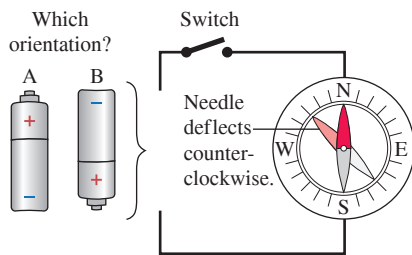
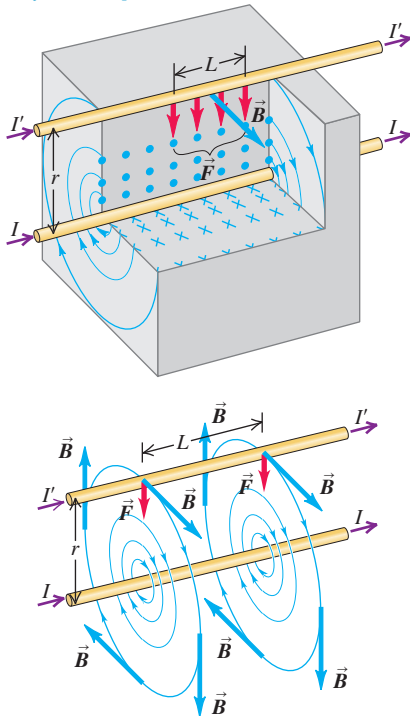


Figure 28.9 Parallel conductors carrying currents in the same direction attract each other. The diagrams show how the magnetic field $\vec{B}$ caused by the current in the lower conductor exerts a force $\vec{F}$ on the upper conductor.

The magnetic field of the lower wire exerts an attractive force on the upper wire. By the same token, the upper wire attracts the lower one.

If the wires had currents in *opposite* directions, they would *repel* each other.



TEST YOUR UNDERSTANDING OF SECTION 28.3 The accompanying figure shows a circuit that lies on a horizontal table. A compass is placed on top of the circuit as shown. A battery is to be connected to the circuit so that when the switch is closed, the compass needle deflects counter-clockwise. In which orientation, A or B, should the battery be placed in the circuit?

ANSWER

A This orientation will cause current to flow clockwise around the circuit. Hence current will flow south through the wire under the compass. From the right-hand rule for the magnetic field produced by a long, straight, current-carrying conductor, this will produce a magnetic field that points to the left at the position of the compass (which lies atop the wire). The combination of the northward magnetic field of the earth and the westward field produced by the current gives a net magnetic field to the north-west, so the compass needle will swing counterclockwise to align with this field.

28.4 FORCE BETWEEN PARALLEL CONDUCTORS

Now that we know how to calculate the magnetic field produced by a long, current-carrying conductor, we can find the *magnetic force* that one such conductor exerts on another. This force plays a role in many practical situations in which current-carrying wires are close to each other. **Figure 28.9** shows segments of two long, straight, parallel conductors separated by a distance r and carrying currents I and I' in the same direction. Each conductor lies in the magnetic field set up by the other, so each experiences a force. The figure shows some of the field lines set up by the current in the lower conductor.

From Eq. (28.9) the lower conductor produces a $\vec{B}$ field that, at the position of the upper conductor, has magnitude

$$B = \frac{\mu_0 I}{2\pi r}$$

From Eq. (27.19) the force that this field exerts on a length L of the upper conductor is $\vec{F} = I'\vec{L} \times \vec{B}$, where the vector $\vec{L}$ is in the direction of the current I' and has magnitude L . Since $\vec{B}$ is perpendicular to the length of the conductor and hence to $\vec{L}$, the magnitude of this force is

$$F = I'LB = \frac{\mu_0 II' L}{2\pi r}$$

and the force *per unit length* F/L is

$$\text{Magnetic force per unit length between two long, straight, parallel, current-carrying conductors} \quad \frac{F}{L} = \frac{\mu_0 II'}{2\pi r} \quad (28.11)$$

μ_0 Magnetic constant
 I Current in first conductor
 I' Current in second conductor
 r Distance between conductors

Applying the right-hand rule to $\vec{F} = I'\vec{L} \times \vec{B}$ shows that the force on the upper conductor is directed *downward*.

The current in the *upper* conductor also sets up a $\vec{B}$ field at the position of the *lower* conductor. Two successive applications of the right-hand rule for vector products (one to find the direction of the $\vec{B}$ field due to the upper conductor, as in Section 28.2, and one to find the direction of the force that this field exerts on the lower conductor, as in Section 27.6) show that the force on the lower conductor is *upward*. Thus *two parallel conductors carrying current in the same direction attract each other*. If the direction of either current is reversed, the forces also reverse. *Parallel conductors carrying currents in opposite directions repel each other*.

Magnetic Forces and the Value of μ_0

We saw in Section 28.1 that the magnetic constant μ_0 is very nearly equal to $4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$. If μ_0 had *exactly* this value, Eq. (28.11) shows that the force per unit length on each of two infinitely long parallel conductors one meter apart, each carrying a current of one ampere, would be

$$\frac{F}{L} = \frac{\mu_0 II'}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(1 \text{ A})(1 \text{ A})}{2\pi(1 \text{ m})} = 2 \times 10^{-7} \text{ T} \cdot \text{A} = 2 \times 10^{-7} \text{ N/m}$$

Prior to 2018 the ampere was defined so that F/L would have exactly this value in this situation, and μ_0 was indeed exactly equal to $4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$. The coulomb was defined to be the amount of charge transferred in one second by a current of one ampere.

As we described in Section 21.3, the coulomb is now defined in terms of e , the magnitude of charge on an electron or proton. Hence the value of μ_0 has changed very slightly from $4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ and has a fractional uncertainty of about 2×10^{-10} . The difference between the old and new values of μ_0 is so miniscule, however, that we can safely use $4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ in calculations.

Mutual forces of attraction exist not only between *wires* carrying currents in the same direction, but also between the elements of a single current-carrying conductor. If the conductor is a liquid or an ionized gas (a plasma), these forces result in a constriction of the conductor called the *pinch effect*. The pinch effect in a plasma has been used in one technique to bring about nuclear fusion.

EXAMPLE 28.5 Forces between parallel wires

WITH  **ARIATION PROBLEMS**

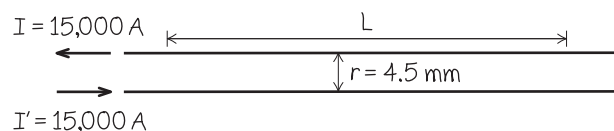
Two straight, parallel, superconducting wires 4.5 mm apart carry equal currents of 15,000 A in opposite directions. What force, per unit length, does each wire exert on the other?

IDENTIFY and SET UP Figure 28.10 shows the situation. We find F/L , the magnetic force per unit length of wire, from Eq. (28.11).

EXECUTE The conductors *repel* each other because the currents are in opposite directions. From Eq. (28.11) the force per unit length is

$$\begin{aligned}\frac{F}{L} &= \frac{\mu_0 I I'}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(15,000 \text{ A})^2}{(2\pi)(4.5 \times 10^{-3} \text{ m})} \\ &= 1.0 \times 10^4 \text{ N/m}\end{aligned}$$

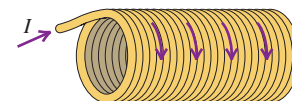
Figure 28.10 Our sketch for this problem.



EVALUATE This is a large force, more than one ton per meter. Currents and separations of this magnitude are used in superconducting electromagnets in particle accelerators, and mechanical stress analysis is a crucial part of the design process.

KEYCONCEPT Two long, straight, parallel, current-carrying conductors attract each other if they carry current in the same direction. They repel each other if they carry current in opposite directions.

TEST YOUR UNDERSTANDING OF SECTION 28.4 A solenoid is a wire wound into a helical coil. The accompanying figure shows a solenoid that carries a current I . (a) Is the *magnetic* force that one turn of the coil exerts on an adjacent turn (i) attractive, (ii) repulsive, or (iii) zero? (b) Is the *electric* force that one turn of the coil exerts on an adjacent turn (i) attractive, (ii) repulsive, or (iii) zero? (c) Is the *magnetic* force between opposite sides of the same turn of the coil (i) attractive, (ii) repulsive, or (iii) zero? (d) Is the *electric* force between opposite sides of the same turn of the coil (i) attractive, (ii) repulsive, or (iii) zero?



ANSWER

(a) (i), (b) (iii), (c) (iii), (d) (iii). Current flows in the same direction in adjacent turns of the coil, so the magnetic forces between these turns are attractive. Current flows in opposite directions on opposite sides of the same turn, so the magnetic forces between these sides are repulsive. Thus the magnetic forces on the solenoid turns squeeze them together in the direction along its axis but push them apart radially. The *electric* forces are zero because the wire is electrically neutral, with as much positive charge as there is negative charge.

Figure 28.11 This electromagnet contains a current-carrying coil with numerous turns of wire. The resulting magnetic field can pick up large quantities of steel bars and other iron-bearing items.

28.5 MAGNETIC FIELD OF A CIRCULAR CURRENT LOOP

If you look inside a doorbell, a transformer, an electric motor, or an electromagnet (Fig. 28.11), you'll find coils of wire with a large number of turns, spaced so closely that each turn is very nearly a planar circular loop. A current in such a coil is used to establish a magnetic field. In Section 27.7 we considered the force and torque on such a current loop placed in an external magnetic field produced by other currents; we are now about to find the magnetic field produced by such a loop or by a collection of closely spaced loops forming a coil.

Figure 28.12 (next page) shows a circular conductor with radius a . A current I is led into and out of the loop through two long, straight wires side by side; the currents in these straight wires are in opposite directions, and their magnetic fields very nearly cancel each other (see Example 28.4 in Section 28.3).



Figure 28.12 Magnetic field on the axis of a circular loop. The current in the segment $d\vec{l}$ causes the field $d\vec{B}$, which lies in the xy -plane. The currents in other $d\vec{l}$'s cause $d\vec{B}$'s with different components perpendicular to the x -axis; these components add to zero. The x -components of the $d\vec{B}$'s combine to give the total $\vec{B}$ field at point P .

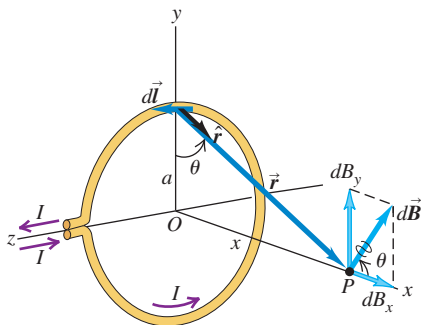


Figure 28.13 The right-hand rule for the direction of the magnetic field produced on the axis of a current-carrying coil.

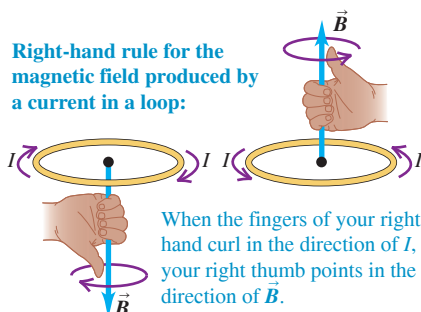
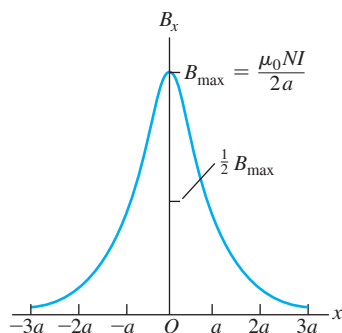


Figure 28.14 Graph of the magnetic field along the axis of a circular coil with N turns. When x is much larger than a , the field magnitude decreases approximately as $1/x^3$.



We can use the law of Biot and Savart, Eq. (28.5) or (28.6), to find the magnetic field at a point P on the axis of the loop, at a distance x from the center. As the figure shows, $d\vec{l}$ and $\vec{r}$ are perpendicular, and the direction of the field $d\vec{B}$ caused by this particular element $d\vec{l}$ lies in the xy -plane. Since $r^2 = x^2 + a^2$, the magnitude dB of the field due to element $d\vec{l}$ is

$$dB = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \quad (28.12)$$

The components of the vector $d\vec{B}$ are

$$dB_x = dB \cos \theta = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \frac{a}{(x^2 + a^2)^{1/2}} \quad (28.13)$$

$$dB_y = dB \sin \theta = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \frac{x}{(x^2 + a^2)^{1/2}} \quad (28.14)$$

The total field $\vec{B}$ at P has only an x -component (it is perpendicular to the plane of the loop). Here's why: For every element $d\vec{l}$ there is a corresponding element on the opposite side of the loop, with opposite direction. These two elements give equal contributions to the x -component of $d\vec{B}$, given by Eq. (28.13), but *opposite* components perpendicular to the x -axis. Thus all the perpendicular components cancel and only the x -components survive.

To obtain the x -component of the total field $\vec{B}$, we integrate Eq. (28.13), including all the $d\vec{l}$'s around the loop. Everything in this expression except dl is constant and can be taken outside the integral, and we have

$$B_x = \int \frac{\mu_0 I}{4\pi} \frac{a dl}{(x^2 + a^2)^{3/2}} = \frac{\mu_0 I a}{4\pi(x^2 + a^2)^{3/2}} \int dl$$

The integral of dl is just the circumference of the circle, $\int dl = 2\pi a$, and so

$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}} \quad (28.15)$$

Magnetic field on axis of a circular current-carrying loop B_x is given by the equation above. The components are: Magnetic constant μ_0 , Current I , Radius of loop a , and Distance along axis from center of loop to field point x .

The direction of this magnetic field is given by a right-hand rule. If you curl the fingers of your right hand around the loop in the direction of the current, your right thumb points in the direction of the field (Fig. 28.13).

Magnetic Field on the Axis of a Coil

Now suppose that instead of the single loop in Fig. 28.12 we have a coil consisting of N loops, all with the same radius. The loops are closely spaced so that the plane of each loop is essentially the same distance x from the field point P . Then the total field is N times the field of a single loop:

$$B_x = \frac{\mu_0 N I a^2}{2(x^2 + a^2)^{3/2}} \quad (\text{on the axis of } N \text{ circular loops}) \quad (28.16)$$

The factor N in Eq. (28.16) is the reason coils of wire, not single loops, are used to produce strong magnetic fields; for a desired field strength, using a single loop might require a current I so great as to exceed the rating of the loop's wire.

Figure 28.14 shows a graph of B_x as a function of x . The maximum value of the field is at $x = 0$, the center of the loop or coil:

$$B_x = \frac{\mu_0 N I}{2a} \quad (28.17)$$

Magnetic field at center of N circular current-carrying loops B_x is given by the equation above. The components are: Magnetic constant μ_0 , Number of loops N , Current I , and Radius of loop a .

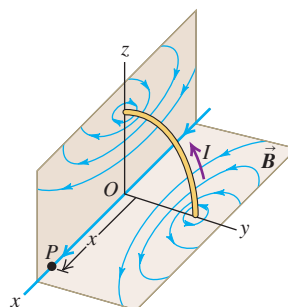
In Section 27.7 we defined the *magnetic dipole moment* μ (or *magnetic moment*) of a current-carrying loop to be equal to IA , where A is the cross-sectional area of the loop. If there are N loops, the total magnetic moment is NIA . The circular loop in Fig. 28.12 has area $A = \pi a^2$, so the magnetic moment of a single loop is $\mu = I\pi a^2$; for N loops, $\mu = NI\pi a^2$. Substituting these results into Eqs. (28.15) and (28.16), we find

$$B_x = \frac{\mu_0 \mu}{2\pi(x^2 + a^2)^{3/2}} \quad (\text{on the axis of any number of circular loops}) \quad (28.18)$$

We described a magnetic dipole in Section 27.7 in terms of its response to a magnetic field produced by currents outside the dipole. But a magnetic dipole is also a *source* of magnetic field; Eq. (28.18) describes the magnetic field *produced* by a magnetic dipole for points along the dipole axis. This field is directly proportional to the magnetic dipole moment μ . Note that the field at all points on the x -axis is in the same direction as the vector magnetic moment $\vec{\mu}$.

Figure 28.15 shows some of the magnetic field lines surrounding a circular current loop (magnetic dipole) in planes through the axis. The directions of the field lines are given by the same right-hand rule as for a long, straight conductor. Grab the conductor with your right hand, with your thumb in the direction of the current; your fingers curl around in the same direction as the field lines. The field lines for the circular current loop are closed curves that encircle the conductor; they are *not* circles, however.

Figure 28.15 Magnetic field lines produced by the current in a circular loop. At points on the axis the $\vec{B}$ field has the same direction as the magnetic moment of the loop.



CAUTION **Magnetic field of a coil** Equations (28.15), (28.16), and (28.18) are valid only on the *axis* of a loop or coil. Don't attempt to apply these equations at other points!

BIO APPLICATION Magnetic Fields for MRI

Magnetic resonance imaging (see Section 27.7) requires a magnetic field of about 1.5 T. In a typical MRI scan, the patient lies inside a coil that produces the intense field. The currents required are very high, so the coils are bathed in liquid helium at a temperature of 4.2 K to keep them from overheating.



EXAMPLE 28.6 Magnetic field of a coil

A coil consisting of 100 circular loops with radius 0.60 m carries a 5.0 A current. (a) Find the magnetic field at a point along the axis of the coil, 0.80 m from the center. (b) Along the axis, at what distance from the center of the coil is the field magnitude $\frac{1}{8}$ as great as it is at the center?

IDENTIFY and SET UP This problem concerns the magnetic-field magnitude B along the axis of a current-carrying coil, so we can use Eq. (28.16). We are given $N = 100$, $I = 5.0$ A, and $a = 0.60$ m. In part (a) our target variable is B_x at a given value of x . In part (b) the target variable is the value of x at which the field has $\frac{1}{8}$ of the magnitude that it has at the origin.

EXECUTE (a) Using $x = 0.80$ m, from Eq. (28.16) we have

$$\begin{aligned} B_x &= \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(100)(5.0 \text{ A})(0.60 \text{ m})^2}{2[(0.80 \text{ m})^2 + (0.60 \text{ m})^2]^{3/2}} \\ &= 1.1 \times 10^{-4} \text{ T} \end{aligned}$$

(b) Considering Eq. (28.16), we seek a value of x such that

$$\frac{1}{(x^2 + a^2)^{3/2}} = \frac{1}{8} \frac{1}{(0^2 + a^2)^{3/2}}$$

To solve this for x , we take the reciprocal of the whole thing and then take the $\frac{2}{3}$ power of both sides; the result is

$$x = \pm \sqrt{3}a = \pm 1.04 \text{ m}$$

EVALUATE We check our answer in part (a) by finding the coil's magnetic moment and substituting the result into Eq. (28.18):

$$\begin{aligned} \mu &= NI\pi a^2 = (100)(5.0 \text{ A})\pi(0.60 \text{ m})^2 = 5.7 \times 10^2 \text{ A} \cdot \text{m}^2 \\ B_x &= \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(5.7 \times 10^2 \text{ A} \cdot \text{m}^2)}{2\pi[(0.80 \text{ m})^2 + (0.60 \text{ m})^2]^{3/2}} = 1.1 \times 10^{-4} \text{ T} \end{aligned}$$

The magnetic moment μ is relatively large, yet it produces a rather small field, comparable to that of the earth. This illustrates how difficult it is to produce strong fields of 1 T or more.

KEYCONCEPT The magnetic field $\vec{B}$ on the axis of a current-carrying coil points along the axis, with a direction given by a right-hand rule. The magnitude of $\vec{B}$ is proportional to the magnetic dipole moment of the coil and is greatest at the center of the coil.

TEST YOUR UNDERSTANDING OF SECTION 28.5 Figure 28.12 shows the magnetic field $d\vec{B}$ produced at point P by a segment $d\vec{l}$ that lies on the positive y -axis (at the top of the loop). This field has components $dB_x > 0$, $dB_y > 0$, $dB_z = 0$. (a) What are the signs of the components of the field $d\vec{B}$ produced at P by a segment $d\vec{l}$ on the negative y -axis (at the bottom of the loop)? (i) $dB_x > 0$, $dB_y > 0$, $dB_z = 0$; (ii) $dB_x > 0$, $dB_y < 0$, $dB_z = 0$; (iii) $dB_x < 0$, $dB_y > 0$, $dB_z = 0$; (iv) $dB_x < 0$, $dB_y < 0$, $dB_z = 0$; (v) none of these. (b) What are the signs of the components of the field $d\vec{B}$ produced at P by a segment $d\vec{l}$ on the negative z -axis (at the right-hand side of the loop)? (i) $dB_x > 0$, $dB_y > 0$, $dB_z = 0$; (ii) $dB_x > 0$, $dB_y < 0$, $dB_z = 0$; (iii) $dB_x < 0$, $dB_y > 0$, $dB_z = 0$; (iv) $dB_x < 0$, $dB_y < 0$, $dB_z = 0$; (v) none of these.

ANSWER

(a) (i) The vector $d\vec{B}$ is in the direction of $d\vec{l} \times \vec{r}$. For a segment on the negative y -axis, $d\vec{l}$ points in the negative y -direction and $\vec{r}$ has a positive x -component, a negative y -component, and a negative z -component. Hence $d\vec{l} \times \vec{r} = 1(a)(-b)\hat{i} - 1(-b)(-c)\hat{j} + 1(a)(-c)\hat{k} = -ab\hat{i} - bc\hat{j} - ac\hat{k}$. Hence $d\vec{B}$ has components $dB_x < 0$, $dB_y < 0$, and $dB_z < 0$. (ii) For a segment on the negative z -axis, $d\vec{l}$ points in the negative z -direction and $\vec{r}$ has a positive x -component, a negative y -component, and zero z -component. Hence $d\vec{l} \times \vec{r} = 1(-b)(-c)\hat{i} - 1(-a)(-c)\hat{j} + 1(-b)(-a)\hat{k} = bc\hat{i} - ac\hat{j} + ab\hat{k}$. Hence $d\vec{B}$ has components $dB_x > 0$, $dB_y < 0$, and $dB_z > 0$. (iii) For a segment on the positive y -axis, $d\vec{l}$ points in the positive y -direction and $\vec{r}$ has a positive x -component, zero y -component, and zero z -component. Hence $d\vec{l} \times \vec{r} = 1(a)(-b)\hat{i} - 1(a)(-c)\hat{j} + 1(b)(-c)\hat{k} = -ab\hat{i} + ac\hat{j} - bc\hat{k}$. Hence $d\vec{B}$ has components $dB_x < 0$, $dB_y > 0$, and $dB_z < 0$. (iv) For a segment on the positive z -axis, $d\vec{l}$ points in the positive z -direction and $\vec{r}$ has a positive x -component, a negative y -component, and a positive z -component. Hence $d\vec{l} \times \vec{r} = 1(a)(-b)\hat{i} - 1(a)(-c)\hat{j} + 1(b)(-c)\hat{k} = -ab\hat{i} + ac\hat{j} - bc\hat{k}$. Hence $d\vec{B}$ has components $dB_x < 0$, $dB_y > 0$, and $dB_z < 0$. (v) none of these.

28.6 AMPERE'S LAW

So far our calculations of the magnetic field due to a current have involved finding the infinitesimal field $d\vec{B}$ due to a current element and then summing all the $d\vec{B}$'s to find the total field. This approach is directly analogous to our *electric*-field calculations in Chapter 21.

For the electric-field problem we found that in situations with a highly symmetric charge distribution, it was often easier to use Gauss's law to find $\vec{E}$. There is likewise a law that allows us to more easily find the *magnetic* fields caused by highly symmetric *current* distributions. But the law that allows us to do this, called *Ampere's law*, is rather different in character from Gauss's law.

Gauss's law for electric fields (Chapter 22) involves the flux of $\vec{E}$ through a closed surface; it states that this flux is equal to the total charge enclosed within the surface, divided by the constant ϵ_0 . Thus this law relates electric fields and charge distributions. By contrast, Gauss's law for *magnetic* fields, Eq. (28.10), is *not* a relationship between magnetic fields and current distributions; it states that the flux of $\vec{B}$ through *any* closed surface is always zero, whether or not there are currents within the surface. So Gauss's law for $\vec{B}$ can't be used to determine the magnetic field produced by a particular current distribution.

Ampere's law is formulated not in terms of magnetic flux, but rather in terms of the *line integral* of $\vec{B}$ around a closed path, denoted by

$$\oint \vec{B} \cdot d\vec{l}$$

We used line integrals to define work in Chapter 6 and to calculate electric potential in Chapter 23. To evaluate this integral, we divide the path into infinitesimal segments $d\vec{l}$, calculate the scalar product of $\vec{B} \cdot d\vec{l}$ for each segment, and sum these products. In general, $\vec{B}$ varies from point to point, and we must use the value of $\vec{B}$ at the location of each $d\vec{l}$. An alternative notation is $\oint B_{\parallel} dl$, where $B_{\parallel}$ is the component of $\vec{B}$ parallel to $d\vec{l}$ at each point. The circle on the integral sign indicates that this integral is always computed for a *closed* path, one whose beginning and end points are the same.

Ampere's Law for a Long, Straight Conductor

To introduce the basic idea of Ampere's law, let's consider again the magnetic field caused by a long, straight conductor carrying a current I . We found in Section 28.3 that the field at a distance r from the conductor has magnitude

$$B = \frac{\mu_0 I}{2\pi r}$$

The magnetic field lines are circles centered on the conductor. Let's take the line integral of $\vec{B}$ around a circle with radius r , as in **Fig. 28.16a**. At every point on the circle, $\vec{B}$ and $d\vec{l}$ are parallel, and so $\vec{B} \cdot d\vec{l} = B dl$; since r is constant around the circle, B is constant as well. Alternatively, we can say that $B_{\parallel}$ is constant and equal to B at every point on the circle. Hence we can take B outside of the integral. The remaining integral is just the circumference of the circle, so

$$\oint \vec{B} \cdot d\vec{l} = \oint B_{\parallel} dl = B \oint dl = \frac{\mu_0 I}{2\pi r} (2\pi r) = \mu_0 I$$

The line integral is thus independent of the radius of the circle and is equal to μ_0 multiplied by the current passing through the area bounded by the circle.

In **Fig. 28.16b** the situation is the same, but the integration path now goes around the circle in the opposite direction. Now $\vec{B}$ and $d\vec{l}$ are antiparallel, so $\vec{B} \cdot d\vec{l} = -B dl$ and the line integral equals $-\mu_0 I$. We get the same result if the integration path is the same as in **Fig. 28.16a**, but the direction of the current is reversed. Thus $\oint \vec{B} \cdot d\vec{l}$ equals μ_0 multiplied by the current passing through the area bounded by the integration path, with a positive or negative sign depending on the direction of the current relative to the direction of integration.

There's a simple rule for the sign of the current; you won't be surprised to learn that it uses your right hand. Curl the fingers of your right hand around the integration path so that they curl in the direction of integration (that is, the direction that you use to evaluate $\oint \vec{B} \cdot d\vec{l}$). Then your right thumb indicates the positive current direction. Currents that pass through the integration path in this direction are positive; those in the opposite direction are negative. Using this rule, convince yourself that the current is positive in **Fig. 28.16a** and negative in **Fig. 28.16b**. Here's another way to say the same thing: Looking at the surface bounded by the integration path, integrate counterclockwise around the path as in **Fig. 28.16a**. Currents moving toward you through the surface are positive, and those going away from you are negative.

An integration path that does *not* enclose the conductor is used in **Fig. 28.16c**. Along the circular arc ab of radius r_1 , $\vec{B}$ and $d\vec{l}$ are parallel and $B_{\parallel} = B_1 = \mu_0 I / 2\pi r_1$; along the circular arc cd of radius r_2 , $\vec{B}$ and $d\vec{l}$ are antiparallel and $B_{\parallel} = -B_2 = -\mu_0 I / 2\pi r_2$. The $\vec{B}$ field is perpendicular to $d\vec{l}$ at each point on the straight sections bc and da , so $B_{\parallel} = 0$ and these sections contribute zero to the line integral. The total line integral is then

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= \oint B_{\parallel} dl = B_1 \int_a^b dl + (0) \int_b^c dl + (-B_2) \int_c^d dl + (0) \int_d^a dl \\ &= \frac{\mu_0 I}{2\pi r_1} (r_1 \theta) + 0 - \frac{\mu_0 I}{2\pi r_2} (r_2 \theta) + 0 = 0 \end{aligned}$$

The magnitude of $\vec{B}$ is greater on arc cd than on arc ab , but the arc length is less, so the contributions from the two arcs exactly cancel. Even though there is a magnetic field everywhere along the integration path, the line integral $\oint \vec{B} \cdot d\vec{l}$ is zero if there is no current passing through the area bounded by the path.

We can also derive these results for more general integration paths, such as the one in **Fig. 28.17a**. At the position of the line element $d\vec{l}$, the angle between $d\vec{l}$ and $\vec{B}$ is ϕ , and

$$\vec{B} \cdot d\vec{l} = B dl \cos \phi$$

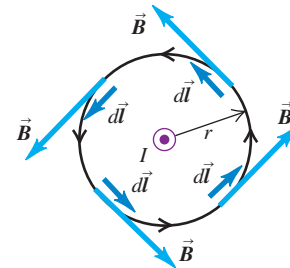
From the figure, $dl \cos \phi = r d\theta$, where $d\theta$ is the angle subtended by $d\vec{l}$ at the position of the conductor and r is the distance of $d\vec{l}$ from the conductor. Thus

$$\oint \vec{B} \cdot d\vec{l} = \oint \frac{\mu_0 I}{2\pi r} (r d\theta) = \frac{\mu_0 I}{2\pi} \oint d\theta$$

Figure 28.16 Three integration paths for the line integral of $\vec{B}$ in the vicinity of a long, straight conductor carrying current I out of the plane of the page (as indicated by the circle with a dot). The conductor is seen end-on.

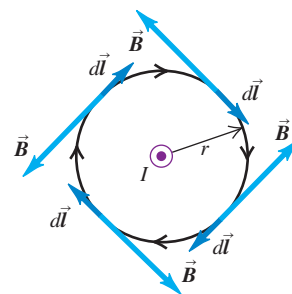
(a) Integration path is a circle centered on the conductor; integration goes around the circle counterclockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$



(b) Same integration path as in (a), but integration goes around the circle clockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = -\mu_0 I$



(c) An integration path that does not enclose the conductor

Result: $\oint \vec{B} \cdot d\vec{l} = 0$

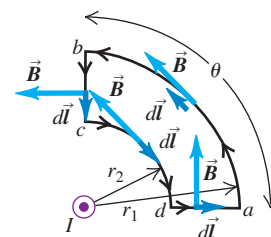


Figure 28.17 (a) A more general integration path for the line integral of $\vec{B}$ around a long, straight conductor carrying current I out of the plane of the page. The conductor is seen end-on. (b) A more general integration path that does not enclose the conductor.

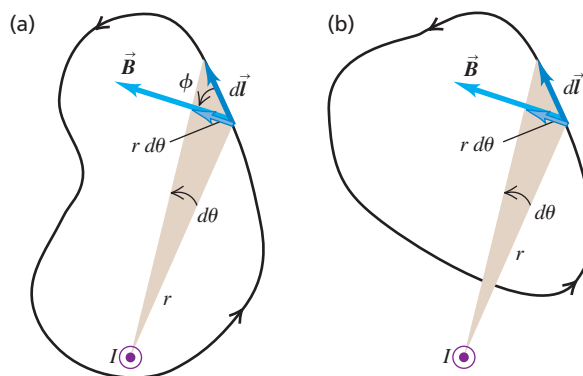
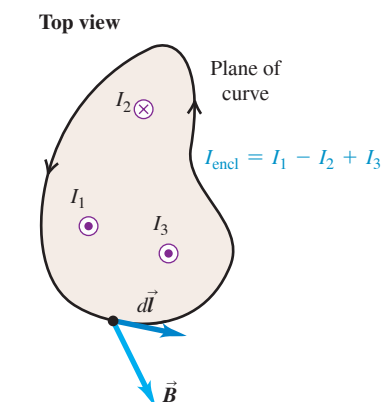
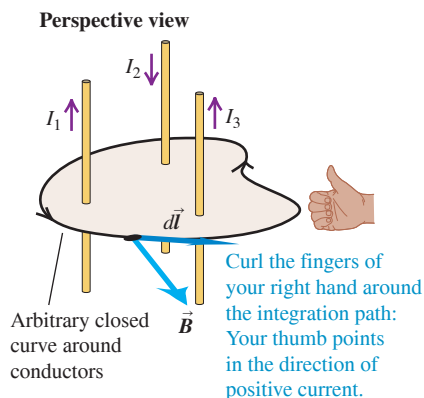
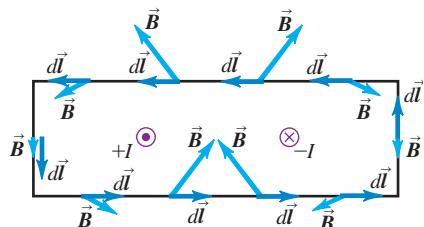


Figure 28.18 Ampere's law.



Ampere's law: If we calculate the line integral of the magnetic field around a closed curve, the result equals μ_0 times the total enclosed current: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$.

Figure 28.19 An end-on view of two long, straight conductors carrying equal currents in opposite directions. The line integral $\oint \vec{B} \cdot d\vec{l}$ gets zero contribution from the upper and lower segments, a positive contribution from the left segment, and a negative contribution from the right segment; the net integral is zero.



But $\oint d\theta$ is just equal to 2π , the total angle swept out by the radial line from the conductor to $d\vec{l}$ during a complete trip around the path. So we get

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \quad (28.19)$$

This result doesn't depend on the shape of the path or on the position of the wire inside it. If the current in the wire is opposite to that shown, the integral has the opposite sign. But if the path doesn't enclose the wire (Fig. 28.17b), then the net change in θ during the trip around the integration path is zero; $\oint d\theta$ is zero instead of 2π and the line integral is zero.

Ampere's Law: General Statement

We can generalize Ampere's law even further. Suppose *several* long, straight conductors pass through the surface bounded by the integration path. The total magnetic field $\vec{B}$ at any point on the path is the vector sum of the fields produced by the individual conductors. Thus the line integral of the total $\vec{B}$ equals μ_0 times the *algebraic sum* of the currents. In calculating this sum, we use the sign rule for currents described above. If the integration path does not enclose a particular wire, the line integral of the $\vec{B}$ field of that wire is zero, because the angle θ for that wire sweeps through a net change of zero rather than 2π during the integration. Any conductors present that are not enclosed by a particular path may still contribute to the value of $\vec{B}$ at every point, but the *line integrals* of their fields around the path are zero.

Thus we can replace I in Eq. (28.19) with I_{encl} , the algebraic sum of the currents *enclosed* or *linked* by the integration path, with the sum evaluated by using the sign rule just described (Fig. 28.18). Then **Ampere's law** says

Line integral around a closed path

Magnetic constant

Net current enclosed by path

Scalar product of magnetic field and vector segment of path

Ampere's law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$ (28.20)

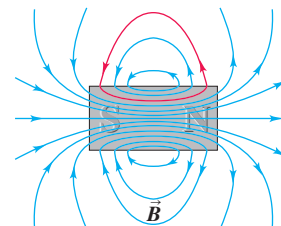
While we have derived Ampere's law only for the special case of the field of several long, straight, parallel conductors, Eq. (28.20) is in fact valid for conductors and paths of *any* shape. The general derivation is no different in principle from what we have presented, but the geometry is more complicated.

If $\oint \vec{B} \cdot d\vec{l} = 0$, it *does not* necessarily mean that $\vec{B} = 0$ everywhere along the path, only that the total current through an area bounded by the path is zero. In Figs. 28.16c and 28.17b, the integration paths enclose no current at all; in **Fig. 28.19** there are positive and negative currents of equal magnitude through the area enclosed by the path. In both cases, $I_{\text{encl}} = 0$ and the line integral is zero.

Equation (28.20) turns out to be valid *only* if the currents are steady and if no time-varying electric fields are present. In Chapter 29 we'll see how to generalize Ampere's law for time-varying fields.

CAUTION **Line integrals of electric and magnetic fields** In Chapter 23 we saw that the line integral of the electrostatic field $\vec{E}$ around any closed path is equal to zero; this is a statement that the electrostatic force $\vec{F} = q\vec{E}$ on a point charge q is conservative, so this force does zero work on a charge that moves around a closed path that returns to the starting point. The value of the line integral $\oint \vec{B} \cdot d\vec{l}$ is not similarly related to the question of whether the *magnetic* force is conservative. Remember that the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on a moving charged particle is always *perpendicular* to $\vec{B}$, so $\oint \vec{B} \cdot d\vec{l}$ is *not* related to the work done by the magnetic force; as stated in Ampere's law, this integral is related only to the total current through a surface bounded by the integration path. In fact, the magnetic force on a moving charged particle is *not* conservative. A conservative force depends on only the position of the object on which the force is exerted, but the magnetic force on a moving charged particle also depends on the *velocity* of the particle. ■

TEST YOUR UNDERSTANDING OF SECTION 28.6 The accompanying figure shows magnetic field lines through the center of a permanent magnet. The magnet is not connected to a source of emf. One of the field lines is colored red. What can you conclude about the currents inside the permanent magnet within the region enclosed by this field line? (i) There are no currents inside the magnet; (ii) there are currents directed out of the plane of the page; (iii) there are currents directed into the plane of the page; (iv) not enough information is given to decide.



ANSWER

(ii) Imagine carrying out the integral $\oint \vec{B} \cdot d\vec{l}$ along an integration path that goes counterclockwise around the red magnetic field line. At each point along the path the magnetic field $\vec{B}$ and the infinitesimal segment $d\vec{l}$ are both tangent to the path, so $\vec{B} \cdot d\vec{l}$ is positive at each point and the integral $\oint \vec{B} \cdot d\vec{l}$ is likewise positive. It follows from Ampere's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$ and the right-hand rule that the integration path encloses a current directed out of the plane of the page. There are no currents in the empty space outside the magnet, so there must be currents inside the magnet (see Section 28.8). ■

28.7 APPLICATIONS OF AMPERE'S LAW

Below are several examples of using Ampere's law to calculate the magnetic field due to a current distribution. Problem-Solving Strategy 28.2 is directly analogous to Problem-Solving Strategy 22.1 (Section 22.4) for using Gauss's law to calculate the electric field due to a charge distribution; we suggest you review that strategy now.

CAUTION **Ampere's law is powerful but limited** Like Gauss's law for $\vec{E}$, Ampere's law for $\vec{B}$ can be used to calculate the field *only* in highly symmetric situations. As the next examples show, in such situations you can exploit the symmetry to calculate the line integral of $\vec{B}$ around a simple path. ■

PROBLEM-SOLVING STRATEGY 28.2 Ampere's Law

IDENTIFY the relevant concepts: Ampere's law, $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$, can yield the magnitude of $\vec{B}$ as a function of position if the geometry of the field-generating electric current is highly symmetric.

SET UP the problem using the following steps:

1. Determine the target variable(s). Usually one will be the magnitude of the $\vec{B}$ field as a function of position.
2. Select the integration path to use with Ampere's law. To determine the magnetic field at a certain point, the path must pass through that point. The integration path doesn't have to be any actual physical boundary; it may be in empty space, embedded in a solid object, or some of each. The integration path has to have enough *symmetry* to make evaluation of the integral possible. Ideally the path will be tangent to $\vec{B}$ in regions of interest; elsewhere the path should be perpendicular to $\vec{B}$ or should run through regions where $\vec{B} = 0$.

EXECUTE the solution as follows:

1. Carry out the integral $\oint \vec{B} \cdot d\vec{l}$ along the chosen path. If $\vec{B}$ is tangent to all or some portion of the path and has the same magnitude B at every point, then its line integral is the product of B and the length of that portion of the path. If $\vec{B}$ is perpendicular

to some portion of the path, or if $\vec{B} = 0$, that portion makes no contribution to the integral.

2. In the integral $\oint \vec{B} \cdot d\vec{l}$, $\vec{B}$ is the *total* magnetic field at each point on the path; it can be caused by currents enclosed *or not enclosed* by the path. If *no* net current is enclosed by the path, the field at points on the path need not be zero, but the integral $\oint \vec{B} \cdot d\vec{l}$ is always zero.
3. Determine the current I_{encl} enclosed by the integration path. A right-hand rule gives the sign of this current: If you curl the fingers of your right hand so that they follow the path in the direction of integration, then your right thumb points in the direction of positive current. If $\vec{B}$ is tangent to the path everywhere and I_{encl} is positive, the direction of $\vec{B}$ is the same as the direction of integration. If instead I_{encl} is negative, $\vec{B}$ is in the direction opposite to that of the integration.
4. Use Ampere's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$ to solve for the target variable.

EVALUATE your answer: If your result is an expression for the field magnitude as a function of position, check it by examining how the expression behaves in different limits.

EXAMPLE 28.7 Field of a long, straight, current-carrying conductor

In Section 28.6 we derived Ampere's law from Eq. (28.9) for the field $\vec{B}$ of a long, straight, current-carrying conductor. Reverse this process, and use Ampere's law to find $\vec{B}$ for this situation.

IDENTIFY and SET UP The situation has cylindrical symmetry, so in Ampere's law we take our integration path to be a circle with radius r centered on the conductor and lying in a plane perpendicular to it, as in Fig. 28.16a. The field $\vec{B}$ is everywhere tangent to this circle and has the same magnitude B everywhere on the circle.

EXECUTE With our choice of the integration path, Ampere's law [Eq. (28.20)] becomes

$$\oint \vec{B} \cdot d\vec{l} = \oint B_{\parallel} dl = B(2\pi r) = \mu_0 I$$

Equation (28.9), $B = \mu_0 I / 2\pi r$, follows immediately.

Ampere's law determines the direction of $\vec{B}$ as well as its magnitude. Since we chose to go counterclockwise around the integration path, the positive direction for current is out of the plane of Fig. 28.16a; this is the same as the actual current direction in the figure, so I is positive and the integral $\oint \vec{B} \cdot d\vec{l}$ is also positive. Since the $d\vec{l}$'s run counterclockwise, the direction of $\vec{B}$ must be counterclockwise as well, as shown in Fig. 28.16a.

EVALUATE Our results are consistent with those in Section 28.6.

KEYCONCEPT Ampere's law is easiest to use if the magnetic field $\vec{B}$ is everywhere tangent to an integration path and has the same magnitude at all points along that path.

EXAMPLE 28.8 Field of a long cylindrical conductor**WITH VARIATION PROBLEMS**

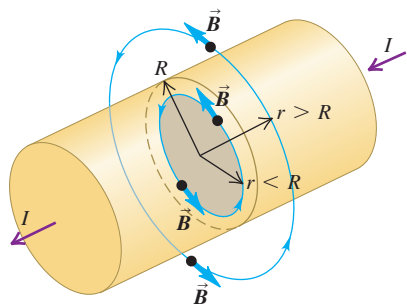
A cylindrical conductor with radius R carries a current I (Fig. 28.20). The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnetic field as a function of the distance r from the conductor axis for points both inside ($r < R$) and outside ($r > R$) the conductor.

IDENTIFY and SET UP As in Example 28.7, the current distribution has cylindrical symmetry, and the magnetic field lines must be circles concentric with the conductor axis. To find the magnetic field inside and outside the conductor, we choose circular integration paths with radii $r < R$ and $r > R$, respectively (see Fig. 28.20).

EXECUTE In either case the field $\vec{B}$ has the same magnitude at every point on the circular integration path and is tangent to the path. Thus the magnitude of the line integral is simply $B(2\pi r)$. To find the current I_{encl} enclosed by a circular integration path inside the conductor ($r < R$), note that the current density (current per unit area) is $J = I/\pi R^2$ so $I_{\text{encl}} = J(\pi r^2) = Ir^2/R^2$. Hence Ampere's law gives $B(2\pi r) = \mu_0 Ir^2/R^2$, or

$$B = \frac{\mu_0 I}{2\pi} \frac{r}{R^2} \quad (\text{inside the conductor, } r < R) \quad (28.21)$$

Figure 28.20 To find the magnetic field at radius $r < R$, we apply Ampere's law to the circle enclosing the gray area. The current through the gray area is $(r^2/R^2)I$. To find the magnetic field at radius $r > R$, we apply Ampere's law to the circle enclosing the entire conductor.



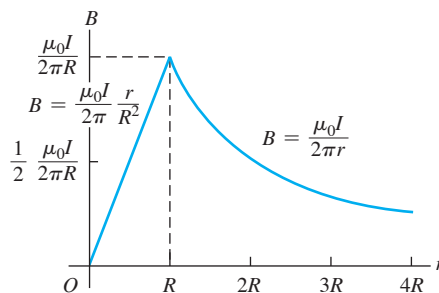
A circular integration path outside the conductor encloses the total current in the conductor, so $I_{\text{encl}} = I$. Applying Ampere's law gives the same equation as in Example 28.7, with the same result for B :

$$B = \frac{\mu_0 I}{2\pi r} \quad (\text{outside the conductor, } r > R) \quad (28.22)$$

Outside the conductor, the magnetic field is the same as that of a long, straight conductor carrying current I , independent of the radius R over which the current is distributed. Indeed, the magnetic field outside *any* cylindrically symmetric current distribution is the same as if the entire current were concentrated along the axis of the distribution. This is analogous to the results of Examples 22.5 and 22.9 (Section 22.4), in which we found that the *electric* field outside a spherically symmetric *charged* object is the same as though the entire charge were concentrated at the center.

EVALUATE At the surface of the conductor ($r = R$), Eqs. (28.21) and (28.22) agree, as they must. **Figure 28.21** shows a graph of B as a function of r .

Figure 28.21 Magnitude of the magnetic field inside and outside a long, straight cylindrical conductor with radius R carrying a current I .



KEYCONCEPT The magnetic field lines produced by a solid, cylindrical, current-carrying conductor are concentric circles around the conductor, with the field direction given by a right-hand rule. Inside the conductor the field magnitude increases with increasing distance from the center; outside the conductor the magnitude decreases with distance.

EXAMPLE 28.9 Field of a solenoid

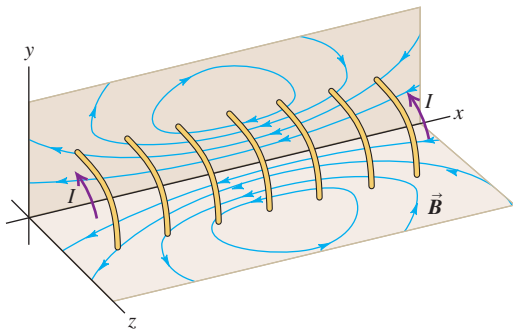
WITH VARIATION PROBLEMS

A **solenoid** consists of a helical winding of wire on a cylinder, usually circular in cross section. There can be thousands of closely spaced turns (often in several layers), each of which can be regarded as a circular loop. For simplicity, **Fig. 28.22** shows a solenoid with only a few turns. All turns carry the same current I , and the total $\vec{B}$ field at every point is the vector sum of the fields caused by the individual turns. The figure shows field lines in the xy - and xz -planes. We draw field lines that are uniformly spaced at the center of the solenoid. Exact calculations show that for a long, closely wound solenoid, half of these field lines emerge from the ends and half “leak out” through the windings between the center and the end, as the figure suggests.

If the solenoid is long in comparison with its cross-sectional diameter and the coils are tightly wound, the field inside the solenoid near its midpoint is very nearly uniform over the cross section and parallel to the axis; the *external* field near the midpoint is very small.

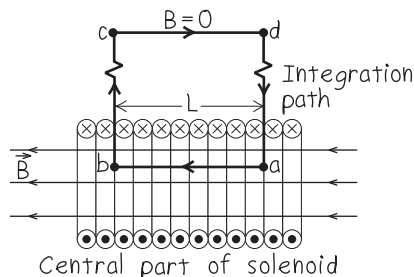
Use Ampere's law to find the field at or near the center of such a solenoid if it has n turns per unit length and carries current I .

Figure 28.22 Magnetic field lines produced by the current in a solenoid. For clarity, only a few turns are shown.



IDENTIFY and SET UP We assume that $\vec{B}$ is uniform inside the solenoid and zero outside. **Figure 28.23** shows the situation and our chosen integration path, rectangle $abcd$. Side ab , with length L , is parallel to the axis of the solenoid. Sides bc and da are taken to be very long so that side cd is far from the solenoid; then the field at side cd is negligibly small.

Figure 28.23 Our sketch for this problem.



EXECUTE Along side ab , $\vec{B}$ is parallel to the path and is constant. Our Ampere's-law integration takes us along side ab in the same direction as $\vec{B}$, so here $B_{\parallel} = +B$ and

$$\int_a^b \vec{B} \cdot d\vec{l} = BL$$

Along sides bc and da , $\vec{B}$ is perpendicular to the path, and so $B_{\parallel} = 0$; along side cd , $\vec{B} = 0$ and so $B_{\parallel} = 0$. Around the entire closed path, then, we have $\oint \vec{B} \cdot d\vec{l} = BL$.

In a length L there are nL turns, each of which passes once through $abcd$ carrying current I . Hence the total current enclosed by the rectangle is $I_{\text{encl}} = nLI$. The integral $\oint \vec{B} \cdot d\vec{l}$ is positive, so from Ampere's law I_{encl} must be positive as well. This means that the current passing through the surface bounded by the integration path must be in the direction shown in **Fig. 28.23**. Ampere's law then gives $BL = \mu_0 nLI$, or

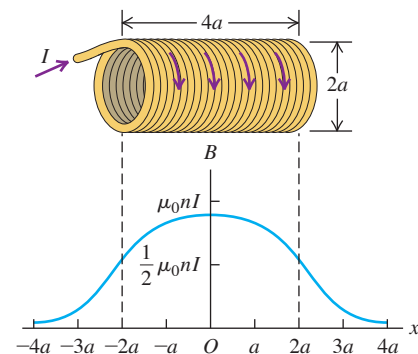
$$B = \mu_0 nI \quad (\text{solenoid}) \quad (28.23)$$

Side ab need not lie on the axis of the solenoid, so this result demonstrates that the field is uniform over the entire cross section at the center of the solenoid's length.

EVALUATE Note that the *direction* of $\vec{B}$ inside the solenoid is in the same direction as the solenoid's vector magnetic moment $\vec{\mu}$, as we found in Section 28.5 for a single current-carrying loop.

For points along the axis, the field is strongest at the center of the solenoid and drops off near the ends. For a solenoid very long in comparison to its diameter, the field magnitude at each end is exactly half that at the center. This is approximately the case even for a relatively short solenoid, as **Fig. 28.24** shows.

Figure 28.24 Magnitude of the magnetic field at points along the axis of a solenoid with length $4a$, equal to four times its radius a . The field magnitude at each end is about half its value at the center. (Compare with **Fig. 28.14** for the field of N circular loops.)



KEYCONCEPT The magnetic field $\vec{B}$ of an ideal solenoid is uniform inside the solenoid and zero outside. The direction of $\vec{B}$ inside the solenoid is given by the same right-hand rule as for the field along the axis of a current-carrying coil (see Example 28.6).

EXAMPLE 28.10 Field of a toroidal solenoid

WITH VARIATION PROBLEMS

Figure 28.25a shows a doughnut-shaped **toroidal solenoid**, tightly wound with N turns of wire carrying a current I . (In a practical solenoid the turns would be much more closely spaced than they are in the figure.) Find the magnetic field at all points.

IDENTIFY and SET UP Ignoring the slight pitch of the helical windings, we can consider each turn of a tightly wound toroidal solenoid as a loop lying in a plane perpendicular to the large, circular axis of the toroid. The symmetry of the situation then tells us that the magnetic field lines must be circles concentric with the toroid axis. Therefore we choose circular integration paths for use with Ampere's law, so that the field $\vec{B}$ (if any) is tangent to each path at all points along the path. Figure 28.25b shows three such paths.

EXECUTE Along each path, $\oint \vec{B} \cdot d\vec{l}$ equals the product of B and the path circumference $l = 2\pi r$. The total current enclosed by path 1 is zero, so from Ampere's law the field $\vec{B} = \vec{0}$ everywhere on this path.

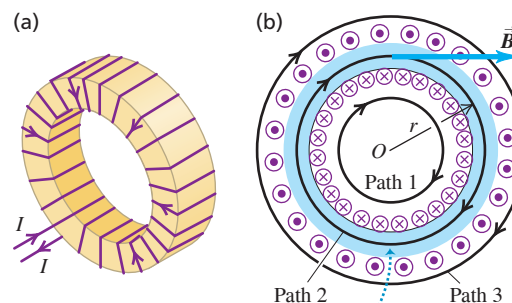
Each turn of the winding passes *twice* through the area bounded by path 3, carrying equal currents in opposite directions. The *net* current enclosed is therefore zero, and hence $\vec{B} = \vec{0}$ at all points on this path as well. We conclude that *the field of an ideal toroidal solenoid is confined to the space enclosed by the windings*. We can think of such a solenoid as a tightly wound, straight solenoid that has been bent into a circle.

For path 2, we have $\oint \vec{B} \cdot d\vec{l} = 2\pi rB$. Each turn of the winding passes *once* through the area bounded by this path, so $I_{\text{encl}} = NI$. We note that I_{encl} is positive for the clockwise direction of integration in Fig. 28.25b, so $\vec{B}$ is in the direction shown. Ampere's law then says that $2\pi rB = \mu_0 NI$, so

$$B = \frac{\mu_0 NI}{2\pi r} \quad (\text{toroidal solenoid}) \quad (28.24)$$

EVALUATE Equation (28.24) indicates that B is *not* uniform over the interior of the core, because different points in the interior are different distances r from the toroid axis. However, if the radial extent of the core is small in comparison to r , the variation is slight. In that case, considering that $2\pi r$ is the circumferential length of the toroid and that $N/2\pi r$

Figure 28.25 (a) A toroidal solenoid. For clarity, only a few turns of the winding are shown. (b) Integration paths (black circles) used to compute the magnetic field $\vec{B}$ set up by the current (shown as dots and crosses).



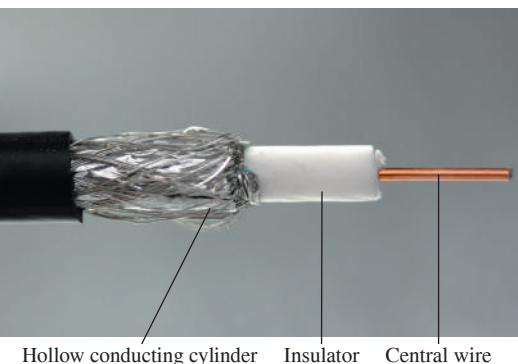
The magnetic field is confined almost entirely to the space enclosed by the windings (in blue).

is the number of turns per unit length n , the field may be written as $B = \mu_0 nI$, just as it is at the center of a long, *straight* solenoid.

In a real toroidal solenoid the turns are not precisely circular loops but rather segments of a bent helix. As a result, the external field is not exactly zero. To estimate its magnitude, we imagine Fig. 28.25a as being *very* roughly equivalent, for points outside the torus, to a *single-turn* circular loop with radius r . At the center of such a loop, Eq. (28.17) gives $B = \mu_0 I/2r$; this is smaller than the field inside the solenoid by the factor N/π .

The equations we have derived for the field in a closely wound straight or toroidal solenoid are strictly correct only for windings in *vacuum*. For most practical purposes, however, they can be used for windings in air or on a core of any nonmagnetic, nonsuperconducting material. In the next section we'll show how these equations are modified if the core is a magnetic material.

KEYCONCEPT Inside the space enclosed by the windings of an ideal toroidal solenoid, the magnetic field lines are circles that follow the curvature of the toroid. Outside the windings, the magnetic field $\vec{B}$ is zero.



TEST YOUR UNDERSTANDING OF SECTION 28.7 In a *coaxial cable* (see photo), a conducting wire runs along the central axis of a hollow conducting cylinder. (The cable that connects a television set to a local cable provider is an example of a coaxial cable.) In such a cable a current I runs in one direction along the hollow conducting cylinder and is spread uniformly over the cylinder's cross-sectional area. An equal current runs in the opposite direction along the central wire. How does the magnitude B of the magnetic field outside such a cable depend on the distance r from the central axis of the cable? (i) B is proportional to $1/r$; (ii) B is proportional to $1/r^2$; (iii) B is zero at all points outside the cable.

ANSWER

(iii) By symmetry, any $\vec{B}$ field outside the cable must circulate around the cable, with circular field lines like those surrounding the solid cylindrical conductor in Fig. 28.20. Choose an integration path like the one shown in Fig. 28.20 with radius $r > R$, so that the path completely encloses the cable. As in Example 28.8, the integral $\oint \vec{B} \cdot d\vec{l}$ for this path has magnitude $B(2\pi r)$. From Ampere's law this is equal to $\mu_0 I_{\text{encl}}$. The net enclosed current I_{encl} is zero because it includes two currents of equal magnitude but opposite direction: one in the central wire and one in the hollow cylinder. Hence $B(2\pi r) = 0$, and so $B = 0$ for any value of r outside the cable. (The field is non-zero *inside* the cable; see Exercise 28.39.)

28.8 MAGNETIC MATERIALS

In discussing how currents cause magnetic fields, we have assumed that the conductors are surrounded by vacuum. But the coils in transformers, motors, generators, and electromagnets nearly always have iron cores to increase the magnetic field and confine it to desired regions. Permanent magnets, magnetic recording tapes, and computer disks depend directly on the magnetic properties of materials; when you store information on a computer disk, you are actually setting up an array of microscopic permanent magnets on the disk. So it is worthwhile to examine some aspects of the magnetic properties of materials. After describing the atomic origins of magnetic properties, we'll discuss three broad classes of magnetic behavior that occur in materials; these are called *paramagnetism*, *diamagnetism*, and *ferromagnetism*.

The Bohr Magneton

As we discussed briefly in Section 27.7, the atoms that make up all matter contain moving electrons, and these electrons form microscopic current loops that produce magnetic fields of their own. In many materials these currents are randomly oriented and cause no net magnetic field. But in some materials an external field (a field produced by currents outside the material) can cause these loops to become oriented preferentially with the field, so their magnetic fields *add* to the external field. We then say that the material is *magnetized*.

Let's look at how these microscopic currents come about. **Figure 28.26** shows a primitive model of an electron in an atom. We picture the electron (mass m , charge $-e$) as moving in a circular orbit with radius r and speed v . This moving charge is equivalent to a current loop. In Section 27.7 we found that a current loop with area A and current I has a magnetic dipole moment μ given by $\mu = IA$; for the orbiting electron the area of the loop is $A = \pi r^2$. To find the current associated with the electron, we note that the orbital period T (the time for the electron to make one complete orbit) is the orbit circumference divided by the electron speed: $T = 2\pi r/v$. The equivalent current I is the total charge passing any point on the orbit per unit time, which is just the magnitude e of the electron charge divided by the orbital period T :

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

The magnetic moment $\mu = IA$ is then

$$\mu = \frac{ev}{2\pi r}(\pi r^2) = \frac{evr}{2} \quad (28.25)$$

It is useful to express μ in terms of the *angular momentum* L of the electron. For a particle moving in a circular path, the magnitude of angular momentum equals the magnitude of momentum mv multiplied by the radius r —that is, $L = mvr$ (see Section 10.5). Comparing this with Eq. (28.25), we can write

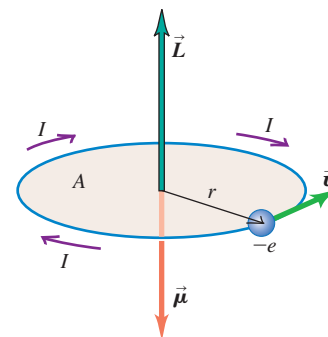
$$\mu = \frac{e}{2m}L \quad (28.26)$$

Equation (28.26) is useful in this discussion because atomic angular momentum is *quantized*; its component in a particular direction is always an integer multiple of $h/2\pi$, where h is a fundamental physical constant called *Planck's constant*. The numerical value of h is

$$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$$

The quantity $h/2\pi$ thus represents a fundamental unit of angular momentum in atomic systems, just as e is a fundamental unit of charge. Associated with the quantization of $\vec{L}$ is a fundamental uncertainty in the *direction* of $\vec{L}$ and therefore of $\vec{\mu}$. In the following discussion, when we speak of the magnitude of a magnetic moment, a more precise statement would be “maximum component in a given direction.” Thus, to say that a magnetic moment $\vec{\mu}$ is aligned with a magnetic field $\vec{B}$ really means that $\vec{\mu}$ has its maximum possible component in the direction of $\vec{B}$; such components are always quantized.

Figure 28.26 An electron moving with speed v in a circular orbit of radius r has an angular momentum $\vec{L}$ and an oppositely directed orbital magnetic dipole moment $\vec{\mu}$. It also has a spin angular momentum and an oppositely directed spin magnetic dipole moment.



CAUTION Planck's constant and SI units

We first introduced Planck's constant h in Section 1.3. Since h has units of $\text{J} \cdot \text{s} = \text{kg} \cdot \text{m}^2/\text{s}$, its *defined* value, combined with the definitions of the meter and the second, serves as the basis for the definition of the kilogram. Physicists define the kilogram this way because the value of h is relatively easy to measure in any physics laboratory, so it provides a useful standard for determining masses in kilograms. **|**

Equation (28.26) shows that associated with the fundamental unit of angular momentum is a corresponding fundamental unit of magnetic moment. If $L = h/2\pi$, then

$$\mu = \frac{e}{2m} \left(\frac{h}{2\pi} \right) = \frac{eh}{4\pi m} \quad (28.27)$$

This quantity is called the **Bohr magneton**, denoted by μ_B . Its numerical value is

$$\mu_B = 9.274 \times 10^{-24} \text{ A} \cdot \text{m}^2 = 9.274 \times 10^{-24} \text{ J/T}$$

You should verify that these two sets of units are consistent. The second set is useful when we compute the potential energy $U = -\vec{\mu} \cdot \vec{B}$ for a magnetic moment in a magnetic field.

Electrons also have an intrinsic angular momentum, called *spin*, that is not related to orbital motion but that can be pictured in a classical model as spinning on an axis. This angular momentum also has an associated magnetic moment, and its magnitude turns out to be almost exactly one Bohr magneton. (Effects having to do with quantization of the electromagnetic field cause the spin magnetic moment to be about $1.001 \mu_B$.)

Paramagnetism

In an atom, most of the various orbital and spin magnetic moments of the electrons add up to zero. However, in some cases the atom has a net magnetic moment that is of the order of μ_B . When such a material is placed in a magnetic field, the field exerts a torque on each magnetic moment, as given by Eq. (27.26): $\vec{\tau} = \vec{\mu} \times \vec{B}$. These torques tend to align the magnetic moments with the field, as we discussed in Section 27.7. In this position, the directions of the current loops are such as to *add* to the externally applied magnetic field.

We saw in Section 28.5 that the $\vec{B}$ field produced by a current loop is proportional to the loop's magnetic dipole moment. In the same way, the additional $\vec{B}$ field produced by microscopic electron current loops is proportional to the total magnetic moment $\vec{\mu}_{\text{total}}$ per unit volume V in the material. We call this vector quantity the **magnetization** of the material, denoted by $\vec{M}$:

$$\vec{M} = \frac{\vec{\mu}_{\text{total}}}{V} \quad (28.28)$$

The additional magnetic field due to magnetization of the material turns out to be equal simply to $\mu_0 \vec{M}$, where μ_0 is the same constant that appears in the law of Biot and Savart and Ampere's law. When such a material completely surrounds a current-carrying conductor, the total magnetic field $\vec{B}$ in the material is

$$\vec{B} = \vec{B}_0 + \mu_0 \vec{M} \quad (28.29)$$

where $\vec{B}_0$ is the field caused by the current in the conductor.

To check that the units in Eq. (28.29) are consistent, note that magnetization $\vec{M}$ is the magnetic moment per unit volume. The units of magnetic moment are current times area ($\text{A} \cdot \text{m}^2$), so the units of magnetization are $(\text{A} \cdot \text{m}^2)/\text{m}^3 = \text{A/m}$. From Section 28.1, the units of the constant μ_0 are $\text{T} \cdot \text{m/A}$. So the units of $\mu_0 \vec{M}$ are the same as the units of $\vec{B}$: $(\text{T} \cdot \text{m/A})(\text{A/m}) = \text{T}$.

A material showing the behavior just described is said to be **paramagnetic**. The result is that the magnetic field at any point in such a material is greater by a dimensionless factor K_m , called the **relative permeability** of the material, than it would be if the material were replaced by vacuum. The value of K_m is different for different materials; for common paramagnetic solids and liquids at room temperature, K_m typically ranges from 1.00001 to 1.003.

All of the equations in this chapter that relate magnetic fields to their sources can be adapted to the situation in which the current-carrying conductor is embedded in a paramagnetic material. All that need be done is to replace μ_0 by $K_m \mu_0$. This product is usually denoted as μ and is called the **permeability** of the material:

$$\mu = K_m \mu_0 \quad (28.30)$$

CAUTION Two meanings of the symbol μ

Equation (28.30) involves dangerous notation because we use μ for magnetic dipole moment as well as for permeability, as is customary. But beware: From now on, every time you see a μ , make sure you know whether it is permeability or magnetic moment. You can usually tell from the context. **|**

The amount by which the relative permeability differs from unity is called the **magnetic susceptibility**, denoted by χ_m :

$$\chi_m = K_m - 1 \quad (28.31)$$

Both K_m and χ_m are dimensionless quantities. **Table 28.1** lists values of magnetic susceptibility for several materials. For example, for aluminum, $\chi_m = 2.2 \times 10^{-5}$ and $K_m = 1.000022$. The first group in the table consists of paramagnetic materials; we'll soon discuss the second group, which contains *diamagnetic* materials.

The tendency of atomic magnetic moments to align themselves parallel to the magnetic field (where the potential energy is minimum) is opposed by random thermal motion, which tends to randomize their orientations. For this reason, paramagnetic susceptibility always decreases with increasing temperature. In many cases it is inversely proportional to the absolute temperature T , and the magnetization M can be expressed as

$$M = C \frac{B}{T} \quad (28.32)$$

This relationship is called *Curie's law*, after its discoverer, Pierre Curie (1859–1906). The quantity C is a constant, different for different materials, called the *Curie constant*.

As we described in Section 27.7, an object with atomic magnetic dipoles is attracted to the poles of a magnet. In most paramagnetic substances this attraction is very weak due to thermal randomization of the atomic magnetic moments. But at very low temperatures the thermal effects are reduced, the magnetization increases in accordance with Curie's law, and the attractive forces are greater.

TABLE 28.1 Magnetic Susceptibilities of Paramagnetic and Diamagnetic Materials at $T = 20^\circ\text{C}$

Material	$\chi_m = K_m - 1 (\times 10^{-5})$
Paramagnetic	
Iron ammonium alum	66
Uranium	40
Platinum	26
Aluminum	2.2
Sodium	0.72
Oxygen gas	0.19
Diamagnetic	
Bismuth	−16.6
Mercury	−2.9
Silver	−2.6
Carbon (diamond)	−2.1
Lead	−1.8
Sodium chloride	−1.4
Copper	−1.0

EXAMPLE 28.11 Magnetic dipoles in a paramagnetic material

Nitric oxide (NO) is a paramagnetic compound. The magnetic moment of each NO molecule has a maximum component in any direction of about one Bohr magneton. Compare the interaction energy of such magnetic moments in a 1.5 T magnetic field with the average translational kinetic energy of molecules at 300 K.

IDENTIFY and SET UP This problem involves the energy of a magnetic moment in a magnetic field and the average thermal kinetic energy. We have Eq. (27.27), $U = -\vec{\mu} \cdot \vec{B}$, for the interaction energy of a magnetic moment $\vec{\mu}$ with a $\vec{B}$ field, and Eq. (18.16), $K = \frac{3}{2}kT$, for the average translational kinetic energy of a molecule at temperature T .

EXECUTE We can write $U = -\mu_{\parallel}B$, where $\mu_{\parallel}$ is the component of the magnetic moment $\vec{\mu}$ in the direction of the $\vec{B}$ field. Here the maximum value of $\mu_{\parallel}$ is about μ_B , so

$$\begin{aligned} |U|_{\max} &\approx \mu_B B = (9.27 \times 10^{-24} \text{ J/T})(1.5 \text{ T}) \\ &= 1.4 \times 10^{-23} \text{ J} = 8.7 \times 10^{-5} \text{ eV} \end{aligned}$$

The average translational kinetic energy K is

$$\begin{aligned} K &= \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K}) \\ &= 6.2 \times 10^{-21} \text{ J} = 0.039 \text{ eV} \end{aligned}$$

EVALUATE At 300 K the magnetic interaction energy is only about 0.2% of the thermal kinetic energy, so we expect only a slight degree of alignment. This is why paramagnetic susceptibilities at ordinary temperature are usually very small.

KEYCONCEPT In a paramagnetic material the magnetic dipole moments of its atoms tend to align with an externally applied magnetic field. Random thermal motions tend to cancel out this alignment, however. Paramagnetic materials are weakly attracted to the poles of a magnet.

Diamagnetism

In some materials the total magnetic moment of all the atomic current loops is zero when no magnetic field is present. But even these materials have magnetic effects because an external field alters electron motions within the atoms, causing additional current loops and induced magnetic dipoles comparable to the induced *electric* dipoles we studied in Section 28.5. In this case the additional field caused by these current loops is always *opposite* in direction to that of the external field. (This behavior is explained by Faraday's law of induction, which we'll study in Chapter 29. An induced current always tends to cancel the field change that caused it.)

Figure 28.27 In this drawing adapted from a magnified photo, the arrows show the directions of magnetization in the domains of a single crystal of nickel. Domains that are magnetized in the direction of an applied magnetic field grow larger.

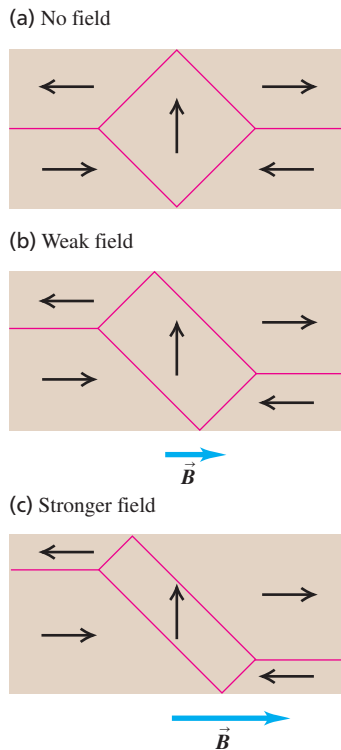
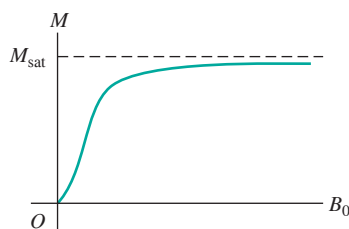


Figure 28.28 A magnetization curve for a ferromagnetic material. The magnetization M approaches its saturation value M_{sat} as the magnetic field B_0 (caused by external currents) becomes large.



Such materials are said to be **diamagnetic**. They always have negative susceptibility, as shown in Table 28.1, and relative permeability K_m slightly *less* than unity, typically of the order of 0.99990 to 0.99999 for solids and liquids. Diamagnetic susceptibilities are very nearly temperature independent.

Ferromagnetism

There is a third class of materials, called **ferromagnetic** materials, that includes iron, nickel, cobalt, and many alloys containing these elements. In these materials, strong interactions between atomic magnetic moments cause them to line up parallel to each other in regions called **magnetic domains**, even when no external field is present. Figure 28.27 shows an example of magnetic domain structure. Within each domain, nearly all of the atomic magnetic moments are parallel.

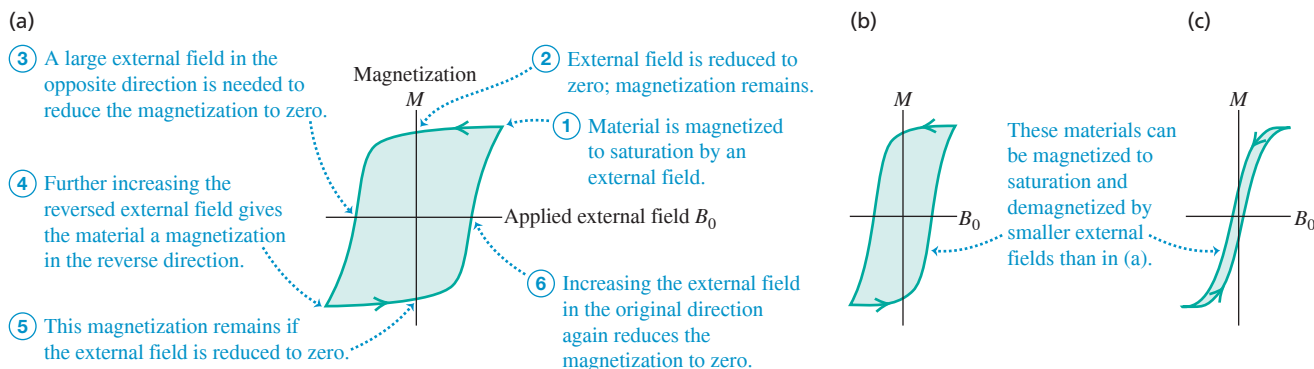
When there is no externally applied field, the domain magnetizations are randomly oriented. But when a field $\vec{B}_0$ (caused by external currents) is present, the domains tend to orient themselves parallel to the field. The domain boundaries also shift; the domains that are magnetized in the field direction grow, and those that are magnetized in other directions shrink. Because the total magnetic moment of a domain may be many thousands of Bohr magnetons, the torques that tend to align the domains with an external field are much stronger than occur with paramagnetic materials. The relative permeability K_m is *much* larger than unity, typically of the order of 1000 to 100,000. As a result, an object made of a ferromagnetic material such as iron is strongly magnetized by the field from a permanent magnet and is attracted to the magnet (see Fig. 27.38). A paramagnetic material such as aluminum is also attracted to a permanent magnet, but K_m for paramagnetic materials is so much smaller for such a material than for ferromagnetic materials that the attraction is very weak. Thus a magnet can pick up iron nails, but not aluminum cans.

As the external field is increased, a point is eventually reached at which nearly *all* the magnetic moments in the ferromagnetic material are aligned parallel to the external field. This condition is called **saturation magnetization**; after it is reached, further increase in the external field causes no increase in magnetization or in the additional field caused by the magnetization.

Figure 28.28 shows a “magnetization curve,” a graph of magnetization M as a function of external magnetic field B_0 , for soft iron. An alternative description of this behavior is that K_m is not constant but decreases as B_0 increases. (Paramagnetic materials also show saturation at sufficiently strong fields. But the magnetic fields required are so large that departures from a linear relationship between M and B_0 in these materials can be observed only at very low temperatures, 1 K or so.)

For many ferromagnetic materials the relationship of magnetization to external magnetic field is different when the external field is increasing from when it is decreasing. Figure 28.29a shows this relationship for such a material. When the material is magnetized to saturation and then the external field is reduced to zero, some magnetization

Figure 28.29 Hysteresis loops. The materials of both (a) and (b) remain strongly magnetized when B_0 is reduced to zero. Since (a) is also hard to demagnetize, it would be good for permanent magnets. Since (b) magnetizes and demagnetizes more easily, it could be used as a computer memory material. The material of (c) would be useful for transformers and other alternating-current devices where zero hysteresis would be optimal.



remains. This behavior is characteristic of permanent magnets, which retain most of their saturation magnetization when the magnetizing field is removed. To reduce the magnetization to zero requires a magnetic field in the reverse direction.

This behavior is called **hysteresis**, and the curves in Fig. 28.29 are called *hysteresis loops*. Magnetizing and demagnetizing a material that has hysteresis involve the dissipation of energy, and the temperature of the material increases during such a process.

Ferromagnetic materials are widely used in electromagnets, transformer cores, and motors and generators, in which it is desirable to have as large a magnetic field as possible for a given current. Because hysteresis dissipates energy, materials that are used in these applications should usually have as narrow a hysteresis loop as possible. Soft iron is often used; it has high permeability without appreciable hysteresis. For permanent magnets a broad hysteresis loop is usually desirable, with large zero-field magnetization and large reverse field needed to demagnetize. Many kinds of steel and many alloys, such as Alnico, are used for permanent magnets. The remaining magnetic field in such a material, after it has been magnetized to near saturation, is typically of the order of 1 T, corresponding to a remaining magnetization $M = B/\mu_0$ of about 800,000 A/m.

EXAMPLE 28.12 A ferromagnetic material

A cube-shaped permanent magnet is made of a ferromagnetic material with a magnetization M of about 8×10^5 A/m. The side length is 2 cm. (a) Find the magnetic dipole moment of the magnet. (b) Estimate the magnetic field due to the magnet at a point 10 cm from the magnet along its axis.

IDENTIFY and SET UP This problem uses the relationship between magnetization M and magnetic dipole moment μ_{total} and the idea that a magnetic dipole produces a magnetic field. We find μ_{total} from Eq. (28.28). To estimate the field, we approximate the magnet as a current loop with this same magnetic moment and use Eq. (28.18).

EXECUTE (a) From Eq. (28.28),

$$\mu_{\text{total}} = MV = (8 \times 10^5 \text{ A/m})(2 \times 10^{-2} \text{ m})^3 = 6 \text{ A} \cdot \text{m}^2$$

(b) From Eq. (28.18), the magnetic field on the axis of a current loop with magnetic moment μ_{total} is

$$B = \frac{\mu_0 \mu_{\text{total}}}{2\pi(x^2 + a^2)^{3/2}}$$

where x is the distance from the loop and a is its radius. We can use this expression here if we take a to refer to the size of the permanent magnet. Strictly speaking, there are complications because our magnet does not have the same geometry as a circular current loop. But because $x = 10$ cm is fairly large in comparison to the 2 cm size of the magnet, the term a^2 is negligible in comparison to x^2 and can be ignored. So

$$B \approx \frac{\mu_0 \mu_{\text{total}}}{2\pi x^3} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(6 \text{ A} \cdot \text{m}^2)}{2\pi(0.1 \text{ m})^3} = 1 \times 10^{-3} \text{ T} = 10 \text{ G}$$

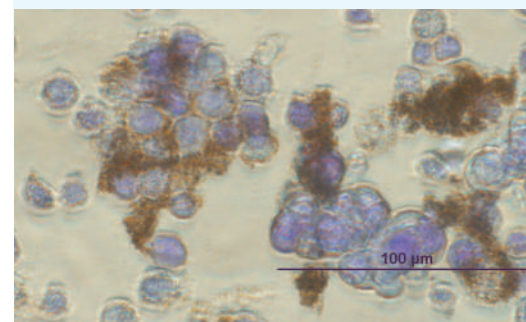
which is about ten times stronger than the earth's magnetic field.

EVALUATE We calculated B at a point *outside* the magnetic material and therefore used μ_0 , not the permeability μ of the magnetic material, in our calculation. You would substitute permeability μ for μ_0 if you were calculating B *inside* a material with relative permeability K_m , for which $\mu = K_m \mu_0$.

KEYCONCEPT In a ferromagnetic material there are strong interactions between the magnetic dipole moments of neighboring atoms, and the material can maintain a large overall magnetic dipole even if there is no externally applied magnetic field.

BIO APPLICATION Ferromagnetic Nanoparticles for Cancer Therapy

The violet blobs in this microscope image are cancer cells that have broken away from a tumor and threaten to spread throughout a patient's body. An experimental technique for fighting these cells uses particles of a ferromagnetic material (shown in brown) injected into the body. These particles are coated with a chemical that preferentially attaches to cancer cells. A magnet outside the patient then "steers" the particles out of the body, taking the cancer cells with them. (Photo courtesy of cancer researcher Dr. Kenneth Scarberry.)



TEST YOUR UNDERSTANDING OF SECTION 28.8 Which of the following materials are attracted to a magnet? (i) Sodium; (ii) bismuth; (iii) lead; (iv) uranium.

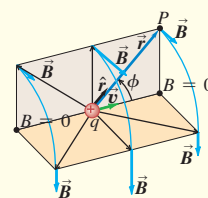
ANSWER

(i), (ii), and (iv) are attracted to a magnet. (iii) is repelled by a magnet. (See Table 28.1.)

CHAPTER 28 SUMMARY

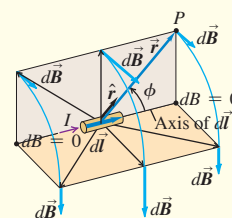
Magnetic field of a moving charge: The magnetic field $\vec{B}$ created by a charge q moving with velocity $\vec{v}$ depends on the distance r from the source point (the location of q) to the field point (where $\vec{B}$ is measured). The $\vec{B}$ field is perpendicular to $\vec{v}$ and to $\hat{r}$, the unit vector directed from the source point to the field point. The principle of superposition of magnetic fields states that the total $\vec{B}$ field produced by several moving charges is the vector sum of the fields produced by the individual charges. (See Example 28.1.)

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2} \quad (28.2)$$



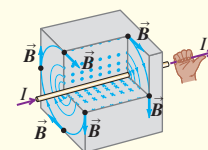
Magnetic field of a current-carrying conductor: The law of Biot and Savart gives the magnetic field $d\vec{B}$ created by an element $d\vec{l}$ of a conductor carrying current I . The field $d\vec{B}$ is perpendicular to both $d\vec{l}$ and $\hat{r}$, the unit vector from the element to the field point. The $\vec{B}$ field created by a finite current-carrying conductor is the integral of $d\vec{B}$ over the length of the conductor. (See Example 28.2.)

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} \quad (28.6)$$



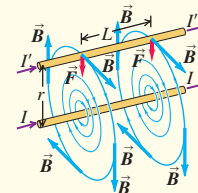
Magnetic field of a long, straight, current-carrying conductor: The magnetic field $\vec{B}$ at a distance r from a long, straight conductor carrying a current I has a magnitude that is inversely proportional to r . The magnetic field lines are circles coaxial with the wire, with directions given by the right-hand rule. (See Examples 28.3 and 28.4.)

$$B = \frac{\mu_0 I}{2\pi r} \quad (28.9)$$



Magnetic force between current-carrying conductors: Two long, parallel, current-carrying conductors attract if the currents are in the same direction and repel if the currents are in opposite directions. The magnetic force per unit length between the conductors depends on their currents I and I' and separation r . The definition of the ampere is based on this relationship. (See Example 28.5.)

$$\frac{F}{L} = \frac{\mu_0 II'}{2\pi r} \quad (28.11)$$



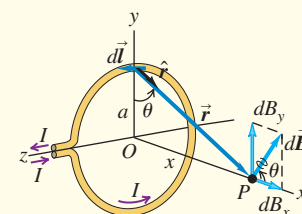
Magnetic field of a current loop: The law of Biot and Savart allows us to calculate the magnetic field produced along the axis of a circular conducting loop of radius a carrying current I . The field depends on the distance x along the axis from the center of the loop to the field point. If there are N loops, the field is multiplied by N . At the center of the loop, $x = 0$. (See Example 28.6.)

$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}} \quad (28.15)$$

(circular loop)

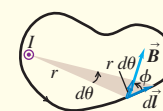
$$B_x = \frac{\mu_0 N I}{2a} \quad (28.17)$$

(center of N circular loops)



Ampere's law: Ampere's law states that the line integral of $\vec{B}$ around any closed path equals μ_0 times the net current through the area enclosed by the path. The positive sense of current is determined by a right-hand rule. (See Examples 28.7–28.10.)

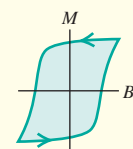
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}} \quad (28.20)$$



Magnetic fields due to current distributions: The table lists magnetic fields caused by several current distributions. In each case the conductor is carrying current I .

Current Distribution	Point in Magnetic Field	Magnetic-Field Magnitude
Long, straight conductor	Distance r from conductor	$B = \frac{\mu_0 I}{2\pi r}$
Circular loop of radius a (for N loops, multiply these expressions for magnetic-field amplitude by N)	On axis of loop	$B = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$
	At center of loop	$B = \frac{\mu_0 I}{2a}$
Long cylindrical conductor of radius R	Inside conductor, $r < R$	$B = \frac{\mu_0 I}{2\pi} \frac{r}{R^2}$
	Outside conductor, $r > R$	$B = \frac{\mu_0 I}{2\pi r}$
Long, closely wound solenoid with n turns per unit length, near its midpoint	Inside solenoid, near center	$B = \mu_0 n I$
	Outside solenoid	$B \approx 0$
Tightly wound toroidal solenoid (toroid) with N turns	Within the space enclosed by the windings, distance r from symmetry axis	$B = \frac{\mu_0 N I}{2\pi r}$
	Outside the space enclosed by the windings	$B \approx 0$

Magnetic materials: When magnetic materials are present, the magnetization of the material causes an additional contribution to $\vec{B}$. For paramagnetic and diamagnetic materials, μ_0 is replaced in magnetic-field expressions by $\mu = K_m \mu_0$, where μ is the permeability of the material and K_m is its relative permeability. The magnetic susceptibility χ_m is defined as $\chi_m = K_m - 1$. Magnetic susceptibilities for paramagnetic materials are small positive quantities; those for diamagnetic materials are small negative quantities. For ferromagnetic materials, K_m is much larger than unity and is not constant. Some ferromagnetic materials are permanent magnets, retaining their magnetization even after the external magnetic field is removed. (See Examples 28.11 and 28.12.)



Chapter 28 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 28.1** (Section 28.1) and **EXAMPLE 28.2** (Section 28.2) before attempting these problems.

VP28.2.1 A proton with charge $+1.60 \times 10^{-19}$ C is at the origin ($x = y = z = 0$) and is moving at 2.00×10^5 m/s in the $+x$ -direction. Find the magnetic field $\vec{B}$ that this proton produces at (a) $x = 0$, $y = 1.00$ mm, $z = 0$; (b) $x = 0$, $y = 0$, $z = 2.00$ mm; (c) $x = 1.00$ mm, $y = 0$, $z = 0$; (d) $x = 1.00$ mm, $y = 1.00$ mm, $z = 0$.

VP28.2.2 A proton (charge $+1.60 \times 10^{-19}$ C) and an electron (charge -1.60×10^{-19} C) are both moving in the xy -plane with the same speed, 4.20×10^5 m/s. The proton is moving in the $+y$ -direction along the line $x = 0$, and the electron is moving in the $-y$ -direction along the line $x = +2.00$ mm. At the instant when the proton and electron are at their closest approach, what are the magnitude and direction of the magnetic force that (a) the proton exerts on the electron and (b) the electron exerts on the proton?

VP28.2.3 A segment of wire centered at the origin ($x = y = z = 0$) is 2.00 mm in length and carries a current of 6.00 A in the $+y$ -direction. You measure the magnetic field due to this segment at the point $x = 3.00$ m, $y = 4.00$ m, $z = 0$. Find (a) the unit vector from the wire segment to this point and (b) the magnetic field at this point.

VP28.2.4 A segment of wire centered at the origin ($x = y = z = 0$) is 8.00 mm in length and carries a current of 4.00 A in the $+z$ -direction. Find the magnetic force that this segment exerts on an electron (charge -1.60×10^{-19} C) at the point $x = 1.25$ m, $y = 0$, $z = 0$ that is moving at 3.00×10^5 m/s in the $-x$ -direction.

Be sure to review **EXAMPLES 28.3 and 28.4** (Section 28.3) and **EXAMPLE 28.5** (Section 28.4) before attempting these problems.

VP28.5.1 How much current must a long, straight wire carry in order for the magnetic field that it produces 1.40 cm from the central axis of the wire to have magnitude 3.20×10^{-5} T?

VP28.5.2 Two long, straight, conducting wires are both perpendicular to the xy -plane. Wire 1 passes through the point $x = 0$, $y = 1.00$ cm, $z = 0$ and carries current 1.00 A in the $+z$ -direction. Wire 2 passes through the point $x = 0$, $y = -1.00$ cm, $z = 0$ and carries current 4.00 A also in the $+z$ -direction. Find the net magnetic field due to both wires at (a) $x = 0$, $y = 0$, $z = 0$; (b) $x = 0$, $y = 2.00$ cm, $z = 0$; (c) $x = 0$, $y = -2.00$ cm, $z = 0$.

VP28.5.3 A long, straight, conducting wire carries a current of 2.00 A. A second, identical wire is parallel to the first and separated from it by 3.00 cm. The magnetic field at a point halfway between the two wires has magnitude 4.00×10^{-5} T. What is the current in the second wire if its direction is (a) the same as in the first wire? (b) Opposite to that in the first wire?

VP28.5.4 Two straight, parallel, superconducting wires are 0.900 m apart. Both wires are oriented straight up and down. One wire carries a current of 1.30×10^4 A upward and experiences a magnetic force per unit length of 95.0 N/m that attracts it to the other wire. What are the magnitude and direction of the current in the other wire?

Be sure to review EXAMPLES 28.8, 28.9, and 28.10 (Section 28.7) before attempting these problems.

VP28.10.1 A solid cylindrical conductor with radius 4.50 cm carries a current of 2.00 A. The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnitude of the magnetic field at the following distances from the conductor axis: (a) 2.50 cm; (b) 4.50 cm; (c) 6.00 cm.

VP28.10.2 A long, hollow cylinder with inner radius R_1 and outer radius R_2 carries current along its length. The current is uniformly distributed over the cross-sectional area of the cylinder and has current

density J . (a) Find the magnetic-field magnitude B as a function of the distance r from the conductor axis for points inside the hollow interior ($r < R_1$), inside the solid conductor ($R_2 > r > R_1$), and outside the conductor ($r > R_2$). (b) For which value of r is B greatest?

VP28.10.3 A very long solenoid with tightly wound coils has 8.00×10^3 turns of wire per meter of length. What must the current be in the solenoid in order for the field inside the solenoid to have magnitude 0.0320 T?

VP28.10.4 A copper wire carries a current of 0.840 A. You wrap this wire around a hollow toroidal core, forming a toroidal solenoid. The inner and outer radii of the core are 6.00 cm and 8.00 cm, respectively. (a) How many turns of wire around the core are required to produce a magnetic field of magnitude 2.00×10^{-3} T at a distance of 7.00 cm from the toroid axis? (b) With this number of turns, what are the maximum and minimum field magnitudes in the interior of the core?

BRIDGING PROBLEM Magnetic Field of a Charged, Rotating Dielectric Disk

A thin dielectric disk with radius a has a total charge $+Q$ distributed uniformly over its surface (**Fig. 28.30**). It rotates n times per second about an axis perpendicular to the surface of the disk and passing through its center. Find the magnetic field at the center of the disk.

SOLUTION GUIDE

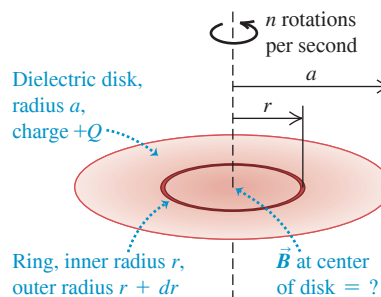
IDENTIFY and SET UP

1. Think of the rotating disk as a series of concentric rotating rings. Each ring acts as a circular current loop that produces a magnetic field at the center of the disk.
2. Use the results of Section 28.5 to find the magnetic field due to a single ring. Then integrate over all rings to find the total field.

EXECUTE

3. Find the charge on a ring with inner radius r and outer radius $r + dr$ (**Fig. 28.30**).
4. How long does it take the charge found in step 3 to make a complete trip around the rotating ring? Use this to find the current of the rotating ring.
5. Use a result from Section 28.5 to determine the magnetic field that this ring produces at the center of the disk.

Figure 28.30 Finding the $\vec{B}$ field at the center of a uniformly charged, rotating disk.



6. Integrate your result from step 5 to find the total magnetic field from all rings with radii from $r = 0$ to $r = a$.

EVALUATE

7. Does your answer have the correct units?
8. Suppose all of the charge were concentrated at the rim of the disk (at $r = a$). Would this increase or decrease the field at the center of the disk?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q28.1 A topic of current interest in physics research is the search (thus far unsuccessful) for an isolated magnetic pole, or magnetic *monopole*. If such an entity were found, how could it be recognized? What would its properties be?

Q28.2 Streams of charged particles emitted from the sun during periods of solar activity create a disturbance in the earth's magnetic field. How does this happen?

Q28.3 The text discussed the magnetic field of an infinitely long, straight conductor carrying a current. Of course, there is no such thing

as an infinitely long *anything*. How do you decide whether a particular wire is long enough to be considered infinite?

Q28.4 Two parallel conductors carrying current in the same direction attract each other. If they are permitted to move toward each other, the forces of attraction do work. From where does the energy come? Does this contradict the assertion in Chapter 27 that magnetic forces on moving charges do no work? Explain.

Q28.5 Pairs of conductors carrying current into or out of the power-supply components of electronic equipment are sometimes twisted together to reduce magnetic-field effects. Why does this help?

Q28.6 Suppose you have three long, parallel wires arranged so that in cross section they are at the corners of an equilateral triangle. Is there any way to arrange the currents so that all three wires attract each other? So that all three wires repel each other? Explain.

Q28.7 In deriving the force on one of the long, current-carrying conductors in Section 28.4, why did we use the magnetic field due to only one of the conductors? That is, why didn't we use the *total* magnetic field due to *both* conductors?

Q28.8 Two concentric, coplanar, circular loops of wire of different diameter carry currents in the same direction. Describe the nature of the force exerted on the inner loop by the outer loop and on the outer loop by the inner loop.

Q28.9 A current was sent through a helical coil spring. The spring contracted, as though it had been compressed. Why?

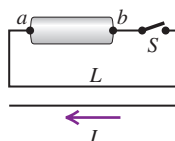
Q28.10 What are the relative advantages and disadvantages of Ampere's law and the law of Biot and Savart for practical calculations of magnetic fields?

Q28.11 Magnetic field lines never have a beginning or an end. Use this to explain why it is reasonable for the field of an ideal toroidal solenoid to be confined entirely to its interior, while a straight solenoid *must* have some field outside.

Q28.12 Two very long, parallel wires carry equal currents in opposite directions. (a) Is there any place that their magnetic fields completely cancel? If so, where? If not, why not? (b) How would the answer to part (a) change if the currents were in the same direction?

Q28.13 In the circuit shown in Fig. Q28.13, Figure Q28.13

when switch S is suddenly closed, the wire L is pulled toward the lower wire carrying current I . Which (a or b) is the positive terminal of the battery? How do you know?



Q28.14 A metal ring carries a current that causes a magnetic field B_0 at the center of the ring and a field B at point P a distance x from the center along the axis of the ring. If the radius of the ring is doubled, find the magnetic field at the center. Will the field at point P change by the same factor? Why?

Q28.15 Show that the units $A \cdot m^2$ and J/T for the Bohr magneton are equivalent.

Q28.16 Why should the permeability of a paramagnetic material be expected to decrease with increasing temperature?

Q28.17 If a magnet is suspended over a container of liquid air, it attracts droplets to its poles. The droplets contain only liquid oxygen; even though nitrogen is the primary constituent of air, it is not attracted to the magnet. Explain what this tells you about the magnetic susceptibilities of oxygen and nitrogen, and explain why a magnet in ordinary, room-temperature air doesn't attract molecules of oxygen *gas* to its poles.

Q28.18 What features of atomic structure determine whether an element is diamagnetic or paramagnetic? Explain.

Q28.19 The magnetic susceptibility of paramagnetic materials is quite strongly temperature dependent, but that of diamagnetic materials is nearly independent of temperature. Why the difference?

Q28.20 A cylinder of iron is placed so that it is free to rotate around its axis. Initially the cylinder is at rest, and a magnetic field is applied to the cylinder so that it is magnetized in a direction parallel to its axis. If the direction of the *external* field is suddenly reversed, the direction of magnetization will also reverse and the cylinder will begin rotating around its axis. (This is called the *Einstein-de Haas effect*.) Explain why the cylinder begins to rotate.

EXERCISES

Section 28.1 Magnetic Field of a Moving Charge

28.1 •• A $+6.00 \mu C$ point charge is moving at a constant 8.00×10^6 m/s in the $+y$ -direction, relative to a reference frame. At the instant when the point charge is at the origin of this reference frame, what is the magnetic-field vector $\vec{B}$ it produces at the following points: (a) $x = 0.500$ m, $y = 0$, $z = 0$; (b) $x = 0$, $y = -0.500$ m, $z = 0$; (c) $x = 0$, $y = 0$, $z = +0.500$ m; (d) $x = 0$, $y = -0.500$ m, $z = +0.500$ m?

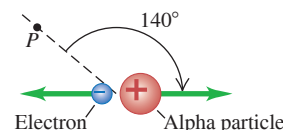
28.2 • **Fields Within the Atom.** In the Bohr model of the hydrogen atom, the electron moves in a circular orbit of radius 5.3×10^{-11} m with a speed of 2.2×10^6 m/s. If we are viewing the atom in such a way that the electron's orbit is in the plane of the paper with the electron moving clockwise, find the magnitude and direction of the electric and magnetic fields that the electron produces at the location of the nucleus (treated as a point).

28.3 • An electron moves at $0.100c$ as shown Figure E28.3

in Fig. E28.3. Find the magnitude and direction of the magnetic field this electron produces at the following points, each $2.00 \mu m$ from the electron: (a) points A and B ; (b) point C ; (c) point D .

28.4 •• An alpha particle (charge $+2e$) and an electron move in opposite directions from the same point, each with the speed of 2.50×10^5 m/s (Fig. E28.4). Find the magnitude and direction of the total magnetic field these charges produce at point P , which is 8.65 nm from each charge.

Figure E28.4

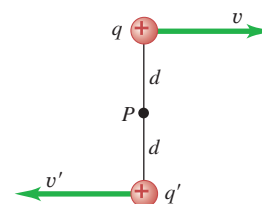
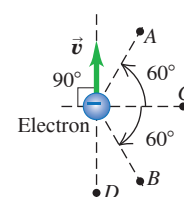


28.5 • A $-4.80 \mu C$ charge is moving at a constant speed of 6.80×10^5 m/s in the $+x$ -direction relative to a reference frame. At the instant when the point charge is at the origin, what is the magnetic-field vector it produces at the following points: (a) $x = 0.500$ m, $y = 0$, $z = 0$; (b) $x = 0$, $y = 0.500$ m, $z = 0$; (c) $x = 0.500$ m, $y = 0.500$ m, $z = 0$; (d) $x = 0$, $y = 0$, $z = 0.500$ m?

28.6 • Positive point charges Figure E28.6

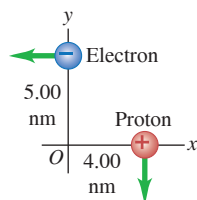
$q = +8.00 \mu C$ and $q' = +3.00 \mu C$ are moving relative to an observer at point P , as shown in Fig. E28.6. The distance d is 0.120 m, $v = 4.50 \times 10^6$ m/s, and $v' = 9.00 \times 10^6$ m/s. (a) When the two charges are at the locations shown in the figure, what are the magnitude and direction of the net magnetic field they produce at point P ? (b) What are the magnitude and direction of the electric and magnetic forces that each charge exerts on the other, and what is the ratio of the magnitude of the electric force to the magnitude of the magnetic force? (c) If the direction of $\vec{v}'$ is reversed, so both charges are moving in the same direction, what are the magnitude and direction of the magnetic forces that the two charges exert on each other?

28.7 • At one instant, point P is $6.60 \mu m$ to the left of a proton that is moving at 3.30 km/s in vacuum. (a) What is the direction of the unit vector in Eq. (28.2)? (b) What is the magnetic field caused by this proton if it is moving to the right or to the left? (c) What is the magnetic field (magnitude and direction) caused by this proton if it is moving toward the top of the page? (d) What is the answer to part (c) for an electron instead of a proton?



28.8 • An electron and a proton are each moving at 735 km/s in perpendicular paths as shown in **Fig. E28.8**. At the instant when they are at the positions shown, find the magnitude and direction of (a) the total magnetic field they produce at the origin; (b) the magnetic field the electron produces at the location of the proton; (c) the total electric force and the total magnetic force that the electron exerts on the proton.

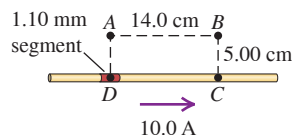
Figure E28.8



Section 28.2 Magnetic Field of a Current Element

28.9 • A straight wire carries a 10.0 A current (**Fig. E28.9**). $ABCD$ is a rectangle with point D in the middle of a 1.10 mm segment of the wire and point C in the wire. Find the magnitude and direction of the magnetic field due to this segment at (a) point A ; (b) point B ; (c) point C .

Figure E28.9



28.10 • A short current element $d\vec{l} = (0.500 \text{ mm})\hat{j}$ carries a current of 5.40 A in the same direction as $d\vec{l}$. Point P is located at $\vec{r} = (-0.730 \text{ m})\hat{i} + (0.390 \text{ m})\hat{k}$. Use unit vectors to express the magnetic field at P produced by this current element.

28.11 • A long, straight wire lies along the z -axis and carries a 4.00 A current in the $+z$ -direction. Find the magnetic field (magnitude and direction) produced at the following points by a 0.500 mm segment of the wire centered at the origin: (a) $x = 2.00 \text{ m}$, $y = 0$, $z = 0$; (b) $x = 0$, $y = 2.00 \text{ m}$, $z = 0$; (c) $x = 2.00 \text{ m}$, $y = 2.00 \text{ m}$, $z = 0$; (d) $x = 0$, $y = 0$, $z = 2.00 \text{ m}$.

28.12 • Two parallel wires are 5.00 cm apart and carry currents in opposite directions, as shown in **Fig. E28.12**. Find the magnitude and direction of the magnetic field at point P due to two 1.50 mm segments of wire that are opposite each other and each 8.00 cm from P .

Figure E28.12

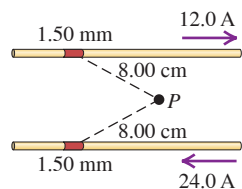
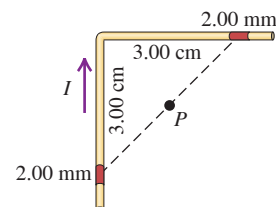


Figure E28.13



28.13 • A wire carrying a 28.0 A current bends through a right angle. Consider two 2.00 mm segments of wire, each 3.00 cm from the bend (**Fig. E28.13**). Find the magnitude and direction of the magnetic field these two segments produce at point P , which is midway between them.

Section 28.3 Magnetic Field of a Straight Current-Carrying Conductor

28.14 • A long, straight wire lies along the x -axis and carries current $I = 60.0 \text{ A}$ in the $+x$ -direction. A small particle with mass $3.00 \times 10^{-6} \text{ kg}$ and charge $8.00 \times 10^{-3} \text{ C}$ is traveling in the vicinity of the wire. At an instant when the particle is on the y -axis at $y = 8.00 \text{ cm}$, its acceleration has components $a_x = -5.00 \times 10^3 \text{ m/s}^2$ and $a_y = +9.00 \times 10^3 \text{ m/s}^2$. At that instant what are the x - and y -components of the velocity of the particle?

28.15 • **The Magnetic Field from a Lightning Bolt.** Lightning bolts can carry currents up to approximately 20 kA. We can model such a current as the equivalent of a very long, straight wire. (a) If you were unfortunate enough to be 5.0 m away from such a lightning bolt, how large a magnetic field would you experience? (b) How does this field compare to one you would experience by being 5.0 cm from a long, straight household current of 10 A?

28.16 • A very long, straight horizontal wire carries a current such that 8.20×10^{18} electrons per second pass any given point going from west to east. What are the magnitude and direction of the magnetic field this wire produces at a point 4.00 cm directly above it?

28.17 • **BIO Currents in the Heart.** The body contains many small currents caused by the motion of ions in the organs and cells. Measurements of the magnetic field around the chest due to currents in the heart give values of about $10 \mu\text{G}$. Although the actual currents are rather complicated, we can gain a rough understanding of their magnitude if we model them as a long, straight wire. If the surface of the chest is 5.0 cm from this current, how large is the current in the heart?

28.18 • Two long, straight wires, one above the other, are separated by a distance $2a$ and are parallel to the x -axis. Let the $+y$ -axis be in the plane of the wires in the direction from the lower wire to the upper wire. Each wire carries current I in the $+x$ -direction. What are the magnitude and direction of the net magnetic field of the two wires at a point in the plane of the wires (a) midway between them; (b) at a distance a above the upper wire; (c) at a distance a below the lower wire?

28.19 • A long, straight wire lies along the y -axis and carries a current $I = 8.00 \text{ A}$ in the $-y$ -direction (**Fig. E28.19**). In addition to the magnetic field due to the current in the wire, a uniform magnetic field $\vec{B}_0$ with magnitude $1.50 \times 10^{-6} \text{ T}$ is in the $+x$ -direction. What is the total field (magnitude and direction) at the following points in the xz -plane:

(a) $x = 0$, $z = 1.00 \text{ m}$; (b) $x = 1.00 \text{ m}$, $z = 0$; (c) $x = 0$, $z = -0.25 \text{ m}$?

28.20 • **BIO Transmission Lines and Health.** Currents in dc transmission lines can be 100 A or higher.

Some people are concerned that the electromagnetic fields from such lines near their homes could pose health dangers. For a line that has current 150 A and a height of 8.0 m above the ground, what magnetic field does the line produce at ground level? Express your answer in teslas and as a percentage of the earth's magnetic field, which is 0.50 G. Is this value cause for worry?

28.21 • Two long, straight, parallel wires, 10.0 cm apart, carry equal 4.00 A currents in the same direction, as shown in **Fig. E28.21**. Find the magnitude and direction of the magnetic field at (a) point P_1 , midway between the wires; (b) point P_2 , 25.0 cm to the right of P_1 ; (c) point P_3 , 20.0 cm directly above P_1 .

28.22 • A rectangular loop with dimensions 4.20 cm by 9.50 cm carries current I . The current in the loop produces a magnetic field at the center of the loop that has magnitude $5.50 \times 10^{-5} \text{ T}$ and direction away from you as you view the plane of the loop. What are the magnitude and direction (clockwise or counterclockwise) of the current in the loop?

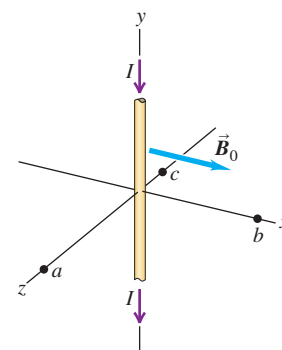
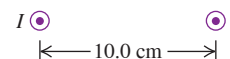
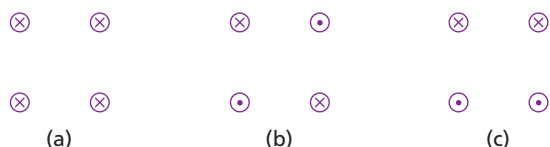


Figure E28.21



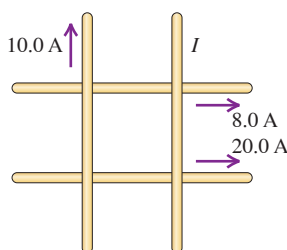
28.23 • Four long, parallel power lines each carry 100 A currents. A cross-sectional diagram of these lines is a square, 20.0 cm on each side. For each of the three cases shown in **Fig. E28.23**, calculate the magnetic field at the center of the square.

Figure E28.23



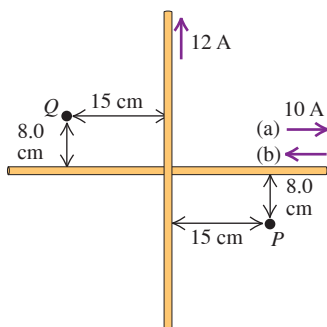
28.24 • Four very long, current-carrying wires in the same plane intersect to form a square 40.0 cm on each side, as shown in **Fig. E28.24**. Find the magnitude and direction of the current I so that the magnetic field at the center of the square is zero.

Figure E28.24



28.25 •• Two very long insulated wires perpendicular to each other in the same plane carry currents as shown in **Fig. E28.25**. Find the magnitude of the *net* magnetic field these wires produce at points P and Q if the 10.0 A current is (a) to the right or (b) to the left.

Figure E28.25

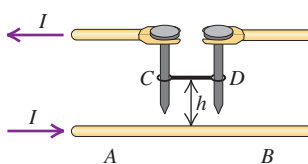


28.26 • A wire of length 20.0 cm lies along the x -axis with the center of the wire at the origin. The wire carries current $I = 8.00$ A in the $-x$ -direction. (a) What is the magnitude B of the magnetic field of the wire at the point $y = 5.00$ cm on the y -axis? (b) What is the percent difference between the answer in (a) and the value you obtain if you assume the wire is infinitely long and use Eq. (28.9) to calculate B ?

Section 28.4 Force Between Parallel Conductors

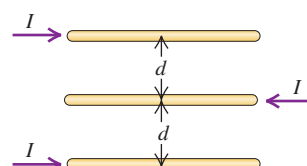
28.27 • A long, horizontal wire AB rests on the surface of a table and carries a current I . Horizontal wire CD is vertically above wire AB and is free to slide up and down on the two vertical metal guides C and D (**Fig. E28.27**). Wire CD is connected through the sliding contacts to another wire that also carries a current I , opposite in direction to the current in wire AB . The mass per unit length of the wire CD is λ . To what equilibrium height h will the wire CD rise, assuming that the magnetic force on it is due entirely to the current in the wire AB ?

Figure E28.27



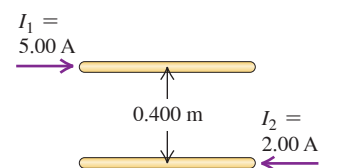
28.28 • Three very long parallel wires each carry current I in the directions shown in **Fig. E28.28**. If the separation between adjacent wires is d , calculate the magnitude and direction of the net magnetic force per unit length on each wire.

Figure E28.28



28.29 • Two long, parallel wires are separated by a distance of 0.400 m (**Fig. E28.29**). The currents I_1 and I_2 have the directions shown. (a) Calculate the magnitude of the force exerted by each wire on a 1.20 m length of the other. Is the force attractive or repulsive? (b) Each current is doubled, so that I_1 becomes 10.0 A and I_2 becomes 4.00 A. Now what is the magnitude of the force that each wire exerts on a 1.20 m length of the other?

Figure E28.29



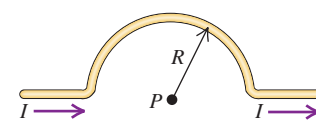
28.30 • Two long, parallel wires are separated by a distance of 2.50 cm. The force per unit length that each wire exerts on the other is 4.00×10^{-5} N/m, and the wires repel each other. The current in one wire is 0.600 A. (a) What is the current in the second wire? (b) Are the two currents in the same direction or in opposite directions?

Section 28.5 Magnetic Field of a Circular Current Loop

28.31 • BIO Currents in the Brain. The magnetic field around the head has been measured to be approximately 3.0×10^{-8} G. Although the currents that cause this field are quite complicated, we can get a rough estimate of their size by modeling them as a single circular current loop 16 cm (the width of a typical head) in diameter. What is the current needed to produce such a field at the center of the loop?

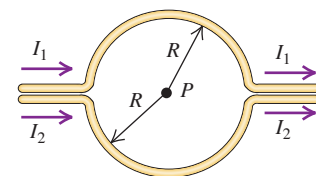
28.32 • Calculate the magnitude and direction of the magnetic field at point P due to the current in the semicircular section of wire shown in **Fig. E28.32**. (Hint: Does the current in the long, straight section of the wire produce any field at P ?)

Figure E28.32



28.33 •• Calculate the magnitude of the magnetic field at point P of **Fig. E28.33** in terms of R , I_1 , and I_2 . What does your expression give when $I_1 = I_2$?

Figure E28.33



28.34 •• A closely wound, circular coil with radius 2.40 cm has 800 turns. (a) What must the current in the coil be if the magnetic field at the center of the coil is 0.0770 T? (b) At what distance x from the center of the coil, on the axis of the coil, is the magnetic field half its value at the center?

28.35 •• Two concentric circular loops of wire lie on a tabletop, one inside the other. The inner wire has a diameter of 20.0 cm and carries a clockwise current of 12.0 A, as viewed from above, and the outer wire has a diameter of 30.0 cm. What must be the magnitude and direction (as viewed from above) of the current in the outer wire so that the net magnetic field due to this combination of wires is zero at the common center of the wires?

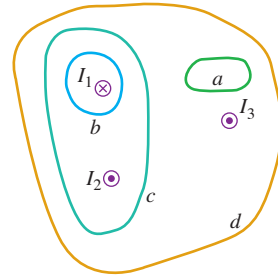
28.36 •• A closely wound coil has a radius of 6.00 cm and carries a current of 2.50 A. How many turns must it have if, at a point on the coil axis 6.00 cm from the center of the coil, the magnetic field is 6.39×10^{-4} T?

Section 28.6 Ampere's Law

28.37 • A closed curve encircles several conductors. The line integral $\oint \vec{B} \cdot d\vec{l}$ around this curve is $3.83 \times 10^{-4} \text{ T} \cdot \text{m}$. (a) What is the net current in the conductors? (b) If you were to integrate around the curve in the opposite direction, what would be the value of the line integral? Explain.

28.38 • Figure E28.38 shows, in cross section, several conductors that carry currents through the plane of the figure. The currents have the magnitudes $I_1 = 4.0 \text{ A}$, $I_2 = 6.0 \text{ A}$, and $I_3 = 2.0 \text{ A}$, and the directions shown. Four paths, labeled a through d , are shown. What is the line integral $\oint \vec{B} \cdot d\vec{l}$ for each path? Each integral involves going around the path in the counterclockwise direction. Explain your answers.

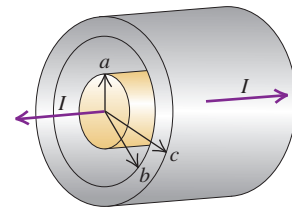
Figure E28.38



Section 28.7 Applications of Ampere's Law

28.39 • **Coaxial Cable.** A solid conductor with radius a is supported by insulating disks on the axis of a conducting tube with inner radius b and outer radius c (Fig. E28.39). The central conductor and tube carry equal currents I in opposite directions. The currents are distributed uniformly over the cross sections of each conductor. Derive an expression for the magnitude of the magnetic field (a) at points outside the central, solid conductor but inside the tube ($a < r < b$) and (b) at points outside the tube ($r > c$).

Figure E28.39



28.40 • As a new electrical technician, you are designing a large solenoid to produce a uniform 0.150 T magnetic field near the center of the solenoid. You have enough wire for 4000 circular turns. This solenoid must be 55.0 cm long and 2.80 cm in diameter. What current will you need to produce the necessary field?

28.41 • Repeat Exercise 28.39 for the case in which the current in the central, solid conductor is I_1 , the current in the tube is I_2 , and these currents are in the same direction rather than in opposite directions.

28.42 • A 15.0-cm -long solenoid with radius 0.750 cm is closely wound with 600 turns of wire. The current in the windings is 8.00 A . Compute the magnetic field at a point near the center of the solenoid.

28.43 • A solenoid is designed to produce a magnetic field of 0.0270 T at its center. It has radius 1.40 cm and length 40.0 cm , and the wire can carry a maximum current of 12.0 A . (a) What minimum number of turns per unit length must the solenoid have? (b) What total length of wire is required?

28.44 • An ideal toroidal solenoid (see Example 28.10) has inner radius $r_1 = 15.0 \text{ cm}$ and outer radius $r_2 = 18.0 \text{ cm}$. The solenoid has 250 turns and carries a current of 8.50 A . What is the magnitude of the magnetic field at the following distances from the center of the torus: (a) 12.0 cm ; (b) 16.0 cm ; (c) 20.0 cm ?

28.45 • A magnetic field of 37.2 T has been achieved at the MIT Francis Bitter Magnet Laboratory. Find the current needed to achieve such a field (a) 2.00 cm from a long, straight wire; (b) at the center of a circular coil of radius 42.0 cm that has 100 turns; (c) near the center of a solenoid with radius 2.40 cm , length 32.0 cm , and 40,000 turns.

Section 28.8 Magnetic Materials

28.46 • A toroidal solenoid with 400 turns of wire and a mean radius of 6.0 cm carries a current of 0.25 A . The relative permeability of the core is 80. (a) What is the magnetic field in the core? (b) What part of the magnetic field is due to the magnetic moments of the atoms in the core?

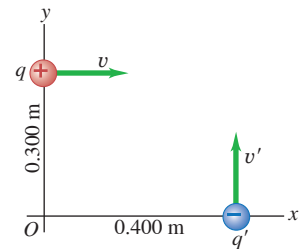
28.47 • A long solenoid with 60 turns of wire per centimeter carries a current of 0.15 A . The wire that makes up the solenoid is wrapped around a solid core of silicon steel ($K_m = 5200$). (The wire of the solenoid is jacketed with an insulator so that none of the current flows into the core.) (a) For a point inside the core, find the magnitudes of (i) the magnetic field $\vec{B}_0$ due to the solenoid current; (ii) the magnetization $\vec{M}$; (iii) the total magnetic field $\vec{B}$. (b) In a sketch of the solenoid and core, show the directions of the vectors $\vec{B}$, $\vec{B}_0$, and $\vec{M}$ inside the core.

28.48 • The current in the windings of a toroidal solenoid is 2.400 A . There are 500 turns, and the mean radius is 25.00 cm . The toroidal solenoid is filled with a magnetic material. The magnetic field inside the windings is found to be 1.940 T . Calculate (a) the relative permeability and (b) the magnetic susceptibility of the material that fills the toroid.

PROBLEMS

28.49 • A pair of point charges, $q = +8.00 \mu\text{C}$ and $q' = -5.00 \mu\text{C}$, are moving as shown in Fig. P28.49 with speeds $v = 9.00 \times 10^4 \text{ m/s}$ and $v' = 6.50 \times 10^4 \text{ m/s}$. When the charges are at the locations shown in the figure, what are the magnitude and direction of (a) the magnetic field produced at the origin and (b) the magnetic force that q' exerts on q ?

Figure P28.49



28.50 • At a particular instant, charge $q_1 = +4.80 \times 10^{-6} \text{ C}$ is at the point $(0, 0.250 \text{ m}, 0)$ and has velocity $\vec{v}_1 = (9.20 \times 10^5 \text{ m/s})\hat{i}$. Charge $q_2 = -2.90 \times 10^{-6} \text{ C}$ is at the point $(0.150 \text{ m}, 0, 0)$ and has velocity $\vec{v}_2 = (-5.30 \times 10^5 \text{ m/s})\hat{j}$. At this instant, what are the magnitude and direction of the magnetic force that q_1 exerts on q_2 ?

28.51 • A long, straight wire lies along the x -axis and carries current $I_1 = 2.00 \text{ A}$ in the $+\hat{x}$ -direction. A second wire lies in the xy -plane and is parallel to the x -axis at $y = +0.800 \text{ m}$. It carries current $I_2 = 6.00 \text{ A}$, also in the $+\hat{x}$ -direction. In addition to $y \rightarrow \pm\infty$, at what point on the y -axis is the resultant magnetic field of the two wires equal to zero?

28.52 • Repeat Problem 28.51 for I_2 in the $-\hat{x}$ -direction, with all the other quantities the same.

28.53 • We can estimate the strength of the magnetic field of a refrigerator magnet in the following way: Imagine the magnet as a collection of current-loop magnetic dipoles. (a) Derive the force between two current loops with radius R and current I separated by distance $d \ll R$. Very close to the wire its magnetic field is about the same as for an infinitely long wire, and Eq.(28.11) can be used. (b) Using Eq. (28.17), express the current I in terms of the magnetic field at the middle of the loop, and express the radius R in terms of the area of the loop. In this way, derive an expression for the force F between two identical current loops separated by a small distance d in terms of their mutual area A and center magnetic field B . (c) Rearrange your result to obtain an expression for the magnetic field of a dipole with area A in terms of the force F from an identical dipole separated by a small distance d . (d) Now notice that the force it takes to separate one magnet from your refrigerator is nearly the same as the force it takes to separate two magnets stuck together. Estimate that force F . (e) Estimate the area of a refrigerator magnet. (f) Assume that when these magnets are stuck together or to the refrigerator, they are separated by an effective distance $d = 25 \mu\text{m}$. Use the formula derived above to estimate the magnetic field strength of the magnet.

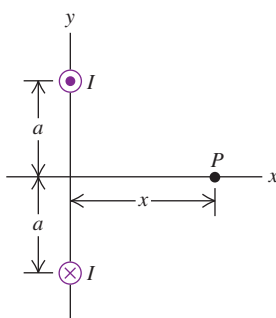
28.54 • A long, straight wire carries a current of 8.60 A. An electron is traveling in the vicinity of the wire. At the instant when the electron is 4.50 cm from the wire and traveling at a speed of 6.00×10^4 m/s directly toward the wire, what are the magnitude and direction (relative to the direction of the current) of the force that the magnetic field of the current exerts on the electron?

28.55 • CP A long, straight wire carries a 13.0 A current. An electron is fired parallel to this wire with a velocity of 250 km/s in the same direction as the current, 2.00 cm from the wire. (a) Find the magnitude and direction of the electron's initial acceleration. (b) What should be the magnitude and direction of a uniform electric field that will allow the electron to continue to travel parallel to the wire? (c) Is it necessary to include the effects of gravity? Justify your answer.

28.56 •• An electron is moving in the vicinity of a long, straight wire that lies along the x -axis. The wire has a constant current of 9.00 A in the $-x$ -direction. At an instant when the electron is at point $(0, 0.200 \text{ m}, 0)$ and the electron's velocity is $\vec{v} = (5.00 \times 10^4 \text{ m/s})\hat{i} - (3.00 \times 10^4 \text{ m/s})\hat{j}$, what is the force that the wire exerts on the electron? Express the force in terms of unit vectors, and calculate its magnitude.

28.57 •• CP (a) Determine the transmission power P of your cell phone. (This information is available online.) (b) A typical cell phone battery supplies a 1.5 V potential. If your phone battery supplies the power P , what is a good estimate of the current supplied by the battery? (c) Estimate the width of your head. (d) Estimate the diameter of the phone speaker that goes next to your ear. Model the current in the speaker as a current loop with the same diameter as the speaker. Use these values to estimate the magnetic field generated by your phone midway between the ears when it is held near one ear. (e) How does your answer compare to the earth's field, which is about $50 \mu\text{T}$?

28.58 •• Figure P28.58 shows an end view of two long, parallel wires perpendicular to the xy -plane, each carrying a current I but in opposite directions. (a) Copy the diagram, and draw vectors to show the $\vec{B}$ field of each wire and the net $\vec{B}$ field at point P . (b) Derive the expression for the magnitude of $\vec{B}$ at any point on the x -axis in terms of the x -coordinate of the point. What is the direction of $\vec{B}$? (c) Graph the magnitude of $\vec{B}$ at points on the x -axis. (d) At what value of x is the magnitude of $\vec{B}$ a maximum? (e) What is the magnitude of $\vec{B}$ when $x \gg a$?



28.59 • Two long, straight, parallel wires are 1.00 m apart (Fig. P28.59). The wire on the left carries a current I_1 of 6.00 A into the plane of the paper. (a) What must the magnitude and direction of the current I_2 be for the net field at point P to be zero? (b) Then what are the magnitude and direction of the net field at Q ? (c) Then what is the magnitude of the net field at S ?

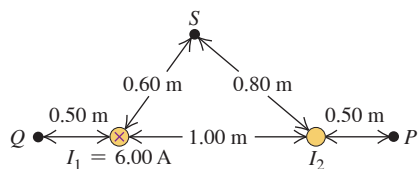
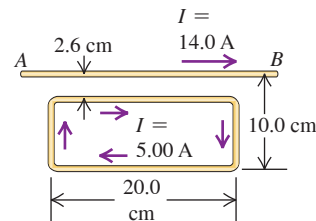


Figure P28.59

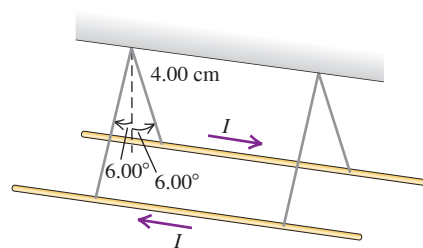
28.60 • The long, straight wire AB shown in Fig. P28.60 carries a current of 14.0 A. The rectangular loop whose long edges are parallel to the wire carries a current of 5.00 A. Find the magnitude and direction of the net force exerted on the loop by the magnetic field of the wire.

Figure P28.60



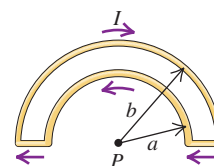
28.61 ••• CP Two long, parallel wires hang by 4.00-cm-long cords from a common axis (Fig. P28.61). The wires have a mass per unit length of 0.0125 kg/m and carry the same current in opposite directions. What is the current in each wire if the cords hang at an angle of 6.00° with the vertical?

Figure P28.61



28.62 • The wire semicircles shown in Fig. P28.62 have radii a and b . Calculate the net magnetic field (magnitude and direction) that the current in the wires produces at point P .

Figure P28.62



28.63 • A long, straight, solid cylinder, oriented with its axis in the z -direction, carries a current whose current density is $\vec{J}$. The current density, although symmetric about the cylinder axis, is not constant and varies according to the relationship

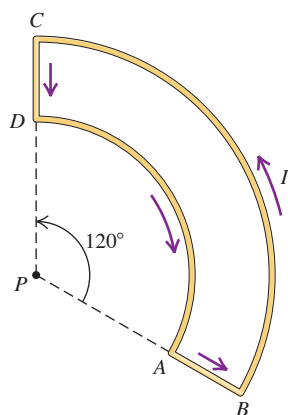
$$\vec{J} = \left(\frac{b}{r}\right)e^{(r-a)/\delta}\hat{k} \quad \text{for } r \leq a$$

$$= 0 \quad \text{for } r \geq a$$

where the radius of the cylinder is $a = 5.00$ cm, r is the radial distance from the cylinder axis, b is a constant equal to 600 A/m, and δ is a constant equal to 2.50 cm. (a) Let I_0 be the total current passing through the entire cross section of the wire. Obtain an expression for I_0 in terms of b , δ , and a . Evaluate your expression to obtain a numerical value for I_0 . (b) Using Ampere's law, derive an expression for the magnetic field $\vec{B}$ in the region $r \geq a$. Express your answer in terms of I_0 rather than b . (c) Obtain an expression for the current I contained in a circular cross section of radius $r \leq a$ and centered at the cylinder axis. Express your answer in terms of I_0 rather than b . (d) Using Ampere's law, derive an expression for the magnetic field $\vec{B}$ in the region $r \leq a$. (e) Evaluate the magnitude of the magnetic field at $r = \delta$, $r = a$, and $r = 2a$.

28.64 •• Calculate the magnetic field (magnitude and direction) at a point P due to a current $I = 12.0$ A in the wire shown in **Fig. P28.64**. Segment BC is an arc of a circle with radius 30.0 cm, and point P is at the center of curvature of the arc. Segment DA is an arc of a circle with radius 20.0 cm, and point P is at its center of curvature. Segments CD and AB are straight lines of length 10.0 cm each.

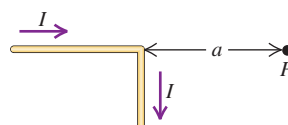
Figure P28.64



28.65 • CALC A long, straight wire with a circular cross section of radius R carries a current I . Assume that the current density is not constant across the cross section of the wire, but rather varies as $J = ar$, where a is a constant. (a) By the requirement that J integrated over the cross section of the wire gives the total current I , calculate the constant a in terms of I and R . (b) Use Ampere's law to calculate the magnetic field $B(r)$ for (i) $r \leq R$ and (ii) $r \geq R$. Express your answers in terms of I .

28.66 • CALC The wire shown in **Fig. P28.66** is infinitely long and carries a current I . Calculate the magnitude and direction of the magnetic field that this current produces at point P .

Figure P28.66



28.67 • CALC A long, straight, solid cylinder, oriented with its axis in the z -direction, carries a current whose current density is $\vec{J}$. The current density, although symmetric about the cylinder axis, is not constant but varies according to the relationship

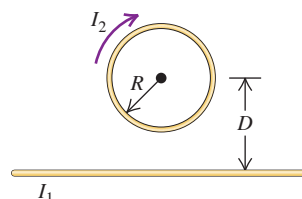
$$\vec{J} = \frac{2I_0}{\pi a^2} \left[1 - \left(\frac{r}{a} \right)^2 \right] \hat{k} \quad \text{for } r \leq a$$

$$= \mathbf{0} \quad \text{for } r \geq a$$

where a is the radius of the cylinder, r is the radial distance from the cylinder axis, and I_0 is a constant having units of amperes. (a) Show that I_0 is the total current passing through the entire cross section of the wire. (b) Using Ampere's law, derive an expression for the magnitude of the magnetic field $\vec{B}$ in the region $r \geq a$. (c) Obtain an expression for the current I contained in a circular cross section of radius $r \leq a$ and centered at the cylinder axis. (d) Using Ampere's law, derive an expression for the magnitude of the magnetic field $\vec{B}$ in the region $r \leq a$. How do your results in parts (b) and (d) compare for $r = a$?

28.68 • A circular loop has radius R and carries current I_2 in a clockwise direction (**Fig. P28.68**). The center of the loop is a distance D above a long, straight wire. What are the magnitude and direction of the current I_1 in the wire if the magnetic field at the center of the loop is zero?

Figure P28.68

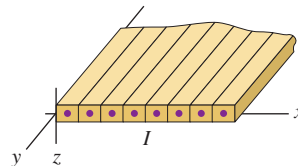


28.69 • An Infinite Current Sheet.

Long, straight conductors with square cross sections and each carrying current I are laid side by side to form an infinite current sheet (**Fig. P28.69**). The conductors lie in the xy -plane, are parallel to the y -axis, and carry current in the $+y$ -direction. There are n conductors per unit length measured along the x -axis.

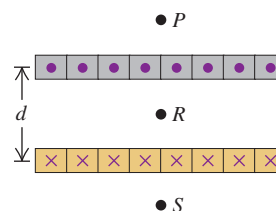
(a) What are the magnitude and direction of the magnetic field a distance a below the current sheet? (b) What are the magnitude and direction of the magnetic field a distance a above the current sheet?

Figure P28.69



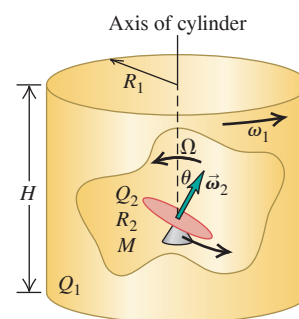
28.70 • Long, straight conductors with square cross section, each carrying current I , are laid side by side to form an infinite current sheet with current directed out of the plane of the page (**Fig. P28.70**). A second infinite current sheet is a distance d below the first and is parallel to it. The second sheet carries current into the plane of the page. Each sheet has n conductors per unit length. (Refer to Problem 28.69.) Calculate the magnitude and direction of the net magnetic field at (a) point P (above the upper sheet); (b) point R (midway between the two sheets); (c) point S (below the lower sheet).

Figure P28.70



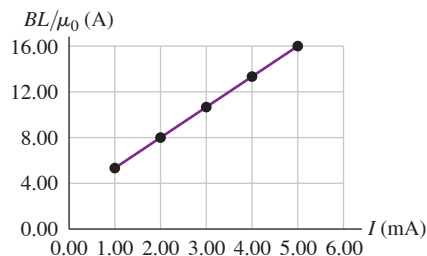
28.71 ••• CP A cylindrical shell with radius R_1 and height H has charge Q_1 and rotates around its axis with angular speed ω_1 , as shown in **Fig. P28.71**. Inside the cylinder, far from its edges, sits a very small disk with radius R_2 , mass M , and charge Q_2 mounted on a pivot, spinning with a large angular velocity $\vec{\omega}_2$ and oriented at angle θ with respect to the axis of the cylinder. The center of the disk is on the axis of the cylinder. The magnetic interaction between the cylinder and the disk causes a precession of the axis of the disk. (a) What is the magnitude of the enclosed current I_{encl} surrounded by a loop that has one vertical side that is along the axis of the cylinder and extends beyond the top and bottom of the cylinder? The other vertical side of the loop is very far outside the cylinder. (b) Assume the field is uniform within the cylinder and use Ampere's law to find the magnetic field at the center of the disk. (c) The magnetic moment of the disk has magnitude $\mu = \frac{1}{4}Q_2\omega_2 R_2^2$. What is the magnitude of the torque exerted on the disk? (d) What is the magnitude of the angular momentum of the disk?

Figure P28.71



28.72 •• DATA As a summer intern at a research lab, you are given a long solenoid that has two separate windings that are wound close together, in the same direction, on the same hollow cylindrical form. You must determine the number of turns in each winding. The solenoid has length $L = 40.0$ cm and diameter 2.80 cm. You let a 2.00 mA current flow in winding 1 and vary the current I in winding 2; both currents flow in the same direction. Then you measure the magnetic-field magnitude B at the center of the solenoid as a function of I . You plot your results as BL/μ_0 versus I . The graph in **Fig. P28.72** shows the best-fit straight line to your data. (a) Explain why the data plotted in this way should fall close to a straight line. (b) Use **Fig. P28.72** to calculate N_1 and N_2 , the number of turns in windings 1 and 2. (c) If the current in winding 1 remains 2.00 mA in its original direction and winding 2 has $I = 5.00$ mA in the opposite direction, what is B at the center of the solenoid?

Figure P28.72



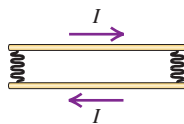
28.73 •• DATA You use a teslameter (a Hall-effect device) to measure the magnitude of the magnetic field at various distances from a long, straight, thick cylindrical copper cable that is carrying a large constant current. To exclude the earth's magnetic field from the measurement, you first set the meter to zero. You then measure the magnetic field B at distances x from the surface of the cable and obtain these data:

x (cm)	2.0	4.0	6.0	8.0	10.0
B (mT)	0.406	0.250	0.181	0.141	0.116

(a) You think you remember from your physics course that the magnetic field of a wire is inversely proportional to the distance from the wire. Therefore, you expect that the quantity Bx from your data will be constant. Calculate Bx for each data point in the table. Is Bx constant for this set of measurements? Explain. (b) Graph the data as x versus $1/B$. Explain why such a plot lies close to a straight line. (c) Use the graph in part (b) to calculate the current I in the cable and the radius R of the cable.

28.74 ••• DATA A pair of long, rigid metal rods, each of length 0.50 m, lie parallel to each other on a frictionless table. Their ends are connected by identical, very lightweight conducting springs with unstretched length l_0 and force constant k (Fig. P28.74). When a current I runs through the circuit consisting of the rods and springs, the springs stretch. You measure the distance x each spring stretches for certain values of I . When $I = 8.05$ A, you measure that $x = 0.40$ cm. When $I = 13.1$ A, you find $x = 0.80$ cm. In both cases the rods are much longer than the stretched springs, so it is accurate to use Eq. (28.11) for two infinitely long, parallel conductors. (a) From these two measurements, calculate l_0 and k . (b) If $I = 12.0$ A, what distance x will each spring stretch? (c) What current is required for each spring to stretch 1.00 cm?

Figure P28.74

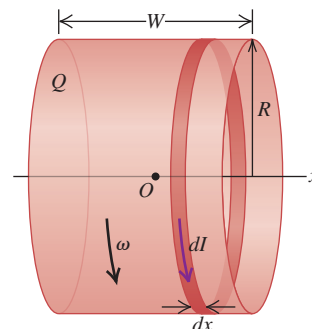


28.75 ••• CP CALC A plasma is a gas of ionized (charged) particles. When plasma is in motion, magnetic effects “squeeze” its volume, inducing inward pressure known as a pinch. Consider a cylindrical tube of plasma with radius R and length L moving with velocity $\vec{v}$ along its axis. If there are n ions per unit volume and each ion has charge q , we can determine the pressure felt by the walls of the cylinder. (a) What is the volume charge density ρ in terms of n and q ? (b) The thickness of the cylinder “surface” is $n^{-1/3}$. What is the surface charge density σ in terms of n and q ? (c) The current density inside the cylinder is $\vec{J} = \rho\vec{v}$. Use this result along with Ampere’s law to determine the magnetic field on the surface of the cylinder. Denote the circumferential unit vector as $\hat{\phi}$. (d) The width of a differential strip of surface current is $R d\phi$. What is the differential current dI_{surface} that flows along this strip? (e) What differential force is felt by this strip due to the magnetic field generated by the volume current? (f) Integrate to determine the total force on the walls of the cylinder; then divide by the wall area to obtain the pressure in terms of n , q , R , and v . (g) If a plasma cylinder with radius 2.0 cm has a charge density of 8.0×10^{16} ions/cm³, where each ion has a charge of $e = 1.6 \times 10^{-19}$ C and is moving axially with a speed of 20.0 m/s, what is its pinch pressure?

CHALLENGE PROBLEMS

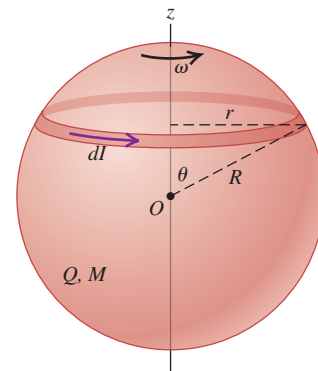
28.76 ••• CALC A cylindrical shell with radius R and length W carries a uniform charge Q and rotates about its axis with angular speed ω . The center of the cylinder lies at the origin O and its axis is coincident with the x -axis, as shown in Fig. P28.76. (a) What is the charge density σ ? (b) What is the differential current dI on a circular strip of the cylinder centered at x and with width dx ? (c) Use Eq. (28.15) to write an expression for the differential magnetic field $d\vec{B}$ at the origin due to this strip. (d) Integrate to determine the magnetic field at the origin.

Figure P28.76



28.77 ••• CALC When a rigid charge distribution with charge Q and mass M rotates about an axis, its magnetic moment $\vec{\mu}$ is linearly proportional to its angular momentum $\vec{L}$, with $\vec{\mu} = \gamma\vec{L}$. The constant of proportionality γ is called the gyromagnetic ratio of the object. We can write $\gamma = g(Q/2M)$, where g is a dimensionless number called the g -factor of the object. Consider a spherical shell with mass M and uniformly distributed charge Q centered on the origin O and rotating about the z -axis with angular speed ω . (a) A thin slice with latitude θ measured with respect to the positive z -axis describes a current loop with width $R d\theta$ and radius $r = R \sin\theta$, as shown in Fig. P28.77. What is the differential current dI carried by this loop, in terms of Q , ω , R , θ , and $d\theta$? (b) The differential magnetic moment contributed by that slice is $d\vec{\mu} = A dI$, where $A = \pi r^2$ is the area enclosed by the loop. Express the differential magnetic moment in terms of Q , ω , R , θ , and $d\theta$. (c) Integrate over θ to determine the magnetic moment $\vec{\mu}$. (d) What is the magnitude of the angular momentum $\vec{L}$? (e) Determine the gyromagnetic ratio γ . (f) What is the g -factor for a spherical shell?

Figure P28.77



28.78 ••• CALC The law of Biot and Savart in Eq. (28.7) generalizes to the case of surface currents as

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{\sigma \vec{v} \times \hat{r}}{r^2} da$$

where σ is the local charge density, $\vec{v}$ is the local velocity, and da is a differential area element. Re-visit Challenge Problem 28.76 and use the above equation as an alternative means to derive the magnetic field at the center of the cylinder. Use the following steps: (a) Write the charge density σ . (b) The origin is at the center of the cylinder. What is the vector $\vec{r}$ that points from the element with coordinates $(x, y, z) = (x, R \cos\phi, R \sin\phi)$ to the origin? (c) What is the velocity $\vec{v}$ of the element? (d) What is the vector product $\vec{v} \times \hat{r}$? (e) An area element on the cylinder may be written as $da = R dx d\phi$. Use this and the previously established information to write the generalized law of Biot and Savart as a double integral. Evaluate the integral to determine the magnetic field $\vec{B}$ at the center of the cylinder. (f) Is your result consistent with your result in Challenge Problem 28.76?

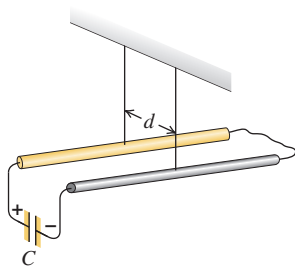
28.79 ••• CP Two long, straight conducting wires with linear mass density λ are suspended from cords so that they are each horizontal, parallel to each other, and a distance d apart. The back ends of the wires are connected to each other by a slack, low-resistance connecting wire. A charged capacitor (capacitance C) is now added to the system; the positive plate of the capacitor (initial charge $+Q_0$) is connected to the front end of one of the wires, and the negative plate of the capacitor (initial charge $-Q_0$) is connected to the front end of the other wire (**Fig. P28.79**). Both of these connections are also made by slack, low-resistance wires. When the connection is made, the wires are pushed aside by the repulsive force between the wires, and each wire has an initial horizontal velocity of magnitude v_0 . Assume that the time constant for the capacitor to discharge is negligible compared to the time it takes for any appreciable displacement in the position of the wires to occur. (a) Show that the initial speed v_0 of either wire is given by

$$v_0 = \frac{\mu_0 Q_0^2}{4\pi\lambda R C d}$$

where R is the total resistance of the circuit. (b) To what height h will each wire rise as a result of the circuit connection?

28.80 ••• A wide, long, insulating belt has a uniform positive charge per unit area σ on its upper surface. Rollers at each end move the belt to the right at a constant speed v . Calculate the magnitude and direction of the magnetic field produced by the moving belt at a point just above its surface. (*Hint:* At points near the surface and far from its edges or ends, the moving belt can be considered to be an infinite current sheet like that in Problem 28.69.)

Figure P28.79



MCAT-STYLE PASSAGE PROBLEMS

BIO Studying Magnetic Bacteria. Some types of bacteria contain chains of ferromagnetic particles parallel to their long axis. The chains act like small bar magnets that align these *magnetotactic* bacteria with the earth's magnetic field. In one experiment to study the response of such bacteria to magnetic fields, a solenoid is constructed with copper wire 1.0 mm in diameter, evenly wound in a single layer to form a helical coil of length 40 cm and diameter 12 cm. The wire has a very thin layer of insulation, and the coil is wound so that adjacent turns are just touching. The solenoid, which generates a magnetic field, is in an enclosure that shields it from other magnetic fields. A sample of magnetotactic bacteria is placed inside the solenoid. The torque on an individual bacterium in the solenoid's magnetic field is proportional to the magnitude of the magnetic field and to the sine of the angle between the long axis of the bacterium and the magnetic-field direction.

28.81 What current is needed in the wire so that the magnetic field experienced by the bacteria has a magnitude of $150 \mu\text{T}$? (a) 0.095 A ; (b) 0.12 A ; (c) 0.30 A ; (d) 14 A .

28.82 To use a larger sample, the experimenters construct a solenoid that has the same length, type of wire, and loop spacing but twice the diameter of the original. How does the maximum possible magnetic torque on a bacterium in this new solenoid compare with the torque the bacterium would have experienced in the original solenoid? Assume that the currents in the solenoids are the same. The maximum torque in the new solenoid is (a) twice that in the original one; (b) half that in the original one; (c) the same as that in the original one; (d) one-quarter that in the original one.

28.83 The solenoid is removed from the enclosure and then used in a location where the earth's magnetic field is $50 \mu\text{T}$ and points horizontally. A sample of bacteria is placed in the center of the solenoid, and the same current is applied that produced a magnetic field of $150 \mu\text{T}$ in the lab. Describe the field experienced by the bacteria: The field (a) is still $150 \mu\text{T}$; (b) is now $200 \mu\text{T}$; (c) is between 100 and $200 \mu\text{T}$, depending on how the solenoid is oriented; (d) is between 50 and $150 \mu\text{T}$, depending on how the solenoid is oriented.

ANSWERS

Chapter Opening Question?

(iv) There would be *no* change in the magnetic field strength. From Example 28.9 (Section 28.7), the field inside a solenoid has magnitude $B = \mu_0 n I$, where n is the number of turns of wire per unit length. Joining two solenoids end to end doubles both the number of turns and the length, so the number of turns per unit length is unchanged.

Key Example VARIATION Problems

VP28.2.1 (a) $(3.20 \times 10^{-15} \text{ T})\hat{k}$ (b) $(-8.00 \times 10^{-16} \text{ T})\hat{j}$ (c) zero (d) $(1.13 \times 10^{-15} \text{ T})\hat{k}$

VP28.2.2 (a) $1.13 \times 10^{-28} \text{ N}$ in the $-x$ -direction (toward the proton) (b) $1.13 \times 10^{-28} \text{ N}$ in the $+x$ -direction (toward the electron)

VP28.2.3 (a) $(0.600)\hat{i} + (0.800)\hat{j}$ (b) $(-2.88 \times 10^{-11} \text{ T})\hat{k}$

VP28.2.4 $(9.83 \times 10^{-23} \text{ N})\hat{k}$

VP28.5.1 2.24 A

VP28.5.2 (a) $(-6.00 \times 10^{-5} \text{ T})\hat{i}$ (b) $(-4.67 \times 10^{-5} \text{ T})\hat{i}$ (c) $(8.67 \times 10^{-5} \text{ T})\hat{i}$

VP28.5.3 (a) 5.00 A (b) 1.00 A

VP28.5.4 $3.29 \times 10^4 \text{ A}$, upward

VP28.10.1 (a) $4.94 \times 10^{-6} \text{ T}$ (b) $8.89 \times 10^{-6} \text{ T}$ (c) $6.67 \times 10^{-6} \text{ T}$

VP28.10.2 (a) $B = 0$ for $r < R_1$, $B = (\mu_0 J/2r)(r^2 - R_1^2)$ for $R_2 > r > R_1$, $B = (\mu_0 J/2r)(R_2^2 - R_1^2)$ for $r > R_2$ (b) $r = R_2$

VP28.10.3 3.18 A

VP28.10.4 (a) 833 (b) $2.33 \times 10^{-3} \text{ T}$ maximum, $1.75 \times 10^{-3} \text{ T}$ minimum

Bridging Problem

$$B = \frac{\mu_0 n Q}{a}$$



? The card reader at your bank's cash machine scans the information that is coded in a magnetic pattern on the back of your card. Why must you remove the card quickly rather than hold it motionless in the card reader's slot? (i) To maximize the magnetic force on the card; (ii) to maximize the magnetic force on the mobile charges in the card reader; (iii) to generate an electric force on the card; (iv) to generate an electric force on the mobile charges in the card reader.

29 Electromagnetic Induction

Almost every modern device or machine, from a computer to a washing machine to a power drill, has electric circuits at its heart. We learned in Chapter 25 that an electromotive force (emf) is required for a current to flow in a circuit; in Chapters 25 and 26 we almost always took the source of emf to be a battery. But for most devices that you plug into a wall socket, the source of emf is *not* a battery but an electric generating station. Such a station produces electrical energy by converting other forms of energy: gravitational potential energy at a hydroelectric plant, chemical energy in a coal-, gas-, or oil-fired plant, nuclear energy at a nuclear plant. But how is this energy conversion done?

The answer is a phenomenon known as *electromagnetic induction*: If the magnetic flux through a circuit changes, an emf and a current are induced in the circuit. In a power-generating station, magnets move relative to coils of wire to produce a changing magnetic flux in the coils and hence an emf.

The central principle of electromagnetic induction is *Faraday's law*. This law relates induced emf to changing magnetic flux in any loop, including a closed circuit. We also discuss Lenz's law, which helps us to predict the directions of induced emfs and currents. These principles will allow us to understand electrical energy-conversion devices such as motors, generators, and transformers.

Electromagnetic induction tells us that a time-varying magnetic field can act as a source of electric field. We'll also see how a time-varying *electric* field can act as a source of *magnetic* field. These remarkable results form part of a neat package of formulas, called *Maxwell's equations*, that describe the behavior of electric and magnetic fields in general. Maxwell's equations pave the way toward an understanding of electromagnetic waves, the topic of Chapter 32.

LEARNING OUTCOMES

In this chapter, you'll learn...

- 29.1 The experimental evidence that a changing magnetic field induces an emf.
- 29.2 How Faraday's law relates the induced emf in a loop to the change in magnetic flux through the loop.
- 29.3 How to determine the direction of an induced emf.
- 29.4 How to calculate the emf induced in a conductor moving through a magnetic field.
- 29.5 How a changing magnetic flux generates a circulating electric field.
- 29.6 How eddy currents arise in a metal that moves in a magnetic field.
- 29.7 The four fundamental equations that completely describe both electricity and magnetism.
- 29.8 The remarkable electric and magnetic properties of superconductors.

You'll need to review...

- 23.1 Conservative electric fields.
- 25.4 Electromotive force (emf).
- 27.3, 27.8, 27.9 Magnetic flux; direct-current motors; Hall effect.
- 28.5–28.7 Magnetic field of a current loop and solenoid; Ampere's law.

BIO APPLICATION Exploring the Brain with Induced emfs Transcranial magnetic stimulation (TMS) is a technique for studying the function of various parts of the brain. A coil held to the subject's head carries a varying electric current, and so produces a varying magnetic field. This field causes an induced emf, and that triggers electric activity in the region of the brain underneath the coil. By observing how the TMS subject responds (for instance, which muscles move as a result of stimulating a certain part of the brain), a physician can test for various neurological conditions.



29.1 INDUCTION EXPERIMENTS

During the 1830s, several pioneering experiments with magnetically induced emf were carried out in England by Michael Faraday and in the United States by Joseph Henry (1797–1878). **Figure 29.1** shows several examples. In Fig. 29.1a, a coil of wire is connected to a galvanometer. When the nearby magnet is stationary, the meter shows no current. This isn't surprising; there is no source of emf in the circuit. But when we *move* the magnet either toward or away from the coil, the meter shows current in the circuit, but *only* while the magnet is moving (Fig. 29.1b). If we keep the magnet stationary and move the coil, we again detect a current during the motion. We call this an **induced current**, and the corresponding emf required to cause this current is called an **induced emf**.

In Fig. 29.1c we replace the magnet with a second coil connected to a battery. When the second coil is stationary, there is no current in the first coil. However, when we move the second coil toward or away from the first or move the first toward or away from the second, there is current in the first coil, but again *only* while one coil is moving relative to the other.

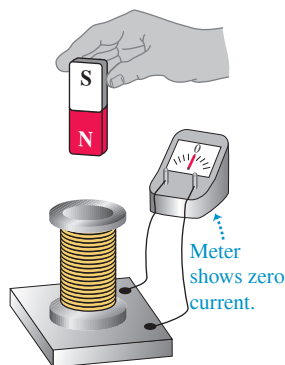
Finally, using the two-coil setup in Fig. 29.1d, we keep both coils stationary and vary the current in the second coil by opening and closing the switch. As we open or close the switch, there is a momentary current pulse in the first coil. The induced current in the first coil is present only while the current in the second coil is changing.

To explore further the common elements in these observations, let's consider a more detailed series of experiments (**Fig. 29.2**). We connect a coil of wire to a galvanometer and then place the coil between the poles of an electromagnet whose magnetic field we can vary. Here's what we observe:

1. When there is no current in the electromagnet, so that $\vec{B} = 0$, the galvanometer shows no current.
2. When the electromagnet is turned on, there is a momentary current through the meter as $\vec{B}$ increases.
3. When $\vec{B}$ levels off at a steady value, the current drops to zero.
4. With the coil in a horizontal plane, we squeeze it so as to decrease the cross-sectional area of the coil. The meter detects current only *during* the deformation, not before or after. When we increase the area to return the coil to its original shape, there is current in the opposite direction, but only while the area of the coil is changing.

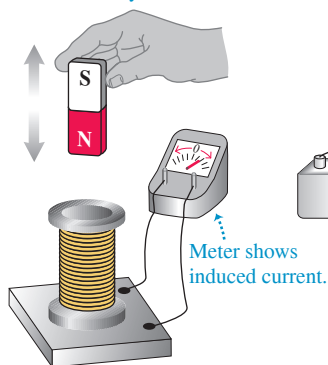
Figure 29.1 Demonstrating the phenomenon of induced current.

(a) A stationary magnet does NOT induce a current in a coil.

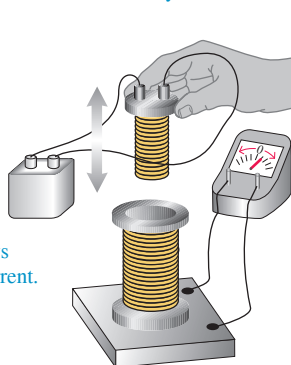


All these actions DO induce a current in the coil. What do they have in common?*

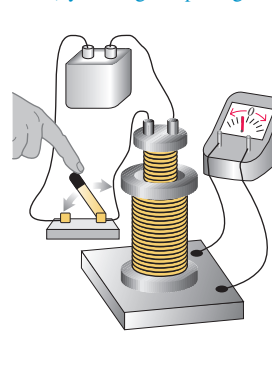
(b) Moving the magnet toward or away from the coil



(c) Moving a second, current-carrying coil toward or away from the coil



(d) Varying the current in the second coil (by closing or opening a switch)



*Answer: They cause the magnetic field through the coil to *change*.

5. If we rotate the coil a few degrees about a horizontal axis, the meter detects current during the rotation, in the same direction as when we decreased the area. When we rotate the coil back, there is a current in the opposite direction during this rotation.
6. If we jerk the coil out of the magnetic field, there is a current during the motion, in the same direction as when we decreased the area.
7. If we decrease the number of turns in the coil by unwinding one or more turns, there is a current during the unwinding, in the same direction as when we decreased the area. If we wind more turns onto the coil, there is a current in the opposite direction during the winding.
8. When the magnet is turned off, there is a momentary current in the direction opposite to the current when it was turned on.
9. The faster we carry out any of these changes, the greater the current.
10. If all these experiments are repeated with a coil that has the same shape but different material and different resistance, the current in each case is inversely proportional to the total circuit resistance. This shows that the induced emfs that are causing the current do not depend on the material of the coil but only on its shape and the magnetic field.

The common element in these experiments is changing *magnetic flux* Φ_B through the coil connected to the galvanometer. In each case the flux changes either because the magnetic field changes with time or because the coil is moving through a nonuniform magnetic field. What's more, in each case the induced emf is proportional to the *rate of change* of magnetic flux Φ_B through the coil. The *direction* of the induced emf depends on whether the flux is increasing or decreasing. If the flux is constant, there is no induced emf.

Induced emfs have a tremendous number of practical applications. If you are reading these words indoors, you are making use of induced emfs right now! At the power plant that supplies your neighborhood, an electric generator produces an emf by varying the magnetic flux through coils of wire. (In the next section we'll see how this is done.) This emf supplies the voltage between the terminals of the wall sockets in your home, and this voltage supplies the power to your reading lamp.

Magnetically induced emfs, just like the emfs discussed in Section 25.4, are the result of *nonelectrostatic* forces. We have to distinguish carefully between the electrostatic electric fields produced by charges (according to Coulomb's law) and the nonelectrostatic electric fields produced by changing magnetic fields. We'll return to this distinction later in this chapter and the next.

29.2 FARADAY'S LAW

The common element in all induction effects is changing magnetic flux through a circuit. Before stating the simple physical law that summarizes all of the kinds of experiments described in Section 29.1, let's first review the concept of magnetic flux Φ_B (which we introduced in Section 27.3). For an infinitesimal-area element $d\vec{A}$ in a magnetic field $\vec{B}$ (Fig. 29.3), the magnetic flux $d\Phi_B$ through the area is

$$d\Phi_B = \vec{B} \cdot d\vec{A} = B_{\perp} dA = B dA \cos \phi$$

where $B_{\perp}$ is the component of $\vec{B}$ perpendicular to the surface of the area element and ϕ is the angle between $\vec{B}$ and $d\vec{A}$. (As in Chapter 27, be careful to distinguish between two quantities named "phi," ϕ and Φ_B .) The total magnetic flux Φ_B through a finite area is the integral of this expression over the area:

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \int B dA \cos \phi \quad (29.1)$$

Figure 29.2 A coil in a magnetic field. When the $\vec{B}$ field is constant and the shape, location, and orientation of the coil do not change, no current is induced in the coil. A current is induced when any of these factors change.

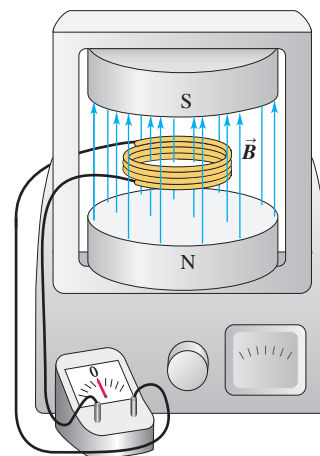
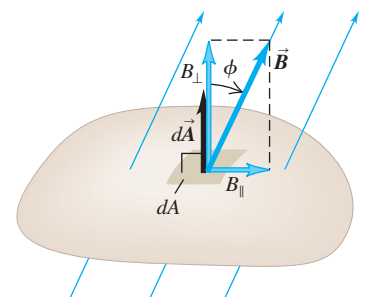


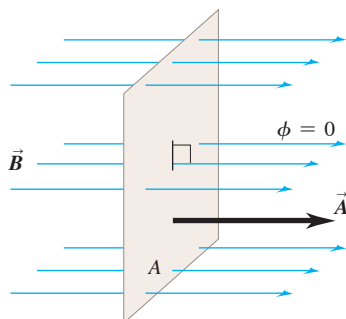
Figure 29.3 Calculating the magnetic flux through an area element.



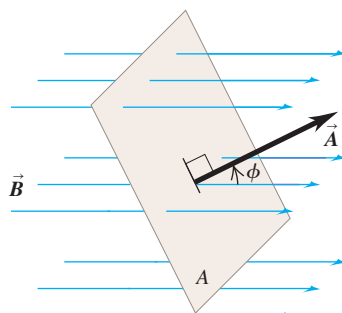
Magnetic flux through element of area $d\vec{A}$:
 $d\Phi_B = \vec{B} \cdot d\vec{A} = B_{\perp} dA = B dA \cos \phi$

Figure 29.4 Calculating the flux of a uniform magnetic field through a flat area. (Compare to Fig. 22.6, which shows the rules for calculating the flux of a uniform *electric* field.)

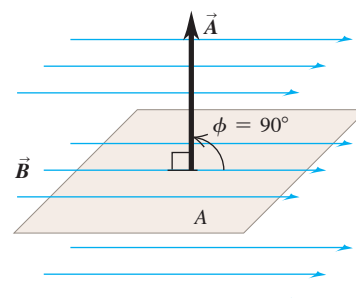
- Surface is face-on to magnetic field:
- $\vec{B}$ and $\vec{A}$ are parallel (the angle between $\vec{B}$ and $\vec{A}$ is $\phi = 0$).
 - The magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = BA$.



- Surface is tilted from a face-on orientation by an angle ϕ :
- The angle between $\vec{B}$ and $\vec{A}$ is ϕ .
 - The magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \phi$.



- Surface is edge-on to magnetic field:
- $\vec{B}$ and $\vec{A}$ are perpendicular (the angle between $\vec{B}$ and $\vec{A}$ is $\phi = 90^\circ$).
 - The magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = BA \cos 90^\circ = 0$.



If $\vec{B}$ is uniform over a flat area $\vec{A}$, then

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \phi \quad (29.2)$$

Figure 29.4 reviews the rules for using Eq. (29.2).

CAUTION Choosing the direction of $d\vec{A}$ or $\vec{A}$ In Eqs. (29.1) and (29.2) we must define the direction of the vector area $d\vec{A}$ or $\vec{A}$ unambiguously. There are always two directions perpendicular to any given area, and the sign of the magnetic flux through the area depends on which one we choose. For example, in Fig. 29.3 we chose $d\vec{A}$ to point upward, so ϕ is less than 90° and $\vec{B} \cdot d\vec{A}$ is positive. We could have chosen $d\vec{A}$ to point downward, in which case ϕ would have been greater than 90° and $\vec{B} \cdot d\vec{A}$ would have been negative. Both choices are equally good, but once we make a choice we must stick with it. ■

Faraday's law of induction states:

Faraday's law:
The induced emf $\mathcal{E}$ in a closed loop ... equals the negative of the time rate of change of magnetic flux through the loop.

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \quad (29.3)$$

To understand the negative sign, we have to introduce a sign convention for the induced emf $\mathcal{E}$. But first let's look at a simple example of this law in action.

EXAMPLE 29.1 Emf and current induced in a loop

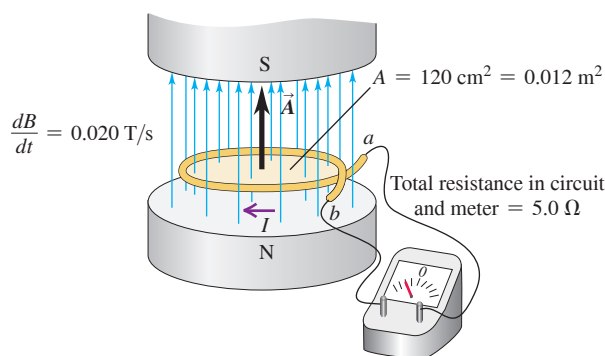
WITH ✓ VARIATION PROBLEMS

The magnetic field between the poles of the electromagnet in Fig. 29.5 is uniform at any time, but its magnitude is increasing at the rate of 0.020 T/s . The area of the conducting loop in the field is 120 cm^2 , and the total circuit resistance, including the meter, is 5.0Ω . (a) Find the induced emf and the induced current in the circuit. (b) If the loop is replaced by one made of an insulator, what effect does this have on the induced emf and induced current?

IDENTIFY and SET UP The magnetic flux Φ_B through the loop changes as the magnetic field changes. Hence there will be an induced emf $\mathcal{E}$ and an induced current I in the loop. We calculate Φ_B from Eq. (29.2), then find $\mathcal{E}$ by using Faraday's law. Finally, we calculate I from $\mathcal{E} = IR$, where R is the total resistance of the circuit that includes the loop.

EXECUTE (a) The area vector $\vec{A}$ for the loop is perpendicular to the plane of the loop; we take $\vec{A}$ to be vertically upward. Then $\vec{A}$ and $\vec{B}$ are

Figure 29.5 A stationary conducting loop in an increasing magnetic field.



parallel, and because $\vec{B}$ is uniform the magnetic flux through the loop is $\Phi_B = \vec{B} \cdot \vec{A} = BA \cos 0 = BA$. The area $A = 0.012 \text{ m}^2$ is constant, so the rate of change of magnetic flux is

$$\begin{aligned}\frac{d\Phi_B}{dt} &= \frac{d(BA)}{dt} = \frac{dB}{dt}A = (0.020 \text{ T/s})(0.012 \text{ m}^2) \\ &= 2.4 \times 10^{-4} \text{ V} = 0.24 \text{ mV}\end{aligned}$$

This, apart from a sign that we haven't discussed yet, is the induced emf $\mathcal{E}$. The corresponding induced current is

$$I = \frac{\mathcal{E}}{R} = \frac{2.4 \times 10^{-4} \text{ V}}{5.0 \, \Omega} = 4.8 \times 10^{-5} \text{ A} = 0.048 \text{ mA}$$

(b) By changing to an insulating loop, we've made the resistance of the loop very high. Faraday's law, Eq. (29.3), does not involve the resistance of the circuit in any way, so the induced *emf* does not change. But the *current* will be smaller, as given by the equation $I = \mathcal{E}/R$.

If the loop is made of a perfect insulator with infinite resistance, the induced current is zero. This situation is analogous to an isolated battery whose terminals aren't connected to anything: An emf is present, but no current flows.

EVALUATE We can verify unit consistency in this calculation by noting that the magnetic-force relationship $\vec{F} = q\vec{v} \times \vec{B}$ implies that the units of $\vec{B}$ are the units of force divided by the units of (charge times velocity): $1 \text{ T} = (1 \text{ N})/(1 \text{ C} \cdot \text{m/s})$. The units of magnetic flux are then $(1 \text{ T})(1 \text{ m}^2) = 1 \text{ N} \cdot \text{s} \cdot \text{m/C}$, and the rate of change of magnetic flux is $1 \text{ N} \cdot \text{m/C} = 1 \text{ J/C} = 1 \text{ V}$. Thus the unit of $d\Phi_B/dt$ is the volt, as required by Eq. (29.3). Also recall from Section 27.3 that the unit of magnetic flux is the weber (Wb): $1 \text{ T} \cdot \text{m}^2 = 1 \text{ Wb}$, so $1 \text{ V} = 1 \text{ Wb/s}$.

KEYCONCEPT When there is a change in the magnetic flux through a curve in space that forms a closed loop, an emf is induced in that loop. If the loop is made of a conductor, an induced current flows in response to the emf.

Direction of Induced emf

We can find the direction of an induced emf or current by using Eq. (29.3) together with some simple sign rules. Here's the procedure:

1. Define a positive direction for the vector area $\vec{A}$.
2. From the directions of $\vec{A}$ and the magnetic field $\vec{B}$, determine the sign of the magnetic flux Φ_B and its rate of change $d\Phi_B/dt$. **Figure 29.6** shows several examples.
3. Determine the sign of the induced emf or current. If the flux is increasing, so $d\Phi_B/dt$ is positive, then the induced emf or current is negative; if the flux is decreasing, $d\Phi_B/dt$ is negative and the induced emf or current is positive.
4. Finally, use your right hand to determine the direction of the induced emf or current. Curl the fingers of your right hand around the $\vec{A}$ vector, with your right thumb in the direction of $\vec{A}$. If the induced emf or current in the circuit is *positive*, it is in the same direction as your curled fingers; if the induced emf or current is *negative*, it is in the opposite direction.

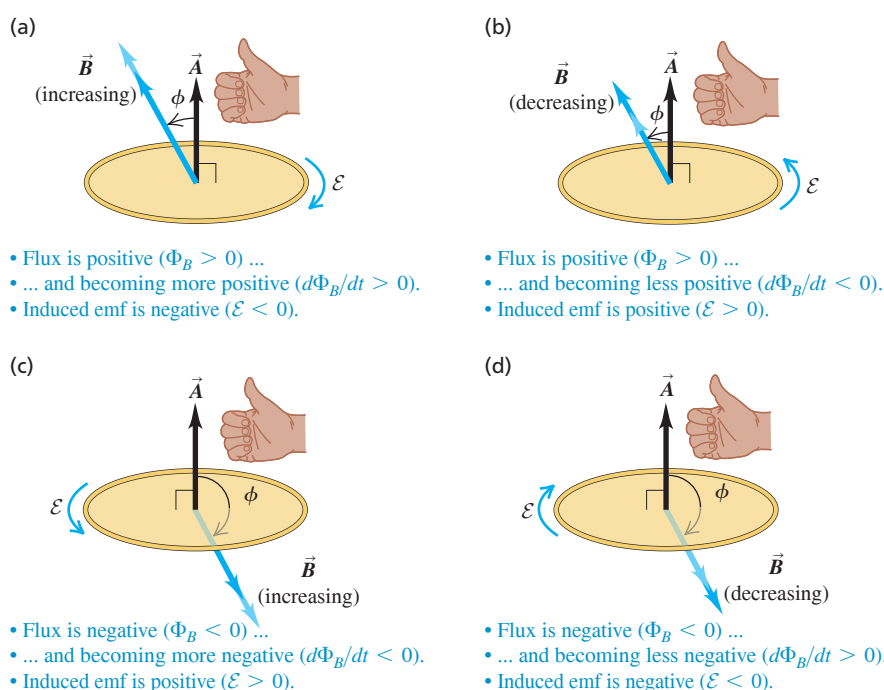


Figure 29.6 The magnetic flux is becoming (a) more positive, (b) less positive, (c) more negative, and (d) less negative. Therefore Φ_B is increasing in (a) and (d) and decreasing in (b) and (c). In (a) and (d) the emfs are negative (they are opposite to the direction of the curled fingers of your right hand when your right thumb points along $\vec{A}$). In (b) and (c) the emfs are positive (in the same direction as the curled fingers).

CAUTION Induced emfs are caused by changes in flux It can be tempting to think that *flux* is the cause of induced emf and that an induced emf will appear in a circuit whenever there is a magnetic field in the region bordered by the circuit. But Eq. (29.3) shows that only a *change* in flux through a circuit, not flux itself, can induce an emf in a circuit. If the flux through a circuit has a constant value, whether positive, negative, or zero, there is no induced emf. Note also that an induced emf will appear in a circuit if the flux is changed for any reason, including rotating the circuit relative to the magnetic field or changing the circuit's shape. **|**

In Example 29.1, in which $\vec{A}$ is upward, a positive $\mathcal{E}$ would be directed counterclockwise around the loop, as seen from above. Both $\vec{A}$ and $\vec{B}$ are upward in this example, so Φ_B is positive; the magnitude B is increasing, so $d\Phi_B/dt$ is positive. Hence by Eq. (29.3), $\mathcal{E}$ in Example 29.1 is *negative*—that is, *clockwise* around the loop, as seen from above.

If the loop in Fig. 29.5 is a conductor, the clockwise induced emf causes a clockwise induced current. This induced current produces an additional magnetic field through the loop, and the right-hand rule described in Section 28.5 shows that this field is *opposite* in direction to the increasing field produced by the electromagnet. This is an example of a general rule called *Lenz's law*, which says that any induction effect tends to oppose the change that caused it; in this case the change is the increase in the flux of the electromagnet's field through the loop. (We'll study Lenz's law in detail in the next section.)

You should check out the signs of the induced emfs and currents for the list of experiments in Section 29.1. For example, when the loop in Fig. 29.2 is in a constant field and we tilt it or squeeze it to *decrease* the flux through it, the induced emf and current are counterclockwise, as seen from above.

If a coil has N identical turns and if the flux varies at the same rate through each turn, the *total* rate of change through all turns is N times that for a single turn. If Φ_B is the flux through each turn, the total emf in a coil with N turns is

$$\mathcal{E} = -N \frac{d\Phi_B}{dt} \quad (29.4)$$

As we discussed in this chapter's introduction, induced emfs play an essential role in the generation of electrical power for commercial use. Several of the following examples explore different methods of producing emfs by changing the flux through a circuit.

PROBLEM-SOLVING STRATEGY 29.1 Faraday's Law

IDENTIFY the relevant concepts: Faraday's law applies when a magnetic flux is changing. To use the law, identify an area through which there is a flux of magnetic field. This will usually be the area enclosed by a loop made of a conducting material (though not always—see part (b) of Example 29.1). Identify the target variables.

SET UP the problem using the following steps:

1. Faraday's law relates the induced emf to the rate of change of magnetic flux. To calculate this rate of change, you first have to understand what is making the flux change. Is the conductor moving or changing orientation? Is the magnetic field changing?
2. The area vector $\vec{A}$ (or $d\vec{A}$) must be perpendicular to the plane of the area. You always have two choices of its direction; for example, if the area is in a horizontal plane, $\vec{A}$ could point up or down. Choose a direction and use it throughout the problem.

EXECUTE the solution as follows:

1. Calculate the magnetic flux from Eq. (29.2) if $\vec{B}$ is uniform over the area of the loop or Eq. (29.1) if it isn't uniform. Remember the direction you chose for the area vector.
2. Calculate the induced emf from Eq. (29.3) or (if your conductor has N turns in a coil) Eq. (29.4). Apply the sign rule (described just after Example 29.1) to determine the positive direction of emf.
3. If the circuit resistance is known, you can calculate the magnitude of the induced current I by using $\mathcal{E} = IR$.

EVALUATE your answer: Check your results for the proper units, and double-check that you have properly implemented the sign rules for magnetic flux and induced emf.

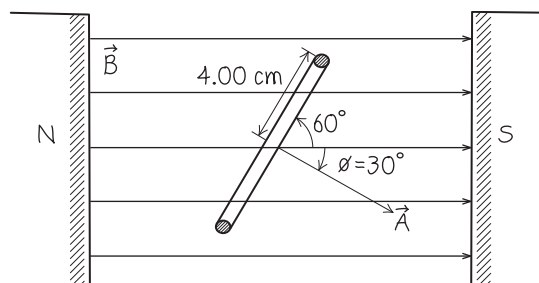
EXAMPLE 29.2 Magnitude and direction of an induced emf

WITH  **ARIATION PROBLEMS**

A 500-loop circular wire coil with radius 4.00 cm is placed between the poles of a large electromagnet. The magnetic field is uniform and makes an angle of 60° with the plane of the coil; it decreases at 0.200 T/s. What are the magnitude and direction of the induced emf?

IDENTIFY and SET UP Our target variable is the emf induced by a varying magnetic flux through the coil. The flux varies because the magnetic field decreases in amplitude. We choose the area vector $\vec{A}$ to be in the direction shown in **Fig. 29.7**. With this choice, the geometry is similar to that of Fig. 29.6b. That figure will help us determine the direction of the induced emf.

Figure 29.7 Our sketch for this problem.



EXECUTE The magnetic field is uniform over the loop, so we can calculate the flux from Eq. (29.2): $\Phi_B = BA \cos \phi$, where $\phi = 30^\circ$. In this expression, the only quantity that changes with time is the magnitude B of the field, so $d\Phi_B/dt = (dB/dt)A \cos \phi$.

CAUTION Remember how ϕ is defined You may have been tempted to say that $\phi = 60^\circ$ in this problem. If so, remember that ϕ is the angle between $\vec{A}$ and $\vec{B}$, *not* the angle between $\vec{B}$ and the plane of the loop. **I**

From Eq. (29.4), the induced emf in the coil of $N = 500$ turns is

$$\begin{aligned}\mathcal{E} &= -N \frac{d\Phi_B}{dt} = -N \frac{dB}{dt} A \cos \phi \\ &= -500(-0.200 \text{ T/s})\pi(0.0400 \text{ m})^2(\cos 30^\circ) = 0.435 \text{ V}\end{aligned}$$

The positive answer means that when you point your right thumb in the direction of area vector $\vec{A}$ (30° below field $\vec{B}$ in Fig. 29.7), the positive direction for $\mathcal{E}$ is in the direction of the curled fingers of your right hand. If you viewed the coil from the left in Fig. 29.7 and looked in the direction of $\vec{A}$, the emf would be clockwise.

EVALUATE If the ends of the wire are connected, the direction of current in the coil is in the same direction as the emf—that is, clockwise as seen from the left side of the coil. A clockwise current increases the magnetic flux through the coil, and therefore tends to oppose the decrease in total flux. This is an example of Lenz's law, which we'll discuss in Section 29.3.

KEYCONCEPT Faraday's law states that the induced emf in a coil is proportional to the rate of change of the magnetic flux through the coil and to the number of turns in the coil. The sign of the induced emf tells you its direction.

EXAMPLE 29.3 Generator I: A simple alternator

WITH VARIATION PROBLEMS

Figure 29.8a shows a simple *alternator*, a device that generates an emf. A rectangular loop is rotated with constant angular speed ω about the axis shown. The magnetic field $\vec{B}$ is uniform and constant. At time $t = 0$, $\phi = 0$. Determine the induced emf.

IDENTIFY and SET UP The magnetic field $\vec{B}$ and the loop area A are constant, but the flux through the loop varies because the loop rotates and so the angle ϕ between $\vec{B}$ and the area vector $\vec{A}$ changes (Fig. 29.8a). Because the angular speed is constant and $\phi = 0$ at $t = 0$, the angle as a function of time is $\phi = \omega t$.

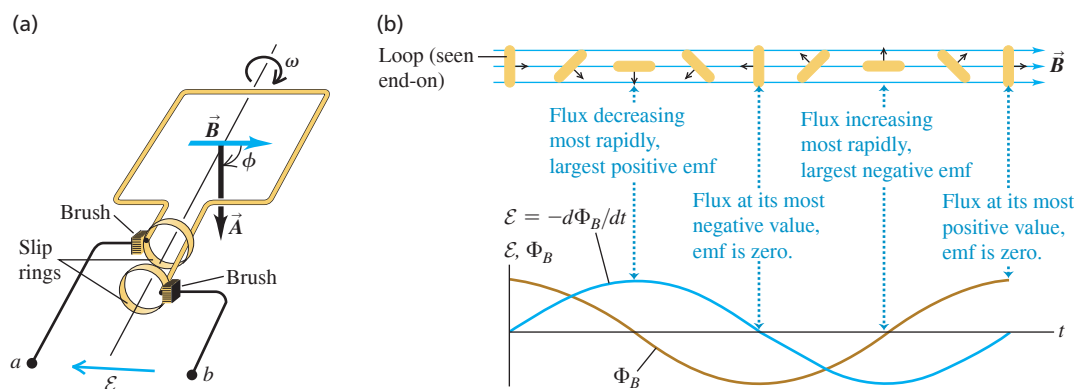
EXECUTE The magnetic field is uniform over the loop, so the magnetic flux is $\Phi_B = BA \cos \phi = BA \cos \omega t$. Hence, by Faraday's law [Eq. (29.3)] the induced emf is

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(BA \cos \omega t) = \omega BA \sin \omega t$$

EVALUATE The induced emf $\mathcal{E}$ varies sinusoidally with time (see Fig. 29.8b). When the plane of the loop is perpendicular to $\vec{B}$ ($\phi = 0$ or 180°), Φ_B reaches its maximum and minimum values. At these times, its instantaneous rate of change is zero and $\mathcal{E}$ is zero. Conversely, $\mathcal{E}$ reaches its maximum and minimum values when the plane of the loop is parallel to $\vec{B}$ ($\phi = 90^\circ$ or 270°) and Φ_B is changing most rapidly. We note that the induced emf does not depend on the *shape* of the loop, but only on its area.

We can use the alternator as a source of emf in an external circuit by using two *slip rings* that rotate with the loop, as shown in Fig. 29.8a. The rings slide against stationary contacts called *brushes*, which are connected to the output terminals a and b . Since the emf varies sinusoidally, the current that results in the circuit is an *alternating* current that also

Figure 29.8 (a) Schematic diagram of an alternator. A conducting loop rotates in a magnetic field, producing an emf. Connections from each end of the loop to the external circuit are made by means of that end's slip ring. The system is shown at the time when the angle $\phi = \omega t = 90^\circ$. (b) Graph of the flux through the loop and the resulting emf between terminals a and b , along with the corresponding positions of the loop during one complete rotation.



Continued

varies sinusoidally in magnitude and direction. The amplitude of the emf can be increased by increasing the rotation speed, the field magnitude, or the loop area or by using N loops instead of one, as in Eq. (29.4).

Alternators are used in automobiles to generate the currents in the ignition, the lights, and the entertainment system. The arrangement is a little different than in this example; rather than having a rotating loop in a magnetic field, the loop stays fixed and an electromagnet rotates. (The rotation is provided by a mechanical connection between the alternator and the engine.) But the result is the same; the flux through the loop varies sinusoidally, producing a sinusoidally varying emf. Larger alternators of this same type are used in electric power plants (**Fig. 29.9**).

KEYCONCEPT In an alternator, a coil rotates relative to a magnetic field. The flux through the coil varies sinusoidally with time, as does the emf induced in the coil. The more rapid the rotation, the greater the magnitudes of the induced emf and current.

Figure 29.9 A commercial alternator uses many loops of wire wound around a barrel-like structure called an armature. The armature and wire remain stationary while electromagnets rotate on a shaft (not shown) through the center of the armature. The resulting induced emf is far larger than would be possible with a single loop of wire.



EXAMPLE 29.4 Generator II: A dc generator and back emf in a motor

The alternator in Example 29.3 produces a sinusoidally varying emf and hence an alternating current. **Figure 29.10a** shows a *direct-current* (dc) *generator* that produces an emf that always has the same sign. The arrangement of split rings, called a *commutator*, reverses the connections to the external circuit at angular positions at which the emf reverses. **Figure 29.10b** shows the resulting emf. Commercial dc generators have a large number of coils and commutator segments, smoothing out the bumps in the emf so that the terminal voltage is not only one-directional but also practically constant. This brush-and-commutator arrangement is the same as that in the direct-current motor discussed in Section 27.8. The motor's *back emf* is just the emf induced by the changing magnetic flux through its rotating coil. Consider a motor with a square, 500-turn coil 10.0 cm on a side. If the magnetic field has magnitude 0.200 T, at what rotation speed is the *average* back emf of the motor equal to 112 V?

IDENTIFY and SET UP As far as the rotating loop is concerned, the situation is the same as in Example 29.3 except that we now have N turns of wire. Without the commutator, the emf would alternate between positive and negative values and have an average value of zero (see **Fig. 29.8b**). With the commutator, the emf is never negative and its average value is positive (**Fig. 29.10b**). We'll use our result from Example 29.3 to obtain an expression for this average value and solve it for the rotational speed ω .

EXECUTE Comparison of **Figs. 29.8b** and **29.10b** shows that the back emf of the motor is just N times the absolute value of the emf found for an alternator in Example 29.3, as in Eq. (29.4):

$|\mathcal{E}| = N\omega BA |\sin \omega t|$. To find the *average* back emf, we must replace $|\sin \omega t|$ by its average value. We find this by integrating $|\sin \omega t|$ over half a cycle, from $t = 0$ to $t = T/2 = \pi/\omega$, and dividing by the elapsed time π/ω . During this half cycle, the sine function is positive, so $|\sin \omega t| = \sin \omega t$, and we find

$$(|\sin \omega t|)_{\text{av}} = \frac{\int_0^{\pi/\omega} \sin \omega t \, dt}{\pi/\omega} = \frac{2}{\pi}$$

The average back emf is then

$$\mathcal{E}_{\text{av}} = \frac{2N\omega BA}{\pi}$$

Solving for ω , we obtain

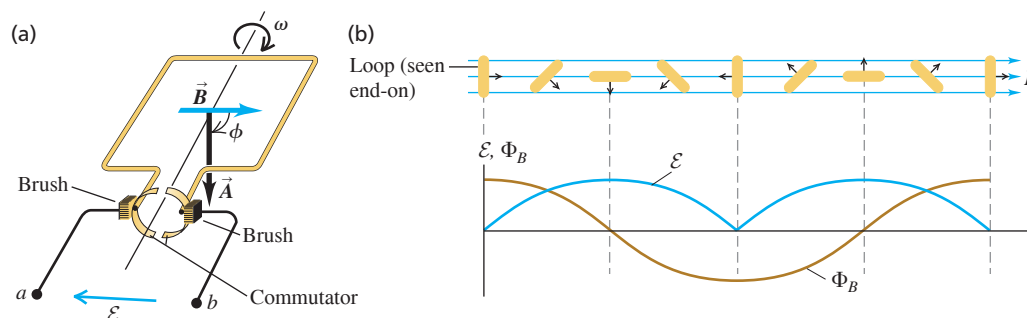
$$\omega = \frac{\pi \mathcal{E}_{\text{av}}}{2NBA} = \frac{\pi(112 \text{ V})}{2(500)(0.200 \text{ T})(0.100 \text{ m})^2} = 176 \text{ rad/s}$$

(Recall from Example 29.1 that $1 \text{ V} = 1 \text{ Wb/s} = 1 \text{ T} \cdot \text{m}^2/\text{s}$.)

EVALUATE The average back emf is directly proportional to ω . Hence the slower the rotation speed, the less the back emf and the greater the possibility of burning out the motor, as we described in Example 27.11 (Section 27.8).

KEYCONCEPT The back emf in a motor is a result of Faraday's law. The average value of the back emf is proportional to the rotation speed of the motor.

Figure 29.10 (a) Schematic diagram of a dc generator, using a split-ring commutator. The ring halves are attached to the loop and rotate with it. (b) Graph of the resulting induced emf between terminals a and b . Compare to **Fig. 29.8b**.



EXAMPLE 29.5 Generator III: The slidewire generator

Figure 29.11 shows a U-shaped conductor in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the figure and directed *into* the page. We lay a metal rod (the “slidewire”) with length L across the two arms of the conductor, forming a circuit, and move it to the right with constant velocity $\vec{v}$. This induces an emf and a current, which is why this device is called a *slidewire generator*. Find the magnitude and direction of the resulting induced emf.

IDENTIFY and SET UP The magnetic flux changes because the area of the loop—bounded on the right by the moving rod—is increasing. Our target variable is the emf $\mathcal{E}$ induced in this expanding loop. The magnetic field is uniform over the area of the loop, so we can find the flux from $\Phi_B = BA \cos \phi$. We choose the area vector $\vec{A}$ to point straight into the page, in the same direction as $\vec{B}$. With this choice a positive emf will be one that is directed clockwise around the loop. (You can check this with the right-hand rule: Using your right hand, point your thumb into the page and curl your fingers as in Fig. 29.6.)

EXECUTE Since $\vec{B}$ and $\vec{A}$ point in the same direction, the angle $\phi = 0$ and $\Phi_B = BA$. The magnetic-field magnitude B is constant, so the induced emf is

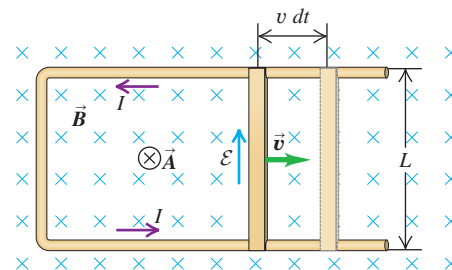
$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -B \frac{dA}{dt}$$

To calculate dA/dt , note that in a time dt the sliding rod moves a distance $v dt$ (Fig. 29.11) and the loop area increases by an amount $dA = Lv dt$. Hence the induced emf is

$$\mathcal{E} = -B \frac{Lv dt}{dt} = -BLv$$

The minus sign tells us that the emf is directed *counterclockwise* around the loop. The induced current is also counterclockwise, as shown in the figure.

Figure 29.11 A slidewire generator. The magnetic field $\vec{B}$ and the vector area $\vec{A}$ are both directed into the figure. The increase in magnetic flux (caused by an increase in area) induces the emf and current.



EVALUATE The emf of a slidewire generator is constant if $\vec{v}$ is constant. Hence the slidewire generator is a *direct-current* generator. It's not a very practical device because the rod eventually moves beyond the U-shaped conductor and loses contact, after which the current stops.

KEYCONCEPT An induced emf can be produced if the area of a circuit in a magnetic field changes. The magnitude of the emf depends on the speed at which the area of the circuit changes.

EXAMPLE 29.6 Work and power in the slidewire generator

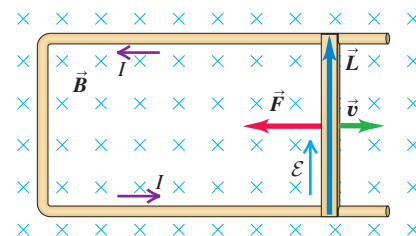
In the slidewire generator of Example 29.5, energy is dissipated in the circuit owing to its resistance. Let the resistance of the circuit (made up of the moving slidewire and the U-shaped conductor that connects the ends of the slidewire) at a given point in the slidewire's motion be R . Find the rate at which energy is dissipated in the circuit and the rate at which work must be done to move the rod through the magnetic field.

IDENTIFY and SET UP Our target variables are the *rates* at which energy is dissipated and at which work is done. Energy is dissipated in the circuit at the rate $P_{\text{dissipated}} = I^2 R$. The current I in the circuit equals $|\mathcal{E}|/R$; we found an expression for the induced emf $\mathcal{E}$ in this circuit in Example 29.5. There is a magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ on the rod, where $\vec{L}$ points along the rod in the direction of the current. **Figure 29.12** shows that this force is opposite to the rod velocity $\vec{v}$; to maintain the motion, whoever is pushing the rod must apply a force of equal magnitude in the direction of $\vec{v}$. This force does work at the rate $P_{\text{applied}} = Fv$.

EXECUTE First we'll calculate $P_{\text{dissipated}}$. From Example 29.5, $\mathcal{E} = -BLv$, so the current in the rod is $I = |\mathcal{E}|/R = BLv/R$. Hence

$$P_{\text{dissipated}} = I^2 R = \left(\frac{BLv}{R}\right)^2 R = \frac{B^2 L^2 v^2}{R}$$

Figure 29.12 The magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ that acts on the rod due to the induced current is to the left, opposite to $\vec{v}$.



To calculate P_{applied} , we first calculate the magnitude of $\vec{F} = I\vec{L} \times \vec{B}$. Since $\vec{L}$ and $\vec{B}$ are perpendicular, this magnitude is

$$F = ILB = \frac{BLv}{R} LB = \frac{B^2 L^2 v}{R}$$

The applied force has the same magnitude and does work at the rate

$$P_{\text{applied}} = Fv = \frac{B^2 L^2 v^2}{R}$$

Continued

CAUTION You can't violate energy conservation You might think that reversing the direction of $\vec{B}$ or of $\vec{v}$ would allow the magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ to be in the *same* direction as $\vec{v}$. This would be a neat trick. Once the rod was moving, the changing magnetic flux would induce an emf and a current, and the magnetic force on the rod would make it move even faster, increasing the emf and current until the rod was moving at tremendous speed and producing electrical power at a prodigious rate. If this seems too good to be true and a violation of energy conservation, that's because it is. Reversing $\vec{B}$ also reverses the sign of the induced emf and current and hence the direction of $\vec{L}$, so the magnetic force still opposes the motion of the rod; a similar result holds true if we reverse $\vec{v}$.

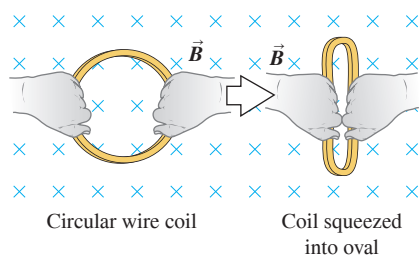
EVALUATE The rate at which work is done is exactly *equal* to the rate at which energy is dissipated in the resistance.

KEYCONCEPT You must do work to move any part of a circuit in a magnetic field. The work that you do is used to make the induced current flow in the presence of the circuit's resistance.

Generators as Energy Converters

Example 29.6 shows that the slidewire generator doesn't produce electrical energy out of nowhere; the energy is supplied by whatever object exerts the force that keeps the rod moving. All that the generator does is *convert* that energy into a different form. The equality between the rate at which *mechanical* energy is supplied to a generator and the rate at which *electrical* energy is generated holds for all types of generators, including the alternator described in Example 29.3. (We are ignoring the effects of friction in the bearings of an alternator or between the rod and the U-shaped conductor of a slidewire generator. The energy lost to friction is not available for conversion to electrical energy, so in real generators the friction is kept to a minimum.)

In Chapter 27 we stated that the magnetic force on moving charges can never do work. You might think, however, that the magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ in Example 29.6 is doing (negative) work on the current-carrying rod as it moves, contradicting our earlier statement. In fact, the work done by the magnetic force is zero. The moving charges that make up the current in the rod in Fig. 29.12 have a vertical component of velocity, causing a horizontal component of force on these charges. As a result, there is a horizontal displacement of charge within the rod, the left side acquiring a net positive charge and the right side a net negative charge. The result is a horizontal component of electric field, perpendicular to the length of the rod (analogous to the Hall effect, described in Section 27.9). It is this field, in the direction of motion of the rod, that does work on the mobile charges in the rod and hence indirectly on the atoms making up the rod.



TEST YOUR UNDERSTANDING OF SECTION 29.2 The accompanying figure shows a wire coil being squeezed in a uniform magnetic field. (a) While the coil is being squeezed, is the induced emf in the coil (i) clockwise, (ii) counterclockwise, or (iii) zero? (b) Once the coil has reached its final squeezed shape, is the induced emf in the coil (i) clockwise, (ii) counterclockwise, or (iii) zero?

ANSWER

ANSWER (a) (i) In (a), initially there is magnetic flux into the plane of the page, which we call positive. While the loop is being squeezed, the flux is becoming less positive ($d\Phi_B/dt < 0$) and so the induced emf is positive as in Fig. 29.6b ($\mathcal{E} = -d\Phi_B/dt > 0$). If you point the thumb of your right hand into the page, your fingers curl clockwise, so this is the direction of positive induced emf. In (b), since the coil's shape is no longer changing, the magnetic flux is not changing and there is no induced emf.

29.3 LENZ'S LAW

Lenz's law is a convenient alternative method for determining the direction of an induced current or emf. Lenz's law, named for the Russian physicist H. F. E. Lenz (1804–1865), is not an independent principle; it can be derived from Faraday's law. It always gives the same results as the sign rules we introduced in connection with Faraday's law, but it is often easier to use. Lenz's law also helps us gain intuitive understanding of various induction effects and of the role of energy conservation.

LENZ'S LAW The direction of any magnetic induction effect is such as to oppose the cause of the effect.

The “cause” may be changing flux through a stationary circuit due to a varying magnetic field, changing flux due to motion of the conductors that make up the circuit, or any combination. If the flux in a stationary circuit changes, as in Examples 29.1 and 29.2, the induced current sets up a magnetic field of its own. Within the area bounded by the circuit, this field is *opposite* to the original field if the original field is *increasing* but is in the *same* direction as the original field if the latter is *decreasing*. That is, the induced current opposes the *change in flux* through the circuit (*not* the flux itself).

If the flux change is due to motion of the conductors, as in Examples 29.3 through 29.6, the direction of the induced current in the moving conductor is such that the direction of the magnetic-field force on the conductor is opposite in direction to its motion. Thus the motion of the conductor, which caused the induced current, is opposed. We saw this explicitly for the slidewire generator in Example 29.6. In all these cases the induced current tries to preserve the *status quo* by opposing motion or a change of flux.

Lenz's law is also directly related to energy conservation. If the induced current in Example 29.6 were in the direction opposite to that given by Lenz's law, the magnetic force on the rod would accelerate it to ever-increasing speed with no external energy source, even though electrical energy is being dissipated in the circuit. This would be a clear violation of energy conservation and doesn't happen in nature.

CONCEPTUAL EXAMPLE 29.7 Lenz's law and the slidewire generator

WITH  **ARIATION PROBLEMS**

In Fig. 29.11, the induced current in the loop causes an additional magnetic field in the area bounded by the loop. The direction of the induced current is counterclockwise, so from the discussion of Section 28.5, this additional magnetic field is directed *out of* the plane of the figure. That direction is opposite that of the original magnetic field, so it tends to cancel the effect of that field. This is just what Lenz's law predicts.

KEYCONCEPT A change in the magnetic flux through a circuit induces a current that produces an additional magnetic field of its own. This induced field always opposes the change in the magnetic flux (Lenz's law).

CONCEPTUAL EXAMPLE 29.8 Lenz's law and the direction of induced current

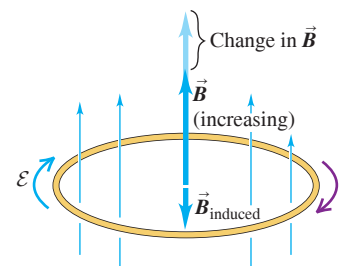
WITH  **ARIATION PROBLEMS**

In **Fig. 29.13** there is a uniform magnetic field $\vec{B}$ through the coil. The magnitude of the field is increasing, so there is an induced emf. Use Lenz's law to determine the direction of the resulting induced current.

SOLUTION This situation is the same as in Example 29.1 (Section 29.2). By Lenz's law the induced current must produce a magnetic field $\vec{B}_{\text{induced}}$ inside the coil that is downward, opposing the change in flux. From the right-hand rule we described in Section 28.5 for the direction of the magnetic field produced by a circular loop, $\vec{B}_{\text{induced}}$ will be in the desired direction if the induced current flows as shown in Fig. 29.13.

Figure 29.14 (next page) shows several applications of Lenz's law to the similar situation of a magnet moving near a conducting loop. In each case, the induced current produces a magnetic field whose direction opposes the change in flux through the loop due to the magnet's motion.

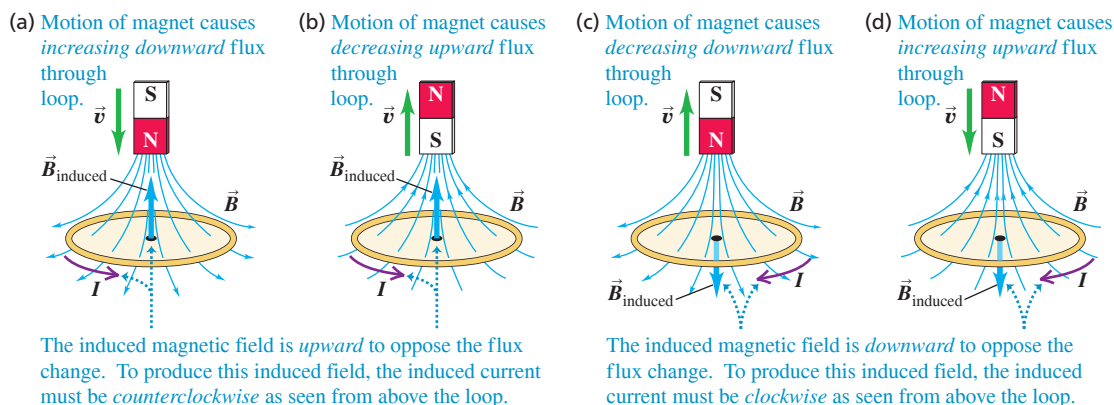
Figure 29.13 The induced current due to the change in $\vec{B}$ is clockwise, as seen from above the loop. The added field $\vec{B}_{\text{induced}}$ that it causes is downward, opposing the change in the upward field $\vec{B}$.



KEYCONCEPT Whenever there is a changing magnetic flux through a circuit, you can use Lenz's law to decide what the induced current direction will be: The current must produce an induced magnetic field that opposes the change in the flux.

Continued

Figure 29.14 Directions of induced currents as a bar magnet moves along the axis of a conducting loop. If the bar magnet is stationary, there is no induced current.



Lenz's Law and the Response to Flux Changes

Since an induced current always opposes any change in magnetic flux through a circuit, how is it possible for the flux to change at all? The answer is that Lenz's law gives only the *direction* of an induced current; the *magnitude* of the current depends on the resistance of the circuit. The greater the circuit resistance, the less the induced current that appears to oppose any change in flux and the easier it is for a flux change to take effect. If the loop in Fig. 29.14 were made out of wood (an insulator), there would be almost no induced current in response to changes in the flux through the loop.

Conversely, the less the circuit resistance, the greater the induced current and the more difficult it is to change the flux through the circuit. If the loop in Fig. 29.14 is a good conductor, an induced current flows as long as the magnet moves relative to the loop. Once the magnet and loop are no longer in relative motion, the induced current very quickly decreases to zero because of the nonzero resistance in the loop.

The extreme case occurs when the resistance of the circuit is *zero*. Then the induced current in Fig. 29.14 will continue to flow even after the induced emf has disappeared—that is, even after the magnet has stopped moving relative to the loop. Thanks to this *persistent current*, it turns out that the flux through the loop is exactly the same as it was before the magnet started to move, so the flux through a loop of zero resistance *never* changes. Exotic materials called *superconductors* do indeed have zero resistance; we discuss these further in Section 29.8.

TEST YOUR UNDERSTANDING OF SECTION 29.3 (a) Suppose the magnet in Fig. 29.14a were stationary and the loop of wire moved upward. Would the induced current in the loop be (i) in the same direction as shown in Fig. 29.14a, (ii) in the direction opposite to that shown in Fig. 29.14a, or (iii) zero? (b) Suppose the magnet and loop of wire in Fig. 29.14a both moved downward at the same velocity. Would the induced current in the loop be (i) in the same direction as shown in Fig. 29.14a, (ii) in the direction opposite to that shown in Fig. 29.14a, or (iii) zero?

ANSWER

(a) (i) In (a), as in the original situation, the magnet and loop are approaching each other and the downward flux through the loop is increasing. Hence the induced emf and induced current are the same. In (b), since the magnet and loop are moving together, the flux through the loop is not changing and no emf is induced.

29.4 MOTIONAL EMF

We've seen several situations in which a conductor moves in a magnetic field, as in the generators discussed in Examples 29.3 through 29.6. We can gain additional insight into the origin of the induced emf in these situations by considering the magnetic forces on mobile charges in the conductor. **Figure 29.15a** shows the same moving rod that we discussed in Example 29.5, separated for the moment from the U-shaped conductor. The magnetic field $\vec{B}$ is uniform and directed into the page, and we move the rod to the right at a constant velocity $\vec{v}$. A charged particle q in the rod then experiences a magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ with magnitude $F = |q|vB$. We'll assume in the following discussion that q is positive; in that case the direction of this force is upward along the rod, from b toward a .

This magnetic force causes the free charges in the rod to move, creating an excess of positive charge at the upper end a and negative charge at the lower end b . This in turn creates an electric field $\vec{E}$ within the rod, in the direction from a toward b (opposite to the magnetic force). Charge continues to accumulate at the ends of the rod until $\vec{E}$ becomes large enough for the downward electric force (with magnitude qE) to cancel exactly the upward magnetic force (with magnitude qvB). Then $qE = qvB$ and the charges are in equilibrium.

The magnitude of the potential difference $V_{ab} = V_a - V_b$ is equal to the electric-field magnitude E multiplied by the length L of the rod. From the above discussion, $E = vB$, so

$$V_{ab} = EL = vBL \quad (29.5)$$

with point a at higher potential than point b .

Now suppose the moving rod slides along a stationary U-shaped conductor, forming a complete circuit (Fig. 29.15b). No magnetic force acts on the charges in the stationary U-shaped conductor, but the charge that was near points a and b redistributes itself along the stationary conductor, creating an electric field within it. This field establishes a current in the direction shown. The moving rod has become a source of emf; within it, charge moves from lower to higher potential, and in the remainder of the circuit, charge moves from higher to lower potential. We call this emf a **motional emf**, denoted by $\mathcal{E}$. From the above discussion, the magnitude of this emf is

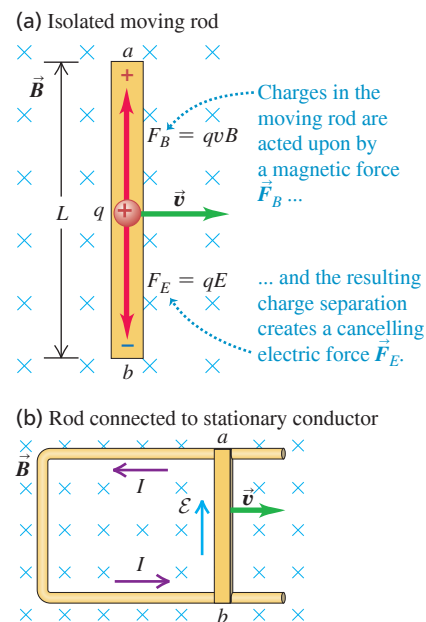
$$\text{Motional emf,} \quad \text{conductor length and velocity perpendicular to uniform } \vec{B} \quad \mathcal{E} = vBL \quad \begin{array}{l} \text{Conductor speed} \\ \text{Conductor length} \\ \text{Magnitude of uniform magnetic field} \end{array} \quad (29.6)$$

This corresponds to a force per unit charge of magnitude vB acting for a distance L along the moving rod. If the total circuit resistance of the U-shaped conductor and the sliding rod is R , the induced current I in the circuit is given by $vBL = IR$. This is the same result we obtained in Section 29.2 by using Faraday's law, and indeed motional emf is a particular case of Faraday's law. Verify that if we express v in meters per second, B in teslas, and L in meters, then $\mathcal{E}$ is in volts. (Recall that $1 \text{ V} = 1 \text{ J/C} = 1 \text{ T} \cdot \text{m}^2/\text{s}$.)

The emf associated with the moving rod in Fig. 29.15b is analogous to that of a battery with its positive terminal at a and its negative terminal at b , although the origins of the two emfs are quite different. In each case a nonelectrostatic force acts on the charges in the device, in the direction from b to a , and the emf is the work per unit charge done by this force when a charge moves from b to a in the device. When the device is connected to an external circuit, the direction of current is from b to a in the device and from a to b in the external circuit. Note that a motional emf is also present in the isolated moving rod in Fig. 29.15a, in the same way that a battery has an emf even when it's not part of a circuit.

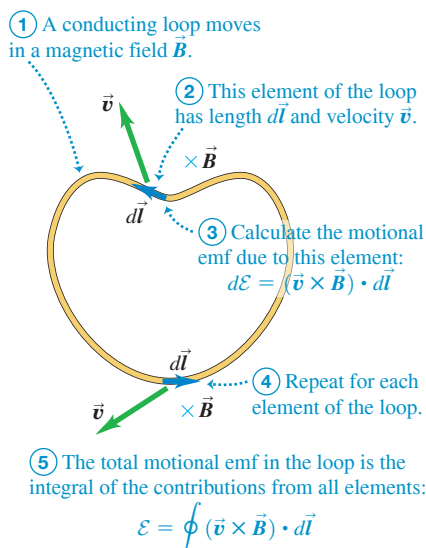
You can determine the direction of the induced emf in Fig. 29.15 by using Lenz's law, even if (as in Fig. 29.15a) the conductor does not form a complete circuit. In this case we can mentally complete the circuit between the ends of the conductor and use Lenz's law to determine the direction of the current. From this we can deduce the polarity of the ends of the open-circuit conductor. The direction from the $-$ end to the $+$ end within the conductor is the direction the current would have if the circuit were complete.

Figure 29.15 A conducting rod moving in a uniform magnetic field. (a) The rod, the velocity, and the field are mutually perpendicular. (b) Direction of induced current in the circuit.



The motional emf $\mathcal{E}$ in the moving rod creates an electric field in the stationary conductor.

Figure 29.16 Calculating the motional emf for a moving current loop. The velocity $\vec{v}$ can be different for different elements if the loop is rotating or changing shape. The magnetic field $\vec{B}$ can also have different values at different points around the loop.



Motional emf: General Form

We can generalize the concept of motional emf for a conductor with *any* shape, moving in any magnetic field, uniform or not, if we assume that the magnetic field at each point does not vary with time (Fig. 29.16). For an element $d\vec{l}$ of the conductor, the contribution $d\mathcal{E}$ to the emf is the magnitude dl multiplied by the component of $\vec{v} \times \vec{B}$ (the magnetic force per unit charge) parallel to $d\vec{l}$; that is,

$$d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

For any closed conducting loop, the total emf is

$$\text{Motional emf, general case} \quad \mathcal{E} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{l} \quad (29.7)$$

Line integral over all elements of closed conducting loop

Length vector of conductor element

Velocity of conductor element

Magnetic field at position of element

This expression looks very different from our original statement of Faraday's law, $\mathcal{E} = -d\Phi_B/dt$ [Eq. (29.3)]. In fact, though, the two statements are equivalent. It can be shown that the rate of change of magnetic flux through a moving conducting loop is always given by the negative of the expression in Eq. (29.7). Thus this equation gives us an alternative formulation of Faraday's law that is often convenient in problems with *moving* conductors. But when we have *stationary* conductors in changing magnetic fields, Eq. (29.7) *cannot* be used; in this case, $\mathcal{E} = -d\Phi_B/dt$ is the only correct way to express Faraday's law.

EXAMPLE 29.9 Motional emf in the slidewire generator

WITH VARIATION PROBLEMS

Suppose the moving rod in Fig. 29.15b is 0.10 m long, the velocity v is 2.5 m/s, the total resistance of the loop is $0.030 \, \Omega$, and B is 0.60 T. Find the motional emf, the induced current, and the force acting on the rod.

IDENTIFY and SET UP We'll find the motional emf $\mathcal{E}$ from Eq. (29.6) and the current from the values of $\mathcal{E}$ and the resistance R . The force on the rod is a *magnetic* force, exerted by $\vec{B}$ on the current in the rod; we'll find this force by using $\vec{F} = I\vec{L} \times \vec{B}$.

EXECUTE From Eq. (29.6) the motional emf is

$$\mathcal{E} = vBL = (2.5 \text{ m/s})(0.60 \text{ T})(0.10 \text{ m}) = 0.15 \text{ V}$$

The induced current in the loop is

$$I = \frac{\mathcal{E}}{R} = \frac{0.15 \text{ V}}{0.030 \, \Omega} = 5.0 \text{ A}$$

In the expression for the magnetic force, $\vec{F} = I\vec{L} \times \vec{B}$, the vector $\vec{L}$ points in the same direction as the induced current in the rod

(from b to a in Fig. 29.15). The right-hand rule for vector products shows that this force is directed *opposite* to the rod's motion. Since $\vec{L}$ and $\vec{B}$ are perpendicular, the force has magnitude

$$F = ILB = (5.0 \text{ A})(0.10 \text{ m})(0.60 \text{ T}) = 0.30 \text{ N}$$

EVALUATE We can check our answer for the direction of $\vec{F}$ by using Lenz's law. If we take the area vector $\vec{A}$ to point into the plane of the loop, the magnetic flux is positive and increasing as the rod moves to the right and increases the area of the loop. Lenz's law tells us that a force appears to oppose this increase in flux. Hence the force on the rod is to the left, opposite its motion.

KEYCONCEPT When a straight conductor moves through a magnetic field, the magnetic forces on the mobile charges give rise to a motional emf. This emf is proportional to the field strength and to the speed and length of the conductor.

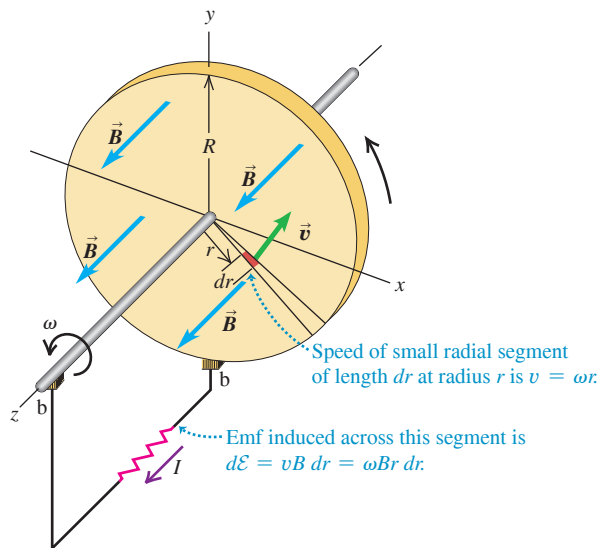
EXAMPLE 29.10 The Faraday disk dynamo

Figure 29.17 shows a conducting disk with radius R that lies in the xy -plane and rotates with constant angular velocity ω about the z -axis. The disk is in a uniform, constant $\vec{B}$ field in the z -direction. Find the induced emf between the center and the rim of the disk.

IDENTIFY and SET UP A motional emf arises because the conducting disk moves relative to $\vec{B}$. The complication is that different parts of the disk move at different speeds v , depending on their distance from the rotation axis. We'll address this by considering small segments of the disk

and integrating their contributions to determine our target variable, the emf between the center and the rim. Consider the small segment of the disk shown in red in Fig. 29.17 and labeled by its velocity vector $\vec{v}$. The magnetic force per unit charge on this segment is $\vec{v} \times \vec{B}$, which points radially outward from the center of the disk. Hence the induced emf tends to make a current flow radially outward, which tells us that the moving conducting path to think about here is a straight line from the center to the rim. We can find the emf from each small disk segment along this line by using $d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l}$ and then integrate to find the total emf.

Figure 29.17 A conducting disk with radius R rotating at an angular speed ω in a magnetic field $\vec{B}$. The emf is induced along radial lines of the disk and is applied to an external circuit through the two sliding contacts labeled b.



EXECUTE The length vector $d\vec{l}$ (of length dr) associated with the segment points radially outward, in the same direction as $\vec{v} \times \vec{B}$. The vectors $\vec{v}$ and $\vec{B}$ are perpendicular, and the magnitude of $\vec{v}$ is $v = \omega r$. The emf from the segment is then $d\mathcal{E} = \omega r B dr$. The total emf is the integral of $d\mathcal{E}$ from the center ($r = 0$) to the rim ($r = R$):

$$\mathcal{E} = \int_0^R \omega B r dr = \frac{1}{2} \omega B R^2$$

EVALUATE We can use this device as a source of emf in a circuit by completing the circuit through two stationary brushes (labeled b in the figure) that contact the disk and its conducting shaft as shown. Such a disk is called a *Faraday disk dynamo* or a *homopolar generator*. Unlike the alternator in Example 29.3, the Faraday disk dynamo is a direct-current generator; it produces an emf that is constant in time. Can you use Lenz's law to show that for the direction of rotation in Fig. 29.17, the current in the external circuit must be in the direction shown?

KEYCONCEPT In general, finding the induced emf in a conductor requires doing an integral. This is because the magnetic force on a mobile charge can be different at different locations in the conductor.

TEST YOUR UNDERSTANDING OF SECTION 29.4 The earth's magnetic field points toward (magnetic) north. For simplicity, assume that the field has no vertical component (as is the case near the earth's equator). (a) If you hold a metal rod in your hand and walk toward the east, how should you orient the rod to get the maximum motional emf between its ends? (i) East-west; (ii) north-south; (iii) up-down; (iv) you get the same motional emf with all of these orientations. (b) How should you hold it to get *zero* emf as you walk toward the east? (i) East-west; (ii) north-south; (iii) up-down; (iv) none of these. (c) In which direction should you travel so that the motional emf across the rod is zero no matter how the rod is oriented? (i) West; (ii) north; (iii) south; (iv) straight up; (v) straight down.

ANSWER (a) (iii); (b) (i) or (ii); (c) (iii) or (iv). You'll get the maximum motional emf if you hold the rod vertically, so that its length is perpendicular to both the magnetic field and the direction of motion. With this orientation, $\vec{L}$ is parallel to $\vec{v} \times \vec{B}$. If you hold the rod in any horizontal orientation, $\vec{L}$ will be perpendicular to $\vec{v} \times \vec{B}$ and no emf will be induced. If you walk due north or south, $\vec{v} \times \vec{B}$ and no emf will be induced for any orientation of the rod.

29.5 INDUCED ELECTRIC FIELDS

When a conductor moves in a magnetic field, we can understand the induced emf on the basis of magnetic forces on charges in the conductor, as described in Section 29.4. But an induced emf also occurs when there is a changing flux through a stationary conductor. What is it that pushes the charges around the circuit in this type of situation?

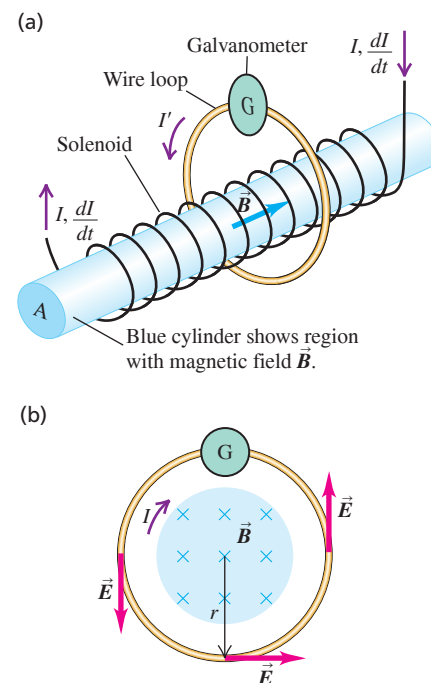
Let's consider the situation shown in **Fig. 29.18**. A long, thin solenoid with cross-sectional area A and n turns per unit length is encircled at its center by a circular conducting loop. The galvanometer G measures the current in the loop. A current I in the winding of the solenoid sets up a magnetic field $\vec{B}$ along the solenoid axis, as shown, with magnitude B as calculated in Section 28.7: $B = \mu_0 n I$, where n is the number of turns per unit length. If we ignore the small field outside the solenoid and take the area vector $\vec{A}$ to point in the same direction as $\vec{B}$, then the magnetic flux Φ_B through the loop is

$$\Phi_B = BA = \mu_0 n I A$$

When the solenoid current I changes with time, the magnetic flux Φ_B also changes, and according to Faraday's law the induced emf in the loop is given by

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\mu_0 n A \frac{dI}{dt} \quad (29.8)$$

Figure 29.18 (a) The windings of a long solenoid carry a current I that is increasing at a rate dI/dt . The magnetic flux in the solenoid is increasing at a rate $d\Phi_B/dt$, and this changing flux passes through a wire loop. An emf $\mathcal{E} = -d\Phi_B/dt$ is induced in the loop, inducing a current I' that is measured by the galvanometer G . (b) Cross-sectional view.



If the total resistance of the loop is R , the induced current in the loop, which we may call I' , is $I' = \mathcal{E}/R$.

But what *force* makes the charges move around the wire loop? It can't be a magnetic force because the loop isn't even *in* a magnetic field. We are forced to conclude that there has to be an **induced electric field** in the conductor *caused by the changing magnetic flux*. Induced electric fields are *very* different from the electric fields caused by charges, which we discussed in Chapter 23. To see this, note that when a charge q goes once around the loop, the total work done on it by the electric field must be equal to q times the emf $\mathcal{E}$. That is, the electric field in the loop is *not conservative*, as we used the term in Section 23.1, because the line integral of $\vec{E}$ around a closed path is not zero. Indeed, this line integral, representing the work done by the induced $\vec{E}$ field per unit charge, is equal to the induced emf $\mathcal{E}$:

$$\oint \vec{E} \cdot d\vec{l} = \mathcal{E} \quad (29.9)$$

From Faraday's law the emf $\mathcal{E}$ is also the negative of the rate of change of magnetic flux through the loop. Thus for this case we can restate Faraday's law as

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt} \quad \text{Negative of the time rate of change of magnetic flux through path} \quad (29.10)$$

Line integral of electric field around path

Faraday's law for a stationary integration path:

Note that Faraday's law is *always* true in the form $\mathcal{E} = -d\Phi_B/dt$; the form given in Eq. (29.10) is valid *only* if the path around which we integrate is *stationary*.

Let's apply Eq. (29.10) to the stationary circular loop in Fig. 29.18b, which we take to have radius r . Because of cylindrical symmetry, $\vec{E}$ has the same magnitude at every point on the circle and is tangent to it at each point. (Symmetry would also permit the field to be *radial*, but then Gauss's law would require the presence of a net charge inside the circle, and there is none.) The line integral in Eq. (29.10) becomes simply the magnitude E times the circumference $2\pi r$ of the loop, $\oint \vec{E} \cdot d\vec{l} = 2\pi rE$, and Eq. (29.10) gives

$$E = \frac{1}{2\pi r} \left| \frac{d\Phi_B}{dt} \right| \quad (29.11)$$

The directions of $\vec{E}$ at points on the loop are shown in Fig. 29.18b. We know that $\vec{E}$ has to have the direction shown when $\vec{B}$ in the solenoid is increasing, because $\oint \vec{E} \cdot d\vec{l}$ has to be negative when $d\Phi_B/dt$ is positive. The same approach can be used to find the induced electric field *inside* the solenoid when the solenoid $\vec{B}$ field is changing; we leave the details to you.

Nonelectrostatic Electric Fields

We've learned that Faraday's law, Eq. (29.3), is valid for two rather different situations. In one, an emf is induced by magnetic forces on charges when a conductor moves through a magnetic field. In the other, a time-varying magnetic field induces an electric field and hence an emf; the $\vec{E}$ field is induced even when no conductor is present. This $\vec{E}$ field differs from an *electrostatic* field in an important way. It is *nonconservative*; the line integral $\oint \vec{E} \cdot d\vec{l}$ around a closed path is not zero, and when a charge moves around a closed path, the field does a nonzero amount of work on it. It follows that for such a field the concept of *potential* has no meaning. We call such a field a **nonelectrostatic field**. In contrast, an *electrostatic* field is *always* conservative, as we discussed in Section 23.1, and always has an associated potential function. Despite this difference, the fundamental effect of *any* electric field is to exert a force $\vec{F} = q\vec{E}$ on a charge q . This relationship is valid whether $\vec{E}$ is conservative and produced by charges or nonconservative and produced by changing magnetic flux.

So a changing magnetic field acts as a source of electric field of a sort that we *cannot* produce with any static charge distribution. What's more, we'll see in Section 29.7 that a changing *electric* field acts as a source of *magnetic* field. We'll explore this symmetry between the two fields in our study of electromagnetic waves in Chapter 32.

If any doubt remains in your mind about the reality of magnetically induced electric fields, consider a few of the many practical applications (**Fig. 29.19**). Pickups in electric

Figure 29.19 Applications of induced electric fields. (a) This car is powered by an electric motor. As the car comes to a halt, the spinning wheels run the motor backward so that it acts as a generator. The resulting induced current is used to recharge the car's batteries. (b) The rotating crankshaft of a piston-engine airplane spins a magnet, inducing an emf in an adjacent coil and generating the spark that ignites fuel in the engine cylinders. This keeps the engine running even if the airplane's other electrical systems fail.



guitars use currents induced in stationary pickup coils by the vibration of nearby ferromagnetic strings. Alternators in most cars use rotating magnets to induce currents in stationary coils. Whether we realize it or not, magnetically induced electric fields play an important role in everyday life.

EXAMPLE 29.11 Induced electric fields

WITH VARIATION PROBLEMS

Suppose the long solenoid in Fig. 29.18a has 500 turns per meter and cross-sectional area 4.0 cm^2 . The current in its windings is increasing at 100 A/s . (a) Find the magnitude of the induced emf in the wire loop outside the solenoid. (b) Find the magnitude of the induced electric field within the loop if its radius is 2.0 cm .

IDENTIFY and SET UP As in Fig. 29.18b, the increasing magnetic field inside the solenoid causes a change in the magnetic flux through the wire loop and hence induces an electric field $\vec{E}$ around the loop. Our target variables are the induced emf $\mathcal{E}$ and the electric-field magnitude E . We use Eq. (29.8) to determine the emf. The loop and the solenoid share the same central axis. Hence, by symmetry, the electric field is tangent to the loop and has the same magnitude E all the way around its circumference. We can therefore use Eq. (29.9) to find E .

EXECUTE (a) From Eq. (29.8), the induced emf is

$$\begin{aligned}\mathcal{E} &= -\frac{d\Phi_B}{dt} = -\mu_0 n A \frac{dI}{dt} \\ &= -(4\pi \times 10^{-7} \text{ Wb/A} \cdot \text{m})(500 \text{ turns/m})(4.0 \times 10^{-4} \text{ m}^2)(100 \text{ A/s}) \\ &= -25 \times 10^{-6} \text{ Wb/s} = -25 \times 10^{-6} \text{ V} = -25 \mu\text{V}\end{aligned}$$

(b) By symmetry the line integral $\oint \vec{E} \cdot d\vec{l}$ has absolute value $2\pi rE$ no matter which direction we integrate around the loop. This is equal to the absolute value of the emf, so

$$E = \frac{|\mathcal{E}|}{2\pi r} = \frac{25 \times 10^{-6} \text{ V}}{2\pi(2.0 \times 10^{-2} \text{ m})} = 2.0 \times 10^{-4} \text{ V/m}$$

EVALUATE In Fig. 29.18b the magnetic flux *into* the plane of the figure is increasing. According to the right-hand rule for induced emf (Fig. 29.6), a positive emf would be clockwise around the loop; the negative sign of $\mathcal{E}$ shows that the emf is in the counterclockwise direction. Can you also show this by using Lenz's law?

KEYCONCEPT A changing magnetic flux through a closed loop produces an induced electric field $\vec{E}$. The line integral of $\vec{E}$ around the loop is equal to the induced emf in the loop.

TEST YOUR UNDERSTANDING OF SECTION 29.5 If you wiggle a magnet back and forth in your hand, are you generating an electric field? If so, is this field conservative?

ANSWER **yes, no** The magnetic field at a fixed position changes as you move the magnet, which induces an electric field. Such induced electric fields are *not* conservative.

29.6 EDDY CURRENTS

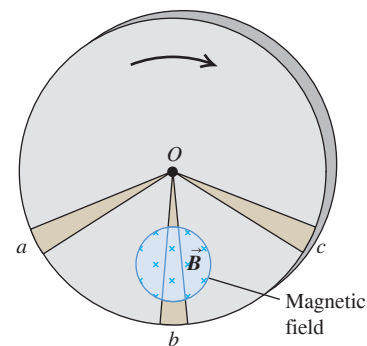
In the examples of induction effects that we have studied, the induced currents have been confined to well-defined paths in conductors and other components forming a circuit. However, many pieces of electrical equipment contain masses of metal moving in magnetic fields or located in changing magnetic fields. In situations like these we can have induced currents that circulate throughout the volume of a material. Because their flow patterns resemble swirling eddies in a river, we call these **eddy currents**.

As an example, consider a metallic disk rotating in a magnetic field perpendicular to the plane of the disk but confined to a limited portion of the disk's area, as shown in Fig. 29.20a. Sector Ob is moving across the field and has an emf induced in it. Sectors Oa and Oc are not in the field, but they provide return conducting paths for charges displaced along Ob to return from b to O . The result is a circulation of eddy currents in the disk, somewhat as sketched in Fig. 29.20b.

We can use Lenz's law to decide on the direction of the induced current in the neighborhood of sector Ob . This current must experience a magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ that *opposes* the rotation of the disk, and so this force must be to the right in Fig. 29.20b. Since $\vec{B}$ is directed into the plane of the disk, the current and hence $\vec{L}$ have downward components. The return currents lie outside the field, so they do not experience magnetic forces. The interaction between the eddy currents and the field causes a braking action on the disk. Such effects can be used to stop the rotation of a circular saw quickly when

Figure 29.20 Eddy currents induced in a rotating metal disk.

(a) Metal disk rotating through a magnetic field



(b) Resulting eddy currents and braking force

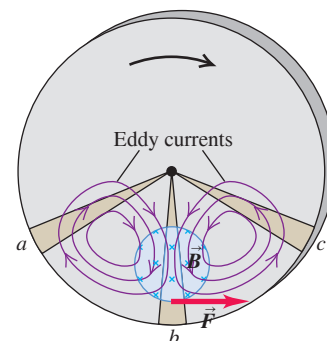
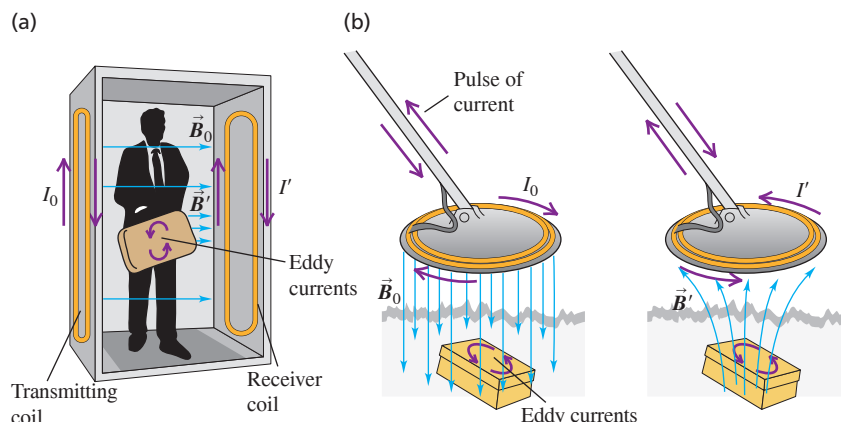
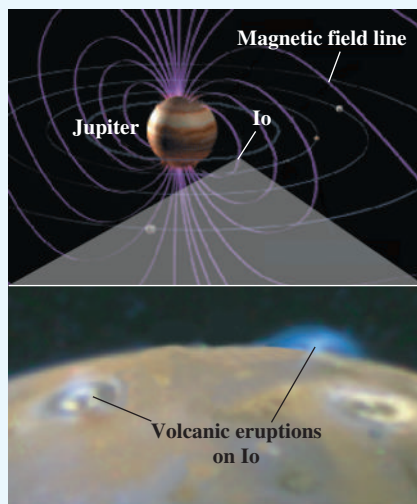


Figure 29.21 (a) A metal detector at an airport security checkpoint generates an alternating magnetic field $\vec{B}_0$. This induces eddy currents in a conducting object carried through the detector. The eddy currents in turn produce an alternating magnetic field $\vec{B}'$, which induces a current in the detector's receiver coil. (b) Portable metal detectors work on the same principle.



APPLICATION Eddy Currents Help Power Io's Volcanoes

Jupiter's moon Io is slightly larger than the earth's moon. It moves at more than 60,000 km/h through Jupiter's intense magnetic field (about ten times stronger than the earth's field), which sets up strong eddy currents within Io that dissipate energy at a rate of 10^{12} W. This dissipated energy helps to heat Io's interior and causes volcanic eruptions on its surface, as shown in the lower close-up image. (Gravitational effects from Jupiter cause even more heating.)



the power is turned off. Eddy current braking is used on some electrically powered rapid-transit vehicles. Electromagnets mounted in the cars induce eddy currents in the rails; the resulting magnetic fields cause braking forces on the electromagnets and thus on the cars.

Eddy currents have many other practical uses. In induction furnaces, eddy currents are used to heat materials in completely sealed containers for processes in which it is essential to avoid the slightest contamination of the materials. The metal detectors used at airport security checkpoints (Fig. 29.21a) operate by detecting eddy currents induced in metallic objects. Similar devices (Fig. 29.21b) are used to find buried treasure such as bottlecaps and lost pennies.

Eddy currents also have undesirable effects. In an alternating-current transformer, coils wrapped around an iron core carry a sinusoidally varying current. The resulting eddy currents in the core waste energy through I^2R heating and set up an unwanted opposing emf in the coils. To minimize these effects, the core is designed so that the paths for eddy currents are as narrow as possible. We'll describe how this is done when we discuss transformers in Section 31.6.

TEST YOUR UNDERSTANDING OF SECTION 29.6 Suppose that the magnetic field in Fig. 29.20 were directed out of the plane of the figure and the disk were rotating counterclockwise. Compared to the directions of the force $\vec{F}$ and the eddy currents shown in Fig. 29.20b, what would the new directions be? (i) The force $\vec{F}$ and the eddy currents would both be in the same direction; (ii) the force $\vec{F}$ would be in the same direction, but the eddy currents would be in the opposite direction; (iii) the force $\vec{F}$ would be in the opposite direction, but the eddy currents would be in the same direction; (iv) the force $\vec{F}$ and the eddy currents would be in the opposite directions.

ANSWER

(iii) By Lenz's law, the force must oppose the motion of the disk through the magnetic field. Since the disk material is now moving to the right through the field region, the force $\vec{F}$ is to the left—that is, in the opposite direction to that shown in Fig. 29.20b. To produce a leftward magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ on currents moving through a magnetic field $\vec{B}$ directed out of the plane of the figure, the eddy currents must be moving downward in the figure—that is, in the same direction shown in Fig. 29.20b.

29.7 DISPLACEMENT CURRENT AND MAXWELL'S EQUATIONS

We have seen that a varying magnetic field gives rise to an induced electric field. In one of the more remarkable examples of the symmetry of nature, it turns out that a varying electric field gives rise to a magnetic field. This effect is of tremendous importance, for it turns out to explain the existence of radio waves, gamma rays, and visible light, as well as all other forms of electromagnetic waves.

Generalizing Ampere's Law

To see the origin of the relationship between varying electric fields and magnetic fields, let's return to Ampere's law as given in Section 28.6, Eq. (28.20):

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

The problem with Ampere's law in this form is that it is *incomplete*. To see why, let's consider the process of charging a capacitor (**Fig. 29.22**). Conducting wires lead current i_C into one plate and out of the other; the charge Q increases, and the electric field $\vec{E}$ between the plates increases. The notation i_C indicates *conduction* current to distinguish it from another kind of current we are about to encounter, called *displacement* current i_D . We use lowercase i 's and v 's to denote instantaneous values of currents and potential differences, respectively, that may vary with time.

Let's apply Ampere's law to the circular path shown. The integral $\oint \vec{B} \cdot d\vec{l}$ around this path equals $\mu_0 I_{\text{encl}}$. For the plane circular area bounded by the circle, I_{encl} is just the current i_C in the left conductor. But the surface that bulges out to the right is bounded by the same circle, and the current through that surface is zero. So $\oint \vec{B} \cdot d\vec{l}$ is equal to $\mu_0 i_C$, and at the same time it is equal to zero! This is a clear contradiction.

However, something else is happening on the bulged-out surface. As the capacitor charges, the electric field $\vec{E}$ and the electric flux Φ_E through the surface are increasing. We can determine their rates of change in terms of the charge and current. The instantaneous charge is $q = Cv$, where C is the capacitance and v is the instantaneous potential difference. For a parallel-plate capacitor, $C = \epsilon_0 A/d$, where A is the plate area and d is the spacing. The potential difference v between plates is $v = Ed$, where E is the electric-field magnitude between plates. (We ignore fringing and assume that $\vec{E}$ is uniform in the region between the plates.) If this region is filled with a material with permittivity ϵ , we replace ϵ_0 by ϵ everywhere; we'll use ϵ in the following discussion.

Substituting these expressions for C and v into $q = Cv$, we can express the capacitor charge q in terms of the electric flux $\Phi_E = EA$ through the surface:

$$q = Cv = \frac{\epsilon A}{d}(Ed) = \epsilon EA = \epsilon \Phi_E \quad (29.12)$$

As the capacitor charges, the rate of change of q is the conduction current, $i_C = dq/dt$. Taking the derivative of Eq. (29.12) with respect to time, we get

$$i_C = \frac{dq}{dt} = \epsilon \frac{d\Phi_E}{dt} \quad (29.13)$$

Stretching our imagination a bit, we invent a fictitious **displacement current** i_D in the region between the plates, defined as

$$i_D = \epsilon \frac{d\Phi_E}{dt} \quad (29.14)$$

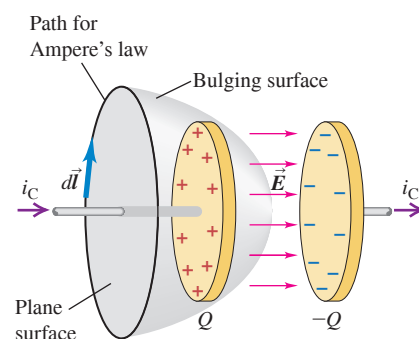
Displacement current through an area Time rate of change of electric flux through area
Permittivity of material in area

That is, we imagine that the changing flux through the curved (bulged-out) surface in Fig. 29.22 is equivalent, in Ampere's law, to a conduction current through that surface. We include this fictitious current, along with the real conduction current i_C , in Ampere's law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_C + i_D)_{\text{encl}} \quad (\text{generalized Ampere's law}) \quad (29.15)$$

Ampere's law in this form is obeyed no matter which surface we use in Fig. 29.22. For the flat surface, i_D is zero; for the curved surface, i_C is zero; and i_C for the flat surface equals i_D for the curved surface. Equation (29.15) remains valid in a magnetic material, provided that the magnetization is proportional to the external field and we replace μ_0 by μ .

Figure 29.22 Parallel-plate capacitor being charged. The conduction current through the plane surface is i_C , but there is no conduction current through the surface that bulges out to pass between the plates. The two surfaces have a common boundary, so this difference in I_{encl} leads to an apparent contradiction in applying Ampere's law.



The fictitious displacement current i_D was invented in 1865 by the Scottish physicist James Clerk Maxwell. There is a corresponding *displacement current density* $j_D = i_D/A$; using $\Phi_E = EA$ and dividing Eq. (29.14) by A , we find

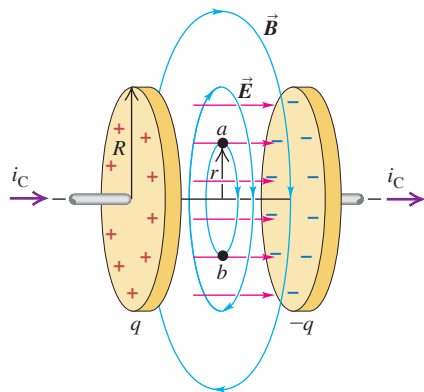
$$j_D = \epsilon \frac{dE}{dt} \quad (29.16)$$

We have pulled the concept out of thin air, as Maxwell did, but we see that it enables us to save Ampere's law in situations such as that in Fig. 29.22.

Another benefit of displacement current is that it lets us generalize Kirchhoff's junction rule, discussed in Section 26.2. Considering the left plate of the capacitor, we have conduction current into it but none out of it. But when we include the displacement current, we have conduction current coming in on one side and an equal displacement current coming out the other side. With this generalized meaning of the term "current," we can speak of current going *through* the capacitor.

The Reality of Displacement Current

Figure 29.23 A capacitor being charged by a current i_C has a displacement current equal to i_C between the plates, with displacement-current density $j_D = \epsilon dE/dt$. This can be regarded as the source of the magnetic field between the plates.



You might well ask at this point whether displacement current has any real physical significance or whether it is just a ruse to satisfy Ampere's law and Kirchhoff's junction rule. Here's a fundamental experiment that helps to answer that question. We take a plane circular area between the capacitor plates (**Fig. 29.23**). If displacement current really plays the role in Ampere's law that we have claimed, then there ought to be a magnetic field in the region between the plates while the capacitor is charging. We can use our generalized Ampere's law, including displacement current, to predict what this field should be.

To be specific, let's picture round capacitor plates with radius R . To find the magnetic field at a point in the region between the plates at a distance r from the axis, we apply Ampere's law to a circle of radius r passing through the point, with $r < R$. This circle passes through points a and b in Fig. 29.23. The total current enclosed by the circle is j_D times its area, or $(i_D/\pi R^2)(\pi r^2)$. The integral $\oint \vec{B} \cdot d\vec{l}$ in Ampere's law is just B times the circumference $2\pi r$ of the circle, and because $i_D = i_C$ for the charging capacitor, Ampere's law becomes

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= 2\pi r B = \mu_0 \frac{r^2}{R^2} i_C \quad \text{or} \\ B &= \frac{\mu_0}{2\pi} \frac{r}{R^2} i_C \end{aligned} \quad (29.17)$$

This result predicts that in the region between the plates $\vec{B}$ is zero at the axis and increases linearly with distance from the axis. A similar calculation shows that *outside* the region between the plates (that is, for $r > R$), $\vec{B}$ is the same as though the wire were continuous and the plates not present at all.

When we *measure* the magnetic field in this region, we find that it really is there and that it behaves just as Eq. (29.17) predicts. This confirms directly the role of displacement current as a source of magnetic field. It is now established beyond reasonable doubt that Maxwell's displacement current, far from being just an artifice, is a fundamental fact of nature.

Maxwell's Equations of Electromagnetism

We are now in a position to wrap up in a single package *all* of the relationships between electric and magnetic fields and their sources. This package consists of four equations, called **Maxwell's equations**. Maxwell did not discover all of these equations single-handedly (though he did develop the concept of displacement current). But he did put them together and recognized their significance, particularly in predicting the existence of electromagnetic waves.

For now we'll state Maxwell's equations in their simplest form, for the case in which we have charges and currents in otherwise empty space. In Chapter 32 we'll discuss how to modify these equations if a dielectric or a magnetic material is present.

Two of Maxwell's equations involve an integral of $\vec{E}$ or $\vec{B}$ over a closed surface. The first is simply Gauss's law for electric fields, Eq. (22.8):

Gauss's law for $\vec{E}$:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (29.18)$$

Flux of electric field through a closed surface
Charge enclosed by surface
Electric constant

The second is the analogous relationship for *magnetic* fields, Eq. (27.8):

Gauss's law for $\vec{B}$:

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (29.19)$$

Flux of magnetic field through any closed surface ...
... equals zero.

This statement means, among other things, that there are no magnetic monopoles (single magnetic charges) to act as sources of magnetic field.

The third and fourth equations involve a line integral of $\vec{E}$ or $\vec{B}$ around a closed path. Faraday's law states that a changing magnetic flux acts as a source of electric field:

Faraday's law for a stationary integration path:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (29.20)$$

Line integral of electric field around path
Negative of the time rate of change of magnetic flux through path

If there is a changing magnetic field, the line integral in Eq. (29.20)—which must be carried out over a *stationary* closed path—is not zero. Thus the $\vec{E}$ field produced by a changing $\vec{B}$ is not conservative.

The fourth and final equation is Ampere's law including displacement current. It states that both a conduction current and a changing electric flux act as sources of magnetic field:

Ampere's law for a stationary integration path:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \quad (29.21)$$

Line integral of magnetic field around path
Electric constant
Time rate of change of electric flux through path
Magnetic constant
Conduction current through path
Displacement current through path

It's worthwhile to look more carefully at the electric field $\vec{E}$ and its role in Maxwell's equations. In general, the total $\vec{E}$ field at a point in space can be the superposition of an electrostatic field $\vec{E}_c$ caused by a distribution of charges at rest and a magnetically induced, nonelectrostatic field $\vec{E}_n$. That is,

$$\vec{E} = \vec{E}_c + \vec{E}_n$$

The electrostatic part $\vec{E}_c$ is *always* conservative, so $\oint \vec{E}_c \cdot d\vec{l} = 0$. This conservative part of the field does not contribute to the integral in Faraday's law, so we can take $\vec{E}$ in Eq. (29.20) to be the *total* electric field $\vec{E}$, including both the part $\vec{E}_c$ due to charges and the magnetically induced part $\vec{E}_n$. Similarly, the nonconservative part $\vec{E}_n$ of the $\vec{E}$ field does not contribute to the integral in Gauss's law, because this part of the field is not caused by static charges. Hence $\oint \vec{E}_n \cdot d\vec{A}$ is always zero. We conclude that in all the Maxwell equations, $\vec{E}$ is the total electric field; these equations don't distinguish between conservative and nonconservative fields.

Figure 29.24 Maxwell's equations in empty space are highly symmetric.

In empty space there are no charges, so the fluxes of $\vec{E}$ and $\vec{B}$ through any closed surface are equal to zero.

$$\begin{aligned}\oint \vec{E} \cdot d\vec{A} &= 0 \\ \oint \vec{B} \cdot d\vec{A} &= 0 \\ \oint \vec{E} \cdot d\vec{l} &= -\frac{d\Phi_B}{dt} \\ \oint \vec{B} \cdot d\vec{l} &= \mu_0\epsilon_0 \frac{d\Phi_E}{dt}\end{aligned}$$

In empty space there are no conduction currents, so the line integrals of $\vec{E}$ and $\vec{B}$ around any closed path are related to the rate of change of flux of the other field.

Symmetry in Maxwell's Equations

There is a remarkable symmetry in Maxwell's four equations. In empty space where there is no charge, the first two equations [Eqs. (29.18) and (29.19)] are identical in form, one containing $\vec{E}$ and the other containing $\vec{B}$ (Fig. 29.24). When we compare the second two equations, Eq. (29.20) says that a changing magnetic flux creates an electric field, and Eq. (29.21) says that a changing electric flux creates a magnetic field. In empty space, where there is no conduction current, $i_C = 0$ and the two equations have the same form, apart from a numerical constant and a negative sign, with the roles of $\vec{E}$ and $\vec{B}$ exchanged.

We can rewrite Eqs. (29.20) and (29.21) in a different but equivalent form by introducing the definitions of magnetic flux, $\Phi_B = \int \vec{B} \cdot d\vec{A}$, and electric flux, $\Phi_E = \int \vec{E} \cdot d\vec{A}$, respectively. In empty space, where there is no charge or conduction current, $i_C = 0$ and $Q_{\text{encl}} = 0$, and we have

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A} \quad (29.22)$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0\epsilon_0 \frac{d}{dt} \int \vec{E} \cdot d\vec{A} \quad (29.23)$$

Again we notice the symmetry between the roles of $\vec{E}$ and $\vec{B}$ in these expressions.

The most remarkable feature of these equations is that a time-varying field of *either* kind induces a field of the other kind in neighboring regions of space. Maxwell recognized that these relationships predict the existence of electromagnetic disturbances consisting of time-varying electric and magnetic fields that travel or *propagate* from one region of space to another, even if no matter is present in the intervening space. Such disturbances, called *electromagnetic waves*, provide the physical basis for light, radio and television waves, infrared, ultraviolet, and x rays. We'll return to this vitally important topic in Chapter 32.

Although it may not be obvious, *all* the basic relationships between fields and their sources are contained in Maxwell's equations. We can derive Coulomb's law from Gauss's law, we can derive the law of Biot and Savart from Ampere's law, and so on. When we add the equation that defines the $\vec{E}$ and $\vec{B}$ fields in terms of the forces that they exert on a charge q , namely,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (29.24)$$

we have *all* the fundamental relationships of electromagnetism!

Maxwell's equations would have even greater symmetry between the $\vec{E}$ and $\vec{B}$ fields if single magnetic charges (magnetic monopoles) existed. The right side of Eq. (29.19) would contain the total *magnetic* charge enclosed by the surface, and the right side of Eq. (29.20) would include a magnetic monopole current term. However, no magnetic monopoles have yet been found.

In conciseness and generality, Maxwell's equations are in the same league with Newton's laws of motion and the laws of thermodynamics. Indeed, a major goal of science is learning how to express very broad and general relationships in a concise and compact form. Maxwell's synthesis of electromagnetism stands as a towering intellectual achievement, comparable to the Newtonian synthesis we described at the end of Section 13.5 and to the development of relativity and quantum mechanics in the 20th century.

TEST YOUR UNDERSTANDING OF SECTION 29.7 (a) Which of Maxwell's equations explains how a credit card reader works? (b) Which one describes how a wire carrying a steady current generates a magnetic field?

ANSWER

(a) **Faraday's law.** A credit card reader works by inducing currents in the reader's coils as the card's magnetized stripe is swiped (see the answer to the chapter opening question). Ampere's law describes how currents of all kinds (both conduction currents and displacement currents) give rise to magnetic fields.

29.8 SUPERCONDUCTIVITY

The most familiar property of a superconductor is the sudden disappearance of all electrical resistance when the material is cooled below a temperature called the *critical temperature*, denoted by T_c . We discussed this behavior and the circumstances of its discovery in Section 25.2. But superconductivity is far more than just the absence of measurable resistance. As we'll see in this section, superconductors also have extraordinary *magnetic* properties.

The first hint of unusual magnetic properties was the discovery that for any superconducting material the critical temperature T_c changes when the material is placed in an externally produced magnetic field $\vec{B}_0$. **Figure 29.25** shows this dependence for mercury, the first element in which superconductivity was observed. As the external field magnitude B_0 increases, the superconducting transition occurs at lower and lower temperature. When B_0 is greater than 0.0412 T, *no* superconducting transition occurs. The minimum magnitude of magnetic field that is needed to eliminate superconductivity at a temperature below T_c is called the *critical field*, denoted by B_c .

The Meissner Effect

Another aspect of the magnetic behavior of superconductors appears if we place a homogeneous sphere of a superconducting material in a uniform applied magnetic field $\vec{B}_0$ at a temperature T greater than T_c . The material is then in the normal phase, not the superconducting phase (**Fig. 29.26a**). Now we lower the temperature until the superconducting transition occurs. (We assume that the magnitude of $\vec{B}_0$ is not large enough to prevent the phase transition.) What happens to the field?

Measurements of the field outside the sphere show that the field lines become distorted as in **Fig. 29.26b**. There is no longer any field inside the material, except possibly in a very thin surface layer a hundred or so atoms thick. If a coil is wrapped around the sphere, the emf induced in the coil shows that during the superconducting transition the magnetic flux through the coil decreases from its initial value to zero; this is consistent with the absence of field inside the material. Finally, if the field is now turned off while the material is still in its superconducting phase, no emf is induced in the coil, and measurements show no field outside the sphere (**Fig. 29.26c**).

We conclude that during a superconducting transition in the presence of the field $\vec{B}_0$, all of the magnetic flux is expelled from the bulk of the sphere, and the magnetic flux Φ_B through the coil becomes zero. This expulsion of magnetic flux is called the *Meissner effect*. As **Fig. 29.26b** shows, this expulsion crowds the magnetic field lines closer together to the side of the sphere, increasing $\vec{B}$ there.

Superconductor Levitation and Other Applications

The diamagnetic nature of a superconductor has some interesting *mechanical* consequences. A paramagnetic or ferromagnetic material is attracted by a permanent magnet because the magnetic dipoles in the material align with the nonuniform magnetic field of the permanent magnet. (We discussed this in Section 27.7.) For a diamagnetic material

Figure 29.25 Phase diagram for pure mercury, showing the critical magnetic field B_c and its dependence on temperature. Superconductivity is impossible above the critical temperature T_c . The curves for other superconducting materials are similar but with different numerical values.

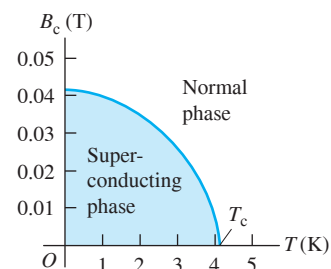


Figure 29.26 A superconducting material (a) above the critical temperature and (b), (c) below the critical temperature.

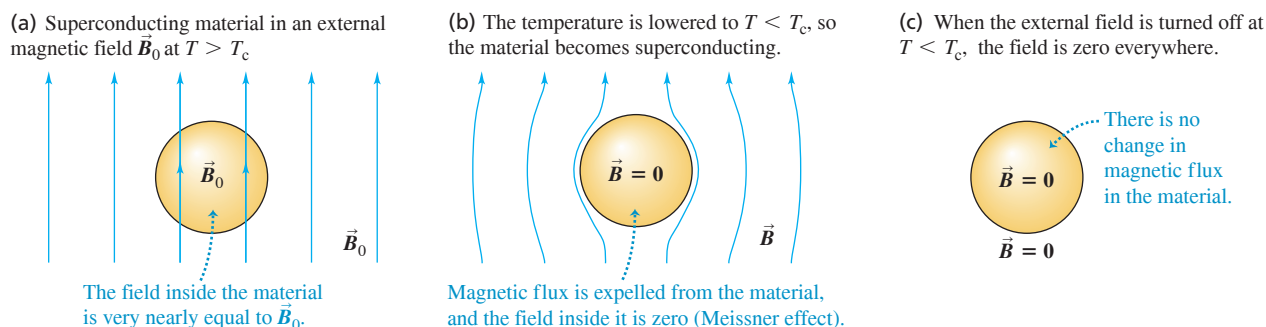
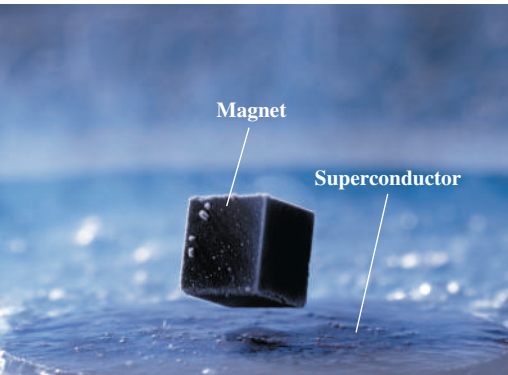


Figure 29.27 A superconductor exerts a repulsive force on a magnet, supporting the magnet in midair.



the magnetization is in the opposite sense, and a diamagnetic material is *repelled* by a permanent magnet. By Newton's third law the magnet is also repelled by the diamagnetic material. **Figure 29.27** shows the repulsion between a specimen of a high-temperature superconductor and a magnet; the magnet is supported ("levitated") by this repulsive magnetic force.

The behavior we have described is characteristic of what are called *type-I superconductors*. There is another class of superconducting materials called *type-II superconductors*. When such a material in the superconducting phase is placed in a magnetic field, the bulk of the material remains superconducting, but thin filaments of material, running parallel to the field, may return to the normal phase. Currents circulate around the boundaries of these filaments, and there *is* magnetic flux inside them. Type-II superconductors are used for electromagnets because they usually have much larger values of B_c than do type-I materials, permitting much larger magnetic fields without destroying the superconducting state. Type-II superconductors have *two* critical magnetic fields: The first, B_{c1} , is the field at which magnetic flux begins to enter the material, forming the filaments just described, and the second, B_{c2} , is the field at which the material becomes normal.

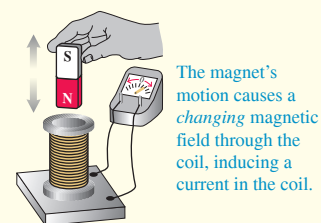
Superconducting electromagnets are in everyday use not only in research laboratories but also in medical MRI (magnetic resonance imaging) scanners. As we described in Section 27.7, scanning a patient through MRI requires a strong magnetic field to align the magnetic dipoles of the patient's atomic nuclei. A steady field of 1.5 T or more is needed, which is very difficult to produce with a conventional electromagnet, since this would require very high currents and hence very high energy losses due to resistance in the electromagnet coils. But with a superconducting electromagnet there is no resistive energy loss, and fields up to 10 T can routinely be attained.

Very sensitive measurements of magnetic fields can be made with superconducting quantum interference devices (SQUIDS), which can detect changes in magnetic flux of less than 10^{-14} Wb; these devices have applications in medicine, geology, and other fields. The number of potential uses for superconductors has increased greatly since the discovery in 1987 of high-temperature superconductors. These materials have critical temperatures that are above the temperature of liquid nitrogen (about 77 K) and so are comparatively easy to attain. Development of practical applications of superconductor science promises to be an exciting chapter in contemporary technology.

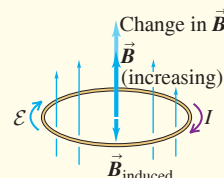
CHAPTER 29 SUMMARY

Faraday's law: Faraday's law states that the induced emf in a closed loop equals the negative of the time rate of change of magnetic flux through the loop. This relationship is valid whether the flux change is caused by a changing magnetic field, motion of the loop, or both. (See Examples 29.1–29.6.)

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \quad (29.3)$$



Lenz's law: Lenz's law states that an induced current or emf always tends to oppose or cancel out the change that caused it. Lenz's law can be derived from Faraday's law and is often easier to use. (See Examples 29.7 and 29.8.)



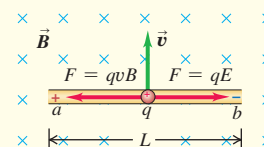
Motional emf: If a conductor moves in a magnetic field, a motional emf is induced. (See Examples 29.9 and 29.10.)

$$\mathcal{E} = vBL \quad (29.6)$$

(conductor with length L moves in uniform $\vec{B}$ field, $\vec{L}$ and $\vec{v}$ both perpendicular to $\vec{B}$ and to each other)

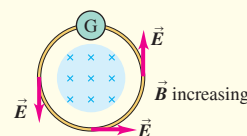
$$\mathcal{E} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{l} \quad (29.7)$$

(all or part of a closed loop moves in a $\vec{B}$ field)



Induced electric fields: When an emf is induced by a changing magnetic flux through a stationary conductor, there is an induced electric field $\vec{E}$ of nonelectrostatic origin. This field is nonconservative and cannot be associated with a potential. (See Example 29.11.)

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (29.10)$$



Displacement current and Maxwell's equations: A time-varying electric field generates displacement current i_D , which acts as a source of magnetic field in exactly the same way as conduction current. The relationships between electric and magnetic fields and their sources can be stated compactly in four equations, called Maxwell's equations. Together they form a complete basis for the relationship of $\vec{E}$ and $\vec{B}$ fields to their sources.

$$i_D = \epsilon \frac{d\Phi_E}{dt} \quad (29.14)$$

(displacement current)

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (29.18)$$

(Gauss's law for $\vec{E}$ fields)

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (29.19)$$

(Gauss's law for $\vec{B}$ fields)

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (29.20)$$

(Faraday's law)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \quad (29.21)$$

(Ampere's law including displacement current)



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 29.1, 29.2, and 29.3** (Section 29.2) and **CONCEPTUAL EXAMPLES 29.7 and 29.8** (Section 29.3) before attempting these problems.

VP29.8.1 A single circular loop of wire with radius 2.40 cm lies in the xy -plane. There is a uniform magnetic field that changes at a steady rate from 0.140 T in the $+z$ -direction at $t = 0$ to 0.110 T in the $-z$ -direction at $t = 2.00$ s. Take the area vector for the loop to be in the $+z$ -direction. Find (a) the magnetic flux through the loop at $t = 0$ and $t = 2.00$ s, and (b) the magnitude of the induced emf in the loop while the field is changing.

VP29.8.2 A circular coil with 455 turns of wire has radius 5.00 cm and resistance 14.5 Ω . The area vector of the coil points in the $+z$ -direction. The coil is in a region of uniform magnetic field that points 20.0° from the $+z$ -direction. The magnitude of the field is initially 0.0600 T and decreases at a rate of -3.00×10^{-3} T/s. Find the magnitude of (a) the induced emf and (b) the induced current in the coil while the field is changing.

VP29.8.3 A square single loop of wire 4.00 cm on a side with 875 turns lies in the xy -plane. The loop is in a uniform magnetic field that changes at a steady rate from $\vec{B} = (0.200 \text{ T})\hat{i} + (0.150 \text{ T})\hat{k}$ at $t = 0$ to $\vec{B} = (0.300 \text{ T})\hat{i} + (-0.200 \text{ T})\hat{k}$ at $t = 3.00$ s. At the time $t = 2.00$ s, find (a) the magnitude of the induced emf and (b) the direction of the induced current in the coil as seen from a point on the $+z$ -axis.

VP29.8.4 A coil with area A and N turns of wire lies in the xy -plane. Its area vector points in the $+z$ -direction. It is exposed to a uniform time-varying magnetic field $\vec{B} = (B_0 \sin \omega t)\hat{k}$, where ω is a positive constant. Find (a) the induced emf in the coil and (b) the direction of the induced current in the coil as seen from a point on the $+z$ -axis at $t = 0$ and at $t = \pi/\omega$.

Be sure to review **EXAMPLE 29.9** (Section 29.4) before attempting these problems.

VP29.9.1 An isolated conducting rod of length 8.00 cm is oriented parallel to the x -axis. It moves in the $+y$ -direction at 3.90 m/s in the presence of a uniform magnetic field of magnitude 0.600 T that points in the $-z$ -direction. Find (a) the magnitude of the motional emf in the rod and (b) the magnitude and direction of the electric field produced in the rod.

VP29.9.2 In the slidewire generator of Fig. 29.15b, you reverse the rod's motion so that it moves to the left rather than to the right. The moving rod is 5.00 cm long and moves at 2.40 m/s, and the uniform

magnetic field has magnitude 0.150 T. If the resistance of the circuit at a given instant is 0.0200 Ω , find the current in the circuit and its direction around the circuit.

VP29.9.3 An isolated conducting rod of length 8.00 cm is oriented parallel to the x -axis. At $t = 0$, end a is at $x = y = z = 0$ and end b is at $x = 8.00$ cm, $y = z = 0$. The rod moves with constant velocity $\vec{v} = (1.50 \text{ m/s})\hat{i} + (2.00 \text{ m/s})\hat{j}$ in a uniform magnetic field $\vec{B} = (-0.120 \text{ T})\hat{i} + (0.150 \text{ T})\hat{k}$. (a) Find the magnitude of the induced emf in the rod. (b) Which end of the rod, a or b , is at higher potential?

VP29.9.4 The magnetic force on the moving rod in a slidewire generator (Fig. 29.15b) has magnitude 0.0600 N at an instant when the resistance of the circuit is 0.750 Ω . The rod is 0.190 m long and the magnetic field (which is perpendicular to the plane of the generator) has magnitude 0.550 T. Find the speed of the rod.

Be sure to review **EXAMPLE 29.11** (Section 29.5) before attempting these problems.

VP29.11.1 A long solenoid has a cross-sectional area of 3.00 cm². The current through the windings is decreasing at a rate of 22.0 A/s. A wire loop of radius 3.10 cm is around the solenoid, parallel with its coils, and centered on the axis of the solenoid. The magnitude of the induced emf is 15.0 μ V. Find (a) the number of turns per meter in the solenoid and (b) the magnitude of the induced electric field within the loop.

VP29.11.2 A circular wire loop of radius 0.360 cm lies in the xz -plane. There is a uniform magnetic field in the y -direction that decreases at 0.0150 T/s. Find the magnitude of the induced electric field in the wire.

VP29.11.3 A long solenoid with cross-sectional area 4.00 cm² and 965 turns per meter is oriented with its axis along the z -axis. The field inside the solenoid points in the $+z$ -direction. A wire loop of radius 5.00 cm is around the solenoid, parallel with its coils, centered on the axis of the solenoid, and lying in the xy -plane. Find the rate of change of the current in the solenoid if the electric field in the loop at the point $x = 5.00$ cm, $y = 0$, $z = 0$ is (a) $\vec{E} = (+1.20 \times 10^{-5} \text{ V/m})\hat{j}$; (b) $\vec{E} = (-1.80 \times 10^{-5} \text{ V/m})\hat{j}$.

VP29.11.4 A long solenoid has 585 turns per meter and a cross-sectional area of 2.70 cm². The current through the windings as a function of time is $(0.600 \text{ A/s}^2)t^2$. A wire loop of resistance 0.600 Ω is around the solenoid, parallel with its coils, and centered on the axis of the solenoid. What is the current induced in the loop at $t = 13.9$ s?

BRIDGING PROBLEM A Falling Square Loop

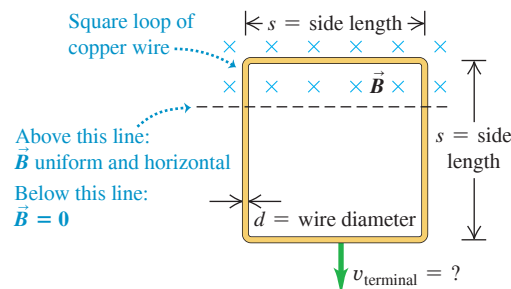
A vertically oriented square loop of copper wire falls from rest in a region in which the field $\vec{B}$ is horizontal, uniform, and perpendicular to the plane of the loop, into a field-free region (Fig. 29.28). The side length of the loop is s and the wire diameter is d . The resistivity of copper is ρ_R and the density of copper is ρ_m . If the loop reaches its terminal speed while its upper segment is still in the magnetic-field region, find an expression for the terminal speed.

SOLUTION GUIDE

IDENTIFY and SET UP

1. The motion of the loop through the magnetic field induces an emf and a current in the loop. The field then gives rise to a magnetic

Figure 29.28 A wire loop falling in a horizontal magnetic field $\vec{B}$. The plane of the loop is perpendicular to $\vec{B}$.



force on this current that opposes the downward force of gravity. The loop reaches terminal speed (it no longer accelerates) when the upward magnetic force balances the downward force of gravity.

2. Consider the case in which the entire loop is in the magnetic-field region. Is there an induced emf in this case? If so, what is its direction?
3. Consider the case in which only the upper segment of the loop is in the magnetic-field region. Is there an induced emf in this case? If so, what is its direction?
4. For the case in which there is an induced emf and hence an induced current, what is the direction of the magnetic force on each of the four sides of the loop? What is the direction of the *net* magnetic force on the loop?

EXECUTE

5. For the case in which the loop is falling at speed v and there is an induced emf, find (i) the emf, (ii) the induced current, and (iii) the magnetic force on the loop in terms of its resistance R .
6. Find R and the mass of the loop in terms of the given information about the loop.
7. Use your results from steps 5 and 6 to find an expression for the terminal speed.

EVALUATE

8. How does the terminal speed depend on the magnetic-field magnitude B ? Explain why this makes sense.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q29.1 A sheet of copper is placed between the poles of an electromagnet with the magnetic field perpendicular to the sheet. When the sheet is pulled out, a considerable force is required, and the force required increases with speed. Explain. Is a force required also when the sheet is inserted between the poles? Explain.

Q29.2 In Fig. 29.8, if the angular speed ω of the loop is doubled, then the frequency with which the induced current changes direction doubles, and the maximum emf also doubles. Why? Does the torque required to turn the loop change? Explain.

Q29.3 Two circular loops lie side by side in the same plane. One is connected to a source that supplies an increasing current; the other is a simple closed ring. Is the induced current in the ring in the same direction as the current in the loop connected to the source, or opposite? What if the current in the first loop is decreasing? Explain.

Q29.4 For Eq. (29.6), show that if v is in meters per second, B in teslas, and L in meters, then the units of the right-hand side of the equation are joules per coulomb or volts (the correct SI units for $\mathcal{E}$).

Q29.5 A long, straight conductor passes through the center of a metal ring, perpendicular to its plane. If the current in the conductor increases, is a current induced in the ring? Explain.

Q29.6 A student asserted that if a permanent magnet is dropped down a vertical copper pipe, it eventually reaches a terminal velocity even if there is no air resistance. Why should this be? Or should it?

Q29.7 An airplane is in level flight over Antarctica, where the magnetic field of the earth is mostly directed upward away from the ground. As viewed by a passenger facing toward the front of the plane, is the left or the right wingtip at higher potential? Does your answer depend on the direction the plane is flying?

Q29.8 Consider the situation in Exercise 29.21. In part (a), find the direction of the force that the large circuit exerts on the small one. Explain how this result is consistent with Lenz's law.

Q29.9 A metal rectangle is close to a long, straight, current-carrying wire, with two of its sides parallel to the wire. If the current in the long wire is decreasing, is the rectangle repelled by or attracted to the wire? Explain why this result is consistent with Lenz's law.

Q29.10 A square conducting loop is in a region of uniform, constant magnetic field. Can the loop be rotated about an axis along one side and no emf be induced in the loop? Discuss, in terms of the orientation of the rotation axis relative to the magnetic-field direction.

Q29.11 Example 29.6 discusses the external force that must be applied to the slidewire to move it at constant speed. If there were a break in the left-hand end of the U-shaped conductor, how much force would be needed to move the slidewire at constant speed? As in the example, you can ignore friction.

Q29.12 In the situation shown in Fig. 29.18, would it be appropriate to ask how much *energy* an electron gains during a complete trip around the wire loop with current I ? Would it be appropriate to ask what *potential difference* the electron moves through during such a complete trip? Explain your answers.

Q29.13 A metal ring is oriented with the plane of its area perpendicular to a spatially uniform magnetic field that increases at a steady rate. If the radius of the ring is doubled, by what factor do (a) the emf induced in the ring and (b) the electric field induced in the ring change?

Q29.14 Small one-cylinder gasoline engines sometimes use a device called a *magneto* to supply current to the spark plug. A permanent magnet is attached to the flywheel, and a stationary coil is mounted adjacent to it. Explain how this device is able to generate current. What happens when the magnet passes the coil?

Q29.15 Does Lenz's law say that the induced current in a metal loop always flows to oppose the magnetic flux through that loop? Explain.

Q29.16 Does Faraday's law say that a large magnetic flux induces a large emf in a coil? Explain.

Q29.17 Can one have a displacement current as well as a conduction current within a conductor? Explain.

Q29.18 Your physics study partner asks you to consider a parallel-plate capacitor that has a dielectric completely filling the volume between the plates. He then claims that Eqs. (29.13) and (29.14) show that the conduction current in the dielectric equals the displacement current in the dielectric. Do you agree? Explain.

Q29.19 Match the mathematical statements of Maxwell's equations as given in Section 29.7 to these verbal statements. (a) Closed electric field lines are evidently produced only by changing magnetic flux. (b) Closed magnetic field lines are produced both by the motion of electric charge and by changing electric flux. (c) Electric field lines can start on positive charges and end on negative charges. (d) Evidently there are no magnetic monopoles on which to start and end magnetic field lines.

Q29.20 If magnetic monopoles existed, the right-hand side of Eq. (29.20) would include a term proportional to the current of magnetic monopoles. Suppose a steady monopole current is moving in a long straight wire. Sketch the *electric* field lines that such a current would produce.

Q29.21 A type-II superconductor in an external field between B_{c1} and B_{c2} has regions that contain magnetic flux and have resistance, and also has superconducting regions. What is the resistance of a long, thin cylinder of such material?

EXERCISES

Section 29.2 Faraday's Law

29.1 • A single loop of wire with an area of 0.0900 m^2 is in a uniform magnetic field that has an initial value of 3.80 T , is perpendicular to the plane of the loop, and is decreasing at a constant rate of 0.190 T/s . (a) What emf is induced in this loop? (b) If the loop has a resistance of 0.600Ω , find the current induced in the loop.

29.2 •• In a physics laboratory experiment, a coil with 200 turns enclosing an area of 12 cm^2 is rotated in 0.040 s from a position where its plane is perpendicular to the earth's magnetic field to a position where its plane is parallel to the field. The earth's magnetic field at the lab location is $6.0 \times 10^{-5} \text{ T}$. (a) What is the magnetic flux through each turn of the coil before it is rotated? After it is rotated? (b) What is the average emf induced in the coil?

29.3 • The magnetic flux through a coil is given by $\Phi_B = \alpha t - \beta t^3$, where α and β are constants. (a) What are the units of α and β ? (b) If the induced emf is zero at $t = 0.500 \text{ s}$, how is α related to β ? (c) If the emf at $t = 0$ is -1.60 V , what is the emf at $t = 0.250 \text{ s}$?

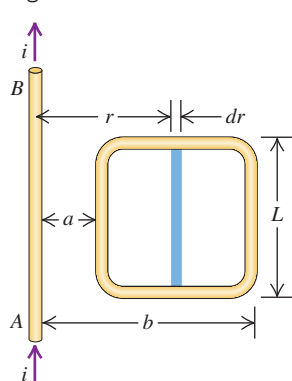
29.4 •• A small, closely wound coil has N turns, area A , and resistance R . The coil is initially in a uniform magnetic field that has magnitude B and a direction perpendicular to the plane of the loop. The coil is then rapidly pulled out of the field so that the flux through the coil is reduced to zero in time Δt . (a) What are the magnitude of the average emf $\mathcal{E}_{\text{av}}$ and average current I_{av} induced in the coil? (b) The total charge Q that flows through the coil is given by $Q = I_{\text{av}}\Delta t$. Derive an expression for Q in terms of N , A , B , and R . Note that Q does not depend on Δt . (c) What is Q if $N = 150$ turns, $A = 4.50 \text{ cm}^2$, $R = 30.0 \Omega$, and $B = 0.200 \text{ T}$?

29.5 • A circular loop of wire with a radius of 12.0 cm and oriented in the horizontal xy -plane is located in a region of uniform magnetic field. A field of 1.5 T is directed along the positive z -direction, which is upward. (a) If the loop is removed from the field region in a time interval of 2.0 ms , find the average emf that will be induced in the wire loop during the extraction process. (b) If the coil is viewed looking down on it from above, is the induced current in the loop clockwise or counterclockwise?

29.6 • CALC A coil 4.00 cm in radius, containing 500 turns, is placed in a uniform magnetic field that varies with time according to $B = (0.0120 \text{ T/s})t + (3.00 \times 10^{-5} \text{ T/s}^4)t^4$. The coil is connected to a 600Ω resistor, and its plane is perpendicular to the magnetic field. You can ignore the resistance of the coil. (a) Find the magnitude of the induced emf in the coil as a function of time. (b) What is the current in the resistor at time $t = 5.00 \text{ s}$?

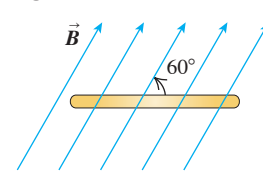
29.7 • CALC The current in the long, straight wire AB shown in Fig. E29.7 is upward and is increasing steadily at a rate di/dt . (a) At an instant when the current is i , what are the magnitude and direction of the field $\vec{B}$ at a distance r to the right of the wire? (b) What is the flux $d\Phi_B$ through the narrow, shaded strip? (c) What is the total flux through the loop? (d) What is the induced emf in the loop? (e) Evaluate the numerical value of the induced emf if $a = 12.0 \text{ cm}$, $b = 36.0 \text{ cm}$, $L = 24.0 \text{ cm}$, and $di/dt = 9.60 \text{ A/s}$.

Figure E29.7



29.8 • CALC A flat, circular, steel loop of radius 75 cm is at rest in a uniform magnetic field, as shown in an edge-on view in Fig. E29.8. The field is changing with time, according to $B(t) = (1.4 \text{ T})e^{-(0.057 \text{ s}^{-1})t}$. (a) Find the emf induced in the loop as a function of time. (b) When is the induced emf equal to $\frac{1}{10}$ of its initial value? (c) Find the direction of the current induced in the loop, as viewed from above the loop.

Figure E29.8



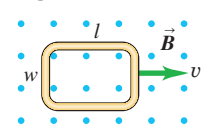
29.9 • Shrinking Loop. A circular loop of flexible iron wire has an initial circumference of 165.0 cm , but its circumference is decreasing at a constant rate of 12.0 cm/s due to a tangential pull on the wire. The loop is in a constant, uniform magnetic field oriented perpendicular to the plane of the loop and with magnitude 0.500 T . (a) Find the emf induced in the loop at the instant when 9.0 s have passed. (b) Find the direction of the induced current in the loop as viewed looking along the direction of the magnetic field.

29.10 • A closely wound rectangular coil of 80 turns has dimensions of 25.0 cm by 40.0 cm . The plane of the coil is rotated from a position where it makes an angle of 37.0° with a magnetic field of 1.70 T to a position perpendicular to the field. The rotation takes 0.0600 s . What is the average emf induced in the coil?

29.11 •• CALC A circular loop of wire with radius 2.00 cm and resistance 0.600Ω is in a region of a spatially uniform magnetic field $\vec{B}$ that is perpendicular to the plane of the loop. At $t = 0$ the magnetic field has magnitude $B_0 = 3.00 \text{ T}$. The magnetic field then decreases according to the equation $B(t) = B_0 e^{-t/\tau}$, where $\tau = 0.500 \text{ s}$. (a) What is the maximum magnitude of the current I induced in the loop? (b) What is the induced current I when $t = 1.50 \text{ s}$?

29.12 • A flat, rectangular coil of dimensions l and w is pulled with uniform speed v through a uniform magnetic field B with the plane of its area perpendicular to the field (Fig. E29.12). (a) Find the emf induced in this coil. (b) If the speed and magnetic field are both tripled, what is the induced emf?

Figure E29.12

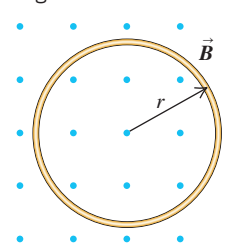


29.13 •• The armature of a small generator consists of a flat, square coil with 120 turns and sides with a length of 1.60 cm . The coil rotates in a magnetic field of 0.0750 T . What is the angular speed of the coil if the maximum emf produced is 24.0 mV ?

Section 29.3 Lenz's Law

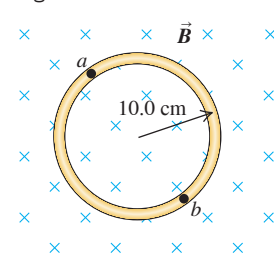
29.14 • A circular loop of wire with radius $r = 0.0480 \text{ m}$ and resistance $R = 0.160 \Omega$ is in a region of spatially uniform magnetic field, as shown in Fig. E29.14. The magnetic field is directed out of the plane of the figure. The magnetic field has an initial value of 8.00 T and is decreasing at a rate of $dB/dt = -0.680 \text{ T/s}$. (a) Is the induced current in the loop clockwise or counterclockwise? (b) What is the rate at which electrical energy is being dissipated by the resistance of the loop?

Figure E29.14



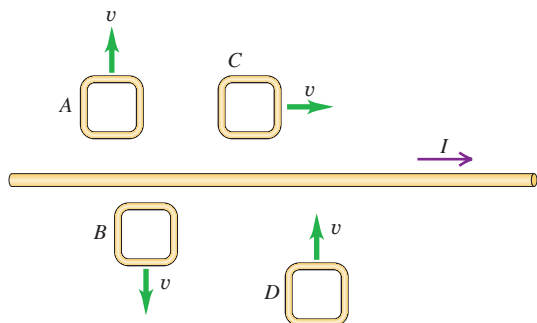
29.15 • A circular loop of wire is in a region of spatially uniform magnetic field, as shown in Fig. E29.15. The magnetic field is directed into the plane of the figure. Determine the direction (clockwise or counterclockwise) of the induced current in the loop when (a) B is increasing; (b) B is decreasing; (c) B is constant with value B_0 . Explain your reasoning.

Figure E29.15



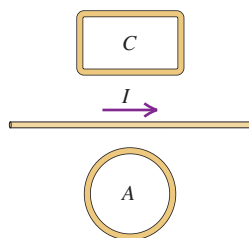
29.16 • The current I in a long, straight wire is constant and is directed toward the right as in **Fig. E29.16**. Conducting loops A , B , C , and D are moving, in the directions shown, near the wire. (a) For each loop, is the direction of the induced current clockwise or counterclockwise, or is the induced current zero? (b) For each loop, what is the direction of the net force that the wire exerts on the loop? Give your reasoning for each answer.

Figure E29.16



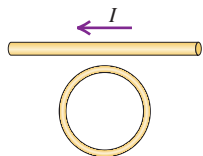
29.17 • Two closed loops A and C are close to a long wire carrying a current I (**Fig. E29.17**). (a) Find the direction (clockwise or counterclockwise) of the current induced in each loop if I is steadily decreasing. (b) While I is decreasing, what is the direction of the net force that the wire exerts on each loop? Explain how you obtain your answer.

Figure E29.17



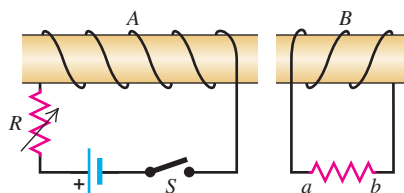
29.18 • The current in **Fig. E29.18** obeys the equation $I(t) = I_0 e^{-bt}$, where $b > 0$. Find the direction (clockwise or counterclockwise) of the current induced in the round coil for $t > 0$.

Figure E29.18



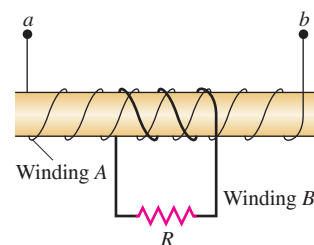
29.19 • Using Lenz's law, determine the direction of the current in resistor ab of **Fig. E29.19** when (a) switch S is opened after having been closed for several minutes; (b) coil B is brought closer to coil A with the switch closed; (c) the resistance of R is decreased while the switch remains closed.

Figure E29.19



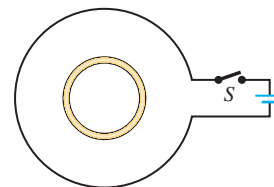
29.20 • A cardboard tube is wrapped with two windings of insulated wire wound in opposite directions, as shown in **Fig. E29.20**. Terminals a and b of winding A may be connected to a battery through a reversing switch. State whether the induced current in the resistor R is from left to right or from right to left in the following circumstances: (a) the current in winding A is from a to b and is increasing; (b) the current in winding A is from b to a and is decreasing; (c) the current in winding A is from b to a and is increasing.

Figure E29.20



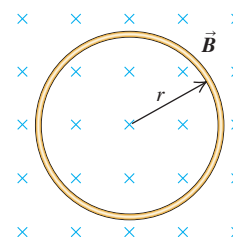
29.21 • A small, circular ring is inside a larger loop that is connected to a battery and a switch (**Fig. E29.21**). Use Lenz's law to find the direction of the current induced in the small ring (a) just after switch S is closed; (b) after S has been closed a long time; (c) just after S has been reopened after it was closed for a long time.

Figure E29.21



29.22 • **CALC** A circular loop of wire with radius $r = 0.0250$ m and resistance $R = 0.390 \Omega$ is in a region of spatially uniform magnetic field, as shown in **Fig. E29.22**. The magnetic field is directed into the plane of the figure. At $t = 0$, $B = 0$. The magnetic field then begins increasing, with $B(t) = (0.380 \text{ T/s}^3)t^3$. What is the current in the loop (magnitude and direction) at the instant when $B = 1.33$ T?

Figure E29.22



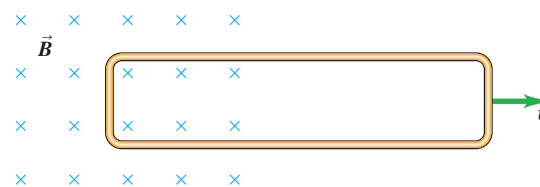
Section 29.4 Motional Electromotive Force

29.23 • A magnetic field of 0.080 T is in the y -direction. The velocity of wire segment S has a magnitude of 78 m/s and components of 18 m/s in the x -direction, 24 m/s in the y -direction, and 72 m/s in the z -direction. The segment has length 0.50 m and is parallel to the z -axis as it moves. (a) Find the motional emf induced between the ends of the segment. (b) What would the motional emf be if the wire segment was parallel to the y -axis?

29.24 • A rectangular loop of wire with dimensions 1.50 cm by 8.00 cm and resistance $R = 0.600 \Omega$ is being pulled to the right out of a region of uniform magnetic field. The magnetic field has magnitude $B = 2.40$ T and is directed into the plane of **Fig. E29.24**. At the instant when the speed of the loop is 3.00 m/s and it is still partially in the field region, what force (magnitude and direction) does the magnetic field exert on the loop?

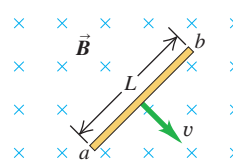
29.25 • In **Fig. E29.25** a conducting rod of length $L = 30.0$ cm moves in a magnetic field $\vec{B}$ of magnitude 0.450 T directed into the plane of the figure.

Figure E29.24



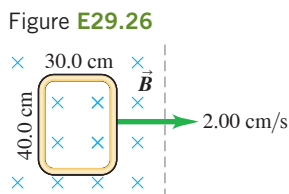
The rod moves with speed $v = 5.00$ m/s in the direction shown.

Figure E29.25



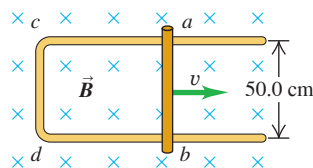
(a) What is the potential difference between the ends of the rod? (b) Which point, a or b , is at higher potential? (c) When the charges in the rod are in equilibrium, what are the magnitude and direction of the electric field within the rod? (d) When the charges in the rod are in equilibrium, which point, a or b , has an excess of positive charge? (e) What is the potential difference across the rod if it moves (i) parallel to ab and (ii) directly out of the page?

29.26 • A rectangle measuring 30.0 cm by 40.0 cm is located inside a region of a spatially uniform magnetic field of 1.25 T, with the field perpendicular to the plane of the coil (**Fig. E29.26**). The coil is pulled out at a steady rate of 2.00 cm/s traveling perpendicular to the field lines. The region of the field ends abruptly as shown. Find the emf induced in this coil when it is (a) all inside the field; (b) partly inside the field; (c) all outside the field.



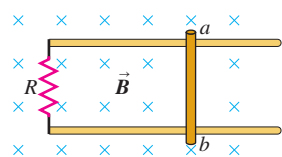
29.27 • The conducting rod ab shown in **Fig. E29.27** makes contact with metal rails ca and db . The apparatus is in a uniform magnetic field of 0.800 T, perpendicular to the plane of the figure. (a) Find the magnitude of the emf induced in the rod when it is moving toward the right with a speed 7.50 m/s. (b) In what direction does the current flow in the rod? (c) If the resistance of the circuit $abdc$ is $1.50\ \Omega$ (assumed to be constant), find the force (magnitude and direction) required to keep the rod moving to the right with a constant speed of 7.50 m/s. You can ignore friction. (d) Compare the rate at which mechanical work is done by the force (Fv) with the rate at which thermal energy is developed in the circuit (I^2R).

Figure E29.27



29.28 • A 0.650-m-long metal bar is pulled to the right at a steady 5.0 m/s perpendicular to a uniform, 0.750 T magnetic field. The bar rides on parallel metal rails connected through a $25.0\ \Omega$ resistor (**Fig. E29.28**), so the apparatus makes a complete circuit. Ignore the resistance of the bar and the rails.

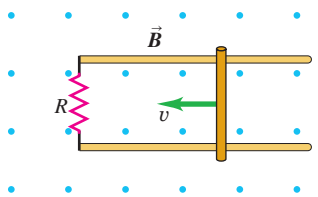
Figure E29.28



(a) Calculate the magnitude of the emf induced in the circuit. (b) Find the direction of the current induced in the circuit by using (i) the magnetic force on the charges in the moving bar; (ii) Faraday's law; (iii) Lenz's law. (c) Calculate the current through the resistor.

29.29 • A 0.360-m-long metal bar is pulled to the left by an applied force F . The bar rides on parallel metal rails connected through a $45.0\ \Omega$ resistor, as shown in **Fig. E29.29**, so the apparatus makes a complete circuit. You can ignore the resistance of the bar and rails. The circuit is in a uniform 0.650 T magnetic field that is directed out of the plane of the figure. At the instant when the bar is moving to the left at 5.90 m/s, (a) is the induced current in the circuit clockwise or counterclockwise and (b) what is the rate at which the applied force is doing work on the bar?

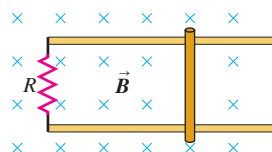
Figure E29.29



29.30 • Consider the circuit shown in **Fig. E29.29**, but with the bar moving to the right with speed v . As in Exercise 29.29, the bar has length 0.360 m, $R = 45.0\ \Omega$, and $B = 0.650$ T. (a) Is the induced current in the circuit clockwise or counterclockwise? (b) At an instant when the $45.0\ \Omega$ resistor is dissipating electrical energy at a rate of 0.840 J/s, what is the speed of the bar?

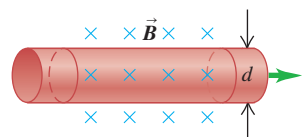
29.31 • A 0.250-m-long bar moves on parallel rails that are connected through a $6.00\ \Omega$ resistor, as shown in **Fig. E29.31**, so the apparatus makes a complete circuit. You can ignore the resistance of the bar and rails. The circuit is in a uniform magnetic field $B = 1.20$ T that is directed into the plane of the figure. At an instant when the induced current in the circuit is counterclockwise and equal to 1.75 A, what is the velocity of the bar (magnitude and direction)?

Figure E29.31



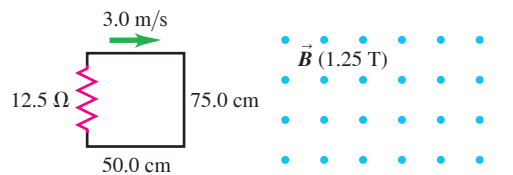
29.32 • **BIO Measuring Blood Flow.** Blood contains positive and negative ions and thus is a conductor. A blood vessel, therefore, can be viewed as an electrical wire. We can even picture the flowing blood as a series of parallel conducting slabs whose thickness is the diameter d of the vessel moving with speed v . (See **Fig. E29.32**.) (a) If the blood vessel is placed in a magnetic field B perpendicular to the vessel, as in the figure, show that the motional potential difference induced across it is $\mathcal{E} = vBd$. (b) If you expect that the blood will be flowing at 15 cm/s for a vessel 5.0 mm in diameter, what strength of magnetic field will you need to produce a potential difference of 1.0 mV? (c) Show that the volume rate of flow (R) of the blood is equal to $R = \pi \mathcal{E} d / 4B$. (Note: Although the method developed here is useful in measuring the rate of blood flow in a vessel, it is limited to use in surgery because measurement of the potential $\mathcal{E}$ must be made directly across the vessel.)

Figure E29.32



29.33 • A rectangular circuit is moved at a constant velocity of 3.0 m/s into, through, and then out of a uniform 1.25 T magnetic field, as shown in **Fig. E29.33**. The magnetic-field region is considerably wider than 50.0 cm. Find the magnitude and direction (clockwise or counterclockwise) of the current induced in the circuit as it is (a) going into the magnetic field; (b) totally within the magnetic field, but still moving; and (c) moving out of the field. (d) Sketch a graph of the current in this circuit as a function of time, including the preceding three cases.

Figure E29.33



Section 29.5 Induced Electric Fields

29.34 • A metal ring 4.50 cm in diameter is placed between the north and south poles of large magnets with the plane of its area perpendicular to the magnetic field. These magnets produce an initial uniform field of 1.12 T between them but are gradually pulled apart, causing this field to remain uniform but decrease steadily at 0.250 T/s. (a) What is the magnitude of the electric field induced in the ring? (b) In which direction (clockwise or counterclockwise) does the current flow as viewed by someone on the south pole of the magnet?

29.35 •• A long, thin solenoid has 400 turns per meter and radius 1.10 cm. The current in the solenoid is increasing at a uniform rate di/dt . The induced electric field at a point near the center of the solenoid and 3.50 cm from its axis is 8.00×10^{-6} V/m. Calculate di/dt .

29.36 •• A long, thin solenoid has 900 turns per meter and radius 2.50 cm. The current in the solenoid is increasing at a uniform rate of 36.0 A/s. What is the magnitude of the induced electric field at a point near the center of the solenoid and (a) 0.500 cm from the axis of the solenoid; (b) 1.00 cm from the axis of the solenoid?

29.37 • A long, straight solenoid with a cross-sectional area of 8.00 cm^2 is wound with 90 turns of wire per centimeter, and the windings carry a current of 0.350 A. A second winding of 12 turns encircles the solenoid at its center. The current in the solenoid is turned off such that the magnetic field of the solenoid becomes zero in 0.0400 s. What is the average induced emf in the second winding?

Section 29.7 Displacement Current and Maxwell's Equations

29.38 • A parallel-plate, air-filled capacitor is being charged as in Fig. 29.23. The circular plates have radius 4.00 cm, and at a particular instant the conduction current in the wires is 0.520 A. (a) What is the displacement current density j_D in the air space between the plates? (b) What is the rate at which the electric field between the plates is changing? (c) What is the induced magnetic field between the plates at a distance of 2.00 cm from the axis? (d) At 1.00 cm from the axis?

29.39 • Displacement Current in a Dielectric. Suppose that the parallel plates in Fig. 29.23 have an area of 3.00 cm^2 and are separated by a 2.50-mm-thick sheet of dielectric that completely fills the volume between the plates. The dielectric has dielectric constant 4.70. (You can ignore fringing effects.) At a certain instant, the potential difference between the plates is 120 V and the conduction current i_C equals 6.00 mA. At this instant, what are (a) the charge q on each plate; (b) the rate of change of charge on the plates; (c) the displacement current in the dielectric?

29.40 • CALC In Fig. 29.23 the capacitor plates have area 5.00 cm^2 and separation 2.00 mm. The plates are in vacuum. The charging current i_C has a constant value of 1.80 mA. At $t = 0$ the charge on the plates is zero. (a) Calculate the charge on the plates, the electric field between the plates, and the potential difference between the plates when $t = 0.500 \mu\text{s}$. (b) Calculate dE/dt , the time rate of change of the electric field between the plates. Does dE/dt vary in time? (c) Calculate the displacement current density j_D between the plates, and from this the total displacement current i_D . How do i_C and i_D compare?

29.41 • CALC The electric flux is $(4.0 \text{ V} \cdot \text{m/s}^5)t^5$ through a certain area of a dielectric that has dielectric constant 2.5. (a) Find the displacement current through that area at $t = 1.5 \text{ s}$. (b) At what time was the displacement current $\frac{1}{6}$ as much?

Section 29.8 Superconductivity

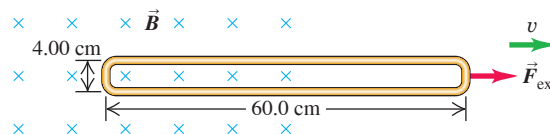
29.42 • At temperatures near absolute zero, B_c approaches 0.142 T for vanadium, a type-I superconductor. The normal phase of vanadium has a magnetic susceptibility close to zero. Consider a long, thin vanadium cylinder with its axis parallel to an external magnetic field $\vec{B}_0$ in the $+x$ -direction. At points far from the ends of the cylinder, by symmetry, all the magnetic vectors are parallel to the x -axis. At temperatures near absolute zero, what are the resultant magnetic field $\vec{B}$ and the magnetization $\vec{M}$ inside and outside the cylinder (far from the ends) for (a) $\vec{B}_0 = (0.130 \text{ T})\hat{i}$ and (b) $\vec{B}_0 = (0.260 \text{ T})\hat{i}$?

PROBLEMS

29.43 •• CP A motor vehicle generates electrical power using an alternator, which employs electromagnetic induction to convert mechanical energy to electrical energy. The alternator acts as a dc generator (Example 29.4). The alternator maintains and replenishes charge on the car's battery and operates headlights, radiator fans, windshield wipers, power windows, computer systems, sensors, sound systems, and other components. (a) A typical car battery provides 70 amp-hours of charge. How many coulombs is that? (b) If headlights each draw 20 A of current, a radiator fan draws 10 A, and windshield wipers each draw 5 A, estimate the peak current needed for a car to operate on a rainy night. (c) A car's alternator supplies an average emf of 14 V as emf induced in a sequence of stator coils in the presence of a magnetic field created by rotor coil electromagnets turned by a pulley system. A stator coil may have 42 windings and a cross-sectional diameter of 5.0 cm, and it rotates at 400 Hz. Estimate the strength of the magnetic field generated by a rotor coil.

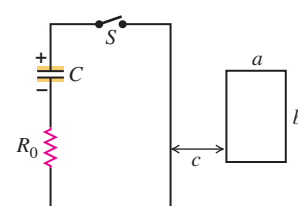
29.44 •• A very long, rectangular loop of wire can slide without friction on a horizontal surface. Initially the loop has part of its area in a region of uniform magnetic field that has magnitude $B = 2.90 \text{ T}$ and is perpendicular to the plane of the loop. The loop has dimensions 4.00 cm by 60.0 cm, mass 24.0 g, and resistance $R = 5.00 \times 10^{-3} \Omega$. The loop is initially at rest; then a constant force $F_{\text{ext}} = 0.180 \text{ N}$ is applied to the loop to pull it out of the field (Fig. P29.44). (a) What is the acceleration of the loop when $v = 3.00 \text{ cm/s}$? (b) What are the loop's terminal speed and acceleration when the loop is moving at that terminal speed? (c) What is the acceleration of the loop when it is completely out of the magnetic field?

Figure P29.44



29.45 •• CP CALC In the circuit shown in Fig. P29.45, the capacitor has capacitance $C = 20 \mu\text{F}$ and is initially charged to 100 V with the polarity shown. The resistor R_0 has resistance 10Ω . At time $t = 0$ the switch S is closed. The small circuit is not connected in any way to the large one. The wire of the small circuit has a resistance of $1.0 \Omega/\text{m}$ and contains 25 loops. The large circuit is a rectangle 2.0 m by 4.0 m, while the small one has dimensions $a = 10.0 \text{ cm}$ and $b = 20.0 \text{ cm}$. The distance c is 5.0 cm. (The figure is not drawn to scale.) Both circuits are held stationary. Assume that only the wire nearest the small circuit produces an appreciable magnetic field through it. (a) Find the current in the large circuit $200 \mu\text{s}$ after S is closed. (b) Find the current in the small circuit $200 \mu\text{s}$ after S is closed. (Hint: See Exercise 29.7.) (c) Find the direction of the current in the small circuit. (d) Justify why we can ignore the magnetic field from all the wires of the large circuit except for the wire closest to the small circuit.

Figure P29.45

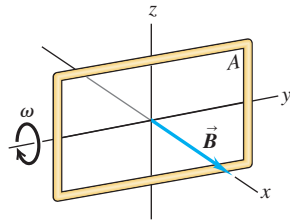


29.46 •• CP CALC In the circuit in Fig. P29.45, an emf of 90.0 V is added in series with the capacitor and the resistor, and the capacitor is initially uncharged. The emf is placed between the capacitor and switch S , with the positive terminal of the emf adjacent to the capacitor. Otherwise, the two circuits are the same as in Problem 29.45. The switch is closed at $t = 0$. When the current in the large circuit is 5.00 A, what are the magnitude and direction of the induced current in the small circuit?

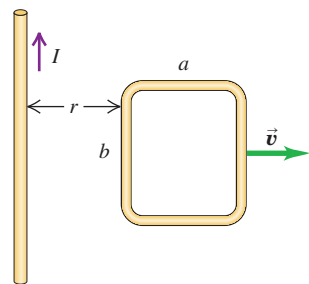
29.47 •• CALC A very long, straight solenoid with a cross-sectional area of 2.00 cm^2 is wound with 90.0 turns of wire per centimeter. Starting at $t = 0$, the current in the solenoid is increasing according to $i(t) = (0.160 \text{ A/s}^2)t^2$. A secondary winding of 5 turns encircles the solenoid at its center, such that the secondary winding has the same cross-sectional area as the solenoid. What is the magnitude of the emf induced in the secondary winding at the instant that the current in the solenoid is 3.20 A ?

29.48 • Suppose the loop in **Fig. P29.48** Figure **P29.48**

is (a) rotated about the y -axis; (b) rotated about the x -axis; (c) rotated about an edge parallel to the z -axis. What is the maximum induced emf in each case if $A = 600 \text{ cm}^2$, $\omega = 35.0 \text{ rad/s}$, and $B = 0.320 \text{ T}$?



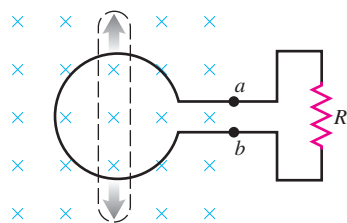
29.49 • In **Fig. P29.49** the loop is being pulled to the right at constant speed v . A constant current I flows in the long wire, in the direction shown. (a) Calculate the magnitude of the net emf $\mathcal{E}$ induced in the loop. Do this two ways: (i) by using Faraday's law of induction (*Hint*: See Exercise 29.7) and (ii) by looking at the emf induced in each segment of the loop due to its motion. (b) Find the direction (clockwise or counterclockwise) of the current induced in the loop. Do this two ways: (i) using Lenz's law and (ii) using the magnetic force on charges in the loop. (c) Check your answer for the emf in part (a) in the following special cases to see whether it is physically reasonable: (i) The loop is stationary; (ii) the loop is very thin, so $a \rightarrow 0$; (iii) the loop gets very far from the wire.

Figure **P29.49**

29.50 •• If you secure a refrigerator magnet about 2 mm from the metallic surface of a refrigerator door and then move the magnet sideways, you can feel a resistive force, indicating the presence of eddy currents in the surface. (a) Estimate the magnetic field strength B of the magnet to be 5 mT (Problem 28.53) and assume the magnet is rectangular with dimensions 4 cm wide by 2 cm high, so its area A is 8 cm^2 . Now estimate the magnetic flux Φ_B into the refrigerator door behind the magnet. (b) If you move the magnet sideways at a speed of 2 cm/s , what is a corresponding estimate of the time rate at which the magnetic flux through an area A fixed on the refrigerator is changing as the magnet passes over? Use this estimate to estimate the emf induced under the rectangle on the door's surface.

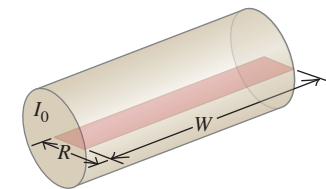
29.51 • A flexible circular loop Figure **P29.51**

6.50 cm in diameter lies in a magnetic field with magnitude 1.35 T , directed into the plane of the page as shown in **Fig. P29.51**. The loop is pulled at the points indicated by the arrows, forming a loop of zero area in 0.250 s . (a) Find the average induced emf in the circuit. (b) What is the direction of the current in R : from a to b or from b to a ? Explain your reasoning.

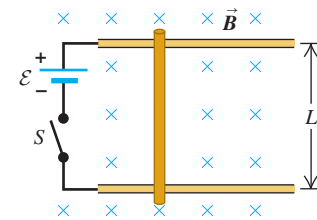


29.52 ••• CALC A conducting rod with length $L = 0.200 \text{ m}$, mass $m = 0.120 \text{ kg}$, and resistance $R = 80.0 \Omega$ moves without friction on metal rails as shown in **Fig. 29.11**. A uniform magnetic field with magnitude $B = 1.50 \text{ T}$ is directed into the plane of the figure. The rod is initially at rest, and then a constant force with magnitude $F = 1.90 \text{ N}$ and directed to the right is applied to the rod. How many seconds after the force is applied does the rod reach a speed of 25.0 m/s ?

29.53 •• CALC A very long, cylindrical wire of radius R carries a current I_0 uniformly distributed across the cross section of the wire. Calculate the magnetic flux through a rectangle that has one side of length W running down the center of the wire and another side of length R , as shown in **Fig. P29.53** (see Exercise 29.7).

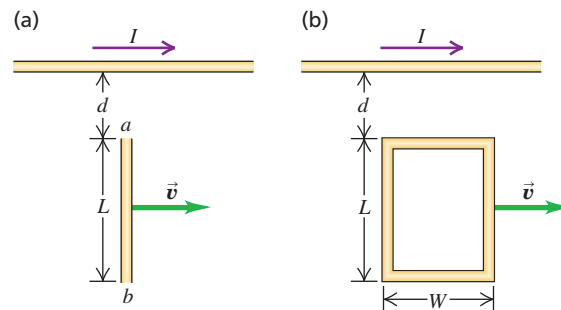
Figure **P29.53**

29.54 •• CP CALC Terminal Speed. A bar of length $L = 0.36 \text{ m}$ is free to slide without friction on horizontal rails as shown in **Fig. P29.54**. A uniform magnetic field $B = 2.4 \text{ T}$ is directed into the plane of the figure.

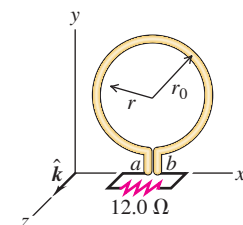
Figure **P29.54**

At one end of the rails there is a battery with emf $\mathcal{E} = 12 \text{ V}$ and a switch S . The bar has mass 0.90 kg and resistance 5.0Ω ; ignore all other resistance in the circuit. The switch is closed at time $t = 0$. (a) Sketch the bar's speed as a function of time. (b) Just after the switch is closed, what is the acceleration of the bar? (c) What is the acceleration of the bar when its speed is 2.0 m/s ? (d) What is the bar's terminal speed?

29.55 • CALC The long, straight wire shown in **Fig. P29.55a** carries constant current I . A metal bar with length L is moving at constant velocity $\vec{v}$, as shown in the figure. Point a is a distance d from the wire. (a) Calculate the emf induced in the bar. (b) Which point, a or b , is at higher potential? (c) If the bar is replaced by a rectangular wire loop of resistance R (**Fig. P29.55b**), what is the magnitude of the current induced in the loop?

Figure **P29.55**

29.56 • CALC A circular conducting ring with radius $r_0 = 0.0420 \text{ m}$ lies in the xy -plane in a region of uniform magnetic field $\vec{B} = B_0[1 - 3(t/t_0)^2 + 2(t/t_0)^3]\hat{k}$. In this expression, $t_0 = 0.0100 \text{ s}$ and is constant, t is time, $\hat{k}$ is the unit vector in the $+z$ -direction, and $B_0 = 0.0800 \text{ T}$ and is constant. At points a and b (**Fig. P29.56**) there is a small gap in the ring with wires leading to an external circuit of resistance $R = 12.0 \Omega$. There is no magnetic field at the location of the external circuit. (a) Derive an expression, as a function of time, for the total magnetic flux Φ_B through the ring. (b) Determine the emf induced in the ring at time $t = 5.00 \times 10^{-3} \text{ s}$. What is the polarity of the emf? (c) Because of the internal resistance of the ring, the current through R at the time given in part (b) is only 3.00 mA . Determine the internal resistance of the ring. (d) Determine the emf in the ring at a time $t = 1.21 \times 10^{-2} \text{ s}$. What is the polarity of the emf? (e) Determine the time at which the current through R reverses its direction.

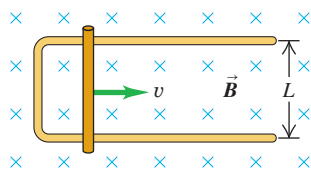
Figure **P29.56**

29.57 • CALC A slender rod, 0.240 m long, rotates with an angular speed of 8.80 rad/s about an axis through one end and perpendicular to the rod. The plane of rotation of the rod is perpendicular to a uniform magnetic field with a magnitude of 0.650 T. (a) What is the induced emf in the rod? (b) What is the potential difference between its ends? (c) Suppose instead the rod rotates at 8.80 rad/s about an axis through its center and perpendicular to the rod. In this case, what is the potential difference between the ends of the rod? Between the center of the rod and one end?

29.58 •• A 25.0-cm-long metal rod lies in the xy -plane and makes an angle of 36.9° with the positive x -axis and an angle of 53.1° with the positive y -axis. The rod is moving in the $+x$ -direction with a speed of 6.80 m/s. The rod is in a uniform magnetic field $\vec{B} = (0.120 \text{ T})\hat{i} - (0.220 \text{ T})\hat{j} - (0.0900 \text{ T})\hat{k}$. (a) What is the magnitude of the emf induced in the rod? (b) Indicate in a sketch which end of the rod is at higher potential.

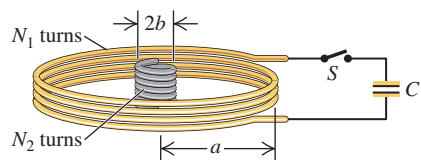
29.59 •• CP CALC A rectangular loop with width L and a slidewire with mass m are as shown in **Fig. P29.59**. A uniform magnetic field $\vec{B}$ is directed perpendicular to the plane of the loop into the plane of the figure. The slidewire is given an initial speed of v_0 and then released. There is no friction between the slidewire and the loop, and the resistance of the loop is negligible in comparison to the resistance R of the slidewire. (a) Obtain an expression for F , the magnitude of the force exerted on the wire while it is moving at speed v . (b) Show that the distance x that the wire moves before coming to rest is $x = mv_0 R / L^2 B^2$.

Figure P29.59



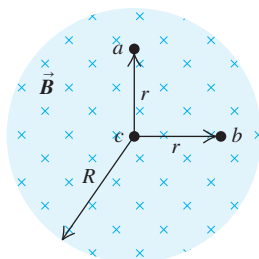
29.60 ••• CP A circular coil with $N_1 = 5000$ turns is made of a conducting material with resistance $0.0100 \Omega/\text{m}$ and radius $a = 40.0 \text{ cm}$. The coil is attached to a $C = 10.00 \mu\text{F}$ capacitor as shown in **Fig. P29.60**. A second coil with radius $b = 4.00 \text{ cm}$, made of the same wire, with $N_2 = 100$ turns, is concentric with the first coil and parallel to it. The capacitor has a charge of $+100 \mu\text{C}$ on its upper plate, and the switch S is open. At time $t = 0$ the switch is closed. (a) What is the magnitude of the current in the larger coil immediately after the switch is closed? (b) What is the magnetic flux through each turn of the smaller coil immediately after the switch is closed? (Since $b \ll a$, we may treat the magnetic field in the smaller coil due to the larger coil as uniform.) (c) What is the direction of the current in the smaller coil immediately after the switch is closed? (d) What is the direction of the current in the smaller coil at $t = 1.26 \text{ ms}$? (e) What is the magnitude of the current in the smaller coil at $t = 1.26 \text{ ms}$?

Figure P29.60



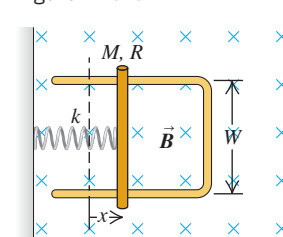
29.61 • The magnetic field $\vec{B}$, at all points within a circular region of radius R , is uniform in space and directed into the plane of the page as shown in **Fig. P29.61**. (The region could be a cross section inside the windings of a long, straight solenoid.) If the magnetic field is increasing at a rate dB/dt , what are the magnitude and direction of the force on a stationary positive point charge q located at points a , b , and c ? (Point a is a distance r above the center of the region, point b is a distance r to the right of the center, and point c is at the center of the region.)

Figure P29.61



29.62 ••• CP A bar with mass $M = 1.20 \text{ kg}$ and resistance $R = 0.500 \Omega$ slides without friction on a horizontal U-shaped rail with width $W = 40.0 \text{ cm}$ and negligible resistance. The bar is attached to a spring with spring constant $k = 90.0 \text{ N/m}$, as shown in **Fig. P29.62**. A constant magnetic field with magnitude 1.00 T points into the plane everywhere in the vicinity. At time $t = 0$ the bar is stretched beyond its equilibrium position by an amount $x = 10.0 \text{ cm}$ and released from rest.

Figure P29.62

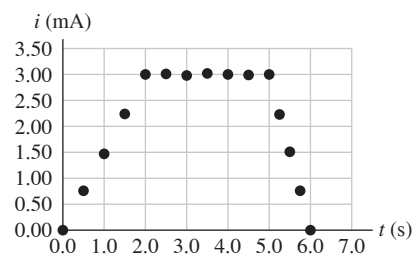


(a) This system behaves like a damped oscillator, described by Eq. (14.41). What is the damping coefficient b ? (b) With what frequency does the bar oscillate around its equilibrium position? (c) What is the amplitude of the motion at time $t = 5.00 \text{ s}$? (d) What is the magnitude of the current in the bar when it passes the equilibrium position for the first time? (e) What is the direction of that current?

29.63 ••• CALC A dielectric of permittivity $3.5 \times 10^{-11} \text{ F/m}$ completely fills the volume between two capacitor plates. For $t > 0$ the electric flux through the dielectric is $(8.0 \times 10^3 \text{ V} \cdot \text{m/s}^3)t^3$. The dielectric is ideal and nonmagnetic; the conduction current in the dielectric is zero. At what time does the displacement current in the dielectric equal $21 \mu\text{A}$?

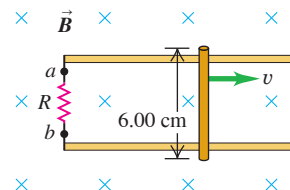
29.64 •• DATA You are evaluating the performance of a large electromagnet. The magnetic field of the electromagnet is zero at $t = 0$ and increases as the current through the windings of the electromagnet is increased. You determine the magnetic field as a function of time by measuring the time dependence of the current induced in a small coil that you insert between the poles of the electromagnet, with the plane of the coil parallel to the pole faces as in **Fig. 29.5**. The coil has 4 turns, a radius of 0.800 cm , and a resistance of 0.250Ω . You measure the current i in the coil as a function of time t . Your results are shown in **Fig. P29.64**. Throughout your measurements, the current induced in the coil remains in the same direction. Calculate the magnetic field at the location of the coil for (a) $t = 2.00 \text{ s}$, (b) $t = 5.00 \text{ s}$, and (c) $t = 6.00 \text{ s}$.

Figure P29.64



29.65 •• DATA You are conducting an experiment in which a metal bar of length 6.00 cm and mass 0.200 kg slides without friction on two parallel metal rails (**Fig. P29.65**). A resistor with resistance $R = 0.800 \Omega$ is connected across one end of the rails so that the bar, rails, and resistor form a complete conducting path. The resistances of the rails and of the bar are much less than R and can be ignored. The entire apparatus is in a uniform magnetic field $\vec{B}$ that is directed into the plane of the figure. You give the bar an initial velocity $v = 20.0 \text{ cm/s}$ to the right and then release it, so that the only force on the bar then is the force exerted by the magnetic field. Using high-speed photography, you measure the magnitude of the acceleration of the bar as a function of its speed. Your results are given in the table (next page):

Figure P29.65

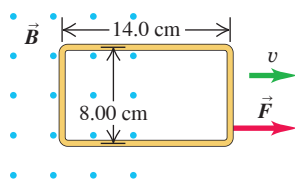


v (cm/s)	20.0	16.0	14.0	12.0	10.0	8.0
a (cm/s ²)	6.2	4.9	4.3	3.7	3.1	2.5

(a) Plot the data as a graph of a versus v . Explain why the data points plotted this way lie close to a straight line, and determine the slope of the best-fit straight line for the data. (b) Use your graph from part (a) to calculate the magnitude B of the magnetic field. (c) While the bar is moving, which end of the resistor, a or b , is at higher potential? (d) How many seconds does it take the speed of the bar to decrease from 20.0 cm/s to 10.0 cm/s?

29.66 ••• DATA You measure the magnitude of the external force $\vec{F}$ that must be applied to a rectangular conducting loop to pull it at constant speed v out of a region of uniform magnetic field $\vec{B}$ that is directed out of the plane of **Fig. P29.66**. The loop has dimensions 14.0 cm by 8.00 cm and resistance $4.00 \times 10^{-3} \Omega$; it does not change shape as it moves. The measurements you collect are listed in the table.

Figure P29.66



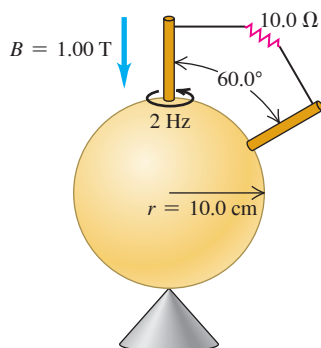
F (N)	0.10	0.21	0.31	0.41	0.52
v (cm/s)	2.0	4.0	6.0	8.0	10.0

(a) Plot the data as a graph of F versus v . Explain why the data points plotted this way lie close to a straight line, and determine the slope of the best-fit straight line for the data. (b) Use your graph from part (a) to calculate the magnitude B of the uniform magnetic field. (c) In **Fig. P29.66**, is the current induced in the loop clockwise or counterclockwise? (d) At what rate is electrical energy being dissipated in the loop when the speed of the loop is 5.00 cm/s?

CHALLENGE PROBLEMS

29.67 ••• CP A conducting spherical shell with radius 10.0 cm spins about a vertical axis twice every second in the presence of a constant magnetic field $\vec{B}$ with magnitude 1.00 T that points downward. Two conducting rods supported by a frame contact the sphere with conducting brushes and extend away from the sphere radially. One rod extends from the top of the sphere and the other forms a 60.0° angle with vertical, as shown in **Fig. P29.67**.

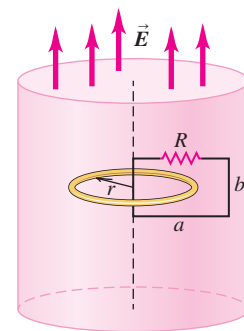
Figure P29.67



The outer ends of the rods are connected to each other by a conducting wire that includes a 10.0Ω resistor. (a) Construct a Cartesian coordinate system with the origin at the center of the sphere, the z -axis pointing upward, and the y -axis pointing rightward so that both rods lie in the yz -plane. What is the velocity $\vec{v}$ of a point on the sphere in the yz -plane at angle θ measured from the positive z -axis? (b) What is the vector product $\vec{v} \times \vec{B}$? (c) What is the line element $d\vec{l}$ along the shortest path on the sphere from the upper rod to the angled rod at angle θ ? (d) What is the magnitude of the current in the wire? (e) What direction does the current flow in the vertical rod?

29.68 ••• CP A uniform electric field is directed axially in a cylindrical region that includes a rectangular loop of wire with total resistance R . This loop has radially oriented width a and axially oriented length b , and sits tight against the cylinder axis, as shown in **Fig. P29.68**. The electric field is zero at time $t = 0$ and then increases in time according to $\vec{E} = \eta t^2 \hat{k}$, where η is a constant with units of $\text{V}/(\text{m} \cdot \text{s}^2)$. (a) What is the magnitude of the displacement current through a circular loop centered on the cylinder axis with radius $r \leq a$, at time t ?

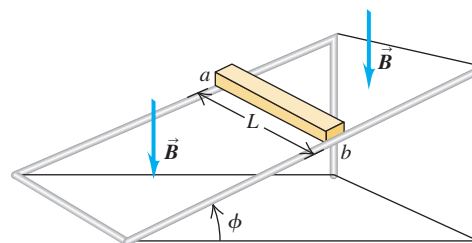
Figure P29.68



(b) Use Ampere's law to determine the magnitude of the magnetic field a distance $r \leq a$ from the cylinder axis at time t . (c) What is the magnetic flux at time t through the rectangular wire loop? (d) What magnitude of current flows in the wire? (e) Does the current flow clockwise or counterclockwise from the perspective shown in the figure?

29.69 ••• A metal bar with length L , mass m , and resistance R is placed on frictionless metal rails that are inclined at an angle ϕ above the horizontal. The rails have negligible resistance. A uniform magnetic field of magnitude B is directed downward as shown in **Fig. P29.69**. The bar is released from rest and slides down the rails. (a) Is the direction of the current induced in the bar from a to b or from b to a ? (b) What is the terminal speed of the bar? (c) What is the induced current in the bar when the terminal speed has been reached? (d) After the terminal speed has been reached, at what rate is electrical energy being converted to thermal energy in the resistance of the bar? (e) After the terminal speed has been reached, at what rate is work being done on the bar by gravity? Compare your answer to that in part (d).

Figure P29.69

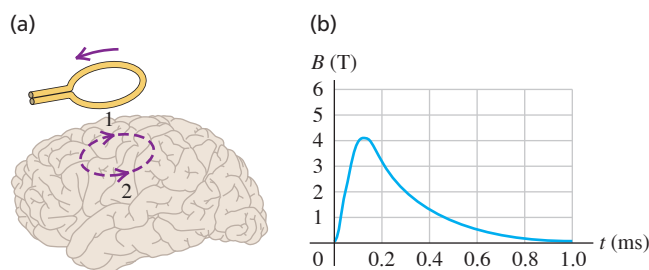


29.70 ••• CP CALC A square, conducting, wire loop of side L , total mass m , and total resistance R initially lies in the horizontal xy -plane, with corners at $(x, y, z) = (0, 0, 0)$, $(0, L, 0)$, $(L, 0, 0)$, and $(L, L, 0)$. There is a uniform, upward magnetic field $\vec{B} = B\hat{k}$ in the space within and around the loop. The side of the loop that extends from $(0, 0, 0)$ to $(L, 0, 0)$ is held in place on the x -axis; the rest of the loop is free to pivot around this axis. When the loop is released, it begins to rotate due to the gravitational torque. (a) Find the *net* torque (magnitude and direction) that acts on the loop when it has rotated through an angle ϕ from its original orientation and is rotating downward at an angular speed ω . (b) Find the angular acceleration of the loop at the instant described in part (a). (c) Compared to the case with zero magnetic field, does it take the loop a longer or shorter time to rotate through 90° ? Explain. (d) Is mechanical energy conserved as the loop rotates downward? Explain.

MCAT-STYLE PASSAGE PROBLEMS

BIO Stimulating the Brain. Communication in the nervous system is based on the propagation of electrical signals called *action potentials* along axons, which are extensions of nerve cells (see the MCAT-style Passage Problems in Chapter 26). Action potentials are generated when the electric potential difference across the membrane of the nerve cell changes: Specifically, the inside of the cell becomes more positive. Researchers in clinical medicine and neurobiology cannot stimulate nerves (even noninvasively) at specific locations in conscious human subjects. Using electrodes to apply current to the skin is painful and requires large currents, which could be dangerous.

Anthony Barker and colleagues at the University of Sheffield in England developed a technique called *transcranial magnetic stimulation* (TMS). In this widely used procedure, a coil positioned near the skull produces a time-varying magnetic field that induces in the conductive tissue of the brain (see part (a) of the figure) electric currents that are sufficient to cause action potentials in nerve cells. For example, if the coil is placed near the motor cortex (the region of the brain that controls voluntary movement), scientists can monitor muscle contraction and assess the connections between the brain and the muscles. Part (b) of the figure is a graph of the typical dependence on time t of the magnetic field B produced by the coil.



29.71 In part (a) of the figure, a current pulse increases to a peak and then decreases to zero in the direction shown in the stimulating coil. What will be the direction of the induced current (dashed line) in the brain tissue? (a) 1; (b) 2; (c) 1 while the current increases in the stimulating coil, 2 while the current decreases; (d) 2 while the current increases in the stimulating coil, 1 while the current decreases.

ANSWERS

Chapter Opening Question ?

(iv) As the magnetic stripe moves through the card reader, the coded pattern of magnetization in the stripe causes a varying magnetic flux. An electric field is induced, which causes a current in the reader's circuits. If the card does not move, there is no induced current and none of the credit card's information is read.

Key Example VARIATION Problems

VP29.8.1 (a) $+2.53 \times 10^{-4} \text{ T} \cdot \text{m}^2$ at $t = 0$, $-1.99 \times 10^{-4} \text{ T} \cdot \text{m}^2$ at $t = 2.00 \text{ s}$ (b) $2.26 \times 10^{-4} \text{ V}$

VP29.8.2 (a) $1.01 \times 10^{-2} \text{ V}$ (b) $6.95 \times 10^{-4} \text{ A}$

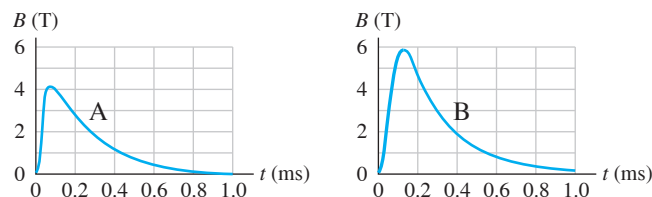
VP29.8.3 (a) 0.163 V (b) counterclockwise

VP29.8.4 (a) $-\omega NB_0 A \cos \omega t$ (b) clockwise at $t = 0$, counterclockwise at $t = \pi/\omega$

29.72 Consider the brain tissue at the level of the dashed line to be a series of concentric circles, each behaving independently of the others. Where will the induced emf be the greatest? (a) At the center of the dashed line; (b) at the periphery of the dashed line; (c) nowhere—it will be the same in all concentric circles; (d) at the center while the stimulating current increases, at the periphery while the current decreases.

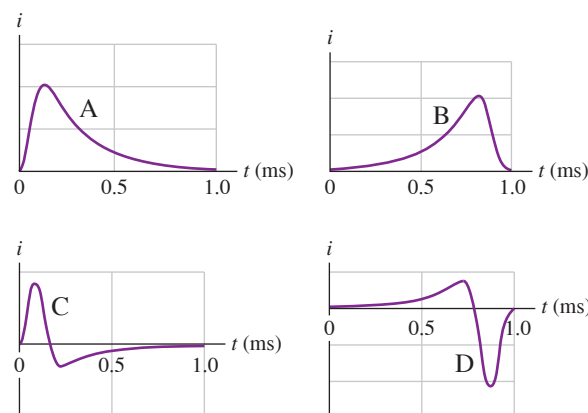
29.73 It may be desirable to increase the maximum induced current in the brain tissue. In **Fig. P29.73**, which time-dependent graph of the magnetic field B in the coil achieves that goal? Assume that everything else remains constant. (a) A; (b) B; (c) either A or B; (d) neither A nor B.

Figure P29.73



29.74 Which graph in **Fig. P29.74** best represents the time t dependence of the current i induced in the brain tissue, assuming that this tissue can be modeled as a resistive circuit? (The units of i are arbitrary.) (a) A; (b) B; (c) C; (d) D.

Figure P29.74



VP29.9.1 (a) 0.187 V (b) 2.34 V/m , $+x$ -direction

VP29.9.2 0.900 A , clockwise

VP29.9.3 (a) $2.40 \times 10^{-2} \text{ V}$ (b) end b

VP29.9.4 4.12 m/s

VP29.11.1 (a) $1.81 \times 10^3 \text{ turns/meter}$ (b) $7.70 \times 10^{-5} \text{ V/m}$

VP29.11.2 $2.70 \times 10^{-5} \text{ V/m}$

VP29.11.3 (a) -7.77 A/s (b) $+11.7 \text{ A/s}$

VP29.11.4 $5.52 \times 10^{-6} \text{ A}$

Bridging Problem

$$v_{\text{terminal}} = 16\rho_m\rho_Rg/B^2$$

? Many traffic lights change when a car rolls up to the intersection. This process works because the car contains (i) conducting material; (ii) insulating material that carries a net electric charge; (iii) ferromagnetic material; (iv) ferromagnetic material that is already magnetized.



30 Inductance

LEARNING OUTCOMES

In this chapter, you'll learn...

- 30.1** How a time-varying current in one coil can induce an emf in a second, unconnected coil.
- 30.2** How to relate the induced emf in a circuit to the rate of change of current in the same circuit.
- 30.3** How to calculate the energy stored in a magnetic field.
- 30.4** How to analyze circuits that include both a resistor and an inductor (coil).
- 30.5** Why electrical oscillations occur in circuits that include both an inductor and a capacitor.
- 30.6** Why oscillations decay in circuits with an inductor, a resistor, and a capacitor.

You'll need to review...

- 14.2, 14.3, 14.7** Simple harmonic motion, damped oscillations.
- 24.1, 24.3** Capacitance, electric-field energy.
- 26.2, 26.4** Kirchhoff's rules, R - C circuits.
- 28.4, 28.7, 28.8** Magnetic forces between conductors; field of a solenoid; permeability.
- 29.2, 29.3, 29.7** Faraday's law; Lenz's law; conservative and nonconservative electric fields.

Take a length of copper wire and wrap it around a pencil to form a coil. If you put this coil in a circuit, the coil behaves quite differently than a straight piece of wire. In an ordinary gasoline-powered car, a coil of this kind makes it possible for the 12 volt car battery to provide thousands of volts to the spark plugs in order for the plugs to fire and make the engine run. Other coils are used to keep fluorescent light fixtures shining. Larger coils placed under city streets are used to control the operation of traffic signals. All of these applications, and many others, involve the *induction* effects that we studied in Chapter 29.

A changing current in a coil induces an emf in an adjacent coil. The coupling between the coils is described by their *mutual inductance*. A changing current in a coil also induces an emf in that same coil. Such a coil is called an *inductor*, and the relationship of current to emf is described by the *inductance* (also called *self-inductance*) of the coil. If a coil is initially carrying a current, energy is released when the current decreases; this principle is used in automotive ignition systems. We'll find that this released energy was stored in the magnetic field caused by the current that was initially in the coil, and we'll look at some of the practical applications of magnetic-field energy.

We'll also take a first look at what happens when an inductor is part of a circuit. In Chapter 31 we'll go on to study how inductors behave in alternating-current circuits, and we'll learn why inductors play an essential role in modern electronics.

30.1 MUTUAL INDUCTANCE

In Section 28.4 we considered the magnetic interaction between two wires carrying *steady* currents; the current in one wire causes a magnetic field, which exerts a force on the current in the second wire. But an additional interaction arises between two circuits when there is a *changing* current in one of the circuits. Consider two neighboring coils of wire, as in **Fig. 30.1**. A current flowing in coil 1 produces a magnetic field $\mathbf{B}$ and hence a magnetic flux through coil 2. If the current in coil 1 changes, the flux through coil 2 changes as well; according to Faraday's law (Section 29.2), this induces an emf in coil 2. In this way, a change in the current in one circuit can induce a current in a second circuit.

Let's analyze the situation shown in Fig. 30.1 in more detail. We'll use lowercase letters to represent quantities that vary with time; for example, a time-varying current is i , often with a subscript to identify the circuit. In Fig. 30.1 a current i_1 in coil 1 sets up a magnetic field $\vec{B}$, and some of the (blue) field lines pass through coil 2. We denote the magnetic flux through *each* turn of coil 2, caused by the current i_1 in coil 1, as Φ_{B2} . (If the flux is different through different turns of the coil, then Φ_{B2} denotes the *average* flux.) The magnetic field is proportional to i_1 , so Φ_{B2} is also proportional to i_1 . When i_1 changes, Φ_{B2} changes; this changing flux induces an emf $\mathcal{E}_2$ in coil 2, given by

$$\mathcal{E}_2 = -N_2 \frac{d\Phi_{B2}}{dt} \quad (30.1)$$

We could represent the proportionality of Φ_{B2} and i_1 in the form $\Phi_{B2} = (\text{constant})i_1$, but instead it is more convenient to include the number of turns N_2 in the relationship. Introducing a proportionality constant M_{21} , called the **mutual inductance** of the two coils, we write

$$N_2\Phi_{B2} = M_{21}i_1 \quad (30.2)$$

where Φ_{B2} is the flux through a *single* turn of coil 2. From this,

$$N_2 \frac{d\Phi_{B2}}{dt} = M_{21} \frac{di_1}{dt}$$

and we can rewrite Eq. (30.1) as

$$\mathcal{E}_2 = -M_{21} \frac{di_1}{dt} \quad (30.3)$$

That is, a change in the current i_1 in coil 1 induces an emf in coil 2 that is directly proportional to the rate of change of i_1 (**Fig. 30.2**).

We may also write the definition of mutual inductance, Eq. (30.2), as

$$M_{21} = \frac{N_2\Phi_{B2}}{i_1}$$

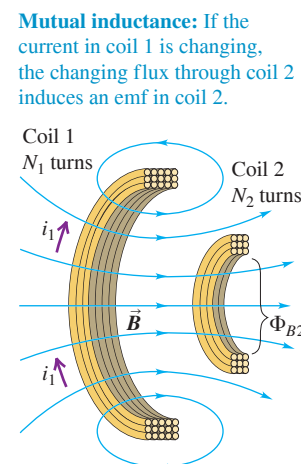
If the coils are in vacuum, the flux Φ_{B2} through each turn of coil 2 is directly proportional to the current i_1 . Then the mutual inductance M_{21} is a constant that depends only on the geometry of the two coils (the size, shape, number of turns, and orientation of each coil and the separation between the coils). If a magnetic material is present, M_{21} also depends on the magnetic properties of the material. If the material has nonlinear magnetic properties—that is, if the relative permeability K_m (defined in Section 28.8) is not constant and magnetization is not proportional to magnetic field—then Φ_{B2} is no longer directly proportional to i_1 . In that case the mutual inductance also depends on the value of i_1 . In this discussion we'll assume that any magnetic material present has constant K_m so that flux *is* directly proportional to current and M_{21} depends on geometry only.

We can repeat our discussion for the opposite case in which a changing current i_2 in coil 2 causes a changing flux Φ_{B1} and an emf $\mathcal{E}_1$ in coil 1. It turns out that the corresponding constant M_{12} is *always* equal to M_{21} , even though in general the two coils are not identical and the flux through them is not the same. We call this common value simply the mutual inductance, denoted by the symbol M without subscripts; it characterizes completely the induced-emf interaction of two coils. Then

$$\text{Mutually induced emfs: } \mathcal{E}_2 = -M \frac{di_1}{dt} \quad \text{and} \quad \mathcal{E}_1 = -M \frac{di_2}{dt} \quad (30.4)$$

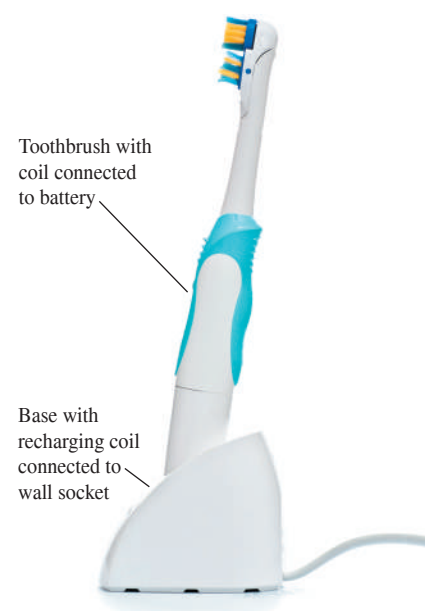
Rate of change of current in coil 1 Rate of change of current in coil 2
Induced emf in coil 2 Induced emf in coil 1
Mutual inductance of coils 1 and 2

Figure 30.1 A current i_1 in coil 1 gives rise to a magnetic flux through coil 2. (For clarity, only one-half of each coil is shown.)



Mutual inductance: If the current in coil 1 is changing, the changing flux through coil 2 induces an emf in coil 2.

Figure 30.2 This electric toothbrush makes use of mutual inductance. The base contains a coil that is supplied with alternating current from a wall socket. Even though there is no direct electrical contact between the base and the toothbrush, this varying current induces an emf in a coil within the toothbrush itself, recharging the toothbrush battery.



The negative signs in Eqs. (30.4) reflect Lenz's law (Section 29.3). The first equation says that a current change in coil 1 causes a flux change through coil 2, inducing an emf in coil 2 that opposes the flux change; in the second equation, the roles of the two coils are interchanged. The mutual inductance M is

$$\text{Mutual inductance of coils 1 and 2} \quad M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_1 \Phi_{B1}}{i_2} \quad (30.5)$$

Labels in diagram:
 Turns in coil 2 (points to N_2)
 Magnetic flux through each turn of coil 2 (points to Φ_{B2})
 Turns in coil 1 (points to N_1)
 Magnetic flux through each turn of coil 1 (points to Φ_{B1})
 Current in coil 1 (causes flux through coil 2) (points to i_1)
 Current in coil 2 (causes flux through coil 1) (points to i_2)

CAUTION Only a time-varying current induces an emf Only a *time-varying* current in a coil can induce an emf and hence a current in a second coil. Equations (30.4) show that the induced emf in each coil is directly proportional to the *rate of change* of the current in the other coil, not to the *value* of the current. A steady current in one coil, no matter how strong, cannot induce a current in a neighboring coil. ■

The SI unit of mutual inductance is called the **henry** (1 H), in honor of the American physicist Joseph Henry (1797–1878), one of the discoverers of electromagnetic induction. From Eq. (30.5), one henry is equal to one weber per ampere. Other equivalent units, obtained by using Eqs. (30.4), are

$$1 \text{ H} = 1 \text{ Wb/A} = 1 \text{ V} \cdot \text{s/A} = 1 \Omega \cdot \text{s} = 1 \text{ J/A}^2$$

Just as the farad is a rather large unit of capacitance (see Section 24.1), the henry is a rather large unit of mutual inductance. Typical values of mutual inductance can be in the millihenry (mH) or microhenry (μH) range.

Drawbacks and Uses of Mutual Inductance

Mutual inductance can be a nuisance in electric circuits, since variations in current in one circuit can induce unwanted emfs in other nearby circuits. To minimize these effects, multiple-circuit systems must be designed so that M is as small as possible; for example, two coils would be placed far apart.

Happily, mutual inductance also has many useful applications. A *transformer*, used in alternating-current circuits to raise or lower voltages, is fundamentally no different from the two coils shown in Fig. 30.1. A time-varying alternating current in one coil of the transformer produces an alternating emf in the other coil; the value of M , which depends on the geometry of the coils, determines the amplitude of the induced emf in the second coil and hence the amplitude of the output voltage. (We'll describe transformers in more detail in Chapter 31.)

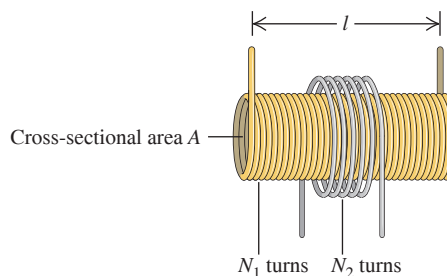
EXAMPLE 30.1 Calculating mutual inductance

WITH VARIATION PROBLEMS

In one form of Tesla coil (a high-voltage generator popular in science museums), a long solenoid with length l and cross-sectional area A is closely wound with N_1 turns of wire. A coil with N_2 turns surrounds it at its center (Fig. 30.3). Find the mutual inductance M .

IDENTIFY and SET UP Mutual inductance occurs here because a current in either coil sets up a magnetic field that causes a flux through the other coil. From Example 28.9 (Section 28.7) we have an expression [Eq. (28.23)] for the field magnitude B_1 at the center of the solenoid (coil 1) in terms of the solenoid current i_1 . This allows us to determine the flux through a cross section of the solenoid. Since there is almost no magnetic field outside a very long solenoid, this is also equal to the flux Φ_{B2} through each turn of the *outer* coil (2). We then use Eq. (30.5), in the form $M = N_2 \Phi_{B2}/i_1$, to determine M .

Figure 30.3 A long solenoid with cross-sectional area A and N_1 turns is surrounded at its center by a coil with N_2 turns.



EXECUTE Equation (28.23) is expressed in terms of the number of turns per unit length, which for solenoid (1) is $n_1 = N_1/L$. So

$$B_1 = \mu_0 n_1 i_1 = \frac{\mu_0 N_1 i_1}{l}$$

The flux through a cross section of the solenoid equals $B_1 A$. As we mentioned above, this also equals the flux Φ_{B2} through each turn of the outer coil, independent of its cross-sectional area. From Eq. (30.5), the mutual inductance M is then

$$M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_2 B_1 A}{i_1} = \frac{N_2}{i_1} \frac{\mu_0 N_1 i_1}{l} A = \frac{\mu_0 A N_1 N_2}{l}$$

EVALUATE The mutual inductance M of any two coils is proportional to the product $N_1 N_2$ of their numbers of turns. Notice that M depends only on the geometry of the two coils, not on the current.

Here's a numerical example to give you an idea of magnitudes. Suppose $l = 0.50$ m, $A = 10$ cm² = 1.0×10^{-3} m², $N_1 = 1000$ turns, and $N_2 = 10$ turns. Then

$$\begin{aligned} M &= \frac{(4\pi \times 10^{-7} \text{ Wb/A} \cdot \text{m})(1.0 \times 10^{-3} \text{ m}^2)(1000)(10)}{0.50 \text{ m}} \\ &= 25 \times 10^{-6} \text{ Wb/A} = 25 \times 10^{-6} \text{ H} = 25 \mu\text{H} \end{aligned}$$

KEYCONCEPT A current in one coil causes a magnetic flux through an adjacent coil. How much flux is caused in the other coil by a given current in the first coil is determined by the mutual inductance M of the two coils. The value of M doesn't depend on the currents in the coils, only on their geometry.

EXAMPLE 30.2 Emf due to mutual inductance

WITH VARIATION PROBLEMS

In Example 30.1, suppose the current i_2 in the outer coil is given by $i_2 = (2.0 \times 10^6 \text{ A/s})t$. (Currents in wires can indeed increase this rapidly for brief periods.) (a) At $t = 3.0 \mu\text{s}$, what is the average magnetic flux through each turn of the solenoid (coil 1) due to the current in the outer coil? (b) What is the induced emf in the solenoid?

IDENTIFY and SET UP In Example 30.1 we found the mutual inductance by relating the current in the solenoid to the flux produced in the outer coil; to do that, we used Eq. (30.5) in the form $M = N_2 \Phi_{B2}/i_1$. Here we are given the current i_2 in the outer coil and want to find the resulting flux Φ_1 in the solenoid. The mutual inductance is the *same* in either case, and we have $M = 25 \mu\text{H}$ from Example 30.1. We use Eq. (30.5) in the form $M = N_1 \Phi_{B1}/i_2$ to determine the average flux Φ_{B1} through each turn of the solenoid caused by current i_2 in the outer coil. We then use Eqs. (30.4) to find the emf induced in the solenoid by the time variation of i_2 .

EXECUTE (a) At $t = 3.0 \mu\text{s} = 3.0 \times 10^{-6}$ s, the current in the outer coil is $i_2 = (2.0 \times 10^6 \text{ A/s})(3.0 \times 10^{-6} \text{ s}) = 6.0$ A. We solve Eq. (30.5) for the flux Φ_{B1} through each turn of coil 1:

$$\Phi_{B1} = \frac{M i_2}{N_1} = \frac{(25 \times 10^{-6} \text{ H})(6.0 \text{ A})}{1000} = 1.5 \times 10^{-7} \text{ Wb}$$

We emphasize that this is an *average* value; the flux can vary considerably between the center and the ends of the solenoid.

(b) We are given $i_2 = (2.0 \times 10^6 \text{ A/s})t$, so $di_2/dt = 2.0 \times 10^6 \text{ A/s}$; then, from Eqs. (30.4), the induced emf in the solenoid is

$$\mathcal{E}_1 = -M \frac{di_2}{dt} = -(25 \times 10^{-6} \text{ H})(2.0 \times 10^6 \text{ A/s}) = -50 \text{ V}$$

EVALUATE This is a substantial induced emf in response to a very rapid current change. In an operating Tesla coil, there is a high-frequency alternating current rather than a continuously increasing current as in this example; both di_2/dt and $\mathcal{E}_1$ alternate as well, with amplitudes that can be thousands of times larger than in this example.

KEYCONCEPT When the current in a coil changes, it causes a changing magnetic flux and hence an induced emf in an adjacent coil. The mutual inductance of the two coils determines how much emf is induced in one coil for a given rate of change of the current in the other coil.

TEST YOUR UNDERSTANDING OF SECTION 30.1 Consider the Tesla coil described in Example 30.1. If you make the solenoid out of twice as much wire, so that it has twice as many turns and is twice as long, how much larger is the mutual inductance? (i) M is four times greater; (ii) M is twice as great; (iii) M is unchanged; (iv) M is $\frac{1}{2}$ as great; (v) M is $\frac{1}{4}$ as great.

ANSWER

(iii) Doubling both the length of the solenoid (l) and the number of turns of wire in the solenoid (N_1) would have *no* effect on the mutual inductance M . Example 30.1 shows that M depends on the ratio of these quantities, which would remain unchanged. This is because the magnetic field produced by the solenoid depends on the number of turns *per unit length*, and the proposed change has no effect on this quantity.

30.2 SELF-INDUCTANCE AND INDUCTORS

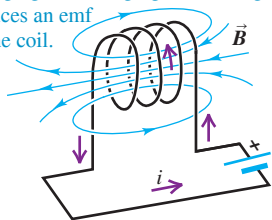
In our discussion of mutual inductance we considered two separate, independent circuits: A current in one circuit creates a magnetic field that gives rise to a flux through the second circuit. If the current in the first circuit changes, the flux through the second circuit changes and an emf is induced in the second circuit.

An important related effect occurs in a *single* isolated circuit. A current in a circuit sets up a magnetic field that causes a magnetic flux through the *same* circuit; this flux changes when the current changes. Thus any circuit that carries a varying current has an emf induced in it by the variation in *its own* magnetic field. Such an emf is called a **self-induced emf**. By Lenz's law, a self-induced emf opposes the change in the current that caused the emf and so tends to make it more difficult for variations in current to occur. Hence self-induced emfs can be of great importance whenever there is a varying current.

Self-induced emfs can occur in *any* circuit, since there is always some magnetic flux through the closed loop of a current-carrying circuit. But the effect is greatly enhanced if the circuit includes a coil with N turns of wire (Fig. 30.4). As a result of the current i , there is an average magnetic flux Φ_B through each turn of the coil. In analogy to Eq. (30.5) we define the **self-inductance** L of the circuit as

Figure 30.4 The current i in the circuit causes a magnetic field $\vec{B}$ in the coil and hence a flux through the coil.

Self-inductance: If the current i in the coil is changing, the changing flux through the coil induces an emf in the coil.



$$\text{Self-inductance (or inductance) of a coil} \quad L = \frac{N\Phi_B}{i} \quad (30.6)$$

Number of turns in coil
Flux due to current through each turn of coil
Current in coil

When there is no danger of confusion with mutual inductance, the self-inductance is called simply the **inductance**. Comparing Eqs. (30.5) and (30.6), we see that the units of self-inductance are the same as those of mutual inductance; the SI unit of self-inductance is the henry.

If the current i changes, so does the flux Φ_B ; after we rearrange Eq. (30.6) and differentiate with respect to time, the rates of change are related by

$$N \frac{d\Phi_B}{dt} = L \frac{di}{dt}$$

From Faraday's law for a coil with N turns, Eq. (29.4), the self-induced emf is $\mathcal{E} = -N d\Phi_B/dt$, so it follows that

$$\text{Self-induced emf in a circuit} \quad \mathcal{E} = -L \frac{di}{dt} \quad (30.7)$$

Inductance of circuit
Rate of change of current in circuit

The minus sign in Eq. (30.7) is a reflection of Lenz's law; it says that the self-induced emf in a circuit opposes any change in the current in that circuit.

Equation (30.7) states that the self-inductance of a circuit is the magnitude of the self-induced emf per unit rate of change of current. This relationship makes it possible to measure an unknown self-inductance: Change the current at a known rate di/dt , measure the induced emf, and take the ratio to determine L .

Inductors as Circuit Elements

A circuit device that is designed to have a particular inductance is called an **inductor**, or a **choke**. The usual circuit symbol for an inductor is



Like resistors and capacitors, inductors are among the indispensable circuit elements of modern electronics. Their purpose is to oppose any variations in the current through the

APPLICATION Inductors, Power Transmission, and Lightning Strikes

If lightning strikes part of an electrical power transmission system, it causes a sudden spike in voltage that can damage the components of the system as well as anything connected to that system (for example, home appliances). To minimize these effects, large inductors are incorporated into the transmission system. These use the principle that an inductor opposes and suppresses any rapid changes in the current.



circuit. An inductor in a direct-current circuit helps to maintain a steady current despite any fluctuations in the applied emf; in an alternating-current circuit, an inductor tends to suppress variations of the current that are more rapid than desired.

To understand the behavior of circuits containing inductors, we need to develop a general principle analogous to Kirchhoff's loop rule (discussed in Section 26.2). To apply that rule, we go around a conducting loop, measuring potential differences across successive circuit elements as we go. The algebraic sum of these differences around any closed loop must be zero because the electric field produced by charges distributed around the circuit is *conservative*. In Section 29.7 we denoted such a conservative field as $\vec{E}_c$.

When an inductor is included in the circuit, the situation changes. The magnetically induced electric field within the coils of the inductor is *not* conservative; as in Section 29.7, we'll denote it by $\vec{E}_n$. We need to think very carefully about the roles of the various fields. Let's assume we are dealing with an inductor whose coils have negligible resistance. Then a negligibly small electric field is required to make charge move through the coils, so the *total* electric field $\vec{E}_c + \vec{E}_n$ within the coils must be zero, even though neither field is individually zero. Because $\vec{E}_c$ is nonzero, there have to be accumulations of charge on the terminals of the inductor and the surfaces of its conductors to produce this field.

Consider the circuit shown in **Fig. 30.5**; the box contains some combination of batteries and variable resistors that enables us to control the current i in the circuit. According to Faraday's law, Eq. (29.10), the line integral of $\vec{E}_n$ around the circuit is the negative of the rate of change of flux through the circuit, which in turn is given by Eq. (30.7). Combining these two relationships, we get

$$\oint \vec{E}_n \cdot d\vec{l} = -L \frac{di}{dt}$$

where we integrate clockwise around the loop (the direction of the assumed current). But $\vec{E}_n$ is different from zero only within the inductor. Therefore the integral of $\vec{E}_n$ around the whole loop can be replaced by its integral only from a to b through the inductor; that is,

$$\int_a^b \vec{E}_n \cdot d\vec{l} = -L \frac{di}{dt}$$

Next, because $\vec{E}_c + \vec{E}_n = \mathbf{0}$ at each point within the inductor coils, $\vec{E}_n = -\vec{E}_c$. So we can rewrite the above equation as

$$\int_a^b \vec{E}_c \cdot d\vec{l} = L \frac{di}{dt}$$

But this integral is just the potential V_{ab} of point a with respect to point b , so

$$V_{ab} = V_a - V_b = L \frac{di}{dt} \quad (30.8)$$

We conclude that there is a genuine potential difference between the terminals of the inductor, associated with conservative, electrostatic forces, even though the electric field associated with magnetic induction is nonconservative. Thus we are justified in using Kirchhoff's loop rule to analyze circuits that include inductors. Equation (30.8) gives the potential difference across an inductor in a circuit.

CAUTION Self-induced emf opposes changes in current Note that the self-induced emf does not oppose the current i itself; rather, it opposes any *change* (di/dt) in the current. Thus the circuit behavior of an inductor is quite different from that of a resistor. **Figure 30.6** compares the behaviors of a resistor and an inductor and summarizes the sign relationships. ■

Figure 30.5 A circuit containing a source of emf and an ideal inductor. The source is variable, so the current i and its rate of change di/dt can be varied.

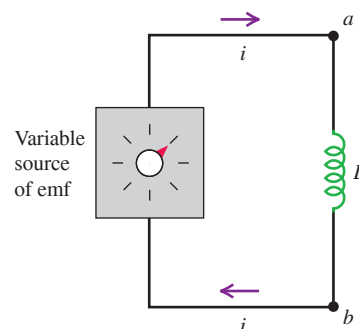
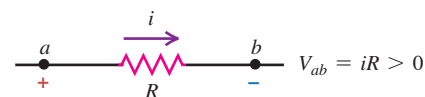
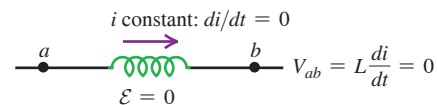


Figure 30.6 (a) The potential difference across a resistor depends on the current, whereas (b), (c), (d) the potential difference across an inductor depends on the rate of change of the current.

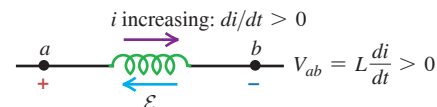
(a) Resistor with current i flowing from a to b : potential drops from a to b .



(b) Inductor with *constant* current i flowing from a to b : no potential difference.



(c) Inductor with *increasing* current i flowing from a to b : potential drops from a to b .



(d) Inductor with *decreasing* current i flowing from a to b : potential increases from a to b .

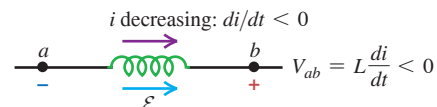


Figure 30.7 These fluorescent light tubes are wired in series with an inductor, or ballast, that helps to sustain the current flowing through the tubes.



Applications of Inductors

Because an inductor opposes changes in current, it plays an important role in fluorescent light fixtures (**Fig. 30.7**). In such fixtures, current flows from the wiring into the gas that fills the tube, ionizing the gas and causing it to glow. However, an ionized gas or *plasma* is a highly nonohmic conductor: The greater the current, the more highly ionized the plasma becomes and the lower its resistance. If a sufficiently large voltage is applied to the plasma, the current can grow so much that it damages the circuitry outside the fluorescent tube. To prevent this problem, an inductor or *magnetic ballast* is put in series with the fluorescent tube to keep the current from growing out of bounds.

The ballast also makes it possible for the fluorescent tube to work with the alternating voltage provided by household wiring. This voltage oscillates sinusoidally with a frequency of 60 Hz, so that it goes momentarily to zero 120 times per second. If there were no ballast, the plasma in the fluorescent tube would rapidly deionize when the voltage went to zero and the tube would shut off. With a ballast present, a self-induced emf sustains the current and keeps the tube lit. Magnetic ballasts are also used for this purpose in streetlights (which obtain their light from a glowing mercury or sodium vapor) and in neon lights. (In compact fluorescent lamps, the magnetic ballast is replaced by a more complicated scheme that utilizes transistors, discussed in Chapter 42.)

The self-inductance of a circuit depends on its size, shape, and number of turns. For N turns close together, it is always proportional to N^2 . It also depends on the magnetic properties of the material enclosed by the circuit. In the following examples we'll assume that the circuit encloses only vacuum (or air, which from the standpoint of magnetism is essentially vacuum). If, however, the flux is concentrated in a region containing a magnetic material with permeability μ , then in the expression for B we must replace μ_0 (the permeability of vacuum) by $\mu = K_m \mu_0$, as discussed in Section 28.8. If the material is diamagnetic or paramagnetic, this replacement makes very little difference, since K_m is very close to 1. If the material is *ferromagnetic*, however, the difference is of crucial importance. A solenoid wound on a soft iron core having $K_m = 5000$ can have an inductance approximately 5000 times as great as that of the same solenoid with an air core. Ferromagnetic-core inductors are very widely used in a variety of electronic and electric-power applications.

With ferromagnetic materials, the magnetization is in general not a linear function of magnetizing current, especially as saturation is approached. As a result, the inductance is not constant but can depend on current in a fairly complicated way. In our discussion we'll ignore this complication and assume always that the inductance is constant. This is a reasonable assumption even for a ferromagnetic material if the magnetization remains well below the saturation level.

Because automobiles contain steel, a ferromagnetic material, driving an automobile over a coil causes an appreciable increase in the coil's inductance. This effect is used in traffic light sensors, which use a large, current-carrying coil embedded under the road surface near an intersection. The circuitry connected to the coil detects the inductance change as a car drives over. When a preprogrammed number of cars have passed over the coil, the light changes to green to allow the cars through the intersection.

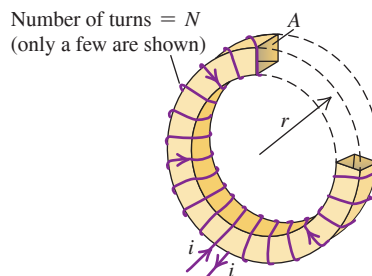
EXAMPLE 30.3 Calculating self-inductance

WITH VARIATION PROBLEMS

Determine the self-inductance of a toroidal solenoid with cross-sectional area A and mean radius r , closely wound with N turns of wire on a nonmagnetic core (**Fig. 30.8**). Assume that B is uniform across a cross section (that is, neglect the variation of B with distance from the toroid axis).

IDENTIFY and SET UP Our target variable is the self-inductance L of the toroidal solenoid. We can find L by using Eq. (30.6), which requires knowing the flux Φ_B through each turn and the current i in the coil. For this, we use the results of Example 28.10 (Section 28.7), in which we found the magnetic field in the interior of a toroidal solenoid as a function of the current.

Figure 30.8 Determining the self-inductance of a closely wound toroidal solenoid. For clarity, only a few turns of the winding are shown. Part of the toroid has been cut away to show the cross-sectional area A and radius r .



EXECUTE From Eq. (30.6), the self-inductance is $L = N\Phi_B/i$. From Example 28.10, the field magnitude at a distance r from the toroid axis is $B = \mu_0 Ni/2\pi r$. If we assume that the field has this magnitude over the entire cross-sectional area A , then

$$\Phi_B = BA = \frac{\mu_0 NiA}{2\pi r}$$

The flux Φ_B is the same through each turn, and

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 A}{2\pi r} \quad (\text{self-inductance of a toroidal solenoid})$$

EVALUATE Suppose $N = 200$ turns, $A = 5.0 \text{ cm}^2 = 5.0 \times 10^{-4} \text{ m}^2$, and $r = 0.10 \text{ m}$; then

$$\begin{aligned} L &= \frac{(4\pi \times 10^{-7} \text{ Wb/A} \cdot \text{m})(200)^2(5.0 \times 10^{-4} \text{ m}^2)}{2\pi(0.10 \text{ m})} \\ &= 40 \times 10^{-6} \text{ H} = 40 \mu\text{H} \end{aligned}$$

KEYCONCEPT The self-inductance L of a coil (or inductor) tells you how much magnetic flux is produced through that coil by the current that flows in it. Like mutual inductance, the value of L depends only on the coil's geometry, not on how much current it carries.

EXAMPLE 30.4 Calculating self-induced emf

WITH VARIATION PROBLEMS

If the current in the toroidal solenoid in Example 30.3 increases uniformly from 0 to 6.0 A in 3.0 μs , find the magnitude and direction of the self-induced emf.

IDENTIFY and SET UP We are given L , the self-inductance, and di/dt , the rate of change of the solenoid current. We find the magnitude of the self-induced emf $\mathcal{E}$ by using Eq. (30.7) and its direction by using Lenz's law.

EXECUTE We have $di/dt = (6.0 \text{ A})/(3.0 \times 10^{-6} \text{ s}) = 2.0 \times 10^6 \text{ A/s}$. From Eq. (30.7), the magnitude of the induced emf is

$$\begin{aligned} |\mathcal{E}| &= L \left| \frac{di}{dt} \right| \\ &= (40 \times 10^{-6} \text{ H})(2.0 \times 10^6 \text{ A/s}) = 80 \text{ V} \end{aligned}$$

The current is increasing, so according to Lenz's law the direction of the emf is opposite to that of the current. This corresponds to the situation in Fig. 30.6c; the emf is in the direction from b to a , like a battery with a as the $+$ terminal and b the $-$ terminal, tending to oppose the current increase from the external circuit.

EVALUATE This example shows that even a small inductance L can give rise to a substantial induced emf if the current changes rapidly.

KEYCONCEPT If the current in a coil changes, the resulting change in magnetic flux through the coil produces an induced emf. The self-inductance of the coil determines how much emf is induced in the coil for a given rate of change of the current in the coil.

TEST YOUR UNDERSTANDING OF SECTION 30.2 Rank the following inductors in order of the potential difference V_{ab} , from most positive to most negative. In each case the inductor has zero resistance and the current flows from point a through the inductor to point b . (i) The current through a 2.0 μH inductor increases from 1.0 A to 2.0 A in 0.50 s; (ii) the current through a 4.0 μH inductor decreases from 3.0 A to 0 in 2.0 s; (iii) the current through a 1.0 μH inductor remains constant at 4.0 A; (iv) the current through a 1.0 μH inductor increases from 0 to 4.0 A in 0.25 s.

ANSWER

For the four cases we find (i) $V_{ab} = (2.0 \mu\text{H})(2.0 \text{ A})/(0.50 \text{ s}) = 4.0 \mu\text{V}$; (ii) $V_{ab} = -6.0 \mu\text{V}$; (iii) $V_{ab} = 0$ because the rate of change of current is zero; and (iv) $V_{ab} = (1.0 \mu\text{H})(4.0 \text{ A})/(0.25 \text{ s}) = 4.0 \mu\text{V}$. **(i), (iv), (iii), (ii)**

30.3 MAGNETIC-FIELD ENERGY

Establishing a current in an inductor requires an input of energy, and an inductor carrying a current has energy stored in it. Let's see how this comes about. In Fig. 30.5, an increasing current i in the inductor causes an emf $\mathcal{E}$ between its terminals and a corresponding potential difference V_{ab} between the terminals of the source, with point a at higher potential than point b . Thus the source must be adding energy to the inductor, and the instantaneous power P (rate of transfer of energy into the inductor) is $P = V_{ab}i$.

Energy Stored in an Inductor

We can calculate the total energy input U needed to establish a final current I in an inductor with inductance L if the initial current is zero. We assume that the inductor has zero resistance, so no energy is dissipated within the inductor. Let the rate of change of the current i at some instant be di/dt ; the current is increasing, so $di/dt > 0$. The voltage between the terminals a and b of the inductor at this instant is $V_{ab} = L di/dt$, and the rate P at which energy is being delivered to the inductor (equal to the instantaneous power supplied by the external source) is

$$P = V_{ab}i = Li \frac{di}{dt}$$

The energy dU supplied to the inductor during an infinitesimal time interval dt is $dU = P dt$, so

$$dU = Li di$$

The total energy U supplied while the current increases from zero to a final value I is

$$\text{Energy stored in an inductor} \rightarrow U = L \int_0^I i di = \frac{1}{2} LI^2 \quad (30.9)$$

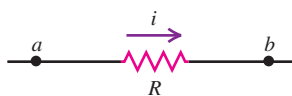
Inductance
Final current
Integral from initial (zero) value of instantaneous current to final value

After the current has reached its final steady value I , $di/dt = 0$ and no more energy is input to the inductor. When there is no current, the stored energy U is zero; when the current is I , the energy is $\frac{1}{2} LI^2$.

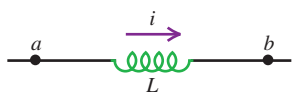
When the current decreases from I to zero, the inductor acts as a source that supplies a total amount of energy $\frac{1}{2} LI^2$ to the external circuit. If we interrupt the circuit suddenly by opening a switch, the current decreases very rapidly, the induced emf is very large, and the energy may be dissipated in an arc across the switch contacts.

Figure 30.9 A resistor is a device in which energy is irrecoverably dissipated. By contrast, energy stored in a current-carrying inductor can be recovered when the current decreases to zero.

Resistor with current i : energy is dissipated.



Inductor with current i : energy is stored.



CAUTION Energy, resistors, and inductors Don't confuse the behavior of resistors and inductors where energy is concerned (**Fig. 30.9**). Energy flows into a resistor whenever a current passes through it, whether the current is steady or varying; this energy is dissipated in the form of heat. By contrast, energy flows into an ideal, zero-resistance inductor only when the current in the inductor *increases*. This energy is not dissipated; it is stored in the inductor and released when the current *decreases*. When a steady current flows through an inductor, there is no energy flow in or out. ■

Magnetic Energy Density

The energy in an inductor is actually stored in the magnetic field of the coil, just as the energy of a capacitor is stored in the electric field between its plates. We can develop relationships for magnetic-field energy analogous to those we obtained for electric-field energy in Section 24.3 [Eqs. (24.9) and (24.11)]. We'll concentrate on one simple case, the ideal toroidal solenoid. This system has the advantage that its magnetic field is confined completely to a finite region of space within its core. As in Example 30.3, we assume that the cross-sectional area A is small enough that we can pretend the magnetic field is uniform over the area. The volume V enclosed by the toroidal solenoid is approximately equal to the circumference $2\pi r$ multiplied by the area A : $V = 2\pi rA$. From Example 30.3, the self-inductance of the toroidal solenoid with vacuum within its coils is

$$L = \frac{\mu_0 N^2 A}{2\pi r}$$

From Eq. (30.9), the energy U stored in the toroidal solenoid when the current is I is

$$U = \frac{1}{2}LI^2 = \frac{1}{2} \frac{\mu_0 N^2 A}{2\pi r} I^2$$

The magnetic field and therefore this energy are localized in the volume $V = 2\pi rA$ enclosed by the windings. The energy *per unit volume*, or *magnetic energy density*, is $u = U/V$:

$$u = \frac{U}{2\pi rA} = \frac{1}{2} \mu_0 \frac{N^2 I^2}{(2\pi r)^2}$$

We can express this in terms of the magnitude B of the magnetic field inside the toroidal solenoid. From Eq. (28.24) in Example 28.10 (Section 28.7), this is

$$B = \frac{\mu_0 NI}{2\pi r}$$

and so

$$\frac{N^2 I^2}{(2\pi r)^2} = \frac{B^2}{\mu_0^2}$$

When we substitute this into the above equation for u , we finally find the expression for **magnetic energy density** in vacuum:

$$\text{Magnetic energy density in vacuum} \quad u = \frac{B^2}{2\mu_0} \quad \text{Magnetic-field magnitude} \quad \text{Magnetic constant} \quad (30.10)$$

This is the magnetic analog of the energy per unit volume in an *electric* field in vacuum, $u = \frac{1}{2}\epsilon_0 E^2$, which we derived in Section 24.3. As an example, the energy density in the 1.5 T magnetic field of an MRI scanner (see Section 27.7) is $u = B^2/2\mu_0 = (1.5 \text{ T})^2/(2 \times 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}) = 9.0 \times 10^5 \text{ J/m}^3$.

When the material inside the toroid is not vacuum but a material with (constant) magnetic permeability $\mu = K_m \mu_0$, we replace μ_0 by μ in Eq. (30.10):

$$\text{Magnetic energy density in a material} \quad u = \frac{B^2}{2\mu} \quad \text{Magnetic-field magnitude} \quad \text{Permeability of material} \quad (30.11)$$

Although we have derived Eq. (30.11) for only one special situation, it turns out to be the correct expression for the energy per unit volume associated with *any* magnetic-field configuration in a material with constant permeability. For vacuum, Eq. (30.11) reduces to Eq. (30.10). We'll use the expressions for electric-field and magnetic-field energy in Chapter 32 when we study the energy associated with electromagnetic waves.

Magnetic-field energy plays an important role in the ignition systems of gasoline-powered automobiles. A primary coil of about 250 turns is connected to the car's battery and produces a strong magnetic field. This coil is surrounded by a secondary coil with some 25,000 turns of very fine wire. When it is time for a spark plug to fire (see Fig. 20.5 in Section 20.3), the current to the primary coil is interrupted, the magnetic field quickly drops to zero, and an emf of tens of thousands of volts is induced in the secondary coil. The energy stored in the magnetic field thus goes into a powerful pulse of current that travels through the secondary coil to the spark plug, generating the spark that ignites the fuel-air mixture in the engine's cylinders (**Fig. 30.10**).

APPLICATION A Magnetic Eruption on the Sun

This composite of two images of the sun shows a coronal mass ejection, a dramatic event in which about 10^{12} kg (a billion tons) of material from the sun's outer atmosphere is ejected into space at speeds of 500 km/s or faster. Such ejections happen at intervals of a few hours to a few days. These immense eruptions are powered by the energy stored in the sun's magnetic field. Unlike the earth's relatively steady magnetic field, the sun's field is constantly changing, and regions of unusually strong field (and hence unusually high magnetic energy density) frequently form. A coronal mass ejection occurs when the energy stored in such a region is suddenly released.

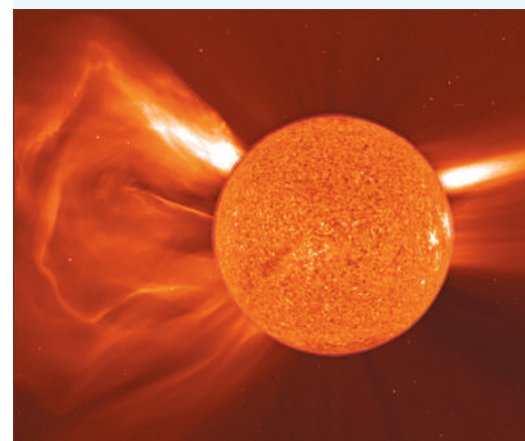


Figure 30.10 The energy required to fire an automobile spark plug is derived from magnetic-field energy stored in the ignition coil.



EXAMPLE 30.5 Storing energy in an inductor

The electric-power industry would like to find efficient ways to store electrical energy generated during low-demand hours to help meet customer requirements during high-demand hours. Could a large inductor be used? What inductance would be needed to store $1.00 \text{ kW} \cdot \text{h}$ of energy in a coil carrying a 200 A current?

IDENTIFY and SET UP We are given the required amount of stored energy U and the current $I = 200 \text{ A}$. We use Eq. (30.9) to find the self-inductance L .

EXECUTE Here we have $I = 200 \text{ A}$ and $U = 1.00 \text{ kW} \cdot \text{h} = (1.00 \times 10^3 \text{ W})(3600 \text{ s}) = 3.60 \times 10^6 \text{ J}$. Solving Eq. (30.9) for L , we find

$$L = \frac{2U}{I^2} = \frac{2(3.60 \times 10^6 \text{ J})}{(200 \text{ A})^2} = 180 \text{ H}$$

EVALUATE The required inductance is more than a million times greater than the self-inductance of the toroidal solenoid of Example 30.3. Conventional wires that are to carry 200 A would have to be of large diameter to keep the resistance low and avoid unacceptable energy losses due to I^2R heating. As a result, a 180 H inductor using conventional wire would be very large (room-size). A superconducting inductor can be much smaller, since the resistance of a superconductor is zero and much thinner wires can be used. However, such wires must be kept at low temperature to remain superconducting, and maintaining this temperature itself requires energy. Despite these limitations, superconducting inductors are very efficient for energy storage and are presently in small-scale use by several electric utilities.

KEYCONCEPT Energy is stored in any magnetic field, including the field of a current-carrying inductor. The magnetic energy of an inductor depends on its self-inductance and current.

TEST YOUR UNDERSTANDING OF SECTION 30.3 The current in a solenoid is reversed in direction while keeping the same magnitude. (a) Does this change the magnetic field within the solenoid? (b) Does this change the magnetic energy density in the solenoid?

ANSWER (a) **no** Reversing the direction of the current has no effect on the magnetic-field magnitude, but it causes the direction of the magnetic field to reverse. It has no effect on the magnetic-field energy density, which is proportional to the square of the magnitude of the magnetic field. (b) **yes**, (a) **yes**, (b) **no**

30.4 THE *R-L* CIRCUIT

Let's look at some examples of the circuit behavior of an inductor. One thing is clear already; an inductor in a circuit makes it difficult for rapid changes in current to occur, thanks to the effects of self-induced emf. Equation (30.7) shows that the greater the rate of change of current di/dt , the greater the self-induced emf and the greater the potential difference between the inductor terminals. This equation, together with Kirchhoff's rules (see Section 26.2), gives us the principles we need to analyze circuits containing inductors.

PROBLEM-SOLVING STRATEGY 30.1 Inductors in Circuits

IDENTIFY *the relevant concepts:* An inductor is just another circuit element, like a source of emf, a resistor, or a capacitor. One key difference is that when an inductor is included in a circuit, all the voltages, currents, and capacitor charges are in general functions of time, not constants as they have been in most of our previous circuit analysis. But even when the voltages and currents vary with time, Kirchhoff's rules (see Section 26.2) hold at each instant of time.

SET UP *the problem* using the following steps:

1. Follow the procedure given in Problem-Solving Strategy 26.2 (Section 26.2). Draw a circuit diagram and label all quantities, known and unknown. Apply the junction rule immediately to express the currents in terms of as few quantities as possible.
2. Determine which quantities are the target variables.

EXECUTE *the solution* as follows:

1. As in Problem-Solving Strategy 26.2, apply Kirchhoff's loop rule to each loop in the circuit.

2. Review the sign rules given in Problem-Solving Strategy 26.2. To get the correct sign for the potential difference between the terminals of an inductor, apply Lenz's law and the sign rule described in Section 30.2 in connection with Eq. (30.7) and Fig. 30.6. In Kirchhoff's loop rule, when we go through an inductor in the *same* direction as the assumed current, we encounter a voltage *drop* equal to $L di/dt$, so the corresponding term in the loop equation is $-L di/dt$. When we go through an inductor in the *opposite* direction from the assumed current, the potential difference is reversed and the term to use in the loop equation is $+L di/dt$.
3. Solve for the target variables.

EVALUATE *your answer:* Check whether your answer is consistent with the behavior of inductors. By Lenz's law, if the current through an inductor is changing, your result should indicate that the potential difference across the inductor opposes the change.

Current Growth in an R-L Circuit

We can learn a great deal about inductor behavior by analyzing the circuit of **Fig. 30.11**. A circuit that includes both a resistor and an inductor, and possibly a source of emf, is called an **R-L circuit**. The inductor helps to prevent rapid changes in current, which can be useful if a steady current is required but the source has a fluctuating emf. The resistor R may be a separate circuit element, or it may be the resistance of the inductor windings; every real-life inductor has some resistance unless it is made of superconducting wire. By closing switch S_1 , we can connect the R-L combination to a source with constant emf $\mathcal{E}$. (We assume that the source has zero internal resistance, so the terminal voltage equals $\mathcal{E}$.)

Suppose both switches are open to begin with, and then at some initial time $t = 0$ we close switch S_1 . The current cannot change suddenly from zero to some final value, since di/dt and the induced emf in the inductor would both be infinite. Instead, the current begins to grow at a rate that depends on the value of L in the circuit.

Let i be the current at some time t after switch S_1 is closed, and let di/dt be its rate of change at that time. The potential differences v_{ab} (across the resistor) and v_{bc} (across the inductor) are

$$v_{ab} = iR \quad \text{and} \quad v_{bc} = L \frac{di}{dt}$$

Note that if the current is in the direction shown in Fig. 30.11 and is increasing, then both v_{ab} and v_{bc} are positive; a is at a higher potential than b , which in turn is at a higher potential than c . (Compare to Figs. 30.6a and 30.6c.) We apply Kirchhoff's loop rule, starting at the negative terminal and proceeding counterclockwise around the loop:

$$\mathcal{E} - iR - L \frac{di}{dt} = 0 \quad (30.12)$$

Solving this for di/dt , we find that the rate of increase of current is

$$\frac{di}{dt} = \frac{\mathcal{E} - iR}{L} = \frac{\mathcal{E}}{L} - \frac{R}{L}i \quad (30.13)$$

At the instant that switch S_1 is first closed, $i = 0$ and the potential drop across R is zero. The initial rate of change of current is

$$\left(\frac{di}{dt} \right)_{\text{initial}} = \frac{\mathcal{E}}{L}$$

The greater the inductance L , the more slowly the current increases.

As the current increases, the term $(R/L)i$ in Eq. (30.13) also increases, and the *rate* of increase of current given by Eq. (30.13) becomes smaller and smaller. This means that the current is approaching a final, steady-state value I . When the current reaches this value, its rate of increase is zero. Then Eq. (30.13) becomes

$$\begin{aligned} \left(\frac{di}{dt} \right)_{\text{final}} = 0 &= \frac{\mathcal{E}}{L} - \frac{R}{L}I \quad \text{and} \\ I &= \frac{\mathcal{E}}{R} \end{aligned}$$

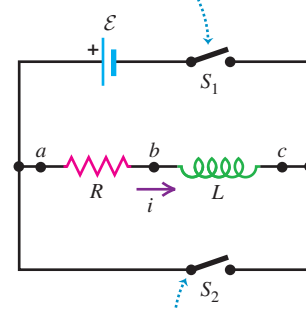
The *final* current I does not depend on the inductance L ; it is the same as it would be if the resistance R alone were connected to the source with emf $\mathcal{E}$.

Figure 30.12 shows the behavior of the current as a function of time. To derive the equation for this curve (that is, an expression for current as a function of time), we proceed just as we did for the charging capacitor in Section 26.4. First we rearrange Eq. (30.13) to the form

$$\frac{di}{i - (\mathcal{E}/R)} = -\frac{R}{L}dt$$

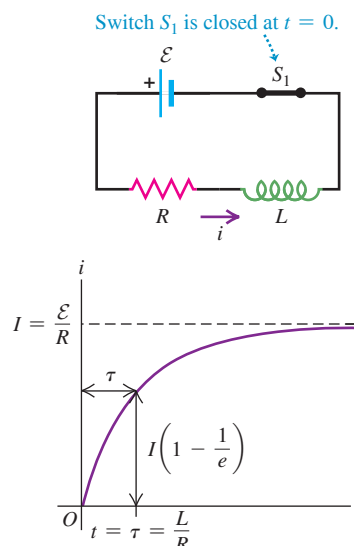
Figure 30.11 An R-L circuit.

Closing switch S_1 connects the R-L combination in series with a source of emf $\mathcal{E}$.



Closing switch S_2 while opening switch S_1 disconnects the combination from the source.

Figure 30.12 Graph of i versus t for growth of current in an R-L circuit with an emf in series. The final current is $I = \mathcal{E}/R$; after one time constant τ , the current is $1 - 1/e$ of this value.



This separates the variables, with i on the left side and t on the right. Then we integrate both sides, renaming the integration variables i' and t' so that we can use i and t as the upper limits. (The lower limit for each integral is zero, corresponding to zero current at the initial time $t = 0$.) We get

$$\int_0^i \frac{di'}{i' - (\mathcal{E}/R)} = - \int_0^t \frac{R}{L} dt'$$

$$\ln\left(\frac{i - (\mathcal{E}/R)}{-\mathcal{E}/R}\right) = -\frac{R}{L}t$$

Now we take exponentials of both sides and solve for i . We leave the details for you to work out; the final result is the equation of the curve in Fig. 30.12:

$$i = \frac{\mathcal{E}}{R}(1 - e^{-(R/L)t}) \quad (\text{current in an } R\text{-}L \text{ circuit with emf}) \quad (30.14)$$

Taking the derivative of Eq. (30.14), we find

$$\frac{di}{dt} = \frac{\mathcal{E}}{L}e^{-(R/L)t} \quad (30.15)$$

At time $t = 0$, $i = 0$ and $di/dt = \mathcal{E}/L$. As $t \rightarrow \infty$, $i \rightarrow \mathcal{E}/R$ and $di/dt \rightarrow 0$, as we predicted.

As Fig. 30.12 shows, the instantaneous current i first rises rapidly, then increases more slowly and approaches the final value $I = \mathcal{E}/R$ asymptotically. At a time equal to L/R , the current has risen to $(1 - 1/e)$, or about 63%, of its final value. The quantity L/R is therefore a measure of how quickly the current builds toward its final value; this quantity is called the **time constant** for the circuit, denoted by τ :

$$\text{Time constant } \tau = \frac{L}{R} \quad \begin{array}{l} \text{Inductance} \\ \text{Resistance} \end{array} \quad (30.16)$$

In a time equal to 2τ , the current reaches 86% of its final value; in 5τ , 99.3%; and in 10τ , 99.995%. (Compare the discussion in Section 26.4 of charging a capacitor of capacitance C that was in series with a resistor of resistance R ; the time constant for that situation was the product RC .)

The graphs of i versus t have the same general shape for all values of L . For a given value of R , the time constant τ is greater for greater values of L . When L is small, the current rises rapidly to its final value; when L is large, it rises more slowly. For example, if $R = 100 \, \Omega$ and $L = 10 \, \text{H}$,

$$\tau = \frac{L}{R} = \frac{10 \, \text{H}}{100 \, \Omega} = 0.10 \, \text{s}$$

and the current increases to about 63% of its final value in 0.10 s. (Recall that $1 \, \text{H} = 1 \, \Omega \cdot \text{s}$.) But if $L = 0.010 \, \text{H}$, $\tau = 1.0 \times 10^{-4} \, \text{s} = 0.10 \, \text{ms}$, and the rise is much more rapid.

Energy considerations offer us additional insight into the behavior of an R - L circuit. The instantaneous rate at which the source delivers energy to the circuit is $P = \mathcal{E}i$. The instantaneous rate at which energy is dissipated in the resistor is i^2R , and the rate at which energy is stored in the inductor is $i v_{bc} = Li \, di/dt$ [or, equivalently, $(d/dt)(\frac{1}{2}Li^2) = Li \, di/dt$]. When we multiply Eq. (30.12) by i and rearrange, we find

$$\mathcal{E}i = i^2R + Li \frac{di}{dt} \quad (30.17)$$

Of the power $\mathcal{E}i$ supplied by the source, part (i^2R) is dissipated in the resistor and part ($Li \, di/dt$) goes to store energy in the inductor. This discussion is analogous to our power analysis for a charging capacitor, given at the end of Section 26.4.

CAUTION **Current and its rate of change in an inductor** Note that the current i cannot change abruptly in a circuit that contains an inductor, so i must be a continuous function of time t . However, the rate of change of the current di/dt can change abruptly (for example, when the switch in Fig. 30.12a is closed and the current suddenly starts to increase, as shown at $t = 0$ in Fig. 30.12b). ■

EXAMPLE 30.6 Analyzing an R-L circuit**WITH VARIATION PROBLEMS**

A sensitive electronic device of resistance $R = 175\ \Omega$ is to be connected to a source of emf (of negligible internal resistance) by a switch. The device is designed to operate with a 36 mA current, but to avoid damage to the device, the current can rise to no more than 4.9 mA in the first $58\ \mu\text{s}$ after the switch is closed. An inductor is therefore connected in series with the device, as in Fig. 30.11; the switch in question is S_1 . (a) What is the required source emf $\mathcal{E}$? (b) What is the required inductance L ? (c) What is the R - L time constant τ ?

IDENTIFY and SET UP This problem concerns current and current growth in an R - L circuit, so we can use the ideas of this section. Figure 30.12 shows the current i versus the time t that has elapsed since closing S_1 . The graph shows that the final current is $I = \mathcal{E}/R$; we are given $R = 175\ \Omega$, so the emf is determined by the requirement that the final current be $I = 36\ \text{mA}$. The other requirement is that the current be no more than $i = 4.9\ \text{mA}$ at $t = 58\ \mu\text{s}$; to satisfy this, we use Eq. (30.14) for the current as a function of time and solve for the inductance, which is the only unknown quantity. Equation (30.16) then tells us the time constant.

EXECUTE (a) We solve $I = \mathcal{E}/R$ for $\mathcal{E}$:

$$\mathcal{E} = IR = (0.036\ \text{A})(175\ \Omega) = 6.3\ \text{V}$$

(b) To find the required inductance, we solve Eq. (30.14) for L . First we multiply through by $(-R/\mathcal{E})$ and add 1 to both sides:

$$1 - \frac{iR}{\mathcal{E}} = e^{-(R/L)t}$$

Then we take natural logs of both sides, solve for L , and substitute:

$$\begin{aligned} L &= \frac{-Rt}{\ln(1 - iR/\mathcal{E})} \\ &= \frac{-(175\ \Omega)(58 \times 10^{-6}\ \text{s})}{\ln[1 - (4.9 \times 10^{-3}\ \text{A})(175\ \Omega)/(6.3\ \text{V})]} = 69\ \text{mH} \end{aligned}$$

(c) From Eq. (30.16),

$$\tau = \frac{L}{R} = \frac{69 \times 10^{-3}\ \text{H}}{175\ \Omega} = 3.9 \times 10^{-4}\ \text{s} = 390\ \mu\text{s}$$

EVALUATE Note that $58\ \mu\text{s}$ is much less than the time constant. In $58\ \mu\text{s}$ the current builds up from zero to 4.9 mA, a small fraction of its final value of 36 mA; after $390\ \mu\text{s}$ the current equals $(1 - 1/e)$ of its final value, or about $(0.63)(36\ \text{mA}) = 23\ \text{mA}$.

KEYCONCEPT The presence of an inductor in a circuit that contains a source of emf and a resistor slows the growth of the current when the circuit is completed. The equilibrium value of the current after a very long time is the same as if the inductor were not present.

Current Decay in an R-L Circuit

Now suppose switch S_1 in the circuit of Fig. 30.11 has been closed for a while and the current has reached the value I_0 . Resetting our stopwatch to redefine the initial time, we close switch S_2 at time $t = 0$, bypassing the battery. (At the same time we should open S_1 to protect the battery.) The current through R and L does not instantaneously go to zero but decays smoothly, as shown in **Fig. 30.13**. The Kirchhoff's-rule loop equation is obtained from Eq. (30.12) by omitting the $\mathcal{E}$ term. We challenge you to retrace the steps in the above analysis and show that the current i varies with time according to

$$i = I_0 e^{-(R/L)t} \quad (30.18)$$

where I_0 is the initial current at time $t = 0$. The time constant, $\tau = L/R$, is the time for current to decrease to $1/e$, or about 37%, of its original value. In time 2τ it has dropped to 13.5%, in time 5τ to 0.67%, and in 10τ to 0.0045%.

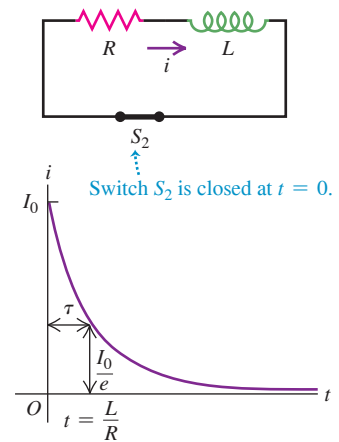
The energy that is needed to maintain the current during this decay is provided by the energy stored in the magnetic field of the inductor. The detailed energy analysis is simpler this time. In place of Eq. (30.17) we have

$$0 = i^2 R + Li \frac{di}{dt} \quad (30.19)$$

Now $Li\,di/dt$ is negative; Eq. (30.19) shows that the energy stored in the inductor *decreases* at the same rate $i^2 R$ at which energy is dissipated in the resistor.

This entire discussion should look familiar; the situation is very similar to that of a charging and discharging capacitor, analyzed in Section 26.4. It would be a good idea to compare that section with our discussion of the R - L circuit.

Figure 30.13 Graph of i versus t for decay of current in an R - L circuit. After one time constant τ , the current is $1/e$ of its initial value.



EXAMPLE 30.7 Energy in an R - L circuit**WITH VARIATION PROBLEMS**

When the current in an R - L circuit is decaying, what fraction of the original energy stored in the inductor has been dissipated after 2.3 time constants?

IDENTIFY and SET UP This problem concerns current decay in an R - L circuit as well as the relationship between the current in an inductor and the amount of stored energy. The current i at any time t is given by Eq. (30.18); the stored energy associated with this current is given by Eq. (30.9), $U = \frac{1}{2}Li^2$.

EXECUTE From Eq. (30.18), the current i at any time t is

$$i = I_0 e^{-(R/L)t}$$

We substitute this into $U = \frac{1}{2}Li^2$ to obtain an expression for the stored energy at any time:

$$U = \frac{1}{2}LI_0^2 e^{-2(R/L)t} = U_0 e^{-2(R/L)t}$$

where $U_0 = \frac{1}{2}LI_0^2$ is the energy at the initial time $t = 0$. When $t = 2.3\tau = 2.3L/R$, we have

$$U = U_0 e^{-2(2.3)} = U_0 e^{-4.6} = 0.010U_0$$

That is, only 0.010 or 1.0% of the energy initially stored in the inductor remains, so 99.0% has been dissipated in the resistor.

EVALUATE To get a sense of what this result means, consider the R - L circuit we analyzed in Example 30.6, for which $\tau = 390 \mu\text{s}$. With $L = 69 \text{ mH}$ and $I_0 = 36 \text{ mA}$, we have

$$\begin{aligned} U_0 &= \frac{1}{2}LI_0^2 = \frac{1}{2}(0.069 \text{ H})(0.036 \text{ A})^2 \\ &= 4.5 \times 10^{-5} \text{ J} \end{aligned}$$

Of this, 99.0% or $4.4 \times 10^{-5} \text{ J}$ is dissipated in $2.3(390 \mu\text{s}) = 9.0 \times 10^{-4} \text{ s} = 0.90 \text{ ms}$. In other words, this circuit can be almost completely powered off (or powered on) in 0.90 ms, so the minimum time for a complete on-off cycle is 1.8 ms. Even shorter cycle times are required for many purposes, such as in fast switching networks for telecommunications. In such cases a smaller time constant $\tau = L/R$ is needed.

KEYCONCEPT The presence of an inductor in a circuit that contains a resistor slows the decay of the current when the source of emf is removed. As the current decays, the magnetic energy stored in the inductor is dissipated in the resistor.

TEST YOUR UNDERSTANDING OF SECTION 30.4 (a) In Fig. 30.11, what are the algebraic signs of the potential differences v_{ab} and v_{bc} when switch S_1 is closed and switch S_2 is open? (i) $v_{ab} > 0, v_{bc} > 0$; (ii) $v_{ab} > 0, v_{bc} < 0$; (iii) $v_{ab} < 0, v_{bc} > 0$; (iv) $v_{ab} < 0, v_{bc} < 0$. (b) What are the signs of v_{ab} and v_{bc} when S_1 is open, S_2 is closed, and current is flowing in the direction shown? (i) $v_{ab} > 0, v_{bc} > 0$; (ii) $v_{ab} > 0, v_{bc} < 0$; (iii) $v_{ab} < 0, v_{bc} > 0$; (iv) $v_{ab} < 0, v_{bc} < 0$.

ANSWER

(a) (i) Recall that v_{ab} is the potential at a minus the potential at b , and similarly for v_{bc} . For either arrangement of the switches, current flows through the resistor from a to b . The upstream end of the resistor is always at the higher potential, so v_{ab} is positive. With S_1 closed and S_2 open, the current through the inductor flows from b to c and is increasing. The self-induced emf opposes this increase and is therefore directed from c toward b , which means that b is at the higher potential. Hence v_{bc} is positive. With S_1 open and S_2 closed, the inductor current again flows from b to c but is now decreasing. The self-induced emf is directed from b to c in an effort to sustain the decaying current, so c is at the higher potential and v_{bc} is negative.

30.5 THE L - C CIRCUIT

A circuit containing an inductor and a capacitor shows an entirely new mode of behavior, characterized by *oscillating* current and charge. This is in sharp contrast to the *exponential* approach to a steady-state situation that we have seen with both R - C and R - L circuits. In the **L - C circuit** in **Fig. 30.14a** we charge the capacitor to a potential difference V_m and initial charge $Q_m = CV_m$ on its left-hand plate and then close the switch. What happens?

The capacitor begins to discharge through the inductor. Because of the induced emf in the inductor, the current cannot change instantaneously; it starts at zero and eventually builds up to a maximum value I_m . During this buildup the capacitor is discharging. At each instant the capacitor potential equals the induced emf, so as the capacitor discharges, the *rate of change* of current decreases. When the capacitor potential becomes zero, the induced emf is also zero, and the current has leveled off at its maximum value I_m .

Figure 30.14 In an oscillating L-C circuit, the charge on the capacitor and the current through the inductor both vary sinusoidally with time. Energy is transferred between magnetic energy in the inductor (U_B) and electrical energy in the capacitor (U_E). As in simple harmonic motion, the total energy E remains constant. (Compare Fig. 14.14 in Section 14.3.)

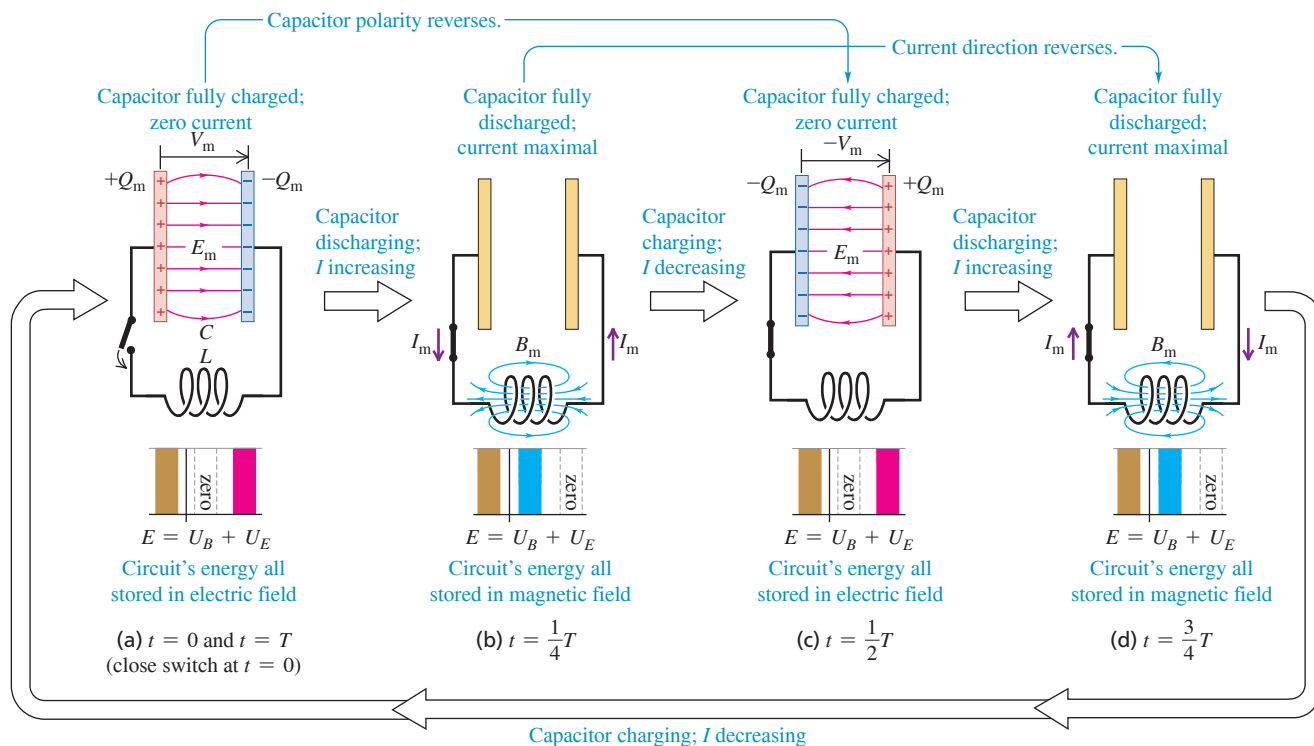


Figure 30.14b shows this situation; the capacitor has completely discharged. The potential difference between its terminals (and those of the inductor) has decreased to zero, and the current has reached its maximum value I_m .

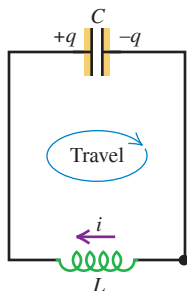
During the discharge of the capacitor, the increasing current in the inductor has established a magnetic field in the space around it, and the energy that was initially stored in the capacitor's electric field is now stored in the inductor's magnetic field.

Although the capacitor is completely discharged in Fig. 30.14b, the current persists (it cannot change instantaneously), and the capacitor begins to charge with polarity opposite to that in the initial state. As the current decreases, the magnetic field also decreases, inducing an emf in the inductor in the *same* direction as the current; this slows down the decrease of the current. Eventually, the current and the magnetic field reach zero, and the capacitor has been charged in the sense *opposite* to its initial polarity (Fig. 30.14c), with potential difference $-V_m$ and charge $-Q_m$ on its left-hand plate.

The process now repeats in the reverse direction; a little later, the capacitor has again discharged, and there is a current in the inductor in the opposite direction (Fig. 30.14d). Still later, the capacitor charge returns to its original value (Fig. 30.14a), and the whole process repeats. If there are no energy losses, the charges on the capacitor continue to oscillate back and forth indefinitely. This process is called an **electrical oscillation**. (Before you read further, review the analogous case of *mechanical* oscillation in Sections 14.2 and 14.3.)

From an energy standpoint the oscillations of an electric circuit transfer energy from the capacitor's electric field to the inductor's magnetic field and back. The *total* energy associated with the circuit is constant. This is analogous to the transfer of energy in an oscillating mechanical system from potential energy to kinetic energy and back, with constant total energy (Section 14.3). As we'll see, this analogy goes much further.

Figure 30.15 Applying Kirchhoff's loop rule to the L - C circuit. The direction of travel around the loop in the loop equation is shown. Just after the circuit is completed and the capacitor first begins to discharge, as in Fig. 30.14a, the current is negative (opposite to the direction shown).



Electrical Oscillations in an L - C Circuit

To study the flow of charge in detail, we proceed just as we did for the R - L circuit. **Figure 30.15** shows our definitions of q and i .

CAUTION Positive current in an L - C circuit After you have examined Fig. 30.14, the positive direction for current in Fig. 30.15 may seem backward. In fact, we chose this direction to simplify the relationship between current and capacitor charge. We define the current at each instant to be $i = dq/dt$, the rate of change of the charge on the left-hand capacitor plate. If the capacitor is initially charged and begins to discharge as in Figs. 30.14a and 30.14b, then $dq/dt < 0$ and the initial current i is negative; the current direction is opposite to the (positive) direction shown in Fig. 30.15. ▮

We apply Kirchhoff's loop rule to the circuit in Fig. 30.15. Starting at the lower-right corner of the circuit and adding voltages as we go clockwise around the loop, we obtain

$$-L \frac{di}{dt} - \frac{q}{C} = 0$$

Since $i = dq/dt$, it follows that $di/dt = d^2q/dt^2$. We substitute this expression into the above equation and divide by $-L$ to obtain

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0 \quad (L\text{-}C \text{ circuit}) \quad (30.20)$$

Equation (30.20) has exactly the same form as the equation we derived for simple harmonic motion in Section 14.2, Eq. (14.4): $d^2x/dt^2 = -(k/m)x$, or

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

In an L - C circuit the capacitor charge q plays the role of the displacement x , and the current $i = dq/dt$ is analogous to the particle's velocity $v_x = dx/dt$. The inductance L is analogous to the mass m , and the reciprocal of the capacitance, $1/C$, is analogous to the force constant k .

Pursuing this analogy, we recall that the angular frequency $\omega = 2\pi f$ of the harmonic oscillator is equal to $(k/m)^{1/2}$ [Eq. (14.10)], and the position is given as a function of time by Eq. (14.13),

$$x = A \cos(\omega t + \phi)$$

where the amplitude A and the phase angle ϕ depend on the initial conditions. In the analogous electrical situation, the capacitor charge q is given by

$$q = Q \cos(\omega t + \phi) \quad (30.21)$$

and the angular frequency ω of oscillation is given by

$$\text{Angular frequency of oscillation in an } L\text{-}C \text{ circuit} \quad \omega = \sqrt{\frac{1}{LC}} \quad \begin{array}{l} \text{Capacitance} \\ \text{Inductance} \end{array} \quad (30.22)$$

Verify that Eq. (30.21) satisfies the loop equation, Eq. (30.20), when ω has the value given by Eq. (30.22). In doing this, you'll find that the instantaneous current $i = dq/dt$ is

$$i = -\omega Q \sin(\omega t + \phi) \quad (30.23)$$

Thus the charge and current in an L - C circuit oscillate sinusoidally with time, with an angular frequency determined by the values of L and C . The ordinary frequency f , the number of cycles per second, is equal to $\omega/2\pi$. The constants Q and ϕ in Eqs. (30.21) and (30.23) are determined by the initial conditions. If at time $t = 0$ the left-hand capacitor plate in Fig. 30.15 has its maximum charge Q and the current i is zero, then $\phi = 0$. If $q = 0$ at $t = 0$, then $\phi = \pm\pi/2$ rad.

Energy in an L - C Circuit

We can also analyze the L - C circuit by using an energy approach. The analogy to simple harmonic motion is equally useful here. In the mechanical problem an object with mass m is attached to a spring with force constant k . Suppose we displace the object a distance A from its equilibrium position and release it from rest at time $t = 0$. The kinetic energy of the system at any later time is $\frac{1}{2}mv_x^2$, and its elastic potential energy is $\frac{1}{2}kx^2$. Because the system is conservative, the sum of these energies equals the initial energy of the system, $\frac{1}{2}kA^2$. We find the velocity v_x at any position x just as we did in Section 14.3, Eq. (14.22):

$$v_x = \pm \sqrt{\frac{k}{m}} \sqrt{A^2 - x^2} \quad (30.24)$$

The L - C circuit is also a conservative system. Again let Q be the maximum capacitor charge. The magnetic-field energy $\frac{1}{2}Li^2$ in the inductor at any time corresponds to the kinetic energy $\frac{1}{2}mv^2$ of the oscillating object, and the electric-field energy $q^2/2C$ in the capacitor corresponds to the elastic potential energy $\frac{1}{2}kx^2$ of the spring. The sum of these energies is the total energy $Q^2/2C$ of the system:

$$\frac{1}{2}Li^2 + \frac{q^2}{2C} = \frac{Q^2}{2C} \quad (30.25)$$

The total energy in the L - C circuit is *constant*; it oscillates between the magnetic and the electric forms, just as the constant total mechanical energy in simple harmonic motion is constant and oscillates between the kinetic and potential forms.

Solving Eq. (30.25) for i , we find that when the charge on the capacitor is q , the current i is

$$i = \pm \sqrt{\frac{1}{LC}} \sqrt{Q^2 - q^2} \quad (30.26)$$

Verify this equation by substituting q from Eq. (30.21) and i from Eq. (30.23). Comparing Eqs. (30.24) and (30.26), we see that current $i = dq/dt$ and charge q are related in the same way as are velocity $v_x = dx/dt$ and position x in the mechanical problem.

Table 30.1 summarizes the analogies between simple harmonic motion and L - C circuit oscillations. The striking parallels shown there are so close that we can solve complicated mechanical problems by setting up analogous electric circuits and measuring the currents and voltages that correspond to the mechanical quantities to be determined. This is the basic principle of many analog computers. This analogy can be extended to *damped* oscillations, which we consider in the next section. In Chapter 31 we'll extend the analogy further to include *forced* electrical oscillations, which occur in all alternating-current circuits.

TABLE 30.1 Oscillation of a Mass-Spring System Compared with Electrical Oscillation in an L - C Circuit

Mass-Spring System

$$\text{Kinetic energy} = \frac{1}{2}mv_x^2$$

$$\text{Potential energy} = \frac{1}{2}kx^2$$

$$\frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$$

$$v_x = \pm \sqrt{k/m} \sqrt{A^2 - x^2}$$

$$v_x = dx/dt$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$x = A \cos(\omega t + \phi)$$

Inductor-Capacitor Circuit

$$\text{Magnetic energy} = \frac{1}{2}Li^2$$

$$\text{Electrical energy} = q^2/2C$$

$$\frac{1}{2}Li^2 + q^2/2C = Q^2/2C$$

$$i = \pm \sqrt{1/LC} \sqrt{Q^2 - q^2}$$

$$i = dq/dt$$

$$\omega = \sqrt{\frac{1}{LC}}$$

$$q = Q \cos(\omega t + \phi)$$

EXAMPLE 30.8 An oscillating circuit

A 300 V dc power supply is used to charge a 25 μF capacitor. After the capacitor is fully charged, it is disconnected from the power supply and connected across a 10 mH inductor. The resistance in the circuit is negligible. (a) Find the frequency and period of oscillation of the circuit. (b) Find the capacitor charge and the circuit current 1.2 ms after the inductor and capacitor are connected.

IDENTIFY and SET UP Our target variables are the oscillation frequency f and period T , as well as the charge q and current i at a particular time t . We are given the capacitance C and the inductance L , with which we can calculate the frequency and period from Eq. (30.22). We find the charge and current from Eqs. (30.21) and (30.23). Initially the capacitor is fully charged and the current is zero, as in Fig. 30.14a, so the phase angle is $\phi = 0$ [see the discussion that follows Eq. (30.23)].

Continued

EXECUTE (a) The natural angular frequency is

$$\omega = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{(10 \times 10^{-3} \text{ H})(25 \times 10^{-6} \text{ F})}} = 2.0 \times 10^3 \text{ rad/s}$$

The frequency f and period T are then

$$f = \frac{\omega}{2\pi} = \frac{2.0 \times 10^3 \text{ rad/s}}{2\pi \text{ rad/cycle}} = 320 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{320 \text{ Hz}} = 3.1 \times 10^{-3} \text{ s} = 3.1 \text{ ms}$$

(b) Since the period of the oscillation is $T = 3.1 \text{ ms}$, $t = 1.2 \text{ ms}$ equals $0.38T$; this corresponds to a situation intermediate between Fig. 30.14b ($t = T/4$) and Fig. 30.14c ($t = T/2$). Comparing those figures with Fig. 30.15, we expect the capacitor charge q to be negative (that is, there will be negative charge on the left-hand plate of the capacitor) and the current i to be negative as well (that is, current will flow counterclockwise).

To find q , we use Eq. (30.21), $q = Q \cos(\omega t + \phi)$. The charge is maximum at $t = 0$, so $\phi = 0$ and $Q = C\mathcal{E} = (25 \times 10^{-6} \text{ F})(300 \text{ V}) = 7.5 \times 10^{-3} \text{ C}$. Hence Eq. (30.21) becomes

$$q = (7.5 \times 10^{-3} \text{ C}) \cos \omega t$$

At time $t = 1.2 \times 10^{-3} \text{ s}$,

$$\omega t = (2.0 \times 10^3 \text{ rad/s})(1.2 \times 10^{-3} \text{ s}) = 2.4 \text{ rad}$$

$$q = (7.5 \times 10^{-3} \text{ C}) \cos(2.4 \text{ rad}) = -5.5 \times 10^{-3} \text{ C}$$

From Eq. (30.23), the current i at any time is $i = -\omega Q \sin \omega t$. At $t = 1.2 \times 10^{-3} \text{ s}$,

$$i = -(2.0 \times 10^3 \text{ rad/s})(7.5 \times 10^{-3} \text{ C}) \sin(2.4 \text{ rad}) = -10 \text{ A}$$

EVALUATE The signs of both q and i are negative, as predicted.

KEYCONCEPT An L - C circuit that contains an inductor and a capacitor (and no resistance) is analogous to an ideal mass-spring system. The capacitor charge and the current in the circuit both oscillate sinusoidally, just like the displacement and the velocity for a mass on a spring.

EXAMPLE 30.9 Energy in an oscillating circuit

WITH VARIATION PROBLEMS

For the L - C circuit of Example 30.8, find the magnetic and electrical energies (a) at $t = 0$ and (b) at $t = 1.2 \text{ ms}$.

IDENTIFY and SET UP We must calculate the magnetic energy U_B (stored in the inductor) and the electrical energy U_E (stored in the capacitor) at two times during the L - C circuit oscillation. From Example 30.8 we know the values of the capacitor charge q and circuit current i for both times. We use them to calculate $U_B = \frac{1}{2}Li^2$ and $U_E = q^2/2C$.

EXECUTE (a) At $t = 0$ there is no current and $q = Q$. Hence there is no magnetic energy, and all the energy in the circuit is in the form of electrical energy in the capacitor:

$$U_B = \frac{1}{2}Li^2 = 0 \quad U_E = \frac{Q^2}{2C} = \frac{(7.5 \times 10^{-3} \text{ C})^2}{2(25 \times 10^{-6} \text{ F})} = 1.1 \text{ J}$$

(b) From Example 30.8, at $t = 1.2 \text{ ms}$ we have $i = -10 \text{ A}$ and $q = -5.5 \times 10^{-3} \text{ C}$. Hence

$$U_B = \frac{1}{2}Li^2 = \frac{1}{2}(10 \times 10^{-3} \text{ H})(-10 \text{ A})^2 = 0.5 \text{ J}$$

$$U_E = \frac{q^2}{2C} = \frac{(-5.5 \times 10^{-3} \text{ C})^2}{2(25 \times 10^{-6} \text{ F})} = 0.6 \text{ J}$$

EVALUATE The magnetic and electrical energies are the same at $t = 3T/8 = 0.375T$, halfway between the situations in Figs. 30.14b and 30.14c. We saw in Example 30.8 that the time considered in part (b), $t = 1.2 \text{ ms}$, equals $0.38T$; this is slightly later than $0.375T$, so U_B is slightly less than U_E . At all times the total energy $E = U_B + U_E$ has the same value, 1.1 J. An L - C circuit without resistance is a conservative system; no energy is dissipated.

KEYCONCEPT In an L - C circuit, the inductor is analogous to the mass in an ideal mass-spring system and the capacitor is analogous to the spring. Magnetic energy in the inductor behaves like the kinetic energy of the mass; electrical energy in the capacitor behaves like the potential energy of the spring.

TEST YOUR UNDERSTANDING OF SECTION 30.5 One way to think about the energy stored in an L - C circuit is to say that the circuit elements do positive or negative work on the charges that move back and forth through the circuit. (a) Between stages (a) and (b) in Fig. 30.14, does the capacitor do positive or negative work on the charges? (b) What kind of force (electric or magnetic) does the capacitor exert on the charges to do this work? (c) During this process, does the inductor do positive or negative work on the charges? (d) What kind of force (electric or magnetic) does the inductor exert on the charges?

ANSWER

(a) positive, (b) electric, (c) negative, (d) electric. The capacitor loses energy between stages (a) and (b), so it does positive work on the charges. It does this by exerting an electric force that pushes current away from the positively charged left-hand capacitor plate and toward the negatively charged right-hand plate. At the same time, the inductor gains energy and does negative work on the moving charges. Although the inductor stores magnetic energy, the force that the inductor exerts is *electric*. This force comes about from the inductor's self-induced emf (see Section 30.2).

30.6 THE L - R - C SERIES CIRCUIT

In our discussion of the L - C circuit we assumed that there was no *resistance* in the circuit. This is an idealization, of course; every real inductor has resistance in its windings, and there may also be resistance in the connecting wires. Because of resistance, the electromagnetic energy in the circuit is dissipated and converted to other forms, such as internal energy of the circuit materials. Resistance in an electric circuit is analogous to friction in a mechanical system.

Suppose an inductor with inductance L and a resistor of resistance R are connected in series across the terminals of a charged capacitor, forming an **L - R - C series circuit**. As before, the capacitor starts to discharge as soon as the circuit is completed. But due to i^2R losses in the resistor, the magnetic-field energy that the inductor acquires when the capacitor is completely discharged is *less* than the original electric-field energy of the capacitor. In the same way, the energy of the capacitor when the magnetic field has decreased to zero is still less and so on.

If the resistance R of the resistor is relatively small, the circuit still oscillates, but with **damped harmonic motion** (Fig. 30.16a), and we say that the circuit is **underdamped**. If we increase R , the oscillations die out more rapidly. When R reaches a certain value, the circuit no longer oscillates; it is **critically damped** (Fig. 30.16b). For still larger values of R , the circuit is **overdamped** (Fig. 30.16c), and the capacitor charge approaches zero even more slowly. We used these same terms to describe the behavior of the analogous mechanical system, the damped harmonic oscillator, in Section 14.7.

Analyzing an L - R - C Series Circuit

To analyze L - R - C series circuit behavior in detail, consider the circuit shown in Fig. 30.17. It is like the L - C circuit of Fig. 30.15 except for the added resistor R ; we also show the source that charges the capacitor initially. The labeling of the positive senses of q and i is the same as for the L - C circuit.

First we close the switch in the upward position, connecting the capacitor to a source of emf $\mathcal{E}$ for a long enough time to ensure that the capacitor acquires its final charge $Q = C\mathcal{E}$ and any initial oscillations have died out. Then at time $t = 0$ we flip the switch to the downward position, removing the source from the circuit and placing the capacitor in series with the resistor and inductor. Note that the initial current is negative, opposite to the direction of i shown in Fig. 30.17.

To find how q and i vary with time, we apply Kirchhoff's loop rule. Starting at point a and going around the loop in the direction $abcda$, we obtain

$$-iR - L \frac{di}{dt} - \frac{q}{C} = 0$$

Replacing i with dq/dt and rearranging, we get

$$\frac{d^2q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC}q = 0 \quad (30.27)$$

Note that when $R = 0$, this reduces to Eq. (30.20) for an L - C circuit.

There are general methods for obtaining solutions of Eq. (30.27). The form of the solution is different for the underdamped (small R) and overdamped (large R) cases. When R^2 is less than $4L/C$, the solution has the form

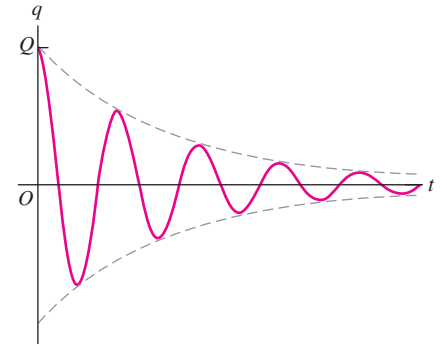
$$q = Ae^{-(R/2L)t} \cos\left(\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}t + \phi\right) \quad (30.28)$$

where A and ϕ are constants. You can take the first and second derivatives of this function and show by direct substitution that it does satisfy Eq. (30.27).

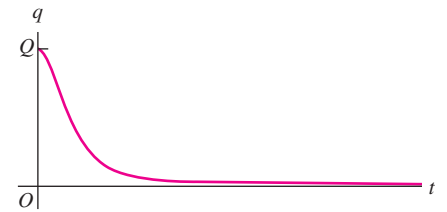
This solution corresponds to the *underdamped* behavior shown in Fig. 30.16a; the function represents a sinusoidal oscillation with an exponentially decaying amplitude. (Note that the exponential factor $e^{-(R/2L)t}$ is *not* the same as the factor $e^{-(R/L)t}$ that we encountered in describing the R - L circuit in Section 30.4.) When $R = 0$, Eq. (30.28) reduces to Eq. (30.21)

Figure 30.16 Graphs of capacitor charge as a function of time in an L - R - C series circuit with initial charge Q .

(a) Underdamped circuit (small resistance R)



(b) Critically damped circuit (larger resistance R)



(c) Overdamped circuit (very large resistance R)

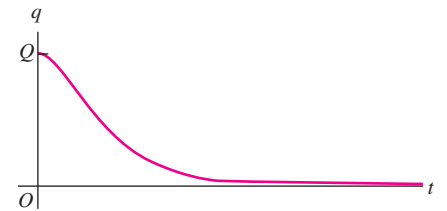
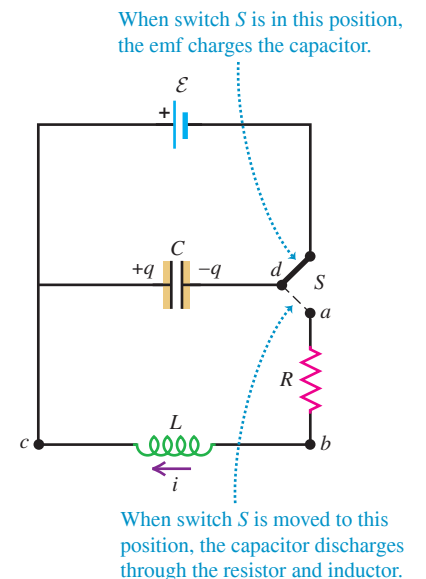


Figure 30.17 An L - R - C series circuit.



for the oscillations in an L - C circuit. If R is not zero, the angular frequency of the oscillation is *less* than $1/(LC)^{1/2}$ because of the term containing R . The angular frequency ω' of the damped oscillations is given by

$$\text{Angular frequency of underdamped oscillations in an } L\text{-}R\text{-}C \text{ series circuit} \rightarrow \omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \quad (30.29)$$

Inductance Capacitance Resistance

When $R = 0$, this reduces to Eq. (30.22), $\omega = (1/LC)^{1/2}$. As R increases, ω' becomes smaller and smaller. When $R^2 = 4L/C$, the quantity under the radical becomes zero; the system no longer oscillates, and the case of *critical damping* (Fig. 30.16b) has been reached. For still larger values of R the system behaves as in Fig. 30.16c. In this case the circuit is *overdamped*, and q is given as a function of time by the sum of two decreasing exponential functions.

In the *underdamped* case the phase constant ϕ in the cosine function of Eq. (30.28) provides for the possibility of both an initial charge and an initial current at time $t = 0$, analogous to an underdamped harmonic oscillator given both an initial displacement and an initial velocity (see Exercise 30.38).

We emphasize once more that the behavior of the L - R - C series circuit is completely analogous to that of the damped harmonic oscillator (see Section 14.7). We invite you to verify, for example, that if you start with Eq. (14.41) and substitute q for x , L for m , $1/C$ for k , and R for the damping constant b , the result is Eq. (30.27). Similarly, the cross-over point between underdamping and overdamping occurs at $b^2 = 4km$ for the mechanical system and at $R^2 = 4L/C$ for the electrical one. Can you find still other aspects of this analogy?

The practical applications of the L - R - C series circuit emerge when we include a sinusoidally varying source of emf in the circuit. This is analogous to the *forced oscillations* that we discussed in Section 14.7, and there are analogous *resonance* effects. Such a circuit is called an *alternating-current (ac) circuit*. The analysis of ac circuits is the principal topic of the next chapter.

EXAMPLE 30.10 An underdamped L - R - C series circuit

WITH VARIATION PROBLEMS

What resistance R is required (in terms of L and C) to give an L - R - C series circuit a frequency that is one-half the undamped frequency?

IDENTIFY and SET UP This problem concerns an underdamped L - R - C series circuit (Fig. 30.16a). We want just enough resistance to reduce the oscillation frequency to one-half of the undamped value. Equation (30.29) gives the angular frequency ω' of an underdamped L - R - C series circuit; Eq. (30.22) gives the angular frequency ω of an undamped L - C circuit. We use these two equations to solve for R .

EXECUTE From Eqs. (30.29) and (30.22), the requirement $\omega' = \omega/2$ yields

$$\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \frac{1}{2} \sqrt{\frac{1}{LC}}$$

We square both sides and solve for R :

$$R = \sqrt{\frac{3L}{C}}$$

For example, adding $35 \, \Omega$ to the circuit of Example 30.8 ($L = 10 \, \text{mH}$, $C = 25 \, \mu\text{F}$) would reduce the frequency from 320 Hz to 160 Hz.

EVALUATE The circuit becomes critically damped with no oscillations when $R = \sqrt{4L/C}$. Our result for R is smaller than that, as it should be; we want the circuit to be *underdamped*.

KEYCONCEPT A series L - R - C circuit is analogous to a mass-spring system with damping. Increasing the resistance reduces the oscillation frequency of the charge and current; if the resistance is too great, the circuit is critically damped or overdamped and there is no oscillation.

TEST YOUR UNDERSTANDING OF SECTION 30.6 An L - R - C series circuit includes a $2.0 \, \Omega$ resistor. At $t = 0$ the capacitor charge is $2.0 \, \mu\text{C}$. For which of the following values of the inductance and capacitance will the charge on the capacitor *not* oscillate? (i) $L = 3.0 \, \mu\text{H}$, $C = 6.0 \, \mu\text{F}$; (ii) $L = 6.0 \, \mu\text{H}$, $C = 3.0 \, \mu\text{F}$; (iii) $L = 3.0 \, \mu\text{H}$, $C = 3.0 \, \mu\text{F}$.

ANSWER

(i) and (iii) There are no oscillations if $R^2 \geq 4L/C$. In each case $R^2 = (2.0 \, \Omega)^2 = 4.0 \, \Omega^2$. In case (i) $4L/C = 4(3.0 \, \mu\text{H})/(6.0 \, \mu\text{F}) = 2.0 \, \Omega^2$, so there are no oscillations (the system is overdamped); in case (ii) $4L/C = 4(6.0 \, \mu\text{H})/(3.0 \, \mu\text{F}) = 8.0 \, \Omega^2$, so there are oscillations (the system is underdamped); and in case (iii) $4L/C = 4(3.0 \, \mu\text{H})/(3.0 \, \mu\text{F}) = 4.0 \, \Omega^2$, so there are no oscillations (the system is critically damped).

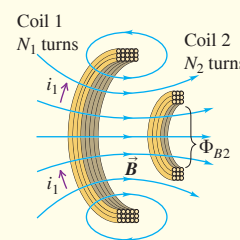
CHAPTER 30 SUMMARY

Mutual inductance: When a changing current i_1 in one circuit causes a changing magnetic flux in a second circuit, an emf $\mathcal{E}_2$ is induced in the second circuit. Likewise, a changing current i_2 in the second circuit induces an emf $\mathcal{E}_1$ in the first circuit. If the circuits are coils of wire with N_1 and N_2 turns, the mutual inductance M can be expressed in terms of the average flux Φ_{B2} through each turn of coil 2 caused by the current i_1 in coil 1, or in terms of the average flux Φ_{B1} through each turn of coil 1 caused by the current i_2 in coil 2. The SI unit of mutual inductance is the henry, abbreviated H. (See Examples 30.1 and 30.2.)

$$\mathcal{E}_2 = -M \frac{di_1}{dt} \quad \text{and} \quad (30.4)$$

$$\mathcal{E}_1 = -M \frac{di_2}{dt} \quad (30.5)$$

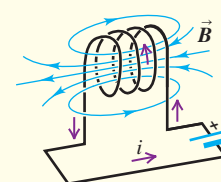
$$M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_1 \Phi_{B1}}{i_2} \quad (30.5)$$



Self-inductance: A changing current i in any circuit causes a self-induced emf $\mathcal{E}$. The inductance (or self-inductance) L depends on the geometry of the circuit and the material surrounding it. The inductance of a coil of N turns is related to the average flux Φ_B through each turn caused by the current i in the coil. An inductor is a circuit device, usually including a coil of wire, intended to have a substantial inductance. (See Examples 30.3 and 30.4.)

$$\mathcal{E} = -L \frac{di}{dt} \quad (30.7)$$

$$L = \frac{N\Phi_B}{i} \quad (30.6)$$

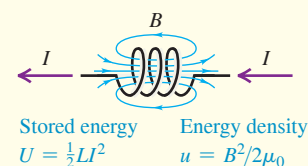


Magnetic-field energy: An inductor with inductance L carrying current I has energy U associated with the inductor's magnetic field. The magnetic energy density u (energy per unit volume) is proportional to the square of the magnetic-field magnitude. (See Example 30.5.)

$$U = \frac{1}{2} LI^2 \quad (30.9)$$

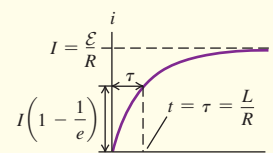
$$u = \frac{B^2}{2\mu_0} \quad (\text{in vacuum}) \quad (30.10)$$

$$u = \frac{B^2}{2\mu} \quad (\text{in a material with magnetic permeability } \mu) \quad (30.11)$$



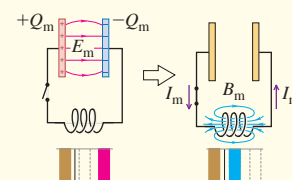
R-L circuits: In a circuit containing a resistor R , an inductor L , and a source of emf, the growth and decay of current are exponential. The time constant τ is the time required for the current to approach within a fraction $1/e$ of its final value. (See Examples 30.6 and 30.7.)

$$\tau = \frac{L}{R} \quad (30.16)$$



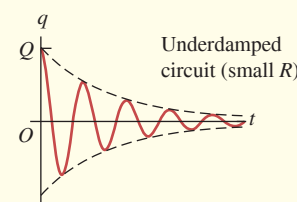
L-C circuits: A circuit that contains inductance L and capacitance C undergoes electrical oscillations with an angular frequency ω that depends on L and C . This is analogous to a mechanical harmonic oscillator, with inductance L analogous to mass m , the reciprocal of capacitance $1/C$ to force constant k , charge q to displacement x , and current i to velocity v_x . (See Examples 30.8 and 30.9.)

$$\omega = \sqrt{\frac{1}{LC}} \quad (30.22)$$



L-R-C series circuits: A circuit that contains inductance, resistance, and capacitance undergoes damped oscillations for sufficiently small resistance. The frequency ω' of damped oscillations depends on the values of L , R , and C . As R increases, the damping increases; if R is greater than a certain value, the behavior becomes overdamped and no longer oscillates. (See Example 30.10.)

$$\omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \quad (30.29)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 30.1 and 30.2** (Section 30.1) and **EXAMPLES 30.3 and 30.4** (Section 30.2) before attempting these problems.

VP30.4.1 A long solenoid (coil 1) is surrounded at its center by a shorter coil (coil 2) as in Fig. 30.3. At an instant when the current in coil 1 is 24.0 A and increasing at a rate of 2.00 A/s, the induced emf in coil 2 is 1.30×10^{-4} V. (a) What is the mutual inductance of the two coils? (b) At this instant, what is the total magnetic flux through all the turns of coil 2 due to the current in coil 1?

VP30.4.2 The mutual inductance of two coils (1 and 2) is 48.0 μ H. Find the magnitude of the emf induced in coil 2 if the current in coil 1 is (a) 35.0 A and increasing at 7.00 A/s; (b) 28.0 A and constant; (c) 26.0 A and decreasing at 3.00 A/s.

VP30.4.3 A toroidal solenoid with self-inductance 76.0 μ H has 465 turns of wire. Find (a) the magnetic flux through each turn when the current in the solenoid is 12.0 A and (b) the magnitude of the induced emf in the solenoid when the current is 12.0 A and decreasing at 55.0 A/s.

VP30.4.4 When a coil is carrying a current of 25.0 A that is increasing at 145 A/s, the induced emf in the coil has magnitude 3.70 mV. Find (a) the self-inductance of the coil and (b) the magnitude and direction of the induced emf at $t = 2.00$ s if the current is $i(t) = (225 \text{ A/s}^2)t^2$.

Be sure to review **EXAMPLES 30.6 and 30.7** (Section 30.4) before attempting these problems.

VP30.7.1 In the circuit shown in Fig. 30.11, the inductance is 4.90 mH and the emf is 24.0 V. Initially both switches are open. You want the current to reach 75.0% of its maximum value 80.0 μ s after switch S_1 is closed. Find (a) the required resistance and (b) the current 80.0 μ s after switch S_1 is closed.

VP30.7.2 In the circuit shown in Fig. 30.11, $\mathcal{E} = 12.0$ V, $R = 18.0 \Omega$, and $L = 37.5$ mH. Initially both switches are open. You close switch S_1

at $t = 0$. At $t = 0.700$ ms, find (a) the current, (b) the rate of change of the current, and (c) the amount of energy stored in the inductor.

VP30.7.3 In the circuit shown in Fig. 30.11, $\mathcal{E} = 12.0$ V, $R = 18.0 \Omega$, and $L = 37.5$ mH. Switch S_1 has been closed for a long time and switch S_2 is open. At $t = 0$ you open switch S_1 and close switch S_2 . At $t = 0.700$ ms, find (a) the current, (b) the rate of change of the current, and (c) the amount of energy stored in the inductor.

VP30.7.4 For the R - L circuit with an increasing current shown in Fig. 30.12, find (a) the time when the voltage across the inductor has the same magnitude as the voltage across the resistor, (b) the current at this time, and (c) the rate of change of the current at this time.

Be sure to review **EXAMPLES 30.8 and 30.9** (Section 30.5) and **EXAMPLE 30.10** (Section 30.6) before attempting these problems.

VP30.10.1 The L - C circuit in Fig. 30.15 contains an inductor with $L = 25.0$ μ H. The charge and current in this circuit oscillate with frequency 445 Hz, and the maximum capacitor charge is 5.00 mC. Find (a) the capacitance of the capacitor and (b) the maximum current in the circuit.

VP30.10.2 You construct an L - C circuit that contains a 30.0 μ H inductor and a 655 μ F capacitor. At $t = 0$ the capacitor has its maximum charge of 0.400 mC. Find (a) the first time after $t = 0$ when the capacitor charge is zero and (b) the magnitude of the current in the circuit at this time.

VP30.10.3 The sum of the electrical and magnetic energies in an L - C circuit is 0.800 J. At a certain instant the energy is exactly half electrical and half magnetic, the capacitor charge is 5.30 mC, and the current is 8.00 A. Find (a) the capacitance, (b) the inductance, and (c) the angular frequency of oscillation.

VP30.10.4 An L - R - C series circuit has inductance 42.0 mH, capacitance C , and resistance R . Without the resistor, the angular frequency of oscillation is 624 rad/s. With the resistor, the angular frequency is 208 rad/s. Find the values of (a) C and (b) R .

BRIDGING PROBLEM Analyzing an L - C Circuit

An L - C circuit like that shown in Fig. 30.14 consists of a 60.0 mH inductor and a 250 μ F capacitor. The initial charge on the capacitor is 6.00 μ C, and the initial current in the inductor is 0.400 mA. (a) What is the maximum energy stored in the inductor? (b) What is the maximum current in the inductor? (c) What is the maximum voltage across the capacitor? (d) When the current in the inductor has half its maximum value, what are the energy stored in the inductor and the voltage across the capacitor?

SOLUTION GUIDE

IDENTIFY and SET UP

1. An L - C circuit is a conservative system—there is no resistance to dissipate energy. The energy oscillates between electrical energy in the capacitor and magnetic energy stored in the inductor.
2. Oscillations in an L - C circuit are analogous to the mechanical oscillations of a particle at the end of an ideal spring (see Table 30.1). Compare this problem to the analogous mechanical problem (see Example 14.3 in Section 14.2 and Example 14.4 in Section 14.3).

3. Which key equations are needed to describe the capacitor? To describe the inductor?

EXECUTE

4. Find the initial total energy in the circuit. Use it to determine the maximum energy stored in the inductor during the oscillation.
5. Use the result of step 4 to find the maximum current in the inductor.
6. Use the result of step 4 to find the maximum energy stored in the capacitor during the oscillation. Then use this to find the maximum capacitor voltage.
7. Find the energy in the inductor and the capacitor charge when the current has half the value that you found in step 5.

EVALUATE

8. Initially, what fraction of the total energy is in the inductor? Is it possible to tell whether this is initially increasing or decreasing?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q30.1 In an electric trolley or bus system, the vehicle's motor draws current from an overhead wire by means of a long arm with an attachment at the end that slides along the overhead wire. A brilliant electric spark is often seen when the attachment crosses a junction in the wires where contact is momentarily lost. Explain this phenomenon.

Q30.2 From Eq. (30.5) $1 \text{ H} = 1 \text{ Wb/A}$, and from Eqs. (30.4) $1 \text{ H} = 1 \Omega \cdot \text{s}$. Show that these two definitions are equivalent.

Q30.3 In Fig. 30.1, if coil 2 is turned 90° so that its axis is vertical, does the mutual inductance increase or decrease? Explain.

Q30.4 The tightly wound toroidal solenoid is one of the few configurations for which it is easy to calculate self-inductance. What features of the toroidal solenoid give it this simplicity?

Q30.5 Two identical, closely wound, circular coils, each having self-inductance L , are placed next to each other, so that they are coaxial and almost touching. If they are connected in series, what is the self-inductance of the combination? What if they are connected in parallel? Can they be connected so that the total inductance is zero? Explain.

Q30.6 Two closely wound circular coils have the same number of turns, but one has twice the radius of the other. How are the self-inductances of the two coils related? Explain your reasoning.

Q30.7 You are to make a resistor by winding a wire around a cylindrical form. To make the inductance as small as possible, it is proposed that you wind half the wire in one direction and the other half in the opposite direction. Would this achieve the desired result? Why or why not?

Q30.8 For the same magnetic field strength B , is the energy density greater in vacuum or in a magnetic material? Explain. Does Eq. (30.11) imply that for a long solenoid in which the current is I the energy stored is proportional to $1/\mu$? And does this mean that for the same current less energy is stored when the solenoid is filled with a ferromagnetic material rather than with air? Explain.

Q30.9 In an R - C circuit, a resistor, an uncharged capacitor, a dc battery, and an open switch are in series. In an R - L circuit, a resistor, an inductor, a dc battery, and an open switch are in series. Compare the behavior of the current in these circuits (a) just after the switch is closed and (b) long after the switch has been closed. In other words, compare the way in which a capacitor and an inductor affect a circuit.

Q30.10 A Differentiating Circuit. Figure **Q30.10**

The current in a resistanceless inductor is caused to vary with time as shown in the graph of **Fig. Q30.10**. (a) Sketch the pattern that would be observed on the screen of an oscilloscope connected to the terminals of the inductor. (The oscilloscope spot sweeps horizontally across the screen at a constant speed, and its vertical deflection is proportional to the potential difference between the inductor terminals.) (b) Explain why a circuit with an inductor can be described as a "differentiating circuit."

Q30.11 In Section 30.5 Kirchhoff's loop rule is applied to an L - C circuit where the capacitor is initially fully charged and the equation $-L(di/dt) - (q/C) = 0$ is derived. But as the capacitor starts to discharge, the current increases from zero. The equation says $L di/dt = -q/C$, so it says $L di/dt$ is negative. Explain how $L di/dt$ can be negative when the current is increasing.

Q30.12 In Section 30.5 the relationship $i = dq/dt$ is used in deriving Eq. (30.20). But a flow of current corresponds to a decrease in the charge on the capacitor. Explain, therefore, why this is the correct equation to use in the derivation, rather than $i = -dq/dt$.

Q30.13 In the R - L circuit shown in Fig. 30.11, when switch S_1 is closed, the potential v_{ac} changes suddenly and discontinuously, but the current does not. Explain why the voltage can change suddenly but the current can't.

Q30.14 In the R - L circuit shown in Fig. 30.11, is the current in the resistor always the same as the current in the inductor? How do you know?

Q30.15 Suppose there is a steady current in an inductor. If you attempt to reduce the current to zero instantaneously by quickly opening a switch, an arc can appear at the switch contacts. Why? Is it physically possible to stop the current instantaneously? Explain.

Q30.16 In an L - R - C series circuit, what criteria could be used to decide whether the system is overdamped or underdamped? For example, could we compare the maximum energy stored during one cycle to the energy dissipated during one cycle? Explain.

EXERCISES

Section 30.1 Mutual Inductance

30.1 • Two coils have mutual inductance $M = 3.25 \times 10^{-4} \text{ H}$. The current i_1 in the first coil increases at a uniform rate of 830 A/s . (a) What is the magnitude of the induced emf in the second coil? Is it constant? (b) Suppose that the current described is in the second coil rather than the first. What is the magnitude of the induced emf in the first coil?

30.2 • Two coils are wound around the same cylindrical form, like the coils in Example 30.1. When the current in the first coil is decreasing at a rate of -0.242 A/s , the induced emf in the second coil has magnitude $1.65 \times 10^{-3} \text{ V}$. (a) What is the mutual inductance of the pair of coils? (b) If the second coil has 25 turns, what is the flux through each turn when the current in the first coil equals 1.20 A ? (c) If the current in the second coil increases at a rate of 0.360 A/s , what is the magnitude of the induced emf in the first coil?

30.3 • Two toroidal solenoids are wound around the same form so that the magnetic field of one passes through the turns of the other. Solenoid 1 has 700 turns, and solenoid 2 has 400 turns. When the current in solenoid 1 is 6.52 A , the average flux through each turn of solenoid 2 is 0.0320 Wb . (a) What is the mutual inductance of the pair of solenoids? (b) When the current in solenoid 2 is 2.54 A , what is the average flux through each turn of solenoid 1?

30.4 • A solenoidal coil with 25 turns of wire is wound tightly around another coil with 300 turns (see Example 30.1). The inner solenoid is 25.0 cm long and has a diameter of 2.00 cm . At a certain time, the current in the inner solenoid is 0.120 A and is increasing at a rate of $1.75 \times 10^3 \text{ A/s}$. For this time, calculate: (a) the average magnetic flux through each turn of the inner solenoid; (b) the mutual inductance of the two solenoids; (c) the emf induced in the outer solenoid by the changing current in the inner solenoid.

Section 30.2 Self-Inductance and Inductors

30.5 • A 2.50 mH toroidal solenoid has an average radius of 6.00 cm and a cross-sectional area of 2.00 cm^2 . (a) How many coils does it have? (Make the same assumption as in Example 30.3.) (b) At what rate must the current through it change so that a potential difference of 2.00 V is developed across its ends?

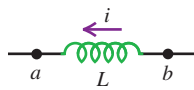
30.6 • A toroidal solenoid has 500 turns, cross-sectional area 6.25 cm^2 , and mean radius 4.00 cm . (a) Calculate the coil's self-inductance. (b) If the current decreases uniformly from 5.00 A to 2.00 A in 3.00 ms , calculate the self-induced emf in the coil. (c) The current is directed from terminal a of the coil to terminal b . Is the direction of the induced emf from a to b or from b to a ?

30.7 • At the instant when the current in an inductor is increasing at a rate of 0.0640 A/s , the magnitude of the self-induced emf is 0.0160 V . (a) What is the inductance of the inductor? (b) If the inductor is a solenoid with 400 turns, what is the average magnetic flux through each turn when the current is 0.720 A ?

30.8 •• When the current in a toroidal solenoid is changing at a rate of 0.0260 A/s , the magnitude of the induced emf is 12.6 mV . When the current equals 1.40 A , the average flux through each turn of the solenoid is 0.00285 Wb . How many turns does the solenoid have?

30.9 • The inductor in **Fig. E30.9** has inductance 0.260 H and carries a current in the direction shown that is decreasing at a uniform rate, $di/dt = -0.0180 \text{ A/s}$. (a) Find the self-induced emf. (b) Which end of the inductor, a or b , is at a higher potential?

Figure E30.9



30.10 • The inductor shown in **Fig. E30.9** has inductance 0.260 H and carries a current in the direction shown. The current is changing at a constant rate. (a) The potential between points a and b is $V_{ab} = 1.04 \text{ V}$, with point a at higher potential. Is the current increasing or decreasing? (b) If the current at $t = 0$ is 12.0 A , what is the current at $t = 2.00 \text{ s}$?

30.11 •• Inductance of a Solenoid. (a) A long, straight solenoid has N turns, uniform cross-sectional area A , and length l . Show that the inductance of this solenoid is given by the equation $L = \mu_0 AN^2/l$. Assume that the magnetic field is uniform inside the solenoid and zero outside. (Your answer is approximate because B is actually smaller at the ends than at the center. For this reason, your answer is actually an upper limit on the inductance.) (b) A metallic laboratory spring is typically 5.00 cm long and 0.150 cm in diameter and has 50 coils. If you connect such a spring in an electric circuit, how much self-inductance must you include for it if you model it as an ideal solenoid?

30.12 • A long, straight solenoid has 800 turns. When the current in the solenoid is 2.90 A , the average flux through each turn of the solenoid is $3.25 \times 10^{-3} \text{ Wb}$. What must be the magnitude of the rate of change of the current in order for the self-induced emf to equal 6.20 mV ?

Section 30.3 Magnetic-Field Energy

30.13 •• When the current in a long, straight, air-filled solenoid is changing at the rate of 2000 A/s , the voltage across the solenoid is 0.600 V . The solenoid has 1200 turns and uniform cross-sectional area 25.0 mm^2 . Assume that the magnetic field is uniform inside the solenoid and zero outside, so the result $L = \mu_0 AN^2/l$ (see Exercise 30.11) applies. What is the magnitude B of the magnetic field in the interior of the solenoid when the current in the solenoid is 3.00 A ?

30.14 • An inductor used in a dc power supply has an inductance of 12.0 H and a resistance of 180Ω . It carries a current of 0.500 A . (a) What is the energy stored in the magnetic field? (b) At what rate is thermal energy developed in the inductor? (c) Does your answer to part (b) mean that the magnetic-field energy is decreasing with time? Explain.

30.15 • An air-filled toroidal solenoid has a mean radius of 15.0 cm and a cross-sectional area of 5.00 cm^2 . When the current is 12.0 A , the energy stored is 0.390 J . How many turns does the winding have?

30.16 • An air-filled toroidal solenoid has 300 turns of wire, a mean radius of 12.0 cm , and a cross-sectional area of 4.00 cm^2 . If the current is 5.00 A , calculate: (a) the magnetic field in the solenoid; (b) the self-inductance of the solenoid; (c) the energy stored in the magnetic field; (d) the energy density in the magnetic field. (e) Check your answer for part (d) by dividing your answer to part (c) by the volume of the solenoid.

30.17 •• A solenoid 25.0 cm long and with a cross-sectional area of 0.500 cm^2 contains 400 turns of wire and carries a current of 80.0 A . Calculate: (a) the magnetic field in the solenoid; (b) the energy density in the magnetic field if the solenoid is filled with air; (c) the total energy contained in the coil's magnetic field (assume the field is uniform); (d) the inductance of the solenoid.

30.18 • It has been proposed to use large inductors as energy storage devices. (a) How much electrical energy is converted to light and thermal energy by a 150 W light bulb in one day? (b) If the amount of energy calculated in part (a) is stored in an inductor in which the current is 80.0 A , what is the inductance?

30.19 •• In a proton accelerator used in elementary particle physics experiments, the trajectories of protons are controlled by bending magnets that produce a magnetic field of 4.80 T . What is the magnetic-field energy in a 10.0 cm^3 volume of space where $B = 4.80 \text{ T}$?

30.20 • A region of vacuum contains both a uniform electric field with magnitude E and a uniform magnetic field with magnitude B . (a) What is the ratio E/B if the energy density for the magnetic field equals the energy density for the electric field? (b) If $E = 500 \text{ V/m}$, what is B , in teslas, if the magnetic-field and electric-field energy densities are equal?

Section 30.4 The R-L Circuit

30.21 • An inductor with an inductance of 2.50 H and a resistance of 8.00Ω is connected to the terminals of a battery with an emf of 6.00 V and negligible internal resistance. Find (a) the initial rate of increase of current in the circuit; (b) the rate of increase of current at the instant when the current is 0.500 A ; (c) the current 0.250 s after the circuit is closed; (d) the final steady-state current.

30.22 • In **Fig. 30.11**, $R = 15.0 \Omega$ and the battery emf is 6.30 V . With switch S_2 open, switch S_1 is closed. After several minutes, S_1 is opened and S_2 is closed. (a) At 2.00 ms after S_1 is opened, the current has decayed to 0.280 A . Calculate the inductance of the coil. (b) How long after S_1 is opened will the current reach 1.00% of its original value?

30.23 • A 35.0 V battery with negligible internal resistance, a 50.0Ω resistor, and a 1.25 mH inductor with negligible resistance are all connected in series with an open switch. The switch is suddenly closed. (a) How long after closing the switch will the current through the inductor reach one-half of its maximum value? (b) How long after closing the switch will the energy stored in the inductor reach one-half of its maximum value?

30.24 •• A resistor and an inductor are connected in series to a battery with emf 240 V and negligible internal resistance. The circuit is completed at time $t = 0$. At a later time $t = T$ the current is 5.00 A and is increasing at a rate of 20.0 A/s . After a long time the current in the circuit is 15.0 A . What is the value of T , the time when the current is 5.00 A ?

30.25 • A resistor with $R = 30.0 \Omega$ and an inductor with $L = 0.600 \text{ H}$ are connected in series to a battery that has emf 50.0 V and negligible internal resistance. At time t after the circuit is completed, the energy stored in the inductor is 0.400 J . At this instant, what is the voltage across the inductor?

30.26 • In the circuit shown in **Fig. 30.11** switch S_1 has been closed a long time while switch S_2 has been left open. Then S_2 is closed at the same instant when S_1 is opened. Just after S_2 is closed, the current through the resistor is 12.0 A and its rate of decrease is $di/dt = -36.0 \text{ A/s}$. How long does it take the current to decrease to 6.00 A , one-half its initial value?

30.27 • In **Fig. 30.11**, suppose that $\mathcal{E} = 60.0 \text{ V}$, $R = 240 \Omega$, and $L = 0.160 \text{ H}$. With switch S_2 open, switch S_1 is left closed until a constant current is established. Then S_2 is closed and S_1 is opened, taking the battery out of the circuit. (a) What is the initial current in the resistor, just after S_2 is closed and S_1 is opened? (b) What is the current in the resistor at $t = 4.00 \times 10^{-4} \text{ s}$? (c) What is the potential difference between points b and c at $t = 4.00 \times 10^{-4} \text{ s}$? Which point is at a higher potential? (d) How long does it take the current to decrease to half its initial value?

30.28 • In Fig. 30.11, suppose that $\mathcal{E} = 60.0 \text{ V}$, $R = 240 \, \Omega$, and $L = 0.160 \text{ H}$. Initially there is no current in the circuit. Switch S_2 is left open, and switch S_1 is closed. (a) Just after S_1 is closed, what are the potential differences v_{ab} and v_{bc} ? (b) A long time (many time constants) after S_1 is closed, what are v_{ab} and v_{bc} ? (c) What are v_{ab} and v_{bc} at an intermediate time when $i = 0.150 \text{ A}$?

30.29 • In Fig. 30.11 switch S_1 is closed while switch S_2 is kept open. The inductance is $L = 0.380 \text{ H}$, the resistance is $R = 48.0 \, \Omega$, and the emf of the battery is 18.0 V . At time t after S_1 is closed, the current in the circuit is increasing at a rate of $di/dt = 7.20 \text{ A/s}$. At this instant what is v_{ab} , the voltage across the resistor?

30.30 •• Consider the circuit in Exercise 30.21. (a) Just after the circuit is completed, at what rate is the battery supplying electrical energy to the circuit? (b) When the current has reached its final steady-state value, how much energy is stored in the inductor? What is the rate at which electrical energy is being dissipated in the resistance of the inductor? What is the rate at which the battery is supplying electrical energy to the circuit?

30.31 • Consider the rate P_L at which energy is being stored in the R - L circuit of Fig. 30.12. Answer these questions, in terms of $\mathcal{E}$, R , and L as needed: (a) What is P_L at $t = 0$, just after the circuit is completed? (b) What is P_L at $t \rightarrow \infty$, a long time after the circuit is completed? (c) What is P_L at the instant when $i = \mathcal{E}/2R$, one-half the current's final value?

Section 30.5 The L - C Circuit

30.32 •• A $15.0 \, \mu\text{F}$ capacitor is charged by a 150.0 V power supply, then disconnected from the power and connected in series with a 0.280 mH inductor. Calculate: (a) the oscillation frequency of the circuit; (b) the energy stored in the capacitor at time $t = 0 \text{ ms}$ (the moment of connection with the inductor); (c) the energy stored in the inductor at $t = 1.30 \text{ ms}$.

30.33 • In an L - C circuit, $L = 85.0 \text{ mH}$ and $C = 3.20 \, \mu\text{F}$. During the oscillations the maximum current in the inductor is 0.850 mA . (a) What is the maximum charge on the capacitor? (b) What is the magnitude of the charge on the capacitor at an instant when the current in the inductor has magnitude 0.500 mA ?

30.34 • A 7.50 nF capacitor is charged to 12.0 V , then disconnected from the power supply and connected in series through a coil. The period of oscillation of the circuit is then measured to be $8.60 \times 10^{-5} \text{ s}$. Calculate: (a) the inductance of the coil; (b) the maximum charge on the capacitor; (c) the total energy of the circuit; (d) the maximum current in the circuit.

30.35 • L - C Oscillations. A capacitor with capacitance $6.00 \times 10^{-5} \text{ F}$ is charged by connecting it to a 12.0 V battery. The capacitor is disconnected from the battery and connected across an inductor with $L = 1.50 \text{ H}$. (a) What are the angular frequency ω of the electrical oscillations and the period of these oscillations (the time for one oscillation)? (b) What is the initial charge on the capacitor? (c) How much energy is initially stored in the capacitor? (d) What is the charge on the capacitor 0.0230 s after the connection to the inductor is made? Interpret the sign of your answer. (e) At the time given in part (d), what is the current in the inductor? Interpret the sign of your answer. (f) At the time given in part (d), how much electrical energy is stored in the capacitor and how much is stored in the inductor?

30.36 • A Radio Tuning Circuit. The minimum capacitance of a variable capacitor in a radio is 4.18 pF . (a) What is the inductance of a coil connected to this capacitor if the oscillation frequency of the L - C circuit is $1600 \times 10^3 \text{ Hz}$, corresponding to one end of the AM radio broadcast band, when the capacitor is set to its minimum capacitance? (b) The frequency at the other end of the broadcast band is $540 \times 10^3 \text{ Hz}$. What is the maximum capacitance of the capacitor if the oscillation frequency is adjustable over the range of the broadcast band?

30.37 •• An L - C circuit containing an 80.0 mH inductor and a 1.25 nF capacitor oscillates with a maximum current of 0.750 A . Calculate: (a) the maximum charge on the capacitor and (b) the oscillation frequency of the circuit. (c) Assuming the capacitor had its maximum charge at time $t = 0$, calculate the energy stored in the inductor after 2.50 ms of oscillation.

Section 30.6 The L - R - C Series Circuit

30.38 • For the circuit of Fig. 30.17, let $C = 15.0 \text{ nF}$, $L = 22 \text{ mH}$, and $R = 75.0 \, \Omega$. (a) Calculate the oscillation frequency of the circuit once the capacitor has been charged and the switch has been connected to point a . (b) How long will it take for the amplitude of the oscillation to decay to 10.0% of its original value? (c) What value of R would result in a critically damped circuit?

30.39 • An L - R - C series circuit has $L = 0.450 \text{ H}$, $C = 2.50 \times 10^{-5} \text{ F}$, and resistance R . (a) What is the angular frequency of the circuit when $R = 0$? (b) What value must R have to give a 5.0% decrease in angular frequency compared to the value calculated in part (a)?

30.40 • An L - R - C series circuit has $L = 0.400 \text{ H}$, $C = 7.00 \, \mu\text{F}$, and $R = 320 \, \Omega$. At $t = 0$ the current is zero and the initial charge on the capacitor is $2.80 \times 10^{-4} \text{ C}$. (a) What are the values of the constants A and ϕ in Eq. (30.28)? (b) How much time does it take for each complete current oscillation after the switch in this circuit is closed? (c) What is the charge on the capacitor after the first complete current oscillation?

PROBLEMS

30.41 •• It is possible to make your own inductor by winding wire around a cylinder, such as a pencil. Assume you have a spool of AWG 20 copper wire, which has a diameter of 0.812 mm . (a) Estimate the diameter of a pencil. (b) Estimate how many times can you tightly wrap AWG 20 copper wire around a pencil to form a solenoid with a length of 4.0 cm . (c) Estimate the inductance of this solenoid by assuming the magnetic field inside is constant. (d) If a current of 1.0 A flows through this solenoid, how much magnetic energy will be stored inside?

30.42 • An inductor is connected to the terminals of a battery that has an emf of 16.0 V and negligible internal resistance. The current is 4.86 mA at 0.940 ms after the connection is completed. After a long time, the current is 6.45 mA . What are (a) the resistance R of the inductor and (b) the inductance L of the inductor?

30.43 •• CP Consider a coil of wire that has radius 3.00 cm and carries a sinusoidal current given by $i(t) = I_0 \sin(2\pi ft)$, where the frequency $f = 60.0 \text{ Hz}$ and the initial current $I_0 = 1.20 \text{ A}$. (a) Estimate the magnetic flux through this coil as the product of the magnetic field at the center of the coil and the area of the coil. Use this magnetic flux to estimate the self-inductance L of the coil. (b) Use the value of L that you estimated in part (a) to calculate the magnitude of the maximum emf induced in the coil.

30.44 •• CALC A coil has 400 turns and self-inductance 7.50 mH . The current in the coil varies with time according to $i = (680 \text{ mA}) \cos(\pi t / 0.0250 \text{ s})$. (a) What is the maximum emf induced in the coil? (b) What is the maximum average flux through each turn of the coil? (c) At $t = 0.0180 \text{ s}$, what is the magnitude of the induced emf?

30.45 •• Solar Magnetic Energy. Magnetic fields within a sunspot can be as strong as 0.4 T . (By comparison, the earth's magnetic field is about $1/10,000$ as strong.) Sunspots can be as large as $25,000 \text{ km}$ in radius. The material in a sunspot has a density of about $3 \times 10^{-4} \text{ kg/m}^3$. Assume μ for the sunspot material is μ_0 . If 100% of the magnetic-field energy stored in a sunspot could be used to eject the sunspot's material away from the sun's surface, at what speed would that material be ejected? Compare to the sun's escape speed, which is about $6 \times 10^5 \text{ m/s}$. (Hint: Calculate the kinetic energy the magnetic field could supply to 1 m^3 of sunspot material.)

30.46 •• CP CALC A Coaxial Cable. A small solid conductor with radius a is supported by insulating, nonmagnetic disks on the axis of a thin-walled tube with inner radius b . The inner and outer conductors carry equal currents i in opposite directions. (a) Use Ampere's law to find the magnetic field at any point in the volume between the conductors. (b) Write the expression for the flux $d\Phi_B$ through a narrow strip of length l parallel to the axis, of width dr , at a distance r from the axis of the cable and lying in a plane containing the axis. (c) Integrate your expression from part (b) over the volume between the two conductors to find the total flux produced by a current i in the central conductor. (d) Show that the inductance of a length l of the cable is

$$L = l \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$$

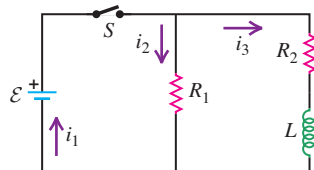
(e) Use Eq. (30.9) to calculate the energy stored in the magnetic field for a length l of the cable.

30.47 • (a) What would have to be the self-inductance of a solenoid for it to store 10.0 J of energy when a 2.00 A current runs through it? (b) If this solenoid's cross-sectional diameter is 4.00 cm, and if you could wrap its coils to a density of 10 coils/mm, how long would the solenoid be? (See Exercise 30.11.) Is this a realistic length for ordinary laboratory use?

30.48 •• CALC Consider the circuit in Fig. 30.11 with both switches open. At $t = 0$ switch S_1 is closed while switch S_2 is left open. (a) Use Eq. (30.14) to derive an equation for the rate P_R at which electrical energy is being consumed in the resistor. In terms of $\mathcal{E}$, R , and L , at what value of t is P_R a maximum? What is that maximum value? (b) Use Eqs. (30.14) and (30.15) to derive an equation for P_L , the rate at which energy is being stored in the inductor. (c) What is P_L at $t = 0$ and as $t \rightarrow \infty$? (d) In terms of $\mathcal{E}$, R , and L , at what value of t is P_L a maximum? What is that maximum value? (e) Obtain an expression for $P_{\mathcal{E}}$, the rate at which the battery is supplying electrical energy to the circuit. In terms of $\mathcal{E}$, R , and L , at what value of t is $P_{\mathcal{E}}$ a maximum? What is that maximum value?

30.49 • An Electromagnetic Car Alarm. Your latest invention is a car alarm that produces sound at a particularly annoying frequency of 3500 Hz. To do this, the car-alarm circuitry must produce an alternating electric current of the same frequency. That's why your design includes an inductor and a capacitor in series. The maximum voltage across the capacitor is to be 12.0 V. To produce a sufficiently loud sound, the capacitor must store 0.0160 J of energy. What values of capacitance and inductance should you choose for your car-alarm circuit?

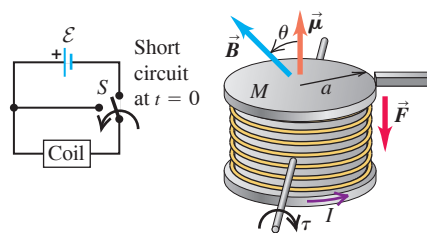
30.50 •• CALC An inductor with inductance $L = 0.300$ H and negligible resistance is connected to a battery, a switch S , and two resistors, $R_1 = 12.0 \Omega$ and $R_2 = 16.0 \Omega$ (Fig. P30.50). The battery has emf 96.0 V and negligible internal resistance. S is closed at $t = 0$. (a) What are the currents i_1 , i_2 , and i_3 just after S is closed? (b) What are i_1 , i_2 , and i_3 after S has been closed a long time? (c) What is the value of t for which i_3 has half of the final value that you calculated in part (b)? (d) When i_3 has half of its final value, what are i_1 and i_2 ?



30.51 ••• CP An alternating-current electric motor includes a thin, hollow, cylindrical spool (similar to a ring) with mass $M = 1.11$ kg and radius $a = 5.00$ cm wrapped $N = 500$ times with a copper wire with resistance $R = 5.00 \Omega$ and inductance $L = 77.0$ mH. Within the spool is a battery that supplies current $I = 1.00$ A, which makes the spool a magnetic dipole with dipole moment $\vec{\mu}$ parallel to the cylinder axis. A constant magnetic field with magnitude $B = 2.00$ T is supplied by an external stator magnet, while the spool turns freely on an axis perpendicular to its own axis. At a certain time, a bar is inserted, stopping the spool's motion (Fig. P30.51). At that instant the angle between the spool's

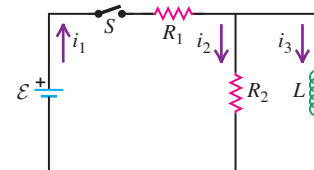
axis and the magnetic field is $\theta = 45^\circ$. (a) What is the magnitude of the downward force $\vec{F}$ applied by the bar onto the spool immediately after the bar is inserted? (b) Later, at time $t = 0$ with spool still at rest, the coil is short-circuited and a constant counter-torque $\tau = 0.500$ N·m is applied. The current subsides, and the magnetic torque decreases exponentially. At what time t does the force applied by the bar vanish? (Hint: Determine when the magnetic torque balances the counter-torque.) (c) After the spool rotates 180° it becomes stuck on the top side of the bar. The counter-torque is no longer applied, and the switch is returned to its original position. After a long time, the bar is removed. What is the angular acceleration of the spool immediately after the bar is removed? The moment of inertia of the spool for an axis along its diameter is $I = \frac{1}{2}Ma^2$.

Figure P30.51



30.52 ••• CALC An inductor with inductance $L = 0.200$ H and negligible resistance is connected to a battery, a switch S , and two resistors, $R_1 = 8.00 \Omega$ and $R_2 = 6.00 \Omega$ (Fig. P30.52). The battery has emf 48.0 V and negligible internal resistance. S is closed at $t = 0$. (a) What are the currents i_1 , i_2 , and i_3 just after S is closed? (b) What are i_1 , i_2 , and i_3 after S has been closed a long time? (c) Apply Kirchhoff's rules to the circuit and obtain a differential equation for $i_3(t)$. Integrate this equation to obtain an equation for i_3 as a function of the time t that has elapsed since S was closed. (d) Use the equation that you derived in part (c) to calculate the value of t for which i_3 has half of the final value that you calculated in part (b). (e) When i_3 has half of its final value, what are i_1 and i_2 ?

Figure P30.52



30.53 ••• CP CALC A cylindrical solenoid with radius 1.00 cm and length 10.0 cm consists of 300 windings of AWG 20 copper wire, which has a resistance per length of $0.0333 \Omega/\text{m}$. This solenoid is connected in series with a $10.0 \mu\text{F}$ capacitor, which is initially uncharged. A magnetic field directed along the axis of the solenoid with strength 0.100 T is switched on abruptly. (a) The solenoid may be considered an inductor and a resistor in series. Use Faraday's law to determine the average emf across the solenoid during the brief switch-on interval, and determine the net charge initially deposited on the capacitor. (See Exercise 29.4.) (b) At time $t = 0$ the capacitor is fully charged and there is no current. How much time does it take for the capacitor to fully discharge the first time? (c) What is the frequency with which the current oscillates? (d) How much energy is stored in the capacitor at $t = 0$? (e) How long does it take for the total energy stored in the circuit to drop to 10% of that value?

30.54 •• A 6.40 nF capacitor is charged to 24.0 V and then disconnected from the battery in the circuit and connected in series with a coil that has $L = 0.0660$ H and negligible resistance. After the circuit has been completed, there are current oscillations. (a) At an instant when the charge of the capacitor is $0.0800 \mu\text{C}$, how much energy is stored in the capacitor and in the inductor, and what is the current in the inductor? (b) At the instant when the charge on the capacitor is $0.0800 \mu\text{C}$, what are the voltages across the capacitor and across the inductor, and what is the rate at which current in the inductor is changing?

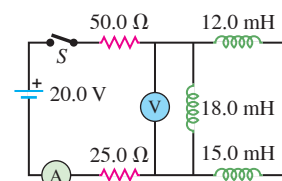
30.55 • An L - C circuit consists of a 60.0 mH inductor and a $250\text{ }\mu\text{F}$ capacitor. The initial charge on the capacitor is $6.00\text{ }\mu\text{C}$, and the initial current in the inductor is zero. (a) What is the maximum voltage across the capacitor? (b) What is the maximum current in the inductor? (c) What is the maximum energy stored in the inductor? (d) When the current in the inductor has half its maximum value, what is the charge on the capacitor and what is the energy stored in the inductor?

30.56 •• A charged capacitor with $C = 590\text{ }\mu\text{F}$ is connected in series to an inductor that has $L = 0.330\text{ H}$ and negligible resistance. At an instant when the current in the inductor is $i = 2.50\text{ A}$, the current is increasing at a rate of $di/dt = 73.0\text{ A/s}$. During the current oscillations, what is the maximum voltage across the capacitor?

30.57 ••• CP In the circuit shown in Figure P30.57

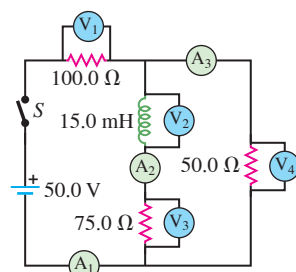
Fig. P30.57, the switch S has been open for a long time and is suddenly closed. Neither the battery nor the inductors have any appreciable resistance. What do the ammeter and the voltmeter read (a) just after S is closed; (b) after S has been closed a very long time; (c) 0.115 ms after S is closed?

Figure P30.57



30.58 •• CP In the circuit shown in Fig. P30.58, find the reading in each ammeter and voltmeter (a) just after switch S is closed and (b) after S has been closed a very long time.

Figure P30.58



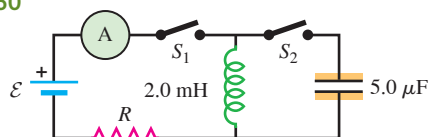
30.59 •• DATA To investigate the properties of a large industrial solenoid, you connect the solenoid and a resistor in series with a battery. Switches allow the battery to be replaced by a short circuit across the solenoid and resistor. Therefore Fig. 30.11 applies, with $R = R_{\text{ext}} + R_L$, where R_L is the resistance of the solenoid and R_{ext} is the resistance of the series resistor. With switch S_2 open, you close switch S_1 and keep it closed until the current i in the solenoid is constant (Fig. 30.11). Then you close S_2 and open S_1 simultaneously, using a rapid-response switching mechanism. With high-speed electronics you measure the time t_{half} that it takes for the current to decrease to half of its initial value. You repeat this measurement for several values of R_{ext} and obtain these results:

$R_{\text{ext}}\text{ (}\Omega\text{)}$	3.0	4.0	5.0	6.0	7.0	8.0	10.0	12.0
$t_{\text{half}}\text{ (s)}$	0.735	0.654	0.589	0.536	0.491	0.453	0.393	0.347

(a) Graph your data in the form of $1/t_{\text{half}}$ versus R_{ext} . Explain why the data points plotted this way fall close to a straight line. (b) Use your graph from part (a) to calculate the resistance R_L and inductance L of the solenoid. (c) If the current in the solenoid is 20.0 A , how much energy is stored there? At what rate is electrical energy being dissipated in the resistance of the solenoid?

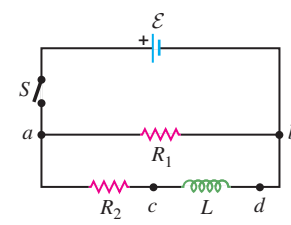
30.60 •• In the circuit shown in Fig. P30.60, switch S_1 has been closed for a long enough time so that the current reads a steady 3.50 A . Suddenly, switch S_2 is closed and S_1 is opened at the same instant. (a) What is the maximum charge that the capacitor will receive? (b) What is the current in the inductor at this time?

Figure P30.60



30.61 •• CP In the circuit shown in Fig. P30.61, $\mathcal{E} = 60.0\text{ V}$, $R_1 = 40.0\text{ }\Omega$, $R_2 = 25.0\text{ }\Omega$, and $L = 0.300\text{ H}$. Switch S is closed at $t = 0$. Just after the switch is closed, (a) what is the potential difference v_{ab} across the resistor R_1 ; (b) which point, a or b , is at a higher potential; (c) what is the potential difference v_{cd} across the inductor L ; (d) which point, c or d , is at a higher potential? The switch is left closed a long time and then opened. Just after the switch is opened, (e) what is the potential difference v_{ab} across the resistor R_1 ; (f) which point, a or b , is at a higher potential; (g) what is the potential difference v_{cd} across the inductor L ; (h) which point, c or d , is at a higher potential?

Figure P30.61

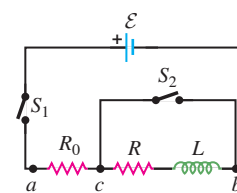


30.62 •• CP In the circuit shown in Fig. P30.61, $\mathcal{E} = 60.0\text{ V}$, $R_1 = 40.0\text{ }\Omega$, $R_2 = 25.0\text{ }\Omega$, and $L = 0.300\text{ H}$. (a) Switch S is closed. At some time t afterward, the current in the inductor is increasing at a rate of $di/dt = 50.0\text{ A/s}$. At this instant, what are the current i_1 through R_1 and the current i_2 through R_2 ? (Hint: Analyze two separate loops: one containing $\mathcal{E}$ and R_1 and the other containing $\mathcal{E}$, R_2 , and L .) (b) After the switch has been closed a long time, it is opened again. Just after it is opened, what is the current through R_1 ?

30.63 •• CALC Consider the circuit shown in Figure P30.63

Fig. P30.63. Let $\mathcal{E} = 36.0\text{ V}$, $R_0 = 50.0\text{ }\Omega$, $R = 150\text{ }\Omega$, and $L = 4.00\text{ H}$. (a) Switch S_1 is closed and switch S_2 is left open. Just after S_1 is closed, what are the current i_0 through R_0 and the potential differences v_{ac} and v_{cb} ? (b) After S_1 has been closed a long time (S_2 is still open) so that the current has reached its final, steady value, what are i_0 , v_{ac} , and v_{cb} ? (c) Find the expressions for i_0 , v_{ac} , and v_{cb} as functions of the time t since S_1 was closed. Your results should agree with part (a) when $t = 0$ and with part (b) when $t \rightarrow \infty$. Graph i_0 , v_{ac} , and v_{cb} versus time.

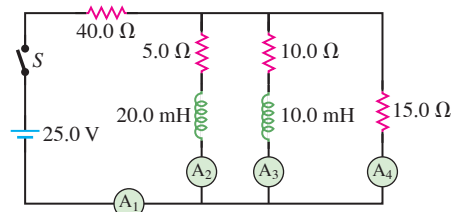
Figure P30.63



30.64 •• After the current in the circuit of Fig. P30.63 has reached its final, steady value with switch S_1 closed and S_2 open, switch S_2 is closed, thus short-circuiting the inductor. (Switch S_1 remains closed. See Problem 30.63 for numerical values of the circuit elements.) (a) Just after S_2 is closed, what are v_{ac} and v_{cb} , and what are the currents through R_0 , R , and S_2 ? (b) A long time after S_2 is closed, what are v_{ac} and v_{cb} , and what are the currents through R_0 , R , and S_2 ? (c) Derive expressions for the currents through R_0 , R , and S_2 as functions of the time t that has elapsed since S_2 was closed. Your results should agree with part (a) when $t = 0$ and with part (b) when $t \rightarrow \infty$. Graph these three currents versus time.

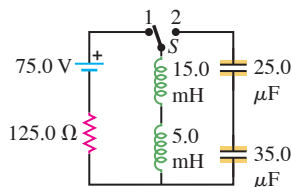
30.65 •• CP In the circuit shown in Fig. P30.65, switch S is closed at time $t = 0$. (a) Find the reading of each meter just after S is closed. (b) What does each meter read long after S is closed?

Figure P30.65



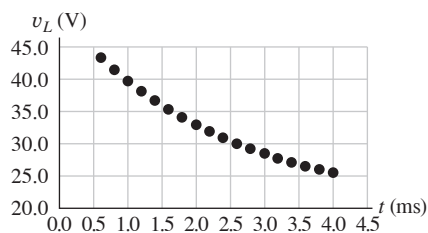
30.66 •• CP In the circuit shown in **Fig. P30.66**, neither the battery nor the inductors have any appreciable resistance, the capacitors are initially uncharged, and the switch S has been in position 1 for a very long time. (a) What is the current in the circuit? (b) The switch is now suddenly flipped to position 2. Find the maximum charge that each capacitor will receive, and how much time after the switch is flipped it will take them to acquire this charge.

Figure P30.66



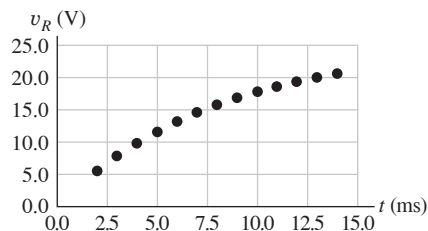
30.67 •• DATA During a summer internship as an electronics technician, you are asked to measure the self-inductance L of a solenoid. You connect the solenoid in series with a $10.0\ \Omega$ resistor, a battery that has negligible internal resistance, and a switch. Using an ideal voltmeter, you measure and digitally record the voltage v_L across the solenoid as a function of the time t that has elapsed since the switch is closed. Your measured values are shown in **Fig. P30.67**, where v_L is plotted versus t . In addition, you measure that $v_L = 50.0\ \text{V}$ just after the switch is closed and $v_L = 20.0\ \text{V}$ a long time after it is closed. (a) Apply the loop rule to the circuit and obtain an equation for v_L as a function of t . [Hint: Use an analysis similar to that used to derive Eq. (30.15).] (b) What is the emf $\mathcal{E}$ of the battery? (c) According to your measurements, what is the voltage amplitude across the $10.0\ \Omega$ resistor as $t \rightarrow \infty$? Use this result to calculate the current in the circuit as $t \rightarrow \infty$. (d) What is the resistance R_L of the solenoid? (e) Use the theoretical equation from part (a), **Fig. P30.67**, and the values of $\mathcal{E}$ and R_L from parts (b) and (d) to calculate L . (Hint: According to the equation, what is v_L when $t = \tau$, one time constant? Use **Fig. P30.67** to estimate the value of $t = \tau$.)

Figure P30.67



30.68 •• DATA You are studying a solenoid of unknown resistance and inductance. You connect it in series with a $50.0\ \Omega$ resistor, a $25.0\ \text{V}$ battery that has negligible internal resistance, and a switch. Using an ideal voltmeter, you measure and digitally record the voltage v_R across the resistor as a function of the time t that has elapsed after the switch is closed. Your measured values are shown in **Fig. P30.68**, where v_R is plotted versus t . In addition, you measure that $v_R = 0$ just after the switch is closed and $v_R = 25.0\ \text{V}$ a long time after it is closed. (a) What is the resistance R_L of the solenoid? (b) Apply the loop rule to the circuit and obtain an equation for v_R as a function of t . (c) According to the equation that you derived in part (b), what is v_R when $t = \tau$, one time constant? Use **Fig. P30.68** to estimate the value of $t = \tau$. What is the inductance of the solenoid? (d) How much energy is stored in the inductor a long time after the switch is closed?

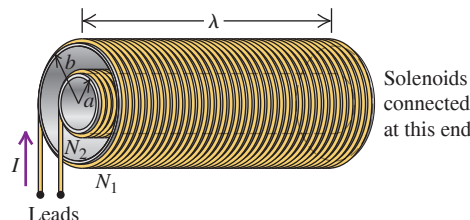
Figure P30.68



CHALLENGE PROBLEMS

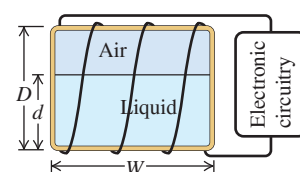
30.69 ••• A A long solenoid with N_1 windings and radius b surrounds a coaxial, narrower solenoid with N_2 windings and radius $a < b$, as shown in **Fig. P30.69**. At the far end, the outer solenoid is attached to the inner solenoid by a wire, so that current flows down the outer coil and then back through the inner coil as shown. (a) The two leads are attached to a supply circuit that includes a battery, supplying current I as indicated. What is the magnetic flux through each turn of the inner coil, taking rightward as the positive direction? (b) What is the magnetic flux through each turn of the outer coil? (c) What is the inductance as seen by the two leads? (d) What would be the inductance if the sense of the inner coil were reversed? (e) In the original configuration, what would be the inductance if $\lambda = 20.0\ \text{cm}$, $a = 1.00\ \text{cm}$, $b = 2.00\ \text{cm}$, $N_1 = 1200$, and $N_2 = 750$? (f) Using the values from part (e), what would be the inductance in the configuration of part (d)?

Figure P30.69



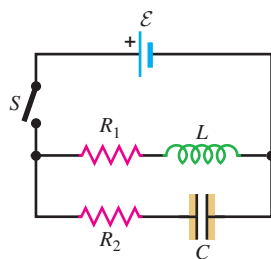
30.70 ••• CP A Volume Gauge. A tank containing a liquid has turns of wire wrapped around it, causing it to act like an inductor. The liquid content of the tank can be measured by using its inductance to determine the height of the liquid in the tank. The inductance of the tank changes from a value of L_0 corresponding to a relative permeability of 1 when the tank is empty to a value of L_f corresponding to a relative permeability of K_m (the relative permeability of the liquid) when the tank is full. The appropriate electronic circuitry can determine the inductance to five significant figures and thus the effective relative permeability of the combined air and liquid within the rectangular cavity of the tank. The four sides of the tank each have width W and height D (**Fig. P30.70**). The height of the liquid in the tank is d . You can ignore any fringing effects and assume that the relative permeability of the material of which the tank is made can be ignored. (a) Derive an expression for d as a function of L , the inductance corresponding to a certain fluid height, L_0 , L_f , and D . (b) What is the inductance (to five significant figures) for a tank $\frac{1}{4}$ full, $\frac{1}{2}$ full, $\frac{3}{4}$ full, and completely full if the tank contains liquid oxygen? Take $L_0 = 0.63000\ \text{H}$. The magnetic susceptibility of liquid oxygen is $\chi_m = 1.52 \times 10^{-3}$. (c) Repeat part (b) for mercury. The magnetic susceptibility of mercury is given in Table 28.1. (d) For which material is this volume gauge more practical?

Figure P30.70



30.71 ••• CP CALC Consider the circuit shown in **Fig. P30.71**. Switch S is closed at time $t = 0$, causing a current i_1 through the inductive branch and a current i_2 through the capacitive branch. The initial charge on the capacitor is zero, and the charge at time t is q_2 . (a) Derive expressions for i_1 , i_2 , and q_2 as functions of time. Express your answers in terms of $\mathcal{E}$, L , C , R_1 , R_2 , and t . For the remainder of the problem let the circuit elements have the following values: $\mathcal{E} = 48 \text{ V}$, $L = 8.0 \text{ H}$, $C = 20 \mu\text{F}$, $R_1 = 25 \Omega$, and $R_2 = 5000 \Omega$. (b) What is the initial current through the inductive branch? What is the initial current through the capacitive branch? (c) What are the currents through the inductive and capacitive branches a long time after the switch has been closed? How long is a “long time”? Explain. (d) At what time t_1 (accurate to two significant figures) will the currents i_1 and i_2 be equal? (*Hint*: You might consider using series expansions for the exponentials.) (e) For the conditions given in part (d), determine i_1 . (f) The total current through the battery is $i = i_1 + i_2$. At what time t_2 (accurate to two significant figures) will i equal one-half of its final value? (*Hint*: The numerical work is greatly simplified if one makes suitable approximations. A sketch of i_1 and i_2 versus t may help you decide what approximations are valid.)

Figure P30.71



of superconducting wire cooled by liquid helium at a temperature just under its boiling point (4.2 K). After a current is established in the wire, the power supply is disconnected and the magnet leads are shorted together through a piece of superconductor so that the current flows without resistance as long as the liquid helium keeps the magnet cold.

Under rare circumstances, a small segment of the magnet's wire may lose its superconducting properties and develop resistance. In this segment, electrical energy is converted to thermal energy, which can boil off some of the liquid helium. More of the wire then warms up and loses its superconducting properties, thus dissipating even more energy as heat. Because the latent heat of vaporization of liquid helium is quite low (20.9 kJ/kg), once the wire begins to warm up, all of the liquid helium may boil off rapidly. This event, called a *quench*, can damage the magnet. Also, a large volume of helium gas is generated as the liquid helium boils off, causing an asphyxiation hazard, and the resulting rapid pressure buildup can lead to an explosion. You can see how important it is to keep the wire resistance in an MRI magnet at zero and to have devices that detect a quench and shut down the current immediately.

30.72 How many turns does this typical MRI magnet have? (a) 1100; (b) 3000; (c) 4000; (d) 22,000.

30.73 If a small part of this magnet loses its superconducting properties and the resistance of the magnet wire suddenly rises from 0 to a constant 0.005Ω , how much time will it take for the current to decrease to half of its initial value? (a) 4.7 min; (b) 10 min; (c) 15 min; (d) 30 min.

30.74 If part of the magnet develops resistance and liquid helium boils away, rendering more and more of the magnet nonsuperconducting, how will this quench affect the time for the current to drop to half of its initial value? (a) The time will be shorter because the resistance will increase; (b) the time will be longer because the resistance will increase; (c) the time will be the same; (d) not enough information is given.

30.75 If all of the magnetic energy stored in this MRI magnet is converted to thermal energy, how much liquid helium will boil off? (a) 27 kg; (b) 38 kg; (c) 60 kg; (d) 110 kg.

MCAT-STYLE PASSAGE PROBLEMS

BIO Quenching an MRI Magnet. Magnets carrying very large currents are used to produce the uniform, large-magnitude magnetic fields that are required for *magnetic resonance imaging* (MRI). A typical MRI magnet may be a solenoid that is 2.0 m long and 1.0 m in diameter, has a self-inductance of 4.4 H, and carries a current of 750 A. A normal wire carrying that much current would dissipate a great deal of electrical power as heat, so most MRI magnets are made with coils

ANSWERS

Chapter Opening Question ?

(iii) As explained in Section 30.2, traffic light sensors work by measuring the change in inductance of a coil embedded under the road surface when a car (which contains ferromagnetic material) drives over it.

Key Example VARIATION Problems

VP30.4.1 (a) $6.50 \times 10^{-5} \text{ H}$ (b) $1.56 \times 10^{-3} \text{ T} \cdot \text{m}^2$

VP30.4.2 (a) $3.36 \times 10^{-4} \text{ V}$ (b) zero (c) $1.44 \times 10^{-4} \text{ V}$

VP30.4.3 (a) $1.96 \times 10^{-6} \text{ T} \cdot \text{m}^2$ (b) $4.18 \times 10^{-3} \text{ V}$

VP30.4.4 (a) 25.5 μH (b) 23.0 mV, opposite to the current

VP30.7.1 (a) 84.9 Ω (b) 0.212 A

VP30.7.2 (a) 0.190 A (b) +229 A/s (c) $6.79 \times 10^{-4} \text{ J}$

VP30.7.3 (a) 0.476 A (b) -229 A/s (c) $4.26 \times 10^{-3} \text{ J}$

VP30.7.4 (a) $(L/R)\ln 2 = 0.693(L/R)$ (b) $\mathcal{E}/2R$ (c) $\mathcal{E}/2L$

VP30.10.1 (a) $5.12 \times 10^{-3} \text{ F}$ (b) 14.0 A

VP30.10.2 (a) $2.20 \times 10^{-4} \text{ s}$ (b) 2.85 A

VP30.10.3 (a) 35.1 μF (b) 0.0125 H (c) $1.51 \times 10^3 \text{ rad/s}$

VP30.10.4 (a) 61.1 μF (b) 49.4 Ω

Bridging Problem

(a) $7.68 \times 10^{-8} \text{ J}$

(b) 1.60 mA

(c) 24.8 mV

(d) $1.92 \times 10^{-8} \text{ J}$, 21.5 mV



? These containers hold solutions of microscopic semiconductor particles of different sizes. The particles glow when exposed to ultraviolet light; the smallest particles glow blue and the largest particles glow red. This is because the energy levels of electrons (i) are spaced farther apart in smaller particles; (ii) are spaced farther apart in larger particles; (iii) have the same spacing in all particles but have higher energies in smaller particles; (iv) have the same spacing in all particles but have higher energies in larger particles; (v) depend on the wavelength of ultraviolet light used.

40 Quantum Mechanics I: Wave Functions

In Chapter 39 we found that particles can behave like waves. In fact, it turns out that we can use the wave picture to completely describe the behavior of a particle. This approach, called *quantum mechanics*, is the key to understanding the behavior of matter on the molecular, atomic, and nuclear scales. In this chapter we'll see how to find the *wave function* of a particle by solving the *Schrödinger equation*, which is as fundamental to quantum mechanics as Newton's laws are to mechanics or as Maxwell's equations are to electromagnetism.

We'll begin with a quantum-mechanical analysis of a *free particle* that moves along a straight line without being acted on by forces of any kind. We'll then consider particles that are acted on by forces and are trapped in *bound states*, just as electrons are bound within an atom. We'll see that solving the Schrödinger equation automatically gives the possible energy levels for the system.

Besides energies, solving the Schrödinger equation gives us the probabilities of finding a particle in various regions. One surprising result is that there is a nonzero probability that microscopic particles will pass through thin barriers, even though such a process is forbidden by Newtonian mechanics.

In this chapter we'll consider the Schrödinger equation for one-dimensional motion only. In Chapter 41 we'll see how to extend this equation to three-dimensional problems such as the hydrogen atom. The hydrogen-atom wave functions will in turn form the foundation for our analysis of more complex atoms, of the periodic table of the elements, of x-ray energy levels, and of other properties of atoms.

40.1 WAVE FUNCTIONS AND THE ONE-DIMENSIONAL SCHRÖDINGER EQUATION

We have now seen compelling evidence that on an atomic or subatomic scale, an object such as an electron cannot be described simply as a classical, Newtonian point particle. Instead, we must take into account its *wave* characteristics. In the Bohr model

LEARNING OUTCOMES

In this chapter, you'll learn...

- 40.1 The wave function that describes the behavior of a particle and the Schrödinger equation that this function must satisfy.
- 40.2 How to calculate the wave functions and energy levels for a particle confined to a box.
- 40.3 How to analyze the quantum-mechanical behavior of a particle in a potential well.
- 40.4 How quantum mechanics makes it possible for particles to go where Newtonian mechanics says they cannot.
- 40.5 How to use quantum mechanics to analyze a harmonic oscillator.
- 40.6 How measuring a quantum-mechanical system can change that system's state.

You'll need to review...

- 7.5 Potential wells.
- 14.2, 14.3 Harmonic oscillators.
- 15.3, 15.7, 15.8 Wave functions for waves on a string; standing waves.
- 32.3 Wave functions for electromagnetic waves.
- 38.1, 38.4 Work function; photon interpretation of interference and diffraction.
- 39.1, 39.3, 39.5, 39.6 De Broglie relationships; Bohr model for hydrogen; Planck radiation law; Heisenberg uncertainty principle.

Figure 40.1 These children are talking over a cup-and-string “telephone.” The displacement of the string is completely described by a wave function $y(x, t)$. In an analogous way, a particle is completely described by a quantum-mechanical wave function $\Psi(x, y, z, t)$.



CAUTION Particle waves vs. mechanical waves Unlike waves on a string or sound waves in air, the wave function for a particle is *not* a mechanical wave that needs a material medium in order to propagate. The wave function describes the particle, but we cannot define the function itself in terms of anything material. We can describe only how it is related to physically observable effects. ■

of the hydrogen atom (Section 39.3) we tried to have it both ways: We pictured the electron as a classical particle in a circular orbit around the nucleus, and used the de Broglie relationship between particle momentum and wavelength to explain why only orbits of certain radii are allowed. As we saw in Section 39.6, however, the Heisenberg uncertainty principle tells us that a hybrid description of this kind can't be wholly correct. In this section we'll explore how to describe the state of a particle by using *only* the language of waves. This new description, called **quantum mechanics**, replaces the classical scheme of describing the state of a particle by its coordinates and velocity components.

Our new quantum-mechanical scheme for describing a particle has a lot in common with the language of classical wave motion. In Section 15.3 of Chapter 15, we described transverse waves on a string by specifying the position of each point in the string at each instant of time by means of a *wave function* $y(x, t)$ that represents the displacement from equilibrium, at time t , of a point on the string at a distance x from the origin (Fig. 40.1). Once we know the wave function for a particular wave motion, we know everything there is to know about the motion. For example, we can find the velocity and acceleration of any point on the string at any time. We worked out specific forms for these functions for *sinusoidal* waves, in which each particle undergoes simple harmonic motion.

We followed a similar pattern for sound waves in Chapter 16. The wave function $p(x, t)$ for a wave traveling along the x -direction represented the pressure variation at any point x and any time t . In Section 32.3 we used *two* wave functions to describe the $\vec{E}$ and $\vec{B}$ fields in an electromagnetic wave.

Thus it's natural to use a wave function as the central element of our new language of quantum mechanics. The customary symbol for this wave function is the Greek letter psi, Ψ or ψ . In general, we'll use an uppercase Ψ to denote a function of all the space coordinates and time, and a lowercase ψ for a function of the space coordinates only—*not* of time. Just as the wave function $y(x, t)$ for mechanical waves on a string provides a complete description of the motion, so the wave function $\Psi(x, y, z, t)$ for a particle contains all the information that can be known about the particle.

Waves in One Dimension: Waves on a String

The wave function of a particle depends in general on all three dimensions of space. For simplicity, however, we'll begin our study of these functions by considering *one-dimensional* motion, in which a particle of mass m moves parallel to the x -axis and the wave function Ψ depends on the coordinate x and the time t only. (In the same way, we studied one-dimensional kinematics in Chapter 2 before going on to study two- and three-dimensional motion in Chapter 3.)

What does a one-dimensional quantum-mechanical wave look like, and what determines its properties? We can answer this question by first recalling the properties of a wave on a string. We saw in Section 15.3 that any wave function $y(x, t)$ that describes a wave on a string must satisfy the *wave equation*:

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2} \quad (\text{wave equation for waves on a string}) \quad (40.1)$$

In Eq. (40.1) v is the speed of the wave, which is the same no matter what the wavelength. As an example, consider the following wave function for a wave of wavelength λ and frequency f moving in the positive x -direction along a string:

$$y(x, t) = A \cos(kx - \omega t) + B \sin(kx - \omega t) \quad \begin{array}{l} \text{(sinusoidal wave} \\ \text{on a string)} \end{array} \quad (40.2)$$

Here $k = 2\pi/\lambda$ is the *wave number* and $\omega = 2\pi f$ is the *angular frequency*. (We used these same quantities for mechanical waves in Chapter 15 and electromagnetic waves in Chapter 32.) The quantities A and B are constants that determine the amplitude and phase of the wave. The expression in Eq. (40.2) is a valid wave function if and only if it satisfies the wave equation, Eq. (40.1). To check this, take the first and second

derivatives of $y(x, t)$ with respect to x and take the first and second derivatives with respect to t :

$$\frac{\partial y(x, t)}{\partial x} = -kA \sin(kx - \omega t) + kB \cos(kx - \omega t) \quad (40.3a)$$

$$\frac{\partial^2 y(x, t)}{\partial x^2} = -k^2 A \cos(kx - \omega t) - k^2 B \sin(kx - \omega t) \quad (40.3b)$$

$$\frac{\partial y(x, t)}{\partial t} = \omega A \sin(kx - \omega t) - \omega B \cos(kx - \omega t) \quad (40.3c)$$

$$\frac{\partial^2 y(x, t)}{\partial t^2} = -\omega^2 A \cos(kx - \omega t) - \omega^2 B \sin(kx - \omega t) \quad (40.3d)$$

If we substitute Eqs. (40.3b) and (40.3d) into the wave equation, Eq. (40.1), we get

$$\begin{aligned} & -k^2 A \cos(kx - \omega t) - k^2 B \sin(kx - \omega t) \\ &= \frac{1}{v^2} [-\omega^2 A \cos(kx - \omega t) - \omega^2 B \sin(kx - \omega t)] \end{aligned} \quad (40.4)$$

For Eq. (40.4) to be satisfied at all coordinates x and all times t , the coefficients of $\cos(kx - \omega t)$ must be the same on both sides of the equation, and likewise for the coefficients of $\sin(kx - \omega t)$. Both of these conditions will be satisfied if

$$k^2 = \frac{\omega^2}{v^2} \quad \text{or} \quad \omega = vk \quad (\text{waves on a string}) \quad (40.5)$$

Since $\omega = 2\pi f$ and $k = 2\pi/\lambda$, Eq. (40.5) is equivalent to

$$2\pi f = v \frac{2\pi}{\lambda} \quad \text{or} \quad v = \lambda f \quad (\text{waves on a string})$$

This equation is just the familiar relationship among wave speed, wavelength, and frequency for waves on a string. So our calculation shows that Eq. (40.2) is a valid wave function for waves on a string for any values of A and B , provided that ω and k are related by Eq. (40.5).

Waves in One Dimension: Particle Waves

What we need is a quantum-mechanical version of the wave equation, Eq. (40.1), valid for particle waves. We expect this equation to involve partial derivatives of the wave function $\Psi(x, t)$ with respect to x and with respect to t . However, this new equation *cannot* be the same as Eq. (40.1) for waves on a string because the relationship between ω and k is different. We can show this by considering a **free particle**, one that experiences no force at all as it moves along the x -axis. For such a particle the potential energy $U(x)$ has the same value for all x (recall from Chapter 7 that $F_x = -dU(x)/dx$, so zero force means the potential energy has zero derivative). For simplicity let $U = 0$ for all x . Then the energy of the free particle is equal to its kinetic energy, which we can express in terms of its momentum p :

$$E = \frac{1}{2}mv^2 = \frac{m^2v^2}{2m} = \frac{(mv)^2}{2m} = \frac{p^2}{2m} \quad (\text{energy of a free particle}) \quad (40.6)$$

The de Broglie relationships (Section 39.1) tell us that the energy E is proportional to the angular frequency ω and the momentum p is proportional to the wave number k :

$$E = hf = \frac{h}{2\pi} 2\pi f = \hbar\omega \quad (40.7a)$$

$$p = \frac{h}{\lambda} = \frac{h}{2\pi} \frac{2\pi}{\lambda} = \hbar k \quad (40.7b)$$

Remember that $\hbar = h/2\pi$. If we substitute Eqs. (40.7) into Eq. (40.6), we find that the relationship between ω and k for a free particle is

$$\hbar\omega = \frac{\hbar^2 k^2}{2m} \quad (\text{free particle}) \quad (40.8)$$

Equation (40.8) is *very* different from the corresponding relationship for waves on a string, Eq. (40.5): The angular frequency ω for particle waves is proportional to the *square* of the wave number, while for waves on a string ω is directly proportional to k . Our task is therefore to construct a quantum-mechanical version of the wave equation whose free-particle solutions satisfy Eq. (40.8).

We'll attack this problem by assuming a sinusoidal wave function $\Psi(x, t)$ of the same form as Eq. (40.2) for a sinusoidal wave on a string. For a wave on a string, Eq. (40.2) represents a wave of wavelength $\lambda = 2\pi/k$ and frequency $f = \omega/2\pi$ propagating in the positive x -direction. By analogy, our sinusoidal wave function $\Psi(x, t)$ represents a free particle of mass m , momentum $p = \hbar k$, and energy $E = \hbar\omega$ moving in the positive x -direction:

$$\Psi(x, t) = A \cos(kx - \omega t) + B \sin(kx - \omega t) \quad \begin{array}{l} \text{(sinusoidal wave} \\ \text{function representing} \\ \text{a free particle)} \end{array} \quad (40.9)$$

The wave number k and angular frequency ω in Eq. (40.9) must satisfy Eq. (40.8). If you look at Eq. (40.3b), you'll see that taking the second derivative of $\Psi(x, t)$ in Eq. (40.9) with respect to x gives us $\Psi(x, t)$ multiplied by $-k^2$. Hence if we multiply $\partial^2 \Psi(x, t)/\partial x^2$ by $-\hbar^2/2m$, we get

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} &= -\frac{\hbar^2}{2m} [-k^2 A \cos(kx - \omega t) - k^2 B \sin(kx - \omega t)] \\ &= \frac{\hbar^2 k^2}{2m} [A \cos(kx - \omega t) + B \sin(kx - \omega t)] \\ &= \frac{\hbar^2 k^2}{2m} \Psi(x, t) \end{aligned} \quad (40.10)$$

Equation (40.10) suggests that $(-\hbar^2/2m)\partial^2 \Psi(x, t)/\partial x^2$ should be one side of our quantum-mechanical wave equation, with the other side equal to $\hbar\omega\Psi(x, t)$ in order to satisfy Eq. (40.8). If you look at Eq. (40.3c), you'll see that taking the *first* time derivative of $\Psi(x, t)$ in Eq. (40.9) brings out a factor of ω . So we'll make the educated guess that the right-hand side of our quantum-mechanical wave equation involves $\hbar = h/2\pi$ times $\partial\Psi(x, t)/\partial t$. So our tentative equation is

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} = C\hbar \frac{\partial \Psi(x, t)}{\partial t} \quad (40.11)$$

At this point we include a constant C as a “fudge factor” to make sure that everything turns out right. Now let's substitute the wave function from Eq. (40.9) into Eq. (40.11). From Eq. (40.10) and Eq. (40.3c), we get

$$\begin{aligned} \frac{\hbar^2 k^2}{2m} [A \cos(kx - \omega t) + B \sin(kx - \omega t)] \\ = C\hbar\omega [A \sin(kx - \omega t) - B \cos(kx - \omega t)] \end{aligned} \quad (40.12)$$

From Eq. (40.8), $\hbar\omega = \hbar^2 k^2/2m$, so we can cancel these factors on the two sides of Eq. (40.12). What remains is

$$\begin{aligned} A \cos(kx - \omega t) + B \sin(kx - \omega t) \\ = CA \sin(kx - \omega t) - CB \cos(kx - \omega t) \end{aligned} \quad (40.13)$$

As in our discussion above of the wave equation for waves on a string, in order for Eq. (40.13) to be satisfied for all values of x and all values of t , the coefficients of $\cos(kx - \omega t)$ must be the same on both sides of the equation, and likewise for the coefficients of $\sin(kx - \omega t)$. Hence we have the following relationships among the coefficients A , B , and C in Eqs. (40.9) and (40.11):

$$A = -CB \quad (40.14a)$$

$$B = CA \quad (40.14b)$$

If we use Eq. (40.14b) to eliminate B from Eq. (40.14a), we get $A = -C^2A$, which means that $C^2 = -1$. Thus C is equal to the *imaginary* number $i = \sqrt{-1}$, and Eq. (40.11) becomes

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} = i\hbar \frac{\partial \Psi(x, t)}{\partial t} \quad \text{(one-dimensional Schrödinger equation for a free particle)} \quad (40.15)$$

Equation (40.15) is the one-dimensional **Schrödinger equation** for a free particle, developed in 1926 by the Austrian physicist Erwin Schrödinger (**Fig. 40.2**). The presence of the imaginary number i in Eq. (40.15) means that the solutions to the Schrödinger equation are complex quantities, with a real part and an imaginary part. (The imaginary part of $\Psi(x, t)$ is a real function multiplied by the imaginary number $i = \sqrt{-1}$.) An example is our free-particle wave function from Eq. (40.9). Since we found that $C = i$ in Eqs. (40.14), it follows from Eq. (40.14b) that $B = iA$. Then Eq. (40.9) becomes

$$\Psi(x, t) = A[\cos(kx - \omega t) + i\sin(kx - \omega t)] \quad \begin{array}{l} \text{(sinusoidal wave} \\ \text{function representing} \\ \text{a free particle)} \end{array} \quad (40.16)$$

The real part of $\Psi(x, t)$ is $\text{Re}\Psi(x, t) = A\cos(kx - \omega t)$ and the imaginary part is $\text{Im}\Psi(x, t) = A\sin(kx - \omega t)$. **Figure 40.3** graphs the real and imaginary parts of $\Psi(x, t)$ at $t = 0$, so $\Psi(x, 0) = A\cos kx + iA\sin kx$.

We can rewrite Eq. (40.16) with *Euler's formula*, which states that for any angle θ ,

$$\begin{aligned} e^{i\theta} &= \cos\theta + i\sin\theta \\ e^{-i\theta} &= \cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta \end{aligned} \quad (40.17)$$

Thus our sinusoidal free-particle wave function becomes

$$\Psi(x, t) = Ae^{i(kx - \omega t)} = Ae^{ikx}e^{-i\omega t} \quad \begin{array}{l} \text{(sinusoidal wave function} \\ \text{representing a free particle)} \end{array} \quad (40.18)$$

If k is positive in Eq. (40.16), the wave function represents a free particle moving in the positive x -direction with momentum $p = \hbar k$ and energy $E = \hbar\omega = \hbar^2 k^2 / 2m$. If k is negative, the momentum and hence the motion are in the negative x -direction. (With a negative value of k , the wavelength is $\lambda = 2\pi/|k|$.)

Interpreting the Wave Function

The complex nature of the wave function for a free particle makes this function challenging to interpret. (We certainly haven't needed imaginary numbers before this point to describe real physical phenomena.) Here's how to think about this function: $\Psi(x, t)$ describes the *distribution* of a particle in space, just as the wave functions for an electromagnetic wave describe the distribution of the electric and magnetic fields. When we worked out interference and diffraction patterns in Chapters 35 and 36, we found that the intensity I of the radiation at any point in a pattern is proportional to the square of the electric-field magnitude—that is, to E^2 . In the photon interpretation of interference and diffraction (see Section 38.4), the intensity at each point is proportional to the number of photons striking around that point or, alternatively, to the *probability* that any individual

Figure 40.2 Erwin Schrödinger (1887–1961) developed the equation that bears his name in 1926, an accomplishment for which he shared (with the British physicist P. A. M. Dirac) the 1933 Nobel Prize in physics. His grave marker is adorned with a version of Eq. (40.15).



Figure 40.3 The spatial wave function $\Psi(x, t) = Ae^{i(kx - \omega t)}$ for a free particle of definite momentum $p = \hbar k$ is a complex function: It has both a real part and an imaginary part. These are graphed here as functions of x for $t = 0$.

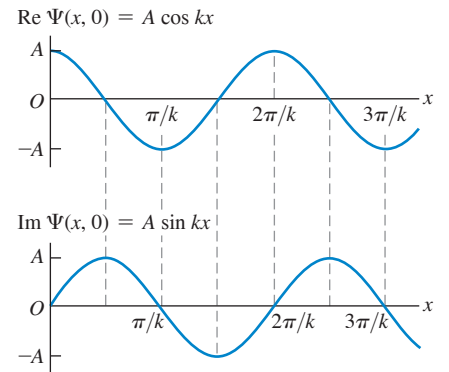


Figure 40.4 In 1926, the German physicist Max Born (1882–1970) devised the interpretation that $|\Psi|^2$ is the probability distribution function for a particle that is described by the wave function Ψ . He also coined the term “quantum mechanics” (in the original German, *Quantenmechanik*). For his contributions, Born shared (with Walther Bothe) the 1954 Nobel Prize in physics.



photon will strike around the point. Thus the square of the electric-field magnitude at each point is proportional to the probability of finding a photon around that point.

In exactly the same way, the square of the wave function of a particle at each point tells us about the probability of finding the particle around that point. More precisely, we should say the square of the *absolute value* of the wave function, $|\Psi|^2$. This is necessary because, as we have seen, the wave function is a complex quantity with real and imaginary parts.

For a particle that can move only along the x -direction, the quantity $|\Psi(x, t)|^2 dx$ is the probability that the particle will be found at time t at a coordinate in the range from x to $x + dx$. The particle is most likely to be found in regions where $|\Psi|^2$ is large, and so on. This interpretation, first made by the German physicist Max Born (Fig. 40.4), requires that the wave function Ψ be *normalized*. That is, the integral of $|\Psi(x, t)|^2 dx$ over all possible values of x must equal exactly 1. In other words, the probability is exactly 1, or 100%, that the particle is *somewhere*.

CAUTION Interpreting $|\Psi|^2$ Note that $|\Psi(x, t)|^2$ itself is *not* a probability. Rather, $|\Psi(x, t)|^2 dx$ is the probability of finding the particle between position x and position $x + dx$ at time t . If the length dx is made smaller, it becomes less likely that the particle will be found within that length, so the probability decreases. A better name for $|\Psi(x, t)|^2$ is the **probability distribution function**, since it describes how the probability of finding the particle at different locations is distributed over space. Another common name for $|\Psi(x, t)|^2$ is the *probability density*.

We can use the probability interpretation of $|\Psi|^2$ to get a better understanding of Eq. (40.18), the wave function for a free particle. This function describes a particle that has a definite momentum $p = \hbar k$ in the x -direction and *no* uncertainty in momentum: $\Delta p_x = 0$. The Heisenberg uncertainty principle for position and momentum, Eqs. (39.29), says that $\Delta x \Delta p_x \geq \hbar/2$. If Δp_x is zero, then Δx must be infinite, and we have no idea where along the x -axis the particle can be found. (We saw a similar result for photons in Section 38.4.) We can show this by calculating the probability distribution function $|\Psi(x, t)|^2$: the product of Ψ and its *complex conjugate* Ψ^* . To find the complex conjugate of a complex number, we replace all i with $-i$. For example, the complex conjugate of $c = a + ib$, where a and b are real, is $c^* = a - ib$, so $|c|^2 = c^* c = (a + ib)(a - ib) = a^2 + b^2$ (recall that $i^2 = -1$). The complex conjugate of Eq. (40.18) is

$$\Psi^*(x, t) = A^* e^{-i(kx - \omega t)} = A^* e^{-ikx} e^{i\omega t}$$

(We have to allow for the possibility that the coefficient A is itself a complex number.) Hence the probability distribution function is

$$\begin{aligned} |\Psi(x, t)|^2 &= \Psi^*(x, t) \Psi(x, t) = (A^* e^{-ikx} e^{i\omega t})(A e^{ikx} e^{-i\omega t}) \\ &= A^* A e^0 = |A|^2 \end{aligned}$$

The probability distribution function doesn't depend on position, which says that we are equally likely to find the particle *anywhere* along the x -axis! Mathematically, this is because the sinusoidal wave function $\Psi(x, t) = A e^{i(kx - \omega t)} = A [\cos(kx - \omega t) + i \sin(kx - \omega t)]$ extends all the way from $x = -\infty$ to $x = +\infty$ with the same amplitude A . This also means that the wave function can't be normalized: The integral of $|\Psi(x, t)|^2$ over all space is infinite for any value of A .

Note also that the wave function in Eq. (40.18) describes a particle with a definite energy $E = \hbar\omega$, so there is zero uncertainty in energy: $\Delta E = 0$. The Heisenberg uncertainty principle for energy and time interval, $\Delta t \Delta E \geq \hbar/2$ [Eq. (39.30)], tells us that the time uncertainty Δt for this particle is infinite. In other words, we can have no idea *when* the particle will pass a given point on the x -axis. That also agrees with our result $|\Psi(x, t)|^2 = |A|^2$; the probability distribution function has the same value at all times.

Since we always have some idea of where a particle is, the wave function given in Eq. (40.18) isn't a realistic description. In our study of light in Section 38.4, we saw that we can make a wave function that's more *localized* in space by superposing two or more sinusoidal functions. (This would be a good time to review that section.) As an illustration, let's calculate $|\Psi(x, t)|^2$ for a wave function of this kind.

EXAMPLE 40.1 A localized free-particle wave function

The wave function $\Psi(x, t) = Ae^{i(k_1x - \omega_1t)} + Ae^{i(k_2x - \omega_2t)}$ is a superposition of *two* free-particle wave functions of the form given by Eq. (40.18). Both k_1 and k_2 are positive. (a) Show that this wave function satisfies the Schrödinger equation for a free particle of mass m . (b) Find the probability distribution function for $\Psi(x, t)$.

IDENTIFY and SET UP Both wave functions $Ae^{i(k_1x - \omega_1t)}$ and $Ae^{i(k_2x - \omega_2t)}$ represent a particle moving in the positive x -direction, but with different momenta and kinetic energies: $p_1 = \hbar k_1$ and $E_1 = \hbar\omega_1 = \hbar^2 k_1^2/2m$ for the first function, $p_2 = \hbar k_2$ and $E_2 = \hbar\omega_2 = \hbar^2 k_2^2/2m$ for the second function. To test whether a superposition of these is also a valid wave function for a free particle, we'll see whether our function $\Psi(x, t)$ satisfies the free-particle Schrödinger equation, Eq. (40.15). We'll use the derivatives of the exponential function: $(d/du)e^{au} = ae^{au}$ and $(d^2/du^2)e^{au} = a^2e^{au}$. The probability distribution function $|\Psi(x, t)|^2$ is the product of $\Psi(x, t)$ and its complex conjugate.

EXECUTE (a) If we substitute $\Psi(x, t)$ into Eq. (40.15), the left-hand side of the equation is

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} &= -\frac{\hbar^2}{2m} \frac{\partial^2 (Ae^{i(k_1x - \omega_1t)} + Ae^{i(k_2x - \omega_2t)})}{\partial x^2} \\ &= -\frac{\hbar^2}{2m} [(ik_1)^2 Ae^{i(k_1x - \omega_1t)} + (ik_2)^2 Ae^{i(k_2x - \omega_2t)}] \\ &= \frac{\hbar^2 k_1^2}{2m} Ae^{i(k_1x - \omega_1t)} + \frac{\hbar^2 k_2^2}{2m} Ae^{i(k_2x - \omega_2t)} \end{aligned}$$

The right-hand side is

$$\begin{aligned} i\hbar \frac{\partial \Psi(x, t)}{\partial t} &= i\hbar \frac{\partial (Ae^{i(k_1x - \omega_1t)} + Ae^{i(k_2x - \omega_2t)})}{\partial t} \\ &= i\hbar [(-i\omega_1)Ae^{i(k_1x - \omega_1t)} + (-i\omega_2)Ae^{i(k_2x - \omega_2t)}] \\ &= \hbar\omega_1 Ae^{i(k_1x - \omega_1t)} + \hbar\omega_2 Ae^{i(k_2x - \omega_2t)} \end{aligned}$$

The two sides *are* equal, provided that $\hbar\omega_1 = \hbar^2 k_1^2/2m$ and $\hbar\omega_2 = \hbar^2 k_2^2/2m$. These are just the relationships that we noted above. So we conclude that $\Psi(x, t) = Ae^{i(k_1x - \omega_1t)} + Ae^{i(k_2x - \omega_2t)}$ is a valid free-particle wave function. In general, if we take any two wave functions that are solutions of the Schrödinger equation and then make a superposition of these to create a third wave function $\Psi(x, t)$, then $\Psi(x, t)$ is also a solution of the Schrödinger equation.

(b) The complex conjugate of $\Psi(x, t)$ is

$$\Psi^*(x, t) = A^* e^{-i(k_1x - \omega_1t)} + A^* e^{-i(k_2x - \omega_2t)}$$

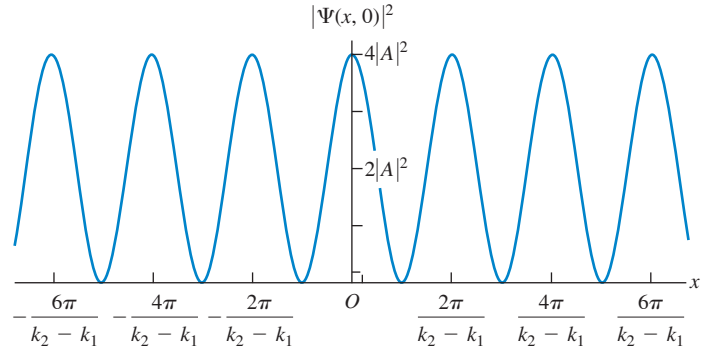
Hence

$$\begin{aligned} |\Psi(x, t)|^2 &= \Psi^*(x, t)\Psi(x, t) \\ &= (A^* e^{-i(k_1x - \omega_1t)} + A^* e^{-i(k_2x - \omega_2t)})(Ae^{i(k_1x - \omega_1t)} + Ae^{i(k_2x - \omega_2t)}) \\ &= A^* A \left[e^{-i(k_1x - \omega_1t)} e^{i(k_1x - \omega_1t)} + e^{-i(k_2x - \omega_2t)} e^{i(k_2x - \omega_2t)} \right. \\ &\quad \left. + e^{-i(k_1x - \omega_1t)} e^{i(k_2x - \omega_2t)} + e^{-i(k_2x - \omega_2t)} e^{i(k_1x - \omega_1t)} \right] \\ &= |A|^2 [e^0 + e^0 + e^{i[(k_2 - k_1)x - (\omega_2 - \omega_1)t]} + e^{-i[(k_2 - k_1)x - (\omega_2 - \omega_1)t]}] \end{aligned}$$

To simplify this expression, recall that $e^0 = 1$. From Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$, so $e^{i\theta} + e^{-i\theta} = 2\cos\theta$. Hence

$$\begin{aligned} |\Psi(x, t)|^2 &= |A|^2 \{ 2 + 2\cos[(k_2 - k_1)x - (\omega_2 - \omega_1)t] \} \\ &= 2|A|^2 \{ 1 + \cos[(k_2 - k_1)x - (\omega_2 - \omega_1)t] \} \end{aligned}$$

Figure 40.5 The probability distribution function at $t = 0$ for $\Psi(x, t) = Ae^{i(k_1x - \omega_1t)} + Ae^{i(k_2x - \omega_2t)}$.



EVALUATE Figure 40.5 is a graph of the probability distribution function $|\Psi(x, t)|^2$ at $t = 0$. The value of $|\Psi(x, t)|^2$ varies between 0 and $4|A|^2$; probabilities can never be negative! The particle has become *some-what* localized: The particle is most likely to be found near a point where $|\Psi(x, t)|^2$ is maximum (where the functions $Ae^{i(k_1x - \omega_1t)}$ and $Ae^{i(k_2x - \omega_2t)}$ interfere constructively) and is very unlikely to be found near a point where $|\Psi(x, t)|^2 = 0$ (where $Ae^{i(k_1x - \omega_1t)}$ and $Ae^{i(k_2x - \omega_2t)}$ interfere destructively).

Note also that the probability distribution function is not stationary: It moves in the positive x -direction like the particle that it represents. To see this, recall from Section 15.3 that a sinusoidal wave given by $y(x, t) = A\cos(kx - \omega t)$ moves in the positive x -direction with speed $v = \omega/k$; since $|\Psi(x, t)|^2$ includes a term $\cos[(k_2 - k_1)x - (\omega_2 - \omega_1)t]$, the probability distribution moves at a speed $v_{av} = (\omega_2 - \omega_1)/(k_2 - k_1)$. The subscript “av” reminds us that v_{av} represents the *average* value of the particle's speed.

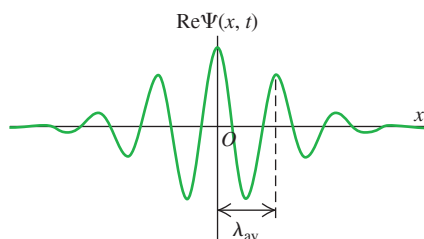
The price we pay for localizing the particle somewhat is that, unlike a particle represented by Eq. (40.18), it no longer has either a definite momentum or a definite energy. That's consistent with the Heisenberg uncertainty principles: If we decrease the uncertainties about where a particle is and when it passes a certain point, the uncertainties in its momentum and energy must increase.

The average momentum of the particle is $p_{av} = (\hbar k_2 + \hbar k_1)/2$, which is the average of the momenta associated with the free-particle wave functions we added to create $\Psi(x, t)$. This corresponds to the particle having an average speed $v_{av} = p_{av}/m = (\hbar k_2 + \hbar k_1)/2m$. Can you show that this is equal to the expression $v_{av} = (\omega_2 - \omega_1)/(k_2 - k_1)$ that we found above?

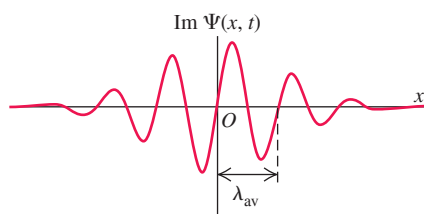
KEYCONCEPT A free particle (one on which no forces act) that moves in the x -direction only is described by a wave function $\Psi(x, t)$ that obeys the free-particle Schrödinger equation [Eq. (40.15)]. A wave function with definite momentum and energy is completely delocalized; to write a wave function that is localized in space, you must superpose wave functions with different values of momentum and energy.

Figure 40.6 Superposing a large number of sinusoidal waves with different wave numbers and appropriate amplitudes can produce a wave pulse that has a wavelength $\lambda_{\text{av}} = 2\pi/k_{\text{av}}$ and is localized within a region of space of length Δx . This localized pulse has aspects of both particle and wave.

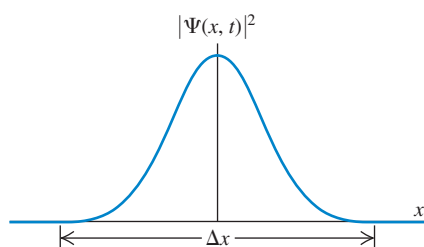
(a) Real part of the wave function at time t



(b) Imaginary part of the wave function at time t



(c) Probability distribution function at time t



Wave Packets

The wave function that we examined in Example 40.1 is not very well localized: The probability distribution function still extends from $x = -\infty$ to $x = +\infty$. Hence this wave function can't be normalized. To make a wave function that's more highly localized, imagine superposing two additional sinusoidal waves with different wave numbers and amplitudes so as to reinforce alternate maxima of $|\Psi(x, t)|^2$ in Fig. 40.5 and cancel out the in-between ones. If we continue this process and superpose a very large number of waves with different wave numbers, we can construct a wave with only *one* maximum of $|\Psi(x, t)|^2$ (Fig. 40.6). Then we have something that begins to look like both a particle and a wave. It is a particle in the sense that it is localized in space; if we look from a distance, it may look like a point. But it also has a periodic structure that is characteristic of a wave.

A localized wave pulse like that shown in Fig. 40.6 is called a **wave packet**. We can represent a wave packet by an expression such as

$$\Psi(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega t)} dk \quad (40.19)$$

This integral represents a superposition of a very large number of waves, each with a different wave number k and angular frequency $\omega = \hbar k^2/2m$, and each with an amplitude $A(k)$ that depends on k .

There is an important relationship between the two functions $\Psi(x, t)$ and $A(k)$, which we show qualitatively in Fig. 40.7. If the function $A(k)$ is sharply peaked, as in Fig. 40.7a, we are superposing only a narrow range of wave numbers. The resulting wave pulse is then relatively broad (Fig. 40.7b). But if we use a wider range of wave numbers, so that the function $A(k)$ is broader (Fig. 40.7c), then the wave pulse is more narrowly localized (Fig. 40.7d). This is simply the uncertainty principle in action. A narrow range of k means a narrow range of $p_x = \hbar k$ and thus a small Δp_x ; the result is a relatively large Δx . A broad range of k corresponds to a large Δp_x , and the resulting Δx is smaller. You can see that the uncertainty principle for position and momentum, $\Delta x \Delta p_x \geq \hbar/2$, is really just a consequence of the properties of integrals like Eq. (40.19).

CAUTION Matter waves versus light waves in vacuum We can regard both a wave packet that represents a particle and a short pulse of light from a laser as superpositions of waves of different wave numbers and angular frequencies. An important difference is that the speed of light in vacuum is the same for all wavelengths λ and hence all wave numbers $k = 2\pi/\lambda$, but the speed of a matter wave is *different* for different wavelengths. You can see this from the formula for the speed of the wave crests in a periodic wave, $v = \lambda f = \omega/k$. For a matter wave, $\omega = \hbar k^2/2m$, so $v = \hbar k/2m = h/2m\lambda$. Hence matter waves with longer wavelengths and smaller wave numbers travel more slowly than those with short wavelengths and large wave numbers. (This shouldn't be too surprising. The de Broglie relationships that we learned in Section 39.1 tell us that shorter wavelength corresponds to greater momentum and hence a greater speed.) Since the individual sinusoidal waves that make up a wave packet travel at different speeds, the shape of the packet changes as it moves. That's why we've specified the time for which the wave packets in Figs. 40.6 and 40.7 are drawn; at later times, the packets become more spread out. By contrast, a pulse of light waves in vacuum retains the same shape at all times because all of its constituent sinusoidal waves travel together at the same speed. ■

The One-Dimensional Schrödinger Equation with Potential Energy

The one-dimensional Schrödinger equation that we presented in Eq. (40.15) is valid only for free particles, for which the potential-energy function is zero: $U(x) = 0$. But for an electron within an atom, a proton within an atomic nucleus, and many other real situations, the potential energy plays an important role. To study the behavior of matter waves in these situations, we need a version of the Schrödinger equation that describes a particle moving in the presence of a nonzero potential-energy function $U(x)$. This equation is

General one-dimensional Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + U(x) \Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t} \quad (40.20)$$

Planck's constant divided by 2π (pointing to $\hbar$)
Particle's mass (pointing to m)
Particle's wave function (pointing to $\Psi(x, t)$)
Potential-energy function (pointing to $U(x)$)

Note that if $U(x) = 0$, Eq. (40.20) reduces to the free-particle Schrödinger equation given in Eq. (40.15).

Here's the motivation behind Eq. (40.20). If $\Psi(x, t)$ is a sinusoidal wave function for a free particle, $\Psi(x, t) = Ae^{i(kx - \omega t)} = Ae^{ikx}e^{-i\omega t}$, the derivative terms in Eq. (40.20) become

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} &= -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} (Ae^{ikx}e^{-i\omega t}) = -\frac{\hbar^2}{2m} (ik)^2 (Ae^{ikx}e^{-i\omega t}) \\ &= \frac{\hbar^2 k^2}{2m} \Psi(x, t) \\ i\hbar \frac{\partial \Psi(x, t)}{\partial t} &= i\hbar \frac{\partial}{\partial t} (Ae^{ikx}e^{-i\omega t}) = i\hbar(-i\omega)(Ae^{ikx}e^{-i\omega t}) = \hbar\omega \Psi(x, t) \end{aligned}$$

In these expressions $(\hbar^2 k^2/2m)\Psi(x, t)$ is just the kinetic energy $K = p^2/2m = \hbar^2 k^2/2m$ multiplied by the wave function, and $\hbar\omega \Psi(x, t)$ is the total energy $E = \hbar\omega$ multiplied by the wave function. So for a wave function of this kind, Eq. (40.20) says that kinetic energy times $\Psi(x, t)$ plus potential energy times $\Psi(x, t)$ equals total energy times $\Psi(x, t)$. That's equivalent to the statement in classical physics that the sum of kinetic energy and potential energy equals total mechanical energy: $K + U = E$.

The observations we've just made certainly aren't a *proof* that Eq. (40.20) is correct. The real reason we know this equation *is* correct is that it works: Predictions made with this equation agree with experimental results. Later in this chapter we'll apply Eq. (40.20) to several physical situations, each with a different form of the function $U(x)$.

Stationary States

We saw in our discussion of wave packets that any free-particle wave function can be built up as a superposition of sinusoidal wave functions of the form $\Psi(x, t) = Ae^{ikx}e^{-i\omega t}$. Each such sinusoidal wave function corresponds to a state of definite energy $E = \hbar\omega = \hbar^2 k^2/2m$ and definite angular frequency $\omega = E/\hbar$, so we can rewrite these functions as $\Psi(x, t) = Ae^{ikx}e^{-iEt/\hbar}$. If the potential-energy function $U(x)$ is nonzero, these sinusoidal wave functions do not satisfy the Schrödinger equation, Eq. (40.20), and so these functions cannot be the basic “building blocks” of more complicated wave functions. However, we can still write the wave function for a state of definite energy E in the form

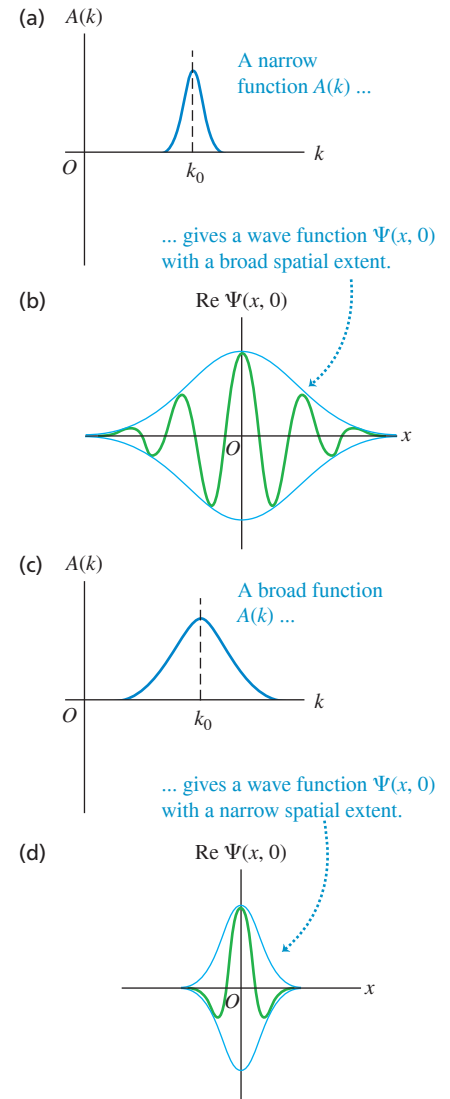
Time-dependent wave function for a state of definite energy

$$\Psi(x, t) = \psi(x) e^{-iEt/\hbar} \quad (40.21)$$

Time-independent wave function (pointing to $\psi(x)$)
Energy of state (pointing to E)
Planck's constant divided by 2π (pointing to $\hbar$)

That is, the wave function $\Psi(x, t)$ for a state of definite energy is the product of a *time-independent* wave function $\psi(x)$ and a factor $e^{-iEt/\hbar}$. (For the free-particle sinusoidal wave function, $\psi(x) = Ae^{ikx}$.) States of definite energy are of tremendous importance in quantum mechanics. For example, for each energy level in a hydrogen atom (Section 39.3) there is a specific wave function. It is possible for an atom to be in a state that does not have a definite energy. The wave function for any such state can be written as a combination of definite-energy wave functions, in precisely the same way that a free-particle wave packet can be written as a superposition of sinusoidal wave functions of definite energy as in Eq. (40.19).

Figure 40.7 How varying the function $A(k)$ in the wave-packet expression, Eq. (40.19), changes the character of the wave function $\Psi(x, t)$ (shown here at a specific time $t = 0$).



A state of definite energy is commonly called a **stationary state**. To see where this name comes from, let's multiply Eq. (40.21) by its complex conjugate to find the probability distribution function $|\Psi|^2$:

$$\begin{aligned} |\Psi(x, t)|^2 &= \Psi^*(x, t)\Psi(x, t) = [\psi^*(x)e^{+iEt/\hbar}] [\psi(x)e^{-iEt/\hbar}] \\ &= \psi^*(x)\psi(x)e^{(+iEt/\hbar)+(-iEt/\hbar)} = |\psi(x)|^2 e^0 \\ &= |\psi(x)|^2 \end{aligned} \quad (40.22)$$

Since $|\psi(x)|^2$ does not depend on time, Eq. (40.22) shows that the same must be true for the probability distribution function $|\Psi(x, t)|^2$. This justifies the term “stationary state” for a state of definite energy.

CAUTION A stationary state does not mean a stationary particle. Saying that a particle is in a stationary state does *not* mean that the particle is at rest. It's the *probability distribution* (that is, the relative likelihood of finding the particle at various positions), not the particle itself, that's stationary. ■

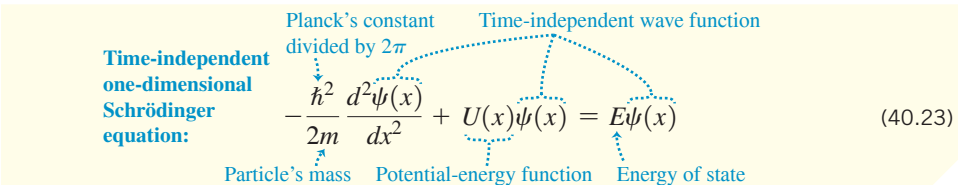
The Schrödinger equation, Eq. (40.20), becomes quite a bit simpler for stationary states. To see this, we substitute Eq. (40.21) into Eq. (40.20):

$$-\frac{\hbar^2}{2m} \frac{\partial^2 [\psi(x)e^{-iEt/\hbar}]}{\partial x^2} + U(x)\psi(x)e^{-iEt/\hbar} = i\hbar \frac{\partial [\psi(x)e^{-iEt/\hbar}]}{\partial t}$$

The derivative in the first term on the left-hand side is with respect to x , so the factor $e^{-iEt/\hbar}$ comes outside of the derivative. Now we take the derivative with respect to t on the right-hand side of the equation:

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} e^{-iEt/\hbar} + U(x)\psi(x)e^{-iEt/\hbar} &= i\hbar \left(\frac{-iE}{\hbar} \right) [\psi(x)e^{-iEt/\hbar}] \\ &= E\psi(x)e^{-iEt/\hbar} \end{aligned}$$

If we divide both sides of this equation by $e^{-iEt/\hbar}$, we get



The diagram shows the equation $-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + U(x)\psi(x) = E\psi(x)$ with several labels and arrows pointing to its parts:

- Time-independent one-dimensional Schrödinger equation:** Points to the entire equation.
- Planck's constant divided by 2π :** Points to $\hbar$ in the first term.
- Particle's mass:** Points to m in the first term.
- Potential-energy function:** Points to $U(x)$.
- Time-independent wave function:** Points to $\psi(x)$ in the second term.
- Energy of state:** Points to E on the right-hand side.

(40.23)

This is called the **time-independent one-dimensional Schrödinger equation**. The time-dependent factor $e^{-iEt/\hbar}$ does not appear, and Eq. (40.23) involves only the time-independent wave function $\psi(x)$. We'll devote much of this chapter to solving this equation to find the definite-energy, stationary-state wave functions $\psi(x)$ and the corresponding values of E —that is, the energies of the allowed levels—for different physical situations.

EXAMPLE 40.2 A stationary state

Consider the wave function $\psi(x) = A_1 e^{ikx} + A_2 e^{-ikx}$, where k is positive. Is this a valid time-independent wave function for a free particle in a stationary state? What is the energy corresponding to this wave function?

IDENTIFY and SET UP A valid stationary-state wave function for a free particle must satisfy the time-independent Schrödinger equation, Eq. (40.23), with $U(x) = 0$. To test the given function $\psi(x)$, we simply substitute it into the left-hand side of the equation. If the result is a constant times $\psi(x)$, then the wave function is indeed a solution and the constant is equal to the particle energy E .

EXECUTE Substituting $\psi(x) = A_1 e^{ikx} + A_2 e^{-ikx}$ and $U(x) = 0$ into Eq. (40.23), we obtain

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} &= -\frac{\hbar^2}{2m} \frac{d^2(A_1 e^{ikx} + A_2 e^{-ikx})}{dx^2} \\ &= -\frac{\hbar^2}{2m} [(ik)^2 A_1 e^{ikx} + (-ik)^2 A_2 e^{-ikx}] \\ &= \frac{\hbar^2 k^2}{2m} (A_1 e^{ikx} + A_2 e^{-ikx}) = \frac{\hbar^2 k^2}{2m} \psi(x) \end{aligned}$$

The result is a constant times $\psi(x)$, so this $\psi(x)$ is indeed a valid stationary-state wave function for a free particle. Comparing with Eq. (40.23) shows that the constant on the right-hand side is the particle energy: $E = \hbar^2 k^2 / 2m$.

EVALUATE Note that $\psi(x)$ is a *superposition* of two different wave functions: one function ($A_1 e^{ikx}$) that represents a particle with magnitude of momentum $p = \hbar k$ moving in the positive x -direction, and one function ($A_2 e^{-ikx}$) that represents a particle with the same magnitude of momentum moving in the negative x -direction. So while the

combined wave function $\psi(x)$ represents a stationary state with a definite energy, this state does *not* have a definite momentum. We'll see in Section 40.2 that such a wave function can represent a *standing wave*, and we'll explore situations in which such standing matter waves can arise.

KEYCONCEPT A quantum-mechanical state of definite energy E has a wave function of the form $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$. It is called a stationary state because the associated probability distribution function $|\Psi(x, t)|^2 = |\psi(x)|^2$ does not depend on time.

TEST YOUR UNDERSTANDING OF SECTION 40.1 Does a wave packet given by Eq. (40.19) represent a stationary state?

ANSWER Equation (40.19) represents a superposition of wave functions with different values of wave number k and hence different values of energy $E = \hbar^2 k^2 / 2m$. The state that this combined wave function represents is not a state of definite energy, and therefore not a stationary state. Another way to see this is to note that there is a factor $e^{-iEt/\hbar}$ inside the integral in Eq. (40.19), with a different value of E for each value of k . This wave function therefore has a very complicated time dependence, and the probability distribution function $|\Psi(x, t)|^2$ does depend on time.

40.2 PARTICLE IN A BOX

An important problem in quantum mechanics is how to use the time-independent Schrödinger equation, Eq. (40.23), to determine the possible energy levels and the corresponding wave functions for various systems. That is, for a given potential-energy function $U(x)$, what are the possible stationary-state wave functions $\psi(x)$, and what are the corresponding energies E ?

In Section 40.1 we solved this problem for the case $U(x) = 0$, corresponding to a *free* particle. The allowed wave functions and corresponding energies are

$$\psi(x) = A e^{ikx} \quad E = \frac{\hbar^2 k^2}{2m} \quad (\text{free particle}) \quad (40.24)$$

The wave number k is equal to $2\pi/\lambda$, where λ is the wavelength. We found that k can have any real value, so the energy E of a free particle can have any value from zero to infinity. Furthermore, the particle can be found with equal probability at any value of x from $-\infty$ to $+\infty$.

Now let's look at a simple model in which a particle is *bound* so that it cannot escape to infinity, but rather is confined to a restricted region of space. Our system consists of a particle confined between two rigid walls separated by a distance L (**Fig. 40.8**). The motion is purely one dimensional, with the particle moving along the x -axis only and the walls at $x = 0$ and $x = L$. The potential energy corresponding to the rigid walls is infinite, so the particle cannot escape; between the walls, the potential energy is zero (**Fig. 40.9**). This situation is often described as a “**particle in a box**.” This model might represent an electron that is free to move within a long, straight molecule or along a very thin wire.

Wave Functions for a Particle in a Box

To solve the Schrödinger equation for this system, we begin with some restrictions on the particle's stationary-state wave function $\psi(x)$. Because the particle is confined to the region $0 \leq x \leq L$, we expect the probability distribution function $|\Psi(x, t)|^2 = |\psi(x)|^2$ and the wave function $\psi(x)$ to be zero outside that region. This agrees with the Schrödinger equation: If the term $U(x)\psi(x)$ in Eq. (40.23) is to be finite, then $\psi(x)$ must be zero where $U(x)$ is infinite.

Furthermore, $\psi(x)$ must be a *continuous* function to be a mathematically well-behaved solution to the Schrödinger equation. This implies that $\psi(x)$ must be zero at the region's boundary, $x = 0$ and $x = L$. These two conditions serve as *boundary conditions* for the

Figure 40.8 The Newtonian view of a particle in a box.

A particle with mass m moves along a straight line at constant speed, bouncing between two rigid walls a distance L apart.

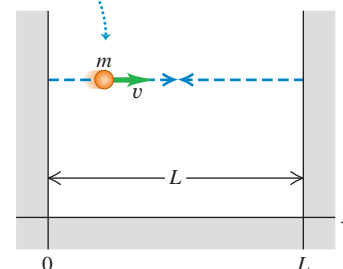


Figure 40.9 The potential-energy function for a particle in a box.

The potential energy U is zero in the interval $0 < x < L$ and is infinite everywhere outside this interval.

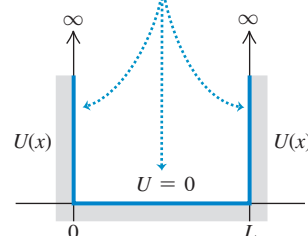
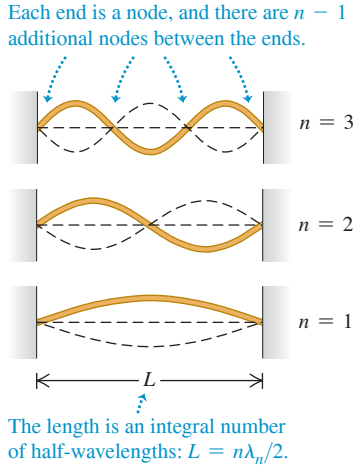


Figure 40.10 Normal modes of vibration for a string with length L , held at both ends.



problem. They should look familiar, because they are the same conditions that we used to find the normal modes of a vibrating string in Section 15.8 (Fig. 40.10); you should review that discussion.

An additional condition is that to calculate the second derivative $d^2\psi(x)/dx^2$ in Eq. (40.23), the *first* derivative $d\psi(x)/dx$ must also be continuous except at points where the potential energy becomes infinite (as it does at the walls of the box). This is analogous to the requirement that a vibrating string, like those shown in Fig. 40.10, can't have any kinks in it (which would correspond to a discontinuity in the first derivative of the wave function) except at the ends of the string.

We now solve for the wave functions in the region $0 \leq x \leq L$ subject to the above conditions. In this region $U(x) = 0$, so $\psi(x)$ in this region must satisfy

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x) \quad (\text{particle in a box}) \quad (40.25)$$

Equation (40.25) is the *same* Schrödinger equation as for a free particle, so it is tempting to conclude that the wave functions and energies are given by Eq. (40.24). It is true that $\psi(x) = Ae^{ikx}$ satisfies the Schrödinger equation with $U(x) = 0$, is continuous, and has a continuous first derivative $d\psi(x)/dx = ikAe^{ikx}$. However, this wave function does *not* satisfy the boundary conditions that $\psi(x)$ must be zero at $x = 0$ and $x = L$: At $x = 0$ the wave function in Eq. (40.24) is equal to $Ae^0 = A$, and at $x = L$ it is equal to Ae^{ikL} . (These would be equal to zero if $A = 0$, but then the wave function would be zero and there would be no particle at all!)

The way out of this dilemma is to recall Example 40.2 (Section 40.1), in which we found that a more general stationary-state solution to the time-independent Schrödinger equation with $U(x) = 0$ is

$$\psi(x) = A_1e^{ikx} + A_2e^{-ikx} \quad (40.26)$$

This wave function is a superposition of two waves: one traveling in the $+x$ -direction of amplitude A_1 , and one traveling in the $-x$ -direction with the same wave number but amplitude A_2 . This is analogous to a standing wave on a string (Fig. 40.10), which we can regard as the superposition of two sinusoidal waves propagating in opposite directions (see Section 15.7). The energy that corresponds to Eq. (40.26) is $E = \hbar^2k^2/2m$, just as for a single wave.

To see whether the wave function given by Eq. (40.26) can satisfy the boundary conditions, let's first rewrite it in terms of sines and cosines by using Euler's formula, Eq. (40.17):

$$\begin{aligned} \psi(x) &= A_1(\cos kx + i\sin kx) + A_2[\cos(-kx) + i\sin(-kx)] \\ &= A_1(\cos kx + i\sin kx) + A_2(\cos kx - i\sin kx) \\ &= (A_1 + A_2)\cos kx + i(A_1 - A_2)\sin kx \end{aligned} \quad (40.27)$$

At $x = 0$ this is equal to $\psi(0) = A_1 + A_2$, which must equal zero to satisfy the boundary condition at that point. Hence $A_2 = -A_1$, and Eq. (40.27) becomes

$$\psi(x) = 2iA_1\sin kx = C\sin kx \quad (40.28)$$

We have simplified the expression by introducing the constant $C = 2iA_1$. (We'll come back to this constant later.) We can also satisfy the second boundary condition that $\psi = 0$ at $x = L$ by choosing values of k such that $kL = n\pi$ ($n = 1, 2, 3, \dots$). Hence Eq. (40.28) does indeed give the stationary-state wave functions for a particle in a box in the region $0 \leq x \leq L$. (Outside this region, $\psi(x) = 0$.) The possible values of k and the wavelength $\lambda = 2\pi/k$ are

$$k = \frac{n\pi}{L} \quad \text{and} \quad \lambda = \frac{2\pi}{k} = \frac{2L}{n} \quad (n = 1, 2, 3, \dots) \quad (40.29)$$

Just as for the string in Fig. 40.10, the length L of the region is an integral number of half-wavelengths.

Energy Levels for a Particle in a Box

The possible energy levels for a particle in a box are given by $E = \hbar^2 k^2 / 2m = p^2 / 2m$, where $p = \hbar k = (h/2\pi)(2\pi/\lambda) = h/\lambda$ is the magnitude of momentum of a free particle with wave number k and wavelength λ . This makes sense, since inside the region $0 \leq x \leq L$ the potential energy is zero and the energy is all kinetic. For each value of n , there are corresponding values of p , λ , and E ; let's call them p_n , λ_n , and E_n . Putting the pieces together, we get

$$p_n = \frac{h}{\lambda_n} = \frac{nh}{2L} \quad (40.30)$$

and so the energy levels for a particle in a box are

$$E_n = \frac{p_n^2}{2m} = \frac{n^2 h^2}{8mL^2} = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad (n = 1, 2, 3, \dots) \quad (40.31)$$

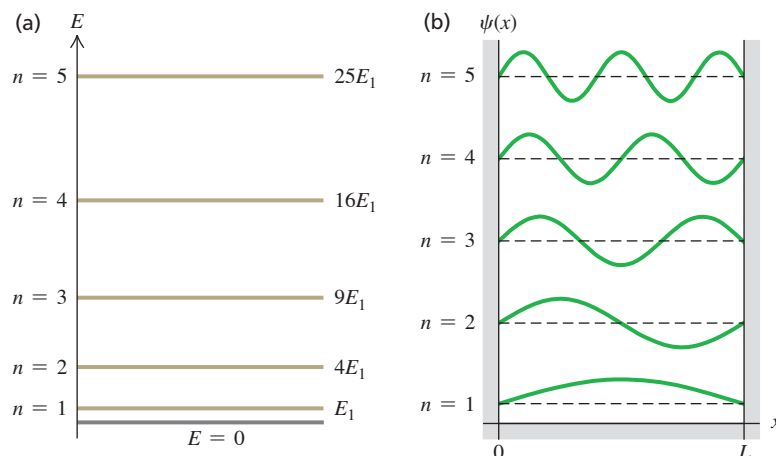
Energy levels for a particle in a box Magnitude of momentum Planck's constant Planck's constant divided by 2π
Particle's mass Width of box Quantum number

Each energy level has its own value of the quantum number n and a corresponding wave function, which we denote by ψ_n . When we replace k in Eq. (40.28) by $n\pi/L$ from Eq. (40.29), we find

$$\psi_n(x) = C \sin \frac{n\pi x}{L} \quad (n = 1, 2, 3, \dots) \quad (40.32)$$

The energy-level diagram in **Fig. 40.11a** shows the five lowest levels for a particle in a box. The energy levels are proportional to n^2 , so successively higher levels are spaced farther and farther apart. There are an infinite number of levels because the walls are perfectly rigid; even a particle of infinitely great kinetic energy is confined within the box. Figure 40.11b shows graphs of the wave functions $\psi_n(x)$ for $n = 1, 2, 3, 4$, and 5. Note that these functions look identical to those for a standing wave on a string (see Fig. 40.10).

CAUTION A particle in a box cannot have zero energy The energy of a particle in a box *cannot* be zero. Equation (40.31) shows that $E = 0$ would require $n = 0$, but substituting $n = 0$ into Eq. (40.32) gives a zero wave function. Since a particle is described by a *nonzero* wave function, there cannot be a particle with $E = 0$. This is a consequence of the Heisenberg uncertainty principle: A particle in a zero-energy state would have a definite value of momentum (precisely zero), so its position uncertainty would be infinite and the particle could be found anywhere along the x -axis. But this is impossible, since a particle in a box can be found only between $x = 0$ and $x = L$. Hence $E = 0$ is not allowed. By contrast, the allowed stationary-state wave functions with $n = 1, 2, 3, \dots$ do not represent states of definite momentum (each is an equal mixture of a state of x -momentum $+p_n = nh/2L$ and a state of x -momentum $-p_n = -nh/2L$). Hence each stationary state has a nonzero momentum uncertainty, consistent with having a finite position uncertainty. ■



APPLICATION **Particles in a Polymer “Box”** Polyacetylene is one of a class of long-chain organic molecules that conduct electricity along their length. The molecule is made up of a large number of (C_2H_2) units, called *monomers* (only three monomers are shown here). Electrons can move freely along the length of the molecule but not perpendicular to the length, so the molecule is like a one-dimensional “box” for electrons. The length L of the molecule depends on the number of monomers. Experiment shows that the allowed energy levels agree well with Eq. (40.31): The greater the number of monomers and the greater the length L , the lower the energy levels and the smaller the spacing between these levels.

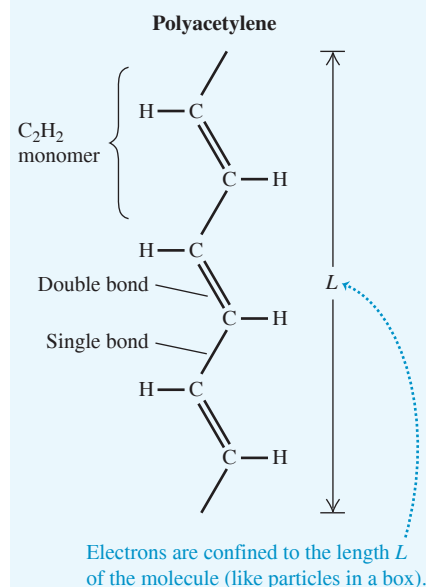


Figure 40.11 (a) Energy-level diagram for a particle in a box. Each energy is $n^2 E_1$, where E_1 is the ground-level energy. (b) Wave functions for a particle in a box, with $n = 1, 2, 3, 4$, and 5. **CAUTION:** The five graphs have been displaced vertically for clarity, as in Fig. 40.10. Each of the horizontal dashed lines represents $\psi = 0$ for the respective wave function.

EXAMPLE 40.3 Electron in an atom-size box

WITH VARIATION PROBLEMS

Find the first two energy levels for an electron confined to a one-dimensional box 5.0×10^{-10} m across (about the diameter of an atom).

IDENTIFY and SET UP This problem uses what we have learned in this section about a particle in a box. The first two energy levels correspond to $n = 1$ and $n = 2$ in Eq. (40.31).

EXECUTE From Eq. (40.31),

$$E_1 = \frac{h^2}{8mL^2} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})^2}{8(9.109 \times 10^{-31} \text{ kg})(5.0 \times 10^{-10} \text{ m})^2} = 2.4 \times 10^{-19} \text{ J} = 1.5 \text{ eV}$$

$$E_2 = \frac{2^2 h^2}{8mL^2} = 4E_1 = 9.6 \times 10^{-19} \text{ J} = 6.0 \text{ eV}$$

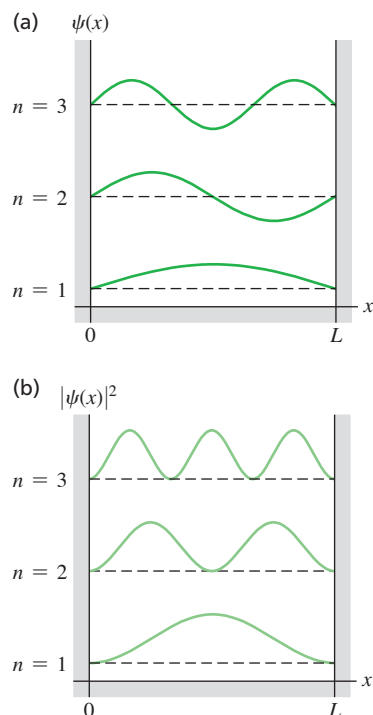
EVALUATE The difference between the first two energy levels is $E_2 - E_1 = 4.5 \text{ eV}$. An electron confined to a box is different from an electron bound in an atom, but it is reassuring that this result is of the same order of magnitude as the difference between actual atomic energy levels.

You can show that for a proton or neutron ($m = 1.67 \times 10^{-27} \text{ kg}$) confined to a box $1.1 \times 10^{-14} \text{ m}$ across (the width of a medium-sized atomic nucleus), the energies of the first two levels are about a million times larger: $E_1 = 1.7 \times 10^6 \text{ eV} = 1.7 \text{ MeV}$, $E_2 = 4E_1 = 6.8 \text{ MeV}$, $E_2 - E_1 = 5.1 \text{ MeV}$. This suggests why nuclear reactions (which involve transitions between energy levels in nuclei) release so much more energy than chemical reactions (which involve transitions between energy levels of electrons in atoms).

Finally, you can show (see Exercise 40.9) that the energy levels of a billiard ball ($m = 0.2 \text{ kg}$) confined to a box 1.3 m across—the width of a billiard table—are separated by about $5 \times 10^{-67} \text{ J}$. Quantum effects won't disturb a game of billiards.

KEYCONCEPT The wave functions for the stationary states of a particle confined to a one-dimensional box are sinusoidal standing waves inside the box with a node at each end; the wave function is zero outside the box. The n th stationary state has a wave function with n half-wavelengths within the length of the box, and has an energy equal to n^2 times the energy of the lowest-energy ($n = 1$) state. There are an infinite number of these stationary states.

Figure 40.12 Graphs of (a) $\psi(x)$ and (b) $|\psi(x)|^2$ for the first three wave functions ($n = 1, 2, 3$) for a particle in a box. The horizontal dashed lines represent $\psi(x) = 0$ and $|\psi(x)|^2 = 0$ for each of the three levels. The value of $|\psi(x)|^2 dx$ at each point is the probability of finding the particle in a small interval dx about the point. As in Fig. 40.11b, the three graphs in each part have been displaced vertically for clarity.



Probability and Normalization

Let's look a bit more closely at the wave functions for a particle in a box, keeping in mind the *probability* interpretation of the wave function ψ that we discussed in Section 40.1. In our one-dimensional situation the quantity $|\psi(x)|^2 dx$ is proportional to the probability that the particle will be found within a small interval dx about x . For a particle in a box,

$$|\psi(x)|^2 dx = C^2 \sin^2 \frac{n\pi x}{L} dx$$

Figure 40.12 shows graphs of both $\psi(x)$ and $|\psi(x)|^2$ for $n = 1, 2$, and 3 . Note that not all positions are equally likely. By contrast, in classical mechanics the particle is equally likely to be found at any position between $x = 0$ and $x = L$. We see from Fig. 40.12b that $|\psi(x)|^2 = 0$ at some points, so there is zero probability of finding the particle at exactly these points. Don't let that bother you; the uncertainty principle has already shown us that we can't measure position exactly. The particle is localized only to be somewhere between $x = 0$ and $x = L$.

The particle must be *somewhere* on the x -axis—that is, somewhere between $x = -\infty$ and $x = +\infty$. So the *sum* of the probabilities for all the dx 's everywhere (the *total* probability of finding the particle) must equal 1. That's the normalization condition that we discussed in Section 40.1:

$$\text{Normalization condition, time-independent wave function:} \quad \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1 \quad \text{Time-independent wave function} \quad (40.33)$$

Probability distribution function Probability that particle is somewhere on x -axis

A wave function is said to be *normalized* if it has a constant such as C in Eq. (40.32) that is calculated to make the total probability equal 1 in Eq. (40.33). For a normalized wave function, $|\psi(x)|^2 dx$ is not merely proportional to, but *equals*, the probability of finding the particle between the coordinates x and $x + dx$. That's why we call $|\psi(x)|^2$ the probability distribution function. (In Section 40.1 we called $|\Psi(x, t)|^2$ the probability distribution function. For the case of a stationary-state wave function, however, $|\Psi(x, t)|^2$ is equal to $|\psi(x)|^2$.)

Let's normalize the particle-in-a-box wave functions $\psi_n(x)$ given by Eq. (40.32). Since $\psi_n(x)$ is zero except between $x = 0$ and $x = L$, Eq. (40.33) becomes

$$\int_0^L C^2 \sin^2 \frac{n\pi x}{L} dx = 1 \quad (40.34)$$

You can evaluate this integral by using the trigonometric identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$; the result is $C^2 L/2$. Thus our probability interpretation of the wave function demands that $C^2 L/2 = 1$, or $C = (2/L)^{1/2}$; the constant C is *not* arbitrary. (This is in contrast to the classical vibrating string problem, in which C represents an amplitude that depends on initial conditions.) Thus the normalized stationary-state wave functions for a particle in a box are

Stationary-state wave functions for a particle in a box

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \quad (n = 1, 2, 3, \dots) \quad (40.35)$$

Quantum number
Width of box

EXAMPLE 40.4 A nonsinusoidal wave function?

WITH VARIATION PROBLEMS

(a) Show that $\psi(x) = Ax + B$, where A and B are constants, is a solution of the Schrödinger equation for an $E = 0$ energy level of a particle in a box. (b) What constraints do the boundary conditions at $x = 0$ and $x = L$ place on the constants A and B ?

IDENTIFY and SET UP To be physically reasonable, a wave function must satisfy both the Schrödinger equation and the appropriate boundary conditions. In part (a) we'll substitute $\psi(x)$ into the Schrödinger equation for a particle in a box, Eq. (40.25), to determine whether it is a solution. In part (b) we'll see what restrictions on $\psi(x)$ arise from applying the boundary conditions that $\psi(x) = 0$ at $x = 0$ and $x = L$.

EXECUTE (a) From Eq. (40.25), the Schrödinger equation for an $E = 0$ energy level of a particle in a box is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x) = 0$$

in the region $0 \leq x \leq L$. Differentiating $\psi(x) = Ax + B$ twice with respect to x gives $d^2\psi(x)/dx^2 = 0$, so the left side of the equation is

zero, and so $\psi(x) = Ax + B$ is a solution of this Schrödinger equation for $E = 0$. (Note that both $\psi(x)$ and its derivative $d\psi(x)/dx = A$ are continuous functions, as they must be.)

(b) Applying the boundary condition at $x = 0$ gives $\psi(0) = B = 0$, and so $\psi(x) = Ax$. Applying the boundary condition at $x = L$ gives $\psi(L) = AL = 0$, so $A = 0$. Hence $\psi(x) = 0$ both inside the box ($0 \leq x \leq L$) and outside: There is *zero* probability of finding the particle anywhere with this wave function, and so $\psi(x) = Ax + B$ is *not* a physically valid wave function.

EVALUATE The moral is that there are many functions that satisfy the Schrödinger equation for a given physical situation, but most of these—including the function considered here—have to be rejected because they don't satisfy the appropriate boundary conditions.

KEYCONCEPT To be physically reasonable, a stationary-state wave function for a given situation must satisfy both the Schrödinger equation and any boundary conditions that apply to that situation.

Time Dependence

The wave functions $\psi_n(x)$ in Eq. (40.35) depend only on the *spatial* coordinate x . Equation (40.21) shows that if $\psi(x)$ is the wave function for a state of definite energy E , the full time-dependent wave function is $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$. Hence the *time-dependent* stationary-state wave functions for a particle in a box are

$$\Psi_n(x, t) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) e^{-iE_n t/\hbar} \quad (n = 1, 2, 3, \dots) \quad (40.36)$$

In this expression the energies E_n are given by Eq. (40.31). The higher the quantum number n , the greater the angular frequency $\omega_n = E_n/\hbar$ at which the wave function oscillates. Note that since $|e^{-iE_n t/\hbar}|^2 = e^{+iE_n t/\hbar} e^{-iE_n t/\hbar} = e^0 = 1$, the probability distribution function $|\Psi_n(x, t)|^2 = (2/L) \sin^2(n\pi x/L)$ is independent of time and does *not* oscillate. (Remember, this is why we say that these states of definite energy are *stationary*.)

TEST YOUR UNDERSTANDING OF SECTION 40.2 If a particle in a box is in the n th energy level, what is the average value of its x -component of momentum p_x ? (i) $nh/2L$; (ii) $(\sqrt{2}/2)nh/L$; (iii) $(1/\sqrt{2})nh/L$; (iv) $[1/(2\sqrt{2})]nh/L$; (v) zero.

ANSWER (v) Our derivation of the stationary-state wave functions for a particle in a box shows that they are superpositions of waves propagating in opposite directions, just like a standing wave on a string. One wave has momentum in the positive x -direction, while the other wave has an equal magnitude of momentum in the negative x -direction. The total x -component of momentum is zero.

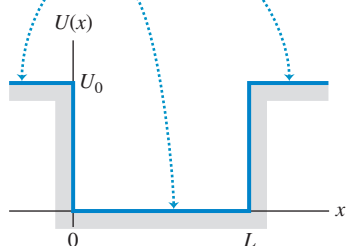
40.3 POTENTIAL WELLS

A **potential well** is a potential-energy function $U(x)$ that has a minimum. We introduced this term in Section 7.5, and we also used it in our discussion of periodic motion in Chapter 14. In Newtonian mechanics a particle trapped in a potential well can vibrate back and forth with periodic motion. Our first application of the Schrödinger equation, the particle in a box, involved a rudimentary potential well with a function $U(x)$ that is zero within a certain interval and infinite everywhere else. As we mentioned in Section 40.2, this function corresponds to a few situations found in nature, but the correspondence is only approximate.

A better approximation to several physical situations is a **finite well**, which is a potential well that has straight sides but *finite* height. **Figure 40.13** shows a potential-energy function that is zero in the interval $0 \leq x \leq L$ and has the value U_0 outside this interval. This function is often called a **square-well potential**. It could serve as a simple model of an electron within a metallic sheet with thickness L , moving perpendicular to the surfaces of the sheet. The electron can move freely inside the metal but has to climb a potential-energy barrier with height U_0 to escape from either surface of the metal. The energy U_0 is related to the *work function* that we discussed in Section 38.1 in connection with the photoelectric effect. In three dimensions, a spherical version of a finite well gives an approximate description of the motions of protons and neutrons within a nucleus.

Figure 40.13 A square-well potential.

The potential energy U is zero within the potential well (in the interval $0 \leq x \leq L$) and has the constant value U_0 outside this interval.



Bound States of a Square-Well Potential

In Newtonian mechanics, the particle is trapped (localized) in a well if the total mechanical energy E is less than U_0 . In quantum mechanics, such a trapped state is often called a **bound state**. All states are bound for an infinitely deep well like the one we described in Section 40.2. For a finite well like that shown in Fig. 40.13, if E is greater than U_0 , the particle is *not* bound.

Let's see how to solve the Schrödinger equation for the bound states of a square-well potential. Our goal is to find the energies and wave functions for which $E < U_0$. It's easiest to consider separately the regions where $U = 0$ and where $U = U_0$. Inside the square well ($0 \leq x \leq L$), where $U = 0$, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x) \quad \text{or} \quad \frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2} \psi(x) \quad (40.37)$$

This is the same as Eq. (40.25) from Section 40.2, which describes a particle in a box. As in Section 40.2, we can express the solutions of this equation as combinations of $\cos kx$ and $\sin kx$, where $E = \hbar^2 k^2 / 2m$. We can rewrite the relationship between E and k as $k = \sqrt{2mE}/\hbar$. Hence inside the square well we have

$$\psi(x) = A \cos\left(\frac{\sqrt{2mE}}{\hbar} x\right) + B \sin\left(\frac{\sqrt{2mE}}{\hbar} x\right) \quad (\text{inside the well}) \quad (40.38)$$

where A and B are constants. So far, this looks a lot like the particle-in-a-box analysis in Section 40.2. The difference is that for the square-well potential, the potential energy outside the well is not infinite, so the wave function $\psi(x)$ outside the well is *not* zero.

For the regions outside the well ($x < 0$ and $x > L$) the potential-energy function in the time-independent Schrödinger equation is $U = U_0$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + U_0\psi(x) = E\psi(x) \quad \text{or} \quad \frac{d^2\psi(x)}{dx^2} = \frac{2m(U_0 - E)}{\hbar^2}\psi(x) \quad (40.39)$$

The quantity $U_0 - E$ is positive, so the solutions of this equation are exponential functions rather than sines or cosines. Using κ (the Greek letter kappa) to represent the quantity $[2m(U_0 - E)]^{1/2}/\hbar$ and taking κ as positive, we can write the solutions as

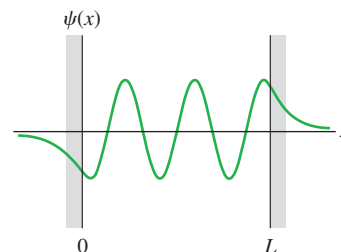
$$\psi(x) = Ce^{\kappa x} + De^{-\kappa x} \quad (\text{outside the well}) \quad (40.40)$$

where C and D are constants with different values in the two regions $x < 0$ and $x > L$. Note that ψ can't be allowed to approach infinity as $x \rightarrow +\infty$ or $x \rightarrow -\infty$. [If it did, we wouldn't be able to satisfy the normalization condition, Eq. (40.33).] This means that in Eq. (40.40), we must have $D = 0$ for $x < 0$ and $C = 0$ for $x > L$.

Our calculations so far show that the bound-state wave functions for a finite well are sinusoidal inside the well [Eq. (40.38)] and exponential outside it [Eq. (40.40)]. We have to *match* the wave functions inside and outside the well so that they satisfy the boundary conditions that we mentioned in Section 40.2: $\psi(x)$ and $d\psi(x)/dx$ must be continuous at the boundary points $x = 0$ and $x = L$. If the wave function $\psi(x)$ or the slope $d\psi(x)/dx$ were to change discontinuously at a point, the second derivative $d^2\psi(x)/dx^2$ would be *infinite* at that point. That would violate the time-independent Schrödinger equation, Eq. (40.23), which says that at every point $d^2\psi(x)/dx^2$ is proportional to $U - E$. For a finite well $U - E$ is finite everywhere, so $d^2\psi(x)/dx^2$ must also be finite everywhere.

Matching the sinusoidal and exponential functions at the boundary points so that they join smoothly is possible only for certain specific values of the total energy E , so this requirement determines the possible energy levels of the finite square well. There is no simple formula for the energy levels as there was for the infinitely deep well. Finding the levels is a fairly complex mathematical problem that requires solving a transcendental equation by numerical approximation; we won't go into the details. **Figure 40.14** shows the general shape of a possible wave function. The most striking features of this wave function are the “exponential tails” that extend outside the well into regions that are forbidden by Newtonian mechanics (because in those regions the particle would have negative kinetic energy). We see that there is some probability for finding the particle *outside* the potential well, which would be impossible in classical mechanics. In Section 40.4 we'll discuss an amazing result of this effect.

Figure 40.14 A possible wave function for a particle in a finite potential well. The function is sinusoidal inside the well ($0 \leq x \leq L$) and exponential outside it. It approaches zero asymptotically at large $|x|$. The functions must join smoothly at $x = 0$ and $x = L$; the wave function and its derivative must be continuous.



EXAMPLE 40.5 Outside a finite well

(a) Show that Eq. (40.40), $\psi(x) = Ce^{\kappa x} + De^{-\kappa x}$, is indeed a solution of the time-independent Schrödinger equation outside a finite well of height U_0 . (b) What happens to $\psi(x)$ in the limit $U_0 \rightarrow \infty$?

IDENTIFY and SET UP In part (a), we try the given function $\psi(x)$ in the time-independent Schrödinger equation for $x < 0$ and for $x > L$, Eq. (40.39). In part (b), we note that in the limit $U_0 \rightarrow \infty$ the finite well becomes an *infinite* well, like that for a particle in a box (Section 40.2). So in this limit the wave functions outside a finite well must reduce to the wave functions outside the box.

EXECUTE (a) We must show that $\psi(x) = Ce^{\kappa x} + De^{-\kappa x}$ satisfies $d^2\psi(x)/dx^2 = [2m(U_0 - E)/\hbar^2]\psi(x)$. We recall that $(d/du)e^{au} = ae^{au}$ and $(d^2/du^2)e^{au} = a^2e^{au}$; the left-hand side of the Schrödinger equation is then

$$\begin{aligned} \frac{d^2\psi(x)}{dx^2} &= \frac{d^2}{dx^2}(Ce^{\kappa x}) + \frac{d^2}{dx^2}(De^{-\kappa x}) \\ &= C\kappa^2 e^{\kappa x} + D(-\kappa)^2 e^{-\kappa x} \\ &= \kappa^2(Ce^{\kappa x} + De^{-\kappa x}) = \kappa^2\psi(x) \end{aligned}$$

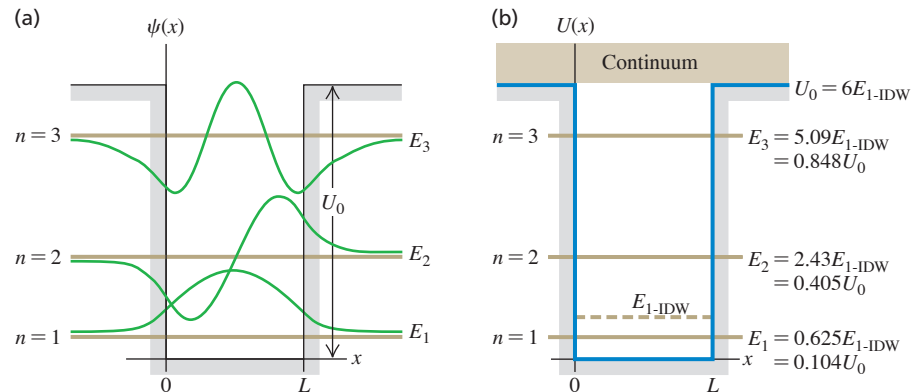
Since from Eq. (40.40) $\kappa^2 = 2m(U_0 - E)/\hbar^2$, this is equal to the right-hand side of the equation. The equation is satisfied, and $\psi(x)$ is a solution.

(b) As U_0 approaches infinity, κ also approaches infinity. In the region $x < 0$, $\psi(x) = Ce^{\kappa x}$; as $\kappa \rightarrow \infty$, $\kappa x \rightarrow -\infty$ (since x is negative) and $e^{\kappa x} \rightarrow 0$, so the wave function approaches zero for all $x < 0$. Likewise, we can show that the wave function also approaches zero for all $x > L$. This is just what we found in Section 40.2; the wave function for a particle in a box must be zero outside the box.

EVALUATE Our result in part (b) shows that the infinite square well is a *limiting case* of the finite well. We've seen many cases in Newtonian mechanics where it's important to consider limiting cases [such as Examples 5.11 (Section 5.2) and 5.13 (Section 5.3)]. Limiting cases are no less important in quantum mechanics.

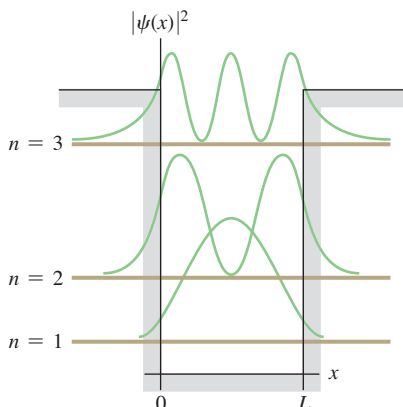
KEYCONCEPT A finite potential well is a region where the potential energy is constant and less than the constant value outside that region. If a particle is in a bound state, so that in Newtonian mechanics it would not have enough energy to escape from the well, its stationary-state wave function is sinusoidal within the well and exponential outside the well.

Figure 40.15 (a) Wave functions for the three bound states for a particle in a finite potential well of depth $U_0 = 6E_{1-\text{IDW}}$. (Here $E_{1-\text{IDW}}$ is the ground-level energy for an infinite well of the same width.) The horizontal brown line for each wave function corresponds to $\psi = 0$; the vertical placement of these lines indicates the energy of each bound state (compare Fig. 40.11). (b) Energy-level diagram for this system. All energies greater than U_0 are possible; states with $E > U_0$ form a continuum.



CAUTION Misconceptions about the finite potential well Note that in our notation the energy of a particle in a finite potential well is given relative to the *bottom* of the well, which we take to be $E = 0$. (We did the same for the particle in a box, or infinite potential well, in Section 40.2.) If the depth of a finite square well is U_0 , a bound state has an energy that is greater than zero but less than U_0 . Note also that how far the exponential part of a bound state wave function extends outside a finite potential well depends on the energy of the bound state. The greater the bound state energy, the farther outside the well the wave function extends and the farther outside the particle is likely to be found (Fig. 40.15). **I**

Figure 40.16 Probability distribution functions $|\psi(x)|^2$ for the square-well wave functions shown in Fig. 40.15. The horizontal brown line for each wave function corresponds to $|\psi|^2 = 0$.



Comparing Finite and Infinite Square Wells

Let's continue the comparison of the finite-depth potential well with the infinitely deep well, which we began in Example 40.5. First, because the wave functions for the finite well don't go to zero at $x = 0$ and $x = L$, the wavelength of the sinusoidal part of each wave function is *longer* than it would be with an infinite well. This increase in λ corresponds to a reduced magnitude of momentum $p = h/\lambda$ and therefore a reduced energy. Thus each energy level, including the ground level, is *lower* for a finite well than for an infinitely deep well with the same width.

Second, a well with finite depth U_0 has only a *finite* number of bound states and corresponding energy levels, compared to the *infinite* number for an infinitely deep well. How many levels there are depends on the magnitude of U_0 in comparison with the ground-level energy for the infinitely deep well (IDW), which we call $E_{1-\text{IDW}}$. From Eq. (40.31),

$$E_{1-\text{IDW}} = \frac{\pi^2 \hbar^2}{2mL^2} \quad (\text{ground-level energy, infinitely deep well}) \quad (40.41)$$

When the well is very deep so U_0 is much larger than $E_{1-\text{IDW}}$, there are many bound states and the energies of the lowest few are nearly the same as the energies for the infinitely deep well. When U_0 is only a few times as large as $E_{1-\text{IDW}}$ there are only a few bound states. (There is always at least *one* bound state, no matter how shallow the well.) As with the infinitely deep well, there is no state with $E = 0$; such a state would violate the uncertainty principle.

Figure 40.15 shows the case $U_0 = 6E_{1-\text{IDW}}$, for which there are three bound states. In the figure, we express the energy levels both as fractions of the well depth U_0 and as multiples of $E_{1-\text{IDW}}$. If the well were infinitely deep, the lowest three levels, as given by Eq. (40.31), would be $E_{1-\text{IDW}}$, $4E_{1-\text{IDW}}$, and $9E_{1-\text{IDW}}$. Figure 40.15 also shows the wave functions for the three bound states.

It turns out that when U_0 is less than $E_{1-\text{IDW}}$, there is only one bound state. In the limit when U_0 is *much smaller* than $E_{1-\text{IDW}}$ (a very shallow well), the energy of this single state is approximately $E = 0.68U_0$.

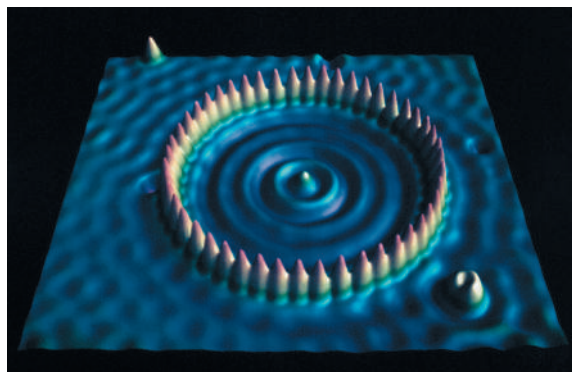
Figure 40.16 shows graphs of the probability distributions—that is, the values of $|\psi|^2$ —for the wave functions shown in Fig. 40.15a. As with the infinite well, not all positions are equally likely. Unlike the infinite well, there is some probability of finding the particle outside the well in the classically forbidden regions.

There are also states for which E is *greater* than U_0 . In these *free-particle states* the particle is not bound but is free to move through all values of x . Any energy E greater than U_0 is possible, so the free-particle states form a *continuum* rather than a discrete set

of states with definite energy levels. The free-particle wave functions are sinusoidal both inside and outside the well. The wavelength is shorter inside the well than outside, corresponding to greater kinetic energy inside the well than outside it.

Figure 40.17 shows particles in a *two-dimensional* finite potential well. Example 40.6 describes another application of the square-well potential.

Figure 40.17 To make this image, 48 iron atoms (shown as yellow peaks) were placed in a circle on a copper surface. The “elevation” at each point inside the circle indicates the electron density within the circle. The standing-wave pattern is very similar to the probability distribution function for a particle in a one-dimensional finite potential well. (This image was made with a scanning tunneling microscope, discussed in Section 40.4.)



EXAMPLE 40.6 An electron in a finite well

WITH VARIATION PROBLEMS

An electron is trapped in a square well 0.50 nm across (roughly five times a typical atomic diameter). (a) Find the ground-level energy E_{1-IDW} if the well is infinitely deep. (b) Find the energy levels if the actual well depth U_0 is six times the ground-level energy found in part (a). (c) Find the wavelength of the photon emitted when the electron makes a transition from the $n = 2$ level to the $n = 1$ level. In what region of the electromagnetic spectrum does the photon wavelength lie? (d) If the electron is in the $n = 1$ (ground) level and absorbs a photon, what is the minimum photon energy that will free the electron from the well? In what region of the spectrum does the wavelength of this photon lie?

IDENTIFY and SET UP Equation (40.41) gives the ground-level energy E_{1-IDW} for an infinitely deep well, and Fig. 40.15b shows the energies for a square well with $U_0 = 6E_{1-IDW}$. The energy of the photon emitted or absorbed in a transition is equal to the difference in energy between two levels involved in the transition; the photon wavelength is given by $E = hc/\lambda$ (see Chapter 38).

EXECUTE (a) From Eq. (40.41),

$$E_{1-IDW} = \frac{\pi^2 \hbar^2}{2mL^2} = \frac{\pi^2 (1.055 \times 10^{-34} \text{ J} \cdot \text{s})^2}{2(9.11 \times 10^{-31} \text{ kg})(0.50 \times 10^{-9} \text{ m})^2} = 2.4 \times 10^{-19} \text{ J} = 1.5 \text{ eV}$$

(b) We have $U_0 = 6E_{1-IDW} = 6(1.5 \text{ eV}) = 9.0 \text{ eV}$. We can read off the energy levels from Fig. 40.15b:

$$\begin{aligned} E_1 &= 0.625E_{1-IDW} = 0.625(1.5 \text{ eV}) = 0.94 \text{ eV} \\ E_2 &= 2.43E_{1-IDW} = 2.43(1.5 \text{ eV}) = 3.6 \text{ eV} \\ E_3 &= 5.09E_{1-IDW} = 5.09(1.5 \text{ eV}) = 7.6 \text{ eV} \end{aligned}$$

(c) The photon energy and wavelength for the $n = 2$ to $n = 1$ transition are

$$\begin{aligned} E_2 - E_1 &= 3.6 \text{ eV} - 0.94 \text{ eV} = 2.7 \text{ eV} \\ \lambda &= \frac{hc}{E} = \frac{(4.136 \times 10^{-15} \text{ eV} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{2.7 \text{ eV}} = 460 \text{ nm} \end{aligned}$$

in the blue region of the visible spectrum.

(d) We see from Fig. 40.15b that the minimum energy needed to free the electron from the well from the $n = 1$ level is $U_0 - E_1 = 9.0 \text{ eV} - 0.94 \text{ eV} = 8.1 \text{ eV}$, which is three times the 2.7 eV photon energy found in part (c). Hence the corresponding photon wavelength is one-third of 460 nm, or (to two significant figures) 150 nm, which is in the ultraviolet region of the spectrum.

EVALUATE As a check, you can calculate the bound-state energies by using the formulas $E_1 = 0.104U_0$, $E_2 = 0.405U_0$, and $E_3 = 0.848U_0$ given in Fig. 40.15b. As an additional check, note that the first three energy levels of an infinitely deep well of the same width are $E_{1-IDW} = 1.5 \text{ eV}$, $E_{2-IDW} = 4E_{1-IDW} = 6.0 \text{ eV}$, and $E_{3-IDW} = 9E_{1-IDW} = 13.5 \text{ eV}$. The energies we found in part (b) are less than these values: As we mentioned earlier, the finite depth of the well lowers the energy levels compared to the levels for an infinitely deep well.

One application of these ideas is *quantum dots*, which are **?** nanometer-sized particles of a semiconductor such as cadmium selenide (CdSe). An electron within a quantum dot behaves much like a particle in a finite potential well of width L equal to the size of the dot. When quantum dots are illuminated with ultraviolet light, the electrons absorb the ultraviolet photons and are excited into high energy levels, such as the $n = 3$ level described in this example. If the electron returns to the ground level ($n = 1$) in two or more steps (for example,

Continued

from $n = 3$ to $n = 2$ and from $n = 2$ to $n = 1$), one step will involve emitting a visible-light photon, as we have calculated here. (We described this process of *fluorescence* in Section 39.3.) Increasing the value of L decreases the energies of the levels and hence the spacing between them, and thus decreases the energy and increases the wavelength of the emitted photons. The photograph that opens this chapter shows quantum dots of different sizes in solution: Each emits a characteristic wavelength that depends on the dot size. Quantum dots can

be injected into living tissue and their fluorescent glow used as a tracer for biological research and for medicine. They may also be the key to a new generation of lasers and ultrafast computers.

KEYCONCEPT A finite square well has a finite number of bound states. The energies of these bound states are determined by the boundary conditions that the wave function and its derivative are both continuous at the edges of the well.

TEST YOUR UNDERSTANDING OF SECTION 40.3 Suppose that the width of the finite potential well shown in Fig. 40.15 is reduced by one-half. How must the value of U_0 change so that there are still just three bound energy levels whose energies are the fractions of U_0 shown in Fig. 40.15b? U_0 must (i) increase by a factor of four; (ii) increase by a factor of two; (iii) remain the same; (iv) decrease by a factor of one-half; (v) decrease by a factor of one-fourth.

ANSWER

(i) The energy levels are arranged as shown in Fig. 40.15b if $U_0 = 6E_{1-IDW}$, where $E_{1-IDW} = \pi^2\hbar^2/2mL^2$ is the ground-level energy of an infinite well. If the well width L is reduced to one-half of its initial value, E_{1-IDW} increases by a factor of four and so U_0 must also increase by a factor of four. The energies E_1 , E_2 , and E_3 shown in Fig. 40.15b are all specific fractions of U_0 , so they will also increase by a factor of four.

Figure 40.18 A potential-energy barrier. According to Newtonian mechanics, if the total energy of the system is E_1 , a particle to the left of the barrier can go no farther than $x = a$. If the total energy is greater than E_2 , the particle can pass over the barrier.

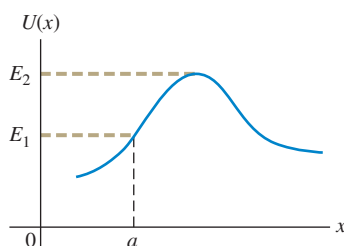
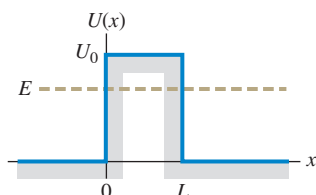


Figure 40.19 A rectangular potential-energy barrier with width L and height U_0 . According to Newtonian mechanics, if the total energy E is less than U_0 , a particle cannot pass over this barrier but is confined to the side where it starts.



40.4 POTENTIAL BARRIERS AND TUNNELING

A **potential barrier** is the opposite of a potential well; it is a potential-energy function with a *maximum*. **Figure 40.18** shows an example. In classical Newtonian mechanics, if a particle (such as a roller coaster) is located to the left of the barrier (which might be a hill), and if the total mechanical energy of the system is E_1 , the particle cannot move farther to the right than $x = a$. If it did, the potential energy U would be greater than the total mechanical energy E and the kinetic energy $K = E - U$ would be negative. This is impossible in classical mechanics since $K = \frac{1}{2}mv^2$ can never be negative.

A quantum-mechanical particle behaves differently: If it encounters a barrier like the one in Fig. 40.18 and has energy less than E_2 , it *may* appear on the other side. This phenomenon is called *tunneling*. In quantum-mechanical tunneling, unlike macroscopic, mechanical tunneling, the particle does not actually push through the barrier and loses no energy in the process.

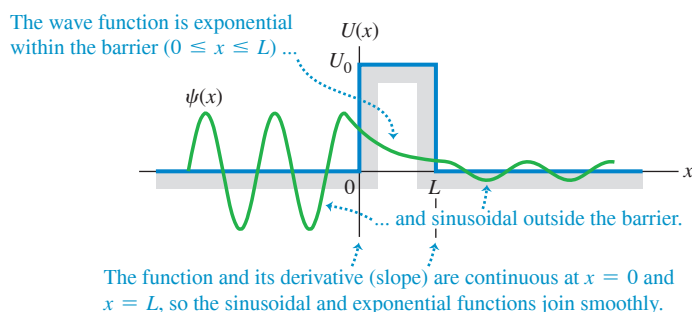
Tunneling Through a Rectangular Barrier

To understand how tunneling can occur, let's look at the potential-energy function $U(x)$ shown in **Fig. 40.19**. It's like Fig. 40.13 turned upside-down; the potential energy is zero everywhere except in the range $0 \leq x \leq L$, where it has the value U_0 . This might represent a simple model for the potential energy of an electron in the presence of two slabs of metal separated by an air gap of thickness L . The potential energy is lower within either slab than in the gap between them.

Let's consider solutions of the Schrödinger equation for this potential-energy function for the case in which E is less than U_0 . We can use our results from Section 40.3. In the regions $x < 0$ and $x > L$, where $U = 0$, the solution is sinusoidal and is given by Eq. (40.38). Within the barrier ($0 \leq x \leq L$), $U = U_0$ and the solution is exponential as in Eq. (40.40). Just as with the finite potential well, the functions have to join smoothly at the boundary points $x = 0$ and $x = L$, which means that both $\psi(x)$ and $d\psi(x)/dx$ have to be continuous at these points.

These requirements lead to a wave function like the one shown in **Fig. 40.20**. The function is *not* zero inside the barrier (the region forbidden by Newtonian mechanics). Even more remarkable, a particle that is initially to the *left* of the barrier has some probability of being found to the *right* of the barrier. How great this probability is depends on the width L of the barrier and the particle's energy E in comparison with the barrier height U_0 .

Figure 40.20 A possible wave function for a particle tunneling through the potential-energy barrier shown in Fig. 40.19.



The **tunneling probability** T that the particle gets through the barrier is proportional to the square of the ratio of the amplitudes of the sinusoidal wave functions on the two sides of the barrier. These amplitudes are determined by matching wave functions and their derivatives at the boundary points, a fairly involved mathematical problem. When T is much smaller than unity, it is given approximately by

$$T = Ge^{-2\kappa L} \text{ where } G = 16 \frac{E}{U_0} \left(1 - \frac{E}{U_0}\right) \text{ and } \kappa = \frac{\sqrt{2m(U_0 - E)}}{\hbar} \quad (40.42)$$

(probability of tunneling)

The probability decreases rapidly with increasing barrier width L . It also depends critically on the energy difference $U_0 - E$, which in Newtonian physics is the additional kinetic energy the particle would need to be able to climb over the barrier.

EXAMPLE 40.7 Tunneling through a barrier

WITH VARIATION PROBLEMS

A 2.0 eV electron encounters a barrier 5.0 eV high. What is the probability that it will tunnel through the barrier if the barrier width is (a) 1.00 nm and (b) 0.50 nm?

IDENTIFY and SET UP This problem uses the ideas of tunneling through a rectangular barrier, as in Figs. 40.19 and 40.20. Our target variable is the tunneling probability T in Eq. (40.42), which we evaluate for the given values $E = 2.0$ eV (electron energy), $U = 5.0$ eV (barrier height), $m = 9.11 \times 10^{-31}$ kg (mass of the electron), and $L = 1.00$ nm or 0.50 nm (barrier width).

EXECUTE First we evaluate G and κ in Eq. (40.42), using $E = 2.0$ eV:

$$G = 16 \left(\frac{2.0 \text{ eV}}{5.0 \text{ eV}} \right) \left(1 - \frac{2.0 \text{ eV}}{5.0 \text{ eV}} \right) = 3.8$$

$$U_0 - E = 5.0 \text{ eV} - 2.0 \text{ eV} = 3.0 \text{ eV} = 4.8 \times 10^{-19} \text{ J}$$

$$\kappa = \frac{\sqrt{2(9.11 \times 10^{-31} \text{ kg})(4.8 \times 10^{-19} \text{ J})}}{1.055 \times 10^{-34} \text{ J} \cdot \text{s}} = 8.9 \times 10^9 \text{ m}^{-1}$$

(a) When $L = 1.00 \text{ nm} = 1.00 \times 10^{-9} \text{ m}$, we have $2\kappa L = 2(8.9 \times 10^9 \text{ m}^{-1})(1.00 \times 10^{-9} \text{ m}) = 17.8$ and $T = Ge^{-2\kappa L} = 3.8e^{-17.8} = 7.1 \times 10^{-8}$.

(b) When $L = 0.50 \text{ nm}$, one-half of 1.00 nm, $2\kappa L$ is one-half of 17.8, or 8.9. Hence $T = 3.8e^{-8.9} = 5.2 \times 10^{-4}$.

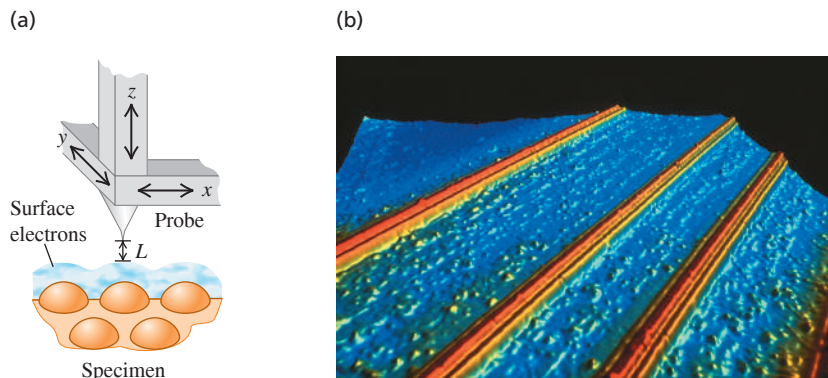
EVALUATE Halving the width of this barrier increases the tunneling probability T by a factor of $(5.2 \times 10^{-4})/(7.1 \times 10^{-8}) = 7.3 \times 10^3$, or nearly ten thousand. The tunneling probability is an *extremely* sensitive function of the barrier width. Note that our results for T in both part (a) and (b) are much less than 1, which is consistent with the assumptions that go into Eq. (40.42).

KEYCONCEPT Quantum-mechanical particles can be found in places where Newtonian mechanics would forbid them to be. An example is tunneling, in which a particle can pass through a barrier where its total mechanical energy is less than the potential energy. The probability that tunneling occurs depends on the width of the barrier and the additional energy the particle would need to “climb” over the barrier.

Applications of Tunneling

Tunneling has a number of practical applications, some of considerable importance. When you twist two copper wires together or close the contacts of a switch, current passes from one conductor to the other despite a thin layer of nonconducting copper oxide between them. The electrons tunnel through this thin insulating layer. A *tunnel diode* is a semiconductor device in which electrons tunnel through a potential barrier. The current can be switched on and off very quickly (within a few picoseconds) by varying the height of the barrier.

Figure 40.21 (a) Schematic diagram of the probe of a scanning tunneling microscope (STM). As the sharp conducting probe is scanned across the surface in the x - and y -directions, it is also moved in the z -direction to maintain a constant tunneling current. The changing position of the probe is recorded and used to construct an image of the surface. (b) This colored STM image shows “quantum wires”: thin strips, just 10 atoms wide, of a conductive rare-earth silicide atop a silicon surface. Such quantum wires may one day be the basis of ultraminiaturized circuits.



BIO APPLICATION Electron Tunneling in Enzymes

Protein molecules play essential roles as enzymes in living organisms. Enzymes like the one shown here are large molecules, and in many cases their function depends on the ability of electrons to tunnel across the space that separates one part of the molecule from another. Without tunneling, life as we know it would be impossible!

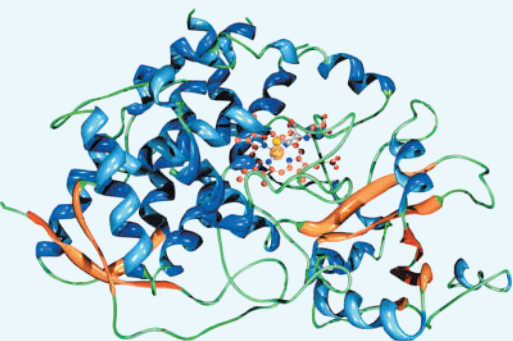
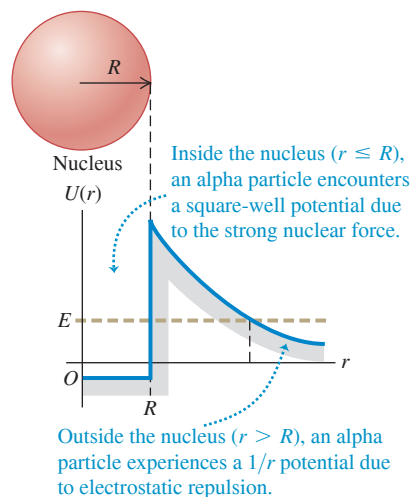


Figure 40.22 Approximate potential-energy function for an alpha particle interacting with a nucleus of radius R . If an alpha particle inside the nucleus has energy E greater than zero, it can tunnel through the barrier and escape from the nucleus.



A *Josephson junction* consists of two superconductors separated by an oxide layer a few atoms (1 to 2 nm) thick. Electron pairs in the superconductors can tunnel through the barrier layer, giving such a device unusual circuit properties. Josephson junctions are useful for establishing precise voltage standards and measuring tiny magnetic fields, and they play a crucial role in the developing field of quantum computing.

The *scanning tunneling microscope* (STM) uses electron tunneling to create images of surfaces down to the scale of individual atoms. An extremely sharp conducting needle is brought very close to the surface, within 1 nm or so (Fig. 40.21a). When the needle is at a positive potential with respect to the surface, electrons can tunnel through the surface potential-energy barrier and reach the needle. As Example 40.7 shows, the tunneling probability and hence the tunneling current are very sensitive to changes in the width L of the barrier (the distance between the surface and the needle tip). In one mode of operation the needle is scanned across the surface while being moved perpendicular to the surface to maintain a constant tunneling current. The needle motion is recorded; after many parallel scans, an image of the surface can be reconstructed. Extremely precise control of needle motion, including isolation from vibration, is essential. Figure 40.21b shows an STM image. (Figure 40.17 is also an STM image.)

Tunneling is also of great importance in nuclear physics. A fusion reaction can occur when two nuclei tunnel through the barrier caused by their electrical repulsion and approach each other closely enough for the attractive nuclear force to cause them to fuse. Fusion reactions occur in the cores of stars, including the sun; without tunneling, the sun wouldn't shine. The emission of alpha particles from unstable nuclei such as radium also involves tunneling. An alpha particle is a cluster of two protons and two neutrons (the same as a nucleus of the most common form of helium). Such clusters form naturally within larger atomic nuclei. An alpha particle trying to escape from a nucleus encounters a potential barrier that results from the combined effect of the attractive nuclear force and the electrical repulsion of the remaining part of the nucleus (Fig. 40.22). To escape, the alpha particle must tunnel through this barrier. Depending on the barrier height and width for a given kind of alpha-emitting nucleus, the tunneling probability can be low or high, and the alpha-emitting material will have low or high radioactivity. Recall from Section 39.2 that Ernest Rutherford used alpha particles from a radioactive source to discover the atomic nucleus. Although Rutherford did not know it, tunneling allowed these alpha particles to escape from their parent nuclei, which made his experiments possible! We'll learn more about alpha decay in Chapter 43.

TEST YOUR UNDERSTANDING OF SECTION 40.4 Is it possible for a particle undergoing tunneling to be found *within* the barrier rather than on either side of it?

ANSWER Yes. Figure 40.20 shows a possible wave function $\psi(x)$ for tunneling. Since $\psi(x)$ is not zero within the barrier ($0 \leq x \leq L$), there is some probability that the particle can be found there.

40.5 THE HARMONIC OSCILLATOR

Systems that *oscillate* are of tremendous importance in the physical world, from the oscillations of your eardrums in response to a sound wave to the vibrations of the ground caused by an earthquake. Oscillations are equally important on the microscopic scale where quantum effects dominate. The molecules of the air around you can be set into vibration when they collide with each other, the protons and neutrons in an excited atomic nucleus can oscillate in opposite directions, and a microwave oven transfers energy to food by making water molecules in the food flip back and forth. In this section we'll look at the solutions of the Schrödinger equation for the simplest kind of vibrating system, the quantum-mechanical harmonic oscillator.

As we learned in Section 14.2, a **harmonic oscillator** is a particle with mass m that moves along the x -axis under the influence of a conservative force $F_x = -k'x$. The constant k' is called the *force constant*. (In Section 14.2 we used the symbol k for the force constant. In this section we'll use the symbol k' instead to minimize confusion with the wave number $k = 2\pi/\lambda$.) The force is proportional to the particle's displacement x from its equilibrium position, $x = 0$. The corresponding potential-energy function is $U = \frac{1}{2}k'x^2$ (**Fig. 40.23**). In Newtonian mechanics, when the particle is displaced from equilibrium, it undergoes sinusoidal motion with frequency $f = (1/2\pi)(k'/m)^{1/2}$ and angular frequency $\omega = 2\pi f = (k'/m)^{1/2}$. The amplitude (that is, the maximum displacement from equilibrium) of these Newtonian oscillations is A , which is related to the energy E of the oscillator by $E = \frac{1}{2}k'A^2$.

Let's make an enlightened guess about the energy levels of a quantum-mechanical harmonic oscillator. In classical physics an electron oscillating with angular frequency ω emits electromagnetic radiation with that same angular frequency. It's reasonable to guess that when an excited quantum-mechanical harmonic oscillator with angular frequency $\omega = (k'/m)^{1/2}$ (according to Newtonian mechanics, at least) makes a transition from one energy level to a lower level, it would emit a photon with this same angular frequency ω . The energy of such a photon is $hf = (2\pi\hbar)(\omega/2\pi) = \hbar\omega$. So we would expect that the spacing between adjacent energy levels of the harmonic oscillator would be

$$hf = \hbar\omega = \hbar \sqrt{\frac{k'}{m}} \quad (40.43)$$

That's the same spacing between energy levels that Planck assumed in deriving his radiation law (see Section 39.5). It was a good assumption; as we'll see, the energy levels are in fact half-integer $(\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots)$ multiples of $\hbar\omega$.

Wave Functions, Boundary Conditions, and Energy Levels

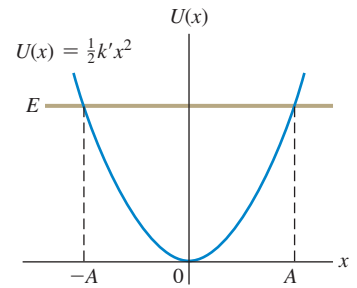
We'll begin our quantum-mechanical analysis of the harmonic oscillator by writing down the one-dimensional time-independent Schrödinger equation, Eq. (40.23), with $\frac{1}{2}k'x^2$ in place of U :

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2}k'x^2\psi(x) = E\psi(x) \quad \text{(Schrödinger equation for the harmonic oscillator)} \quad (40.44)$$

The solutions of this equation are wave functions for the physically possible states of the system.

In the discussion of square-well potentials in Section 40.2 we found that the energy levels are determined by boundary conditions at the walls of the well. However, the harmonic-oscillator potential has no walls as such; what, then, are the appropriate boundary conditions? Classically, $|x|$ cannot be greater than the amplitude A given by $E = \frac{1}{2}k'A^2$. Quantum mechanics does allow some penetration into classically forbidden regions, but the probability decreases as that penetration increases. Thus the wave functions must approach zero as $|x|$ grows large.

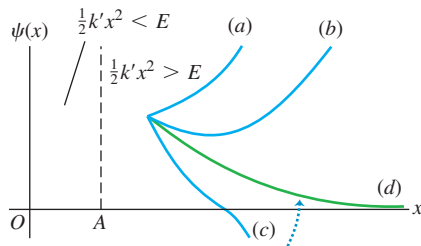
Figure 40.23 Potential-energy function for the harmonic oscillator. In Newtonian mechanics the amplitude A is related to the total energy E by $E = \frac{1}{2}k'A^2$, and the particle is restricted to the range from $x = -A$ to $x = A$. In quantum mechanics the particle can be found at $x > A$ or $x < -A$.



Satisfying the requirement that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$ is not as trivial as it may seem. To see why this is, let's rewrite Eq. (40.44) in the form

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2} \left(\frac{1}{2}k'x^2 - E \right) \psi(x) \quad (40.45)$$

Figure 40.24 Possible behaviors of harmonic-oscillator wave functions in the region $\frac{1}{2}k'x^2 > E$. In this region, $\psi(x)$ and $d^2\psi(x)/dx^2$ have the same sign. The curve is concave upward when $d^2\psi(x)/dx^2$ is positive and concave downward when $d^2\psi(x)/dx^2$ is negative.



Only curve *d*, which approaches the *x*-axis asymptotically for large *x*, is an acceptable wave function for this system.

Equation (40.45) shows that when x is large enough (either positive or negative) to make the quantity $(\frac{1}{2}k'x^2 - E)$ positive, the function $\psi(x)$ and its second derivative $d^2\psi(x)/dx^2$ have the same sign. **Figure 40.24** shows four possible kinds of behavior of $\psi(x)$ beginning at a point where x is greater than the classical amplitude A , so that $\frac{1}{2}k'x^2 - \frac{1}{2}k'A^2 = \frac{1}{2}k'x^2 - E > 0$. Let's look at these four cases more closely. If $\psi(x)$ is positive as shown in Fig. 40.24, Eq. (40.45) tells us that $d^2\psi(x)/dx^2$ is also positive and the function is *concave upward*. Note also that $d^2\psi(x)/dx^2$ is the rate of change of the *slope* of $\psi(x)$; this will help us understand how our four possible wave functions behave.

- **Curve *a*:** The slope of $\psi(x)$ is positive at point x . Since $d^2\psi(x)/dx^2 > 0$, the function curves upward increasingly steeply and goes to infinity. This violates the boundary condition that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$, so this isn't a viable wave function.
- **Curve *b*:** The slope of $\psi(x)$ is negative at point x , and $d^2\psi(x)/dx^2$ has a large positive value. Hence the slope changes rapidly from negative to positive and keeps on increasing—so, again, the wave function goes to infinity. This wave function isn't viable either.
- **Curve *c*:** As for curve *b*, the slope is negative at point x . However, $d^2\psi(x)/dx^2$ now has a *small* positive value, so the slope increases only gradually as $\psi(x)$ decreases to zero and crosses over to negative values. Equation (40.45) tells us that once $\psi(x)$ becomes negative, $d^2\psi(x)/dx^2$ also becomes negative. Hence the curve becomes *concave downward* and heads for *negative* infinity. This wave function, too, fails to satisfy the requirement that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$ and thus isn't viable.
- **Curve *d*:** If the slope of $\psi(x)$ at point x is negative, and the positive value of $d^2\psi(x)/dx^2$ at this point is neither too large nor too small, the curve bends just enough to glide in asymptotically to the x -axis. In this case $\psi(x)$, $d\psi(x)/dx$, and $d^2\psi(x)/dx^2$ all approach zero at large x . This case offers the only hope of satisfying the boundary condition that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$, and it occurs only for certain very special values of the energy E .

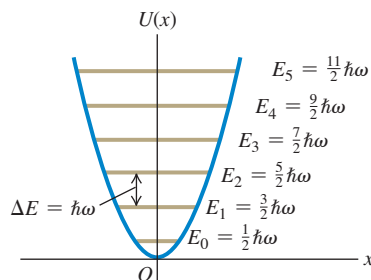
This qualitative discussion suggests how the boundary conditions as $|x| \rightarrow \infty$ determine the possible energy levels for the quantum-mechanical harmonic oscillator. It turns out that these boundary conditions are satisfied only if the energy E is equal to one of the values E_n :

$$E_n = \left(n + \frac{1}{2} \right) \hbar \sqrt{\frac{k'}{m}} = \left(n + \frac{1}{2} \right) \hbar \omega \quad (n = 0, 1, 2, \dots) \quad (40.46)$$

Quantum number
Force constant
Oscillation angular frequency

Planck's constant divided by 2π
Particle's mass

Figure 40.25 Energy levels for the harmonic oscillator. The spacing between any two adjacent levels is $\Delta E = \hbar\omega$. The energy of the ground level is $E_0 = \frac{1}{2}\hbar\omega$.



Note that the ground level of energy $E_0 = \frac{1}{2}\hbar\omega$ is denoted by the quantum number $n = 0$, not $n = 1$.

Equation (40.46) confirms our guess [Eq. (40.43)] that adjacent energy levels are separated by a constant interval of $\hbar\omega = hf$, as Planck assumed in 1900. There are infinitely many levels; this shouldn't be surprising because we are dealing with an infinitely deep potential well. As $|x|$ increases, $U = \frac{1}{2}k'x^2$ increases without bound.

Figure 40.25 shows the lowest six energy levels and the potential-energy function $U(x)$. For each level n , the value of $|x|$ at which the horizontal line representing the total energy E_n intersects $U(x)$ gives the amplitude A_n of the corresponding Newtonian oscillator.

EXAMPLE 40.8 Vibration in a crystal

A sodium atom of mass 3.82×10^{-26} kg vibrates within a crystal. The potential energy increases by 0.0075 eV when the atom is displaced 0.014 nm from its equilibrium position. Treat the atom as a harmonic oscillator. (a) Find the angular frequency of the oscillations according to Newtonian mechanics. (b) Find the spacing (in electron volts) of adjacent vibrational energy levels according to quantum mechanics. (c) What is the wavelength of a photon emitted as the result of a transition from one level to the next lower level? In what region of the electromagnetic spectrum does this lie?

IDENTIFY and SET UP We'll find the force constant k' from the expression $U = \frac{1}{2}k'x^2$ for potential energy. We'll then find the angular frequency $\omega = (k'/m)^{1/2}$ and use this in Eq. (40.46) to find the spacing between adjacent energy levels. We'll calculate the wavelength of the emitted photon as in Example 40.6.

EXECUTE We are given that $U = 0.0075$ eV $= 1.2 \times 10^{-21}$ J when $x = 0.014 \times 10^{-9}$ m, so we can solve $U = \frac{1}{2}k'x^2$ for k' :

$$k' = \frac{2U}{x^2} = \frac{2(1.2 \times 10^{-21} \text{ J})}{(0.014 \times 10^{-9} \text{ m})^2} = 12.2 \text{ N/m}$$

(a) The Newtonian angular frequency is

$$\omega = \sqrt{\frac{k'}{m}} = \sqrt{\frac{12.2 \text{ N/m}}{3.82 \times 10^{-26} \text{ kg}}} = 1.79 \times 10^{13} \text{ rad/s}$$

(b) From Eq. (40.46) and Fig. 40.25, the spacing between adjacent energy levels is

$$\begin{aligned}\hbar\omega &= (1.055 \times 10^{-34} \text{ J}\cdot\text{s})(1.79 \times 10^{13} \text{ s}^{-1}) \\ &= 1.89 \times 10^{-21} \text{ J} \left(\frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} \right) = 0.0118 \text{ eV}\end{aligned}$$

(c) The energy E of the emitted photon is equal to the energy lost by the oscillator in the transition, 0.0118 eV. Then

$$\begin{aligned}\lambda &= \frac{hc}{E} = \frac{(4.136 \times 10^{-15} \text{ eV}\cdot\text{s})(3.00 \times 10^8 \text{ m/s})}{0.0118 \text{ eV}} \\ &= 1.05 \times 10^{-4} \text{ m} = 105 \mu\text{m}\end{aligned}$$

This photon wavelength is in the infrared region of the spectrum.

EVALUATE This example shows us that interatomic force constants are a few newtons per meter, about the same as those of household springs or spring-based toys such as the Slinky®. It also suggests that we can learn about the vibrations of molecules by measuring the radiation that they emit in transitioning to a lower vibrational state. We'll explore this idea further in Chapter 42.

KEYCONCEPT As in Newtonian physics, in a quantum-mechanical harmonic oscillator a particle of mass m moves under the influence of the potential-energy function of an ideal spring with force constant k' . The lowest-energy stationary state of the harmonic oscillator has energy $\frac{1}{2}\hbar\omega$, where $\omega = \sqrt{k'/m}$ is the Newtonian angular frequency; each excited state has $\hbar\omega$ more energy than the state below it.

Comparing Quantum and Newtonian Oscillators

The wave functions for the levels $n = 0, 1, 2, \dots$ of the harmonic oscillator are called *Hermite functions*; they aren't encountered in elementary calculus courses but are well known to mathematicians. Each Hermite function is an exponential function multiplied by a polynomial in x . The harmonic-oscillator wave function corresponding to $n = 0$ and $E = E_0$ (the ground level) is

$$\psi(x) = Ce^{-\sqrt{mk'}x^2/2\hbar} \quad (40.47)$$

The constant C is chosen to normalize the function—that is, to make $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$. (We're using C rather than A as a normalization constant in this section, since we've already appropriated the symbol A to denote the Newtonian amplitude of a harmonic oscillator.) You can find C by using the following result from integral tables:

$$\int_{-\infty}^{\infty} e^{-a^2x^2} dx = \frac{\sqrt{\pi}}{a}$$

To confirm that $\psi(x)$ as given by Eq. (40.47) really is a solution of the Schrödinger equation for the harmonic oscillator, you can calculate the second derivative of this wave function, substitute it into Eq. (40.44), and verify that the equation is satisfied if the energy E is equal to $E_0 = \frac{1}{2}\hbar\omega$. It's a little messy, but the result is satisfying and worth the effort.

Figure 40.26 (next page) shows the the first four harmonic-oscillator wave functions. Each graph also shows the amplitude A of a Newtonian harmonic oscillator with the same energy—that is, the value of A determined from

$$\frac{1}{2}k'A^2 = \left(n + \frac{1}{2}\right)\hbar\omega \quad (40.48)$$

In each case there is some penetration of the wave function into the regions $|x| > A$ that are forbidden by Newtonian mechanics. This is similar to the effect that we noted in Section 40.3 for a particle in a finite square well.

Figure 40.26 The first four wave functions for the harmonic oscillator. The amplitude A of a Newtonian oscillator with the same total energy is shown for each. Each wave function penetrates somewhat into the classically forbidden regions $|x| > A$. The total number of finite maxima and minima for each function is $n + 1$, one more than the quantum number.

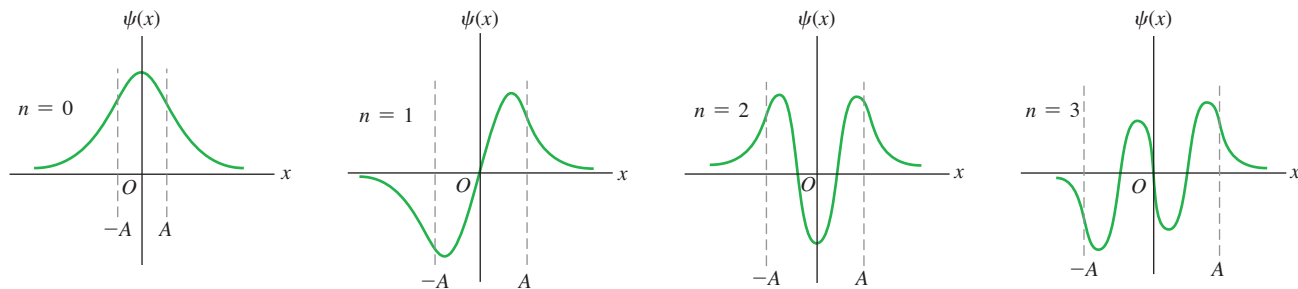


Figure 40.27 Probability distribution functions $|\psi(x)|^2$ for the harmonic-oscillator wave functions shown in Fig. 40.26. The amplitude A of the Newtonian motion with the same energy is shown for each. The blue lines show the corresponding probability distributions for the Newtonian motion. As n increases, the averaged-out quantum-mechanical functions resemble the Newtonian curves more and more.

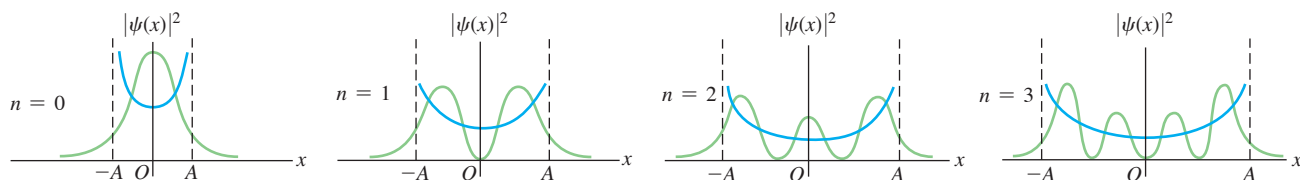


Figure 40.28 Newtonian and quantum-mechanical probability distribution functions for a harmonic oscillator for the state $n = 10$. The Newtonian amplitude A is also shown.

The larger the value of n , the more closely the quantum-mechanical probability distribution (green) matches the Newtonian probability distribution (blue).

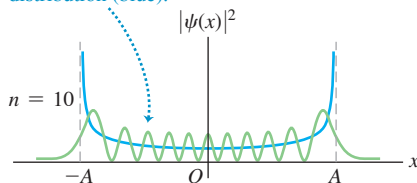


Figure 40.27 shows the probability distributions $|\psi(x)|^2$ for these states. Each graph also shows the probability distribution determined from Newtonian physics, in which the probability of finding the particle near a randomly chosen point is inversely proportional to the particle's speed at that point. If we average out the wiggles in the quantum-mechanical probability curves, the results for $n > 0$ resemble the Newtonian predictions. This agreement improves with increasing n ; Fig. 40.28 shows the classical and quantum-mechanical probability functions for $n = 10$. Notice that the spacing between zeros of $|\psi(x)|^2$ in Fig. 40.28 increases with increasing distance from $x = 0$. This makes sense from the Newtonian perspective: As a particle moves away from $x = 0$, its kinetic energy K and the magnitude p of its momentum both decrease. Thinking quantum-mechanically, this means that the wavelength $\lambda = h/p$ increases, so the spacing between zeros of $\psi(x)$ (and hence of $|\psi(x)|^2$) also increases.

In the Newtonian analysis of the harmonic oscillator the minimum energy is zero, with the particle at rest at its equilibrium position $x = 0$. This is not possible in quantum mechanics; no solution of the Schrödinger equation has $E = 0$ and satisfies the boundary conditions. Furthermore, such a state would violate the Heisenberg uncertainty principle because there would be no uncertainty in either position or momentum. The energy must be at least $\frac{1}{2}\hbar\omega$ for the system to conform to the uncertainty principle. To see qualitatively why this is so, consider a Newtonian oscillator with total energy $\frac{1}{2}\hbar\omega$. We can find the amplitude A and the maximum velocity just as we did in Section 14.3. When the particle is at its maximum displacement ($x = \pm A$) and instantaneously at rest, $K = 0$ and $E = U = \frac{1}{2}k'A^2$. When the particle is at equilibrium ($x = 0$) and moving at its maximum speed, $U = 0$ and $E = K = \frac{1}{2}mv_{\max}^2$. Setting $E = \frac{1}{2}\hbar\omega$, we find

$$E = \frac{1}{2}k'A^2 = \frac{1}{2}\hbar\omega = \frac{1}{2}\hbar\left(\frac{k'}{m}\right)^{1/2} \quad \text{so} \quad A = \frac{\hbar^{1/2}}{k'^{1/4}m^{1/4}}$$

$$E = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}k'A^2 \quad \text{so} \quad v_{\max} = A\left(\frac{k'}{m}\right)^{1/2} = \frac{\hbar^{1/2}k'^{1/4}}{m^{3/4}}$$

The maximum *momentum* of the particle is

$$p_{\max} = mv_{\max} = \hbar^{1/2}k'^{1/4}m^{1/4}$$

Here's where the Heisenberg uncertainty principle comes in. It turns out that the uncertainties in the particle's position and momentum (calculated as standard deviations) are, respectively, $\Delta x = A/\sqrt{2} = A/2^{1/2}$ and $\Delta p_x = p_{\max}/\sqrt{2} = p_{\max}/2^{1/2}$. Then the product of the two uncertainties is

$$\Delta x \Delta p_x = \left(\frac{\hbar^{1/2}}{2^{1/2} k^{1/4} m^{1/4}} \right) \left(\frac{\hbar^{1/2} k^{1/4} m^{1/4}}{2^{1/2}} \right) = \frac{\hbar}{2}$$

This product equals the minimum value allowed by Eq. (39.29), $\Delta x \Delta p_x \geq \hbar/2$, and thus satisfies the uncertainty principle. If the energy had been less than $\frac{1}{2}\hbar\omega$, the product $\Delta x \Delta p_x$ would have been less than $\hbar/2$, and the uncertainty principle would have been violated.

Even when a potential-energy function isn't precisely parabolic in shape, we may be able to approximate it by the harmonic-oscillator potential for sufficiently small displacements from equilibrium. **Figure 40.29** shows a typical potential-energy function for an interatomic force in a molecule. At large separations the curve of $U(r)$ versus r levels off, corresponding to the absence of force at great distances. But the curve is approximately parabolic near the minimum of $U(r)$ (the equilibrium separation of the atoms). Near equilibrium the molecular vibration is approximately simple harmonic with energy levels given by Eq. (40.46), as we assumed in Example 40.8.

TEST YOUR UNDERSTANDING OF SECTION 40.5 A quantum-mechanical system initially in its ground level absorbs a photon and ends up in the first excited state. The system then absorbs a second photon and ends up in the second excited state. For which of the following systems does the second photon have a longer wavelength than the first one? (i) A harmonic oscillator; (ii) a hydrogen atom; (iii) a particle in a box.

ANSWER

(iii) If the second photon has a longer wavelength and hence lower energy than the first photon, the difference in energy between the first and second excited levels must be less than the difference between the ground level and the first excited level. This is the case for the hydrogen atom, for which the energy difference between levels decreases as the energy increases (see Fig. 39.24). By contrast, the energy difference between successive levels increases for a particle in a box (see Fig. 40.11b) and is constant for a harmonic oscillator (see Fig. 40.25).

40.6 MEASUREMENT IN QUANTUM MECHANICS

We've seen how to use the Schrödinger equation to calculate the stationary-state wave functions and energy levels for various potential-energy functions $U(x)$. We've also seen how to interpret the wave function $\Psi(x, t)$ of a particle in terms of the probability distribution function $|\Psi(x, t)|^2$. We'll conclude with a brief discussion of what happens when we try to *measure* the properties of a quantum-mechanical particle. As we'll see, the consequences of such a measurement can be startlingly different from what happens when we measure the properties of a familiar Newtonian particle, such as a marble or billiard ball.

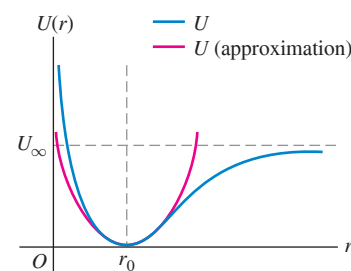
Let's consider a "particle in a box"—that is, a particle in an infinite square well of width L , as described in Section 40.2. This particle of mass m is free to move along the x -axis in the region $0 \leq x \leq L$ but cannot move beyond this region. Let's suppose the particle is in a stationary state with definite energy E , equal to one of the energy levels E_n given by Eq. (40.31). If we measure the x -component of momentum of this particle, what is the result?

First let's consider the answer to that question for a Newtonian particle in a box (see Fig. 40.8). This could be a hockey puck sliding on frictionless ice and bouncing back and forth between two parallel walls. The energy E of the puck is equal to its kinetic energy $p^2/2m$, so the magnitude of its momentum is $p = \sqrt{2mE}$. The x -component of its momentum p_x is therefore

$$p_x = +\sqrt{2mE} \quad \text{or} \quad p_x = -\sqrt{2mE} \quad (40.49)$$

Whether p_x is positive or negative depends on whether the hockey puck is moving in the $+x$ -direction (then $p_x = +\sqrt{2mE}$) or the $-x$ -direction (then $p_x = -\sqrt{2mE}$). To

Figure 40.29 A potential-energy function describing the interaction of two atoms in a diatomic molecule. The distance r is the separation between the centers of the atoms, and the equilibrium separation is $r = r_0$. The energy needed to dissociate the molecule is U_∞ .



When r is near r_0 , the potential-energy curve is approximately parabolic (as shown by the red curve) and the system behaves approximately like a harmonic oscillator.

determine which value of p_x is correct at a given time, we need only look at the puck to see in which direction it's moving.

We can't make such an observation in the dark; we need to shine some light on the hockey puck. We know from Section 38.1 that light comes in the form of photons and that a photon of wavelength λ has momentum $p = h/\lambda$. When we shine light on the puck to observe it, the photons collide with the puck and *change* its momentum. The mere act of measuring the puck's momentum can affect the quantity that we're trying to measure! The good news is that this change is minuscule: A hockey puck of mass $m = 0.165$ kg moving at speed $v = 1.00$ m/s has momentum $p = mv = 0.165$ kg · m/s, while a visible-light photon of wavelength 500 nm has momentum $p = h/\lambda = 1.33 \times 10^{-27}$ kg · m/s. Even if we directed all of the photons from a 100 W light source onto the puck for a 1.00 s burst of light, the total momentum in this burst would be only 3.33×10^{-7} kg · m/s, and the resulting change in the momentum of the puck would be negligible. In general, we can measure any of the properties of a Newtonian particle—its momentum, position, energy, and so on—without appreciably changing the quantity that we are measuring.

The situation is very different for a quantum-mechanical particle in a box. From Eq. (40.21) the state of such a particle with energy $E = E_n$ is described by the wave function

$$\Psi(x, t) = \psi_n(x)e^{-iE_nt/\hbar} = \psi_n(x)e^{-i\omega_nt} \quad (40.50)$$

In Eq. (40.50) the angular frequency is $\omega_n = E_n/\hbar$ and the time-independent, stationary-state wave function $\psi_n(x)$ is given by Eq. (40.35):

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \quad (n = 1, 2, 3, \dots) \quad (40.51)$$

This is a state of definite *energy*, but it is *not* a state of definite momentum: It represents a standing wave with equal amounts of momentum in the $+x$ -direction and the $-x$ -direction. To make this more explicit, recall from Eqs. (40.30) and (40.31) that the magnitude of momentum in a state of energy E_n is $p_n = \sqrt{2mE_n} = nh/2L = n\pi\hbar/L$, and the corresponding wave number is $k_n = p_n/\hbar = n\pi/L$. So we can replace $n\pi/L$ in Eq. (40.51) with k_n :

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin k_n x$$

Recall also Euler's formula from Eq. (40.17): $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$. Hence $\sin\theta = (e^{i\theta} - e^{-i\theta})/2i$, and we can write

$$\psi_n(x) = \sqrt{\frac{2}{L}} \left(\frac{e^{ik_n x} - e^{-ik_n x}}{2i} \right) = \frac{1}{i\sqrt{2L}} (e^{ik_n x} - e^{-ik_n x}) \quad (40.52)$$

Now we substitute Eq. (40.52) into Eq. (40.50) and distribute the factors $1/i\sqrt{2L}$ and $e^{-i\omega_nt}$:

$$\begin{aligned} \Psi(x, t) &= \frac{1}{i\sqrt{2L}} (e^{ik_n x} - e^{-ik_n x}) e^{-i\omega_nt} \\ &= \frac{1}{i\sqrt{2L}} e^{ik_n x} e^{-i\omega_nt} - \frac{1}{i\sqrt{2L}} e^{-ik_n x} e^{-i\omega_nt} \end{aligned} \quad (40.53)$$

In Eq. (40.53) the $e^{ik_n x} e^{-i\omega_nt}$ term is a wave function for a free particle with energy $E_n = \hbar\omega_n$ and a *positive* x -component of momentum $p_x = p_n = \hbar k_n$. In the $e^{-ik_n x} e^{-i\omega_nt}$ term, k_n is replaced by $-k_n$, so this term is a wave function for a free particle with the same energy $E_n = \hbar\omega_n$ but a *negative* x -component of momentum $p_x = -p_n = -\hbar k_n$. These two possible values of p_x are the same as for a Newtonian particle in a box, Eq. (40.49). The difference is that as the Newtonian particle bounces back and forth between the walls of the box, it has positive p_x half of the time and negative p_x half of the time. Only its time-averaged value of p_x is zero. But because both terms for positive p_x and negative p_x are present in Eq. (40.53), the quantum-mechanical particle has *both* signs of the x -component of momentum present at *all* times. As we stated earlier, this stationary state for a quantum-mechanical particle in a box has a definite energy [both terms in Eq. (40.53) have the same value of ω_n

and hence the same value of $E_n = \hbar\omega_n$] but does not have a definite momentum. Because the $e^{ik_n x}e^{-i\omega_n t}$ and $e^{-ik_n x}e^{-i\omega_n t}$ terms in Eq. (40.53) have coefficients of the same magnitude, $1/\sqrt{2L}$, the *instantaneous* average value of p_x for the quantum-mechanical particle is zero (the average of $\hbar k_n$ and $-\hbar k_n$) at *all* times.

What value do we get if we *measure* the momentum of the quantum-mechanical particle in a box? As for the Newtonian particle, we can measure the momentum by shining a light on it. Let's fire a single photon, moving in the $-y$ -direction, at the particle and allow the photon and particle to collide (**Fig. 40.30**). Before the collision the total x -component of momentum of the system of photon and particle is zero. Momentum is conserved in the collision, so the same is true after the collision. After the collision, whichever sign of p_x the photon has, the x -component of momentum of the particle will have the opposite sign. If the photon is detected in detector A, we conclude that the particle has $p_x = \hbar k_n$; if instead the photon is detected in detector B, we conclude that the particle has $p_x = -\hbar k_n$.

In this experiment, we need to be even more concerned about how the photon changes the momentum of the particle than in the Newtonian case. For an electron in a box of width $L = 1.00 \times 10^{-6} \text{ m} = 1.00 \mu\text{m}$, the electron momentum has a minimum magnitude of $p = 3.31 \times 10^{-28} \text{ kg} \cdot \text{m/s}$ (corresponding to the $n = 1$ energy level), which is only about one-quarter that of a visible-light photon of wavelength 500 nm. To minimize the change in the electron's magnitude of momentum due to the collision, we should use a photon of much longer wavelength (say, a radio-wave photon) and hence much smaller momentum.

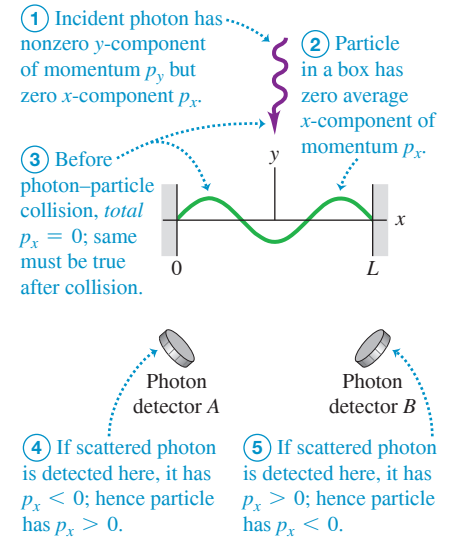
Even when we use a photon with the lowest possible momentum, however, we find that the state of the particle in the box *must* change as a result of the experiment. Here's a summary of the results:

1. If the measurement shows that the particle has positive $p_x = \hbar k_n$, the wave function *changes* from that given in Eq. (40.53) to one with an $e^{ik_n x}e^{-i\omega_n t}$ term *only*. The other term, which corresponds to $p_x = -\hbar k_n$, disappears. We say that the wave function, which was a combination of two terms with different values of p_x , has undergone *wave-function collapse*—it has collapsed to one term with $p_x = \hbar k_n$ as a consequence of measuring the value of p_x . To test this result, we fire a second photon immediately after the first. The second photon scatters from the particle as we would expect if the particle had the value $p_x = \hbar k_n$.
2. If the measurement shows that the particle has negative $p_x = -\hbar k_n$, the wave function collapses in the opposite way: It changes to one with an $e^{-ik_n x}e^{-i\omega_n t}$ term only. The $p_x = \hbar k_n$ term disappears.
3. If we repeat the experiment many times, each time starting with the particle described by the wave function in Eq. (40.53), 50% of the time we measure the particle to have $p_x = \hbar k_n$ and 50% of the time we measure the particle to have $p_x = -\hbar k_n$. For any given time that we try the experiment, there is no way to predict which outcome will occur. We can state only that there is equal probability of either outcome.

These results reveal a fact of quantum-mechanical life: *Measuring a physical property of a system can change the wave function of that system.* By measuring the value of p_x for a particle in a box, we changed the wave function from one that was a combination of two wave functions, one for $p_x = \hbar k_n$ and one for $p_x = -\hbar k_n$, to one with a definite value of p_x . This change in the wave function is not described by the time-dependent Schrödinger equation [Eq. (40.20)] but is a consequence of the measurement process. It is also independent of how the measurement is carried out: No matter how small the momentum of the incident photon shown in Fig. 40.30, the same collapse of the wave function takes place. Indeed, *any* experiment to measure p_x for a particle in a box in a steady state, no matter how the experiment is designed, will have the results that we described earlier.

(After the measurement, the wave function will undergo further change that is described by the Schrödinger equation. Neither $e^{ik_n x}e^{-i\omega_n t}$ nor $e^{-ik_n x}e^{-i\omega_n t}$ by itself satisfies the boundary conditions for a particle in a box—namely, that the wave function vanishes at $x = 0$ and $x = L$. The wave function must evolve to satisfy these conditions.)

Figure 40.30 Using photon scattering to measure the x -component of momentum of a particle in a box.



CAUTION Quantum measurement misconceptions If we measure the particle to have $p_x = \hbar k_n$, does that mean it had $p_x = \hbar k_n$ before the measurement? No; the particle acquired that value as a result of the measurement. If we measure the particle to have $p_x = \hbar k_n$ instead of $p_x = -\hbar k_n$, does that mean there was some bias in the way we did the measurement? Again, no; the result of any given experiment is random. All quantum mechanics can do is predict the probability that this experiment will give us a certain result. ■

Note that not every measurement of a quantum-mechanical system causes a change in the wave function. If we perform an experiment that measures only the *energy* of a particle given by the wave function in Eq. (40.53), the wave function does *not* change. That's because the wave function already corresponds to a state of definite energy $E_n = \hbar\omega_n$, so there is a 100% probability that we'll measure that value of energy.

You may ask, Does the wave function really collapse? Many physicists would answer yes, but some theorists have devised alternative models of what happens in a quantum-mechanical measurement. One model, called the *many-worlds interpretation*, asserts that there is a *universal* wave function that describes all particles in the universe. Whenever a measurement of any sort takes place, whether of human origin (like our experiment) or natural origin (for example, a photon of sunlight scattering from an electron in an atom in the atmosphere), this universal wave function does not collapse. Instead, every measurement causes the universe to branch into alternative timelines. So, when we carry out the experiment depicted in Fig. 40.30, the universe splits into one timeline in which the photon goes into detector A and a second timeline in which the photon goes into detector B. These two timelines then no longer communicate.

As weird as these aspects of quantum mechanics are, others are far weirder. We'll investigate these in Chapter 41 after we have learned more about the nature of the electron.

TEST YOUR UNDERSTANDING OF SECTION 40.6 A particle in a box is described by a wave function that is a combination of the $n = 1$ and $n = 2$ stationary states: $\Psi(x, t) = C\psi_1(x)e^{-iE_1t/\hbar} + D\psi_2(x)e^{-iE_2t/\hbar}$, where $\psi_1(x)$ and $\psi_2(x)$ are given by Eq. (40.35), E_1 and E_2 are given by Eq. (40.31), and C and D are nonzero constants. If you carry out an experiment to measure the energy of this particle, the result is *guaranteed* to be (i) E_1 ; (ii) E_2 ; (iii) $(E_1 + E_2)/2$; (iv) intermediate between E_1 and E_2 , with a value that depends on the values of C and D ; (v) none of these.

ANSWER

(v) The value of the energy of a particle in a box must be equal to one of the allowed energy levels, so the measured value will be either E_1 or E_2 . *Neither* of these results is guaranteed. If $|C| = |D|$, then E_1 and E_2 are of equal probability. E_1 is the more likely result if $|C| > |D|$, and E_2 is the more likely result if $|C| < |D|$.

CHAPTER 40 SUMMARY

Wave functions: The wave function for a particle contains all of the information about that particle. If the particle moves in one dimension in the presence of a potential-energy function $U(x)$, the wave function $\Psi(x, t)$ obeys the one-dimensional Schrödinger equation. (For a *free* particle on which no forces act, $U(x) = 0$.) The quantity $|\Psi(x, t)|^2$, called the probability distribution function, determines the relative probability of finding a particle near a given position at a given time. If the particle is in a state of definite energy, called a stationary state, $\Psi(x, t)$ is a product of a function $\psi(x)$ that depends on only spatial coordinates and a function $e^{-iEt/\hbar}$ that depends on only time. For a stationary state, the probability distribution function is independent of time.

A spatial stationary-state wave function $\psi(x)$ for a particle that moves in one dimension in the presence of a potential-energy function $U(x)$ satisfies the time-independent Schrödinger equation. More complex wave functions can be constructed by superposing stationary-state wave functions. These can represent particles that are localized in a certain region, thus representing both particle and wave aspects. (See Examples 40.1 and 40.2.)

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + U(x)\Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t} \quad (40.20)$$

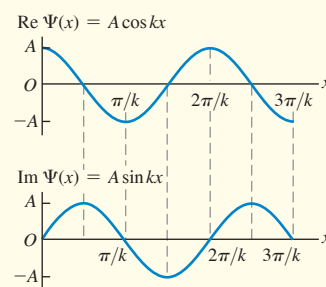
(general 1-D Schrödinger equation)

$$\Psi(x, t) = \psi(x)e^{-iEt/\hbar} \quad (40.21)$$

(time-dependent wave function for a state of definite energy)

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + U(x)\psi(x) = E\psi(x) \quad (40.23)$$

(time-independent 1-D Schrödinger equation)



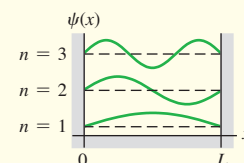
Particle in a box: The energy levels for a particle of mass m in a box (an infinitely deep square potential well) with width L are given by Eq. (40.31). The corresponding normalized stationary-state wave functions of the particle are given by Eq. (40.35). (See Examples 40.3 and 40.4.)

$$E_n = \frac{p_n^2}{2m} = \frac{n^2 h^2}{8mL^2} = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad (40.31)$$

$$(n = 1, 2, 3, \dots)$$

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \quad (40.35)$$

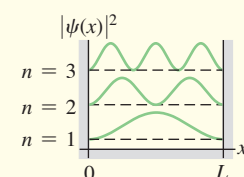
$$(n = 1, 2, 3, \dots)$$



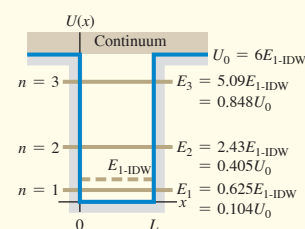
Wave functions and normalization: To be a solution of the Schrödinger equation, the wave function $\psi(x)$ and its derivative $d\psi(x)/dx$ must be continuous everywhere except where the potential-energy function $U(x)$ has an infinite discontinuity. Wave functions are usually normalized so that the total probability of finding the particle somewhere is unity.

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1 \quad (40.33)$$

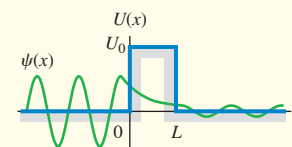
(normalization condition)



Finite potential well: In a potential well with finite depth U_0 , the energy levels are lower than those for an infinitely deep well with the same width, and the number of energy levels corresponding to bound states is finite. The levels are obtained by matching wave functions at the well walls to satisfy the continuity of $\psi(x)$ and $d\psi(x)/dx$. (See Examples 40.5 and 40.6.)



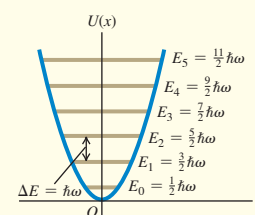
Potential barriers and tunneling: There is a certain probability that a particle will penetrate a potential-energy barrier even though its initial energy is less than the barrier height. This process is called tunneling. (See Example 40.7.)



Quantum harmonic oscillator: The energy levels for the harmonic oscillator (for which $U(x) = \frac{1}{2}k'x^2$) are given by Eq. (40.46). The spacing between any two adjacent levels is $\hbar\omega$, where $\omega = \sqrt{k'/m}$ is the oscillation angular frequency of the corresponding Newtonian harmonic oscillator. (See Example 40.8.)

$$E_n = \left(n + \frac{1}{2}\right) \hbar \sqrt{\frac{k'}{m}} = \left(n + \frac{1}{2}\right) \hbar \omega \quad (40.46)$$

$$(n = 0, 1, 2, 3, \dots)$$



Measurement in quantum mechanics: If the wave function of a particle does not correspond to a definite value of a certain physical property (such as momentum or energy), the wave function changes when we measure that property. This phenomenon is called wave-function collapse.

Chapter 40 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 40.3 and 40.4** (Section 40.2) before attempting these problems.

VP40.4.1 The lowest energy level for an electron confined to a one-dimensional box is 2.00×10^{-19} J. Find (a) the width of the box and (b) the energies of the $n = 2$ and $n = 3$ energy levels.

VP40.4.2 A proton (mass 1.67×10^{-27} kg) is confined to a one-dimensional box of width 5.00×10^{-15} m. Find the energy difference between (a) the $n = 2$ and $n = 1$ energy levels and (b) the $n = 3$ and $n = 2$ energy levels.

VP40.4.3 A photon is emitted when an electron in a one-dimensional box transitions from the $n = 2$ energy level to the $n = 1$ energy level.

The wavelength of this photon is 655 nm. Find (a) the energy of this photon, (b) the width of the box, and (c) the wavelength of the photon emitted when the electron transitions from the $n = 3$ level to the $n = 2$ level.

VP40.4.4 A wave function for a particle in a box with energy E is

$$\psi(x) = A \cos\left(\frac{x\sqrt{2mE}}{\hbar} + \phi\right)$$

(a) Find the second derivative of $\psi(x)$ and show that this function satisfies the Schrödinger equation. (b) Use the boundary condition at $x = 0$ to determine the value of ϕ . (c) Use the boundary condition at $x = L$ to determine the value of E .

Be sure to review EXAMPLE 40.6 (Section 40.3) **before attempting these problems.**

VP40.6.1 An electron is confined to a potential well of width 0.350 nm. (a) Find the ground-level energy E_{1-IDW} if the well is infinitely deep. If instead the depth U_0 of the well is six times the value of E_{1-IDW} , find (b) the ground-level energy and (c) the minimum energy required to free the electron from the well.

VP40.6.2 A finite potential well has a depth U_0 that is six times greater than the ground-level energy E_{1-IDW} of an electron in an infinitely deep well of the same width. The photon emitted when an electron in the finite potential well transitions from the $n = 3$ level to the $n = 2$ level has energy 2.50×10^{-19} J. Find (a) the value of E_{1-IDW} , (b) the value of U_0 , and (c) the width of the potential well.

VP40.6.3 An electron in a finite potential well can emit a photon of only one of three different wavelengths when it makes a transition from one energy level to a lower energy level. The depth of this potential well

is six times greater than the ground-level energy E_{1-IDW} that the electron would have if the well had the same width but was infinitely deep. (a) If the shortest of the three wavelengths is 355 nm, find the quantum numbers n for the initial and final electron energy levels for this transition. (b) Find the other two wavelengths and the quantum numbers n of the initial and final energy levels for each transition.

VP40.6.4 An electron is in a potential well that is 0.400 nm wide. (a) Find the ground-level energy of the electron E_{1-IDW} , in eV, if the potential well is infinitely deep. (b) If the depth of the potential is only 0.015 eV, much less than E_{1-IDW} , there is only one bound state. Find the energy, in eV, of this bound state.

Be sure to review EXAMPLE 40.7 (Section 40.4) **before attempting these problems.**

VP40.7.1 An electron of energy 3.75 eV encounters a potential barrier 6.10 eV high. Find the probability that the electron will tunnel through the barrier if the barrier width is (a) 0.750 nm and (b) 0.500 nm.

VP40.7.2 A 3.50 eV electron encounters a barrier with height 4.00 eV. Find (a) the probability that the electron will tunnel through the barrier if its width is 0.800 nm and (b) the barrier width that gives twice the tunneling probability you found in part (a).

VP40.7.3 A potential barrier is 5.00 eV high and 0.900 nm wide. Find the probability that an electron will tunnel through this barrier if its energy is (a) 4.00 eV, (b) 4.30 eV, and (c) 4.60 eV.

VP40.7.4 There is a 1-in-417 probability that an electron with energy 3.00 eV will tunnel through a barrier with height 5.00 eV. Find the width of the barrier.

BRIDGING PROBLEM A Packet in a Box

A particle of mass m in an infinitely deep well (see Fig. 40.9) has the following wave function in the region from $x = 0$ to $x = L$:

$$\Psi(x, t) = \frac{1}{\sqrt{2}}\psi_1(x)e^{-iE_1t/\hbar} + \frac{1}{\sqrt{2}}\psi_2(x)e^{-iE_2t/\hbar}$$

Here $\psi_1(x)$ and $\psi_2(x)$ are the normalized stationary-state wave functions for the first two levels ($n = 1$ and $n = 2$), given by Eq. (40.35). E_1 and E_2 , given by Eq. (40.31), are the energies of these levels. The wave function is zero for $x < 0$ and for $x > L$. (a) Find the probability distribution function for this wave function. (b) Does $\Psi(x, t)$ represent a stationary state of definite energy? How can you tell? (c) Show that the wave function $\Psi(x, t)$ is normalized. (d) Find the angular frequency of oscillation of the probability distribution function. What is the interpretation of this oscillation? (e) Suppose instead that $\Psi(x, t)$ is a combination of the wave functions of the two lowest levels of a finite well of length L and height U_0 equal to six times the energy of the lowest-energy bound state of an infinite well of length L . What would be the angular frequency of the probability distribution function in this case?

SOLUTION GUIDE

IDENTIFY and SET UP

1. In Section 40.1 we saw how to interpret a combination of two free-particle wave functions of different energies. In this problem you need to apply these same ideas to a combination of wave functions for the infinite well (Section 40.2) and the finite well (Section 40.3).

EXECUTE

2. Write down the full time-dependent wave function $\Psi(x, t)$ and its complex conjugate $\Psi^*(x, t)$ by using the functions $\psi_1(x)$ and $\psi_2(x)$ from Eq. (40.35). Use these to calculate the probability

distribution function, and decide whether or not this function depends on time.

3. To check for normalization, you'll need to verify that when you integrate the probability distribution function from step 2 over all values of x , the integral is equal to 1. [Hint: The trigonometric identities $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ and $\sin\theta \sin\phi = \cos(\theta - \phi) - \cos(\theta + \phi)$ may be helpful.]
4. To find the answer to part (d) you'll need to identify the oscillation angular frequency ω_{osc} in your expression from step 2 for the probability distribution function. To interpret the oscillations, draw graphs of the probability distribution functions at times $t = 0$, $t = T/4$, $t = T/2$, and $t = 3T/4$, where $T = 2\pi/\omega_{\text{osc}}$ is the oscillation period of the probability distribution function.
5. For the finite well you do not have simple expressions for the first two stationary-state wave functions $\psi_1(x)$ and $\psi_2(x)$. However, you can still find the oscillation angular frequency ω_{osc} , which is related to the energies E_1 and E_2 in the same way as for the infinite-well case. (Can you see why?)

EVALUATE

6. Why are the factors of $1/\sqrt{2}$ in the wave function $\Psi(x, t)$ important?
7. Why do you suppose the oscillation angular frequency for a finite well is lower than for an infinite well of the same width?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q40.1 If quantum mechanics replaces the language of Newtonian mechanics, why don't we have to use wave functions to describe the motion of macroscopic objects such as baseballs and cars?

Q40.2 A student remarks that the relationship of ray optics to the more general wave picture is analogous to the relationship of Newtonian mechanics, with well-defined particle trajectories, to quantum mechanics. Comment on this remark.

Q40.3 As Eq. (40.21) indicates, the time-dependent wave function for a stationary state is a complex number having a real part and an imaginary part. How can this function have any physical meaning, since part of it is *imaginary*?

Q40.4 Why must the wave function of a particle be normalized?

Q40.5 If a particle is in a stationary state, does that mean that the particle is not moving? If a particle moves in empty space with constant momentum $\vec{p}$ and hence constant energy $E = p^2/2m$, is it in a stationary state? Explain your answers.

Q40.6 For the particle in a box, we chose $k = n\pi/L$ with $n = 1, 2, 3, \dots$ to fit the boundary condition that $\psi = 0$ at $x = L$. However, $n = 0, -1, -2, -3, \dots$ also satisfy that boundary condition. Why didn't we also choose those values of n ?

Q40.7 If ψ is normalized, what is the physical significance of the area under a graph of $|\psi|^2$ versus x between x_1 and x_2 ? What is the total area under the graph of $|\psi|^2$ when all x are included? Explain.

Q40.8 For a particle in a box, what would the probability distribution function $|\psi|^2$ look like if the particle behaved like a classical (Newtonian) particle? Do the actual probability distributions approach this classical form when n is very large? Explain.

Q40.9 In Chapter 15 we represented a standing wave as a superposition of two waves traveling in opposite directions. Can the wave functions for a particle in a box also be thought of as a combination of two traveling waves? Why or why not? What physical interpretation does this representation have? Explain.

Q40.10 A particle in a box is in the ground level. What is the probability of finding the particle in the right half of the box? (Refer to Fig. 40.12, but don't evaluate an integral.) Is the answer the same if the particle is in an excited level? Explain.

Q40.11 The wave functions for a particle in a box (see Fig. 40.12a) are zero at certain points. Does this mean that the particle can't move past one of these points? Explain.

Q40.12 For a particle confined to an infinite square well, is it correct to say that each state of definite energy is also a state of definite wavelength? Is it also a state of definite momentum? Explain. (*Hint*: Remember that momentum is a vector.)

Q40.13 For a particle in a finite potential well, is it correct to say that each bound state of definite energy is also a state of definite wavelength? Is it a state of definite momentum? Explain.

Q40.14 In Fig. 40.12b, the probability function is zero at the points $x = 0$ and $x = L$, the "walls" of the box. Does this mean that the particle never strikes the walls? Explain.

Q40.15 A particle is confined to a finite potential well in the region $0 < x < L$. How does the area under the graph of $|\psi|^2$ in the region $0 < x < L$ compare to the total area under the graph of $|\psi|^2$ when including all possible x ?

Q40.16 Compare the wave functions for the first three energy levels for a particle in a box of width L (see Fig. 40.12a) to the corresponding

wave functions for a finite potential well of the same width (see Fig. 40.15a). How does the wavelength in the interval $0 \leq x \leq L$ for the $n = 1$ level of the particle in a box compare to the corresponding wavelength for the $n = 1$ level of the finite potential well? Use this to explain why E_1 is less than E_{1-IDW} in the situation depicted in Fig. 40.15b.

Q40.17 It is stated in Section 40.3 that a finite potential well always has at least one bound level, no matter how shallow the well. Does this mean that as $U_0 \rightarrow 0$, $E_1 \rightarrow 0$? Does this violate the Heisenberg uncertainty principle? Explain.

Q40.18 Figure 40.15a shows that the higher the energy of a bound state for a finite potential well, the more the wave function extends outside the well (into the intervals $x < 0$ and $x > L$). Explain why this happens.

Q40.19 In classical (Newtonian) mechanics, the total energy E of a particle can never be less than the potential energy U because the kinetic energy K cannot be negative. Yet in barrier tunneling (see Section 40.4) a particle passes through regions where E is less than U . Is this a contradiction? Explain.

Q40.20 Figure 40.17 shows the scanning tunneling microscope image of 48 iron atoms placed on a copper surface, the pattern indicating the density of electrons on the copper surface. What can you infer about the potential-energy function inside the circle of iron atoms?

Q40.21 Qualitatively, how would you expect the probability for a particle to tunnel through a potential barrier to depend on the height of the barrier? Explain.

Q40.22 The wave function shown in Fig. 40.20 is nonzero for both $x < 0$ and $x > L$. Does this mean that the particle splits into two parts when it strikes the barrier, with one part tunneling through the barrier and the other part bouncing off the barrier? Explain.

Q40.23 The probability distributions for the harmonic-oscillator wave functions (see Figs. 40.27 and 40.28) begin to resemble the classical (Newtonian) probability distribution when the quantum number n becomes large. Would the distributions become the same as in the classical case in the limit of very large n ? Explain.

Q40.24 In Fig. 40.28, how does the probability of finding a particle in the center half of the region $-A < x < A$ compare to the probability of finding the particle in the outer half of the region? Is this consistent with the physical interpretation of the situation?

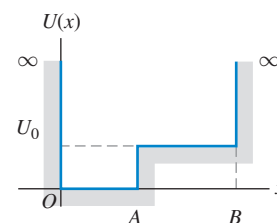
Q40.25 Compare the allowed energy levels for the hydrogen atom, the particle in a box, and the harmonic oscillator. What are the values of the quantum number n for the ground level and the second excited level of each system?

Q40.26 Sketch the wave function for Figure **Q40.26**

the potential-energy well shown in **Fig. Q40.26** when E_1 is less than U_0 and when E_3 is greater than U_0 .

Q40.27 (a) A particle in a box has wave function $\Psi(x, t) = \psi_2(x)e^{-iE_2t/\hbar}$, where ψ_n and E_n are given by Eqs. (40.35) and (40.31), respectively. If the energy of the particle is measured, what is the result?

(b) If instead the particle has wave function $\Psi(x, t) = (1/\sqrt{2})(\psi_1(x)e^{-iE_1t/\hbar} + \psi_2(x)e^{-iE_2t/\hbar})$ and the energy of the particle is measured, what is the result? (c) If we had many identical particles with the wave function of part (b) and measured the energy of each, what would be the average value of all of the measurements? Can we say that, before the measurement was made, each particle had this average energy? Explain.



EXERCISES

Section 40.1 Wave Functions and the One-Dimensional Schrödinger Equation

40.1 • An electron is moving as a free particle in the $-x$ -direction with momentum that has magnitude $4.50 \times 10^{-24} \text{ kg} \cdot \text{m/s}$. What is the one-dimensional time-dependent wave function of the electron?

40.2 • A free particle moving in one dimension has wave function

$$\Psi(x, t) = A[e^{i(kx - \omega t)} - e^{i(2kx - 4\omega t)}]$$

where k and ω are positive real constants. (a) At $t = 0$ what are the two smallest positive values of x for which the probability function $|\Psi(x, t)|^2$ is a maximum? (b) Repeat part (a) for time $t = 2\pi/\omega$. (c) Calculate v_{av} as the distance the maxima have moved divided by the elapsed time. Compare your result to the expression $v_{av} = (\omega_2 - \omega_1)/(k_2 - k_1)$ from Example 40.1.

40.3 • Consider the free-particle wave function of Example 40.1. Let $k_2 = 3k_1 = 3k$. At $t = 0$ the probability distribution function $|\Psi(x, t)|^2$ has a maximum at $x = 0$. (a) What is the smallest positive value of x for which the probability distribution function has a maximum at time $t = 2\pi/\omega$, where $\omega = \hbar k^2/2m$? (b) From your result in part (a), what is the average speed with which the probability distribution is moving in the $+x$ -direction? Compare your result to the expression $v_{av} = (\omega_2 - \omega_1)/(k_2 - k_1)$ from Example 40.1.

40.4 • A particle is described by a wave function $\psi(x) = Ae^{-\alpha x^2}$, where A and α are real, positive constants. If the value of α is increased, what effect does this have on (a) the particle's uncertainty in position and (b) the particle's uncertainty in momentum? Explain your answers.

40.5 • Consider a wave function given by $\psi(x) = A \sin kx$, where $k = 2\pi/\lambda$ and A is a real constant. (a) For what values of x is there the highest probability of finding the particle described by this wave function? Explain. (b) For which values of x is the probability zero? Explain.

40.6 • **CALC** Let ψ_1 and ψ_2 be two solutions of Eq. (40.23) with energies E_1 and E_2 , respectively, where $E_1 \neq E_2$. Is $\psi = A\psi_1 + B\psi_2$, where A and B are nonzero constants, a solution to Eq. (40.23)? Explain your answer.

Section 40.2 Particle in a Box

40.7 • An electron is in a one-dimensional box. When the electron is in its ground state, the longest-wavelength photon it can absorb is 420 nm. What is the next longest-wavelength photon it can absorb, again starting in the ground state?

40.8 • **CALC** A particle moving in one dimension (the x -axis) is described by the wave function

$$\psi(x) = \begin{cases} Ae^{-bx}, & \text{for } x \geq 0 \\ Ae^{bx}, & \text{for } x < 0 \end{cases}$$

where $b = 2.00 \text{ m}^{-1}$, $A > 0$, and the $+x$ -axis points toward the right. (a) Determine A so that the wave function is normalized. (b) Sketch the graph of the wave function. (c) Find the probability of finding this particle in each of the following regions: (i) within 50.0 cm of the origin, (ii) on the left side of the origin (can you first guess the answer by looking at the graph of the wave function?), (iii) between $x = 0.500 \text{ m}$ and $x = 1.00 \text{ m}$.

40.9 • **Ground-Level Billiards**. (a) Find the lowest energy level for a particle in a box if the particle is a billiard ball ($m = 0.20 \text{ kg}$) and the box has a width of 1.3 m, the size of a billiard table. (Assume that the billiard ball slides without friction rather than rolls; that is, ignore the rotational kinetic energy.) (b) Since the energy in part (a) is all kinetic, to what speed does this correspond? How much time would it take at this speed for the ball to move from one side of the table to the other? (c) What is the difference in energy between the $n = 2$ and $n = 1$ levels? (d) Are quantum-mechanical effects important for the game of billiards?

40.10 • A proton is in a box of width L . What must the width of the box be for the ground-level energy to be 5.0 MeV, a typical value for the energy with which the particles in a nucleus are bound? Compare your result to the size of a nucleus—that is, on the order of 10^{-14} m .

40.11 • Find the width L of a one-dimensional box for which the ground-state energy of an electron in the box equals the absolute value of the ground state of a hydrogen atom.

40.12 • When a hydrogen atom undergoes a transition from the $n = 2$ to the $n = 1$ level, a photon with $\lambda = 122 \text{ nm}$ is emitted. (a) If the atom is modeled as an electron in a one-dimensional box, what is the width of the box in order for the $n = 2$ to $n = 1$ transition to correspond to emission of a photon of this energy? (b) For a box with the width calculated in part (a), what is the ground-state energy? How does this correspond to the ground-state energy of a hydrogen atom? (c) Do you think a one-dimensional box is a good model for a hydrogen atom? Explain. (*Hint*: Compare the spacing between adjacent energy levels as a function of n .)

40.13 • A certain atom requires 3.0 eV of energy to excite an electron from the ground level to the first excited level. Model the atom as an electron in a box and find the width L of the box.

40.14 • An electron in a one-dimensional box has ground-state energy 2.00 eV. What is the wavelength of the photon absorbed when the electron makes a transition to the second excited state?

40.15 • A particle with mass m is in a one-dimensional box with width L . If the energy of the particle is $9\pi^2\hbar^2/2mL^2$, (a) what is the linear momentum of the particle and (b) what is the ratio of the width of the box to the de Broglie wavelength λ of the particle?

40.16 • Recall that $|\psi|^2 dx$ is the probability of finding the particle that has normalized wave function $\psi(x)$ in the interval x to $x + dx$. Consider a particle in a box with rigid walls at $x = 0$ and $x = L$. Let the particle be in the ground level and use ψ_n as given in Eq. (40.35). (a) For which values of x , if any, in the range from 0 to L is the probability of finding the particle zero? (b) For which values of x is the probability highest? (c) In parts (a) and (b) are your answers consistent with Fig. 40.12? Explain.

40.17 • Consider two different particles in two different boxes. Particle A has nine times the mass of particle B , while particle B 's box has twice the width of particle A 's box. What are the two lowest quantum numbers for the two systems for which particles A and B have equal energies?

40.18 • (a) Find the excitation energy from the ground level to the third excited level for an electron confined to a box of width 0.360 nm. (b) The electron makes a transition from the $n = 1$ to $n = 4$ level by absorbing a photon. Calculate the wavelength of this photon.

40.19 • An electron is in a box of width $3.0 \times 10^{-10} \text{ m}$. What are the de Broglie wavelength and the magnitude of the momentum of the electron if it is in (a) the $n = 1$ level; (b) the $n = 2$ level; (c) the $n = 3$ level? In each case how does the wavelength compare to the width of the box?

40.20 • When an electron in a one-dimensional box makes a transition from the $n = 1$ energy level to the $n = 2$ level, it absorbs a photon of wavelength 426 nm. What is the wavelength of that photon when the electron undergoes a transition (a) from the $n = 2$ to the $n = 3$ energy level and (b) from the $n = 1$ to the $n = 3$ energy level? (c) What is the width L of the box?

40.21 • A particle is in a box with infinitely rigid walls. The walls are at $x = -L/2$ and $x = +L/2$. (a) Show that $\psi_n = A \cos k_n x$ is a possible solution. What must k_n equal? (b) Show that $\psi_n = A \sin k_n x$ is a possible solution. What must k_n equal? (c) What are the allowed energy levels for $\psi_n = A \sin k_n x$ and for $\psi_n = A \cos k_n x$? (d) How does the set of energy levels you found in part (c) compare to the energy levels given by Eq. (40.31)?

Section 40.3 Potential Wells

40.22 •• An electron is moving past the square well shown in Fig. 40.13. The electron has energy $E = 3U_0$. What is the ratio of the de Broglie wavelength of the electron in the region $x > L$ to the wavelength for $0 < x < L$?

40.23 • An electron is bound in a square well of depth $U_0 = 6E_{1-IDW}$. What is the width of the well if its ground-state energy is 2.00 eV?

40.24 •• An electron is in the ground state of a square well of width $L = 4.00 \times 10^{-10}$ m. The depth of the well is six times the ground-state energy of an electron in an infinite well of the same width. What is the kinetic energy of this electron after it has absorbed a photon of wavelength 72 nm and moved away from the well?

40.25 •• An electron is bound in a square well of width 1.50 nm and depth $U_0 = 6E_{1-IDW}$. If the electron is initially in the ground level and absorbs a photon, what maximum wavelength can the photon have and still liberate the electron from the well?

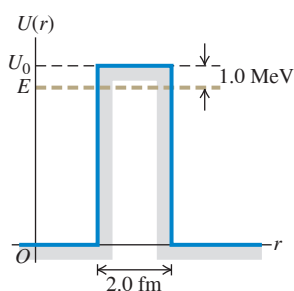
40.26 •• An electron is bound in a square well that has a depth equal to six times the ground-level energy E_{1-IDW} of an infinite well of the same width. The longest-wavelength photon that is absorbed by this electron has a wavelength of 582 nm. Determine the width of the well.

40.27 •• A proton is bound in a square well of width 4.0 fm $= 4.0 \times 10^{-15}$ m. The depth of the well is six times the ground-level energy E_{1-IDW} of the corresponding infinite well. If the proton makes a transition from the level with energy E_1 to the level with energy E_3 by absorbing a photon, find the wavelength of the photon.

Section 40.4 Potential Barriers and Tunneling

40.28 •• Alpha Decay. In a simple model for a radioactive nucleus, an alpha particle ($m = 6.64 \times 10^{-27}$ kg) is trapped by a square barrier that has width 2.0 fm and height 30.0 MeV. (a) What is the tunneling probability when the alpha particle encounters the barrier if its kinetic energy is 1.0 MeV below the top of the barrier (Fig. E40.28)? (b) What is the tunneling probability if the energy of the alpha particle is 10.0 MeV below the top of the barrier?

Figure E40.28



40.29 •• (a) An electron with initial kinetic energy 32 eV encounters a square barrier with height 41 eV and width 0.25 nm. What is the probability that the electron will tunnel through the barrier? (b) A proton with the same kinetic energy encounters the same barrier. What is the probability that the proton will tunnel through the barrier?

40.30 • An electron with initial kinetic energy 5.0 eV encounters a barrier with height U_0 and width 0.60 nm. What is the transmission coefficient if (a) $U_0 = 7.0$ eV; (b) $U_0 = 9.0$ eV; (c) $U_0 = 13.0$ eV?

40.31 • An electron with initial kinetic energy 6.0 eV encounters a barrier with height 11.0 eV. What is the probability of tunneling if the width of the barrier is (a) 0.80 nm and (b) 0.40 nm?

40.32 •• An electron is moving past the square barrier shown in Fig. 40.19, but the energy of the electron is *greater* than the barrier height. If $E = 2U_0$, what is the ratio of the de Broglie wavelength of the electron in the region $x > L$ to the wavelength for $0 < x < L$?

Section 40.5 The Harmonic Oscillator

40.33 • A wooden block with mass 0.250 kg is oscillating on the end of a spring that has force constant 110 N/m. Calculate the ground-level energy and the energy separation between adjacent levels. Express your results in joules and in electron volts. Are quantum effects important?

40.34 •• A harmonic oscillator with mass m and force constant k' is in an excited state that has quantum number n . (a) Let $p_{\max} = mv_{\max}$, where $v_{\max}$ is the maximum speed calculated in the Newtonian analysis of the oscillator. Derive an expression for $p_{\max}$ in terms of n , $\hbar$, k' , and m . (b) Derive an expression for the classical amplitude A in terms of n , $\hbar$, k' , and m . (c) If $\Delta x = A/\sqrt{2}$ and $\Delta p_x = p_{\max}/\sqrt{2}$, what is the uncertainty product $\Delta x \Delta p_x$? How does the uncertainty product depend on n ?

40.35 • Chemists use infrared absorption spectra to identify chemicals in a sample. In one sample, a chemist finds that light of wavelength 5.8 μm is absorbed when a molecule makes a transition from its ground harmonic oscillator level to its first excited level. (a) Find the energy of this transition. (b) If the molecule can be treated as a harmonic oscillator with mass 5.6×10^{-26} kg, find the force constant.

40.36 • A harmonic oscillator absorbs a photon of wavelength 6.35 μm when it undergoes a transition from the ground state to the first excited state. What is the ground-state energy, in electron volts, of the oscillator?

40.37 •• The ground-state energy of a harmonic oscillator is 5.60 eV. If the oscillator undergoes a transition from its $n = 3$ to $n = 2$ level by emitting a photon, what is the wavelength of the photon?

40.38 •• While undergoing a transition from the $n = 1$ to the $n = 2$ energy level, a harmonic oscillator absorbs a photon of wavelength 6.50 μm . What is the wavelength of the absorbed photon when this oscillator undergoes a transition (a) from the $n = 2$ to the $n = 3$ energy level and (b) from the $n = 1$ to the $n = 3$ energy level? (c) What is the value of $\sqrt{k'/m}$, the angular oscillation frequency of the corresponding Newtonian oscillator?

40.39 •• For the sodium atom of Example 40.8, find (a) the ground-state energy; (b) the wavelength of a photon emitted when the $n = 4$ to $n = 3$ transition occurs; (c) the energy difference for any $\Delta n = 1$ transition.

40.40 •• For the ground-level harmonic oscillator wave function $\psi(x)$ given in Eq. (40.47), $|\psi|^2$ has a maximum at $x = 0$. (a) Compute the ratio of $|\psi|^2$ at $x = +A$ to $|\psi|^2$ at $x = 0$, where A is given by Eq. (40.48) with $n = 0$ for the ground level. (b) Compute the ratio of $|\psi|^2$ at $x = +2A$ to $|\psi|^2$ at $x = 0$. In each case is your result consistent with what is shown in Fig. 40.27?

PROBLEMS

40.41 •• A particle of mass m in a one-dimensional box has the following wave function in the region $x = 0$ to $x = L$:

$$\Psi(x, t) = \frac{1}{\sqrt{2}}\psi_1(x)e^{-iE_1t/\hbar} + \frac{1}{\sqrt{2}}\psi_3(x)e^{-iE_3t/\hbar}$$

Here $\psi_1(x)$ and $\psi_3(x)$ are the normalized stationary-state wave functions for the $n = 1$ and $n = 3$ levels, and E_1 and E_3 are the energies of these levels. The wave function is zero for $x < 0$ and for $x > L$. (a) Find the value of the probability distribution function at $x = L/2$ as a function of time. (b) Find the angular frequency at which the probability distribution function oscillates.

40.42 •• Hydrogen emits radiation with four prominent visible wavelengths—one red, one cyan, one blue, and one violet. The respective frequencies are 656 nm, 486 nm, 434 nm, and 410 nm. We can model the hydrogen atom as an electron in a one-dimensional box, and attempt to match four adjacent emission lines in the predicted spectrum to the visible part of the hydrogen spectrum. (a) Determine the photon energies associated with the visible part of the hydrogen spectrum. (b) The electron-in-a-box emission spectrum is $E_{n_i \rightarrow n_f} = (n_i^2 - n_f^2)\epsilon$, where n_i and n_f are the initial and final quantum numbers of the electron when it drops to a lower energy level and ϵ is the energy determined

by the Schrödinger equation. What is the smallest possible value of n_i that can accommodate four emission lines? (c) Using the value from part (b) for n_i , estimate the order of magnitude of ϵ by dividing the four photon energies by the relevant differences $n_i^2 - n_f^2$ for transitions in the possible sets of (n_i, n_f) pairings. (d) Using Eq. (40.31) to identify ϵ , and using the mass of the electron, use your result from part (c) to estimate the length L of the box. (e) What is the ratio of your estimate of L to twice the Bohr radius? (Note: The hydrogen atom is better modeled using the Coulomb potential rather than as a particle in a box.)

40.43 •• CALC Consider a beam of free particles that move with velocity $v = p/m$ in the x -direction and are incident on a potential-energy step $U(x) = 0$, for $x < 0$, and $U(x) = U_0 < E$, for $x > 0$. The wave function for $x < 0$ is $\psi(x) = Ae^{ik_1x} + Be^{-ik_1x}$, representing incident and reflected particles, and for $x > 0$ is $\psi(x) = Ce^{ik_2x}$, representing transmitted particles. Use the conditions that both ψ and its first derivative must be continuous at $x = 0$ to find the constants B and C in terms of k_1 , k_2 , and A .

40.44 • CALC A particle is in the ground level of a box that extends from $x = 0$ to $x = L$. (a) What is the probability of finding the particle in the region between 0 and $L/4$? Calculate this by integrating $|\psi(x)|^2 dx$, where ψ is normalized, from $x = 0$ to $x = L/4$. (b) What is the probability of finding the particle in the region $x = L/4$ to $x = L/2$? (c) How do the results of parts (a) and (b) compare? Explain. (d) Add the probabilities calculated in parts (a) and (b). (e) Are your results in parts (a), (b), and (d) consistent with Fig. 40.12b? Explain.

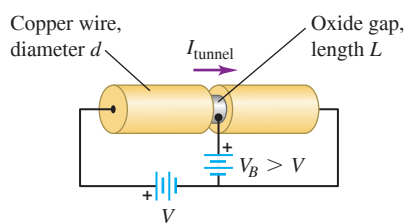
40.45 • Photon in a Dye Laser. An electron in a long, organic molecule used in a dye laser behaves approximately like a particle in a box with width 4.18 nm. What is the wavelength of the photon emitted when the electron undergoes a transition (a) from the first excited level to the ground level and (b) from the second excited level to the first excited level?

40.46 •• Consider a particle in a box with rigid walls at $x = 0$ and $x = L$. Let the particle be in the ground level. Calculate the probability $|\psi|^2 dx$ that the particle will be found in the interval x to $x + dx$ for (a) $x = L/4$; (b) $x = L/2$; (c) $x = 3L/4$.

40.47 •• Repeat Problem 40.46 for a particle in the first excited level.

40.48 •• Suppose we have two copper wires that meet on either side of an oxide gap with

Figure P40.48



length $L = 1$ nm, as shown in **Fig. P40.48**. If the left side is at a potential V relative to the right side, then we can estimate the tunneling current as follows: The wires have diameter $d = 1$ mm, and copper has $n = 8.5 \times 10^{28}$ free electrons per m^3 . (a) Assume that the oxide gap provides a rectangular potential barrier of height 12 eV and that the energy of the free electrons incident on this barrier is 5.0 eV. Assume the gap has length $L = 1$ nm. What is the probability T that an electron will tunnel through the barrier? (b) A length x of wire on the left of the barrier will have nAx free electrons, where $A = \pi(d/2)^2$ is the cross-sectional area of the wire. In a similar length of wire on the right side of the barrier, the number of tunneled electrons will be T times that number. If these electrons move with effective velocity $v = dx/dt$, then what will be the tunneling current, expressed symbolically in terms of T , n , A , e , and v ? (c) Define their effective speed v by $\frac{1}{2}mv^2 = eV$. The tunneled electrons have energy eV . What is their effective speed v , expressed symbolically in terms of e , V , and m , where m is the mass of an electron? (f) Put these elements together to estimate the tunneling current.

40.49 •• CALC What is the probability of finding a particle in a box of length L in the region between $x = L/4$ and $x = 3L/4$ when the particle is in (a) the ground level and (b) the first excited level? (Hint: Integrate $|\psi(x)|^2 dx$, where ψ is normalized, between $L/4$ and $3L/4$.) (c) Are your results in parts (a) and (b) consistent with Fig. 40.12b? Explain.

40.50 •• The *penetration distance* η in a finite potential well is the distance at which the wave function has decreased to $1/e$ of the (b) wave function at the classical turning point:

$$\psi(x = L + \eta) = \frac{1}{e}\psi(L)$$

The penetration distance can be shown to be

$$\eta = \frac{\hbar}{\sqrt{2m(U_0 - E)}}$$

The probability of finding the particle beyond the penetration distance is nearly zero. (a) Find η for an electron having a kinetic energy of 13 eV in a potential well with $U_0 = 20$ eV. (b) Find η for a 20.0 MeV proton trapped in a 30.0-MeV-deep potential well.

40.51 •• CALC A fellow student proposes that a possible wave function for a free particle with mass m (one for which the potential-energy function $U(x)$ is zero) is

$$\psi(x) = \begin{cases} e^{+\kappa x}, & x < 0 \\ e^{-\kappa x}, & x \geq 0 \end{cases}$$

where κ is a positive constant. (a) Graph this proposed wave function. (b) Show that the proposed wave function satisfies the Schrödinger equation for $x < 0$ if the energy is $E = -\hbar^2\kappa^2/2m$ —that is, if the energy of the particle is *negative*. (c) Show that the proposed wave function also satisfies the Schrödinger equation for $x \geq 0$ with the same energy as in part (b). (d) Explain why the proposed wave function is nonetheless *not* an acceptable solution of the Schrödinger equation for a free particle. (Hint: What is the behavior of the function at $x = 0$?) It is in fact impossible for a free particle (one for which $U(x) = 0$) to have an energy less than zero.

40.52 • An electron with initial kinetic energy 5.5 eV encounters a square potential barrier of height 10.0 eV. What is the width of the barrier if the electron has a 0.50% probability of tunneling through the barrier?

40.53 ••• CP Consider a one-dimensional lattice of carbon nuclei with interatomic spacing $d = 0.500$ nm. The potential energy of one nucleus due to electrostatic interaction with its two nearest neighbors is

$$V(x) = \frac{q^2}{4\pi\epsilon_0} \left(\frac{1}{d-x} + \frac{1}{d+x} \right)$$

where $q = 6e$ is the charge of each carbon nucleus and x is its position relative to its equilibrium position. (a) Sketch the potential $V(x)$ between $x = -d$ and $x = d$. (b) At low energy, the nucleus remains near its equilibrium position at $x = 0$ and the potential energy is well approximated as a harmonic oscillator. Use the power series in Appendix B to write $V(x)$ as a quadratic function in x , dropping contributions at higher order in x/d . (c) What is the spring constant k' ? (d) What is the energy of the classical ground state, in eV? (e) What is the energy of the lowest-energy photons emitted when this system transitions between energy levels? (f) What is the wavelength of the corresponding radiation?

40.54 • CP A harmonic oscillator consists of a 0.020 kg mass on a spring. The oscillation frequency is 1.50 Hz, and the mass has a speed

of 0.480 m/s as it passes the equilibrium position. (a) What is the value of the quantum number n for its energy level? (b) What is the difference in energy between the levels E_n and E_{n+1} ? Is this difference detectable?

40.55 • For small amplitudes of oscillation the motion of a pendulum is simple harmonic. For a pendulum with a period of 0.500 s, find the ground-level energy and the energy difference between adjacent energy levels. Express your results in joules and in electron volts. Are these values detectable?

40.56 •• CALC (a) Show by direct substitution in the Schrödinger equation for the one-dimensional harmonic oscillator that the wave function $\psi_1(x) = A_1 x e^{-\alpha^2 x^2/2}$, where $\alpha^2 = m\omega/\hbar$, is a solution with energy corresponding to $n = 1$ in Eq. (40.46). (b) Find the normalization constant A_1 . (c) Show that the probability density has a minimum at $x = 0$ and maxima at $x = \pm 1/\alpha$, corresponding to the classical turning points for the ground state $n = 0$.

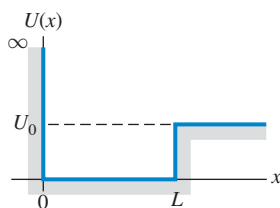
40.57 •• CP (a) The wave nature of particles results in the quantum-mechanical situation that a particle confined in a box can assume only wavelengths that result in standing waves in the box, with nodes at the box walls. Use this to show that an electron confined in a one-dimensional box of length L will have energy levels given by

$$E_n = \frac{n^2 h^2}{8mL^2}$$

(Hint: Recall that the relationship between the de Broglie wavelength and the speed of a nonrelativistic particle is $mv = h/\lambda$. The energy of the particle is $\frac{1}{2}mv^2$.) (b) If a hydrogen atom is modeled as a one-dimensional box with length equal to the Bohr radius, what is the energy (in electron volts) of the lowest energy level of the electron?

40.58 ••• Consider a potential Figure P40.58

well defined as $U(x) = \infty$ for $x < 0$, $U(x) = 0$ for $0 < x < L$, and $U(x) = U_0 > 0$ for $x > L$ (Fig. P40.58). Consider a particle with mass m and kinetic energy $E < U_0$ that is trapped in the well. (a) The boundary condition at the infinite wall ($x = 0$) is $\psi(0) = 0$.



What must the form of the function $\psi(x)$ for $0 < x < L$ be in order to satisfy both the Schrödinger equation and this boundary condition? (b) The wave function must remain finite as $x \rightarrow \infty$. What must the form of the function $\psi(x)$ for $x > L$ be in order to satisfy both the Schrödinger equation and this boundary condition at infinity? (c) Impose the boundary conditions that ψ and $d\psi/dx$ are continuous at $x = L$. Show that the energies of the allowed levels are obtained from solutions of the equation $k \cot kL = -\kappa$, where $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(U_0 - E)}/\hbar$.

40.59 •• DATA In your research on new solid-state devices, you are studying a solid-state structure that can be modeled accurately as an electron in a one-dimensional infinite potential well (box) of width L . In one of your experiments, electromagnetic radiation is absorbed in transitions in which the initial state is the $n = 1$ ground state. You measure that light of frequency $f = 9.0 \times 10^{14}$ Hz is absorbed and that the next higher absorbed frequency is 16.9×10^{14} Hz. (a) What is quantum number n for the final state in each of the transitions that leads to the absorption of photons of these frequencies? (b) What is the width L of the potential well? (c) What is the longest wavelength in air of light that can be absorbed by an electron if it is initially in the $n = 1$ state?

40.60 •• DATA As an intern at a research lab, you study the transmission of electrons through a potential barrier. You know the height of the barrier, 8.0 eV, but must measure the width L of the barrier. When you

measure the tunneling probability T as a function of the energy E of the electron, you get the results shown in the table.

E (eV)	4.0	5.0	6.0	7.0	7.6
T	2.4×10^{-6}	1.5×10^{-5}	1.2×10^{-4}	1.3×10^{-3}	8.1×10^{-3}

(a) For each value of E , calculate the quantities G and κ that appear in Eq. (40.42). Graph $\ln(T/G)$ versus κ . Explain why your data points, when plotted this way, fall close to a straight line. (b) Use the slope of the best-fit straight line to the data in part (a) to calculate L .

40.61 •• DATA When low-energy electrons pass through an ionized gas, electrons of certain energies pass through the gas as if the gas atoms weren't there and thus have transmission coefficients (tunneling probabilities) T equal to unity. The gas ions can be modeled approximately as a rectangular barrier. The value of $T = 1$ occurs when an integral or half-integral number of de Broglie wavelengths of the electron as it passes over the barrier equal the width L of the barrier. You are planning an experiment to measure this effect. To assist you in designing the necessary apparatus, you estimate the electron energies E that will result in $T = 1$. You assume a barrier height of 10 eV and a width of 1.8×10^{-10} m. Calculate the three lowest values of E for which $T = 1$.

40.62 ••• CP A particle is confined to move on a circle with radius R but is otherwise free. We can parameterize points on this circle by using the distance x from a reference point or by using the angle $\theta = x/R$. Since $x = 0$ and $x = 2\pi R$ describe the same point, the wave function must satisfy $\psi(x) = \psi(x + 2\pi R)$ and $\psi'(x) = \psi'(x + 2\pi R)$.

(a) Solve the free-particle time-independent Schrödinger equation subject to these boundary conditions. You should find solutions $\psi_n^\pm$, where n is a positive integer and where lower n corresponds to lower energy. Express your solutions in terms of θ using unspecified normalization constants A_n^+ and A_n^- corresponding, respectively, to modes that move "counterclockwise" toward higher x and "clockwise" toward lower x . (b) Normalize these functions to determine $A_n^\pm$. (c) What are the energy levels E_n ? (d) Write the time-dependent wave functions $\Psi_n^\pm(x, t)$ corresponding to $\Psi_n^\pm(x)$. Use the symbol ω for $E_1/\hbar$. (e) Consider the nonstationary state defined at $t = 0$ by $\Psi(x, 0) = \frac{1}{\sqrt{2}}[\Psi_1^+(x) + \Psi_2^+(x)]$. Determine the probability density $|\Psi(x, t)|^2$ in terms of R , ω , and t . Simplify your result using the identity $1 + \cos \alpha = 2\cos^2(\alpha/2)$. (f) With what angular speed does the density peak move around the circle? (g) If the particle is an electron and the radius is the Bohr radius, then with what speed does its probability peak move?

CHALLENGE PROBLEMS

40.63 ••• CALC The Schrödinger equation for the quantum-mechanical harmonic oscillator in Eq. (40.44) may be written as

$$-\frac{\hbar^2}{2m}\psi'' + \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

where $\omega = \sqrt{k'/m}$ is the angular frequency for classical oscillations. Its solutions may be cleverly addressed using the following observations: The difference in applying two operations A and B , in opposite order, called a commutator, is denoted by $[A, B] = AB - BA$.

For example, if A represents differentiation, so that $Af = \frac{df}{dx}$, where $f = f(x)$ is a function, and B represents multiplication by x , then $[A, B] = \left[\frac{d}{dx}, x\right]f = \frac{d}{dx}(xf) - x\frac{df}{dx}$. Applying the product rule for differentiation on the first term, we determine $\left[\frac{d}{dx}, x\right]f = f$, which

we summarize by writing $\left[\frac{d}{dx}, x\right] = 1$. Similarly, consider the two operators a_+ and a_- defined by $a_{\pm} = 1/\sqrt{2m\hbar\omega} \left(\mp \hbar \frac{d}{dx} + m\omega x\right)$.

(a) Determine the commutator $[a_-, a_+]$. (b) Compute $a_+ a_- \psi$ in terms of ψ and ψ'' , where $\psi = \psi(x)$ is a wave function. Be mindful that $\frac{d}{dx}(x\psi) = \psi + x\psi'$. (c) The Schrödinger equation above may be written as $H\psi = E\psi$, where H is a differential operator. Use your previous result to write H in terms of a_+ and a_- . (d) Determine the commutators $[H, a_{\pm}]$. (e) Consider a wave function ψ_n with definite energy E_n , whereby $H\psi_n = E_n\psi_n$. If we define a new function $\psi_{n+1} = a_+\psi_n$, then we can readily determine its energy by computing $H\psi_{n+1} = Ha_+\psi_n = (a_+H + [H, a_+])\psi_n$ and comparing the result with $E_{n+1}\psi_{n+1}$. Finish this calculation by inserting what we have determined to find E_{n+1} in terms of E_n and n . (f) We can prove that the lowest-energy wave function ψ_0 is annihilated by a_- , meaning that $a_-\psi_0 = 0$. Using your expression for H in terms of $a_{\pm}$, determine the ground-state energy E_0 . (g) By applying a_+ repeatedly, we can generate all higher states. The energy of the n th excited state can be determined by considering $Ha_+^n\psi_0 = E_0\psi_0$. Using the above results, determine E_n . (Note: We did not have to solve any differential equations or take any derivatives to determine the spectrum of this theory.)

40.64 ••• CALC The WKB Approximation. It can be a challenge to solve the Schrödinger equation for the bound-state energy levels of an arbitrary potential well. An alternative approach that can yield good approximate results for the energy levels is the *WKB approximation* (named for the physicists Gregor Wentzel, Hendrik Kramers, and Léon Brillouin, who pioneered its application to quantum mechanics). The WKB approximation begins from three physical statements: (i) According to de Broglie, the magnitude of momentum p of a quantum-mechanical particle is $p = h/\lambda$. (ii) The magnitude of momentum is related to the kinetic energy K by the relationship $K = p^2/2m$. (iii) If there are no nonconservative forces, then in Newtonian mechanics the energy E for a particle is constant and equal at each point to the sum of the kinetic and potential energies at that point: $E = K + U(x)$, where x is the coordinate. (a) Combine these three relationships to show that the wavelength of the particle at a coordinate x can be written as

$$\lambda(x) = \frac{h}{\sqrt{2m[E - U(x)]}}$$

Thus we envision a quantum-mechanical particle in a potential well $U(x)$ as being like a free particle, but with a wavelength $\lambda(x)$ that is a function of position. (b) When the particle moves into a region of increasing potential energy, what happens to its wavelength? (c) At a point where $E = U(x)$, Newtonian mechanics says that the particle has zero kinetic energy and must be instantaneously at rest. Such a point is called a *classical turning point*, since this is where a Newtonian particle must stop its motion and reverse direction. As an example, an object oscillating in simple harmonic motion with amplitude A moves back and forth between the points $x = -A$ and $x = +A$; each of these is a classical turning point, since there the potential energy $\frac{1}{2}k'x^2$ equals the total energy $\frac{1}{2}k'A^2$. In the WKB expression for $\lambda(x)$, what is the wavelength at a classical turning point? (d) For a particle in a box with length L , the walls of the box are classical turning points (see Fig. 40.8). Furthermore, the number of wavelengths that fit within the box must be a half-integer (see Fig. 40.10), so that $L = (n/2)\lambda$ and hence $L/\lambda = n/2$, where $n = 1, 2, 3, \dots$ [Note that this is a restatement of Eq. (40.29).] The WKB scheme for finding the allowed bound-state energy levels of an *arbitrary* potential well is an extension of these

observations. It demands that for an allowed energy E , there must be a half-integer number of wavelengths between the classical turning points for that energy. Since the wavelength in the WKB approximation is not a constant but depends on x , the number of wavelengths between the classical turning points a and b for a given value of the energy is the integral of $1/\lambda(x)$ between those points:

$$\int_a^b \frac{dx}{\lambda(x)} = \frac{n}{2} \quad (n = 1, 2, 3, \dots)$$

Using the expression for $\lambda(x)$ you found in part (a), show that the *WKB condition for an allowed bound-state energy* can be written as

$$\int_a^b \sqrt{2m[E - U(x)]} dx = \frac{nh}{2} \quad (n = 1, 2, 3, \dots)$$

(e) As a check on the expression in part (d), apply it to a particle in a box with walls at $x = 0$ and $x = L$. Evaluate the integral and show that the allowed energy levels according to the WKB approximation are the same as those given by Eq. (40.31). (Hint: Since the walls of the box are infinitely high, the points $x = 0$ and $x = L$ are classical turning points for *any* energy E . Inside the box, the potential energy is zero.) (f) For the finite square well shown in Fig. 40.13, show that the WKB expression given in part (d) predicts the *same* bound-state energies as for an infinite square well of the same width. (Hint: Assume $E < U_0$. Then the classical turning points are at $x = 0$ and $x = L$.) This shows that the WKB approximation does a poor job when the potential-energy function changes discontinuously, as for a finite potential well. In the next two problems we consider situations in which the potential-energy function changes gradually and the WKB approximation is much more useful.

40.65 ••• CALC The WKB approximation (see Challenge Problem 40.64) can be used to calculate the energy levels for a harmonic oscillator. In this approximation, the energy levels are the solutions to the equation

$$\int_a^b \sqrt{2m[E - U(x)]} dx = \frac{nh}{2} \quad n = 1, 2, 3, \dots$$

Here E is the energy, $U(x)$ is the potential-energy function, and $x = a$ and $x = b$ are the classical turning points (the points at which E is equal to the potential energy, so the Newtonian kinetic energy would be zero).

(a) Determine the classical turning points for a harmonic oscillator with energy E and force constant k' . (b) Carry out the integral in the WKB approximation and show that the energy levels in this approximation are $E_n = \hbar\omega$, where $\omega = \sqrt{k'/m}$ and $n = 1, 2, 3, \dots$ (Hint: Recall that $\hbar = h/2\pi$. A useful standard integral is

$$\int \sqrt{A^2 - x^2} dx = \frac{1}{2} \left[x\sqrt{A^2 - x^2} + A^2 \arcsin\left(\frac{x}{A}\right) \right]$$

where $\arcsin$ denotes the inverse sine function. Note that the integrand is even, so the integral from $-x$ to x is equal to twice the integral from 0 to x .) (c) How do the approximate energy levels found in part (b) compare with the true energy levels given by Eq. (40.46)? Does the WKB approximation give an underestimate or an overestimate of the energy levels?

40.66 ••• CALC Protons, neutrons, and many other particles are made of more fundamental particles called *quarks* and *antiquarks* (the antimatter equivalent of quarks). A quark and an antiquark can form a bound state with a variety of different energy levels, each of which corresponds to a different particle observed in the laboratory. As an example, the ψ particle is a low-energy bound state of a so-called charm quark and its antiquark, with a rest energy of 3097 MeV; the $\psi(2S)$ particle is

an excited state of this same quark–antiquark combination, with a rest energy of 3686 MeV. A simplified representation of the potential energy of interaction between a quark and an antiquark is $U(x) = A|x|$, where A is a positive constant and x represents the distance between the quark and the antiquark. You can use the WKB approximation (see Challenge Problem 40.64) to determine the bound-state energy levels for this potential-energy function. In the WKB approximation, the energy levels are the solutions to the equation

$$\int_a^b \sqrt{2m[E - U(x)]} dx = \frac{n\hbar}{2} \quad (n = 1, 2, 3, \dots)$$

Here E is the energy, $U(x)$ is the potential-energy function, and $x = a$ and $x = b$ are the classical turning points (the points at which E is equal to the potential energy, so the Newtonian kinetic energy would be zero). (a) Determine the classical turning points for the potential $U(x) = A|x|$ and for an energy E . (b) Carry out the above integral and show that the allowed energy levels in the WKB approximation are given by

$$E_n = \frac{1}{2m} \left(\frac{3mA\hbar}{4} \right)^{2/3} n^{2/3} \quad (n = 1, 2, 3, \dots)$$

(Hint: The integrand is even, so the integral from $-x$ to x is equal to twice the integral from 0 to x .) (c) Does the difference in energy between successive levels increase, decrease, or remain the same as n increases? How does this compare to the behavior of the energy levels for the harmonic oscillator? For the particle in a box? Can you suggest a simple rule that relates the difference in energy between successive levels to the shape of the potential-energy function?

MCAT-STYLE PASSAGE PROBLEMS

Quantum Dots. A *quantum dot* is a type of crystal so small that quantum effects are significant. One application of quantum dots is in fluorescence imaging, in which a quantum dot is bound to a molecule or structure of interest. When the quantum dot is illuminated with light, it absorbs photons and then re-emits photons at a different wavelength. This phenomenon is called *fluorescence*. The wavelength that a quantum dot emits when

stimulated with light depends on the dot's size, so the synthesis of quantum dots with different photon absorption and emission properties may be possible. We can understand many quantum-dot properties via a model in which a particle of mass M (roughly the mass of the electron) is confined to a two-dimensional rigid square box of sides L . In this model, the quantum-dot energy levels are given by $E_{m,n} = (m^2 + n^2)(\pi^2\hbar^2)/2ML^2$, where m and n are integers 1, 2, 3, ...

40.67 According to this model, which statement is true about the energy-level spacing of dots of different sizes? (a) Smaller dots have equally spaced levels, but larger dots have energy levels that get farther apart as the energy increases. (b) Larger dots have greater spacing between energy levels than do smaller dots. (c) Smaller dots have greater spacing between energy levels than do larger dots. (d) The spacing between energy levels is independent of the dot size.

40.68 When a given dot with side length L makes a transition from its first excited state to its ground state, the dot emits green (550 nm) light. If a dot with side length $1.1L$ is used instead, what wavelength is emitted in the same transition, according to this model? (a) 600 nm; (b) 670 nm; (c) 500 nm; (d) 460 nm.

40.69 Dots that are the same size but made from different materials are compared. In the same transition, a dot of material 1 emits a photon of longer wavelength than the dot of material 2 does. Based on this model, what is a possible explanation? (a) The mass of the confined particle in material 1 is greater. (b) The mass of the confined particle in material 2 is greater. (c) The confined particles make more transitions per second in material 1. (d) The confined particles make more transitions per second in material 2.

40.70 One advantage of the quantum dot is that, compared to many other fluorescent materials, excited states have relatively long lifetimes (10 ns). What does this mean for the spread in the energy of the photons emitted by quantum dots? (a) Quantum dots emit photons of more well-defined energies than do other fluorescent materials. (b) Quantum dots emit photons of less well-defined energies than do other fluorescent materials. (c) The spread in the energy is affected by the size of the dot, not by the lifetime. (d) There is no spread in the energy of the emitted photons, regardless of the lifetime.

ANSWERS

Chapter Opening Question ?

(i) When an electron in one of these particles—called *quantum dots*—makes a transition from an excited level to a lower level, it emits a photon whose energy is equal to the difference in energy between the levels. The smaller the quantum dot, the larger the energy spacing between levels and hence the shorter (bluer) the wavelength of the emitted photons. See Example 40.6 (Section 40.3) for more details.

Key Example VARIATION Problems

VP40.4.1 (a) 0.549 nm (b) $E_2 = 8.00 \times 10^{-19}$ J, $E_3 = 1.80 \times 10^{-18}$ J

VP40.4.2 (a) 3.94×10^{-12} J (b) 6.57×10^{-12} J

VP40.4.3 (a) 3.03×10^{-19} J (b) 0.772 nm (c) 393 nm

VP40.4.4 (a) $\frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2} \text{Acos}\left(\frac{x\sqrt{2mE}}{\hbar} + \phi\right) = -\frac{2mE}{\hbar^2} \psi(x)$,

so $-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x)$ (b) $\phi = \pm\pi/2$ (c) $\frac{L\sqrt{2mE}}{\hbar} = n\pi$, so

$$E = \frac{n^2\pi^2\hbar^2}{2mL^2}, \text{ where } n = 1, 2, 3, \dots$$

VP40.6.1 (a) 4.92×10^{-19} J (b) 3.07×10^{-19} J (c) 2.64×10^{-18} J

VP40.6.2 (a) 9.40×10^{-20} J (b) 5.64×10^{-19} J (c) 0.800 nm

VP40.6.3 (a) initial $n = 3$, final $n = 1$ (b) 596 nm for initial $n = 3$, final $n = 2$; 878 nm for initial $n = 2$, final $n = 1$

VP40.6.4 (a) 2.35 eV (b) 0.010 eV

VP40.7.1 (a) 2.90×10^{-5} (b) 1.47×10^{-3}

VP40.7.2 (a) 5.33×10^{-3} (b) 0.704 nm

VP40.7.3 (a) 2.54×10^{-4} (b) 8.59×10^{-4} (c) 3.46×10^{-3}

VP40.7.4 0.509 nm

Bridging Problem

$$(a) |\Psi(x, t)|^2 = \frac{1}{L} \left[\sin^2 \frac{\pi x}{L} + \sin^2 \frac{2\pi x}{L} + 2 \sin \frac{\pi x}{L} \sin \frac{2\pi x}{L} \cos \left(\frac{(E_2 - E_1)t}{\hbar} \right) \right]$$

(b) no

$$(d) \frac{3\pi^2\hbar}{2mL^2}$$

$$(e) \frac{0.903\pi^2\hbar}{mL^2}$$

? Lithium (with three electrons per atom) is a metal that burns spontaneously in water, while helium (with two electrons per atom) is a gas that undergoes almost no chemical reactions. The additional electron makes lithium behave very differently from helium primarily because (i) the third electron is strongly repelled by electric forces from the other two electrons; (ii) the third electron and larger nucleus make the lithium atom more massive than the helium atom; (iii) there is a limit on the number of electrons that can occupy a given quantum-mechanical state; (iv) the lithium nucleus has more positive charge than a helium nucleus has.



41 Quantum Mechanics II: Atomic Structure

LEARNING OUTCOMES

In this chapter, you'll learn...

- 41.1 How to extend quantum-mechanical calculations to three-dimensional problems.
- 41.2 How to solve the Schrödinger equation for a particle trapped in a cubical box.
- 41.3 How to describe the states of a hydrogen atom in terms of quantum numbers.
- 41.4 How magnetic fields affect the orbital motion of atomic electrons.
- 41.5 How we know that electrons have their own intrinsic angular momentum.
- 41.6 How to analyze the structure of many-electron atoms.
- 41.7 How x rays emitted by atoms reveal their inner structure.
- 41.8 What happens when the quantum-mechanical states of two particles become entangled.

You'll need to review...

- 22.3 Gauss's law.
- 27.7 Magnetic dipole moment.
- 32.5 Standing electromagnetic waves.
- 38.2 X-ray production.
- 39.2, 39.3 Atoms and the Bohr model.
- 40.1, 40.2, 40.5, 40.6 One-dimensional Schrödinger equation; particle in a box; harmonic-oscillator wave functions; measuring a quantum-mechanical system.

Some physicists claim that all of chemistry is contained in the Schrödinger equation. This is somewhat of an exaggeration, but this equation can teach us a great deal about the chemical behavior of elements, the periodic table, and the nature of chemical bonds.

In order to learn about the quantum-mechanical structure of atoms, we'll first construct a three-dimensional version of the Schrödinger equation. We'll try this equation out by looking at a three-dimensional version of a particle in a box: a particle confined to a cubical volume.

We'll then see that we can learn a great deal about the structure and properties of *all* atoms from the solutions to the Schrödinger equation for the hydrogen atom. These solutions have quantized values of orbital angular momentum; we don't need to impose quantization as we did with the Bohr model. We label the states with a set of quantum numbers, which we'll use later with many-electron atoms as well. We'll find that the electron also has an intrinsic *spin* angular momentum with its own set of quantized values.

We'll also encounter the exclusion principle, a kind of microscopic zoning ordinance that is the key to understanding many-electron atoms. This principle says that no two electrons in an atom can have the same quantum-mechanical state. We'll then use the principles of this chapter to explain the characteristic x-ray spectra of atoms. Finally, we'll end our discussion of quantum mechanics with a look at the curious concept of quantum entanglement and its application to the new science of quantum computing.

41.1 THE SCHRÖDINGER EQUATION IN THREE DIMENSIONS

We have discussed the Schrödinger equation and its applications only for *one-dimensional* problems, the analog of a Newtonian particle moving along a straight line. The straight-line model is adequate for some applications, but to understand atomic structure, we need a three-dimensional generalization.

It's not difficult to guess what the three-dimensional Schrödinger equation should look like. First, the wave function Ψ is a function of time and all three space coordinates (x, y, z) . In general, the potential-energy function also depends on all three coordinates and can be written as $U(x, y, z)$. Next, recall from Section 40.1 that the term $-(\hbar^2/2m)\partial^2\Psi/\partial x^2$ in the one-dimensional Schrödinger equation, Eq. (40.20), is related to the kinetic energy of the particle in the state described by the wave function Ψ . For example, if we insert into this term the wave function $\Psi(x, t) = Ae^{ikx}e^{-i\omega t}$ for a free particle with magnitude of momentum $p = \hbar k$ and kinetic energy $K = p^2/2m$, we obtain $-(\hbar^2/2m)(ik)^2 Ae^{ikx}e^{-i\omega t} = (\hbar^2 k^2/2m) Ae^{ikx}e^{-i\omega t} = (p^2/2m)\Psi(x, t) = K\Psi(x, t)$. If the particle can move in three dimensions, its momentum has three components (p_x, p_y, p_z) and its kinetic energy is

$$K = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} \quad (41.1)$$

These observations, taken together, suggest that the correct generalization of the Schrödinger equation to three dimensions is

$$\begin{aligned} -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \Psi(x, y, z, t)}{\partial x^2} + \frac{\partial^2 \Psi(x, y, z, t)}{\partial y^2} + \frac{\partial^2 \Psi(x, y, z, t)}{\partial z^2} \right) \\ + U(x, y, z) \Psi(x, y, z, t) = i\hbar \frac{\partial \Psi(x, y, z, t)}{\partial t} \end{aligned} \quad (41.2)$$

(general three-dimensional Schrödinger equation)

The three-dimensional wave function $\Psi(x, y, z, t)$ has a similar interpretation as in one dimension. The wave function itself is a complex quantity with both a real part and an imaginary part, but $|\Psi(x, y, z, t)|^2$ —the square of its absolute value, equal to the product of $\Psi(x, y, z, t)$ and its complex conjugate $\Psi^*(x, y, z, t)$ —is real and either positive or zero at every point in space. We interpret $|\Psi(x, y, z, t)|^2 dV$ as the *probability* of finding the particle within a small volume dV centered on the point (x, y, z) at time t , so $|\Psi(x, y, z, t)|^2$ is the *probability distribution function* in three dimensions. The *normalization condition* on the wave function is that the probability that the particle is *somewhere* in space is exactly 1. Hence the integral of $|\Psi(x, y, z, t)|^2$ over all space must equal 1:

$$\int |\Psi(x, y, z, t)|^2 dV = 1 \quad \begin{array}{l} \text{(normalization condition} \\ \text{in three dimensions)} \end{array} \quad (41.3)$$

If the wave function $\Psi(x, y, z, t)$ represents a state of a definite energy E —that is, a stationary state—we can write it as the product of a spatial wave function $\psi(x, y, z)$ and a function of time $e^{-iEt/\hbar}$:

$$\Psi(x, y, z, t) = \psi(x, y, z) e^{-iEt/\hbar} \quad \begin{array}{l} \text{(time-dependent wave function} \\ \text{for a state of definite energy)} \end{array} \quad (41.4)$$

(Compare this to Eq. (40.21) for a one-dimensional state of definite energy.) If we substitute Eq. (41.4) into Eq. (41.2), the right-hand side of the equation becomes $i\hbar\psi(x, y, z)(-iE/\hbar)e^{-iEt/\hbar} = E\psi(x, y, z)e^{-iEt/\hbar}$. We can then divide both sides by the factor $e^{-iEt/\hbar}$, leaving the *time-independent* Schrödinger equation in three dimensions for a stationary state:

Time-independent three-dimensional Schrödinger equation:

$$\begin{aligned} -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x, y, z)}{\partial x^2} + \frac{\partial^2 \psi(x, y, z)}{\partial y^2} + \frac{\partial^2 \psi(x, y, z)}{\partial z^2} \right) \\ + U(x, y, z) \psi(x, y, z) = E \psi(x, y, z) \end{aligned} \quad (41.5)$$

Planck's constant divided by 2π Time-independent wave function
Particle's mass Potential-energy function Energy of state

The probability distribution function for a stationary state is just the square of the absolute value of the spatial wave function: $|\psi(x, y, z)e^{-iEt/\hbar}|^2 = \psi^*(x, y, z)e^{+iEt/\hbar}\psi(x, y, z)e^{-iEt/\hbar} = |\psi(x, y, z)|^2$. Note that this doesn't depend on time. (As we discussed in Section 40.1, that's why we call these states *stationary*.) Hence for a stationary state the wave function normalization condition, Eq. (41.3), becomes

$$\int |\psi(x, y, z)|^2 dV = 1 \quad \text{(normalization condition for a stationary state in three dimensions)} \quad (41.6)$$

We won't pretend that we have *derived* Eqs. (41.2) and (41.5). Like their one-dimensional versions, these equations have to be tested by comparison of their predictions with experimental results. Happily, Eqs. (41.2) and (41.5) both pass this test with flying colors, so we are confident that they *are* the correct equations.

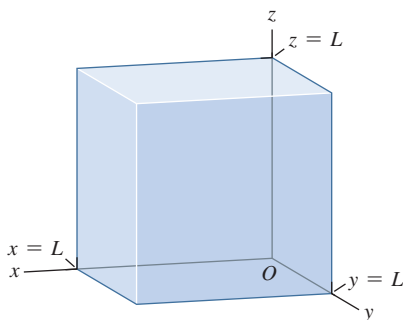
An important topic that we'll address in this chapter is the solutions for Eq. (41.5) for the stationary states of the hydrogen atom. The potential-energy function for an electron in a hydrogen atom is *spherically symmetric*; it depends only on the distance $r = (x^2 + y^2 + z^2)^{1/2}$ from the origin of coordinates. To take advantage of this symmetry, it's best to use *spherical coordinates* rather than the Cartesian coordinates (x, y, z) to solve the Schrödinger equation for the hydrogen atom. Before introducing these new coordinates and investigating the hydrogen atom, it's useful to look at the three-dimensional version of the particle in a box that we considered in Section 40.2. Solving this simpler problem will give us insight into the more complicated stationary states found in atomic physics.

TEST YOUR UNDERSTANDING OF SECTION 41.1 In a certain region of space the potential-energy function for a quantum-mechanical particle is zero. In this region the wave function $\psi(x, y, z)$ for a certain stationary state is real and satisfies $\partial^2\psi/\partial x^2 > 0$, $\partial^2\psi/\partial y^2 > 0$, and $\partial^2\psi/\partial z^2 > 0$. The particle has a definite energy E that is positive. What can you conclude about $\psi(x, y, z)$ in this region? (i) It must be positive; (ii) it must be negative; (iii) it must be zero; (iv) not enough information given to decide.

ANSWER Since $E > 0$, the quantity $-2mE/\hbar^2$ is negative, and so $\psi(x, y, z)$ must be negative. We are told that all of the second derivatives of $\psi(x, y, z)$ are positive in this region, so the left-hand side of the Schrödinger equation [Eq. (41.5)] for that region as $\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 + \partial^2\psi/\partial z^2 = (-2mE/\hbar^2)\psi$. Hence the right-hand side $(-2mE/\hbar^2)\psi$ must also be positive.

41.2 PARTICLE IN A THREE-DIMENSIONAL BOX

Figure 41.1 A particle is confined in a cubical box with walls at $x = 0$, $x = L$, $y = 0$, $y = L$, $z = 0$, and $z = L$.



Consider a particle enclosed within a cubical box of side L . This could represent an electron that's free to move anywhere within the interior of a solid metal cube but cannot escape the cube. We'll choose the origin to be at one corner of the box, with the x -, y -, and z -axes along edges of the box. Then the particle is confined to the region $0 \leq x \leq L$, $0 \leq y \leq L$, $0 \leq z \leq L$ (**Fig. 41.1**). What are the stationary states of this system?

As for the model of a particle in a one-dimensional box that we considered in Section 40.2, we'll say that the potential energy is zero inside the box but infinite outside. Hence the spatial wave function $\psi(x, y, z)$ must be zero outside the box in order that the term $U(x, y, z)\psi(x, y, z)$ in the time-independent Schrödinger equation, Eq. (41.5), not be infinite. Consequently the probability distribution function $|\psi(x, y, z)|^2$ is zero outside the box, and the probability that the particle will be found there is zero. Inside the box, the spatial wave function for a stationary state obeys the time-independent Schrödinger equation, Eq. (41.5), with $U(x, y, z) = 0$:

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2\psi(x, y, z)}{\partial x^2} + \frac{\partial^2\psi(x, y, z)}{\partial y^2} + \frac{\partial^2\psi(x, y, z)}{\partial z^2} \right) = E\psi(x, y, z) \quad (41.7)$$

(particle in a three-dimensional box)

In order for the wave function to be continuous from the inside to the outside of the box, $\psi(x, y, z)$ must equal zero on the walls. Hence our boundary conditions are that $\psi(x, y, z) = 0$ at $x = 0$, $x = L$, $y = 0$, $y = L$, $z = 0$, and $z = L$.

Guessing a solution to a complicated partial differential equation like Eq. (41.7) seems like quite a challenge. To make progress, recall that we wrote the time-*dependent* wave function for a stationary state as the product of one function that depends on only the spatial coordinates x , y , and z and a second function that depends on only the time t : $\Psi(x, y, z, t) = \psi(x, y, z)e^{-iEt/\hbar}$. In the same way, let's try a technique called *separation of variables*: We'll write the spatial wave function $\psi(x, y, z)$ as a product of one function X that depends on only x , a second function Y that depends on only y , and a third function Z that depends on only z :

$$\psi(x, y, z) = X(x)Y(y)Z(z) \quad (41.8)$$

If we substitute Eq. (41.8) into Eq. (41.7), we get

$$\begin{aligned} & -\frac{\hbar^2}{2m} \left(Y(y)Z(z) \frac{d^2X(x)}{dx^2} + X(x)Z(z) \frac{d^2Y(y)}{dy^2} + X(x)Y(y) \frac{d^2Z(z)}{dz^2} \right) \\ & = EX(x)Y(y)Z(z) \end{aligned} \quad (41.9)$$

The partial derivatives in Eq. (41.7) have become ordinary derivatives since they act on functions of a single variable. Now we divide both sides of Eq. (41.9) by the product $X(x)Y(y)Z(z)$:

$$\left(-\frac{\hbar^2}{2m} \frac{1}{X(x)} \frac{d^2X(x)}{dx^2} \right) + \left(-\frac{\hbar^2}{2m} \frac{1}{Y(y)} \frac{d^2Y(y)}{dy^2} \right) + \left(-\frac{\hbar^2}{2m} \frac{1}{Z(z)} \frac{d^2Z(z)}{dz^2} \right) = E \quad (41.10)$$

The right-hand side of Eq. (41.10) is the energy of the stationary state. Since E is a constant that does not depend on the values of x , y , and z , the left-hand side of the equation must also be independent of the values of x , y , and z . Hence the first term in parentheses on the left-hand side of Eq. (41.10) must equal a constant that doesn't depend on x , the second term in parentheses must equal another constant that doesn't depend on y , and the third term in parentheses must equal a third constant that doesn't depend on z . Let's call these constants E_X , E_Y , and E_Z , respectively. We then have a separate equation for each of the three functions $X(x)$, $Y(y)$, and $Z(z)$:

$$-\frac{\hbar^2}{2m} \frac{d^2X(x)}{dx^2} = E_X X(x) \quad (41.11a)$$

$$-\frac{\hbar^2}{2m} \frac{d^2Y(y)}{dy^2} = E_Y Y(y) \quad (41.11b)$$

$$-\frac{\hbar^2}{2m} \frac{d^2Z(z)}{dz^2} = E_Z Z(z) \quad (41.11c)$$

To satisfy the boundary conditions that $\psi(x, y, z) = X(x)Y(y)Z(z)$ be equal to zero on the walls of the box, we demand that $X(x) = 0$ at $x = 0$ and $x = L$, $Y(y) = 0$ at $y = 0$ and $y = L$, and $Z(z) = 0$ at $z = 0$ and $z = L$.

How can we interpret the three constants E_X , E_Y , and E_Z in Eqs. (41.11)? From Eq. (41.10), they are related to the energy E by

$$E_X + E_Y + E_Z = E \quad (41.12)$$

Equation (41.12) should remind you of Eq. (41.1) in Section 41.1, which states that the kinetic energy of a particle is the sum of contributions coming from its x -, y -, and z -components of momentum. Hence the constants E_X , E_Y , and E_Z tell us how much of the particle's energy is due to motion along each of the three coordinate axes. (Inside the box the potential energy is zero, so the particle's energy is purely kinetic.)

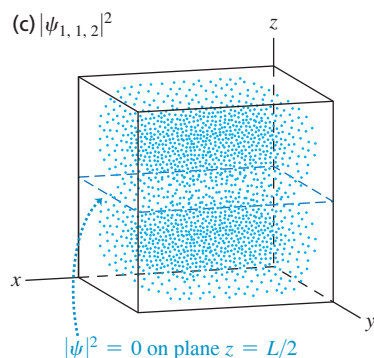
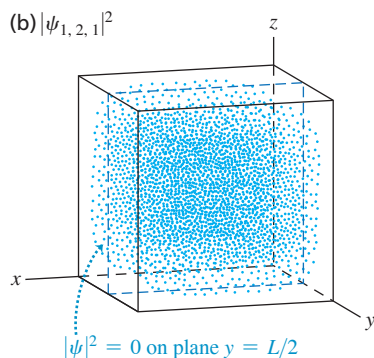
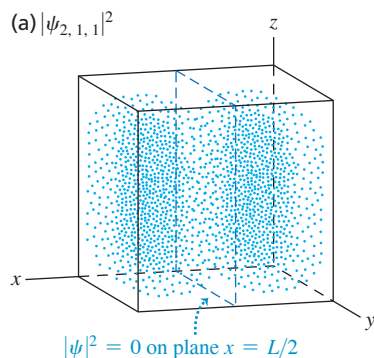
Equations (41.11) represent an enormous simplification; we've reduced the problem of solving a fairly complex *partial* differential equation with three independent variables to the much simpler problem of solving three separate *ordinary* differential equations with one independent variable each. What's more, each of these ordinary differential equations is the same as the time-independent Schrödinger equation for a particle in a *one-dimensional* box, Eq. (40.25), and with exactly the same boundary conditions at 0 and L . (The only differences are that some of the quantities are labeled by different symbols.) By comparing with our work in Section 40.2, you can see that the solutions to Eqs. (41.11) are

$$X_{n_X}(x) = C_X \sin \frac{n_X \pi x}{L} \quad (n_X = 1, 2, 3, \dots) \quad (41.13a)$$

$$Y_{n_Y}(y) = C_Y \sin \frac{n_Y \pi y}{L} \quad (n_Y = 1, 2, 3, \dots) \quad (41.13b)$$

$$Z_{n_Z}(z) = C_Z \sin \frac{n_Z \pi z}{L} \quad (n_Z = 1, 2, 3, \dots) \quad (41.13c)$$

Figure 41.2 Probability distribution function $|\psi_{n_X, n_Y, n_Z}(x, y, z)|^2$ for (n_X, n_Y, n_Z) equal to (a) (2, 1, 1), (b) (1, 2, 1), and (c) (1, 1, 2). The value of $|\psi|^2$ is proportional to the density of dots. The wave function is zero on the walls of the box and on a midplane of the box, so $|\psi|^2 = 0$ at these locations.



where C_X , C_Y , and C_Z are constants. The corresponding values of E_X , E_Y , and E_Z are

$$E_X = \frac{n_X^2 \pi^2 \hbar^2}{2mL^2} \quad (n_X = 1, 2, 3, \dots) \quad (41.14a)$$

$$E_Y = \frac{n_Y^2 \pi^2 \hbar^2}{2mL^2} \quad (n_Y = 1, 2, 3, \dots) \quad (41.14b)$$

$$E_Z = \frac{n_Z^2 \pi^2 \hbar^2}{2mL^2} \quad (n_Z = 1, 2, 3, \dots) \quad (41.14c)$$

There is only one quantum number n for the one-dimensional particle in a box, but *three* quantum numbers n_X , n_Y , and n_Z for the three-dimensional box. If we substitute Eqs. (41.13) back into Eq. (41.8) for the total spatial wave function, $\psi(x, y, z) = X(x)Y(y)Z(z)$, we get the following stationary-state wave functions for a particle in a three-dimensional cubical box:

$$\psi_{n_X, n_Y, n_Z}(x, y, z) = C \sin \frac{n_X \pi x}{L} \sin \frac{n_Y \pi y}{L} \sin \frac{n_Z \pi z}{L} \quad (41.15)$$

$$(n_X = 1, 2, 3, \dots; n_Y = 1, 2, 3, \dots; n_Z = 1, 2, 3, \dots)$$

where $C = C_X C_Y C_Z$. The value of the constant C is determined by the normalization condition, Eq. (41.6).

In Section 40.2 we saw that the stationary-state wave functions for a particle in a one-dimensional box were analogous to standing waves on a string. In a similar way, the *three-dimensional* wave functions given by Eq. (41.15) are analogous to standing electromagnetic waves in a cubical cavity like the interior of a microwave oven (see Section 32.5). In a microwave oven there are “dead spots” where the wave intensity is zero, corresponding to the nodes of the standing wave. (The moving platform in a microwave oven ensures even cooking by making sure that no part of the food sits at any “dead spot.”) In a similar fashion, the probability distribution function corresponding to Eq. (41.15) can have “dead spots” where there is zero probability of finding the particle. As an example, consider the case $(n_X, n_Y, n_Z) = (2, 1, 1)$. From Eq. (41.15), the probability distribution function for this case is

$$|\psi_{2,1,1}(x, y, z)|^2 = |C|^2 \sin^2 \frac{2\pi x}{L} \sin^2 \frac{\pi y}{L} \sin^2 \frac{\pi z}{L}$$

As Fig. 41.2a shows, this probability distribution function is zero on the plane $x = L/2$, where $\sin^2(2\pi x/L) = \sin^2 \pi = 0$. The particle is most likely to be found near where

all three of the sine-squared functions are greatest, at $(x, y, z) = (L/4, L/2, L/2)$ or $(x, y, z) = (3L/4, L/2, L/2)$. Figures 41.2b and 41.2c show the similar cases $(n_X, n_Y, n_Z) = (1, 2, 1)$ and $(n_X, n_Y, n_Z) = (1, 1, 2)$. For higher values of the quantum numbers n_X , n_Y , and n_Z there are additional planes on which the probability distribution function equals zero, just as the probability distribution function $|\psi(x)|^2$ for a one-dimensional box has more zeros for higher values of n (see Fig. 40.12).

EXAMPLE 41.1 Probability in a three-dimensional box

WITH  **ARIATION PROBLEMS**

(a) Find the value of the constant C that normalizes the wave function of Eq. (41.15). (b) Find the probability that the particle will be found somewhere in the region $0 \leq x \leq L/4$ (**Fig. 41.3**) for the cases (i) $(n_X, n_Y, n_Z) = (1, 2, 1)$, (ii) $(n_X, n_Y, n_Z) = (2, 1, 1)$, and (iii) $(n_X, n_Y, n_Z) = (3, 1, 1)$.

IDENTIFY and SET UP Equation (41.6) tells us that to normalize the wave function, we have to choose the value of C so that the integral of the probability distribution function $|\psi_{n_X, n_Y, n_Z}(x, y, z)|^2$ over the volume within the box equals 1. (The integral is actually over *all* space, but the particle-in-a-box wave functions are zero outside the box.)

The probability of finding the particle within a certain volume within the box equals the integral of the probability distribution function over that volume. Hence in part (b) we'll integrate $|\psi_{n_X, n_Y, n_Z}(x, y, z)|^2$ for the given values of (n_X, n_Y, n_Z) over the volume $0 \leq x \leq L/4, 0 \leq y \leq L, 0 \leq z \leq L$.

EXECUTE (a) From Eq. (41.15),

$$|\psi_{n_X, n_Y, n_Z}(x, y, z)|^2 = |C|^2 \sin^2 \frac{n_X \pi x}{L} \sin^2 \frac{n_Y \pi y}{L} \sin^2 \frac{n_Z \pi z}{L}$$

Hence the normalization condition is

$$\begin{aligned} & \int |\psi_{n_X, n_Y, n_Z}(x, y, z)|^2 dV \\ &= |C|^2 \int_{x=0}^{x=L} \int_{y=0}^{y=L} \int_{z=0}^{z=L} \sin^2 \frac{n_X \pi x}{L} \sin^2 \frac{n_Y \pi y}{L} \sin^2 \frac{n_Z \pi z}{L} dx dy dz \\ &= |C|^2 \left(\int_{x=0}^{x=L} \sin^2 \frac{n_X \pi x}{L} dx \right) \left(\int_{y=0}^{y=L} \sin^2 \frac{n_Y \pi y}{L} dy \right) \\ & \quad \times \left(\int_{z=0}^{z=L} \sin^2 \frac{n_Z \pi z}{L} dz \right) \\ &= 1 \end{aligned}$$

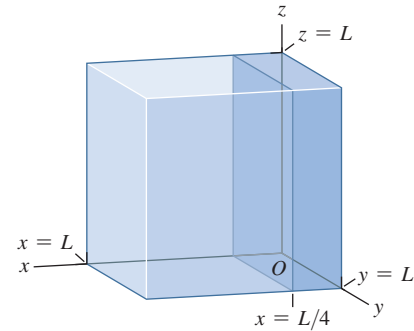
We can use the identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ and the variable substitution $\theta = n_X \pi x/L$ to show that

$$\begin{aligned} \int \sin^2 \frac{n_X \pi x}{L} dx &= \frac{L}{2n_X \pi} \left[\frac{n_X \pi x}{L} - \frac{1}{2} \sin \left(\frac{2n_X \pi x}{L} \right) \right] \\ &= \frac{x}{2} - \frac{L}{4n_X \pi} \sin \left(\frac{2n_X \pi x}{L} \right) \end{aligned}$$

If we evaluate this integral between $x = 0$ and $x = L$, the result is $L/2$ (recall that $\sin 0 = 0$ and $\sin 2n_X \pi = 0$ for any integer n_X). The y - and z -integrals each yield the same result, so the normalization condition is

$$|C|^2 \left(\frac{L}{2} \right) \left(\frac{L}{2} \right) \left(\frac{L}{2} \right) = |C|^2 \left(\frac{L}{2} \right)^3 = 1$$

Figure 41.3 What is the probability that the particle is in the dark-colored quarter of the box?



or $|C|^2 = (2/L)^3$. If we choose C to be real and positive, then $C = (2/L)^{3/2}$.

(b) We have the same y - and z -integrals as in part (a), but now the limits of integration on the x -integral are $x = 0$ and $x = L/4$:

$$\begin{aligned} P &= \int_{0 \leq x \leq L/4} |\psi_{n_X, n_Y, n_Z}|^2 dV \\ &= |C|^2 \left(\int_{x=0}^{x=L/4} \sin^2 \frac{n_X \pi x}{L} dx \right) \left(\int_{y=0}^{y=L} \sin^2 \frac{n_Y \pi y}{L} dy \right) \\ & \quad \times \left(\int_{z=0}^{z=L} \sin^2 \frac{n_Z \pi z}{L} dz \right) \end{aligned}$$

The x -integral is

$$\begin{aligned} \int_{x=0}^{x=L/4} \sin^2 \frac{n_X \pi x}{L} dx &= \left[\frac{x}{2} - \frac{L}{4n_X \pi} \sin \left(\frac{2n_X \pi x}{L} \right) \right] \Big|_{x=0}^{x=L/4} \\ &= \frac{L}{8} - \frac{L}{4n_X \pi} \sin \left(\frac{n_X \pi}{2} \right) \end{aligned}$$

Hence the probability of finding the particle somewhere in the region $0 \leq x \leq L/4$ is

$$\begin{aligned} P &= \left(\frac{2}{L} \right)^3 \left[\frac{L}{8} - \frac{L}{4n_X \pi} \sin \left(\frac{n_X \pi}{2} \right) \right] \left(\frac{L}{2} \right) \left(\frac{L}{2} \right) \\ &= \frac{1}{4} - \frac{1}{2n_X \pi} \sin \left(\frac{n_X \pi}{2} \right) \end{aligned}$$

This depends only on the value of n_X , not on n_Y or n_Z . For each of the three cases we have

Continued

$$\begin{aligned}
 \text{(i) } n_X = 1: P &= \frac{1}{4} - \frac{1}{2(1)\pi} \sin\left(\frac{\pi}{2}\right) = \frac{1}{4} - \frac{1}{2\pi}(1) \\
 &= \frac{1}{4} - \frac{1}{2\pi} = 0.091 \\
 \text{(ii) } n_X = 2: P &= \frac{1}{4} - \frac{1}{2(2)\pi} \sin\left(\frac{2\pi}{2}\right) = \frac{1}{4} - \frac{1}{4\pi} \sin \pi \\
 &= \frac{1}{4} - 0 = 0.250 \\
 \text{(iii) } n_X = 3: P &= \frac{1}{4} - \frac{1}{2(3)\pi} \sin\left(\frac{3\pi}{2}\right) = \frac{1}{4} - \frac{1}{6\pi}(-1) \\
 &= \frac{1}{4} + \frac{1}{6\pi} = 0.303
 \end{aligned}$$

EVALUATE You can see why the probabilities in part (b) are different by looking at part (b) of Fig. 40.12, which shows $\sin^2 n_X \pi x/L$ for $n_X = 1, 2$, and 3 . For $n_X = 2$ the area under the curve between $x = 0$ and $x = L/4$ (equal to the integral between these two points) is exactly $\frac{1}{4}$ of the total area between $x = 0$ and $x = L$. For $n_X = 1$ the area between $x = 0$ and $x = L/4$ is less than $\frac{1}{4}$ of the total area, and for $n_X = 3$ it is greater than $\frac{1}{4}$ of the total area.

KEYCONCEPT If a particle is in a three-dimensional stationary state described by wave function $\psi(x, y, z)$, the probability of finding that particle in an infinitesimal volume dV centered on (x, y, z) is $|\psi(x, y, z)|^2 dV$. The probability of finding the particle somewhere in a finite volume equals the integral $\int |\psi(x, y, z)|^2 dV$ evaluated over that volume.

Energy Levels, Degeneracy, and Symmetry

From Eqs. (41.12) and (41.14), the allowed energies for a particle of mass m in a cubical box of side L are

$$E_{n_X, n_Y, n_Z} = \frac{(n_X^2 + n_Y^2 + n_Z^2) \pi^2 \hbar^2}{2mL^2} \quad (41.16)$$

Quantum numbers n_X, n_Y, n_Z can each equal 1, 2, 3, ...

Planck's constant divided by 2π

Energy levels, particle in a three-dimensional cubical box

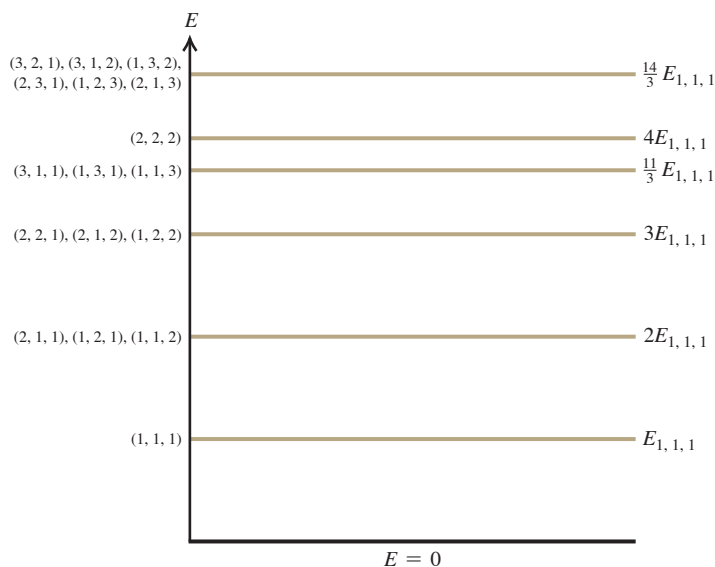
Particle's mass

Length of each side of box

Figure 41.4 shows the six lowest energy levels given by Eq. (41.16). Note that most energy levels correspond to more than one set of quantum numbers (n_X, n_Y, n_Z) and hence to more than one quantum state. Having two or more distinct quantum states with the same energy is called **degeneracy**, and states with the same energy are said to be **degenerate**. For example, Fig. 41.4 shows that the states $(n_X, n_Y, n_Z) = (2, 1, 1), (1, 2, 1),$ and $(1, 1, 2)$ are degenerate. By comparison, for a particle in a one-dimensional box there is just one state for each energy level (see Fig. 40.11a) and no degeneracy.

The reason the cubical box exhibits degeneracy is that it is *symmetric*: All sides of the box have the same dimensions. As an illustration, Fig. 41.2 shows the probability distribution functions for the three states $(n_X, n_Y, n_Z) = (2, 1, 1), (1, 2, 1),$ and $(1, 1, 2)$. You can

Figure 41.4 Energy-level diagram for a particle in a three-dimensional cubical box. We label each level with the quantum numbers of the states (n_X, n_Y, n_Z) with that energy. Several of the levels are degenerate (more than one state has the same energy). The lowest (ground) level, $(n_X, n_Y, n_Z) = (1, 1, 1)$, has energy $E_{1,1,1} = (1^2 + 1^2 + 1^2) \pi^2 \hbar^2 / 2mL^2 = 3\pi^2 \hbar^2 / 2mL^2$; we show the energies of the other levels as multiples of $E_{1,1,1}$.



transform any one of these three states into a different one by simply rotating the cubical box by 90° . This rotation doesn't change the energy, so the three states are degenerate.

Since degeneracy is a consequence of symmetry, we can remove the degeneracy by making the box asymmetric. We do this by giving the three sides of the box different lengths L_X , L_Y , and L_Z . If we repeat the steps that we followed to solve the time-independent Schrödinger equation, we find that the energy levels are given by

$$E_{n_X, n_Y, n_Z} = \left(\frac{n_X^2}{L_X^2} + \frac{n_Y^2}{L_Y^2} + \frac{n_Z^2}{L_Z^2} \right) \frac{\pi^2 \hbar^2}{2m} \quad \begin{matrix} (n_X = 1, 2, 3, \dots; \\ n_Y = 1, 2, 3, \dots; \\ n_Z = 1, 2, 3, \dots) \end{matrix} \quad (41.17)$$

(energy levels, particle in a three-dimensional box with sides
of length L_X , L_Y , and L_Z)

If L_X , L_Y , and L_Z are all different, the states $(n_X, n_Y, n_Z) = (2, 1, 1)$, $(1, 2, 1)$, and $(1, 1, 2)$ have different energies and hence are no longer degenerate. Note that Eq. (41.17) reduces to Eq. (41.16) if $L_X = L_Y = L_Z = L$.

Returning to a particle in a three-dimensional cubical box, let's summarize the differences from the one-dimensional case that we examined in Section 40.2:

- We can write the wave function for a three-dimensional stationary state as a product of three functions, one for each spatial coordinate. Only a single function of the coordinate x is needed in one dimension.
- In the three-dimensional case, three quantum numbers are needed to describe each stationary state. Only one quantum number is needed in the one-dimensional case.
- Most of the energy levels for the three-dimensional case are degenerate: More than one stationary state has this energy. There is no degeneracy in the one-dimensional case.
- For a stationary state of the three-dimensional case, there are surfaces on which the probability distribution function $|\psi|^2$ is zero. In the one-dimensional case, there are positions on the x -axis where $|\psi|^2$ is zero.

We'll see these same features in the following section for a three-dimensional situation that's more realistic than a particle in a cubical box: a hydrogen atom in which a negatively charged electron orbits a positively charged nucleus.

TEST YOUR UNDERSTANDING OF SECTION 41.2 Rank the following states of a particle in a cubical box of side L in order from highest to lowest energy: (i) $(n_X, n_Y, n_Z) = (2, 3, 2)$; (ii) $(n_X, n_Y, n_Z) = (4, 1, 1)$; (iii) $(n_X, n_Y, n_Z) = (2, 2, 3)$; (iv) $(n_X, n_Y, n_Z) = (1, 3, 3)$.

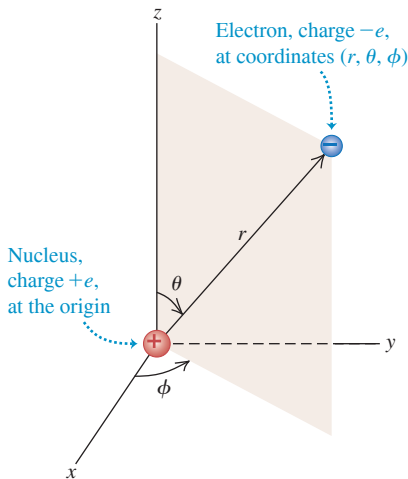
ANSWER

Equation (41.16) shows that the energy levels for a cubical box are proportional to the quantity $n_X^2 + n_Y^2 + n_Z^2$. Hence ranking in order of this quantity is the same as ranking in order of energy. For the four cases we are given, we have (i) $n_X^2 + n_Y^2 + n_Z^2 = 2^2 + 3^2 + 2^2 = 17$; (ii) $n_X^2 + n_Y^2 + n_Z^2 = 4^2 + 1^2 + 1^2 = 18$; (iii) $n_X^2 + n_Y^2 + n_Z^2 = 2^2 + 2^2 + 3^2 = 17$; and (iv) $n_X^2 + n_Y^2 + n_Z^2 = 1^2 + 3^2 + 3^2 = 19$. The states $(n_X, n_Y, n_Z) = (2, 3, 2)$ and $(2, 2, 3)$ have the same energy (they are degenerate).

41.3 THE HYDROGEN ATOM

Let's continue the discussion of the hydrogen atom that we began in Chapter 39. In the Bohr model, electrons move in circular orbits like Newtonian particles, but with quantized values of angular momentum. While this model gave the correct energy levels of the hydrogen atom, as deduced from spectra, it had many conceptual difficulties. It mixed classical physics with new and seemingly contradictory concepts. It provided no insight into the process by which photons are emitted and absorbed. It could not be generalized to atoms with more than one electron. It predicted the wrong magnetic properties for the hydrogen atom. And perhaps most important, its picture of the electron as a localized point particle was inconsistent with the more general view we developed in Chapters 39 and 40. To go beyond the Bohr model, let's apply the Schrödinger equation to find the

Figure 41.5 The Schrödinger equation for the hydrogen atom can be solved most readily by using spherical coordinates.



wave functions for stationary states (states of definite energy) of the hydrogen atom. As in Section 39.3, we include the motion of the nucleus by simply replacing the electron mass m with the reduced mass m_r .

The Schrödinger Equation for the Hydrogen Atom

We discussed the three-dimensional version of the Schrödinger equation in Section 41.1. The potential-energy function is *spherically symmetric*: It depends only on the distance $r = (x^2 + y^2 + z^2)^{1/2}$ from the origin of coordinates:

$$U(r) = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r} \quad (41.18)$$

The hydrogen-atom problem is best formulated in spherical coordinates (r, θ, ϕ) , shown in **Fig. 41.5**; the spherically symmetric potential-energy function depends only on r , not on θ or ϕ . The Schrödinger equation with this potential-energy function can be solved exactly; the solutions are combinations of familiar functions. Without going into a lot of detail, we can describe the most important features of the procedure and the results.

First, we find the solutions by using the same method of separation of variables that we employed for a particle in a cubical box in Section 41.2. We express the wave function $\psi(r, \theta, \phi)$ as a product of three functions, each one a function of only one of the three coordinates:

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi) \quad (41.19)$$

That is, the function $R(r)$ depends on only r , $\Theta(\theta)$ depends on only θ , and $\Phi(\phi)$ depends on only ϕ . Just as for a particle in a three-dimensional box, when we substitute Eq. (41.19) into the Schrödinger equation, we get three separate ordinary differential equations. One equation involves only r and $R(r)$, a second involves only θ and $\Theta(\theta)$, and a third involves only ϕ and $\Phi(\phi)$:

$$-\frac{\hbar^2}{2m_r r^2} \frac{d}{dr} \left(r^2 \frac{dR(r)}{dr} \right) + \left(\frac{\hbar^2 l(l+1)}{2m_r r^2} + U(r) \right) R(r) = ER(r) \quad (41.20a)$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta(\theta)}{d\theta} \right) + \left(l(l+1) - \frac{m_l^2}{\sin^2 \theta} \right) \Theta(\theta) = 0 \quad (41.20b)$$

$$\frac{d^2 \Phi(\phi)}{d\phi^2} + m_l^2 \Phi(\phi) = 0 \quad (41.20c)$$

In Eqs. (41.20) E is the energy of the stationary state and l and m_l are constants that we'll discuss later.

CAUTION Two uses of the symbol m Don't confuse the constant m_l in Eqs. (41.20b) and (41.20c) with the similar symbol m_r for the reduced mass of the electron and nucleus (see Section 39.3). The constant m_l is a dimensionless number; the reduced mass m_r has units of kilograms. ■

We won't attempt to solve this set of three equations, but we can describe how it's done. As for the particle in a cubical box, the physically acceptable solutions of these three equations are determined by boundary conditions. The radial function $R(r)$ in Eq. (41.20a) must approach zero at large r , because we are describing *bound states* of the electron that are localized near the nucleus. This is analogous to the requirement that the harmonic-oscillator wave functions (see Section 40.5) must approach zero at large x . The angular functions $\Theta(\theta)$ and $\Phi(\phi)$ in Eqs. (41.20b) and (41.20c) must be *finite* for all relevant values of the angles. For example, there are solutions of the Θ equation that become infinite at $\theta = 0$ and $\theta = \pi$; these are unacceptable, since $\psi(r, \theta, \phi)$ must be normalizable. Furthermore, the angular function $\Phi(\phi)$ in Eq. (41.20c) must be *periodic*. For example, (r, θ, ϕ) and $(r, \theta, \phi + 2\pi)$ describe the same point, so $\Phi(\phi + 2\pi)$ must equal $\Phi(\phi)$.

The allowed radial functions $R(r)$ turn out to be an exponential function $e^{-\alpha r}$ (where α is positive) multiplied by a polynomial in r . The functions $\Theta(\theta)$ are polynomials containing various powers of $\sin \theta$ and $\cos \theta$, and the functions $\Phi(\phi)$ are simply proportional to $e^{im_l \phi}$, where $i = \sqrt{-1}$ and m_l is an integer that may be positive, zero, or negative.

In the process of finding solutions that satisfy the boundary conditions, we also find the corresponding energy levels. We denote the energies of these levels [E in Eq. (41.20a)] by E_n ($n = 1, 2, 3, \dots$). These turn out to be *identical* to those from the Bohr model, as given by Eq. (39.15), with the electron rest mass m replaced by the reduced mass m_r . Rewriting that equation with $\hbar = h/2\pi$, we have

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2} \frac{m_r e^4}{2n^2 \hbar^2} = -\frac{13.60 \text{ eV}}{n^2} \quad (41.21)$$

Energy levels of hydrogen $\rightarrow$ E_n
Reduced mass $\rightarrow$ m_r
Magnitude of electron charge $\rightarrow$ e^4
Electric constant $\rightarrow$ $(4\pi\epsilon_0)^2$
Principal quantum number $\rightarrow$ n ($n = 1, 2, 3, \dots$)
 Planck's constant divided by 2π $\rightarrow$ $\hbar^2$

As in Section 39.3, we call n the **principal quantum number**.

Equation (41.21) is an important validation of our Schrödinger-equation analysis of the hydrogen atom. The Schrödinger analysis is quite different from the Bohr model, both mathematically and conceptually, yet both yield the same energy-level scheme—a scheme that agrees with the energies determined from spectra. As we'll see, the Schrödinger analysis can explain many more aspects of the hydrogen atom than can the Bohr model.

Quantization of Orbital Angular Momentum

The solutions to Eqs. (41.20) that satisfy the boundary conditions mentioned above also have quantized values of *orbital angular momentum*. That is, only certain discrete values of the magnitude and components of orbital angular momentum are permitted. In discussing the Bohr model in Section 39.3, we mentioned that quantization of angular momentum was a result with little fundamental justification. With the Schrödinger equation it appears automatically.

The possible values of the magnitude L of orbital angular momentum $\vec{L}$ are determined by the requirement that the $\Theta(\theta)$ function in Eq. (41.20b) must be finite at $\theta = 0$ and $\theta = \pi$. In a level with energy E_n and principal quantum number n , the possible values of L are

$$L = \sqrt{l(l+1)}\hbar \quad (l = 0, 1, 2, \dots, n-1) \quad (41.22)$$

Magnitude of orbital angular momentum, hydrogen atom $\rightarrow$ L
Orbital quantum number $\rightarrow$ l
Planck's constant divided by 2π $\rightarrow$ $\hbar$
Principal quantum number $\rightarrow$ n ($n = 1, 2, 3, \dots$)

The **orbital quantum number** l in Eq. (41.22) is the same l that appears in Eqs. (41.20a) and (41.20b). In the Bohr model, each energy level corresponded to a single value of angular momentum. Equation (41.22) shows that in fact there are n different possible values of L for the n th energy level.

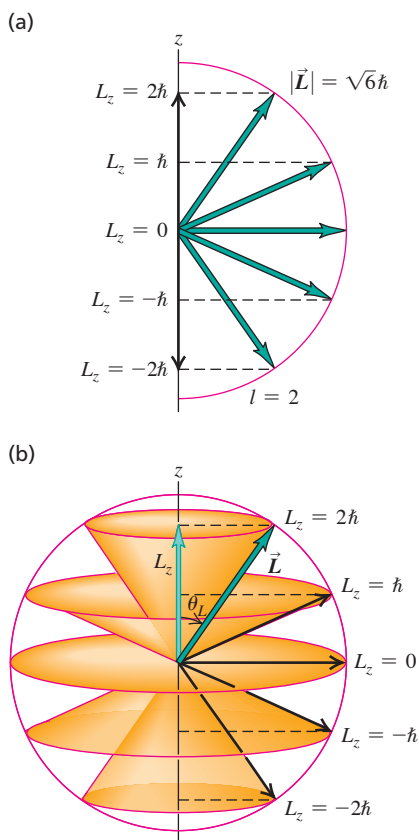
An interesting feature of Eq. (41.22) is that the orbital angular momentum is *zero* for $l = 0$ states. This result disagrees with the Bohr model, in which the electron always moved in a circle of definite radius and L was never zero. The $l = 0$ wave functions ψ depend only on r ; for these states, the functions $\Theta(\theta)$ and $\Phi(\phi)$ are constants. Thus the wave functions for $l = 0$ states are spherically symmetric. There is nothing in their probability distribution $|\psi|^2$ to favor one direction over any other, and there is no orbital angular momentum.

The permitted values of the *component* of $\vec{L}$ in a given direction, say the z -component L_z , are determined by the requirement that the $\Phi(\phi)$ function must equal $\Phi(\phi + 2\pi)$. The possible values of L_z are

$$L_z = m_l \hbar \quad (m_l = 0, \pm 1, \pm 2, \dots, \pm l) \quad (41.23)$$

z-component of orbital angular momentum, hydrogen atom $\rightarrow$ L_z
Orbital magnetic quantum number $\rightarrow$ m_l
Planck's constant divided by 2π $\rightarrow$ $\hbar$
Orbital quantum number $\rightarrow$ l

Figure 41.6 (a) When $l = 2$, the magnitude of the angular momentum vector $\vec{L}$ is $\sqrt{6}\hbar = 2.45\hbar$, but $\vec{L}$ does not have a definite direction. In this semiclassical vector picture, $\vec{L}$ makes an angle of 35.3° with the z -axis when the z -component has its maximum value of $2\hbar$. (b) These cones show the possible directions of $\vec{L}$ for different values of L_z .



The quantum number m_l is the same as that in Eqs. (41.20b) and (41.20c). We see that m_l can be zero or a positive or negative integer up to, but no larger in magnitude than, l . That is, $|m_l| \leq l$. For example, if $l = 1$, m_l can equal 1, 0, or -1 . For reasons that will emerge later, we call m_l the *orbital magnetic quantum number*, or **magnetic quantum number** for short.

The component L_z can never be quite as large as L (unless both are zero). For example, when $l = 2$, the largest possible value of m_l is also 2; then Eqs. (41.22) and (41.23) give

$$L = \sqrt{2(2+1)}\hbar = \sqrt{6}\hbar = 2.45\hbar$$

$$L_z = 2\hbar$$

Figure 41.6 shows the situation. The minimum value of the angle θ_L between the vector $\vec{L}$ and the z -axis is

$$\begin{aligned}\theta_L &= \arccos \frac{L_z}{L} \\ &= \arccos \frac{2}{2.45} = 35.3^\circ\end{aligned}$$

That $|L_z|$ is always less than L is also required by the uncertainty principle. Suppose we could know the precise *direction* of the orbital angular momentum vector. Then we could let that be the direction of the z -axis, and L_z would equal L . This corresponds to a particle moving in the xy -plane only, in which case the z -component of the linear momentum $\vec{p}$ would be zero with no uncertainty Δp_z . Then the uncertainty principle $\Delta z \Delta p_z \geq \hbar$ requires infinite uncertainty Δz in the coordinate z . This is impossible for a localized state; we conclude that we can't know the direction of $\vec{L}$ precisely. Thus, as we've already stated, the component of $\vec{L}$ in a given direction can never be quite as large as its magnitude L . Also, if we can't know the direction of $\vec{L}$ precisely, we can't determine the components L_x and L_y precisely. Thus we show *cones* of possible directions for $\vec{L}$ in Fig. 41.6b.

You may wonder why we have singled out the z -axis. We can't determine all three components of orbital angular momentum with certainty, so we arbitrarily pick one as the component we want to measure. When we discuss interactions of the atom with a magnetic field, we'll consistently choose the positive z -axis to be in the direction of $\vec{B}$.

Quantum Number Notation

The wave functions for the hydrogen atom are determined by the values of three quantum numbers n , l , and m_l . (Compare this to the particle in a three-dimensional box that we considered in Section 41.2. There, too, three quantum numbers were needed to describe each stationary state.) The energy E_n is determined by the principal quantum number n according to Eq. (41.21). The magnitude of orbital angular momentum is determined by the orbital quantum number l , as in Eq. (41.22). The component of orbital angular momentum in a specified axis direction (customarily the z -axis) is determined by the magnetic quantum number m_l , as in Eq. (41.23). The energy does not depend on the values of l or m_l (Fig. 41.7), so for each energy level E_n given by Eq. (41.21), there is more than one distinct state having the same energy but different quantum numbers. That is, these states are *degenerate*, just like most of the states of a particle in a three-dimensional box. As for the three-dimensional box, degeneracy arises because the hydrogen atom is symmetric: If you rotate the atom through any angle, the potential-energy function at a distance r from the nucleus has the same value.

States with various values of the orbital quantum number l are often labeled with letters, according to the following scheme:

$l = 0$: s states	$l = 3$: f states
$l = 1$: p states	$l = 4$: g states
$l = 2$: d states	$l = 5$: h states

and so on alphabetically. This seemingly irrational choice of the letters s , p , d , and f originated in the early days of spectroscopy and has no fundamental significance. In an important form of *spectroscopic notation* that we'll use often, a state with $n = 2$ and $l = 1$ is called a $2p$ state; a state with $n = 4$ and $l = 0$ is a $4s$ state; and so on. Only s states ($l = 0$) are spherically symmetric.

Here's another bit of notation. The radial extent of the wave functions increases with the principal quantum number n , and we can speak of a region of space associated with a particular value of n as a **shell**. Especially in discussions of many-electron atoms, these shells are denoted by capital letters:

$n = 1$: K shell
$n = 2$: L shell
$n = 3$: M shell
$n = 4$: N shell

and so on alphabetically. For each n , different values of l correspond to different *subshells*. For example, the L shell ($n = 2$) contains the $2s$ and $2p$ subshells.

Table 41.1 shows some of the possible combinations of the quantum numbers n , l , and m_l for hydrogen-atom wave functions. The spectroscopic notation and the shell notation for each are also shown.

TABLE 41.1 Quantum States of the Hydrogen Atom

n	l	m_l	Spectroscopic Notation	Shell
1	0	0	$1s$	K
2	0	0	$2s$	L
2	1	$-1, 0, 1$	$2p$	
3	0	0	$3s$	M
3	1	$-1, 0, 1$	$3p$	
3	2	$-2, -1, 0, 1, 2$	$3d$	
4	0	0	$4s$	N

and so on

Figure 41.7 The energy for an orbiting satellite such as the Hubble Space Telescope depends on the average distance between the satellite and the center of the earth. It does *not* depend on whether the orbit is circular (with a large orbital angular momentum L) or elliptical (in which case L is smaller). In the same way, the energy of a hydrogen atom does not depend on the orbital angular momentum.



PROBLEM-SOLVING STRATEGY 41.1 Atomic Structure

IDENTIFY *the relevant concepts:* Many problems in atomic structure can be solved simply by reference to the quantum numbers n , l , and m_l that describe the total energy E , the magnitude of the orbital angular momentum $\vec{L}$, the z -component of $\vec{L}$, and other properties of an atom.

SET UP *the problem:* Identify the target variables and choose the appropriate equations, which may include Eqs. (41.21), (41.22), and (41.23).

EXECUTE *the solution* as follows:

1. Be sure you understand the possible values of the quantum numbers n , l , and m_l for the hydrogen atom. They are all integers; n is always greater than zero, l can be zero or positive up to $n - 1$,

and m_l can range from $-l$ to l . You should know how to count the number of (n, l, m_l) states in each shell (K , L , M , and so on) and subshell ($3s$, $3p$, $3d$, and so on). Be able to *construct* Table 41.1, not just to write it from memory.

2. Solve for the target variables.

EVALUATE *your answer:* It helps to be familiar with typical magnitudes in atomic physics. For example, the electric potential energy of a proton and electron 0.10 nm apart (typical of atomic dimensions) is about -15 eV. Visible light has wavelengths around 500 nm and frequencies around 5×10^{14} Hz. Problem-Solving Strategy 39.1 (Section 39.1) gives other typical magnitudes.

EXAMPLE 41.2 Counting hydrogen states**WITH VARIATION PROBLEMS**

How many distinct (n, l, m_l) states of the hydrogen atom with $n = 3$ are there? What are their energies?

IDENTIFY and SET UP This problem uses the relationships among the principal quantum number n , orbital quantum number l , magnetic quantum number m_l , and energy of a state for the hydrogen atom. We use the rule that l can have n integer values, from 0 to $n - 1$, and that m_l can have $2l + 1$ values, from $-l$ to l . Equation (41.21) gives the energy of any particular state.

EXECUTE When $n = 3$, l can be 0, 1, or 2. When $l = 0$, m_l can be only 0 (1 state). When $l = 1$, m_l can be $-1, 0$, or 1 (3 states). When $l = 2$, m_l can be $-2, -1, 0, 1$, or 2 (5 states). The total number of (n, l, m_l) states with $n = 3$ is therefore $1 + 3 + 5 = 9$. (In Section 41.5 we'll find that the total number of $n = 3$ states is in fact twice this, or 18, because of electron spin.)

The energy of a hydrogen-atom state depends only on n , so all 9 of these states have the same energy. From Eq. (41.21),

$$E_3 = \frac{-13.60 \text{ eV}}{3^2} = -1.51 \text{ eV}$$

EVALUATE For a given value of n , the total number of (n, l, m_l) states turns out to be n^2 . In this case $n = 3$ and there are $3^2 = 9$ states. Remember that the ground level of hydrogen has $n = 1$ and $E_1 = -13.6 \text{ eV}$; the $n = 3$ excited states have a higher (less negative) energy.

KEYCONCEPT Because the potential-energy function for the hydrogen atom is spherically symmetric, its energy levels are degenerate. Stationary states with the same principal quantum number n have the same energy, even though they have different values of the orbital quantum number l and the orbital magnetic quantum number m_l .

EXAMPLE 41.3 Angular momentum in an excited level of hydrogen**WITH VARIATION PROBLEMS**

Consider the $n = 4$ states of hydrogen. (a) What is the maximum magnitude L of the orbital angular momentum? (b) What is the maximum value of L_z ? (c) What is the minimum angle between $\vec{L}$ and the z -axis? Give your answers to parts (a) and (b) in terms of $\hbar$.

IDENTIFY and SET UP We again need to relate the principal quantum number n and the orbital quantum number l for a hydrogen atom. We also need to relate the value of l and the magnitude and possible directions of the orbital angular momentum vector. We'll use Eq. (41.22) in part (a) to determine the maximum value of L ; then we'll use Eq. (41.23) in part (b) to determine the maximum value of L_z . The angle between $\vec{L}$ and the z -axis is minimum when L_z is maximum (so that $\vec{L}$ is most nearly aligned with the positive z -axis).

EXECUTE (a) When $n = 4$, the maximum value of the orbital quantum number l is $(n - 1) = (4 - 1) = 3$; from Eq. (41.22),

$$L_{\max} = \sqrt{3(3 + 1)} \hbar = \sqrt{12} \hbar = 3.464\hbar$$

(b) For $l = 3$ the maximum value of m_l is 3. From Eq. (41.23),

$$(L_z)_{\max} = 3\hbar$$

(c) The *minimum* allowed angle between $\vec{L}$ and the z -axis corresponds to the *maximum* allowed values of L_z and m_l (Fig. 41.6b shows an $l = 2$ example). For the state with $l = 3$ and $m_l = 3$,

$$\theta_{\min} = \arccos \frac{(L_z)_{\max}}{L} = \arccos \frac{3\hbar}{3.464\hbar} = 30.0^\circ$$

EVALUATE As a check, you can verify that θ is greater than 30.0° for all states with smaller values of l .

KEYCONCEPT For each value of the principal quantum number $n = 1, 2, 3, \dots$ for the stationary states of the hydrogen atom, there are n possible values of the orbital quantum number l that range from 0 to $n - 1$. For each value of l , there are $2l + 1$ possible values of the orbital magnetic quantum number m_l that range from $-l$ to l .

Electron Probability Distributions

Rather than picturing the electron as a point particle moving in a precise circle, the Schrödinger equation gives an electron *probability distribution* surrounding the nucleus. The hydrogen-atom probability distributions are three-dimensional, so they are harder to visualize than the two-dimensional circular orbits of the Bohr model. It's helpful to look at the *radial probability distribution function* $P(r)$ —that is, the probability per radial length for the electron to be found at various distances from the proton. From Section 41.1 the probability for finding the electron in a small volume element dV is $|\psi|^2 dV$. (We assume that ψ is normalized in accordance with Eq. (41.6)—that is, that the integral of $|\psi|^2 dV$ over all space equals unity so that there is 100% probability of finding the electron somewhere in the universe.) Let's take as our volume element a thin spherical shell with inner radius r and outer radius $r + dr$. The volume dV of this shell is approximately its area $4\pi r^2$ multiplied by its thickness dr :

$$dV = 4\pi r^2 dr \quad (41.24)$$

We denote by $P(r) dr$ the probability of finding the particle within the radial range dr ; then, using Eq. (41.24), we get

$$P(r) dr = |\psi|^2 dV = |\psi|^2 4\pi r^2 dr \quad (41.25)$$

Labels in the diagram:
 - Probability that the electron is between r and $r + dr$ (points to $P(r) dr$)
 - Radial probability distribution function (points to $P(r)$)
 - Probability distribution function (points to $|\psi|^2$)
 - Wave function (points to ψ)
 - Volume of spherical shell with inner radius r , outer radius $r + dr$ (points to dV)

For wave functions that depend on θ and ϕ as well as r , we use the value of $|\psi|^2$ averaged over all angles in Eq. (41.25).

Figure 41.8 shows graphs of $P(r)$ for several hydrogen-atom wave functions. The r scales are labeled in multiples of a , the smallest distance between the electron and the nucleus in the Bohr model:

$$a = \frac{\epsilon_0 \hbar^2}{\pi m_e e^2} = \frac{4\pi \epsilon_0 \hbar^2}{m_e e^2} = 5.29 \times 10^{-11} \text{ m} \quad (41.26)$$

Labels in the diagram:
 - Radius of smallest orbit in Bohr model (points to a)
 - Electric constant (points to ϵ_0)
 - Planck's constant (points to $\hbar$)
 - Planck's constant divided by 2π (points to $\hbar$)
 - Reduced mass (points to m_e)
 - Magnitude of electron charge (points to e)

As for a particle in a cubical box (see Section 41.2), there are some locations where the probability is zero. These surfaces are planes for a particle in a box; for a hydrogen atom these are spherical surfaces (that is, surfaces of constant r). Note that for the states that have the largest possible l for each n (such as $1s$, $2p$, $3d$, and $4f$ states), $P(r)$ has a single maximum at $n^2 a$. For these states, the electron is most likely to be found at the distance from the nucleus that is predicted by the Bohr model, $r = n^2 a$.

Figure 41.8 shows *radial* probability distribution functions $P(r) = 4\pi r^2 |\psi|^2$, which indicate the relative probability of finding the electron within a thin spherical shell of radius r . By contrast, **Figs. 41.9** and **41.10** (next page) show the *three-dimensional* probability distribution functions $|\psi|^2$, which indicate the relative probability of finding the electron within a small box at a given position. The darker the blue “cloud,” the greater the value of $|\psi|^2$. (These are similar to the “clouds” shown in Fig. 41.2.) Figure 41.9 shows cross sections of the spherically symmetric probability clouds for the lowest three s subshells, for which $|\psi|^2$ depends only on the radial coordinate r . Figure 41.10 shows cross sections of the clouds for other electron states for which $|\psi|^2$ depends on both r and θ . For these states the probability distribution function is zero for certain values of θ as well as for certain values of r . In *any* stationary state of the hydrogen atom, $|\psi|^2$ is independent of ϕ .

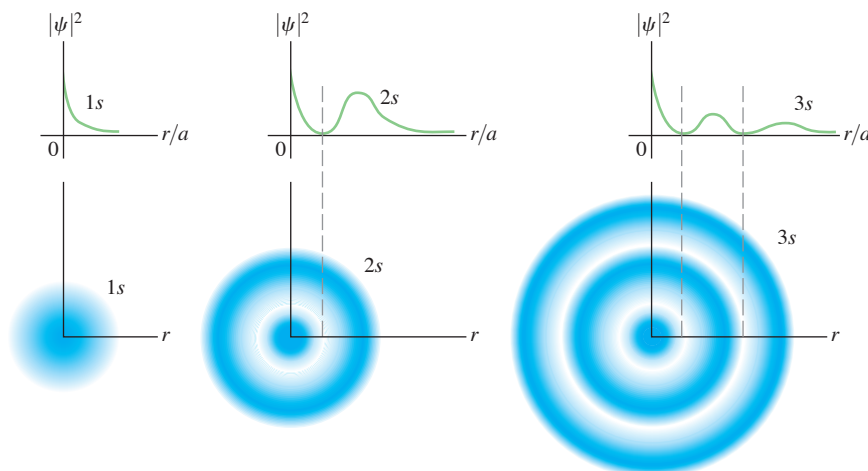


Figure 41.8 Radial probability distribution functions $P(r)$ for several hydrogen-atom wave functions, plotted as functions of the ratio r/a [see Eq. (41.26)]. For each function, the number of maxima is $(n - l)$. The curves for which $l = n - 1$ ($1s$, $2p$, $3d$, ...) have only one maximum, located at $r = n^2 a$.

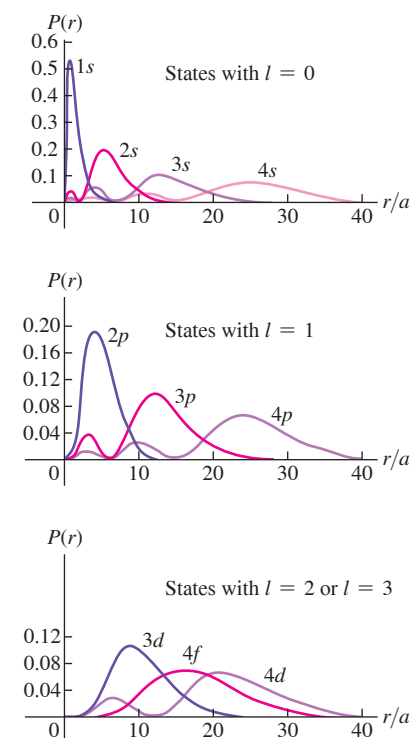
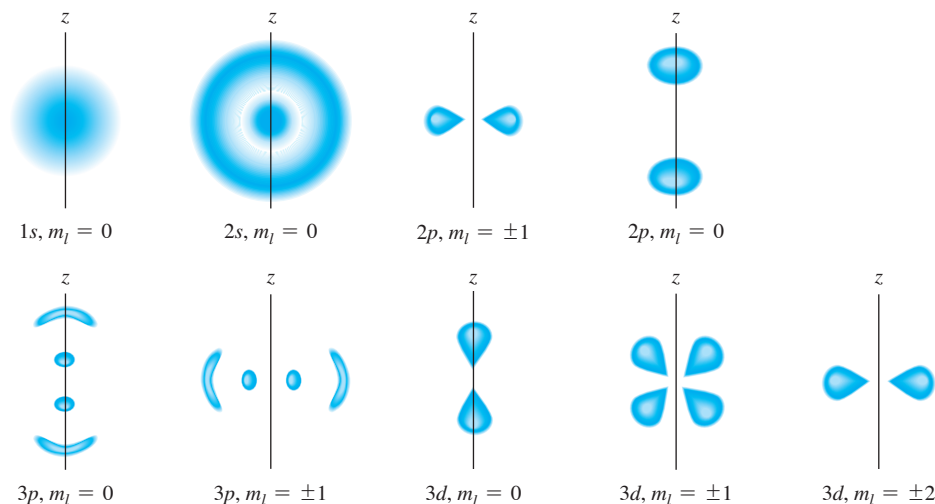


Figure 41.9 Three-dimensional probability distribution functions $|\psi|^2$ for the spherically symmetric $1s$, $2s$, and $3s$ hydrogen-atom wave functions.

Figure 41.10 Cross sections of three-dimensional probability distributions for a few quantum states of the hydrogen atom. They are not to the same scale. Mentally rotate each drawing about the z -axis to obtain the three-dimensional representation of $|\psi|^2$. For example, the $2p$, $m_l = \pm 1$ probability distribution looks like a fuzzy donut.



EXAMPLE 41.4 A hydrogen wave function

WITH VARIATION PROBLEMS

The ground-state wave function for hydrogen (a $1s$ state) is

$$\psi_{1s}(r) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$$

(a) Verify that this function is normalized. (b) What is the probability that the electron will be found at a distance less than a from the nucleus?

IDENTIFY and SET UP This example is similar to Example 41.1 in Section 41.2. We need to show that this wave function satisfies the condition that the probability of finding the electron *somewhere* is 1. We then need to find the probability that it will be found in the region $r < a$. In part (a) we'll carry out the integral $\int |\psi|^2 dV$ over all space; if it is equal to 1, the wave function is normalized. In part (b) we'll carry out the same integral over a spherical volume that extends from the origin (the nucleus) out to a distance a from the nucleus.

EXECUTE (a) Since the wave function depends only on the radial coordinate r , we can choose our volume elements to be spherical shells of radius r , thickness dr , and volume dV given by Eq. (41.24). We then have

$$\int_{\text{all space}} |\psi_{1s}|^2 dV = \int_0^\infty \frac{1}{\pi a^3} e^{-2r/a} (4\pi r^2 dr) = \frac{4}{a^3} \int_0^\infty r^2 e^{-2r/a} dr$$

You can find the following indefinite integral in a table of integrals or by integrating by parts:

$$\int r^2 e^{-2r/a} dr = \left(-\frac{ar^2}{2} - \frac{a^2 r}{2} - \frac{a^3}{4} \right) e^{-2r/a}$$

Evaluating this between the limits $r = 0$ and $r = \infty$ is simple; it is zero at $r = \infty$ because of the exponential factor, and at $r = 0$ only the

last term in the parentheses survives. Thus the value of the definite integral is $a^3/4$. Putting it all together, we find

$$\int_0^\infty |\psi_{1s}|^2 dV = \frac{4}{a^3} \int_0^\infty r^2 e^{-2r/a} dr = \frac{4}{a^3} \frac{a^3}{4} = 1$$

The wave function *is* normalized.

(b) To find the probability P that the electron is found within $r < a$, we carry out the same integration but with the limits 0 and a . We'll leave the details to you. From the upper limit we get $-5e^{-2}a^3/4$; the final result is

$$\begin{aligned} P &= \int_0^a |\psi_{1s}|^2 4\pi r^2 dr = \frac{4}{a^3} \left(-\frac{5a^3 e^{-2}}{4} + \frac{a^3}{4} \right) \\ &= (-5e^{-2} + 1) = 1 - 5e^{-2} = 0.323 \end{aligned}$$

EVALUATE Our results tell us that in a ground state we expect to find the electron at a distance from the nucleus less than a about $\frac{1}{3}$ of the time and at a greater distance about $\frac{2}{3}$ of the time. It's hard to tell, but in Fig. 41.8, about $\frac{2}{3}$ of the area under the $1s$ curve is at distances greater than a (that is, $r/a > 1$).

KEYCONCEPT If a particle is in a stationary state described by the wave function ψ , the probability of finding the particle at a radial coordinate between r and $r + dr$ is $P(r)dr$, where $P(r) = 4\pi r^2 |\psi|^2$. If the wave function depends on the angle as well as on the radial coordinate, you must first average $|\psi|^2$ over all angles.

Hydrogenlike Atoms

Two generalizations that we discussed with the Bohr model in Section 39.3 are equally valid in the Schrödinger analysis. First, if the “atom” is not composed of a single proton and a single electron, using the reduced mass m_r of the system in Eqs. (41.21) and (41.26) will lead to changes that are substantial for some exotic systems. One example is *positronium*, in which a positron and an electron orbit each other; another is a *muonic atom*,

in which the electron is replaced by an unstable particle called a muon that has the same charge as an electron but is 207 times more massive. Second, our analysis is applicable to single-electron ions, such as He^+ , Li^{2+} , and so on. For such ions we replace e^2 by Ze^2 in Eqs. (41.21) and (41.26), where Z is the number of protons (the **atomic number**).

TEST YOUR UNDERSTANDING OF SECTION 41.3 Rank the following states of the hydrogen atom in order from highest to lowest probability of finding the electron in the vicinity of $r = 5a$: (i) $n = 1, l = 0, m_l = 0$; (ii) $n = 2, l = 1, m_l = +1$; (iii) $n = 2, l = 1, m_l = 0$.

ANSWER (i) $n = 1, l = 0, m_l = 0$ is most likely to be found at $r = 5a$. (ii) $n = 2, l = 1, m_l = +1$ is more likely to be found at $r = 5a$. (iii) $n = 2, l = 1, m_l = 0$ is most likely to be found at $r = 5a$. This result is independent of the values of the quantum numbers l and m_l . Hence

41.4 THE ZEEMAN EFFECT

The **Zeeman effect** is the splitting of atomic energy levels and the associated spectral lines when the atoms are placed in a magnetic field (Fig. 41.11). This effect confirms experimentally the quantization of angular momentum. In this section we'll assume that the only angular momentum is the *orbital* angular momentum of a single electron and learn why we call m_l the magnetic quantum number.

Atoms contain charges in motion, so it should not be surprising that magnetic forces cause changes in that motion and in the energy levels. In 1896 the Dutch physicist Pieter Zeeman was the first to show that in the presence of a magnetic field, some spectral lines were split into groups of closely spaced lines (Fig. 41.12). This effect now bears his name.

Magnetic Moment of an Orbiting Electron

Let's begin our analysis of the Zeeman effect by reviewing the concept of *magnetic dipole moment* or *magnetic moment*, introduced in Section 27.7. A plane current loop with vector area $\vec{A}$ carrying current I has a magnetic moment $\vec{\mu}$ given by

$$\vec{\mu} = I\vec{A} \quad (41.27)$$

When a magnetic dipole of moment $\vec{\mu}$ is placed in a magnetic field $\vec{B}$, the field exerts a torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on the dipole. The potential energy U associated with this interaction is given by Eq. (27.27):

$$U = -\vec{\mu} \cdot \vec{B} \quad (41.28)$$

Now let's use Eqs. (41.27) and (41.28) and the Bohr model to look at the interaction of a hydrogen atom with a magnetic field. The orbiting electron with speed v is equivalent to a current loop with radius r and area πr^2 . The average current I is the average charge per unit time that passes a given point of the orbit. This is equal to the charge magnitude e divided by the time T for one revolution, given by $T = 2\pi r/v$. Thus $I = ev/2\pi r$, and from Eq. (41.27) the magnitude μ of the magnetic moment is

$$\mu = IA = \frac{ev}{2\pi r} \pi r^2 = \frac{evr}{2} \quad (41.29)$$

We can also express this in terms of the magnitude L of the orbital angular momentum. From Eq. (10.28) the angular momentum of a particle in a circular orbit is $L = mvr$, so Eq. (41.29) becomes

$$\mu = \frac{e}{2m} L \quad (41.30)$$

The ratio of the magnitude of $\vec{\mu}$ to the magnitude of $\vec{L}$ is $\mu/L = e/2m$ and is called the *gyromagnetic ratio*.

Figure 41.11 Magnetic effects on the spectrum of sunlight. (a) The slit of a spectrograph is positioned along the black line crossing a portion of a sunspot. (b) The 0.4 T magnetic field in the sunspot (a thousand times greater than the earth's field) splits the middle spectral line into three lines.

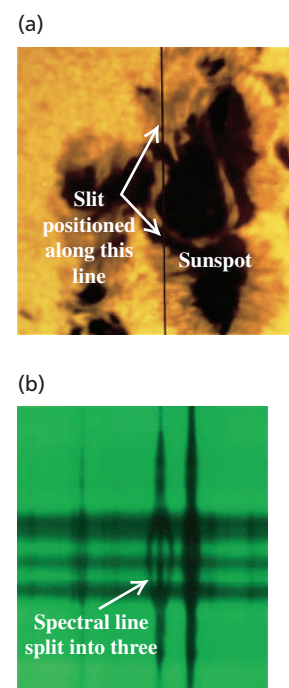
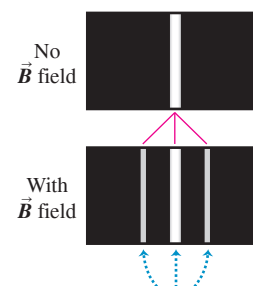


Figure 41.12 The normal Zeeman effect. Compare this to the magnetic splitting in the solar spectrum shown in Fig. 41.11b.



When an excited gas is placed in a magnetic field, the interaction of orbital magnetic moments with the field splits individual spectral lines of the gas into sets of three lines.

In the Bohr model, $L = nh/2\pi = n\hbar$, where $n = 1, 2, \dots$. For an $n = 1$ state (a ground state), Eq. (41.30) becomes $\mu = (e/2m)\hbar$. This quantity is a natural unit for magnetic moment; it is called one **Bohr magneton**, denoted by μ_B :

$$\mu_B = \frac{e\hbar}{2m} \quad (\text{definition of the Bohr magneton}) \quad (41.31)$$

(We defined this quantity in Section 28.8.) Evaluating Eq. (41.31) gives

$$\mu_B = 5.788 \times 10^{-5} \text{ eV/T} = 9.274 \times 10^{-24} \text{ J/T or A} \cdot \text{m}^2$$

Note that the units J/T and $\text{A} \cdot \text{m}^2$ are equivalent.

While the Bohr model suggests that the orbital motion of an atomic electron gives rise to a magnetic moment, this model does *not* give correct predictions about magnetic interactions. As an example, the Bohr model predicts that an electron in a hydrogen-atom ground state has an orbital magnetic moment of magnitude μ_B . But the Schrödinger picture tells us that such a ground-state electron is in an s state with zero angular momentum, so the orbital magnetic moment must be *zero*! To get the correct results, we must describe the states by using Schrödinger wave functions.

It turns out that in the Schrödinger formulation, electrons have the same ratio of μ to L (gyromagnetic ratio) as in the Bohr model—namely, $e/2m$. Suppose the magnetic field $\vec{B}$ is directed along the $+z$ -axis. From Eq. (41.28) the interaction energy U of the atom's magnetic moment with the field is

$$U = -\mu_z B \quad (41.32)$$

where μ_z is the z -component of the vector $\vec{\mu}$.

Now we use Eq. (41.30) to find μ_z , recalling that e is the *magnitude* of the electron charge and that the actual charge is $-e$. Because the electron charge is negative, the orbital angular momentum and magnetic moment vectors are opposite. We find

$$\mu_z = -\frac{e}{2m}L_z \quad (41.33)$$

For the Schrödinger wave functions, $L_z = m_l\hbar$, with $m_l = 0, \pm 1, \pm 2, \dots, \pm l$, so Eq. (41.33) becomes

$$\mu_z = -\frac{e}{2m}L_z = -m_l \frac{e\hbar}{2m} \quad (41.34)$$

Finally, using Eq. (41.31) for the Bohr magneton, we can express the interaction energy from Eq. (41.32) as

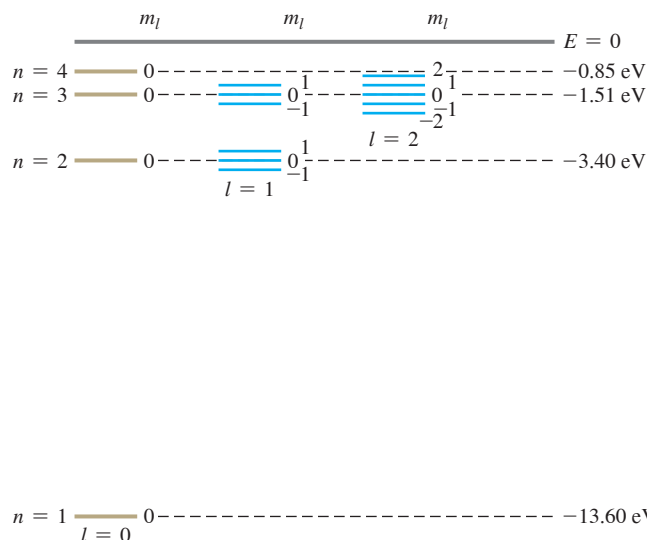
CAUTION Again, two uses of the symbol m . As in Section 41.3, the symbol m is used in two ways in Eq. (41.34). Don't confuse the electron mass m with the orbital magnetic quantum number m_l .

$$U = -\mu_z B = m_l \frac{e\hbar}{2m} B = m_l \mu_B B \quad (41.35)$$

The magnetic field shifts the energy of each orbital state by an amount U . The interaction energy U depends on the value of m_l because m_l determines the orientation of the orbital magnetic moment relative to the magnetic field. This dependence is the reason m_l is called the magnetic quantum number.

The values of m_l range from $-l$ to $+l$ in steps of one, so an energy level with a particular value of the orbital quantum number l contains $(2l + 1)$ different orbital states. Without a magnetic field these states all have the same energy; that is, they are degenerate. The magnetic field removes this degeneracy. In the presence of a magnetic field they are split into $2l + 1$ distinct energy levels; adjacent levels differ in energy by $(e\hbar/2m)B = \mu_B B$. We can understand this in terms of the connection between degeneracy and symmetry. With a

Figure 41.13 This energy-level diagram for hydrogen shows how the levels are split when the electron's orbital magnetic moment interacts with an external magnetic field. The values of m_l are shown adjacent to the various levels. The relative magnitudes of the level splittings are exaggerated for clarity. The $n = 4$ splittings are not shown; can you draw them in?



magnetic field applied along the z -axis, the atom is no longer completely symmetric under rotation: There is a preferred direction in space. By removing the symmetry, we remove the degeneracy of states.

Figure 41.13 shows the effect on the energy levels of hydrogen. Spectral lines corresponding to transitions from one set of levels to another set are correspondingly split and appear as a series of three closely spaced spectral lines replacing a single line. As the following example shows, the splitting of spectral lines is quite small because the value of $\mu_B B$ is small even for substantial magnetic fields.

EXAMPLE 41.5 An atom in a magnetic field

An atom in a state with $l = 1$ emits a photon with wavelength 600.000 nm as it decays to a state with $l = 0$. If the atom is placed in a magnetic field with magnitude $B = 2.00$ T, what are the shifts in the energy levels and in the wavelength that result from the interaction between the atom's orbital magnetic moment and the magnetic field?

IDENTIFY and SET UP This problem concerns the splitting of atomic energy levels by a magnetic field (the Zeeman effect). We use Eq. (41.35) to determine the energy-level shifts. The relationship $E = hc/\lambda$ between the energy and wavelength of a photon then lets us calculate the wavelengths emitted during transitions from the $l = 1$ states to the $l = 0$ state.

EXECUTE The energy of a 600 nm photon is

$$E = \frac{hc}{\lambda} = \frac{(4.14 \times 10^{-15} \text{ eV} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{600 \times 10^{-9} \text{ m}} = 2.07 \text{ eV}$$

If there is no external magnetic field, that is the difference in energy between the $l = 0$ and $l = 1$ levels.

With a 2.00 T field present, Eq. (41.35) shows that there is no shift of the $l = 0$ state (which has $m_l = 0$). For the $l = 1$ states, the splitting of levels is given by

$$\begin{aligned} U &= m_l \mu_B B = m_l (5.788 \times 10^{-5} \text{ eV/T})(2.00 \text{ T}) \\ &= m_l (1.16 \times 10^{-4} \text{ eV}) = m_l (1.85 \times 10^{-23} \text{ J}) \end{aligned}$$

The possible values of m_l for $l = 1$ are $-1, 0$, and $+1$, and the three corresponding levels are separated by equal intervals of $1.16 \times 10^{-4} \text{ eV}$. This is a small fraction of the 2.07 eV photon energy:

$$\frac{\Delta E}{E} = \frac{1.16 \times 10^{-4} \text{ eV}}{2.07 \text{ eV}} = 5.60 \times 10^{-5}$$

The corresponding wavelength shifts are about $(5.60 \times 10^{-5}) \times (600 \text{ nm}) = 0.034 \text{ nm}$. The original 600.000 nm line is split into a triplet with wavelengths 599.966, 600.000, and 600.034 nm.

EVALUATE Even though 2.00 T would be a strong field in most laboratories, the wavelength splittings are extremely small. Nonetheless, modern spectrographs have more than enough chromatic resolving power to measure these splittings (see Section 36.5).

KEYCONCEPT An electron in an atom has a nonzero magnetic moment if it is in a state with a nonzero orbital angular momentum. If this atom is in a magnetic field $\vec{B}$, the electron energy depends on the electron's orbital magnetic quantum number m_l , which describes how the magnetic moment vector is oriented with respect to $\vec{B}$.

Figure 41.14 This figure shows how the splitting of the energy levels of a d state ($l = 2$) depends on the magnitude B of an external magnetic field, assuming only an orbital magnetic moment.

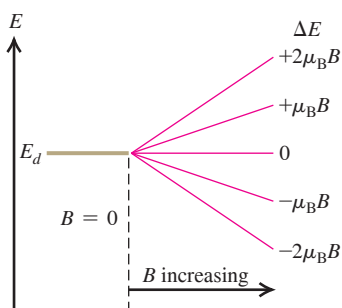
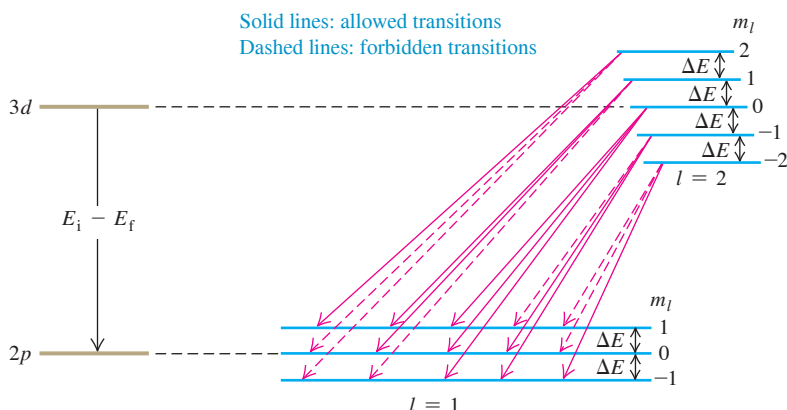


Figure 41.15 The cause of the normal Zeeman effect. The magnetic field splits the levels, but selection rules allow transitions with only three different energy changes, giving three different photon frequencies and wavelengths.



Selection Rules

Figure 41.14 shows what happens to a set of d states ($l = 2$) as the magnetic field increases. With zero field the five states $m_l = -2, -1, 0, 1,$ and 2 are degenerate (have the same energy), but the applied field spreads the states out. Figure 41.15 shows the splittings of both the $3d$ and $2p$ states. Equal energy differences $(e\hbar/2m)B = \mu_B B$ separate adjacent levels. In the absence of a magnetic field, a transition from a $3d$ to a $2p$ state would yield a single spectral line with photon energy $E_i - E_f$. With the levels split as shown, it might seem that there are five possible photon energies.

In fact, there are only three possibilities. Not all combinations of initial and final levels are possible because of a restriction associated with conservation of angular momentum. The photon ordinarily carries off one unit ($\hbar$) of angular momentum, which leads to the requirements that in a transition l must change by 1 and m_l must change by 0 or ± 1 . These requirements are called **selection rules**. Transitions that obey these rules are called *allowed transitions*; those that don't are *forbidden transitions*. In Fig. 41.15 we show the allowed transitions by solid arrows. You should count the possible transition energies to convince yourself that the nine solid arrows give only three possible energies; the zero-field value $E_i - E_f$, and that value plus or minus $\Delta E = (e\hbar/2m)B = \mu_B B$. Figure 41.12 shows the corresponding spectral lines.

What we have described is called the *normal Zeeman effect*. It is based entirely on the orbital angular momentum of the electron. However, it leaves out a very important consideration: the electron *spin* angular momentum, the subject of the next section.

TEST YOUR UNDERSTANDING OF SECTION 41.4 In this section we assumed that the magnetic field points in the positive z -direction. Would the results be different if the magnetic field pointed in the positive x -direction?

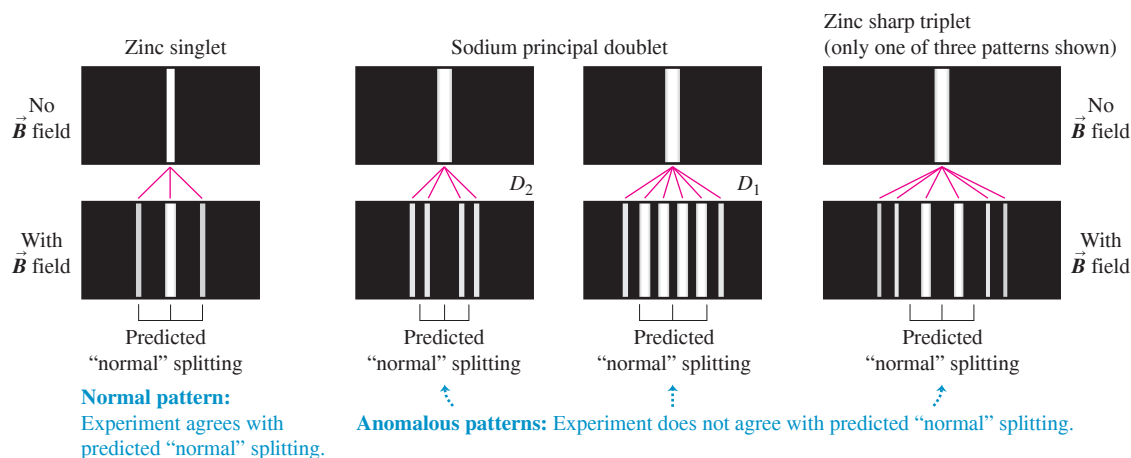
ANSWER

no All that matters is the component of the electron's orbital magnetic moment along the direction of $\mathbf{B}$. We called this quantity μ_z in Eq. (41.32) because we defined the positive z -axis to be in the direction of $\mathbf{B}$. In reality, the names of the axes are arbitrary.

41.5 ELECTRON SPIN

Despite the success of the Schrödinger equation in predicting the energy levels of the hydrogen atom, experimental observations indicate that it doesn't tell the whole story of the behavior of electrons in atoms. First, spectroscopists have found magnetic-field splitting into other than the three equally spaced lines that we explained in Section 41.4 (see Fig. 41.12). Before this effect was understood, it was called the *anomalous Zeeman*

Figure 41.16 Illustrations of the normal and anomalous Zeeman effects for two elements, zinc and sodium. The brackets under each illustration show the “normal” splitting predicted by ignoring the effect of electron spin.



effect to distinguish it from the “normal” effect discussed in the preceding section.

Figure 41.16 shows both kinds of splittings.

Second, some energy levels show splittings that resemble the Zeeman effect even when there is *no* external magnetic field. For example, when the lines in the hydrogen spectrum are examined with a high-resolution spectrograph, some lines are found to consist of sets of closely spaced lines called *multiplets*. Similarly, the orange-yellow line of sodium, corresponding to the transition $4p \rightarrow 3s$ of the outer electron, is found to be a doublet ($\lambda = 589.0, 589.6$ nm), suggesting that the $4p$ level might in fact be two closely spaced levels. The Schrödinger equation in its original form didn't predict any of this.

The Stern–Gerlach Experiment

Similar anomalies appeared in 1922 in atomic-beam experiments performed in Germany by Otto Stern and Walter Gerlach. When they passed a beam of neutral atoms through a nonuniform magnetic field (Fig. 41.17), atoms were deflected according to the orientation of their magnetic moments with respect to the field. These experiments demonstrated the quantization of angular momentum in a very direct way. If there were only orbital angular momentum, the deflections would split the beam into an odd number ($2l + 1$) of different components. However, some atomic beams were split into an *even* number of components. If we use a different symbol j for an angular momentum quantum number, setting

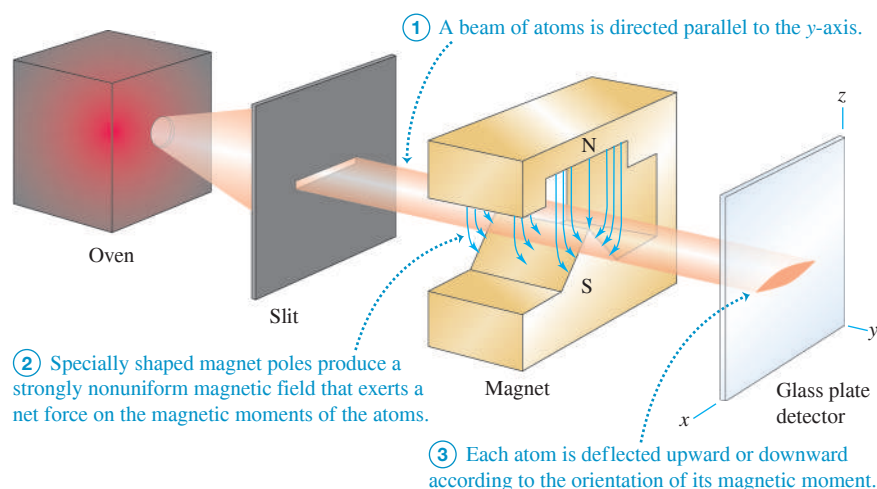


Figure 41.17 The Stern–Gerlach experiment.

$2j + 1$ equal to an even number gives $j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$, suggesting a half-integer angular momentum. This can't be understood on the basis of the Bohr model and similar pictures of atomic structure.

In 1925 two graduate students in the Netherlands, Samuel Goudsmit and George Uhlenbeck, proposed that the electron might have some additional motion. Using a semi-classical model, they suggested that the electron might behave like a spinning sphere of charge instead of a particle. If so, it would have an additional *spin* angular momentum and magnetic moment. If these were quantized in much the same way as *orbital* angular momentum and magnetic moment, they might help explain the observed energy-level anomalies.

An Analogy for Electron Spin

To introduce the concept of **electron spin**, let's start with an analogy. The earth travels in a nearly circular orbit around the sun, and at the same time it *rotates* on its axis. Each motion has its associated angular momentum, which we call the *orbital* and *spin* angular momentum, respectively. The total angular momentum of the earth is the vector sum of the two. If we were to model the earth as a single point, it would have no moment of inertia about its spin axis and thus no spin angular momentum. But when our model includes the finite size of the earth, spin angular momentum becomes possible.

In the Bohr model, suppose the electron is not just a point charge but a small spinning sphere that orbits the nucleus. Then the electron has not only orbital angular momentum but also spin angular momentum associated with the rotation of its mass about its axis. The sphere carries an electric charge, so the spinning motion leads to current loops and to a magnetic moment, as we discussed in Section 27.7. In a magnetic field, the *spin* magnetic moment has an interaction energy in addition to that of the *orbital* magnetic moment (the normal Zeeman-effect interaction that we discussed in Section 41.4). We should see additional Zeeman shifts due to the spin magnetic moment.

As we mentioned, such shifts *are* indeed observed in precise spectroscopic analysis. This and a variety of other experimental evidence have shown conclusively that the electron *does* have a spin angular momentum and a spin magnetic moment that do not depend on its orbital motion but are intrinsic to the electron itself. The origin of this spin angular momentum is fundamentally quantum-mechanical, so it's not correct to model the electron as a spinning charged sphere. But just as the Bohr model can be a useful conceptual picture for the motion of an electron in an atom, the spinning-sphere analogy can help you visualize the intrinsic spin angular momentum of an electron.

Spin Quantum Numbers

Like orbital angular momentum, the spin angular momentum of an electron (denoted by $\vec{S}$) is found to be quantized. Suppose we have an apparatus that measures a particular component of $\vec{S}$, say the z -component S_z . We find that the only possible values are

$$\begin{array}{l} \text{z-component of} \\ \text{spin angular momentum} \\ \text{of electron} \end{array} \cdots S_z = m_s \hbar \cdots \begin{array}{l} \text{Spin magnetic quantum number} = \pm \frac{1}{2} \\ \downarrow \\ \text{Planck's constant} \\ \text{divided by } 2\pi \end{array} \quad (41.36)$$

This relationship is reminiscent of the expression $L_z = m_l \hbar$ for the z -component of orbital angular momentum, except that $|S_z|$ is *one-half* of $\hbar$ instead of an *integer* multiple. In analogy to the orbital magnetic quantum number m_l , we call the quantum number m_s the **spin magnetic quantum number**. Since m_s has only two possible values, $+\frac{1}{2}$ and $-\frac{1}{2}$, it follows that the spin angular momentum vector $\vec{S}$ can have only two orientations in space relative to the z -axis: “*spin up*” with a z -component of $+\frac{1}{2}\hbar$ and “*spin down*” with a z -component of $-\frac{1}{2}\hbar$.

Equation (41.36) also suggests that the magnitude S of the spin angular momentum is given by an expression analogous to Eq. (41.22) with the orbital quantum number l replaced by the **spin quantum number** $s = \frac{1}{2}$:

$$\text{Magnitude of spin angular momentum of electron} \quad S = \sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right)}\hbar = \sqrt{\frac{3}{4}}\hbar \quad (41.37)$$

Maximum value of spin magnetic quantum number = $\frac{1}{2}$
 Planck's constant divided by 2π

The electron is often called a “spin-one-half particle” or “spin- $\frac{1}{2}$ particle.”

We see that to label the state of the electron in a hydrogen atom completely, we need *four* quantum numbers: n , l , and m_l (described in Section 41.3) to specify the electron's motion relative to the nucleus, plus the spin magnetic quantum number m_s to specify the electron spin orientation.

To visualize the quantized spin of an electron in a hydrogen atom, think of the electron probability distribution function $|\psi|^2$ as a cloud surrounding the nucleus like those shown in Figs. 41.9 and 41.10. Then imagine many tiny spin arrows distributed throughout the cloud, either all with components in the $+z$ -direction or all with components in the $-z$ -direction. But don't take this picture too seriously.

Just as the orbital magnetic moment of the electron is proportional to its orbital angular momentum $\vec{L}$ (see Section 41.4), the electron's spin magnetic moment is proportional to its spin angular momentum $\vec{S}$. The z -component of the spin magnetic moment (μ_z) turns out to be related to S_z by

$$\mu_z = -(2.00232) \frac{e}{2m} S_z \quad (41.38)$$

where $-e$ and m are (as usual) the charge and mass of the electron. When the atom is placed in a magnetic field, the interaction energy $-\vec{\mu} \cdot \vec{B}$ of the spin magnetic dipole moment with the field causes further splittings in energy levels and in the corresponding spectral lines.

Equation (41.38) shows that the gyromagnetic ratio for electron spin is approximately *twice* as great as the value $e/2m$ for *orbital* angular momentum and magnetic dipole moment. This result has no classical analog. But in 1928 Paul Dirac developed a relativistic generalization of the Schrödinger equation for electrons. His equation gave a spin gyromagnetic ratio of exactly $2(e/2m)$. It took another two decades to develop the area of physics called *quantum electrodynamics*, abbreviated QED, which predicts the value we've given to “only” six significant figures as 2.00232. QED now predicts a value that agrees with the currently accepted experimental value of 2.00231930436182(52), making QED the most precise theory in all science.

EXAMPLE 41.6 Energy of electron spin in a magnetic field

WITH VARIATION PROBLEMS

Calculate the interaction energy for an electron in an $l = 0$ state in a magnetic field with magnitude 2.00 T.

IDENTIFY and SET UP For $l = 0$ the electron has zero orbital angular momentum and zero orbital magnetic moment. Hence the only magnetic interaction is that between the $\vec{B}$ field and the spin magnetic moment $\vec{\mu}$. From Eq. (41.28), the interaction energy is $U = -\vec{\mu} \cdot \vec{B}$. As in Section 41.4, we take $\vec{B}$ to be in the positive z -direction so that $U = -\mu_z B$ [Eq. (41.32)]. Equation (41.38) gives μ_z in terms of S_z , and Eq. (41.36) gives S_z .

EXECUTE Combining Eqs. (41.36) and (41.38), we have

$$\begin{aligned} \mu_z &= -(2.00232) \left(\frac{e}{2m} \right) \left(\pm \frac{1}{2} \hbar \right) \\ &= \mp \frac{1}{2} (2.00232) \left(\frac{e\hbar}{2m} \right) = \mp (1.00116) \mu_B \end{aligned}$$

$$\begin{aligned} &= \mp (1.00116)(9.274 \times 10^{-24} \text{ J/T}) \\ &= \mp 9.285 \times 10^{-24} \text{ J/T} = \mp 5.795 \times 10^{-5} \text{ eV/T} \end{aligned}$$

Then from Eq. (41.32),

$$\begin{aligned} U &= -\mu_z B = \pm (9.285 \times 10^{-24} \text{ J/T})(2.00 \text{ T}) \\ &= \pm 1.86 \times 10^{-23} \text{ J} = \pm 1.16 \times 10^{-4} \text{ eV} \end{aligned}$$

The positive value of U and the negative value of μ_z correspond to $S_z = +\frac{1}{2}\hbar$ (spin up); the negative value of U and the positive value of μ_z correspond to $S_z = -\frac{1}{2}\hbar$ (spin down).

EVALUATE Let's check the *signs* of our results. If the electron is spin down, $\vec{S}$ points generally opposite to $\vec{B}$. Then the magnetic moment $\vec{\mu}$ (which is opposite to $\vec{S}$ because the electron charge is negative)

BIO APPLICATION Electron Spins and Dating Human Origins

In many atoms, the net spin of all of the electrons is zero (as many electrons are “spin up” as are “spin down”). If these atoms are ionized and lose an electron, however, the net spin of the ion that remains is nonzero. This happens naturally in tooth enamel, where ionization is caused by radioactivity in the environment. The longer a tooth is exposed, the more ions are present. To find the age of fossil teeth, such as those in this skull of *Homo neanderthalensis*, a sample of the enamel is placed in a strong magnetic field. The ion spins align opposite to this field (become “spin down”). The sample is then illuminated with microwave photons of just the right energy to flip the spins to the higher-energy configuration aligned with the field (“spin up”). The amount of microwave energy absorbed in this process (called *electron spin resonance*) indicates the number of ions present and hence the age of the enamel.



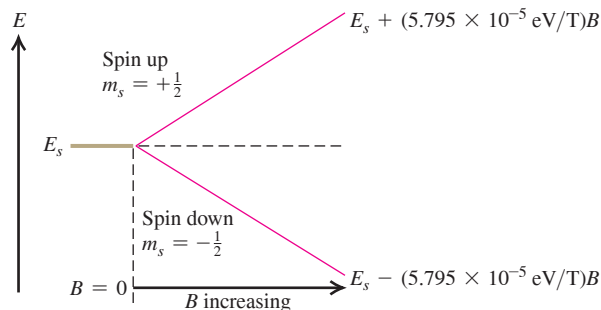
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points generally parallel to $\vec{B}$, and μ_z is positive. From Eq. (41.28), $U = -\vec{\mu} \cdot \vec{B}$, the interaction energy is negative if $\vec{\mu}$ and $\vec{B}$ are parallel. Our results show that U is indeed negative in this case. We can similarly confirm that U must be positive and μ_z negative for a spin-up electron.

The red lines in Fig. 41.18 show how the interaction energies for the two spin states vary with the magnetic-field magnitude B . The graphs are straight lines because, from Eq. (41.32), U is proportional to B .

KEYCONCEPT In addition to any orbital angular momentum, an electron has an intrinsic angular momentum $\vec{S}$ called spin. The component of $\vec{S}$ along a given axis can have only two possible values, $+\frac{1}{2}\hbar$ (“spin up”) or $-\frac{1}{2}\hbar$ (“spin down”). An electron also has a spin magnetic moment $\vec{\mu}$ directed opposite to $\vec{S}$, so when it is placed in a magnetic field $\vec{B}$, the electron’s energy depends on the component of $\vec{S}$ along the direction of $\vec{B}$.

Figure 41.18 An $l = 0$ level of a single electron is split by interaction of the spin magnetic moment with an external magnetic field. The greater the magnitude B of the magnetic field, the greater the splitting. The quantity $5.795 \times 10^{-5} \text{ eV/T}$ is just $(1.00116)\mu_B$.



Spin-Orbit Coupling

We mentioned earlier that the spin magnetic dipole moment also gives splitting of energy levels even when there is *no* external field. One cause involves the orbital motion of the electron. In the Bohr model, observers moving with the electron would see the positively charged nucleus revolving around them (just as to earthbound observers the sun seems to be orbiting the earth). This apparent motion of charge causes a magnetic field at the location of the electron, as measured in the electron’s moving frame of reference. The resulting interaction with the spin magnetic moment causes a twofold splitting of this level, corresponding to the two possible orientations of electron spin.

Discussions based on the Bohr model can’t be taken too seriously, but a similar result can be derived from the Schrödinger equation. The interaction energy U can be expressed in terms of the scalar product of the angular momentum vectors $\vec{L}$ and $\vec{S}$. This effect is called **spin-orbit coupling**; it is responsible for the small energy difference between the two closely spaced, lowest excited levels of sodium shown in Fig. 39.19a and for the corresponding doublet (589.0, 589.6 nm) in the spectrum of sodium.

EXAMPLE 41.7 An effective magnetic field

WITH VARIATION PROBLEMS

To six significant figures, the wavelengths of the two spectral lines that make up the sodium doublet are $\lambda_1 = 588.995 \text{ nm}$ and $\lambda_2 = 589.592 \text{ nm}$. Calculate the effective magnetic field experienced by the electron in the $3p$ levels of the sodium atom.

IDENTIFY and SET UP The two lines in the sodium doublet result from transitions from the two $3p$ levels, which are split by spin-orbit coupling, to the $3s$ level, which is *not* split because it has $L = 0$. We picture the spin-orbit coupling as an interaction between the electron spin magnetic moment and an effective magnetic field due to the nucleus. This example is like Example 41.6 in reverse: There we were given B and found the difference between the energies of the two spin states, while here we use the energy difference to find the target variable B . The difference in energy between the two $3p$ levels is equal to the difference in energy between the two photons of the sodium doublet. We use this relationship and the results of Example 41.6 to determine B .

EXECUTE The energies of the two photons are $E_1 = hc/\lambda_1$ and $E_2 = hc/\lambda_2$. Here $E_1 > E_2$ because $\lambda_1 < \lambda_2$, so the difference in their energies is

$$\begin{aligned}\Delta E &= \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2} = hc \left(\frac{\lambda_2 - \lambda_1}{\lambda_2 \lambda_1} \right) \\ &= (4.136 \times 10^{-15} \text{ eV} \cdot \text{s}) (2.998 \times 10^8 \text{ m/s})\end{aligned}$$

$$\begin{aligned}&\times \frac{(589.592 \times 10^{-9} \text{ m}) - (588.995 \times 10^{-9} \text{ m})}{(589.592 \times 10^{-9} \text{ m})(588.995 \times 10^{-9} \text{ m})} \\ &= 0.00213 \text{ eV} = 3.41 \times 10^{-22} \text{ J}\end{aligned}$$

This equals the energy difference between the two $3p$ levels. The spin-orbit interaction raises one level by $1.70 \times 10^{-22} \text{ J}$ (one-half of $3.41 \times 10^{-22} \text{ J}$) and lowers the other by $1.70 \times 10^{-22} \text{ J}$. From Example 41.6, the amount each state is raised or lowered is $|U| = (1.00116)\mu_B B$, so

$$B = \left| \frac{U}{(1.00116)\mu_B} \right| = \frac{1.70 \times 10^{-22} \text{ J}}{9.28 \times 10^{-24} \text{ J/T}} = 18.0 \text{ T}$$

EVALUATE The electron experiences a *very* strong effective magnetic field. To produce a steady, macroscopic field of this magnitude in the laboratory requires state-of-the-art electromagnets.

KEYCONCEPT An atomic electron with an orbital angular momentum around the nucleus experiences an effective magnetic field. This field interacts with the spin magnetic moment of the electron, so the electron energy depends on the relative orientation of its spin angular momentum and orbital angular momentum. This is called spin-orbit coupling.

Combining Orbital and Spin Angular Momenta

The orbital and spin angular momenta ($\vec{L}$ and $\vec{S}$, respectively) can combine in various ways. The vector sum of $\vec{L}$ and $\vec{S}$ is the *total angular momentum* $\vec{J}$:

$$\vec{J} = \vec{L} + \vec{S} \quad (41.39)$$

The possible values of the magnitude J are given in terms of a quantum number j , called the **total angular momentum quantum number**:

$$J = \sqrt{j(j+1)}\hbar \quad (41.40)$$

We can then have states in which $j = |l \pm \frac{1}{2}|$. The $l + \frac{1}{2}$ states correspond to the case in which the vectors $\vec{L}$ and $\vec{S}$ have parallel z -components; for the $l - \frac{1}{2}$ states, $\vec{L}$ and $\vec{S}$ have antiparallel z -components. For example, when $l = 1$, j can be $\frac{1}{2}$ or $\frac{3}{2}$. In another spectroscopic notation these p states are labeled $^2P_{1/2}$ and $^2P_{3/2}$, respectively. The superscript is the number of possible spin orientations, the letter P (now capitalized) indicates states with $l = 1$, and the subscript is the value of j . We used this scheme to label the energy levels of the sodium atom in Fig. 39.19a.

In addition to shifts in energy levels due to magnetic effects within the atom, there are shifts of the same magnitude due to relativistic corrections to the kinetic energy of the electron. (In the Bohr model, an electron in the $n = 1$ orbit of hydrogen moves at about 1% of the speed of light.) The term “fine structure” refers to the energy-level shifts caused by magnetic and relativistic effects together, as well as to the line splittings that result from these shifts. Including these effects, the energy levels of the hydrogen atom are

Energy levels
of hydrogen,
including fine
structure

$$E_{n,j} = -\frac{13.60 \text{ eV}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right]$$

Principal quantum number
($n = 1, 2, 3, \dots$)

Fine-structure constant

Total angular momentum
quantum number

(41.41)

The *fine-structure constant* α that appears in Eq. (41.41) is a dimensionless number:

$$\alpha = \frac{1}{4\pi\epsilon_0} \frac{e^2}{\hbar c} = 7.2973525664(17) \times 10^{-3} \quad (\text{fine-structure constant}) \quad (41.42)$$

To five significant figures, $\alpha = 7.2974 \times 10^{-3} = 1/137.04$.

In Section 41.3 we found that the energy levels of the hydrogen atom are degenerate: All states that have the same principal quantum number n have the same energy. Our more complete treatment including fine structure shows that this degeneracy is removed: States with the same n but different values of the total angular momentum quantum number j have different energies. Example 41.8 illustrates this for the $n = 2$ levels of hydrogen.

EXAMPLE 41.8 Fine structure and spectral-line splitting

WITH VARIATION PROBLEMS

For an electron with orbital quantum number $l = 0$, the angular momentum is due to spin alone and the only possible value of the total angular momentum quantum number is $j = \frac{1}{2}$. If $l = 1$, two values are possible: $j = \frac{3}{2}$ (the spin and orbital angular momentum vectors are in roughly the same direction and so add together) and $j = \frac{1}{2}$ (the spin and orbital angular momentum vectors are in roughly opposite directions and so partially cancel). (a) Find the energies of a state of the electron in a hydrogen atom with $n = 2$, $l = 1$, $j = \frac{3}{2}$ (a $^2P_{3/2}$ state) and a state with $n = 2$, $l = 1$, $j = \frac{1}{2}$ (a $^2P_{1/2}$ state), and calculate the difference between the two energies. Which state has the higher energy? (b) Find the difference in wavelengths between (i) a photon emitted in a transition from a state with $n = 2$, $l = 1$, $j = \frac{3}{2}$ to a state with $n = 1$, $l = 0$, $j = \frac{1}{2}$ and (ii) a photon

emitted in a transition from a state with $n = 2$, $l = 1$, $j = \frac{1}{2}$ to a state with $n = 1$, $l = 0$, $j = \frac{1}{2}$. Which photon has the longer wavelength?

IDENTIFY and SET UP In part (a) we use Eq. (41.41) to find the difference in energy between these two states, which have the same n value but different j values. The difference between the two energies is due to fine structure, so we expect this difference to be small. In part (b) both transitions end in the same state with $n = 1$, so we recognize from Section 39.3 that both are members of the Lyman series. If there were no fine structure, the two initial states would have the same energies and both photons would have the same energy E and hence the same wavelength $\lambda = hc/E$. But because the two initial states differ slightly in energy, the photons in the two transitions will have slightly different wavelengths.

Continued

EXECUTE (a) From Eq. (41.41), the energies of the two states are

$$\begin{aligned} E_{n=2, j=3/2} &= -\frac{13.60 \text{ eV}}{2^2} \left[1 + \frac{\alpha^2}{2^2} \left(\frac{2}{\frac{3}{2} + \frac{1}{2}} - \frac{3}{4} \right) \right] \\ &= -3.40 \text{ eV} \left(1 + \frac{\alpha^2}{16} \right) \\ E_{n=2, j=1/2} &= -\frac{13.60 \text{ eV}}{2^2} \left[1 + \frac{\alpha^2}{2^2} \left(\frac{2}{\frac{1}{2} + \frac{1}{2}} - \frac{3}{4} \right) \right] \\ &= -3.40 \text{ eV} \left(1 + \frac{5\alpha^2}{16} \right) \end{aligned}$$

The fine-structure terms involving α^2 cause both states to have lower (more negative) energies than in the Bohr model, in which both states would have energy $E_2 = -3.40 \text{ eV}$. The fine-structure term is five times greater for the $j = \frac{1}{2}$ state, so the $j = \frac{3}{2}$ state has the higher (less negative) energy. Using the value of the fine-structure constant α from Eq. (41.42), we get the difference in energy between the two states:

$$\begin{aligned} E_{n=2, j=3/2} - E_{n=2, j=1/2} &= \left[-3.40 \text{ eV} \left(1 + \frac{\alpha^2}{16} \right) \right] - \left[-3.40 \text{ eV} \left(1 + \frac{5\alpha^2}{16} \right) \right] \\ &= 3.40 \text{ eV} \left(\frac{4\alpha^2}{16} \right) = (3.40 \text{ eV}) \left(\frac{4}{16} \right) \left(\frac{1}{137.04} \right)^2 \\ &= 4.53 \times 10^{-5} \text{ eV} \end{aligned}$$

(b) The photon energy in each case equals the difference between the energies of the initial and final states of the electron. The final electron state for both transitions has $n = 1$ and $j = \frac{1}{2}$, which from Eq. (41.41) has energy

$$\begin{aligned} E_{n=1, j=1/2} &= -\frac{13.60 \text{ eV}}{1^2} \left[1 + \frac{\alpha^2}{1^2} \left(\frac{1}{\frac{1}{2} + \frac{1}{2}} - \frac{3}{4} \right) \right] \\ &= -13.60 \text{ eV} \left(1 + \frac{\alpha^2}{4} \right) \end{aligned}$$

Note that as for the two $n = 2$ states in part (a), the fine-structure correction to the $n = 1$ state makes the energy more negative. The photon energies for the two transitions are then

$$\begin{aligned} E_{\text{photon}}(n = 2, l = 1, j = \frac{3}{2} \text{ to } n = 1, l = 0, j = \frac{1}{2}) &= E_{n=2, j=3/2} - E_{n=1, j=1/2} \\ &= \left[-3.40 \text{ eV} \left(1 + \frac{\alpha^2}{16} \right) \right] - \left[-13.60 \text{ eV} \left(1 + \frac{\alpha^2}{4} \right) \right] \\ &= 10.20 \text{ eV} + (3.40 \text{ eV}) \left(\frac{15\alpha^2}{16} \right) = 10.20 \text{ eV} + 1.70 \times 10^{-4} \text{ eV} \end{aligned}$$

$$\begin{aligned} E_{\text{photon}}(n = 2, l = 1, j = \frac{1}{2} \text{ to } n = 1, l = 0, j = \frac{1}{2}) &= E_{n=2, j=1/2} - E_{n=1, j=1/2} \\ &= \left[-3.40 \text{ eV} \left(1 + \frac{5\alpha^2}{16} \right) \right] - \left[-13.60 \text{ eV} \left(1 + \frac{\alpha^2}{4} \right) \right] \\ &= 10.20 \text{ eV} + (3.40 \text{ eV}) \left(\frac{11\alpha^2}{16} \right) = 10.20 \text{ eV} + 1.24 \times 10^{-4} \text{ eV} \end{aligned}$$

The photon emitted when the initial state is $n = 2, l = 1, j = \frac{1}{2}$ has a lower energy E_{photon} and hence will have a longer wavelength, as given by the equation $\lambda = hc/E_{\text{photon}}$. If you plug the two photon energies into this equation, your calculator will tell you that $\lambda = 1.216 \times 10^{-7} \text{ m} = 121.6 \text{ nm}$ in both cases because the energy difference is so small. To find the wavelength difference $\Delta\lambda$, we instead take the differential of both sides of $\lambda = hc/E_{\text{photon}}$:

$$\begin{aligned} d\lambda &= d\left(\frac{hc}{E_{\text{photon}}}\right) = -\frac{hc}{(E_{\text{photon}})^2} dE_{\text{photon}} \\ &= -\left(\frac{hc}{E_{\text{photon}}}\right) \left(\frac{1}{E_{\text{photon}}}\right) dE_{\text{photon}} = -\frac{\lambda}{E_{\text{photon}}} dE_{\text{photon}} \end{aligned}$$

The minus sign means that a *decrease* in photon energy corresponds to an *increase* in photon wavelength. Replacing $d\lambda$ with $\Delta\lambda$ (the wavelength difference that we seek) and dE_{photon} with ΔE_{photon} , we get the difference between the two photon wavelengths:

$$\Delta\lambda = -\frac{\lambda}{E_{\text{photon}}} \Delta E_{\text{photon}}$$

To four significant digits, we have $\lambda = 121.6 \text{ nm}$ and $E_{\text{photon}} = 10.20 \text{ eV}$. We find the photon energy difference ΔE_{photon} from the two expressions above, subtracting the larger energy from the smaller one so that $\Delta\lambda$ is positive:

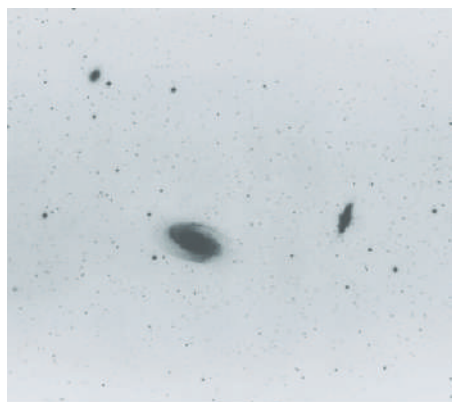
$$\begin{aligned} \Delta\lambda &= -\frac{121.6 \text{ nm}}{10.20 \text{ eV}} \left\{ \left[10.20 \text{ eV} + (3.40 \text{ eV}) \left(\frac{11\alpha^2}{16} \right) \right] \right. \\ &\quad \left. - \left[10.20 \text{ eV} + (3.40 \text{ eV}) \left(\frac{15\alpha^2}{16} \right) \right] \right\} \\ &= -\frac{121.6 \text{ nm}}{10.20 \text{ eV}} (3.40 \text{ eV}) \left(-\frac{4\alpha^2}{16} \right) \\ &= \frac{121.6 \text{ nm}}{10.20 \text{ eV}} (3.40 \text{ eV}) \left(\frac{4}{16} \right) \left(\frac{1}{137.04} \right)^2 \\ &= 5.40 \times 10^{-4} \text{ nm} \end{aligned}$$

EVALUATE This line splitting is very small, as predicted. Fine structure is fine indeed! It is nonetheless observable with a diffraction grating that has a sufficient number of lines (see Section 36.5). The measured wavelengths are 121.567364 nm for the transition that begins in the $j = \frac{1}{2}$ state and 121.566824 nm for the transition that begins in the $j = \frac{3}{2}$ state. These are ultraviolet wavelengths.

There are also states of the hydrogen atom with $n = 2, l = 0, j = \frac{1}{2}$. [From Eq. (41.41), these states have the same energy as those with $n = 2, l = 1, j = \frac{1}{2}$; the energy $E_{n,j}$ depends on n and j but not on l .] However, an electron in an $n = 2, l = 0, j = \frac{1}{2}$ state *cannot* emit a photon and transition to an $n = 1, l = 0, j = \frac{1}{2}$ state. Such a transition is forbidden by the selection rule that l must change by 1 when a photon is emitted (see Section 41.4).

KEYCONCEPT When both magnetic and relativistic effects are included, the energy levels of the hydrogen atom [Eq. (41.41)] depend on both the principal quantum number $n = 1, 2, 3, \dots$ and the quantum number j associated with the electron's total angular momentum (the vector sum of the orbital and spin angular momenta). If the electron has orbital quantum number l , the value of j is either $l + \frac{1}{2}$ or $|l - \frac{1}{2}|$. Transitions between states that involve emitting or absorbing a photon are allowed only if l changes by 1 in the transition.

(a) Galaxies in visible light (negative image; galaxies appear dark)



(b) Radio image of the same galaxies at wavelength 21 cm

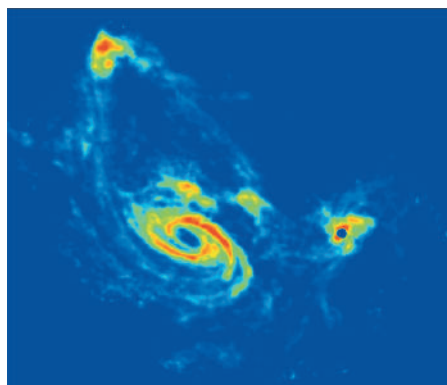


Figure 41.19 (a) In a visible-light image, these three distant galaxies appear to be unrelated. But in fact these galaxies are connected by immense streamers of hydrogen gas. This is revealed in (b), the false-color image made with a radio telescope tuned to the 21 cm wavelength emitted by hydrogen atoms.

Additional, much smaller splittings are associated with the fact that the *nucleus* of the atom has a magnetic dipole moment that interacts with the orbital and/or spin magnetic dipole moments of the electrons. These effects are called *hyperfine structure*. For example, the ground level of hydrogen is split into two states, separated by only 5.9×10^{-6} eV. The photon that is emitted in the transitions between these states has a wavelength of 21 cm. Radio astronomers use this wavelength to map clouds of interstellar hydrogen gas that are too cold to emit visible light (**Fig. 41.19**).

TEST YOUR UNDERSTANDING OF SECTION 41.5 In which of the following situations is the magnetic moment of an electron perfectly aligned with a magnetic field that points in the positive z -direction? (i) $m_s = +\frac{1}{2}$; (ii) $m_s = -\frac{1}{2}$; (iii) both (i) and (ii); (iv) neither (i) nor (ii).

ANSWER

(iv) For the magnetic moment to be perfectly aligned with the z -direction, the z -component of the spin vector $\vec{S}$ would have to have the same absolute value as S . However, the possible values of S_z are $\pm \frac{1}{2}\hbar$ [Eq. (41.36)], while the magnitude of the spin vector is $S = \sqrt{\frac{3}{4}}\hbar$ [Eq. (41.37)]. Hence $\vec{S}$ can never be perfectly aligned with any one direction in space.

41.6 MANY-ELECTRON ATOMS AND THE EXCLUSION PRINCIPLE

So far our analysis of atomic structure has concentrated on the hydrogen atom. That's natural; neutral hydrogen, with only one electron, is the simplest atom. Let's now take what we've learned about the hydrogen atom and apply that knowledge to the more complicated case of many-electron atoms.

An atom in its normal (electrically neutral) state has Z electrons and Z protons. Recall from Section 41.3 that we call Z the *atomic number*. The total electric charge of such an atom is exactly zero because the neutron has no charge while the proton and electron charges have the same magnitude but opposite sign.

A complete understanding of such a general atom requires that we know the wave function that describes the behavior of all Z of its electrons. This wave function depends on $3Z$ coordinates (three for each electron), so its complexity increases very rapidly with increasing Z . What's more, each of the Z electrons interacts not only with the nucleus but also with every other electron. The potential energy is therefore a complicated function of all $3Z$ coordinates, and the Schrödinger equation contains second derivatives with respect to all of them. Finding exact solutions to such equations is such a complex task that it has not been successfully achieved even for the neutral helium atom, which has only two electrons.

Fortunately, various approximation schemes are available. The simplest approximation is to ignore all interactions between electrons and consider each electron as moving under the action only of the nucleus (considered to be a point charge). In this approximation, we

write a separate wave function for each *individual* electron. Each such function is like that for the hydrogen atom, specified by four quantum numbers (n, l, m_l, m_s) . The nuclear charge is Ze instead of e , so we replace every factor of e^2 in the wave functions and the energy levels by Ze^2 . In particular, the energy levels are given by Eq. (41.21) with e^4 replaced by Z^2e^4 :

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2} \frac{m_e Z^2 e^4}{2n^2 \hbar^2} = -\frac{Z^2}{n^2} (13.6 \text{ eV}) \quad (41.43)$$

This approximation is fairly drastic; when there are many electrons, their interactions with each other are as important as the interaction of each with the nucleus. So this model isn't very useful for quantitative predictions.

The Central-Field Approximation

A less drastic and more useful approximation is to think of all the electrons together as making up a charge cloud that is, on average, *spherically symmetric*. We can then think of each individual electron as moving in the total electric field due to the nucleus and this averaged-out cloud of all the other electrons. There is a corresponding spherically symmetric potential-energy function $U(r)$. This picture is called the **central-field approximation**; it provides a useful starting point for understanding atomic structure.

In the central-field approximation we can again deal with one-electron wave functions. The Schrödinger equation for these functions differs from the equation for hydrogen, which we discussed in Section 41.3, only in that the $1/r$ potential-energy function is replaced by a different function $U(r)$. Now, Eqs. (41.20) show that $U(r)$ does not appear in the differential equations for $\Theta(\theta)$ and $\Phi(\phi)$. So those angular functions are exactly the same as for hydrogen, and the orbital angular momentum *states* are also the same as before. The quantum numbers l, m_l , and m_s have the same meanings as before, and Eqs. (41.22) and (41.23) again give the magnitude and z -component of the orbital angular momentum.

The radial wave functions and probabilities are different than for hydrogen because of the change in $U(r)$, so the energy levels are no longer given by Eq. (41.21). We can still label a state by using the four quantum numbers (n, l, m_l, m_s) . In general, the energy of a state now depends on both n and l , rather than just on n as with hydrogen. (Due to fine-structure effects, the energy can also depend on the total angular momentum quantum number j . These effects are generally small, however, so we ignore them for this discussion.) The restrictions on the values of the quantum numbers are the same as before:

$$\begin{array}{llll} \text{Allowed values of} & \text{Principal} & \text{Orbital magnetic} & \text{Spin magnetic} \\ \text{quantum numbers} & \text{quantum number} & \text{quantum number} & \text{quantum number} \\ \text{for one-electron} & & & \\ \text{wave functions:} & n \geq 1 & 0 \leq l \leq n-1 & |m_l| \leq l \\ & & \text{Orbital quantum number} & m_s = \pm \frac{1}{2} \end{array} \quad (41.44)$$

The Exclusion Principle

To understand the structure of many-electron atoms, we need an additional principle, the *exclusion principle*. To see why this principle is needed, let's consider the lowest-energy state, or *ground state*, of a many-electron atom. In the one-electron states of the central-field model, there is a lowest-energy state (corresponding to an $n = 1$ state of hydrogen). We might expect that in the ground state of a complex atom, *all* the electrons should be in this lowest state. If so, then we should see only gradual changes in physical and chemical properties when we look at the behavior of atoms with increasing numbers of electrons (Z).

Such gradual changes are *not* what is observed. Instead, properties of elements vary widely from one to the next, with each element having its own distinct personality. For example, the elements fluorine, neon, and sodium have 9, 10, and 11 electrons, respectively,

per atom. Fluorine ($Z = 9$) is a *halogen*; it tends strongly to form compounds in which each fluorine atom acquires an extra electron. Sodium ($Z = 11$) is an *alkali metal*; it forms compounds in which each sodium atom *loses* an electron. Neon ($Z = 10$) is a *noble gas*, forming no naturally occurring compounds at all. Such observations show that in the ground state of a complex atom the electrons *cannot* all be in the lowest-energy states. But why not?

The key to this puzzle, discovered by the Austrian physicist Wolfgang Pauli (Fig. 41.20) in 1925, is called the **exclusion principle**. This principle states that **no two electrons can occupy the same quantum-mechanical state** in a given system. That is, **no two electrons in an atom can have the same values of all four quantum numbers (n, l, m_l, m_s)**. Each quantum state corresponds to a certain distribution of the electron “cloud” in space. Therefore the principle also says, in effect, that no more than two electrons with opposite values of the quantum number m_s can occupy the same region of space. We shouldn’t take this last statement too seriously because the electron probability functions don’t have sharp, definite boundaries. But the exclusion principle limits the amount by which electron wave functions can overlap. Think of it as the quantum-mechanical analog of a university rule that allows only one student per desk. This same exclusion principle applies to all spin- $\frac{1}{2}$ particles, not just electrons. (We’ll see in Chapter 43 that protons and neutrons are also spin- $\frac{1}{2}$ particles. As a result, the exclusion principle plays an important role in the structure of atomic nuclei.)

CAUTION The meaning of the exclusion principle Don’t confuse the exclusion principle with the electrical repulsion between electrons. While both effects tend to keep electrons within an atom separated from each other, they are very different in character. Two electrons can always be pushed closer together by adding energy to combat electrical repulsion, but *nothing* can overcome the exclusion principle and force two electrons into the same quantum-mechanical state. ■

Table 41.2 lists some of the sets of quantum numbers for electron states in an atom. It’s similar to Table 41.1 (Section 41.3), but we’ve added the number of states in each subshell and shell. Because of the exclusion principle, the “number of states” is the same as the *maximum* number of electrons that can be found in those states. For each state, m_s can be either $+\frac{1}{2}$ or $-\frac{1}{2}$.

As with the hydrogen wave functions, different states correspond to different spatial distributions; electrons with larger values of n are concentrated at larger distances from the nucleus. Figure 41.8 (Section 41.3) shows this effect. When an atom has more than two electrons, they can’t all huddle down in the low-energy $n = 1$ states nearest to the nucleus because there are only two of these states; the exclusion principle forbids multiple occupancy of a state. Some electrons are forced into states farther away, with higher energies. Each value of n corresponds roughly to a region of space around the nucleus in the form of a spherical *shell*. Hence we speak of the *K* shell as the region that is occupied by the electrons in the $n = 1$ states, the *L* shell as the region of the $n = 2$ states, and so on. States with the same n but different l form *subshells*, such as the $3p$ subshell.

Figure 41.20 The key to understanding the periodic table of the elements was the discovery by Wolfgang Pauli (1900–1958) of the exclusion principle. Pauli received the 1945 Nobel Prize in physics for his accomplishment. This photo shows Pauli (on the left) and Niels Bohr watching the physics of a toy top spinning on the floor—a macroscopic analog of a microscopic electron with spin.



TABLE 41.2 Quantum States of Electrons in the First Four Shells

n	l	m_l	Spectroscopic Notation	Number of States	Shell
1	0	0	1s	2	<i>K</i>
2	0	0	2s	2	<i>L</i>
2	1	-1, 0, 1	2p	6	
3	0	0	3s	2	<i>M</i>
3	1	-1, 0, 1	3p	6	
3	2	-2, -1, 0, 1, 2	3d	10	
4	0	0	4s	2	<i>N</i>
4	1	-1, 0, 1	4p	6	
4	2	-2, -1, 0, 1, 2	4d	10	
4	3	-3, -2, -1, 0, 1, 2, 3	4f	14	

The Periodic Table

We can use the exclusion principle to derive the most important features of the structure and chemical behavior of multielectron atoms, including the periodic table of the elements. Let's imagine constructing a neutral atom by starting with a bare nucleus with Z protons and then adding Z electrons, one by one. To obtain the ground state of the atom as a whole, we fill the lowest-energy electron states (those closest to the nucleus, with the smallest values of n and l) first, and we use successively higher states until all the electrons are in place. The chemical properties of an atom are determined principally by interactions involving the outermost, or *valence*, electrons, so we particularly want to learn how these electrons are arranged.

Let's look at the ground-state electron configurations for the first few atoms (in order of increasing Z). For hydrogen the ground state is $1s$; the single electron is in a state $n = 1$, $l = 0$, $m_l = 0$, and $m_s = \pm \frac{1}{2}$. In the helium atom ($Z = 2$), *both* electrons are in $1s$ states, with opposite spins; one has $m_s = -\frac{1}{2}$ and the other has $m_s = +\frac{1}{2}$. We denote the helium ground state as $1s^2$. (The superscript 2 is not an exponent; the notation $1s^2$ tells us that there are two electrons in the $1s$ subshell. Also, the superscript 1 is understood, as in $2s$.) For helium the K shell is completely filled, and all others are empty. Helium is a noble gas; it has no tendency to gain or lose an electron, and it forms no naturally occurring compounds.

Lithium ($Z = 3$) has three electrons. In its ground state, two are in $1s$ states and one is in a $2s$ state, so we denote the lithium ground state as $1s^2 2s$. On average, the $2s$ electron is considerably farther from the nucleus than are the $1s$ electrons (Fig. 41.21). According to Gauss's law, the *net* charge Q_{encl} attracting the $2s$ electron is nearer to $+e$ than to the value $+3e$ it would have without the two $1s$ electrons present. As a result, the $2s$ electron is loosely bound; only 5.4 eV is required to remove it, compared with the 30.6 eV given by Eq. (41.43) with $Z = 3$ and $n = 2$. Chemically, lithium is an *alkali metal*. It forms ionic compounds in which each lithium atom loses an electron and has a valence of $+1$.

Next is beryllium ($Z = 4$); its ground-state configuration is $1s^2 2s^2$, with its two valence electrons filling the s subshell of the L shell. Beryllium is the first of the *alkaline earth* elements, forming ionic compounds in which the valence of the atoms is $+2$.

Table 41.3 shows the ground-state electron configurations of the first 30 elements. The L shell can hold eight electrons. At $Z = 10$, both the K and L shells are filled, and there are no electrons in the M shell. We expect this to be a particularly stable configuration, with little tendency to gain or lose electrons. This element is neon, a noble gas with no naturally occurring compounds. The next element after neon is sodium ($Z = 11$), with filled K and L shells and one electron in the M shell. Its “noble-gas-plus-one-electron” structure resembles that of lithium; both are alkali metals. The element *before* neon is fluorine, with $Z = 9$. It has a vacancy in the L shell and has an affinity for an extra electron to fill the shell. Fluorine forms ionic compounds in which it has a valence of -1 . This behavior is characteristic of the *halogens* (fluorine, chlorine, bromine, iodine, and astatine), all of which have “noble-gas-minus-one” configurations (Fig. 41.22).

Proceeding down the list, we can understand the regularities in chemical behavior displayed by the **periodic table of the elements** (Appendix D) on the basis of electron configurations. The similarity of elements in each *group* (vertical column) of the periodic table is the result of similarity in outer-electron configuration. All the noble gases (helium, neon, argon, krypton, xenon, and radon) have filled-shell or filled-shell plus filled p subshell configurations. All the alkali metals (lithium, sodium, potassium, rubidium, cesium, and francium) have “noble-gas-plus-one” configurations. All the alkaline earth metals (beryllium, magnesium, calcium, strontium, barium, and radium) have “noble-gas-plus-two” configurations, and, as we just mentioned, all the halogens (fluorine, chlorine, bromine, iodine, and astatine) have “noble-gas-minus-one” structures.

A slight complication occurs with the M and N shells because the $3d$ and $4s$ subshell levels ($n = 3$, $l = 2$ and $n = 4$, $l = 0$, respectively) have similar energies. (We'll discuss in the next subsection why this happens.) Argon ($Z = 18$) has all the $1s$, $2s$, $2p$, $3s$, and $3p$ subshells filled, but in potassium ($Z = 19$) the additional electron goes into a $4s$ energy state rather than a $3d$ state (because the $4s$ state has slightly lower energy).

Figure 41.21 Schematic representation of the charge distribution in a lithium atom. The nucleus has a charge of $+3e$.

On average, the $2s$ electron is considerably farther from the nucleus than the $1s$ electrons. Therefore, it experiences a net nuclear charge of approximately $+3e - 2e = +e$ (rather than $+3e$).

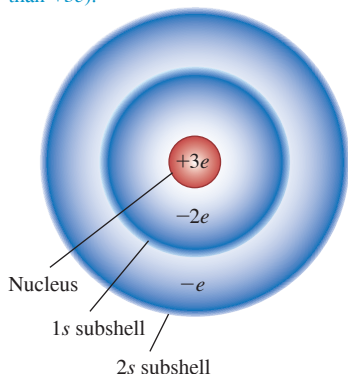


Figure 41.22 Salt (sodium chloride, NaCl) dissolves readily in water, making seawater salty. This is due to the electron configurations of sodium and chlorine: Sodium can easily lose an electron to form an Na^+ ion, and chlorine can easily gain an electron to form a Cl^- ion. These ions are held in solution because they are attracted to the polar ends of water molecules (see Fig. 21.30a).



TABLE 41.3 Ground-State Electron Configurations

Element	Symbol	Atomic Number (<i>Z</i>)	Electron Configuration
Hydrogen	H	1	1s
Helium	He	2	1s ²
Lithium	Li	3	1s ² 2s
Beryllium	Be	4	1s ² 2s ²
Boron	B	5	1s ² 2s ² 2p
Carbon	C	6	1s ² 2s ² 2p ²
Nitrogen	N	7	1s ² 2s ² 2p ³
Oxygen	O	8	1s ² 2s ² 2p ⁴
Fluorine	F	9	1s ² 2s ² 2p ⁵
Neon	Ne	10	1s ² 2s ² 2p ⁶
Sodium	Na	11	1s ² 2s ² 2p ⁶ 3s
Magnesium	Mg	12	1s ² 2s ² 2p ⁶ 3s ²
Aluminum	Al	13	1s ² 2s ² 2p ⁶ 3s ² 3p
Silicon	Si	14	1s ² 2s ² 2p ⁶ 3s ² 3p ²
Phosphorus	P	15	1s ² 2s ² 2p ⁶ 3s ² 3p ³
Sulfur	S	16	1s ² 2s ² 2p ⁶ 3s ² 3p ⁴
Chlorine	Cl	17	1s ² 2s ² 2p ⁶ 3s ² 3p ⁵
Argon	Ar	18	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶
Potassium	K	19	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s
Calcium	Ca	20	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ²
Scandium	Sc	21	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d
Titanium	Ti	22	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ²
Vanadium	V	23	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ³
Chromium	Cr	24	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ¹ 3d ⁵
Manganese	Mn	25	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁵
Iron	Fe	26	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁶
Cobalt	Co	27	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁷
Nickel	Ni	28	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁸
Copper	Cu	29	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ¹ 3d ¹⁰
Zinc	Zn	30	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ¹⁰

The next several elements have one or two electrons in the 4s subshell and increasing numbers in the 3d subshell. These elements are all metals with rather similar chemical and physical properties; they form the first *transition series*, starting with scandium ($Z = 21$) and ending with zinc ($Z = 30$), for which all the 3d and 4s subshells are filled.

Something similar happens with $Z = 57$ through $Z = 71$, which have one or two electrons in the 6s subshell but only partially filled 4f and 5d subshells. These *rare earth* elements all have very similar physical and chemical properties. Another such series, called the *actinide* series, starts with $Z = 91$.

Screening

We have mentioned that in the central-field model, the energy levels depend on l as well as n . Let's take sodium ($Z = 11$) as an example. If 10 of its electrons fill its K and L shells, the energies of some of the states for the remaining electron are found experimentally to be

- 3s states: -5.138 eV
- 3p states: -3.035 eV
- 3d states: -1.521 eV
- 4s states: -1.947 eV

BIO APPLICATION Electron Configurations and Bone Cancer Radiotherapy

The orange spots in this colored x-ray image are bone cancer tumors. One method of treating bone cancer is to inject a radioactive isotope of strontium (^{89}Sr) into a patient's vein. Strontium is chemically similar to calcium because in both atoms the two outer electrons are in an s state (the structures are $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2$ for strontium and $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ for calcium). Hence the strontium is readily taken up by the tumors, where calcium turnover is more rapid than in healthy bone. Radiation from the strontium helps destroy the tumors.



The $3s$ states are the lowest (most negative); one is the ground state for the 11th electron in sodium. The energy of the $3d$ states is quite close to the energy of the $n = 3$ state in hydrogen. The surprise is that the $4s$ state energy is 0.426 eV *below* the $3d$ state, even though the $4s$ state has larger n .

We can understand these results by using Gauss's law (Section 22.3). For any spherically symmetric charge distribution, the electric-field magnitude at a distance r from the center is $Q_{\text{encl}}/4\pi\epsilon_0 r^2$, where Q_{encl} is the total charge enclosed within a sphere with radius r . Mentally remove the outer (valence) electron atom from a sodium atom. What you have left is a spherically symmetric collection of 10 electrons (filling the K and L shells) and 11 protons, so $Q_{\text{encl}} = -10e + 11e = +e$. If the 11th electron is completely outside this collection of charges, it is attracted by an effective charge of $+e$, not $+11e$. This is a more extreme example of the effect depicted in Fig. 41.21.

This effect is called **screening**; the 10 electrons *screen* 10 of the 11 protons in the sodium nucleus, leaving an effective net charge of $+e$. From the viewpoint of the 11th electron, this is equivalent to reducing the number of protons in the nucleus from $Z = 11$ to a smaller *effective atomic number* Z_{eff} . If the 11th electron is *completely* outside the charge distribution of the other 10 electrons, then $Z_{\text{eff}} = 1$. Since the probability distribution of the 11th electron does extend somewhat into those of the other electrons, in fact Z_{eff} is greater than 1 (but still much less than 11). In general, an electron that spends all its time completely outside a positive charge $Z_{\text{eff}}e$ has energy levels given by the hydrogen expression with e^2 replaced by $Z_{\text{eff}}e^2$. From Eq. (41.43) this is

$$E_n = -\frac{Z_{\text{eff}}^2}{n^2} (13.6 \text{ eV}) \quad (41.45)$$

Energy levels of an electron with screening Effective (screened) atomic number Principal quantum number

CAUTION Different equations for different atoms Equations (41.21), (41.43), and (41.45) all give values of E_n in terms of $(13.6 \text{ eV})/n^2$, but they don't apply in general to the same atoms. Equation (41.21) is for hydrogen *only*. Equation (41.43) is for only the case in which there is no interaction with any other electron (and is thus accurate only when the atom has just one electron). Equation (41.45) is useful when one electron is screened from the nucleus by other electrons. I

Now let's use the radial probability functions shown in Fig. 41.8 to explain why the energy of a sodium $3d$ state is approximately the same as the $n = 3$ value of hydrogen, -1.51 eV . The distribution for the $3d$ state (for which l has the maximum value $n - 1$) has one peak, and its most probable radius is *outside* the positions of the electrons with $n = 1$ or 2. (Those electrons also are pulled closer to the nucleus than in hydrogen because they are less effectively screened from the positive charge $11e$ of the nucleus.) Thus in sodium a $3d$ electron spends most of its time well outside the $n = 1$ and $n = 2$ states (the K and L shells). The 10 electrons in these shells screen about ten-elevenths of the charge of the 11 protons, leaving a net charge of about $Z_{\text{eff}}e = (1)e$. Then, from Eq. (41.45), the corresponding energy is approximately $-(1)^2(13.6 \text{ eV})/3^2 = -1.51 \text{ eV}$. This approximation is very close to the experimental value of -1.521 eV .

Looking again at Fig. 41.8, we see that the radial probability density for the $3p$ state (for which $l = n - 2$) has two peaks and that for the $3s$ state ($l = n - 3$) has three peaks. For sodium the first small peak in the $3p$ distribution gives a $3p$ electron a higher probability (compared to the $3d$ state) of being *inside* the charge distributions for the electrons in the $n = 2$ states. That is, a $3p$ electron is less completely screened from the nucleus than is a $3d$ electron because it spends some of its time within the filled K and L shells. Thus for the $3p$ electrons, Z_{eff} is greater than unity. From Eq. (41.45) the $3p$ energy is lower (more negative) than the $3d$ energy of -1.521 eV . The actual value is -3.035 eV . A $3s$ electron spends even more time within the inner electron shells than a $3p$ electron does, giving an even larger Z_{eff} and an even more negative energy.

This discussion shows that the energy levels given by Eq. (41.45) depend on both the principal quantum number n and the orbital quantum number l . That's because the value of Z_{eff} is different for the $3s$ state ($n = 3, l = 0$), the $3p$ state ($n = 3, l = 1$), and the $3d$ state ($n = 3, l = 2$).

EXAMPLE 41.9 Determining Z_{eff} experimentally

The measured energy of a $3s$ state of sodium is -5.138 eV . Calculate the value of Z_{eff} .

IDENTIFY and SET UP Sodium has a single electron in the M shell outside filled K and L shells. The ten K and L electrons partially screen the single M electron from the $+11e$ charge of the nucleus; our goal is to determine the extent of this screening. We are given $n = 3$ and $E_n = -5.138 \text{ eV}$, so we can use Eq. (41.45) to determine Z_{eff} .

EXECUTE Solving Eq. (41.45) for Z_{eff} , we have

$$Z_{\text{eff}}^2 = -\frac{n^2 E_n}{13.6 \text{ eV}} = -\frac{3^2(-5.138 \text{ eV})}{13.6 \text{ eV}} = 3.40$$

$$Z_{\text{eff}} = 1.84$$

EVALUATE The effective charge attracting a $3s$ electron is $1.84e$. Sodium's 11 protons are screened by an average of $11 - 1.84 = 9.16$

electrons instead of 10 electrons because the $3s$ electron spends some time within the inner (K and L) shells.

Each alkali metal (lithium, sodium, potassium, rubidium, and cesium) has one more electron than the corresponding noble gas (helium, neon, argon, krypton, and xenon). This extra electron is mostly outside the other electrons in the filled shells and subshells. Therefore all the alkali metals behave similarly to sodium.

KEYCONCEPT If a many-electron atom has just one outer valence electron, the valence electron's energy levels are like those of a single-electron atom but with the atomic number Z of the nucleus replaced by an effective atomic number Z_{eff} . The value of Z_{eff} is less than Z because the inner electrons partially "screen" the electric field of the nucleus.

EXAMPLE 41.10 Energies for a valence electron

The valence electron in potassium has a $4s$ ground state. Calculate the approximate energy of the $n = 4$ state having the smallest Z_{eff} , and discuss the relative energies of the $4s$, $4p$, $4d$, and $4f$ states.

IDENTIFY and SET UP The state with the smallest Z_{eff} is the one in which the valence electron spends the most time outside the inner filled shells and subshells, so that it is most effectively screened from the charge of the nucleus. Once we have determined which state has the smallest Z_{eff} , we can use Eq. (41.45) to determine the energy of this state.

EXECUTE A $4f$ state has $n = 4$ and $l = 3 = 4 - 1$. Thus it is the state of greatest orbital angular momentum for $n = 4$, and thus the state in which the electron spends the most time outside the electron charge clouds of the inner filled shells and subshells. This makes Z_{eff} for a $4f$ state close to unity. Equation (41.45) then gives

$$E_4 = -\frac{Z_{\text{eff}}^2}{n^2}(13.6 \text{ eV}) = -\frac{1}{4^2}(13.6 \text{ eV}) = -0.85 \text{ eV}$$

This approximation agrees with the measured energy of the sodium $4f$ state to the precision given.

An electron in a $4d$ state spends a bit more time within the inner shells, and its energy is therefore a bit more negative (measured to be -0.94 eV). For the same reason, a $4p$ state has an even lower energy (measured to be -2.73 eV) and a $4s$ state has the lowest energy (measured to be -4.339 eV).

EVALUATE We can extend this analysis to the *singly ionized alkaline earth elements*: Be^+ , Mg^+ , Ca^+ , Sr^+ , and Ba^+ . For any allowed value of n , the highest- l state ($l = n - 1$) of the one remaining outer electron sees an effective charge of almost $+2e$, so for these states, $Z_{\text{eff}} = 2$. A $3d$ state for Mg^+ , for example, has an energy of about $-2^2(13.6 \text{ eV})/3^2 = -6.0 \text{ eV}$.

KEYCONCEPT For a many-electron atom with just one outer valence electron, if that electron is in a state of high orbital angular momentum it "sees" the effective atomic number Z_{eff} of the nucleus to be 1. In such a state the valence electron is almost always outside of the other $Z - 1$ electrons, which "screen" $(Z - 1)e$ of the nuclear charge $+Ze$.

TEST YOUR UNDERSTANDING OF SECTION 41.6 If electrons did *not* obey the exclusion principle, would it be easier or more difficult to remove the first electron from sodium?

ANSWER

more difficult If there were no exclusion principle, all 11 electrons in the sodium atom would be in the level of lowest energy (the $1s$ level) and the configuration would be $1s^{11}$. Consequently, it would be more difficult to remove the first electron. (In a real sodium atom the valence electron is in a screened $3s$ state, which has a comparatively high energy.)

41.7 X-RAY SPECTRA

X-ray spectra provide an example of the richness and power of the model of atomic structure that we derived in the preceding section. In Section 38.2 we discussed how x-ray photons are produced when electrons strike a metal target (see Fig. 38.7). In this section we'll see how the spectrum of x rays produced in this way depends on the type of metal used in the target and how the ideas of atomic energy levels and screening help us understand this dependence.

Characteristic X Rays and Atomic Energy Levels

X-ray diffraction techniques (see Section 36.6) make it possible to measure x-ray wavelengths quite precisely (to within 0.1% or less). **Figure 41.23** shows the spectrum of x rays produced when fast-moving electrons strike a target of the metal molybdenum. This spectrum has two features:

1. There is a *continuous* spectrum of wavelengths (see Fig. 38.8 in Section 38.2), with a minimum wavelength (corresponding to a maximum frequency and a maximum photon energy) that is determined by the voltage V_{AC} used to accelerate the electrons. As we saw in Section 38.2, this continuous spectrum is due to *bremsstrahlung*, in which the electrons slow down as they interact with the metal atoms in the target and convert their kinetic energy into photons. The minimum wavelength λ_{min} corresponds to all of the kinetic energy eV_{AC} of the electron being converted into the energy of a single photon of energy hc/λ_{min} , so

$$\lambda_{min} = \frac{hc}{eV_{AC}} \quad (41.46)$$

This continuous-spectrum radiation is nearly independent of the target material in the x-ray tube.

2. Depending on the accelerating voltage, sharp peaks may be superimposed on the continuous bremsstrahlung spectrum, as in Fig. 41.23. These peaks are caused when the target atoms are struck by high-energy electrons and emit x rays of very definite wavelengths. Unlike the continuous spectrum, the wavelengths of the peaks are *different* for different target elements; they form what is called a *characteristic x-ray spectrum* for each target element.

In 1913 the English physicist Henry G. J. Moseley undertook a careful experimental study of characteristic x-ray spectra. He found that the most intense short-wavelength line in the characteristic x-ray spectrum from a particular target element, called the K_α line, varied smoothly with that element's atomic number Z (**Fig. 41.24**). This is in sharp contrast to optical spectra, in which elements with adjacent Z -values have spectra that often bear no resemblance to each other.

Figure 41.23 Graph of intensity per unit wavelength as a function of wavelength for x rays produced with an accelerating voltage of 35 kV and a molybdenum target. The curve is a smooth function similar to the bremsstrahlung spectra in Fig. 38.8 (Section 38.2), but with two sharp spikes corresponding to part of the characteristic x-ray spectrum for molybdenum.

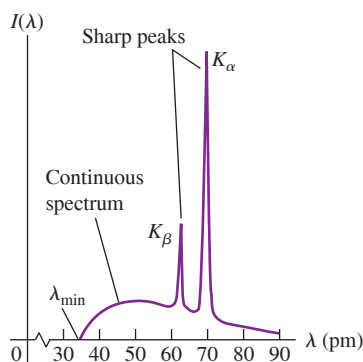
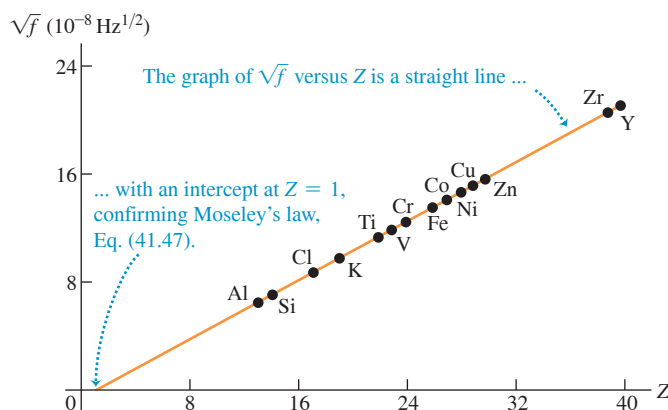


Figure 41.24 The square root of Moseley's measured frequencies of the K_α line for 14 elements.



Moseley found that the relationship could be expressed in terms of x-ray frequencies f by a simple formula called *Moseley's law*:

$$\text{Moseley's law: } f = (2.48 \times 10^{15} \text{ Hz})(Z - 1)^2 \quad (41.47)$$

Frequency of K_α line in characteristic x-ray spectrum of an element
Atomic number of element

Moseley went far beyond this empirical relationship; he showed how characteristic x-ray spectra could be understood on the basis of energy levels of atoms in the target. His analysis was based on the Bohr model, published in the same watershed year of 1913. We'll recast it somewhat, using the ideas of atomic structure that we discussed in Section 41.6. First recall that the *outer* electrons of an atom are responsible for optical spectra. Their excited states are usually only a few electron volts above their ground state. In transitions from excited states to the ground state, they usually emit photons in or near the visible region.

Characteristic x rays, by contrast, are emitted in transitions involving the *inner* shells of a complex atom. In an x-ray tube the electrons may strike the target with enough energy to knock electrons out of the inner shells of the target atoms. These inner electrons are much closer to the nucleus than are the electrons in the outer shells; they are much more tightly bound, and hundreds or thousands of electron volts may be required to remove them.

Suppose one electron is knocked out of the K shell. This process leaves a vacancy, which we'll call a *hole*. (One electron remains in the K shell.) The hole can then be filled by an electron falling in from one of the outer shells, such as the L , M , N , . . . shell. This transition is accompanied by a decrease in the energy of the atom (because *less* energy would be needed to remove an electron from an L , M , N , . . . shell), and an x-ray photon is emitted with energy equal to this decrease. Each state has definite energy, so the emitted x rays have definite wavelengths; the emitted spectrum is a *line* spectrum.

We can estimate the energy and frequency of K_α x-ray photons by using the concept of screening from Section 41.6. A K_α x-ray photon is emitted when an electron in the L shell ($n = 2$) drops down to fill a hole in the K shell ($n = 1$). As the electron drops down, it is attracted by the Z protons in the nucleus screened by the one remaining electron in the K shell. We therefore approximate the energy by Eq. (41.45), with $Z_{\text{eff}} = Z - 1$, $n_i = 2$, and n_f . The energy before the transition is

$$E_i \approx -\frac{(Z - 1)^2}{2^2}(13.6 \text{ eV}) = -(Z - 1)^2(3.4 \text{ eV})$$

and the energy after the transition is

$$E_f \approx -\frac{(Z - 1)^2}{1^2}(13.6 \text{ eV}) = -(Z - 1)^2(13.6 \text{ eV})$$

$E_{K_\alpha} = E_i - E_f \approx (Z - 1)^2(-3.4 \text{ eV} + 13.6 \text{ eV})$ is the energy of the K_α x-ray photon. That is,

$$E_{K_\alpha} \approx (Z - 1)^2(10.2 \text{ eV}) \quad (41.48)$$

The frequency of the photon is its energy divided by Planck's constant:

$$f = \frac{E_{K_\alpha}}{h} \approx \frac{(Z - 1)^2(10.2 \text{ eV})}{4.136 \times 10^{-15} \text{ eV} \cdot \text{s}} = (2.47 \times 10^{15} \text{ Hz})(Z - 1)^2$$

This relationship agrees almost exactly with Moseley's experimental law, Eq. (41.47). Indeed, considering the approximations we have made, the agreement is better than we have a right to expect. But our calculation does show how Moseley's law can be understood on the bases of screening and transitions between energy levels.

APPLICATION X Rays in Forensic Science

X-ray spectra are an important tool in crime analysis. When a handgun is fired, a cloud of gunshot residue (GSR) is ejected from the barrel. The x-ray emission spectrum of GSR includes characteristic peaks from lead (Pb), antimony (Sb), and barium (Ba). If a sample taken from a suspect's skin or clothing has an x-ray emission spectrum with these characteristics, it indicates that the suspect recently fired a gun.

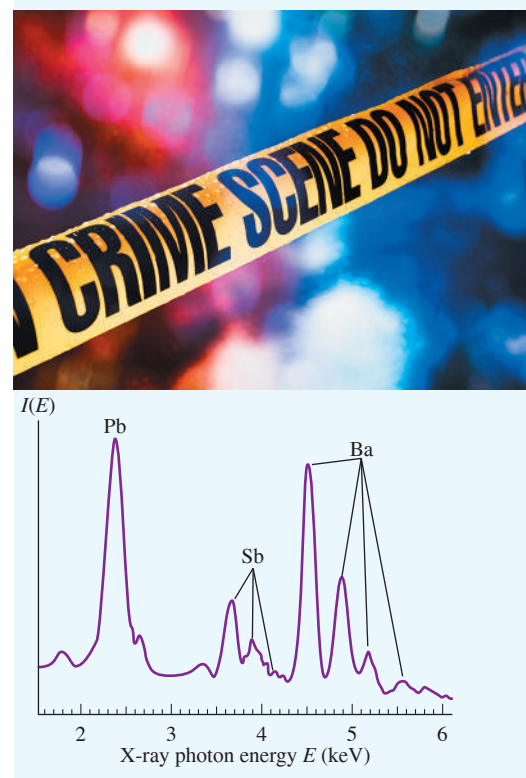
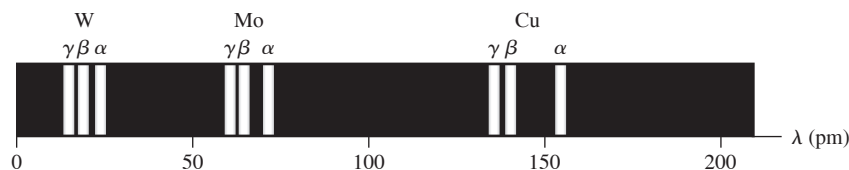


Figure 41.25 Wavelengths of the K_α , K_β , and K_γ lines of tungsten (W), molybdenum (Mo), and copper (Cu).



The three lines in each series are called the K_α , K_β , and K_γ lines. The K_α line is produced by the transition of an L electron to the vacancy in the K shell, the K_β line by an M electron, and the K_γ line by an N electron.

The hole in the K shell may also be filled by an electron falling from the M or N shell, assuming that these are occupied. If so, the x-ray spectrum of a large group of atoms of a single element shows a series, named the K series, of three lines, called the K_α , K_β , and K_γ lines. These three lines result from transitions in which the K -shell hole is filled by an L , M , or N electron, respectively. **Figure 41.25** shows the K series for tungsten ($Z = 74$), molybdenum ($Z = 42$), and copper ($Z = 29$).

There are other series of x-ray lines, called the L , M , and N series, that are produced after the ejection of electrons from the L , M , and N shells rather than the K shell. Electrons in these outer shells are farther away from the nucleus and are not held as tightly as are those in the K shell, so removing these outer electrons requires less energy. Hence the x-ray photons that are emitted when these vacancies are filled have lower energy than those in the K series.

EXAMPLE 41.11 Chemical analysis by x-ray emission

You measure the K_α wavelength for an unknown element, obtaining the value 0.0709 nm. What is the element?

IDENTIFY and SET UP To determine which element this is, we need to know its atomic number Z . We can find this by using Moseley's law, which relates the frequency of an element's K_α x-ray emission line to that element's atomic number Z . We'll use the relationship $f = c/\lambda$ to calculate the frequency for the K_α line, and then use Eq. (41.47) to find the corresponding value of the atomic number Z . We'll then consult the periodic table (Appendix D) to determine which element has this atomic number.

EXECUTE The frequency is

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m/s}}{0.0709 \times 10^{-9} \text{ m}} = 4.23 \times 10^{18} \text{ Hz}$$

Solving Moseley's law for Z , we get

$$Z = 1 + \sqrt{\frac{f}{2.48 \times 10^{15} \text{ Hz}}} = 1 + \sqrt{\frac{4.23 \times 10^{18} \text{ Hz}}{2.48 \times 10^{15} \text{ Hz}}} = 42.3$$

We know that Z has to be an integer; we conclude that $Z = 42$, corresponding to the element molybdenum.

EVALUATE If you're worried that our calculation did not give an integer for Z , remember that Moseley's law is an empirical relationship. There are slight variations from one atom to another due to differences in the structure of the electron shells. Nonetheless, this example suggests the power of Moseley's law.

Niels Bohr commented that it was Moseley's observations, not the alpha-particle scattering experiments of Rutherford, Geiger, and Marsden (see Section 39.2), that truly convinced physicists that the atom consists of a positive nucleus surrounded by electrons in motion. Unlike Bohr or Rutherford, Moseley did not receive a Nobel Prize for his important work; these awards are given to living scientists only, and Moseley (at age 27) was killed in combat during the First World War.

KEYCONCEPT If an electron is kicked out of the innermost, or K , shell of an atom of atomic number Z , then an electron from the next shell (L) drops down to the K shell to fill in the vacancy and emits an x-ray photon in the process. The frequency of this photon is proportional to $(Z - 1)^2$ (Moseley's law).

X-Ray Absorption Spectra

We can also observe x-ray *absorption* spectra. Unlike optical spectra, the absorption wavelengths are usually not the same as those for emission, especially in many-electron atoms, and do not give simple line spectra. For example, the K_α emission line results from a transition from the L shell to a hole in the K shell. The reverse transition doesn't occur in atoms with $Z \geq 10$ because in the atom's ground state, there is no vacancy in the L shell. To be absorbed, a photon must have enough energy to move an electron to an empty state. Since empty states are only a few electron volts in energy below the free-electron continuum, the minimum absorption energies in many-electron atoms are about the same as the

minimum energies that are needed to remove an electron from its shell. Experimentally, if we gradually increase the accelerating voltage and hence the maximum photon energy, we observe sudden increases in absorption when we reach these minimum energies. These sudden jumps of absorption are called *absorption edges* (Fig. 41.26).

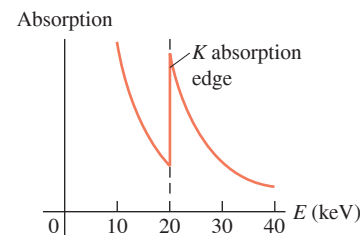
Characteristic x-ray spectra provide a very useful analytical tool. Satellite-borne x-ray spectrometers are used to study x-ray emission lines from highly excited atoms in distant astronomical sources. X-ray spectra are also used in air-pollution monitoring and in studies of the abundance of various elements in rocks.

TEST YOUR UNDERSTANDING OF SECTION 41.7 A beam of photons is passed through a sample of high-temperature atomic hydrogen. At what photon energy would you expect there to be an absorption edge like that shown in Fig. 41.26? (i) 13.60 eV; (ii) 3.40 eV; (iii) 1.51 eV; (iv) all of these; (v) none of these.

ANSWER

(iv) An absorption edge appears if the photon energy is just high enough to remove an electron from the atom. In a sample of high-temperature hydrogen we expect to find atoms whose electron is in the ground level ($n = 1$), the first excited level ($n = 2$), and the second excited level ($n = 3$). From Eq. (41.21) these levels have energies $E_n = -13.60 \text{ eV}/n^2 = -13.60 \text{ eV}$, -3.40 eV , and -1.51 eV (see Fig. 39.24b).

Figure 41.26 When a beam of x rays is passed through a slab of molybdenum, the extent to which the beam is absorbed depends on the energy E of the x-ray photons. A sharp increase in absorption occurs at the K absorption edge at 20 keV. The increase occurs because photons with energies above this value can excite an electron from the K shell of a molybdenum atom into an empty state.



41.8 QUANTUM ENTANGLEMENT

We've seen that quantum mechanics is very successful at correctly predicting the results of experiments. As we'll see in the remaining chapters, quantum mechanics is the basis of all electronic devices and is essential to our understanding of atomic nuclei and subatomic particles. But even though the central ideas of quantum mechanics have been established for decades, some aspects of the theory continue to baffle physicists and remain topics of ongoing research. We close this chapter with a discussion of one of these topics, *quantum entanglement*.

The Wave Function for Two Identical Particles

To understand what is meant by “quantum entanglement,” let's consider how to write the wave function for two identical particles, such as two electrons (the electrons in the neutral helium atom, for instance). We'll use the subscripts 1 and 2 to refer to these particles.

As we discussed in Section 41.6, a wave function that describes both particles is a function of both the coordinates (x_1, y_1, z_1) of particle 1 and the coordinates (x_2, y_2, z_2) of particle 2. As a shorthand, we'll use the position vectors $\vec{r}_1 = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ and $\vec{r}_2 = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$. In terms of these vectors, we can write the time-dependent two-particle wave function as $\Psi(\vec{r}_1, \vec{r}_2, t)$. Just as for a single particle, if the system of two particles has total energy E , we can write this time-dependent wave function as the product of a time-independent two-particle wave function $\psi(\vec{r}_1, \vec{r}_2)$ and a factor that depends on time only:

$$\Psi(\vec{r}_1, \vec{r}_2, t) = \psi(\vec{r}_1, \vec{r}_2)e^{-iEt/\hbar} \quad (41.49)$$

We saw in Section 41.2 that a technique called *separation of variables* is useful for expressing a wave function that depends on several variables as a product of functions of the individual variables. Let's see whether we can express the time-independent two-particle wave function $\psi(\vec{r}_1, \vec{r}_2)$ in Eq. (41.49) as a product of a function of $\vec{r}_1$ and a function of $\vec{r}_2$. We interpret the function of $\vec{r}_1$ to be the *single-particle* wave function for particle 1 and the function of $\vec{r}_2$ to be the single-particle wave function for particle 2. Suppose particle 1 is in a state A , for which the single-particle wave function is ψ_A , and particle 2 is in a state B , for which the single-particle wave function is ψ_B . [In the helium atom, with two electrons, one electron could be in the spin-up state ($n = 1, l = 0, m_l = 0$, and $m_s = +\frac{1}{2}$) and the other could be in the spin-down state ($n = 1, l = 0, m_l = 0$, and $m_s = -\frac{1}{2}$).] Using separation of variables, we would write

$$\psi(\vec{r}_1, \vec{r}_2) = \psi_A(\vec{r}_1)\psi_B(\vec{r}_2) \quad (41.50)$$

(first guess at the two-particle wave function)

However, Eq. (41.50) *cannot* be correct, because particles 1 and 2 are *identical* and *indistinguishable*. We may be able to state with confidence that one particle is in state *A* and the other is in state *B*, but it's impossible to specify which particle is in which state. (There's no way even in principle to “tag” the particles.)

To account for this, let's make an improved guess for the two-particle wave function: a combination of two terms like Eq. (41.50)—one term for which particle 1 is in state *A* and particle 2 is in state *B*, and one term for which particle 1 is in state *B* and particle 2 is in state *A*. Our improved guess is then

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}[\psi_A(\vec{r}_1)\psi_B(\vec{r}_2) \pm \psi_B(\vec{r}_1)\psi_A(\vec{r}_2)] \quad (41.51)$$

(second guess at the two-particle wave function)

The factor $1/\sqrt{2}$ in Eq. (41.51) ensures that if ψ_A and ψ_B are normalized, then $\psi(\vec{r}_1, \vec{r}_2)$ will be normalized as well. Note that the terms $\psi_A(\vec{r}_1)\psi_B(\vec{r}_2)$ and $\psi_B(\vec{r}_1)\psi_A(\vec{r}_2)$ appear with equal magnitudes, so that the two possibilities (particle 1 in *A* and particle 2 in *B*, or particle 1 in *B* and particle 2 in *A*) are equally probable.

How can we decide whether the $\pm$ sign in Eq. (41.51) should be a plus or a minus? If the particles are two electrons, or two of any other type of spin- $\frac{1}{2}$ particle, the Pauli exclusion principle (Section 41.6) tells us that we must use the minus sign:

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}[\psi_A(\vec{r}_1)\psi_B(\vec{r}_2) - \psi_B(\vec{r}_1)\psi_A(\vec{r}_2)] \quad (41.52)$$

(two-particle wave function, spin- $\frac{1}{2}$ particles)

To check this, suppose we demand that *both* particles be in the same state *A*. Then we would replace ψ_B in Eq. (41.52) with ψ_A :

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}[\psi_A(\vec{r}_1)\psi_A(\vec{r}_2) - \psi_A(\vec{r}_1)\psi_A(\vec{r}_2)] = 0$$

The zero value of the wave function says that our demand cannot be met. This is in agreement with the Pauli exclusion principle: No two electrons, and indeed no two identical spin- $\frac{1}{2}$ particles of any kind, can occupy the same quantum-mechanical state.

Measurement and “Spooky Action at a Distance”

The wave function in Eq. (41.52) shares features with that of the quantum-mechanical particle in a box that we discussed in Section 40.6. That particle's wave function is also a combination of two terms of equal magnitude representing different situations: one in which the momentum of the particle is in the $+x$ -direction, so $p_x > 0$, and one in which its momentum is in the $-x$ -direction, so $p_x < 0$. We saw in Section 40.6 that if we *measured* the momentum of the particle, we would get either $p_x > 0$ or $p_x < 0$. Making such a measurement causes the wave function to *collapse*, and only the term corresponding to the measured value of p_x survives. The other term disappears from the wave function.

The same sort of wave-function collapse happens in our two-particle system. Suppose we make a measurement of one particle—call it particle 1—and determine that it is in state *A*. The measurement causes the wave function in Eq. (41.52) to collapse to $\psi(\vec{r}_1, \vec{r}_2) = \psi_A(\vec{r}_1)\psi_B(\vec{r}_2)$. This wave equation corresponds to particle 1 being in state *A* but also corresponds to particle 2 being in state *B*. It follows that, after the measurement, particle 2 *must* be in state *B*, even though we have not directly measured the state of particle 2. In other words, making a measurement on *one* particle affects the state of the *other* particle. Schrödinger described this situation by saying that the two particles are *entangled*.

If the two entangled particles are the two electrons within a helium atom, the idea that their states are entangled may not seem troubling. After all, these electrons are in very close proximity (a helium atom is only about 0.1 nm in diameter) and exert substantial

electric forces on each other. You might imagine that when we measure electron 1 to be in state *A* (say, spin up with $m_s = +\frac{1}{2}$), it exerts forces on electron 2 that require electron 2 to be in state *B* (say, spin down with $m_s = -\frac{1}{2}$).

But suppose we arrange for two identical particles to be in an entangled state in which the particles are *not* close to each other, so they cannot exert forces on each other. When the same kind of measurement experiment is done on such a distant pair of entangled particles, the result is the same as if they are close together: If we measure particle 1 to be in state *A* and subsequently make a measurement on particle 2, we always find that particle 2 is in state *B*. If instead we measure particle 1 to be in state *B* and then make a measurement on particle 2, we always find that particle 2 is in state *A*. So measuring the state of one particle affects the state of the other particle, even when the two particles *cannot* exert forces on each other (**Fig. 41.27**). This finding has been confirmed with entangled particles that are *more than 1200 km apart!* (These experiments with very large distances are done with photons rather than electrons. Like electrons, photons have spin, and the “spin-up” and “spin-down” states correspond to left and right circular polarization. The only difference is that photons are spin-1 particles, not spin- $\frac{1}{2}$, and do not obey the Pauli exclusion principle. As a result, we must use the plus sign rather than the minus sign in Eq. (41.51) to describe two entangled photons. The rest of the physics is identical, however.)

These results contradict the idea of *locality*—the notion that a particle responds to forces or fields that act at its position only, not at some other point in space. We used locality in Chapters 4 and 13 when we expressed the gravitational force on a particle of mass m as $\vec{F}_g = m\vec{g}$, where $\vec{g}$ is the acceleration due to gravity at the point in space where the particle is located. We used locality again in Chapters 21 and 27 when we wrote the electric force $\vec{F}_E$ and the magnetic force $\vec{F}_B$ on a particle of charge q moving with velocity $\vec{v}$ as $\vec{F}_E = q\vec{E}$ and $\vec{F}_B = q\vec{v} \times \vec{B}$, where $\vec{E}$ and $\vec{B}$, respectively, are the electric and magnetic fields at the position of the particle. But the interaction between two entangled, widely separated particles seems *not* to obey locality. For this reason, Albert Einstein referred to the results of an experiment like that shown in Fig. 41.27 as “spooky action at a distance.” Spooky or not, quantum mechanics appears to be intrinsically nonlocal.

What makes these results even more striking is that no matter how far apart the two entangled particles are, there appears to be *zero delay* between the time that we make a measurement on one particle and the time that the state of the other particle changes as a result. At first glance this seems to violate a key idea of Einstein’s special theory of

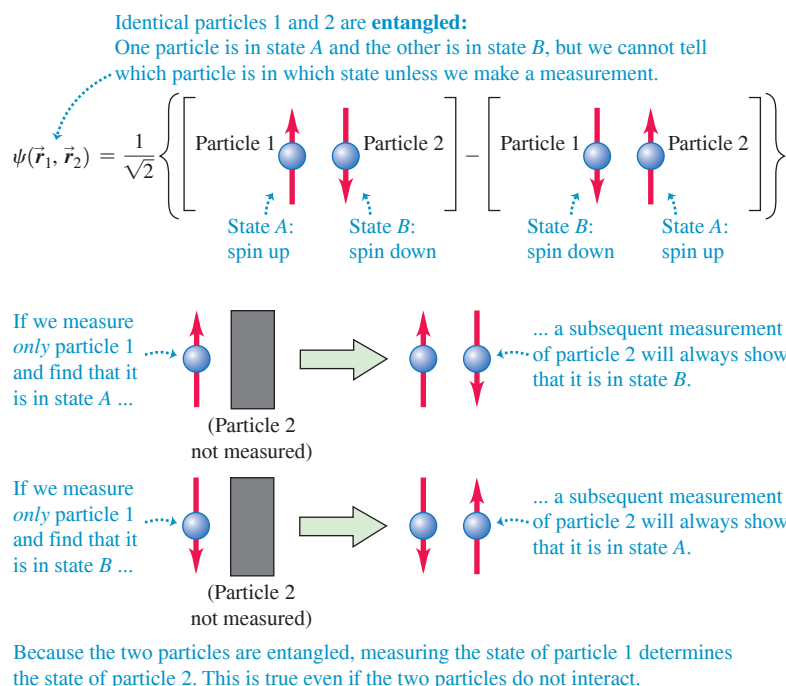


Figure 41.27 If two particles are in an entangled state, making a measurement of one particle determines the result of a subsequent measurement of the other particle.

relativity: that signals of any kind—radio waves, light signals, or beams of particles—cannot travel faster than the speed of light in vacuum, c . If measuring the state of particle 1 in Fig. 41.27 causes the state of particle 2 to change instantaneously, couldn't we make a “quantum radio” that sends signals faster than c , with particle 1 as the transmitter and particle 2 as the receiver?

The answer is no. The “message” in our quantum radio would be the result of a measurement of particle 1 by a physicist (call her Primo) at that particle's position. Primo's measurement collapses the wave function of the two particles, and her result would be that particle 1 is in either state A or state B . Another physicist (call him Secondo) at the position of particle 2 would measure particle 2 to be in state B if Primo measured particle 1 to be in state A , and to be in state A if Primo measured particle 1 to be in state B . But Secondo would have no way of knowing whether his result was caused by Primo making a measurement first or by *Secondo* himself making an independent measurement of particle 2 *without* Primo having made any measurement. (Secondo could determine this later by, for instance, sending text messages back and forth with Primo. But that method of communication involves signals that travel at the speed of light, not instantaneously.) So our quantum radio would transmit no information at all and would not allow us to communicate at speeds faster than c .

A remarkable practical application of quantum entanglement is *quantum computing*. In a conventional (“classical”) computer, the memory is made up of *bits*. Each bit has only two possible values (say, 0 or 1), so a computer memory with N bits can have any of 2^N different configurations. (This is analogous to coins that can be either heads up or tails up. Figure 20.21 in Section 20.8 shows the possible configurations of four coins; the number of possibilities is $2^4 = 16$.) In a quantum computer, bits are replaced with *qubits* (short for “quantum bits”). An example is a spin- $\frac{1}{2}$ electron that can be in a spin-up state ($m_s = +\frac{1}{2}$) or a spin-down state ($m_s = -\frac{1}{2}$), as usual, but can also be in *any combination* of these states. The wave function of N entangled qubits can correspond to any of 2^N configurations (like ordinary bits or coins that can be heads up or tails up) or to an entangled state in which the qubits are in any combination of these configurations. So, unlike a classical computer memory, which can be in only one of its 2^N configurations at a time, a quantum computer memory can essentially be in *all* of these configurations simultaneously. This holds the promise of the ability to do certain types of computations, such as those involved in breaking codes in cryptography, much more rapidly than a classical computer could. As of this writing, the quest to build a fully quantum computer is still in its early stages, but intensive research is under way and rapid progress is being made.

TEST YOUR UNDERSTANDING OF SECTION 41.8 Particle 1 is an electron that can be in state C or D . Particle 2 is a proton that can be in state E or F . Is $\psi(\vec{r}_1, \vec{r}_2) = (1/\sqrt{2})[\psi_C(\vec{r}_1)\psi_E(\vec{r}_2) + \psi_C(\vec{r}_1)\psi_F(\vec{r}_2)]$ a possible wave function for this two-particle system? If so, does it represent an entangled state?

ANSWER

Yes; no This wave function says that it is equally possible that the electron (particle 1) is in state C and the proton (particle 2) is in state E or that particle 1 is in state C and particle 2 is in state F . Since particles 1 and 2 are not identical and are distinguishable, there is no reason we can't know that particle 1 is in state C independent of which state particle 2 is in. So this is a valid wave function for the system, and the two particles are not entangled. If we measure the state of the electron alone, we are guaranteed to get C as a result; a subsequent measurement of the proton's state will give either E or F with equal probability, the same as if we had not first measured the state of the electron. Similarly, if we first measure the state of the proton, the result of that measurement will not affect a subsequent measurement of the state of the electron (for which the result is guaranteed to be C).

CHAPTER 41 SUMMARY

Three-dimensional problems: The time-independent Schrödinger equation for three-dimensional problems is given by Eq. (41.5).

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x, y, z)}{\partial x^2} + \frac{\partial^2 \psi(x, y, z)}{\partial y^2} + \frac{\partial^2 \psi(x, y, z)}{\partial z^2} \right) + U(x, y, z) \psi(x, y, z) = E \psi(x, y, z) \quad (41.5)$$

(three-dimensional time-independent Schrödinger equation)

Particle in a three-dimensional box: The wave function for a particle in a cubical box is the product of a function of x only, a function of y only, and a function of z only. Each stationary state is described by three quantum numbers (n_x, n_y, n_z). Most of the energy levels given by Eq. (41.16) exhibit degeneracy: More than one quantum state has the same energy. (See Example 41.1.)

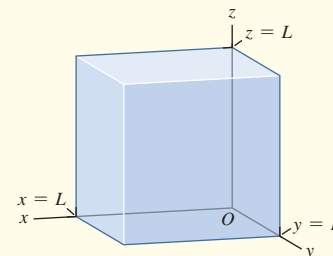
$$E_{n_x, n_y, n_z} = \frac{(n_x^2 + n_y^2 + n_z^2) \pi^2 \hbar^2}{2mL^2} \quad (41.16)$$

$$(n_x = 1, 2, 3, \dots;$$

$$n_y = 1, 2, 3, \dots;$$

$$n_z = 1, 2, 3, \dots)$$

(energy levels, particle in a three-dimensional cubical box)



The hydrogen atom: The Schrödinger equation for the hydrogen atom gives the same energy levels as the Bohr model. If the nucleus has charge Ze , there is an additional factor of Z^2 in the numerator of Eq. (41.21). The possible magnitudes L of orbital angular momentum are given by Eq. (41.22), and the possible values of the z -component of orbital angular momentum are given by Eq. (41.23). (See Examples 41.2 and 41.3.)

The probability that an atomic electron is between r and $r + dr$ from the nucleus is $P(r) dr$, given by Eq. (41.25). Atomic distances are often measured in units of a , the smallest distance between the electron and the nucleus in the Bohr model. (See Example 41.4.)

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2} \frac{m_e e^4}{2n^2 \hbar^2} = -\frac{13.60 \text{ eV}}{n^2} \quad (41.21)$$

(energy levels of hydrogen)

$$L = \sqrt{l(l+1)} \hbar \quad (41.22)$$

$$(l = 0, 1, 2, \dots, n-1)$$

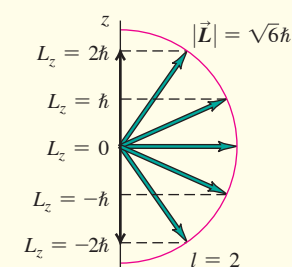
$$L_z = m_l \hbar \quad (41.23)$$

$$(m_l = 0, \pm 1, \pm 2, \dots, \pm l)$$

$$P(r) dr = |\psi|^2 dV = |\psi|^2 4\pi r^2 dr \quad (41.25)$$

$$a = \frac{\epsilon_0 \hbar^2}{\pi m_e e^2} = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} \quad (41.26)$$

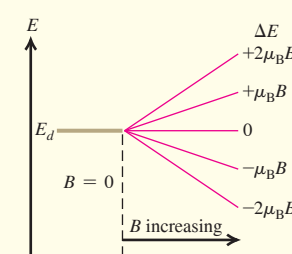
$$= 5.29 \times 10^{-11} \text{ m}$$



The Zeeman effect: The interaction energy of an electron (mass m) with magnetic quantum number m_l in a magnetic field $\vec{B}$ along the $+z$ -direction is given by Eq. (41.35), where $\mu_B = e\hbar/2m$ is called the Bohr magneton. (See Example 41.5.)

$$U = -\mu_z B = m_l \frac{e\hbar}{2m} B = m_l \mu_B B \quad (41.35)$$

$$(m_l = 0, \pm 1, \pm 2, \dots, \pm l)$$



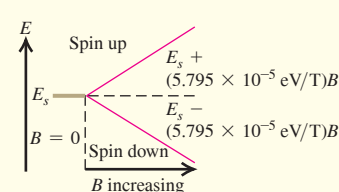
Electron spin: An electron has an intrinsic spin angular momentum of magnitude S , given by Eq. (41.37). The possible values of the z -component of the spin angular momentum are $S_z = m_s \hbar$, where $m_s = \pm \frac{1}{2}$. (See Examples 41.6 and 41.7.)

An orbiting electron experiences an interaction between its spin and the effective magnetic field produced by the relative motions of electron and nucleus. This spin-orbit coupling, along with relativistic effects, splits the energy levels according to their total angular momentum quantum number j . (See Example 41.8.)

$$S = \sqrt{\frac{1}{2}(\frac{1}{2} + 1)} \hbar = \sqrt{\frac{3}{4}} \hbar \quad (41.37)$$

$$S_z = m_s \hbar \quad (m_s = \pm \frac{1}{2}) \quad (41.36)$$

$$E_{n,j} = -\frac{13.60 \text{ eV}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] \quad (41.41)$$

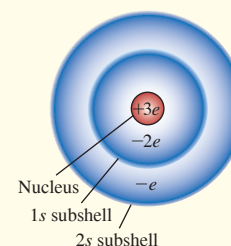


Many-electron atoms: In a hydrogen atom, the quantum numbers n , l , m_l , and m_s of the electron have certain allowed values given by Eq. (41.44). In a many-electron atom, the allowed quantum numbers for each electron are the same as in hydrogen, but the energy levels depend on both n and l because of screening, the partial cancellation of the field of the nucleus by the inner electrons. If the effective (screened) charge attracting an electron is $Z_{\text{eff}}e$, the energies of the levels are given approximately by Eq. (41.45). (See Examples 41.9 and 41.10.)

$$n \geq 1 \quad 0 \leq l \leq n - 1 \quad (41.44)$$

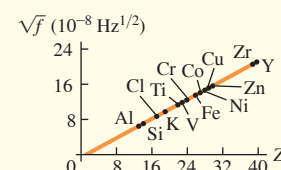
$$|m_l| \leq l \quad m_s = \pm \frac{1}{2}$$

$$E_n = -\frac{Z_{\text{eff}}^2}{n^2} (13.6 \text{ eV}) \quad (41.45)$$



X-ray spectra: Moseley's law states that the frequency of a K_α x ray from a target with atomic number Z is given by Eq. (41.47). Characteristic x-ray spectra result from transitions to a hole in an inner energy level of an atom. (See Example 41.11.)

$$f = (2.48 \times 10^{15} \text{ Hz})(Z - 1)^2 \quad (41.47)$$



Quantum entanglement: The wave function of two identical particles can be such that neither particle is itself in a definite state. For example, the wave function could be a combination of one term with particle 1 in state A and particle 2 in state B and one term with particle 1 in state B and particle 2 in state A . The two particles are said to be entangled, since measuring the state of one particle automatically determines the results of subsequent measurements of the other particle.

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}} \left\{ \begin{bmatrix} \uparrow & \downarrow \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \downarrow & \uparrow \\ 1 & 2 \end{bmatrix} \right\}$$

Chapter 41 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 41.1** (Section 41.2) before attempting these problems.

VP41.1.1 A particle in the three-dimensional box shown in Fig. 41.1 is in the state $n_x = 2$, $n_y = 1$, $n_z = 3$. Find (a) the planes (other than the walls of the box) on which the probability distribution function is zero and (b) the probability that the particle will be found somewhere in the region $0 \leq x \leq L/3$.

VP41.1.2 Half of the volume of the three-dimensional box shown in Fig. 41.1 is in the region $L/4 \leq x \leq 3L/4$. Find the probability that a particle in the box will be found in this region if the state of the particle is (a) $n_x = 1$, $n_y = 1$, $n_z = 1$; (b) $n_x = 2$, $n_y = 1$, $n_z = 2$; (c) $n_x = 3$, $n_y = 2$, $n_z = 3$; (d) $n_x = 4$, $n_y = 1$, $n_z = 1$.

VP41.1.3 The region $0 \leq x \leq L/4$, $0 \leq y \leq L/4$ makes up $\frac{1}{16}$ of the volume of the three-dimensional box shown in Fig. 41.1. Find the probability that a particle in the box will be found in this region if the state of the particle is (a) $n_x = 1$, $n_y = 1$, $n_z = 1$; (b) $n_x = 2$, $n_y = 1$, $n_z = 2$; (c) $n_x = 3$, $n_y = 2$, $n_z = 3$; (d) $n_x = 4$, $n_y = 1$, $n_z = 1$.

VP41.1.4 Consider the cubical region given by $L/4 \leq x \leq 3L/4$, $L/4 \leq y \leq 3L/4$, $L/4 \leq z \leq 3L/4$ at the center of the three-dimensional box shown in Fig. 41.1. (a) What fraction of the total volume of the box is inside this cubical region? (b) If a particle in the box is in the state $n_x = 1$, $n_y = 1$, $n_z = 1$, find the probability that it will be found somewhere in this cubical region at the center of the box.

Be sure to review **EXAMPLES 41.2, 41.3, and 41.4** (Section 41.3) before attempting these problems.

VP41.4.1 Consider the $n = 6$ states of the hydrogen atom. (a) How many distinct (l, m_l) states are there? (b) In terms of $\hbar$, what is the maximum magnitude of the orbital angular momentum L ? (c) In terms of $\hbar$, what is the maximum value of the z -component of orbital angular momentum?

VP41.4.2 (a) List all the possible combinations of values of l and m_l for the $n = 3$ states of the hydrogen atom. (b) For which of these states is the angle between the orbital angular momentum vector and the negative z -axis a minimum, and what is that angle?

VP41.4.3 A photon is emitted when a hydrogen atom transitions from one energy level to a lower energy level. Find the energy of this photon, in eV, for each transition: (a) $n = 3$, $l = 2$, $m_l = -2$ to $n = 2$, $l = 1$, $m_l = -1$; (b) $n = 4$, $l = 2$, $m_l = 1$ to $n = 2$, $l = 1$, $m_l = 0$; (c) $n = 2$, $l = 1$, $m_l = 1$ to $n = 1$, $l = 0$, $m_l = 0$.

VP41.4.4 The wave function for an electron in a $1s$ state in a hydrogen atom is $\psi_{1s}(r) = 1/\sqrt{\pi a^3} e^{-r/a}$, where r is the distance from the nucleus. Find the probability that the electron will be found in the region (a) $0 \leq r \leq 2a$; (b) $a \leq r \leq 3a$; (c) $r \geq 4a$.

Be sure to review **EXAMPLES 41.6, 41.7, and 41.8** (Section 41.5) before attempting these problems.

VP41.8.1 An isolated electron is placed in a magnetic field $\vec{B} = (3.14 \text{ T})\hat{k}$. (a) Find the difference in energy between the $S_z = +\frac{1}{2}\hbar$ and $S_z = -\frac{1}{2}\hbar$ states of the electron. (b) Which state has the higher energy?

VP41.8.2 The outermost electron in a potassium atom is in an $l = 0$ state. If you place a potassium atom in a magnetic field of magnitude 2.36 T and illuminate it with monochromatic electromagnetic radiation, what must be the frequency and wavelength of the radiation to cause a transition between the spin-up and spin-down states of the outermost electron?

VP41.8.3 When the outermost electron in a potassium atom makes a transition from a $4p$ level to a $4s$ level, the wavelength of the photon it emits can be either 766.490 nm or 769.896 nm depending on the initial spin orientation of the electron. Find (a) the energy difference between

the two $4p$ levels and (b) the effective magnetic field that the electron experiences in the $4p$ levels.

VP41.8.4 (a) Find the energy in terms of α of a state of the electron in a hydrogen atom with $n = 3$, $l = 1$, $j = \frac{3}{2}$ and the energy in terms of α of a state with $n = 3$, $l = 1$, $j = \frac{1}{2}$. (b) Find the difference between these energies, in eV. Which state has the higher energy? (c) Find the difference in wavelengths of the photons emitted in transitions from each of the states in part (a) to the state $n = 2$, $l = 0$, $j = \frac{1}{2}$. For which initial state is the wavelength longer?

BRIDGING PROBLEM A Many-Electron Atom in a Box

An atom of titanium (Ti) has 22 electrons and has a radius of 1.47×10^{-10} m. As a simple model of this atom, imagine putting 22 electrons into a cubical box that has the same volume as a titanium atom. (a) What is the length of each side of the box? (b) What will be the configuration of the 22 electrons? (c) Find the energies of each of the levels occupied by the electrons. (Ignore the electric forces that the electrons exert on each other.) (d) You remove one of the electrons from the lowest level. As a result, one of the electrons from the highest occupied level drops into the lowest level to fill the hole, emitting a photon in the process. What is the energy of this photon? How does this compare to the energy of the K_α photon for titanium as predicted by Moseley's law?

SOLUTION GUIDE

IDENTIFY and SET UP

1. In this problem you'll use ideas from Section 41.2 about a particle in a cubical box. You'll also apply the exclusion principle from Section 41.6 to find the electron configuration of this cubical "atom." The ideas about x-ray spectra from Section 41.7 are also important.
2. The target variables are (a) the dimensions of the box, (b) the electron configurations (like those given in Table 41.3 for real atoms), (c) the occupied energy levels of the cubical box, and (d) the energy of the emitted photon.

EXECUTE

3. Use your knowledge of geometry to find the length of each side of the box.
4. Each electron state is described by four quantum numbers: n_x , n_y , and n_z as described in Section 41.2 and the spin magnetic quantum number m_s described in Section 41.5. Use the exclusion principle to determine the quantum numbers of each of the 22 electrons in the "atom." (Hint: Figure 41.4 in Section 41.2 shows the first several energy levels of a cubical box relative to the ground level $E_{1,1,1}$.)
5. Use your results from steps 3 and 4 to find the energies of each of the occupied levels.
6. Use your result from step 5 to find the energy of the photon emitted when an electron makes a transition from the highest occupied level to the ground level. Compare this to the energy that we calculated for titanium by using Moseley's law.

EVALUATE

7. Is this cubical "atom" a useful model for titanium? Why or why not?
8. In this problem you ignored the electrical interactions between electrons. To estimate how large these are, find the electrostatic potential energy of two electrons separated by half the length of the box. How does this compare to the energy levels you calculated in step 5? Is it a good approximation to ignore these interactions?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q41.1 Particle A is described by the wave function $\psi(x, y, z)$. Particle B is described by the wave function $\psi(x, y, z)e^{i\phi}$, where ϕ is a real constant. How does the probability of finding particle A within a volume dV around a certain point in space compare with the probability of finding particle B within this same volume?

Q41.2 What are the most significant differences between the Bohr model of the hydrogen atom and the Schrödinger analysis? What are the similarities?

Q41.3 For an object orbiting the sun, such as a planet, comet, or asteroid, is there any restriction on the z -component of its orbital angular momentum such as there is with the z -component of the electron's orbital angular momentum in hydrogen? Explain.

Q41.4 Why is the analysis of the helium atom much more complex than that of the hydrogen atom, either in a Bohr type of model or using the Schrödinger equation?

Q41.5 The Stern–Gerlach experiment is always performed with beams of *neutral* atoms. Wouldn't it be easier to form beams using *ionized* atoms? Why won't this work?

Q41.6 (a) If two electrons in hydrogen atoms have the same principal quantum number, can they have different orbital angular momenta? How? (b) If two electrons in hydrogen atoms have the same orbital quantum number, can they have different principal quantum numbers? How?

Q41.7 In the Stern–Gerlach experiment, why is it essential for the magnetic field to be *inhomogeneous* (that is, nonuniform)?

Q41.8 In the ground state of the helium atom one electron must have "spin down" and the other "spin up." Why?

Q41.9 An electron in a hydrogen atom is in an s level, and the atom is in a magnetic field $\vec{B} = B\hat{k}$. Explain why the "spin up" state ($m_s = +\frac{1}{2}$) has a higher energy than the "spin down" state ($m_s = -\frac{1}{2}$).

Q41.10 The central-field approximation is more accurate for alkali metals than for transition metals such as iron, nickel, or copper. Why?